

ENHANCED DISTILLATION

In distillation operations, separation results from differences in vapor- and liquid-phase compositions arising from the partial vaporization of a liquid mixture or the partial condensation of a vapor mixture. The vapor phase becomes enriched in the more volatile components while the liquid phase is depleted of those same components. In many situations, however, the change in composition between the vapor and liquid phases in equilibrium becomes small (so-called pinched condition), and a large number of successive partial vaporizations and partial condensations are required to achieve the desired separation. Alternatively, the vapor and liquid phases may have identical compositions, because of the formation of an azeotrope, and no separation by simple distillation is possible.

Several enhanced distillation-based separation techniques have been developed for close-boiling or low-relative-volatility systems, and for systems exhibiting azeotropic behavior. All these special techniques are ultimately based on the same differences in the vapor and liquid compositions as ordinary distillation; but, in addition, they rely on some additional mechanism to further modify the vapor-liquid behavior of the key components. These enhanced techniques can be classified according to their effect on the relationship between the vapor and liquid compositions:

1. *Azeotropic distillation and pressure-swing distillation.* Methods that cause or exploit azeotrope formation or behavior to alter the boiling characteristics and separability of the mixture.
2. *Extractive distillation and salt distillation.* Methods that primarily modify liquid-phase behavior to alter the relative volatility of the components of the mixture.
3. *Reactive distillation.* Methods that use chemical reaction to modify the composition of the mixture or, alternatively, use existing vapor-liquid differences between reaction products and reactants to enhance the performance of a reaction.

AZEOTROPY

At low to moderate pressure ranges typical of most industrial applications, the fundamental composition relationship between the vapor and liquid phases in equilibrium can be expressed as a function of the total system pressure, the vapor pressure of each pure component, and the liquid-phase activity coefficient of each component i in the mixture:

$$y_i P = x_i \gamma_i P_i^{\text{sat}} \quad (13-114)$$

In systems that exhibit ideal liquid-phase behavior, the activity coefficients γ_i are equal to unity and Eq. (13-114) simplifies to Raoult's law. For nonideal liquid-phase behavior, a system is said to show negative deviations from Raoult's law if $\gamma_i < 1$ and, conversely, positive deviations from Raoult's law if $\gamma_i > 1$. In sufficiently nonideal systems, the deviations may be so large that the temperature-composition phase diagrams exhibit extrema, as shown in Figs. 13-6, 13-7, and 13-8. At such maxima or minima, the equilibrium vapor and liquid compositions are identical. Thus,

$$y_i = x_i \quad \text{for all } i = 1, \dots, c \quad (13-115)$$

and the system is said to form an azeotrope (from the Greek word meaning "to boil unchanged"). Azeotropic systems show a minimum in the T vs. x , y diagram when the deviations from Raoult's law are positive (Fig. 13-6) and a maximum in the T vs. x , y diagram when the deviations from Raoult's law are negative (Fig. 13-7). If, at these two conditions, a single liquid phase is in equilibrium with the vapor phase, the azeotrope is homogeneous. If multiple-liquid-phase behavior is exhibited at the azeotropic condition, the azeotrope is heterogeneous. For heterogeneous azeotropes, the vapor-phase composition is equal to the overall composition of the two (or more) liquid phases (Fig. 13-8). These conditions are consequences of the general definition of an azeotrope in any kind of mixture (i.e., homogeneous, heterogeneous, reactive, or in any combination), which is as follows: An *azeotropic state* is one in which mass transfer occurs between phases in a closed system while the composition of each phase remains constant, but not necessarily equal (see Prigogine and Defay, *Chemical Thermodynamics*, 4th ed., Longmans Green and Co., London, 1967;

Rowlinson, *Liquids and Liquid Mixtures*, 2d ed., Butterworths, London, 1969; Doherty and Malone, *Conceptual Design of Distillation Systems*, McGraw-Hill, 2001, chaps. 5, 8, App. C).

Mixtures with only small deviations from Raoult's law (i.e., ideal or nearly ideal mixtures) may form an azeotrope but only if the saturated vapor pressure curves of the two pure components cross each other (such a point is called a Bancroft point). In such a situation, the azeotrope occurs at the temperature and pressure where the curves cross, and perhaps also in the vicinity close to the Bancroft point [e.g., cyclohexane (n.b.p. 80.7°C) and benzene (n.b.p. 80.1°C) form an almost ideal mixture yet exhibit a minimum-boiling azeotrope with roughly equal proportions of each component]. As the boiling point difference between the components increases, the composition of the azeotrope shifts closer to one of the pure components (toward the lower-boiling pure component for minimum-boiling azeotropes, and toward the higher-boiling pure component for maximum-boiling azeotropes). For example, the minimum-boiling azeotrope between methanol (n.b.p. 64.5°C) and toluene (n.b.p. 110.6°C) occurs at ≈ 90 mol % methanol, and the minimum-boiling azeotrope between methyl acetate (n.b.p. 56.9°C) and water (n.b.p. 100°C) occurs at ≈ 90 mol % methyl acetate. Mixtures of components whose boiling points differ by more than about 50°C generally do not exhibit azeotropes distinguishable from the pure components even if large deviations from Raoult's law are present. As a qualitative guide to liquid-phase activity coefficient behavior, Robbins [*Chem. Eng. Prog.*, **76**(10), 58 (1980)] developed a matrix of chemical families, shown in Table 13-17, which indicates expected deviations from Raoult's law.

The formation of two liquid phases within some boiling temperature range is generally an indication that the system will also exhibit a minimum-boiling azeotrope, since two liquid phases may form when deviations from Raoult's law are large and positive. The fact that immiscibility does occur, however, does not guarantee that the azeotrope will be heterogeneous since the azeotropic composition may not necessarily fall within the composition range of the two-liquid phase region, as is the case for the methyl acetate–water and tetrahydrofuran–water systems. Since strong positive deviations from Raoult's law are required for liquid-liquid phase splitting, maximum-boiling azeotropes ($\gamma_i < 1$) are never heterogeneous.

Additional general information on the thermodynamics of phase equilibria and azeotropy is available in Malesinski (*Azeotropy and Other Theoretical Problems of Vapour-Liquid Equilibrium*, Interscience, London, 1965), Swietoslawski (*Azeotropy and Polyazeotropy*, Pergamon, London, 1963), Van Winkle (*Distillation*, McGraw-Hill, New York, 1967), Smith and Van Ness (*Introduction to Chemical Engineering Thermodynamics*, McGraw-Hill, New York, 1975), Wiziak [*Chem. Eng. Sci.*, **38**, 969 (1983)], and Walas (*Phase Equilibria in Chemical Engineering*, Butterworths, Boston, 1985). Horsley (*Azeotropic Data-III*, American Chemical Society, Washington, 1983) compiled an extensive list of experimental azeotropic boiling point and composition data for binary and some multicomponent mixtures. Another source for azeotropic data and activity coefficient model parameters is the multivolume *Vapor-Liquid Equilibrium Data Collection* (DECHEMA, Frankfurt, 1977), a compendium of published experimental VLE data. Most of the data have been tested for thermodynamic consistency and have been fitted to the Wilson, UNIQUAC, Van Laar, Margules, and NRTL equations. An extensive two-volume compilation of azeotropic data for 18,800 systems involving 1700 compounds, entitled *Azeotropic Data* by Gmehling et al., was published in 1994 by VCH Publishers, Deerfield Beach, Fla. A computational method for determining the temperatures and compositions of all homogeneous azeotropes of a multicomponent mixture, from liquid-phase activity coefficient correlations, by a differential arclength homotopy continuation method is given by Fidkowski, Malone, and Doherty [*Computers and Chem. Eng.*, **17**, 1141 (1993)]. The method was generalized to determine all homogeneous and heterogeneous azeotropes by Wasylkiewicz, Doherty, and Malone [*Ind. Eng. Chem. Res.*, **38**, 4901 (1999)].

RESIDUE CURVE MAPS AND DISTILLATION REGION DIAGRAMS

The simplest form of distillation involves boiling a multicomponent liquid mixture in an open evaporation from a single-stage batch still. As the liquid is boiled, the vapor generated is removed from contact with the liquid as soon as it is formed. Because the vapor is richer in the more volatile components than the liquid, the composition and boiling temperature of the liquid remaining in the still change continuously over time and move progressively toward less volatile compositions and higher temperatures until the last drop is vaporized. This last composition may be a pure-component species, or a maximum-boiling azeotrope, and it may depend on the initial composition of the mixture charged to the still.

The trajectory of liquid compositions starting from some initial composition is called a residue curve, the collection of all such curves for a given mixture is called a residue curve map. Arrows are usually added to these curves, pointing in the direction of increasing time, which corresponds to increasing temperature, and decreasing volatility. If the liquid is well mixed and the vaporization is slow, such that the escaping vapor is in phase equilibrium with the residual liquid, then residue curve maps contain exactly the same information as the corresponding phase equilibrium diagram for the mixture, but they represent it in a way that is much more useful for understanding distillation systems. Composition changes taking place in simple batch distillation can be described mathematically by the following ordinary differential equation

$$\frac{dx_i}{d\xi} = x_i - y_i \quad \text{for all } i = 1, \dots, c \quad (13-116)$$

where ξ is a dimensionless nonlinear time scale. Normally, y_i and x_i are related by an isobaric VLE model. Integrating these equations forward in time leads to the less volatile final compositions; integrating them backward in time leads to the more volatile compositions which would produce a residue curve passing through the specified initial composition. A *residue curve map* (RCM) is generated by varying the initial composition and integrating Eq. (13-116) both forward and backward in time [Doherty and Perkins, *Chem. Eng. Sci.*, **33**, 281

(1978); Doherty and Malone, op. cit., chap. 5]. Unlike a binary y - x plot, relative-volatility information is not represented on an RCM. Therefore, it is difficult to determine the ease of separation from a residue curve map alone. The steady states of Eq. (13-116) are the constant-composition trajectories corresponding to $dx_i/d\xi = 0$ for all $i = 1, \dots, c$. The steady states therefore correspond to *all* the pure components and *all* the azeotropes in the mixture.

Residue curve maps can be constructed for mixtures of any number of components, but can be pictured graphically only up to four components. For binary mixtures, a T vs. x , y diagram or a y - x diagram suffices; the system is simple enough that vapor-phase information can be included with liquid-phase information without confusion. For ternary mixtures, liquid-phase compositions are plotted on a triangular diagram, similar to that used in liquid-liquid extraction. Four-component systems can be plotted in a three-dimensional tetrahedron. The vertices of the triangular diagram or tetrahedron represent the pure components. Any binary, ternary, and quaternary azeotropes are placed at the appropriate compositions on the edges and/or interior of the triangle and tetrahedron.

The simplest form of ternary RCM, as exemplified for the ideal normal-paraffin system of pentane-hexane-heptane, is illustrated in Fig. 13-78a, using a right-triangle diagram. Maps for all other nonazeotropic ternary mixtures are qualitatively similar. Each of the infinite number of possible residue curves originates at the pentane vertex, travels toward and then away from the hexane vertex, and terminates at the heptane vertex. The family of all residue curves that originate at one composition and terminate at another composition defines a *distillation region*. Systems that do not involve azeotropes have only one region—the entire composition space. However, for many systems, not all residue curves originate or terminate at the same two compositions. Such systems will have more than one distillation region. The residue curve that divides two distillation regions in which adjacent residue curves originate from different compositions or terminate at different compositions is called a *simple batch distillation boundary* or *separatrix*. Distillation boundaries are related to the existence of azeotropes. In the composition space for a binary system, the distillation boundary is a point (the azeotropic composition). For three components, the distillation boundary is a curve; for four components, the boundary is a surface; and so on.

TABLE 13-17 Solute-Solvent Group Interactions

Solute class	Group	Solvent class											
		1	2	3	4	5	6	7	8	9	10	11	12
H-donor													
1	Phenol	0	0	—	0	—	—	—	—	—	—	—	—
2	Acid, thiol	0	0	—	0	—	—	0	0	0	—	—	—
3	Alcohol, water	—	—	0	+	+	0	—	—	—	—	—	—
4	Active-H on multihalo paraffin	0	0	+	0	—	—	—	—	—	—	0	—
H-acceptor													
5	Ketone, amide with no H on N, sulfone, phosphine oxide	—	—	+	—	0	+	—	—	—	+	—	—
6	Tertamine	—	—	0	—	+	0	—	—	0	+	0	0
7	Secamine	—	0	—	—	+	+	0	0	0	0	0	—
8	Priamine, ammonia, amide with 2H on N	—	0	—	—	+	+	0	0	—	+	—	—
9	Ether, oxide, sulfoxide	—	0	+	—	+	0	0	—	0	+	0	—
10	Ester, aldehyde, carbonate, phosphate, nitrate, nitrite, nitrile, intramolecular bonding, e.g., <i>o</i> -nitro phenol	—	0	+	—	+	+	0	—	—	0	—	—
11	Aromatic, olefin, halogen aromatic, multihalo paraffin without active H, monohalo paraffin	+	+	+	0	+	0	0	—	0	+	0	0
Non-H-bonding													
12	Paraffin, carbon disulfide	+	+	+	+	+	0	+	+	+	+	0	0

SOURCE: Robbins, L. A., *Chem. Eng. Prog.*, **76**(10), 58–61 (1980), by permission.

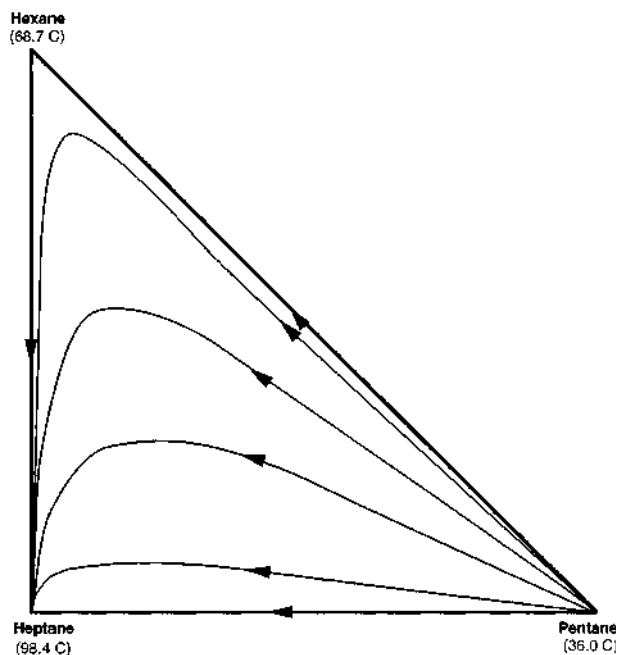


FIG. 13-78a Residue curve map: Nonazeotropic pentane-hexane-heptane system at 1 atm.

The boundaries of the composition diagram (e.g., the edges of a composition triangle) also form region boundaries since they divide physically realistic residue curves with positive compositions from unrealistic curves with negative compositions. All pure components and azeotropes in a system lie on region boundaries. Within each region, the most volatile composition on the boundary (either a pure

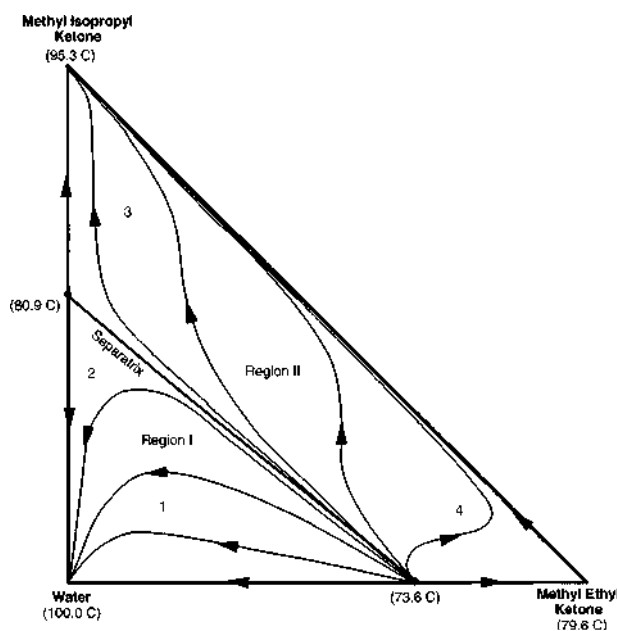


FIG. 13-78b Residue curve map: MEK-MIPK-water system at 1 atm containing two minimum-boiling binary azeotropes.

component or a minimum-boiling azeotrope, and the origin of all residue curves in that region) is called the *low-boiling node*. The least-volatile composition on the boundary (either a pure component or a maximum-boiling azeotrope, and the terminus of all residue curves in that region) is called the *high-boiling node*. All other pure components and azeotropes are called *intermediate-boiling saddles*. Adjacent regions may share some (but not all) nodes and saddles. Pure components and azeotropes are labeled as nodes and saddles as a result of the boiling points of all the components and azeotropes in a system. If one species is removed, the labeling of all remaining pure components and azeotropes, particularly those that were saddles, may change. Distillation boundaries always originate or terminate at saddle azeotropes, but never at pure component saddles—distillation boundaries can be calculated by using the method proposed by Lucia and Taylor [AIChE J., **52**, 582 (2006)]. Ternary saddle azeotropes are particularly interesting because they are more difficult to detect experimentally (being neither minimum-boiling nor maximum-boiling). However, their presence in a mixture implies the existence of distillation boundaries which may have an important impact on the design of a separation system. The first ternary saddle azeotrope to be detected experimentally was reported by Ewell and Welch [Ind. Eng. Chem., **37**, 1224 (1945)], and a particularly comprehensive set of experimental residue curves were reported by Bushmakina and Kish [J. Appl. Chem. USSR (Engl. Trans.), **30**, 205 (1957)] for a ternary mixture with a ternary saddle azeotrope (reproduced as fig. 5.9 in Doherty and Malone, op. cit.). More ternary saddle azeotropes are reported in Gmehling et al. (Azeotropic Data, 1994).

Both methylethylketone (MEK) and methylisopropylketone (MIPK) form minimum-boiling homogeneous azeotropes with water (Fig. 13-78b). In this ternary system, a distillation boundary connects the binary azeotropes and divides the RCM into two distillation regions, I and II. The high-boiling node of region I is pure water, while the low-boiling node is the MEK-water azeotrope. In region II, the high- and low-boiling nodes are MIPK and the MEK-water azeotrope, respectively. These two regions, however, have a different number of saddles—one in region I and two in region II. This leads to region I having three sides, while region II has four sides. The more complicated cyclohexane-ethanol-water system (Fig. 13-78c) has three boundaries and three regions, all of which are four-sided and share the ternary azeotrope as the low-boiling node.

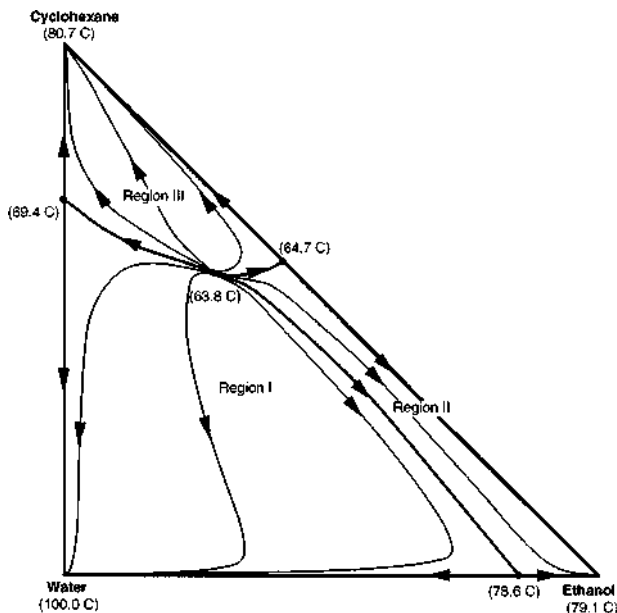


FIG. 13-78c Residue curve map: Ethanol-cyclohexane-water system at 1 atm containing four minimum-boiling azeotropes (three binary and one ternary) and three distillation regions.

The liquid composition profiles in continuous staged or packed distillation columns operating at infinite reflux and boil-up are closely approximated by simple distillation residue curves [Van Dongen and Doherty, *Ind. Eng. Chem. Fundam.*, **24**, 454 (1985)]. Residue curves are also indicative of many aspects of the general behavior of continuous columns operating at more practical reflux ratios. For example, to a first approximation, the stage-to-stage liquid compositions (along with the distillate and bottoms compositions) of a single-feed, two-product, continuous distillation column lie on the same residue curve. Therefore, for systems with distillation boundaries and multiple regions, distillation composition profiles are constrained to lie in specific regions. The precise boundaries of these distillation regions are a function of reflux ratio, but they are closely approximated by the RCM distillation boundaries. If a RCM distillation boundary exists in a system, a corresponding continuous distillation boundary will also exist. Both types of boundaries correspond exactly at all pure components and azeotropes.

Residue curves can be constructed from experimental data or can be calculated by integrating Eq. (13-116) if equation-of-state or activity-coefficient expressions are available (e.g., Wilson binary-interaction parameters, UNIFAC groups). However, considerable information on system behavior can still be deduced from a simple qualitative sketch of the RCM distillation boundaries based only on pure-component and azeotrope boiling point data and approximate azeotrope compositions. Rules for constructing such qualitative *distillation region diagrams* (DRDs) are given by Foucher et al. [*Ind. Eng. Chem. Res.*, **30**, 760, 2364 (1991)]. For ternary systems containing no more than one ternary azeotrope and no more than one binary azeotrope between each pair of components, 125 such DRDs are mathematically possible [Matsuyama and Nishimura, *J. Chem. Eng. Japan*, **10**, 181 (1977); Doherty and Calderola, *Ind. Eng. Chem. Fundam.*, **24**, 474 (1985); Peterson and Partin, *Ind. Eng. Chem. Res.*, **36**, 1799 (1997)], although only a dozen or so represent most systems commonly encountered in practice.

Figure 13-79 illustrates all the 125 possible DRDs for ternary systems [see Peterson and Partin, *Ind. Eng. Chem. Res.*, **36**, 1799 (1997)]. Azeotropes are schematically depicted generally to have equimolar composition, distillation boundaries are shown as straight lines, and the arrows on the distillation boundaries indicate increasing temperature. These DRDs are indexed in Table 13-18 according to a temperature profile sequence of position numbers, defined in a keyed triangular diagram at the bottom of the table, arranged by increasing boiling point. Positions 1, 3, and 5 are the pure components in order of decreasing volatility. Positions 2, 4, and 6 are binary azeotropes at the positions shown in the keyed triangle, and position 7 is the ternary azeotrope. Azeotrope position numbers are deleted from the temperature profile if the corresponding azeotrope is known not to exist. Note that not every conceivable temperature profile corresponds to a thermodynamically consistent system, and such combinations have been excluded from the index. As is evident from the index, some DRDs are consistent with more than one temperature profile. Also some temperature profiles are consistent with more than one DRD. In such cases, the correct diagram for a system must be determined from residue curves obtained from experimental or calculated data.

Schematic DRDs are particularly useful in determining the implications of possibly unknown ternary saddle azeotropes by postulating position 7 at interior positions in the temperature profile. Also note that some combinations of binary azeotropes require the existence of a ternary saddle azeotrope. As an example, consider the system acetone (56.4°C), chloroform (61.2°C), and methanol (64.7°C) at 1-atm pressure. Methanol forms minimum-boiling azeotropes with both acetone (54.6°C) and chloroform (53.5°C), and acetone-chloroform forms a maximum-boiling azeotrope (64.5°C). Experimentally there are no data for maximum- or minimum-boiling ternary azeotropes for this mixture. Assuming no ternary azeotrope, the temperature profile for this system is 461325, which from Table 13-18 is consistent with DRD 040 and DRD 042. However, Table 13-18 also indicates that the pure-component and binary azeotrope data are consistent with three temperature profiles involving a ternary saddle azeotrope, namely, 4671325, 4617325, and 4613725. All three of these temperature profiles correspond to DRD 107. Calculated residue curve trajectories for the acetone-chloroform-methanol system at 1-atm pressure, as

shown in Fig. 13-80, suggest the existence of a ternary saddle azeotrope and DRD 107 as the correct approximation of the distillation regions. Ewell and Welch [*Ind. Eng. Chem.*, **37**, 1224 (1945)] confirmed experimentally such a ternary saddle at 57.5°C.

APPLICATIONS OF RCM AND DRD

Residue curve maps and distillation region diagrams are very powerful tools for understanding all types of batch and continuous distillation operations, particularly when combined with other information such as liquid-liquid binodal curves. Applications include

1. *System visualization.* Location of distillation boundaries, azeotropes, distillation regions, feasible products, and liquid-liquid regions.
2. *Evaluation of laboratory data.* Location and confirmation of saddle ternary azeotropes and a thermodynamic consistency check of data.
3. *Process synthesis.* Concept development, construction of flow sheets for new processes, and redesign or modification of existing process flow sheets.
4. *Process modeling.* Identification of infeasible or problematic column specifications that could cause simulation convergence difficulties or failure, and determination of initial estimates of column parameters including feed-stage location, number of stages in the stripping and enriching sections, reflux ratio, and product compositions.
5. *Control analysis/design.* Analysis of column balances and profiles to aid in control system design and operation.

6. *Process trouble shooting.* Analysis of separation system operation and malfunction, examination of composition profiles, and tracking of trace impurities with implications for corrosion and process specifications. Material balances for mixing or continuous separation operations are represented graphically on triangular composition diagrams such as residue curve maps or distillation region diagrams by straight lines connecting pertinent compositions. The straight lines are exact representations of the compositions due to the lever rule. Overall flow rates are found by the inverse-lever-arm rule. Distillation material balance lines are governed by two constraints:

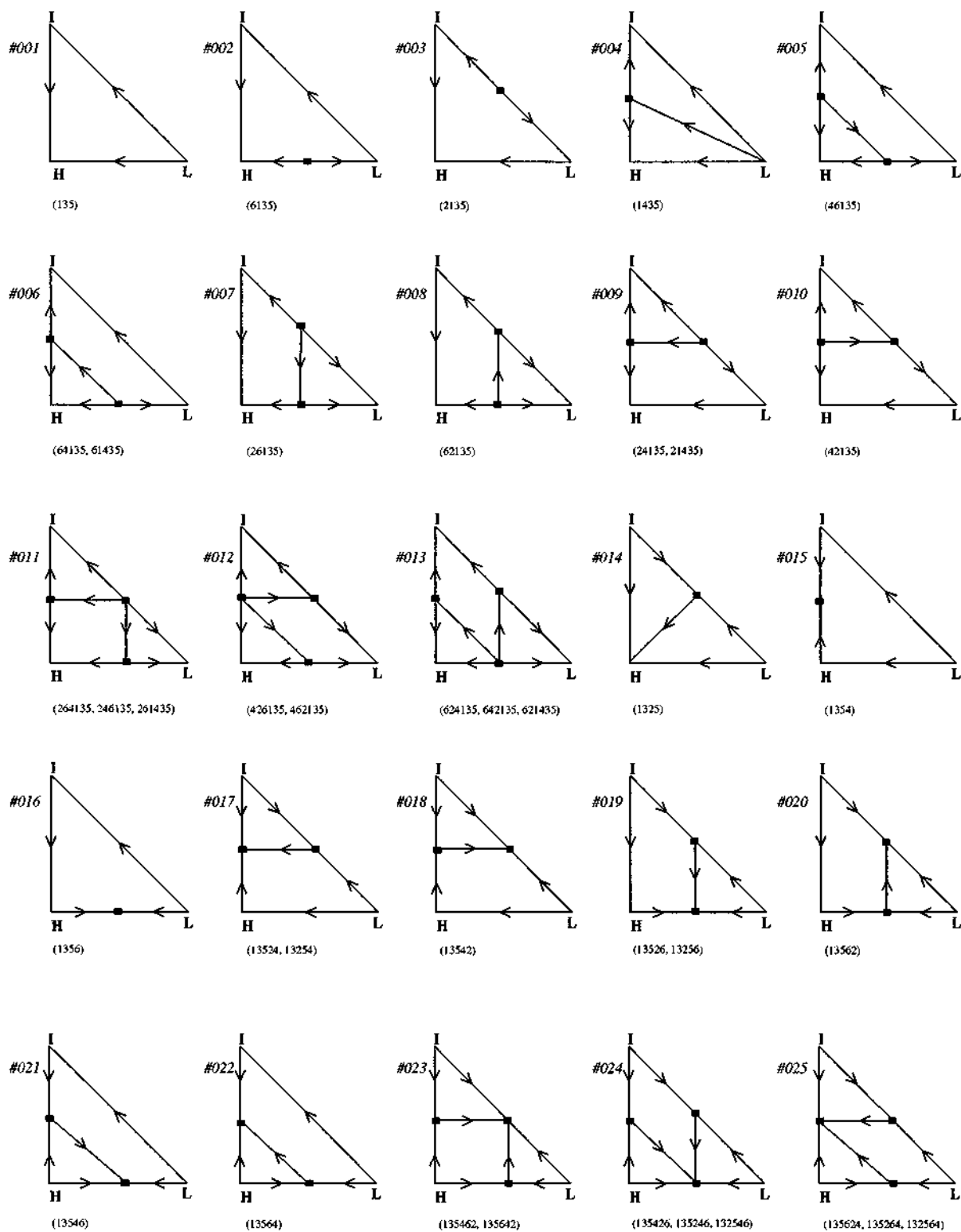
1. The bottoms, distillate, and overall feed compositions must lie on the same straight line.
2. The bottoms and distillate compositions must lie (to a very close approximation) on the same residue curve.

Since residue curves do not cross simple batch distillation boundaries, the distillate and bottoms compositions must be in the same distillation region with the mass balance line intersecting a residue curve in two places. Mass balance lines for mixing and for other separations not involving vapor-liquid equilibria, such as extraction and decantation, are of course not limited by distillation boundaries.

For a given multicomponent mixture, a single-feed, two-product distillation column (simple column) can be designed with sufficient stages, reflux, and material balance control to produce separations ranging from the *direct-split* mode of operation (low-boiling node taken as distillate) to the *indirect-split* mode (high-boiling node taken as bottoms). The bow tie shaped set of reachable product compositions for a simple distillation column is roughly bounded by the (straight) material balance lines connecting the feed composition to the sharpest direct separation and the sharpest indirect separation possible (see Fig. 13-81). A more accurate approximation involves replacing two of the straight-line segments of the bow tie with the residue curve through the feed composition [Stichlmair and Herguijuela, *AIChE J.*, **38**, 1523 (1992)]. The exact shape of the reachable product composition regions involves replacing two of the straight-line segments of the bow tie with a locus of pinch points, as explained by Wahnschafft et al. [*Ind. Eng. Chem. Res.*, **31**, 2345 (1992)] and Fidkowski, Doherty, and Malone [*AIChE J.*, **39**, 1303 (1993)]. Since residue curves are deflected by saddles, it is generally not possible to obtain a saddle product (pure component or azeotrope) from a simple distillation column.

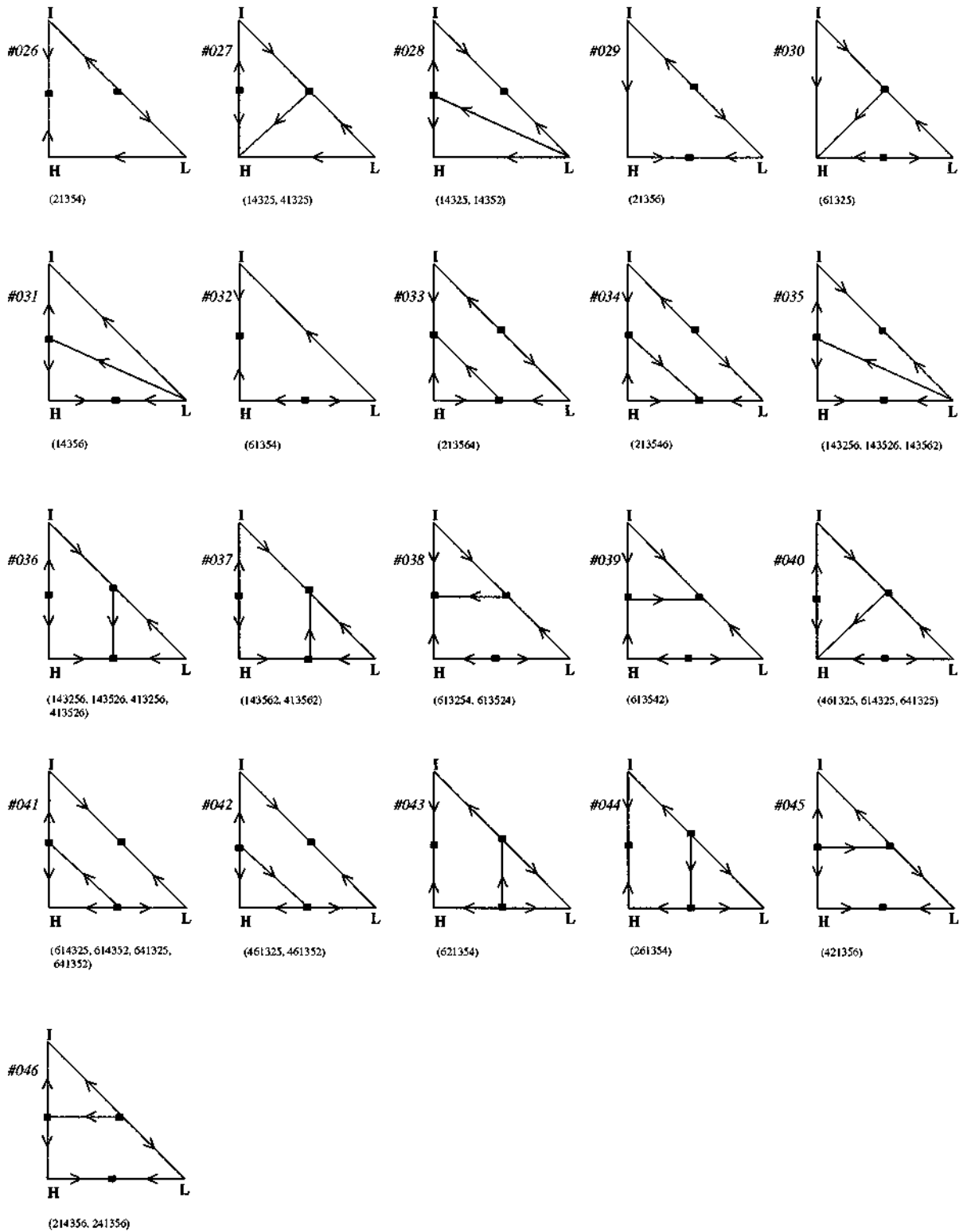
Consider the recovery of MIPK from an MEK-MIPK-water mixture. The approximate bow tie regions of product compositions for three different feeds are shown in Fig 13-81. From feed F3, which is situated in a different distillation region than the desired product, pure MIPK cannot be obtained at all. Feed F2 is more favorable, with the upper edge of the bow tie region along the MEK-MIPK (water-free) face of the composition triangle and part of the lower

13-72 DISTILLATION



(a)

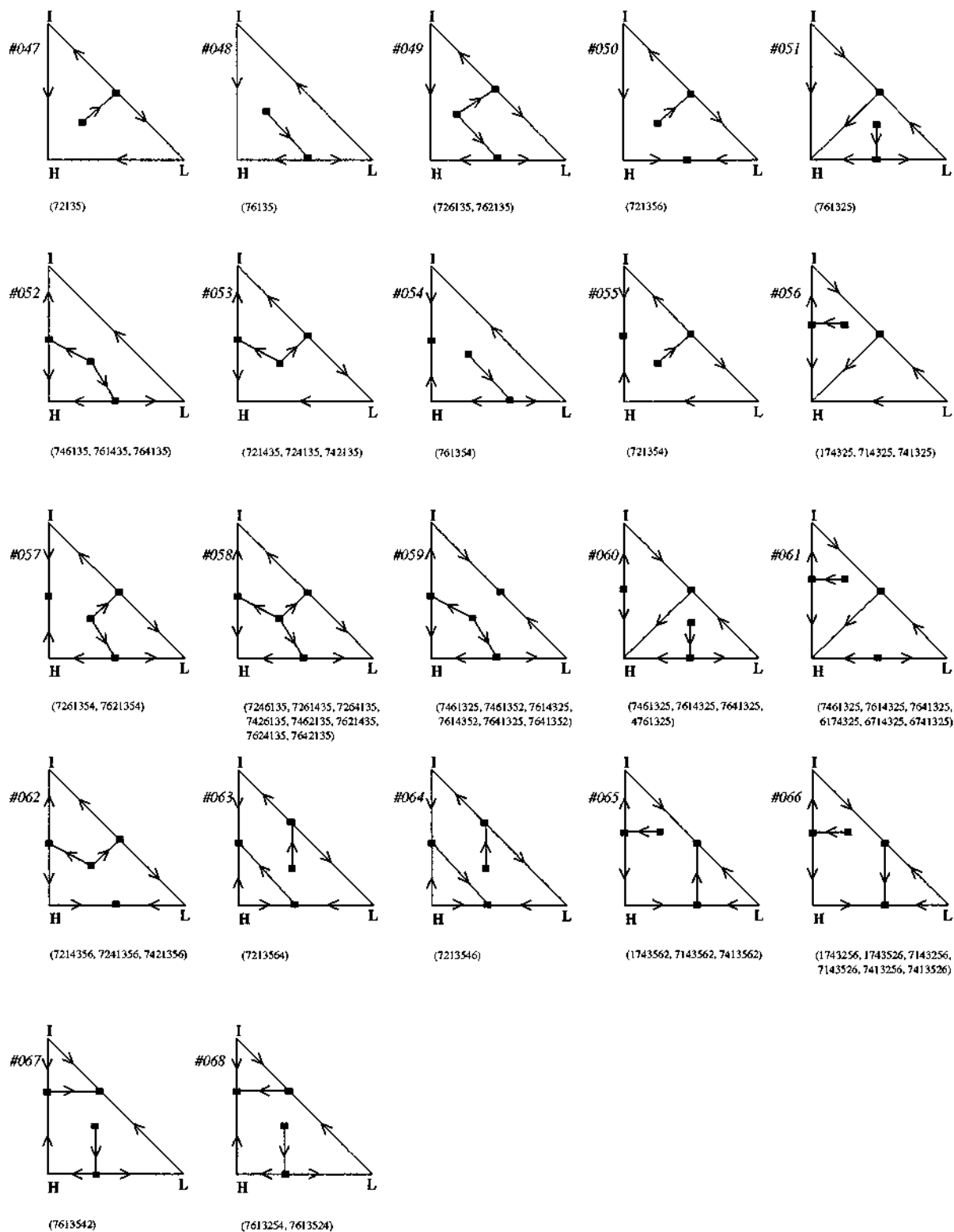
FIG. 13-79 Distillation region diagrams for ternary mixtures.



(b)

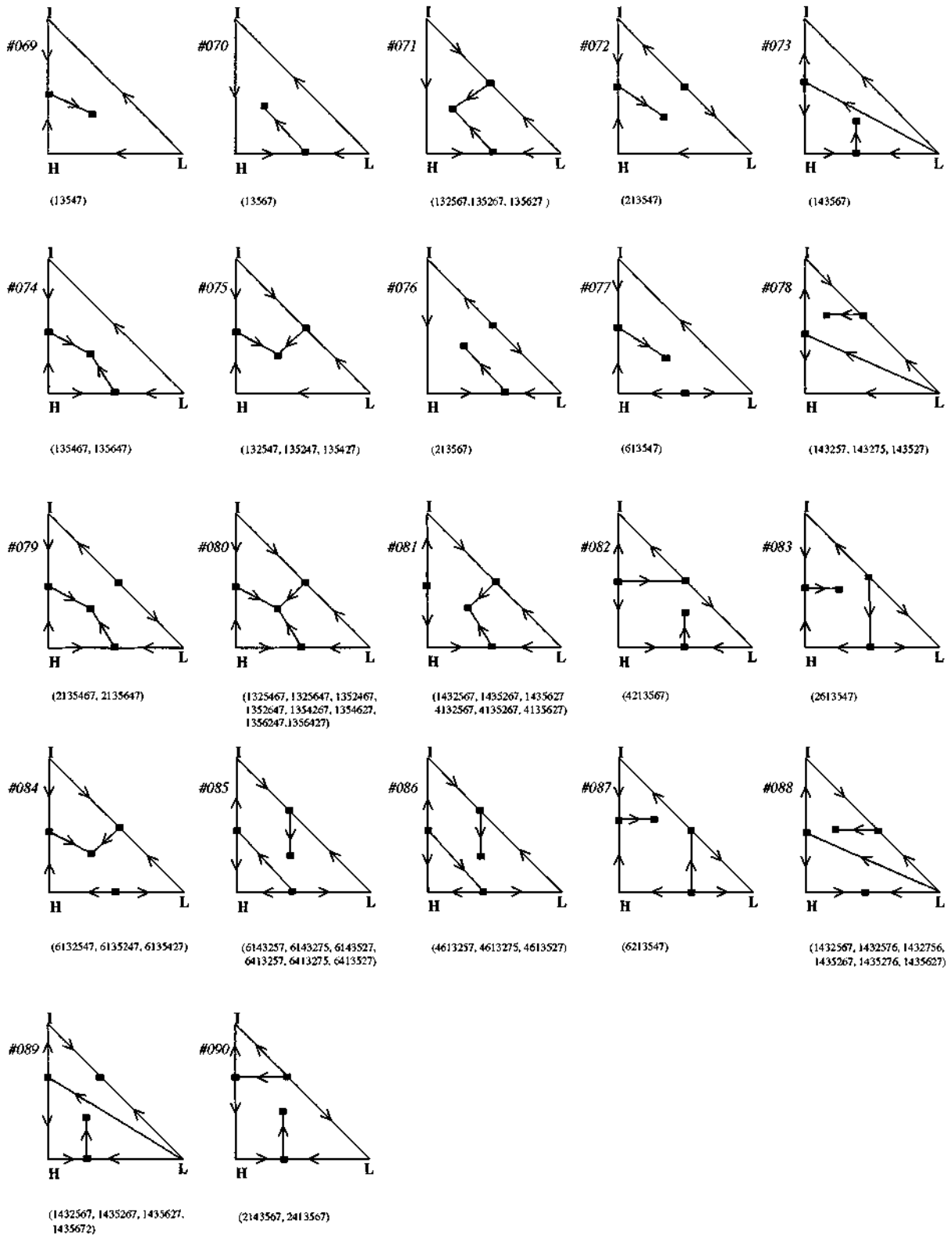
FIG. 13-79 (Continued)

13-74 DISTILLATION



(c)

FIG. 13-79 (Continued)



(d)

FIG. 13-79 (Continued)

13-76 DISTILLATION

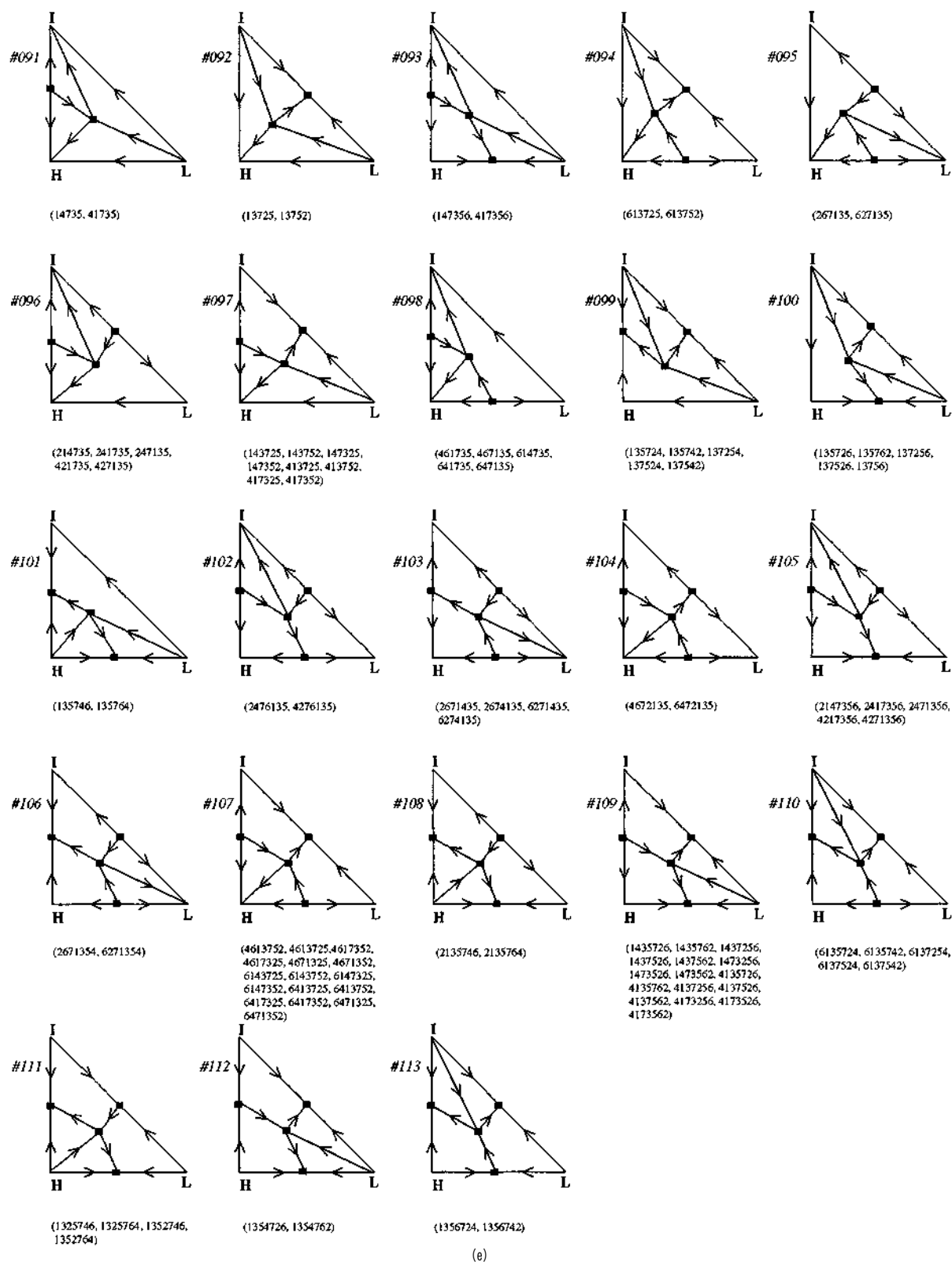
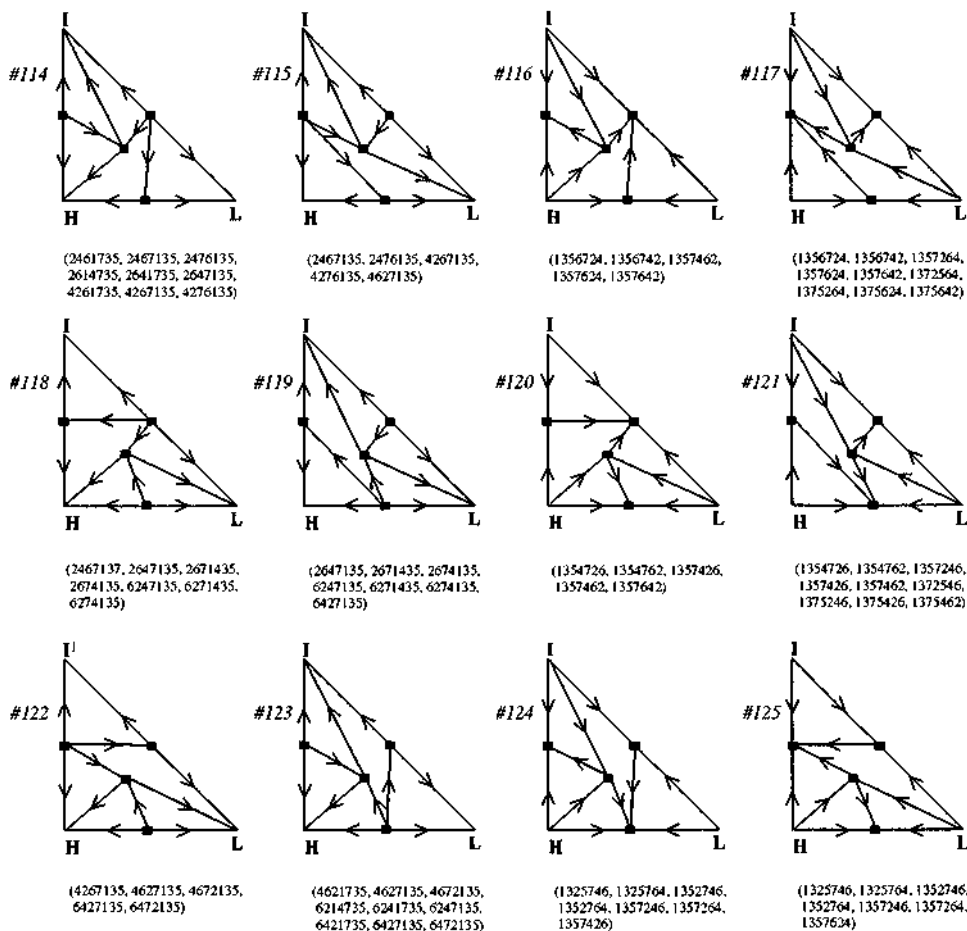


FIG. 13-79 (Continued)



(f)

FIG. 13-79 (Continued)

edge along the MEK-water (MIPK-free) face. There are conditions under which both the water in the MIPK bottoms product can be driven to low levels (high-product purity) and MIPK in the distillate can be driven to low levels (high-product recovery), although achieving such an operation depends on having an adequate number of stages and reflux ratio.

Although feed F2 lies in the same distillation region as F1, the bow tie region for feed F2 is significantly different than that for F1, with the upper edge along the water-MIPK (MEK-free) face of the triangle and the lower edge along the distillation boundary. From this feed it is not possible to simultaneously achieve a high-purity MIPK specification while obtaining high MIPK recovery. If the column is operated to get a high purity of MIPK, then the material balance line runs into the distillation boundary. Alternatively, if the column is operated to obtain a high recovery of MIPK (by removing the MEK-water azeotrope as distillate), the material balance requires the bottoms to lie on the water-MIPK face of the triangle.

The number of saddles in a particular distillation region can have significant impact on column profile behavior, process stability, and convergence behavior in process simulation of the system. Referring to the MIPK-MEK-water system in Fig. 13-78b, region I contains one saddle (MIPK-water azeotrope), while region II contains two saddles (pure MEK and the MIPK-water azeotrope). These are three- and four-sided regions, respectively. In a three-sided region, all residue

curves track toward the solitary saddle. However, in a four- (or more) sided region with saddles on either side of a node, some residue curves will tend to track toward one saddle, while others track toward another saddle. For example, residue curve 1 in region I originates from the MEK-water azeotrope low-boiling node and travels first toward the single saddle of the region (MIPK-water azeotrope) before ending at the water high-boiling node. Likewise, residue curve 2 and all other residue curves in region I follow the same general path.

In region II, residue curve 3 originates from the MEK-water azeotrope, travels toward the MIPK-water saddle azeotrope, and ends at pure MIPK. However, residue curve 4 follows a completely different path, traveling toward the pure MEK saddle before ending at pure MIPK. Some multicomponent columns have been designed for operation in four-sided regions with the feed composition adjusted so that both the high-boiling and low-boiling nodes can be obtained simultaneously as products. However, small perturbations in feed composition or reflux can result in feasible operation on many different residue curves that originate and terminate at these product compositions. Multiple steady states and composition profiles that shift dramatically from tracking toward one saddle to the other are possible [Kovach and Seider, *AIChE J.*, **33**, 1300 (1987); Pham, Ryan, and Doherty, *AIChE J.*, **35**, 1585 (1989)]. Consider a column operating in region II of the MIPK-MEK-water diagram. Figure 13-82 shows the composition and temperature profiles for the column operating at three different sets of

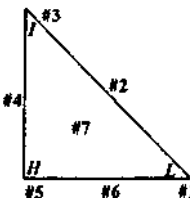
TABLE 13-18 Temperature Profile—DRD # Table*

Temp. Profile	DRD #	Temp. Profile	DRD #	Temp. Profile	DRD #	Temp. Profile	DRD #	Temp. Profile	DRD #	Temp. Profile	DRD #
135	001	137524	099	624135	013	1357462	120	2671435	103	6247135	118
1325	014	137526	100	627135	095		121		119		119
1354	015	137542	099	641325	041		116		118		123
1356	016	137562	100		040	1357624	117	2674135	118	6271354	106
1435	004	143256	035	641352	041		116		119	6271435	118
2135	003		036	641735	098		125		103		103
6135	002	143257	078	642135	013	1357642	120	4132567	081		119
13254	017	143275	078	647135	098		117	4135267	081	6274135	103
13256	019	143526	036	714325	056		116	4135627	081		118
13524	017		035	721354	055	1372546	121	4135726	109		119
13526	019	143527	078	721356	050	1372564	117	4135762	109	6413257	085
13542	018	143562	035	721435	053	1375246	121	4137256	109	6413275	085
13546	021		037	724135	053	1375264	117	4137526	109	6413527	085
13547	069	143567	073	726135	049	1375426	121	4137562	109	6413725	107
13562	020	143725	097	741325	056	1375462	121	4173256	109	6413752	107
13564	022	143752	097	742135	053	1375624	117	4173526	109	6417325	107
13567	070	147325	097	746135	052	1375642	117	4173562	109	6417352	107
13725	092	147352	097	761325	051	1432567	089	4213567	082	6421735	123
13752	092	147356	093	761354	054		088	4217356	105	6427135	122
14325	028	174325	056	761435	052		081	4261735	114		123
027		213546	034	762135	049	1432576	088	4267135	122		119
14352	028	213547	072	764135	052	1432756	088		115	6471325	107
14356	031	213564	033	1325467	080	1435267	081		114	6471352	107
14735	091	213567	076	1325647	080		089	4271356	105	6472135	104
21354	026	214356	046	1325746	111		088	4276135	114		122
21356	029	214735	096		124	1435276	088		102		123
21435	009	241356	046		125	1435627	089		115	6714325	061
24135	009	241735	096	1325764	111		081	4613257	086	6741325	061
26135	007	246135	011		124		088	4613275	086	7143256	066
41325	027	247135	096		125	1435672	089	4613527	086	7143526	066
41735	091	261354	044	1352467	080	1435726	109	4613725	107	7143562	065
42135	010	261435	011	1352647	080	1435762	109	4613752	107	7213546	064
46135	005	264135	011	1352746	125	1437256	109	4617325	107	7213564	063
61325	030	267135	095		111	1437526	109	4617352	107	7214356	062
61354	032	413256	036		124	1437562	109	4621735	123	7241356	062
61435	006	413526	036	1352764	124	1473256	109	4627135	123	7246135	058
62135	008	413562	037		111	1473526	109		115	7261354	057
64135	006	413725	097		125	1473562	109		122	7261435	058
72135	047	413752	097	1354267	080	1743256	066	4671325	107	7264135	058
76135	048	417325	097	1354627	080	1743526	066	4671352	107	7413256	066
132546	024	417352	097	1354726	120	1743562	065	4672135	104	7413526	066
132547	075	417356	093		121	2135467	079		122	7413562	065
132564	025	421356	045		112	2135647	079		123	7421356	062
132567	071	421735	096	1354762	121	2135746	108	4761325	060	7426135	058
135246	024	426135	012		120	2135764	108	6132547	084	7461325	060
135247	075	427135	096		112	2143567	090	6135247	084		059
135264	025	461325	040	1356247	080	2147356	105	6135427	084		061
135267	071		042	1356427	080	2413567	090	6135724	110	7461352	059
135426	024	461352	042	1356724	113	2417356	105	6135742	110	7462135	058
135427	075	461735	098		117	2461735	114	6137254	110	7613254	068
135462	023	462135	012		116	2467135	114	6137524	110	7613524	068
135467	074	467135	098	1356742	116		115	6137542	110	7613542	067
135624	025	613254	038		117		118	6143257	085	7614325	061
135627	071	613524	038		113	2471356	105	6143275	085		060
135642	023	613542	039	1357246	121	2476135	115	6143527	085		059
135647	074	613547	077		125		114	6143725	107	7614352	059
135724	099	613725	094		124		102	6143752	107	7621354	057
135726	100	613752	094	1357264	117	2613547	083	6147325	107	7621435	058
135742	099	614325	041		125	2614735	114	6147352	107	7624135	058
135746	101		040		124	2641735	114	6174325	061	7641325	061
135762	100	614735	098	1357426	121	2647135	118	6213547	087		060
135764	101	621354	043		120		119	6214735	123		059
137254	099	614352	041		124		114	6214735	123	7641352	059
137256	100	621435	013			2671354	106	6241735	123	7642135	058

Ternary DRD table lookup procedure:

- Classify a system by writing down each position number in ascending order of boiling points.
 - A position number is not written down if there is no azeotrope at that position.
 - The resulting sequence of numbers is known as the *temperature profile*.
 - Each temperature profile will have a minimum of three numbers and a maximum of seven numbers.
 - List multiple temperature profiles when you have incomplete azeotropic data.
 - All seven position numbers are shown on the diagram.
- Using the table, look up the temperature profile(s) to find the corresponding DRD #.

*Table 13-18 and Fig. 13-79 developed by Eric J. Peterson, Eastman Chemical Co.



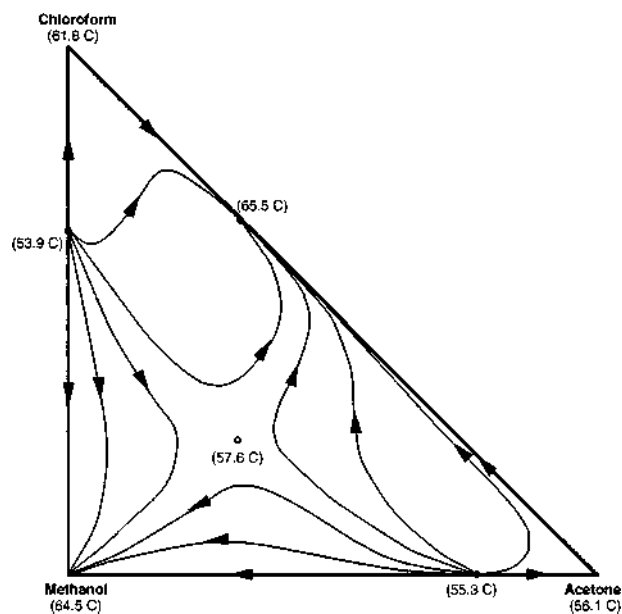


FIG. 13-80 Residue curves for acetone-chloroform-methanol system at 1-atm pressure suggesting a ternary saddle azeotrope.

operating conditions and two feed locations, as given in Table 13-19. The desired product specification is 97 mol % MIPK, no more than 3 mol % MEK, and less than 10 ppm residual water. For case A (Fig. 13-82a), the column profile tracks up the water-free side of the diagram. A pinched zone (i.e., section of little change in tray temperature and composition) occurs above the feed between the feed tray (tray 4) and tray 18. The temperature remains constant at about 93°C throughout the pinch zone. Product specifications are met.

When the feed composition becomes slightly enriched in water, as with case B, the column profile changes drastically (Fig. 13-82b). At

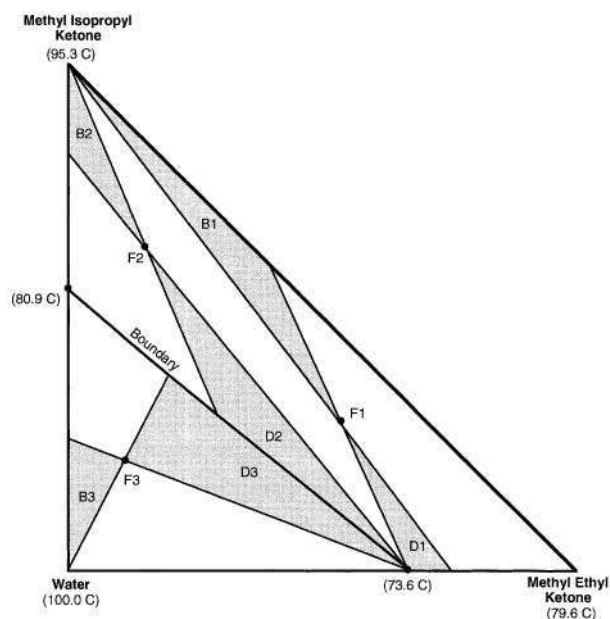


FIG. 13-81 MEK-MIPK-water system. Approximate product composition regions for a simple distillation column.

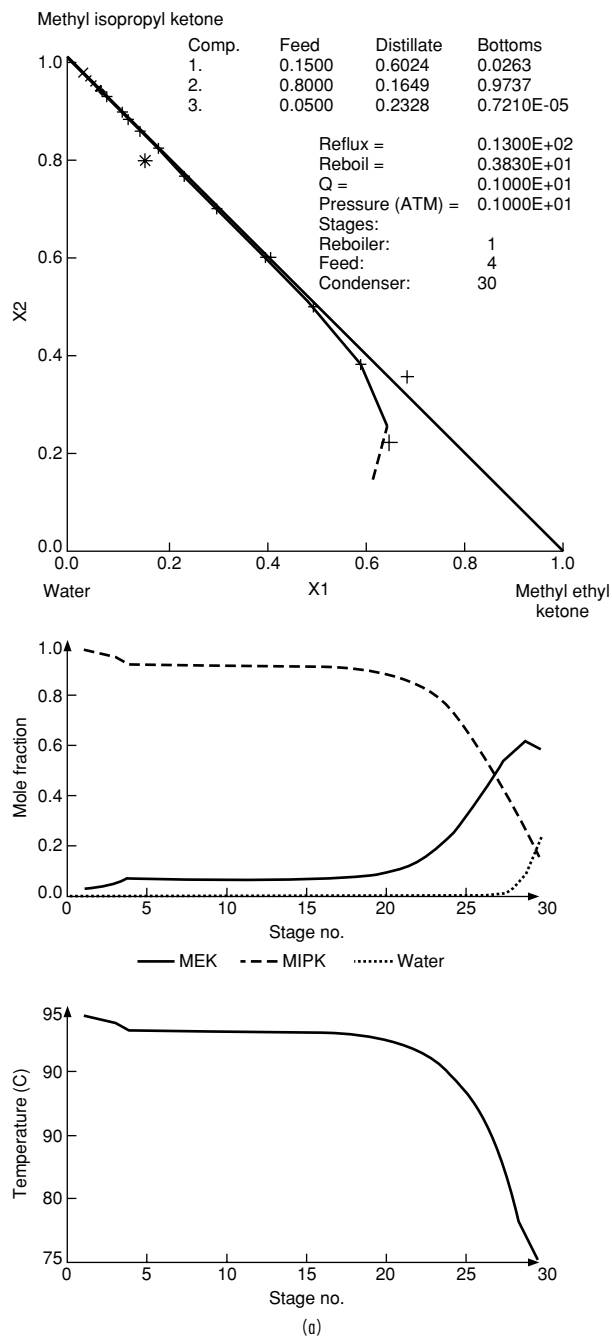


FIG. 13-82 Sensitivity of composition and temperature profiles for MEK-MIPK-water system at 1 atm.

the same reflux and boil-up, the column no longer meets specifications. The MIPK product is lean in MIPK and too rich in water. The profile now tracks generally up the left side of region II. Note also the dramatic change in the temperature profile. A pinched zone still exists above the feed between trays 4 and 18, but the tray temperature in the zone has dropped to 80°C (from 93°C). Most of the trays are required to move through the vicinity of the saddle. Typically, pinches (if they exist) occur close to saddles and nodes.

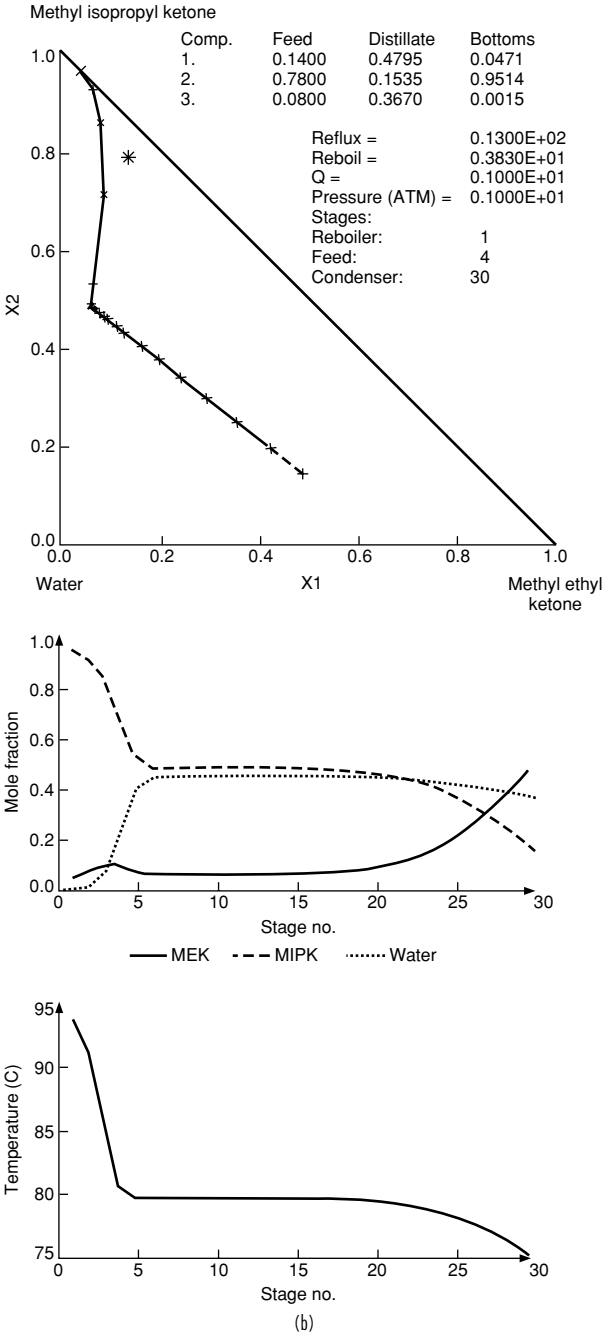


FIG. 13-82 (Continued)

In case C (Fig. 13-82c), increasing the boil-up ratio to 6 brings the MIPK product back within specifications, but the production rate and recovery have dropped off. In addition, the profile has switched back to the right side of the region; and the temperatures on trays in the pinched zone (trays 4 through 18) are back to 93°C. Such a drastic fluctuation in tray temperature with a relatively minor adjustment of the manipulated variable (boil-up in this case) can make control diffi-

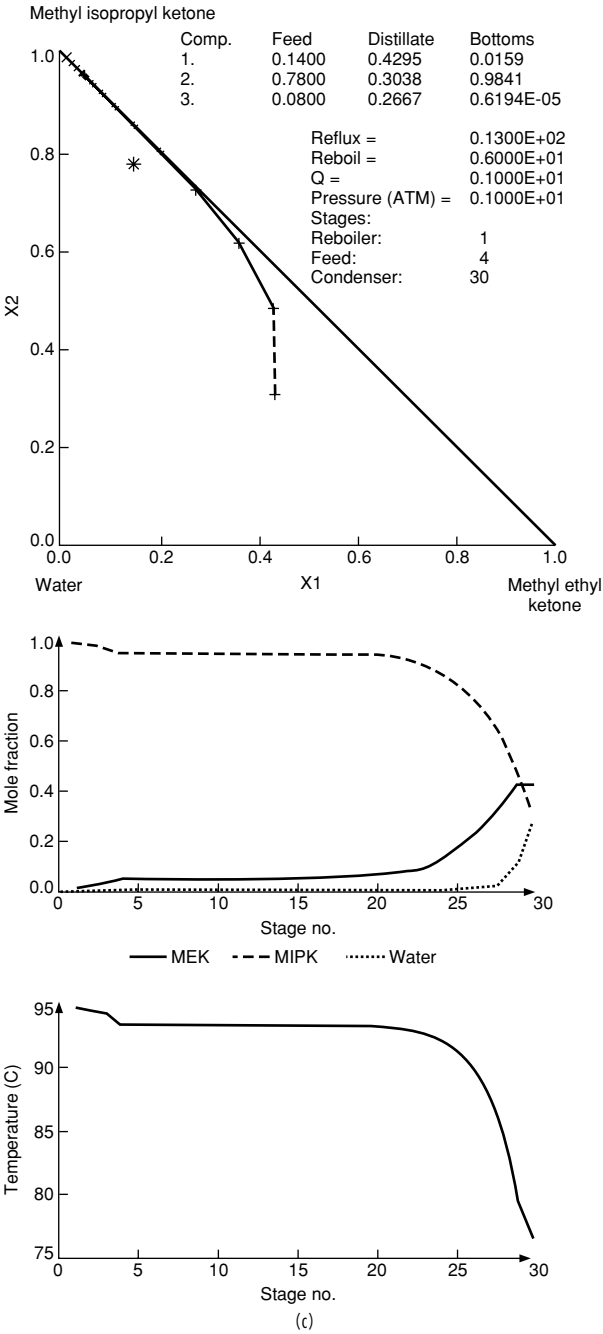


FIG. 13-82 (Continued)

cult. This is especially true if the control strategy involves maintaining a constant temperature on one of the trays between trays 4 and 18. If a tray is selected that exhibits wide temperature swings, the control system may have a difficult time compensating for disturbances. Such columns are also often difficult to model with a process simulator. Design algorithms often rely on perturbation of a variable (such as reflux or reboil) while checking for convergence of column heat and

TABLE 13-19 Sets of Operating Conditions for Fig. 13-82

Case	Reflux ratio	Reboil ratio		Feed composition	Distillate composition	Bottoms composition
A	13	3.8	MEK	0.15	0.60	0.03
			MIPK	0.80	0.16	0.97
			water	0.05	0.24	7 ppm
B	13	3.8	MEK	0.14	0.48	0.05
			MIPK	0.78	0.15	0.95
			water	0.08	0.37	20,000 ppm
C	13	6	MEK	0.14	0.43	0.02
			MIPK	0.78	0.30	0.98
			water	0.08	0.27	6.5 ppm

material balances. In situations where the column profile is altered drastically by minor changes in the perturbed variable, the simulator may be close to a feasible solution, but successive iterations may appear to be very far apart. The convergence routine may continue to oscillate between column profiles and never reach a solution. Likewise, when an attempt is made to design a column to obtain product compositions in different distillation regions, the simulation will never converge.

AZEOTROPIC DISTILLATION

The term *azeotropic distillation* has been applied to a broad class of fractional distillation-based separation techniques when specific azeotropic behavior is exploited to effect a separation. The agent that causes the specific azeotropic behavior, often called the *entrainer*, may already be present in the feed mixture (a self-entraining mixture) or may be an added mass separation agent. Azeotropic distillation techniques are used throughout the petrochemical and chemical processing industries for the separation of close-boiling, pinched, or azeotropic systems for which simple distillation is either too expensive or impossible. With an azeotropic feed mixture, presence of the azeotroping agent results in the formation of a more favorable azeotropic pattern for the desired separation. For a close-boiling or pinched feed mixture, the azeotroping agent changes the dimensionality of the system and allows separation to occur along a less pinched path. Within the general heading of azeotropic distillation techniques, several approaches have been followed in devising azeotropic distillation flow sheets including

1. Choosing an entrainer to give a residue curve map with specific distillation regions and node temperatures
2. Exploiting changes in azeotropic composition with total system pressure
3. Exploiting curvature of distillation region boundaries
4. Choosing an entrainer to cause azeotrope formation in combination with liquid-liquid immiscibility

The first three of these are solely VLE-based approaches, involving a series of simple distillation column operations and recycles. The final approach relies on distillation (VLE), but also exploits another physical phenomenon, liquid-liquid phase formation (phase splitting), to assist in entrainer recovery. This approach is the most powerful and versatile. Examples of industrial uses of azeotropic distillation grouped by method are given in Table 13-20.

The choice of the appropriate azeotropic distillation method and the resulting flow sheet for the separation of a particular mixture are strong functions of the separation objective. For example, it may be desirable to recover all constituents of the original feed mixture as pure components, or only some as pure components and others as azeotropic mixtures suitable for recycle. Not every objective may be obtainable by azeotropic distillation for a given mixture and portfolio of candidate entrainers.

Exploiting Homogeneous Azeotropes Homogeneous azeotropic distillation refers to a flow sheet structure in which azeotrope formation is exploited or avoided in order to accomplish the desired separation in one or more distillation columns. Either the azeotropes in the system

do not exhibit two-liquid-phase behavior, or the liquid-phase behavior is not or cannot be exploited in the separation sequence. The structure of a particular sequence will depend on the geometry of the residue curve map or distillation region diagram for the feed mixture-entrainer system. Two approaches are possible:

1. Selection of an entrainer such that the desired products all lie within the *same* distillation region (the products may be pure components or azeotropic mixtures)
2. Selection of an entrainer such that some type of distillation boundary-crossing mechanism is employed to separate desired products that lie in *different* regions.

As mentioned previously, ternary mixtures can be represented by 125 different residue curve maps or distillation region diagrams. However, feasible distillation sequences using the first approach can be developed for breaking homogeneous binary azeotropes by the addition of a third component only for those more restricted systems that do not have a distillation boundary connected to the azeotrope and for which one of the original components is a node. For example, from Fig. 13-79, the following eight residue curve maps are suitable for breaking homogeneous minimum-boiling azeotropes: DRD 002, 027, 030, 040, 051, 056, 060, and 061 as collected in Fig. 13-83. To produce the necessary distillation region diagrams, an entrainer must be found that is either: (1) an intermediate boiler that forms no azeotropes (DRD 002), or (2) lowest-boiling or intermediate-boiling and forms a maximum-boiling azeotrope with the lower-boiling original component (A). In these cases, the entrainer may also optionally form a minimum-boiling azeotrope with the higher boiling of the original components or a minimum-boiling ternary azeotrope. In all cases, after the addition of the entrainer, the higher-boiling original component (B) is a high-boiling node and is removed as bottoms product from a first column operated in the indirect-split mode with the lower-boiling original component recovered as distillate in a second column; see flow sheet in Fig. 13-83.

The seven residue curve maps suitable for breaking homogeneous maximum-boiling binary azeotropes (DRD 028, 031, 035, 073, 078, 088, 089) are shown in Fig. 13-84. In this case, the entrainer must form a minimum-boiling azeotrope with the higher-boiling original component and either a maximum-boiling azeotrope or no azeotrope with the lower-boiling original component. In all cases, after the addition of the entrainer, the lower-boiling original component is a low-boiling node and is removed as distillate from a first column operated in the direct-split mode with the higher-boiling original component recovered as bottoms product in a second column; see flow sheet in Fig. 13-84.

The restrictions on the boiling point and azeotrope formation of the entrainer act as efficient screening criteria for entrainer selection. Entrainers that do not show appropriate boiling point characteristics can be discarded without detailed analysis. However, the entrainers in Fig. 13-83 do suffer from serious drawbacks that limit their practical application. DRD 002 requires that the entrainer be an intermediate-boiling component that forms no azeotropes. Unfortunately these are often difficult criteria to meet, as any intermediate boiler will be closer-boiling to both of the original components and, therefore, will be more likely to be at least pinched or even form azeotropes. The remaining feasible distillation region diagrams require that the entrainer form a maximum-boiling azeotrope with the lower-boiling original component. Because maximum-boiling azeotropes are relatively rare, finding a suitable entrainer may be difficult.

For example, the dehydration of organics that form homogeneous azeotropes with water is a common industrial problem. It is extremely difficult to find an intermediate-boiling entrainer that also does not form an azeotrope with water. Furthermore, the resulting separation is likely to be close-boiling or pinched throughout most of the column, requiring a large number of stages. For example, consider the separation of valeric acid (187.0°C) and water. This system exhibits a minimum-boiling azeotrope (99.8°C) with a composition and boiling point close to those of pure water. Ignoring for the moment potentially severe corrosion problems, formic acid (100.7°C), which is an intermediate boiler and which forms a maximum-boiling azeotrope with water (107.1°C), is a candidate entrainer (DRD 030, Fig. 13-85a). In the conceptual sequence shown in Fig. 13-85b, a recycle of the formic acid-water maximum-boiling azeotrope is added to the

TABLE 13-20 Examples of Azeotropic Distillation

System	Type	Entrainer(s)	Remark
Exploitation of homogeneous azeotropes			
No known industrial examples			
Exploitation of pressure sensitivity			
THF-water Methyl acetate-methanol	Minimum-boiling azeotrope Minimum-boiling azeotrope	None None	Alternative to extractive distillation Element of recovery system for alternative to production of methyl acetate by reactive distillation; alternative to azeotropic, extractive distillation
Alcohol-ketone systems Ethanol-water	Minimum-boiling azeotropes Minimum-boiling azeotrope	None None	Alternative to extractive distillation, salt extractive distillation, heterogeneous azeotropic distillation; must reduce pressure to less than 11.5 kPa for azeotrope to disappear
Exploitation of boundary curvature			
Hydrochloric acid-water	Maximum-boiling azeotrope	Sulfuric acid	Alternative to salt extractive distillation
Nitric acid-water	Maximum-boiling azeotrope	Sulfuric acid	Alternative to salt extractive distillation
Exploitation of azeotropy and liquid phase immiscibility			
Ethanol-water	Minimum-boiling azeotrope	Cyclohexane, benzene, heptane, hexane, toluene, gasoline, diethyl ether	Alternative to extractive distillation, pressure-swing distillation
Acetic acid-water	Pinched system	Ethyl acetate, propyl acetate, diethyl ether, dichloroethane, butyl acetate	
Butanol-water Acetic acid-water-vinyl acetate Methyl acetate-methanol	Minimum-boiling azeotrope Pinched, azeotropic system Minimum-boiling azeotrope	Self-entraining Self-entraining Toluene, methyl isobutyl ketone	Element of recovery system for alternative to production of methyl acetate by reactive distillation; alternative to extractive pressure-swing distillation
Diethoxymethanol-water-ethanol Pyridine-water Hydrocarbon-water	Minimum-boiling azeotropes Minimum-boiling azeotrope Minimum-boiling azeotrope	Self-entraining Benzene Self-entraining	Alternative to extractive distillation

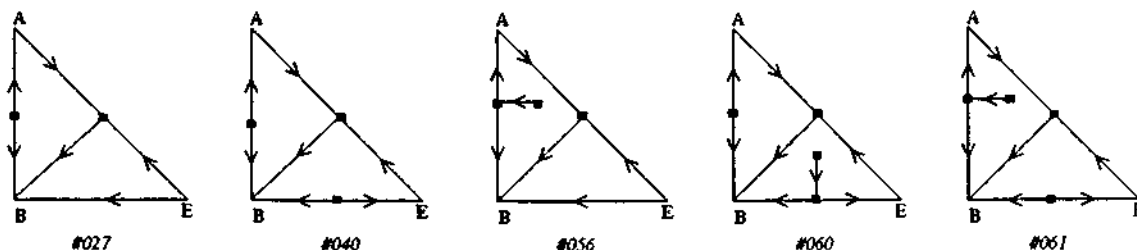
original valeric acid–water feed, which may be of any composition. Using the indirect-split mode of operation, the high-boiling node valeric acid is removed in high purity and high recovery as bottoms in a first column, which by mass balance produces a formic acid–water distillate. This binary mixture is fed to a second column that produces pure water as distillate and the formic acid–water azeotrope as bottoms for recycle to the first column. The inventory of formic acid is an important optimization variable in this theoretically feasible but difficult separation scheme.

Exploiting Pressure Sensitivity Breaking a homogeneous azeotrope that is part of a distillation boundary (i.e., the desired products lie in different distillation regions on either side of the boundary) requires that the boundary be “crossed” by the separation system. This may be done by mixing some external stream with the original feed stream in one region such that the resulting composition is in another region for further processing. However, the external stream must be completely regenerated, and mass balance must be preserved. For example, it is not possible to break a homogeneous binary azeotrope simply by adding one of the products to cross the azeotropic composition.

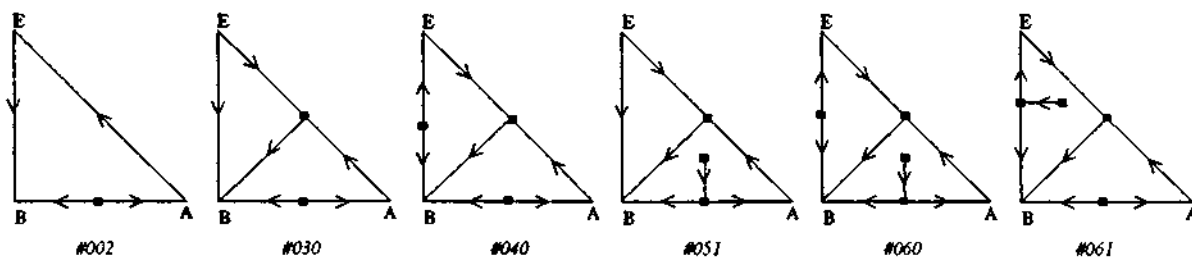
The composition of many azeotropes varies with the system pressure (Horsley, *Azeotropic Data-III*, American Chemical Society, Washington, 1983; Gmehling et al., *Azeotropic Data*, VCH Publishers, Deerfield Beach, Fla., 1994). This effect can be exploited to separate azeotropic mixtures by so-called pressure-swing distillation if at some pressure the azeotrope simply disappears, such as does the ethanol–water azeotrope at pressures below 11.5 kPa. However, pressure sensitivity can still be exploited if the azeotropic composition and related distillation boundary change sufficiently over a moderate

change in total system pressure. A composition in one distillation region under one set of conditions could be in a different region under a different set of conditions. A two-column sequence for separating a binary maximum-boiling azeotrope is shown in Fig. 13-86 for a system in which the azeotropic composition at pressure P1 is richer in component B than the azeotropic composition at pressure P2. The first column, operating at pressure P1, is fed a mixture of fresh feed plus recycle stream from the second column such that the overall composition lies on the A-rich side of the azeotropic composition at P1. Pure component A is recovered as distillate, and a mixture near the azeotropic composition is produced as bottoms. The pressure of this bottoms stream is changed to P2 and fed to the second column. This feed is on the B-rich side of the azeotropic composition at P2. Pure component B is now recovered as the distillate, and the azeotropic bottoms composition is recycled to the first column. An analogous flow sheet can be used for separating binary homogeneous minimum-boiling azeotropes. In this case the pure components are recovered as bottoms in both columns, and the distillate from each column is recycled to the other column.

For pressure-swing distillation to be practical, the azeotropic composition must vary at least 5 percent (preferably 10 percent or more) over a moderate pressure range (not more than 10 atm between the two pressures). With a very large pressure range, refrigeration may be required for condensation of the low-pressure distillate, or an impractically high reboiler temperature may result in the high-pressure column. The smaller the variation of azeotrope composition over the pressure range, the larger the recycle flow rates between the two columns. In particular, for minimum-boiling azeotropes, the pressure-swing distillation approach requires high energy usage and high capital



LOW-BOILING ENTRAINER.



INTERMEDIATE-BOILING ENTRAINER.

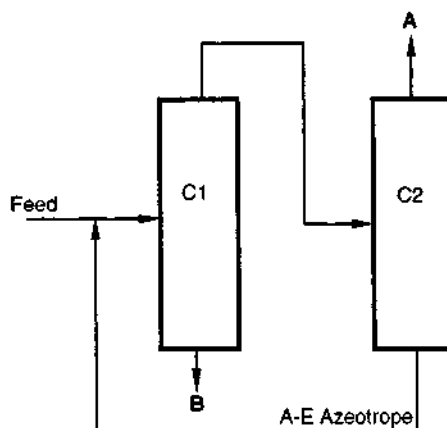


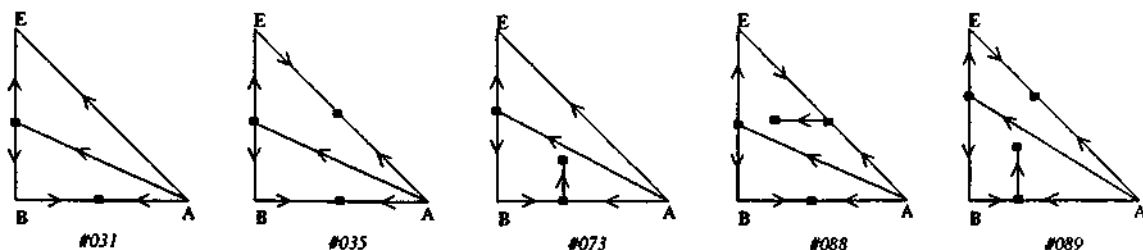
FIG. 13-83 Feasible distillation region diagrams and associated distillation system for breaking a homogeneous minimum-boiling binary azeotrope A-B. Component B boils at a higher temperature than does A.

costs (large-diameter columns) because both recycled azeotropic compositions must be taken overhead. Moreover, one lobe of an azeotropic VLE diagram is often pinched regardless of pressure; and, therefore, one of the columns will require a large number of stages to produce the corresponding pure-component product.

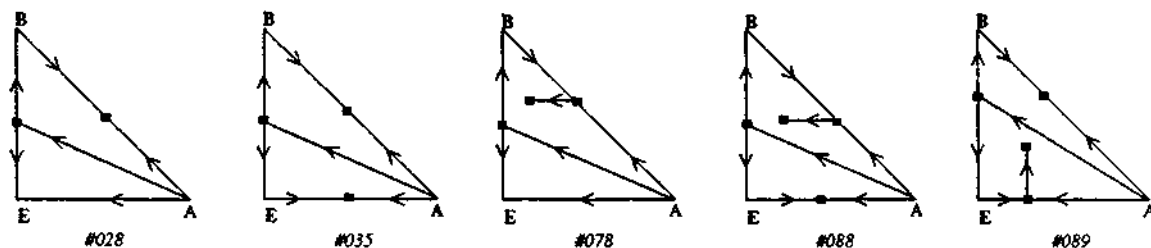
General information on pressure-swing distillation can be found in Van Winkle (*Distillation*, McGraw-Hill, New York, 1967), Wankat (*Equilibrium-Staged Separations*, Elsevier, New York, 1988), and Knapp and Doherty [*Ind. Eng. Chem. Res.*, **31**, 346 (1992)]. Only a relatively small fraction of azeotropes are sufficiently pressure-sensitive for a pressure-swing process to be economical. Some applications include the minimum-boiling azeotrope tetrahydrofuran-water [Tanabe et al.,

U.S. Patent 4,093,633 (1978)], and maximum-boiling azeotropes of hydrogen chloride-water and formic acid-water (Horsley, *Azeotropic Data-III*, American Chemical Society, Washington, 1983). Since distillation boundaries move with pressure-sensitive azeotropes, the pressure-swing principle can also be used for overcoming distillation boundaries in multicomponent azeotropic mixtures.

Exploiting Boundary Curvature A second approach to boundary crossing exploits boundary curvature to produce compositions in different distillation regions. When distillation boundaries exhibit extreme curvature, it may be possible to design a column such that the distillate and bottoms compositions are on the same residue curve in one distillation region, while the feed composition (which is not



INTERMEDIATE-BOILING ENTRAINER.



HIGH-BOILING ENTRAINER.

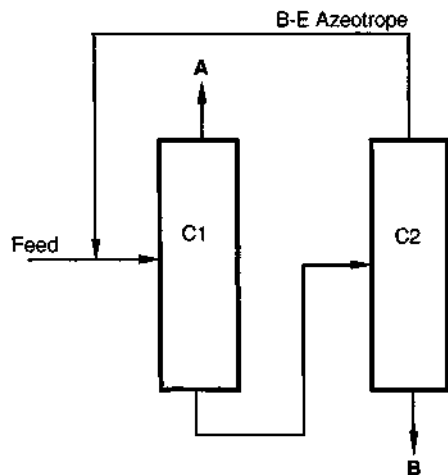
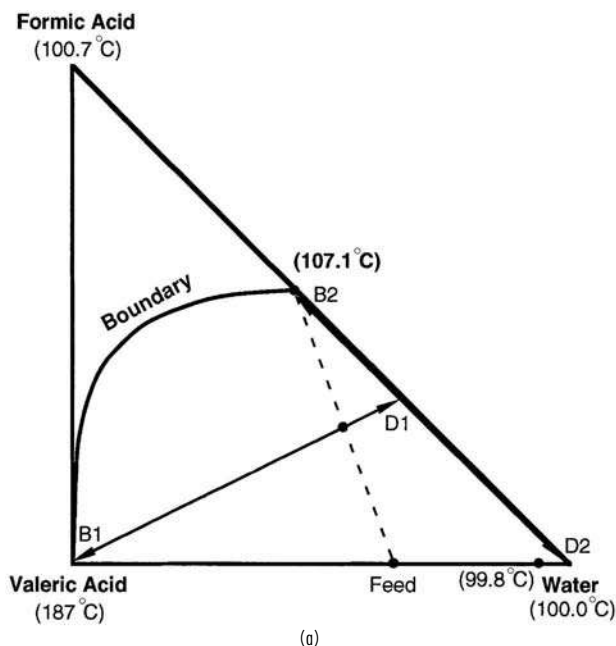


FIG. 13-84 Feasible distillation region diagrams and associated distillation system for breaking a homogeneous maximum-boiling binary azeotrope A-B. Component B boils at a higher temperature than does A.

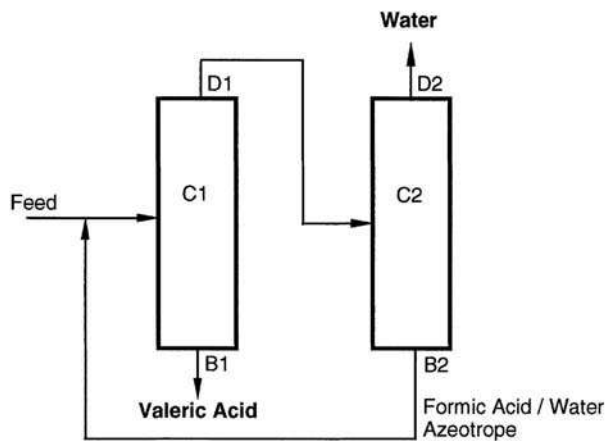
required to lie on the column composition profile) is in another distillation region. For such a column to meet material balance constraints (i.e., bottom, distillate, feed compositions on a straight line), the feed must be located in a region where the boundary is concave.

As an example, Van Dongen (Ph.D. Thesis, University of Massachusetts, 1983) considered the separation of a methanol-methyl acetate mixture, which forms a homogeneous azeotrope, using *n*-hexane as an entrainer. The distillation boundaries for this system (Fig. 13-87a) are somewhat curved. A separation sequence that exploits this boundary curvature is shown in Fig. 13-87b. Recycled methanol-methyl acetate binary azeotrope and methanol-methyl acetate-hexane ternary azeotrope are added to the original feed F0 to produce a net feed com-

position F1 for column C1 designed to lie on a line between pure methanol and the curved part of the boundary between regions I and II. Column C1 is operated in the indirect-split mode, producing the high-boiling node methanol as a bottoms product, and by mass balance, a distillate near the curved boundary. The distillate, although in region I, becomes feed F2 to column C2 which is operated in the direct-split mode entirely in region II, producing the low-boiling node ternary azeotrope as distillate and by mass balance, a methanol-methyl acetate mixture as bottoms (B2). This bottoms mixture is on the opposite side of the methanol-methyl acetate azeotrope than the original feed F0. The bottoms from C2 is finally fed to binary distillation column C3, which produces pure methyl acetate as bottoms product (B3)



(a)



(b)

FIG. 13-85 Valeric acid-water separation with formic acid. (a) Mass balances on distillation region diagram. (b) Conceptual sequence.

and the methanol-methyl acetate azeotrope as distillate (D3). The distillates from columns C2 and C3 are recycled to column C1. The distillate and bottoms compositions for column C2 lie on the same residue curve, and the composition profile lies entirely within region II, even though its feed composition is in region I. Additional information on exploiting boundary curvature, including the useful concept of a *pitchfork distillation boundary*, can be found in Doherty and Malone (*Conceptual Design of Distillation Systems*, McGraw-Hill, 2001, sec. 5.4).

Exploiting boundary curvature for breaking azeotropes is very similar to exploiting pressure sensitivity from a mass balance point of view, and suffers from the same disadvantages. Separation schemes have large recycle flows, and in the case of minimum-boiling azeotropes, the recycle streams are distillates. However, in the case of maximum-boiling azeotropes, these recycles are bottoms products, and the economics are improved. One such application, illustrated in Fig. 13-88, is the separation of the maximum-boiling nitric acid-water azeotrope by adding sulfuric acid. Recycled sulfuric acid is added to a

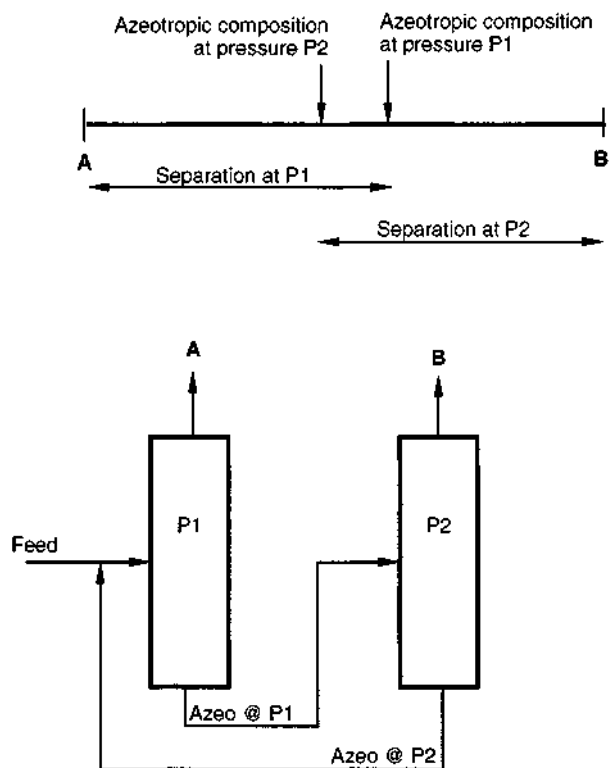


FIG. 13-86 Conceptual sequence for separating maximum-boiling binary azeotrope with pressure-swing distillation.

nitric acid-water mixture near the azeotropic composition to produce a net feed F1 in region II. The first column, operated in the direct-split mode, produces a nitric acid distillate and a bottoms product, by mass balance, near the distillation boundary. In this case, sulfuric acid associates with water so strongly and the distillation boundary is so curved and nearly tangent to the water-sulfuric acid edge of the composition diagram that the second column operating in the indirect-split mode in region I, producing sulfuric acid as bottoms product, also produces a distillate close enough to the water specification that a third column is not required [Thiemann et al., in *Ullmann's Encyclopedia of Industrial Chemistry*, 5th ed., vol. A17, VCH Verlagsgesellschaft mbH, Weinheim, (1991)].

Exploiting Azeotropy and Liquid-Phase Immiscibility One powerful and versatile separation approach exploits several physical phenomena simultaneously including nonideal vapor-liquid behavior, where possible, and liquid-liquid behavior to bypass difficult distillation separations. For example, the overall separation of close-boiling mixtures can be made easier by the addition of an entrainer that introduces liquid-liquid immiscibility and forms a heterogeneous minimum-boiling azeotrope with one (generally the lower-boiling) of the key components. Two-liquid-phase formation provides a means of breaking this azeotrope, thus simplifying the entrainer recovery and recycle process. Moreover, since liquid-liquid tie lines are unaffected by distillation boundaries (and the separate liquid phases are often located in different distillation regions), liquid-liquid phase splitting is a powerful mechanism for crossing distillation boundaries. The phase separator is usually a simple decanter, but sometimes a multistage extractor is substituted. The decanter or extractor can also be replaced by some other non-VLE-based separation technique such as membrane permeation, chromatography, adsorption, or crystallization. Also sequences may include additional separation operations (distillations or other methods) for preconcentration of the feed mixture, entrainer recovery, and final-product purification.

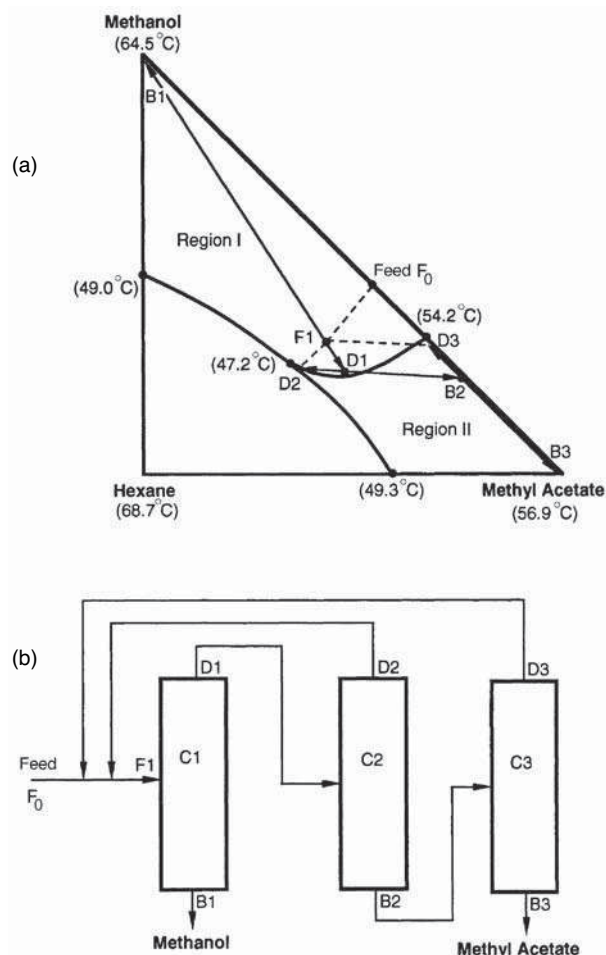


FIG. 13-87 Separation of methanol-methyl acetate by exploitation of distillation boundary curvature.

The simplest case of combining VLE and LLE is the separation of a binary heterogeneous azeotropic mixture. One example is the dehydration of 1-butanol, a self-entraining system, in which butanol (117.7°C) and water form a minimum-boiling heterogeneous azeotrope (93.0°C). As shown in Fig. 13-89, the fresh feed may be added to either column C1 or C2, depending on whether the feed is on the organic-rich side or the water-rich side of the azeotrope. The feed may also be added into the decanter directly if it does not move the overall composition of the decanter outside of the two-liquid phase region. Column C1 produces anhydrous butanol as a bottoms product and a composition close to the butanol-water azeotrope as the distillate. After condensation, the azeotrope rapidly phase-separates in the decanter. The upper layer, consisting of 78 wt % butanol, is refluxed totally to column C1 for further butanol recovery. The water layer, consisting of 92 wt % water, is fed to column C2. This column produces pure water as a bottoms product and, again, a composition close to the azeotrope as distillate for recycle to the decanter. Sparged steam may be used in C2, saving the cost of a reboiler. A similar flow sheet can be used for dehydration of hydrocarbons and other species that are largely immiscible with water.

A second example of the use of liquid-liquid immiscibilities in an azeotropic distillation sequence is the separation of the ethanol-water minimum-boiling homogeneous azeotrope. For this separation, a number of entrainers have been proposed, which are usually chosen to be immiscible with water and form a ternary minimum-boiling

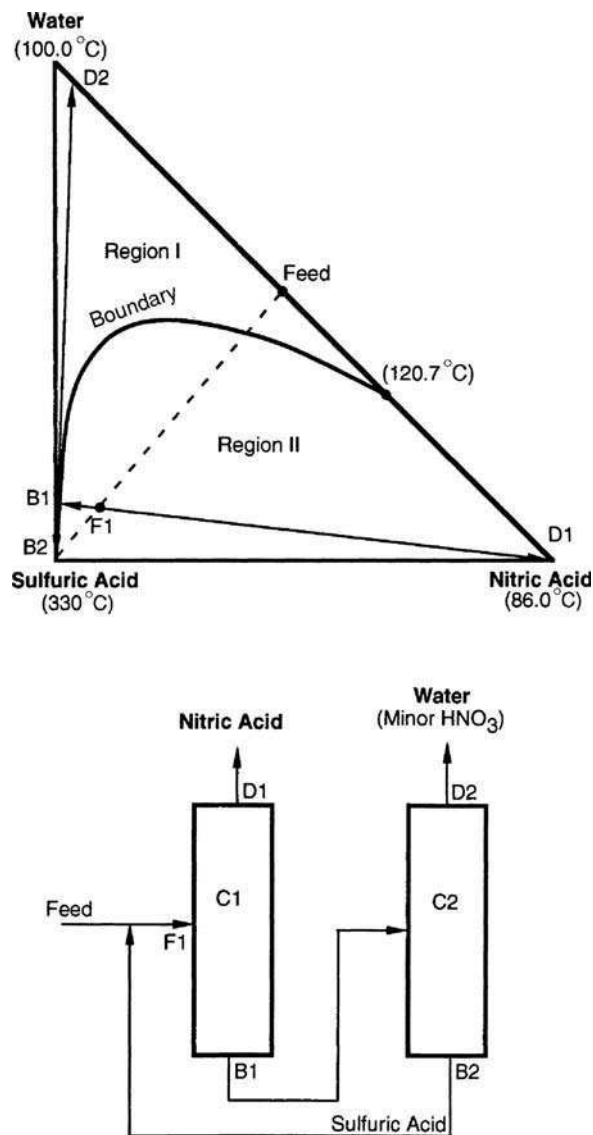


FIG. 13-88 Separation of nitric acid-water system with sulfuric acid in a two-column sequence exploiting extreme boundary curvature.

(preferably heterogeneous) azeotrope with ethanol and water (and, therefore, usually also binary minimum-boiling azeotropes with both ethanol and water). All such systems correspond to DRD 058, although the labeling of the vertices depends on whether the entrainer is lower-boiling than ethanol, intermediate-boiling, or higher-boiling than water. The residue curve map for the case of cyclohexane as entrainer was illustrated in Fig. 13-78c. One three-column distillation sequence is shown in Fig. 13-90. Other two-, three-, or four-column sequences have been described by Knapp and Doherty (*Kirk-Othmer Encyclopedia of Chemical Technology*, 5th ed., vol. 8, p. 786, Wiley, New York, 2004).

Fresh aqueous ethanol feed is first preconcentrated to nearly the azeotropic composition in column C3, while producing a water bottoms product. The distillate from C3 is sent to column C1, which is refluxed with the entire organic (entrainer-rich) layer, recycled from a decanter. Mixing of these two streams is the key to this sequence as it allows the overall feed composition to cross the distillation boundary

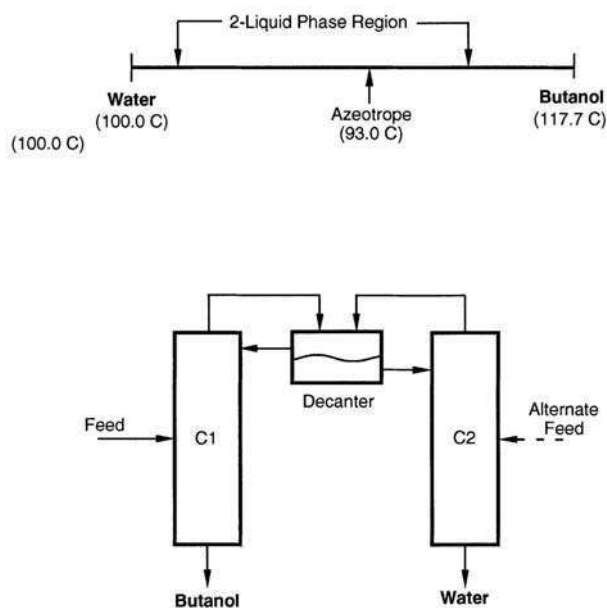


FIG. 13-89 Separation of butanol-water with heterogeneous azeotropic distillation.

into region II. Column C1 is operated to recover pure high-boiling node ethanol as a bottoms product and to produce a distillate close to the ternary azeotrope. If the ternary azeotrope is heterogeneous (as it is in this case), it is sent to the decanter for phase separation. If the ternary azeotrope is homogeneous (as it is in the alternative case of ethyl acetate as the entrainer), the distillate is first mixed with water before being sent to the decanter. The inventory of entrainer is adjusted to allow column C1 to operate essentially between two nodes, although such practice, as discussed previously, is relatively susceptible to instabilities from minor feed or reflux perturbations. Refluxing a fraction of the water-rich decanter layer results in an additional degree of freedom to mitigate against variability in the feed composition. The remaining portion of the water layer from the decanter is stripped of residual cyclohexane in column C2, which may be operated either in the direct-split mode (producing low-boiling node ternary azeotrope as distillate and, by mass balance, an ethanol-water bottoms for recycle to C3) or in the indirect-split mode (producing high-boiling node water as bottoms and, by mass balance, a ternary distillate near the distillation boundary). (The distillate may be recycled to the decanter, the top of column C1, or the C1 feed.) The indirect-split mode alternatives are discussed in greater detail by Knapp and Doherty (*Kirk-Othmer Encyclopedia of Chemical Technology*, 5th ed., vol. 8, p. 786, Wiley, New York, 2004).

Design and Operation of Azeotropic Distillation Columns Simulation and design of azeotropic distillation columns are a difficult computational problem, but one that is readily handled, in most cases, by widely available commercial computer process simulation packages [Glasscock and Hale, *Chem. Eng.*, **101**(11), 82 (1994)]. Most simulators are capable of modeling the steady-state and dynamic behavior of both homogeneous azeotropic distillation systems and those systems involving two-liquid phase behavior within the column, if accurate thermodynamic data and activity coefficient or equation-of-state models are available. However, VLE and VLLE estimated or extrapolated from binary data or predicted from such methods as UNIFAC may not be able to accurately locate boundaries and predict the extent of liquid immiscibilities. Moreover, different activity coefficient models fit to the same experimental data often give very different results for the shape of distillation boundaries and liquid-liquid regions. Therefore the design of separation schemes relying on boundary curvature should not be attempted unless accurate, reliable experimental equilibrium data are available.

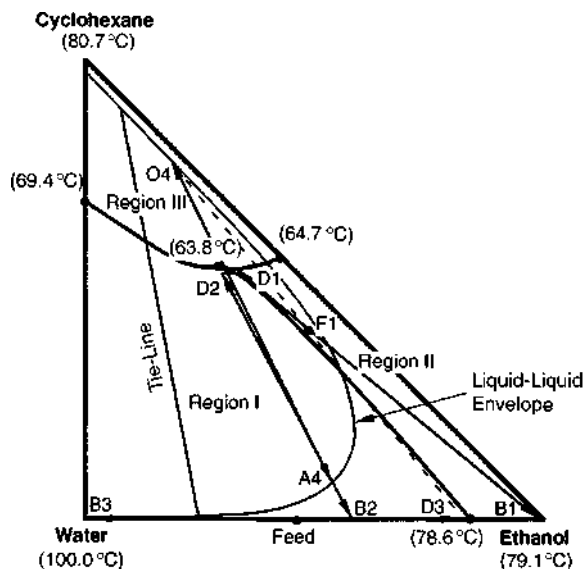


FIG. 13-90 Three-column sequence for ethanol dehydration with cyclohexane (operating column C2 in the direct-split mode).

Two liquid phases can occur within a column in the distillation of heterogeneous systems. Older references, e.g., Robinson and Gilliland (*Elements of Fractional Distillation*, McGraw-Hill, New York, 1950), state that the presence of two liquid phases in a column should be avoided as much as possible because performance may be reduced. However, more recent studies indicate that problems with two-phase flow have been overstated [Herron et al., *AIChE J.*, **34**, 1267 (1988); and Harrison, *Chem. Eng. Prog.*, **86**(11), 80 (1990)]. Based on case history data and experimental evidence, there is no reason to expect unusual capacity or pressure-drop limitations, and standard correlations for these parameters should give acceptable results. Because of the intense nature of the gas-liquid-liquid mixing on trays, mass-transfer efficiencies are relatively unaffected by liquid-liquid phase behavior. The falling-film nature of gas-liquid-liquid contact in packing, however, makes that situation more uncertain. Reduced efficiencies may be expected in systems where one of the keys distributes between the phases.

EXTRACTIVE DISTILLATION

Extractive distillation is a partial vaporization process in the presence of a miscible, high-boiling, nonvolatile mass separation agent, normally

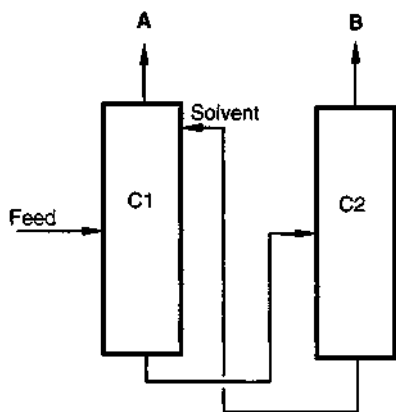


FIG. 13-91 Typical extractive distillation sequence. Component A is less associated with the solvent.

called the *solvent*, which is added to an azeotropic or nonazeotropic feed mixture to alter the volatilities of the key components without the formation of any additional azeotropes. Extractive distillation is used throughout the petrochemical and chemical processing industries for the separation of close-boiling, pinched, or azeotropic systems for which simple single-feed distillation is either too expensive or impossible. It can also be used to obtain products which are residue curve saddles, a task not generally possible with single-feed distillation.

Figure 13-91 illustrates the classical implementation of an extractive distillation process for the separation of a binary mixture. The configuration consists of a double-feed extractive column (C1) and a solvent recovery column (C2). The components A and B may have a low relative volatility or form a minimum-boiling azeotrope. The solvent is introduced into the extractive column at a high concentration a few stages below the condenser, but above the primary-feed stage. Since the solvent is chosen to be nonvolatile, it remains at a relatively high concentration in the liquid phase throughout the sections of the column below the solvent-feed stage.

One of the components, A (not necessarily the most volatile species of the original mixture), is withdrawn as an essentially pure distillate stream. Because the solvent is nonvolatile, at most a few stages above the solvent-feed stage are sufficient to rectify the solvent from the distillate. The bottoms product, consisting of B and the solvent, is sent to the recovery column. The distillate from the recovery column is pure B, and the solvent-bottoms product is recycled to the extractive column.

Extractive distillation works by the exploitation of the selective solvent-induced enhancements or moderations of the liquid-phase nonidealities of the original components to be separated. The solvent selectively alters the activity coefficients of the components being separated. To do this, a high concentration of solvent is necessary. Several features are essential:

1. The solvent must be chosen to affect the liquid-phase behavior of the key components differently; otherwise, no enhancement in separability will occur.
2. The solvent must be higher-boiling than the key components of the separation and must be relatively nonvolatile in the extractive column, in order to remain largely in the liquid phase.
3. The solvent should not form additional azeotropes with the components in the mixture to be separated.
4. The extractive column *must* be a double-feed column, with the solvent feed above the primary feed. The column must have an extractive section (middle section) between the rectifying section and the stripping section.

As a consequence of these restrictions, separation of binary mixtures by extractive distillation corresponds to only two possible three-component distillation region diagrams, depending on whether the binary mixture is pinched or close-boiling (DRD 001), or forms a minimum-boiling azeotrope (DRD 003). The addition of high-boiling solvents

can also facilitate the breaking of maximum-boiling azeotropes (DRD 014), for example, splitting the nitric acid–water azeotrope with sulfuric acid. However, as explained in the subsection on azeotropic distillation, this type of separation might better be characterized as exploiting extreme boundary curvature rather than extractive distillation, as the important liquid-phase activity coefficient modification occurs in the bottom of the column. Although many references show sulfuric acid being introduced high in the column, in fact two separate feeds are not required.

Examples of industrial uses of extractive distillation grouped by distillation region diagram type are given in Table 13-21. Achievable product compositions in double-feed extractive distillation columns are very different from the bow tie regions for single-feed columns. For a given solvent, only one of the pure components in the original binary mixture can be obtained as distillate from the extractive column (the higher-boiling of which is a saddle for close-boiling systems, and both of which are saddles for minimum-boiling azeotropic systems). However, different solvents are capable of selecting either A or B as distillate (but not both). Simple tests are available for determining which component is the distillate, as discussed below.

Extractive distillation is generally only applicable to systems in which the components to be separated contain one or more different functional groups. Extractive distillation is usually uneconomical for separating stereoisomers, homologs, or structural isomers containing the same functional groups, unless the differences in structure also contribute to significantly different polarity, dipole moment, or hydrophobic character. One such counterexample is the separation of ethanol from isopropanol, where the addition of methyl benzoate raises the relative volatility from 1.09 to 1.27 [Berg et al., *Chem. Eng. Comm.*, **66**, 1 (1988)].

Solvent Effects in Extractive Distillation In the ordinary distillation of ideal or nonazeotropic mixtures, the component with the lowest pure-component boiling point is always recovered primarily in the distillate, while the highest boiler is recovered primarily in the bottoms. The situation is not as straightforward for an extractive distillation operation. With some solvents, the key component with the lower pure-component boiling point in the original mixture will be recovered in the distillate as in ordinary distillation. For another solvent, the expected order is reversed, and the component with the higher pure-component boiling point will be recovered in the distillate. The possibility that the expected relative volatility may be reversed by the addition of solvent is entirely a function of the way the solvent interacts with and modifies the activity coefficients and, thus, the volatility of the components in the mixture.

In normal applications of extractive distillation (i.e., pinched, close-boiling, or azeotropic systems), the relative volatilities between the light and heavy key components will be unity or close to unity. Assuming an ideal vapor phase and subcritical components, the relative volatility between the light and heavy keys of the desired separation can be written as the product of the ratios of the pure-component vapor pressures and activity coefficients whether the solvent is present or not:

$$\alpha_{L,H} = \left(\frac{P_L^{\text{sat}}}{P_H^{\text{sat}}} \right) \left(\frac{\gamma_L}{\gamma_H} \right) \quad (13-117)$$

where L and H denote the lower-boiling and higher-boiling key pure component, respectively.

The addition of the solvent has an indirect effect on the vapor-pressure ratio. Because the solvent is high-boiling and is generally added at a relatively high mole ratio to the primary-feed mixture, the temperature of an extractive distillation process tends to increase over that of a simple distillation of the original mixture (unless the system pressure is lowered). The result is a corresponding increase in the vapor pressure of both key components. However, the rise in operating temperature generally *does not* result in a significant modification of the relative volatility, because the ratio of vapor pressures often remains approximately constant, unless the slopes of the vapor-pressure curves differ significantly. The ratio of the vapor pressures typically remains greater than unity, following the “natural” volatility of the system.

TABLE 13-21 Examples of Extractive Distillation, Salt Extractive Distillation

System	Type	Solvent(s)	Remark
Ethanol-water	Minimum-boiling azeotrope	Ethylene glycol, acetate salts for salt process	Alternative to azeotropic distillation, pressure swing distillation
Benzene-cyclohexane	Minimum-boiling azeotrope	Aniline	Process similar for other alcohol-ester systems
Ethyl acetate-ethanol	Minimum-boiling azeotrope	Higher esters or alcohols, aromatics	
THF-water	Minimum-boiling azeotrope	Propylene glycol	Alternative to pressure swing distillation
Acetone-methanol	Minimum-boiling azeotrope	Water, aniline, ethylene glycol Furfural, DMF, acetonitrile Bisphenol Ethylene glycol monomethyl ether	Element of recovery system for alternative to production of methyl acetate by reactive distillation; alternative to azeotropic, pressure, swing distillation
Isoprene-pentane	Minimum-boiling azeotrope		
Pyridine-water	Minimum-boiling azeotrope		
Methyl acetate-methanol	Minimum-boiling azeotrope		
C4 alkenes/C4 alkanes/ C4 dienes	Close-boiling and minimum-boiling azeotropes	Furfural, DMF, acetonitrile, <i>n</i> -methylpyrrolidone	
C5 alkenes/C5 alkanes/ C5 dienes	Close-boiling and minimum-boiling azeotropes	Furfural, DMF, acetonitrile, <i>n</i> -methylpyrrolidone	
Heptane isomers-cyclohexane	Close-boiling	Aniline, phenol	
Heptane isomers-toluene	Close-boiling and minimum-boiling azeotropes	Aniline, phenol	
Vinyl acetate-ethyl acetate	Close-boiling	Phenol, aromatics Acrylonitrile	Alternative to simple distillation Alternative to simple distillation, adsorption
Propane-propylene	Close-boiling		
Ethanol-isopropanol	Close-boiling	Methyl benzoate Sulfuric acid, calcium chloride for salt process Sulfuric acid, magnesium nitrate for salt process	Alternative to simple distillation Sulfuric acid process relies heavily on boundary curvature Sulfuric acid process relies heavily on boundary curvature
Hydrochloric acid-water	Maximum-boiling azeotrope		
Nitric acid-water	Maximum-boiling azeotrope		

Since activity coefficients have a strong dependence on composition, the effect of the solvent on the activity coefficients is generally more pronounced. However, the magnitude and direction of change are highly dependent on the solvent concentration as well as on the liquid-phase interactions between the solvent and the key components. The solvent acts to lessen the nonidealities of the key component whose liquid-phase behavior is similar to that of the solvent, while enhancing the nonideal behavior of the dissimilar key. The solvent and the key component that show most similar liquid-phase behavior tend to exhibit weak molecular interactions. These components form an ideal or nearly ideal liquid solution. The activity coefficient of this key approaches unity, or may even show negative deviations from Raoult's law if solvating or complexing interactions occur. On the other hand, the dissimilar key and the solvent demonstrate unfavorable molecular interactions, and the activity coefficient of this key increases. The positive deviations from Raoult's law are further enhanced by the diluting effect of the high-solvent concentration, and the value of the activity coefficient of this key may approach the infinite dilution value, often a very large number.

The natural relative volatility of the system is enhanced when the activity coefficient of the lower-boiling pure component is increased by the solvent addition (γ_L/γ_H increases and $P_L^{\text{sat}}/P_H^{\text{sat}} > 1$). In this case, the lower-boiling pure component will be recovered in the distillate as expected. For the higher-boiling pure component to be recovered in the distillate, the addition of the solvent must decrease the ratio γ_L/γ_H such that the product of γ_L/γ_H and $P_L^{\text{sat}}/P_H^{\text{sat}}$ (that is, α_{LH}) in the presence of the solvent is less than unity. Generally, the latter is more difficult to achieve and requires higher solvent-to-feed ratios. It is normally better to select a solvent that forces the lower-boiling component overhead.

The effect of solvent concentration on the activity coefficients of the key components is shown in Fig. 13-92 for the system methanol-acetone with either water or methylisopropylketone (MIPK) as solvent. For an initial feed mixture of 50 mol % methanol and 50 mol % acetone (no solvent present), the ratio of activity coefficients of methanol and acetone is close to unity. With water as the solvent, the

activity coefficient of the similar key (methanol) rises slightly as the solvent concentration increases, while the coefficient of acetone approaches the relatively large infinite dilution value. With methylisopropylketone as the solvent, acetone is the similar key and its activity coefficient drops toward unity as the solvent concentration increases, while the activity coefficient of the methanol increases.

Several methods are available for determining whether the lower- or higher-boiling pure component will be recovered in the distillate. For a series of solvent concentrations, the binary y - x phase diagram for the low-boiling and high-boiling keys can be plotted on a solvent-free basis. At a particular solvent concentration (dependent on the selected solvent and keys), the azeotropic point in the binary plot disappears at one of the pure-component corners. The component corresponding to the corner where the azeotrope disappears is recovered in the distillate (Knapp and Doherty, *Kirk-Othmer Encyclopedia of Chemical Technology*, 5th ed., vol. 8, p. 786, Wiley, New York, 2004). LaRoche et al. [*Can. J. Chem. Eng.*, **69**, 1302 (1991)] present a related method in which the $\alpha_{LH} = 1$ line is plotted on the ternary composition diagram. If this line intersects the lower-boiling pure component + solvent binary face, then the lower-boiling component will be recovered in the distillate, and vice versa if the $\alpha_{LH} = 1$ line intersects the higher-boiling pure component + solvent face. A very simple method, if a rigorous residue curve map is available, is to examine the shape and inflection of the residue curves as they approach the pure solvent vertex. Whichever solvent-key component face the residue curves predominantly tend toward as they approach the solvent vertex is the key component that will be recovered in the bottoms with the solvent (see property 6, p. 193, in Doherty and Malone, *op. cit.*). In Fig. 13-93a, all residue curves approaching the water (solvent) vertex are inflected toward the methanol-water face, with the result that methanol will be recovered in the bottoms and acetone in the distillate. Alternatively, with MIPK as the solvent, all residue curves show inflection toward the acetone-MIPK face (Fig. 13-93b), indicating that acetone will be recovered in the bottoms and methanol in the distillate.

Extractive Distillation Design and Optimization Extractive distillation column composition profiles have a very characteristic

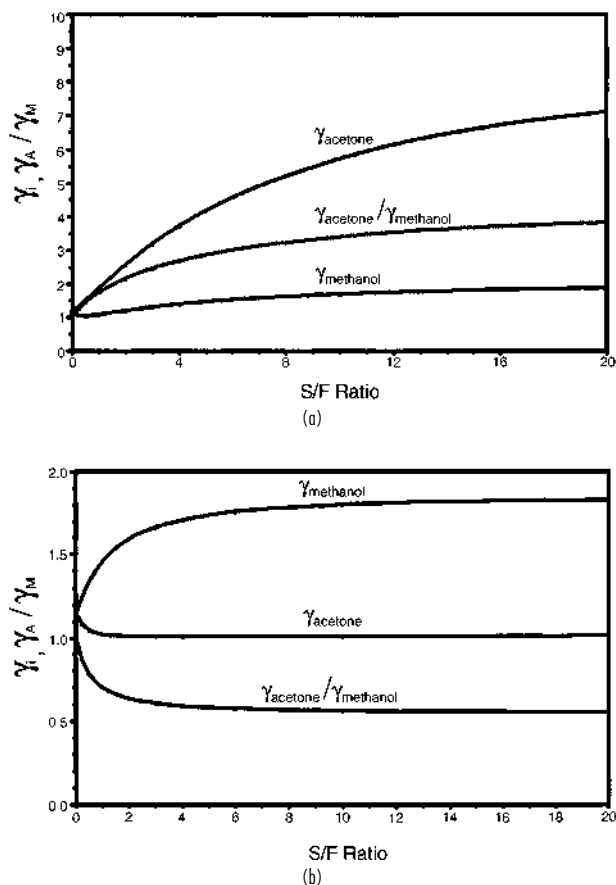


FIG. 13-92 Effect of solvent concentration on activity coefficients for acetone-methanol system. (a) Water solvent. (b) MIPK solvent.

shape on a ternary diagram. The composition profile for the separation of methanol-acetone with water is given in Fig. 13-94. Stripping and rectifying profiles start at the bottoms and distillate compositions, respectively, track generally along the faces of the composition triangle, and then turn toward the high-boiling (solvent) node and low-boiling node, respectively. For a feasible single-feed design, these profiles must cross at some point. However, in an extractive distillation they cannot cross. The extractive section profile acts at the bridge between these two sections. Most of the key component separation occurs in this section in the presence of high-solvent composition.

The variable that has the most significant impact on the economics of an extractive distillation is the solvent-to-feed flow rate ratio S/F . For close-boiling or pinched nonazeotropic mixtures, no minimum-solvent flow rate is required to effect the separation, as the separation is always theoretically possible (if not economical) in the absence of the solvent. However, the extent of enhancement of the relative volatility is largely determined by the solvent composition in the lower column sections and hence the S/F ratio. The relative volatility tends to increase as the S/F ratio increases. Thus, a given separation can be accomplished in fewer equilibrium stages. As an illustration, the total number of theoretical stages required as a function of S/F ratio is plotted in Fig. 13-95a for the separation of the nonazeotropic mixture of vinyl acetate and ethyl acetate using phenol as the solvent.

For the separation of a minimum-boiling binary azeotrope by extractive distillation, there is clearly a minimum-solvent flow rate below which the separation is impossible (due to the azeotrope). For azeotropic separations, the number of equilibrium stages is infinite at or below $(S/F)_{\min}$ and decreases rapidly with increasing solvent feed

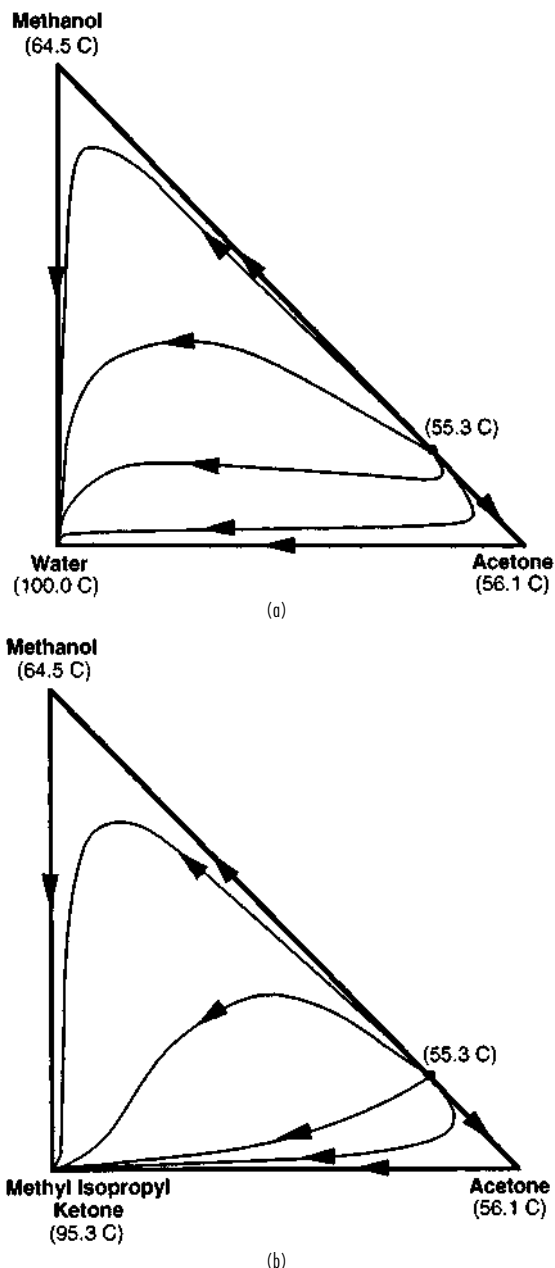


FIG. 13-93 Residue curve maps for acetone-methanol systems. (a) With water. (b) With MIPK.

flow, and then may asymptote, or rise slowly. The relationship between the total number of stages and the S/F ratio for a given purity and recovery for the azeotropic acetone-methanol system with water as solvent is shown in Fig. 13-95b. A rough idea of $(S/F)_{\min}$ can be determined from a pseudobinary diagram or by plotting the $\alpha_{L,H} = 1$ line on a ternary diagram. The solvent composition at which the azeotrope disappears in a corner of the pseudobinary diagram is an indication of $(S/F)_{\min}$ [LaRoche et al., *Can. J. Chem. Eng.*, **69**, 1302 (1991)]. An exact method for calculating $(S/F)_{\min}$ is given by Knapp and Doherty [*AIChE J.*, **40**, 243 (1994)]. Typically, operating S/F ratios for economically acceptable solvents are between 2 and 5. Higher

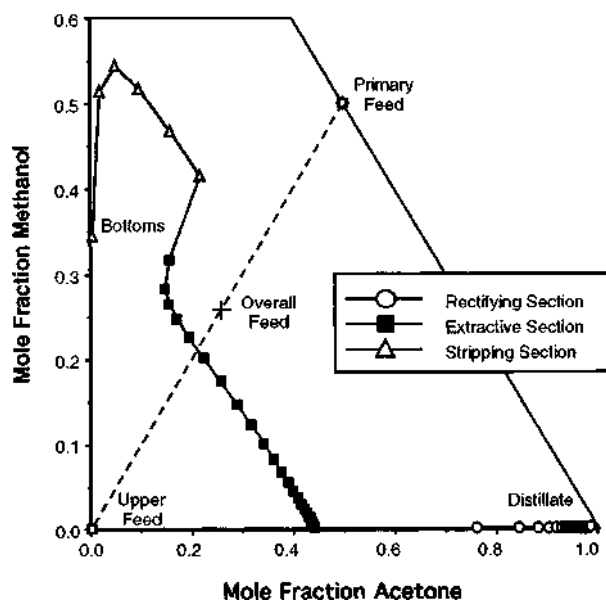


FIG. 13-94 Extractive distillation column composition profile for the separation of acetone-methanol with water.

S/F ratios tend to increase the diameter of both the extractive column and the solvent recovery columns, but reduce the required number of equilibrium stages and minimum reflux ratio. Moreover, higher S/F ratios lead to higher reboiler temperatures, resulting in the use of higher-cost utilities, higher utility usages, and greater risk of degradation.

Knight and Doherty [Ind. Eng. Chem. Fundam., 28, 564 (1989)] have published rigorous methods for computing minimum reflux for extractive distillation; they found that an operating reflux ratio of 1.2 to 1.5 times the minimum value is usually acceptable. Interestingly, unlike other forms of distillation, in extractive distillation the distillate purity or recovery does not increase monotonically with increasing reflux ratio for a given number of stages. Above a maximum reflux ratio the separation can no longer be achieved, and the distillate purity actually decreases for a given number of stages [LaRoche et al., AIChE J., 38, 1309 (1992)]. The difference between $R_{\min}$ and $R_{\max}$ increases as the S/F ratio increases. Large amounts of reflux lowers the solvent composition in the upper section of the column, degrading rather than enhancing column performance. Because the reflux ratio goes through a maximum, the conventional control strategy of increasing reflux to maintain purity can be detrimental rather than beneficial. However, $R_{\max}$ generally occurs at impractically high reflux ratios and is typically not of major concern.

The thermal quality of the solvent feed has no effect on the value of $S/F_{\min}$, but does affect the minimum reflux to some extent, especially as the S/F ratio increases. The maximum reflux ratio $R_{\max}$ occurs at higher values of the reflux ratio as the upper-feed quality decreases; a subcooled upper feed provides additional refluxing capacity and less external reflux is required for the same separation. It is also sometimes advantageous to introduce the primary feed to the extractive distillation column as a vapor to help maintain a higher solvent composition on the feed tray and the trays immediately below.

Robinson and Gilliland (Elements of Fractional Distillation, McGraw-Hill, New York, 1950), Smith (Design of Equilibrium Stage Processes, McGraw-Hill, New York, 1963), Van Winkle (Distillation, McGraw-Hill, New York, 1967), and Walas (Chemical Process Equipment, Butterworths, Boston, 1988) discuss rigorous stage-to-stage design techniques as well as shortcut and graphical methods for determining minimum stages, $(S/F)_{\min}$, minimum reflux, and the optimum locations of the solvent and primary feed points. Knapp and Doherty [AIChE J., 40, 243 (1994)] have published column design methods

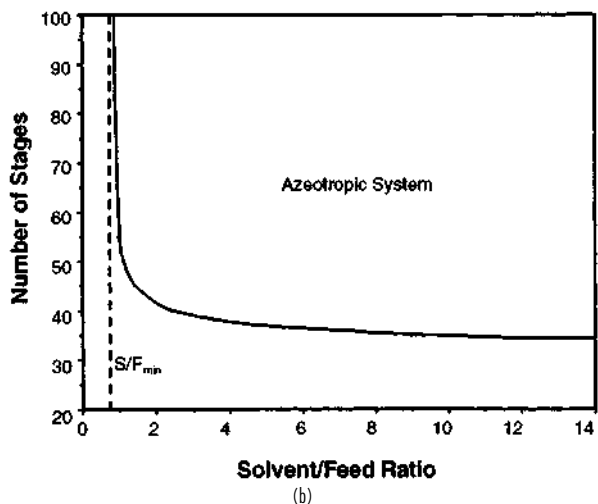
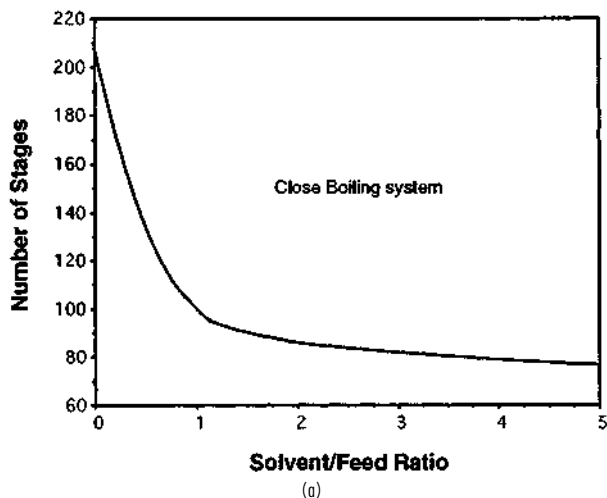


FIG. 13-95 Number of theoretical stages versus solvent-to-feed ratio for extractive distillation. (a) Close-boiling vinyl acetate-ethyl acetate system with phenol solvent. (b) Azeotropic acetone-methanol system with water solvent.

based on geometric arguments and fixed-point analysis that are capable of calculating $(S/F)_{\min}$, as well as the minimum and maximum reflux ratio. Most commercial simulators are capable of solving multiple-feed extractive distillation heat and material balances, but do not include straightforward techniques for calculating $(S/F)_{\min}$, or the minimum and maximum reflux ratio.

Solvent Screening and Selection Choosing an effective solvent can have the most profound effect on the economics of an extractive distillation process. The approach most often adopted is to first generate a short list of potential solvents by using simple qualitative screening and selection methods. Experimental verification is best undertaken only after a list of promising candidate solvents has been generated and some chance at economic viability has been demonstrated via preliminary process modeling.

Solvent selection and screening approaches can be divided into two levels of analysis. The first level focuses on identification of functional groups or chemical families that are likely to give favorable solvent-key component molecular interactions. The second level of analysis identifies and compares individual candidate solvents. The

various methods of analysis are described briefly and illustrated with an example of choosing a solvent for the methanol-acetone separation.

First Level: Broad Screening by Functional Group or Chemical Family

Homologous series. Select candidate solvents from the high-boiling homologous series of both light and heavy key components. Favor homologs of the heavy key, as this tends to enhance the natural relative volatility of the system. Homologous components tend to form ideal solutions and are unlikely to form azeotropes [Scheibel, *Chem. Eng. Prog.*, **44**(12), 927 (1948)].

Robbins chart. Select candidate solvents from groups in the Robbins chart (Table 13-17) that tend to give positive (or no) deviations from Raoult's law for the key component desired in the distillate and negative (or no) deviations for the other key.

Hydrogen-bonding characteristics. Select candidate solvents from groups that are likely to cause the formation of hydrogen bonds with the key component to be removed in the bottoms, or disruption of hydrogen bonds with the key to be removed in the distillate. Formation and disruption of hydrogen bonds are often associated with strong negative and positive deviations, respectively, from Raoult's law. Several authors have developed charts indicating expected hydrogen bonding interactions between families of compounds [Ewell et al., *Ind. Eng. Chem.*, **36**, 871 (1944); Gilmont et al., *Ind. Eng. Chem.*, **53**, 223 (1961); and Berg, *Chem. Eng. Prog.*, **65**(9), 52 (1969)]. Table 13-22 presents a hydrogen bonding classification of chemical families and a summary of deviations from Raoult's law.

fication of chemical families and a summary of deviations from Raoult's law.

Polarity characteristics. Select candidate solvents from chemical groups that tend to show higher polarity than one key component or lower polarity than the other key. Polarity effects are often cited as a factor in causing deviations from Raoult's law [Hopkins and Fritsch, *Chem. Eng. Prog.*, **51**(8), (1954); Carlson et al., *Ind. Eng. Chem.*, **46**, 350 (1954); and Prausnitz and Anderson, *AIChE J.*, **7**, 96 (1961)]. The general trend in polarity based on the functional group of a molecule is given in Table 13-23. The chart is best for molecules of similar size. A more quantitative measure of the polarity of a molecule is the polarity contribution to the three-term Hansen solubility parameter. A tabulation of calculated three-term solubility parameters is provided by Barton (*CRC Handbook of Solubility Parameters and Other Cohesion Parameters*, CRC Press, Boca Raton, Fla., 1991), along with a group contribution method for calculating the three-term solubility parameters of compounds not listed in the reference.

Second Level: Identification of Individual Candidate Solvents

Boiling point characteristics. Select only candidate solvents that boil at least 30 to 40°C above the key components to ensure that the solvent is relatively nonvolatile and remains largely in the liquid phase. With this boiling point difference, the solvent should also not form azeotropes with the other components.

Selectivity at infinite dilution. Rank candidate solvents according to their selectivity at infinite dilution. The selectivity at infinite

TABLE 13-22 Hydrogen Bonding Classification of Chemical Families

Class	Chemical family			
H-Bonding, Strongly Associative (HBSA)	Water Primary amides Secondary amides	Polyacids Dicarboxylic acids Monohydroxy acids	Polyphenols Oximes Hydroxylamines	Amino alcohols Polyols
H-Bond Acceptor-Donor (HBAD)	Phenols Aromatic acids Aromatic amines Alpha H nitriles	Imines Monocarboxylic acids Other monoacids Peracids	Alpha H nitros Azines Primary amines Secondary amines	n-alcohols Other alcohols Ether alcohols
H-Bond Acceptor (HBA)	Acyl chlorides Acyl fluorides Hetero nitrogen aromatics Hetero oxygen aromatics	Tertiary amides Tertiary amines Other nitriles Other nitros Isocyanates Peroxides	Aldehydes Anhydrides Cyclo ketones Aliphatic ketones Esters Ethers	Aromatic esters Aromatic nitriles Aromatic ethers Sulfones Sulfolanes
π -Bonding Acceptor (π -HBA)	Alkynes Alkenes	Aromatics Unsaturated esters		
H-Bond Donor (HBD)	Inorganic acids Active H chlorides	Active H fluorides Active H iodides	Active H bromides	
Non-Bonding (NB)	Paraffins Nonactive H chlorides	Nonactive H fluorides Sulfides	Nonactive H iodides Disulfides	Nonactive H bromides Thiols

Deviations from Raoult's Law

H-Bonding classes	Type of deviations	Comments
HBSA + NB HBAD + NB	Always positive dev., HBSA + NB often limited miscibility	H-bonds broken by interactions
HBA + HBD	Always negative dev.	H-bonds formed by interactions
HBSA + HBD HBAD + HBD	Always positive deviations, HBSA + HBD often limited miscibility	H-bonds broken and formed; dissociation of HBSA or HBAD liquid most important effect
HBSA + HBSA HBSA + HBAD HBSA + HBA HBAD + HBAD HBAD + HBA	Usually positive deviations; some give maximum-boiling azeotropes	H-bonds broken and formed
HBA + HBA HBA + NB HBD + HBD HBD + NB NB + NB	Ideal, quasi-ideal systems; always positive or no deviations; azeotropes, if any, minimum-boiling	No H-bonding involved

NOTE: π -HBA is enhanced version of HBA.

TABLE 13-23 Relative Polarities of Functional Groups

MOST POLAR	Water
	Organic acids
	Amines
	Polyols
	Alcohols
	Esters
	Ketones
	Aldehydes
	Ethers
	Aromatics
	Olefins
	Paraffins
LEAST POLAR	
Effect of branching	
MOST POLAR	Normal
	Secondary
LEAST POLAR	Tertiary

dilution is defined as the ratio of the activity coefficients at infinite dilution of the two key components in the solvent. Since solvent effects tend to increase as solvent concentration increases, the infinite-dilution selectivity gives an upper bound on the efficacy of a solvent. Infinite-dilution activity coefficients can be predicted by using such methods as UNIFAC, ASOG, MOSCED (Reid et al., *Properties of Gases and Liquids*, 4th ed., McGraw-Hill, New York, 1987). They can be found experimentally by using a rapid gas-liquid chromatography method based on relative retention times in candidate solvents (Tassios, in *Extractive and Azeotropic Distillation*, Advances in Chemistry Series 115, American Chemical Society, Washington, 1972), and they can be correlated to bubble point data [Kojima and Ochi, *J. Chem. Eng. Japan*, 7(2), 71 (1974)]. DECHEMA (*Vapor-Liquid Equilibrium Data Collection*, Frankfurt, 1977) has also published a compilation of experimental infinite-dilution activity coefficients.

Experimental measurement of relative volatility. Rank candidate solvents by the increase in relative volatility caused by the addition of the solvent. One technique is to experimentally measure the relative volatility of a fixed-composition, key component + solvent mixture (often a 1/1 ratio of each key, with a 1/1 to 3/1 solvent/key ratio) for various solvents [Carlson et al., *Ind. Eng. Chem.*, 46, 350 (1954)]. The Othmer equilibrium still is the apparatus of choice for these measurements [Zudkevitch, *Chem. Eng. Comm.*, 116, 41 (1992)].

At atmospheric pressure, methanol and acetone boil at 64.5 and 56.1°C, respectively, and form a minimum-boiling azeotrope at 55.3°C. The natural volatility of the system is acetone > methanol, so the favored solvents most likely will be those that cause acetone to be recovered in the distillate. However, for the purposes of the example, a solvent that reverses the natural volatility will also be identified. First, by examining the polarity of ketones and alcohols (Table 13-23), solvents favored for the recovery of methanol in the bottoms would come from groups more polar than methanol, such as acids, water, and polyols. Turning to the Robbins chart (Table 13-17), we see favorable groups are amines, alcohols, polyols, and water since these show expected positive deviations for acetone and zero or negative deviations for methanol. For reversing the natural volatility, solvents should be chosen that are less polar than acetone, such as ethers, hydrocarbons, and aromatics. Unfortunately, both ethers and hydrocarbons are expected to give positive deviations for both acetone and methanol, so should be discarded. Halohydrocarbons and ketones are expected to give positive deviations for methanol and either negative or no deviations for acetone. The other qualitative indicators show that both homologous series (ketones and alcohols) look promising. Thus, after discounting halohydrocarbons for environmental reasons, the best solvents will probably come from alcohols, polyols, and water for recovering methanol in the bottoms and ketones for recovering acetone in the bottoms. Table 13-24 shows the boiling points and experimental or estimated infinite-dilution activity coefficients for several candidate solvents from the aforementioned groups. Methyl ethyl ketone boils too low, as does ethanol, and also forms an azeotrope with methanol. These two candidates can be discarded. Other members of the homologous series, along with water and ethylene glycol, have acceptable

TABLE 13-24 Comparison of Candidate Solvents for Methanol/Acetone Extractive Distillation

Solvent	Boiling pt. (°C)	Azeotrope formation	$\gamma_{\text{Acetone}}^{\infty}$	$\gamma_{\text{MeOH}}^{\infty}$	$\gamma_{\text{Acetone}}^{\infty}/\gamma_{\text{MeOH}}^{\infty}$
MEK	79.6	With MeOH	1.01	1.88	0.537
MIPK	102.0	No	1.01	1.89	0.534
MIBK	115.9	No	1.06	2.05	0.517
Ethanol	78.3	No	1.85	1.04	1.78
1-Propanol	97.2	No	1.90	1.20	1.58
1-Butanol	117.8	No	1.93	1.33	1.45
Water	100.0	No	11.77	2.34	5.03
EG	197.2	No	3.71	1.25	2.97

$\gamma_{\text{Acetone}}^{\infty} = 1.79$ (in MeOH)

$\gamma_{\text{MeOH}}^{\infty} = 1.81$ (in acetone)

boiling points (at least 30°C higher than those of the keys). Of these, water (the solvent used industrially) clearly has the largest effect on the activity coefficients, followed by ethylene glycol. Although inferior to water or ethylene glycol, both MIPK and MIBK would probably be acceptable for reversing the natural volatility of the system.

Extractive Distillation by Salt Effects A second method of modifying the liquid-phase behavior (and thus the relative volatility) of a mixture to effect a separation is by the addition of a nonvolatile, soluble, ionic salt. The process is analogous to extractive distillation with a high-boiling liquid. In the simplest case, for the separation of a binary mixture, the salt is fed at the top of the column by dissolving it in the hot reflux stream before introduction into the column. To function effectively, the salt must be adequately soluble in both components throughout the range of compositions encountered in the column. Since the salt is completely nonvolatile, it remains in the liquid phase on each tray and alters the relative volatility throughout the length of the column. No rectification section is needed above the salt feed. The bottoms product is recovered from the salt solution by evaporation or drying, and the salt is recycled. The ions of a salt are typically capable of causing much larger and more selective effects on liquid-phase behavior than the molecules of a liquid solvent. As a result, salt-to-feed ratios of less than 0.1 are typical.

The use of a salting agent presents a number of problems not associated with a liquid solvent, such as the difficulty of transporting and metering a solid or saturated salt solution, slow mixing or dissolution rate of the salt, limits to solubility in the feed components, and potential for corrosion. However, in the limited number of systems for which an effective salt can be found, the energy usage, equipment size, capital investment, and ultimate separation cost can be significantly reduced compared to that for extractive distillation using a liquid solvent [Furter, *Chem. Eng. Commun.*, 116, 35 (1992)]. Applications of salt extractive distillation include acetate salts to produce absolute ethanol, magnesium nitrate for the production of concentrated nitric acid as an alternative to the sulfuric acid solvent process, and calcium chloride to produce anhydrous hydrogen chloride. Other examples are noted by Furter [*Can. J. Chem. Eng.*, 55, 229 (1977)].

One problem limiting the consideration of salt extractive distillation is the fact that the performance and solubility of a salt in a particular system are difficult to predict without experimental data. Some recent advances have been made in modeling the VLE behavior of organic aqueous salt solutions using modified UNIFAC, NRTL, UNIQUAC, and other approaches [Kumar, *Sep. Sci. Tech.*, 28(1), 799 (1993)].

REACTIVE DISTILLATION

Reactive distillation is a unit operation in which chemical reaction and distillation are carried out simultaneously within a fractional distillation apparatus. Reactive distillation may be advantageous for liquid-phase reaction systems when the reaction must be carried out with a large excess of one or more of the reactants, when a reaction

can be driven to completion by removal of one or more of the products as they are formed, or when the product recovery or by-product recycle scheme is complicated or made infeasible by azeotrope formation.

For consecutive reactions in which the desired product is formed in an intermediate step, excess reactant can be used to suppress additional series reactions by keeping the intermediate-species concentration low. A reactive distillation can achieve the same result by removing the desired intermediate from the reaction zone as it is formed. Similarly, if the equilibrium constant of a reversible reaction is small, high conversions of one reactant can be achieved by use of a large excess of the other reactant. Alternatively, by Le Chatelier's principle, the reaction can be driven to completion by removal of one or more of the products as they are formed. Typically, reactants can be kept much closer to stoichiometric proportions in a reactive distillation.

When a reaction mixture exhibits azeotropes, the recovery of products and recycle of excess reagents can be quite complicated and expensive. Reactive distillation can provide a means of breaking azeotropes by altering or eliminating the condition for azeotrope formation in the reaction zone through the combined effects of vaporization-condensation and consumption-production of the species in the mixture. Alternatively, a reaction may be used to convert the species to components that are more easily distilled. In each of these situations, the conversion and selectivity often can be improved markedly, with much lower reactant inventories and recycle rates, and much simpler recovery schemes. The capital savings can be quite dramatic. A list of applications of reactive distillation appearing in the literature is given in Table 13-25. Additional industrial applications are described by Sharma and Mahajani (chap. 1 in Sundmacher and Kienle, eds., *Reactive Distillation*, Wiley-VCH, 2003).

Although reactive distillation has many potential applications, it is not appropriate for all situations. Since it is in essence a distillation process, it has the same range of applicability as other distillation operations. Distillation-based equipment is not designed to effectively handle solids, supercritical components (where no separate vapor and liquid phases exist), gas-phase reactions, or high-temperature or high-pressure reactions such as hydrogenation, steam reforming, gasification, and hydrodealkylation.

Simulation, Modeling, and Design Feasibility Because reaction and separation phenomena are closely coupled in a reactive distillation process, simulation and design are significantly more complex than those of sequential reaction and separation processes. In spite of the complexity, however, most commercial computer process modeling packages offer reliable and flexible routines for simulating steady-state reactive distillation columns, with either equilibrium or kinetically controlled reaction models

[Venkataraman et al., *Chem. Eng. Prog.*, **86**(6), 45 (1990)]. As with other enhanced distillation processes, the results are very sensitive to the thermodynamic models chosen and the accuracy of the VLE data used to generate model parameters. Of equal, if not greater significance is the accuracy of data and models for reaction rate as a function of catalyst concentration, temperature, and composition. Very different conclusions can be drawn about the feasibility of a reactive distillation if the reaction is assumed to reach chemical equilibrium on each stage of the column or if the reaction is assumed to be kinetically controlled [Barbosa and Doherty, *Chem. Eng. Sci.*, **43**, 541 (1988); Chadda, Malone, and Doherty, *AIChE J.*, **47**, 590 (2001)]. Tray holdup and stage requirements are two important variables directly affected by the reaction time relative to the residence time inside the column. Unlike distillation without reaction, product feasibility can be quite sensitive to changes in tray holdup and production rate.

When an equilibrium reaction occurs in a vapor-liquid system, the phase compositions depend not only on the relative volatility of the components in the mixture, but also on the consumption (and production) of species. Thus, the condition for azeotropy in a nonreactive system ($y_i = x_i$ for all i) no longer holds true in a reactive system and must be modified to include reaction stoichiometry:

$$\frac{y_i - x_i}{v_i - v_T x_i} = \frac{y_i - x_i}{v_i - v_T x_i} \quad \text{for all } i = 1, \dots, c \quad (13-118)$$

where

$$v_T = \sum_{i=1}^c v_i$$

and v_i represents the stoichiometric coefficient of component i (negative for reactants, positive for products).

Phase compositions that satisfy Eq. (13-118) are stationary points on a phase diagram and have been labeled *reactive azeotropes* by Barbosa and Doherty [*Chem. Eng. Sci.*, **43**, 529 (1988)]. At a reactive azeotrope the mass exchange between the vapor and liquid phases and the generation (or consumption) of each species are balanced such that the composition of neither phase changes. Reactive azeotropes show the same distillation properties as ordinary azeotropes and therefore affect the achievable products. Reactive azeotropes are predicted to exist in numerous reacting mixtures and have been confirmed experimentally in the reactive boiling mixture of acetic acid + isopropanol + isopropyl acetate + water [Song et al., *Nature*, **388**, 561 (1997); Huang et al., *Chem. Eng. Sci.*, **60**, 3363 (2005)].

Reactive azeotropes are not easily visualized in conventional y - x coordinates but become apparent upon a transformation of coordinates which depends on the number of reactions, the order of each reaction (for example, $A + B \rightleftharpoons C$ or $A + B \rightleftharpoons C + D$), and the presence of nonreacting components. The general vector-matrix form of

TABLE 13-25 Applications of Reactive Distillation

Process	Reaction type	Reference
Methyl acetate from methanol and acetic acid General process for ester formation	Esterification Esterification	Agreda et al., <i>Chem. Eng. Prog.</i> , 86 (2), 40 (1990) Simons, "Esterification" in <i>Encyclopedia of Chemical Processing and Design</i> , Vol 19, Dekker, New York, 1983 Oyevaar et al., U.S. Patent 6,093,842 (2000)
Diphenyl carbonate from dimethyl carbonate and phenol	Esterification	Berman et al., <i>Ind. Eng. Chem.</i> , 40 , 2139 (1948)
Dibutyl phthalate from butanol and phthalic acid	Esterification	Davies and Jeffreys, <i>Trans. Inst. Chem. Eng.</i> , 51 , 275 (1973)
Ethyl acetate from ethanol and butyl acetate	Transesterification	Fuchigami, <i>J. Chem. Eng. Jap.</i> , 23 , 354 (1990)
Recovery of acetic acid and methanol from methyl acetate by-product of vinyl acetate production	Hydrolysis	
Nylon 6,6 prepolymer from adipic acid and hexamethylenediamine	Amidation	Jaswal and Pugi, U.S. Patent 3,900,450 (1975)
MTBE from isobutene and methanol	Etherification	DeGarmo et al., <i>Chem. Eng. Prog.</i> , 88 (3), 43 (1992)
TAME from pentenes and methanol	Etherification	Brockwell et al., <i>Hyd. Proc.</i> , 70 (9), 133 (1991)
Separation of close boiling 3- and 4-picoline by complexation with organic acids	Acid-base	Duprat and Gau, <i>Can. J. Chem. Eng.</i> , 69 , 1320 (1991)
Separation of close-boiling meta and para xylenes by formation of tert-butyl meta-xylene	Transalkylation	Saito et al., <i>J. Chem. Eng. Jap.</i> , 4 , 37 (1971)
Cumene from propylene and benzene	Alkylation	Shoemaker and Jones, <i>Hyd. Proc.</i> , 67 (6), 57 (1987)
General process for the alkylation of aromatics with olefins	Alkylation	Crossland, U.S. Patent 5,043,506 (1991)
Production of specific higher and lower alkenes from butenes	Diproporationation	Jung et al., U.S. Patent 4,709,115 (1987)
4-Nitrochlorobenzene from chlorobenzene and nitric acid	Nitration	Belson, <i>Ind. Eng. Chem. Res.</i> , 29 , 1562 (1990)
Production of methylal and high purity formaldehyde		Masamoto and Matsuzaki, <i>J. Chem. Eng. Jap.</i> , 27 , 1 (1994)

the transform for c reacting components, with R reactions, and I non-reacting components, has been derived by Ung and Doherty [*Chem. Eng. Sci.*, **50**, 23 (1995)]. For the transformed mole fraction of component i in the liquid phase X_i , they give

$$X_i = \left[\frac{x_i - \mathbf{v}_i^T (\mathbf{v}_{\text{Ref}})^{-1} \mathbf{x}_{\text{Ref}}}{1 - \mathbf{v}_{\text{TOT}}^T (\mathbf{v}_{\text{Ref}})^{-1} \mathbf{x}_{\text{Ref}}} \right] \quad i = 1, \dots, c - R \quad (13-119)$$

where $\mathbf{v}_i^T$ = row vector of stoichiometric coefficients of component i for each reaction

$\mathbf{v}_{\text{Ref}}$ = square matrix of stoichiometric coefficients for R reference components in R reactions

$\mathbf{x}_{\text{Ref}}$ = column vector of mole fractions for R reference components in liquid phase

$\mathbf{v}_{\text{TOT}}^T$ = row vector composed of sum of stoichiometric coefficients for each reaction

An equation identical to (13-119) defines the transformed mole fraction of component i in the vapor phase Y_i , where the terms in x are replaced by terms in y .

The transformed variables describe the system composition with or without reaction and sum to unity as do x_i and y_i . The condition for reactive azeotropy becomes $X_i = Y_i$. Barbosa and Doherty have shown that phase diagrams and distillation diagrams constructed by using the transformed composition coordinates have the same properties as phase and distillation diagrams for nonreactive systems and similarly can be used to assist in design feasibility and operability studies [*Chem. Eng. Sci.*, **43**, 529, 541, 1523, and 2377 (1988)]. Residue curve maps in transformed coordinates for the reactive system methanol–acetic acid–methyl acetate–water are shown in Fig. 13-96. Note that the nonreactive azeotrope between water and methyl acetate has disappeared, while the methyl acetate–methanol azeotrope remains intact. Only those azeotropes containing all the reactants or products will be altered by the reaction (water and methyl acetate can back-react to form acetic acid and methanol, whereas methanol and methyl acetate cannot further react in the absence of either water or acetic acid). This reactive system consists of only one distillation region in which the methanol–methyl acetate azeotrope is the low-boiling node and acetic acid is the high-boiling node.

The situation becomes more complicated when the reaction is kinetically controlled and does not come to complete chemical equilibrium under the conditions of temperature, liquid holdup, and rate of vaporization in the column reactor. Venimadhavan et al. [*AIChE J.*, **40**, 1814

(1994); **45**, 546 (1999)] and Rev [*Ind. Eng. Chem. Res.*, **33**, 2174 (1994)] show that the concept of a reactive azeotrope generalizes to the concept of a *reactive fixed point*, whose existence and location are a function of approach to equilibrium as well as the evaporation rate [see also Frey and Stichlmair, *Trans IChemE*, **77**, Part A, 613 (1999); Chadda, Malone, and Doherty, *AIChE J.*, **47**, 590 (2001); Chiplunkar et al., *AIChE J.*, **51**, 464 (2005)]. In the limit of simultaneous phase and reaction equilibrium, a reactive fixed point becomes identical to the thermodynamic concept of a reactive azeotrope.

These ideas have been extended to reacting systems with (1) multiple chemical reactions [Ung and Doherty, *Ind. Eng. Chem. Res.*, **34**, 3195, 2555 (1995)], (2) multiple liquid phases [Ung and Doherty, *Chem. Eng. Sci.*, **50**, 3201 (1995); Qi, Kolah, and Sundmacher, *Chem. Eng. Sci.*, **57**, 163 (2002); Qi and Sundmacher, *Comp. Chem. Eng.*, **26**, 1459 (2002)], (3) membrane separations [Huang et al., *Chem. Eng. Sci.*, **59**, 2863 (2004)], (4) finite rates of vapor-liquid mass transfer [Baur, Taylor, and Krishna, *Chem. Eng. Sci.*, **56**, 2085 (2001); Nisoli, Doherty, and Malone, *AIChE J.*, **50**, 1795 (2004)], (5) column design and multiple steady-states [Guttinger, Dorn, and Morari, *Ind. Eng. Chem. Res.*, **36**, 794 (1997); Hauan, Hertzberg, and Lien, *Comput. Chem. Eng.*, **21**, 1117 (1997); Sneesby et al., *Ind. Eng. Chem. Res.*, **36**, 1855 (1997); Bessling et al., *Chem. Eng. Technol.*, **21**, 393 (1998); Okasinski and Doherty, *Ind. Eng. Chem. Res.*, **37**, 2821 (1998); Sneesby, Tade, and Smith, *Trans IChemE*, **76**, Part A, 525 (1998); Guttinger and Morari, *Ind. Eng. Chem. Res.*, **38**, 1633, 1649 (1999); Higler, Taylor, and Krishna, *Chem. Eng. Sci.*, **54**, 1389 (1999); Mohl et al., *Chem. Eng. Sci.*, **54**, 1029 (1999); Chen et al., *Comput. Chem. Eng.*, **26**, 81 (2002)]. Much useful information is available in Taylor and Krishna [*Chem. Eng. Sci.*, **55**, 5183 (2000)] and Sundmacher and Kienle [*Reactive Distillation*, Wiley-VCH, 2003].

Mechanical Design and Implementation Issues The choice of catalyst has a significant impact on the mechanical design and operation of the reactive column. The catalyst must allow the reaction to occur at reasonable rates at the relatively low temperatures and pressures common in distillation operations (typically less than 10 atm and between 50 and 250°C). Selection of a homogeneous catalyst, such as a high-boiling mineral acid, allows the use of more traditional tray designs and internals (albeit designed with exotic materials of construction to avoid corrosion, and allowance for high-liquid holdups to achieve suitable reaction contact times). With a homogeneous catalyst, lifetime is not a problem, as it is added (and withdrawn) continuously. Alternatively, heterogeneous solid catalysts

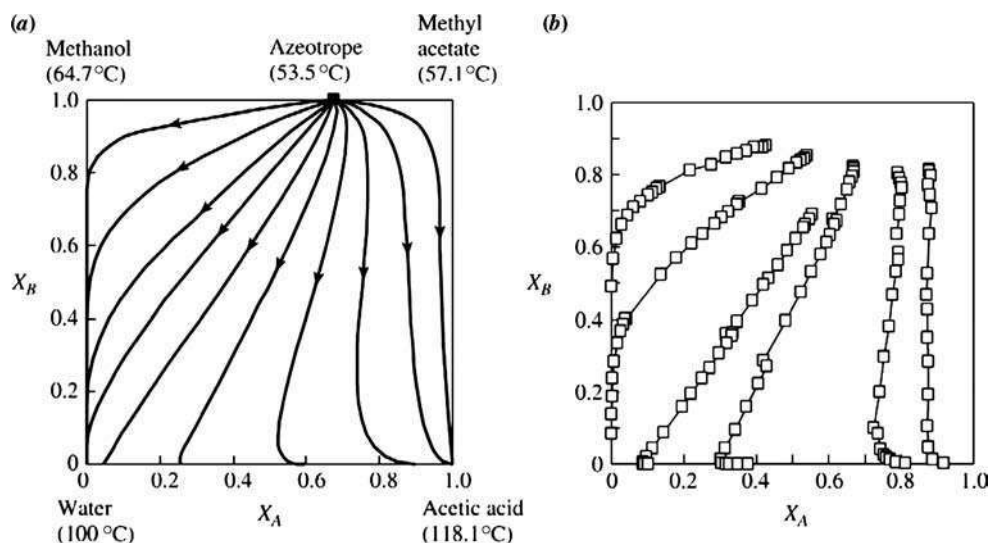


FIG. 13-96 Residue curve maps for the reactive system methanol–acetic acid–methyl acetate–water in phase and chemical equilibrium at 1-atm pressure. (a) Calculated by Barbosa and Doherty [*Chem. Eng. Sci.*, **43**, 1523 (1988)]. (b) Measured by Song et al. [*Ind. Eng. Chem. Res.*, **37**, 1917 (1998)].

TABLE 13-26 Catalyst Systems for Reactive Distillation

Description	Application	Reference
<i>Homogeneous catalysis</i>		
Liquid-phase mineral-acid catalyst added to column or reboiler	Esterifications Dibutyl phthalate Methyl acetate	Keyes, <i>Ind. Eng. Chem.</i> , 24 , 1096 (1932) Berman et al., <i>Ind. Eng. Chem.</i> , 40 , 2139 (1948) Agreda et al., U.S. Patent 4,435,595 (1984)
<i>Heterogeneous catalysis</i>		
Catalyst-resin beads placed in cloth bags attached to fiberglass strip. Strip wound around helical stainless steel mesh spacer	Etherifications Cumene	Smith et al., U.S. Patent 4,443,559 (1981) Shoemaker and Jones, <i>Hyd.</i> 57 (6), 57 (1987)
Ion exchange resin beads used as column packing	Hydrolysis of methyl acetate	Fuchigami, <i>J. Chem. Eng. Jap.</i> , 23 , 354 (1990)
Molecular sieves placed in bags or porous containers	Alkylation of aromatics	Crossland, U.S. Patent 5,043,506 (1991)
Ion exchange resins formed into Raschig rings	MTBE	Flato and Hoffman, <i>Chem. Eng. Tech.</i> , 15 , 193 (1992)
Granular catalyst resin loaded in corrugated sheet casings	Dimethyl acetals of formaldehyde	Zhang et al., Chinese Patent 1,065,412 (1992)
Trays modified to hold catalyst bed	MTBE	Sanfilippo et al., Eur. Pat. Appl. EP 470,625 (1992)
Distillation trays constructed of porous catalytically active material and reinforcing resins	None specified	Wang et al., Chinese Patent 1,060,228 (1992)
Method described for removing or replacing catalyst on trays as a liquid slurry	None specified	Jones, U.S. Patent, 5,133,942 (1992)
Catalyst bed placed in downcomer, designed to prevent vapor flow through bed	Etherifications, alkylations	Asselineau, Eur. Pat. Appl. EP 547,939 (1993)
Slotted plate for catalyst support designed with openings for vapor flow	None specified	Evans and Stark, Eur. Pat. Appl. EP 571,163 (1993)
Ion exchanger fibers (reinforced ion exchange polymer) used as solid-acid catalyst	Hydrolysis of methyl acetate	Hirata et al., Jap. Patent 05,212,290 (1993)
High-liquid holdup trays designed with catalyst bed extending below tray level, perforated for vapor-liquid contact	None specified	Yeoman et al., Int. Pat. Appl., WO 9408679 (1994)
Catalyst bed placed in downcomer, in-line withdrawal/addition system	None specified	Carland, U.S. Patent, 5,308,451 (1994)

require either complicated mechanical means for continuous replenishment or relatively long lifetimes to avoid constant maintenance. As with other multiphase reactors, use of a solid catalyst adds an additional resistance to mass transfer from the bulk liquid (or vapor) to the catalyst surface, which may be the limiting resistance. The catalyst containment system must be designed to ensure adequate liquid-solid contacting and minimize bypassing. A number of specialized column internal designs, catalyst containment methods, and catalyst replenishment systems have been proposed for both homogeneous and heterogeneous catalysts. A partial list of these methods is given in Table 13-26; see also the useful ideas presented by Krishna [*Chem. Eng. Sci.*, **57**, 1491 (2002); and chap. 7 in Sundmacher and Kienle, eds., *Reactive Distillation*, Wiley-VCH, 2003].

Heat management is another important consideration in the implementation of a reactive distillation process. Conventional reactors for highly exothermic or endothermic reactions are often designed as modified shell-and-tube heat exchangers for efficient heat transfer. However, a trayed or packed distillation column is a rather poor mechanical design for the management of the heat of reaction. Although heat can be removed or added in the condenser or reboiler easily, the only mechanism for heat transfer in the column proper is through vaporization (or condensation). For highly exothermic reactions, a large excess of reactants may be required as a heat sink, necessitating high-reflux rates and larger-diameter columns to return the vaporized reactants back to the reaction zone. Often a prereactor of conventional design is used to accomplish most of the reaction and heat removal before feeding to the reactive column for final conversion, as exemplified in most processes for the production of tertiary amyl methyl ether (TAME) [Brockwell et al., *Hyd. Proc.*, **70**(9), 133 (1991)]. Highly endothermic reactions may require intermediate reboilers. None of these heat management issues preclude the use of reactive distillation, but must be taken into account during the design phase. Comparison of heat of reaction and average heat of vaporization data for a system, as in Fig. 13-97, gives some indication of potential heat imbalances [Sundmacher, Rihko, and Hoffmann, *Chem. Eng. Comm.*, **127**, 151 (1994)]. The heat-neutral systems [$-\Delta H_{\text{react}} \approx \Delta H_{\text{vap}}$ (avg)]

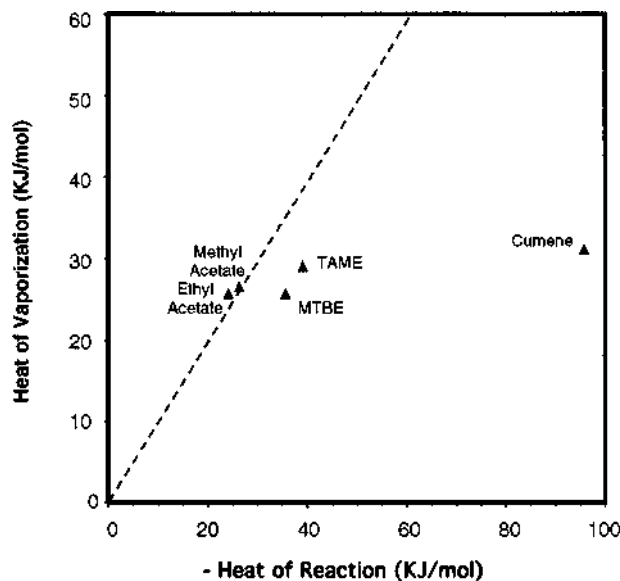


FIG. 13-97 Similarity of heats of reaction and vaporization for compounds made by reactive distillation.

such as methyl acetate and other esters can be accomplished in one reactive column, whereas the MTBE and TAME processes, with higher heats of reaction than that of vaporization, often include an additional prereactor. One exception is the catalytic distillation process for cumene production, which is accomplished without a prereactor. Three moles of benzene reactant are vaporized (and refluxed) for every mole of cumene produced. The relatively high heat of reaction is advantageous in this case as it reduces the overall

heat duty of the process by about 30 percent [Shoemaker and Jones, *Hyd. Proc.*, **57**(6), 57 (1987)].

Distillation columns with multiple conventional side reactors were first suggested by Schoenmakers and Buehler [*German Chem. Eng.*, **5**, 292 (1982)] and have the potential to accommodate gas-phase reactions, highly exo- or endothermic reactions, catalyst deactivation, and operating conditions outside the normal range suitable for distillation (e.g., short contact times, high temperature and pressure, etc). Krishna (chap. 7 in Sundmacher and Kienle, eds., *Reactive Distillation*, Wiley-VCH, 2003).

Process Applications The production of esters from alcohols and carboxylic acids illustrates many of the principles of reactive distillation as applied to equilibrium-limited systems. The true thermodynamic equilibrium constants for esterification reactions are usually in the range of 5 to 20. Large excesses of alcohols must be used to obtain acceptable yields, resulting in large recycle flow rates. In a reactive distillation scheme, the reaction is driven to completion by removal of the water of esterification. The method used for removal of the water depends on the boiling points, compositions, and liquid-phase behavior of any azeotropes formed between the products and reactants and largely dictates the structure of the reactive distillation flow sheet.

When the ester forms a binary low-boiling azeotrope with water or a ternary alcohol-ester-water azeotrope and that azeotrope is heterogeneous (or can be moved into the two-liquid phase region), the flow sheet illustrated in Fig. 13-98 can be used. Such a flow sheet works for the production of ethyl acetate and higher homologs. In this process scheme, acetic acid and the alcohol are continuously fed to the reboiler of the esterification column, along with a homogeneous strong-acid catalyst. Since the catalyst is largely nonvolatile, the reboiler acts as the primary reaction section. The alcohol is usually fed in slight excess to ensure complete reaction of the acid and to compensate for alcohol losses through distillation of the water-ester-(alcohol) azeotrope. The esterification column is operated such that the low-boiling, water-laden azeotrope is taken as the distillation product. Upon cooling, the distillate separates into two liquid phases. The aqueous layer is steam-stripped, with the organics recycled to the decanter or reactor. The ester layer from the decanter contains some water and possibly alcohol. Part of this layer may be refluxed to the esterification column. The remainder is fed to a low-boiler column where the water-ester and alcohol-ester azeotropes are removed overhead and recycled to the decanter or reactor. The dry, alcohol-free ester is then optionally taken overhead in a final refining column. Additional literature on the application of reactive distillation to ester production includes papers by Hanika, Kolenka, and Smejkal [*Chem. Eng. Sci.*, **54**, 5205 (1999)], Schwarzer and Hoffmann [*Chem. Eng. Technol.*, **25**, 975 (2002)], Steingeweg and Gmehling [*Ind. Eng.*

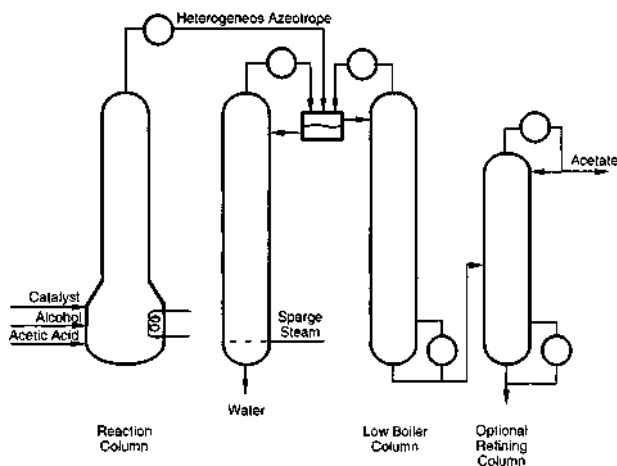


FIG. 13-98 Flow sheet for making esters which form a heterogeneous minimum-boiling azeotrope with water.

Chem. Res., **41**, 5483 (2002)], and Omata, Dimian, and Blik [*Chem. Eng. Sci.*, **58**, 3159, 3175 (2003)].

Methyl acetate cannot be produced in high purity by using the simple esterification scheme described above. The methyl acetate-methanol-water system does not exhibit a ternary minimum-boiling azeotrope, the methyl acetate-methanol azeotrope is lower-boiling than the water-methyl acetate azeotrope, a distillation boundary extends between these two binary azeotropes, and the heterogeneous region does not include either azeotrope, nor does it cross the distillation boundary. Consequently, the water of esterification cannot be removed effectively, and methyl acetate cannot be separated from the methanol and water azeotropes by a simple decantation in the same manner as outlined above. Conventional sequential reaction-separation processes rely on large excesses of acetic acid to drive the reaction to higher conversion to methyl acetate, necessitating a capital- and energy-intensive acetic acid-water separation and large recycle streams. The crude methyl acetate product, contaminated with water and methanol, can be purified by a number of enhanced distillation techniques such as pressure-swing distillation (Harrison, U.S. Patent 2,704,271, 1955), extractive distillation with ethylene glycol monomethylether as the solvent (Kumerle, German Patent 1,070,165, 1959), or azeotropic distillation with an aromatic or ketone entrainer (Yeomans, Eur. Patent Appl. 060717 and 060719, 1982). The end result is a capital- and energy-intensive process typically requiring multiple reactors and distillation columns.

The reactive distillation process in Fig. 13-99 provides a mechanism for overcoming both the limitations on conversion due to chemical

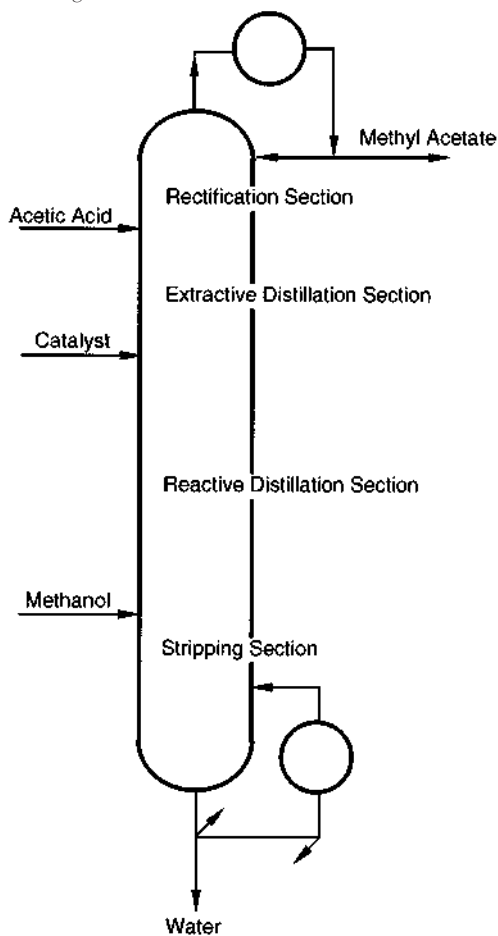


FIG. 13-99 Integrated reactive-extractive distillation column for the production of methyl acetate.

equilibrium as well as the difficulties in purification imposed by the water–methyl acetate and methanol–methyl acetate azeotropes [Agreda and Partin, U.S. Patent 4,435,595, 1984; Agreda, Partin, and Heise, *Chem. Eng. Prog.*, **86**(2), 40 (1990)]. Conceptually, this flow sheet can be thought of as four heat-integrated distillation columns (one of which is also a reactor) stacked on top of each other. The primary reaction zone consists of a series of countercurrent flashing stages in the middle of the column. Adequate residence time for the reaction is provided by high-liquid-holdup bubble cap trays with specially designed downcomer sumps to further increase tray holdup. A nonvolatile homogeneous catalyst is fed at the top of the reactive section and exits with the underflow water by-product. The extractive distillation section, immediately above the reactive section, is critical in achieving high methyl acetate purity. As shown in Fig. 13-96, simultaneous reaction and distillation eliminates the water–methyl acetate azeotrope (and the distillation boundary of the nonreactive system). However, pure methyl acetate remains a saddle in the reactive system and cannot be obtained as a pure component by simple reactive distillation. The acetic acid feed acts as a solvent in an extractive-distillation section placed above the reaction section, breaking the methanol–methyl acetate azeotrope, and yielding a pure methyl acetate distillate product. The uppermost rectification stages serve to remove any acetic acid from the methyl acetate product, and the bottommost stripping section removes any methanol and methyl acetate from the water by-product. The countercurrent flow of the reactants results in high local excesses at each end of the reactive section, even though the overall feed to the reactive column is stoichiometric. Therefore, the large excess of acetic acid at the top of the reactive section prevents methanol from reaching the distillate; similarly, methanol at the bottom of the reactive section keeps acetic acid from the water bottoms. Temperature and composition profiles for this reactive extractive distillation column are shown in Fig. 13-100*a* and *b*, respectively.

Much has been written about this reactive distillation scheme, including works by Bessling et al. [*Chem. Eng. Tech.*, **21**, 393 (1998)], Song et al., [*Ind. Eng. Chem. Res.*, **37**, 1917 (1998)], Huss et al. [*Comput. Chem. Eng.*, **27**, 1855 (2003)], Siirola ("An Industrial Perspective on Process Synthesis," pp. 222–233 in Biegler and Doherty, eds., *Foundations of Computer-Aided Process Design*, AIChE, New York, 1995), and Krishna (chap. 7 in Sundmacher and Kienle, eds., *Reactive Distillation*, Wiley-VCH, 2003).

SYNTHESIS OF MULTICOMPONENT SEPARATION SYSTEMS

The sequencing of distillation columns and other types of equipment for the separation of multicomponent mixtures has received much attention in recent years. Although one separator of complex design can sometimes be devised to produce more than two products, more often a sequence of two-product separators is preferable. Often, the sequence includes simple distillation columns. A summary of sequencing methods, prior to 1977, that can lead to optimal or near-optimal designs, is given by Henley and Seader (op. cit.). Modern methods for distillation column sequencing are reviewed by Modi and Westerberg [*Ind. Eng. Chem. Res.*, **31**, 839 (1992)], who also present a more generally applicable method based on a marginal price that is the change in price of a separation operation when the separation is carried out in the absence of nonkey components. The synthesis of sequences that consider a wide range of separation operations in a knowledge-based approach is given by Barnicki and Fair for liquid mixtures [*Ind. Eng. Chem. Res.*, **29**, 421 (1990)] and for gas/vapor mixtures [*Ind. Eng. Chem. Res.*, **31**, 1679 (1992)]. The problem decomposition approach of Wahnschafft, Le Rudulier, and Westerberg [*Ind. Eng. Chem. Res.*, **32**, 1121 (1993)] is directed to the synthesis of complex separation sequences that involve nonsharp splits and recycle, including azeotropic distillation. The method is applied by using a computer-aided separation process designer called *SPLIT*. An expert system, called *EXSEP*, for the synthesis of solvent-based separation trains is presented by Brunet and Liu [*Ind. Eng. Chem. Res.*, **32**, 315 (1993)]. The use of ternary composition diagrams and residue curve maps is reviewed and evaluated for application to the synthesis of complex separation sequences by Fien and Liu [*Ind. Eng. Chem. Res.*, **33**, 2506 (1994)]. In recent years many optimization-based process synthesis schemes have been proposed for distillation systems. The key research groups are led by Grossmann at Carnegie Mellon University, Manousiouthakis at UCLA, and Pistikopoulos at Imperial College, London. Further information about their methods and availability of computer programs can be obtained from the principals.

Synthesis schemes for reactive distillation have been proposed by Ismail, Proios, and Pistikopoulos [*AIChE J.*, **47**, 629 (2001)], Jackson and Grossmann [*Comput. Chem. Eng.*, **25**, 1661 (2001)], Schembecker and Tlatlik [*Chem. Eng. Process.*, **42**, 179 (2003)], and Burri and Manousiouthakis [*Comput. Chem. Eng.*, **28**, 2509 (2004)].

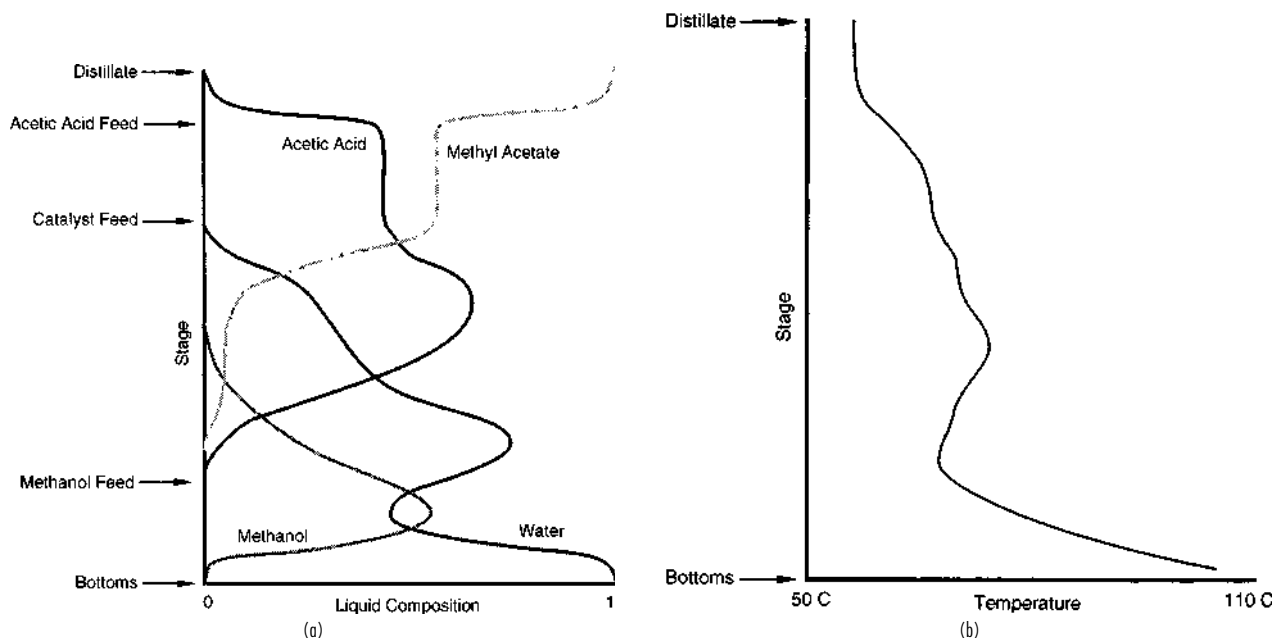


FIG. 13-100 Reactive extractive distillation for methyl acetate production. (a) Composition profile. (b) Temperature profile.

PETROLEUM AND COMPLEX-MIXTURE DISTILLATION

Although the principles of multicomponent distillation apply to petroleum, synthetic crude oil, and other complex mixtures, this subject warrants special consideration for the following reasons:

1. Such feedstocks are of exceedingly complex composition, consisting of, in the case of petroleum, many different types of hydrocarbons and perhaps of inorganic and other organic compounds. The number of carbon atoms in the components may range from 1 to more than 50, so that the compounds may exhibit atmospheric-pressure boiling points from -162°C (-259°F) to more than 538°C (1000°F). In a given boiling range, the number of different compounds that exhibit only small differences in volatility multiplies rapidly with increasing boiling point. For example, 16 of the 18 octane isomers boil within a range of only 12°C (22°F).

2. Products from the distillation of complex mixtures are in themselves complex mixtures. The character and yields of these products vary widely, depending upon the source of the feedstock. Even crude oils from the same locality may exhibit marked variations.

3. The scale of petroleum-distillation operations is generally large, and as discussed in detail by Nelson (*Petroleum Refinery Engineering*, 4th ed., McGraw-Hill, New York, 1958) and Watkins (*Petroleum Refinery Distillation*, 2d ed., Gulf, Houston, 1979), such operations are common in several petroleum refinery processes including atmospheric distillation of crude oil, vacuum distillation of bottoms residuum obtained from atmospheric distillation, main fractionation of gaseous effluent from catalytic cracking of various petroleum fractions, and main fractionation of effluent from thermal coking of various petroleum fractions. These distillation operations are conducted in large pieces of equipment that can consume large quantities of energy. Therefore, optimization of design and operation is very important and frequently leads to a relatively complex equipment configuration.

CHARACTERIZATION OF PETROLEUM AND PETROLEUM FRACTIONS

Although much progress has been made in identifying the chemical species present in petroleum, it is generally sufficient for purposes of design and analysis of plant operation of distillation to characterize petroleum and petroleum fractions by gravity, laboratory distillation curves, component analysis of light ends, and hydrocarbon-type analysis of middle and heavy ends. From such data, as discussed in the *Technical Data Book—Petroleum Refining* [American Petroleum Institute (API), Washington], five different average boiling points and an index of paraffinicity can be determined. These are then used to predict the physical properties of complex mixtures by a number of well-accepted correlations, whose use will be explained in detail and illustrated with examples. Many other characterizing properties or attributes such as sulfur content, pour point, water and sediment content, salt content, metals content, Reid vapor pressure, Saybolt Universal viscosity, aniline point, octane number, freezing point, cloud point, smoke point, diesel index, refractive index, cetane index, neutralization number, wax content, carbon content, and penetration are generally measured for a crude oil or certain of its frac-

tions according to well-specified ASTM tests. But these attributes are of much less interest here even though feedstocks and products may be required to meet certain specified values of the attributes.

Gravity of a crude oil or petroleum fraction is generally measured by the ASTM D 287 test or the equivalent ASTM D 1298 test and may be reported as specific gravity (SG) $60/60^{\circ}\text{F}$ [measured at 60°F (15.6°C) and referred to water at 60°F (15.6°C)] or, more commonly, as API gravity, which is defined as

$$\text{API gravity} = 141.5/(\text{SG } 60/60^{\circ}\text{F}) - 131.5 \quad (13-120)$$

Water, thus, has an API gravity of 10.0, and most crude oils and petroleum fractions have values of API gravity in the range of 10 to 80. Light hydrocarbons (*n*-pentane and lighter) have values of API gravity ranging upward from 92.8.

The volatility of crude oil and petroleum fractions is characterized in terms of one or more laboratory distillation tests that are summarized in Table 13-27. The ASTM D 86 and D 1160 tests are reasonably rapid batch laboratory distillations involving the equivalent of approximately one equilibrium stage and no reflux except for that caused by heat losses. Apparatus typical of the D 86 test is shown in Fig. 13-101 and consists of a heated 100-mL or 125-mL Engler flask containing a calibrated thermometer of suitable range to measure the temperature of the vapor at the inlet to the condensing tube, an inclined brass condenser in a cooling bath using a suitable coolant, and a graduated cylinder for collecting the distillate. A stem correction is not applied to the temperature reading. Related tests using similar apparatus are the D 216 test for natural gasoline and the Engler distillation.

In the widely used ASTM D 86 test, 100 mL of sample is charged to the flask and heated at a sufficient rate to produce the first drop of distillate from the lower end of the condenser tube in 5 to 15 min, depending on the nature of the sample. The temperature of the vapor at that instant is recorded as the initial boiling point (IBP). Heating is continued at a rate such that the time from the IBP to 5 vol % recovered of the sample in the cylinder is 60 to 75 s. Again, vapor temperature is recorded. Then successive vapor temperatures are recorded for 10 to 90 percent recovered in intervals of 10, and at 95 percent recovered, with the heating rate adjusted so that 4 to 5 mL is collected per minute. At 95 percent recovered, the burner flame is increased if necessary to achieve a maximum vapor temperature, referred to as the endpoint (EP) in 3 to 5 additional min. The percent recovery is reported as the maximum percent recovered in the cylinder. Any residue remaining in the flask is reported as percent residue, and percent loss is reported as the difference between 100 mL and the sum of the percent recovery and percent residue. If the atmosphere test pressure P is other than 101.3 kPa (760 torr), temperature readings may be adjusted to that pressure by the Sidney Young equation, which for degrees Fahrenheit is

$$T_{760} = T_p + 0.00012(760 - P)(460 + T_p) \quad (13-121)$$

Another pressure correction for percent loss can also be applied, as described in the ASTM test method.

TABLE 13-27 Laboratory Distillation Tests

Test name	Reference	Main applicability
ASTM (atmospheric)	ASTM D 86	Petroleum fractions or products, including gasolines, turbine fuels, naphthas, kerosines, gas oils, distillate fuel oils, and solvents that do not tend to decompose when vaporized at 760 mmHg
ASTM [vacuum, often 10 torr (1.3 kPa)]	ASTM D 1160	Heavy petroleum fractions or products that tend to decompose in the ASTM D 86 test but can be partially or completely vaporized at a maximum liquid temperature of 750°F (400°C) at pressures down to 1 torr (0.13 kPa)
TBP [atmospheric or 10 torr (1.3 kPa)]	Nelson,* ASTM D 2892	Crude oil and petroleum fractions
Simulated TBP (gas chromatography)	ASTM D 2887	Crude oil and petroleum fractions
EFV (atmospheric, superatmospheric, or subatmospheric)	Nelson†	Crude oil and petroleum fractions

*Nelson, *Petroleum Refinery Engineering*, 4th ed., McGraw-Hill, New York, 1958, pp. 95–99.

†Ibid., pp. 104–105.

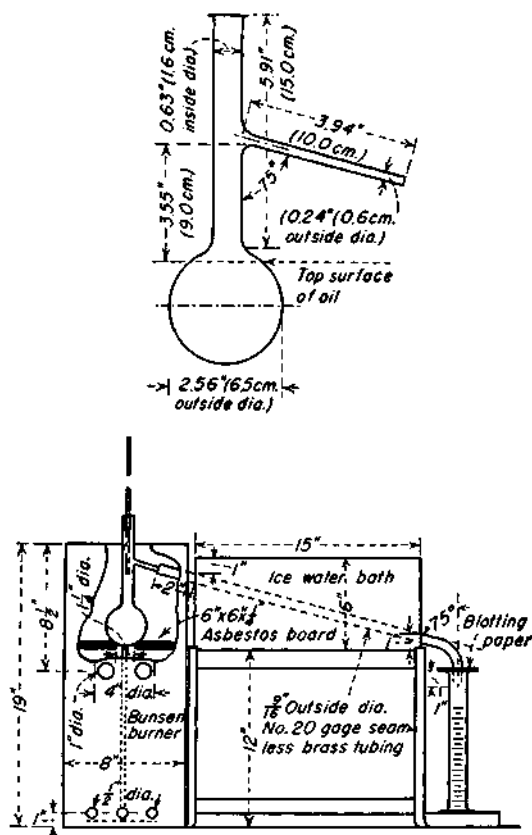


FIG. 13-101 ASTM distillation apparatus; detail of distilling flask is shown in the upper figure.

Results of a typical ASTM distillation test for an automotive gasoline are given in Table 13-28, in which temperatures have already been corrected to a pressure of 101.3 kPa (760 torr). It is generally assumed that percent loss corresponds to volatile noncondensables that are distilled off at the beginning of the test. In that case, the percent recovered values in Table 13-28 do not correspond to percent evaporated values, which are of greater scientific value. Therefore, it

TABLE 13-28 Typical ASTM D 86 Test Results for Automobile Gasoline Pressure, 760 torr (101.3 kPa)

Percent recovered basis (as measured)			Percent evaporated basis (as corrected)		
Percent recovered	T, °F	Percent evaporated	Percent evaporated	T, °F	Percent recovered
0 (IBP)	98	1.5	1.5	98	(IBP)
5	114	6.5	5	109	3.5
10	120	11.5	10	118	8.5
20	150	21.5	20	146	18.5
30	171	31.5	30	168	28.5
40	193	41.5	40	190	38.5
50	215	51.5	50	212	48.5
60	243	61.5	60	239	58.5
70	268	71.5	70	264	68.5
80	300	81.5	80	295	78.5
90	340	91.5	90	334	88.5
95	368	96.5	95	360	93.5
EP	408			408	(EP)

NOTE: Percent recovery = 97.5; percent residue = 1.0; percent loss = 1.5. To convert degrees Fahrenheit to degrees Celsius, $^{\circ}\text{C} = (^{\circ}\text{F} - 32)/1.8$.

is common to adjust the reported temperatures according to a linear interpolation procedure given in the ASTM test method to obtain corrected temperatures in terms of percent evaporated at the standard intervals as included in Table 13-28. In the example, the corrections are not large because the loss is only 1.5 vol %.

Although most crude petroleum can be heated to 600°F (316°C) without noticeable cracking, when ASTM temperatures exceed 475°F (246°C), fumes may be evolved, indicating decomposition, which may cause thermometer readings to be low. In that case, the following correction attributed to S. T. Hadden may be applied:

$$\Delta T_{\text{corr}} = 10^{-1.587 + 0.004735T} \quad (13-122)$$

where

T = measured temperature, °F

ΔT_{corr} = correction to be added to T , °F

At 500 and 600°F (260 and 316°C), the corrections are 6 and 18°F (3.3 and 10°C), respectively.

As discussed by Nelson (op. cit.), virtually no fractionation occurs in an ASTM distillation. Thus, components in the mixture do distill one by one in the order of their boiling points but as mixtures of successively higher boiling points. The IBP, EP, and intermediate points have little theoretical significance, and, in fact, components boiling below the IBP and above the EP are present in the sample. Nevertheless, because ASTM distillations are quickly conducted, have been successfully automated, require only a small sample, and are quite reproducible, they are widely used for comparison and as a basis for specifications on a large number of petroleum intermediates and products, including many solvents and fuels. Typical ASTM curves for several such products are shown in Fig. 13-102.

Data from a true boiling point (TBP) distillation test provide a much better theoretical basis for characterization. If the sample contains compounds that have moderate differences in boiling points such as in

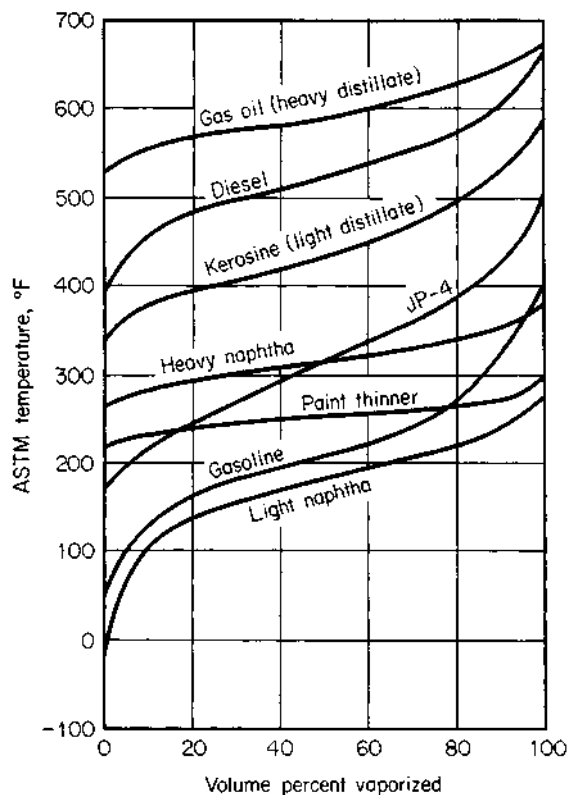


FIG. 13-102 Representative ASTM D 86 distillation curves.

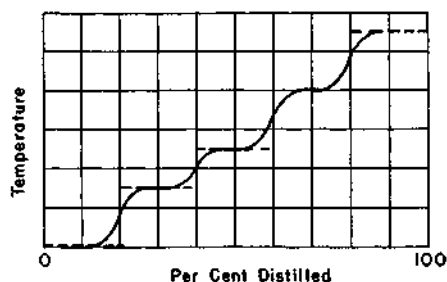


FIG. 13-103 Variation of boiling temperature with percent distilled in TBP distillation of light hydrocarbons.

a light gasoline containing light hydrocarbons (e.g., isobutane, *n*-butane, isopentane), a plot of overhead vapor distillate temperature versus percent distilled in a TBP test would appear in the form of steps as in Fig. 13-103. However, if the sample has a higher average boiling range when the number of close-boiling isomers increases, the steps become indistinct and a TBP curve such as that in Fig. 13-104 results. Because the degree of separation for a TBP distillation test is much higher than that for an ASTM distillation test, the IBP is lower and the EP is higher for the TBP method as compared with the ASTM method, as shown in Fig. 13-104.

A standard TBP laboratory distillation test method has not been well accepted. Instead, as discussed by Nelson (op. cit., pp. 95-99), batch distillation equipment that can achieve a good degree of fractionation is usually considered suitable. In general, TBP distillations are conducted in columns with 15 to 100 theoretical stages at reflux ratios of 5 or greater. Thus, the new ASTM D 2892 test method, which involves a column with 14 to 17 theoretical stages and a reflux ratio of 5, essentially meets the minimum requirements. Distillate may be collected at a constant or a variable rate. Operation may be at 101.3 kPa (760-torr) pressure or at a vacuum at the top of the column as low as 0.067 kPa (0.5 torr) for high-boiling fractions, with 1.3 kPa (10 torr) being common. Results from vacuum operation are extrapolated to 101.3 kPa (760 torr) by the vapor-pressure correlation of

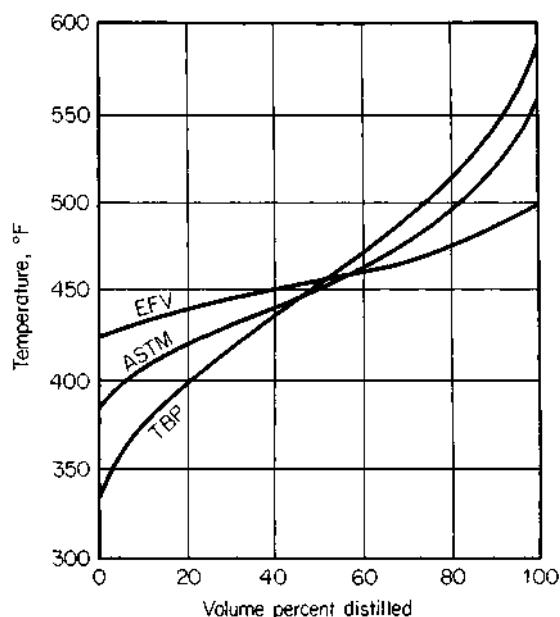


FIG. 13-104 Comparison of ASTM, TBP, and EFV distillation curves for kerosine.

Maxwell and Bonner [*Ind. Eng. Chem.*, **49**, 1187 (1957)], which is given in great detail in the *API Technical Data Book—Petroleum Refining* (op. cit.) and in the ASTM D 2892 test method. It includes a correction for the nature of the sample (paraffin, olefin, naphthene, and aromatic content) in terms of the UOP characterization factor, UOP-K, as given by

$$\frac{\text{UOP-K} = (T_B)^{1/3}}{\text{SG}} \quad (13-123)$$

where T_B is the mean average boiling point in degrees Rankine, which is the arithmetic average of the molal average boiling point and the cubic volumetric average boiling point. Values of UOP-K for *n*-hexane, 1-hexene, cyclohexene, and benzene are 12.82, 12.49, 10.99, and 9.73, respectively. Thus, paraffins with their lower values of specific gravity tend to have high values, and aromatics tend to have low values of UOP-K. A movement toward an international TBP standard is discussed by Vercier and Mouton [*Oil Gas J.*, **77**(38), 121 (1979)].

A crude oil assay always includes a whole crude API gravity and a TBP curve. As discussed by Nelson (op. cit., pp. 89-90) and as shown in Fig. 13-105, a reasonably consistent correlation (based on more than 350 distillation curves) exists between whole crude API gravity and the TBP distillation curve at 101.3 kPa (760 torr). Exceptions not correlated by Fig. 13-105 are highly paraffinic or naphthenic crude oils.

An alternative to TBP distillation is simulated distillation by gas chromatography. As described by Green, Schmauch, and Worman [*Anal. Chem.*, **36**, 1512 (1965)] and Worman and Green [*Anal. Chem.*, **37**, 1620 (1965)], the method is equivalent to a 100-theoretical-plate TBP distillation; is very rapid, reproducible, and easily automated; requires only a small microliter sample; and can better define initial and final boiling points. The ASTM D 2887 standard test method is based on such a simulated distillation and is applicable to samples having a boiling range greater than 55°C (100°F) for temperature determinations as high as 538°C (1000°F). Typically, the test is conducted with a gas chromatograph having a thermal conductivity detector, a programmed temperature capability, helium or hydrogen carrier gas, and column packing of silicone gum rubber on a crushed firebrick or diatomaceous earth support.

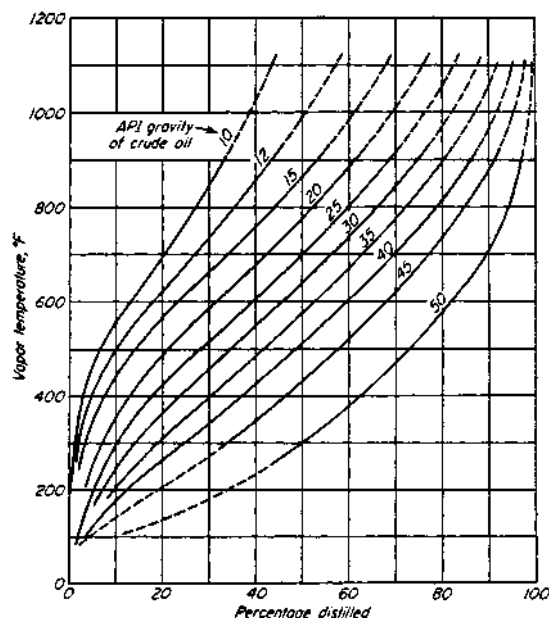


FIG. 13-105 Average true-boiling-point distillation curves of crude oils. (From W. E. Edmister, *Applied Hydrocarbon Thermodynamics*, vol. 1, 1st ed., 1961 Gulf Publishing Company, Houston, Texas, Used with permission. All rights reserved.)

It is important to note that simulated distillation does not always separate hydrocarbons in the order of their boiling points. For example, high-boiling multiple-ring-type compounds may be eluted earlier than normal paraffins (used as the calibration standard) of the same boiling point. Gas chromatography is also used in the ASTM D 2427 test method to determine quantitatively ethane through pentane hydrocarbons.

A third fundamental type of laboratory distillation, which is the most tedious to perform of the three types of laboratory distillations, is equilibrium flash vaporization (EFV), for which no standard test exists. The sample is heated in such a manner that the total vapor produced remains in contact with the total remaining liquid until the desired temperature is reached at a set pressure. The volume percent vaporized at these conditions is recorded. To determine the complete flash curve, a series of runs at a fixed pressure is conducted over a range of temperatures sufficient to cover the range of vaporization from 0 to 100 percent. As seen in Fig. 13-104, the component separation achieved by an EFV distillation is much less than that by the ASTM or TBP distillation tests. The initial and final EFV points are the bubble point and the dew point, respectively, of the sample. If desired, EFV curves can be established at a series of pressures.

Because of the time and expense involved in conducting laboratory distillation tests of all three basic types, it has become increasingly common to use empirical correlations to estimate the other two distillation curves when the ASTM, TBP, or EFV curve is available. Preferred correlations given in the *API Technical Data Book—Petroleum Refining* (op. cit.) are based on the work of Edmister and Pollock [*Chem. Eng. Prog.*, **44**, 905 (1948)], Edmister and Okamoto [*Pet. Refiner*, **38**(8), 117 (1959); **38**(9), 271 (1959)], Maxwell (*Data Book on Hydrocarbons*, Van Nostrand, Princeton, N.J., 1950), and Chu and Staffell [*J. Inst. Pet.*, **41**, 92 (1955)]. Because of the lack of sufficiently precise and consistent data on which to develop the correlations, they are, at best, first approximations and should be used with caution. Also, they do not apply to mixtures containing only a few components of widely different boiling points. Perhaps the most useful correlation of the group is Fig. 13-106 for converting between ASTM D 86 and

TBP distillations of petroleum fractions at 101.3 kPa (760 torr). The ASTM D 2889 test method, which presents a standard method for calculating EFV curves from the results of an ASTM D 86 test for a petroleum fraction having a 10 to 90 vol % boiling range of less than 55°C (100°F), is also quite useful.

APPLICATIONS OF PETROLEUM DISTILLATION

Typical equipment configurations for the distillation of crude oil and other complex hydrocarbon mixtures in a crude unit, a catalytic cracking unit, and a delayed coking unit of a petroleum refinery are shown in Figs. 13-107, 13-108, and 13-109. The initial separation of crude oil into fractions is conducted in two main columns, shown in Fig. 13-107. In the first column, called the atmospheric tower or topping still, partially vaporized crude oil, from which water, sediment, and salt have been removed, is mainly rectified, at a feed tray pressure of no more than about 276 kPa (40 psia), to yield a noncondensable light-hydrocarbon gas, a light naphtha, a heavy naphtha, a light distillate (kerosine), a heavy distillate (diesel oil), and a bottoms residual of components whose TBP exceeds approximately 427°C (800°F). Alternatively, other fractions, shown in Fig. 13-102, may be withdrawn. To control the IBP of the ASTM D 86 curves, each of the sidestreams of the atmospheric tower and the vacuum and main fractionators of Figs. 13-107, 13-108, and 13-109 may be sent to side-cut strippers, which use a partial reboiler or steam stripping. Additional stripping by steam is commonly used in the bottom of the atmospheric tower as well as in the vacuum tower and other main fractionators.

Additional distillate in the TBP range of approximately 427 to 593°C (800 to 1100°F) is recovered from bottoms residuum of the atmospheric tower by rectification in a vacuum tower, also shown in Fig. 13-107, at the minimum practical overhead condenser pressure, which is typically 1.3 kPa (10 torr). Use of special low-pressure-drop trays or column packing permits the feed tray pressure to be approximately 5.3 to 6.7 kPa (40 to 50 torr) to obtain the maximum degree of vaporization. Vacuum towers may be designed or operated to produce several different products including heavy distillates, gas-oil feedstocks for catalytic cracking, lubricating oils, bunker fuel, and bottoms residua of asphalt (5 to 8 API gravity) or pitch (0 to 5 API gravity). The catalytic cracking process of Fig. 13-108 produces a superheated vapor at approximately 538°C (1000°F) and 172 to 207 kPa (25 to 30 psia) of a TBP range that covers hydrogen to compounds with normal boiling points above 482°C (900°F). This gas is sent directly to a main fractionator for rectification to obtain products that are typically gas and naphtha [204°C (400°F) ASTM EP approximately], which are often fractionated further to produce relatively pure light hydrocarbons and gasoline; a light cycle oil [typically 204 to 371°C (400 to 700°F) ASTM D 86 range], which may be used for heating oil, hydrocracked, or recycled to the catalytic cracker; an intermediate cycle oil [typically 371 to 482°C (700 to 900°F) ASTM D 86 range], which is generally recycled to the catalytic cracker to extinction; and a heavy gas oil or bottom slurry oil.

Vacuum-column bottoms, bottoms residuum from the main fractionation of a catalytic cracker, and other residua can be further processed at approximately 510°C (950°F) and 448 kPa (65 psia) in a delayed-coker unit, as shown in Fig. 13-109, to produce petroleum coke and gas of TBP range that covers methane (with perhaps a small amount of hydrogen) to compounds with normal boiling points that may exceed 649°C (1200°F). The gas is sent directly to a main fractionator that is similar to the type used in conjunction with a catalytic cracker, except that in the delayed-coking operation the liquid to be coked first enters into and passes down through the bottom trays of the main fractionator to be preheated by and to scrub coker vapor of entrained coke particles and condensables for recycling to the delayed coker. Products produced from the main fractionator are similar to those produced in a catalytic cracking unit, except for more unsaturated cyclic compounds, and include gas and coker naphtha, which are further processed to separate out light hydrocarbons and a coker naphtha that generally needs hydrotreating; and light and heavy coker gas oils, both of which may require hydrocracking to become suitable blending stocks.

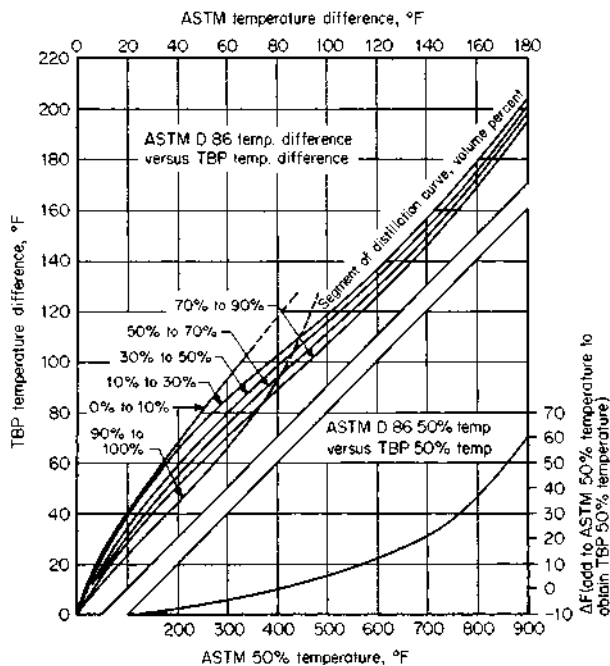


FIG. 13-106 Relationship between ASTM and TBP distillation curves. (From W. C. Edmister, *Applied Hydrocarbon Thermodynamics*, vol. 1, 1st ed., 1961 Gulf Publishing Company, Houston, Tex. Used with permission. All rights reserved.)

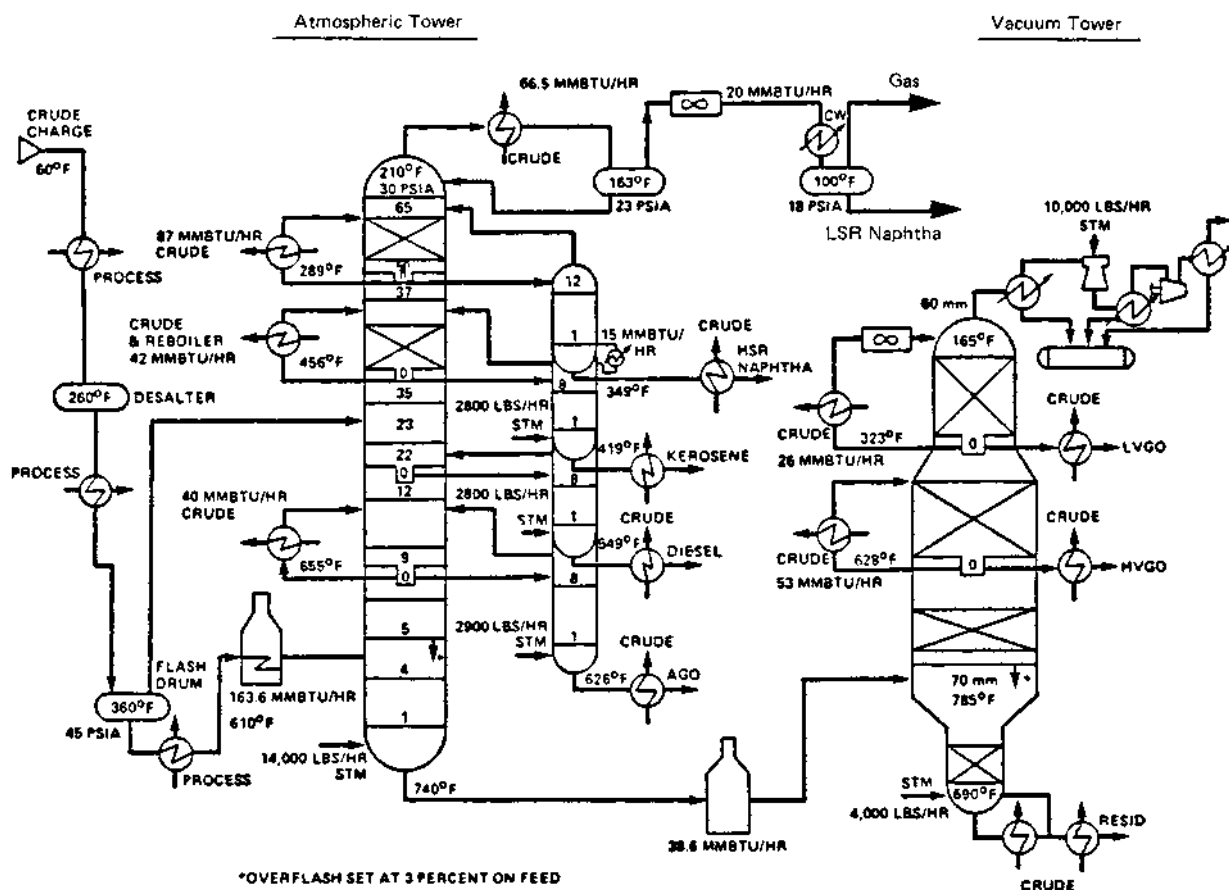


FIG. 13-107 Crude unit with atmospheric and vacuum towers. [Kleinschrodt and Hammer, Chem. Eng. Prog., 79(7), 33 (1983).]

DESIGN PROCEDURES

Two general procedures are available for designing fractionators that process petroleum, synthetic crude oils, and complex mixtures. The first, which was originally developed for crude units by Packie [Trans. Am. Inst. Chem. Eng. J., 37, 51 (1941)], extended to main fractionators by Houghland, Lemieux, and Schreiner [Proc. API, sec. III, Refining, 385 (1954)], and further elaborated and described in great detail by Watkins (op. cit.), uses material and energy balances, with empirical correlations to establish tray requirements, and is essentially a hand calculation procedure that is a valuable learning experience and is suitable for preliminary designs. Also, when backed by sufficient experience from previous designs, this procedure is adequate for final design.

In the second procedure, which is best applied with a digital computer, the complex mixture being distilled is represented by actual components at the light end and by perhaps 30 pseudocomponents (e.g., petroleum fractions) over the remaining portion of the TBP distillation curve for the column feed. Each of the pseudocomponents is characterized by a TBP range, an average normal boiling point, an average API gravity, and an average molecular weight. Rigorous material balance, energy balance, and phase equilibrium calculations are then made by an appropriate equation-tearing method, as shown by Cecchetti et al. [Hydrocarbon Process., 42(9), 159 (1963)] or a simultaneous-correction procedure as shown, e.g., by Goldstein and Stanfield [Ind. Eng. Chem. Process Des. Dev., 9, 78 (1970)] and Hess et al. [Hydrocarbon Process., 56(5), 241 (1977)]. Highly developed procedures of the latter type, suitable for preliminary or final design, are

included in most computer-aided steady-state process design and simulation programs as a special case of interlinked distillation, wherein the crude tower or fractionator is converged simultaneously with the sidecut stripper columns.

Regardless of the procedure used, certain initial steps must be taken for the determination or specification of certain product properties and yields based on the TBP distillation curve of the column feed, method of providing column reflux, column-operating pressure, type of condenser, and type of sidecut strippers and stripping requirements. These steps are developed and illustrated with several detailed examples by Watkins (op. cit.). Only one example, modified from one given by Watkins, is considered briefly here to indicate the approach taken during the initial steps.

For the atmospheric tower shown in Fig. 13-110, suppose distillation specifications are as follows:

- Feed: 50,000 bbl (at 42 U.S. gal each) per stream day (BPSD) of 31.6 API crude oil.
- Measured light-ends analysis of feed:

Component	Volume percent of crude oil
Ethane	0.04
Propane	0.37
Isobutane	0.27
n-Butane	0.89
Isopentane	0.77
n-Pentane	1.13
	3.47

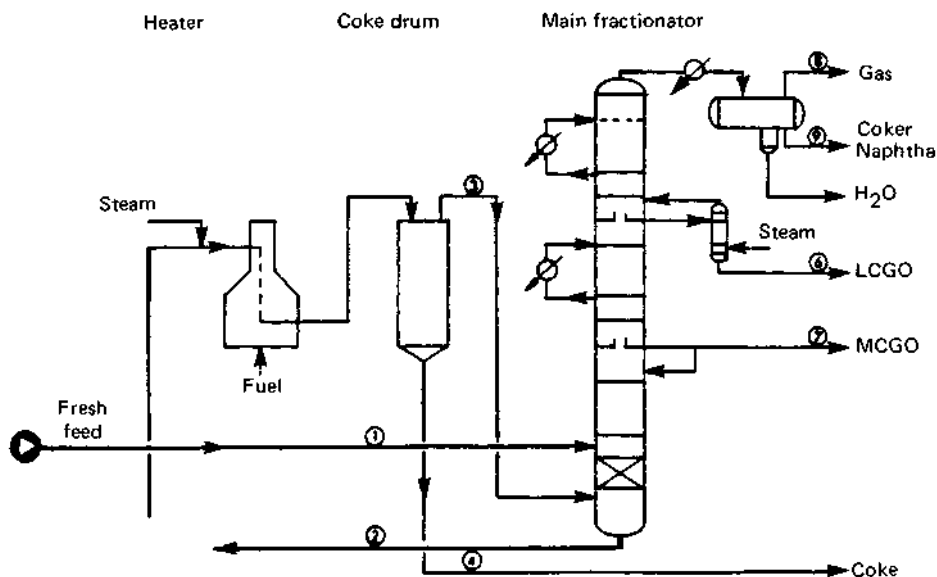


FIG. 13-109 Delayed-coking unit. (Watkins, *Petroleum Refinery Distillation*, 2d ed., Gulf, Houston, Tex., 1979).

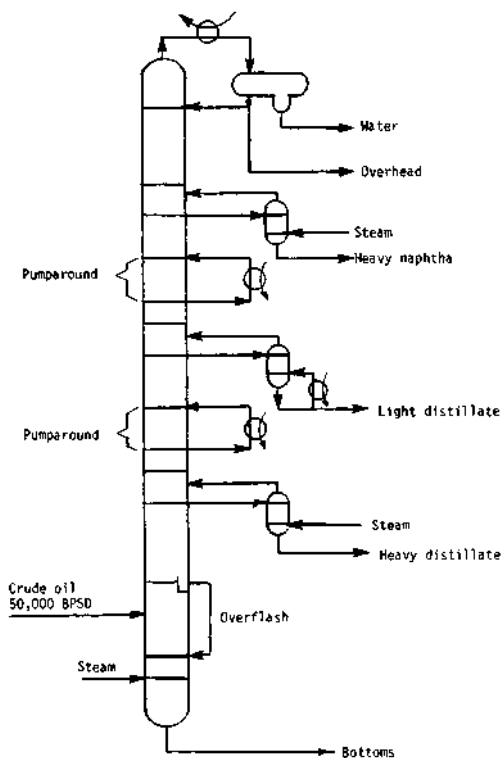


FIG. 13-110 Crude atmospheric tower.

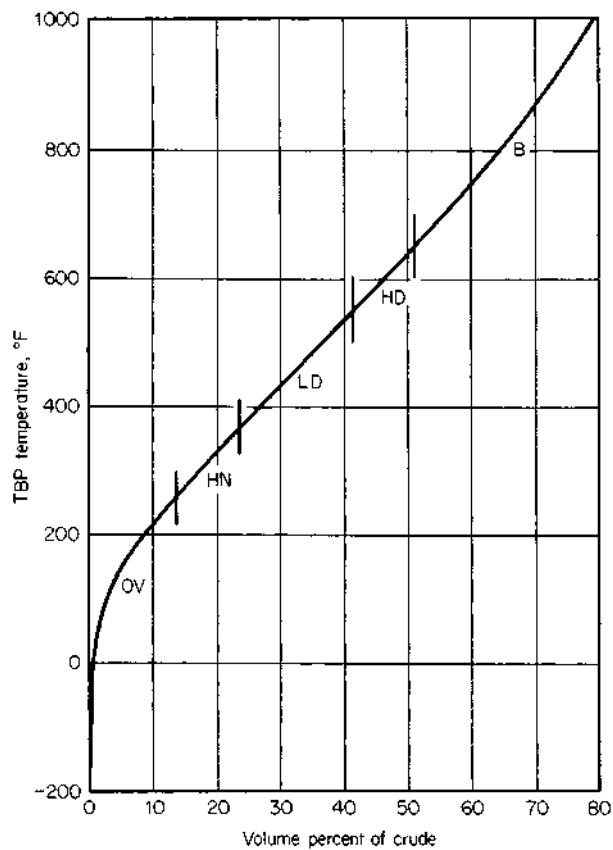


FIG. 13-111 Example of crude oil TBP cut points.

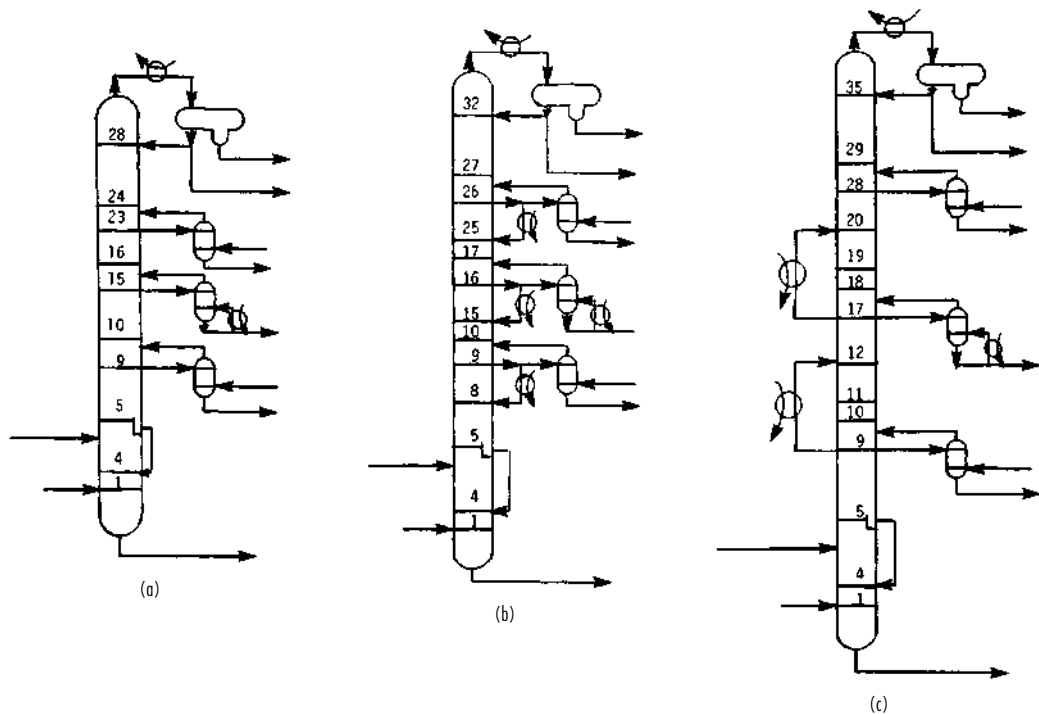


FIG. 13-112 Methods of providing reflux to crude units. (a) Top reflux. (b) Pump-back reflux. (c) Pump-around reflux.

rates of product cuts are readily obtained:

Desired cut	Volume percent of crude oil	BPSD
Overhead (OV)	13.4	6,700
Heavy naphtha (HN)	10.3	5,150
Light distillate (LD)	17.4	8,700
Heavy distillate (HD)	10.0	5,000
Bottoms (B)	48.9	24,450
	100.0	50,000

As shown in Fig. 13-112, methods of providing column reflux include (a) conventional top-tray reflux, (b) pump-back reflux from sidecut strippers, and (c) pump-around reflux. The latter two methods essentially function as intercondenser schemes that reduce the top-tray reflux requirement. As shown in Fig. 13-113 for the example being considered,

the internal-reflux flow rate decreases rapidly from the top tray to the feed-flash zone for case *a*. The other two cases, particularly case *c*, result in better balancing of the column-reflux traffic. Because of this and the opportunity provided to recover energy at a moderate- to high-temperature level, pump-around reflux is the most commonly used technique. However, not indicated in Fig. 13-113 is the fact that in cases *b* and *c* the smaller quantity of reflux present in the upper portion of the column increases the tray requirements. Furthermore, the pump-around circuits, which extend over three trays each, are believed to be equivalent for mass-transfer purposes to only one tray each. Representative tray requirements for the three cases are included in Fig. 13-112. In case *c*, heat-transfer rates associated with the two pump-around circuits account for approximately 40 percent of the total heat removed in the overhead condenser and from the two pump-around circuits combined.

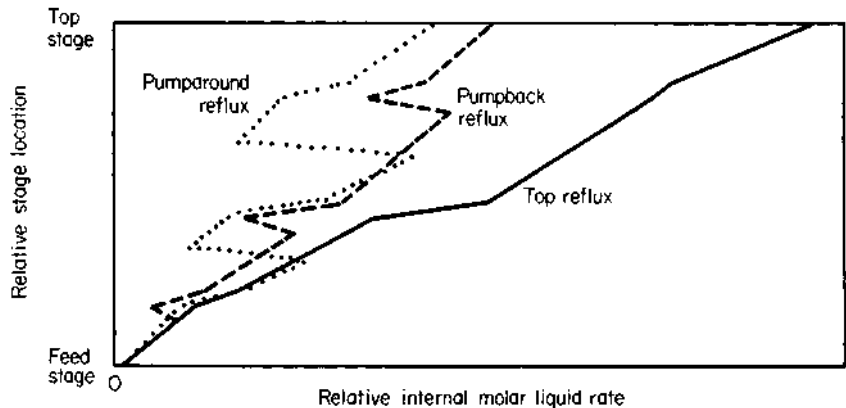


FIG. 13-113 Comparison of internal reflux rates for three methods of providing reflux.

Bottoms and three sidestream strippers remove light ends from products and may use steam or reboilers. In Fig. 13-112 a reboiled stripper is used on the light distillate, which is the largest sidestream withdrawn. Steam-stripping rates in sidestream strippers and at the bottom of the atmospheric column may vary from 0.45 to 4.5 kg (1 to 10 lb) of steam per barrel of stripped liquid, depending on the fraction of stripper feed liquid that is vaporized.

Column pressure at the reflux drum is established so as to condense totally the overhead vapor or some fraction thereof. Flash-zone pressure is approximately 69 kPa (10 psia) higher. Crude oil feed temperature at flash-zone pressure must be sufficient to vaporize the total distillates plus the overflash, which is necessary to provide reflux between the lowest sidestream-product drawoff tray and the flash zone. Calculations are made by using the crude oil EFV curve corrected for pressure. For the example being considered, percent vaporized at the flash zone must be 53.1 percent of the feed.

Tray requirements depend on internal reflux ratios and ASTM 5-95 gaps or overlaps and may be estimated by the correlation of Packie (op. cit.) for crude units and the correlation of Houghland, Lemieux, and Schreiner (op. cit.) for main fractionators.

Example 15: Simulation Calculation of an Atmospheric Tower The ability of a rigorous calculation procedure to simulate operation of an atmospheric tower with its accompanying sidestream strippers may be illustrated by comparing commercial-test data from an actual operation with results computed with the REFINE program of ChemShare Corporation, Houston, Texas. (See also DESIGN II program from WinSim, Inc., Sugar Land, Texas; <http://www.winsim.com>.) The tower configuration and plant operating conditions are shown in Fig. 13-114.

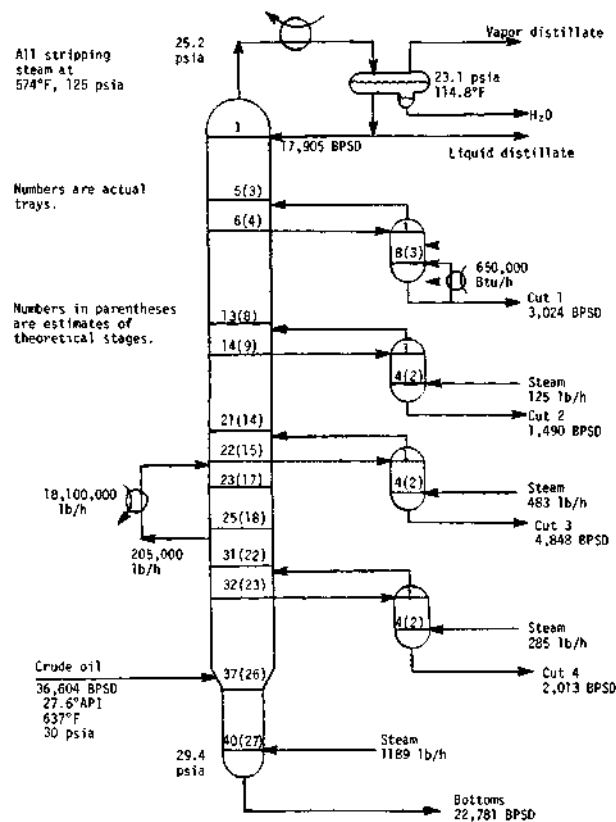


TABLE 13-29 Light-Component Analysis and TBP Distillation of Feed for the Atmospheric Crude Tower of Fig. 13-114

Light-component analysis		
Component	Volume percent	
Methane	0.073	
Ethane	0.388	
Propane	0.618	
n-Butane	0.817	
n-Pentane	2.05	

TBP distillation of feed		
API gravity	TBP, °F	Volume percent
80	-160.	0.1
70	155.	5.
57.5	242.	10.
45.	377.	20.
36.	499.	30.
29.	609.	40.
26.5	707.	50.
23.	805.	60.
20.5	907.	70.
17.	1054.	80.
10.	1210.	90.
-4.	1303.	95.
-22.	1467	100.

NOTE: To convert degrees Fahrenheit to degrees Celsius, °C = (°F - 32)/1.8.

TABLE 13-30 Pseudo-Component Representation of Feed for the Atmospheric Crude Tower of Fig. 13-114

No.	Component name	Molecular weight	Specific gravity	API gravity	(lb-mol)/h
1	Water	18.02	1.0000	10.0	.00
2	Methane	16.04	.3005	339.5	7.30
3	Ethane	30.07	.3561	265.8	24.54
4	Propane	44.09	.5072	147.5	37.97
5	n-Butane	58.12	.5840	110.8	43.84
6	n-Pentane	72.15	.6308	92.8	95.72
7	131 ABP	83.70	.6906	73.4	74.31
8	180 ABP	95.03	.7152	66.3	66.99
9	210 ABP	102.23	.7309	62.1	65.83
10	240 ABP	109.78	.7479	57.7	70.59
11	270 ABP	118.52	.7591	54.9	76.02
12	300 ABP	127.69	.7706	52.1	71.62
13	330 ABP	137.30	.7824	49.4	67.63
14	360 ABP	147.33	.7946	46.6	64.01
15	390 ABP	157.97	.8061	44.0	66.58
16	420 ABP	169.37	.8164	41.8	63.30
17	450 ABP	181.24	.8269	39.6	59.92
18	480 ABP	193.59	.8378	37.4	56.84
19	510 ABP	206.52	.8483	35.3	59.05
20	540 ABP	220.18	.8581	33.4	56.77
21	570 ABP	234.31	.8682	31.5	53.97
22	600 ABP	248.30	.8804	29.2	52.91
23	630 ABP	265.43	.8846	28.5	54.49
24	660 ABP	283.37	.8888	27.7	51.28
25	690 ABP	302.14	.8931	26.9	48.33
26	742 ABP	335.94	.9028	25.2	109.84
27	817 ABP	387.54	.9177	22.7	94.26
28	892 ABP	446.02	.9288	20.8	74.10
29	967 ABP	509.43	.9398	19.1	50.27
30	1055 ABP	588.46	.9531	17.0	57.12
31	1155 ABP	665.13	.9829	12.5	50.59
32	1255 ABP	668.15	1.0658	1.3	45.85
33	1355 ABP	643.79	1.1618	-9.7	29.39
34	1436 ABP	597.05	1.2533	-18.6	21.19
		246.90	.8887	27.7	1922.43

NOTE: To convert (lb-mol)/h to (kg-mol)/h, multiply by 0.454.

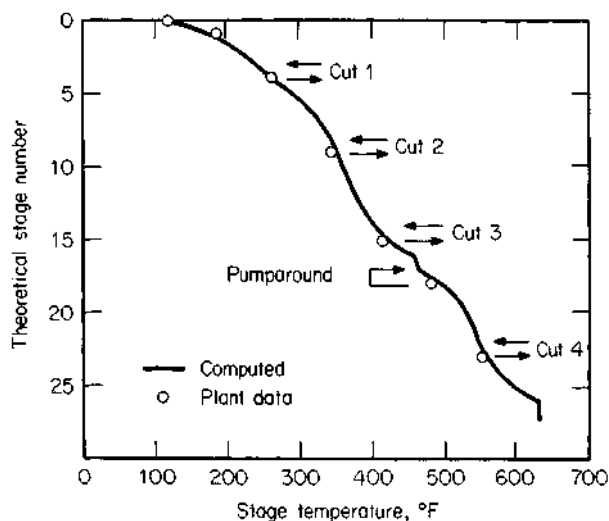


FIG. 13-115 Comparison of computed stage temperatures with plant data for the example of Fig. 13-114.

Light-component analysis and the TBP and API gravity for the feed are given in Table 13-29. Representation of this feed by pseudocomponents is given in Table 13-30 based on 16.7°C (30°F) cuts from 82 to 366°C (180 to 690°F), followed by 41.7°C (75°F) and then 55.6°C (100°F) cuts. Actual tray numbers are shown in Fig. 13-114. Corresponding theoretical-stage numbers, which were determined by trial and error to obtain a reasonable match of computed- and measured-product TBP distillation curves, are shown in parentheses. Overall tray efficiency appears to be approximately 70 percent for the tower and 25 to 50 percent for the sidestream strippers.

Results of rigorous calculations and comparison to plant data, when possible, are shown in Figs. 13-115, 13-116, and 13-117. Plant temperatures are in good agreement with computed values in Fig. 13-115. Computed sidestream-product TBP distillation curves are in reasonably good agreement with values converted from plant ASTM distillations, as shown in Fig. 13-116. Exceptions are the initial points of all four cuts and the higher-boiling end of

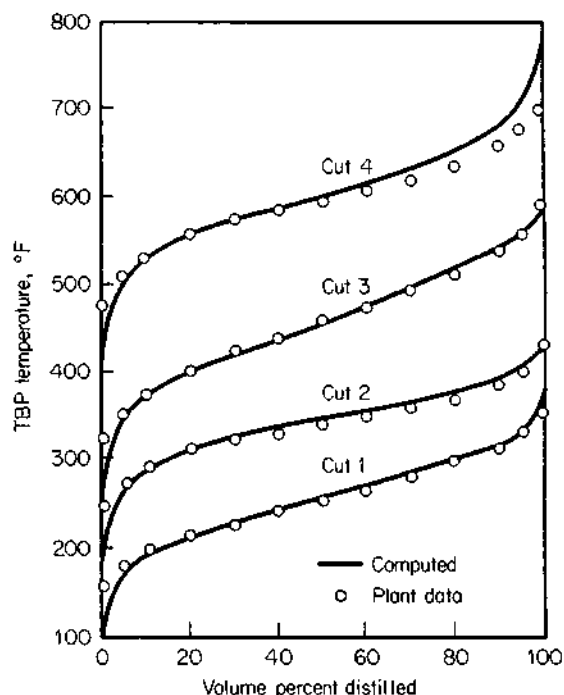


FIG. 13-116 Comparison of computed TBP curves with plant data for the example of Fig. 13-114.

the heavy-distillate curve. This would seem to indicate that more theoretical stripping stages should be added and that either the percent vaporization of the tower feed in the simulation is too high or the internal reflux rate at the lower draw-off tray is too low. The liquid-rate profile in the tower is shown in Fig. 13-117. The use of two or three pump-around circuits instead of one would result in a better traffic pattern than that shown.

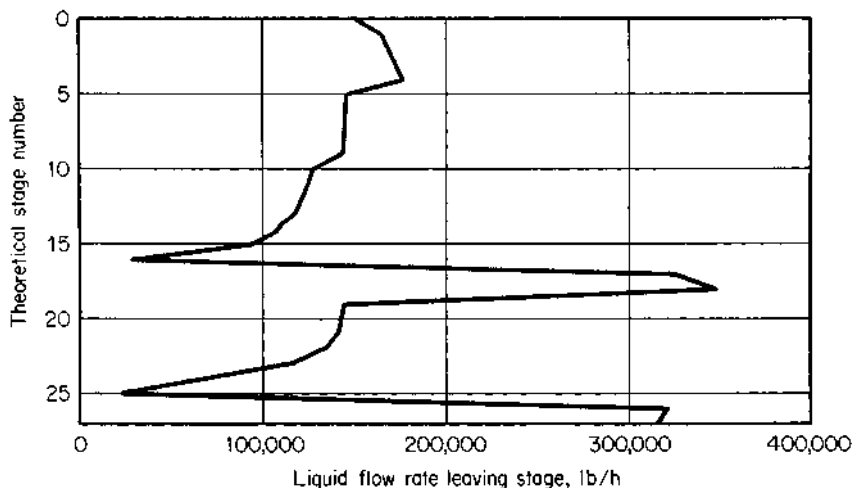


FIG. 13-117 Liquid rate profile for the example of Fig. 13-114.

BATCH DISTILLATION

Batch distillation, which is the process of separating a specific quantity (the charge) of a liquid mixture into products, is used extensively in the laboratory and in small production units that may have to serve for many mixtures. When there are C components in the feed, one batch column will often suffice where $C - 1$ simple continuous distillation columns would be required.

Many larger installations also feature a batch still. The material to be separated may be high in solids content, or it might contain tars or resins that would plug or foul a continuous unit. Use of a batch unit can keep solids separated and permit convenient removal at the termination of the process.

SIMPLE BATCH DISTILLATION

The simplest form of batch distillation consists of a heated vessel (pot or boiler), a condenser, and one or more receiving tanks. No trays or packing is provided. Feed is charged into the vessel and brought to boiling. Vapors are condensed and collected in a receiver. No reflux is returned. The rate of vaporization is sometimes limited to prevent "bumping" the charge and to avoid overloading the condenser, but other controls are minimal. This process is often referred to as a Rayleigh distillation.

If we represent the moles of vapor by V , the moles of liquid in the pot by H , the mole fraction of the more volatile component in this liquid by x , and the mole fraction of the same component in the vapor by y , a material balance yields

$$-y dV = d(Hx) \quad (13-124)$$

Since $dV = -dH$, substitution and expansion give

$$y dH = H dx + x dH \quad (13-125)$$

Rearranging and integrating give

$$\ln \frac{H_i}{H_f} = \int_{x_f}^{x_i} \frac{dx}{y - x} \quad (13-126)$$

where subscript i represents the initial condition and f the final condition of the liquid in the pot. The integration limits have been reversed to obtain a positive integral. Equation (13-126) is equivalent to an integrated form of the defining expression for residue curves in Eq. (13-116), with appropriate substitutions for the variable ξ (see below).

If phase equilibrium is assumed between liquid and vapor, the right-hand side of Eq. (13-126) may be evaluated from the area under a curve of $1/(y - x)$ versus x between the limits x_i and x_f . If the mixture is a binary system for which the relative volatility α can be approximated as a constant over the range considered, then the VLE relationship

$$y = \frac{\alpha x}{1 + (\alpha - 1)x} \quad (13-127)$$

can be substituted into Eq. (13-126) and a direct integration can be made:

$$\ln \left(\frac{H_f}{H_i} \right) = \frac{1}{\alpha - 1} \ln \left[\frac{x_f/(1 - x_f)}{x_i/(1 - x_i)} \right] + \ln \left(\frac{1 - x_i}{1 - x_f} \right) \quad (13-128)$$

For any two components A and B of a multicomponent mixture, if constant α values can be assumed for all pairs of components, then $dH_A/dH_B = y_A/y_B = \alpha_{A,B} (x_A/x_B)$. When this is integrated, we obtain

$$\ln \left(\frac{H_{A,f}}{H_{A,i}} \right) = \alpha_{A,B} \ln \left(\frac{H_{B,f}}{H_{B,i}} \right) \quad (13-129)$$

where $H_{A,i}$ and $H_{A,f}$ are the moles of component A in the pot before and after distillation and $H_{B,i}$ and $H_{B,f}$ are the corresponding moles of component B . Mixtures that cannot be accurately described by using a constant relative volatility require some form of numerical or graphical integration for the solution of Eq. (13-126).

As an example, consider the distillation of an ethanol-water mixture at 101.3 kPa (1 atm). The initial charge is 100 mol of liquid containing 18 mol % ethanol, and the mixture must be reduced to a maximum ethanol concentration in the still of 6 mol %. By using equilibrium data interpolated from Gmehling and Onken [*Vapor-Liquid Equilibrium Data Collection*, DECHEMA Chemistry Data Ser., vol. 1, Part 1, Frankfurt (1977)], we get the following:

x	y	$y - x$	$1/(y - x)$
0.18	0.517	0.337	2.97
0.16	0.502	0.342	2.91
0.14	0.485	0.345	2.90
0.12	0.464	0.344	2.90
0.10	0.438	0.338	2.97
0.08	0.405	0.325	3.08
0.06	0.353	0.293	3.41

The area under a curve of $1/(y - x)$ versus x between $x = 0.06$ and 0.18 is $0.358 = \ln (H_i/H_f)$, so that $H_f = 100/1.43 = 70.0$ mol. The liquid remaining consists of $(70.0)(0.06) = 4.2$ mol of ethanol and 65.8 mol of water. By material balance, the total accumulated distillate must contain $18.0 - 4.2 = 13.8$ mol of alcohol and $82.0 - 65.8 = 16.2$ mol of water. The total distillate is 30 mol, and the average distillate composition is $13.8/30 = 0.46$ mole fraction ethanol. The time, rate of heating, and vapor rate required to carry out the process are related by the energy balance and operating policy, which can be considered separately.

Graphical solutions of models lend significant insight, but there are many cases where such solutions are not possible or where repeated solutions are desired for different conditions. Progress in computer-based models, ranging from specialized simulation software to more general-purpose tools, now permits rapid solutions of most models. One solution of the example above using a general-purpose modeling tool Mathematica® is shown in Fig. 13-118.

The simple batch still provides only one theoretical plate of separation. Its use is usually restricted to laboratory work or preliminary manufacturing in which the products will be held for additional separation at a later time, when most of the volatile component must be removed from the batch before it is processed further, for separation of the batch from heavy undesired components.

BATCH DISTILLATION WITH RECTIFICATION

To obtain products with a narrow composition range, a batch rectifying still is commonly used. The *batch rectifier* consists of a pot (or reboiler) as in simple distillation, plus a rectifying column, a condenser, some means of accumulating and splitting off a portion of the condensed vapor (distillate) for reflux, and one or more product receivers (Fig. 13-119).

The temperature of the distillate is controlled near the bubble point, and reflux is returned at or near the upper column temperature to permit a true indication of reflux quantity and to improve the column operation. A heat exchanger is used to subcool the remainder of the distillate, which is sent to a product receiver. The column may operate at an elevated pressure or at vacuum, in which case appropriate additional devices must be included to obtain the desired pressure. Equipment design methods for batch still components, except for the pot, typically follow the same principles as those presented for continuous distillation under the assumption of conditions close to a steady state (but see the comments below on the effects of holdup). The design should be checked for each mixture if several mixtures are to be processed. The design should be checked at more than one point for each mixture, since the compositions in the pot and in the column change as the distillation proceeds. The pot design is based on the batch size and the vaporization rate, which are related to the time and rate of heating and cooling available. For existing equipment, the pot size will determine the size of the batch or at least a range of feasible sizes H_i .

In operation, a batch of liquid is charged to the pot, and the system is first brought to steady state under total reflux. A portion of the overhead condensate is then continuously withdrawn in accordance with


```

In[1]:= ydata = {{0.18, 0.517}, {0.16, 0.502}, {0.14, 0.485},
               {0.12, 0.464}, {0.10, 0.438}, {0.08, 0.405},
               {0.06, 0.353}};

In[2]:= y = Fit[ydata, {1, x, x^2}, x]

Out[2]= 0.190167 + 3.27321 x - 8.18452 x^2

In[3]:= curve = Plot[y, {x, 0.06, 0.18},
                    {DisplayFunction -> Identity}];

In[4]:= points = ListPlot[ydata,
                          {PlotStyle -> PointSize[0.025],
                           DisplayFunction -> Identity}];

In[5]:= Show [curve, points,
              {DisplayFunction -> $DisplayFunction,
               AxesOrigin -> Last[ydata],
               AxesLabel -> {"x", "y"}}];

In[6]:= Integrate[1 / {y - x}, {x, 0.06, 0.18}]

Out[6]= {{0.35856}}

```

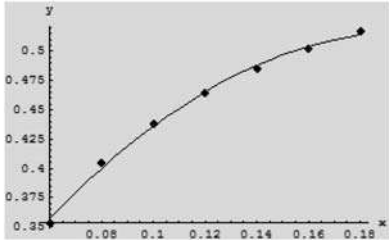


FIG. 13-118 Solution for a simple distillation example using Mathematica,[®] version 5.0.1.

the established reflux policy. "Cuts" are made by switching to alternate receivers, at which time the operating conditions, e.g., reflux rate, may also be altered. The entire column operates as an enriching or rectifying section. As time proceeds, the composition of the liquid in the pot becomes less rich in the more volatile components, and distillation of a cut is stopped when the accumulated distillate attains the desired average composition.

OPERATING METHODS

A batch distillation can be operated in several ways:

1. *Constant reflux, varying overhead composition.* The reflux is set at a predetermined value at which it is maintained for the entire run. Since the pot liquid composition is changing, the instantaneous composition of the distillate also changes. The progress of the distillate and pot compositions in a particular binary separation is illustrated in Fig. 13-120. The variation of the distillate composition for a multicomponent batch distillation is shown in Fig. 13-121 (these distillate product cuts have relatively low purity). The shapes of the curves are functions of volatility, reflux ratio, and number of theoretical plates. The distillation is continued until the average distillate

composition is at the desired value. In the case of a binary mixture, the overhead is then typically diverted to another receiver, and an intermediate or "slop" cut is withdrawn until the remaining pot liquid meets the required specification. The intermediate cut is usually added to the next batch, which can therefore have a somewhat different composition from the previous batch. For a multicomponent mixture, two or more intermediate cuts may be taken between the product cuts. It is preferred to limit the size of the intermediate cuts as far as practical because they reduce the total amount of feed that can be processed.

2. *Constant overhead composition, varying reflux.* If it is desired to maintain a constant overhead composition in the case of a binary mixture, the amount of reflux returned to the column must be constantly increased throughout the run. As time proceeds, the pot is gradually depleted of the lighter component. The increase in reflux is typically gradual at first and more rapid near the end of a cut. Finally, a point is reached at which there is little of the lighter component remaining in the pot and the reflux ratio has attained a very high value. The receivers are then changed, the reflux is reduced, and an intermediate cut is taken as before. This technique can also be extended to a multicomponent mixture.

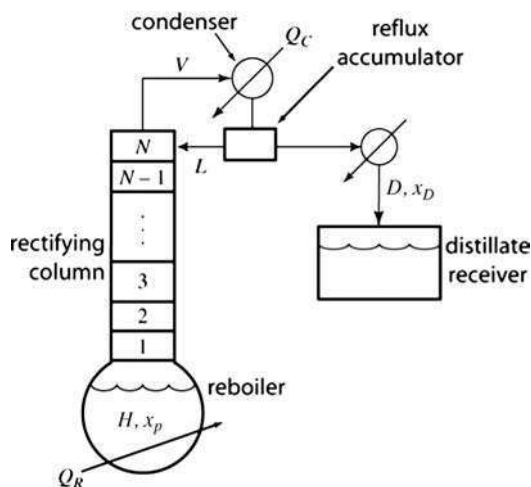


FIG. 13-119 Schematic of a batch rectifier.

3. *Other methods.* A cycling procedure can also be used for the column operation. The unit operates at total reflux until a steady state is established. The distillate is then taken as total drawoff for a short time, after which the column is returned to total reflux operation. This cycle is repeated throughout the course of distillation. Another possibility is to optimize the reflux ratio to achieve the desired separation in a minimum time. More complex operations may involve withdrawal of sidestreams, provision for intercondensers, addition of feeds to trays, and periodic feed additions to the pot.

APPROXIMATE CALCULATION PROCEDURES FOR BINARY MIXTURES

A useful analysis for a binary mixture employs the McCabe-Thiele graphical method. In addition to the usual assumptions of an adiabatic column and constant molar overflow on the trays, the following procedure assumes that the holdup of liquid on the trays, in the column,

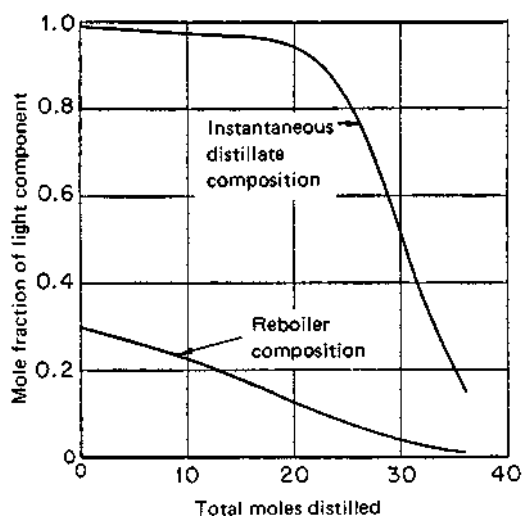


FIG. 13-120 Variation in distillate and reboiler compositions with the amount distilled in binary batch distillation at a constant reflux ratio.

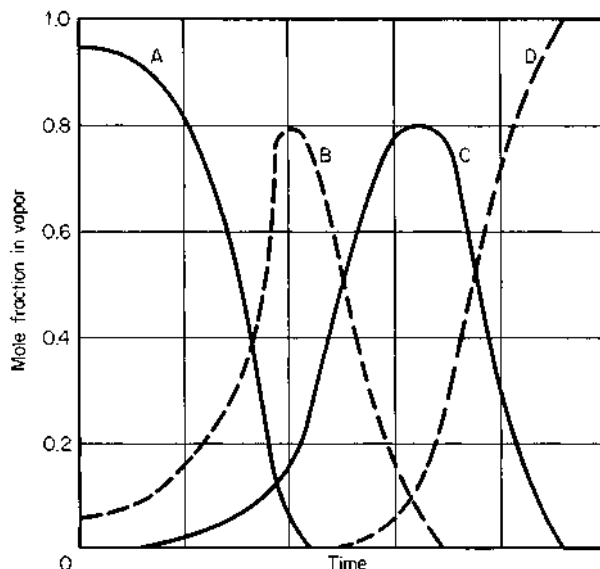


FIG. 13-121 Distillate composition for a batch distillation of a four-component mixture at a constant reflux ratio.

and in the condenser is negligible compared to the holdup in the pot. (The effects of holdup can be significant and are discussed in a later section.)

As a first step, the minimum reflux ratio should be determined. Point D in Fig. 13-122 represents the desired distillate composition and is located on the diagonal since a total condenser is assumed and $x_D = y_D$. Point F represents the initial composition in the pot x_{pi} and for the vapor entering the bottom of the rectifying column y_{pi} . The minimum internal reflux is found from the slope of the line DF .

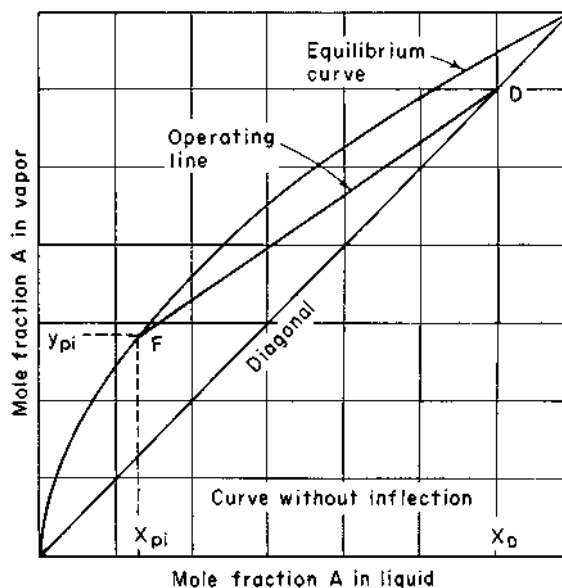


FIG. 13-122 Determination of the minimum reflux for a relatively ideal equilibrium curve.

$$\left(\frac{L}{V}\right)_{\min} = \frac{y_D - y_{pi}}{x_D - x_{pi}} \quad (13-130)$$

where L is the liquid flow rate and V is the vapor rate, both in moles per hour. Since $V = L + D$ (where D is distillate rate) and the external reflux ratio R is defined as $R = L/D$,

$$\frac{L}{V} = \frac{R}{R + 1} \quad (13-131)$$

or

$$R_{\min} = \frac{(L/V)_{\min}}{1 - (L/V)_{\min}} \quad (13-132)$$

The condition of minimum reflux for an equilibrium curve with an inflection point P is shown in Fig. 13-123. In this case the minimum internal reflux is

$$\left(\frac{L}{V}\right)_{\min} = \frac{y_D - y_P}{x_D - x_P} \quad (13-133)$$

The operating reflux ratio is usually 1.5 to 10 times the minimum. By using the ethanol-water equilibrium curve for 101.3-kPa (1-atm) pressure shown in Fig. 13-123 but extending the line to a convenient point for readability, $(L/V)_{\min} = (0.800 - 0.695)/(0.800 - 0.600) = 0.52$ and $R_{\min} = 1.083$.

Batch Rectification at Constant Reflux Using an analysis similar to the simple batch still, Smoker and Rose [*Trans. Am. Inst. Chem. Eng.*, **36**, 285 (1940)] developed the following equation:

$$\ln \frac{H_i}{H_f} = \int_{x_{pf}}^{x_{pi}} \frac{dx_p}{x_D - x_p} \quad (13-134)$$

An overall material balance on the light component gives the average or accumulated distillate composition $x_{D,avg}$:

$$x_{D,avg} = \frac{H_i x_{pi} - H_f x_{pf}}{H_i - H_f} \quad (13-135)$$

If the integral on the right side of Eq. (13-134) is denoted by ξ , the time θ for distillation can be found by

$$\theta = (R + 1) \frac{H_i(e^{\xi} - 1)}{V e^{\xi}} \quad (13-136)$$

An alternative equation is

$$\theta = \frac{R + 1}{V} (H_i - H_f) \quad (13-137)$$

Development of these equations is given by Block [*Chem. Eng.*, **68**, 88 (Feb. 6, 1961)]. The calculation process is illustrated schematically in Fig. 13-124. Operating lines are drawn with the same slope but intersecting the 45° line at different points. The number of theoretical plates under consideration is stepped off to find the corresponding bottoms composition (i.e., still pot composition) for each distillate composition. In Fig. 13-124, operating line $L-1$ with slope L/V drawn from point D_1 where the distillate composition is x_{D1} and the pot composition is x_{p1-3} for three theoretical plates, x_{D2} has a corresponding pot composition of x_{p2-3} , etc. By using these pairs of distillate and pot compositions, the right-hand side of Eq. (13-134) can be evaluated and $x_{D,avg}$ can be found from Eq. (13-135). An iterative calculation is required to find the value of H_f that corresponds to a specified $x_{D,avg}$.

To illustrate the use of these equations, consider a charge of 520 mol of an ethanol-water mixture containing 18 mol % ethanol to be distilled at 101.3 kPa (1 atm). Suppose that the vaporization rate is 75 mol/h, and the product specification is 80 mol % ethanol. Let $L/V = 0.75$, corresponding to a reflux ratio $R = 3.0$. If the column section has six theoretical plates and the pot provides an additional seventh, find how many moles of product will be obtained, what the composition of the pot residue will be, and the time that the distillation will take.

Using the vapor-liquid equilibrium data, plot a y - x diagram. Draw a number of operating lines at a slope of 0.75. Note the composition at the 45° intersection, and step off seven stages on each to find the equilibrium value of the bottoms pot composition. Some of the results are tabulated in the following table:

x_D	x_p	$x_D - x_p$	$1/(x_D - x_p)$
0.800	0.323	0.477	2.097
0.795	0.245	0.550	1.820
0.790	0.210	0.580	1.725
0.785	0.180	0.605	1.654
0.780	0.107	0.673	1.487
0.775	0.041	0.734	1.362

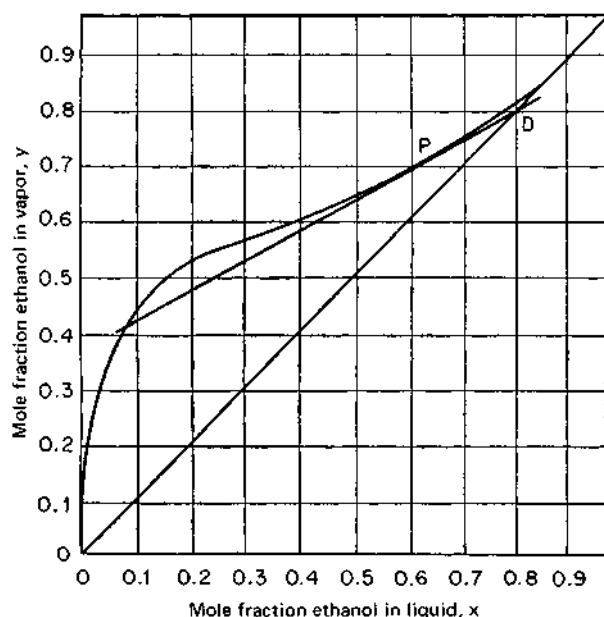


FIG. 13-123 Determination of minimum reflux for an equilibrium curve with an inflection point.

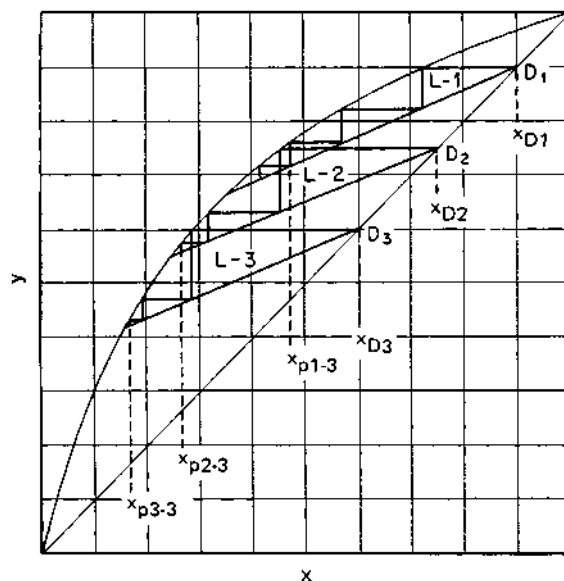


FIG. 13-124 Graphical method for constant-reflux operation.

By using an iterative procedure, integrating between x_{pi} of 0.18 and various lower limits, it is found that $x_{D,avg} = 0.80$ when $x_{pf} = 0.04$, at which time the value of the integral $= 0.205 = \ln(H_i/H_f)$, so that $H_f = 424$ mol. The product collected $= H_i - H_f = 520 - 424 = 96$ mol. From Eq. (13-136),

$$\theta = \frac{(4)(520)(e^{0.205} - 1)}{75(e^{0.205})} = 5.2 \text{ h} \quad (13-138)$$

Batch Rectification at Constant Distillate Composition
Bogart [Trans. Am. Inst. Chem. Eng., **33**, 139 (1937)] developed the following equation for constant distillate composition with the column holdup assumed to be negligible:

$$\theta = \frac{H_i(x_D - x_{pi})}{V} \int_{x_{pf}}^{x_{pi}} \frac{dx_p}{(1 - L/V)(x_D - x_p)^2} \quad (13-139)$$

and where the terms are defined as before. The quantity distilled can then be found by material balance once the initial and final pot compositions are known.

$$H_i - H_f = \frac{H_i(x_{pi} - x_{pf})}{x_D - x_{pf}} \quad (13-140)$$

A schematic example is shown in Fig. 13-125. The distillate composition is held constant by increasing the reflux as the pot composition becomes more dilute. Operating lines with varying slopes ($= L/V$) are drawn from the known distillate composition, and the given number of stages is stepped off to find the corresponding bottoms (still pot) compositions.

As an example, consider the same ethanol-water mixture used above to illustrate constant reflux but now with a constant distillate composition of $x_D = 0.90$. The following table is compiled:

L/V	R	x_p	$x_D - x_p$	$1/(1 - L/V)(x_D - x_p)^2$
0.600	1.50	0.654	0.147	115.7
0.700	2.33	0.453	0.348	27.5
0.750	3.00	0.318	0.483	17.2
0.800	4.00	0.143	0.658	11.5
0.850	5.67	0.054	0.747	11.9
0.900	9.00	0.021	0.780	16.4

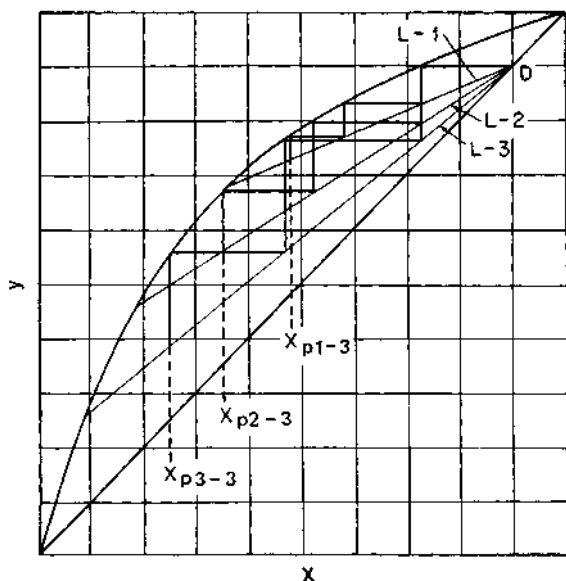


FIG. 13-125 Schematic of constant distillate composition operation.

If the right-hand side of Eq. (13-139) is integrated by using a limit for x_{pf} of 0.04, the value of the integral is 1.615 and the time is

$$\theta = \frac{(520)(0.800 - 0.180)(1.615)}{75} = 7.0 \text{ h} \quad (13-141)$$

The quantity distilled can be found from Eq. (13-140):

$$H_i - H_f = \frac{(520)(0.180 - 0.040)}{0.800 - 0.040} = 96 \text{ mol} \quad (13-142)$$

Other Operating Methods and Optimization A useful control method for difficult industrial or laboratory distillations is *cycling operation*. The most common form of cycling control is to operate the column at total reflux until steady state is established, take off the complete distillate for a short time, and then return to total reflux. An alternative scheme is to interrupt vapor flow to the column periodically by the use of a solenoid-operated butterfly valve in the vapor line from the pot. In both cases, the equations necessary to describe the system are complex, as shown by Schrodt et al. [Chem. Eng. Sci., **22**, 759 (1967)]. The most reliable method for establishing the cycle relationships is by experimental trial on an operating column. Several investigators have also proposed that batch distillation be programmed to attain *time optimization* by proper variation of the reflux ratio. A comprehensive discussion was first presented by Coward [Chem. Eng. Sci., **22**, 503 (1967)] and reviewed and updated by Kim and Diwekar [Rev. Chem. Eng., **17**, 111 (2001)].

The *choice of operating mode* depends upon characteristics of the specific system, the product specifications, and the engineer's preference in setting up a control sequence. Probably the most direct and most common method is *constant reflux*. Operation can be regulated by a timed reflux splitter, a ratio controller, or simply a pair of flowmeters. Since composition is changing with time, some way must be found to estimate the average accumulated distillate composition in order to define the end-point. This is no problem when the specification is not critical or the change in distillate composition is sharply defined. However, when the composition of the distillate changes slowly with time, the cut point is more difficult to determine. Operating with *constant composition* (varying reflux), the specification is automatically achieved if control can be linked to composition or some composition-sensitive physical variable. The relative advantage of the two modes depends upon the materials being separated and upon the number of theoretical plates in the column. A comparison of distillation rates using the same initial and final pot composition for the system benzene-toluene is given in Fig. 13-126. Typical control instrumentation is described by Block [Chem. Eng., **74**, 147 (Jan. 16, 1967)]. Control procedures for *reflux and vapor cycling* operation and for the *time-optimal* process are largely a matter of empirical trial.

Effects of Column Holdup When the holdup of liquid on the trays and in the condenser and reflux accumulator is not negligible compared with the holdup in the pot, the distillate composition at constant reflux ratio changes with time at a different rate than when the column holdup is negligible because of two separate effects.

First, with an appreciable column holdup, the composition of the charge to the pot will be higher in the light component than the pot composition at the start of the distillation. The reason is that before product takeoff begins, the column holdup must be supplied, and due to the rectification, its average composition is higher in the lighter component than that of the liquid charged as feed to the pot. Thus, when overhead takeoff begins, the pot composition is lower than it would be if there were negligible column holdup and the separation is more difficult than expected based on the composition of the feed. The second effect of column holdup is to slow the rate of exchange of the components; the holdup exerts an inertial effect, which prevents compositions from changing as rapidly as they would otherwise, and the degree of separation is usually improved.

Both these effects occur at the same time and change in importance during the course of distillation. Although a number of studies were made and approximate methods developed for predicting the effect of liquid holdup during the 1950s and 1960s (summarized in the 6th edition of Perry's *Chemical Engineers' Handbook*), it is now best to use simulation methods to determine the effect of holdup on a case-by-case basis.

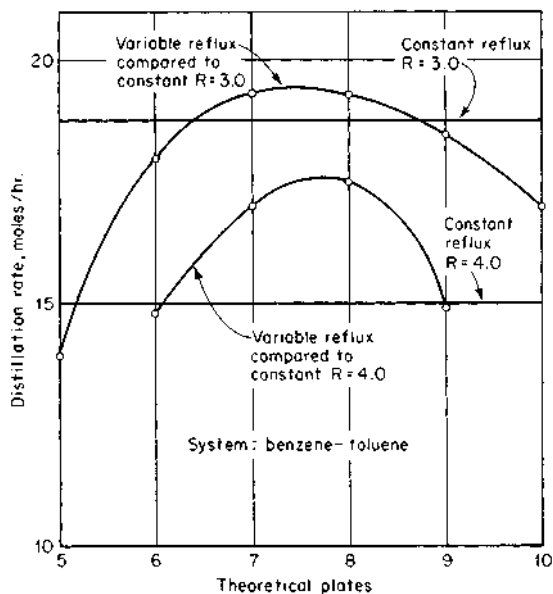


FIG. 13-126 Comparison of operating modes for a batch column.

As an example, consider a batch rectifier fed with a 1:1 mixture of ethanol and *n*-propanol. The rectifier has eight theoretical stages in the column and is operated at a reflux ratio of 19. The distillate and pot compositions are shown in Fig. 13-127 for various values of the holdups.

In Fig. 13-127*a*, the holdup on each stage is 0.01 percent of the initial pot holdup, and in the reflux accumulator it is 0.1 percent of the initial pot holdup (for a total of 0.108 percent). Because this model calculation does not begin with a total reflux period, there is a small initial distillate cut with relatively low ethanol purity. This is followed by a high-purity distillate cut. An intermediate cut of approximately 10 percent of the initial batch size can be collected, leaving the pot with a high purity of *n*-propanol. The column holdup for the case shown in Fig. 13-127*b* is 1 percent of the initial batch size on each stage while the reflux accumulator holdup remains small at 0.1 percent (for a total of 8.1 percent). In this case, both the first low-purity cut and the intermediate cut are somewhat larger for the same purity specifications. These effects are substantially larger when the reflux accumulator has a more significant holdup, as shown in Fig. 13-127*c*, corresponding to a holdup of 1 percent on each stage and 5 percent in the reflux accumulator (for a total of 13 percent). Similar effects are found for multicomponent mixtures. The impact of column and condenser holdup is most important when a high-purity cut is desired for a component that is present in relatively small amounts in the feed.

SHORTCUT METHODS FOR MULTICOMPONENT BATCH RECTIFICATION

For preliminary studies of batch rectification of multicomponent mixtures, shortcut methods that assume constant molar overflow and negligible vapor and liquid holdup are useful in some cases (see the discussion above concerning the effects of holdup). The method of Diwekar and Madhavan [*Ind. Eng. Chem. Res.*, **30**, 713 (1991)] can be used for constant reflux or constant overhead rate. The method of Sundaram and Evans [*Ind. Eng. Chem. Res.*, **32**, 511 (1993)] applies only to the case of constant reflux, but is easy to implement. Both methods employ the Fenske-Underwood-Gilliland (FUG) shortcut procedure at successive time steps. Thus, batch rectification is treated as a sequence of continuous, steady-state rectifications.

CALCULATION METHODS AND SIMULATION

Model predictions such as those shown in Fig. 13-126 or 13-127 are relatively straightforward to obtain by using modern simulation mod-

els and software tools. As discussed in earlier editions of this handbook, such models and algorithms for their solutions have been the subject of intensive study since the early 1960s when digital computing became practical. Detailed calculation procedures for binary and multicomponent batch distillation were initially focused on binary mixtures of constant relative volatility. For example, Huckaba and Danly [*AIChE J.*, **6**, 335 (1960)] developed a simulation model that incorporated more details than can be included in the simple analytical models described above. They assumed constant-mass tray holdups, adiabatic tray operation, and linear enthalpy relationships, but did include energy balances around each tray and permitted the use of nonequilibrium trays by means of specified tray efficiencies. Experimental data were provided to validate the simulation. Meadows [*Chem. Eng. Prog. Symp. Ser.* **46**, 59, 48 (1963)] presented a multicomponent batch distillation model that included equations for energy, material, and volume balances around theoretical trays. The only assumptions made were perfect mixing on each tray, negligible vapor holdup, adiabatic operation, and constant-volume tray holdup. Distefano [*AIChE J.*, **14**, 190 (1968)] extended the model and developed a procedure that was used to simulate several commercial batch distillation columns successfully. Boston et al. (*Foundations of Computer-Aided Chemical Process Design*, vol. 2, Mah and Seider, eds., American Institute of Chemical Engineers, New York, 1981, p. 203) further extended the model, provided a variety of practical sets of specifications, and utilized modern numerical procedures and equation formulations to handle efficiently the nonlinear and often stiff nature of the multicomponent batch distillation problem.

It is important to note that in using computer-aided models for batch distillation, the various assumptions of the model can have a significant impact on the accuracy of the results; e.g., see the discussion of the effects of holdup above. Uncertainties in the physical and chemical parameters in the models can be addressed most effectively by a combination of sensitivity calculations using simulation tools, along with comparison to data. The mathematical treatment of stiffness in the model equations can also be very important, and there is often a substantial advantage in using simulation tools that take special account of this stiffness. (See the 7th edition of *Perry's Chemical Engineers' Handbook* for a more detailed discussion of this aspect).

The availability of detailed models and solution methods has enabled many new studies of complex, mixtures, configurations, and operating and control strategies for batch distillation.

CONSTANT-LEVEL DISTILLATION

Manipulation of the operating conditions such as reflux ratio or pressure during a batch distillation can be useful. In addition, the feed to the batch distillation may vary during the process. A common application is to replace one solvent with another in the presence of a heavy nonvolatile product, as may be encountered in pharmaceutical production. One option for switching solvents is to use simple distillation repeatedly. Initially, a portion of the first solvent is removed by boiling. Then the second solvent is added, and a simple distillation removes more of the first solvent along with some of the second. Repetition of the latter step can be used to reduce the concentration of the first solvent to very small levels.

Gentilcore [*Chem. Eng. Progr.*, **98**(1), 56 (Jan. 2002)] describes an alternative strategy of "constant-level" batch distillation where the replacement solvent is added at a rate to keep the volume of liquid in the pot constant. For simple distillation without rectification the analog of Eq. (13-126) is

$$\frac{S}{H} = \int_{x_f}^{x_i} \frac{dx}{y} \quad (13-143)$$

and the analog of Eq. (13-128) is

$$\frac{S}{H} = \frac{1}{\alpha} \ln \frac{x_i}{x_f} + \frac{\alpha - 1}{\alpha} (x_i - x_f) \quad (13-144)$$

where the mole fractions refer to the compositions of the original solvent and *S* is the amount of the second solvent added to the batch. The amount of solute, a nonvolatile heavy product, is small compared to the size of the batch (alternatively, the analysis can be done on a

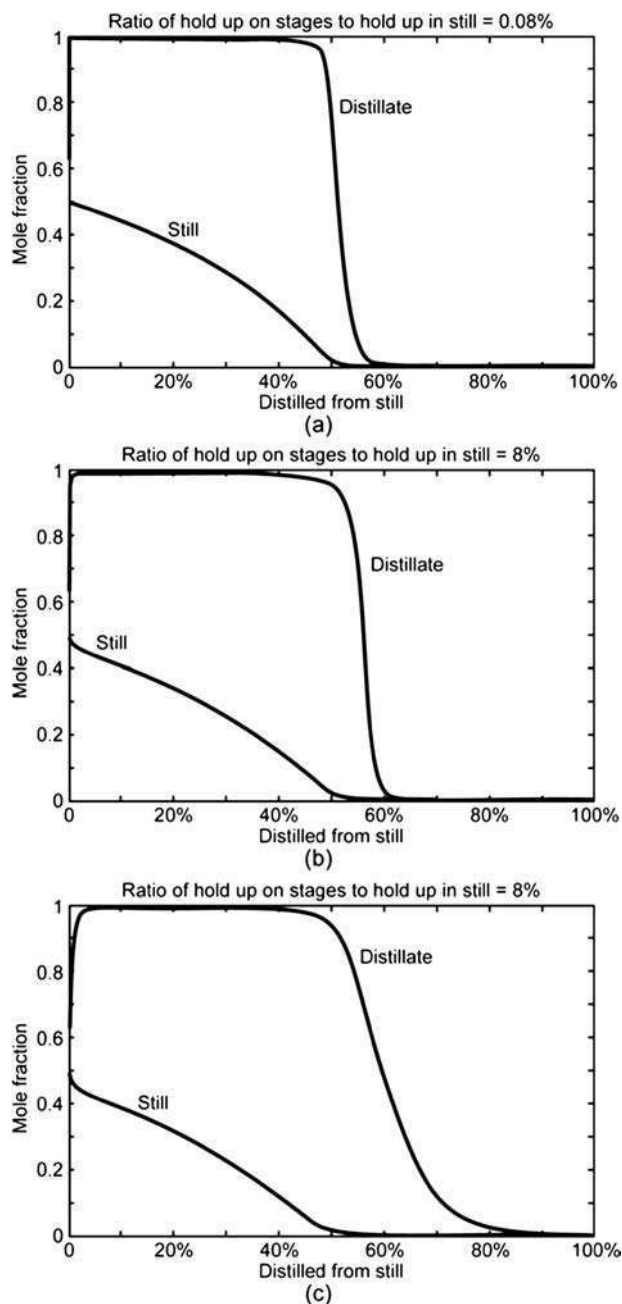


FIG. 13-127 Effects of holdup on batch rectifier.

solute-free basis). The second solvent is assumed to be pure, and the rate of addition is manipulated to keep a constant level in the pot. Compared to the repeated application of simple distillation, this semi-batch operation can typically reduce solvent use by one-half or more depending on the volatility and the desired compositions. This is also a more efficient use of equipment at the expense of a somewhat more complex operation.

An example provided by Gentilcore considers a simple batch still that operates with an initial charge of 80 mol and a minimum of 20 mol. The original solvent use has a volatility $\alpha = 3$ relative to that of the replacement solvent. If simple distillation is used, 60 mol of the original solvent is initially boiled off and then 60 mol of the second solvent is added. A

second distillation of this mixture begins with a composition $x_i = 0.25$ of the original solvent. The solution of Eq. (13-128) by trial and error or root finding gives $x_j = 0.03$ for $H_i = 80$ and $H_j = 20$. Of the 60 mol removed as distillate in this second distillation, $20 - 0.03 = 19.97$ mol is the original solvent and $60 - 19.97 = 40.03$ mol is the replacement. An alternative constant-level batch distillation in the same equipment according to Eq. (13-144) with $H = 20$, $x_i = 0.25$, and $x_j = 0.03$ requires the addition of $S = 35.6$ mol of replacement solvent. The still contains 20 mol of the solvent; 16.2 mol is distilled compared to 40.3 mol in the simple distillation. This 60 percent savings in the use of replacement solvent arises because the distillation takes place beginning with a higher concentration of the original solvent for the second step.

ALTERNATIVE EQUIPMENT CONFIGURATIONS

The batch rectifier shown schematically in Fig. 13-119 is by far the most common configuration of equipment. Several alternative special-purpose configurations have been studied and offer potential advantages in particular applications. Also see Doherty and Malone (*Conceptual Design of Distillation Systems*, McGraw-Hill, 2001, pp. 407–409, 417–419).

For instance, a simple batch distillation can be combined with a stripping column to give the batch stripper shown in Fig. 13-128. The pot holds the batch charge and provides liquid reflux into the stripping section. The reboiler provides vapor to the column and has relatively small holdup. The product stream B in the bottom is concentrated in the higher-boiling compound, and the pot gradually becomes more concentrated in the lighter component. Multiple “cuts” can be taken as products, and the reboil rate either can be constant or can be adjusted by analogy with the reflux ratio in the batch rectifier.

For mixtures containing large concentrations of a heavy component, the batch stripper can be advantageous.

The more complex “middle vessel” column combines aspects of both the batch rectifier and the batch stripper, as shown in Fig. 13-129. The middle vessel arrangement was described qualitatively by Robinson and Gilliland (*Elements of Fractional Distillation*, McGraw-Hill, 1950, p. 388) and analyzed by Bortolini and Guirase [*Quad. Ing. Chim. Ital.*, **6**, 150 (1970)]. This configuration requires more equipment and is more complex, but can produce both distillate and bottoms product cuts simultaneously. Barolo and Botteon [*AIChE J.*, **43**, 2601 (1997)] pointed out that the middle vessel configuration at total reflux and reboil and with the appropriate collection equipment for distillate and bottoms products (not shown in Fig. 13-129) can concentrate a ternary mixture into its three pure fractions. This and analogous configurations for mixtures with more components have been studied by Hasebe et al. [*J. Chem. Eng. Japan*, **29**, 1000 (1996); *Computers Chem. Engng.*, **23**,

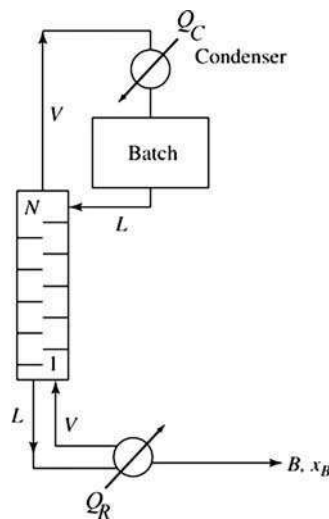


FIG. 13-128 Schematic of a batch stripper.

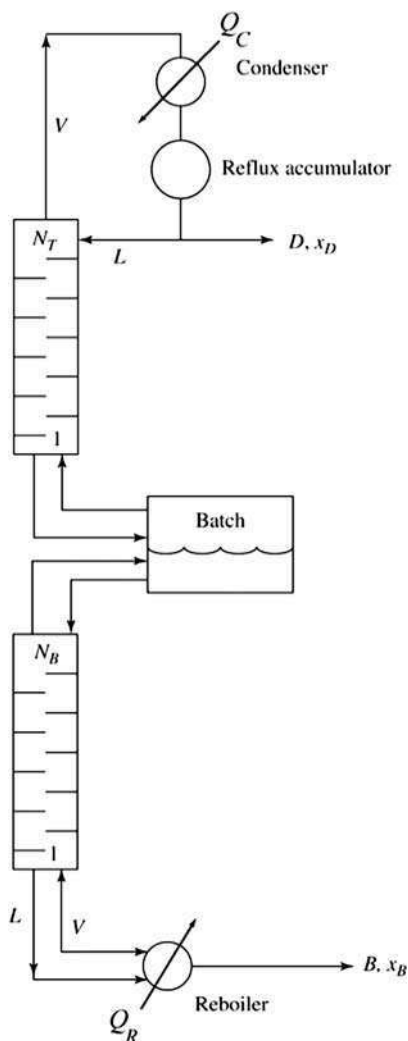


FIG. 13-129 Middle vessel batch distillation.

523 (1999)] and experimentally by Wittgens and Skogestad [*ICHEME Symp Ser.*, **142**, 239 (1997).]

The batch stripper and the middle vessel configurations provide the capability to make separations for certain azeotropic mixtures that are not possible or that cannot be done efficiently in the batch rectifier.

BATCH DISTILLATION OF AZEOTROPIC MIXTURES

Although azeotropic distillation is covered in an earlier subsection, it is appropriate to consider the application of residue curve maps to batch distillation here. (See the subsection Enhanced Distillation for a discussion of residue curve maps.) An essential point is that the sequence, number, and limiting composition of each cut from a batch distillation depend on the form of the residue curve map and the composition of the initial charge to the still. As with continuous distillation operation, the set of reachable products (cuts) for a given charge to a batch distillation is constrained by the residue curve-map distillation boundaries. Furthermore, some pure components can be produced as products from the batch stripper, but not the batch rectifier and vice versa. Doherty and Malone (*Conceptual Design of Distillation Systems*, chap. 9, McGraw-Hill, 2001) give more details, but the main points are the following.

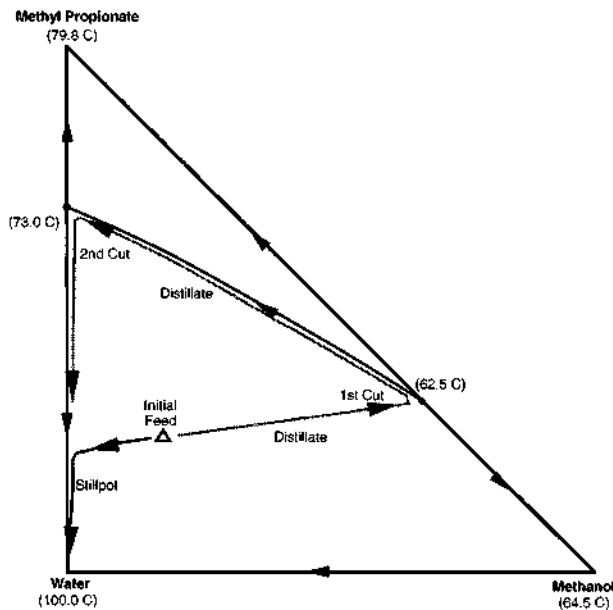


FIG. 13-130 Residue curve map and batch rectifier paths for methanol, methyl propionate, and water.

In the batch rectifier, the limiting cuts, obtainable with a sufficiently large number of stages and reflux, begin with the low-boiling node that defines the distillation region containing the feed composition. For the batch stripper, the first limiting cut is the high-boiling node. In either case, the subsequent cuts depend on the structure of the residue curve map.

For the batch rectifier, as the low-boiling component or azeotrope is removed, the still composition moves along a straight material balance line through the initial feed composition and the low-boiling node and away from the initial composition until it reaches the edge of the composition triangle or a distillation boundary. The path then follows the edge or distillation boundary to the high-boiling node of the region.

As an example, consider the residue curve map structure shown in Fig. 13-130 for a mixture of methanol, methyl propionate, and water at a pressure of 1 atm. There are two minimum-boiling binary azeotropes joined by a distillation boundary that separates the compositions into two distillation regions. Feeds in the upper and lower regions will have different distillate products. For the sample feed shown, and with a sufficient number of theoretical stages and reflux, the distillate will rapidly approach the low-boiling azeotrope of methanol and methyl propionate at 62.5°C. The still pot composition changes along the straight-line segment as shown until it is nearly free of methanol. At that point, the distillate composition changes rapidly along the distillation boundary to a composition for the second cut at or near the methyl propionate–water azeotrope. The still pot composition eventually approaches pure water. The rate of change and the precise approach to these compositions require more detailed study.

For the same feed, a batch stripper can be used to remove a bottoms product that approaches pure water. The pot composition (overhead) will contain all three components near the point of intersection of the distillation boundary with a straight line extended from the water vertex through the feed composition.

For this mixture it is not possible to isolate the pure components in a batch rectifier or batch stripper. The use of additional equipment such as a decanter to exploit liquid-liquid phase behavior or the addition of a fourth component or chemical reactions can sometimes be used to effect the separation.

The product cuts for azeotropic mixtures are also sensitive to the curvature of the distillation boundaries; see Doherty and Malone (*Conceptual Design of Distillation Systems*, McGraw-Hill, 2001; pp. 403–404) and additional references there.

SECTION 14

Equipment for
Distillation, Gas
Absorption, Phase
Dispersion, and
Phase Separation

PERRY'S CHEMICAL ENGINEERS' HANDBOOK

8TH EDITION



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Equipment for Distillation, Gas Absorption, Phase Dispersion, and Phase Separation

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Nomenclature

a, a_e	Effective interfacial area	m^2/m^3	ft^2/ft^3	E_g	Point efficiency, gas phase only, fractional	-/-	-/-
a_p	Packing surface area per unit volume	m^2/m^3	ft^2/ft^3	E_{oc}	Overall column efficiency, fractional	-/-	-/-
A	Absorption factor $L_M/(mG_M)$	-/-	-/-	E_{OG}	Overall point efficiency, gas concentrations, fractional	-/-	-/-
A	Cross-sectional area	m^2	ft^2	E_{mv}, E_{MV}	Murphree tray efficiency, gas concentrations, fractional	-/-	-/-
A_a	Active area, same as bubbling area	m^2	ft^2	E_s	Entrainment, kg entrained liquid per kg gas upflow	kg/kg	lb/lb
A_B	Bubbling (active) area	m^2	ft^2	f	Fractional approach to flood	-/-	-/-
A_D	Downcomer area (straight vertical downcomer)	m^2	ft^2	f	Liquid maldistribution fraction	-/-	-/-
A_{da}	Downcomer apron area	m^2	ft^2	f_{max}	Maximum value of f above which separation cannot be achieved	-/-	-/-
A_{DB}	Area at bottom of downcomer	m^2	ft^2	f_w	Weep fraction, Eq. (14-121)	-/-	-/-
A_{DT}	Area at top of downcomer	m^2	ft^2	F	Fraction of volume occupied by liquid phase, system limit correlation, Eq. (14-170)	-/-	-/-
A_e, A'	Effective absorption factor (Edmister)	-/-	-/-	F	F-factor for gas loading Eq. (14-76)	$m/s(kg/m^3)^{0.5}$	$ft/s(lb/ft^3)^{0.5}$
A_f	Fractional hole area	-/-	-/-	F_{LG}	Flow parameter, Eq. (14-89) and Eq. (14-141)	-/-	-/-
A_h	Hole area	m^2	ft^2	F_p	Packing factor	m^{-1}	ft^{-1}
A_N	Net (free) area	m^2	ft^2	F_{pd}	Dry packing factor	m^{-1}	ft^{-1}
A_S	Slot area	m^2	ft^2	FPL	Flow path length	m	ft
A_{SO}	Open slot area	m^2	ft^2	Fr	Froude number, clear liquid height correlation, Eq. (14-120)	-/-	-/-
A_T	Tower cross-section area	m^2	ft^2	Fr_h	Hole Froude number, Eq. (14-114)	-/-	-/-
c	Concentration	kg-mol/ m^3	lb-mol/ ft^3	F_w	Weir constriction correction factor, Fig. 14-38	-/-	-/-
c'	Stokes-Cunningham correction factor for terminal settling velocity	-/-	-/-	g	Gravitational constant	m/s^2	ft/s^2
C	C-factor for gas loading, Eq. (14-77)	m/s	ft/s	g_c	Conversion factor	1.0 kg-m/(N·s ²)	32.2 lb-ft/(lb·s ²)
C_1	Coefficient in regime transition correlation, Eq. (14-129)	-/-	-/-	G	Gas phase mass velocity	kg/(s·m ²)	lb/(hr·ft ²)
C_1, C_2	Parameters in system limit equation	m/s	ft/s	G_f	Gas loading factor in Robbins' packing pressure drop correlation	kg/(s·m ²)	lb/(h·ft ²)
C_3, C_4	Constants in Robbins' packing pressure drop correlation	-/-	-/-	G_M	Gas phase molar velocity	kg-mol/(s·m ²)	lb-mol/(h·ft ²)
CAF	Flood C-factor, Eq. (14-88)	m/s	ft/s	GPM	Liquid flow rate	—	gpm
CAF ₀	Uncorrected flood C-factor, Fig. 14-30	—	ft/s	h	Pressure head	mm	in
C_d	Coefficient in clear liquid height correlation, Eq. (14-116)	-/-	-/-	h_{dc}	Froth height in downcomer	mm	in
C_G	Gas C-factor; same as C	m/s	ft/s	h_L	Pressure drop through aerated mass on tray	mm	in
C_L	Liquid loading factor, Eq. (14-144)	m/s	ft/s	h_c	Clear liquid height on tray	mm	in
C_{LG}	A constant in packing pressure drop correlation, Eq. (14-143)	(m/s) ^{0.5}	(ft/s) ^{0.5}	h_{cl}	Clearance under downcomer	mm	in
CP	Capacity parameter (packed towers), Eq. (14-140)	-/-	-/-	h_{ct}	Clear liquid height at spray to froth transition	mm	in
C_{SB}, C_{sb}	C-factor at entrainment flood, Eq. (14-80)	m/s	ft/s	h_d	Dry pressure drop across tray	mm	in
C_{sbf}	Capacity parameter corrected for surface tension	m/s	ft/s	h_{da}	Head loss due to liquid flow under downcomer apron	mm	in
C_v, C_V	Discharge coefficient, Fig. 14-35	-/-	-/-	h_{dc}	Clear liquid height in downcomer	mm	in
C_w	A constant in weep rate equation, Eq. (14-123)	-/-	-/-	h_{ds}	Calculated clear liquid height, Eq. (14-108)	mm	in
C_{XY}	Coefficient in Eq. (14-159) reflecting angle of inclination	-/-	-/-	h_f	Height of froth	mm	in
d	Diameter	m	ft	h_{fow}	Froth height over the weir, Eq. (14-117)	mm	in
d_b	Bubble diameter	m	ft	h_{hg}	Hydraulic gradient	mm	in
d_h, d_H	Hole diameter	mm	in	h_{Lo}	Packing holdup in preloading regime, fractional	-/-	-/-
d_o	Orifice diameter	m	ft	h_{La}	Clear liquid height at froth to spray transition, corrected for effect of weir height, Eq. (14-96)	mm	in
d_{pc}	Cut size of a particle collected in a device, 50% mass efficiency	μm	ft	h_{oc}	Height of crest over weir	mm	in
d_{psd}	Mass median size particle in the pollutant gas	μm	ft	h_T	Height of contacting	m	ft
d_{ps50}	Aerodynamic diameter of a real median size particle	μm	ft	h_t	Total pressure drop across tray	mm	in
d_w	Weir diameter, circular weirs	mm	in	h_w	Weir height	mm	in
D	Diffusion coefficient	m ² /s	ft ² /s	H	Height of a transfer unit	m	ft
D	Tube diameter (wetted-wall columns)	m	ft	H	Henry's law constant	kPa/mol fraction	atm/mol fraction
D_{32}	Sauter mean diameter	m	ft	H'	Henry's law constant	kPa/(kmol·m ³)	psi/(lb-mol·ft ³)
D_g	Diffusion coefficient	m ² /s	ft ² /h	H_C	Height of a gas phase transfer unit	m	ft
D_p	Packing particle diameter	m	ft	H_L	Height of a liquid phase transfer unit	m	ft
D_T	Tower diameter	m	ft	H_{OC}	Height of an overall transfer unit, gas phase concentrations	m	ft
D_{tube}	Tube inside diameter	m	ft	H_{OL}	Height of an overall transfer unit, liquid phase concentrations	m	ft
D_{vm}	Volume mean diameter	m	ft				
e	Absolute entrainment of liquid	kg-mol/h	lb-mol/h				
e	Entrainment, mass liquid/mass gas	kg/kg	lb/lb				
E	Plate or stage efficiency, fractional	-/-	-/-				
E	Power dissipation per mass	W	Btu/lb				
E_a	Murphree tray efficiency, with entrainment, gas concentrations, fractional	-/-	-/-				

Nomenclature (Continued)

H'	Henry's law coefficient	kPa/mol frac	atm/mol frac	S	Length of corrugation side, structured packing	m	ft
HETP	Height equivalent to a theoretical plate or stage	m	ft	S	Stripping factor mG_M/L_M	-/-	-/-
J_G^*	Dimensionless gas velocity, weep correlation, Eq. (14-124)	-/-	-/-	S	Tray spacing	mm	in
J_L^*	Dimensionless liquid velocity, weep correlation, Eq. (14-125)	-/-	-/-	S_e, S'	Effective stripping factor (Edmister)	-/-	-/-
k	Individual phase mass transfer coefficient	kmol/(s·m ² · mol frac)	lb·mol/(s·ft ² · mol frac)	SF	Derating (system) factor, Table 14-9	-/-	-/-
k_1	First order reaction velocity constant	1/s	1/s	t_t	Tray thickness	mm	in
k_2	Second order reaction velocity constant	m ³ /(s·kmol)	ft ³ /(h·lb·mol)	t_v	Valve thickness	mm	in
k_g	Gas mass-transfer coefficient, wetted-wall columns [see Eq. (14-171) for unique units]	-/-	-/-	T	Absolute temperature	K	°R
k_G	gas phase mass transfer coefficient	kmol/(s·m ² · mol frac)	lb·mol/(s·ft ² · mol frac)	TS	Tray spacing; same as S	mm	in
k_L	liquid phase mass transfer coefficient	kmol/(s·m ² · mol frac)	lb·mol/(s·ft ² · mol frac)	U, u	Linear velocity of gas	m/s	ft/s
K	Constant in trays dry pressure drop equation	mm·s ² /m ²	in·s ² /ft ²	U_a	Velocity of gas through active area	m/s	ft/s
K	Vapor-liquid equilibrium ratio	-/-	-/-	U_a^*	Gas velocity through active area at froth to spray transition	m/s	ft/s
K_C	Dry pressure drop constant, all valves closed	mm·s ² /m ²	in·s ² /ft ²	U_{h, u_h}	Gas hole velocity	m/s	ft/s
K_D	Orifice discharge coefficient, liquid distributor	-/-	-/-	U_L, u_L	Liquid superficial velocity based on tower cross-sectional area	m/s	ft/s
K_g	Overall mass-transfer coefficient	kg·mol/ (s·m ² ·atm)	lb·mol/ (h·ft ² ·atm)	U_n	Velocity of gas through net area	m/s	ft/s
K_O	Dry pressure drop constant, all valves open	mm·s ² /m ²	in·s ² /ft ²	U_{nf}	Gas velocity through net area at flood	-/-	-/-
K_{OG}, K_C	Overall mass transfer coefficient, gas concentrations	kmol / (s·m ² ·mol frac)	lb·mol/ (s·ft ² ·mol frac)	U_t	Superficial velocity of gas	m/s	ft/s
K_{OL}	Overall mass transfer coefficient, liquid concentrations	kmol/ (s·m ² ·mol frac)	lb·mol/ (s·ft ² ·mol frac)	v_H	Horizontal velocity in trough	m/s	ft/s
L	Liquid mass velocity	kg/(m ² ·s)	lb/ft ² ·h	V	Linear velocity	m/s	ft/s
L_f	Liquid loading factor in Robbins' packing pressure drop correlation	kg/(s·m ²)	lb/(h·ft ²)	V	Molar vapor flow rate	kg·mol/s	lb·mol/h
L_m	Molar liquid downflow rate	kg·mol/h	lb·mol/h	W	Weep rate	m ³ /s	gpm
L_M	Liquid molar mass velocity	kmol/(m ² ·s)	lb·mol/(ft ² ·h)	x	Mole fraction, liquid phase (note 1)	-/-	-/-
L_S	Liquid velocity, based on superficial tower area	m/s	ft/s	x'	Mole fraction, liquid phase, column 1 (note 1)	-/-	-/-
L_w	Weir length	m	in	x''	Mole fraction, liquid phase, column 2 (note 1)	-/-	-/-
m	An empirical constant based on Wallis' countercurrent flow limitation equation, Eqs. (14-123) and (14-143)	-/-	-/-	x°, x°	Liquid mole fraction at equilibrium (note 1)	-/-	-/-
m	Slope of equilibrium curve = dy^*/dx	-/-	-/-	y	Mole fraction, gas or vapor phase (note 1)	-/-	-/-
M	Molecular weight	kg/kmol	lb/(lb·mol)	y'	Mole fraction, vapor phase, column 1 (note 1)	-/-	-/-
n	Parameter in spray regime clear liquid height correlation, Eq. (14-84)	mm	in	y''	Mole fraction, vapor phase, column 2 (note 1)	-/-	-/-
n_A	Rate of solute transfer	kmol/s	lb·mol/s	y°, y°	Gas mole fraction at equilibrium (note 1)	-/-	-/-
n_D	Number of holes in orifice distributor	-/-	-/-	Z	Characteristic length in weep rate equation, Eq. (14-126)	m	ft
N_a	Number of actual trays	-/-	-/-	Z_p	Total packed height	m	ft
N_A, N_t	Number of theoretical stages	-/-	-/-	Greek Symbols			
N_{OG}	Number of overall gas-transfer units	-/-	-/-	α	Relative volatility	-/-	-/-
N_p	Number of tray passes	-/-	-/-	β	Tray aeration factor, Fig. (14-37)	-/-	-/-
p	Hole pitch (center-to-center hole spacing)	mm	in	ϵ	Void fraction	-/-	-/-
p	Partial pressure	kPa	atm	ϕ	Contact angle	deg	deg
P_{BM}	Logarithmic mean partial pressure of inert gas	kPa	atm	ϕ	Relative froth density	-/-	-/-
P, p_T	Total pressure	kPa	atm	γ	Activity coefficient	-/-	-/-
P^v	Vapor pressure	kPa	atm	Γ	Flow rate per length	kg/(s·m)	lb/(s·ft)
Q, q	Volumetric flow rate of liquid	m ³ /s	ft ³ /s	δ	Effective film thickness	m	ft
Q'	Liquid flow per serration of serrated weir	m ³ /s	ft ³ /s	η	Collection efficiency, fractional	-/-	-/-
Q_D	Downcomer liquid load, Eq. (14-79)	m/s	ft/s	η	Factor used in froth density correlation, Eq. (14-118)	-/-	-/-
Q_L	Weir load, Eq. (14-78)	m ³ /(h·m)	gpm/in	λ	Stripping factor = $m/(L_M/G_M)$	-/-	-/-
Q_{MW}	Minimum wetting rate	m ³ /(h·m ²)	gpm/ft ²	μ	Absolute viscosity	Pa·s	cP or lb/(ft·s)
R	Reflux flow rate	kg·mol/h	lb·mol/h	μm	Micrometers	m	-/-
R	Gas constant	-/-	-/-	ν	Kinematic viscosity	m ² /s	cS
R_h	Hydraulic radius	m	ft	π	3.1416 . . .	-/-	-/-
R_{wc}	Ratio of valve weight with legs to valve weight without legs, Table (14-11)	-/-	-/-	θ	Residence time	s	s
				θ	Angle of serration in serrated weir	deg	deg
				ρ	Density	kg/m ³	lb/ft ³
				ρ_m	Valve metal density	kg/m ³	lb/ft ³
				σ	Surface tension	mN/m	dyn/cm
				χ	Parameter used in entrainment correlation, Eq. (14-95)	-/-	-/-
				Ψ	Fractional entrainment, moles liquid entrained per mole liquid downflow	k·mol/ k·mol	lb·mol/ lb·mol
				Φ	Fractional approach to entrainment flood	-/-	-/-
				ΔP	Pressure drop per length of packed bed	mmH ₂ O/m	inH ₂ O/ft
				$\Delta \rho$	$\rho_L - \rho_G$	kg/m ³	lb/ft ³
				Subscripts			
				A	Species A		
				AB	Species A diffusing through species B		

Nomenclature (Concluded)

Subscripts		Subscripts	
<i>B</i>	Species B	<i>n, N</i>	On stage <i>n</i>
<i>B</i>	Based on the bubbling area	<i>N</i>	At the inlet nozzle
<i>d</i>	Dry	<i>NF, nf</i>	Based on net area at flood
<i>da</i>	Downcomer apron	<i>p</i>	Particle
<i>dc</i>	Downcomer	<i>S</i>	Superficial
dry	Uncorrected for entrainment and weeping	<i>t</i>	Total
<i>e</i>	Effective value	ult	At system limit (ultimate capacity)
<i>f</i>	Froth	<i>V</i>	Vapor
<i>Fl</i>	Flood	<i>w</i>	Water
flood	At flood	1	Tower bottom
<i>G, g</i>	Gas or vapor	2	Tower top
<i>h</i>	Based on hole area (or slot area)	Dimensionless Groups	
<i>H₂O</i>	Water	<i>N_{Fr}</i>	Froude number = $(U_L^2)/(Sg)$
<i>i</i>	Interface value	<i>N_{Re}</i>	Reynolds number = $(D_{\text{tube}} U_{\text{sc}} \rho_C)/(\mu_C)$
<i>L, l</i>	Liquid	<i>N_{Sc}</i>	Schmidt number = $\mu/(\rho D)$
<i>m</i>	Mean	<i>N_{We}</i>	Weber number = $(U_L^2 \rho_L S)/(\sigma g_c)$
min	Minimum		
MOC	At maximum operational capacity		

NOTE: 1. Unless otherwise specified, refers to concentration of more volatile component (distillation) or solute (absorption).

GENERAL REFERENCES: Astarita, G., *Mass Transfer with Chemical Reaction*, Elsevier, New York, 1967. Astarita, G., D. W. Savage and A. Bisio, *Gas Treating with Chemical Solvents*, Wiley, New York, 1983. Billet, R., *Distillation Engineering*, Chemical Publishing Co., New York, 1979. Billet, R., *Packed Column Analysis and Design*, Ruhr University, Bochum, Germany, 1989. Danckwerts, P. V., *Gas-Liquid Reactions*, McGraw-Hill, New York, 1970. *Distillation and Absorption 1987*, Rugby, U.K., Institution of Chemical Engineers. *Distillation and Absorption 1992*, Rugby, U.K., Institution of Chemical Engineers. *Distillation and Absorption 1997*, Rugby, U.K., Institution of Chemical Engineers. *Distillation and Absorption 2002*, Rugby, U.K., Institution of Chemical Engineers. *Distillation and Absorption 2006*, Rugby, U.K., Institution of Chemical Engineers. *Distillation Topical Conference Proceedings*, AIChE Spring Meetings (separate Proceedings Book for each Topical Conference): Houston, Texas, March 1999; Houston, Texas, April 22–26, 2001; New Orleans, La., March 10–14, 2002; New Orleans, La., March 30–April 3, 2003; Atlanta, Ga., April 10–13, 2005. Hines, A. L., and R. N. Maddox, *Mass Transfer—Fundamentals and Applications*, Prentice Hall, Englewood Cliffs, New Jersey, 1985. Hobbler,

T., *Mass Transfer and Absorbers*, Pergamon Press, Oxford, 1966. Kister, H. Z., *Distillation Operation*, McGraw-Hill, New York, 1990. Kister, H. Z., *Distillation Design*, McGraw-Hill, New York, 1992. Kister, H. Z., and G. Nalven (eds.), *Distillation and Other Industrial Separations*, Reprints from CEP, AIChE, 1998. Kister, H. Z., *Distillation Troubleshooting*, Wiley, 2006. Kohl, A. L., and R. B. Nielsen, *Gas Purification*, 5th ed., Gulf, Houston, 1997. Lockett, M. J., *Distillation Tray Fundamentals*, Cambridge, U.K., Cambridge University Press, 1986. Maćkowiak, J., “Fluiddynamik von Kolonnen mit Modernen Füllkörpern und Packungen für Gas/Flüssigkeitssysteme,” Otto Salle Verlag, Frankfurt am Main und Verlag Sauerländer Aarau, Frankfurt am Main, 1991. Schweitzer, P. A. (ed.), *Handbook of Separation Techniques for Chemical Engineers*, 3d. ed., McGraw-Hill, New York, 1997. Sherwood, T. K., R. L. Pigford, C. R. Wilke, *Mass Transfer*, McGraw-Hill, New York, 1975. Stichlmair, J., and J. R. Fair, *Distillation Principles and Practices*, Wiley, New York, 1998. Strigle, R. F., Jr., *Packed Tower Design and Applications*, 2d ed., Gulf Publishing, Houston, 1994. Treybal, R. E., *Mass Transfer Operations*, McGraw-Hill, New York, 1980.

INTRODUCTION

Definitions Gas absorption is a unit operation in which soluble components of a gas mixture are dissolved in a liquid. The inverse operation, called stripping or desorption, is employed when it is desired to transfer volatile components from a liquid mixture into a gas. Both absorption and stripping, in common with distillation (Sec. 13), make use of special equipment for bringing gas and liquid phases into intimate contact. This section is concerned with the design of gas-liquid contacting equipment, as well as with the design of absorption and stripping processes.

Equipment Absorption, stripping, and distillation operations are usually carried out in vertical, cylindrical columns or towers in which devices such as plates or packing elements are placed. The gas and liquid normally flow countercurrently, and the devices serve to provide the contacting and development of interfacial surface through which mass transfer takes place. Background material on this mass transfer process is given in Sec. 5.

Design Procedures The procedures to be followed in specifying the principal dimensions of gas absorption and distillation equipment are described in this section and are supported by several worked-out examples. The experimental data required for executing the designs

are keyed to appropriate references or to other sections of the handbook.

For absorption, stripping, and distillation, there are three main steps involved in design:

1. *Data on the gas-liquid or vapor-liquid equilibrium for the system at hand.* If absorption, stripping, and distillation operations are considered equilibrium-limited processes, which is the usual approach, these data are critical for determining the maximum possible separation. In some cases, the operations are considered rate-based (see Sec. 13) but require knowledge of equilibrium at the phase interface. Other data required include physical properties such as viscosity and density and thermodynamic properties such as enthalpy. Section 2 deals with sources of such data.

2. *Information on the liquid- and gas-handling capacity of the contacting device chosen for the particular separation problem.* Such information includes pressure drop characteristics of the device, in order that an optimum balance between capital cost (column cross section) and energy requirements might be achieved. Capacity and pressure drop characteristics of the available devices are covered later in this Sec. 14.

3. *Determination of the required height of contacting zone for the separation to be made as a function of properties of the fluid mixtures and mass-transfer efficiency of the contacting device.* This determination involves the calculation of mass-transfer parameters such as heights of transfer units and plate efficiencies as well as equilibrium or rate parameters such as theoretical stages or numbers of transfer units. An additional consideration for systems in which chemical reaction occurs is the provision of adequate residence time for desired reactions to occur, or minimal residence time to prevent undesired reactions from occurring. For equilibrium-based operations, the parameters for required height are covered in the present section.

Data Sources in the Handbook Sources of data for the analysis or design of absorbers, strippers, and distillation columns are manifold, and a detailed listing of them is outside the scope of the presentation in this section. Some key sources within the handbook are shown in Table 14-1.

Equilibrium Data Finding reliable gas-liquid and vapor-liquid equilibrium data may be the most time-consuming task associated with the design of absorbers and other gas-liquid contactors, and yet it may be the most important task at hand. For gas solubility, an important data source is the set of volumes edited by Kertes et al., *Solubility Data Series*, published by Pergamon Press (1979 ff.). In the introduction to each volume, there is an excellent discussion and definition of the various methods by which gas solubility data have been reported, such as the Bunsen coefficient, the Kuenen coefficient, the Ostwald coefficient, the absorption coefficient, and the Henry's law coefficient. The fifth edition of *The Properties of Gases and Liquids* by Poling, Prausnitz, and O'Connell (McGraw-Hill, New York, 2000) provides data and recommended estimation methods for gas solubility as well as the broader area of vapor-liquid equilibrium. Finally, the Chemistry Data Series by Gmehling et al., especially the title *Vapor-Liquid Equilibrium Collection* (DECHEMA, Frankfurt, Germany, 1979 ff.), is a rich source of data evaluated

against the various models used for interpolation and extrapolation. Section 13 of this handbook presents a good discussion of equilibrium K values.

TABLE 14-1 Directory to Key Data for Absorption and Gas-Liquid Contactor Design

Type of data	Section
Phase equilibrium data	
Gas solubilities	2
Pure component vapor pressures	2
Equilibrium K values	13
Thermal data	
Heats of solution	2
Specific heats	2
Latent heats of vaporization	2
Transport property data	
Diffusion coefficients	
Liquids	2
Gases	2
Viscosities	
Liquids	2
Gases	2
Densities	
Liquids	2
Gases	2
Surface tensions	2
Packed tower data	
Pressure drop and flooding	14
Mass transfer coefficients	5
HTU, physical absorption	5
HTU with chemical reaction	14
Height equivalent to a theoretical plate (HETP)	
Plate tower data	
Pressure drop and flooding	14
Plate efficiencies	14
Costs of gas-liquid contacting equipment	14

DESIGN OF GAS ABSORPTION SYSTEMS

General Design Procedure The design engineer usually is required to determine (1) the best solvent; (2) the best gas velocity through the absorber, or, equivalently, the vessel diameter; (3) the height of the vessel and its internal members, which is the height and type of packing or the number of contacting trays; (4) the optimum solvent circulation rate through the absorber and stripper; (5) temperatures of streams entering and leaving the absorber and stripper, and the quantity of heat to be removed to account for the heat of solution and other thermal effects; (6) pressures at which the absorber and stripper will operate; and (7) mechanical design of the absorber and stripper vessels (predominantly columns or towers), including flow distributors and packing supports. This section covers these aspects.

The problem presented to the designer of a gas absorption system usually specifies the following quantities: (1) gas flow rate; (2) gas composition of the component or components to be absorbed; (3) operating pressure and allowable pressure drop across the absorber; (4) minimum recovery of one or more of the solutes; and, possibly, (5) the solvent to be employed. Items 3, 4, and 5 may be subject to economic considerations and therefore are left to the designer. For determination of the number of variables that must be specified to fix a unique solution for the absorber design, one may use the same phase-rule approach described in Sec. 13 for distillation systems.

Recovery of the solvent, occasionally by chemical means but more often by distillation, is almost always required and is considered an integral part of the absorption system process design. A more complete solvent-stripping operation normally will result in a less costly absorber because of a lower concentration of residual solute in the regenerated (lean) solvent, but this may increase the overall cost of the entire absorption system. A more detailed discussion of these and other economical considerations is presented later in this section.

The design calculations presented in this section are relatively simple and usually can be done by using a calculator or spreadsheet. In many cases, the calculations are explained through design diagrams. It is recognized that most engineers today will perform rigorous, detailed calculations using process simulators. The design procedures presented in this section are intended to be complementary to the rigorous computerized calculations by presenting approximate estimates and insight into the essential elements of absorption and stripping operations.

Selection of Solvent and Nature of Solvents When a choice is possible, preference is given to solvents with high solubilities for the target solute and high selectivity for the target solute over the other species in the gas mixture. A high solubility reduces the amount of liquid to be circulated. The solvent should have the advantages of low volatility, low cost, low corrosive tendencies, high stability, low viscosity, low tendency to foam, and low flammability. Since the exit gas normally leaves saturated with solvent, solvent loss can be costly and can cause environmental problems. The choice of the solvent is a key part of the process economic analysis and compliance with environmental regulations.

Typically, a solvent that is chemically similar to the target solute or that reacts with it will provide high solubility. Water is often used for polar and acidic solutes (e.g., HCl), oils for light hydrocarbons, and special chemical solvents for acid gases such as CO_2 , SO_2 , and H_2S . Solvents are classified as physical and chemical. A chemical solvent forms complexes or chemical compounds with the solute, while physical solvents have only weaker interactions with the solute. Physical and chemical solvents are compared and contrasted by examining the solubility of CO_2 in propylene carbonate (representative physical solvent) and aqueous monoethanolamine (MEA; representative chemical solvent).

Figures 14-1 and 14-2 present data for the solubility of CO_2 in the two representative solvents, each at two temperatures: 40 and 100°C.

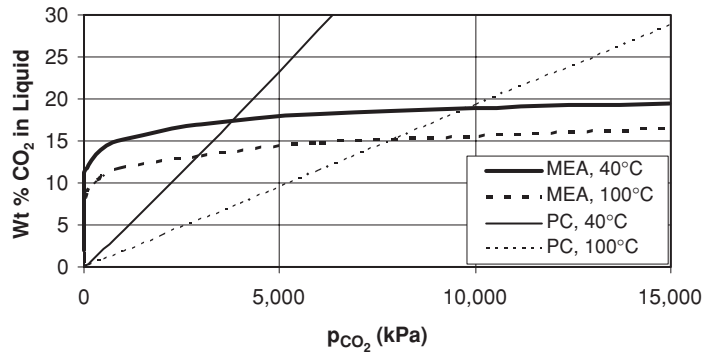


FIG. 14-1 Solubility of CO₂ in 30 wt% MEA and propylene carbonate. Linear scale.

The propylene carbonate data are from Zubchenko et al. [*Zhur. Priklad. Khim.*, **44**, 2044–2047 (1971)], and the MEA data are from Jou, Mather, and Otto [*Can. J. Chem. Eng.*, **73**, 140–147 (1995)]. The two figures have the same content, but Fig. 14-2 focuses on the low-pressure region by converting both composition and pressure to the logarithm scale. Examination of the two sets of data reveals the following characteristics and differences of physical and chemical solvents, which are summarized in the following table:

Characteristic	Physical solvent	Chemical solvent
Solubility variation with pressure	Relatively linear	Highly nonlinear
Low-pressure solubility	Low	High
High-pressure solubility	Continues to increase	Levels off
Heat of solution—related to variation of solubility with temperature at fixed pressure	Relatively low and approximately constant with loading	Relatively high and decreases somewhat with increased solute loading

Chemical solvents are usually preferred when the solute must be reduced to very low levels, when high selectivity is needed, and when the solute partial pressure is low. However, the strong absorption at low solute partial pressures and the high heat of solution are disadvantages for stripping. For chemical solvents, the strong nonlinearity of the absorption makes it necessary that accurate absorption data for the conditions of interest be available.

Selection of Solubility Data Solubility values are necessary for design because they determine the liquid rate necessary for complete or economic solute recovery. Equilibrium data generally will be found in one of three forms: (1) solubility data expressed either as weight or mole percent or as Henry's law coefficients; (2) pure-component vapor pressures; or (3) equilibrium distribution coefficients (*K* values).

Data for specific systems may be found in Sec. 2; additional references to sources of data are presented in this section.

To define completely the solubility of gas in a liquid, it is generally necessary to state the temperature, equilibrium partial pressure of the solute gas in the gas phase, and the concentration of the solute gas in the liquid phase. Strictly speaking, the total pressure of the system should also be identified, but for low pressures (less than about 507 kPa or 5 atm), the solubility for a particular partial pressure of the solute will be relatively independent of the total pressure.

For many physical systems, the equilibrium relationship between solute partial pressure and liquid-phase concentration is given by Henry's law:

$$p_A = Hx_A \tag{14-1}$$

or

$$p_A = H'c_A \tag{14-2}$$

where *H* is Henry's law coefficient expressed in kPa per mole fraction solute in liquid and *H'* is Henry's law coefficient expressed in kPa·m³/kmol.

Figure 14-1 indicates that Henry's law is valid to a good approximation for the solubility CO₂ in propylene carbonate. In general, Henry's law is a reasonable approximation for physical solvents. If Henry's law holds, the solubility is defined by knowing (or estimating) the value of the constant *H* (or *H'*).

Note that the assumption of Henry's law will lead to incorrect results for solubility of chemical systems such as CO₂-MEA (Figs. 14-1 and 14-2) and HCl-H₂O. Solubility modeling for chemical systems requires the use of a *speciation model*, as described later in this section.

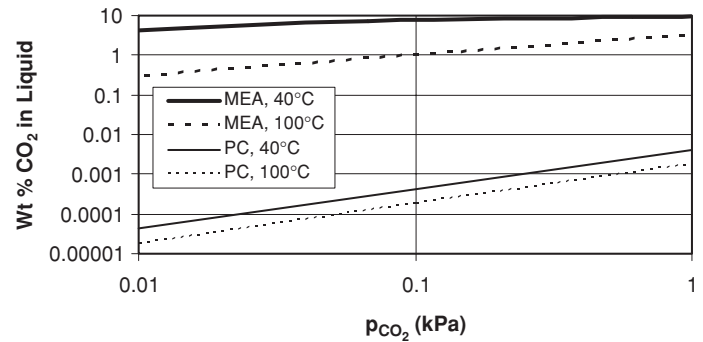


FIG. 14-2 Solubility of CO₂ in 30 wt% MEA and propylene carbonate. Logarithm scale and focus on low-pressure region.

For quite a number of physically absorbed gases, Henry's law holds very well when the partial pressure of the solute is less than about 101 kPa (1 atm). For partial pressures above 101 kPa, H may be independent of the partial pressure (Fig. 14-1), but this needs to be verified for the particular system of interest. The variation of H with temperature is a strongly nonlinear function of temperature as discussed by Poling, Prausnitz, and O'Connell (*The Properties of Gases and Liquids*, 5th ed., McGraw-Hill, New York, 2000). Consultation of this reference is recommended when temperature and pressure extrapolations of Henry's law data are needed.

The use of Henry's law constants is illustrated by the following example.

Example 1: Gas Solubility It is desired to find out how much hydrogen can be dissolved in 100 weights of water from a gas mixture when the total pressure is 101.3 kPa (760 torr; 1 atm), the partial pressure of the H_2 is 26.7 kPa (200 torr), and the temperature is 20°C. For partial pressures up to about 100 kPa the value of H is given in Sec. 3 as 6.92×10^6 kPa (6.83×10^4 atm) at 20°C. According to Henry's law,

$$x_{H_2} = p_{H_2}/H_{H_2} = 26.7/6.92 \times 10^6 = 3.86 \times 10^{-6}$$

The mole fraction x is the ratio of the number of moles of H_2 in solution to the total moles of all constituents contained. To calculate the weights of H_2 per 100 weights of H_2O , one can use the following formula, where the subscripts A and w correspond to the solute (hydrogen) and solvent (water):

$$\begin{aligned} \left(\frac{x_A}{1 - x_A} \right) \frac{M_A}{M_w} 100 &= \left(\frac{3.86 \times 10^{-6}}{1 - 3.86 \times 10^{-6}} \right) \frac{2.02}{18.02} 100 \\ &= 4.33 \times 10^{-5} \text{ weights } H_2/100 \text{ weights } H_2O \\ &= 0.43 \text{ parts per million weight} \end{aligned}$$

Pure-component vapor pressure can be used for predicting solubilities for systems in which **Raoult's law** is valid. For such systems $p_A = p_A^0 x_A$, where p_A^0 is the pure-component vapor pressure of the solute and p_A is its partial pressure. Extreme care should be exercised when using pure-component vapor pressures to predict gas absorption behavior. Both vapor-phase and liquid-phase nonidealities can cause significant deviations from Raoult's law, and this is often the reason particular solvents are used, i.e., because they have special affinity for particular solutes. The book by Poling, Prausnitz, and O'Connell (op. cit.) provides an excellent discussion of the conditions where Raoult's law is valid. Vapor-pressure data are available in Sec. 3 for a variety of materials.

Whenever data are available for a given system under similar conditions of temperature, pressure, and composition, **equilibrium distribution coefficients** ($K = y/x$) provide a much more reliable tool for predicting vapor-liquid distributions. A detailed discussion of equilibrium K values is presented in Sec. 13.

Calculation of Liquid-to-Gas Ratio The minimum possible liquid rate is readily calculated from the composition of the entering gas and the solubility of the solute in the exit liquor, with equilibrium being assumed. It may be necessary to estimate the temperature of the exit liquid based upon the heat of solution of the solute gas. Values of latent heat and specific heat and values of heats of solution (at infinite dilution) are given in Sec. 2.

The actual liquid-to-gas ratio (solvent circulation rate) normally will be greater than the minimum by as much as 25 to 100 percent, and the estimated factor may be arrived at by economic considerations as well as judgment and experience. For example, in some packed-tower applications involving very soluble gases or vacuum operation, the minimum quantity of solvent needed to dissolve the solute may be insufficient to keep the packing surface thoroughly wet, leading to poor distribution of the liquid stream.

When the solvent concentration in the inlet gas is low and when a significant fraction of the solute is absorbed (this often the case), the approximation

$$y_1 G_M = x_1 L_M = (y_1^0/m) L_M \quad (14-3)$$

leads to the conclusion that the ratio mG_M/L_M represents the fractional approach of the exit liquid to saturation with the inlet gas, i.e.,

$$mG_M/L_M = y_1^0/y_1 \quad (14-4)$$

Optimization of the liquid-to-gas ratio in terms of total annual costs often suggests that the molar liquid-to-gas ratio L_M/G_M should be about 1.2 to 1.5 times the theoretical minimum corresponding to equilibrium at the rich end of the tower (infinite height or number of trays), provided flooding is not a problem. This, for example, would be an alternative to assuming that $L_M/G_M \approx m/0.7$.

When the exit-liquor temperature rises because of the heat of absorption of the solute, the value of m changes through the tower, and the liquid-to-gas ratio must be chosen to give reasonable values of $m_1 G_M/L_M$ and $m_2 G_M/L_M$, where the subscripts 1 and 2 refer to the bottom and top of the absorber, respectively. For this case, the value of $m_2 G_M/L_M$ will be taken to be somewhat less than 0.7, so that the value of $m_1 G_M/L_M$ will not approach unity too closely. This rule-of-thumb approach is useful only when the solute concentration is low and heat effects are negligible.

When the solute has a large heat of solution or when the feed gas contains high concentrations of the solute, one should consider the use of internal cooling coils or intermediate liquid withdrawal and cooling to remove the heat of absorption.

Selection of Equipment Trays and random packings have been extensively used for gas absorption; structured packings are less common. Compared to trays, random packings have the advantages of availability in low-cost, corrosion-resistant materials (such as plastics and ceramics), low pressure drop (which can be an advantage when the tower is in the suction of a fan or compressor), easy and economic adaptability to small-diameter (less than 0.6-m or 2-ft) columns, and excellent handling of foams. Trays are much better for handling solids and fouling applications, offer greater residence time for slow absorption reactions, can better handle high L/G ratios and intermediate cooling, give better liquid turndown, and are more robust and less prone to reliability issues such as those resulting from poor distribution. Details on the operating characteristics of tray and packed towers are given later in this section.

Column Diameter and Pressure Drop Flooding determines the minimum possible diameter of the absorber column, and the usual design is for 60 to 80 percent of the flooding velocity. In near-atmospheric applications, pressure drop usually needs to be minimized to reduce the cost of energy for compression of the feed gas. For systems having a significant tendency to foam, the maximum allowable velocity will be lower than the estimated flooding velocity. Methods for predicting flooding velocities and pressure drops are given later in this section.

Computation of Tower Height The required height of a gas absorption or stripping tower for physical solvents depends on (1) the phase equilibria involved; (2) the specified degree of removal of the solute from the gas; and (3) the mass-transfer efficiency of the device. These three considerations apply to both tray and packed towers. Items 1 and 2 dictate the required number of theoretical stages (tray tower) or transfer units (packed tower). Item 3 is derived from the tray efficiency and spacing (tray tower) or from the height of one transfer unit (packed tower). Solute removal specifications are usually derived from economic considerations.

For tray towers, the approximate design methods described below may be used in estimating the number of theoretical stages, and the tray efficiencies and spacings for the tower can be specified on the basis of the information given later. Considerations involved in the rigorous design of theoretical stages for tray towers are treated in Sec. 13.

For packed towers, the continuous differential nature of the contact between gas and liquid leads to a design procedure involving the solution of differential equations, as described in the next subsection. Note that the design procedures discussed in this section are not applicable to reboiled absorbers, which should be designed according to the procedures described in Sec. 13.

Caution is advised in distinguishing between systems involving pure physical absorption and those in which chemical reactions can significantly affect design procedures. Chemical systems require additional procedures, as described later in this section.

Selection of Stripper Operating Conditions Stripping involves the removal of one or more components from the solvent through the application of heat or contacting it with a gas such as steam, nitrogen,

or air. The operating conditions chosen for stripping normally result in a low solubility of solute (i.e., high value of m), so that the ratio mG_M/L_M will be larger than unity. A value of 1.4 may be used for rule-of-thumb calculations involving pure physical absorption. For tray-tower calculations, the stripping factor $S = KG_M/L_M$, where $K = y^0/x$ usually is specified for each tray.

When the solvent from an absorption operation must be regenerated, one may employ a "pressure-swing" or "temperature-swing" concept, or a combination of the two, in specifying the stripping operation. In pressure-swing operation, the temperature of the stripper is about the same as that of the absorber, but the stripping pressure is much lower. In temperature-swing operation, the pressures are about equal, but the stripping temperature is much higher than the absorption temperature.

In pressure-swing operation, a portion of the gas may be "sprung" from the liquid by the use of a flash drum upstream of the stripper feed point. This type of operation has been discussed by Burrows and Preece [*Trans. Inst. Chem. Eng.*, **32**, 99 (1954)] and by Langley and Haselden [*Inst. Chem. Eng. Symp. Ser. (London)*, no. 28 (1968)]. If the flashing of the liquid takes place inside the stripping tower, this effect must be accounted for in the design of the upper section in order to avoid overloading and flooding near the top of the tower.

Often the rate at which residual absorbed gas can be driven from the liquid in a stripping tower is limited by the rate of a chemical reaction, in which case the liquid-phase residence time (and hence the tower liquid holdup) becomes the most important design factor. Thus, many stripper regenerators are designed on the basis of liquid holdup rather than on the basis of mass-transfer rate.

Approximate design equations applicable only to the case of pure physical desorption are developed later in this section for both packed and tray stripping towers. A more rigorous approach using distillation concepts may be found in Sec. 13. A brief discussion of desorption with chemical reaction is given in the subsection "Absorption with Chemical Reaction."

Design of Absorber-Stripper Systems The solute-rich liquor leaving a gas absorber normally is distilled or stripped to regenerate the solvent for recirculation back to the absorber, as depicted in Fig. 14-3. It is apparent that the conditions selected for the absorption step

(e.g., temperature, pressure, L_M/G_M) will affect the design of the stripping tower, and conversely, a selection of stripping conditions will affect the absorber design. The choice of optimum operating conditions for an absorber-stripper system therefore involves a combination of economic factors and practical judgments as to the operability of the system within the context of the overall process flow sheet. In Fig. 14-3, the stripping vapor is provided by a reboiler; alternately, an extraneous stripping gas may be used.

An appropriate procedure for executing the design of an absorber-stripper system is to set up a carefully selected series of design cases and then evaluate the investment costs, the operating costs, and the operability of each case. Some of the economic factors that need to be considered in selecting the optimum absorber-stripper design are discussed later in the subsection "Economic Design of Absorption Systems."

Importance of Design Diagrams One of the first things a designer should do is to lay out a carefully constructed equilibrium curve $y^0 = F(x)$ on an xy diagram, as shown in Fig. 14-4. A horizontal line corresponding to the inlet-gas composition y_1 is then the locus of feasible outlet-liquor compositions, and a vertical line corresponding to the inlet-solvent-liquor composition x_2 is the locus of outlet-gas compositions. These lines are indicated as $y = y_1$ and $x = x_2$, respectively on Fig. 14-4.

For gas absorption, the region of feasible operating lines lies above the equilibrium curve; for stripping, the feasible region for operating lines lies below the equilibrium curve. These feasible regions are bounded by the equilibrium curve and by the lines $x = x_2$ and $y = y_1$. By inspection, one should be able to visualize those operating lines that are feasible and those that would lead to "pinch points" within the tower. Also, it is possible to determine if a particular proposed design for solute recovery falls within the feasible envelope.

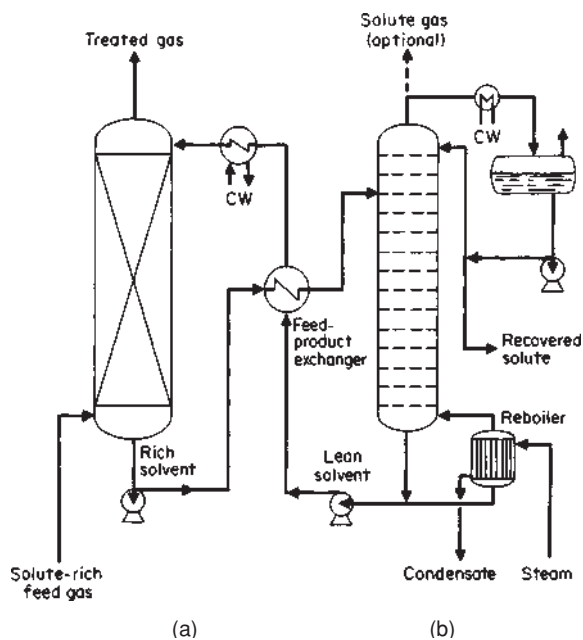


FIG. 14-3 Gas absorber using a solvent regenerated by stripping. (a) Absorber. (b) Stripper.

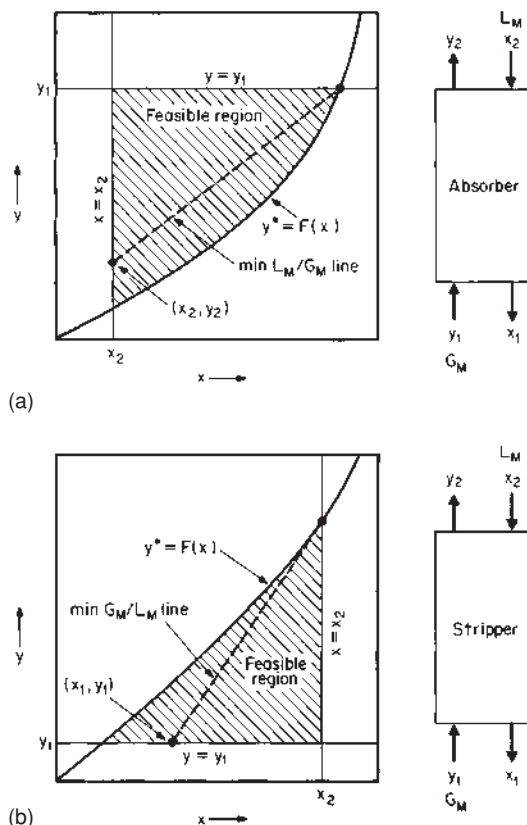


FIG. 14-4 Design diagrams for (a) absorption and (b) stripping.

Once the design recovery for an absorber has been established, the operating line can be constructed by first locating the point x_2, y_2 on the diagram. The intersection of the horizontal line corresponding to the inlet gas composition y_1 with the equilibrium curve $y^0 = F(x)$ defines the theoretical minimum liquid-to-gas ratio for systems in which there are no intermediate pinch points. This operating line which connects this point with the point x_2, y_2 corresponds to the minimum value of L_M/G_M . The actual design value of L_M/G_M should normally be around 1.2 to 1.5 times this minimum value. Thus, the actual design operating line for a gas absorber will pass through the point x_2, y_2 and will intersect the line $y = y_1$ to the left of the equilibrium curve.

For stripping one begins by using the design specification to locate the point x_1, y_1 ; then the intersection of the vertical line $x = x_2$ with the equilibrium curve $y^0 = F(x)$ defines the theoretical minimum gas-to-liquid ratio. The actual value of G_M/L_M is chosen to be about 20 to 50 percent higher than this minimum, so the actual design operating line will intersect the line $x = x_2$ at a point somewhat below the equilibrium curve.

PACKED-TOWER DESIGN

Methods for estimating the height of the active section of counterflow differential contactors such as packed towers, spray towers, and falling-film absorbers are based on rate expressions representing mass transfer at a point on the gas-liquid interface and on material balances representing the changes in bulk composition in the two phases that flow past each other. The rate expressions are based on the interphase mass-transfer principles described in Sec. 5. Combination of such expressions leads to an integral expression for the number of transfer units or to equations related closely to the number of theoretical stages. The paragraphs which follow set forth convenient methods for using such equations, first in a general case and then for cases in which simplifying assumptions are valid.

Use of Mass-Transfer-Rate Expression Figure 14-5 shows a section of a packed absorption tower together with the nomenclature that will be used in developing the equations that follow. In a differential section dh , we can equate the rate at which solute is lost from the gas phase to the rate at which it is transferred through the gas phase to the interface as follows:

$$-d(G_M y) = -G_M dy - y dG_M = N_A a dh \quad (14-5)$$

In Eq. (14-5), G_M is the gas-phase molar velocity [kmol/(s·m²)], N_A is the mass-transfer flux [kmol/(s·m²)], and a is the effective interfacial area (m²/m³).

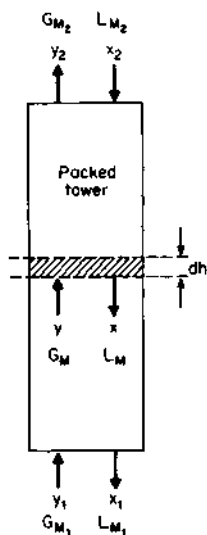


FIG. 14-5 Nomenclature for material balances in a packed-tower absorber or stripper.

When only one component is transferred,

$$dG_M = -N_A a dh \quad (14-6)$$

Substitution of this relation into Eq. (14-5) and rearranging yield

$$dh = -\frac{G_M dy}{N_A a (1-y)} \quad (14-7)$$

For this derivation we use the gas-phase rate expression $N_A = k_G(y - y_i)$ and integrate over the tower to obtain

$$h_T = \int_{y_2}^{y_1} \frac{G_M dy}{k_G a (1-y)(y - y_i)} \quad (14-8)$$

Multiplying and dividing by y_{BM} place Eq. (14-8) into the $H_G N_G$ format

$$\begin{aligned} h_T &= \int_{y_2}^{y_1} \left(\frac{G_M}{k_G a y_{BM}} \right) \frac{y_{BM} dy}{(1-y)(y - y_i)} \\ &= H_{G,av} \int_{y_2}^{y_1} \frac{y_{BM} dy}{(1-y)(y - y_i)} = H_{G,av} N_G \end{aligned} \quad (14-9)$$

The general expression given by Eq. (14-8) is more complex than normally is required, but it must be used when the mass-transfer coefficient varies from point to point, as may be the case when the gas is not dilute or when the gas velocity varies as the gas dissolves. The values of y_i to be used in Eq. (14-8) depend on the local liquid composition x_i and on the temperature. This dependency is best represented by using the operating and equilibrium lines as discussed later.

Example 2 illustrates the use of Eq. (14-8) for scrubbing chlorine from air with aqueous caustic solution. For this case one can make the simplifying assumption that y_i , the interfacial partial pressure of chlorine over the caustic solution, is zero due to the rapid and complete reaction of the chlorine after it dissolves. We note that the feed gas is not dilute.

Example 2: Packed Height Requirement Let us compute the height of packing needed to reduce the chlorine concentration of 0.537 kg/(s·m²), or 396 lb/(h·ft²), of a chlorine-air mixture containing 0.503 mole-fraction chlorine to 0.0403 mole fraction. On the basis of test data described by Sherwood and Pigford (*Absorption and Extraction*, McGraw-Hill, 1952, p. 121) the value of $k_G a y_{BM}$ at a gas velocity equal to that at the bottom of the packing is equal to 0.1175 kmol/(s·m³), or 26.4 lb-mol/(h·ft³). The equilibrium back pressure y_i can be assumed to be negligible.

Solution. By assuming that the mass-transfer coefficient varies as the 0.8 power of the local gas mass velocity, we can derive the following relation:

$$\hat{K}_G a = k_G a y_{BM} = 0.1175 \left[\frac{71y + 29(1-y)}{71y_1 + 29(1-y_1)} \left(\frac{1-y_1}{1-y} \right) \right]^{0.8}$$

where 71 and 29 are the molecular weights of chlorine and air respectively. Noting that the inert-gas (air) mass velocity is given by $G'_M = G_M(1-y) = 5.34 \times 10^{-3}$ kmol/(s·m²), or 3.94 lb-mol/(h·ft²), and introducing these expressions into the integral gives

$$h_T = 1.82 \int_{0.0403}^{0.503} \left[\frac{1-y}{29 + 42y} \right]^{0.8} \frac{dy}{(1-y)^2 \ln [1/(1-y)]}$$

This definite integral can be evaluated numerically by the use of Simpson's rule to obtain $h_T = 0.305$ m (1 ft).

Use of Operating Curve Frequently, it is not possible to assume that $y_i = 0$ as in Example 2, due to diffusional resistance in the liquid phase or to the accumulation of solute in the liquid stream. When the backpressure cannot be neglected, it is necessary to supplement the equations with a material balance representing the operating line or curve. In view of the countercurrent flows into and from the differential section of packing shown in Fig. 14-5, a steady-state material balance leads to the following equivalent relations:

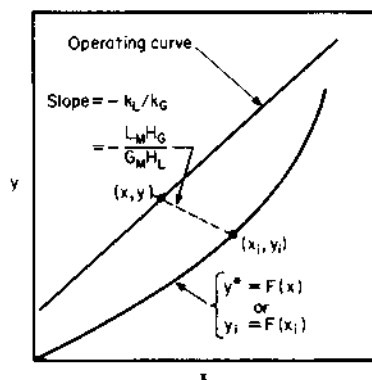


FIG. 14-6 Relationship between equilibrium curve and operating curve in a packed absorber; computation of interfacial compositions.

$$d(G_M y) = d(L_M x) \quad (14-10)$$

$$G'_M \frac{dy}{(1-y)^2} = L'_M \frac{dx}{(1-x)^2} \quad (14-11)$$

where L'_M = molar mass velocity of the inert-liquid component and G'_M = molar mass velocity of the inert gas; L_M , L'_M , G_M , and G'_M are superficial velocities based upon the total tower cross section.

Equation (14-11) is the differential equation of the operating curve, and its integral around the upper portion of the packing is the equation for the operating curve.

$$G'_M \left[\frac{y}{1-y} - \frac{y_2}{1-y_2} \right] = L'_M \left[\frac{x}{1-x} - \frac{x_2}{1-x_2} \right] \quad (14-12)$$

For dilute solutions in which the mole fractions of x and y are small, the total molar flows G_M and L_M will be nearly constant, and the operating-curve equation is

$$G_M(y - y_2) = L_M(x - x_2) \quad (14-13)$$

This equation gives the relation between the bulk compositions of the gas and liquid streams at each height in the tower for conditions in which the operating curve can be approximated as a straight line.

Figure 14-6 shows the relationship between the operating curve and the equilibrium curve $y_i = F(x_i)$ for a typical example involving solvent recovery, where y_i and x_i are the interfacial compositions (assumed to be in equilibrium). Once y is known as a function of x along the operating curve, y_i can be found at corresponding points on the equilibrium curve by

$$(y - y_i)/(x_i - x) = k_L/k_G = L_M H_G / G_M H_L \quad (14-14)$$

where L_M = molar liquid mass velocity, G_M = molar gas mass velocity, H_L = height of one transfer unit based upon liquid-phase resistance, and H_G = height of one transfer unit based upon gas-phase resistance. Using this equation, the integral in Eq. (14-8) can be evaluated.

Calculation of Transfer Units In the general case, the equations described above must be employed in calculating the height of packing required for a given separation. However, if the local mass-transfer coefficient $k_G a y_{BM}$ is approximately proportional to the first power of the local gas velocity G_M , then the height of one gas-phase transfer unit, defined as $H_G = G_M / k_G a y_{BM}$, will be constant in Eq. (14-9). Similar considerations lead to an assumption that the height of one overall gas-phase transfer unit H_{OG} may be taken as constant. The height of packing required is then calculated according to the relation

$$h_T = H_G N_G = H_{OG} N_{OG} \quad (14-15)$$

where N_G = number of gas-phase transfer units and N_{OG} = number of overall gas-phase transfer units. When H_G and H_{OG} are not constant, it

may be valid to employ averaged values between the top and bottom of the tower and the relation

$$h_T = H_{G,av} N_G = H_{OG,av} N_{OG} \quad (14-16)$$

In these equations, the terms N_G and N_{OG} are defined by Eqs. (14-17) and (14-18).

$$N_G = \int_{y_2}^{y_1} \frac{y_{BM} dy}{(1-y)(y-y_i)} \quad (14-17)$$

$$N_{OG} = \int_{y_2}^{y_1} \frac{y_{BM}^o dy}{(1-y)(y-y^o)} \quad (14-18)$$

Equation (14-18) is the more useful one in practice. It requires either actual experimental H_{OG} data or values estimated by combining individual measurements of H_G and H_L by Eq. (14-19). Correlations for H_G , H_L , and H_{OG} in nonreacting systems are presented in Sec. 5.

$$H_{OG} = \frac{y_{BM}}{y_{BM}^o} H_G + \frac{m G_M}{L_M} \frac{x_{BM}}{y_{BM}^o} H_L \quad (14-19a)$$

$$H_{OL} = \frac{x_{BM}}{x_{BM}^o} H_L + \frac{L_M}{m G_M} \frac{y_{BM}}{x_{BM}^o} H_G \quad (14-19b)$$

On occasion, the changes in gas flow and in the mole fraction of inert gas can be neglected so that inclusion of terms such as $1-y$ and y_{BM}^o can be approximated, as is shown below.

One such simplification was suggested by Wiegand [*Trans. Am. Inst. Chem. Eng.*, **36**, 679 (1940)], who pointed out that the logarithmic-mean mole fraction of inert gas y_{BM}^o (or y_{BM}) is often very nearly equal to the arithmetic mean. Thus, substitution of the relation

$$\frac{y_{BM}^o}{(1-y)} = \frac{(1-y^o) + (1-y)}{2(1-y)} = \frac{y-y^o}{2(1-y)} + 1 \quad (14-20)$$

into the equations presented above leads to the simplified forms

$$N_G = \frac{1}{2} \ln \frac{1-y_2}{1-y_1} + \int_{y_2}^{y_1} \frac{dy}{y-y_i} \quad (14-21)$$

$$N_{OG} = \frac{1}{2} \ln \frac{1-y_2}{1-y_1} + \int_{y_2}^{y_1} \frac{dy}{y-y^o} \quad (14-22)$$

The second (integral) terms represent the numbers of transfer units for an infinitely dilute gas. The first terms, usually only a small correction, give the effect of a finite level of gas concentration.

The procedure for applying Eqs. (14-21) and (14-22) involves two steps: (1) evaluation of the integrals and (2) addition of the correction corresponding to the first (logarithmic) term. The discussion that follows deals only with the evaluation of the integral term (first step).

The simplest possible case occurs when (1) both the operating and equilibrium lines are straight (i.e., the solutions are dilute); (2) Henry's law is valid ($y/x = y_i/x_i = m$); and (3) absorption heat effects are negligible. Under these conditions, the integral term in Eq. (14-21) may be computed by Colburn's equation [*Trans. Am. Inst. Chem. Eng.*, **35**, 211 (1939)]:

$$N_{OG} = \frac{1}{1 - m G_M / L_M} \ln \left[\left(1 - \frac{m G_M}{L_M} \right) \left(\frac{y_1 - m x_2}{y_2 - m x_2} \right) + \frac{m G_M}{L_M} \right] \quad (14-23)$$

Figure 14-7 is a plot of Eq. (14-23) from which the value of N_{OG} can be read directly as a function of $m G_M / L_M$ and the ratio of concentrations. This plot and Eq. (14-23) are equivalent to the use of a logarithmic mean of terminal driving forces, but they are more convenient because one does not need to compute the exit-liquor concentration x_1 .

In many practical situations involving nearly complete cleanup of the gas, an approximate result can be obtained from the equations just presented even when the simplifications are not valid, i.e., solutions are concentrated and heat effects occur. In such cases the driving forces in the upper part of the tower are very much smaller than those at the bottom, and the value of $m G_M / L_M$ used in the equations should be the ratio of the operating line L_M / G_M in the low-concentration region near the top of the tower.

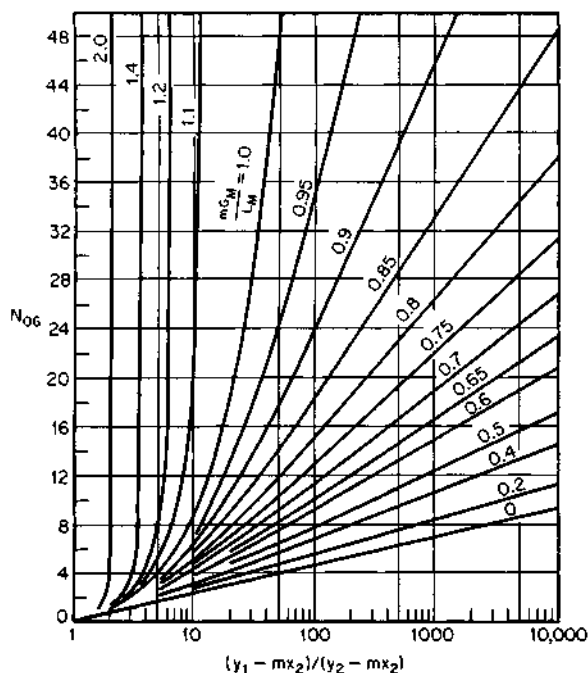


FIG. 14-7 Number of overall gas-phase mass-transfer units in a packed absorption tower for constant mG_M/L_M ; solution of Eq. (14-23). (From Sherwood and Pigford, Absorption and Extraction, McGraw-Hill, New York, 1952.)

Another approach is to divide the tower arbitrarily into a lean section (near the top) where approximate methods are valid, and to deal with the rich section separately. If the heat effects in the rich section are appreciable, consideration should be given to installing cooling units near the bottom of the tower. In any event, a design diagram showing the operating and equilibrium curves should be prepared to check the applicability of any simplified procedure. Figure 14-10, presented in Example 6, is one such diagram for an adiabatic absorption tower.

Stripping Equations Stripping or desorption involves the removal of a volatile component from the liquid stream by contact with an inert gas such as nitrogen or steam or the application of heat. Here the change in concentration of the liquid stream is of prime importance, and it is more convenient to formulate the rate equation analogous to Eq. (14-6) in terms of the liquid composition x . This leads to the following equations defining the number of transfer units and height of transfer units based on liquid-phase resistance:

$$h_T = H_L \int_{x_2}^{x_1} \frac{x_{BM} dx}{(1-x)(x_1-x)} = H_L N_L \quad (14-24)$$

$$h_T H_{OL} \int_{x_2}^{x_1} \frac{x_{BM}^0 dx}{(1-x)(x^0-x)} = H_{OL} N_{OL} \quad (14-25)$$

where, as before, subscripts 1 and 2 refer to the bottom and top of the tower, respectively (see Fig. 14-5).

In situations where one cannot assume that H_L and H_{OL} are constant, these terms need to be incorporated inside the integrals in Eqs. (14-24) and (14-25), and the integrals must be evaluated numerically (using Simpson's rule, for example). In the normal case involving stripping without chemical reactions, the liquid-phase resistance will dominate, making it preferable to use Eq. (14-25) together with the approximation $H_L \approx H_{OL}$.

The Weigand approximations of the above integrals, in which arithmetic means are substituted for the logarithmic means (x_{BM} and x_{BM}^0), are

$$N_L = \frac{1}{2} \ln \frac{1-x_1}{1-x_2} + \int_{x_1}^{x_2} \frac{dx}{x-x_1} \quad (14-26)$$

$$N_{OL} = \frac{1}{2} \ln \frac{1-x_1}{1-x_2} + \int_{x_1}^{x_2} \frac{dx}{x-x^0} \quad (14-27)$$

In these equations, the first term is a correction for finite liquid-phase concentrations, and the integral term represents the numbers of transfer units required for dilute solutions. In most practical stripper applications, the first (logarithmic) term is relatively small.

For dilute solutions in which both the operating and the equilibrium lines are straight and in which heat effects can be neglected, the integral term in Eq. (14-27) is

$$N_{OL} = \frac{1}{1-L_M/mG_M} \ln \left[\left(1 - \frac{L_M}{mG_M} \right) \left(\frac{x_2-y_1/m}{x_1-y_1/m} \right) + \frac{L_M}{mG_M} \right] \quad (14-28)$$

This equation is analogous to Eq. (14-23). Thus, Fig. 14-7 is applicable if the concentration ratio $(x_2-y_1/m)/(x_1-y_1/m)$ is substituted for the abscissa and the parameter on the curves is identified as L_M/mG_M .

Example 3: Air Stripping of VOCs from Water A 0.45-m diameter packed column was used by Dvorack et al. [Environ. Sci. Tech. 20, 945 (1996)] for removing trichloroethylene (TCE) from wastewater by stripping with atmospheric air. The column was packed with 25-mm Pall rings, fabricated from polypropylene, to a height of 3.0 m. The TCE concentration in the entering water was 38 parts per million by weight (ppmw). A molar ratio of entering water to entering air was kept at 23.7. What degree of removal was to be expected? The temperatures of water and air were 20°C. Pressure was atmospheric.

Solution. For TCE in water, the Henry's law coefficient may be taken as 417 atm/mf at 20°C. In this low-concentration region, the coefficient is constant and equal to the slope of the equilibrium line m . The solubility of TCE in water, based on $H = 417$ atm, is 2390 ppm. Because of this low solubility, the entire resistance to mass transfer resides in the liquid phase. Thus, Eq. (14-25) may be used to obtain N_{OL} , the number of overall liquid phase transfer units.

In the equation, the ratio $x_{BM}/(1-x)$ is unity because of the very dilute solution. It is necessary to have a value of H_L for the packing used, at given flow rates of liquid and gas. Methods for estimating H_L may be found in Sec. 5. Dvorack et al. found $H_{OL} = 0.8$ m. Then, for $h_T = 3.0$ m, $N_L = N_{OL} = 3.0/0.8 = 3.75$ transfer units.

Transfer units may be calculated from Eq. 14-25, replacing mole fractions with ppm concentrations, and since the operating and equilibrium lines are straight,

$$N_{OL} = \frac{38 - (\text{ppm})_{\text{exit}}}{\ln 38/(\text{ppm})_{\text{exit}}} = 3.75$$

Solving, $(\text{ppm})_{\text{exit}} = 0.00151$. Thus, the stripped water would contain 1.51 parts per billion of TCE.

Use of HTU and $K_G a$ Data In estimating the size of a commercial gas absorber or liquid stripper it is desirable to have data on the overall mass-transfer coefficients (or heights of transfer units) for the system of interest, and at the desired conditions of temperature, pressure, solute concentration, and fluid velocities. Such data should best be obtained in an apparatus of pilot-plant or semiworks size to avoid the abnormalities of scale-up. Within the packing category, there are both random and ordered (structured) packing elements. Physical characteristics of these devices will be described later.

When no $K_G a$ or HTU data are available, their values may be estimated by means of a generalized model. A summary of useful models is given in Sec. 5. The values obtained may then be combined by use of Eq. (14-19) to obtain values of H_{OG} and H_{OL} . This simple procedure is not valid when the rate of absorption is limited by chemical reaction.

Use of HETP Data for Absorber Design Distillation design methods (see Sec. 13) normally involve determination of the number of theoretical equilibrium stages N . Thus, when packed towers are employed in distillation applications, it is common practice to rate the efficiency of tower packings in terms of the height of packing equivalent to one theoretical stage (HETP).

The HETP of a packed-tower section, valid for either distillation or dilute-gas absorption and stripping systems in which constant molal overflow can be assumed and in which no chemical reactions occur, is related to the height of one overall gas-phase mass-transfer unit H_{OG} by the equation

$$HETP = H_{OG} \frac{\ln(mG_M/L_M)}{(mG_M/L_M - 1)} \quad (14-29)$$

For gas absorption systems in which the inlet gas is concentrated, the corrected equation is

$$HETP = \left(\frac{y_{BM}^0}{1 - y} \right)_{av} H_{OG} \frac{\ln(mG_M/L_M)}{mG_M/L_M - 1} \quad (14-30)$$

where the correction term $y_{BM}^0/(1 - y)$ is averaged over each individual theoretical stage. The equilibrium compositions corresponding to each theoretical stage may be estimated by the methods described in the next subsection, "Tray-Tower Design." These compositions are used in conjunction with the local values of the gas and liquid flow rates and the equilibrium slope m to obtain values for H_G , H_L , and H_{OG} corresponding to the conditions on each theoretical stage, and the local values of the HETP are then computed by Eq. (14-30). The total height of packing required for the separation is the summation of the individual HETPs computed for each theoretical stage.

TRAY-TOWER DESIGN

The design of a tray tower for gas absorption and gas-stripping operations involves many of the same principles employed in distillation calculations, such as the determination of the number of theoretical trays needed to achieve a specified composition change (see Sec. 13). Distillation differs from absorption because it involves the separation of components based upon the distribution of the various substances between a vapor phase and a liquid phase when all components are present in both phases. In distillation, the new phase is generated from the original phase by the vaporization or condensation of the volatile components, and the separation is achieved by introducing reflux to the top of the tower.

In gas absorption, the new phase consists of a relatively nonvolatile solvent (absorption) or a relatively insoluble gas (stripping), and normally no reflux is involved. This section discusses some of the considerations peculiar to gas absorption calculations for tray towers and some of the approximate design methods that can be applied (when simplifying assumptions are valid).

Graphical Design Procedure Construction of design diagrams (xy curves showing the equilibrium and operating curves) should be an integral part of any design involving the distribution of a single solute between an inert solvent and an inert gas. The number of theoretical trays can be stepped off rigorously, provided the curvatures of the operating and equilibrium lines are correctly represented in the diagram. The procedure is valid even though an inert solvent is present in the liquid phase and an inert gas is present in the vapor phase.

Figure 14-8 illustrates the graphical method for a three theoretical stage system. Note that in gas absorption the operating line is above the equilibrium curve, whereas in distillation this does not happen. In gas stripping, the operating line will be below the equilibrium curve.

On Fig. 14-8, note that the stepping-off procedure begins on the operating line. The starting point x_f, y_f represents the compositions of the entering lean wash liquor and of the gas exiting from the top of the tower, as defined by the design specifications. After three steps one reaches the point x_b, y_b representing the compositions of the solute-rich feed gas y_f and of the solute-rich liquor leaving the bottom of the tower x_b .

Algebraic Method for Dilute Gases By assuming that the operating and equilibrium curves are straight lines and that heat effects are negligible, Souders and Brown [Ind. Eng. Chem., 24, 519 (1932)] developed the following equation:

$$(y_1 - y_2)/(y_1 - y_2^0) = (A^{N+1} - A)/(A^{N+1} - 1) \quad (14-31)$$

where N = number of theoretical trays, y_1 = mole fraction of solute in the entering gas, y_2 = mole fraction of solute in the leaving gas, $y_2^0 = mx_2$ = mole fraction of solute in equilibrium with the incoming solvent

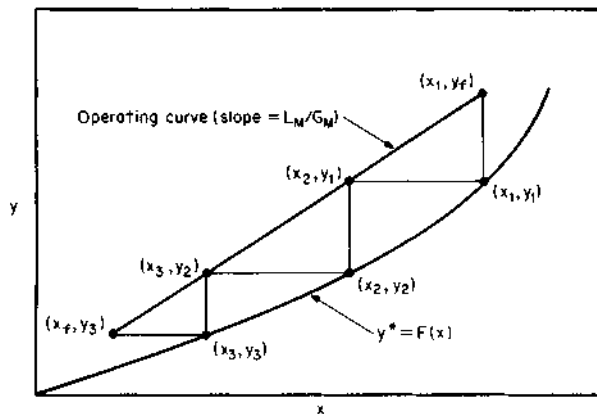


FIG. 14-8 Graphical method for a three-theoretical-plate gas-absorption tower with inlet-liquor composition x_f and inlet-gas composition y_f .

(zero for a pure solvent), and A = absorption factor = L_M/mG_M . Note that the absorption factor is the reciprocal of the expression given in Eq. (14-4) for packed columns.

Note that for the limiting case of $A = 1$, the solution is given by

$$(y_1 - y_2)/(y_1 - y_2^0) = N/(N + 1) \quad (14-32)$$

Although Eq. (14-31) is convenient for computing the composition of the exit gas as a function of the number of theoretical stages, an alternative equation derived by Colburn [Trans. Am. Inst. Chem. Eng., 35, 211 (1939)] is more useful when the number of theoretical plates is the unknown:

$$N = \frac{\ln[(1 - A^{-1})(y_1 - y_2^0)/(y_2 - y_2^0) + A^{-1}]}{\ln(A)} \quad (14-33)$$

The numerical results obtained by using either Eq. (14-31) or Eq. (14-33) are identical. Thus, the two equations may be used interchangeably as the need arises.

Comparison of Eqs. (14-33) and (14-23) shows that

$$N_{OG}/N = \ln(A)/(1 - A^{-1}) \quad (14-34)$$

thus revealing the close relationship between theoretical stages in a plate tower and mass-transfer units in a packed tower. Equations (14-23) and (14-33) are related to each other by virtue of the relation

$$h_T = H_{OG}N_{OG} = (HETP)N \quad (14-35)$$

Algebraic Method for Concentrated Gases When the feed gas is concentrated, the absorption factor, which is defined in general as $A = L_M/KG_M$ and where $K = y^0/x$, can vary throughout the tower due to changes in temperature and composition. An approximate solution to this problem can be obtained by substituting the "effective" adsorption factors A_e and A' derived by Edmister [Ind. Eng. Chem. 35, 837 (1943)] into the equation

$$\frac{y_1 - y_2}{y_1} = \left[1 - \frac{1}{A'} \frac{(L_M x_2)}{(G_M y_1)} \right] \frac{A_e^{N+1} - A_e}{A_e^{N+1} - 1} \quad (14-36)$$

where subscripts 1 and 2 refer to the bottom and top of the tower, respectively, and the absorption factors are defined by the equations

$$A_e = \sqrt{A_1(A_2 + 1) + 0.25} - 0.5 \quad (14-37)$$

$$A' = A_1(A_2 + 1)/(A_1 + 1) \quad (14-38)$$

This procedure has been applied to the absorption of C_5 and lighter hydrocarbon vapors into a lean oil, for example.

Stripping Equations When the liquid feed is dilute and the operating and equilibrium curves are straight lines, the stripping equations analogous to Eqs. (14-31) and (14-33) are

$$(x_2 - x_1)/(x_2 - x_1^0) = (S^{N+1} - S)/(S^{N+1} - 1) \quad (14-39)$$

where $x_1^0 = y_1/m$; $S = mG_M/L_M = A^{-1}$; and

$$N = \frac{\ln [(1-A)(x_2 - x_1^0)/(x_1 - x_1^0) + A]}{\ln (S)} \quad (14-40)$$

For systems in which the concentrations are large and the stripping factor S may vary along the tower, the following Edmister equations [Ind. Eng. Chem., **35**, 837 (1943)] are applicable:

$$\frac{x_2 - x_1}{x_2} = \left[1 - \frac{1}{S'} \frac{(G_M y)_1}{(L_M x)_2} \right] \frac{S_e^{N+1} - S_e}{S_e^{N+1} - 1} \quad (14-41)$$

$$\text{where} \quad S_e = \sqrt{S_2(S_1 + 1) + 0.25} - 0.5 \quad (14-42)$$

$$S' = S_2(S_1 + 1)/(S_2 + 1) \quad (14-43)$$

and the subscripts 1 and 2 refer to the bottom and top of the tower respectively.

Equations (14-37) and (14-42) represent two different ways of obtaining an effective factor, and a value of A_e obtained by taking the reciprocal of S_e from Eq. (14-42) will not check exactly with a value of A_e derived by substituting $A_1 = 1/S_1$ and $A_2 = 1/S_2$ into Eq. (14-37). Regardless of this fact, the equations generally give reasonable results for approximate design calculations.

It should be noted that throughout this section the subscripts 1 and 2 refer to the bottom and to the top of the apparatus respectively regardless of whether it is an absorber or a stripper. This has been done to maintain internal consistency among all the equations and to prevent the confusion created in some derivations in which the numbering system for an absorber is different from the numbering system for a stripper.

Tray Efficiencies in Tray Absorbers and Strippers Computations of the theoretical trays N assume that the liquid on each tray is completely mixed and that the vapor leaving the tray is in equilibrium with the liquid. In practice, complete equilibrium cannot exist since interphase mass transfer requires a finite driving force. This leads to the definition of an overall tray efficiency

$$E = N_{\text{theoretical}}/N_{\text{actual}} \quad (14-44)$$

which can be correlated with the system design variables.

Mass-transfer theory indicates that for trays of a given design, the factors that have the biggest influence on E in absorption and stripping towers are the physical properties of the fluids and the dimensionless ratio mG_M/L_M . Systems in which mass transfer is gas-film-controlled may be expected to have efficiencies as high as 50 to 100 percent, whereas tray efficiencies as low as 1 percent have been reported for the absorption of low-solubility (large- m) gases into solvents of high viscosity.

The fluid properties of interest are represented by the Schmidt numbers of the gas and liquid phases. For gases, the Schmidt numbers are normally close to unity and independent of temperature and pressure. Thus, gas-phase mass-transfer coefficients are relatively independent of the system.

By contrast, the liquid-phase Schmidt numbers range from about 10^2 to 10^4 and depend strongly on temperature. The temperature dependence of the liquid-phase Schmidt number derives primarily from the strong dependence of the liquid viscosity on temperature.

Consideration of the preceding discussion in connection with the relationship between mass-transfer coefficients (see Sec. 5)

$$1/K_G = 1/k_G + m/k_L \quad (14-45)$$

indicates that the variations in the overall resistance to mass transfer in absorbers and strippers are related primarily to variations in the liquid-phase viscosity μ and the slope m . O'Connell [Trans. Am. Inst. Chem. Eng., **42**, 741 (1946)] used the above findings and correlated the tray efficiency in terms of the liquid viscosity and the gas solubility. The O'Connell correlation for absorbers (Fig. 14-9) has Henry's law constant in lb-mol/(atm-ft³), the pressure in atmospheres, and the liquid viscosity in centipoise.

The best procedure for making tray efficiency corrections (which can be quite significant, as seen in Fig. 14-9) is to use experimental

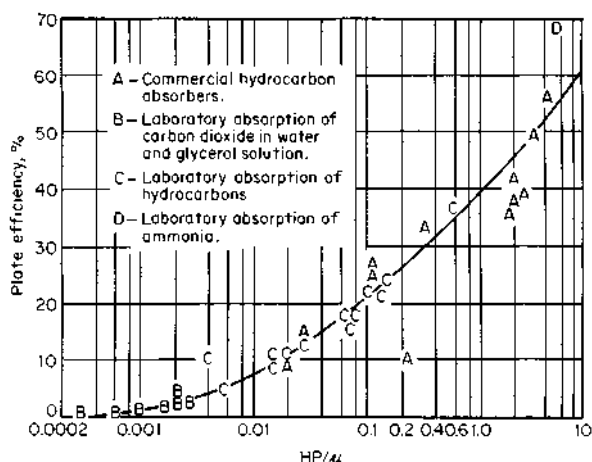


FIG. 14-9 O'Connell correlation for overall column efficiency E_{oc} for absorption. H is in lb-mol/(atm-ft³), P is in atm, and μ is in cP. To convert HP/μ in pound-moles per cubic foot-centipoise to kilogram-moles per cubic meter-pascal-second, multiply by 1.60×10^4 . [O'Connell, Trans. Am. Inst. Chem. Eng., **42**, 741 (1946).]

data from a prototype system that is large enough to be representative of the actual commercial tower.

Example 4: Actual Trays for Steam Stripping The number of actual trays required for steam-stripping an acetone-rich liquor containing 0.573 mole percent acetone in water is to be estimated. The design overhead recovery of acetone is 99.9 percent, leaving 18.5 ppm weight of acetone in the stripper bottoms. The design operating temperature and pressure are 101.3 kPa and 94°C respectively, the average liquid-phase viscosity is 0.30 cP, and the average value of $K = y^0/x$ for these conditions is 33.

By choosing a value of $mG_M/L_M = S = A^{-1} = 1.4$ and noting that the stripping medium is pure steam (i.e., $x_2^0 = 0$), the number of theoretical trays according to Eq. (14-40) is

$$N = \frac{\ln [(1 - 0.714)/(1000) + 0.714]}{\ln (1.4)} = 16.8$$

The O'Connell parameter for gas absorbers is $\rho_L/KM\mu_L$, where ρ_L is the liquid density, lb/ft³; μ_L is the liquid viscosity, cP; M is the molecular weight of the liquid; and $K = y^0/x$. For the present design

$$\rho_L/KM\mu_L = 60.1/(33 \times 18 \times 0.30) = 0.337$$

and according to the O'Connell graph for absorbers (Fig. 14-7) the overall tray efficiency for this case is estimated to be 30 percent. Thus, the required number of actual trays is $16.8/0.3 = 56$ trays.

HEAT EFFECTS IN GAS ABSORPTION

Overview One of the most important considerations involved in designing gas absorption towers is to determine whether temperatures will vary along the height of the tower due to heat effects; note that the solute solubility usually depends strongly on temperature. The simplified design procedures described earlier in this section become more complicated when heat effects cannot be neglected. The role of this section is to enable understanding and design of gas absorption towers where heat effects are important and cannot be ignored.

Heat effects that cause temperatures to vary from point to point in a gas absorber are (1) the heat of solution (including heat of condensation, heat of mixing, and heat of reaction); (2) the heat of vaporization or condensation of the solvent; (3) the exchange of sensible heat between the gas and liquid phases; and (4) the loss of sensible heat from the fluids to internal or external coils.

There are a number of systems where heat effects definitely cannot be ignored. Examples include the absorption of ammonia in

water, dehumidification of air using concentrated H_2SO_4 , absorption of HCl in water, absorption of SO_3 in H_2SO_4 , and absorption of CO_2 in alkanolamines. Even for systems where the heat effects are mild, they may not be negligible; an example is the absorption of acetone in water.

Thorough and knowledgeable discussions of the problems involved in gas absorption with significant heat effects have been presented by Coggan and Bourne [*Trans. Inst. Chem. Eng.*, **47**, T96, T160 (1969)]; Bourn, von Stockar, and Coggan [*Ind. Eng. Chem. Proc. Des. Dev.*, **13**, 115, 124 (1974)]; and von Stockar and Wilke [*Ind. Eng. Chem. Fundam.*, **16**, 89 (1977)]. The first two of these references discuss tray-tower absorbers and include experimental studies of the absorption of ammonia in water. The third reference discusses the design of packed-tower absorbers and includes a shortcut design method based on a semitheoretical correlation of rigorous design calculations. All these authors demonstrate that when the solvent is volatile, the temperature inside an absorber can go through a maximum. They note that the least expensive and most common of solvents—water—is capable of exhibiting this “hot-spot” behavior.

Several approaches may be used in modeling absorption with heat effects, depending on the job at hand: (1) treat the process as isothermal by assuming a particular temperature, then add a safety factor; (2) employ the classical adiabatic method, which assumes that the heat of solution manifests itself only as sensible heat in the liquid phase and that the solvent vaporization is negligible; (3) use semitheoretical shortcut methods derived from rigorous calculations; and (4) employ rigorous methods available from a process simulator.

While simpler methods are useful for understanding the key effects involved, rigorous methods are recommended for final designs. This subsection also discusses the range of safety factors that are required if simpler methods are used.

Effects of Operating Variables Conditions that give rise to significant heat effects are (1) an appreciable heat of solution and/or (2) absorption of large amounts of solute in the liquid phase. The second condition is favored when the solute concentration in the inlet gas is large, when the liquid flow rate is relatively low (small L_M/G_M), when the solubility of the solute in the liquid is high, and/or when the operating pressure is high.

If the solute-rich gas entering the bottom of an absorber tower is cold, the liquid phase may be cooled somewhat by transfer of sensible heat to the gas. A much stronger cooling effect can occur when the solute is volatile and the entering gas is not saturated with respect to the solvent. It is possible to experience a condition in which solvent is being evaporated near the bottom of the tower and condensed near the top. Under these conditions a pinch point may develop in which the operating and equilibrium curves approach each other at a point inside the tower.

In the references previously cited, the authors discuss the influence of operating variables upon the performance of towers when large heat effects are involved. Some key observations are as follows:

Operating Pressure Raising the pressure may increase the separation effectiveness considerably. Calculations for the absorption of methanol in water from water-saturated air showed that doubling the pressure doubles the allowable concentration of methanol in the feed gas while still achieving the required concentration specification in the off gas.

Temperature of Lean Solvent The temperature of the entering (lean) solvent has surprisingly little influence upon the temperature profile in an absorber since any temperature changes are usually caused by the heat of solution or the solvent vaporization. In these cases, the temperature profile in the liquid phase is usually dictated solely by the internal-heat effects.

Temperature and Humidity of the Rich Gas Cooling and consequent dehumidification of the feed gas to an absorption tower can be very beneficial. A high humidity (or relative saturation with the solvent) limits the capacity of the gas to take up latent heat and hence is unfavorable to absorption. Thus dehumidification of the inlet gas is worth considering in the design of absorbers with large heat effects.

Liquid-to-Gas Ratio The L/G ratio can have a significant influence on the development of temperature profiles in gas

absorbers. High L/G ratios tend to result in less strongly developed temperature profiles due to the increased heat capacity of the liquid phase. As the L/G ratio is increased, the operating line moves away from the equilibrium line and more solute is absorbed per stage or packing segment. However, there is a compensating effect; since more heat is liberated in each stage or packing segment, the temperatures will rise, which causes the equilibrium line to shift up. As the L/G ratio is decreased, the concentration of solute tends to build up in the upper part of the absorber, and the point of highest temperature tends to move upward in the tower until finally the maximum temperature occurs at the top of the tower. Of course, the capacity of the liquid to absorb solute falls progressively as L/G is reduced.

Number of Stages or Packing Height When the heat effects combine to produce an extended zone in the tower where little absorption takes place (i.e., a pinch zone), the addition of trays or packing height will have no useful effect on separation efficiency. In this case, increases in absorption may be obtained by increasing solvent flow, introducing strategically placed coolers, cooling and dehumidifying the inlet gas, and/or raising the tower pressure.

Equipment Considerations When the solute has a large heat of solution and the feed gas contains a high concentration of solute, as in absorption of HCl in water, the effects of heat release during absorption may be so pronounced that the installation of heat-transfer surface to remove the heat of absorption may be as important as providing sufficient interfacial area for the mass-transfer process itself. The added heat-transfer area may consist of internal cooling coils on the trays, or the liquid may be withdrawn from the tower, cooled in an external heat exchanger, and then returned to the tower.

In many cases the rate of heat liberation is largest near the bottom of the tower, where the solute absorption is more rapid, so that cooling surfaces or intercoolers are required only at the lower part of the column. Coggan and Bourne [*Trans. Inst. Chem. Eng.*, **47**, T96, T160 (1969)] found, however, that the optimal position for a single interstage cooler does not necessarily coincide with the position of the maximum temperature of the center of the pinch. They found that in a 12-tray tower, two strategically placed interstage coolers tripled the allowable ammonia feed concentration for a given off-gas specification. For a case involving methanol absorption, it was found that greater separation was possible in a 12-stage column with two intercoolers than in a simple column with 100 stages and no intercoolers.

In the case of HCl absorption, a shell-and-tub heat exchanger often is employed as a cooled wetted-wall vertical-column absorber so that the exothermic heat of reaction can be removed continuously as it is released into a liquid film.

Installation of heat-exchange equipment to precool and dehumidify the feed gas to an absorber also deserves consideration, in order to take advantage of the cooling effects created by vaporization of solvent in the lower sections of the tower.

Classical Isothermal Design Method When the feed gas is sufficiently dilute, the exact design solution may be approximated by the isothermal one over the broad range of L/G ratios, since heat effects are generally less important when washing dilute-gas mixtures. The problem, however, is one of defining the term *sufficiently dilute* for each specific case. For a new absorption duty, the assumption of isothermal operation must be subjected to verification by the use of a rigorous design procedure.

When heat-exchange surface is being provided in the design of an absorber, the isothermal design procedure can be rendered valid by virtue of the exchanger design specification. With ample surface area and a close approach, isothermal operation can be guaranteed.

For preliminary screening and feasibility studies or for rough estimates, one may wish to employ a version of the isothermal design method which assumes that the liquid temperatures in the tower are everywhere equal to the inlet-liquid temperature. In their analysis of packed-tower designs, von Stockar and Wilke [*Ind. Eng. Chem. Fundam.*, **16**, 89 (1977)] showed that the isothermal method tended to underestimate the required height of packing by a factor of as much as

1.5 to 2. Thus, for rough estimates one may wish to employ the assumption that the absorber temperature is equal to the inlet-liquid temperature and then apply a design factor to the result.

Another instance in which the constant-temperature method is used involved the direct application of experimental $K_G a$ values obtained at the desired conditions of inlet temperatures, operating pressures, flow rates, and feed-stream compositions. The assumption here is that, regardless of any temperature profiles that may exist within the actual tower, the procedure of "working the problem in reverse" will yield a correct result. One should, however, be cautious about extrapolating such data from the original basis and be careful to use compatible equilibrium data.

Classical Adiabatic Design Method The classical adiabatic design method assumes that the heat of solution serves only to heat up the liquid stream and there is no vaporization of the solvent. This assumption makes it feasible to relate increases in the liquid-phase temperature to the solute concentration x by a simple enthalpy balance. The equilibrium curve can then be adjusted to account for the corresponding temperature rise on an xy diagram. The adjusted equilibrium curve will be concave upward as the concentration increases, tending to decrease the driving forces near the bottom of the tower, as illustrated in Fig. 14-10 in Example 6.

Colburn [*Trans. Am. Inst. Chem. Eng.*, **35**, 211 (1939)] has shown that when the equilibrium line is straight near the origin but curved slightly at its upper end, N_{OG} can be computed approximately by assuming that the equilibrium curve is a parabolic arc of slope m_2 near the origin and passing through the point $x_1, K_1 x_1$ at the upper end. The Colburn equation for this case is

$$N_{OG} = \frac{1}{1 - m_2 G_M/L_M} \times \ln \left[\frac{(1 - m_2 G_M/L_M)^2}{1 - K_1 G_M/L_M} \left(\frac{y_1 - m_2 x_2}{y_2 - m_2 x_2} \right) + \frac{m_2 G_M}{L_M} \right] \quad (14-46)$$

Comparison by von Stockar and Wilke [*Ind. Eng. Chem. Fundam.*, **16**, 89 (1977)] between the rigorous and the classical adiabatic design methods for packed towers indicates that the simple adiabatic design methods underestimate packing heights by as much as a factor of 1.25

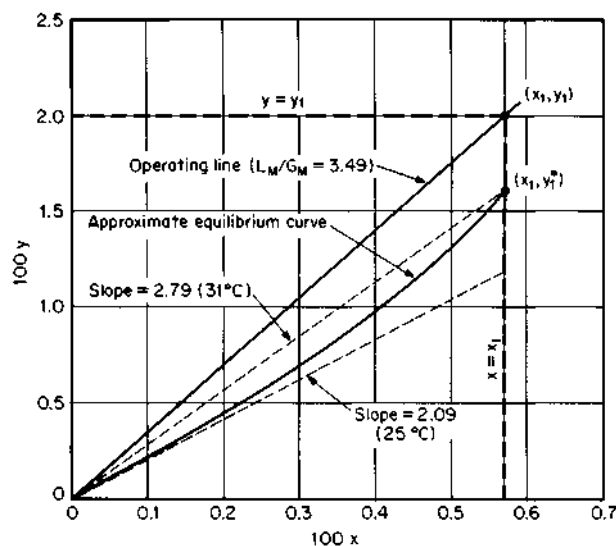


FIG. 14-10 Design diagram for adiabatic absorption of acetone in water, Example 6.

to 1.5. Thus, when using the classical adiabatic method, one should probably apply a design safety factor.

A slight variation of the above method accounts for increases in the solvent content of the gas stream between the inlet and the outlet of the tower and assumes that the evaporation of solvent tends to cool the liquid. This procedure offsets a part of the temperature rise that would have been predicted with no solvent evaporation and leads to the prediction of a shorter tower.

Rigorous Design Methods A detailed discussion of rigorous methods for the design of packed and tray absorbers when large heat effects are involved is beyond the scope of this subsection. In principle, material and energy balances may be executed under the same constraints as for rigorous distillation calculations (see Sec. 13). Further discussion on this subject is given in the subsection "Absorption with Chemical Reaction."

Direct Comparison of Design Methods The following problem, originally presented by Sherwood, Pigford, and Wilke (*Mass Transfer*, McGraw-Hill, New York, 1975, p. 616) was employed by von Stockar and Wilke (op. cit.) as the basis for a direct comparison between the isothermal, adiabatic, semitheoretical shortcut, and rigorous design methods for estimating the height of packed towers.

Example 5: Packed Absorber, Acetone into Water Inlet gas to an absorber consists of a mixture of 6 mole percent acetone in air saturated with water vapor at 15°C and 101.3 kPa (1 atm). The scrubbing liquor is pure water at 15°C, and the inlet gas and liquid rates are given as 0.080 and 0.190 kmol/s respectively. The liquid rate corresponds to 20 percent over the theoretical minimum as calculated by assuming a value of x_1 corresponding to complete equilibrium between the exit liquor and the incoming gas. H_G and H_L are given as 0.42 and 0.30 m respectively, and the acetone equilibrium data at 15°C are $p_A^0 = 19.7$ kPa (147.4 torr), $\gamma_A = 6.46$, and $m_A = 6.46 \times 19.7/101.3 = 1.26$. The heat of solution of acetone is 7656 cal/gmol (32.05 kJ/gmol), and the heat of vaporization of solvent (water) is 10,755 cal/gmol (45.03 kJ/gmol). The problem calls for determining the height of packing required to achieve a 90 percent recovery of the acetone.

The following table compares the results obtained by von Stockar and Wilke (op. cit.) for the various design methods:

Design method used	N_{OG}	Packed height, m	Design safety factor
Rigorous	5.56	3.63	1.00
Shortcut rigorous	5.56	3.73	0.97
Classical adiabatic	4.01	2.38	1.53
Classical isothermal	3.30	1.96	1.85

It should be clear from this example that there is considerable room for error when approximate design methods are employed in situations involving large heat effects, even for a case in which the solute concentration in the inlet gas is only 6 mole percent.

Example 6: Solvent Rate for Absorption Let us consider the absorption of acetone from air at atmospheric pressure into a stream of pure water fed to the top of a packed absorber at 25°C. The inlet gas at 35°C contains 2 percent by volume of acetone and is 70 percent saturated with water vapor (4 percent H_2O by volume). The mole-fraction acetone in the exit gas is to be reduced to 1/400 of the inlet value, or 50 ppmv. For 100 kmol of feed-gas mixture, how many kilomoles of fresh water should be fed to provide a positive-driving force throughout the packing? How many transfer units will be needed according to the classical adiabatic method? What is the estimated height of packing required if $H_{OG} = 0.70$ m?

The latent heats at 25°C are 7656 kcal/kmol for acetone and 10,490 kcal/kmol for water, and the differential heat of solution of acetone vapor in pure water is given as 2500 kcal/kmol. The specific heat of air is 7.0 kcal/(kmol·K).

Acetone solubilities are defined by the equation

$$K = y^0/x = \gamma_a p_a/p_T \quad (14-47)$$

where the vapor pressure of pure acetone in mmHg (torr) is given by (Sherwood et al., *Mass Transfer*, McGraw-Hill, New York, 1975, p. 537):

$$p_A^0 = \exp(18.1594 - 3794.06/T) \quad (14-48)$$

and the liquid-phase-activity coefficient may be approximated for low concentrations ($x \leq 0.01$) by the equation

$$\gamma_e = 6.5 \exp(2.0803 - 601.2/T) \quad (14-49)$$

Typical values of acetone solubility as a function of temperature at a total pressure of 760 mmHg are shown in the following table:

$t, ^\circ\text{C}$	25	30	35	40
γ_e	6.92	7.16	7.40	7.63
p_a, mmHg	229	283	346	422
$K = \gamma_e p_a^0 / 760$	2.09	2.66	3.37	4.23

For dry gas and liquid water at 25°C , the following enthalpies are computed for the inlet- and exit-gas streams (basis, 100 kmol of gas entering):

Entering gas:	
Acetone	$2(2500 + 7656) = 20,312 \text{ kcal}$
Water vapor	$4(10,490) = 41,960$
Sensible heat	$(100)(7.0)(35 - 25) = 7,000$
	69,272 kcal

Exit gas (assumed saturated with water at 25°C):

Acetone	$(2/400)(94/100)(2500) =$	12 kcal
Water vapor	$94\left(\frac{23.7}{760 - 23.7}\right)(10,490) =$	31,600
		31,612 kcal

Enthalpy change of liquid = $69,272 - 31,612 = 37,660 \text{ kcal/100 kmol gas}$. Thus, $\Delta t = t_1 - t_2 = 37,660/18L_M$, and the relation between L_M/G_M and the liquid-phase temperature rise is

$$L_M/G_M = (37,660)/(18)(100) \Delta t = 20.92/\Delta t$$

The following table summarizes the critical values for various assumed temperature rises:

$\Delta t, ^\circ\text{C}$	L_M/G_M	K_1	$K_1 G_M/L_M$	$m_2 G_M/L_M$
0		2.09	0.	0.
2	10.46	2.31	0.221	0.200
3	6.97	2.42	0.347	0.300
4	5.23	2.54	0.486	0.400
5	4.18	2.66	0.636	0.500
6	3.49	2.79	0.799	0.599
7	2.99	2.93	0.980	0.699

Evidently a temperature rise of 7°C would not be a safe design because the equilibrium line nearly touches the operating line near the bottom of the tower, creating a pinch. A temperature rise of 6°C appears to give an operable design, and for this case $L_M = 349 \text{ kmol per 100 kmol of feed gas}$.

The design diagram for this case is shown in Fig. 14-10, in which the equilibrium curve is drawn so that the slope at the origin m_2 is equal to 2.09 and passes through the point $x_1 = 0.02/3.49 = 0.00573$ at $y_1^0 = 0.00573 \times 2.79 = 0.0160$.

The number of transfer units can be calculated from the adiabatic design equation, Eq. (14-46):

$$N_{OG} = \frac{1}{1 - 0.599} \ln \left[\frac{(1 - 0.599)^2}{(1 - 0.799)} (400) + 0.599 \right] = 14.4$$

The estimated height of tower packing by assuming $H_{OG} = 0.70 \text{ m}$ and a design safety factor of 1.5 is

$$h_T = (14.4)(0.7)(1.5) = 15.1 \text{ m (49.6 ft)}$$

For this tower, one should consider the use of two or more shorter packed sections instead of one long section.

Another point to be noted is that this calculation would be done more easily today by using a process simulator. However, the details are presented here to help the reader gain familiarity with the key assumptions and results.

MULTICOMPONENT SYSTEMS

When no chemical reactions are involved in the absorption of more than one soluble component from an insoluble gas, the design conditions (temperature, pressure, liquid-to-gas ratio) are normally determined by the volatility or physical solubility of the least soluble component for which the recovery is specified.

The more volatile (i.e., less soluble) components will only be partially absorbed even for an infinite number of trays or transfer units. This can be seen in Fig. 14-9, in which the asymptotes become vertical for values of mG_M/L_M greater than unity. If the amount of volatile component in the fresh solvent is negligible, then the limiting value of y_1/y_2 for each of the highly volatile components is

$$y_1/y_2 = S/(S - 1) \quad (14-50)$$

where $S = mG_M/L_M$ and the subscripts 1 and 2 refer to the bottom and top of the tower, respectively.

When the gas stream is dilute, absorption of each constituent can be considered separately as if the other components were absent. The following example illustrates the use of this principle.

Example 7: Multicomponent Absorption, Dilute Case Air entering a tower contains 1 percent acetaldehyde and 2 percent acetone. The liquid-to-gas ratio for optimum acetone recovery is $L_M/G_M = 3.1 \text{ mol/mol}$ when the fresh-solvent temperature is 31.5°C . The value of y^0/x for acetaldehyde has been measured as 50 at the boiling point of a dilute solution, 93.5°C . What will the percentage recovery of acetaldehyde be under conditions of optimal acetone recovery?

Solution. If the heat of solution is neglected, y^0/x at 31.5°C is equal to $50(1200/7300) = 8.2$, where the factor in parentheses is the ratio of pure-acetaldehyde vapor pressures at 31.5 and 93.5°C respectively. Since L_M/G_M is equal to 3.1, the value of S for the aldehyde is $S = mG_M/L_M = 8.2/3.1 = 2.64$, and $y_1/y_2 = S/(S - 1) = 2.64/1.64 = 1.61$. The acetaldehyde recovery is therefore equal to $100 \times 0.61/1.61 = 38 \text{ percent recovery}$.

In concentrated systems the change in gas and liquid flow rates within the tower and the heat effects accompanying the absorption of all the components must be considered. A trial-and-error calculation from one theoretical stage to the next usually is required if accurate results are to be obtained, and in such cases calculation procedures similar to those described in Sec. 13 normally are employed. A computer procedure for multicomponent adiabatic absorber design has been described by Feintuch and Treybal [*Ind. Eng. Chem. Process Des. Dev.*, **17**, 505 (1978)]. Also see Holland, *Fundamentals and Modeling of Separation Processes*, Prentice Hall, Englewood Cliffs, N.J., 1975.

In concentrated systems, the changes in the gas and liquid flow rates within the tower and the heat effects accompanying the absorption of all components must be considered. A trial-and-error calculation from one theoretical stage to the next is usually required if accurate and reliable results are to be obtained, and in such cases calculation procedures similar to those described in Sec. 13 need to be employed.

When two or more gases are absorbed in systems involving chemical reactions, the system is much more complex. This topic is discussed later in the subsection "Absorption with Chemical Reaction."

Graphical Design Method for Dilute Systems The following notation for multicomponent absorption systems has been adapted from Sherwood, Pigford, and Wilke (*Mass Transfer*, McGraw-Hill, New York, 1975, p. 415):

L_M^S = moles of solvent per unit time

G_M^0 = moles of rich feed gas to be treated per unit time

X = moles of one solute per mole of solute-free solvent fed to top of tower

Y = moles of one solute in gas phase per mole of rich feed gas

Subscripts 1 and 2 refer to the bottom and the top of the tower, respectively, and the material balance for any one component may be written as

$$L_M^S(X - X_2) = G_M^0(Y - Y_2) \quad (14-51)$$

or else as

$$L_M^S(X_1 - X) = G_M^0(Y_1 - Y) \quad (14-52)$$

For the special case of absorption from lean gases with relatively large amounts of solvent, the equilibrium lines are defined for each component by the relation

$$Y^0 = K'X \quad (14-53)$$

Thus, the equilibrium line for each component passes through the origin with slope K' , where

$$K' = K(G_M^0/G_M^0)/(L_M/L_M^0) \quad (14-54)$$

and $K = y^0/x$. When the system is sufficiently dilute, $K' = K$.

The liquid-to-gas ratio is chosen on the basis of the least soluble component in the feed gas that must be absorbed completely. Each component will then have its own operating line with slope equal to L_M^0/G_M^0 (i.e., the operating lines for the various components will be parallel).

A typical diagram for the complete absorption of pentane and heavier components is shown in Fig. 14-11. The oil used as solvent is assumed to be solute-free (i.e., $X_2 = 0$), and the "key component," butane, was identified as that component absorbed in appreciable amounts whose equilibrium line is most nearly parallel to the operating lines (i.e., the K value for butane is approximately equal to L_M^0/G_M^0).

In Fig. 14-11, the composition of the gas with respect to components more volatile than butane will approach equilibrium with the liquid phase at the bottom of the tower. The gas compositions of the components less volatile (heavier) than butane will approach equilibrium with the oil entering the tower, and since $X_2 = 0$, the components heavier than butane will be completely absorbed.

Four theoretical trays have been stepped off for the key component (butane) on Fig. 14-11, and are seen to give a recovery of 75 percent of the butane. The operating lines for the other components have been drawn with the same slope and placed so as to give approximately the same number of theoretical trays. Figure 14-11 shows that equilibrium is easily achieved in fewer than four theoretical trays and that for the heavier components nearly complete recovery is obtained in four theoretical trays. The diagram also shows that absorption of the light components takes place in the upper part of the tower, and the final recovery of the heavier components takes place in the lower section of the tower.

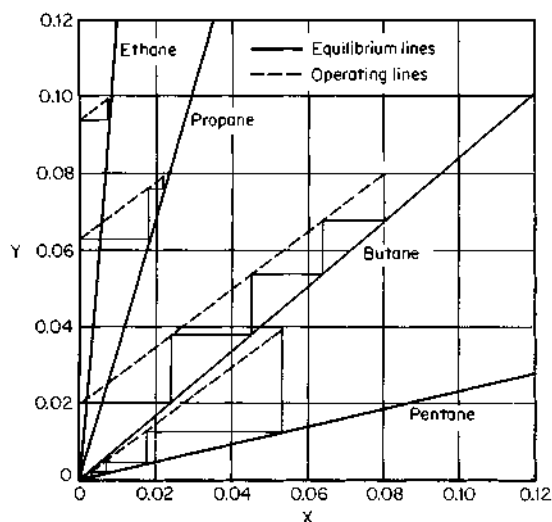


FIG. 14-11 Graphical design method for multicomponent systems; absorption of butane and heavier components in a solute-free lean oil.

Algebraic Design Method for Dilute Systems The design method described above can be performed algebraically by employing the following modified version of the Kremser formula:

$$\frac{Y_1 - Y_2}{Y_1 - mX_2} = \frac{(A^0)^{N+1} - A^0}{(A^0)^{N+1} - 1} \quad (14-55)$$

where for dilute gas absorption $A^0 = L_M^0/mG_M^0$ and $m \approx K = y^0/x$.

The left side of Eq. (14-55) represents the efficiency of absorption of any one component of the feed gas mixture. If the solvent is solute-free so that $X_2 = 0$, the left side is equal to the fractional absorption of the component from the rich feed gas. When the number of theoretical trays N and the liquid and gas feed rates L_M^0 and G_M^0 have been fixed, the fractional absorption of each component may be computed directly, and the operating lines need not be placed by trial and error as in the graphical method described above.

According to Eq. (14-55), when A^0 is less than unity and N is large,

$$(Y_1 - Y_2)/(Y_1 - mX_2) = A^0 \quad (14-56)$$

Equation (14-56) may be used to estimate the fractional absorption of more volatile components when A^0 of the component is greater than A^0 of the key component by a factor of 3 or more.

When A^0 is much larger than unity and N is large, the right side of Eq. (14-55) becomes equal to unity. This signifies that the gas will leave the top of the tower in equilibrium with the incoming oil, and when $X_2 = 0$, it corresponds to complete absorption of the component in question. Thus, the least volatile components may be assumed to be at equilibrium with the lean oil at the top of the tower.

When $A^0 = 1$, the right side of Eq. (14-56) simplifies as follows:

$$(Y_1 - Y_2)/(Y_1 - mX_2) = N/(N + 1) \quad (14-57)$$

For systems in which the absorption factor A^0 for each component is not constant throughout the tower, an effective absorption factor for use in the equations just presented can be estimated by the Edmister formula

$$A_e = \sqrt{A_1^0(A_2^0 + 1) + 0.25} - 0.5 \quad (14-58)$$

This procedure is a reasonable approximation only when no pinch points exist within the tower and when the absorption factors vary in a regular manner between the bottom and the top of the tower.

Example 8: Multicomponent Absorption, Concentrated Case

A hydrocarbon feed gas is to be treated in an existing four-theoretical-tray absorber to remove butane and heavier components. The recovery specification for the key component, butane, is 75 percent. The composition of the exit gas from the absorber and the required liquid-to-gas ratio are to be estimated. The feed-gas composition and the equilibrium K values for each component at the temperature of the (solute-free) lean oil are presented in the following table:

Component	Mole %	K value
Methane	68.0	74.137
Ethane	10.0	12.000
Propane	8.0	3.429
Butane	8.0	0.833
Pentane	4.0	0.233
C ₆ plus	2.0	0.065

For $N = 4$ and $Y_2/Y_1 = 0.25$, the value of A^0 for butane is found to be equal to 0.89 from Eq. (14-55) by using a trial-and-error method. The values of A^0 for the other components are then proportional to the ratios of their K values to that of butane. For example, $A^0 = 0.89(0.833/12.0) = 0.062$ for ethane. The values of A^0 for each of the other components and the exit-gas composition as computed from Eq. (14-55) are shown in the following table:

Component	A^0	Y_2 , mol/mol feed	Exit gas, mole %
Methane	0.010	67.3	79.1
Ethane	0.062	9.4	11.1
Propane	0.216	6.3	7.4
Butane	0.890	2.0	2.4
Pentane	3.182	0.027	0.03
C ₆ plus	11.406	0.0012	0.0014

The molar liquid-to-gas ratio required for this separation is computed as $L_0/G_0 = A^0 \times K = 0.89 \times 0.833 = 0.74$.

We note that this example is the analytical solution to the graphical design problem shown in Fig. 14-11, which therefore is the design diagram for this system.

The simplified design calculations presented in this section are intended to reveal the key features of gas absorption involving multicomponent systems. It is expected that rigorous computations, based upon the methods presented in Sec. 13, will be used in design practice. Nevertheless, it is valuable to study these simplified design methods and examples since they provide insight into the key elements of multicomponent absorption.

ABSORPTION WITH CHEMICAL REACTION

Introduction Many present-day commercial gas absorption processes involve systems in which chemical reactions take place in the liquid phase; an example of the absorption of CO_2 by MEA has been presented earlier in this section. These reactions greatly increase the capacity of the solvent and enhance the rate of absorption when compared to physical absorption systems. In addition, the selectivity of reacting solutes is greatly increased over that of nonreacting solutes. For example, MEA has a strong selectivity for CO_2 compared to chemically inert solutes such as CH_4 , CO , or N_2 . Note that the design procedures presented here are theoretically and practically related to biofiltration, which is discussed in Sec. 25 (Waste Management).

A necessary prerequisite to understanding the subject of absorption with chemical reaction is the development of a thorough understanding of the principles involved in physical absorption, as discussed earlier in this section and in Sec. 5. Excellent references on the subject of absorption with chemical reactions are the books by Dankwerts (*Gas-Liquid Reactions*, McGraw-Hill, New York, 1970) and Astarita et al. (*Gas Treating with Chemical Solvents*, Wiley, New York, 1983).

Recommended Overall Design Strategy When one is considering the design of a gas absorption system involving chemical reactions, the following procedure is recommended:

1. Consider the possibility that the physical design methods described earlier in this section may be applicable.
2. Determine whether commercial design overall $K_G a$ values are available for use in conjunction with the traditional design method, being careful to note whether the conditions under which the $K_G a$ data were obtained are essentially the same as for the new design. Contact the various tower-packing vendors for information as to whether $K_G a$ data are available for your system and conditions.
3. Consider the possibility of scaling up the design of a new system from experimental data obtained in a laboratory bench-scale or small pilot-plant unit.
4. Consider the possibility of developing for the new system a rigorous, theoretically based design procedure which will be valid over a wide range of design conditions. Note that commercial software is readily available today to develop a rigorous model in a relatively small amount of time. These topics are further discussed in the subsections that follow.

Dominant Effects in Absorption with Chemical Reaction When the solute is absorbing into a solution containing a reagent that chemically reacts with it, diffusion and reaction effects become closely coupled. It is thus important for the design engineer to understand the key effects. Figure 14-12 shows the concentration profiles that occur when solute A undergoes an irreversible second-order reaction with component B, dissolved in the liquid, to give product C.



The rate equation is

$$r_A = -k_2 C_A C_B \quad (14-60)$$

Figure 14-12 shows that the fast reaction takes place entirely in the liquid film. In such instances, the dominant mass-transfer mechanism is physical absorption, and physical design methods are applicable but the resistance to mass transfer in the liquid phase is lower due to the reaction. On the other extreme, a slow reaction occurs in the bulk of the liquid, and its rate has little dependence on the resistance to dif-

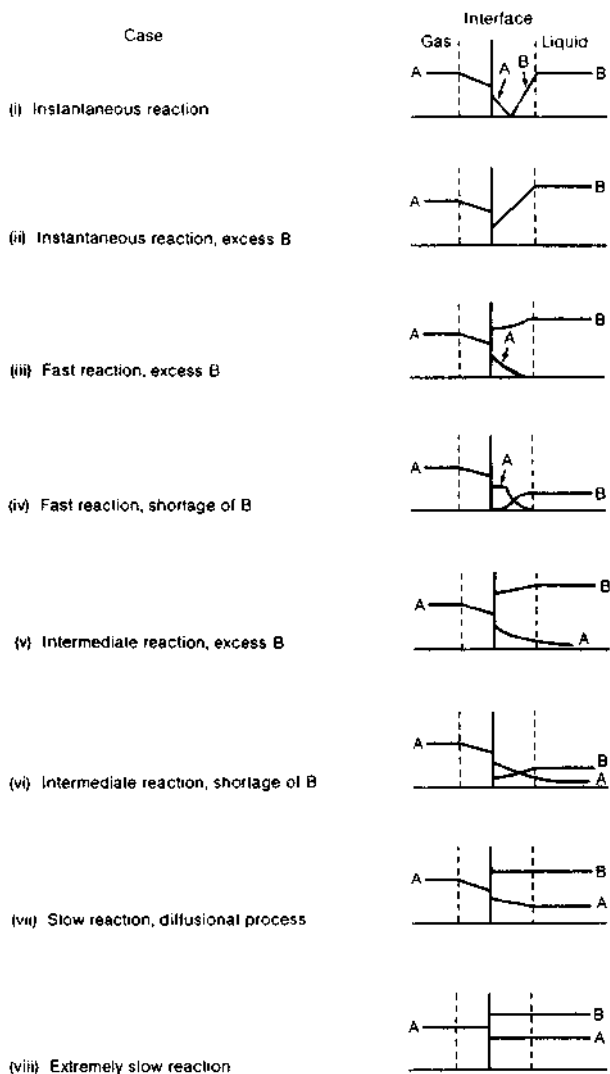


FIG. 14-12 Vapor- and liquid-phase concentration profiles near an interface for absorption with chemical reaction.

fusion in either the gas or the liquid films. Here the mass-transfer mechanism is that of chemical reaction, and holdup in the bulk liquid is the determining factor.

The Hatta number is a dimensionless group used to characterize the importance of the speed of reaction relative to the diffusion rate.

$$N_{\text{Ha}} = \frac{\sqrt{D_A k_2 C_{B0}}}{k_L^0} \quad (14-61)$$

As the Hatta number increases, the effective liquid-phase mass-transfer coefficient increases. Figure 14-13, which was first developed by Van Krevelen and Hoftyzer [*Rec. Trav. Chim.*, **67**, 563 (1948)] and later refined by Perry and Pigford and by Brian et al. [*AIChE J.*, **7**, 226 (1961)], shows how the enhancement (defined as the ratio of the effective liquid-phase mass-transfer coefficient to its physical equivalent $\phi = k_L/k_L^0$) increases with N_{Ha} for a second-order, irreversible reaction of the kind defined by Eqs. (14-60) and (14-61). The various curves in Fig. 14-13 were developed based upon penetration theory and

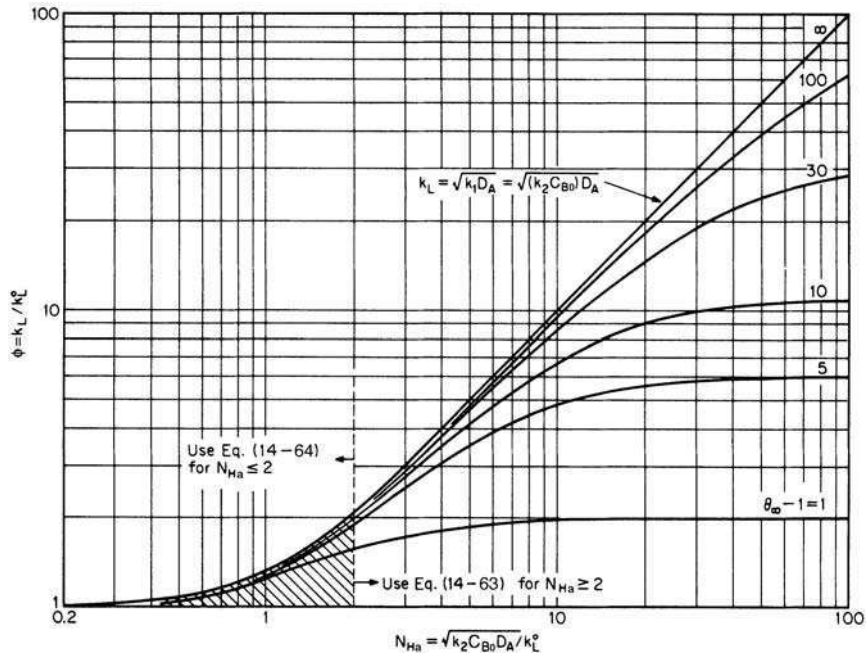


FIG. 14-13 Influence of irreversible chemical reactions on the liquid-phase mass-transfer coefficient k_L . [Adapted from Van Krevelen and Hoftyzer, *Rec. Trav. Chim.*, **67**, 563 (1948).]

depend on the parameter $\phi_\infty - 1$, which is related to the diffusion coefficients and reaction coefficients, as shown below.

$$\phi_\infty = \sqrt{\frac{D_A}{D_B}} + \sqrt{\frac{D_A}{D_B}} \left(\frac{C_B}{C_{Ab}} \right) \quad (14-62)$$

For design purposes, the entire set of curves in Fig. 14-13 may be represented by the following two equations:
For, $N_{Ha} \geq 2$:

$$k_L/k_L^0 = 1 + (\phi_\infty - 1) \{1 - \exp [-(N_{Ha} - 1)/(\phi_\infty - 1)]\} \quad (14-63)$$

For, $N_{Ha} \leq 2$:

$$k_L/k_L^0 = 1 + (\phi_\infty - 1) \{1 - \exp [-(\phi_\infty - 1)^{-1}]\} \exp [1 - 2/N_{Ha}] \quad (14-64)$$

Equation (14-64) was originally reported by Porter [*Trans. Inst. Chem. Eng.*, **44**, T25 (1966)], and Eq. (14-64) was derived by Edwards and first reported in the 6th edition of this handbook.

The Van Krevelen-Hoftyzer (Fig. 14-13) relationship was tested by Nijssing et al. [*Chem. Eng. Sci.*, **10**, 88 (1959)] for the second-order system in which CO_2 reacts with either NaOH or KOH solutions. Nijssing's results are shown in Fig. 14-14 and can be seen to be in excellent

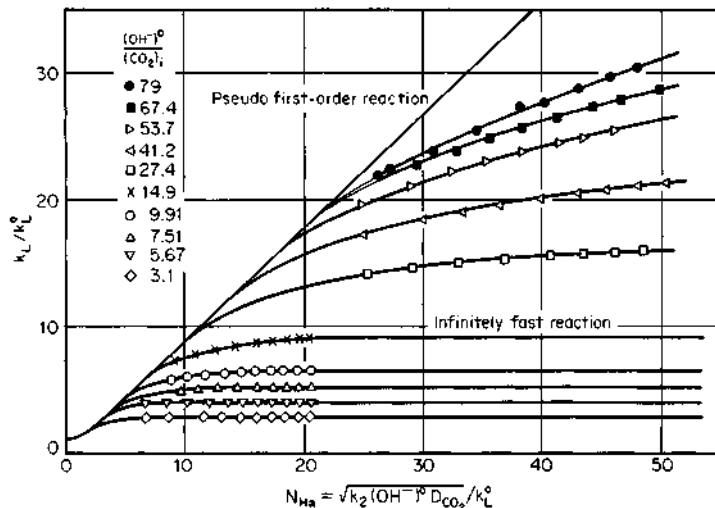


FIG. 14-14 Experimental values of k_L/k_L^0 for absorption of CO_2 into NaOH solutions at 20°C . [Data of Nijssing et al., *Chem. Eng. Sci.*, **10**, 88 (1959).]

agreement with the second-order-reaction theory. Indeed, these experimental data are well described by Eqs. (14-62) and (14-63) when values of $b = 2$ and $D_A/D_B = 0.64$ are employed in the equations.

Applicability of Physical Design Methods Physical design models such as the classical isothermal design method or the classical adiabatic design method may be applicable for systems in which chemical reactions are either extremely fast or extremely slow, or when chemical equilibrium is achieved between the gas and liquid phases.

If the chemical reaction is extremely fast and irreversible, the rate of absorption may in some cases be completely governed by gas-phase resistance. For practical design purposes, one may assume, e.g., that this gas-phase mass-transfer-limited condition will exist when the ratio y/y_i is less than 0.05 everywhere in the apparatus.

From the basic mass-transfer flux relationship for species A (Sec. 5)

$$N_A = k_G(y - y_i) = k_L(x_i - x) \quad (14-65)$$

one can readily show that this condition on y/y_i requires that the ratio x/x_i be negligibly small (i.e., a fast reaction) and that the ratio $mk_C/k_L = mk_C/k_L^0\phi$ be less than 0.05 everywhere in the apparatus. The ratio $mk_C/k_L^0\phi$ will be small if the equilibrium backpressure of the solute over the liquid is small (i.e., small m or high reactant solubility), or the reaction enhancement factor $\phi = k_1/k_L^0$ is very large, or both. The reaction enhancement factor ϕ will be large for all extremely fast pseudo-first-order reactions and will be large for extremely fast second-order irreversible reaction systems in which there is sufficiently large excess of liquid reagent.

Figure 14-12, case (ii), illustrates the gas-film and liquid-film concentration profiles one might find in an extremely fast (gas-phase mass-transfer-limited), second-order irreversible reaction system. The solid curve for reagent B represents the case in which there is a large excess of bulk liquid reagent B^0 . Figure 14-12, case (iv), represents the case in which the bulk concentration B^0 is not sufficiently large to prevent the depletion of B near the liquid interface.

Whenever these conditions on the ratio y/y_i apply, the design can be based upon the physical rate coefficient k_C or upon the height of one gas-phase mass-transfer unit H_G . The gas-phase mass-transfer-limited condition is approximately valid for the following systems: absorption of NH_3 into water or acidic solutions, absorption of H_2O into concentrated sulfuric acid, absorption of SO_2 into alkali solutions, absorption of H_2S from a gas stream into a strong alkali solution, absorption of HCl into water or alkaline solutions, or absorption of Cl_2 into strong alkali solutions.

When the liquid-phase reactions are extremely slow, the gas-phase resistance can be neglected and one can assume that the rate of reaction has a predominant effect upon the rate of absorption. In this case the differential rate of transfer is given by the equation

$$dn_A = R_A f_{H,S} dh = (k_L^0 a / \rho_L)(c_i - c) S dh \quad (14-66)$$

where n_A = rate of solute transfer, R_A = volumetric reaction rate (function of c and T), $f_{H,S}$ = fractional liquid volume holdup in tower or apparatus, S = tower cross-sectional area, h = vertical distance, k_L^0 = liquid-phase mass-transfer coefficient for pure physical absorption, a = effective interfacial mass-transfer area per unit volume of tower or apparatus, ρ_L = average molar density of liquid phase, c_i = solute concentration in liquid at gas-liquid interface, and c = solute concentration in bulk liquid.

Although the right side of Eq. (14-66) remains valid even when chemical reactions are extremely slow, the mass-transfer driving force may become increasingly small, until finally $c \approx c_i$. For extremely slow first-order irreversible reactions, the following rate expression can be derived from Eq. (14-66):

$$R_A = k_1 c = k_1 c / (1 + k_1 \rho_L f_{H,S} / k_L^0 a) \quad (14-67)$$

where k_1 = first-order reaction rate coefficient.

For dilute systems in countercurrent absorption towers in which the equilibrium curve is a straight line (i.e., $y_i = mx_i$), the differential relation of Eq. (14-66) is formulated as

$$dn_A = -G_M S dy = k_1 c f_{H,S} S dh \quad (14-68)$$

where G_M = molar gas-phase mass velocity and y = gas-phase solute mole fraction.

Substitution of Eq. (14-67) into Eq. (14-68) and integration lead to the following relation for an extremely slow first-order reaction in an absorption tower:

$$y_2 = y_1 \exp \left[- \frac{k_1 \rho_L f_{H,S} h_T / (m G_M)}{1 + k_1 \rho_L f_{H,S} / (k_L^0 a)} \right] \quad (14-69)$$

In Eq. (14-69) subscripts 1 and 2 refer to the bottom and top of the tower, respectively.

As discussed above, the Hatta number N_{Ha} usually is employed as the criterion for determining whether a reaction can be considered extremely slow. A reasonable criterion for slow reactions is

$$N_{Ha} = \sqrt{k_1 D_A / k_L^0} \leq 0.3 \quad (14-70)$$

where D_A = liquid-phase diffusion coefficient of the solute in the solvent. Figure 14-12, cases (vii) and (viii), illustrates the concentration profiles in the gas and liquid films for the case of an extremely slow chemical reaction.

Note that when the second term in the denominator of the exponential in Eq. (14-69) is very small, the liquid holdup in the tower can have a significant influence upon the rate of absorption if an extremely slow chemical reaction is involved.

When chemical equilibrium is achieved quickly throughout the liquid phase, the problem becomes one of properly defining the physical and chemical equilibria for the system. It is sometimes possible to design a tray-type absorber by assuming chemical equilibrium relationships in conjunction with a stage efficiency factor, as is done in distillation calculations. Rivas and Prausnitz [*AIChE J.*, **25**, 975 (1979)] have presented an excellent discussion and example of the correct procedures to be followed for systems involving chemical equilibria.

Traditional Design Method The traditional procedure for designing packed-tower gas absorption systems involving chemical reactions makes use of overall mass-transfer coefficients as defined by the equation

$$K_G a = n_A / (h_T S p_T \Delta y_{i,m}^o) \quad (14-71)$$

where $K_G a$ = overall volumetric mass-transfer coefficient, n_A = rate of solute transfer from the gas to the liquid phase, h_T = total height of tower packing, S = tower cross-sectional area, p_T = total system pressure, and $\Delta y_{i,m}^o$ is defined by the equation

$$\Delta y_{i,m}^o = \frac{(y - y^o)_1 - (y - y^o)_2}{\ln [(y - y^o)_1 / (y - y^o)_2]} \quad (14-72)$$

in which subscripts 1 and 2 refer to the bottom and top of the absorption tower respectively, y = mole-fraction solute in the gas phase, and y^o = gas-phase solute mole fraction in equilibrium with bulk-liquid-phase solute concentration x . When the equilibrium line is straight, $y^o = mx$.

The traditional design method normally makes use of overall $K_G a$ values even when resistance to transfer lies predominantly in the liquid phase. For example, the CO_2 -NaOH system which is most commonly used for comparing $K_G a$ values of various tower packings is a liquid-phase-controlled system. When the liquid phase is controlling, extrapolation to different concentration ranges or operating conditions is not recommended since changes in the reaction mechanism can cause k_L to vary unexpectedly and the overall $K_G a$ do not capture such effects.

Overall $K_G a$ data may be obtained from tower-packing vendors for many of the established commercial gas absorption processes. Such data often are based either upon tests in large-diameter test units or upon actual commercial operating data. Since application to untried operating conditions is not recommended, the preferred procedure for applying the traditional design method is equivalent to duplicating a previously successful commercial installation. When this is not possible, a commercial demonstration at the new operating conditions may be required, or else one could consider using some of the more rigorous methods described later.

While the traditional design method is reported here because it has been used extensively in the past, it should be used with extreme

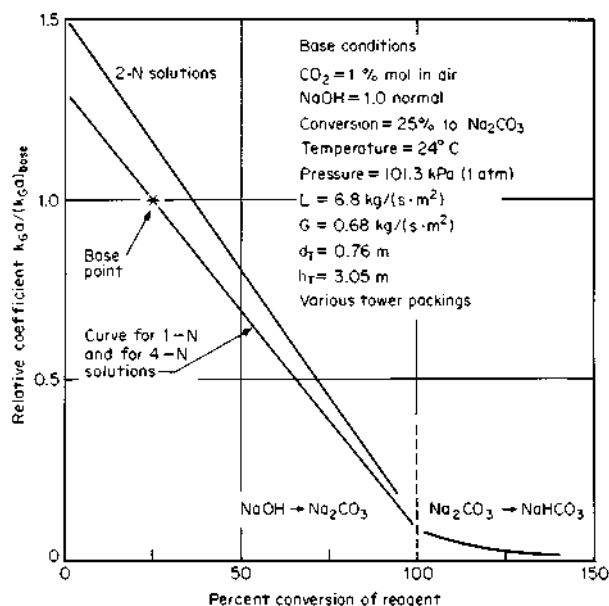


FIG. 14-15 Effects of reagent-concentration and reagent-conversion level upon the relative values of K_{Ga} in the CO_2 - NaOH - H_2O system. [Adapted from Eckert et al., Ind. Eng. Chem., 59(2), 41 (1967).]

caution. In addition to the lack of an explicit liquid-phase resistance term, the method has other limitations. Equation (14-71) assumes that the system is dilute ($y_{BM} \approx 1$) and that the operating and equilibrium lines are straight, which are weak assumptions for reacting systems. Also, Eq. (14-65) is strictly valid only for the temperature and solute partial pressure at which the original test was done even though the total pressure p_T appears in the denominator.

In using Eq. (14-71), therefore, it should be understood that the numerical values of K_{Ga} will be a complex function of pressure, temperature, the type and size of packing employed, the liquid and gas mass flow rates, and the system composition (e.g., the degree of conversion of the liquid-phase reactant).

Figure 14-15 illustrates the influence of system composition and degree of reactant conversion upon the numerical values of K_{Ga} for the absorption of CO_2 into sodium hydroxide at constant conditions of temperature, pressure, and type of packing. An excellent experimental study of the influence of operating variables upon overall K_{Ga} values is that of Field et al. (*Pilot-Plant Studies of the Hot Carbonate*

Process for Removing Carbon Dioxide and Hydrogen Sulfide, U.S. Bureau of Mines Bulletin 597, 1962).

Table 14-2 illustrates the observed variations in K_{Ga} values for different packing types and sizes for the CO_2 - NaOH system at a 25 percent reactant conversion for two different liquid flow rates. The lower rate of $2.7 \text{ kg}/(\text{s}\cdot\text{m}^2)$ or $2000 \text{ lb}/(\text{h}\cdot\text{ft}^2)$ is equivalent to $4 \text{ U.S. gal}/(\text{min}\cdot\text{ft}^2)$ and is typical of the liquid rates employed in fume scrubbers. The higher rate of $13.6 \text{ kg}/(\text{s}\cdot\text{m}^2)$ or $10,000 \text{ lb}/(\text{h}\cdot\text{ft}^2)$ is equivalent to $20 \text{ U.S. gal}/(\text{min}\cdot\text{ft}^2)$ and is more typical of absorption towers such as used in CO_2 removal systems, for example. We also note that two gas velocities are represented in the table, corresponding to superficial velocities of 0.59 and 1.05 m/s (1.94 and 3.44 ft/s).

Table 14-3 presents a typical range of K_{Ga} values for chemically reacting systems. The first two entries in the table represent systems that can be designed by the use of purely physical design methods, because they are completely gas-phase mass-transfer-limited. To ensure a negligible liquid-phase resistance in these two tests, the HCl was absorbed into a solution maintained at less than $8 \text{ wt } \%$ HCl , and the NH_3 was absorbed into a water solution maintained below $\text{pH } 7$ by the addition of acid. The last two entries in Table 14-3 represent liquid-phase mass-transfer-limited systems.

Scaling Up from Laboratory Data Laboratory experimental techniques offer an efficient and cost-effective route to develop commercial absorption designs. For example, Ouwerkerk (*Hydrocarbon Process.*, April 1978, 89-94) revealed that both laboratory and small-scale pilot plant data were employed as the basis for the design of an 8.5-m (28-ft) diameter commercial Shell Claus off-gas treating (SCOT) tray-type absorber. Ouwerkerk claimed that the cost of developing comprehensive design procedures can be minimized, especially in the development of a new process, by the use of these modern techniques.

In a 1966 paper that is considered a classic, Dankwerts and Gillham [*Trans. Inst. Chem. Eng.*, 44, T42 (1966)] showed that data taken in a small stirred-cell laboratory apparatus could be used in the design of a packed-tower absorber when chemical reactions are involved. They showed that if the packed-tower mass-transfer coefficient in the absence of reaction (k_L^0) can be reproduced in the laboratory unit, then the rate of absorption in the laboratory apparatus will respond to chemical reactions in the same way as in the packed column, even though the means of agitating the liquid in the two systems may be quite different.

According to this method, it is not necessary to investigate the kinetics of the chemical reactions in detail; nor is it necessary to determine the solubilities or diffusivities of the various reactants in their unreacted forms. To use the method for scaling up, it is necessary to independently obtain data on the values of the interfacial area per unit volume a and the physical mass-transfer coefficient k_L^0 for the commercial packed tower. Once these data have been measured and tabulated, they can be used directly for scaling up the experimental laboratory data for any new chemically reacting system.

Dankwerts and Gillham did not investigate the influence of the gas-phase resistance in their study (for some processes, gas-phase resistance

TABLE 14-2 Typical Effects of Packing Type, Size, and Liquid Rate on K_{Ga} in a Chemically Reacting System, K_{Ga} , $\text{kmol}/(\text{h}\cdot\text{m}^3)$

Packing size, mm	$L = 2.7 \text{ kg}/(\text{s}\cdot\text{m}^2)$				$L = 13.6 \text{ kg}/(\text{s}\cdot\text{m}^2)$			
	25	38	50	75-90	25	38	50	75-90
Berl-saddle ceramic	30	24	21		45	38	32	
Raschig-ring ceramic	27	24	21		42	34	30	
Raschig-ring metal	29	24	19		45	35	27	
Pall-ring plastic	29	27	26*	16	45	42	38*	24
Pall-ring metal	37	32	27	21*	56	51	43	27*
Intalox-saddle ceramic	34	27	22	16*	56	43	34	26*
Super-Intalox ceramic	37*		26*		59*		40*	
Intalox-saddle plastic	40*		24*	16*	56*		37*	26*
Intalox-saddle metal	43*	35*	30*	24*	66*	58*	48*	37*
Hy-Pak metal	35	32*	27*	18*	54	50*	42*	27*

Data courtesy of the Norton Company.

Operating conditions: CO_2 , 1 percent mole in air; NaOH , 4 percent weight (1 normal); 25 percent conversion to sodium carbonate; temperature, 24°C (75°F); pressure, 98.6 kPa (0.97 atm); gas rate $= 0.68 \text{ kg}/(\text{s}\cdot\text{m}^2) = 0.59 \text{ m/s} = 500 \text{ lb}/(\text{h}\cdot\text{ft}^2) = 1.92 \text{ ft/s}$ except for values with asterisks, which were run at $1.22 \text{ kg}/(\text{s}\cdot\text{m}^2) = 1.05 \text{ m/s} = 900 \text{ lb}/(\text{h}\cdot\text{ft}^2) = 3.46 \text{ ft/s}$ superficial velocity; packed height, 3.05 m (10 ft); tower diameter, 0.76 m (2.5 ft). To convert table values to units of $(\text{lb}\cdot\text{mol})/(\text{h}\cdot\text{ft}^3)$, multiply by 0.0624 .

TABLE 14-3 Typical $K_G a$ Values for Various Chemically Reacting Systems, $\text{kmol}/(\text{h}\cdot\text{m}^3)$

Gas-phase reactant	Liquid-phase reactant	$K_G a$	Special conditions
HCl	H ₂ O	353	Gas-phase limited
NH ₃	H ₂ O	337	Gas-phase limited
Cl ₂	NaOH	272	8% weight solution
SO ₂	Na ₂ CO ₃	224	11% weight solution
HF	H ₂ O	152	
Br ₂	NaOH	131	5% weight solution
HCN	H ₂ O	114	
HCHO	H ₂ O	114	Physical absorption
HBr	H ₂ O	98	
H ₂ S	NaOH	96	4% weight solution
SO ₂	H ₂ O	59	
CO ₂	NaOH	38	4% weight solution
Cl ₂	H ₂ O	8	Liquid-phase limited

Data courtesy of the Norton Company.

Operating conditions (see text): 38-mm ceramic Intalox saddles; solute gases, 0.5–1.0 percent mole; reagent conversions = 33 percent; pressure, 101 kPa (1 atm); temperature, 16–24°C; gas rate = $1.3 \text{ kg}/(\text{s}\cdot\text{m}^2) = 1.1 \text{ m/s}$; liquid rates = 3.4 to $6.8 \text{ kg}/(\text{s}\cdot\text{m}^2)$; packed height, 3.05 m; tower diameter, 0.76 m. Multiply table values by 0.0624 to convert to $(\text{lb}\cdot\text{mol})/(\text{h}\cdot\text{ft}^3)$.

may be neglected). However, in 1975 Dankwerts and Alper [*Trans. Inst. Chem. Eng.*, 53, T42 (1975)] showed that by placing a stirrer in the gas space of the stirred-cell laboratory absorber, the gas-phase mass-transfer coefficient k_G in the laboratory unit could be made identical to that in a packed-tower absorber. When this was done, laboratory data for chemically reacting systems having a significant gas-side resistance could successfully be scaled up to predict the performance of a commercial packed-tower absorber.

If it is assumed that the values for k_G , k_L^0 , and a have been measured for the commercial tower packing to be employed, the procedure for using the laboratory stirred-cell reactor is as follows:

1. The gas-phase and liquid-phase stirring rates are adjusted so as to produce the same values of k_G and k_L^0 as will exist in the commercial tower.

2. For the reaction system under consideration, experiments are made at a series of bulk-liquid and bulk-gas compositions representing the compositions to be expected at different levels in the commercial absorber (on the basis of material balance).

3. The ratios of $r_A(c_b, B^0)$ are measured at each pair of gas and liquid compositions.

For the dilute-gas systems, one form of the equation to be solved in conjunction with these experiments is

$$h_T = \frac{G_M}{a} \int_{y_2}^{y_1} \frac{dy}{r_A} \quad (14-73)$$

where h_T = height of commercial tower packing, G_M = molar gas-phase mass velocity, a = effective mass-transfer area per unit volume in the commercial tower, y = mole fraction solute in the gas phase, and r_A = experimentally determined rate of absorption per unit of exposed interfacial area.

By using the series of experimentally measured rates of absorption, Eq. (14-73) can be integrated numerically to determine the height of packing required in the commercial tower.

A number of different types of experimental laboratory units could be used to develop design data for chemically reacting systems. Charpentier [*ACS Symp. Ser.*, 72, 223–261 (1978)] has summarized the state of the art with respect to methods of scaling up laboratory data and has tabulated typical values of the mass-transfer coefficients, interfacial areas, and contact times to be found in various commercial gas absorbers, as well as in currently available laboratory units.

The laboratory units that have been employed to date for these experiments were designed to operate at a total system pressure of about 101 kPa (1 atm) and at near-ambient temperatures. In practical situations, it may become necessary to design a laboratory absorption unit that can be operated either under vacuum or at elevated pressure

and over a range of temperatures in order to apply the Dankwerts method.

It would be desirable to reinterpret existing data for commercial tower packings to extract the individual values of the interfacial area a and the mass-transfer coefficients k_G and k_L^0 to facilitate a more general usage of methods for scaling up from laboratory experiments. Some progress has already been made, as described later in this section. In the absence of such data, it is necessary to operate a pilot plant or a commercial absorber to obtain k_G , k_L^0 , and a as described by Ouwkerk (op. cit.).

Modern techniques use rigorous modeling computer-based methods to extract fundamental parameters from laboratory-scale measurements and then apply them to the design of commercial absorption towers. These techniques are covered next.

Rigorous Computer-Based Absorber Design While the techniques described earlier in this section are very useful to gain an understanding of the key effects in commercial absorbers, current design methods used in industrial practice for chemically reactive systems are increasingly often based upon computerized rigorous methods, which are commercially available from software vendors. The advantages of these rigorous methods are as follows: (1) Approximations do not have to be made for special cases (e.g., fast chemical reactions or mass-transfer resistance dominated by the gas or liquid phase), and all effects can be simultaneously modeled. (2) Fundamental quantities such as kinetic parameters and mass-transfer coefficients can be extracted from laboratory equipment and applied to commercial absorber towers. (3) Integrated models can be developed for an entire absorption process flowsheet (e.g., the absorber-stripper system with heat integration presented in Fig. 14-3), and consequently the entire system may be optimized.

Computer programs for chemically reacting systems are available from several vendors, notably the following:

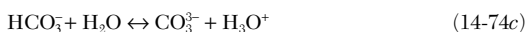
Program	Vendor	Reference
AMSIM	Schlumberger Limited	Zhang and Ng, <i>Proc. Ann. Conv.—Gas Proc. Assoc.</i> , Denver, Colo.; 1996, p. 22.
ProTreat	Sulphur Experts	Weiland and Dingman, <i>Proc. Ann. Conv., Gas Proc. Assoc.</i> , Houston, Tex., 2001, p. 80.
TSWEET	Bryan Research and Engineering	Polasek, Donnelly, and Bullin, <i>Proc. 71st GPA Annual Conv.</i> , 1992, p. 58.
RateSep	Aspen Technology	Chen et al., <i>AIChE Annual Meeting</i> , San Francisco, Nov. 12–17, 2006.

The specific approaches used to model the chemically reacting absorption system are slightly different among the different vendors. The general approach used and the benefits obtained are highlighted

by considering a specific example: removal of CO_2 from flue gases discharged by a power plant using aqueous monoethanolamine (MEA), as presented by Freguia and Rochelle [AIChE J., **49**, 1676 (2003)].

The development and application of a rigorous model for a chemically reactive system typically involves four steps: (1) development of a thermodynamic model to describe the physical and chemical equilibrium; (2) adoption and use of a modeling framework to describe the mass transfer and chemical reactions; (3) parameterization of the mass-transfer and kinetic models based upon laboratory, pilot-plant, or commercial-plant data; and (4) use of the integrated model to optimize the process and perform equipment design.

Development of Thermodynamic Model for Physical and Chemical Equilibrium The first and perhaps most important step in the development of the thermodynamic model is the *speciation*, or representation of the set of chemical reactions. For CO_2 absorption in aqueous MEA solutions, the set of reactions is



In addition, a model is needed that can describe the nonideality of a system containing molecular and ionic species. Freguia and Rochelle adopted the model developed by Chen et al. [AIChE J., **25**, 820 (1979)] and later modified by Mock et al. [AIChE J., **32**, 1655 (1986)] for mixed-electrolyte systems. The combination of the speciation set of reactions [Eqs. (14-74a) to (14-74e)] and the nonideality model is capable of representing the solubility data, such as presented in Figs. 14-1 and 14-2, to good accuracy. In addition, the model accurately and correctly represents the actual species present in the aqueous phase, which is important for faithful description of the chemical kinetics and species mass transfer across the interface. Finally, the thermodynamic model facilitates accurate modeling of the heat effects, such as those discussed in Example 6.

Rafal et al. (Chapter 7, "Models for Electrolyte Solutions," in *Models for Thermodynamic and Phase Equilibria Calculations*, S. I. Sandler, ed., Marcel Dekker, New York, 1994, p. 686) have provided a comprehensive discussion of speciation and thermodynamic models.

Adoption and Use of Modeling Framework The rate of diffusion and species generation by chemical reaction can be described by film theory, penetration theory, or a combination of the two. The most popular description is in terms of a two-film theory, which is

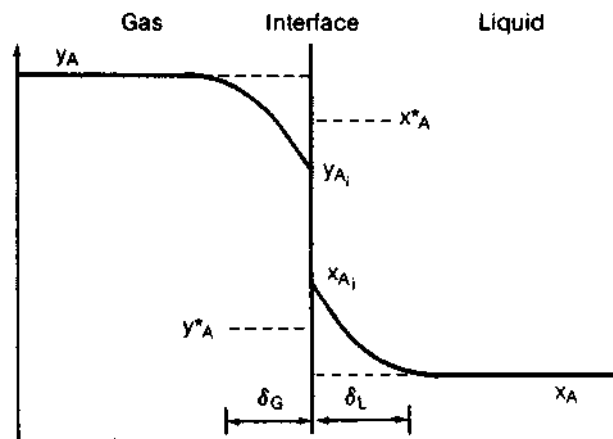


FIG. 14-16 Concentration profiles in the vapor and liquid phases near an interface.

diagrammed in Fig. 14-16 for absorption. Accordingly, there exists a stable interface separating the gas and the liquid. A certain distance from the interface, large fluid motions exist, and these distribute the material rapidly and equally so that no concentration gradients develop. Next to the interface, however, there are regions in which the fluid motion is slow; in these regions, termed *films*, material is transferred by diffusion alone. At the gas-liquid interface, material is transferred instantaneously, so that the gas and liquid are in physical equilibrium at the interface. The rate of diffusion in adsorption is therefore the rate of diffusion in the gas and liquid films adjacent to the interface. The model framework is completed by including terms for species generation (chemical equilibrium and chemical kinetics) in the gas and liquid film and bulk regions. Taylor, Krishna, and Kooijman (*Chem. Eng. Progress*, July 2003, p. 28) have provided an excellent discussion of rate-based models; these authors emphasize that the diffusion flux for multicomponent systems must be based upon the Maxwell-Stefan approach. The book by Taylor and Krishna (*Multicomponent Mass Transfer*, Wiley, New York, 1993) provides a detailed discussion of the Maxwell-Stefan approach. More details and discussion have been provided by the program vendors listed above.

Parameterization of Mass-Transfer and Kinetic Models The mass-transfer and chemical kinetic rates required in the rigorous model are typically obtained from the literature, but must be carefully evaluated; and fine-tuning through pilot-plant and commercial data is highly recommended.

Mass-transfer coefficient models for the vapor and liquid coefficients are of the general form

$$k_{t,j}^L = a \rho_L f(D_{t,j}^m, \mu_L, \rho_V, a, \text{internal characteristics}) \quad (14-75a)$$

$$k_{t,j}^V = a P f(D_{t,j}^m, \mu_V, \rho_V, a, \text{internal characteristics}) \quad (14-75b)$$

where a = effective interfacial area per unit volume, $D_{t,j}^m$ are the Stefan-Maxwell diffusion coefficients, P = pressure, ρ = molar density, and μ = viscosity. The functions in Eqs. (14-75a) and (14-75b) are correlations that depend on the column internals. Popular correlations in the literature are those by Onda et al. [*J. Chem. Eng. Jap.*, **1**, 56 (1968)] for random packing, Bravo and Fair [*Ind. Eng. Chem. Proc. Des. Dev.*, **21**, 162 (1982)] for structured packing, Chan and Fair [*Ind. Eng. Chem. Proc. Des. Dev.*, **23**, 814 (1984)] for sieve trays, Scheffe and Weiland [*Ind. Eng. Chem. Res.*, **26**, 228 (1987)] for valve trays, and Hughmark [AIChE J., **17**, 1295 (1971)] for bubble-cap trays.

It is highly recommended that the mass-transfer correlations be tested and improved by using laboratory, pilot-plant, or commercial data for the specific application. Commercial software generally provides the capability for correction factors to adjust generalized correlations to the particular application.

Kinetic models are usually developed by replacing a subset of the speciation reactions by kinetically reversible reactions. For example, Freguia and Rochelle replaced equilibrium reactions (14-74a) and (14-74b) with kinetically reversible reactions and retained the remaining three reactions as very fast and hence effectively at equilibrium. The kinetic constants were tuned using wetted-wall column data from Dang (M.S. thesis, University of Texas, Austin, 2001) and field data from a commercial plant.

Modern commercial software provides powerful capability to deploy literature correlations and to customize models for specific applications.

Deployment of Rigorous Model for Process Optimization and Equipment Design Techniques similar to those described above may be used to develop models for the stripper as well as other pieces of plant equipment, and thus an integrated model for the entire absorption system may be produced. The value of integrated models is that they can be used to understand the combined effects of many variables that determine process performance and to rationally optimize process performance. Freguia and Rochelle have shown that the reboiler duty (the dominant source of process operating costs) may be reduced by 10 percent if the absorber height is increased by 20 percent and by 4 percent if the absorber is intercooled with a duty equal to one-third of the reboiler duty. They also show that the power plant lost work is affected by varying stripper

pressure, but not significantly, so any convenient pressure can be chosen to operate the stripper.

In this section, we have used the example of CO₂ removal from flue gases using aqueous MEA to demonstrate the development and application of a rigorous model for a chemically reactive system. Modern software enables rigorous description of complex chemically reactive systems, but it is very important to carefully evaluate the models and to tune them using experimental data.

Use of Literature for Specific Systems A large body of experimental data obtained in bench-scale laboratory units and in small-diameter packed towers has been published since the early 1940s. One might wish to consider using such data for a particular chemically reacting system as the basis for scaling up to a commercial design. Extreme caution is recommended in interpreting such data for the purpose of developing commercial designs, as extrapolation of the data can lead to serious errors. Extrapolation to temperatures, pressures, or liquid-phase reagent conversions different from those that were employed by the original investigators definitely should be regarded with caution.

Bibliographies presented in the General References listed at the beginning of this section are an excellent source of information on

specific chemically reacting systems. *Gas-Liquid Reactions* by Dankwerts (McGraw-Hill, New York, 1970) contains a tabulation of references to specific chemically reactive systems. *Gas Treating with Chemical Solvents* by Astarita et al. (Wiley, New York, 1983) deals with the absorption of acid gases and includes an extensive listing of patents. *Gas Purification* by Kohl and Nielsen (Gulf Publishing, Houston, 1997) provides a practical description of techniques and processes in widespread use and typically also sufficient design and operating data for specific applications.

In searching for data on a particular system, a computerized search of *Chemical Abstracts*, *Engineering Index*, and *National Technical Information Service* (NTIS) databases is recommended. In addition, modern search engines will rapidly uncover much potentially valuable information.

The experimental data for the system CO₂-NaOH-Na₂CO₃ are unusually comprehensive and well known as the result of the work of many experimenters. A serious study of the data and theory for this system therefore is recommended as the basis for developing a good understanding of the kind and quality of experimental information needed for design purposes.

EQUIPMENT FOR DISTILLATION AND GAS ABSORPTION: TRAY COLUMNS

Distillation and gas absorption are the prime and most common gas-liquid mass-transfer operations. Other operations that are often performed in similar equipment include stripping (often considered part of distillation), direct-contact heat transfer, flashing, washing, humidification, and dehumidification.

The most common types of contactors by far used for these are tray and packed towers. These are the focus of this subsection. Other contactors used from time to time and their applications are listed in Table 14-4.

In this subsection, the terms *gas* and *vapor* are used interchangeably. *Vapor* is more precise for distillation, where the gas phase is at equilibrium. Also, the terms *tower* and *column* are used interchangeably.

A crossflow tray (Fig. 14-17) consists of the *bubbling area* and the *downcomer*. Liquid descending the downcomer from the tray above enters the bubbling area. Here, the liquid contacts gas ascending through the tray perforations, forming froth or spray. An *outlet weir* on the downstream side of the bubbling area helps maintain liquid level on the tray. Froth overflowing the weir enters the outlet downcomer. Here, gas disengages from the liquid, and the liquid descends to the tray below. The bubbling area can be fitted with numerous types of tray hardware. The most common types by far are:

Sieve trays (Fig. 14-18a) are perforated plates. The velocity of upflowing gas keeps the liquid from descending through the perforations (weeping). At low gas velocities, liquid weeps through the perforations, bypassing part of the tray and reducing tray efficiency. Because of this, sieve trays have relatively poor turndown.

Fixed valve trays (Fig. 14-18b) have the perforations covered by a fixed cover, often a section of the tray floor pushed up. Their performance is similar to that of sieve trays.

Moving valve trays (Fig. 14-18c) have the perforations covered by movable disks (valves). Each valve rises as the gas velocity increases. The upper limit of the rise is controlled by restricting legs on the bottom of the valve (Fig. 14-18c) or by a cage structure around the valve. As the gas velocity falls, some valves close completely, preventing weeping. This gives the valve tray good turndown.

Table 14-5 is a general comparison of the three main tray types, assuming proper design, installation, and operation. Sieve and valve trays are comparable in capacity, efficiency, entrainment, and pressure drop. The turndown of moving valve trays is much better than that of sieve and fixed valve trays. Sieve trays are least expensive; valve trays cost only slightly more. Maintenance, fouling tendency, and effects of corrosion are least troublesome in fixed valve and sieve trays (provided the perforations or fixed valves are large enough) and most troublesome with moving valve trays.

Fixed valve and sieve trays prevail when fouling or corrosion is expected, or if turndown is unimportant. Valve trays prevail when high turndown is required. The energy saved, even during short turndown periods, usually justifies the small additional cost of the moving valve trays.

DEFINITIONS

Tray Area Definitions Some of these are illustrated in Fig. 14-17. **Total tower cross-section area A_T** The inside cross-sectional area of the empty tower (without trays or downcomers).

Net area A_N (also called *free area*) The total tower cross-sectional area A_T minus the area at the top of the downcomer A_{DT} . The

TABLE 14-4 Equipment for Liquid-Gas Systems

Equipment designation	Mode of flow	Gross mechanism	Continuous phase	Primary process applications
Tray column	Cross-flow, countercurrent	Integral	Liquid and/or gas	Distillation, absorption, stripping, DCHT, washing
Packed column	Countercurrent, cocurrent	Differential	Liquid and/or gas	Distillation, absorption, stripping, humidification, dehumidification, DCHT, washing
Wetted-wall (falling-film) column	Countercurrent, cocurrent	Differential	Liquid and/or gas	Distillation, absorption, stripping, evaporation
Spray chamber	Cocurrent, cross-flow, countercurrent	Differential	Gas	Absorption, stripping, humidification, dehumidification
Agitated vessel	Complete mixing	Integral	Liquid	Absorption
Line mixer	Cocurrent	Differential	Liquid or gas	Absorption, stripping

DCHT = direct contact heat transfer.

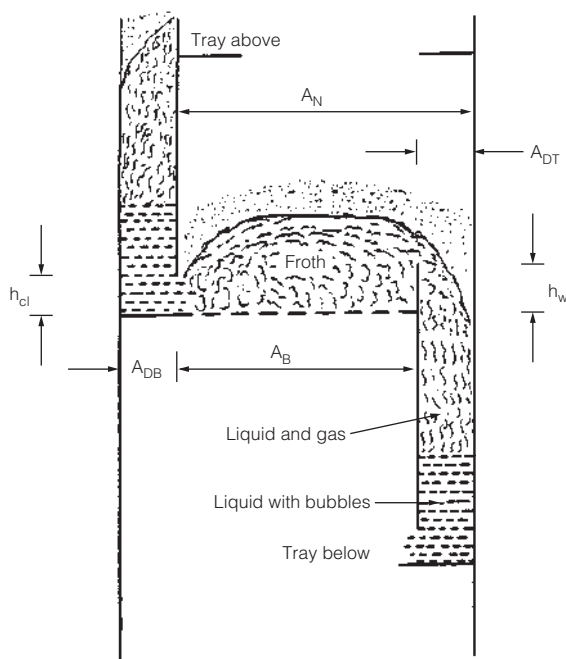


FIG. 14-17 Schematic of a tray operating in the froth regime. (Based on H. Z. Kister, *Distillation Design*, copyright © 1992 by McGraw-Hill; reprinted by permission.)

net area represents the smallest area available for vapor flow in the intertray spacing.

Bubbling area A_B (also called active area) The total tower cross-sectional area minus the sum of downcomer top area A_{DT} , downcomer seal area A_{DB} , and any other nonperforated areas on the tray. The bubbling area represents the area available for vapor flow just above the tray floor.

Hole area A_h The total area of the perforations on the tray. The hole area is the smallest area available for vapor passage on a sieve tray.

Slot area A_s The total (for all open valves) vertical curtain area through which vapor passes in a horizontal direction as it leaves the valves. It is a function of the narrowest opening of each valve and the number of valves that are open. The slot area is normally the smallest area available for vapor flow on a valve tray.

Open slot area A_{so} The slot area when all valves are open.

Fractional hole area A_f The ratio of hole area to bubbling area (sieve trays) or slot area to bubbling area (valve trays).

Vapor and Liquid Load Definitions

F-factor F This is the square root of the kinetic energy of the gas, defined by Eq. (14-76). The velocity in Eq. (14-76) is usually (not always) based on the tower cross-sectional area A_T , the net area A_N , or the bubbling area A_B . The user should beware of any data for which the area basis is not clearly specified.

$$F = u\sqrt{\rho_G} \quad (14-76)$$

C-factor C The C-factor, defined in Eq. (14-77), is the best gas load term for comparing capacities of systems of different physical properties. It has the same units as velocity (m/s or ft/s) and is directly related to droplet entrainment. As with the F-factor, the user should beware of any data for which the area basis is not clearly specified.

$$C = u\sqrt{\frac{\rho_G}{\rho_L - \rho_G}} \quad (14-77)$$

Weir load For trays (as distinct from downcomers), liquid load is normally defined as

$$Q_L = \frac{\text{volume of liquid}}{\text{length of outlet weir}} = \frac{Q}{L_w} \quad (14-78)$$

This definition describes the flux of liquid horizontally across the tray. Units frequently used are $\text{m}^3/(\text{h}\cdot\text{m})$, $\text{m}^3/(\text{s}\cdot\text{m})$, and gpm/in.

Downcomer liquid load For downcomer design, the liquid load is usually defined as the liquid velocity at the downcomer entrance (m/s or ft/s):

$$Q_D = \frac{\text{volume of liquid}}{\text{downcomer entrance area}} = \frac{Q}{A_{DT}} \quad (14-79)$$

FLOW REGIMES ON TRAYS

Three main flow regimes exist on industrial distillation trays. These regimes may all occur on the same tray under different liquid and gas flow rates (Fig. 14-19). Excellent discussion of the fundamentals and modeling of these flow regimes was presented by Lockett (*Distillation Tray Fundamentals*, Cambridge University Press, Cambridge, 1986). An excellent overview of these as well as of less common flow regimes was given by Prince (*PACE*, June 1975, p. 31; July 1975, p. 18).

Froth regime (or mixed regime; Fig. 14-20a). This is the most common operating regime in distillation practice. Each perforation bubbles vigorously. The bubbles circulate rapidly through the liquid, are of nonuniform sizes and shapes, and travel at varying velocities. The froth surface is mobile and not level, and is generally covered by droplets. Bubbles are formed at the tray perforations and are swept away by the froth.

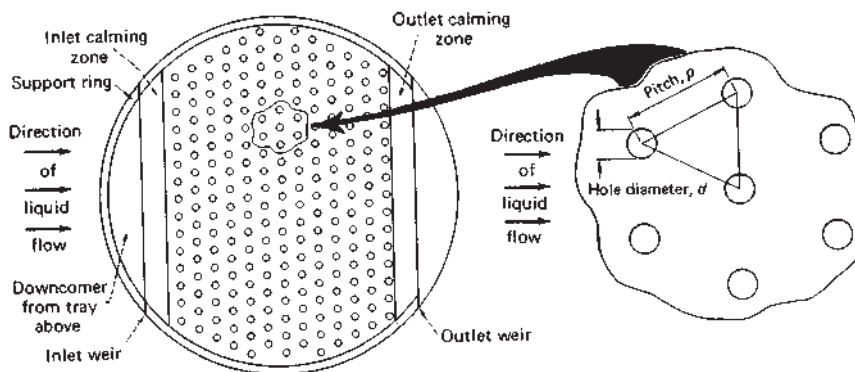
As gas load increases in the froth regime, jetting begins to replace bubbling in some holes. The fraction of holes that is jetting increases with gas velocity. When jetting becomes the dominant mechanism, the dispersion changes from froth to spray. Prado et al. [*Chemical Engineering Progr.* **83**(3), p. 32, (1987)] showed the transition from froth to spray takes place gradually as jetting replaces bubbling in 45 to 70 percent of the tray holes.

Emulsion regime (Fig. 14-20b). At high liquid loads and relatively low gas loads, the high-velocity liquid bends the swarms of gas bubbles leaving the orifices, and tears them off, so most of the gas becomes emulsified as small bubbles within the liquid. The mixture behaves as a uniform two-phase fluid, which obeys the Francis weir formula [see the subsection "Pressure Drop" and Eq. (14-109) (Hofhuis and Zuideweg, *ICHEME Symp. Ser.* **56**, p. 2.2/1 (1979); Zuideweg, *Int. Chem. Eng.* **26**(1), 1 (1986)]. In industrial practice, the emulsion regime is the most common in high-pressure and high-liquid-rate operation.

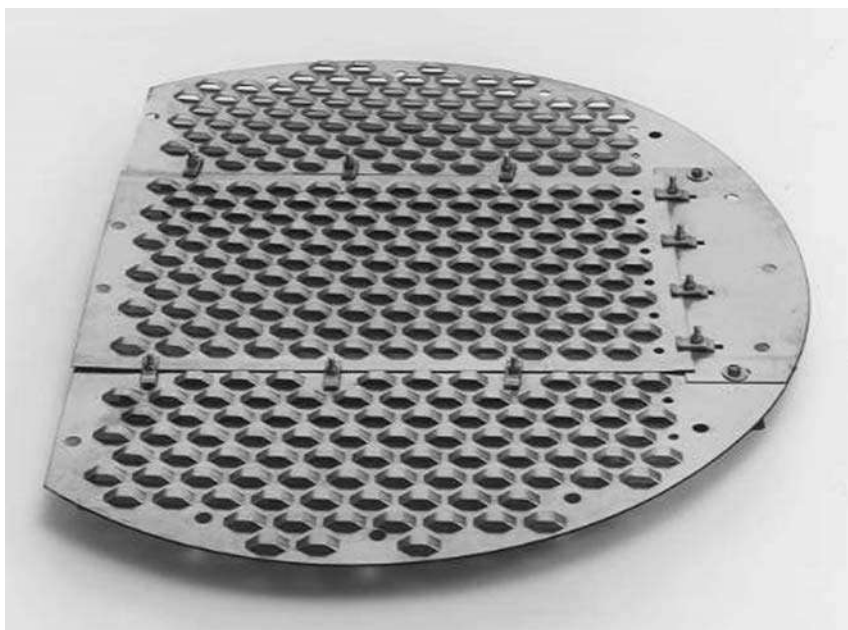
Spray regime (or drop regime, Fig. 14-20c). At high gas velocities and low liquid loads, the liquid pool on the tray floor is shallow and easily atomized by the high-velocity gas. The dispersion becomes a turbulent cloud of liquid droplets of various sizes that reside at high elevations above the tray and follow free trajectories. Some droplets are entrained to the tray above, while others fall back into the liquid pools and become reatomized. In contrast to the liquid-continuous froth and emulsion regimes, the phases are reversed in the spray regime: here the gas is the continuous phase, while the liquid is the dispersed phase.

The spray regime frequently occurs where gas velocities are high and liquid loads are low (e.g., vacuum and rectifying sections at low liquid loads).

Three-layered structure. Van Sinderen, Wijn, and Zanting [*Trans. IChemE*, **81**, Part A, p. 94 (January 2003)] postulate a tray dispersion consisting of a bottom liquid-rich layer where jets/bubbles form; an intermediate liquid-continuous froth layer where bubbles erupt, generating drops; and a top gas-continuous layer of drops. The intermediate layer that dampens the bubbles and



(a)



(b)



(c)

FIG. 14-18 Common tray types. (a) Sieve. (b) Fixed valve. (c) Moving valve with legs. [Part a, from Henry Z. Kister, Chem. Eng., September 8, 1980; reprinted courtesy of Chemical Engineering. Part b, Courtesy of Sulzer Chemtech and Fractionation Research Inc. (FRI). Part c, courtesy of Koch-Glitsch LP.]

TABLE 14-5 Comparison of the Common Tray Types

	Sieve trays	Fixed valve tray	Moving valve tray
Capacity	High	High	High to very high
Efficiency	High	High	High
Turndown	About 2:1. Not generally suitable for operation under variable loads	About 2.5:1. Not generally suitable for operation under variable loads	About 4:1 to 5:1. Some special designs achieve 8:1 or more
Entrainment	Moderate	Moderate	Moderate
Pressure drop	Moderate	Moderate	Slightly higher
Cost	Low	Low	About 20 percent higher
Maintenance	Low	Low	Moderate
Fouling tendency	Low to very low	Low to very low	Moderate
Effects of corrosion	Low	Very low	Moderate
Main applications	(1) Most columns when turndown is not critical (2) High fouling and corrosion potential	(1) Most columns when turndown is not critical (2) High fouling and corrosion potential	(1) Most columns (2) Services where turndown is important

jets disappears at low liquid rates, and the drop layer approaches the tray floor, similar to the classic spray regime.

PRIMARY TRAY CONSIDERATIONS

Number of Passes Tray liquid may be split into two or more flow passes to reduce tray liquid load Q_L (Fig. 14-21). Each pass carries $1/N_p$ fraction of the total liquid load (e.g., $\frac{1}{4}$ in four-pass trays). Liquid in each pass reverses direction on alternate trays. Two-pass trays have perfect symmetry with full remixing in the center downcomers. Four-pass trays are symmetric along the centerline, but the side and central passes are nonsymmetric. Also, the center and off-center downcomers only partially remix the liquid, allowing any maldistribution to propagate. Maldistribution can cause major loss of efficiency and capacity in four-pass trays. Three-pass trays are even more prone to maldistribution due to their complete nonsymmetry. Most designers avoid three-pass trays altogether, jumping from two to four passes. Good practices for liquid and vapor balancing and for avoiding maldistribution in multipass trays were described by Pilling [*Chemical Engineering Progr.*, p. 22 (June 2005)], Bolles [*AIChE J.*, 22(1), p. 153 (1976)], and Kister (*Distillation Operation*, McGraw-Hill, New York, 1990).

Common design practice is to minimize the number of passes, resorting to a larger number only when the liquid load exceeds 100 to 140 $\text{m}^3/(\text{h}\cdot\text{m})$ (11 to 15 gpm/in) of outlet weir length [Davies and Gordon, *Petro/Chem Eng.*, p. 228 (December 1961)]. Trays smaller than 1.5-m (5-ft) diameter seldom use more than a single pass; those with 1.5- to 3-m (5- to 10-ft) diameters seldom use more than two passes. Four-pass trays are common in high liquid services with towers larger than 5-m (16-ft) diameter.

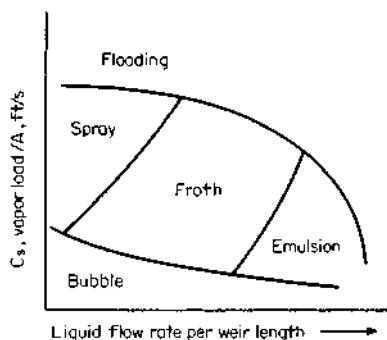


FIG. 14-19 The flow regime likely to exist on a distillation tray as a function of vapor and liquid loads. (From H. Z. Kister, *Distillation Design*, copyright ©1992 by McGraw-Hill; reprinted by permission.)

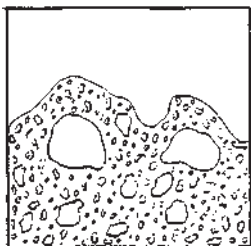
Tray Spacing Taller spacing between successive trays raises capacity, leading to a smaller tower diameter, but also raises tower height. There is an economic tradeoff between tower height and diameter. As long as the tradeoff exists, tray spacing has little effect on tower economics and is set to provide adequate access. In towers with larger than 1.5-m (5-ft) diameter, tray spacing is typically 600 mm (24 in), large enough to permit a worker to crawl between trays. In very large towers (>6-m or 20-ft diameter), tray spacings of 750 mm (30 in) are often used. In chemical towers (as distinct from petrochemical, refinery, and gas plants), 450 mm (18 in) has been a popular tray spacing. With towers smaller than 1.5 m (5 ft), tower walls are reachable from the manways, there is no need to crawl, and it becomes difficult to support thin and tall columns, so smaller tray spacing (typically 380 to 450 mm or 15 to 18 in) is favored. Towers taller than 50 m (160 ft) also favor smaller tray spacings (400 to 450 mm or 16 to 18 in). Finally, cryogenic towers enclosed in cold boxes favor very small spacings, as small as 150 to 200 mm (6 to 8 in), to minimize the size of the cold box.

More detailed considerations for setting tray spacing were discussed by Kister (*Distillation Operation*, McGraw-Hill, New York, 1990) and Mukherjee [*Chem. Eng.*, p. 53 (September 2005)].

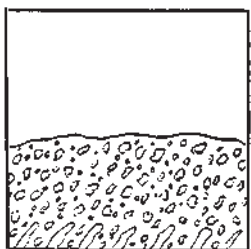
Outlet Weir The outlet weir should maintain a liquid level on the tray high enough to provide sufficient gas-liquid contact without causing excessive pressure drop, downcomer backup, or a capacity limitation. Weir heights are usually set at 40 to 80 mm (1.5 to 3 in). In this range, weir heights have little effect on distillation efficiency [Van Winkle, *Distillation*, McGraw-Hill, New York, 1967; Kreis and Raab, *ICHEME Symp. Ser.* 56, p. 3.2/63 (1979)]. In operations where long residence times are necessary (e.g., chemical reaction, absorption, stripping) taller weirs do improve efficiency, and weirs 80 to 100 mm (3 to 4 in) are more common (Lockett, *Distillation Tray Fundamentals*, Cambridge University Press, Cambridge, England, 1986).

Adjustable weirs (Fig. 14-22a) are used to provide additional flexibility. They are uncommon with conventional trays, but are used with some proprietary trays. Swept-back weirs (Fig. 14-22b) are used to extend the effective length of side weirs, either to help balance liquid flows to nonsymmetric tray passes or/and to reduce the tray liquid loads. Picket fence weirs (Fig. 14-22c) are used to shorten the effective length of a weir, either to help balance multipass trays' liquid flows (they are used in center and off-center weirs) or to raise tray liquid load and prevent drying in low-liquid-load services. To be effective, the pickets need to be tall, typically around 300 to 400 mm (12 to 16 in) above the top of the weir. An excellent discussion of weir picketing practices was provided by Summers and Sloley (*Hydroc. Proc.*, p. 67, January 2007).

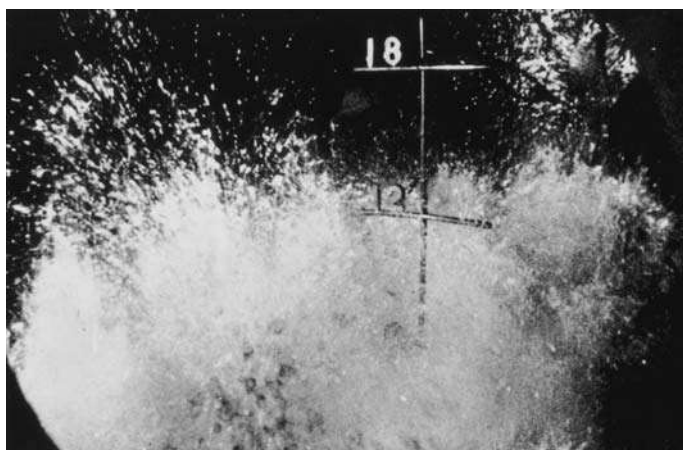
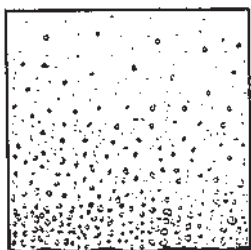
Downcomers A downcomer is the drainpipe of the tray. It conducts liquid from one tray to the tray below. The fluid entering the downcomer is far from pure liquid; it is essentially the froth on the tray, typically 20 to 30 percent liquid by volume, with the balance being gas. Due to the density difference, most of this gas disengages in the downcomer and vents back to the tray from the downcomer entrance. Some gas bubbles usually remain in the liquid even at the bottom of the downcomer, ending on the tray below [Lockett and Gharani, *ICHEME Symp. Ser.* 56, p. 2.3/43 (1979)].



(a)



(b)



(c)

FIG. 14-20 Distillation flow regimes: schematics and photos. (a) Froth. (b) Emulsion. (c) Spray. [Schematics from H. Z. Kister, *Distillation Design*, copyright © 1992 by McGraw-Hill, Inc.; reprinted by permission. Photographs courtesy of Fractionation Research Inc. (FRI).]

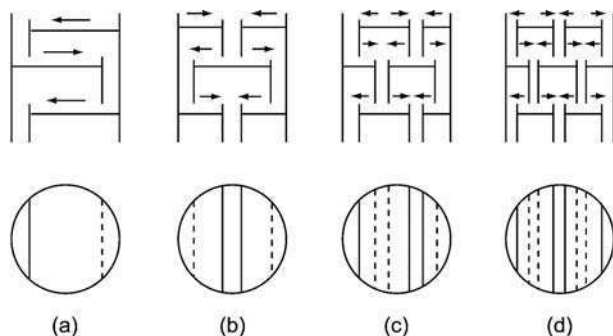


FIG. 14-21 Flow passes on trays. (a) Single-pass. (b) Two-pass. (c) Three-pass. (d) Four-pass.

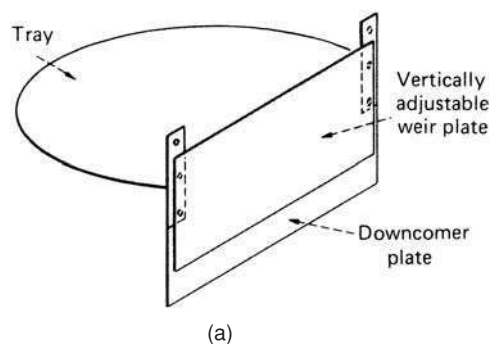
The straight, segmental vertical downcomer (Fig. 14-23a) is the most common downcomer geometry. It is simple and inexpensive and gives good utilization of tower area for downflow. Circular downcomers (downpipes) (Fig. 14-23b), are cheaper, but poorly utilize tower area and are only suitable for very low liquid loads. Sloped downcomers (Fig. 14-23c, d) improve tower area utilization for downflow. They provide sufficient area and volume for gas-liquid disengagement at the top of the downcomer, gradually narrowing as the gas disengages, minimizing the loss of bubbling area at the foot of the downcomer. Sloped downcomers are invaluable when large downcomers are required such as at high liquid loads, high pressures, and foaming systems. Typical ratios of downcomer top to bottom areas are 1.5 to 2.

Antijump baffles (Fig. 14-24) are sometimes installed just above center and off-center downcomers of multipass trays to prevent liquid from one pass skipping across the downcomer onto the next pass. Such liquid jump adds to the liquid load on each pass, leading to premature flooding. These baffles are essential with proprietary trays that induce forward push (see below).

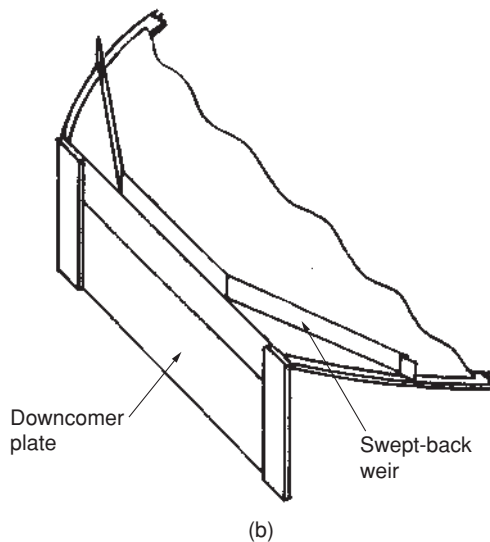
Clearance under the Downcomer Restricting the downcomer bottom opening prevents gas from the tray from rising up the downcomer and interfering with its liquid descent (*downcomer unsealing*). A common design practice makes the downcomer clearance 13 mm (0.5 in) lower than the outlet weir height (Fig. 14-25) to ensure submergence at all times [Davies and Gordon, *Petro/Chem Eng.*, p. 250 (November 1961)]. This practice is sound in the froth and emulsion regimes, where tray dispersions are liquid-continuous, but is ineffective in the spray regime where tray dispersions are gas-continuous and there is no submergence. Also, this practice can be unnecessarily restrictive at high liquid loads where high crests over the weirs sufficiently protect the downcomers from gas rise. Generally, downcomer clearances in the spray regime need to be smaller, while those in the emulsion regime can be larger, than those set by the above practice. Seal pans and inlet weirs are devices sometimes used to help with downcomer sealing while keeping downcomer clearances large. Details are in Kister's book (*Distillation Operation*, McGraw-Hill, New York, 1990).

Hole Sizes Small holes slightly enhance tray capacity when limited by entrainment flood. Reducing sieve hole diameters from 13 to 5 mm ($\frac{1}{2}$ to $\frac{3}{16}$ in) at a fixed hole area typically enhances capacity by 3 to 8 percent, more at low liquid loads. Small holes are effective for reducing entrainment and enhancing capacity in the spray regime ($Q_L < 20$ m³/hm of weir). Hole diameter has only a small effect on pressure drop, tray efficiency, and turndown.

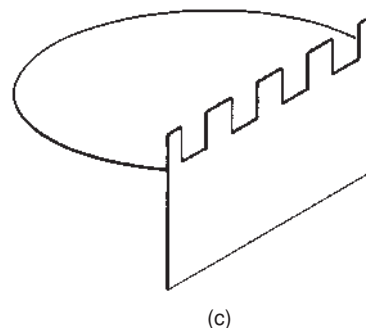
On the debit side, the plugging tendency increases exponentially as hole diameters diminish. Smaller holes are also more prone to corrosion. While 5-mm ($\frac{3}{16}$ -in) holes easily plug even by scale and rust, 13-mm ($\frac{1}{2}$ -in) holes are quite robust and are therefore very common. The small holes are only used in clean, noncorrosive services. Holes smaller than 5 mm are usually avoided because they require drilling (larger holes are punched), which is much more expensive. For highly fouling services, 19- to 25-mm ($\frac{3}{4}$ - to 1-in) holes are preferred.



(a)



(b)



(c)

FIG. 14-22 Unique outlet weir types. (a) Adjustable. (b) Swept back. (c) Picket fence. (Parts a, c, from H. Z. Kister, *Distillation Operation*, copyright © 1990 by McGraw-Hill; reprinted by permission. Part b, courtesy of Koch-Clitsch LP.)

Similar considerations apply to fixed valves. Small fixed valves have a slight capacity advantage, but are far more prone to plugging than larger fixed valves.

For round moving valves, common orifice size is 39 mm ($1\frac{1}{2}$ in). The float opening is usually of the order of 8 to 10 mm (0.3 to 0.4 in).

In recent years there has been a trend toward minivalves, both fixed and moving. These are smaller and therefore give a slight capacity advantage while being more prone to plugging.

Fractional Hole Area Typical sieve and fixed valve tray hole areas are 8 to 12 percent of the bubbling areas. Smaller fractional hole

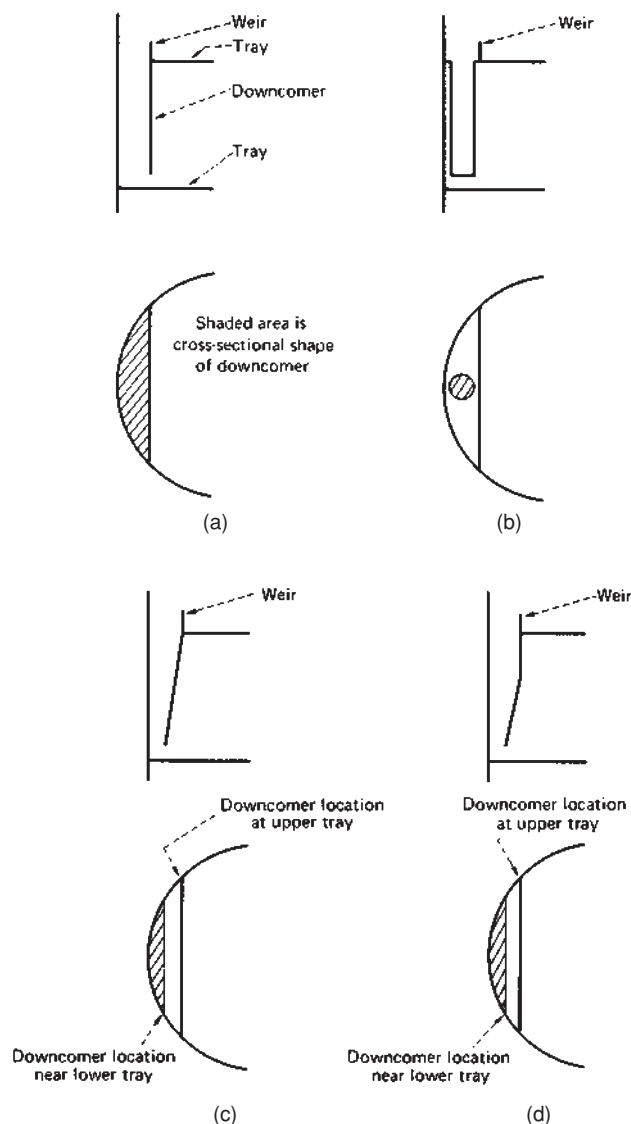


FIG. 14-23 Common downcomer types. (a) Segmental. (b) Circular. (c, d) Sloped. (From Henry Z. Kister, *Chem. Eng.*, December 29, 1980; reprinted courtesy of Chemical Engineering.)

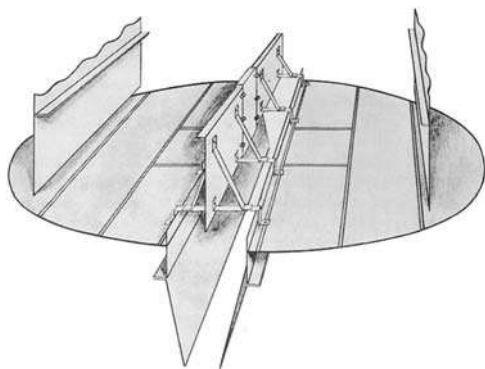


FIG. 14-24 Antijump baffle. (Reprinted courtesy of Koch-Glitsch LP.)

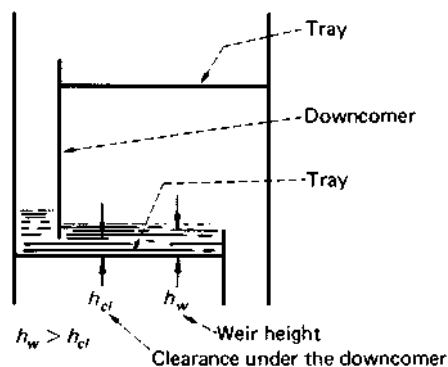


FIG. 14-25 A common design practice of ensuring a positive downcomer seal. (From Henry Z. Kister, *Chem. Eng.*, December 29, 1980; reprinted courtesy of Chemical Engineering.)

areas bring about a capacity reduction when limited by entrainment or downcomer backup flood or by excessive pressure drop. At above 12 percent of the bubbling areas, the capacity gains from higher hole areas become marginal while weeping and, at high liquid loads also channeling, escalate.

Typical open-slot areas for moving valve trays are 14 to 15 percent of the bubbling area. Here the higher hole areas can be afforded due to the high turndown of the valves.

Moving valves can have a sharp or a smooth ("venturi") orifice. The venturi valves have one-half the dry pressure drop of the sharp-orifice valves, but are far more prone to weeping and channeling than the sharp-orifice valves. Sharp orifices are almost always preferred.

Multipass Balancing There are two balancing philosophies: equal bubbling areas and equal flow path lengths. Equal bubbling areas means that all active area panels on Fig. 14-21d are of the same area, and each panel has the same hole (or open-slot) area. In a four-pass tray, one-quarter of the gas flows through each panel. To equalize the L/G ratio on each panel, the liquid needs to be split equally to each panel. Since the center weirs are longer than the side weirs, more liquid tends to flow toward the center weir. To equalize, side weirs are often swept back (Fig. 14-22b) while center weirs often contain picket fences (Fig. 14-22c).

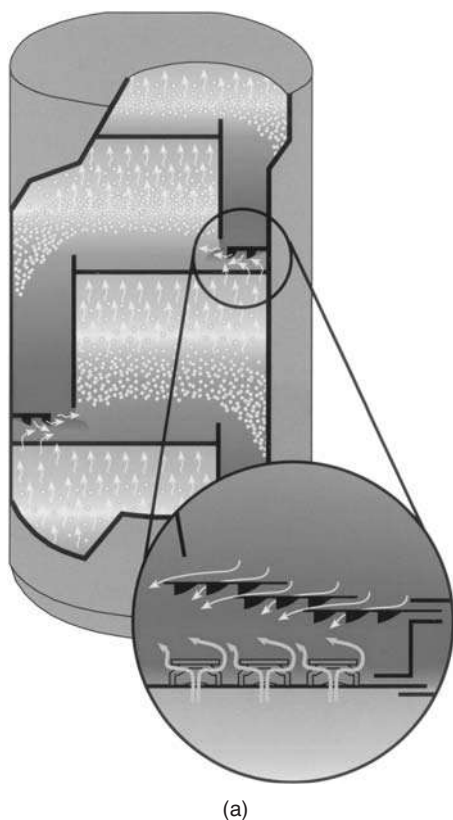
The alternative philosophy (equal flow path lengths) provides more bubbling and perforation areas in the central panels of Fig. 14-21d and less in the side panels. To equalize the L/G ratio, less liquid needs to flow toward the sides, which is readily achieved, as the center weirs are naturally longer than the side weirs. Usually there is no need for swept-back weirs, and only minimal picket-fencing is required at the center weir.

Equal flow path panels are easier to fabricate and are cheaper, while equal bubbling areas have a robustness and reliability advantage due to the ease of equally splitting the fluids. The author had good experience with both when well-designed. Pass balancing is discussed in detail by Pilling [*Chem. Eng. Prog.*, p. 22 (June 2005)] and by Jaguste and Kelkar [*Hydroc. Proc.*, p. 85 (March 2006)].

TRAY CAPACITY ENHANCEMENT

High-capacity trays evolved from conventional trays by including one or more capacity enhancement features such as those discussed below. These features enhance not only the capacity but usually also the complexity and cost. These features have varying impact on the efficiency, turndown, plugging resistance, pressure drop, and reliability of the trays.

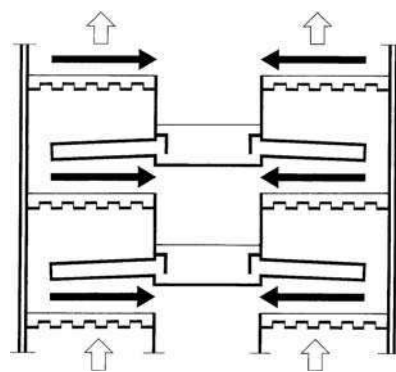
Truncated Downcomers/Forward Push Trays Truncated downcomers/forward push trays include the Nye™ Tray, Maxfrac™ (Fig. 14-26a), Triton™, and MVGT™. In all these, the downcomer from the tray above terminates about 100 to 150 mm (4 to 6 in) above the tray floor. Liquid from the downcomer issues via holes or slots,



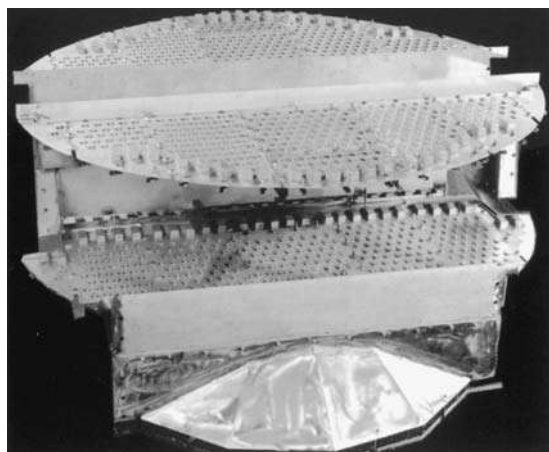
(a)



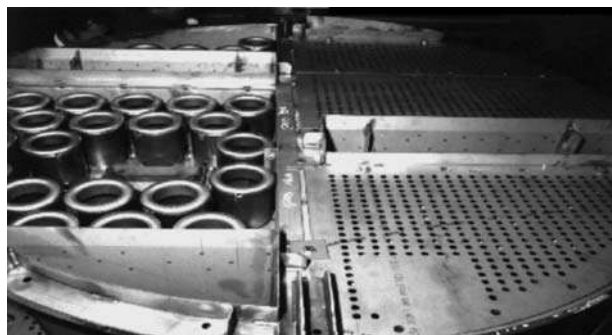
(c)



(d)



(b)



(e)

FIG. 14-26 Tray capacity enhancement. (a) Truncated downcomer/forward-push principle illustrated with a schematic of the Maxfrac™ tray. (b) High top-to-bottom area ratio illustrated with a two-pass Superfrac™ tray. Note the baffle in the front side downcomer that changes the side downcomer shape from segmental to multichordal. Also note the bubble promoters on the side of the upper tray and in the center of the lower tray, which give forward push to the tray liquid. (c) Top view of an MD™ tray with four downcomers. The decks are perforated. The holes in the downcomer lead the liquid to the active area of the tray below, which is rotated 90°. (d) Schematic of the Slit™ tray, type A, showing distribution pipes. Heavy arrows depict liquid movement; open arrows, gas movement. (e) The ConSep™ tray. The right-hand side shows sieve panels. On the left-hand side, these sieve panels were removed to permit viewing the contact cyclones that catch the liquid from the tray below. (Parts a, b, courtesy of Koch-Glitsch LP; part c, courtesy of UOP LLC; part d, courtesy of Kühni AG; part e, courtesy of Sulzer Chemtech Ltd. and Shell Global Solutions International BV.)

directed downward or in the direction of liquid flow. The tray floor under each downcomer is equipped with fixed valves or side perforations. Gas issuing in this region, typically 10 to 20 percent of the total tray gas, is deflected horizontally in the direction of liquid flow by the downcomer floor. This horizontal gas flow pushes liquid droplets toward the tower wall directly above the outlet downcomer. The tower wall catches this liquid, and directs it downward into the downcomer. This deentrains the gas space. In multipass trays, antijump baffles (Fig. 14-24), typically 300 mm or taller, are installed above center and off-center downcomers to catch the liquid and prevent its jumping from pass to pass. The rest of the tray features are similar to those of conventional trays. The tray floor may contain fixed valves, moving valves, or sieve holes.

Trays from this family are proprietary, and have been extensively used in the last two to three decades with great success. Compared to equivalent conventional trays, the truncated downcomer/forward push trays give about 8 to 12 percent more gas-handling capacity at much the same efficiency.

High Top-to-Bottom Downcomer Area and Forward Push Sloping downcomers from top to bottom raises the available tray bubbling area and, therefore, the gas-handling capacity (see "Downcomers"). As long as the ratio of top to bottom areas is not excessive, sloping does not lower downcomer capacity. Downcomer choke flood restricts the downcomer entrance, not exit, because there is much less gas at the downcomer bottom. However, a high top-to-bottom area ratio makes the downcomer bottom a very short chord, which makes distribution of liquid to the tray below difficult. To permit high top-to-bottom area ratios, some trays use a special structure (Fig. 14-26b) to change the downcomer shape from segmental to semiarc or multichordal. This high ratio of top to bottom areas, combined with forward push (above) imparted by bubble and directional fixed or moving valves, and sometimes directional baffles, is used in trays including Superfrac™ III (Fig. 14-26b) and IV and V-Grid Plus™. When the downcomer inlet areas are large, these trays typically gain 15 to 20 percent capacity compared to equivalent conventional trays at much the same efficiency. Trays from this family are proprietary, and have been used successfully for about a decade.

Large Number of Truncated Downcomers These include the MD™ (Fig. 14-26c) and Hi-Fi™ trays. The large number of downcomers raises the total weir length, moving tray operation toward the peak capacity point of 20 to 30 m³/hm (2 to 3 gpm/in) of outlet weir (see Fig. 14-29). The truncated downcomers extend about halfway to the tray below, discharging their liquid via holes or slots at the downcomer floor. The area directly under the downcomers is perforated or valved, and there is enough open height between the tray floor and the bottom of the downcomer for this perforated or valved area to be effective in enhancing the tray bubbling area.

Trays from this family are proprietary and have been successfully used for almost four decades. Their strength is in high-liquid-load services where reducing weir loads provides major capacity gains. Compared to conventional trays, they can gain as much as 20 to 30 percent capacity but at an efficiency loss. The efficiency loss is of the order of 10 to 20 percent due to the large reduction in flow path length (see "Efficiency"). When using these trays, the separation is maintained by either using more trays (typically at shorter spacing) or raising reflux and boilup. This lowers the net capacity gains to 10 to 20 percent above conventional trays. In some variations, forward push slots and antijump baffles are incorporated to enhance the capacity by another 10 percent.

Radial Trays These include the Slit™ tray and feature radial flow of liquid. In the efficiency-maximizing A variation (Fig. 14-26d), a multipipe distributor conducts liquid from each center downcomer to the periphery of the tray below, so liquid flow is from periphery to center on each tray. The capacity-maximizing B variation has central and peripheral (ring) downcomers on alternate trays, with liquid flow alternating from center-to-periphery to periphery-to-center on successive trays. The trays are arranged at small spacing (typically, 200 to 250 mm, or 8 to 10 in) and contain small fixed valves. Slit trays are used in chemical and pharmaceutical low-liquid-rate applications (<40 m³/hm or 4 gpm/in of outlet weir), typically at pressures ranging from moderate vacuum to slight superatmospheric.

Centrifugal Force Deentrainment These trays use a contact step similar to that in conventional trays, followed by a separation step that disentrains the tray dispersion by using centrifugal force. Separation of entrained liquid before the next tray allows very high gas velocities, as high as 25 percent above the system limit (see "System Limit"), to be achieved. The capacity of these trays can be 40 percent above that of conventional trays. The efficiency of these trays can be 10 to 20 percent less than that of conventional trays due to their typical short flow paths (see "Efficiency").

These trays include the Ultrfrac™, the ConSep™ (Fig. 14-26e), and the Swirl Tube™ trays. This technology has been sporadically used in eastern Europe for quite some time. It is just beginning to make inroads into distillation in the rest of the world, and looks very promising.

OTHER TRAY TYPES

Bubble-Cap Trays (Fig. 14-27a) These are flat perforated plates with risers (chimneylike pipes) around the holes, and caps in the form of inverted cups over the risers. The caps are usually (but not always) equipped with slots through which some of the gas comes out, and may be round or rectangular. Liquid and froth are trapped on the tray to a depth at least equal to the riser or weir height, giving the bubble-cap tray a unique ability to operate at very low gas and liquid rates.

The bubble-cap tray was the workhorse of distillation before the 1960s. It was superseded by the much cheaper (as much as 10 times) sieve and valve trays. Compared to the bubble-cap trays, sieve and valve trays also offer slightly higher capacity and efficiency and lower entrainment and pressure drop, and are less prone to corrosion and fouling. Today, bubble-cap trays are only used in special applications where liquid or gas rates are very low. A large amount of information on bubble-cap trays is documented in several texts (e.g., Bolles in B. D. Smith, *Design of Equilibrium Stage Processes*, McGraw-Hill, 1963; Bolles, *Pet. Proc.*, February 1956, p. 65; March 1956, p. 82; April 1956, p. 72; May 1956, p. 109; Ludwig, *Applied Process Design for Chemical and Petrochemical Plants*, 2d ed., vol. 2, Gulf Publishing, Houston, 1979).

Dual-Flow Trays These are sieve trays with no downcomers (Fig. 14-27b). Liquid continuously weeps through the holes, hence their low efficiency. At peak loads they are typically 5 to 10 percent less efficient than sieve or valve trays, but as the gas rate is reduced, the efficiency gap rapidly widens, giving poor turndown. The absence of downcomers gives dual-flow trays more area, and therefore greater capacity, less entrainment, and less pressure drop, than conventional trays. Their pressure drop is further reduced by their large fractional hole area (typically 18 to 30 percent of the tower area). However, this low pressure drop also renders dual-flow trays prone to gas and liquid maldistribution.

In general, gas and liquid flows pulsate, with a particular perforation passing both gas and liquid intermittently, but seldom simultaneously. In large-diameter (>2.5-m, or 8-ft) dual-flow trays, the pulsations sometimes develop into sloshing, instability, and vibrations. The Ripple Tray™ is a proprietary variation in which the tray floor is corrugated to minimize this instability.

With large holes (16 to 25 mm), these trays are some of the most fouling-resistant and corrosion-resistant devices in the industry. This defines their main application: highly fouling services, slurries, and corrosive services. Dual-flow trays are also the least expensive and easiest to install and maintain.

A wealth of information for the design and rating of dual-flow trays, much of it originating from FRI data, was published by Garcia and Fair [*Ind. Eng. Chem. Res.* **41**:1632 (2002)].

Baffle Trays Baffle trays ("shed decks," "shower decks") (Fig. 14-28a) are solid half-circle plates, sloped slightly in the direction of outlet flow, with weirs at the end. Gas contacts the liquid as it showers from the plate. This contact is inefficient, typically giving 30 to 40 percent of the efficiency of conventional trays. This limits their application mainly to heat-transfer and scrubbing services. The capacity is high and pressure drop is low due to the high open area (typically 50 percent of the tower cross-sectional area). Since there is not much



(a)



(b)

FIG. 14-27 Other trays. (a) Bubble-cap tray. (b) Dual-flow tray. [Part a, courtesy of Koch-Glitsch LP; part b, courtesy of Fractionation Research Inc. (FRI).]

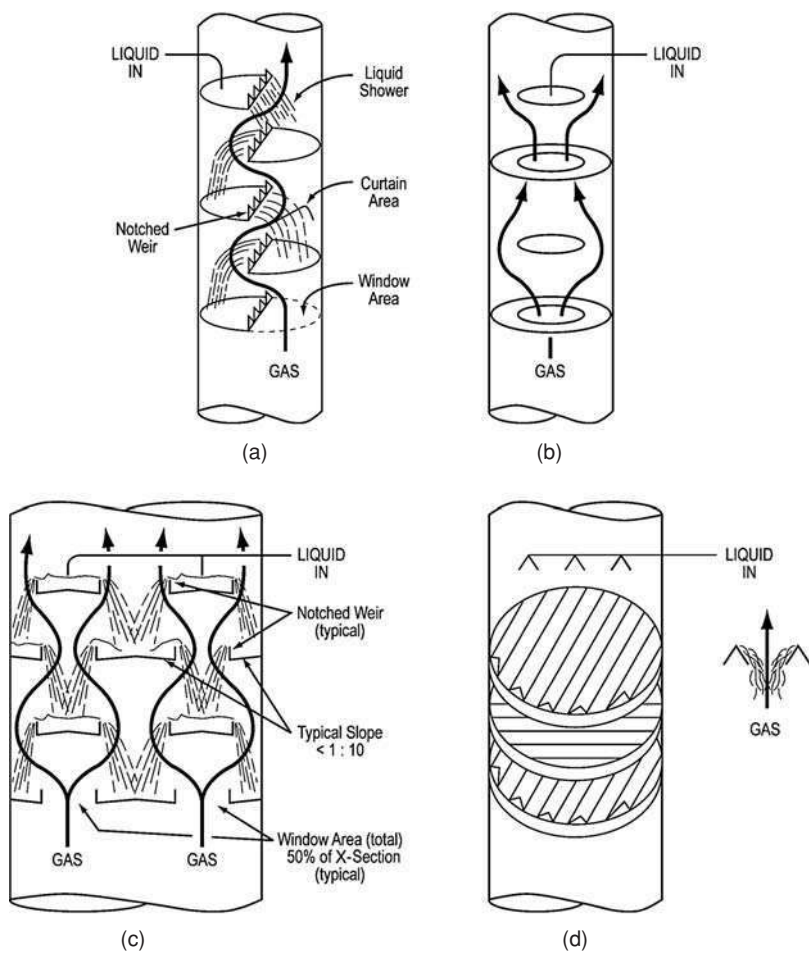


FIG. 14-28 Baffle tray variations. (a) Segmental. (b) Disk and doughnut. (c) Multipass. (d) Angle irons.

that can plug up, the baffle trays are perhaps the most fouling-resistant device in the industry, and their main application is in extremely fouling services. To be effective in these services, their liquid rate needs to exceed 20 m³/hm (2 gpm/in) of outlet weir and dead spots formed due to poor support design (Kister, *Distillation Troubleshooting*, Wiley, 2006) eliminated.

There are several geometric variations. The disk and doughnut trays (Fig. 14-28b) replace the half-circle segmental plates by alternate plates shaped as disks and doughnuts, each occupying about 50 percent of the tower cross-sectional area. In large towers, multipass baffle trays (Fig. 14-28c) are common. Another variation uses angle irons, with one layer oriented at 90° to the one below (Fig. 14-28d). Multipass baffle trays, as well as angle irons, require good liquid (and to a lesser extent, also good gas) distribution, as has been demonstrated from field heat-transfer measurements [Kister and Schwartz, *Oil & Gas J.*, p. 50 (May 20, 2002)]. Excellent overviews of the fundamentals and design of baffle trays were given by Fair and Lemieux [Fair, *Hydro. Proc.*, p. 75 (May 1993); Lemieux, *Hydroc. Proc.*, p. 106 (September 1983)]. Mass-transfer efficiency data with baffle trays by Fractionation Research Inc. (FRI) have been released and presented together with their correlation (Fair, Paper presented at the AIChE Annual Meeting, San Francisco, November 2003).

FLOODING

Flooding is by far the most common upper capacity limit of a distillation tray. Column diameter is set to ensure the column can achieve the required throughput without flooding. Towers are usually designed to operate at 80 to 90 percent of the flood limit.

Flooding is an excessive accumulation of liquid inside a column. Flood symptoms include a rapid rise in pressure drop (the accumulating liquid increases the liquid head on the trays), liquid carryover from the column top, reduction in bottom flow rate (the accumulating liquid does not reach the tower bottom), and instability (accumulation is non-steady-state). This liquid accumulation is generally induced by one of the following mechanisms.

Entrainment (Jet) Flooding Froth or spray height rises with gas velocity. As the froth or spray approaches the tray above, some of the liquid is aspirated into the tray above as entrainment. Upon a further increase in gas flow rate, massive entrainment of the froth or spray begins, causing liquid accumulation and flood on the tray above.

Entrainment flooding can be subclassified into *spray entrainment flooding* (common) and *froth entrainment flooding* (uncommon). Froth entrainment flooding occurs when the froth envelope approaches the tray above, and is therefore only encountered with small tray spacings (<450 mm or 18 in) in the froth regime. At larger (and often even lower) tray spacing, the froth breaks into spray well before the froth envelope approaches the tray above.

The entrainment flooding prediction methods described here are based primarily on spray entrainment flooding. Considerations unique to froth entrainment flooding can be found elsewhere (Kister, *Distillation Design*, McGraw-Hill, New York, 1992).

Spray Entrainment Flooding Prediction Most entrainment flooding prediction methods derive from the original work of Souders and Brown [*Ind. Eng. Chem.* **26**(1), 98 (1934)]. Souders and Brown theoretically analyzed entrainment flooding in terms of droplet settling velocity. Flooding occurs when the upward vapor velocity is high enough to suspend a liquid droplet, giving

$$C_{SB} = u_{s,\text{flood}} \sqrt{\frac{\rho_C}{\rho_L - \rho_C}} \quad (14-80)$$

The Souders and Brown constant C_{SB} is the C -factor [Eq. (14-77)] at the entrainment flood point. Most modern entrainment flooding correlations retain the Souders and Brown equation (14-80) as the basis, but depart from the notion that C_{SB} is a constant. Instead, they express C_{SB} as a weak function of several variables, which differ from one correlation to another. Depending on the correlation, C_{SB} and $u_{s,\text{flood}}$ are based on either the net area A_N or on the bubbling area A_B .

The constant C_{SB} is roughly proportional to the tray spacing to a power of 0.5 to 0.6 (Kister, *Distillation Design*, McGraw-Hill, New

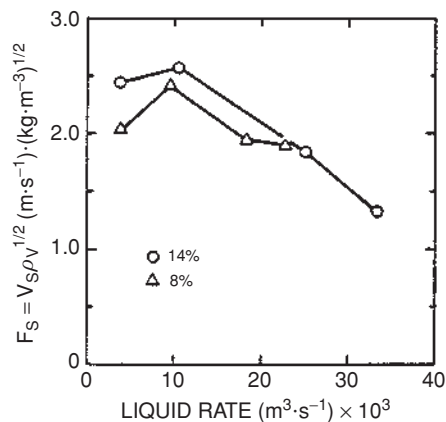


FIG. 14-29 Effect of liquid rate and fractional hole area on flood capacity. FRI sieve tray test data, cyclohexane/n-heptane, 165 kPa (24 psia), $D_T = 1.2$ m (4 ft), $S = 610$ mm (24 in), $h_w = 51$ mm (2 in), $d_H = 12.7$ mm (0.5 in), straight downcomers, $A_D/A_T = 0.13$. (From T. Yanagi and M. Sakata, *Ind. Eng. Chem. Proc. Des. Dev.* **21**, 712; copyright © 1982, American Chemical Society, reprinted by permission.)

York, 1992). Figure 14-29 demonstrates the effect of liquid rate and fractional hole area on C_{SB} . As liquid load increases, C_{SB} first increases, then peaks, and finally declines. Some interpret the peak as the transition from the froth to spray regime [Porter and Jenkins, *I. Chem. E. Symp. Ser.* **56**, Summary Paper, London (1979)]. C_{SB} increases slightly with fractional hole area at lower liquid rates, but there is little effect of fractional hole area on C_{SB} at high liquid rates. C_{SB} slightly increases as hole diameter is reduced.

For sieve trays, the entrainment flood point can be predicted by using the method by Kister and Haas [*Chem. Eng. Progr.*, **86**(9), 63 (1990)]. The method is said to reproduce a large database of measured flood points to within ± 15 percent. $C_{SB,\text{flood}}$ is based on the net area. The equation is

$$C_{SB,\text{flood}} = 0.0277(d_H^2 \sigma / \rho_L)^{0.125} (\rho_C / \rho_L)^{0.1} (TS / h_{ct})^{0.5} \quad (14-81)$$

where d_H = hole diameter, mm

σ = surface tension, mN/m (dyn/cm)

ρ_C, ρ_L = vapor and liquid densities, kg/m³

TS = tray spacing, mm

h_{ct} = clear liquid height at the froth-to-spray transition, mm; obtained from:

$$h_{ct} = h_{ct,H_2O} (996 / \rho_L)^{0.5(1-n)} \quad (14-82)$$

$$h_{ct,H_2O} = \frac{0.497 A_f^{-0.791} d_H^{0.833}}{1 + 0.013 Q_L^{-0.59} A_f^{-1.79}} \quad (14-83)$$

$$n = 0.00091 d_H / A_f \quad (14-84)$$

In Eq. (14-83), Q_L = m³ liquid downflow/(h-m weir length) and A_f = fractional hole area based on active ("bubbling") area; for instance, $A_f = A_N / A_N$.

The Kister and Haas method can also be applied to valve trays, but the additional approximations reduce its data prediction accuracy for valve trays to within ± 20 percent. For valve trays, adaptations of Eqs. (14-81) to (14-84) are required:

$$d_h = \frac{4 \times (\text{area of opening of one fully open valve})}{\text{wetted perimeter of opening of one fully open valve}} \quad (14-85)$$

$$A_f = \frac{\text{no. valves} \times (\text{area of opening of one fully open valve})}{\text{active (bubbling) area}} \quad (14-86)$$

A correlation for valve tray entrainment flooding that has gained respect and popularity throughout the industry is the Glitsch "Equation 13" (Glitsch, Inc., *Ballast Tray Design Manual*, 6th ed., 1993;

available from Koch-Glitsch, Wichita, Kans.). This equation has been applied successfully for valve trays from different manufacturers, as well as for sieve trays with large fractional hole areas (12 to 15 percent). With tray spacings of 600 mm and higher, its flood prediction accuracy for valve trays has generally been within ± 10 percent in the author's experience. The Glitsch correlation is

$$\frac{\% \text{ flood}}{100} = \frac{C_B}{\text{CAF}} + 1.359 \frac{Q \text{ FPL}}{A_B \text{ CAF}} \quad (14-87)$$

where

$$\text{CAF} = 0.3048 \text{ CAF}_0 \text{ SF} \quad (14-88)$$

C_B is the operating C -factor based on the bubbling area, m/s; Q is the liquid flow rate, m^3/s ; A_B is the bubbling area, m^2 ; FPL is the flow path length, m, i.e., the horizontal distance between the inlet downcomer and the outlet weir. The flow path length becomes shorter as the number of passes increases. CAF_0 and CAF are the flood C -factors. CAF_0 is obtained from Fig. 14-30 in English units (ft/s). Equation (14-88) converts CAF_0 to the metric CAF (m/s), and corrects it by using a system factor SF. Values of SF are given in Table 14-9.

The Fair correlation [Pet/Chem Eng. 33(10), 45 (September 1961)] for decades has been the standard of the industry for entrainment flood prediction. It uses a plot (Fig. 14-31) of surface-tension-corrected Souders and Brown flood factor C_{SB} against the dimensionless flow parameter shown in Fig. 14-31. The flow parameter represents a ratio of liquid to vapor kinetic energies:

$$F_{LG} = \frac{L}{G} \left(\frac{\rho_C}{\rho_L} \right)^{0.5} \quad (14-89)$$

Low values of F_{LG} indicate vacuum operation, high values indicate operation at higher pressures or at high liquid/vapor loadings. The liquid/gas ratio L/G is based on mass flow rates. For multipass trays, the ratio needs to be divided by the number of passes. The strength of the correlation is at the lower flow parameters. At higher flow parameters (high L/G ratios, high pressures, emulsion flow), Fig. 14-31 gives excessively conservative predictions, with the low values of C_{sb} to the right likely to result from downcomer flow restrictions rather than excessive entrainment. The curves may be expressed in equation form as [Lygeros and Magoulas, *Hydrocarbon Proc.* 65(12), 43 (1986)]:

$$C_{sb} = 0.0105 + 8.127(10^{-4})(\text{TS}^{0.755}) \exp[-1.463 F_{LG}^{0.842}] \quad (14-90)$$

where TS = plate spacing, mm.

Figure 14-31 or Eq. (14-90) may be used for sieve, valve, or bubble-cap trays. The value of the capacity parameter (ordinate term in Fig. 14-31) may be used to calculate the maximum allowable vapor velocity through the net area of the plate:

$$U_{nf} = C_{sb} \left(\frac{\sigma}{20} \right)^{0.2} \left(\frac{\rho_L - \rho_g}{\rho_g} \right)^{0.5} \quad (14-91)$$

where U_{nf} = gas velocity through net area at flood, m/s

C_{sb} = capacity parameter corrected for surface tension, m/s

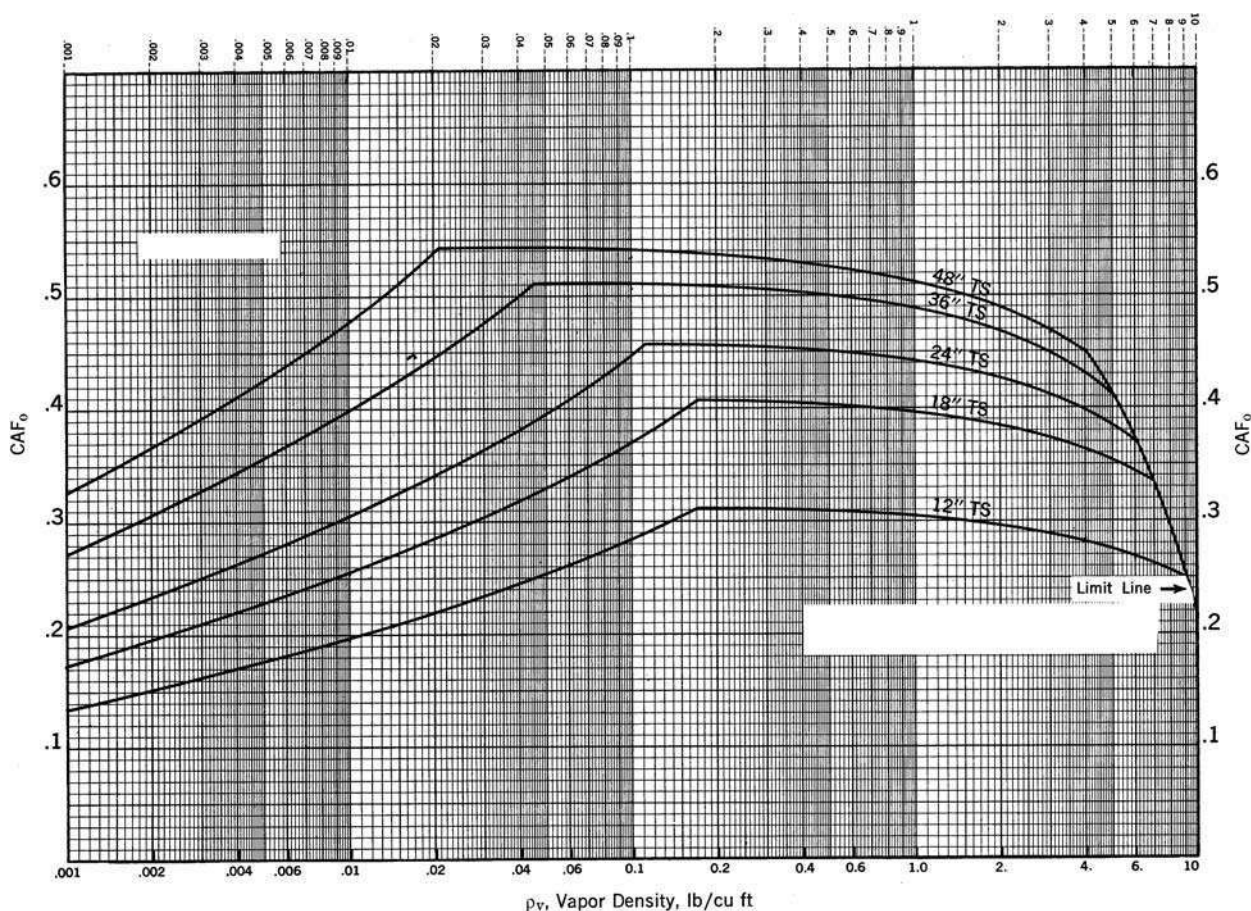


FIG. 14-30 Flood capacity of moving valve trays. (Courtesy of Koch-Glitch LP.)

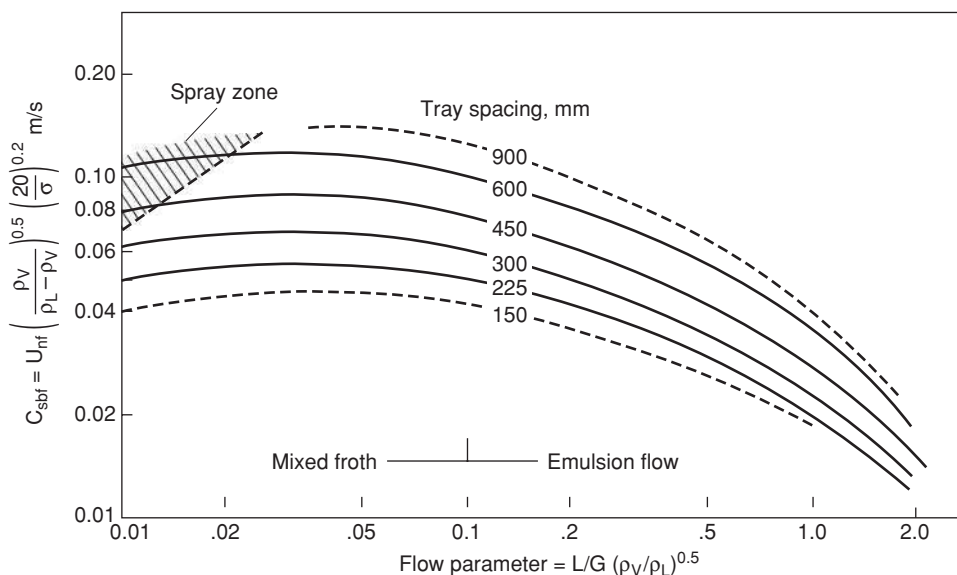


FIG. 14-31 Fair's entrainment flooding correlation for columns with crossflow trays (sieve, valve, bubble-cap). [Fair, *Pet/Chem Eng* 33(10), 45 (September 1961).]

σ = liquid surface tension, mN/m (dyn/cm)
 ρ_L = liquid density, kg/m³
 ρ_G = gas density, kg/m³

The application of the correlation is subject to the following restrictions:

1. System is low or nonfoaming.
2. Weir height is less than 15 percent of tray spacing.
3. Sieve-tray perforations are 13 mm (1/2 in) or less in diameter.
4. Ratio of slot (bubble cap), perforation (sieve), or full valve opening (valve plate) area A_h to active area A_a is 0.1 or greater. Otherwise the value of U_{nf} obtained from Fig. 14-31 should be corrected:

A_h/A_a	$U_{nf}/U_{nf, \text{Fig. 14-31}}$
0.10	1.00
0.08	0.90
0.06	0.80

where A_h = total slot, perforated, or open-valve area on tray.

Example 9: Flooding of a Distillation Tray An available sieve tray column of 2.5-m diameter is being considered for an ethylbenzene/styrene separation. An evaluation of loading at the top tray will be made. Key dimensions of the single-pass tray are:

Column cross section, m ²	4.91
Downcomer area, m ²	0.25
Net area, m ²	4.66
Active area, m ²	4.41
Hole area, m ²	0.617
Hole diameter, mm	4.76
Weir length, m	1.50
Weir height, mm	38
Tray spacing, mm	500

Conditions and properties at the top tray are:

Temperature, °C	78
Pressure, torr	100
Vapor flow, kg/h	25,500
Vapor density, kg/m ³	0.481
Liquid flow, kg/h	22,000
Liquid density, kg/m ³	841
Surface tension, mN/m	25

Solution. The method of Kister and Haas gives:

$$Q_L = \frac{22,000}{841 \times 1.50} = 17.44 \text{ m}^3/\text{h-m weir}$$

$$A_f = \frac{0.617}{4.41} = 0.14$$

By Eq. (14-83), $h_{et, H_2O} = 7.98 \text{ mm}$

$$\text{Eq. (14-84): } n = 0.0309$$

$$\text{Eq. (14-82): } h_{et} = 8.66 \text{ mm}$$

Finally, by Eq. (14-81),

$$C_{SB, \text{flood}} = 0.0277[(4.76^2)(25/841)]^{0.125} \times (0.481/841)^{0.1}(500/8.66)^{0.5} = 0.0947 \text{ m/s}$$

Alternatively, applying the Fair correlation:

The flow parameter $F_{LG} = 0.021$ [Eq. (14-89)]. From Fig. 14-31, $C_{sbf} = 0.095 \text{ m/s}$. Then, based on the net area,

$$C_{SB} = 0.095(25/20)^{0.2} = 0.0993 \text{ m/s}$$

about 5 percent higher than the answer obtained from Kister and Haas.

For the design condition, the C -factor based on the net area is

$$C = \frac{25,500}{3600(0.481)(4.66)} \sqrt{\frac{0.481}{841 - 0.481}} = 0.0756 \text{ m/s}$$

or about 80 percent of flood. The proposed column is entirely adequate for the service required.

System Limit (Ultimate Capacity) This limit is discussed later under "System Limit."

Downcomer Backup Flooding Aerated liquid backs up in the downcomer because of tray pressure drop, liquid height on the tray, and frictional losses in the downcomer apron (Fig. 14-32). All these increase with increasing liquid rate. Tray pressure drop also increases as the gas rate rises. When the backup of aerated liquid exceeds the tray spacing, liquid accumulates on the tray above, causing downcomer backup flooding.

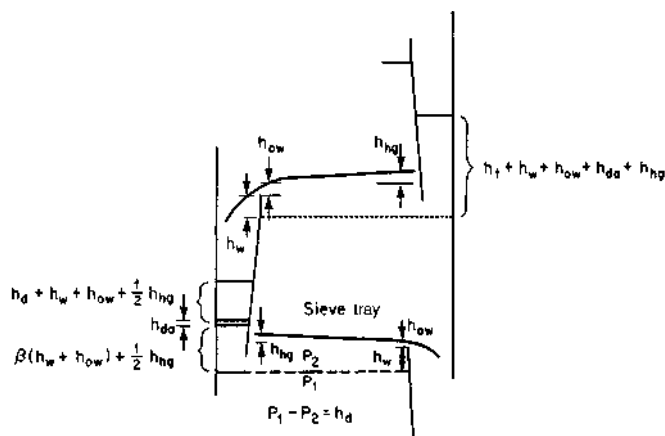


FIG. 14-32 Pressure-drop contributions for trays. h_d = pressure drop through cap or sieve, equivalent height of tray liquid; h_w = height of weir; h_{ow} = weir crest; h_{hg} = hydraulic gradient; h_{da} = loss under downcomer.

Downcomer backup is calculated from the pressure balance

$$h_{dc} = h_t + h_w + h_{ow} + h_{da} + h_{hg} \quad (14-92)$$

where h_{dc} = clear liquid height in downcomer, mm liquid

h_t = total pressure drop across the tray, mm liquid

h_w = height of weir at tray outlet, mm liquid

h_{ow} = height of crest over weir, mm liquid

h_{da} = head loss due to liquid flow under downcomer apron, mm liquid

h_{hg} = liquid gradient across tray, mm liquid

The heights of head losses in Eq. (14-92) should be in consistent units, e.g., millimeters or inches of liquid under operating conditions on the tray.

As noted, h_{dc} is calculated in terms of equivalent clear liquid. Actually, the liquid in the downcomer is aerated and actual backup is

$$h'_{dc} = \frac{h_{dc}}{\phi_{dc}} \quad (14-93)$$

where ϕ_{dc} is an average *relative* froth density (ratio of froth density to liquid density) in the downcomer. Design must not permit h_{dc} to

exceed the value of tray spacing plus weir height; otherwise, flooding can be precipitated.

The value of ϕ_{dc} depends upon the tendency for gas and liquid to disengage (froth to collapse) in the downcomer. For cases favoring rapid bubble rise (low gas density, low liquid viscosity, low system foamability) collapse is rapid, and fairly clear liquid fills the bottom of the downcomer (Fig. 14-17). For such cases, it is usual practice to employ a higher value of ϕ_{dc} . For cases favoring slow bubble rise (high gas density, high liquid viscosity, high system foamability), lower values of ϕ_{dc} should be used. As the critical point is approached in high-pressure distillations and absorptions, special precautions with downcomer sizing are mandatory. Table 14-6 lists values of ϕ_{dc} commonly used by the industry.

Downcomer Choke Flooding This is also called downcomer entrance flood or downcomer velocity flood. A downcomer must be sufficiently large to transport all the liquid downflow. Excessive friction losses in the downcomer entrance, and/or excessive flow rate of gas venting from the downcomer in counterflow, will impede liquid downflow, initiating liquid accumulation (termed downcomer choke flooding) on the tray above. The prime design parameter is the downcomer top area. Further down the downcomer, gas disengages from the liquid and the volumes of aerated liquid downflow and vented gas

TABLE 14-6 Criteria for Downcomer Aeration Factors

Foaming tendency	Bolles' criterion ^a		Glitsch's criterion ¹		Fair et al.'s criterion ¹	
	Examples	ϕ_{dc}	Examples	ϕ_{dc}	Examples	ϕ_{dc}
Low	Low-molecular-weight hydrocarbons ³ and alcohols	0.6	$\rho_G < 1.0 \text{ lb/ft}^3$	0.6	Rapid bubble rise systems, such as low gas density, low liquid viscosity	0.5
Moderate	Distillation of medium-molecular-weight hydrocarbons	0.5	$1.0 < \rho_G < 3.0 \text{ lb/ft}^3$	0.5		
High	Mineral oil absorbers	0.4	$\rho_G > 3.0 \text{ lb/ft}^3$	0.4	Slow bubble rise systems, such as high gas density, high liquid viscosity, foaming systems	0.2 – 0.3
Very high	Amines, glycols	0.3				

^a“Distillation Theory and Practice—an Intensive Course,” University of New South Wales/University of Sydney, August 9–11, 1977.

¹ Glitsch, Inc., *Ballast Tray Design Manual*, 6th ed., 1993; available from Koch-Glitsch LP, Wichita, Kans.

³ R. H. Perry and D. W. Green (eds.), *Perry's Chemical Engineers' Handbook*, 7th ed., McGraw-Hill, 1997.

⁵ The author believes that low-molecular-weight hydrocarbons refers to light hydrocarbons at near atmospheric pressure or under vacuum. The foam stability of light-hydrocarbon distillation at medium and high pressure is best inferred from the Glitsch criterion.

To convert from lb/ft³ to kg/m³, multiply by 16.0.

SOURCE: From H. Z. Kister, *Distillation Design*, copyright © 1992 by McGraw-Hill, Inc.; reprinted by permission.

upflow are greatly reduced. With sloped downcomers, the downcomer bottom area is normally set at 50 to 60 percent of the top downcomer area. This taper is small enough to keep the downcomer top area the prime choke variable.

There is no satisfactory published correlation for downcomer choke. The best that can be done in the absence of data or correlation is to apply the criteria for maximum velocity of clear liquid at the downcomer entrance. Kister (*Distillation Operation*, McGraw-Hill, New York, 1990) surveyed the multitude of published criteria for maximum downcomer velocity and incorporated them into a single set of guidelines (Table 14-7). The values for 30-in spacing were revised to reflect the author's recent experiences. The values given in Table 14-7 are not conservative. For a conservative design, multiply the values from Table 14-7 by a safety factor of 0.75. For very highly foaming systems, where antifoam application is undesirable, there are benefits for reducing downcomer design velocities down to 0.1 to 0.15 ft/s.

Another criterion sometimes used is to provide sufficient residence time in the downcomer to allow adequate disengagement of gas from the descending liquid, so that the liquid is relatively gas-free by the time it enters the tray below. Inadequate removal of gas from the liquid may choke the downcomer. Kister (loc. cit.) reviewed various published criteria for downcomer residence times and recommended those by Bolles (private communication, 1977) and Erbar and Maddox (Maddox, *Process Engineer's Absorption Pocket Handbook*, Gulf Publishing, Houston, 1985). Both sets of guidelines are similar and are summarized in Table 14-8. The residence times in Table 14-8 are *apparent residence times*, defined as the ratio of the total downcomer volume to the clear liquid flow in the downcomer.

As a segmental downcomer becomes smaller, its width decreases faster than its length, turning the downcomer into a long, narrow slot. This geometry increases the resistance to liquid downflow and to the upflow of disengaging gas. Small downcomers are also extremely sensitive to foaming, fouling, construction tolerances, and the introduction of debris. Generally, segmental downcomers smaller than 5 percent of the column cross-sectional area should be avoided. Additional discussion of small downcomers is available (Kister, *Distillation Operation*, McGraw-Hill, New York, 1990).

Derating ("System") Factors With certain systems, traditional flooding equations consistently give optimistic predictions. To allow for this discrepancy, an empirical derating or system factor ($SF < 1.0$) is applied. To obtain the actual or derated flood load, the flood gas load (entrainment flooding) or flood liquid load (downcomer choke) obtained from the traditional equations is multiplied by the derating factor. In the case of downcomer backup flood, the froth height from the traditional flood equation is divided by the derating factor.

Derating factors are vaguely related to the foaming tendency, but are also applied to nonfoaming systems where standard flooding equations consistently predict too high. Sometimes, derating factors are used solely as overdesign factors. Brierley (*Chem. Eng. Prog.*, July 1994, p. 68) states that some derating factors actually evolved from plant misoperation or from misinterpretation of plant data. Kister (loc. cit.) compiled the derating factors found in the literature into Table 14-9.

The application of derating factors is fraught with inconsistent practices and confusion. Caution is required. The following need to be carefully specified:

1. The flooding mechanism to which the derating factor applies (entrainment, downcomer backup, downcomer choke, or all these) must be specified.

2. Avoiding double derating. For instance, the values in Table 14-9 may apply with Eq. (14-81) because Eq. (14-81) does not take foaminess into account. However, they will double-derate a flood calculation that is made with a correlation or criteria that already take foaminess into account, such as the criteria for downcomer choke in Tables 14-7 and 14-8. Similarly, two different factors from Table 14-9 may apply to a single system; only one should be used.

3. Derating factors vary from source to source, and may depend on the correlation used as well as the system. For instance, some caustic wash applications have a track record of foaming more severely than other caustic wash applications (see note in Table 14-9). The derating factors in Table 14-9 are a useful guide, but are far from absolute.

ENTRAINMENT

Entrainment (Fig. 14-33) is liquid transported by the gas to the tray above. As the lower tray liquid is richer with the less-volatile components, entrainment counteracts the mass-transfer process, reducing tray efficiency. At times entrainment may transport nonvolatile impurities upward to contaminate the tower overhead product, or damage rotating machinery located in the path of the overhead gas.

Effect of Gas Velocity Entrainment increases with gas velocity to a high power. Generally, smaller powers, indicative of a relatively gradual change, are typical of low-pressure systems. Higher powers, which indicate a steep change, are typical of high-pressure systems.

Due to the steep change of entrainment with gas velocity at high pressure, the gas velocity at which entrainment becomes significant tends to coincide with the flood point. At low pressure, the rate of change of entrainment with gas velocity is much slower, and entrainment can be significant even if the tray is operating well below the flood point. For this reason, excessive entrainment is a common problem in low-pressure and vacuum systems, but is seldom troublesome with high-pressure systems. If encountered at high pressure, entrainment usually indicates flooding or abnormality.

Effect of Liquid Rate As the liquid rate is raised at constant gas rate, entrainment first diminishes, then passes through a minimum, and finally increases [Sakata and Yanagi, *I. Chem. E. Symp. Ser.* **56**, 3.2/21 (1979); Porter and Jenkins, *I. Chem. E. Symp. Ser.* **56**, Summary Paper, 1979; Friend, Lemieux, and Schreiner, *Chem. Eng.*, October 31, 1960, p. 101]. The entrainment minima coincide with the maxima in plots of entrainment flood F -factor against liquid load (Fig. 14-29). At the low liquid loads (spray regime), an increase in liquid load suppresses atomization, drop formation, and consequently entrainment. At higher liquid loads, an increase in liquid load reduces the effective tray spacing, thereby increasing entrainment. The entrainment minima have been interpreted by many workers as the tray dispersion change from predominantly spray to the froth regime [Porter and Jenkins, loc. cit.; Kister and Haas, *I. Chem. E. Symp. Ser.* **104**, p. A483 (1987)].

Effect of Other Variables Entrainment diminishes with higher tray spacing and increases with hole diameter [Kister and Haas, *I.*

TABLE 14-7 Maximum Downcomer Velocities

Foaming tendency	Example	Clear liquid velocity in downcomer, ft/s		
		18-in spacing	24-in spacing	30-in spacing
Low	Low-pressure (<100 psia) light hydrocarbons, stabilizers, air-water simulators	0.4–0.5	0.5–0.6	0.5–0.6*
Medium	Oil systems, crude oil distillation, absorbers, midpressure (100–300 psia) hydrocarbons	0.3–0.4	0.4–0.5	0.4–0.5*
High	Amines, glycerine, glycols, high-pressure (>300 psia) light hydrocarbons	0.2–0.25	0.2–0.25	0.2–0.3

*Revised from previous versions.

To convert from ft/s to m/s, multiply by 0.3048; from in to mm, multiply by 25.4; from psia to bar, multiply by 0.0689.

SOURCE: From H. Z. Kister, *Distillation Operation*, copyright 1990 by McGraw-Hill, Inc.; reprinted by permission.

TABLE 14-8 Recommended Minimum Residence Time in the Downcomer

Foaming tendency	Example	Residence time, s
Low	Low-molecular-weight hydrocarbons,* alcohols	3
Medium	Medium-molecular-weight hydrocarbons	4
High	Mineral oil absorbers	5
Very high	Amines and glycols	7

*The author believes that low-molecular-weight hydrocarbons refers to light hydrocarbons at atmospheric conditions or under vacuum. The foaming tendency of light-hydrocarbon distillation at medium pressure [>7 bar (100 psia)] is medium; at high pressure [>21 bar (300 psia)], it is high.

SOURCE: W. L. Bolles (Monsanto Company), private communication, 1977.

Chem. E. Symp. Ser. **104**, p. A483 (1987); *Ind. Eng. Chem. Res.* **27**, p. 2331 (1988); Lemieux and Scotti, *Chem. Eng. Prog.* **65**(3), 52 (1969)]. The hole diameter effect is large in the spray regime but small in the froth regime. In the spray regime, entrainment also increases as the fractional hole area is lowered, but this variable has little effect in the froth regime [Yanagi and Sakata, *Ind. Eng. Chem. Proc. Des. Dev.* **21**, 712 (1982); and Kister and Haas, loc. cit.].

Entrainment Prediction For spray regime entrainment, the Kister and Haas correlation was shown to give good predictions to a wide commercial and pilot-scale data bank [*I. Chem. E. Symp. Ser.* **104**, A483 (1987)]. The correlation is

$$E_s = 4.742^{(10/\sigma)^{0.64}} \chi^{(10/\sigma)} \quad (14-94)$$

$$\text{where} \quad \chi = 872 \left(\frac{u_B h_{Lt}}{\sqrt{d_H S}} \right)^4 \left(\frac{\rho_G}{Q_L \rho_L} \right) \left(\frac{\rho_L - \rho_G}{\sigma} \right)^{0.25} \quad (14-95)$$

$$\text{and} \quad h_{Lt} = \frac{h_{et}}{1 + 0.00262 h_w} \quad (14-96)$$

The terms in Eqs. (14-94) through (14-96) are in the metric units described in the Nomenclature table at the beginning of this section.

The recommended range of application of the correlation is given in Table 14-10. The clear liquid height at the froth-to-spray transition h_{et} is calculated using the corrected Jeronimo and Sawistowski [*Trans. Inst. Chem. Engrs.* **51**, 265 (1973)] correlation as per Eqs. (14-82) to (14-84).

For decades, the Fair correlation [*Pet/Chem. Eng.*, **33**(10), 45 (September 1961)] has been used for entrainment prediction. In the spray regime the Kister and Haas correlation was shown to be more accurate [Kozioł and Mackowiak, *Chem. Eng. Process.*, **27**, p. 145 (1990)]. In the froth regime, the Kister and Haas correlation does not apply, and Fair's correlation remains the standard of the industry. Fair's correlation (Fig. 14-34) predicts entrainment in terms of the flow parameter [Eq. (14-89)] and the ratio of gas velocity to entrainment flooding gas velocity. The ordinate values Ψ are fractions of gross liquid downflow, defined as follows:

$$\Psi = \frac{e}{L_m + e} \quad (14-97)$$

where e = absolute entrainment of liquid, mol/time

L_m = liquid downflow rate without entrainment, mol/time

Figure 14-34 also accepts the validity of the Colburn equation [*Ind. Eng. Chem.*, **28**, 526 (1936)] for the effect of entrainment on efficiency:

$$\frac{E_a}{E_{mc}} = \frac{1}{1 + E_{mc}[\Psi/(1 - \Psi)]} \quad (14-98)$$

where E_{mc} = Murphree vapor efficiency [see Eq. (14-134)]

E_a = Murphree vapor efficiency, corrected for recycle of liquid entrainment

The Colburn equation is based on complete mixing on the tray. For incomplete mixing, e.g., liquid approaching plug flow on the tray, Rahman and Lockett [*I. Chem. E. Symp. Ser. No. 61*, 111 (1981)] and Lockett et al. [*Chem. Eng. Sci.*, **38**, 661 (1983)] have provided corrections.

TABLE 14-9 Derating ("System") Factors

System	Factor	Reference	Notes
Nonfoaming regular systems	1.0	1–4	
High pressure ($\rho_G > 1.8$ lb/ft ³)	$2.94/\rho_G^{0.32}$	2	Do not double-derate.
Low-foaming			
Depropanizers	0.9	4	
H ₂ S strippers	0.9	3, 4	
	0.85	2	
Fluorine systems (freon, BF ₃)	0.9	1, 4	
Hot carbonate regenerators	0.9	2, 4	
Moderate-foaming			
Deethanizers			
Absorbing type, top section	0.85	1–4	
Absorbing type, bottom section	1.0	3	
	0.85	1, 2, 4	
Refrigerated type, top section	0.85	4	
	0.8	3	
Refrigerated type, bottom section	1.0	1, 3	
	0.85	4	
Demethanizers			
Absorbing type, top section	0.85	1–3	
Absorbing type, bottom section	1.0	3	
	0.85	1, 2	
Refrigerated type, top section	0.8	3	
Refrigerated type, bottom section	1.0	3	
Oil absorbers			
Above 0°F	0.85	1–4	Ref. 2 proposes these for “absorbers” rather than “oil absorbers.”
Below 0°F	0.95	3	
	0.85	1, 4	
	0.8	2	
Crude towers	1.0	3	
	0.85	4	
Crude vacuum towers	1.0	3	
	0.85	2	
Furfural refining towers	0.85	2	
	0.8	4	
Sulfolane systems	1.0	3	
	0.85	4	
Amine regenerators	0.85	1–4	
Glycol regenerators	0.85	1, 4	
	0.8	3	
	0.65	2	
Hot carbonate absorbers	0.85	1, 4	
Caustic wash	0.65	2	The author suspects that this low factor refers only to some caustic wash applications but not to others.
Heavy-foaming			
Amine absorbers			
	0.8	2	
	0.75	3, 4	
	0.73	1	
Glycol contactors	0.73	1	Ref. 2 recommends 0.65 for glycol contactors in glycol synthesis gas service, 0.5 for others.
	0.65	3, 4	
	0.50	2	
Sour water strippers	0.5–0.7	3	
	0.6	2	
Oil reclaimer	0.7	2	
MEK units	0.6	1, 4	
Stable foam			
Caustic regenerators	0.6	2	
	0.3	1, 4	
Alcohol synthesis absorbers	0.35	2, 4	

References:

- Glitsch, Inc., *Ballast Tray Design Manual*, Bulletin 4900, 6th ed., 1993. Available from Koch-Glitsch, Wichita, Kans.
- Koch Engineering Co., Inc., *Design Manual—Flexitray*, Bulletin 960-1, Wichita, Kans., 1982.
- Nutter Engineering, *Float Valve Design Manual*, 1976. Available from Sulzer ChemTech, Tulsa, Okla.
- M. J. Lockett, *Distillation Tray Fundamentals*, Cambridge University Press, Cambridge, England, 1986.

To convert lb/ft³ to kg/m³, multiply by 16.0.

SOURCE: H. Z. Kister, *Distillation Design*, copyright © 1992 by McGraw-Hill, Inc. Reprinted by permission.

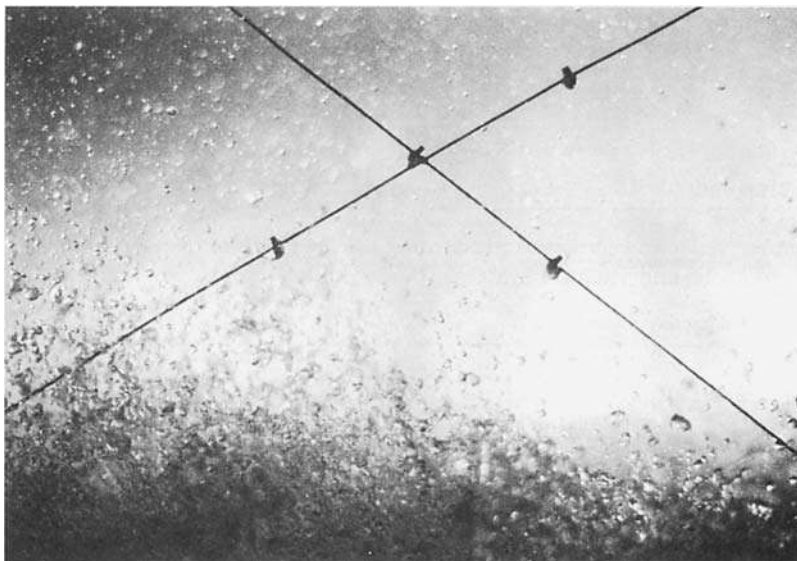


FIG. 14-33 Entrainment. [Reprinted courtesy of Fractionation Research Inc. (FRI).]

Fair (paper presented at the AIChE Annual Meeting, Chicago, Ill., November 1996) correlated data for efficiency reduction due to the rise of entrainment near entrainment flood, getting

$$\ln \psi = A + B\Phi + C\Phi^2 \quad (14-99)$$

where Φ is the fractional approach to entrainment flood and A , B , and C are constants given by

	A	B	C
Highest likely efficiency loss	-3.1898	-4.7413	7.5312
Median (most likely) efficiency loss	-3.2108	-8.9049	11.6291
Lowest likely efficiency loss	4.0992	-29.9141	25.3343

Either the Kister and Haas or the Fair method can be used to evaluate Φ . The correlation has been tested with sieve trays in the flow parameter range of 0.024 to 0.087.

Example 10: Entrainment Effect on Tray Efficiency For the column in Example 9, estimate the efficiency loss should the operation be pushed from the design 80 percent of flood to 90 percent of flood. The midrange dry Murphree tray efficiency is 70 percent.

Solution The vapor and liquid densities and L/V ratio remain unchanged from Example 9, and so is the flow parameter (calculated 0.021 in Example 9). At 80 and 90 percent of flood, respectively, Fig. 14-34 gives $\psi = 0.15$ and 0.24. The respective efficiency reductions are calculated from Eq. (14-98),

TABLE 14-10 Recommended Range of Application for the Kister and Haas Spray Regime Entrainment Correlation

Flow regime	Spray only
Pressure	20–1200 kPa (3–180 psia)
Gas velocity	0.4–5 m/s (1.3–15 ft/s)
Liquid flow rate	3–40 m ³ /(m·h) (0.5–4.5 gpm/in)
Gas density	0.5–30 kg/m ³ (0.03–2 lb/ft ³)
Liquid density	450–1500 kg/m ³ (30–90 lb/ft ³)
Surface tension	5–80 mN/m
Liquid viscosity	0.05–2 cP
Tray spacing	400–900 mm (15–36 in)
Hole diameter	3–15 mm (0.125–0.75 in)
Fractional hole area	0.07–0.16
Weir height	10–80 mm (0.5–3 in)

$$\frac{E_a}{E_{mv}} = \frac{1}{1 + 0.70[0.15/(1 - 0.15)]} = 0.89$$

$$\frac{E_a}{E_{mv}} = \frac{1}{1 + 0.70[0.24/(1 - 0.24)]} = 0.82$$

signifying an efficiency loss from 62 to 57 percent.

Alternatively, at 80 and 90 percent of entrainment flood, the median value of ψ from Eq. (14-99) is

$$\ln \psi = -3.2108 - 8.9049(0.80) + 11.6291(0.80^2)$$

giving $\psi = 0.056$ and from Eq. (14-98) $E_o/E_{mv} = 0.96$

$$\ln \psi = -3.2108 - 8.9049(0.90) + 11.6291(0.90^2)$$

giving $\psi = 0.164$ and from Eq. (14-98) $E_o/E_{mv} = 0.88$

signifying an efficiency reduction from 67 to 62 percent.

PRESSURE DROP

In vacuum distillation, excessive pressure drop causes excessive bottom temperatures which, in turn, increase degradation, polymerization, coking, and fouling, and also loads up the column, vacuum system, and reboiler. In the suction of a compressor, excessive pressure drop increases the compressor size and energy usage. Such services attempt to minimize tray pressure drop. Methods for estimating pressure drops are similar for most conventional trays. The total pressure drop across a tray is given by

$$h_t = h_d + h'_L \quad (14-100)$$

where h_t = total pressure drop, mm liquid

h_d = pressure drop across the dispersion unit (dry hole for sieve trays; dry valve for valve trays), mm liquid

h'_L = pressure drop through aerated mass over and around the disperser, mm liquid

It is convenient and consistent to relate all of these pressure-drop terms to height of equivalent clear liquid (deaerated basis) on the tray, in either millimeters or inches of liquid.

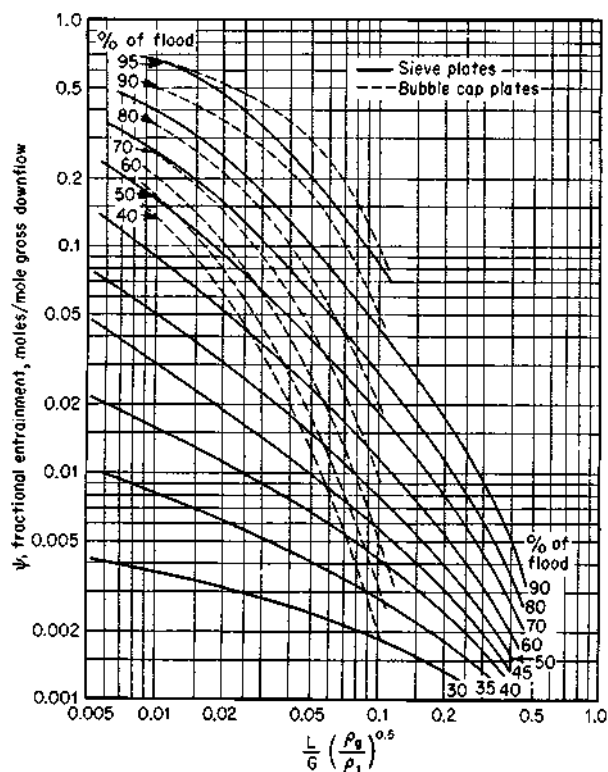


FIG. 14-34 Entrainment correlation. L/G = liquid-gas mass ratio; and ρ_l and ρ_g = liquid and gas densities. [Fair, *Pet./Chem. Eng.*, **33**(10), 45 (September 1961).]

Pressure drop across the disperser is calculated by variations of the standard orifice equation:

$$h_d = K \left(\frac{\rho_G}{\rho_L} \right) U_h^2 \quad (14-101)$$

where U_h = linear gas velocity through slots (valve trays) or perforations (sieve tray), m/s.

For sieve trays, $K = 50.8/C_v^2$. Values of C_v are taken from Fig. 14-35. Values from Fig. 14-35 may be calculated from

$$C_v = 0.74(A_h/A_a) + \exp[0.29(t/d_h) - 0.56] \quad (14-102)$$

For Sulze's fixed valve trays, Summers and van Sinderen (*Distillation 2001: Topical Conference Proceedings*, AIChE Spring National Meeting, p. 444, Houston, April 22–26, 2001) provided the following equation for K :

$$K = 58 + 386A_f \quad \text{for MVG fixed valves} \quad (14-103a)$$

$$K = 58 + 461A_f \quad \text{for SVG and LVG fixed valves} \quad (14-103b)$$

Figure 14-36 illustrates the pressure drop of a typical moving valve tray as a function of gas velocity. At low velocities, all valves are closed. Gas rises through the crevices between the valves and the tray deck, with increasing pressure drop as the gas velocity rises. Once point A, the closed balance point (CBP), is reached, some valves begin to open. Upon further increase in gas velocity, more valves open until point B, the open balance point (OBP), is reached. Between points A and B, gas flow area increases with gas velocity, keeping pressure drop constant. Further increases in gas velocity increase pressure drop similar to that in a sieve tray.

The term K in Eq. (14-101) depends on valve slot area, orifice geometry, deck thickness, and the type, shape, and weight of the valves. These are best obtained from the manufacturer's literature,

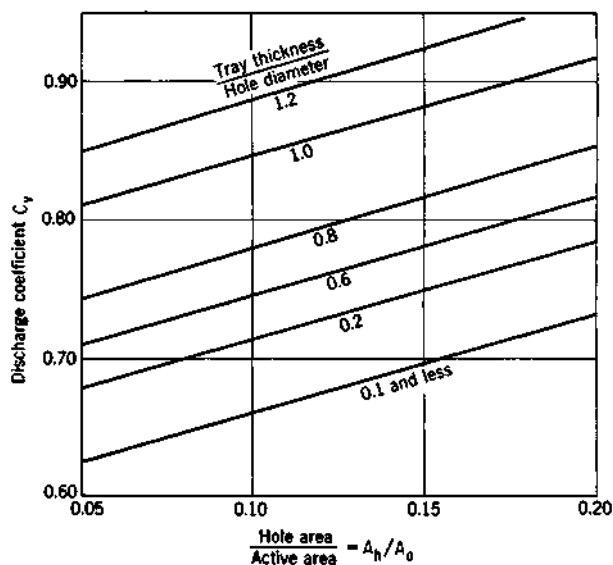


FIG. 14-35 Discharge coefficients for gas flow, sieve trays. [Liebson, Kelley, and Bullington, *Pet. Refiner*, **36**(3), 288 (1957).]

but can also be calculated from Bolles' [*Chem. Eng. Prog.*, **72**(9), 43 (1976)], Lockett's (*Distillation Tray Fundamentals*, Cambridge University Press, Cambridge, England, 1986), and Klein's (*Chem. Eng.*, May 3, 1982, p. 81) methods.

For valve trays, Klein gives the following values for K (in $\text{s}^2\text{-mm/m}^2$) in Eq. (14-101), when based on the total hole area (not slot area):

	Sharp orifice	Venturi valve
All valves open (K_o)	$254.5(2.64/t_i)^{0.5}$	122
All valves closed (K_c)	1683	841

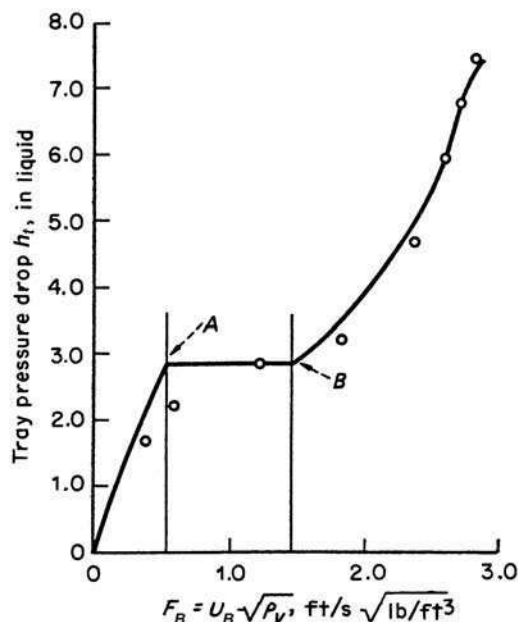


FIG. 14-36 Typical moving valve tray pressure-drop profile. (From G. F. Klein, *Chem. Eng.*, May 3, 1982, p. 81; reprinted courtesy of Chemical Engineering.)

The velocity at which the valves start to open (point A) is given by

$$U_{h,\text{closed}} = 1.14[t_v(R_{vw}/K_C)(\rho_M/\rho_G)]^{0.5} \quad (14-104)$$

where $U_{h,\text{closed}}$ = hole area at point A, m/s; t_v = valve thickness, mm; R_{vw} = ratio of valve weight with legs to valve weight without legs, given in Table 14-11; K_C = orifice coefficient with all valves closed (see above), $\text{s}^2\cdot\text{mm}/\text{m}^2$; ρ_M = valve metal density, kg/m^3 (about $8000 \text{ kg}/\text{m}^3$ for steel); ρ_G = gas density, kg/m^3 .

The velocity at which all the valves are open $U_{h,\text{open}}$ can be calculated from

$$U_{h,\text{open}} = U_{h,\text{closed}}(K_C/K_O)^{0.5} \quad (14-105)$$

Pressure drop through the aerated liquid [h'_L , in Eq. (14-100)] is calculated by

$$h'_L = \beta h_{ds} \quad (14-106)$$

where β = aeration factor, dimensionless

h_{ds} = calculated height of clear liquid, mm (dynamic seal)

The aeration factor β has been determined from Fig. 14-37 for valve and sieve trays. For sieve trays, values of β in the figure may be calculated from

$$\beta = 0.0825 \ln \frac{Q}{L_w} - 0.269 \ln F_h + 1.679 \quad (14-107)$$

where L_w = weir length, m

F_h = F-factor for flow through holes, $F_h = U_h \rho_G^{0.5}$, m/s (kg/m^3)^{0.5}

For sieve and valve trays,

$$h_{ds} = h_w + h_{ow} + 0.5h_{hg} \quad (14-108)$$

where h_w = weir height, mm

h_{ow} = height of crest over weir, mm clear liquid

h_{hg} = hydraulic gradient across tray, mm clear liquid

The value of weir crest h_{ow} may be calculated from the Francis weir equation and its modifications for various weir types. For a segmental weir and for height in millimeters of clear liquid,

$$h_{ow} = 664 \left(\frac{Q}{L_w} \right)^{2/3} \quad (14-109)$$

where Q = liquid flow, m^3/s

L_w = weir length, m

For serrated weirs,

$$h_{ow} = 851 \left(\frac{Q'}{\tan \theta/2} \right)^{0.4} \quad (14-110)$$

where Q' = liquid flow, m^3/s per serration

θ = angle of serration, deg

For circular weirs,

$$h_{ow} = 44,300 \left(\frac{Q}{d_w} \right)^{0.704} \quad (14-111)$$

where q = liquid flow, m^3/s

d_w = weir diameter, mm

For most sieve and valve trays, the hydraulic gradient is small and can be dropped from Eq. (14-108). Some calculation methods are

available and are detailed in previous editions of this handbook. A rule of thumb by the author is 17 mm/m (0.2 in/ft) of flow path length. This rule only applies in the liquid-loaded froth and emulsion regimes ($Q_L > 50 \text{ m}^3/\text{hr}$ or $> 5.5 \text{ gpm/in}$ of outlet weir length). At lower liquid loads, the hydraulic gradient is less.

As noted, the weir crest h_{ow} is calculated on an equivalent clear-liquid basis. A more realistic approach is to recognize that in general a froth or spray flows over the outlet weir (settling can occur upstream of the weir if a large "calming zone" with no dispersers is used). Bennett et al. [AIChE J., **29**, 434 (1983)] allowed for froth overflow in a comprehensive study of pressure drop across sieve trays; their correlation for residual pressure drop h'_L in Eq. (14-100) is presented in detail in the previous (seventh) edition of this handbook, including a worked example. Although more difficult to use, the method of Bennett et al. was recommended when determination of pressure drop is of critical importance.

Example 11: Pressure Drop, Sieve Tray For the conditions of Example 9, estimate the pressure drop for flow across one tray. The thickness of the tray metal is 2 mm. The superficial F-factor is $2.08 \text{ m}/(\text{kg}/\text{m}^3)^{1/2}$.

Solution Equations (14-100), (14-106), and (14-107), where $h_t = h_d + \beta(h_w + h_{ow})$, are used. For $F_s = 2.08$, $F_B = 2.32$ and $F_d = 16.55$. From Example 9, $L_w = 1.50$ m and $h_w = 38$ mm. For a liquid rate of $22,000 \text{ kg/hr}$, $Q = 7.27(10^{-3}) \text{ m}^3/\text{s}$, and $Q/L_w = 4.8(10^{-3})$. By Eq. (14-107) or Fig. 14-37, $\beta = 0.48$. From Eq. (14-102) or Fig. 14-35, $C_c = 0.75$. Then, by Eq. (14-101), $h_d = 29.0$ mm liquid. Using Eq. (14-109), $h_{ow} = 18.9$ mm. Finally, $h_t = h_d + \beta(h_w + h_{ow}) = 29.0 + 0.48(38 + 18.9) = 56.4$ mm liquid.

When straight or serrated segmental weirs are used in a column of circular cross section, a correction may be needed for the distorted pattern of flow at the ends of the weirs, depending on liquid flow rate. The correction factor F_w from Fig. 14-38 is used directly in Eq. (14-109). Even when circular downcomers are utilized, they are often fed by the overflow from a segmental weir.

Loss under Downcomer The head loss under the downcomer apron, as millimeters of liquid, may be estimated from

$$h_{da} = 165.2 \left(\frac{Q}{A_{da}} \right)^2 \quad (14-112)$$

where Q = volumetric flow of liquid, m^3/s and A_{da} = most restrictive (minimum) area of flow under the downcomer apron, m^2 . Equation (14-112) was derived from the orifice equation with an orifice coefficient of 0.6. Although the loss under the downcomer is small, the clearance is significant from the aspect of tray stability and liquid distribution.

The term A_{da} should be taken as the most restrictive area for liquid flow in the downcomer outlet. Usually, this is the area under the downcomer apron (i.e., the downcomer clearance times the length of the segmental downcomer), but not always. For instance, if an inlet weir is used and the area between the segmental downcomer and the inlet weir is smaller than the area under the downcomer apron, the smaller area should be used.

OTHER HYDRAULIC LIMITS

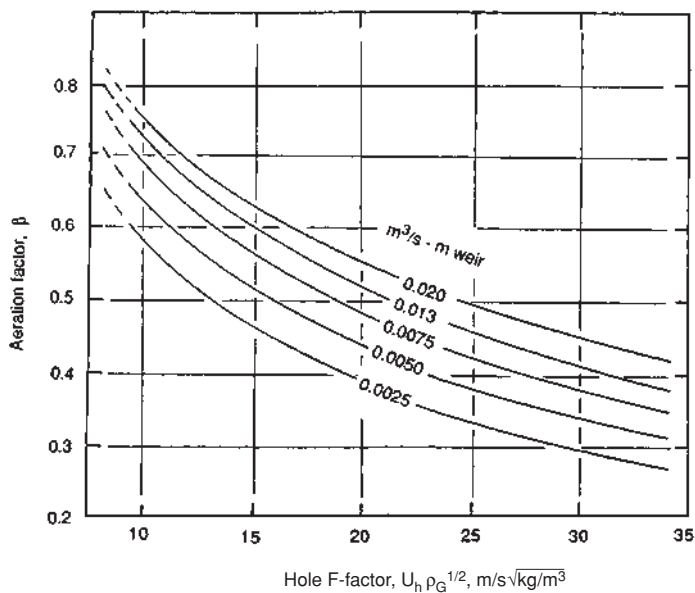
Weeping Weeping is liquid descending through the tray perforations, short-circuiting the contact zone, which lowers tray efficiency. At the tray floor, liquid static head that acts to push liquid down the perforations is counteracted by the gas pressure drop that acts to hold the liquid on the tray. When the static head overcomes the gas pressure drop, weeping occurs.

Some weeping usually takes place under all conditions due to sloshing and oscillation of the tray liquid. Generally, this weeping is too small to appreciably affect tray efficiency. The *weep point* is the gas velocity at which weeping first becomes noticeable. At this point, little efficiency is lost. As gas velocity is reduced below the weep point, the weep rate increases. When the weep rate becomes large enough to significantly reduce tray efficiency, the lower tray operating limit is reached.

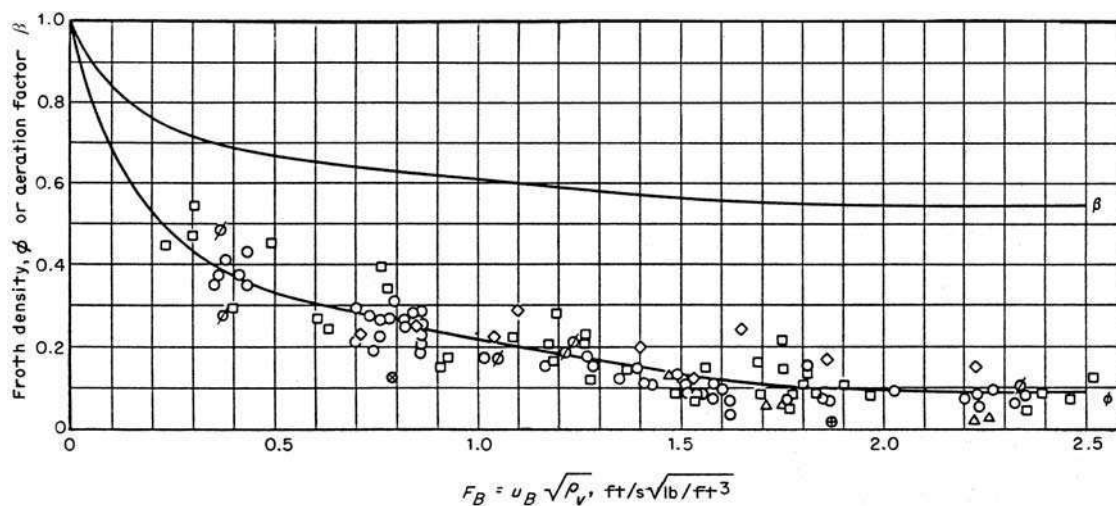
The main factor that affects weeping is the fractional hole area. The larger it is, the smaller the gas pressure drop and the greater the weeping tendency. Larger liquid rates and taller outlet weirs increase

TABLE 14-11 R_{vw} Values for Eq. (14-104)

Valve type	Sharp	Venturi
Three-leg	1.23	1.29
Four-leg	1.34	1.45
Cages (no legs)	1.00	1.00



(a)



(b)

FIG. 14-37 Aeration factor for pressure drop calculation. (a) Sieve trays. [Bolles and Fair, *Encyclopedia of Chemical Processing and Design*, vols. 16, 86, J. M. McKetta (ed.), Marcel Dekker, New York, 1982.] (b) Valve trays. (From G. F. Klein, *Chem. Eng.*, May 3, 1982, p. 81; reprinted courtesy of Chemical Engineering.)

liquid heads and therefore weeping. Hole diameter has a complex effect on weeping, detailed by Lockett and Banik [*Ind. Eng. Chem. Proc. Des. Dev.* **25**, 561 (1986)].

Tests by Lockett and Banik (loc. cit.) show that weeping is often nonuniform, with some hydraulic conditions favoring weeping from the tray inlet and others from the tray outlet. Weeping from the tray inlet is particularly detrimental to tray efficiency because the weeping liquid bypasses two trays.

Weep Rate Prediction Lockett and Banik (loc. cit.) and Hsieh and McNulty (*Chem. Eng. Progr.*, July 1993, p. 71) proposed correlations for predicting weep rates from sieve trays. Colwell and O'Bara (Paper presented at the AIChE Meeting, Houston, April 1989) recommended the Lockett and Banik correlation for low pressures

(<1100 kPa or 165 psia), and the Hsieh and McNulty correlation for high pressures (>1100 kPa or 165 psia). They also corrected the Lockett and Banik correlation to improve its accuracy near the weep point.

The Lockett and Banik correlation (as corrected by Colwell and O'Bara) is

$$\frac{W}{A_h} = \frac{29.45}{\sqrt{Fr_h}} - 44.18 \quad Fr_h < 0.2 \quad (14-113a)$$

$$\frac{W}{A_h} = \frac{1.841}{Fr_h^{1.533}} \quad Fr_h > 0.2 \quad (14-113b)$$

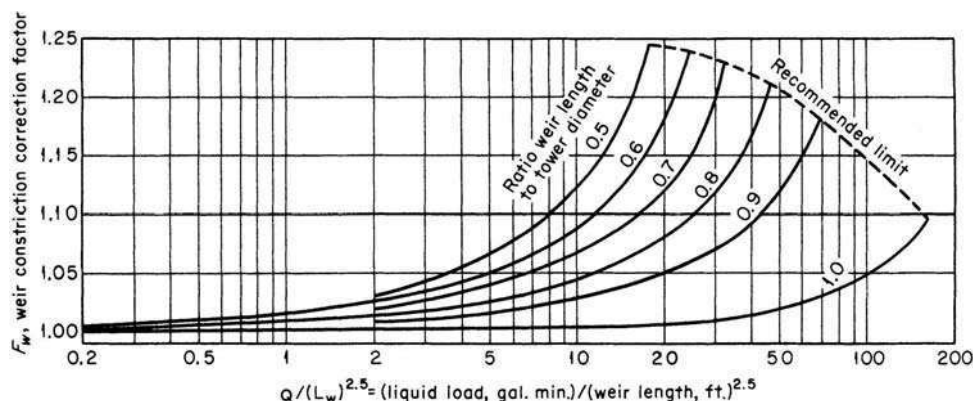


FIG. 14-38 Correction for effective weir length. To convert gallons per minute to cubic meters per second, multiply by 6.309×10^{-5} ; to convert feet to meters, multiply by 0.3048. [Bolles, *Pet. Refiner*, **25**, 613 (1946).]

where

$$Fr_h = 0.373 \frac{u_h^2}{h_c} \frac{\rho_v}{\rho_L - \rho_v} \quad (14-114)$$

Equations (14-113) and (14-114) use English units: W is the weep rate, gpm; A_h is the hole area, ft²; u_h is the hole velocity, ft/s; and h_c is the clear liquid height, in. Colwell's [*Ind. Eng. Chem. Proc. Des. Dev.* **20**(2), 298 (1981)] method below has been recommended for obtaining the clear liquid height h_c in Eq. (14-114).

$$h_c = \phi_f \left\{ 0.527 \left(\frac{Q_L(1-f_w)}{C_d \phi_f} \right)^{2/3} + h_w \right\} \quad (14-115)$$

where ϕ_f is given by Eq. (14-119) and C_d is given by Eq. (14-116)

$$\left. \begin{aligned} C_d &= 0.61 + 0.08 \frac{h_{fow}}{h_w} & \frac{h_{fow}}{h_w} < 8.135 \\ C_d &= 1.06 \left(1 + \frac{h_w}{h_{fow}} \right)^{1.5} & \frac{h_{fow}}{h_w} > 8.135 \end{aligned} \right\} \quad (14-116)$$

$$h_{fow} = h_f - h_w \quad (14-117)$$

where h_f is given by Eq. (14-122). The froth density ϕ_f is calculated from

$$\eta = 12.6 Fr^{0.4} \left(\frac{A_B}{A_h} \right)^{0.25} \quad (14-118)$$

$$\phi_f = \frac{1}{\eta + 1} \quad (14-119)$$

$$Fr = 0.37 \frac{\rho_v u_B^2}{h_c(\rho_L - \rho_v)} \quad (14-120)$$

The term f_w in Eq. (14-115) is the ratio of weep rate from the tray to the total liquid flow entering the tray, calculated as follows:

$$f_w = W/GPM \quad (14-121)$$

Some trial and error is required in this calculation because the clear liquid height h_c and the froth density ϕ_f depend on each other, and the weep fraction f_w depends on the clear liquid height h_c . Clear liquid height is related to froth height and froth density by

$$h_c = \phi_f h_f \quad (14-122)$$

The terms in Eqs. (14-115) to (14-122) are in the English units and are explained in the Nomenclature.

With large-diameter trays and low liquid loads, a small ratio of W/A_h corresponds to a large fractional weep. Under these conditions, the Lockett and Banik correlation is inaccurate. The correlation is unsuitable for trays with very small (<3-mm or $\frac{1}{8}$ -in) holes. The correlation appears to fit most data points to an accuracy of ± 15 to ± 30 percent. The Hsieh and McNulty correlation (loc. cit.) is

$$\sqrt{J_C^*} + m \sqrt{J_L^*} = C_w \quad (14-123)$$

where

$$J_C^* = u_h \left[\frac{\rho_C}{gZ(\rho_L - \rho_C)} \right]^{0.5} \quad (14-124)$$

and

$$J_L^* = \frac{W}{448.83 A_h} \left[\frac{\rho_L}{gZ(\rho_L - \rho_C)} \right]^{0.5} \quad (14-125)$$

$$Z = h_c^{1.5} / (12 d_H^{0.5}) \quad (14-126)$$

The terms in Eqs. (14-123) to (14-126) are in English units and are explained in the Nomenclature. For sieve trays, $m = 1.94$ and $C_w = 0.79$. Note that the constants are a slight revision of those presented in the original paper (C. L. Hsieh, private communication, 1991). Clear liquid height is calculated from Colwell's correlation [Eqs. (14-115) to (14-122)]. The Hsieh and McNulty correlation applies to trays with 9 percent and larger fractional hole area. For trays with smaller hole area, Hsieh and McNulty expect the weeping rate to be smaller than predicted.

Weeping from Valve Trays An analysis of weeping from valve trays [Bolles, *Chem. Eng. Progs.* **72**(9), 43 (1976)] showed that in a well-designed valve tray, the weep point is below the gas load at which the valves open; and throughout the valve opening process, the operating point remains above the weep point. In contrast, if the tray contains too many valves, or the valves are too light, excessive valve opening occurs before the gas pressure drop is high enough to counter weeping. In this case, weeping could be troublesome.

Weep point correlations for valve trays were presented by Bolles (loc. cit.) and by Klein (*Chem. Eng.*, Sept. 17, 1984, p. 128). Hsieh and McNulty (loc. cit.) gave a complex extension of their weep rate correlation to valve trays.

Dumping As gas velocity is lowered below the weep point, the fraction of liquid weeping increases until all the liquid fed to the tray weeps through the holes and none reaches the downcomer. This is the *dump point*, or the *seal point*. The dump point is well below the range of acceptable operation of distillation trays. Below the dump point, tray efficiency is slashed, and mass transfer is extremely poor. Operation below the dump point can be accompanied by severe hydraulic instability due to unsealing of downcomers.

Extensive studies on dumping were reported by Prince and Chan [*Trans. Inst. Chem. Engr.* **43**, T49 (1965)]. The Chan and Prince

dump-point correlation was recommended and is presented in detail elsewhere (Kister, *Distillation Design*, McGraw-Hill, 1992). Alternatively, the dump point can be predicted by setting the weep rate equal to 100 percent of the liquid entering the tray in the appropriate weep correlation.

Turndown The turndown ratio is the ratio of the normal operating (or design) gas throughput to the minimum allowable gas throughput. The minimum allowable throughput is usually set by excessive weeping, while normal operating throughput is a safe margin away from the relevant flooding limit.

Sieve and fixed valve trays have a poor turndown ratio (about 2:1). Their turndown can be improved by blanking some rows of tray holes, which reduces the tendency to weep, but will also reduce the tray's maximum capacity. Turndown of *moving valve trays* is normally between about 4:1 to 5:1. Special valve designs can achieve even better turndown ratios, between 6:1 and 10:1, and even more. Turndown can also be enhanced by blanking strips (which require valve removal) or valve leg crimping. Sloley and Fleming (*Chem. Eng. Progr.*, March 1994, p. 39) stress that correct implementation of turndown enhancement is central to achieving a desired turndown. When poorly implemented, turndown may be restricted by poor vapor-liquid contact rather than by weeping.

Vapor Channeling All the correlations in this section assume an evenly distributed tray vapor. When the vapor preferentially channels through a tray region, premature entrainment flood and excessive entrainment take place due to a high vapor velocity in that region. At the same time, other regions become vapor-deficient and tend to weep, which lowers tray efficiency.

Work by Davies [*Pet. Ref.* **29**(8), p. 93, and **29**(9), p. 121 (1950)] based on bubble-cap tray studies suggests that the vapor pressure drop of the tray (the *dry pressure drop*) counteracts channeling. The higher the dry tray pressure drop, the greater the tendency for vapor to spread uniformly over the bubbling area. If the dry tray pressure drop is too small compared with the channeling potential, channeling prevails.

Perhaps the most common vapor channeling mechanism is *vapor crossflow channeling* (VCFC, Fig. 14-39). The hydraulic gradient on the tray induces preferential vapor rise at the outlet and middle of the tray, and a vapor-deficient region near the tray inlet. The resulting high vapor velocities near the tray outlet step up entrainment, while the low vapor velocities near the tray inlet induce weeping. Interaction between adjacent trays (Fig. 14-39) accelerates both the outlet entrainment and the inlet weeping. The net result is excessive entrainment and premature flooding at the tray middle and outlet, simultaneous with weeping from the tray inlet, accompanied by a loss of efficiency and turndown.

VCFC takes place when the following four conditions exist simultaneously [Kister, Larson, and Madsen, *Chem. Eng. Progr.*, p. 86 (Nov. 1992); Kister, *The Chemical Engineer*, **544**, p. 18 (June 10, 1993)]:

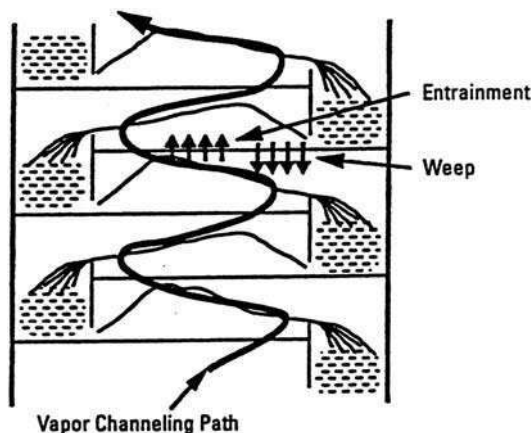


FIG. 14-39 Vapor crossflow channeling. Note entrainment near the tray middle and outlet, and weep near the tray inlet. (Kister, H. Z., K. F. Larson, and P. Madsen, *Chem. Eng. Progr.*, Nov. 1992, p. 86; reproduced with permission.)

1. Absolute pressure < 500 kPa (70 psia).
2. High liquid rates [$>50 \text{ m}^3/(\text{m} \cdot \text{h})$ or 6 gpm/in of outlet weir].
3. High ratio ($>2:1$) of flow path length to tray spacing.
4. Low dry tray pressure drop. On sieve and fixed valve trays, this means high (>11 percent) fractional hole area. On moving valve trays, this means venturi valves (smooth orifices) or long-legged valves (>15 percent slot area). On all trays, the channeling tendency and severity escalate rapidly as the dry pressure drop diminishes (e.g., as fractional hole area increases).

Hartman (*Distillation 2001: Topical Conference Proceedings*, AIChE Spring National Meeting, p. 108, Houston, Tex. (April 22–26, 2001)) reports VCFC even with conventional valve trays (14 percent slot area) at very high ratio (3.6:1) of flow path length to tray spacing and tray truss obstruction.

VCFC is usually avoided by limiting fractional hole areas, avoiding venturi valves, and using forward-push devices. Resitaris and Papademos [Paper presented at the AIChE Annual Meeting, Reno, Nev. (November 2001)] cited tray inlet inactivity as a contributor to VCFC, and advocate inlet forward-push devices to counter it.

TRANSITION BETWEEN FLOW REGIMES

Froth-Spray Froth-spray transition has been investigated for sieve trays using a variety of techniques. The gradual nature of this transition bred a multitude of criteria for defining it, and made its correlation difficult. Excellent overviews were given by Lockett (*Distillation Tray Fundamentals*, Cambridge University Press, Cambridge, England, 1986) and Prado, Johnson, and Fair [*Chem. Eng. Progr.* **83**(3), p. 32 (1987)]. Porter and Jenkins [*I. Chem. E. Symp. Ser.* **56**, Summary Paper (1979)] presented a simple correlation for the froth-to-spray transition.

$$F_{LC} \frac{N_p A_B}{L_w} = 0.0191 \quad (14-127)$$

The terms of this equation are in English units and are explained in the Nomenclature. This correlation is based on the premise that froth-to-spray transition occurs when the entrainment vs. liquid load relationship passes through a minimum (see "Entrainment"). Alternatively, it was argued that the minimum represents a transition from the froth regime to a partially developed spray region (Kister, Pinczewski, and Fell, Paper presented in the 90th National AIChE Meeting, Houston, April 1981). If this alternative argument is valid, then when the correlation predicts froth, it is highly unlikely that the column operates in the spray regime; but when it predicts spray, the column may still be operating in the froth regime. Recent entrainment studies by Ohe [*Distillation 2005: Topical Conference Proceedings*, AIChE Spring National Meeting, p. 283, Atlanta (April 10–13, 2005)] argue that the entrainment minima represent minimum liquid residence times on the tray, and are unrelated to the froth-spray transition.

A second correlation is by Pinczewski and Fell [*Ind. Eng. Chem. Proc. Des. Dev.* **21**, p. 774 (1982)]

$$u_B \sqrt{\rho_G} = 2.25 \left(\frac{Q_L \sqrt{\rho_L}}{100} \right)^n \quad (14-128)$$

The terms of Eq. (14-128) are in English units and are explained in the Nomenclature. The exponent n is calculated from Eq. (14-84). Equation (14-128) is based on transition data obtained from orifice jetting measurements for the air-water system and on entrainment minimum data for some hydrocarbon systems.

A third recent correlation by Johnson and Fair (loc. cit.) is

$$U_a^* = C_1 \rho_G^{-0.50} \rho_L^{0.692} \sigma^{0.06} A_f^{0.25} \left(\frac{q}{L_w} \right)^{0.05} d_h^{-0.1} \quad (14-129)$$

where U_a^* = gas velocity through active area at inversion, m/s

ρ_G = gas density, kg/m^3

ρ_L = liquid density, kg/m^3

σ = surface tension, mN/m

A_f = hole/active area ratio

$$\begin{aligned}
 q/L_w &= \text{liquid flow, m}^3/(\text{s}\cdot\text{m}) \text{ weir} \\
 d_h &= \text{hole diameter, mm} \\
 C_1 &= 0.0583 \text{ for 25.4-mm overflow weirs} \\
 &= 0.0568 \text{ for 50.4-mm overflow weirs} \\
 &= 0.0635 \text{ for 101.6-mm overflow weirs}
 \end{aligned}$$

Froth-Emulsion Froth-emulsion transition occurs [Hofhuis and Zuideweg, *I. Chem. E. Symp. Ser.* **56**, p. 2, 2/1 (1979)] when the aerated mass begins to obey the Francis weir formula. Using this criterion, the latest version of this transition correlation is

$$F_{LG} \frac{N_p A_B}{L_w h_c} = 0.0208 \quad (14-130)$$

The terms of this equation are in English units and are explained in the Nomenclature; h_c is calculated from the Hofhuis and Zuideweg (loc. cit.) equation.

$$h_c = 2.08 \left(F_{LG} \frac{N_p A_B}{L_w} p \right)^{0.25} h_w^{0.5} \quad (14-131)$$

An inspection of the experimental data correlated shows that this, too, is a gradual transition, which occurs over a range of values rather than at a sharp point.

Valve Trays The amount of work reported thus far on valve tray regime transition is small and entirely based on air-water tests. Correlations proposed to date require the knowledge of liquid holdup at transition, which is generally not available, and are therefore of limited application for commercial columns.

TRAY EFFICIENCY

Definitions

Overall Column Efficiency This is the ratio of the number of theoretical stages to the number of actual stages

$$E_{OC} = N_t/N_a \quad (14-132)$$

Since tray efficiencies vary from one section to another, it is best to apply Eq. (14-132) separately for the rectifying and stripping sections. In practice, efficiency data and prediction methods are often too crude to give a good breakdown between the efficiencies of different sections, and so Eq. (14-132) is applied over the entire column.

Point Efficiency This is defined by Eq. (14-133) (Fig. 14-40a):

$$E_{OC} = \left(\frac{y_n - y_{n-1}}{y_n^* - y_{n-1}} \right)_{\text{point}} \quad (14-133)$$

where y_n^* is the composition of vapor in equilibrium with the liquid at point n . The term y_n is actual vapor composition at that point. The point efficiency is the ratio of the change of composition at a point to the change that would occur on a theoretical stage. As the vapor composition at a given point cannot exceed the equilibrium composition, fractional point efficiencies are always lower than 1. If there is a composition gradient on the tray, point efficiency will vary between points on the tray.

Murphree Tray Efficiency [Ind. Eng. Chem. **17**, 747 (1925)] This is the same as point efficiency, except that it applies to the entire tray instead of to a single point (Fig. 14-40b):

$$E_{MV} = \left(\frac{y_n - y_{n-1}}{y_n^* - y_{n-1}} \right)_{\text{tray}} \quad (14-134)$$

If both liquid and vapor are perfectly mixed, liquid and vapor compositions on the tray are uniform, and the Murphree tray efficiency will coincide with the point efficiency at any point on the tray. In practice, a concentration gradient exists in the liquid, and x_n at the tray outlet is lower than x_n^* on the tray (see Fig. 14-40b). This frequently lowers y_n^* relative to y_n , thus enhancing tray efficiency [Eq. (14-134)] compared with point efficiency. The value of y_n^* may even drop below y_n . In this case, E_{MV} exceeds 100 percent [Eq. (14-134)].

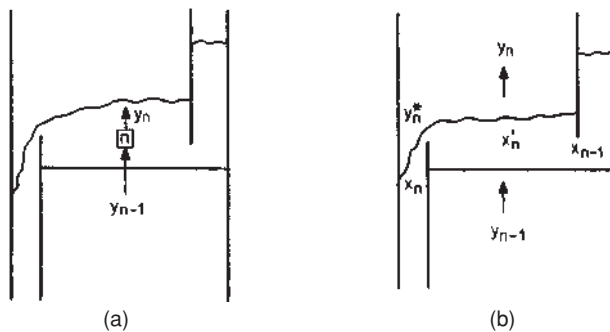


FIG. 14-40 Point and Murphree efficiencies. (a) Point. (b) Murphree. (From H. Z. Kister, *Distillation Design*, copyright © 1992 by McGraw-Hill; reprinted by permission.)

Overall column efficiency can be calculated from the Murphree tray efficiency by using the relationship developed by Lewis [Ind. Eng. Chem. **28**, 399 (1936)].

$$E_{OC} = \frac{\ln[1 + E_{MV}(\lambda - 1)]}{\ln \lambda} \quad (14-135)$$

where
$$\lambda = m \frac{G_M}{L_M} \quad (14-136)$$

Equation (14-135) is based on the assumption of constant molar overflow and a constant value of E_{MV} from tray to tray. It needs to be applied separately to each section of the column (rectifying and stripping) because G_M/L_M , and therefore λ , varies from section to section. Where molar overflow or Murphree efficiencies vary throughout a section of column, the section needs to be divided into subsections small enough to render the variations negligible.

The point and Murphree efficiency definitions above are expressed in terms of vapor concentrations. Analogous definitions can be made in terms of liquid concentrations. Further discussion is elsewhere (Lockett, *Distillation Tray Fundamentals*, Cambridge University Press, Cambridge, England, 1986).

Fundamentals Figure 14-41 shows the sequence of steps for converting phase resistances to a tray efficiency. Gas and liquid film resistances are added to give the point efficiency. Had both vapor and liquid on the tray been perfectly mixed, the Murphree tray efficiency would have equaled the point efficiency. Since the phases are not perfectly mixed, a model of the vapor and liquid mixing patterns is needed for converting point efficiency to tray efficiency. Liquid mixture patterns are plug flow, backmixing, and stagnant zones, while vapor-mixing patterns are perfect mixing and plug flow.

Lewis (loc. cit.) was the first to derive quantitative relationships between the Murphree and the point efficiency. He derived three mixing cases, assuming plug flow of liquid in all. The Lewis cases give the maximum achievable tray efficiency. In practice, efficiency is lower due to liquid and vapor nonuniformities and liquid mixing.

Most tray efficiency models are based on Lewis case 1 with vapor perfectly mixed between trays. For case 1, Lewis derived the following relationship:

$$E_{MV, \text{dry}} = \frac{\exp(\lambda E_{OC}) - 1}{\lambda} \quad (14-137)$$

The “dry” Murphree efficiency calculated thus far takes into account the vapor and liquid resistances and the vapor-liquid contact patterns, but is uncorrected for the effects of entrainment and weeping. This correction converts the dry efficiency to a “wet” or actual Murphree tray efficiency. Colburn [Eq. (14-98), under “Entrainment”] incorporated the effect of entrainment on efficiency, assuming perfect mixing of liquid on the tray.

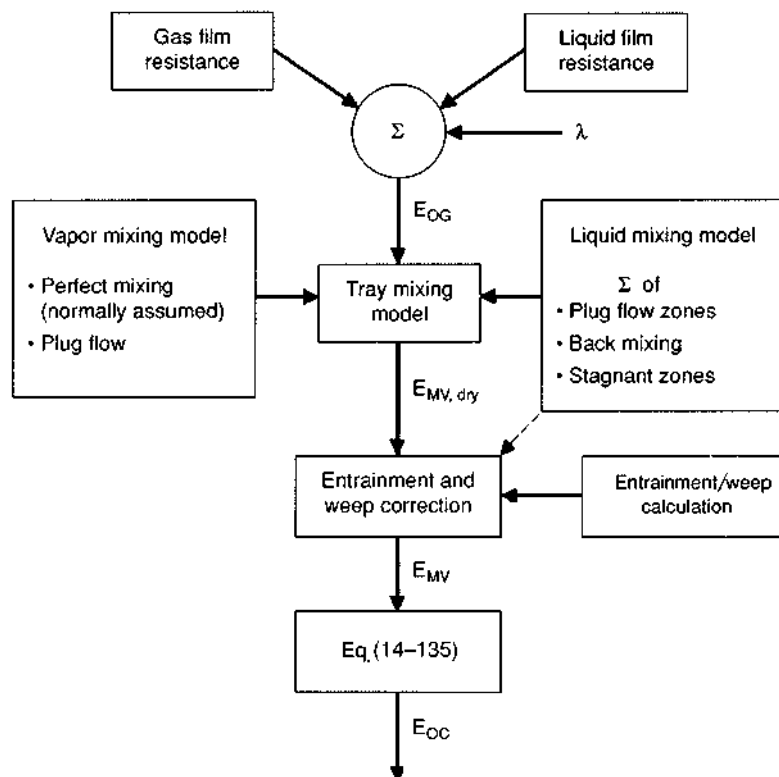


FIG. 14-41 Sequence of steps for theoretical prediction of tray efficiency. (From H. Z. Kister, *Distillation Design*, copyright © 1992 by McGraw-Hill; reprinted by permission.)

Factors Affecting Tray Efficiency Below is a summary based on the industry's experience. A detailed discussion of the fundamentals is found in Lockett's book (*Distillation Tray Fundamentals*, Cambridge University Press, Cambridge, England, 1986). A detailed discussion of the reported experience, and the basis of statements made in this section are in Kister's book (*Distillation Design*, McGraw-Hill, New York, 1992).

Errors in Vapor-Liquid Equilibrium (VLE) Errors in relative volatility are the most underrated factor affecting tray efficiency. Figure 14-42 shows the direct effect of the errors [Deibele and Brandt, *Chem. Ing. Tech.* **57**(5), p. 439 (1985); Roy P. and G. K. Hobson, *I. Chem. E. Symp. Ser.* **104**, p. A273 (1987)]. At very low relative volatilities ($\alpha < 1.2$), small errors in VLE have a huge impact on tray efficiency. For instance, at $\alpha = 1.1$, a -3 percent error gives a tray efficiency 40 to 50 percent higher than its true value (Fig. 14-42). Since VLE errors are seldom lower than ± 2 to 3 percent, tray efficiencies of low-volatility systems become meaningless unless accompanied by VLE data. Likewise, comparing efficiencies derived for a low-volatility system by different sources is misleading unless one is using identical VLE.

Figure 14-42 shows that errors in relative volatility are a problem only at low relative volatilities; for $\alpha > 1.5$ to 2.0, VLE errors have negligible direct impact on tray efficiency.

Most efficiency data reported in the literature are obtained at total reflux, and there are no indirect VLE effects. For measurements at finite reflux ratios, the indirect effects below compound the direct effect of Fig. 14-42. Consider a case where $\alpha_{\text{apparent}} < \alpha_{\text{true}}$ and test data at a finite reflux are analyzed to calculate tray efficiency. Due to the volatility difference $R_{\text{min,apparent}} > R_{\text{min,true}}$. Since the test was conducted at a fixed reflux flow rate, $(R/R_{\text{min}})_{\text{apparent}} < (R/R_{\text{min}})_{\text{true}}$. A calculation based on the apparent R/R_{min} will give more theoretical stages than a calculation based on the true R/R_{min} . This means a higher apparent efficiency than the true value.

The indirect effects add to those of Fig. 14-42, widening the gap between true and apparent efficiency. The indirect effects exponentially escalate as minimum reflux is approached. Small errors in VLE or reflux ratio measurement (this includes column material balance as well as reflux rate) alter R/R_{min} . Near minimum reflux, even small R/R_{min} errors induce huge errors in the number of stages, and therefore in tray efficiency. Efficiency data obtained near minimum reflux are therefore meaningless and potentially misleading.

Liquid Flow Patterns on Large Trays The most popular theoretical models (below) postulate that liquid crosses the tray in plug flow with superimposed backmixing, and that the vapor is perfectly mixed. Increasing tray diameter promotes liquid plug flow and suppresses backmixing.

The presence of stagnant zones on large-diameter distillation trays is well established, but the associated efficiency loss is poorly understood; in some cases, significant efficiency losses, presumably due to stagnant zones, were reported [Weiler, Kirkpatrick, and Lockett, *Chem. Eng. Progr.* **77**(1), 63 (1981)], while in other cases, no efficiency difference was observed [Yanagi and Scott, *Chem. Eng. Progr.*, **69**(10), 75 (1973)]. Several techniques are available for eliminating stagnant regions (see Kister, *Distillation Design*, McGraw-Hill, New York, 1992, for some), but their effectiveness for improving tray efficiency is uncertain.

Weir Height Taller weirs raise the liquid level on the tray in the froth and emulsion regimes. This increases interfacial area and vapor contact time, which should theoretically enhance efficiency. In the spray regime, weir height affects neither liquid level nor efficiency. In distillation systems, the improvement of tray efficiency due to taller weirs is small, often marginal.

Length of Liquid Flow Path Longer liquid flow paths enhance the liquid-vapor contact time, the significance of liquid plug flow, and therefore raise efficiency. Typically, doubling the flow path length

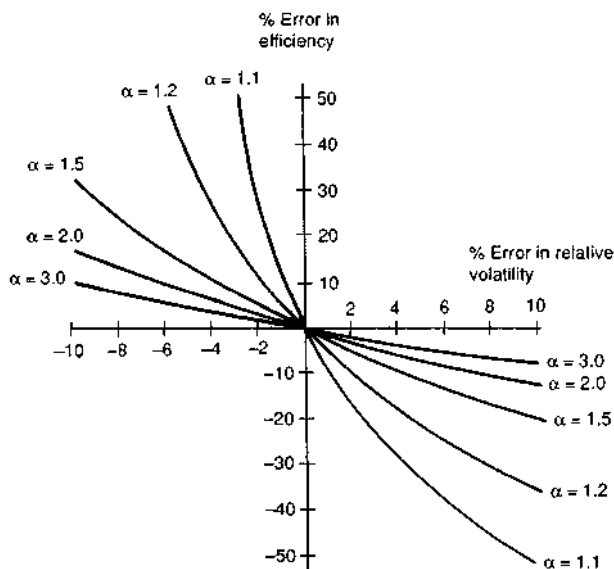


FIG. 14-42 Direct effect of errors in relative volatility on error in tray efficiency. (From H. Z. Kister, *Distillation Design*, copyright © 1992 by McGraw-Hill; reprinted by permission.)

(such as going from two-pass to one-pass trays at a constant tower diameter) raises tray efficiency by 5 to 15 percent.

Fractional Hole Area Efficiency increases with a reduction in fractional hole area. Yanagi and Sakata [*Ind. Eng. Chem. Proc. Des. Dev.* **21**, 712 (1982)] tests in commercial-scale towers show a 5 to 15 percent increase in tray efficiency when fractional hole area was lowered from 14 to 8 percent (Fig. 14-43).

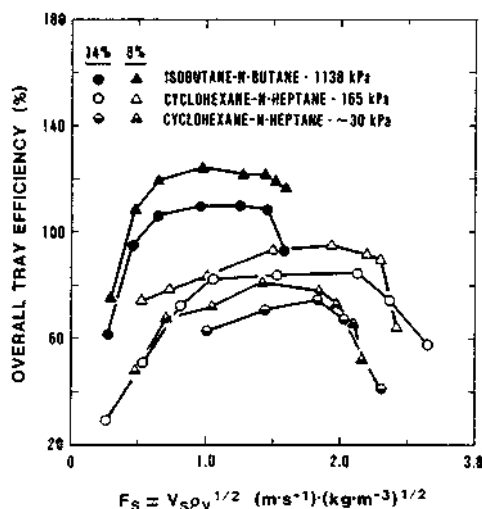


FIG. 14-43 Efficiency reduction when fractional hole area is increased, also showing little effect of vapor and liquid loads on efficiency in the normal operating range (between excessive weeping and excessive entrainment). Also shown is the small increase in efficiency with pressure. FRI data, total reflux, $D_T = 1.2$ m, $S = 610$ mm, $h_w = 50.8$ mm, $d_H = 12.7$ mm. (Reprinted with permission from T. Yanagi and M. Sakata, *Ind. Eng. Chem. Proc. Des. Dev.* **21**, 712; copyright © 1982, American Chemical Society.)

Hole Diameter The jury is out on the effect of hole diameter on tray efficiency. There is, however, a consensus that the effect of hole diameter on efficiency is small, often negligible.

Vapor-Liquid Loads and Reflux Ratio Vapor and liquid loads, as well as the reflux ratio, have a small effect on tray efficiency (Fig. 14-43) as long as no capacity or hydraulic limits (flood, weep, channeling, etc.) are violated.

Viscosity, Relative Volatility Efficiency increases as liquid viscosity and relative volatility diminish. These effects are reflected in the O'Connell correlation (below).

Surface Tension There is uncertainty regarding the effect of surface tension on tray efficiency. Often, it is difficult to divorce the surface tension effects from those of other physical properties.

Pressure Tray efficiency slightly increases with pressure (Fig. 14-43), reflecting the rise of efficiency with a reduction in liquid viscosity and in relative volatility, which generally accompany a distillation pressure increase.

At pressures exceeding 10 to 20 bar (150 to 300 psia), and especially at high liquid rates, vapor entrainment into the downcomer liquid becomes important, and tray efficiency decreases with further increases in pressure [Zuiderveg, *Int. Chem. Eng.* **26**(1), 1 (1986)].

Maldistribution Maldistribution can cause major efficiency reduction in multipass trays (>two passes). Further discussion is given under "Number of Passes."

OBTAINING TRAY EFFICIENCY

Efficiency prediction methods are listed here in decreasing order of reliability.

Rigorous Testing Rigorous testing of a plant column is generally the most reliable method of obtaining tray efficiency. Test procedures can be found elsewhere (AIChE Equipment Testing Procedures Committee, *AIChE Equipment Testing Procedure—Tray Distillation Columns*, 2d ed., 1987; Kister, *Distillation Operation*, McGraw-Hill, New York, 1990).

Scale-up from an Existing Commercial Column As long as data are for the same system under similar process conditions, loadings, and operating regime, data obtained in one column directly extend to another. Fractional hole area and the number of tray passes will have a small but significant effect on efficiency, and any changes in these parameters need to be allowed for during scale-up. The empirical information in the section "Factors Affecting Tray Efficiency" can be used to estimate the magnitude of the changes on efficiency.

Scale-up from Existing Commercial Column to Different Process Conditions During scale-up, test data are analyzed by computer simulation. The number of theoretical stages is varied until the simulated product compositions and temperature profile match the test data. Tray efficiency is determined by the ratio of theoretical stages to actual trays. In this procedure, errors in VLE are offset by compensating errors in tray efficiency. For instance, if the relative volatility calculated by the simulation is too high, fewer stages will be needed to match the measured data, i.e., "apparent" tray efficiency will be lower. Scale-up will be good as long as the VLE and efficiency errors continue to offset each other equally. This requires that process conditions (feed composition, feed temperature, reflux ratio, etc.) remain unchanged during scale-up.

When process conditions change, the VLE and efficiency errors no longer offset each other equally. If the true relative volatility is higher than simulated, then the scale-up will be conservative. If the true relative volatility is lower than simulated, scale-up will be optimistic. A detailed discussion is found in Kister, *Distillation Design*, McGraw-Hill, New York, 1992.

Experience Factors These are tabulations of efficiencies previously measured for various systems. Tray efficiency is insensitive to tray geometry (above), so in the absence of hydraulic anomalies and issues with VLE data, efficiencies measured in one tower are extensible to others distilling the same system. A small allowance to variations in tray geometry as discussed above is in order. Caution is required with mixed aqueous-organic systems, where concentration may have a marked effect on physical properties, relative volatility, and efficiency. Table 14-12 shows typical tray efficiencies reported in the literature.

TABLE 14-12 Representative Tray Efficiencies

Tray	System	Column diameter, ft	Tray spacing, in	Pressure, psia	Efficiency, %	% hole (slot) area	Ref.
Sieve	Methanol-water	3.2	15.7	14.7	70–90	10.8	2
	Ethanol-water	2.5	14	14.7	75–85	10.4	1
	Methanol-water	3.2	15.7	14.7	90–100	4.8	2
	Ethylbenzene-styrene	2.6	19.7	1.9	70	12.3	5
	Benzene-toluene	1.5	15.7	14.7	60–75	8	10
	Methanol- <i>n</i> propanol-sec butanol	6.0	18	18	64°		6
	Mixed xylene + C ₈ -C ₁₀ paraffins and naphthenes	13.0	21	25	86°		4
	Cyclohexane- <i>n</i> -heptane	4.0	24	5	60–70	14	9
		4.0	24	24	80	14	9
				5	70–80	8	8
				24	90	8	8
	Isobutane- <i>n</i> -butane	4.0	24	165	110	14	9
		4.0	24	165	120	8	8
		4.0	24	300	110	8	8
		4.0	24	400	100	8	8
	<i>n</i> -Heptane-toluene	1.5	15.7	14.7	60–75	8	10
	Methanol-water	2.0	13.6	14.7	68–72	10	11
	Isopropanol-water	2.0	13.6	14.7	59–63		11
	Toluene-methylcyclohexane	2.0	13.6	14.7	70–82		11
	Toluene stripping from water	4	24	14.7	31–42	8	13
Valve	Methanol-water	3.2	15.7	14.7	70–80	14.7	2
	Ethanol-water	2.5	14	14.7	75–85		1
	Ethyl benzene-styrene	2.6	19.7	1.9	75–85		3
	Cyclohexane- <i>n</i> -heptane	4.0	24	24	70–96°		7
	Isobutane- <i>n</i> -butane	4.0	24	165	108–121°		7
	Cyclohexane- <i>n</i> -heptane	4.0	24	24	77–93 [†]	14.7	12
				5	57–86 [†]	14.7	12
	Isobutane- <i>n</i> -butane	4.0	24	165	110–123 [†]	14.7	12
	C ₃ -C ₄ splitter	5.6	24	212	65–67 [†]	12	14

References:

1. Kirschbaum, *Destillier-Rektifiziertchnik*, 4th ed., Springer-Verlag, Berlin and Heidelberg, 1969.
2. Kastanek and Standart, *Sep. Sci.*, **2**, 439 (1967).
3. Billet and Raichle, *Chem. Ing. Tech.*, **38**, 825 (1966); **40**, 377 (1968).
4. AIChE Research Committee, *Tray Efficiency in Distillation Columns*, final report, University of Delaware, Newark, 1958.
5. Billet R., *ICHEME, Symp. Ser.* **32**, p. 4:42 (1969).
6. Mayfield et al., *Ind. Eng. Chem.*, **44**, 2238 (1952).
7. Fractionation Research, Inc. "Report of Tests of Nutter Type B Float Valve Tray," July 2, 1964 from Sulzer Chem Tech.
8. Sakata and Yanagi, *ICHEME, Eng. Symp. Ser.*, no. 56, 3.2/21 (1979).
9. Yanagi and Sakata, *Ind. Eng. Chem. Process Des. Dev.*, **21**, 712 (1982).
10. Zuiderweg and Van der Meer, *Chem. Tech. (Leipzig)*, **24**, 10 (1972).
11. Korchinsky, *Trans. I. Chem. E.*, **72**, Part A, 472 (1994).
12. Glitsch, Inc. "Glitsch Ballast Trays," Bulletin 159/160 (FRI Topical Report 15, 1958). Available from Koch-Glitsch LP, Wichita, Kans.
13. Kunesht et al., Paper presented at the AIChE Spring National Meeting, Atlanta, Ga., 1994.
14. Remesat, Chuang, and Svrcek, *Trans. I Chem. E.*, Vol. 83, Part A, p. 508, May 2005.

Notes:

*Rectangular Sulzer BDP valves.

[†]Glitsch V-1 round valves (Koch-Glitsch).

[‡]Two-pass trays, short path length.

To convert feet to meters, multiply by 0.3048; to convert inches to centimeters, multiply by 2.54; and to convert psia to kilopascals, multiply by 6.895.

Vital, Grossel and Olsen [*Hydroc. Proc.* **63**, 11, p. 147 (1984)] and Garcia and Fair [*Ind. Eng. Chem. Res.* **39**, p. 1809 (2000)] present an extensive tabulation of tray efficiency data collected from the published literature.

The *GPSA Engineering Data Book* (10th ed., Gas Processors Association, 1987) and Kaes (*Refinery Process Modeling—A Practical Guide to Steady State Modeling of Petroleum Processes Using Commercial Simulators*, Athens Printing Co., Athens, Ga., 2000) tabulate typical efficiencies in gas plant and refinery columns, respectively. Pilling (Paper presented at the 4th Topical Conference on Separations Science and Technology, November 1999, available from Sulzer Chemtech, Tulsa, Okla.) tabulated more typical efficiencies. Similar information is often available from simulation guide manuals. The quality and reliability of efficiencies from these sources vary and are generally lower than the reliability of actual measured data.

Scale-up from a Pilot- or Bench-Scale Column This is a very common scale-up. No reduction in efficiency on scale-up is expected as long as several precautions are observed. These precautions, generally relevant to pilot- or bench-scale columns, are spelled out with specific reference to the Oldershaw column.

Scale-up from Oldershaw Columns One laboratory-scale device that found wide application in efficiency investigations is the Oldershaw column [Fig. 14-44, Oldershaw, *Ind. Eng. Chem. Anal. Ed.* **13**, 265 (1941)]. This column is available from a number of laboratory supply houses and can be constructed from glass for atmospheric operation or from metal for higher pressures. Typical column diameters are 25 to 100 mm (1 to 4 in), with tray spacing the same as the column diameter.

Fair, Null, and Bolles [*Ind. Eng. Chem. Process Des. Dev.* **22**, 53 (1983)] found that efficiency measurements in Oldershaw columns closely approach the point efficiencies [Eq. (14-133)] measured in

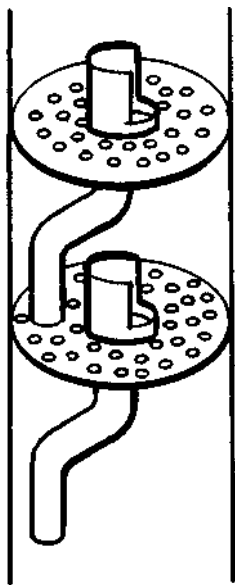


FIG. 14-44 An Oldershaw column. (From H. Z. Kister, *Distillation Design*, copyright © 1992 by McGraw-Hill; reprinted by permission.)

commercial sieve-tray columns (Fig. 14-45) providing (1) the systems being distilled are the same, (2) comparison is made at the same relative approach to the flood point, (3) operation is at total reflux, and (4) a standard Oldershaw device is used in the laboratory experimentation. Fair et al. compared several systems, utilizing as large-scale information the published efficiency studies of Fractionation Research, Inc. (FRI).

A mixing model can be used to convert the Oldershaw point efficiencies to overall column efficiencies. This enhances the commercial column efficiency estimates. A conservative approach suggested by Fair et al. is to apply the Oldershaw column efficiency as the estimate for the overall column efficiency of the commercial column, taking no credit for the greater plug-flow character upon scale-up. The author prefers this conservative approach, considering the poor reliability of mixing models.

Previous work with Oldershaw columns [Ellis, Barker, and Contractor, *Trans. Instn. Chem. Engrs.* **38**, 21 (1960)], spells an additional note of caution. Cellular (i.e., wall-supported) foam may form in pilot or Oldershaw columns, but is rare in commercial columns. For a

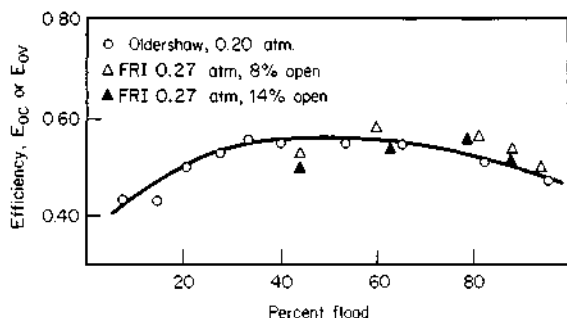


FIG. 14-45 Overall column efficiency of 25-mm Oldershaw column compared with point efficiency of 1.22-m-diameter-sieve-plate column of Fractionation Research, Inc. System = cyclohexane-*n*-heptane. [Fair, Null, and Bolles, *Ind. Eng. Chem. Process Des. Dev.*, **22**, 53 (1982).]

given system, higher Oldershaw column efficiencies were measured under cellular foam conditions than under froth conditions. For this reason, Gerster [*Chem. Eng. Progr.* **59**(3), 35 (1963)] warned that when cellular foam can form, scale-up from an Oldershaw column may be dangerous. The conclusions presented by Fair et al. do not extend to Oldershaw columns operating in the cellular foam regime. Cellular foam can be identified by lower pilot column capacity compared to a standard mixture that is visualized not to form cellular foam.

Heat losses are a major issue in pilot and Oldershaw columns and can lead to optimistic scale-up. Special precautions are needed to keep these at a minimum. Vacuum jackets with viewing ports are commonly used.

Uses of Oldershaw columns to less conventional systems and applications were described by Fair, Reeves, and Seibert [Topical Conference on Distillation, AIChE Spring Meeting, New Orleans, p. 27 (March 10–14, 2002)]. The applications described include scale-up in the absence of good VLE, steam stripping efficiencies, individual component efficiencies in multicomponent distillation, determining component behavior in azeotropic separation, and foam testing.

Empirical Efficiency Prediction Two empirical correlations which have been the standard of the industry for distillation tray efficiency prediction are the Drickamer and Bradford, in Fig. 14-46 [*Trans. Am. Inst. Chem. Eng.* **39**, 319 (1943)] and a modification of it by O'Connell [*Trans. Am. Inst. Chem. Eng.* **42**, 741 (1946)], in Fig. 14-47. The Drickamer-Bradford plot correlates efficiency as a function of liquid viscosity only, which makes it useful for petroleum cuts. O'Connell added the relative volatility to the x axis.

Lockett (*Distillation Tray Fundamentals*, Cambridge University Press, Cambridge, England, 1986) noted some theoretical sense in O'Connell's correlation. Higher viscosity usually implies lower diffusivity, and therefore greater liquid-phase resistance and lower efficiency. Higher relative volatility increases the significance of the liquid-phase resistance, thus reducing efficiency. Lockett expresses the O'Connell plot in equation form:

$$E_{OC} = 0.492(\mu_L \alpha)^{-0.245} \quad (14-138)$$

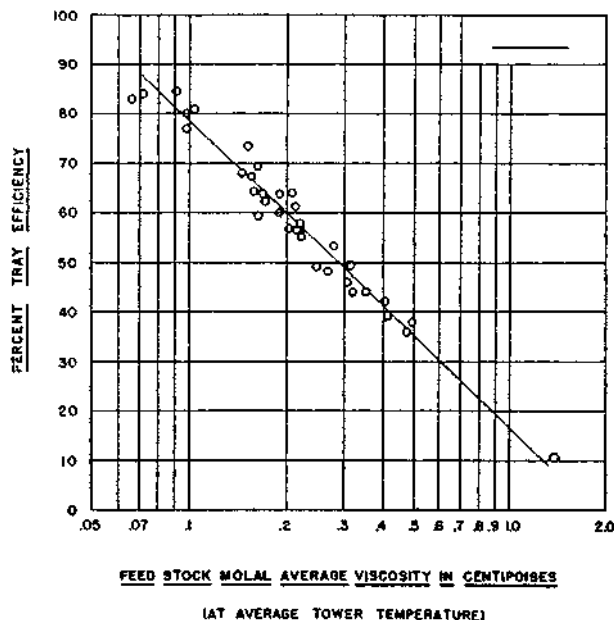


FIG. 14-46 The Drickamer and Bradford tray efficiency correlation for refinery towers. To convert centipoise to pascal-seconds, multiply by 0.0001. [From Drickamer and Bradford, *Trans. Am. Inst. Chem. Eng.* **39**, 319 (1943). Reprinted courtesy of the AIChE.]

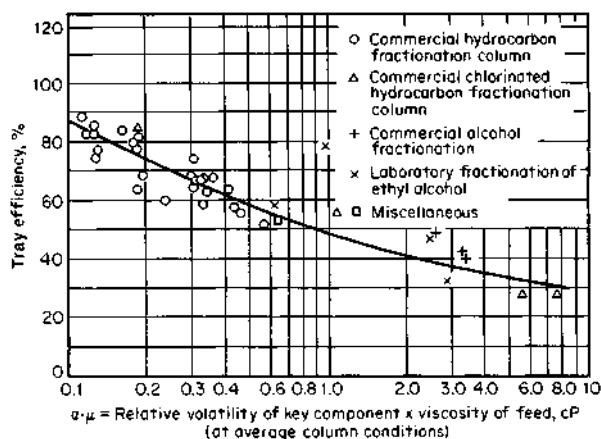


FIG. 14-47 O'Connell correlation for overall column efficiency E_{OC} for distillation. To convert centipoises to pascal-seconds, multiply by 10^{-3} . [O'Connell, Trans. Am. Inst. Chem. Eng., **42**, 741 (1946).]

(The viscosity is in cP and E_{OC} is fractional.) The volatility and viscosity are evaluated at the average arithmetic temperature between the column top and bottom temperatures. The relative volatility is between the key components.

The O'Connell correlation was based on data for bubble-cap trays. For sieve and valve trays, its predictions are likely to be slightly conservative.

Theoretical Efficiency Prediction Theoretical tray efficiency prediction is based on the two-film theory and the sequence of steps in Fig. 14-41. Almost all methods evolved from the AIChE model (AIChE Research Committee, *Bubble Tray Design Manual*, New York, 1958). This model was developed over 5 years in the late 1950s in three universities. Since then, several aspects of the AIChE model have been criticized, corrected, and modified. Reviews are given by Lockett (*Distillation Tray Fundamentals*, Cambridge University Press, Cambridge, England, 1986) and Chan and Fair [*Ind. Eng. Chem. Proc.*

Des. Dev. **23**, 814 (1984)]. An improved version of the AIChE model, which alleviated several of its shortcomings and updated its hydraulic and mass-transfer relationships, was produced by Chan and Fair.

The Chan and Fair correlation generally gave good predictions when tested against a wide data bank, but its authors also observed some deviations. Its authors described it as "tentative until more data become available." The Chan and Fair correlation is considered the most reliable fundamental correlation for tray efficiency, but even this correlation has been unable to rectify several theoretical and practical limitations inherited from the AIChE correlation (see Kister, *Distillation Design*, McGraw-Hill, New York, 1992). Recently, Garcia and Fair (*Ind. Eng. Chem. Res.* **39**, 1818, 2000) proposed a more fundamental and accurate model that is also more complicated to apply.

The prime issue that appears to plague fundamental tray efficiency methods is their tendency to predict efficiencies of 80 to 100 percent for distillation columns larger than 1.2 m (4 ft) in diameter. In the real world, most columns run closer to 60 percent efficiency. Cai and Chen (*Distillation 2003: Topical Conference Proceedings*, AIChE Spring National Meeting, New Orleans, La., March 30–April 3, 2003) show that published eddy diffusivity models, which are based on small-column work, severely underestimate liquid backmixing and overestimate plug flow in commercial-scale columns, leading to optimistic efficiency predictions. Which other limitations (if any) in the theoretical methods contribute to the mismatch, and to what degree, is unknown. For this reason, the author would not recommend any currently published theoretical tray efficiency correlation for obtaining design efficiencies.

Example 12: Estimating Tray Efficiency For the column in Example 9, estimate the tray efficiency, given that at the relative volatility near the feed point is 1.3 and the viscosity is 0.25 cP.

Solution Table 14-12 presents measurements by Billet (loc. cit.) for ethylbenzene-styrene under similar pressure with sieve and valve trays. The column diameter and tray spacing in Billet's tests were close to those in Example 9. Since both have single-pass trays, the flow path lengths are similar. The fractional hole area (14 percent in Example 9) is close to that in Table 14-12 (12.3 percent for the tested sieve trays, 14 to 15 percent for standard valve trays). So the values in Table 14-12 should be directly applicable, that is, 70 to 85 percent. So a conservative estimate would be 70 percent. The actual efficiency should be about 5 to 10 percent higher.

Alternatively, using Eq. (14-138) or Fig. 14-47, $E_{OC} = 0.492(0.25 \times 1.3)^{-0.245} = 0.65$ or 65 percent. As stated, the O'Connell correlation tends to be slightly conservative. This confirms that the 70 percent above will be a good estimate.

EQUIPMENT FOR DISTILLATION AND GAS ABSORPTION: PACKED COLUMNS

Packings are generally divided into three classes:

1. *Random or dumped packings* (Figs. 14-48 and 14-49) are discrete pieces of packing, of a specific geometric shape, that are "dumped" or randomly packed into the column shell.
2. *Structured or systematically arranged packings* (Fig. 14-50) are crimped layers of corrugated sheets (usually) or wire mesh. Sections of these packings are stacked in the column.
3. *Grids*. These are also systematically arranged packings, but instead of wire mesh or corrugated sheets, these use an open-lattice structure.

Random and structured packings are common in commercial practice. The application of grids is limited primarily to heat-transfer and wash services and/or where a high fouling resistance is required. Grids are discussed in detail elsewhere (Kister, *Distillation Design*, McGraw-Hill, New York, 1992).

Figure 14-51 is an illustrative cutaway of a packed tower, depicting typical internals. This tower has a structured-packed top bed and a random-packed bottom bed. Each bed rests on a support grid or plate. The lower bed has a holddown grid at its top to restrict packing uplift. Liquid to each of the beds is supplied by a liquid distributor. An intermediate distributor, termed a *redistributor*, is used to introduce feed and/or to remix liquid at regular height intervals. The intermediate distributor in Fig. 14-51 is not self-collecting, so a chevron collector is

used to collect the liquid from the bed above. An internal pipe passes this liquid to the distributor below. The collected liquid is mixed with the fresh feed (not shown) before entering the distributor. The reboiler return enters behind a baffle above the bottom sump.

As illustrated, the packing needs to be interrupted and a distributor added at each point where a feed enters or a product leaves. A simple distillation tower with a single feed will have a minimum of two beds, a rectifying bed and a stripping bed.

Packing Objectives The objective of any packing is to maximize efficiency for a given capacity, at an economic cost. To achieve these goals, packings are shaped to

1. *Maximize the specific surface area, i.e., the surface area per unit volume.* This maximizes vapor-liquid contact area, and, therefore, efficiency. A corollary is that efficiency generally increases as the random packing size is decreased or as the space between structured packing layers is decreased.

2. *Spread the surface area uniformly.* This improves vapor-liquid contact, and, therefore, efficiency. For instance, a Raschig ring (Fig. 14-48a) and a Pall® ring (Fig. 14-48c) of an identical size have identical surface areas per unit volume, but the Pall® ring has a superior spread of surface area and therefore gives much better efficiency.

3. *Maximize the void space per unit column volume.* This minimizes resistance to gas upflow, thereby enhancing packing capacity.

A corollary is that capacity increases with random packing size or with the space between structured packing layers. Comparing with the first objective, a tradeoff exists; the ideal size of packing is a compromise between maximizing efficiency and maximizing capacity.

4. *Minimize friction.* This favors an open shape that has good aerodynamic characteristics.

5. *Minimize cost.* Packing costs, as well as the requirements for packing supports and column foundations, generally rise with the weight per unit volume of packing. A corollary is that packings become cheaper as the size increases (random packing) and as the space between layers increases (structured packing).

Random Packings Historically, there were three generations of evolution in random packings. The first generation (1907 to the 1950s) produced two basic simple shapes—the Raschig ring and the Berl saddle (Fig. 14-48a, b) that became the ancestors of modern random packings. These packings have been superseded by more modern packing and are seldom used in modern distillation practice.

The second generation (late 1950s to the early 1970s) produced two popular geometries—the Pall® ring, which evolved from the Raschig ring, and the Intalox® saddle (Fig. 14-48c–f), which evolved from the Berl saddle.

BASF developed the Pall® ring by cutting windows in the Raschig ring and bending the window tongues inward. This opened up the ring, lowering the aerodynamic resistance and dramatically enhancing capacity. The bent tongues improved area distribution around the particle, giving also better efficiency. These improvements made the first generation Raschig rings obsolete for distillation.

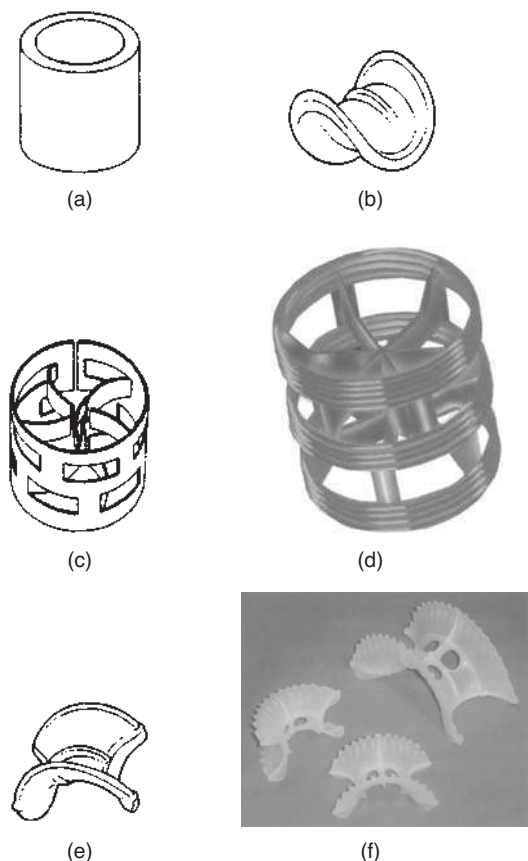


FIG. 14-48 Common first- and second-generation random packings. (a) Raschig ring (metal, plastic, ceramic). (b) Berl saddle (ceramic). (c) Pall ring (metal). (d) Pall ring (plastic). (e) Intalox saddle (ceramic). (f) Super Intalox saddle (plastic). (Parts d, f, courtesy of Koch-Glitsch LP.)

Berl saddles (ceramics) are still used due to their good breakage resistance.

The second-generation packings are still popular and extensively used in modern distillation practice. The third generation (the mid-1970s until present) has produced a multitude of popular geometries, most of which evolved from the Pall® ring and Intalox® saddle. Some are shown in Fig. 14-49. A more comprehensive description of the various packings is given elsewhere (Kister, *Distillation Design*, McGraw-Hill, New York, 1992).

The third generation of packing was a significant, yet not large, improvement over the second generation, so second-generation packings are still commonly used.

Structured Packings Structured packings have been around since as early as the 1940s. First-generation structured packings, such as Panapak, never became popular, and are seldom used nowadays.

The second generation of structured packings began in the late 1950s with high-efficiency wire-mesh packings such as Goodloe®, Hyperfil®, and the Sulzer® (wire-mesh) packings. By the early 1970s, these packings had made substantial inroads into vacuum distillation, where their low pressure drop per theoretical stage is a major advantage. In these services, they are extensively used today. Their high cost, high sensitivity to solids, and low capacity hindered their application outside vacuum distillation.

The corrugated-sheet packing, first introduced by Sulzer in the late 1970s, started a third generation of structured packings. With a high capacity, lower cost, and lower sensitivity to solids, while still retaining a high efficiency, these corrugated-sheet packings became competitive with conventional internals, especially for revamps. The 1980s saw an accelerated rise in popularity of structured packings, to the point of their becoming one of the most popular column internals in use today.

Corrugated structured packings are fabricated from thin, corrugated (crimped) metal sheets, arranged parallel to one another. The corrugated sheets are assembled into an element (Figs. 14-50a, c and 14-51). The sheets in each element are arranged at a fixed angle to the vertical. Table 14-14 contains geometric data for several corrugated packings.

Geometry (Fig. 14-52) The crimp size defines the opening between adjacent corrugated layers. Smaller B , h , and S yield narrower openings, more sheets (and, therefore, greater surface area) per unit volume, and more efficient packing, but higher resistance to gas upflow, lower capacity, and enhanced sensitivity to plugging and fouling.

The corrugations spread gas and liquid flow through a single element in a series of parallel planes. To spread the gas and liquid uniformly in all radial planes, adjacent elements are rotated so that sheets of one element are at a fixed angle to the layer below (Fig. 14-51). For good spread, element height s is relatively short (typically 200 to 300 mm, 8 to 12 in) and the angle of rotation is around 90° .

The surfaces of a few structured packings (especially those used in highly fouling environments) are smooth. Most structured packings have a roughened or enhanced surface that assists the lateral spread of liquid, promotes film turbulence, and enhances the area available for mass transfer. Texturing commonly employed is embossing and grooving (Fig. 14-50a, b).

The surfaces of most (but not all) structured packings contain holes that serve as communication channels between the upper and lower surfaces of each sheet. If the holes are too small, or nonexistent, both sides of a sheet will be wet only at low liquid rates. At high liquid rates, *sheeting* or *blanking* will cause liquid to run down the top surface with little liquid wetting the bottom surface [Chen and Chuang, *Hydroc. Proc.* 68(2), 37 (1989)], which may lower efficiency. Usually, but not always, the holes are circular (Fig. 14-50a, b), about 4 mm in diameter. Olujic et al. (Distillation 2003: Topical Conference Proceedings, p. 523, AIChE, 2003, Spring National Meeting, New Orleans, La.) showed that the hole diameter has a complex effect, strongly dependent on packing size, on both capacity and efficiency.

Inclination Angle In each element, corrugated sheets are most commonly inclined at about 45° to the vertical (typically indicated by the letter 'Y' following the packing size). This angle is large enough for good drainage of liquid, avoiding stagnant pockets and regions of liquid accumulation, and small enough to prevent gas from bypassing the metal surfaces. In some packings, the inclination angle to the vertical is steepened to 30° (typically indicated by the letter X following the

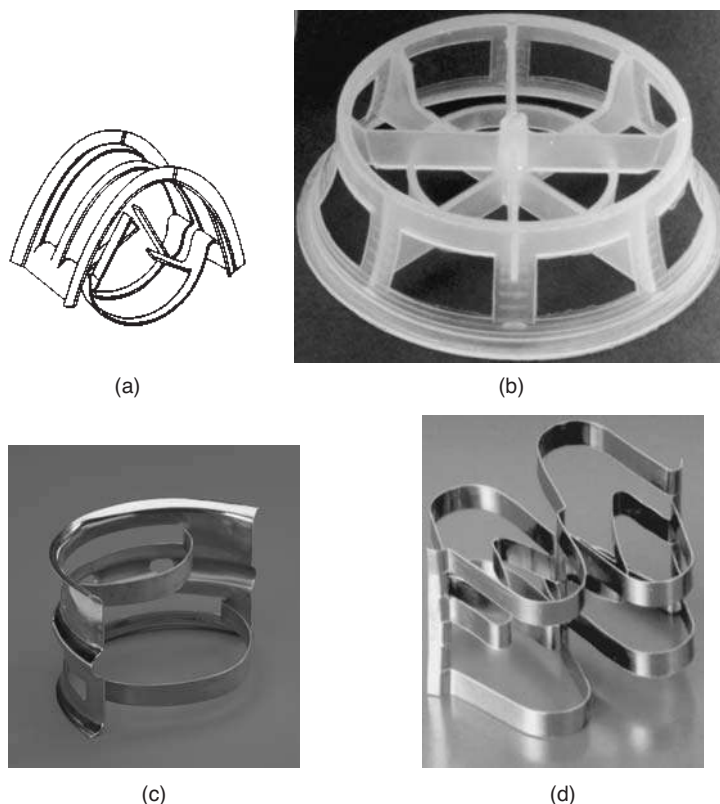


FIG. 14-49 Common third-generation random packings. (a) Intalox metal tower packing (IMTP). (b) Cascade mini-ring (CMR) (plastic). (c) Nutter ring (metal). (d) Raschig Super-Ring (metal). (Parts a, b, courtesy of Koch-Glitsch LP; part c, courtesy of Sulzer Chemtech; part d, courtesy of Raschig AG.)

packing size). This improves drainage, and therefore capacity, but at the expense of reduced gas-liquid contact, and therefore efficiency.

A recent development followed the realization that liquid drainage was restricted at the element-to-element transition rather than inside elements (Lockett and Billingham, *ICHEME Symp. Ser.* **152**, London, 2006). This means that the liquid accumulation leading to flood initiates at the transition region. A fourth generation of structured packing started, in which the main body of each element has layers inclined at 45° , but the ends of each element are almost vertical to permit drainage at this end region (Fig. 14-50d; but keep in mind that successive elements are rotated 90° rather than continuous, as shown in Fig. 14-50d). These S-shaped or high-capacity packings offer greater capacity compared to equivalent 45° inclined packings with efficiency the same with some (Pilling and Haas, Topical Conference Proceedings, p. 132, AIChE Spring Meeting, New Orleans, March 10–14, 2002; McNulty and Sommerfeldt in “Distillation: Horizons for the New Millennium,” Topical Conference Proceedings, p. 89, AIChE Spring Meeting, Houston, Tex., March 1999) and lower with others [Olujic et al., *Chem. Eng. and Proc.*, **42**, p. 55 (2003)].

PACKED-COLUMN FLOOD AND PRESSURE DROP

Pressure drop of a gas flowing upward through a packing countercurrently to liquid flow is characterized graphically in Fig. 14-53. At very low liquid rates, the effective open cross section of the packing is not appreciably different from that of dry packing, and pressure drop is due to flow through a series of variable openings in the bed. Thus, pressure drop is proportional approximately to the square of the gas

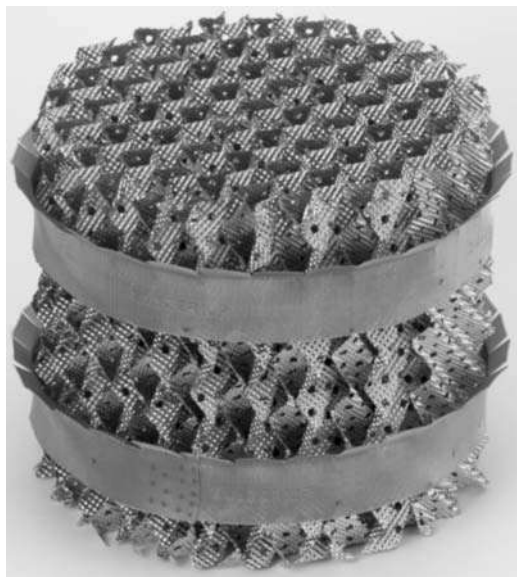
velocity, as indicated in the region AB. At higher liquid rates, the effective open cross section is smaller because of the presence of liquid (region A'B'). The pressure drop is higher, but still proportional to the square of the gas velocity.

At higher gas rates, a portion of the energy of the gas stream is used to support an increasing quantity of liquid in the column. For all liquid rates, a zone is reached where pressure drop is proportional to a gas flow rate to a power distinctly higher than 2; this zone is called the *loading zone*. The increase in pressure drop is due to the liquid accumulation in the packing voids (region BC or B'C')

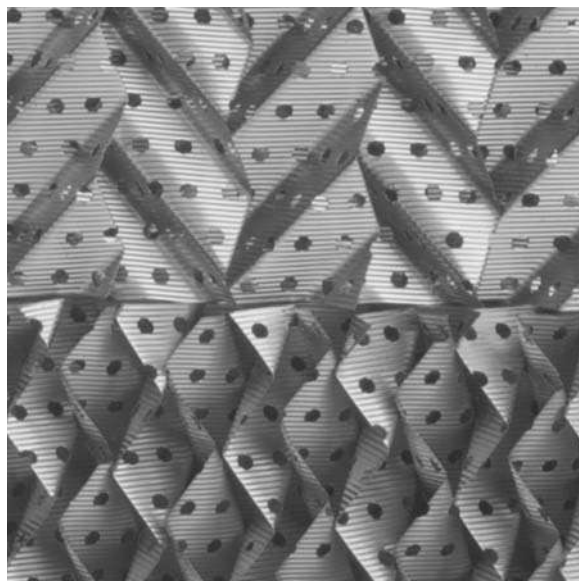
As the liquid holdup increases, the effective orifice diameter may become so small that the liquid surface becomes continuous across the cross section of the column. Column instability occurs concomitantly with a rising continuous-phase liquid body in the column. Pressure drop shoots up with only a slight change in gas rate (condition C or C'). The phenomenon is called *flooding* and is analogous to entrainment flooding in a tray column.

Alternatively, a phase inversion occurs, and gas bubbles through the liquid. The column is not unstable and can be brought back to gas-phase continuous operation by merely reducing the gas rate. A stable operating condition beyond flooding (region CD or C'D') may form with the liquid as the continuous phase and the gas as the dispersed phase [Lerner and Grove, *Ind. Eng. Chem.* **43**, 216 (1951); Teller, *Chem. Eng.* **61**(9), 168 (1954); Leung et al., *Ind. Eng. Chem. Fund.* **14** (1), 63 (1975); Buchanan, *Ind. Eng. Chem. Fund.* **15** (1), 87 (1976)].

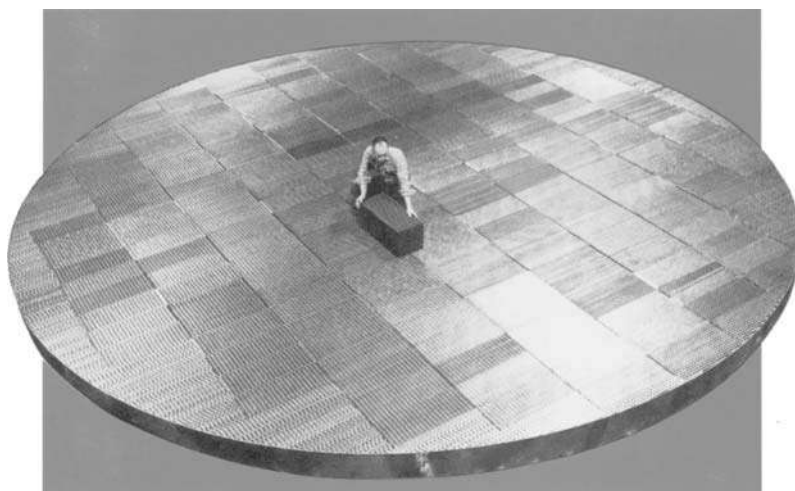
For total-reflux distillation in packed columns, regions of loading and flooding are identified by their effects on mass-transfer efficiency, as shown in Fig. 14-54. Gas and liquid rate increase together, and a



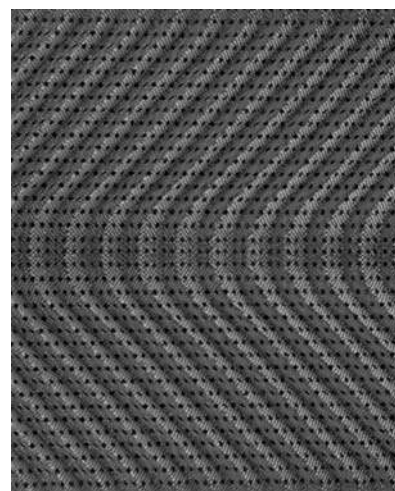
(a)



(b)



(c)



(d)

FIG. 14-50 Common structured packings. (a) A small element of Mellapak™ showing embossed surface, holes, and corrugated-sheet arrangement. (b) A closeup of the surface of Flexipac™ showing grooved surface and holes. (c) Fitting structured packing elements to a large-diameter tower. (d) Mellapak Plus™, a fourth-generation structured packing, showing a 45° inclination angle in the element and near-vertical inclination at the element-to-element transition. Note that in the tower, the successive layers will be oriented 90° to each other as in part b. (Parts a, d, courtesy of Sulzer Chemtech; parts b, c, courtesy of Koch-Glitsch LP.)

point is reached at which liquid accumulates rapidly (point B) and effective surface for mass transfer decreases rapidly.

Flood-Point Definition In 1966, Silvey and Keller [*Chem. Eng. Progr.* 62(1), 68 (1966)] listed 10 different flood point definitions that have been used by different literature sources. A later survey (Kister and Gill, *Proceedings of Chemeca* 92, p. 185-2, Canberra, Australia, 1992) listed twice that many. As Silvey and Keller pointed out, the existence of so many definitions puts into question what constitutes flooding in a packed tower, and at what gas rate it occurs. Symptoms used to identify flood in these definitions include appearance of liquid on top of the bed, excessive entrainment, a sharp rise

in pressure drop, a sharp rise in liquid holdup, and a sharp drop in efficiency. The survey of Kister and Gill suggests that most flood point definitions describe the point of flooding initiation (*incipient flooding*; point C or C' on Figs. 14-53 and 14-54). The different incipient flooding definitions gave surprisingly little scatter of flood point data (for a given packing under similar operating conditions). It follows that any definition describing flooding initiation should be satisfactory.

The author believes that due to the variations in the predominant symptom with the system and the packing, the use of multiple symptoms is most appropriate. The author prefers the following definition

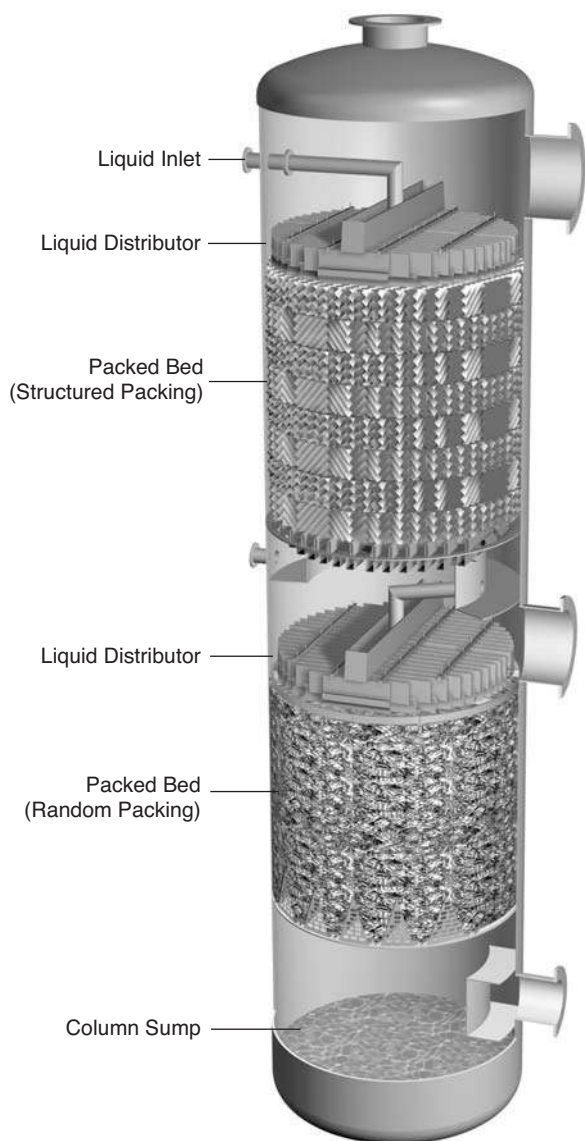


FIG. 14-51 Illustrative cutaway of a packed tower, depicting an upper bed of structured packing and a lower bed of random packing. (Courtesy of Sulzer Chemtech.)

by Fair and Bravo [*Chem. Eng. Symp. Ser.* **104**, A183 (1987)]: "A region of rapidly increasing pressure drop with simultaneous loss of mass transfer efficiency. Heavy entrainment is also recognized as a symptom of this region." An almost identical definition was presented earlier by Billet (*Distillation Engineering*, Chem. Publishing Co., New York, 1979).

The maximum operational capacity or throughput (often also referred to as maximum efficient capacity) is defined (Strigle, *Packed Tower Design and Applications*, 2d ed., Gulf Publishing, Houston, Tex., 1994) as the "Maximum vapor rate that provides normal efficiency of a packing" (i.e., point B in Fig. 14-54). The MOC is clear-cut in Fig. 14-54. On the other hand, locating the MOC in other cases is difficult and leaves a lot of room for subjectivity.

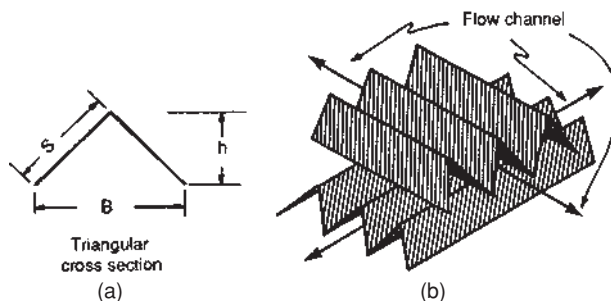


FIG. 14-52 Crimp geometry in structured packings. (a) Flow channel cross section. (b) Flow channel arrangement. (From J. R. Fair and J. L. Bravo, *Chem. Eng. Progr.*, Jan. 1990, p. 19; reproduced courtesy of the American Institute of Chemical Engineers.)

In most cases, [Kister and Gill, *Chem. Eng. Progr.* **87**(2), 32 (1991)], the velocity at which MOC is reached is related to the flood point velocity by

$$u_{s,MOC} = 0.95 u_{s,FI} \quad (14-139)$$

Flood and Pressure Drop Prediction The first generalized correlation of packed-column flood points was developed by Sherwood, Shipley, and Holloway [*Ind. Eng. Chem.*, **30**, 768 (1938)] on the basis of laboratory measurements primarily on the air-water system with random packing. Later work with air and liquids other than water led to modifications of the Sherwood correlation, first by Leva [*Chem. Eng. Progr. Symp. Ser.*, **50**(1), 51 (1954)], who also introduced the pressure drop curves, and later in a series of papers by Eckert. The generalized flooding-pressure drop chart by Eckert [*Chem. Eng. Progr.* **66**(3), 39 (1970)], included in previous editions of this handbook, was modified and simplified by Strigle (*Packed Tower Design and Applications*, 2d ed., Gulf Publishing, Houston, Tex., 1994) (Fig. 14-55). It is often called the generalized pressure drop correlation (GPDC). The ordinate is a capacity parameter [Eq. (14-140)] related to the Souders-Brown coefficient used for tray columns.

$$CP = C_s F_p^{0.5} v^{0.05} = U_s \left(\frac{\rho_G}{\rho_L - \rho_G} \right)^{0.50} F_p^{0.5} v^{0.05} \quad (14-140)$$

where U_s = superficial gas velocity, ft/s
 ρ_G , ρ_L = gas and liquid densities, lb/ft³ or kg/m³

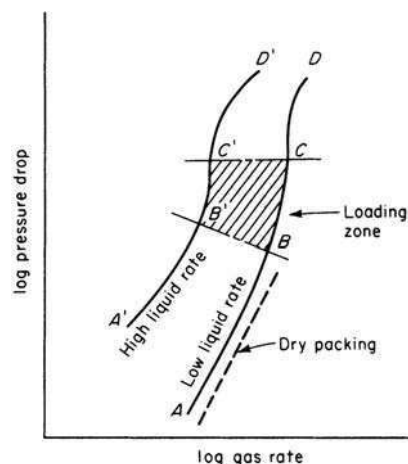


FIG. 14-53 Pressure-drop characteristics of packed columns.

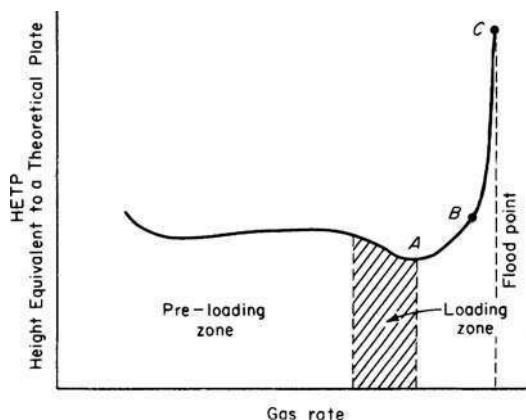


FIG. 14-54 Efficiency characteristics of packed columns (total-reflux distillation.)

F_p = packing factor, ft^{-1}

ν = kinematic viscosity of liquid, cS

C_s = C-factor, Eq. (14-77), based on tower superficial cross-sectional area, ft^2/s

CP = capacity factor, dimensional [units consistent with Eq. (14-140) and its symbols]

The abscissa scale term is the same flow parameter used for tray (dimensionless):

$$F_{LG} = \frac{L}{G} \left(\frac{\rho_G}{\rho_L} \right)^{0.5} \quad (14-141)$$

For structured packing, Kister and Gill [*Chem. Eng. Symp. Ser.* **128**, A109 (1992)] noticed a much steeper rise of pressure drop with flow parameter than that predicted from Fig. 14-55, and presented a modified chart (Fig. 14-56).

The GPDC charts in Figs. 14-55 and 14-56 do not contain specific flood curves. Both Strigle and Kister and Gill recommend calculating the flood point from the flood pressure drop, given by the Kister and Gill equation

$$\Delta P_{\text{flood}} = 0.12 F_p^{0.7} \quad (14-142)$$

Equation (14-142) permits finding the pressure drop curve in Fig. 14-55 or 14-56 at which incipient flooding occurs.

For low-capacity random packings, such as the small first-generation packings and those smaller than 1-in diameter ($F_p > 60 \text{ ft}^{-1}$), calculated flood pressure drops are well in excess of the upper pressure drop curve in Fig. 14-55. For these packings only, the original Eckert flood correlation [*Chem. Eng. Prog.* **66**(3), 39 (1970)] found in pre-1997 editions of this handbook and other major distillation texts is suitable.

The packing factor F_p is empirically determined for each packing type and size. Values of F_p , together with general dimensional data for individual packings, are given for random packings in Table 14-13 (to go with Fig. 14-55) and for structured packings in Table 14-14 (to go with Fig. 14-56).

Packing flood and pressure drop correlations should always be used with caution. Kister and Gill [*Chem. Eng. Progr.*, **87**(2), 32 (1991)] showed that deviations from the GPDC predictions tend to be systematic and not random. To avoid regions in which the systematic deviations lead to poor prediction, they superimposed experimental data points for each individual packing on the curves of the GPDC. Figure 14-57 is an example. This method requires a single chart for each packing type and size. It provides the highest possible accuracy as it interpolates measured data and identifies uncertain regions. A set of charts is in Chapter 10 of Kister's book (*Distillation Design*, McGraw-Hill, New York, 1992) with

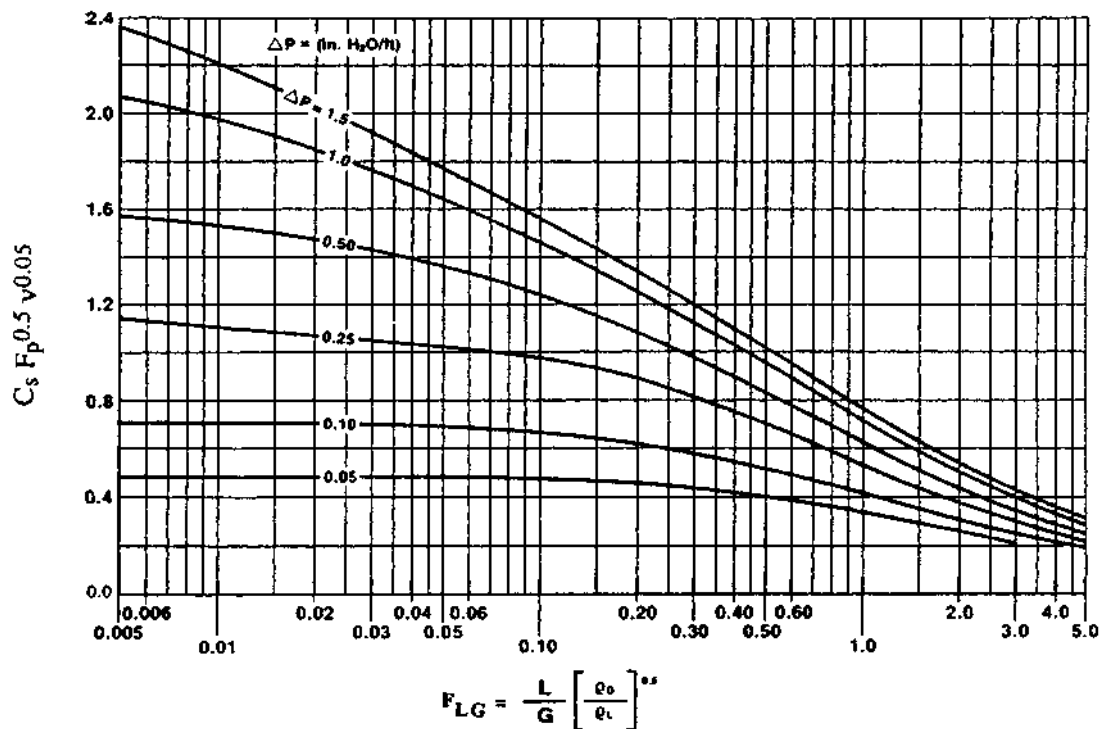


FIG. 14-55 Generalized pressure drop correlation of Eckert as modified by Strigle. To convert inches H_2O to mm H_2O , multiply by 83.31. (From *Packed Tower Design and Applications* by Ralph E. Strigle, Jr. Copyright © 1994 by Gulf Publishing Co., Houston, Texas. Used with permission. All rights reserved.)

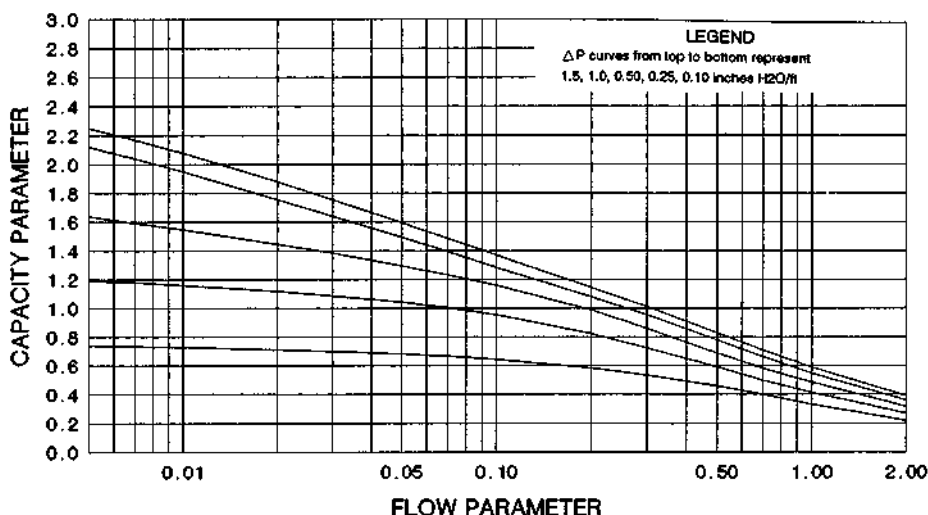


FIG. 14-56 The Kister and Gill GPDC (SP) chart for structured packings only. Abscissa and ordinate same as in Fig. 14-55. (From Kister, H. Z., and D. R. Gill, *ICHEME Symp. Ser.* **128**, p. A109, 1992. Reprinted courtesy of IChemE.)

updates in Kister, Lason, and Gill, Paper presented at the AIChE Spring National Meeting, Houston, Tex., March 19–23, 1995; and in Kister, Scherffius, Afshar, and Abkar, in *Distillation 2007: Topical Conference Proceedings*, 2007 AIChE Spring National Meeting, Houston, Texas. The latter reference also discusses correct and incorrect applications of those interpolation charts.

There are many alternative methods for flood and pressure drop prediction. The Billet and Schultes [*ICHEME Symp. Ser.* **104**, pp. A171 and B255 (1987)] and the Maćkowiak (“*Fluiddynamik von Kolonnen mit Modernen Füllkörpern und Packungen für Gas/Flüssigkeitssysteme*,” Otto Salle Verlag, Frankfurt am Main und Verlag Sauerländer Aarau, Frankfurt am Main, 1991) correlations are versions of the GPDC that take the liquid holdup into account. The Eiden and Bechtel correlation [*ICHEME Symp. Ser.* **142**, p. 757 (1997)] is a version of the GPDC in which accuracy is improved by using constants representative of packing shape instead of packing factors. The Lockett and Billingham correlation (*ICHEME Symp. Ser.* **152**, p. 400, London, 2006) uses a Wallis correlation

$$C_G^{0.5} + mC_L^{0.5} = C_{LG} \quad (14-143)$$

$$\text{where} \quad C_L = u_L [\rho_L / (\rho_L - \rho_G)]^{0.5} \quad (14-144)$$

and was shown to work well for high-surface-area ($>400 \text{ m}^2/\text{m}^3$) structured packings. Here C_G is the gas C -factor, Eq. (14-77), based on the tower superficial cross-sectional area, and m and C_{LG} are constants, available from the cited reference for some packing.

A drawback of most of these correlations (except that of Eiden and Bechtel) is the unavailability of constants for many, often most, of the modern popular packings.

The above methods apply to nonfoaming systems. Foaming systems can be handled either by applying additional derating (system) factors to the flood correlation (see Table 14-9) or by limiting the calculated pressure drop to 0.25 in of water per foot of packing (Hausch, “*Distillation Tools for the Practicing Engineer*,” Topical Conference Proceedings, p. 119, AIChE Spring Meeting, New Orleans, March 10–14, 2002).

Pressure Drop The GPDC discussed above (Figs. 14-55 and 14-56) and the Kister and Gill interpolation charts provide popular methods for calculating packing pressure drops. An alternative popular method that is particularly suitable for lower liquid loads was presented by Robbins (below).

For gas flow through dry packings, pressure drop may be estimated by use of an orifice equation. For irrigated packings, pressure drop

increases because of the presence of liquid, which effectively decreases the available cross section for gas flow (Fig. 14-53). In principle, there should be a method for correcting the dry pressure drop for the presence of liquid. This approach was used by Leva [*Chem. Eng. Progr. Symp. Ser. No. 10*, **50**, 51 (1954)]. A more recent method by Robbins [*Chem. Eng. Progr.*, p. 87 (May 1991)] utilizes the same approach and is described here. The total pressure drop is

$$\Delta P_t = \Delta P_d + \Delta P_L \quad (14-145)$$

where ΔP_t = total pressure drop, inches H_2O per foot of packing

$$\Delta P_d = \text{dry pressure drop} = C_3 G_f^2 10^{(C_4 L_f)} \quad (14-146)$$

$$\Delta P_L = \text{pressure drop due to liquid presence} \\ = 0.4 [L_f / 20,000]^{0.1} [C_3 G_f^2 10^{(C_4 L_f)}]^4 \quad (14-147)$$

$$G_f = \text{gas loading factor} = 986 F_s (F_{pd} / 20)^{0.5} \quad (14-148)$$

$$L_f = \text{liquid loading factor} = L (62.4 / \rho_L) (F_{pd} / 20)^{0.5} \mu_L^{0.1} \quad (14-149)$$

The term F_{pd} is a dry packing factor, specific for a given packing type and size. Values of F_{pd} are given in Tables 14-13 and 14-14. For operating pressures above atmospheric, and for certain packing sizes, L_f and G_f are calculated differently:

$$G_f = 986 F_s (F_{pd} / 20)^{0.5} 10^{0.3 p_c} \quad (14-150)$$

$$L_f = L (62.4 / \rho_L) (F_{pd} / 20)^{0.5} \mu_L^{0.2} \quad F_{pd} > 200 \quad (14-151a)$$

$$L_f = L (62.4 / \rho_L) (20 / F_{pd})^{0.5} \mu_L^{0.1} \quad F_{pd} < 15 \quad (14-151b)$$

The Robbins equations require careful attention to dimensions. However, use of the equations has been simplified through the introduction of Fig. 14-58. The terms L_f and G_f are evaluated, and the ΔP_L is obtained directly from the chart. Basic nomenclature for the Robbins method follows:

$$C_3 = 7.4(10)^{-8}$$

$$C_4 = 2.7(10)^{-5}$$

$$F_{pd} = \text{dry packing factor, ft}^{-1}$$

$$F_s = \text{superficial } F\text{-factor for gas, } U_f \rho_g^{0.5}, \text{ ft/s}(\text{lb/ft}^3)^{0.5}$$

$$G = \text{gas mass velocity, lb/hr-ft}^2$$

$$G_f = \text{gas loading factor, lb/hr-ft}^2$$

$$L = \text{liquid mass velocity, lb/hr-ft}^2$$

$$L_f = \text{liquid loading factor, lb/hr-ft}^2$$

$$\Delta P = \text{pressure drop, inches } \text{H}_2\text{O/ft packing} (\times 83.3 = \text{mm } \text{H}_2\text{O/m packing})$$

TABLE 14-13 Characteristics of Random Packings

Name	Size, mm, or no. (#)	Bed density, ^o kg/m ³	Area, m ² /m ³	% voids	Packing factor, m ⁻¹		Vendor
					Normal <i>F_p</i> [†]	Dry <i>F_{pd}</i> [†]	
Metals							
Pall rings (also Flexi-rings, Ballast rings, P-rings)	16	510	360	92	256	262	Various
	25	325	205	94	183	174	
	38	208	130	95	131	91	
	50	198	105	96	89	79	
	90	135	66	97	59	46	
Metal Intalox (IMTP) [also I-rings, AHPP, RSMR, MSR] [§]	25	224	207	97	134	141	Koch-Glitsch [Sulzer, Amistco, Rauschert, Montz] [§]
	40	153	151	97	79	85	
	50	166	98	98	59	56	
	70	141	60	98	39	—	
Nutter rings	#0.7	177	226	98	—	128	Sulzer
	#1	179	168	98	98	89	
	#1.5	181	124	98	79	66	
	#2	144	96	98	59	56	
	#2.5	121	83	98	52	49	
	#3.0	133	66	98	43	36	
Raschig Super-ring	#0.5	275	250	98	—	—	Raschig
	#0.7	185	175	98	—	—	
	#1	220	160	98	82	—	
	#1.5	170	115	98	59	—	
	#2	155	98	99	49	—	
	#3	150	80	98	36	—	
Cascade mini-rings (CMR)	#1	389	250	96	131	102	Koch-Glitsch
	#1.5	285	190	96	95	—	
	#2	234	151	97	72	79	
	#2.5	195	121	97	62	—	
	#3	160	103	98	46	43	
	#4	125	71	98	33	32	
	#5	108	50	98	26 [†]	—	
Fleximax	#300	—	141	98	85	—	Koch-Glitsch
	#400	—	85	98	56	—	
Jaeger Tripacks (Top-Pak)	#1	223	118	96	85	—	Raschig
	#2	170	75	98	46	—	
VSP	25	352	206	98	105 [†]	—	Raschig
	50	296	112	96	69	—	
Ralu-rings	25	310	215	98	157 [†]	—	Raschig
	38	260	135	97	92 [†]	—	
	50	200	105	98	66 [†]	—	
Hiflow	25	298	203	96	—	—	Rauschert
	50	175	92	98	52	—	
Hy-Pak, K-Pak, AIPR	#1	262	174	97	148	—	Koch-Glitsch, Amistco
	#1.5	180	118	98	95	—	
	#2	161	92	98	85	—	
	#3	181	69	98	52	—	
Raschig rings (¹ / ₁₆ in wall)	19	1500	245	80	722	—	Various
	25	1140	185	86	472	492	
	50	590	95	92	187	223	
	75	400	66	95	105	—	
Ceramics							
Berl saddles	6	900	900	60	—	2950	Various
	13	865	465	62	790	900	
	25	720	250	68	360	308	
	38	640	150	71	215	154	
	50	625	105	72	150	102	
Intalox, Flexi-saddles, Torus-saddles, Novalox	6	864	984	65	—	2720	Various
	13	736	623	71	660	613	
	25	672	256	73	302	308	
	50	608	118	76	131	121	
	75	576	92	79	72	66	
Raschig rings	6	960	710	62	—	5250	Various
	13	880	370	64	1900	1705	
	25	670	190	74	587	492	
	50	660	92	74	213	230	
	75	590	62	75	121	—	
Pall ring	25	620	220	75	350	—	Raschig
	38	540	164	78	180	—	
	50	550	121	78	142	—	
	80	520	82	78	85 [†]	—	

TABLE 14-13 Characteristics of Random Packings (Concluded)

Name	Size, mm, or no. (#)	Bed density ^a kg/m ³	Area, m ² /m ³	% voids	Packing factor, m ⁻¹		Vendor
					Normal F_p [†]	Dry F_{pd} [‡]	
Ceramics							
Hiflow rings	38	409	108	83	121	—	Rauschert
	50	405	89	82	95	—	
	70	333	52	86	49	—	
Plastics							
Pall rings	15	95	350	87	320	348	Various
	25	71	206	90	180	180	
	40	70	131	91	131	131	
	50	60	102	92	85	82	
	90	43	85	95	56	39	
Super Intalox, Flexi-saddles, Super-Torus, Novalox	25	83	207	90	131	131	Various
	50	60	108	93	92	85	
	75	48	89	96	59	46	
Cascade mini-rings (CMR)	#1A	53	185	97	98	92	Koch-Glitsch
	#2A	46	115	97	59	—	
	#3A	40	74	98	39	33	
Raschig Super-ring	#0.6	62	205	96	105 [¶]	—	Raschig
	#2	55	100	96	49	—	
Ralu-ring	15	80	320	94	230 [¶]	—	Raschig
	25	56	190	94	135	—	
	38	65	150	95	80	—	
	50	60	110	95	55	—	
	90	40	75	96	38	—	
	125	30	60	97	30 [¶]	—	
Snowflake	—	51	92	97	43	—	Koch-Glitsch
Nor-Pac	25	72	180	92	102	—	NSW
	38	61	144	93	69	—	
	50	53	102	94	46	—	
Tri-Packs (Hacketten)	#1	67	158	93	53 [¶]	—	Raschig
	#2	53	125	95	39 [¶]	43	
Ralu-Flow	#1	55	165	95	67 [¶]	—	Raschig
	#2	54	100	95	38 [¶]	—	
Hiflow	25	63	192	92	138	—	Rauschert
	50	59	110	93	66	—	
	90	34	65	97	30	—	
Lanpac Impac	90	67	148	93	46	—	Lantec
	#3	83	213	91	49	—	
Tellerettes	25	112	180	87	—	131	Ceilcote
	50	59	125	93	—	—	

^aValues are approximate and may not be consistent. Actual number depends on wall thickness and material.

[†]Packing factor F_p from Kister, *Distillation Design*, McGraw-Hill, 1992; Kister, Larson, and Gill, paper presented at the Houston AIChE meeting, March 19–23, 1995; Strigle, *Packed Tower Design and Applications*, 2d ed., Gulf Publishing Co., Houston, Tex., 1994; Kister et al., in *Distillation 2007*, Topical Conference Proceedings, 2007 AIChE Spring National Meeting, Houston, Tex.

[‡]Dry packing factor F_{pd} from Robbins, *Chem. Eng. Progr.*, **87**(1), 19 (1990).

[§]The bracketed packings are similar to, but not the same as, the Koch-Glitsch IMTP. Some of them have holes that the IMTP do not have, and others have sizes that are different and are made by different dies.

^{††}Packing factor supplied by packing vendor.

Vendors: Koch-Glitsch LLP, Wichita, Kansas; Raschig GmbH, Ludwigshafen/Rhein, Germany; Sulzer Chemtech Ltd., Winterthur, Switzerland; Rauschert Verfahrenstechnik GmbH, Steinwiesen, Germany; Amisco Separation Products Inc., Alvin, Texas; Julius Montz GmbH, Hilden, Germany; Ceilcote Co., Berea, Ohio; NSW Corp., Roanoke, Virginia; Lantec Products Inc., Agoura Hills, California.

ρ_G = gas density, lb/ft³
 ρ_L = liquid density, lb/ft³
 μ_L = liquid viscosity, cP

The Robbins correlation applies near atmospheric pressure and under vacuum, but is not suitable above 3 bar absolute. For high (>0.3) flow parameters [Eq. (14-141)], the correlation has only been tested with air-water data.

For flood and MOC predictions, Robbins recommends his pressure drop method together with Eqs. (14-142) (flood) and (14-139) (MOC).

The GPDC and Robbins correlations are empirical. Fundamental correlations are also available. Most of these use the channel model, which attributes the pressure drop to the resistance to flow in a multitude of parallel channels. The channels may have bends, expansions, and contractions. Popular application of this approach are the Rocha et al. correlation [Rocha, Bravo, and Fair, *Ind. Eng. Chem. Res.* **32**, 641 (1993)] for structured packing and the Mackowiak ("Fluiddynamik von Kolonnen mit Modernen Füllkörpern und Packungen für Gas/Flüssigkeitssysteme," Otto Salle Verlag, Frankfurt am Main und Verlag Sauerländer Aarau, Frankfurt am Main, 1991) and Billet (*Packed Column*

TABLE 14-14 Characteristics of Structured Packings

Name	Size or number	Area, m ² /m ³	% voids ^a	Packing factor, m ⁻¹		Vendor	
				Normal $F_p^†$	Dry $F_{pd}^†$		
Metals, corrugated sheets							
Mellapak	125Y	125	99	33		Sulzer	
	170Y	170	99	39			
	2Y	223	99	46			
	250Y	250	98	66			
	350Y	350	98	75			
	500Y	500	98	112			
	750Y	750	97				
	125X	125	99	16			
	170X	170	99	20			
	2X	223	99	23			
	250X	250	98	26			
	500X	500	98	82			
Mellapak Plus	202Y		99			Sulzer	
	252Y	250	98	39			
	352Y		98				
	452Y	350	98	69			
	752Y	500	98	131			
Flexipac	700Y	710	96			Koch-Glitsch	
	500Y	495	97				
	1Y	420	98	98	(105)		
	350Y	350	98				
	1.6Y	290	98	59			
	250Y	250	99				
	2Y	220	99	49	(36)		
	2.5Y	150	99				
	3.5Y	80	99	30	(15)		
	4Y	55	99	23	(10.5)		
	1X	420	98	52			
	350X	350	98				
	1.6X	290	98	33			
	250X	250	99				
	2X	220	99	23			
	2.5X	150	99				
	3X	110	99	16			
	3.5X	80	99				
	4X	55	99				
	Flexipac High-capacity	700	710	96	223		
500Z		495	97				
1Y		420	98	82			
350Y		350	98				
1.6Y		290	99	56			
250Y		250	99				
	2Y	220	99	43			
Intalox	1T	310	98	66		Koch-Glitsch	
	1.5T	250	99				
	2T	215	99	56			
	3T	170	99	43			

Name	Size or number	Area, m ² /m ³	% voids ^a	Packing factor, m ⁻¹		Vendor
				Normal $F_p^†$	Dry $F_{pd}^†$	
	4T	135	99			Raschig
	5T	90	99			
	5TX	90	99			
Super-Pak	250	250	98	55 [‡]		Raschig
	350	350	98	70 [‡]		
Ralu-Pak	250YC	250	98	66		Raschig
Rhombopac	4M	151				Kuhni
	6M	230		59		
	9M	351				
Max-Pak	0.5-in	229	98	39		Raschig
Montz-Pak	B1-125	125	97			Montz
	B1-200	200				
	B1-250	250	95	66		
	B1-350	350	93			
	BSH-250 [‡]	250	95			
	BSH-500 [‡]	500	91			
	B1-250M	250		43		
	B1-350M	350				
	B1-500M	500				
Wire Mesh						
Sulzer	AX	250	95			Sulzer
	BX	492	90	69	(52.5)	
	CY	700	85			
	BX Plus					
Wire gauze	BX	495	93			Koch-Glitsch
Montz-Pak	A3-500	500	91			Montz
Goodloe	765	1010	96			Koch-Glitsch
	773	1920	95			
	779	2640	92			
Hyperfil	2300	2300	93.6	394/230 ^{§,}	460	Knit Mesh
	1900	1900	94.8	312/180 ^{§,}		
	1400	1400	96.0	180/131 ^{§,}		
Ceramic						
Flexeramic	28	260	66	131		Koch-Glitsch
	48	160	77	79		
	88	100	86	49		
Plastic						
Mellapak	125Y	125				Sulzer
	250Y	250		72		
	125X	125				
	250X	250				
Ralu-Pak	30/160	160	92			Raschig
Multifil plastics	P1500	1500	88.5			Knit Mesh

^a% voids vary with material thickness and values may not be consistent.

[†]Packing factors from Kister, *Distillation Design*, McGraw-Hill, 1992; Kister, Larson, and Gill, paper presented at the Houston AIChE Meeting, March 19–23, 1995; and Kister et al., in *Distillation 2007: Proceedings of Topical Conference*, AIChE Spring Meeting, Houston, Tex., April 22–26, 2007.

[‡]Dry packing factors from Robbins, *Chem. Eng. Prog.*, p. 87, May 1991.

[§]These packing factors supplied by the packing vendor.

^{||}These are expanded metal packings.

^{||}First figure is for hydrocarbon service, second figure for aqueous service.

Vendors: Sulzer Chemtech Ltd., Winterthur, Switzerland; Koch-Glitsch LLP, Wichita, Kansas; Raschig GmbH, Ludwigshafen/Rhein, Germany; Julius Montz GmbH, Hilden, Germany; Knit Mesh/Enhanced Separation Technologies, Houston, Texas; Kuhni Ltd., Allschwil, Switzerland.

Analysis and Design, Ruhr University, Bochum, Germany, 1989) methods. Stichlmair et al. (*Distillation Principles and Practices*, Wiley, New York, 1998; *Gas Sep. Purif.* **3**, March 1989, p. 19) present alternative correlations using the particle model, that attributes packing pressure drop to friction losses due to drag of a particle. This is similar to the Ergun model for single-phase flow [*Chem. Eng. Prog.* **48**(2), 89 (1952)].

Example 13: Packed-Column Pressure Drop Air and water are flowing countercurrently through a bed of 2-inch metal Pall rings. The air mass velocity is 2.03 kg/s·m² (1500 lbs/hr·ft²), and the liquid mass velocity is 12.20 kg/s·m² (9000 lbs/hr·ft²). Calculate the pressure drop by the generalized pressure drop (GPDC, Fig. 14-55) and the Robbins methods. Properties: $\rho_G = 0.074$ lbs/ft³; $\rho_L = 62.4$ lbs/ft³, $\mu_L = 1.0$ cP, $v = 1.0$ cS. The packing factor $F_p = 27$ ft⁻¹. For Robbins, $F_{pd} = 24$ ft⁻¹. The flow parameter $F_{LG} = L/G (\rho_G/\rho_L)^{0.5} = (9000/1500) (0.074/62.4)^{0.5} = 0.207$. The F -factor = $F_p = U_p \rho_G^{0.5} = G/(\rho_G^{0.5} 3600) = 1500/[(0.074)^{0.5} (3600)] = 1.53$ ft/s(lb/ft³)^{0.5}.

Using the GPDC method, the capacity parameter [by Eq. (14-140)] = $U_L[\rho_C/(\rho_L - \rho_C)]^{0.5} F_p^{0.5} v^{0.05}$, which is roughly equivalent to

$$\frac{F_L}{\rho_L^{0.5}} F_p^{0.5} v^{0.05} = \frac{1.53}{62.4^{0.5}} 27^{0.5} (1.0) \\ = 1.01$$

Referring to Fig. 14-55, the intersection of the capacity parameter and the flow parameter lines gives a pressure drop of 0.38 inches H_2O/ft packing.

Using the Robbins method, $G_f = 986 F_L (F_{pL}/20)^{0.5} = 986 (1.53) (24/20)^{0.5} = 1653$. $L_f = L (62.4/\rho_L) (F_{pL}/20)^{0.5} \mu^{0.1} = 9000 (1.0) (1.095) (1.0) = 9859$. $L_f/G_f = 5.96$.

From Fig. 14-58, pressure drop = 0.40 in. H_2O/ft packing.

PACKING EFFICIENCY

HETP vs. Fundamental Mass Transfer The two-film model gives the following transfer unit relationship:

$$H_{OC} = H_G + \lambda H_L \quad (14-152)$$

where H_{OC} = height of an overall transfer unit, gas concentration basis, m

H_G = height of a gas-phase transfer unit, m

H_L = height of a liquid-phase transfer unit, m

$\lambda = m/(L_M/G_M)$ = slope of equilibrium line/slope of operating line

In design practice, a less rigorous parameter, HETP, is used to express packing efficiency. The HETP is the height of packed bed required to achieve a theoretical stage. The terms H_{OC} and HETP may be related under certain conditions:

$$\text{HETP} = H_{OC} \left[\frac{\ln \lambda}{(\lambda - 1)} \right] \quad (14-153)$$

$$\text{and since } Z_p = (H_{OC})(N_{OC}) = (\text{HETP})(N_t) \quad (14-154)$$

$$N_{OC} = N_t [\ln \lambda / (\lambda - 1)] \quad (14-155)$$

Equations (14-153) and (14-155) have been developed for binary mixture separations and hold for cases where the operating line and equilibrium line are straight. Thus, when there is curvature, the equations should be used for sections of the column where linearity can be assumed. When the equilibrium line and operating line have the same slope, $\text{HETP} = H_{OC}$ and $N_{OC} = N_t$ (theoretical stages).

An alternative parameter popular in Europe is the NTSM (number of theoretical stages per meter) which is simply the reciprocal of the HETP.

Factors Affecting HETP: An Overview Generally, packing efficiency increases (HETP decreases) when the following occur.

- Packing surface area per unit volume increases. Efficiency increases as the particle size decreases (random packing, Fig. 14-59) or as the channel size narrows (structured packing, Fig. 14-60).
- The packing surface is better distributed around a random packing element.
- Y structured packings (45° inclination) give better efficiencies than X structured packings (60° inclination to the horizontal) of the same surface areas (Fig. 14-60).
- For constant L/V operation in the preloading regime, generally liquid and vapor loads have little effect on random and most corrugated sheet structured packings HETP (Figs. 14-59 and 14-60). HETP increases with loadings in some wire-mesh structured packing.
- Liquid and vapor are well distributed. Both liquid and vapor maldistribution have a major detrimental effect on packing efficiency.

- Other. These include L/V ratio (λ), pressure, and physical properties. These come into play in some systems and situations, as discussed below.

HETP Prediction HETP can be predicted from mass-transfer models, rules of thumb, and data interpolation.

Mass-Transfer Models Development of a reliable mass-transfer model for packing HETP prediction has been inhibited by a lack of understanding of the complex two-phase flow that prevails in packings, by the shortage of commercial-scale efficiency data for the newer packings, and by difficulty in quantifying the surface generation in modern packings. Bennett and Ludwig (*Chem. Eng. Prog.*, p. 72, April 1994) point out that the abundant air-water data cannot be reliably used for assessing real system mass-transfer resistance due to variations in turbulence, transport properties, and interfacial areas. More important, the success and reliability of rules of thumb for predicting packing efficiency made it difficult for mass-transfer models to compete.

For random packings, the Bravo and Fair correlation [*Ind. Eng. Chem. Proc. Des. Dev.* **21**, 162 (1982)] has been one of the most popular theoretical correlations. It was shown (e.g., McDougall, *Chem SA*, p. 255, October 1985) to be better than other theoretical correlations, yet produced large discrepancies when compared to test data [Shariat and Kunes, *Ind. Eng. Chem. Res.* **34**(4), 1273 (1995)]. For structured packings, the Bravo, Fair, and Rocha correlation [*Chem. Eng. Progr.* **86**(1), 19 (1990); *Ind. Eng. Chem. Res.* **35**, 1660 (1996)] is one of the most popular theoretical correlations. This correlation is based on the two-film theory. Interfacial areas are calculated from the packing geometry and an empirical wetting parameter.

Alternate popular theoretical correlations for random packings, structured packings, or both (e.g., Billet and Schultes, "Beitrage zur Verfahrens- und Umwelttechnik," p. 88, Ruhr Universität, Bochum, Germany, 1991) are also available.

Rules of Thumb Since in most circumstances packing HETP is sensitive to only few variables, and due to the unreliability of even the best mass-transfer model, it has been the author's experience that rules of thumb for HETP are more accurate and more reliable than mass-transfer models. A similar conclusion was reached by Porter and Jenkins (*ICHEME Symp. Ser.* **56**, Summary paper, London, 1979).

The majority of published random packing rules of thumb closely agree with one another. They are based on second- and third-generation random packings and should not be applied to the obsolete first-generation packings. Porter and Jenkins's (loc. cit.), Frank's (*Chem. Eng.*, p. 40, March 14, 1977), Harrison and France's (*Chem. Eng.*, p. 121, April 1989), Chen's (*Chem. Eng.*, p. 40, March 5, 1984), and Walas' (*Chem. Eng.*, p. 75, March 16, 1987) general rules of thumb are practically the same, have been successfully tested against an extensive data bank, and are slightly conservative, and therefore suitable for design.

For small-diameter columns, the rules of thumb presented by Frank (loc. cit.), Ludwig (*Applied Process Design for Chemical and Petrochemical Plants*, vol. 2, 2d ed., Gulf Publishing, Houston, Tex., 1979), and Vital et al. [*Hydrocarbon Processing*, **63**(12), 75 (1984)] are identical. The author believes that for small columns, the more conservative value predicted from either the Porter and Jenkins or the Frank-Ludwig-Vital rule should be selected for design. Summarizing:

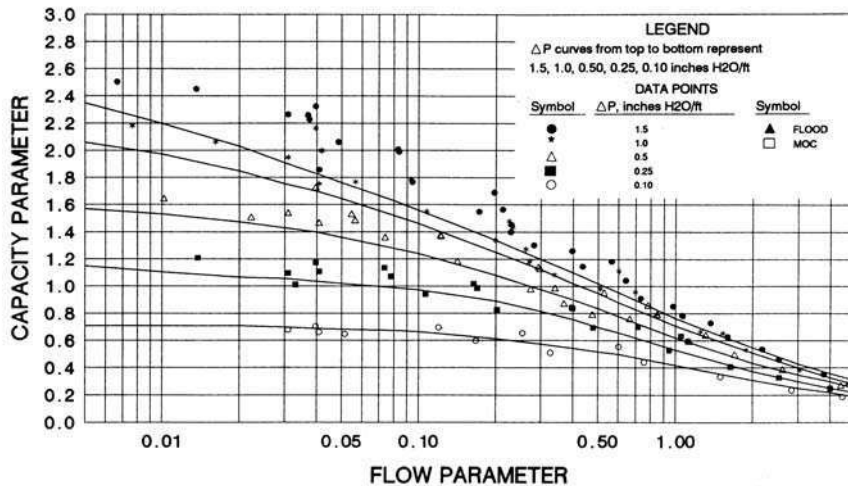
$$\text{HETP} = 18D_p \quad (14-156a)$$

$$\text{HETP} > D_T \quad \text{for } D_T < 0.67 \text{ m} \quad (14-156b)$$

where D_p and D_T are the packing and tower diameters, m, respectively, and the HETP is in meters. In high-vacuum columns (<100 mbar), and when $\lambda = mG_M/L_M$ is outside the range of 0.5 to 2.0, the above rules may be optimistic (see below).

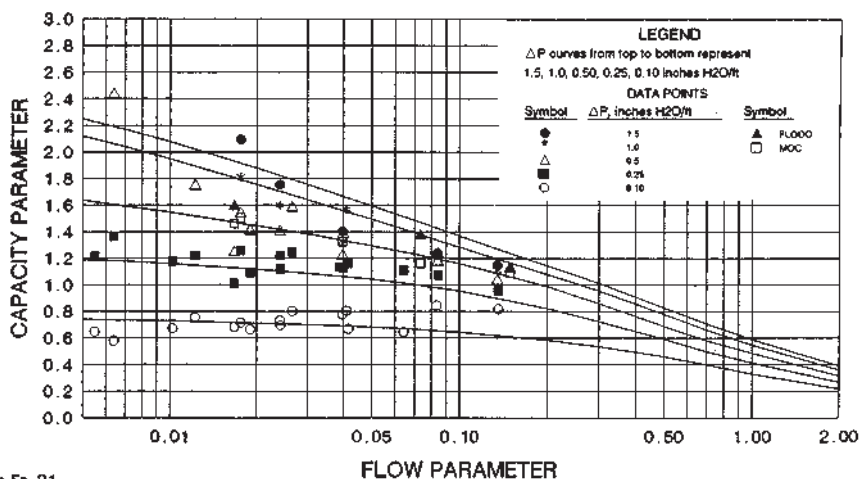
The above rules of thumb were based on experience with Pall rings. The packing diameter may be difficult to establish for some of the modern packings, especially those of saddle or flat shape. For these,

1" (M) PALL RINGS PRESSURE DROP - AQUEOUS SYSTEMS

Basis: $F_p=56$ Pressure drop measured in inches H₂O/ft

(a)

KOCH - SULZER BX PACKING FLOOD & PRESSURE DROP

Basis: $F_p=21$ Pressure drop measured in inches H₂O/ft

Large symbols represent non-aqueous data

(b)

FIG. 14-57 Superimposing experimental pressure-drop data for a given packing generates a GPDC interpolation chart for this packing. (a) A random packing; chart is based on Eckert's GPDC, Fig. 14-55. (b) A structured packing; chart is based on Kister and Gill's GPDC (SP), Fig. 14-56. (From Kister, H. Z., *Distillation Design*, copyright © McGraw-Hill, 1992; used with permission.)

Kister and Larson (in Schweitzer, *Handbook of Separation Techniques for Chemical Engineers*, 3d ed., McGraw-Hill, 1997) extended Eq. (14-156a) by expressing the packing diameter in terms of the more fundamental surface area per unit volume a_p , m²/m³. For Pall rings, it can be shown that

$$a_p = 5.2/D_p \quad (14-157)$$

and Eq. (14-156a) becomes

$$\text{HETP} = 93/a_p \quad (14-158)$$

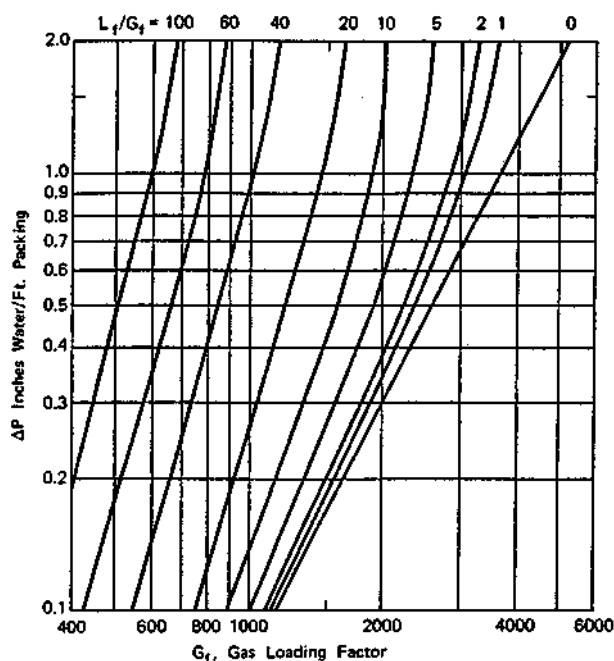


FIG. 14-58 The Robbins generalized pressure-drop correlation. (From L. A. Robbins, *Chem Eng. Progr.*, May 1991, p. 87, reprinted courtesy of the American Institute of Chemical Engineers.)

Harrison and France (loc. cit.) presented the only published rule of thumb for structured packings efficiency as a function of packing crimp. Kister and Larson reexpressed it in terms of the surface area per unit volume to accommodate a wider range of packing geometries. The final expression is

$$\text{HETP} = 100C_{XY}/a_p + 0.10 \quad (14-159)$$

Specific surface areas are listed in Tables 14-13 and 14-14.

The above rules of thumb apply to organic and hydrocarbon systems, whose surface tensions are relatively low ($\sigma < 25$ mN/m). For higher surface tensions, the liquid does not adhere well to the packing surfaces (*underwetting*), causing higher HETPs. In a water-rich system ($\sigma = 70$ mN/m or so) HETPs obtained from Eqs. (14-156), (14-158), and (14-159) need to be doubled. For intermediate surface tension systems (some amines and glycols, whose surface tension at column conditions is 40 to 50 mN/m), HETPs obtained from Eqs. (14-156), (14-158), and (14-159) need to be multiplied by 1.5.

For random packings, Eqs. (14-156) and (14-158) apply for packings of 25-mm diameter and larger. For smaller packings, use of the a_p at 25-mm often gives a slightly conservative HETP estimate. For structured packing, C_{XY} in Eq. (14-159) reflects the effect of the inclination angle (Fig. 14-60). $C_{XY} = 1$ for Y-type, S-type, or high-capacity packings, and $C_{XY} = 1.45$ for the larger (< 300 m²/m³) X-type packings. There are insufficient data to determine C_{XY} for high specific area X-type packings, but Fig. 14-60 suggests it is somewhat lower than 1.45.

Compared to experimental data, the above rules of thumb are slightly conservative. Since packing data are usually measured with perfect distribution, a slight conservative bias is necessary to extend

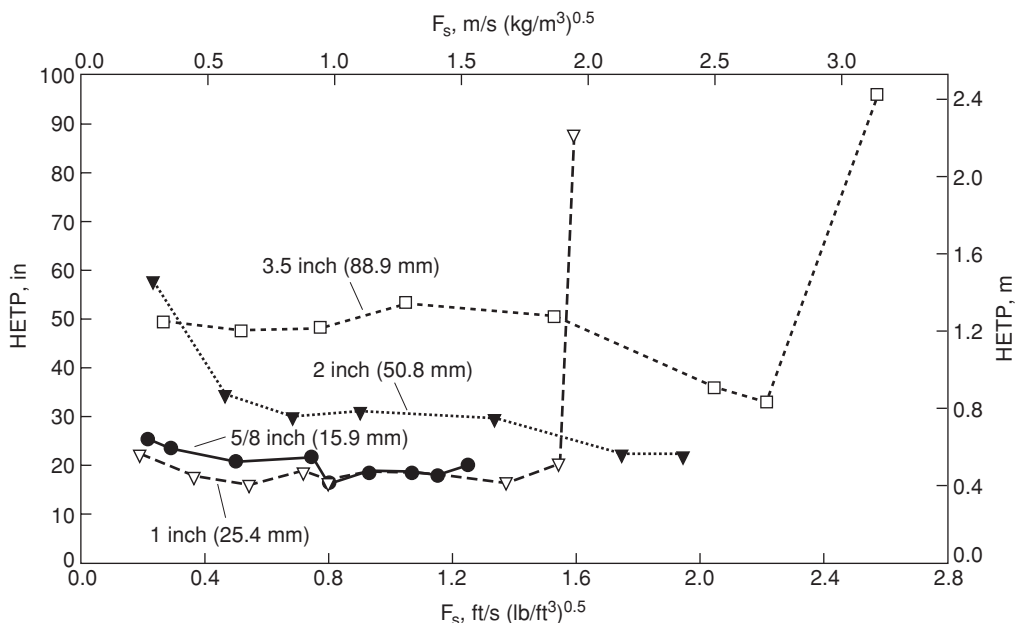


FIG. 14-59 HETP values for four sizes of metal pall rings, vacuum operation. Cyclohexane/*n*-heptane system, total reflux, 35 kPa (5.0 psia). Column diameter = 1.2 m (4.0 ft). Bed height = 3.7 m (12 ft). Distributor = tubed drip pan, 100 streams/m². [Adapted from Shariat and Kunes, *Ind. Eng. Chem. Res.*, **34**, 1273 (1995). Reproduced with permission. Copyright © 1995 American Chemical Society.]

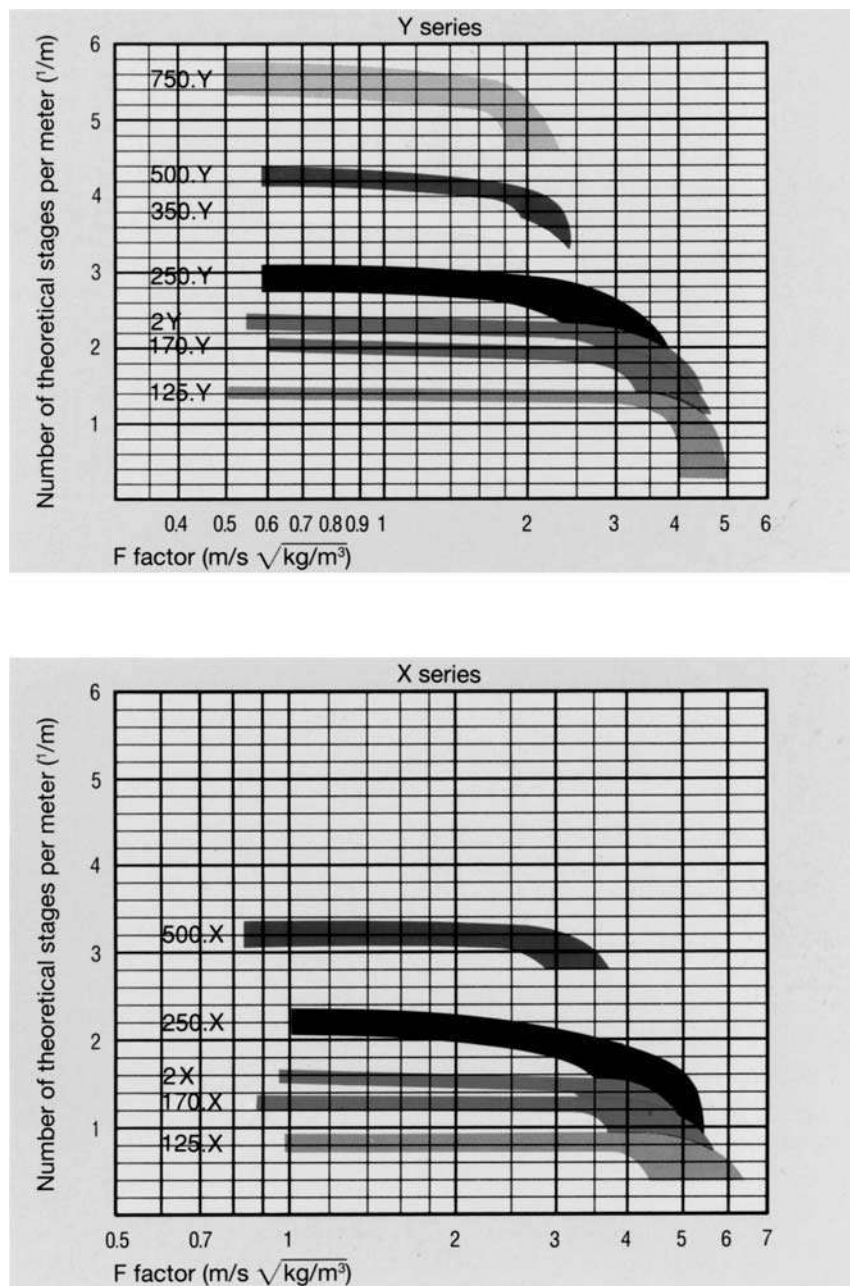


FIG. 14-60 Effect of structured packing surface areas, loads, and inclination angle on packing efficiency. Efficiency expressed as number of theoretical stages per meter, the reciprocal of HETP. Sulzer data, chlorobenzene-ethylbenzene, 100 mbar, at total reflux; 250-mm-diameter test column. (Reprinted courtesy of Sulzer Chemtech.)

these data to the good, yet imperfect, distributors used in the real world. For poor distributors, the above rules of thumb will usually predict well below the HETPs measured in the field.

Lockett (*Chem. Eng. Progr.*, p. 60, January 1998) simplified the fundamental Bravo-Fair-Rocha correlation [*Ind. Eng. Chem. Res.* **35**, p. 1660 (1996)] to derive an alternative rule of thumb for structured packing efficiency. This rule of thumb predicts HETPs under perfect distribution conditions. Lockett recommends caution when applying

this rule of thumb for aqueous systems as it does not predict the effects of underwetting (below).

Service-Oriented Rules of Thumb Strigle (*Packed Tower Design and Applications*, 2d ed., Gulf Publishing, Houston, Tex., 1994) proposed a multitude of rules of thumb as a function of the service, column pressure, and physical properties. These rules are based on the extensive experience of Strigle and the Norton Company (now merged with Koch-Glitsch LP).

Data Interpolation Interpolation of experimental HETP data is the most reliable means of obtaining design HETP values. This is hardly surprising in an area where our understanding of the theory is so poor that rules of thumb can do better than theoretical models. The author believes that it is best to derive HETP from experimental data, and to check it against a rule of thumb.

Eckert [*Chem. Eng. Progr.* **59**(5), 76 (1963)], Chen [*Chem. Eng. p.* 40, March 5, 1984], and Vital et al. [*Hydroc. Proc.* **63**(12), 75 (1984)] tabulated experimental HETP data for various random packings. Kister (*Distillation Design*, McGraw-Hill, 1992) extended these tabulations and included published HETP data and a detailed procedure for interpolating such HETP data. A prerequisite to any interpolation of packing data is thorough familiarity with the factors that affect HETP. Overlooking any of the factors listed can easily lead to poor interpolation and grossly incorrect design. In particular, it is imperative to recognize that the quality of distribution in pilot towers is generally superior to the quality of distribution in commercial towers.

Underwetting Laboratory- and pilot-scale distillation experiments with systems that exhibit large differences in surface tension along the column such as methanol-water showed a sharp drop in efficiency at the high-surface-tension end of the column [Ponter et al., *Trans. Instn. Chem. Engineers [London]*, **45**, T345 (1967)]. There appeared to be a critical methanol composition below which performance deteriorated rapidly. The poor performance at the low-methanol-concentration end appeared independent of the type and size of packing. Visual observations with disk columns attributed these effects to underwetting.

Underwetting is a packing surface phenomenon, which breaks up liquid film. The tendency of the liquid film to break (the degree of wetting) is expressed by the contact angle (Fig. 14-61). A contact angle of 0° indicates perfect wetting; an angle of 180° indicates no wetting. Mersmann and Deixler [*Chem. Ing. Tech.* **58**(1), 19 (1986)] provide a preliminary chart for estimating contact angles. The contact angle depends on both the surface and the liquid and is a strong function of composition. In systems with large surface tension gradients, both contact angles and minimum wetting rates may vary rapidly with changes of composition or surface tension. Extensive studies by Ponter et al. [loc. cit.; also, Ponter and Au-Yeung, *Chem. Ing. Tech.*, **56**(9), 701 (1984)] showed that

- Underwetting is most significant in aqueous-organic systems, and tends to occur at the high-surface-tension (aqueous) end of the composition range. Liquid viscosity may also have an effect.
- Underwetting may be alleviated by changing the material and surface roughness of the packing.
- In systems susceptible to underwetting, column efficiency can sometimes (but not always) be improved by the addition of small amounts of surfactants.

Effect of Lambda Most packed-column efficiency testing has been at total reflux. Some tests for both random and structured packings [Billet, "Packed Towers Analysis and Design," Ruhr University, Bochum, Germany, 1989; Meier, Hunkeler, and Stocker, *ICHEME Symp. Ser.* **56**, 3.3/1 (1979); Eckert and Walter, *Hydroc. Proc.* **43**(2), 107 (1964)] suggest that efficiencies at finite reflux are similar to those at total reflux when lambda ($\lambda = mG_M/L_M$, which is the ratio of the slope of the equilibrium curve to the slope of the operating line) ranges between 0.5 and 2.0. This range is typical for most distillation systems.

Koshy and Rukovena [*Hydroc. Proc.*, **65**(5), 64 (1986)], experimenting with methanol-water and water-DMF using #25 IMTP pack-

ing in a pilot-scale column, observed a sharp efficiency drop when the group λ was greater than 2 or lower than 0.5. The efficiency loss escalated as λ deviated more from this range. Koshy and Rukovena recognized that surface tension gradients and underwetting may have influenced some of their findings, but argue that the lambda effect is the major cause for the efficiency differences observed in their tests. High-relative-volatility systems are those most likely to be affected by λ , because at low volatility, λ ranges from 0.5 to 2. Strigle (loc. cit.) quantified the lambda effect on HETP using the following equation:

$$\text{Actual HETP/standard HETP} = 1 + 0.278[\text{ABS}(\ln \lambda)^3] \quad (14-160)$$

For $0.5 < \lambda < 2$, Eq. (14-160) gives a ratio of less than 1.1; that is, it has a little influence on HETP.

Pressure Generally, pressure has little effect on HETP of both random and structured packing, at least above 100 mbar abs. At deep vacuum (<100 mbar), there are data to suggest that efficiency decreases as pressure is lowered for random packings [Zelvinski, Titov, and Shalygin, *Khim. Tekhnol. Topl. Masel.* **12**(10) (1966)], but most of these data can also be explained by poor wetting or maldistribution.

At high pressures (>15 to 20 bar), structured packing efficiency diminishes as pressure is raised [Fitz, Shariat, and Spiegel, Paper presented at the AIChE Spring National Meeting, Houston, Tex., March 1995]. Zuidervog and Nutter [*ICHEME Symp. Ser.* **128**, A481 (1992)] report the same phenomenon, but to a much smaller extent, also in random packings. They explain the efficiency reduction at higher pressure by vapor backmixing. Nooijen et al. [*ICHEME Symp. Ser.* **142**, 885 (1997)] concur, bringing a supporting case study in which high-pressure distillation efficiency improved by splitting a packed bed.

With structured packings (only) FRI's high-pressure (10 to 30 bar; flow parameters >0.25) distillation tests measured maxima, termed *humps* in the HETP vs. load plot, typically at 65 to 90 percent of flood [Fitz, Shariat, and Kunesh, *ICHEME Symp. Ser.* **142**, 829 (1997); Cai et al., *Trans. IChemE* **81**, Part A, p. 85 (2003)]. These humps (Fig. 14-62) were not observed with lower pressure distillation (flow parameters <0.2) and appeared to intensify with higher pressure. The humps did not always occur; some tests at different distributor positioning and with larger packing showed no humps. Zuidervog et al. [*ICHEME Symp. Ser.* **142**, 865 (1997); *Trans. IChemE* **81**, Part A, p. 85 (January 2003)] and Nooijen et al. [*ICHEME Symp. Ser.* **142**, 885 (1997)] explain the humps by two-phase backmixing. At the high liquid loads in high-pressure distillation, flow unevenness prematurely floods some of the packing channels, carrying vapor bubbles downward and recirculating liquid upward.

Physical Properties Data presented by a number of workers [e.g., Vital, Grossel, and Olsen, *Hydroc. Proc.* **63**(12), 75 (1984)] suggest that, generally, random packing HETP is relatively insensitive to system properties for nonaqueous systems. A survey of data in Chapter

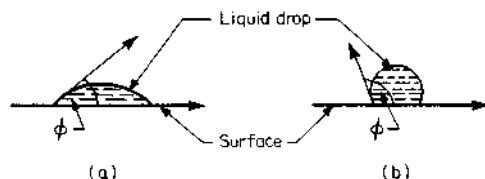


FIG. 14-61 Contact angles. (a) Acute, good wetting. (b) Obtuse, poor wetting.

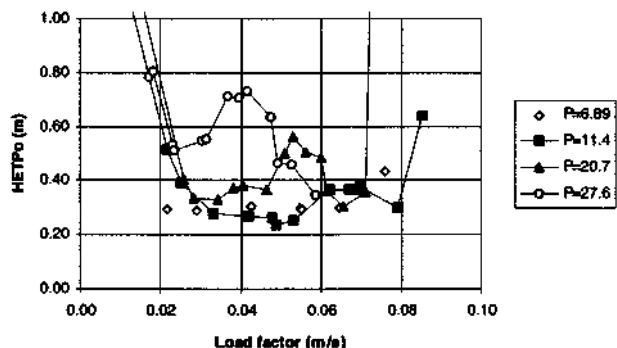


FIG. 14-62 HETP data as measured in the FRI column for the iC_4/nC_4 system at different pressures (bara), showing efficiency humps. (From J. L. Nooijen, K. A. Kusters, and J. J. B. Pek, *ICHEME Symp. Ser.* **142**, p. 885, 1997. Reprinted courtesy of IChemE.)

11 of Kister's *Distillation Design* (McGraw-Hill, New York, 1992) leads to a similar conclusion for structured packings. For water-rich systems, packing HETPs tend to be much higher than for nonaqueous systems due to their high lambda or surface underwetting, as discussed above. High hydrogen concentrations (>30 percent or so in the gas) have also led to low packing efficiencies (Kister et al., *Proc. 4th Ethylene Producers Conference*, AIChE, New Orleans, La., p. 283, 1992), possibly due to the fast-moving hydrogen molecule dragging heavier molecules with it as it diffuses from a liquid film into the vapor.

Errors in VLE These affect packing HETP in the same way as they affect tray efficiency. The discussions and derivation earlier in this subsection apply equally to tray and packed towers.

Comparison of Various Packing Efficiencies for Absorption and Stripping In past editions of this handbook, extensive data on absorption/stripping systems were given. Emphasis was given to the following systems:

Ammonia-air-water	Liquid and gas phases contributing; chemical reaction contributing
Air-water	Gas phase controlling
Sulfur dioxide-air-water	Liquid and gas phase controlling
Carbon dioxide-air-water	Liquid phase controlling

The reader may refer to the data in the 5th edition. For the current work, emphasis will be given to one absorption system, carbon dioxide-air-caustic.

Carbon Dioxide-Air-Caustic System The vendors of packings have adopted this system as a "standard" for comparing the performance of different packing types and sizes for absorption/stripping. For tests, air containing 1.0 mol % CO₂ is passed countercurrently to a circulating stream of sodium hydroxide solution. The initial concentration of NaOH in water is 1.0 N (4.0 wt %), and as the circulating NaOH is converted to sodium carbonate it is necessary to make a mass-transfer correction because of reduced mass-transfer rate in the liquid phase. The procedure has been described by Eckert et al. [*Ind. Eng. Chem.*, **59**(2), 41 (1967); *Chem. Eng. Progr.*, **54**(1), 790 (1958)]. An overall coefficient is measured using gas-phase (CO₂) concentrations:

$$K_{OGa_e} = \frac{\text{moles CO}_2 \text{ absorbed}}{\text{time-bed volume-partial pressure CO}_2 \text{ driving force}} \quad (14-161)$$

The coefficients are usually corrected to a hydroxide conversion of 25 percent at 24°C. For other conversions, Fig. 14-14 may be used. Reported values of K_{OGa} for representative random packings are given in Table 14-15. The effect of liquid rate on the coefficient is shown in Fig. 14-63.

While the carbon dioxide/caustic test method has become accepted, one should use the results with caution. The chemical reaction masks

TABLE 14-15 Overall Coefficients for Representative Packings

CO ₂ -air-caustic system		
	Nominal size, mm	Overall coefficient K_{OGa} , kg-moles/(hr-m ³ -atm)
Ceramic raschig rings	25	37.0
	50	26.1
Ceramic Intalox saddles	25	45.1
	50	30.1
Metal pall rings	25	49.6
	50	34.9
Metal Intalox saddles (IMTP®)	25	54.8
	50	39.1

NOTE: Basis for reported values: CO₂ concentration in inlet gas, 1.0 vol %; 1N NaOH solution in water, 25 percent NaOH conversion; temperature = 24°C; atmospheric pressure: gas rate = 1.264 kg/(s-m²); liquid rate = 6.78 kg/(s-m²).

SOURCE: Strigle, R. L., *Packed Tower Design and Applications*, 2d ed., Gulf Publ. Co., Houston, 1994.

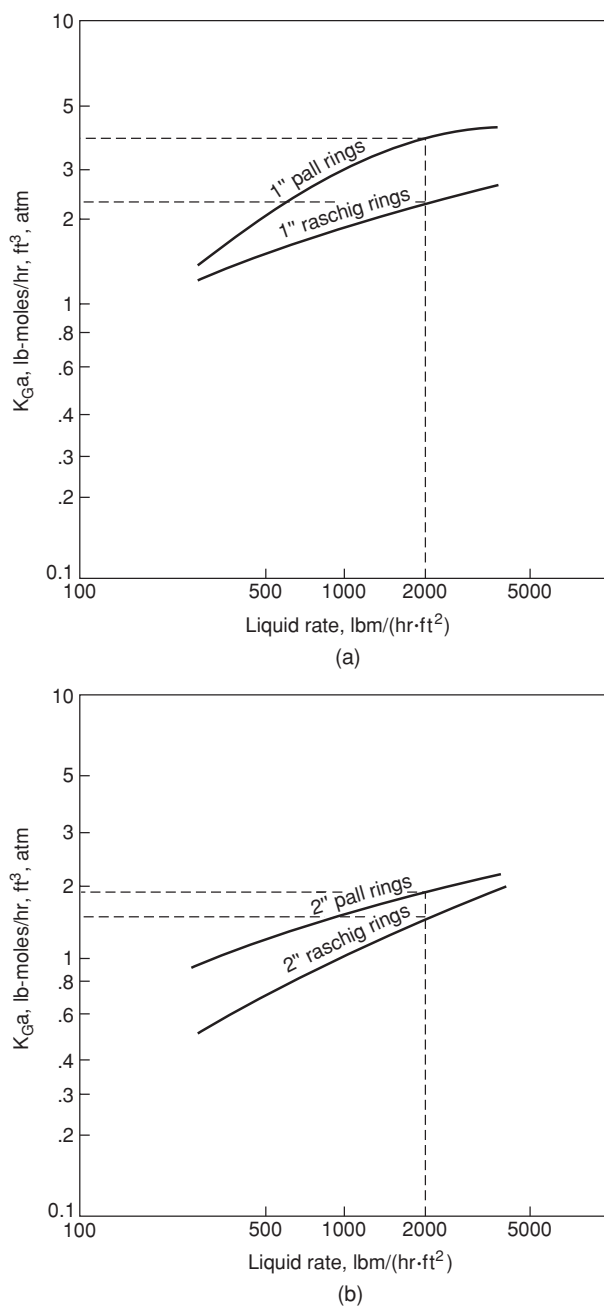


FIG. 14-63 Overall mass transfer coefficients for carbon dioxide absorbed from air by 1N caustic solution. (a) 1-in Pall rings and Raschig rings. (b) 2-in Pall rings and Raschig rings. Air rate = 0.61 kg/s-m² (450 lb/hr-ft²). To convert from lb/hr-ft² to kg/s-m², multiply by 0.00136. To convert from lb-moles/hr-ft³-atm to kg-moles/s-m³-atm, multiply by 0.0045. [Eckert et al., *Chem. Eng. Progr.*, **54**(1), 70 (1958).]

the effect of physical absorption, and the relative values in the table may not hold for other cases, where much of the resistance to mass transfer is in the gas phase. Background on this combination of physical and chemical absorption may be found earlier in the present section, under "Absorption with Chemical Reaction."

Summary In the preloading regime, packing size, type, and distribution affect HETP. With aqueous-organic systems, HETP may be sensitive to underwetting and composition. A lambda value ($\lambda = mG_M/L_M$) outside the range of 0.5 to 2.0 causes HETP to rise, and so does a high hydrogen concentration. HETP of structured packings may also be affected by pressure (at high pressure), and vapor and liquid loads.

MALDISTRIBUTION AND ITS EFFECTS ON PACKING EFFICIENCY

Modeling and Prediction Maldistribution may drastically reduce packing efficiency. HETP may increase by a factor as high as 2 or 3 due to maldistribution. Shariat and Kunesch [*Ind. Eng. Chem. Res.*, **34**(4), 1273 (1995)] provide a good demonstration.

Early models [Mullins, *Ind. Chem. Mfr.*, **33**, 408 (1957); Manning and Cannon, *Ind. Eng. Chem.*, **49**(3), 347 (1957)] expressed the effect of liquid maldistribution on packing efficiency in terms of a simple channeling model. A portion of the liquid bypasses the bed, undergoing negligible mass transfer, and then rejoins and contaminates the rest of the liquid. Huber et al. [*Chem. Ing. Tech.*, **39**, 797 (1967); *Chem. Eng. Sci.*, **21**, 819 (1966)] and Zuideweg et al. [*ICHEME Symp. Ser.*, **104**, A217 (1987)] replaced the simple bypassing by variations in the local L/V ratios. The overirrigated parts have a high L/V ratio, the underirrigated parts a low L/V ratio. Regions with low L/V ratios experience pinching, and, therefore, produce poor separation.

Huber et al. (loc. cit.) and Yuan and Spiegel [*Chem. Ing. Tech.*, **54**, 774 (1982)] added lateral mixing to the model. Lateral deflection of liquid by the packing particles tends to homogenize the liquid, thus counteracting the channeling and pinching effect.

A third factor is the nonuniformity of the flow profile through the packing. This nonuniformity was observed as far back as 1935 [Baker, Chilton, and Vernon, *Trans. Instn. Chem. Engrs.*, **31**, 296 (1935)] and was first modeled by Cihla and Schmidt [*Coll. Czech. Chem. Commun.*, **22**, 896 (1957)]. Hoek (Ph.D. Thesis, The University of Delft, The Netherlands, 1983) combined all three factors into a single model, leading to the zone-stage model below.

The Zone-Stage Model Zuideweg et al. [*ICHEME Symp. Ser.*, **104**, A217, A233 (1987)] extended Hoek's work combining the effects of local L/V ratio, lateral mixing, and flow profile into a model describing the effect of liquid maldistribution on packing efficiency. This work was performed at Fractionation Research Inc. (FRI) and at The University of Delft in The Netherlands. The model postulates that, in the absence of maldistribution, there is a "basic" (or "true" or "inherent") HETP which is a function of the packing and the system only. This HETP can be inferred from data for small towers, in which lateral mixing is strong enough to offset any pinching. For a given initial liquid distribution, the model uses a diffusion-type equation to characterize the splitting and recombining of liquid streams in the horizontal and vertical directions. The mass transfer is then calculated by integrating the liquid flow distribution at each elevation and the basic HETP. Kunesch et al. successfully applied the model to predict measured effects of maldistribution on packing efficiency. However, this model is difficult to use and has not gained industrywide acceptance.

Empirical Prediction Moore and Rukovena [*Chemical Plants and Processing* (European edition), p. 11, August 1987] proposed the empirical correlation in Fig. 14-64 for efficiency loss due to liquid maldistribution in packed towers containing Pall® rings or Metal Intalox® packing. This correlation was shown to work well for several case studies (Fig. 14-64), is simple to use, and is valuable, at least as a preliminary guide.

To quantify the quality of liquid irrigation, the correlation uses the distribution quality rating index. Typical indexes are 10 to 70 percent for most standard commercial distributors, 75 to 90 percent for intermediate-quality distributors, and over 90 percent for high-performance distributors. Moore and Rukovena present a method for calculating a distribution-quality rating index from distributor geometry. Their method is described in detail in their paper as well as in Kister's book (*Distillation Operation*, McGraw-Hill, New York, 1990).

Maximum Liquid Maldistribution Fraction f_{max} . To characterize the sensitivity of packed beds to maldistribution, Lockett and

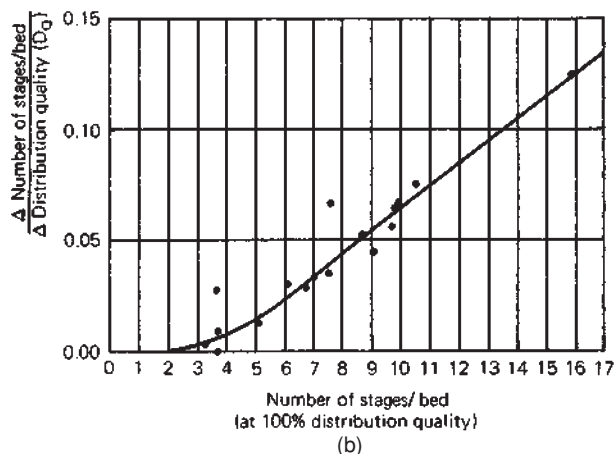
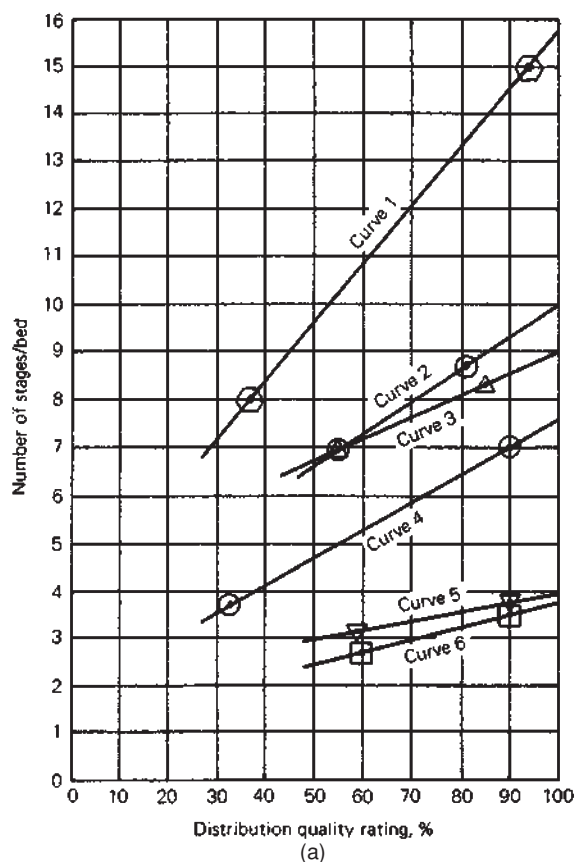


FIG. 14-64 Effect of irrigation quality on packing efficiency. (a) Case histories demonstrating efficiency enhancement with higher distribution quality rating. (b) Correlation of the effect of irrigation quality on packing efficiency. (From F. Moore and F. Rukovena, *Chemical Plants and Processing*, Europe edition, Aug. 1987; reprinted courtesy of Chemical Plants and Processing.)

Billingham (*Trans. IChemE.*, **80**, Part A, p. 373, May 2002; *Trans. IChemE.*, **81**, Part A, p. 134, January 2003) modeled maldistribution as two parallel columns, one receiving more liquid $(1 + f)L$, the other receiving less $(1 - f)L$. The vapor was assumed equally split (Fig. 14-65)

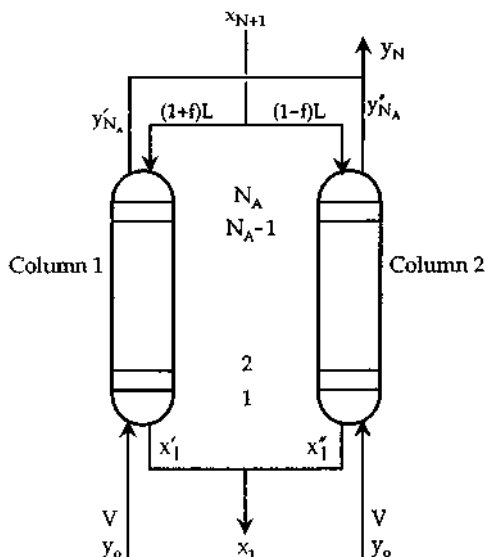


FIG. 14-65 Parallel-columns model. (From Lockett and Billingham, Trans. IChemE **80**, Part A, p. 373, May 2002; reprinted courtesy of IChemE.)

without lateral mixing. Because of the different L/V ratios, the overall separation is less than is obtained at uniform distribution. A typical calculated result (Fig. 14-66) shows the effective number of stages from the combined two-column system decreasing as the maldistribution fraction f increases. Figure 14-66a shows that the decrease is minimal in short beds (e.g., 10 theoretical stages) or when the maldistribution fraction is small. Figure 14-66a shows that there is a limiting fraction $f_{\max}$ which characterizes the maximum maldistribution that still permits achieving the required separation. Physically, $f_{\max}$ represents the maldistribution fraction at which one of the two parallel columns in the model becomes pinched. Figure 14-66b highlights the steep drop in packing efficiency upon the onset of this pinch. Billingham and Lockett derived the following equation for $f_{\max}$ in a binary system:

$$f_{\max} = \frac{y_{N+1} - y_N}{y_N - y_o} + \frac{x_1 - x_o}{x_{N+1} - x_o} - \left(\frac{y_{N+1} - y_N}{y_N - y_o} \right) \left(\frac{x_1 - x_o}{x_{N+1} - x_o} \right) \quad (14-162)$$

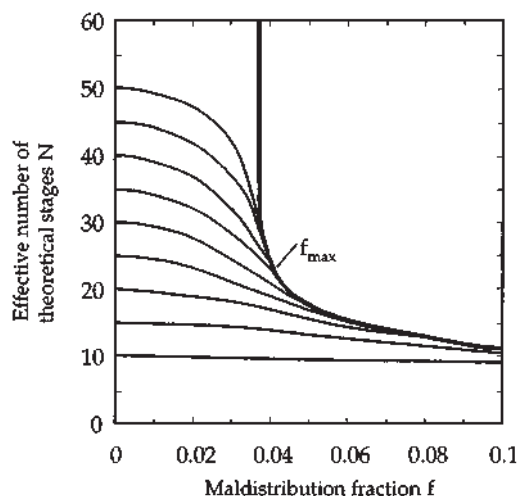
This equation can be used to calculate $f_{\max}$ directly without the need for a parallel column model. Billingham and Lockett show that the various terms in Eq. (14-162) can be readily calculated from the output of a steady-state computer simulation. Multicomponent systems are represented as binary mixtures, either by lumping components together to form a binary mixture of pseudolight and pseudoheavy components, or by normalizing the mole fractions of the two key components. Once $f_{\max}$ is calculated, Billingham and Lockett propose the following guidelines:

- $f_{\max} < 0.05$, extremely sensitive to maldistribution. The required separation will probably not be achieved.
- $0.05 < f_{\max} < 0.10$, sensitive to maldistribution, but separation can probably be achieved.
- $0.10 < f_{\max} < 0.20$, not particularly sensitive to maldistribution.
- $f_{\max} > 0.20$ insensitive to maldistribution.

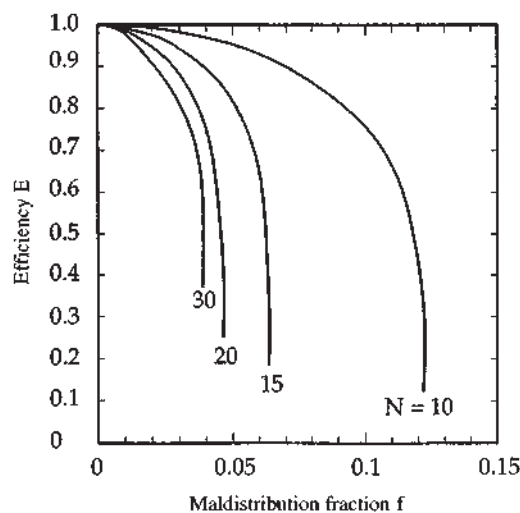
Figure 14-66b shows that shortening the bed can increase $f_{\max}$. Relative volatility and L/V ratio also affect $f_{\max}$. The bed length and L/V ratio can often be adjusted to render the bed less sensitive to maldistribution.

Implications of Maldistribution to Packing Design Practice

These are discussed at length with extensive literature citation in Kister's book *Distillation Operation*, McGraw-Hill, New York, 1990. Below are the highlights:



(a)



(b)

FIG. 14-66 Loss of efficiency due to maldistribution as a function of maldistribution fraction f and the number of stages per bed for a given case study. (a) $f_{\max}$ and reduction in number of stages. (b) Showing larger efficiency loss at higher number of stages per bed and higher f . The steep drops indicate pinching associated with $f_{\max}$. (From Lockett and Billingham, Trans. IChemE **80**, Part A, p. 373, May 2002; reprinted courtesy of IChemE.)

1. Three factors appear to set the effect of maldistribution on efficiency:

a. *Pinching*. Regional changes in L/V ratio cause regional composition pinches.

b. *Lateral mixing*. Packing particles deflect both liquid and vapor laterally. This promotes mixing of vapor and liquid and counteracts the pinching effect.

c. *Liquid nonuniformity*. Liquid flows unevenly through the packing and tends to concentrate at the wall.

2. At small tower-to-packing diameter ratios ($D_T/D_p < 10$), the lateral mixing cancels out the pinching effect, and a greater degree of maldistribution can be tolerated without a serious efficiency loss. At high ratios ($D_T/D_p > 40$), the lateral mixing becomes too small to offset

the pinching effect. The effects of maldistribution on efficiency are therefore most severe in large-diameter columns and small-diameter packings.

A good design practice is to seek a packing size that gives a D_T/D_p between 10 and 40. This is often impractical, and higher ratios are common. When D_T/D_p exceeds 100, avoiding efficiency loss due to maldistribution is difficult. Either ratios exceeding 100 should be avoided, or a special allowance should be made for loss of efficiency due to maldistribution.

3. Wall flow effects become large when D_T/D_p falls below about 10. Packing diameter should be selected such that D_T/D_p exceeds 10.

4. Columns containing less than five theoretical stages per bed are relatively insensitive to liquid maldistribution. With 10 or more stages per bed, efficiency can be extremely sensitive to maldistribution (Strigle, *Packed Tower Design and Applications*, 2d ed., Gulf Publishing, Houston, Tex., 1994) (Fig. 14-66). Beds consisting of small packings or structured packings, which develop more theoretical stages per bed, are therefore more sensitive to maldistribution than equal-depth beds of larger packings. This is clearly demonstrated by FRI's experiments [Shariat and Kunes, *Ind. Eng. Chem. Res.* **34**(4), 1273 (1995)]. Lockett and Billingham (*Trans. IChemE*, vol. 81, Part A, p. 131, January 2003) concur with these comments when their procedure (above) indicates high sensitivity to maldistribution, but allow a higher number of stages per bed when the sensitivity is low.

5. Maldistribution tends to be a greater problem at low liquid flow rates than at high liquid flow rates [Zuiderweg, Hoek, and Lahm, *I. ChemE Symp. Ser.* **104**, A217 (1987)]. The tendency to pinch and to spread unevenly is generally higher at the lower liquid flow rates.

6. A packed column has reasonable tolerance for a uniform or smooth variation in liquid distribution and for a variation that is totally random (*small-scale maldistribution*). The impact of discontinuities or zonal flow (*large-scale maldistribution*) is much more severe [Zuiderweg et al., loc. cit.; Kunes, *Chem. Eng.*, p. 101, Dec. 7, 1987; Kunes, Lahm, and Yanagi, *Ind. Eng. Chem. Res.* **26**(9), 1845 (1987)]. This is so because the local pinching of small-scale maldistribution is evened out by the lateral mixing, and therefore causes few ill effects. In contrast, the lateral mixing either is powerless to rectify a large-scale maldistribution or takes considerable bed length to do so (meanwhile, efficiency is lost).

Figure 14-67 shows HETPs measured in tests that simulate various types of maldistribution in FRI's 1.2-m column containing a 3.6-m bed of 1-in Pall® rings. The y axis is the ratio of measured HETP in the maldistribution tests to the HETP obtained with an excellent distributor. Analogous measurements with structured packing were reported by Fitz, King, and Kunes [Trans. IChemE **77**, Part A, p. 482 (1999)]. Generally, the response of the structured packings resembled that of the Pall® rings, except as noted below.

Figure 14-67a shows virtually no loss of efficiency when a distributor uniformly tilts, such that the ratio of highest to lowest flow is 1.25 (i.e., a "1.25 tilt"). In contrast, an 11 percent chordal blank of a level distributor causes packing HETP to rise by 50 percent.

Figure 14-67b compares continuous tilts with ratios of highest to lowest flow of 1.25 and 1.5 to a situation where one-half of the distributor passes 25 percent more liquid than the other half. The latter ("zonal") situation causes a much greater rise in HETP than a "uniform" maldistribution with twice as much variation from maximum to minimum.

Figure 14-67c shows results of tests in which flows from individual distributor drip points were varied in a gaussian pattern (maximum/mean = 2). When the pattern was randomly assigned, there was no efficiency loss. When the variations above the mean were assigned to a "high zone," and those below the mean to a "low zone," HETP rose by about 20 percent. With structured packing, both random and zonal maldistribution caused about the same loss of efficiency at the same degree of maldistribution.

7. A packed bed appears to have a "natural distribution," which is an inherent and stable property of the packings. An initial distribution which is better than natural will rapidly degrade to it, and one that is worse will finally achieve it, but sometimes at a slow rate. If the rate is extremely slow, recovery from a maldistributed pattern may not be observed in practice (Zuiderweg et al., loc. cit.). Even though the

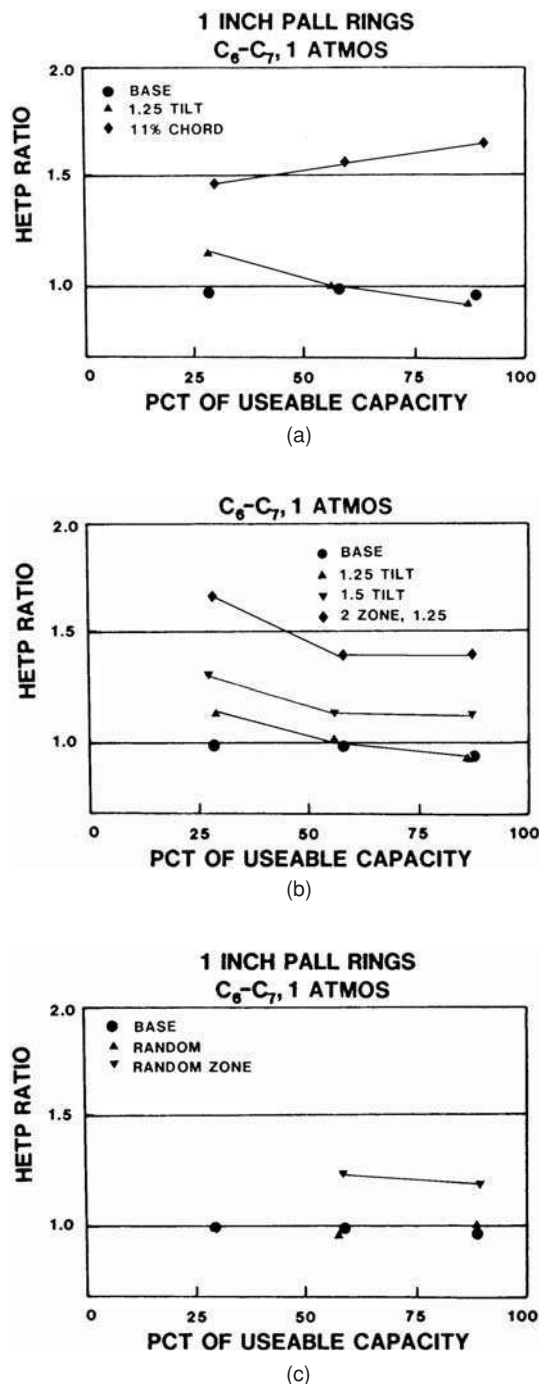


FIG. 14-67 Comparing the effects of "small-scale" and "large-scale" maldistribution on packing HETP. (a) Comparing the effect of a simulated continuous tilt (max/min flow ratio = 1.25) with the simulated effect of blanking a chordal area equal to 11 percent of the tower area. (b) Comparing the effects of simulated continuous tilts (max/min flow ratios of 1.25 and 1.5) with the effects of a situation where one-half of the distributor passes 25 percent more liquid to the other half. (c) Comparing the effects of random maldistribution with those of zonal maldistribution. (Reprinted with permission from J. G. Kunes, L. Lahm, and T. Yanagi, *Ind. Eng. Chem. Res.*, **26**, p. 1845; copyright © 1987, American Chemical Society.)

volumetric distribution improves along the bed, the concentration profile could have already been damaged, and pinching occurs (Bonilla, *Chem. Eng. Prog.*, p. 47, March 1993).

8. Liquid maldistribution lowers packing turndown. The 2-in Pall rings curve in Fig. 14-59 shows HETP rise upon reaching the distributor turndown limit.

9. The major source of gas maldistribution is undersized gas inlet and reboiler return nozzles, leading to the entry of high-velocity gas jets into the tower. These jets persist through low-pressure-drop devices such as packings. Installing gas distributors and improving gas distributor designs, even inlet baffles, have alleviated many of these problems. Vapor distribution is most troublesome in large-diameter columns. Strigle (*Packed Tower Design and Applications*, 2d ed., Gulf Publishing, Houston, Tex., 1994) recommends considering a gas distributing device whenever the gas nozzle F -factor ($F_N = u_N p_c^{0.5}$) exceeds 27 m/s (kg/m^3)^{0.5}, or the kinetic energy of the inlet gas exceeds 8 times the pressure drop through the first foot of packing, or the pressure drop through the bed is less than 0.65 mbar/m. Gas maldistribution is best tackled at the source by paying attention to the gas inlet arrangements.

10. A poor initial liquid maldistribution may cause gas maldistribution in the loading region, i.e., at high gas rates [Stoter, Olujic, and de Graauw, *ICHEME Symp. Ser.* **128**, A201 (1992); Kouri and Sohlo, *ICHEME Symp. Ser.* **104**, B193 (1987)]. At worst, initial liquid maldistribution may induce local flooding, which would channel the gas. The segregation tends to persist down the bed. Outside the loading region, the influence of the liquid flow on gas maldistribution is small or negligible. Similarly, in high-gas-velocity situations, the liquid distribution pattern in the bottom structured packing layers is significantly influenced by a strongly maldistributed inlet gas flow [Olujic et al., *Chem. Eng. and Processing*, **43**, 465 (2004)]. Duss [*ICHEME Symp. Ser.* **152**, 418 (2006)] suggests that high liquid loads such as those experienced in high-pressure distillation also increase the susceptibility to gas maldistribution.

11. The effect of gas maldistribution on packing performance is riddled with unexplained mysteries. FRI's (Cai, Paper presented at the AIChE Annual Meeting, Reno, Nev., 2001) commercial-scale tests show little effect of gas maldistribution on both random and structured packing efficiencies. Cai et al. (*Trans IChE* **81**, Part A, p. 85, 2003) distillation tests in a 1.2-m-diameter tower showed that blocking the central 50 percent or the chordal 30 percent of the tower cross-sectional area beneath a 1.7-m-tall bed of 250 m²/m³ structured packing had no effect on packing efficiency, pressure drop, or capacity. The blocking did not permit gas passage but allowed collection of the descending liquid. Simulator tests with similar blocking with packing heights ranging from 0.8 to 2.4 m [Olujic et al., *Chemical Engineering and Processing*, **43**, p. 465, 2004; *Distillation 2003: Topical Conference Proceedings*, AIChE Spring National Meeting, New Orleans, La., AIChE, p. 567, 2003] differed, showing that a 50 percent chordal blank raised pressure drop, gave a poorer gas pattern, and prematurely loaded the packing. They explain the difference by the ability of liquid to drain undisturbed from the gas in the blocked segment in the FRI tests. Olujic et al. found that while gas maldistribution generated by collectors and by central blockage of 50 percent of the cross-sectional areas was smoothed after two to three layers of structured packing, a chordal blockage of 30 to 50 percent of cross-sectional area generated maldistribution that penetrated deeply into the bed.

12. Computational fluid dynamics (CFD) has been demonstrated effective for analyzing the effects of gas inlet geometry on gas maldistribution in packed beds. Using CFD, Wehrli et al. (*Trans. IChE* **81**, Part A, p. 116, January 2003) found that a very simple device such as the V-baffle (Fig. 14-70) gives much better distribution than a bare nozzle, while a more sophisticated vane device such as a Schoepentoeter (Fig. 14-71c) is even better. Implications of the gas inlet geometry to gas distribution in refinery vacuum towers was studied by Vaidyanathan et al. (*Distillation 2001*, Topical Conference Proceedings, AIChE Spring National Meeting, Houston, Tex., p. 287, April 22–26, 2001); Paladino et al. (*Distillation 2003: Topical Conference Proceedings*, AIChE Spring National Meeting, New Orleans, La., p. 241, 2003); Torres et al. (*ibid.*, p. 284); Waintraub et al. (*Distillation*

2005: Topical Conference Proceedings, AIChE Spring National Meeting, Atlanta, Ga., p. 79, 2005); and Wehrli et al. (*ICHEME Symp. Ser.* **152**, London, 2006). Vaidyanathan et al. and Torres et al. examined the effect of the geometry of a chimney tray (e.g., Fig. 14-72) above the inlet on gas distribution and liquid entrainment. Paladino et al. demonstrated that the presence of liquid in the feed affects the gas velocity profile, and must be accounted for in modeling. Paladino et al. and Waintraub et al. used their two-fluid model to study the velocity distributions and entrainment generated by different designs of vapor horns (e.g., Fig. 14-71). Wehrli et al. produced pilot-scale data simulating a vacuum tower inlet, which can be used in CFD model validation. Ali et al. (*Trans. IChE*, vol. 81, Part A, p. 108, January 2003) found that the gas velocity profile obtained using a commercial CFD package compared well to those measured in a 1.4-m simulator equipped with structured packing together with commercial distributors and collectors. Their CFD model effectively pointed them to a collector design that minimizes gas maldistribution.

PACKED-TOWER SCALE-UP

Diameter For random packings there are many reports [Billet, *Distillation Engineering*, Chem Publishing Co., New York, 1979; Chen, *Chem. Eng.*, p. 40, March 5, 1984; Zuiderweg, Hoek, and Lahm, *ICHEME Symp. Ser.* **104**, A217 (1987)] of an increase in HETP with column diameter. Billet and Mackowiak's (Billet, *Packed Column Analysis and Design*, Ruhr University, Bochum, Germany, 1989) scale-up chart for Pall® rings implies that efficiency decreases as column diameter increases.

Practically all sources explain the increase of HETP with column diameter in terms of enhanced maldistribution or issues with the scale-up procedure. Lab-scale and pilot columns seldom operate at column-to-packing diameter ratios (D_T/D_p) larger than 20; under these conditions, lateral mixing effectively offsets loss of efficiency due to maldistribution pinch. In contrast, industrial-scale columns usually operate at D_T/D_p ratios of 30 to 100; under these conditions, lateral mixing is far less effective for offsetting maldistribution pinch.

To increase D_T/D_p , it may appear attractive to perform the bench-scale tests using a smaller packing size than will be used in the prototype. Deibele, Goedecke, and Schoenmaker [*ICHEME Symp. Ser.* **142**, 1021 (1997)], Goedecke and Alig (Paper presented at the AIChE Spring National Meeting, Atlanta, Ga., April 1994), and Gann et al. [*Chem. Ing. Tech.*, **64**(1), 6 (1992)] studied the feasibility of scaling up from 50- to 75-mm-diameter packed columns directly to industrial columns. Deibele et al. and Gann et al. provide an extensive list of factors that can affect this scale-up, including test mixture, packing pretreatment, column structure, packing installation, snug fit at the wall, column insulation and heat losses, vacuum tightness, measurement and control, liquid distribution, reflux subcooling, prewetting, sampling, analysis, adjusting the number of stages to avoid pinches and analysis issues, evaluation procedure, and more. Data from laboratory columns can be particularly sensitive to some of these factors. Goedecke and Alig show that for wire-mesh structured packing, bench-scale efficiency tends to be better than large-column efficiency, while for corrugated-sheets structured packing, the converse occurs, possibly due to excessive wall flow. For some packings, variation of efficiency with loads at bench scale completely differs from its variation in larger columns. For one structured packing, Kuhn Rom-bopak 9M, there was little load effect and there was good consistency between data obtained from different sources—at least for one test mixture. Deibele et al. present an excellent set of practical guidelines to improve scale-up reliability. So, it appears that great caution is required for packing data scale-up from bench-scale columns.

Height Experimental data for random packings show that HETP slightly increases with bed depth [Billet, *Distillation Engineering*, Chemical Publishing Co., New York, 1979; "Packed Tower Analysis and Design," Ruhr University, Bochum, Germany, 1989; Eckert and Walter, *Hydrocarbon Processing*, **43**(2), 107 (1964)].

For structured packing, some tests with Mellapak 250Y [Meier, Hunkeler, and Stöcker, *ICHEME Symp. Ser.* **56**, p. 3, 3/1 (1979)] showed no effect of bed height on packing efficiency, while others (Cai et al., *Trans IChE*, vol. 81, Part A, p. 89, January 2003) did show a significant effect.

The effect of bed depth on packing HETP is attributed to liquid maldistribution. Zuiderweg et al. [*ICHEME Symp. Ser.* **104**, A217 (1987)] suggest that the uneven irrigation generates an uneven concentration profile and localized pinching near the bottom of the beds. The tests by Martin, Bravo, and Fair (Paper presented at the National AIChE Meeting, New Orleans, La., 1988) confirm that the problem area is near the bottom. According to the zone-stage and Lockett and Billingham models (above), as well as the empirical correlation by Moore and Rukovena (Fig. 14-64), the more stages per bed, the greater is the rise in HETP with bed depth. The presence and extent of maldistribution play an important role in determining the bed-depth effect.

As the bed depth increases, end effects (i.e., mass transfer in the region of liquid introduction and in the region where liquid drips from the packing supports) become less important. Such end effects tend to lower the HETP observed in short columns, such as pilot-plant columns.

In summary, bed depth may significantly influence HETP. This adds uncertainty to scale-up. Shallow test beds should be avoided. Most investigators use beds at least 1.5 m tall, and often more than 3 m tall. The FRI sampling technique (below) can detect maldistribution along the bed height.

Loadings For many random and corrugated-sheet structured packings, HETP is independent of vapor and liquid loadings (Figs. 14-59 and 14-60). For wire-mesh and some corrugated-sheet structured packings, HETP changes with gas and liquid loads.

Wu and Chen [*ICHEME Symp. Ser.* **104**, B225 (1987)] recommend pilot testing over the entire range between the expected minimum and maximum operating rates, and taking the highest measured HETP as the basis for scale-up. The author concurs. With structured packings, the load effect may be due to liquid rather than gas loads, and the pilot tests should cover the range of liquid loads (i.e., m/s based on column cross section) that is expected in the prototype.

Wetting For operation at low liquid loads, the onset of minimum wetting can adversely affect scale-up, particularly with random packings and aqueous systems. Scale-up reliability at low liquid loads can be improved by pilot-testing at the composition range expected in the prototype, and by using identical packing materials and surface treatment in the pilot tests and in the prototype.

Undervetting At the aqueous end of aqueous-organic columns, undervetting is important. Rapid changes of concentration profiles and physical properties in organic-water separations complicate scale-up [Eiden and Kaiser, *ICHEME Symp. Ser.* **142**, 757 (1997)]. Near the onset of undervetting, HETP becomes strongly dependent on composition, packing material and surface roughness, and the presence of surfactants. Scale-up reliability can be enhanced by pilot-testing at the composition range expected in the prototype, and by using identical packing material and surface treatment in the pilot tests and in the prototype.

Preflooding For one structured packing test with an aqueous system, Billet ("Packed Column Analysis and Design," Ruhr University, Bochum, Germany, 1989) measured higher efficiency for a preflooded bed compared with a non-preflooded bed. Presumably, the preflooding improved either wetting or distribution. Billet recommends preflooding the packing, both in the prototype and in the pilot column, to ensure maximum efficiency.

Sampling Fractionation Research Inc. (FRI) developed a sampling technique that eliminates the influence of "end effects" and detects a maldistributed composition profile. This technique [Silvey and Keller, *ICHEME Symp. Ser.* **32**, p. 4:18 (1969)] samples the bed at frequent intervals, typically every 0.6 m or so. HETP is determined from a plot of these interbed samples rather than from the top and bottom compositions.

It is imperative that the interbed samplers catch representative samples, which are an average through the bed cross section. Caution is required when the liquid is highly aerated and turbulent (e.g., above 1300 kPa psia or above 1 m/min). The author highly recommends the FRI sampling technique for all other conditions.

Aging Billet (loc. cit.) showed that for some plastic packings in aqueous systems, the efficiency after one week's operation was almost double the efficiency of new packings. Little further change was

observed after one week. Billet explains the phenomenon by improved wetting. He recommends that data for plastic packings should only be used for scale-up after being in operation for an adequately long period.

DISTRIBUTORS

Liquid Distributors A liquid distributor (or redistributor) should be used in any location in a packed column where an external liquid stream is introduced. Liquid redistributors are also used between packed beds to avoid excessive bed lengths that may impair packing efficiency or mechanical strength. It is best to have the packing supplier also supply the distributor, with the user critically reviewing the design. The user must provide the supplier with concise information about the plugging, corrosive, and foaming tendencies of the service as well as the range of liquid flow rates that it needs to handle and the physical properties of the liquid.

Olsson (*Chem. Eng. Progr.*, p. 57, October 1999) discussed the key for successful distributor design and operation. He states that it is critical to correctly evaluate the fouling potential of the service and design for it (e.g., preventing small holes, filtering the feed, etc.); to avoid gas entry into liquid distributors (e.g., no flashing feed into a liquid distributor); to systematically check the irrigation pattern using a method such as the circle analysis of Moore and Rukovena [*Chem. Plants and Process* (European ed.), p. 11, August 1987; described in detail in Kister's *Distillation Operation*, McGraw-Hill, New York, 1990]; to water-test any liquid distributor (major suppliers have dedicated test stands that are available for those purposes at cost); to ensure correct entry of a feed into a liquid distributor; and to thoroughly inspect a distributor. Kister [*Trans. ICHEME*, vol. 81, Part A, p. 5 (January 2003)] found that between 80 and 90 percent of the distributor failures reported in the literature in the last 50 years could have been prevented if users and suppliers had followed Olsson's measures.

A minimum of 40 irrigation points per square meter has been recommended, with 60 to 100 per square meter being ideal [Strigle, *Packed Tower Design and Applications*, 2d ed., Gulf Publishing, Houston, Tex., 1994; Kister, *Distillation Operation*, McGraw-Hill, New York, 1990; Norton Company (now Koch-Glitsch LP), *Packed Column Internals*, Bulletin TA-80R, 1974]. Commercial-scale tests with both random and structured packings showed no improvement in packing efficiency by increasing the irrigation point density above 40 per square meter [Fitz, King, and Kunesch, *Trans. ICHEME* **77**, Part A, p. 482 (1999)]. So going to larger numbers of irrigation points per square meter provides little improvement while leading to smaller holes which increases the plugging tendency. In orifice-type distributors, which are the most common type, the head-flow relationship is given by the orifice equation

$$Q = 3.96 \times 10^{-4} K_D n_D d_h^{0.5} h^{0.5} \quad (14-163)$$

where Q is the liquid flow rate, m³/h; K_D is the orifice discharge coefficient, with a recommended value of 0.707 (Chen, *Chem. Eng.*, p. 40, March 5, 1984); n_D is the number of holes; d_h is the hole diameter, mm; and h is the liquid head, mm. Equation (14-163) shows that at a given Q , increasing n leads to either smaller d or smaller h .

Figures 14-68 and 14-69 show common distributor types used for distillation and absorption. An excellent detailed discussion of the types of distributors and their pros and cons was given by Bonilla (*Chem. Eng. Progr.*, p. 47, March 1993). The *perforated pipe* (or *ladder pipe*) distributor (Fig. 14-68a) has holes on the underside of the pipes. It is inexpensive, provides a large open area for vapor flow, and does not rely on gravity. On the debit side, it is typically designed for high velocity heads, 500 to 1000 mm of water, which is 5 to 10 times more than gravity distributors, requiring [per Eq. (14-163)] either fewer irrigation points or the use of plugging-prone smaller holes. The high hole velocities make it prone to corrosion and erosion. These disadvantages make it relatively unpopular. A gravity variation of this distributor uses a liquid drum above the distributor that gravity-feeds it.

Spray distributors (Fig. 14-68b) are pipe headers with spray nozzles fitted on the underside. The spray nozzles are typically wide-angle (often 120°) full-cone. Spray distributors are unpopular in distillation but are common in heat transfer, washing and scrubbing services

(especially in refinery towers), and in small-diameter towers where a single spray nozzle can be used. They are inexpensive and offer a large open area for vapor flow and a robustness for handling of fouling fluids when correctly designed, and the sprays themselves contribute to mass and heat transfer. On the debit side, the spray cones often generate regions of over- and underirrigation, the sprays may not be homogeneous, the spray nozzles are prone to corrosion, erosion, and damage. With highly subcooled liquids, the spray angle may collapse when pushed at high pressure drops (above 100 to 150 kPa) (Fractionation Research Inc., "A Spray Collapse Study," motion picture 919, Stillwater, Okla., 1985). The design and spray pattern are highly empirical. Sprays also generate significant entrainment to the section above [Trompiz and Fair, *Ind. Eng. Chem. Res.*, **39**(6), 1797 (2000)].

Orifice pan distributors (Fig. 14-69a) and *orifice tunnel* distributors (Fig. 14-69b) have floor holes for liquid flow and circular (Fig. 14-69a) or rectangular (Fig. 14-69b) risers for vapor passages. When they are used as redistributors, a hat is installed above each riser to prevent liquid from the bed above from dripping into the risers. Unlike the ladder pipe and spray distributors that operate by pressure drop, orifice distributors operate by gravity, and therefore use a much smaller liquid head, typically 100 to 150 mm at maximum rates. Using Eq. (14-163), the lower head translates to either more distributions points (n_D), which helps irrigation quality, or larger hole diameters, which resists plugging. However, the low liquid velocities, large residence times, and open pans (or troughs) make them more prone to plugging than the pressure distributors. A good hole pattern and avoidance of oversized risers are essential. Orifice distributors are self-collecting, a unique advantage for redistributors. Orifice distributors are one of the most popular types and are favored whenever the liquid loads are high enough to afford hole diameters large enough to resist plugging (>12 mm).

Orifice trough (or *orifice channel*) distributors (Fig. 14-69c-f) are some of the most popular types. The trough construction does away with the multitude of joints in the orifice pans, making them far more leak-resistant, a major advantage in large towers and low-liquid-rate applications. Liquid from a central parting box (Fig. 14-69c, e) or middle channel (Fig. 14-69d) is metered into each trough. The troughs can have floor holes, but elevating the holes above the floor (Fig. 14-69c-g) is preferred as it enhances plugging resistance. Tubes (Fig. 14-69c, d, f) or baffles (Fig. 14-69e) direct the liquid issuing from the elevated holes downward onto the packings. Orifice trough distributors are not self-collecting. When used for redistribution, they require a liquid collector to be installed above them.

Turndown of orifice distributors is constrained to about 2 : 1 by Eq. (14-163). For example, a 100-mm liquid head at the design drops to 25 mm when the liquid rate is halved. Lower heads give poor irrigation and high sensitivity to levelness. Turndown is often enhanced by using two rows of side tubes (in the Fig. 14-69c type) or of side holes (in the Fig. 14-69d or e types). *Perforated drip tubes* (as in Fig. 14-69d) are popular in either orifice trough or orifice pan distributors. The lower, smaller hole is active at low liquid rates, with the larger upper hole becoming active at higher liquid rates. Use of perforated drip tubes is not recommended when the vapor dew point is much higher than the liquid bubble point, because liquid may boil in the tubes, causing dryout underneath [Kister, Stupin, and Oude Lenferink, *ICHEME. Symp. Ser.* **152**, p. 409, London (2006)].

A popular type of the orifice trough distributor is the *splash plate* distributor (Fig. 14-69e). The splash plates spread the issuing liquid over their lengths, making it possible to reduce the number of irrigation points. This is a special advantage with small liquid rates, as fewer irrigation points (at a given head) translate to larger, more fouling-resistant

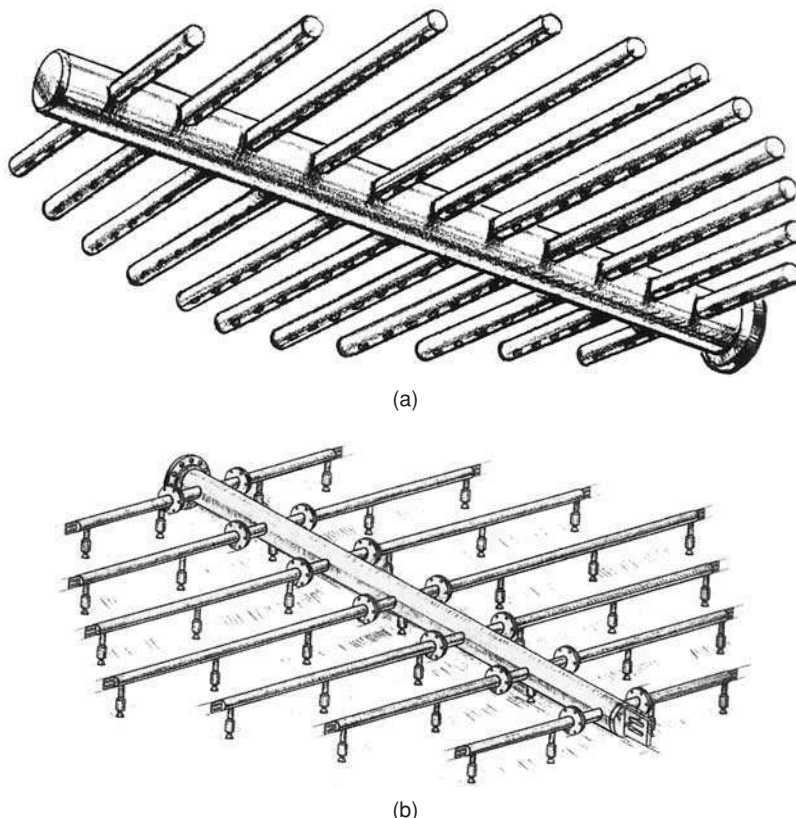


FIG. 14-68 Pressure liquid distributors. (a) Ladder pipe. (b) Spray. (Courtesy of Koch-Glitsch LP.)

hole diameters [Eq. (14-163)]. Lack of the drip tubes eliminates the possible in-tube boiling issue (above).

Multistage orifice trough distributors (Fig. 14-69f) also attempt to provide good irrigation at low liquid rates without resorting to plugging-prone small holes. The primary stage uses fewer irrigation points. Liquid from the primary stage is further split at the secondary stage. The secondary stage is of small size, so leveling and small flow variations are not of great concern. The secondary stage may use the same or a different liquid splitting principle as the primary stage. Even short layers of structured packings have been used as a secondary distribution stage.

Notched trough distributors (Fig. 14-69g) consist of parallel troughs with side V notches. This distributor obeys the triangular notch equation instead of the orifice equation, which makes the flow proportional to $h^{2.5}$ [instead of $h^{0.5}$ in Eq. (14-163)]. This high power renders the distributor highly sensitive to out-of-levelness and

hydraulic gradients and makes it difficult to incorporate a large number of distribution points. Since the liquid issues sideways, it is difficult to predict where the liquid will hit the packings. Baffles are sometimes used to direct the liquid downward. Overall, the quality of distribution is inferior to that of orifice distributors, making notched-trough distributors unpopular. Their strength is their insensitivity to fouling and corrosive environments and their ability to handle high liquid rates at good turndown.

With any trough distributor, and especially those with V notches, excessive hydraulic gradients must be avoided. This is often achieved by using more parting boxes.

The hydraulic gradient is highest where the liquid enters the troughs, approaching zero at the end of the trough. The hydraulic gradient (between entry point and trough end) can be calculated from [Moore and Rukovená, *Chemical Plants and Processing* (European ed.), p. 11, August 1987]

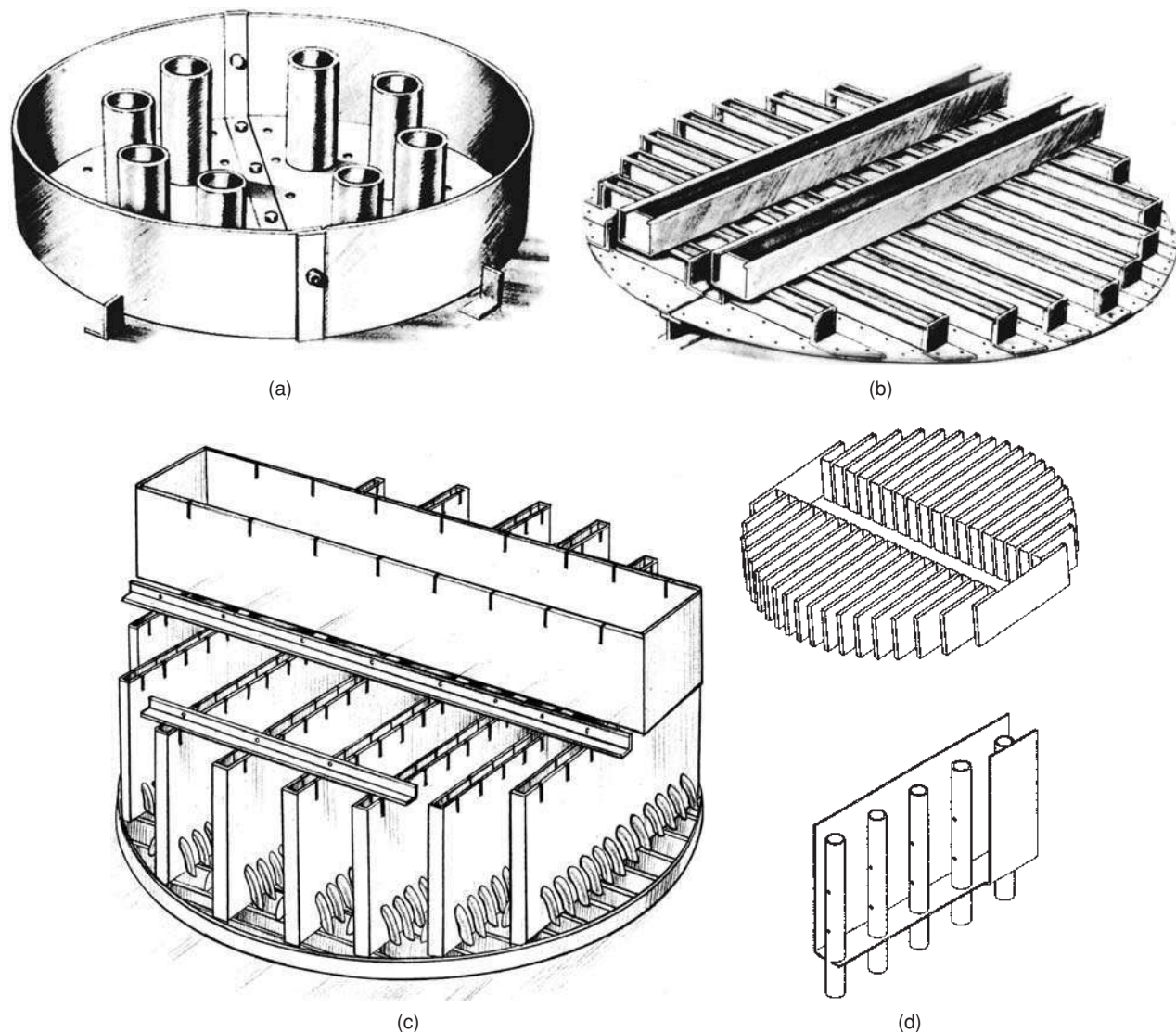


FIG. 14-69 Gravity liquid distributors. (a) Orifice pan. (b) Orifice tunnel. (c) Orifice tube, using external drip tubes. (d) Orifice trough, using internal drip tubes. (e) Splash plate orifice trough. (f) Two-stage orifice trough. (g) Notched trough. (Parts a–c, g, courtesy of Koch-Glitsch LP; parts d–f, courtesy of Sulzer Chemtech.)

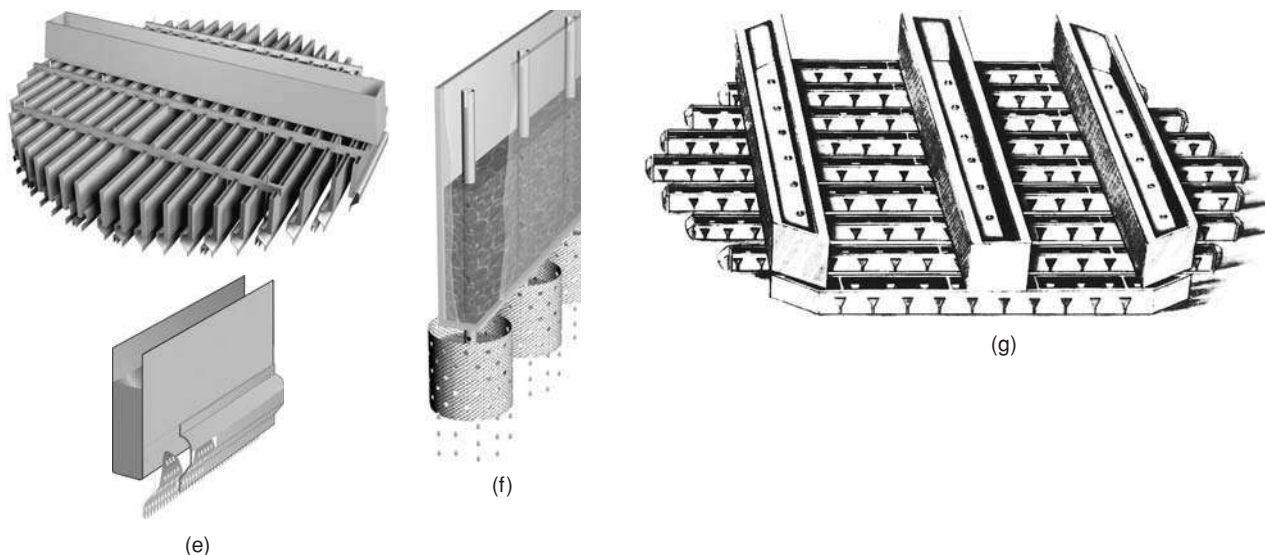


FIG. 14-69 (Continued)

$$h_{hg} = 51v_H^2 \quad (14-164)$$

where h_{hg} is the hydraulic gradient head, mm, and v_H is the horizontal velocity in the troughs, m/s.

Flashing Feed and Vapor Distributors When the feed or reflux is a flashing feed, the vapor must be separated out of the liquid before the liquid enters a liquid distributor. At low velocities (only), this can be achieved by a *bare nozzle* (Fig. 14-70a). A *V baffle* (Fig. 14-70b) is sometimes installed as a primitive flashing feed or vapor distributor.

For better vapor-liquid separation and distribution, with smaller-diameter towers (<1.5 m), a *flash chamber* (Fig. 14-70c) separates the liquid from the vapor, with the collected liquid descending via downpipes to a liquid distributor below. The flash chamber can be peripheral (Fig. 14-70c) or central. In larger towers, *gallery distributors* (Fig. 14-70d) are preferred. The flashing feed enters the peripheral section of the upper plate (the gallery) where vapor disengages and flows up, with liquid descending through holes (Fig. 14-70d) or down pipes onto the liquid distributor below. Alternatively, an external knockout pot is sometimes used to give separate vapor and liquid feeds.

The *vapor horn* (Fig. 14-71a) is unique for high-velocity feeds in which vapor is the continuous phase with liquid present as suspended drops in the feed vapor. This is common when the feed makes up the bulk of the vapor traffic in the tower section above. Typical examples are feeds to refinery vacuum and crude towers and rich solution feeds to hot carbonate regenerators. A tangential helical baffle or vapor horn, covered at the top, open at the bottom, and spiraling downward, is used at the feed entry. This baffle forces the vapor to follow the contour of the vessel as it expands and decreases in velocity. Liquid droplets, due to their higher mass, tend to collide with the tower wall, which deflects them downward, thus reducing upward entrainment. Large forces, generated by hurricane-force winds and vapor flashing, are absorbed by the entire tower wall rather than by a small area. A wear plate is required at the tower wall. Some designs have vane openings on the inside wall.

Alternatively, multivane triangular diffusers (Fig. 14-71b, c) such as the Schoepentoeter have been successful for high-velocity vapor-rich feeds. These are used with radial (as distinct from tangential) nozzles. The vanes knock out the liquid and direct it downward while the vapor expands to the tower diameter.

Pilot-scale tests by Fan et al. [*ICHEME Symp. Ser.* **142**, 899 (1997)] compared vapor distribution and entrainment from sparger, vapor

horns, and multivane triangular diffusers. Vapor horns gave the best overall performance considering vapor distribution, entrainment, and pressure drop, with multivane distributors doing well too. The designs of the inlets compared, however, were not optimized so the comparison could have reflected deviations from optimum rather than real differences between the devices.

Low-velocity vapor-only feeds often enter via bare nozzles or V baffles (above). At higher velocities, perforated vapor spargers are used. At high velocities, vapor horns and Schoepentoeters are often preferred. Alternatively or additionally, a vapor distributor may be mounted above the feed. The vapor distributor is a chimney tray (Fig. 14-72) where liquid is collected on the deck and flows via downcomers or is drawn out while vapor passes through the chimneys. To be effective as a vapor distributor, the pressure drop through the chimneys needs to be high enough to counter the maldistributed vapor. It was recommended to make the pressure drop through the chimneys at least equal to the velocity head at the tower inlet nozzle (Strigle, *Random Packings and Packed Towers*, Gulf Publishing, Houston, Tex., 1987), with common pressure drops ranging from 25 to 200 mm water.

OTHER PACKING CONSIDERATIONS

Liquid Holdup Liquid holdup is the liquid present in the void spaces of the packing. Reasonable liquid holdup is necessary for good mass transfer and efficient tower operation, but beyond that, it should be kept low. High holdup increases tower pressure drop, the weight of the packing, the support load at the bottom of the packing and tower, and the tower drainage time. Most important, when distilling thermally unstable materials, excessive holdup raises product degradation and fouling, and with hazardous chemicals, increases undesirable inventories.

The effect of liquid and gas rates on the operating holdup is shown in Figs. 14-73 and 14-74. In the preloading regime, holdup is essentially independent of gas velocity, but is a strong function of liquid flow rate and packing size. Smaller packings and high liquid rates tend to have greater holdup.

Holdup can be estimated by using Buchanan's correlation [*Ind. Eng. Chem. Fund.* **6**, 400 (1967)], as recommended in previous editions of this handbook. More recent correlations by Billet and Schultes [*ICHEME Symp. Ser.* **104**, A159 (1987)], by Maćkowiak

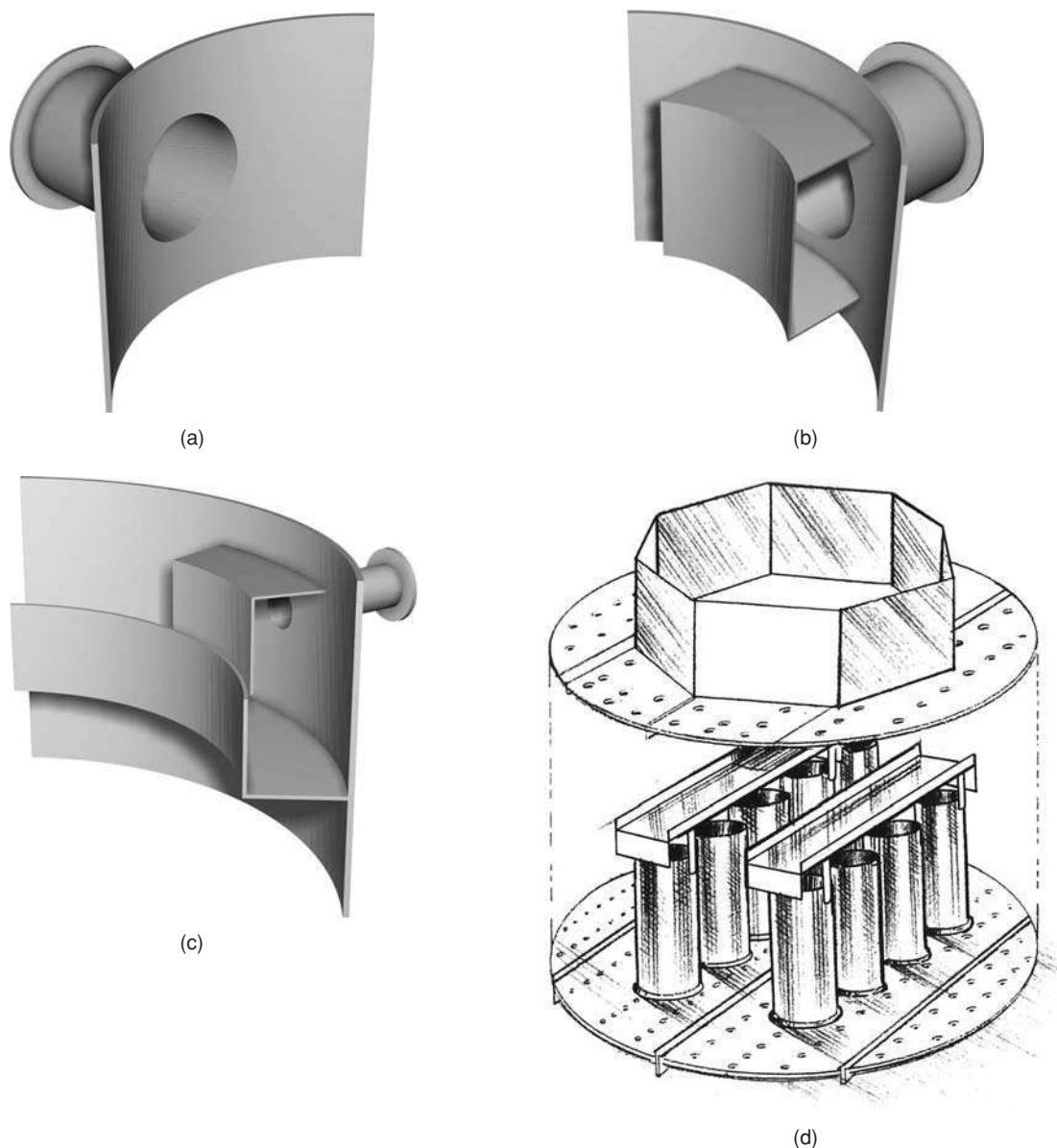


FIG. 14-70 Flashing feed and vapor distributors. (a) Bare nozzle. (b) Rounded V baffle. (c) Peripheral flash box—the box extends right around the tower wall, with the collected liquid descending via downpipes to a liquid distributor below. (d) Gallery distributor—the feed enters the gallery area (upper plate). (Parts a–c, courtesy of Sulzer Chemtech; part d, courtesy of Koch-Glitsch LP.)

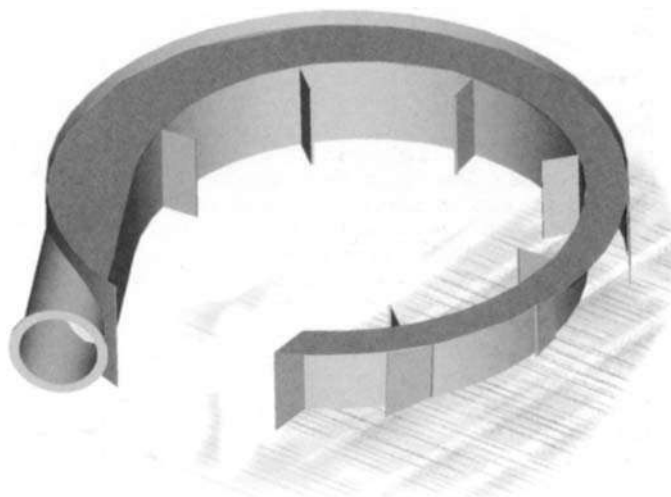
("Fluiddynamik von Kolonnen mit Modernen Füllkörpern und Packungen für Gas/Flüssigkeitssysteme, Otto Salle Verlag, Frankfurt am Main, and Verlag Sauerländer, Aarau, 1991), and by Mersmann and Deixler [*Chem. Ing. Tech.* **58**(1), 19 (1986)] apply to second- and third-generation random packings as well as to some structured packings.

Stichlmair and Fair (*Distillation Principles and Practice*, Wiley-VCH, New York, 1998) show that liquid holdup is primarily a function of the liquid Froude number, specific surface area of the packing, and physical properties. They recommend the correction by Engel, Stichlmair, and Geipel [*Ind. Eng. Chem. Symp. Ser.* **142**, 939 (1997)].

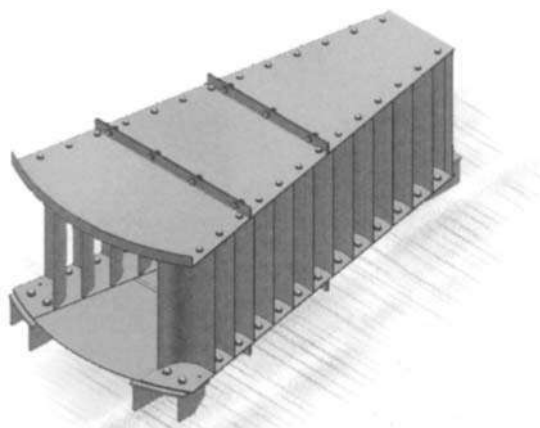
$$h_{L0} = 0.93 \left(\frac{U_L^2 a_p}{g} \right)^{1/6} \left(\frac{\mu_L^2 a_p^3}{\rho_L^2 g} \right)^{1/10} \left(\frac{\sigma a_p^2}{1000 \rho_L g} \right)^{1/8} \quad (14-165)$$

where h_{L0} = liquid holdup, fractional
 U_L = liquid superficial velocity, m/s
 a_p = packing specific surface area, m²/m³
 g = acceleration due to gravity, m/s²
 μ_L = liquid viscosity, kg/(m·s)
 σ = surface tension, mN/m

The Engel, Stichlmair, and Geipel correlation applies only in the preloading regime. The packing geometry is represented in the



(a)



(b)



(c)

FIG. 14-71 High-velocity flashing feed and vapor distributors. (a) Vapor horn. (b) Radial vane distributor. (c) Schoepentoeter. (Parts a, b, courtesy of Koch-Glitsch LP; part c, courtesy of Sulzer Chemtech.)

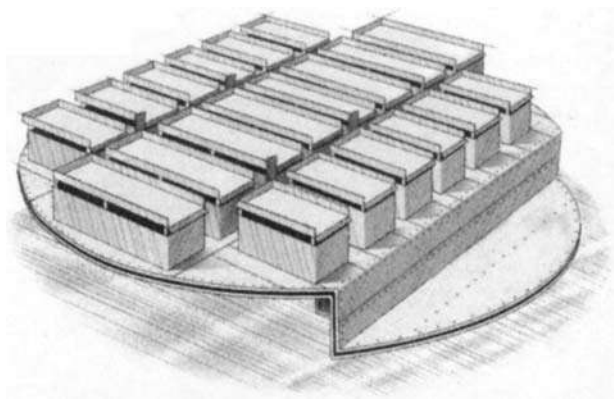


FIG. 14-72 Chimney tray vapor distributor. (Reprinted courtesy of Koch-Glitsch LP.)

correlation solely by the readily available specific surface area (Tables 14-13 and 14-14).

Minimum Wetting Rate The minimum wetting rate (MWR) is the lower stability limit of packings. Below this liquid load the liquid film on the packing surfaces breaks up and dewetting occurs. The area available for mass transfer diminishes, and efficiency drops.

Schmidt [*ICHEME Symp. Ser.* **56**, 3.1/1 (1979)] described the MWR in terms of a force balance at a dry patch along the path of a falling liquid film. While the gravity and viscous forces resist dewetting, the surface tension and vapor shear forces tend to dewet the falling film. The MWR therefore rises with an increase in surface tension and liquid density, and with a decrease in liquid viscosity. Large packing sizes and poor surface wetting characteristics also contribute to higher MWR.

Schmidt presented a fundamental correlation to predict minimum wetting for Raschig and Pall® rings. More popular have been the Glitsch rules of thumb [Table 14-16; Glitsch, Inc. (now Koch Glitsch), Bulletin 345, Dallas, Tex., 1986] for CMR® random packings with packing surface areas around 200 m²/m³. To extend these rules to other random packings, Kister (*Distillation Design*, McGraw-Hill, New York, 1992) applied Schmidt's model to give

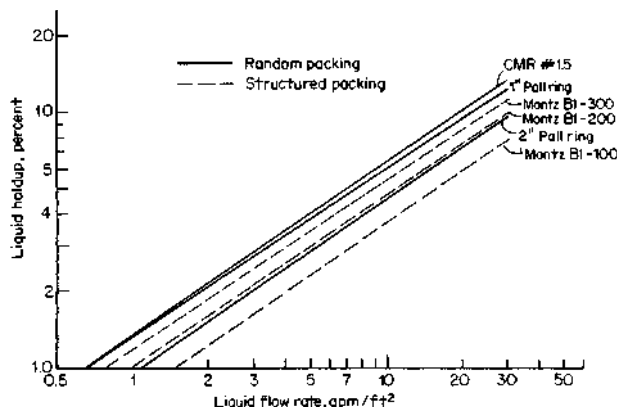


FIG. 14-73 Liquid holdup, air-water data by Billet ("Packed Column Design and Analysis," Ruhr University, Bochum, Germany), preloading regime. (From Kister, H. Z., *Distillation Design*, copyright © by McGraw-Hill; reprinted with permission.)

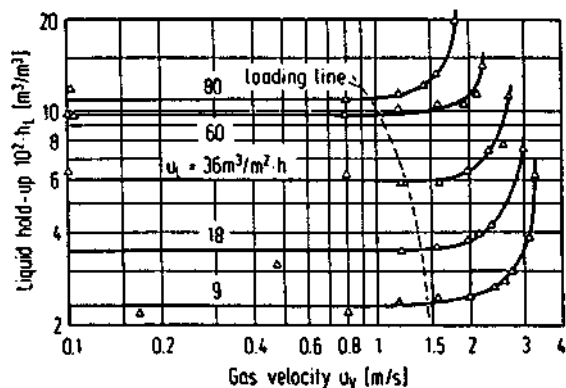


FIG. 14-74 Effect of liquid and gas rates on the operating holdup of modern random packings (25-mm NorPac®). [From R. Billet and M. Schultes, *ICHEME Symp. Ser.* **104**, p. A159, 1987. Reprinted courtesy of the Institution of Chemical Engineers (UK).]

$$Q_{MW} \approx (Q_{MW} \text{ from Table 14-16}) \times (200/a_p)^{0.5} \quad (14-166)$$

The Glitsch brochure did not state the surface tension bases of Table 14-16. The author believes that they conservatively apply to organic and hydrocarbon systems ($\sigma < 25$ mN/m). For water ($\sigma = 70$ mN/m) the author believes that double the values from Table 14-16 is reasonable.

Some surface treatments of the packing (e.g., oxidizing, sandblasting, etching) can substantially reduce the MWR. Chuang and Miller [*Can. J. Chem. Eng.* **66**(6), 377 (1988)] tested a metallic random packing with an aqueous system at low liquid rates (about 0.4 m³/hm²). They used two alternative techniques for oxidizing the packing surfaces. The packings oxidized with the more effective technique gave a column efficiency twice as high as those oxidized by the others.

Superior wetting characteristics characterize structured packings. Satisfactory performance was reported down to 0.25 and 0.12 m³/hm² in corrugated-sheet and wire-mesh metal structured packings, respectively. FRI's commercial-scale tests (Fitz and Kunesh, Paper presented at the AIChE Annual Meeting, Chicago, Nov. 1996) demonstrated good efficiencies down to 0.5 m³/hm² (no lower limit reached) with good liquid and vapor distribution.

Two Liquid Phases Two liquid phases often occur in heterogeneous distillation, in steam stripping, and in solvent recovery. Harrison (*Chem. Eng. Progr.*, p. 80, November 1990) and Meier et al. [*ICHEME Symp. Ser.* **152**, 267 (2006)] saw no reason to expect unusual capacity or pressure drop limitations due to the presence of the second liquid phase, suggesting that standard correlations for these should apply.

In contrast, both expressed uncertainty in predicting packing efficiency. Harrison presented two case studies: In one, adding water to two water-insoluble organics had no effect on HETP; in another, a key

TABLE 14-16 Glitsch's Rule of Thumb for Minimum Wetting

Basis: CMR with $a_p > 150$ m²/m³

Material	Minimum wetting rate, m ³ /(m ² ·h)
Unglazed ceramic (chemical stoneware)	0.5
Oxidized metal (carbon steel, copper)	0.7
Surface-treated metal (etched stainless steel)	1.0
Glazed ceramic	2.0
Glass	2.5
Bright metal (stainless steel, tantalum, other alloys)	3.0
PVC-CPVC	3.5
Polypropylene	4.0
Fluoropolymers (PTFE type)	5.0

component was soluble in both liquid phases, and HETP was about 50 percent above normal. Harrison argued that a second liquid phase leads to lower efficiency only when it impairs diffusion of the key species. On this basis, Harrison expects efficiency loss also when an "inert" liquid or vapor represents a large fraction of the liquid or vapor phase. Meier et al. recommend obtaining efficiencies by scaling up laboratory-scale data using a similar type of packing.

Both Harrison and Meier et al. emphasize adequately distributing each liquid phase to the packing. Harrison noted that a well-designed ladder pipe distributor can maintain high velocities and low residence times that provide good mixing. With trough distributors that separate the phases and then distribute each to the packing, a light-to-heavy phase maldistribution may occur, especially when the phase ratio and separation vary. Meier et al. noted the existence of a cloudy two-liquid

layer between the clear light and heavy liquid and recommend an additional middle distribution point for this layer. They also noticed that phase separation unevenness can have a large influence on the phase ratio irrigated to the packing.

High Viscosity and Surface Tension Bravo (Paper presented at the AIChE Spring National Meeting, Houston, Tex., 1995) studied a system that had 425-cP viscosity, 350 mN/m surface tension, and a high foaming tendency. He found that efficiencies were liquid-phase-controlled and could be estimated from theoretical HTU models. Capacity was less than predicted by conventional methods which do not account for the high viscosity. Design equations for orifice distributors extended well to the system once the orifice coefficient was calculated as a function of the low Reynolds number and the surface tension head was taken into account.

OTHER TOPICS FOR DISTILLATION AND GAS ABSORPTION EQUIPMENT

COMPARING TRAYS AND PACKINGS

Most separations can be performed either with trays or with packings. The factors below represent economic pros and cons that favor each and may be overridden. For instance, column complexity is a factor favoring trays, but gas plant demethanizers that often use one or more interreboilers are traditionally packed.

Factors Favoring Packings

Vacuum systems. Packing pressure drop is much lower than that of trays because the packing open area approaches the tower cross-sectional area, while the tray's open area is only 8 to 15 percent of the tower cross-sectional area. Also, the tray liquid head, which incurs substantial pressure drop (typically about 50 mm of the liquid per tray), is absent in packing. Typically, tray pressure drop is of the order of 10 mbar per theoretical stage, compared to 3 to 4 mbar per theoretical stage with random packings and about one-half of that with structured packings.

Consider a vacuum column with 10 theoretical stages, operating at 70-mbar top pressure. The bottom pressure will be 170 mbar with trays, but only 90 to 110 mbar with packings. The packed tower will have a much better relative volatility in the lower parts, thus reducing reflux and reboil requirements and bottom temperature. These translate to less product degradation, greater capacity, and smaller energy consumption, giving packings a major advantage.

Lower-pressure-drop applications. When the gas is moved by a fan through the tower, or when the tower is in the suction of a compressor, the smaller packing pressure drop is often a controlling consideration. This is particularly true for towers operating close to atmospheric pressure. Here excessive pressure drop in the tower increases the size of the fan or compressor (new plant), bottlenecks them (existing plant), and largely increases power consumption. Due to the compression ratio, pressure drop at the compressor discharge is far less important and seldom a controlling consideration.

Revamps. The pressure drop advantage is invaluable in vacuum column revamps, can be translated to a capacity gain, an energy gain, a separation improvement, or various combinations of these benefits. Likewise, for towers in the suction of compressors, replacing trays by packings reduces the compression ratio and helps debottleneck the compressor.

Packings also offer an easy tradeoff between capacity and separation. In the loaded sections of the tower, larger packings can overcome capacity bottlenecks at the expense of loss in separation. The separation loss can often be regained by retrofitting with smaller packings in sections of the tower that are not highly loaded. In tray towers, changing tray spacing gives similar results, but is more difficult to do.

Foaming (and emulsion). The low gas and liquid velocities in packing suppress foam formation. The large open area of the larger random packing promotes foam dispersal. Both attributes make

random packing excellent for handling foams. In many cases recurrent foaming was alleviated by replacing trays by random packing, especially when tray downcomers were poorly designed.

Switching from trays to structured packing can aggravate foaming. While the low gas and liquid velocities help, the solid walls restrict lateral movement of foams and give support to the foams.

Small-diameter columns. Columns with diameter less than 1 m (3 ft) are difficult to access from inside to install and maintain the trays. "Cartridge" trays or an oversized diameter are often used. Either option is expensive. Cartridge trays also run into problems with sealing to the tower wall and matching tower to tray hardware [Sands, *Chem. Eng.*, p. 86 (April 2006)]. Packing is normally a cheaper and more desirable alternative.

Corrosive systems. The practical range of packing materials is wider. Ceramic and plastic packings are cheap and effective. Trays can be manufactured in nonmetals, but packing is usually a cheaper and more desirable alternative.

Low liquid holdup. Packings have lower liquid holdup than do trays. This is often advantageous for reducing polymerization, degradation, or the inventory of hazardous materials.

Batch distillation. Because of the smaller liquid holdup of packing, a higher percentage of the liquid can be recovered as top product.

Factors Favoring Trays

Solids. Trays handle solids much more easily than packing. Both gas and liquid velocities on trays are often an order of magnitude higher than through packing, providing a sweeping action that keeps tray openings clear. Solids tend to accumulate in packing voids. There are fewer locations on trays where solids can be deposited. Plugging in liquid distributors has been a common trouble spot. Cleaning trays is much easier than cleaning packings.

Not all trays are fouling-resistant. Floats on moving valve trays tend to "stick" to deposits on the tray deck. Fouling-resistant trays have large sieve holes or large fixed valves, and these should be used when plugging and fouling are the primary considerations.

There is much that can be done to alleviate plugging with random packing. Large, open packing with minimal pockets offers good plugging resistance. Distributors that resist plugging have large holes (> 13-mm diameter). Such large holes are readily applied with high liquid flow rates, but often not practical for small liquid flow rates.

Maldistribution. The sensitivity of packing to liquid and gas maldistribution has been a common cause of failures in packed towers. Maldistribution issues are most severe in large-diameter towers, long beds, small liquid flow rates, and smaller packing. Structured packing is generally more prone to maldistribution than random packing. While good distributor design, water testing, and inspection can eliminate most maldistribution issues, it only takes a few small details that fall through the cracks to turn success into failure. Due to maldistribution, there are far more failures experienced

with packing than in trays, and it takes more trials "to get it right" than with trays. This makes trays more robust.

Complex towers. Interboilers, intercondensers, cooling coils, and side drawoffs are more easily incorporated in trays than in packed towers. In packed towers, every complexity requires additional distribution and/or liquid collection equipment.

Feed composition variation. One way of allowing for design uncertainties and feedstock variation is by installing alternate feed points. In packed towers, every alternate feed point requires expensive distribution equipment.

Performance prediction. Due to their sensitivity to maldistribution there is greater uncertainty in predicting packed column performance.

Chemical reaction, absorption. Here the much higher liquid holdup on trays provides greater residence time for absorption or chemical reaction than does packing.

Turndown. Moving valve and bubble-cap trays normally give better turndown than packings. Unless very expensive distributors are used, packed tower turndown is usually limited by distributor turndown.

Weight. Tray towers usually weigh less than packed towers, saving on the cost of foundations, supports, and column shell.

Trays vs. Random Packings The following factors generally favor trays compared to random packings, but not compared to structured packings.

Low liquid rates. With the aid of serrated weirs, splash baffles, reverse-flow trays, and bubble-cap trays, low liquid rates can be handled better in trays. Random packings suffer from liquid dewetting and maldistribution sensitivity at low liquid rates.

Process surges. Random packings are usually more troublesome than trays in services prone to process surges (e.g., those caused by slugs of water entering a hot oil tower, relief valve lifting, compressor surges, or instability of liquid seal loops). Structured packings are usually less troublesome than trays in such services.

Trays vs. Structured Packings The following factors generally favor trays compared to structured packings, but not compared to random packings.

Packing fires. The thin sheets of structured packing (typically 0.1 mm) poorly dissipate heat away from hot spots. Also, cleaning, cooling, and washing can be difficult, especially when distributors or packing plug up. Many incidents of packing fires during turnarounds (while towers with structured packings were open to atmosphere) have been reported. Most of these fires were initiated by pyrophoric deposits, hot work (e.g., welding) above the packing, opening the tower while hot organics were still present, and packing metallurgy that was not fire-resistant. Detailed discussion can be found in Fractionation Research Inc. (FRI) Design Practices Committee, "Causes and Prevention of Packing Fires," *Chem. Eng.*, July 2007.

Materials of construction. Due to the thin sheets of structured packings, their materials of construction need to have better resistance to oxidation or corrosion. For a service in which carbon steel is usually satisfactory with trays, stainless steel is usually required with structured packings.

Column wall inspection. Due to their snug fit, structured packings are easily damaged during removal. This makes it difficult to inspect the column wall (e.g., for corrosion).

Washing and purging. Thorough removal of residual liquid, wash water, air, or process gas trapped in structured packings at start-up and shutdown is more difficult than with trays. Inadequate removal of these fluids may be hazardous.

High liquid rates. Multipass trays effectively lower the liquid load "seen" by each part of the tray. A similar trick cannot be applied with packings. The capacity of structured packings tends to rapidly fall off at high liquid rates.

Capacity and Efficiency Comparison Kister et al. [*Chem. Eng. Progr.*, **90**(2), 23 (1994)] reported a study of the relative capacity and efficiency of conventional trays, modern random packings, and conventional structured packings. They found that, for each device optimally designed for the design requirements, a rough guide could be developed on the basis of flow parameter L/G (ρ_G/ρ_L)^{0.5} (abscissa in

Figs. 14-31, 14-55, and 14-56) and the following tentative conclusions could be drawn:

Flow Parameter 0.02–0.1

1. Trays and random packings have much the same efficiency and capacity.

2. Structured packing efficiency is about 1.5 times that of trays or random packing.

3. At a parameter of 0.02, the structured packing has a 1.3–1.4 capacity advantage over random packing and trays. This advantage disappears as the parameter approaches 0.1.

Flow Parameter 0.1–0.3

1. Trays and random packings have about the same efficiency and capacity.

2. Structured packing has about the same capacity as trays and random packings.

3. The efficiency advantage of structured packing over random packings and trays decreases from 1.5 to 1.2 as the parameter increases from 0.1 to 0.3.

Flow Parameter 0.3–0.5

1. The loss of capacity of structured packing is greatest in this range.

2. The random packing appears to have the highest capacity and efficiency with conventional trays just slightly behind. Structured packing has the least capacity and efficiency.

Experience indicates that use of structured packings has capacity/efficiency disadvantages in the higher-pressure (higher-flow-parameter) region.

Zuidervog and Nutter [*ICHEME Symp. Ser.* **128**, A481 (1992)] explain the loss of capacity/efficiency by a large degree of backmixing and vapor recycle at high flow parameters, promoted by the solid walls of the corrugated packing layers.

SYSTEM LIMIT: THE ULTIMATE CAPACITY OF FRACTIONATORS

Liquid drops of various sizes form in the gas-liquid contact zones of tray or packed towers. Small drops are easily entrained upward, but their volume is usually too small to initiate excessive liquid accumulation (flooding). When the gas velocity is high enough to initiate a massive carryover of the larger drops to the tray above, or upward in a packed bed, liquid accumulation (entrainment flooding) takes place. This flood can be alleviated by increasing the tray spacing or using more hole areas on trays or by using larger, more open packings.

Upon further increase of gas velocity, a limit is reached when the superficial gas velocity in the gas-liquid contact zone exceeds the settling velocity of large liquid drops. At gas velocities higher than this, ascending gas lifts and carries over much of the tray or packing liquid, causing the tower to flood. This flood is termed *system limit* or *ultimate capacity*. This flood cannot be debottlenecked by improving packing size or shape, tray hole area, or tray spacing. The system limit gas velocity is a function only of physical properties and liquid flow rate. Once this limit is reached, the liquid will be blown upward. This is analogous to spraying water against a strong wind and getting drenched (Yangai, *Chem. Eng.*, p. 120, November 1990). The system limit represents the ultimate capacity of the vast majority of existing trays and packings. In some applications, where very open packings (or trays) are used, such as in refinery vacuum towers, the system limit is the actual capacity limit.

The original work of Souders and Brown [*Ind. Eng. Chem.* **26**(1), 98 (1934), Eq. (14-80)] related the capacity of fractionators due to entrainment flooding to the settling velocity of drops. The concept of system limit was advanced by Fractionation Research Inc. (FRI), whose measurements and model have recently been published (Fitz and Kunes, *Distillation 2001: Proceedings of Topical Conference, AIChE Spring National Meeting, Houston, Tex., 2001*; Stupin, FRI Topical Report 34, 1965 available through Special Collection Section, Oklahoma State University Library, Stillwater, Okla.). Figure 14-75 is a plot of FRI system limit data (most derived from tests with dual-flow trays with 29 percent hole area and 1.2- to 2.4-m tray spacing) against liquid superficial velocity for a variety of systems (Stupin, loc. cit.,

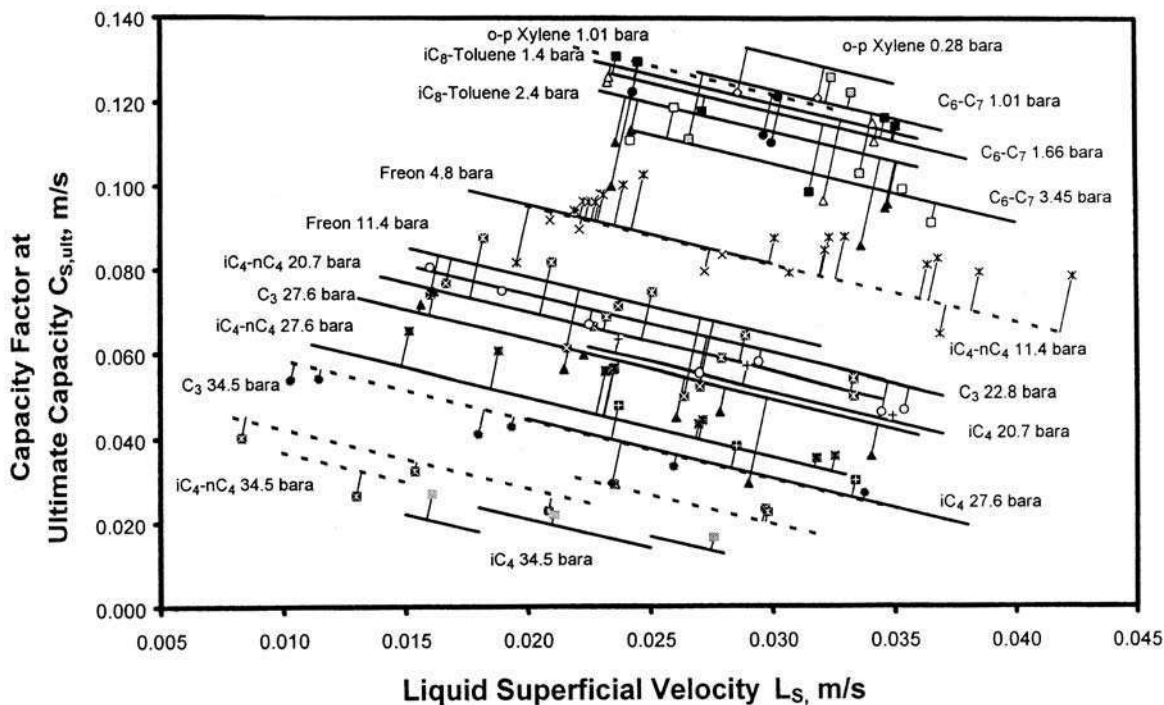


FIG. 14-75 Effect of liquid rate on ultimate capacity at higher liquid rates. (From Stupin, W. J., and H. Z. Kister, *Trans. IChemE*, vol. 81, Part A, p. 136, January 2003. Reprinted courtesy of IChemE.)

1965). The data show a constant-slope linear dependence of the system limit C -factor on the liquid load. There was a shortage of data at low liquid loads. Later data (Fig. 14-76) showed that as the liquid load was reduced, the system limit $C_{s,ult}$ stopped increasing and reached a limiting value. Based on this observation, Stupin and Kister [*Trans. IChemE* 81, Part A, p. 136 (January 2003)] empirically revised the earlier Stupin/FRI correlation to give

$$C_1 = 0.445(1 - F) \left(\frac{\sigma}{\Delta\rho} \right)^{0.25} - 1.4L_s \quad (14-167)$$

$$C_2 = 0.356(1 - F) \left(\frac{\sigma}{\Delta\rho} \right)^{0.25} \quad (14-168)$$

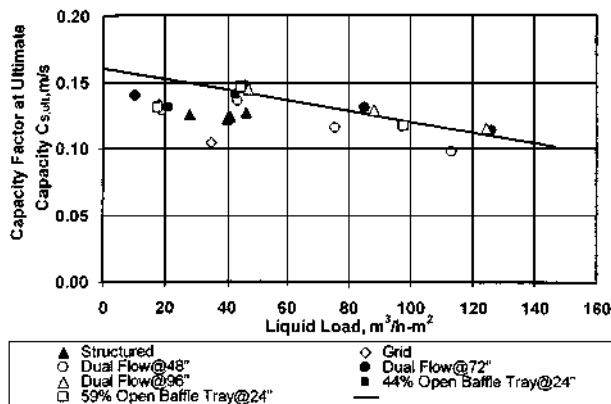


FIG. 14-76 Comparison of original ultimate capacity correlation to test data, C_6/C_7 , 1.66 bar. (From Stupin, W. J., and H. Z. Kister, *Trans. IChemE*, vol. 81, Part A, p. 136, January 2003. Reprinted courtesy of IChemE.)

$$C_{s,ult} = \text{smaller of } C_1 \text{ and } C_2 \quad (14-169)$$

where

$$F = \frac{1}{1 + 1.4(\Delta\rho/\rho_G)^{1/2}} \quad (14-170)$$

In Eqs. (14-167) through (14-170), $C_{s,ult}$ is the system limit C -factor based on the tower superficial area [see Eq. (14-77) for C -factor definition]; L_s is the liquid superficial velocity, m/s; σ is the surface tension, mN/m; $\Delta\rho$ is the difference between the liquid and gas densities, kg/m³; and ρ_G is the gas density, kg/m³.

Stupin and Kister (loc. cit.) relate the flattening of the curve in Fig. 14-76 at low liquid loads to the formation of more, smaller, easier-to-entrain liquid drops when the liquid load is lowered beyond the limiting liquid load. It follows that devices that can restrict the formation of smaller drops may be able to approach the system limit capacity predicted by Stupin's original equation [Eq. (14-167)] even at low liquid loads.

The only devices capable of debottlenecking a tray system-limit device are those that introduce a new force that helps disentrain the vapor space. Devices that use centrifugal force (see "Centrifugal Force Deentrainment") are beginning to make inroads into commercial distillation and have achieved capacities as high as 25 percent above the system limit. Even the horizontal vapor push (see "Truncated Downcomers/Forward-Push Trays") can help settle the entrained drops, but to a much lesser extent. It is unknown whether the horizontal push alone can achieve capacities exceeding the system limit.

WETTED-WALL COLUMNS

Wetted-wall or falling-film columns have found application in mass-transfer problems when high-heat-transfer-rate requirements are concomitant with the absorption process. Large areas of open surface

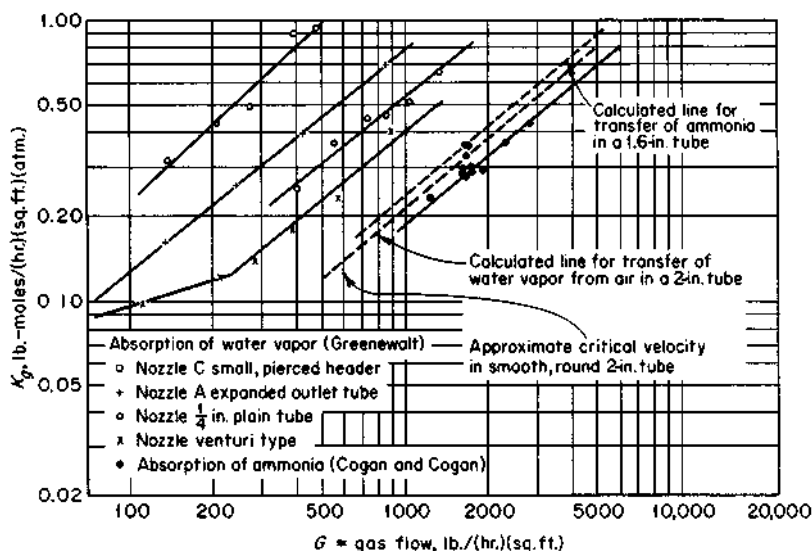


FIG. 14-77 Mass-transfer rates in wetted-wall columns having turbulence promoters. To convert pound-moles per hour-square foot-atmosphere to kilogram-moles per second-square meter-atmosphere, multiply by 0.00136; to convert pounds per hour-square foot to kilograms per second-square meter, multiply by 0.00136; and to convert inches to millimeters, multiply by 25.4. (Data of Greenewalt and Cogan and Cogan, *Sherwood, and Pigford, Absorption and Extraction*, 2d ed., McGraw-Hill, New York, 1952.)

are available for heat transfer for a given rate of mass transfer in this type of equipment because of the low mass-transfer rate inherent in wetted-wall equipment. In addition, this type of equipment lends itself to annular-type cooling devices.

Gilliland and Sherwood [*Ind. Eng. Chem.*, **26**, 516 (1934)] found that, for vaporization of pure liquids in air streams for streamline flow,

$$\frac{k_g D_{\text{tube}}}{D_g} \frac{P_{BM}}{P} = 0.023 N_{Re}^{0.83} N_{Sc}^{0.44} \quad (14-171)$$

where D_g = diffusion coefficient, ft^2/h
 D_{tube} = inside diameter of tube, ft
 k_g = mass-transfer coefficient, gas phase,
 $\text{lb.-mol}/(\text{h-ft}^2)$ ($\text{lb.-mol}/\text{ft}^2$)
 P_{BM} = logarithmic mean partial pressure of inert gas, atm
 P = total pressure, atm
 N_{Re} = Reynolds number, gas phase
 N_{Sc} = Schmidt number, gas phase

Note that the group on the left side of Eq. (14-171) is dimensionless. When turbulence promoters are used at the inlet-gas section, an improvement in gas mass-transfer coefficient for absorption of water vapor by sulfuric acid was observed by Greenewalt [*Ind. Eng. Chem.*, **18**, 1291 (1926)]. A falling off of the rate of mass transfer below that indicated in Eq. (14-171) was observed by Cogan and Cogan (thesis, Massachusetts Institute of Technology, 1932) when a calming zone preceded the gas inlet in ammonia absorption (Fig. 14-77).

In work with the hydrogen chloride-air-water system, Dobratz, Moore, Barnard, and Meyer [*Chem. Eng. Prog.*, **49**, 611 (1953)] using a cocurrent-flow system found that $K_g \propto G^{1.5}$ (Fig. 14-78) instead of the 0.8 power as indicated by the Gilliland equation. Heat-transfer coefficients were also determined in this study. The radical increase in heat-transfer rate in the range of $G = 30 \text{ kg}/(\text{s-m}^2)$ [$20,000 \text{ lb}/(\text{h-ft}^2)$] was similar to that observed by Tepe and Mueller [*Chem. Eng. Prog.*, **43**, 267 (1947)] in condensation inside tubes.

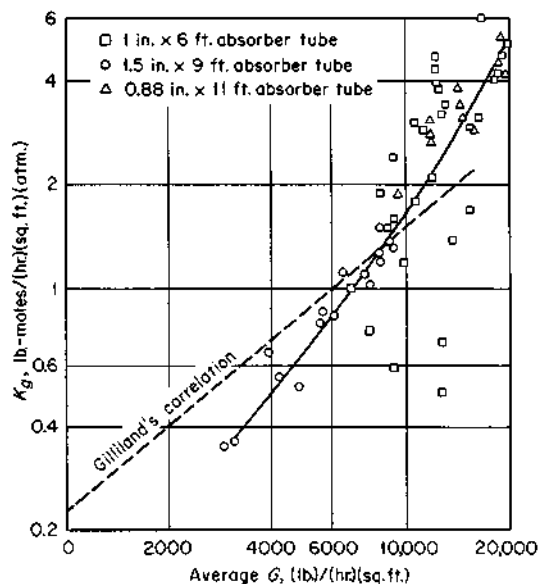


FIG. 14-78 Mass-transfer coefficients versus average gas velocity—HCl absorption, wetted-wall column. To convert pound-moles per hour-square foot-atmosphere to kilogram-moles per second-square meter-atmosphere, multiply by 0.00136; to convert pounds per hour-square foot to kilograms per second-square meter, multiply by 0.00136; to convert feet to meters, multiply by 0.305; and to convert inches to millimeters, multiply by 25.4. [Dobratz et al., *Chem. Eng. Prog.*, **49**, 611 (1953).]

Gaylord and Miranda [*Chem. Eng. Prog.*, **53**, 139M (1957)] using a multitube cocurrent-flow falling-film hydrochloric acid absorber for hydrogen chloride absorption found

$$K_g = \frac{1.66(10^{-5})}{M_m^{1.75}} \left(\frac{D_{\text{tube}} G}{\mu} \right) \quad (14.172)$$

where K_g = overall mass-transfer coefficient, (kg·mol)/(s·m²·atm)

M_m = mean molecular weight of gas stream at inlet to tube

D_{tube} = diameter of tube, m

G = mass velocity of gas at inlet to tube, kg/(s·m²)

μ = viscosity of gas, Pa·s

Note that the group $D_{\text{tube}} G / \mu$ is dimensionless. This relationship also satisfied the data obtained for this system, with a single-tube falling-film unit, by Coull, Bishop, and Gaylor [*Chem. Eng. Prog.*, **45**, 506 (1949)].

The rate of mass transfer in the liquid phase in wetted-wall columns is highly dependent on surface conditions. When laminar-flow conditions prevail without the presence of wave formation, the laminar-penetration theory prevails. When, however, ripples form at the surface, and they

may occur at a Reynolds number exceeding 4, a significant rate of surface regeneration develops, resulting in an increase in mass-transfer rate.

If no wave formations are present, analysis of behavior of the liquid-film mass transfer as developed by Hatta and Katori [*J. Soc. Chem. Ind.*, **37**, 280B (1934)] indicates that

$$k_l = 0.422 \sqrt{\frac{D_l \Gamma}{\rho B_F}} \quad (14.173)$$

where $B_F = (3u\Gamma/\rho^2 g)^{1/3}$

D_l = liquid-phase diffusion coefficient, m²/s

ρ = liquid density, kg/m³

Z = length of surface, m

k_l = liquid-film-transfer coefficient,

(kg·mol)/[(s·m²)(kg·mol)/m³]

Γ = liquid-flow rate, kg/(s·m) based on wetted perimeter

μ = viscosity of liquid, Pa·s

g = gravity acceleration, 9.81 m/s²

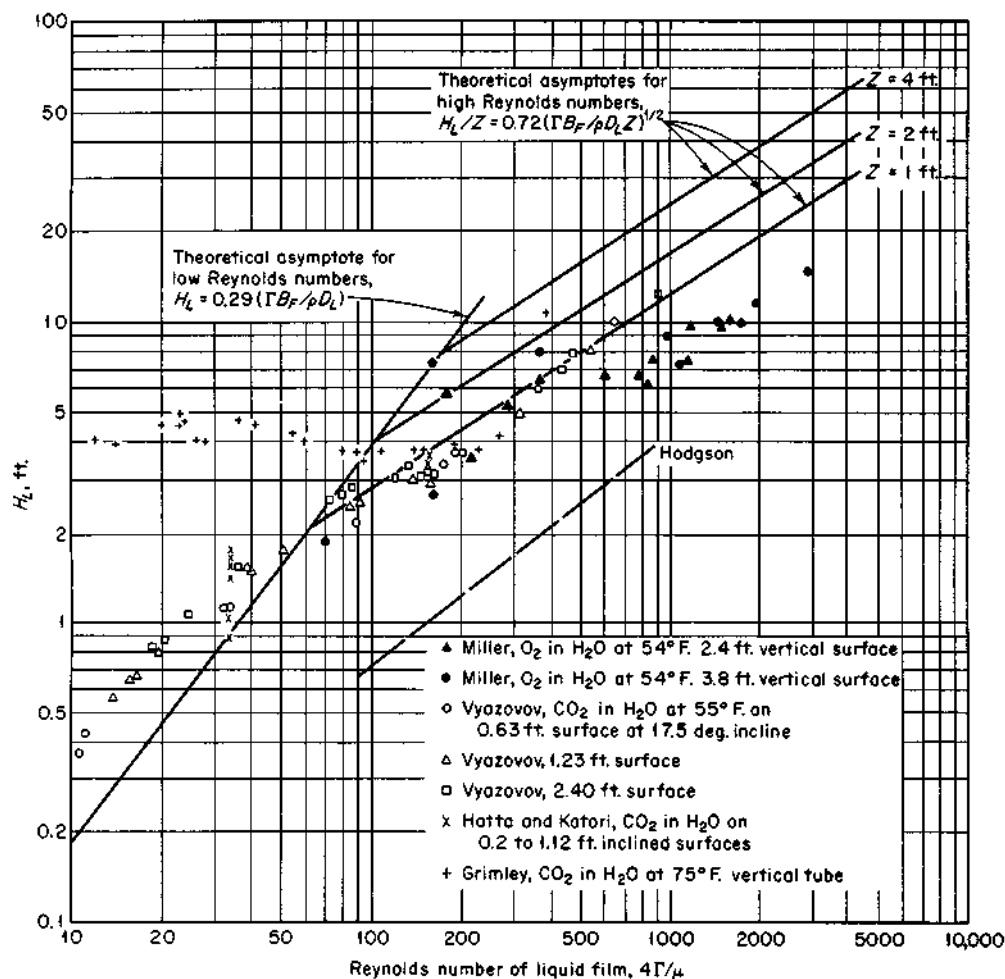


FIG. 14-79 Liquid-film resistance in absorption of gases in wetted-wall columns. Theoretical lines are calculated for oxygen absorption in water at 55°F. To convert feet to meters, multiply by 0.3048; °C = $\frac{5}{9}(\text{°F} - 32)$. (Sherwood and Pigford, *Absorption and Extraction*, 2d ed., McGraw-Hill, New York, 1952.)

When Z is large or $\Gamma/\rho B_F$ is so small that liquid penetration is complete,

$$k_\ell = 11.800 D_\ell / B_F \quad (14-174)$$

and

$$H_\ell = 0.95 \Gamma B_F / D_\ell \quad (14-175)$$

A comparison of experimental data for carbon dioxide absorption obtained by Hatta and Katori (op. cit.), Grimley [*Trans. Inst. Chem. Eng.*, **23**, 228 (1945)], and Vyazov [*Zh. Tekh. Fiz.* (U.S.S.R.), **10**, 1519 (1940)] and for absorption of oxygen and hydrogen by Hodgson (S.M. thesis, Massachusetts Institute of Technology, 1949), Henley (B.S. thesis, University of Delaware, 1949), Miller (B.S. thesis, University of Delaware, 1949), and Richards (B.S. thesis, University of Delaware, 1950) was made by Sherwood and Pigford (*Absorption and Extraction*, McGraw-Hill, New York, 1952) and is indicated in Fig. 14-79.

In general, the observed mass-transfer rates are greater than those predicted by theory and may be related to the development of surface rippling, a phenomenon which increases in intensity with increasing liquid path.

Vivian and Peaceman [*Am. Inst. Chem. Eng. J.*, **2**, 437 (1956)] investigated the characteristics of the $\text{CO}_2\text{-H}_2\text{O}$ and $\text{Cl}_2\text{-HCl}$, H_2O system in a wetted-wall column and found that gas rate had no effect on the liquid-phase coefficient at Reynolds numbers below 2200. Beyond this rate, the effect of the resulting rippling was to increase significantly the liquid-phase transfer rate. The authors proposed a behavior relationship based on a dimensional analysis but suggested caution in its application concomitant with the use of this type of relationship. Cognizance was taken by the authors of the effects of column length, one to induce rippling and increase of rate of transfer, one to increase time of exposure which via the penetration theory decreases the average rate of mass transfer in the liquid phase. The equation is

$$\frac{k_\ell h}{D_\ell} = 0.433 \left(\frac{\mu_\ell}{\rho_\ell D_\ell} \right)^{1/2} \left(\frac{\rho_\ell^2 g h^3}{\mu_\ell^2} \right)^{1/6} \left(\frac{4\Gamma}{\mu_\ell} \right)^{0.4} \quad (14-176)$$

where D_ℓ = diffusion coefficient of solute in liquid, ft^2/h
 g = gravity-acceleration constant, $4.17 \times 10^8 \text{ ft}/\text{h}^2$
 h = length of wetted wall, ft
 k_ℓ = mass-transfer coefficient, liquid phase, ft/h

TABLE 14-17 Relative Fabricated Cost for Metals Used in Tray-Tower Construction*

Materials of construction	Relative cost per ft ² of tray area (based on carbon steel = 1)
Sheet-metal trays	
Steel	1
4-6% chrome—1/2 moly alloy steel	2.1
11-13% chrome type 410 alloy steel	2.6
Red brass	3
Stainless steel type 304	4.2
Stainless steel type 347	5.1
Monel	7.0
Stainless steel type 316	5.5
Inconel	8.2
Cast-iron trays	2.8

*Peters and Timmerhaus, *Plant Design and Economics for Chemical Engineers*, 4th ed., McGraw-Hill, New York, 1991. To convert cost per square foot to cost per square meter, multiply by 10.76.

Γ = mass rate of flow of liquid, $\text{lb}/(\text{h})(\text{ft of periphery})$
 μ_ℓ = viscosity of liquid, $\text{lb}/(\text{ft})(\text{h})$
 ρ_ℓ = density of liquid, lb/ft^3

The equation is dimensionless.

The effect of chemical reaction in reducing the effect of variation of the liquid rate on the rate of absorption in the laminar-flow regime was illustrated by the evaluation of the rate of absorption of chlorine in ferrous chloride solutions in a wetted-wall column by Gilliland, Baddour, and White [*Am. Inst. Chem. Eng. J.*, **4**, 323 (1958)].

Flooding in Wetted-Wall Columns When gas and liquid are in counterflow in wetted-wall columns, flooding can occur at high gas rates. Methods for calculating this flood are given in "Upper Limit Flooding in Vertical Tubes." In the author's experience, Eq. (14-204) has had an excellent track record for calculating flooding in these columns.

COLUMN COSTS

Estimation of column costs for preliminary process evaluations requires consideration not only of the basic type of internals but also of their effect on overall system cost. For a distillation system, for example, the overall system can include the vessel (column), attendant structures, supports, and foundations; auxiliaries such as reboiler, condenser, feed heater, and control instruments; and connecting piping. The choice of internals influences all these costs, but other factors influence them as well. A complete optimization of the system requires a full-process simulation model that can cover all pertinent variables influencing economics.

Cost of Internals Installed costs of trays may be estimated from Fig. 14-80, with corrections for tray material taken from Table 14-17. For two-pass trays the cost is 15 to 20 percent higher. Approximate costs of random packing materials may be obtained from Table 14-18, but it should be recognized that, because of competition, there can be significant variations in these costs from vendor to vendor. Also, packings sold in very large quantities carry discounts. In 1995, costs of structured packings, made from sheet metal, averaged \$90 to \$110 per cubic foot, but the need for special distributors and redistributors can double the cost of structured-packings on a volumetric basis. Note

TABLE 14-18 Costs of Random Packings, Uninstalled, January 1990

Prices in dollars per ft³, 100 ft³ orders, f.o.b. manufacturing plant

	Size, in, \$/ft ³			
	1	1½	2	3
Raschig rings				
Chemical porcelain	12.8	10.3	9.4	7.8
Carbon steel	36.5	23.9	20.5	16.8
Stainless steel	155	117	87.8	—
Carbon	52	46.2	33.9	31.0
Intalox saddles				
Chemical stoneware	17.6	13.0	11.8	10.7
Chemical porcelain	18.8	14.1	12.9	11.8
Polypropylene	21.2	—	13.1	7.0
Berl saddles				
Chemical stoneware	27.0	21.0	—	—
Chemical porcelain	33.5	21.5	15.6	—
Pall rings				
Carbon steel	29.3	19.9	18.2	—
Stainless steel	131	99.0	86.2	—
Polypropylene	21.2	14.4	13.1	—

Peters and Timmerhaus, *Plant Design and Economics for Chemical Engineers*, 4th ed., McGraw-Hill, New York, 1991. To convert cubic feet to cubic meters, multiply by 0.0283; to convert inches to millimeters, multiply by 25.4; and to convert dollars per cubic foot to dollars per cubic meter, multiply by 35.3.

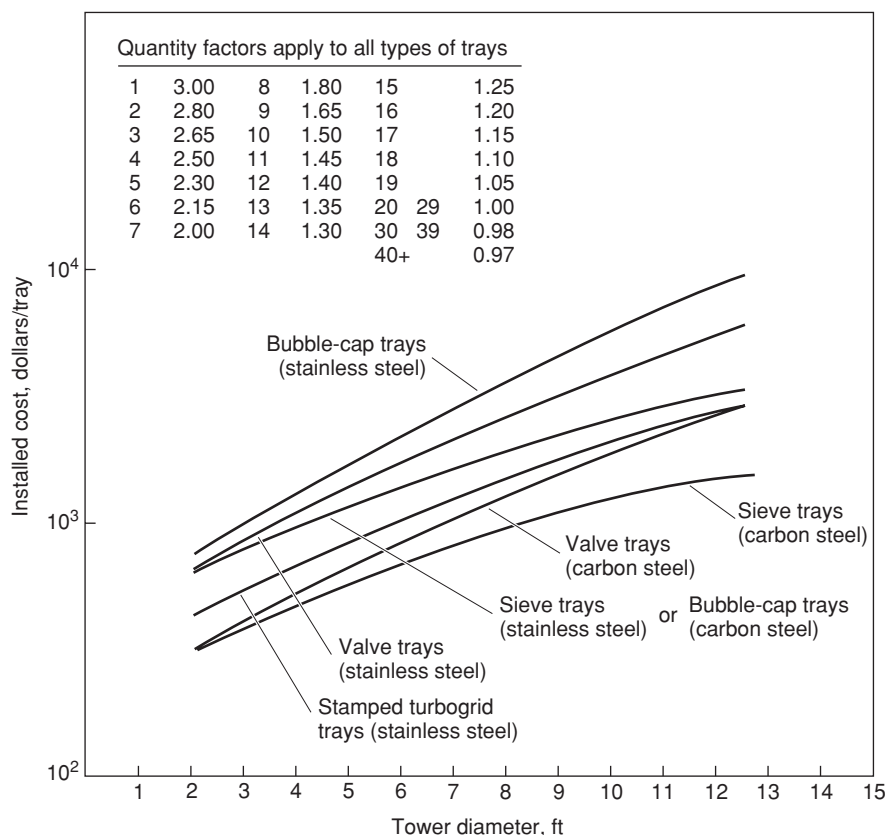


FIG. 14-80 Cost of trays. Price includes tray deck, valves, bubble caps, risers, downcomers, and structural-steel parts. The stainless steel designated is type 410 (*Peters and Timmerhaus, Plant Design and Economics for Chemical Engineers, 4th ed., McGraw-Hill, New York, 1991*).

that for Fig. 14-80 and Table 14-17, the effective cost date is January 1990, with the Marshall and Swift cost index being taken as 904.

As indicated above, packed column internals include liquid distributors, packing support plates, redistributors (as needed), and holddown plates (to prevent movement of packing under flow conditions). Costs of these internals for columns with random packing are given in Fig. 14-81, based on early 1976 prices, and a Marshall and Swift cost index of 460.

Cost of Column The cost of the vessel, including heads, skirt, nozzles, and ladderways, is usually estimated on the basis of weight. Figure 14-82 provides early 1990 cost data for the shell and heads, and Fig. 14-83 provides 1990 cost data for connections. For very approximate estimates of complete columns, including internals, Fig. 14-84 may be used. As for Figs. 14-82 and 14-83, the cost index is 904.

PHASE DISPERSION

GENERAL REFERENCES: For an overall discussion of gas-liquid breakup processes, see Brodkey, *The Phenomena of Fluid Motions*, Addison-Wesley, Reading, Massachusetts, 1967. For a discussion of atomization devices and how they work, see Masters, *Spray Drying Handbook*, 5th ed., Wiley, New York, 1991; and Lefebvre, *Atomization and Sprays*, Hemisphere, New York, 1989. A beautifully illustrated older source is Dombrowski and Munday, *Biochemical and Biological Engineering Science*, vol. 2, Academic Press, London, 1968, pp. 209–320. Steinmeyer [*Chem. Engr. Prog.*, **91**(7), 72–80 (1995)] built on Hinze's work with turbulence and showed that several atomization processes follow the common theme of breakup of large droplets due to turbulence in the gas phase. Turbulence in turn correlates with power dissipation per unit mass. In the text below, correlations referring to *power/mass* are taken from this source. For a survey on fog formation, see Amelin, *Theory of Fog Formation*, Israel Program for Scientific Translations, Jerusalem, 1967.

BASICS OF INTERFACIAL CONTACTORS

Steady-State Systems: Bubbles and Droplets Bubbles are made by injecting vapor below the liquid surface. In contrast, droplets are commonly made by atomizing nozzles that inject liquid into a vapor. Bubble and droplet systems are fundamentally different, mainly because of the enormous difference in density of the injected phase. There are situations where each is preferred. Bubble systems tend to have much higher interfacial area as shown by Example 16 contrasted with Examples 14 and 15. Because of their higher area, bubble systems will usually give a closer approach to equilibrium.

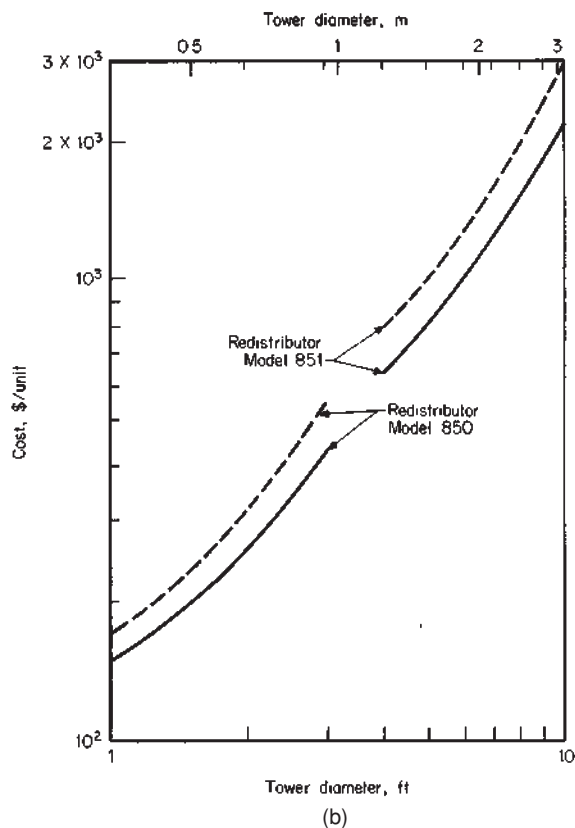
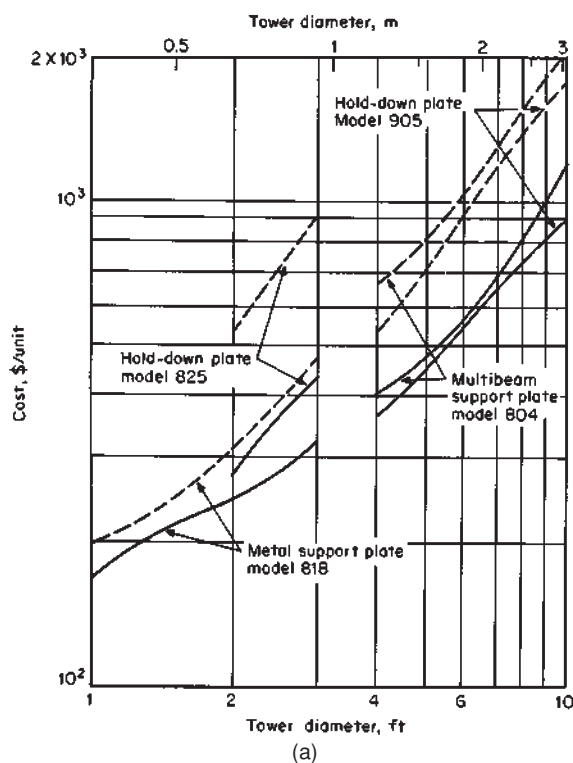


FIG. 14-81 Cost of internal devices for columns containing random packings. (a) Hold-down plates and support plates. (b) Redistributors. [Pikulik and Diaz, Chem. Eng., 84(21), 106 (Oct. 10, 1977).]

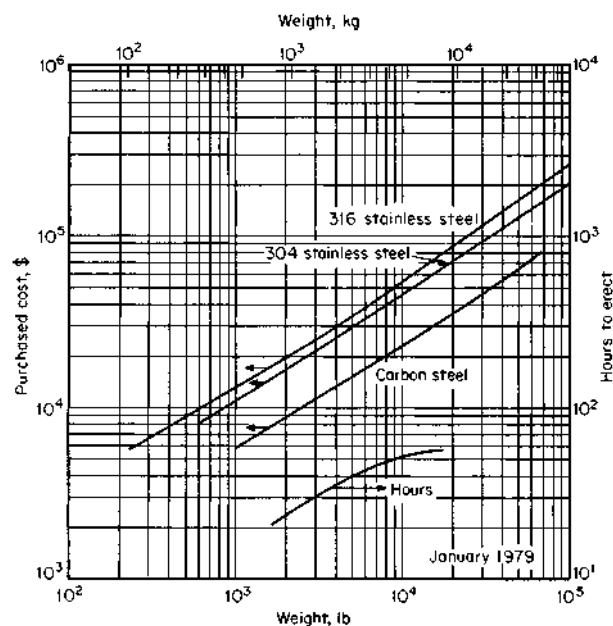


FIG. 14-82 Fabricated costs and installation time of towers. Costs are for shell with two heads and skirt, but without trays, packing, or connections. (Peters and Timmerhaus, Plant Design and Economics for Chemical Engineers, 4th ed., McGraw-Hill, New York, 1991.)

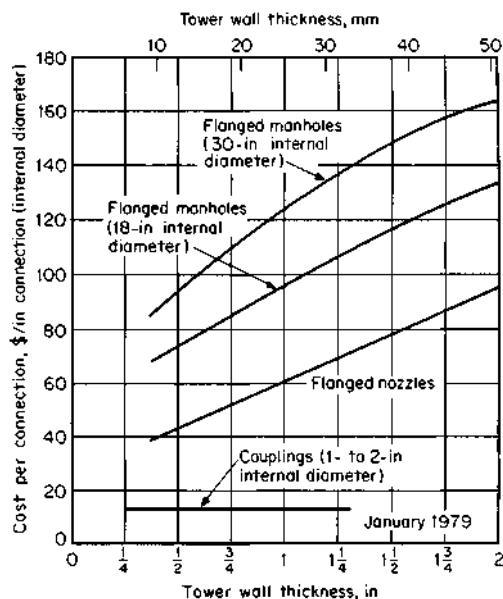


FIG. 14-83 Approximate installed cost of steel-tower connections. Values apply to 2070-kPa connections. Multiply costs by 0.9 for 1035-kPa (150-lb) connections and by 1.2 for 4140-kPa (600-lb) connections. To convert inches to millimeters, multiply by 25.4; to convert dollars per inch to dollars per centimeter, multiply by 0.394. (Peters and Timmerhaus, Plant Design and Economics for Chemical Engineers, 4th ed., New York, McGraw-Hill, 1991.)

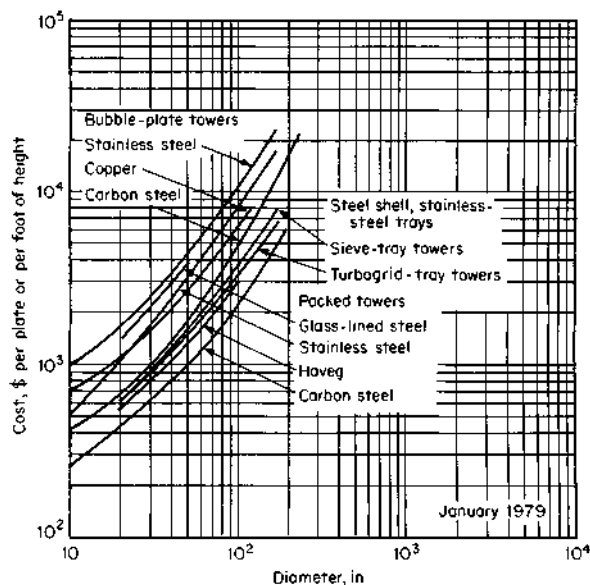


FIG. 14-84 Cost of towers including installation and auxiliaries. To convert inches to millimeters, multiply by 25.4; to convert feet to meters, multiply by 0.305; and to convert dollars per foot to dollars per meter, multiply by 3.28. (Peters and Timmerhaus, *Plant Design and Economics for Chemical Engineers*, 4th ed., McGraw-Hill, New York, 1991.)

However, droplet systems can enable much higher energy input (via gas phase pressure drop in cocurrent systems) and, as a result, dominate applications where a quick quench is needed. See Examples 21 and 22. Conversely, droplet systems can also be designed for very low pressure drop which is advantageous in applications such as vacuum condensers.

Unstable Systems: Froths and Hollow Cone Atomizing Nozzles We usually think of interfacial contact as a steady-state system of raining droplets or rising bubbles, but some of the most efficient interfacial contactors take advantage of unstable interfacial geometry. The most common is the distillation tray which operates with a wild mix of bubbles, jets, films, and droplets. The mix is often described as *froth*. Gas pressure drop provides the energy to create the froth.

A variant on the froth contact is the reverse jet contactor (Example 22), which can be considered as an upside-down distillation tray, operated above the flooding velocity in cocurrent flow of gas and liquid. It is limited to one stage.

An entirely different unstable contactor involves the thin expanding liquid film produced by a hollow cone spray nozzle. Because of fresh surface and the thinness of the film, this can give very high transfer for liquid-limited systems. Two applications are direct contact condensation and removal of volatile components from a high-boiling residual liquid.

Surface Tension Makes Liquid Sheets and Liquid Columns Unstable Surface tension is the energy required to make an increment of interfacial surface. A sheet or column of liquid has more surface than a sphere, hence surface tension converts sheets and columns to droplets. See Fig. 14-86.

There are many different atomizers, but the underlying principle of all is the same—to first generate a flat sheet or a liquid column. Liquid sheets and columns are unstable, a small surface disturbance on either will propagate, and the liquid will reshape itself into droplets. The key property in controlling this process is surface tension. Surface tension gets a high exponent in all the atomization correlations.

Little Droplets and Bubbles vs. Big Droplets and Bubbles—Coalescence vs. Breakup When big drops are subjected to shear forces, as in falling rain, the droplets are distorted; and if the distur-

tions are great enough, the big droplets break into little ones. This is why raindrops never exceed a certain size. A variant on this is breakup in highly turbulent systems such as in high-velocity quench systems or pneumatic nozzles. Here the droplets are distorted by the energy of the turbulent eddies.

But little droplets and bubbles have more surface per unit of liquid than big ones. Hence little droplets tend to coalesce into big ones, and will grow in size if given enough quiet time.

While the primary difficulty is estimating the interfacial area due to the unstable interface, a secondary problem is that freshly made, unstable surface gives higher transfer than older, more stable surface.

Empirical Design Tempered by Operating Data The net of these is that interfacial area is difficult to predict and interfacial contactors are difficult to design.

Prediction methods are given below but should always be tempered by operating experience.

INTERFACIAL AREA—IMPACT OF DROPLET OR BUBBLE SIZE

Transfer is aided by increased interfacial area. Interfacial area per unit volume a_D of a single droplet or bubble is inversely proportional to the diameter of the droplet or bubble D .

$$a_D = 6/D \quad (14-177)$$

To estimate the total interfacial area in a given volume, the a_D value is multiplied by the fractional holdup of dispersed phase in the total volume.

$$a = a_D(\Phi_D) \quad (14-178)$$

where a = interfacial area/volume and Φ_D = fraction of volume in dispersed phase = holdup. Fractional holdup in a continuous process depends on the velocities of the two phases, as if they were flowing by themselves.

$$\Phi_D = (\text{dispersed phase volume})/(\text{volume of dispersed and continuous phases})$$

Example 14: Interfacial Area for Droplets/Gas in Cocurrent Flow For equal mass flow of gas and liquid and with gas density 0.001 of liquid density, the gas velocity in cocurrent flow will be 1000 times the liquid velocity. This sets Φ_D .

$$\Phi_D = 1/(1 + 1000) = 0.00099$$

If the droplets are 500 μm in diameter, Eqs. (14-177) and (14-178) give

$$a = (6/0.0005)(0.00099) = 12 \text{ m}^2/\text{m}^3$$

If the droplets are 100 μm in diameter, Eqs. (14-177) and (14-178) give

$$a = (6/0.0001)(0.00099) = 60 \text{ m}^2/\text{m}^3$$

Example 15: Interfacial Area for Droplets Falling in a Vessel Droplet systems rarely exceed a Φ_D value of 0.01. At this low level, Φ_D in a low-velocity countercurrent contactor can be approximated by Eq. (14-179).

$$\Phi_D = U_L/(U_L - U_G) \quad (14-179)$$

where U_L = liquid superficial velocity
 U_t = terminal velocity of droplet
 U_G = gas superficial velocity

With a gas superficial velocity of 1.5 m/s, for equal mass flow of gas and liquid, with gas density 0.001 of liquid density, and with 500- μm -diameter droplets falling at a terminal settling of 2.5 m/s, Eq. (14-179) gives a fractional holdup of liquid of

$$\Phi_D = (0.001)1.5/(2.5 - 1.5) = 0.0015$$

Equations (14-177) and (14-178) then give

$$a = (6/0.0005)(0.0015) = 18 \text{ m}^2/\text{m}^3$$

Example 16: Interfacial Area for Bubbles Rising in a Vessel For bubble systems (gases dispersed in liquids) fractional holdup can approach 0.5 as shown by Fig. 14-104. However, before reaching this holdup, the bubble systems shift to an unstable mix of bubbles and vapor jets. Hence an exact

comparison to Example 14 isn't possible because at the 1.5 m/s velocity of Example 14, the system becomes a froth. But at about one-fifth the velocity of Example 14, an estimate of interfacial area is possible.

If the bubble size is 10,000 μm and fractional holdup is 0.4, Eqs. (14-177) and (14-178) give an interfacial area of

$$a = (6/0.01)(0.4) = 240 \text{ m}^2/\text{m}^3$$

Measured interfacial area in distillation trays is consistent with this high value.

Note the much higher interfacial area than in the droplet systems of Examples 14 and 15. The higher interfacial area when the gas is dispersed explains why bubbling and froth systems often give better performance than droplet systems. The big difference in interfacial area stems from the much larger volume per unit of mass of gas, i.e., lower density of the gas than the liquid.

RATE MEASURES, TRANSFER UNITS, APPROACH TO EQUILIBRIUM, AND BYPASSING

What Controls Mass/Heat Transfer: Liquid or Gas Transfer or Bypassing Either gas side or liquid side of the interface can be controlling.

Liquid-Controlled In fractionation systems with high viscosity or component relative volatility that greatly exceeds 1, the liquid side will be controlling. This is clearly illustrated by Fig. 14-47 which shows a sharp decline in efficiency with either a rise in liquid viscosity or a rise in component relative volatility.

Note that *high component relative volatility* means the same thing as *sparingly soluble*. Oxygen dissolving in a fermentation reactor is an example of a system being liquid-controlled due to a *sparingly soluble gas*. Another application that is liquid-controlled is the removal of *high relative volatility components* out of residual oil.

Still another case where liquid controls is in condensing a pure vapor, as in Example 23, or absorbing a pure gas, as in Example 24.

Gas-Controlled The gas side dominates in gas cooling applications. An example is the quenching of a furnace effluent with a vaporizing liquid. In this application the liquid is nearly uniform in temperature. Restated, the reduction in driving force across the liquid side of the interface is unimportant.

Other applications that are gas-side-controlled include removal of a component such as NH_3 from a gas by using an acidic liquid, or removing a component such as SO_2 from a gas with a basic liquid. See Examples 19 through 22.

Bypassing-Controlled Trayed or packed columns operate with countercurrent flow and can achieve many equilibrium stages in series by good distribution of gas and liquid, and careful control of details. Other devices such as sprays are vulnerable to bypassing and are limited to one equilibrium stage.

Rate Measures for Interfacial Processes Terminology used for reporting rate data can be confusing. Normally rate data are reported on a volumetric basis with transfer rate and effective area combined. For example, $k_L a$ denotes mass-transfer data per unit volume. The L subscript means it is referenced to the molar concentration difference between the interface and the bulk liquid. This is commonly used on data involving a sparingly soluble (high relative volatility) component. Note that the lowercase k means the data deal only with the resistance in the liquid phase.

Less commonly, data are given as $k_G a$. The G subscript means it is referenced to the molar concentration difference between the interface and the gas. This might be used for data on absorbing a gas such as NH_3 by a highly acidic liquid. Note that $k_G a$ only deals with the resistance in the gas phase.

When one is dealing with direct contact heat transfer, the corresponding terms are $h_L a$ and $h_G a$. Here the driving force is the temperature difference. The L subscript means that we are dealing with a liquid-limited process such as condensing a pure liquid. How to convert $k_L a$ data to an $h_L a$ value is illustrated by Example 23.

There are ways to combine the liquid and gas resistance to get an overall transfer rate such as $K_G a$ (as denoted by the uppercase K). However, data are rarely reported in this form.

Approach to Equilibrium Although rate measures such as $k_G a$ and $h_G a$ are often cited in the literature, they are often not as useful to designers as the simpler concept of approach to equilibrium. *Approach to equilibrium* compares the transfer between liquid and gas phases to the best possible that could be achieved in a single backmixed equilibrium stage.

Approach to equilibrium is easy to understand and easy to apply. Examples 17 through 23 illustrate its use.

Example 17: Approach to Equilibrium—Perfectly Mixed, Complete Exchange This would be approximated by a very long pipeline contactor where an acidic aqueous stream is injected to cool the gas and remove NH_3 .

If the adiabatic saturation temperature of the gas is 70°C , at the exit of the contactor, the gas would be cooled to 70°C .

Similarly, at the exit of the contactor, the NH_3 in the gas would be zero, regardless of the initial concentration.

Example 18: Approach to Equilibrium—Complete Exchange but with 10 Percent Gas Bypassing A spray column is used, and an acidic liquid rains down on the gas of Example 17. If the initial NH_3 is 1000 ppm and 10 percent of the gas bypasses, the NH_3 in the exit gas would be

$$0.1(1000) = 100 \text{ ppm}$$

Similarly, if the gas enters at 120°C , at the exit we would find 10 percent of the differential above the adiabatic saturation temperature. For an adiabatic saturation temperature of 70°C , the exit gas temperature would be

$$70 + 0.1(120 - 70) = 75^\circ\text{C}$$

Approach to Equilibrium—Finite Contactor with No Bypassing When there is no bypassing, the measure that sets the approach is the ratio of change to driving force. This ratio is called the *number of transfer units* N_G . It is dimensionless. For heat-transfer applications, it can be envisioned as a conventional heat exchanger where a vaporizing liquid cools a gas:

$$\text{No. of gas-phase transfer units} = \frac{T_{G,\text{out}} - T_{G,\text{in}}}{(T_G - T_L)_{\text{av}}} = N_G \quad (14-180)$$

Where T_G = gas temperature and T_L = liquid temperature. The number of transfer units N_G can also be calculated as the capability for change divided by the thermal capacitance of the flowing streams.

$$\begin{aligned} N_G &= \frac{(\text{system volume})(h_G a)}{(\text{volumetric flow rate})\rho_G c_G} \\ &= \frac{(\text{gas contact time})(h_G a)}{\rho_G c_G} \end{aligned} \quad (14-181)$$

where a = interfacial area per unit volume

h_G = heat-transfer coefficient from interface to gas

ρ_G = gas density

c_G = gas specific heat

Note that in the above, performance and properties all refer to the gas, which is appropriate when dealing with a gas-limited transfer process.

This leads to a way to estimate the approach to equilibrium.

$$E = 1 - e^{-N_G} \quad (14-182)$$

where E = "approach to equilibrium" fractional removal of NH_3 or fractional approach to adiabatic liquid temperature

N_G = number of transfer units calculated relative to gas flow

Example 19: Finite Exchange, No Bypassing, Short Contactor A short cocurrent horizontal pipeline contactor gives 86 percent removal of NH_3 . There is no bypassing because of the highly turbulent gas flow and injection of liquid into the center of the pipe. What would we expect the exit gas temperature to be?

14-90 EQUIPMENT FOR DISTILLATION, GAS ABSORPTION, PHASE DISPERSION, AND PHASE SEPARATION

Equation (14-182) says that the backcalculated N_G is 2:

$$N_G = -\ln(1 - 0.86) = 2$$

For diffusing gases of similar molecular weight, the properties that control heat transfer follow the same rules as those that control mass transfer. As a result, the NH_3 scrubbing and gas cooling processes achieve similar approaches to equilibrium.

For an entry temperature of 120°C and an adiabatic saturation temperature of 70°C , the expected outlet temperature would be

$$70 + (1 - 0.86)(120 - 70) = 77^\circ\text{C}$$

This looks like a powerful concept, but its value is limited due to uncertainty in estimating $h_G a$. Both h_G and a are difficult to estimate due to dependence on power dissipation as discussed below. The primary value of the N_G concept is in estimating an expected change from baseline data as in the comparison of Example 19 with Example 20.

Example 20: A Contactor That Is Twice as Long, No Bypassing If we double the length of the pipeline contactor, double the effective contact area, and double the number of transfer units to 4, what do we expect for performance?

For $N_G = 4$,

$$E = 1 - e^{-4} = 0.982$$

The NH_3 in the exit gas would be expected to drop to

$$(1 - 0.982)/(1000) = 18 \text{ ppm}$$

and the expected outlet temperature would be

$$70 + (1 - 0.982)(120 - 70) = 70.9^\circ\text{C}$$

If we double the length again, we increase the number of transfer units to 8 and achieve an approach of

$$E = 1 - e^{-8} = 0.9997$$

The outlet temperature would be

$$70 + (1 - 0.9997)(120 - 70) = 70.015^\circ\text{C}$$

Similarly the NH_3 in the exit gas would be

$$(1 - 0.9997)/(1000) = 0.3 \text{ ppm}$$

Note that this approximates the exit condition of Example 17.

Transfer Coefficient—Impact of Droplet Size The transfer coefficients increase as the size of droplets decreases. This is so because the transfer process is easier if it only has to move mass or heat a shorter distance (i.e., as the bubble or droplet gets smaller).

In the limiting case of quiescent small bubbles or droplets, the transfer coefficients vary inversely with average bubble or droplet diameter. For example, in heat transfer from a droplet interface to a gas, the minimum value is

$$h_{G,\min} = \text{heat transfer coefficient from interface to gas} = 2k_G/D \quad (14-183)$$

where k_G = gas thermal conductivity and
 D = droplet diameter.

IMPORTANCE OF TURBULENCE

The designer usually has control over the size of a droplet. As discussed below, several of the correlations show that droplet diameter varies with turbulent energy dissipation. For example, Eqs. (14-190) and (14-201) suggest that in droplet systems

$$D \propto [1/(\text{gas velocity})]^{1.2}$$

and hence from Eq. (14-178)

$$a \propto 1/D \propto (\text{gas velocity})^{1.2} \quad (14-184)$$

However, just looking at the impact of velocity on droplet size underestimates the velocity impact because turbulence gives higher transfer

than Eq. (14-183) predicts. Transfer coefficients increase as the mixing adjacent to the surface increases. This mixing depends on the energy dissipated into the phases. To a first approximation this transfer from droplets increases with local power dissipation raised to the 0.2 power.

$$h_{G,\text{turbulent}} \propto (\text{power dissipated})^{0.2}$$

and since power dissipation per unit volume increases with (velocity)³,

$$h_{G,\text{turbulent}} \propto (\text{velocity})^{0.6} \quad (14-185)$$

The combined effect on interfacial area and on the transfer coefficient is that the effective transfer increases greatly with gas velocity. From Eqs. (14-178) and (14-185)

$$h_G a_{\text{turbulent}} \propto (\text{velocity})^{1.8} \quad (14-186)$$

For quenching operations, this means that even though residence time is cut as gas velocity goes up, the effective approach to equilibrium increases. Since the volume for a given length of pipe falls with (velocity)⁻¹, the expected number of transfer units N_G in a given length of pipe increases with (velocity)^{0.8}.

$$N_{G,\text{turbulent}} \propto (h_G a_{\text{turbulent}})(\text{volume}) \propto (\text{velocity})^{0.8} \quad (14-187)$$

See Example 21.

EXAMPLES OF CONTACTORS

High-Velocity Pipeline Contactors High-velocity cocurrent flow can give more power input than any other approach. This is critical when extremely high rates of reaction quenching are needed.

Example 21: Doubling the Velocity in a Horizontal Pipeline Contactor—Impact on Effective Heat Transfer Velocity in pipeline quench systems often exceeds 62 m/s (200 ft/s). Note that this is far above the flooding velocity in distillation packing, distillation trays, or gas-sparged reactors. There are few data available to validate performance even though liquid injection into high-velocity gas streams has been historically used in quenching reactor effluent systems. However, the designer knows the directional impact of parameters as given by Eq. (14-187).

For example, if a 10-ft length of pipe gives a 90 percent approach to equilibrium in a quench operation, Eq. (14-182) says that the backcalculated N_G is 2.303:

$$N_G - \ln(1 - 0.9) = 2.303$$

Equation (14-187) says if we double velocity but retain the same length, we would expect an increase of N_G to 4.0.

$$N_G = 2.303(2)^{0.8} = 4$$

and

$$E = 1 - e^{-4} = 0.982$$

Restated, the approach to equilibrium rises from 90 percent to greater than 98 percent even though the contact time is cut in half.

Vertical Reverse Jet Contactor A surprisingly effective modification of the liquid injection quench concept is to inject the liquid countercurrent *upward* into a gas flowing *downward*, with the gas velocity at some multiple of the flooding velocity defined by Eq. (14-203). The reverse jet contactor can be envisioned as an upside-down distillation tray. For large gas volumes, multiple injection nozzles are used. One advantage of this configuration is that it minimizes the chance of liquid or gas bypassing. Another advantage is that it operates in the froth region which generates greater area per unit volume than the higher-velocity cocurrent pipeline quench.

The concept was first outlined in U.S. Patent 3,803,805 (1974) and was amplified in U.S. Patent 6,339,169 (2002). The 1974 patent presents data which clarify that the key power input is from the gas stream.

A more recent article discusses use of the reverse jet in refinery offgas scrubbing for removal of both SO_2 and small particles [*Hydrocarbon*

Processing, 84,(9), 99–106 (2005)]. This article cites downward gas velocities in the range of 10 to 37 m/s and notes gas pressure drop in the range of 6 to 20 in of water. Removals of SO₂ and fine particles were both close to 99 percent. The froth produced by the contactor reverses direction, flows down, and is largely disengaged in a vessel mounted below.

Example 22: The Reverse Jet Contactor, U.S. Patent 6,339,169

This patent deals with rapid cooling and removal of NH₃ from gas exiting an acrylonitrile reactor. Liquid is injected upward. The claims suggest downward-flowing gas velocity is between 20 and 25 m/s.

Gas cooling is reported to be largely complete in 0.1 s. NH₃ removal at the exit of the contactor is reported to be greater than 99 percent. The gas is cooled by water vaporizing from the injected liquid, with total water circulated being in the range of 100 times that evaporated.

Since the gas cooling and NH₃ scrubbing move in parallel, they would be expected to achieve nearly the same approach to equilibrium as long as the pH of all the liquid stays below a key threshold. The great excess of liquid enables this.

The key is high froth interfacial area per unit volume.

Simple Spray Towers The other extreme to the pipeline and reverse jet contactors is an open vessel where spray is injected down into upflowing gas to form a rain of liquid. The advantage of simple spray towers is that they give low gas pressure drop and also tend to be nonfouling.

Even though gas velocity is well below flooding velocity, the finer droplets of the spray will be entrained. Note the wide spectrum of particle sizes shown by Fig. 14-88.

However, as shown by Examples 23 and 24, they can be extremely effective in liquid-limited systems.

Bypassing Limits Spray Tower Performance in Gas Cooling

As shown by Example 18, only modest performance is achieved in gas-limited systems. The modest efficiency is due to gas bypassing. Tall spray towers are not effective countercurrent devices. Even with nominally falling droplets, there is a great deal of backmixing because there is no stabilizing pressure drop as there would be in a column filled with packing or trays. A packet of droplets weighs more than a gas-filled space. The result is that the volume that is filled with the most droplets moves down relative to all other volumes. Similarly the gas volume that has fewest droplets moves up more quickly than other volumes. This generates bypassing of liquid and gas. The flows are driven by the rain of droplets themselves. Anything less than perfect distribution of liquid and gas will induce a dodging action between the flowing streams. Most designers limit expectations for spray contactors to some fraction of a single equilibrium stage regardless of height.

One approach that has been employed to get better distribution in spray systems is to mount a single large-capacity nozzle in the center of the vessel with radial discharge of large droplets. The droplets are discharged with enough velocity to penetrate to the vessel walls.

Spray Towers in Liquid-Limited Systems—Hollow Cone Atomizing Nozzles If we follow an element of liquid leaving a hollow cone hydraulic spray nozzle, the sequence is a rapidly thinning cone followed by wave development, followed by shedding of ligaments, followed by breakage of the ligaments into droplets. See Fig. 14-86. The sequence gives high transfer for liquid-limited systems. This results from the thin sheet of the hollow cone as well as the creation of fresh surface in the breakup process.

Devolatilizers Devolatilization systems are liquid-limited due to the combination of high liquid viscosity and removal of a component with high relative volatility. Simpson and Lynn [AIChE J., 23 (5), 666–673 (1977)] reported oxygen stripping from water at 98 percent complete, in less than 1 ft of contact. The concept has been employed for residual devolatilization in refineries.

Spray Towers as Direct Contact Condensers Similarly spray contactors can be highly effective for direct contact condensers, which are also liquid-limited. The high transfer rate in the initial formation of sprays is the key. Kunes [Ind. Engr. Chem. Res., 32, 2387–2389 (1993)] reported a 97 percent approach to equilibrium in a hydrocarbon system in the 6-in space below the discharge of a row of hollow cone spray nozzles.

Other results on heat transfer in a large spray condenser are given by Waintraub et al. ("Removing Packings from Heat Transfer Sections of Vacuum Towers," AIChE 2005 Spring National Meeting, Proceed-

ings of Topical Conference, April 10, 2005, Atlanta, p. 79). The paper highlights the importance of good gas and liquid distribution.

Converting Liquid Mass-Transfer Data to Direct Contact Heat Transfer Liquid-limited performance measures are much more commonly given for mass-transfer than for heat transfer. Often mass-transfer data are reported as $k_L a$ with units of h⁻¹. This can be converted to $h_{L,a}$ with units of Btu/(h·°F·ft³) by Eq. (14-188).

$$h_{L,a} = 187(k_L a) (c_L) (\rho_L) (\mu_L/T)^{0.5} \quad (14-188)$$

where μ_L = liquid, cP

T = temperature, °R

ρ_L = liquid density, lb/ft³

c_L = liquid specific heat, Btu/(lb·°F)

Calculation of transfer units for heat transfer is relatively simple. For a liquid

$$N_L = \frac{(\text{liquid contact time})(h_{L,a})}{\rho_L c_L} \quad (14-189)$$

where ρ_L = liquid density and c_L = liquid specific heat. [See parallel gas expression, Eq. (14-181).]

Unlike gases, the liquid properties that control mass and heat transfer differ greatly. The key term is diffusivity which for liquids drops with viscosity.

The resulting values for $h_{L,a}$ and N_L can be surprisingly large when a pure vapor such as steam is condensed. See Example 23.

Example 23: Estimating Direct Contact Condensing Performance Based on $k_L a$ Mass-Transfer Data If an aqueous system at 560°R gives a $k_L a$ of 60 h⁻¹, what does Eq. (14-188) predict for $h_{L,a}$ in a direct contact steam condenser?

For an aqueous system

$$\mu_L = 1 \text{ cP}$$

$$\rho_L = 62 \text{ lb/ft}^3$$

$$c_L = 1 \text{ Btu/(lb·°F)}$$

and Eq. (14-189) predicts

$$h_{L,a} = 187(60)(1)(62)/(1/560)^{0.5} = 29,400 \text{ Btu/(h·°F·ft}^3\text{)}$$

When a pure gas such as HCl is absorbed by low viscosity liquid such as water, simple spray systems can also be highly effective. See Example 24.

Example 24: HCl Vent Absorber (Kister, *Distillation Troubleshooting*, Wiley, 2006, p. 95.) A 6-in-diameter, 8-ft-tall packed bed was giving major problems due to failure of the packing support. Water was the scrubbing fluid.

The liquid distributors were replaced with carefully positioned spray nozzles, and the packing was removed. HCl in the vent was removed to a level one-fortieth of the original design.

LIQUID-IN-GAS DISPERSIONS

Liquid Breakup into Droplets There are four basic mechanisms for breakup of liquid into droplets:

- Droplets in a field of high turbulence (i.e., high power dissipation per unit mass)
- Simple jets at low velocity
- Expanding sheets of liquid at relatively low velocity
- Droplets in a steady field of high relative velocity

These mechanisms coexist, and the one that gives the smallest drop size will control. The four mechanisms follow distinctly different velocity dependencies:

1. *Breakup in a highly turbulent field* (1/velocity)^{1.2}. This appears to be the dominant breakup process in distillation trays in the spray regime, pneumatic atomizers, and high-velocity pipeline contactors.

2. *Breakup of a low-velocity liquid jet* (1/velocity)⁰. This governs in special applications like prilling towers and is often an intermediate step in liquid breakup processes.

3. *Breakup of a sheet of liquid* (1/velocity)^{0.67}. This governs drop size in most hydraulic spray nozzles.

4. *Single-droplet breakup at very high velocity* $(1/\text{velocity})^2$. This governs drop size in free fall as well as breakup when droplets impinge on solid surfaces.

Droplet Breakup—High Turbulence This is the dominant breakup mechanism for many process applications. Breakup results from local variations in turbulent pressure that distort the droplet shape. Hinze [*Am. Inst. Chem. Eng. J.*, **1**, 289–295 (1953)] applied turbulence theory to obtain the form of Eq. (14-190) and took liquid-liquid data to define the coefficient:

$$D_{\max} = 0.725(\sigma/\rho_G)^{0.6}/E^{0.4} \quad (14-190)$$

where $E = (\text{power dissipated})/\text{mass} \quad \text{length}^2/\text{time}^3$
 $\sigma = \text{surface tension} \quad \text{mass}/\text{time}^2$
 $\rho_G = \text{gas density} \quad \text{mass}/\text{length}^3$

Note that $D_{\max}$ comes out with units of length. Since E typically varies with $(\text{gas velocity})^3$, this results in drop size dependence with $(1/\text{velocity})^{1.2}$.

The theoretical requirement for use of Eq. (14-190) is that the microscale of turbulence $\ll D_{\max}$. This is satisfied in most gas systems. For example, in three cases,

	(microscale of turbulence)/ $D_{\max}$
distillation tray in spray regime	0.007
pipeline @ 40 m/s and atmospheric pressure	0.012
two-fluid atomizer using 100 m/s air	0.03

Many applications involve a three-step process with high velocity first tearing wave crests away from liquid sheets, followed by breakup of ligaments into large droplets, followed by breakup of the large droplets. The prediction of final droplet size based on *power/mass* works surprisingly well, as shown by Eqs. (14-198), (14-201), (14-202), and (14-203).

Liquid-Column Breakup Because of increased pressure at points of reduced diameter, the liquid column is inherently unstable. As a result, it breaks into small drops with no external energy input. Ideally, it forms a series of uniform drops with the size of the drops set by the fastest-growing wave. This yields a dominant droplet diameter about 1.9 times the initial diameter of the jet as shown by

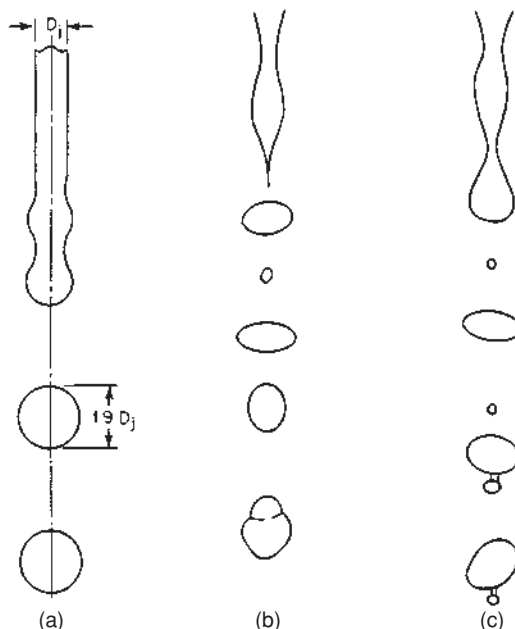


FIG. 14-85 (a) Idealized jet breakup suggesting uniform drop diameter and no satellites. (b) and (c) Actual breakup of a water jet as shown by high-speed photographs. [From W. R. Marshall, "Atomization and Spray Drying," *Chem. Eng. Prog. Monogr. Ser.*, no. 2 (1954).]

Fig. 14-85. As shown, the actual breakup is quite close to prediction, although smaller satellite drops are also formed. The prime advantage of this type of breakup is the greater uniformity of drop size.

For high-viscosity liquids, the drops are larger, as shown by Eq. (14-191):

$$D = 1.9D_j \left[1 + \frac{3\mu_L}{(\sigma\rho_L D_j)^{1/2}} \right] \quad (14-191)$$

where $D = \text{diameter of droplet}$

$D_j = \text{diameter of jet}$

$\mu_L = \text{viscosity of liquid}$

$\rho_L = \text{density of liquid}$

$\sigma = \text{surface tension of liquid}$

These units are dimensionally consistent; any set of consistent units can be used.

As the velocity of the jet is increased, the breakup process changes and ultimately becomes a mix of various competing effects, such as the capture of small drops by bigger ones in the slowing jet and the "turbulent breakup" of the bigger drops. The high-velocity jet is occasionally used in process applications because of the very narrow spray angle ($5\text{--}20^\circ$) and the high penetration into a gas it can give. The focused stream also aids erosion of a surface.

Liquid-Sheet Breakup The basic principle of most hydraulic atomizers is to form a thin sheet that breaks via a variety of mechanisms to form ligaments of liquid which in turn yield chains of droplets. See Fig. 14-86.

For a typical nozzle, the drop size varies with $1/(\text{pressure drop})^{1/3}$. When $(\text{velocity})^2$ is substituted for pressure drop, droplet size is seen to vary with $(\text{velocity})^{-2/3}$.

Isolated Droplet Breakup—in a Velocity Field Much effort has focused on defining the conditions under which an isolated drop will break in a velocity field. The criterion for the largest stable drop size is the ratio of aerodynamic forces to surface-tension forces defined by the Weber number, N_{We} (dimensionless):

$$N_{We \text{ crit}} = \text{constant} = [\rho_G (\text{velocity})^2 (D_{\max}) / (\sigma)] \quad (14-192)$$

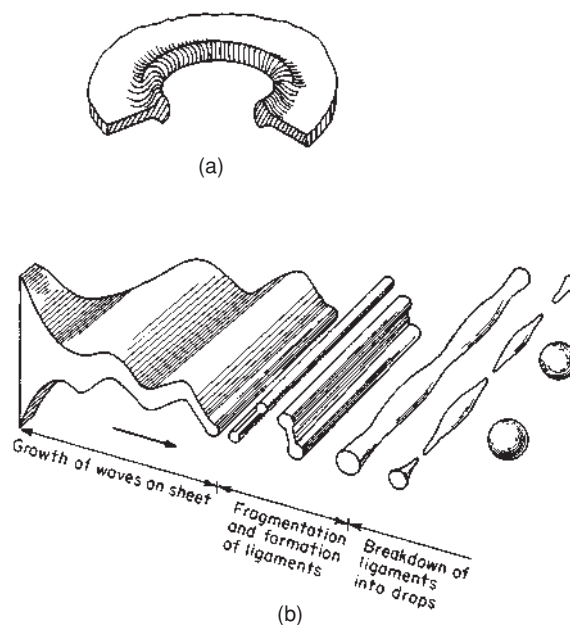


FIG. 14-86 Sheet breakup. (a) By perforation. [After Fraser et al., *Am. Inst. Chem. Eng. J.*, **8**(5), 672 (1962).] (b) By sinusoidal wave growth. [After Dombrowski and Johns, *Chem. Eng. Sci.*, **18**, 203 (1963).]

$N_{We\text{ crit}}$ for low-viscosity fluids commonly ranges from 10 to 20, with the larger value for a free-fall condition and the smaller for a sudden acceleration. High liquid viscosity also increases $N_{We\text{ crit}}$.

Droplet breakup via impingement appears to follow a similar relationship, but much less data is available. This type of breakup can result from impingement on equipment walls or compressor blades. In general, there is less tendency to shatter on wetted surfaces.

Droplet Size Distribution Instead of the single droplet size implied by the discussion above, a spectrum of droplet sizes is produced. The most common ways to characterize this spectrum are:

- **Volume median (mass median) D_{vm} .** This has no fundamental meaning but is easy to determine since it is at the midpoint of a cumulative-volume plot.
- **Sauter mean D_{32} .** This has the same ratio of surface to volume as the total drop population. It is typically 70 to 90 percent of D_{vm} . It is frequently used in transport processes and is used here to characterize drop size.
- **Maximum D_{max} .** This is the largest-sized particle in the population. It is typically 3 to 4 times D_{32} in turbulent breakup processes, per Walzel [*International Chemical Engineering*, **33**, 46, (1993)]. It is the size directly calculated from the power/mass relationship. D_{32} is estimated from D_{max} by

$$D_{32} = 0.3 \cdot D_{max} \quad (14-193)$$

and D_{vm} is estimated from it by

$$D_{vm} = 0.4 \cdot D_{max} \quad (14-194)$$

However, any average drop size is fictitious, and none is completely satisfactory. For example, there is no way in which the high surface and transfer coefficients in small drops can be made available to the larger drops. Hence, a process calculation based on a given droplet size describes only what happens to that size and gives at best an approximation to the total mass.

There are a variety of ways to describe the droplet population. Figures 14-88 and 14-90 illustrate one of the most common methods, the plot of cumulative volume against droplet size on log-normal graph paper. This satisfies the restraint of not extrapolating to a negative drop size. Its other advantages are that it is easy to plot, the results are easy to visualize, and it yields a nearly straight line at lower drop sizes.

Cumulative volume over the range of 1 to 50 percent can also be shown to vary approximately as D^2 . This is equivalent to finding that the number of droplets of a given size is inversely proportional to the droplet area or the surface energy of the droplet.

Atomizers The common need to disperse a liquid into a gas has spawned a large variety of mechanical devices. The different designs emphasize different advantages such as freedom from plugging, pattern of spray, small droplet size, uniformity of spray, high turndown ratio, and low power consumption.

As shown in Table 14-19, most atomizers fall into three categories:

1. Pressure nozzles (hydraulic)
2. Two-fluid nozzles (pneumatic)
3. Rotary devices (spinning cups, disks, or vaned wheels)

These share certain features such as relatively low efficiency and low cost relative to most process equipment. The energy required to produce the increase in area is typically less than 0.1 percent of the total energy consumption. This is because atomization is a secondary process resulting from high interfacial shear or turbulence. As droplet sizes decrease, this efficiency drops lower.

Other types are available that use sonic energy (from gas streams), ultrasonic energy (electronic), and electrostatic energy, but they are less commonly used in process industries. See Table 14-19 for a summary of the advantages/disadvantages of the different type units. An expanded discussion is given by Masters [*Spray Drying Handbook*, Wiley, New York (1991)].

Special requirements such as size uniformity in prilling towers can dictate still other approaches to dispersion. Here plates are drilled with many holes to develop nearly uniform columns.

Commonly, the most important feature of a nozzle is the size of droplet it produces. Since the heat or mass transfer that a given dispersion can produce is often proportional to $(1/D_d)^2$, fine drops are usually favored. On the other extreme, drops that are too fine will not

settle, and a concern is the amount of liquid that will be entrained from a given spray operation. For example, if sprays are used to contact atmospheric air flowing at 1.5 m/s, drops smaller than 350 μm [terminal velocity = 1.5 m/s (4.92 ft/s)] will be entrained. Even for the relative coarse spray of the hollow-cone nozzle shown in Fig. 14-88, 7.5 percent of the total liquid mass will be entrained.

Hydraulic (Pressure) Nozzles Manufacturers' data such as shown by Fig. 14-88 are available for most nozzles for the air-water system. In Fig. 14-88, note the much coarser solid-cone spray. The coarseness results from the less uniform discharge.

Effect of Physical Properties on Drop Size Because of the extreme variety of available geometries, no attempt to encompass this variable is made here. The suggested predictive route starts with air-water droplet size data from the manufacturer at the chosen flow rate. This drop size is then corrected by Eq. (14-195) for different viscosity and surface tension:

$$\frac{D_{cm, \text{system}}}{D_{cm, \text{water}}} = \left(\frac{\sigma_{\text{system}}}{73} \right)^{0.25} \left(\frac{\mu_L}{1.0} \right)^{0.25} \quad (14-195)$$

where D_{cm} = volume median droplet diameter

σ = surface tension, mN/m (dyn/cm)

μ_L = liquid viscosity, mPa·s (cP)

The exponential dependencies in Eq. (14-195) represent averages of values reported by a number of studies with particular weight given to Lefebvre [*Atomization and Sprays*, Hemisphere, New York (1989)]. Since viscosity can vary over a much broader range than surface tension, it has much more leverage on drop size. For example, it is common to find an oil with 1000 times the viscosity of water, while most liquids fall within a factor of 3 of its surface tension. Liquid density is generally even closer to that of water, and since the data are not clear that a liquid density correction is needed, none is shown in Eq. (14-195). Vapor density also has an impact on drop size but the impact is complex, involving conflicts of a number of effects, and vapor density is commonly omitted in atomizer drop size correlations.

Effect of Pressure Drop and Nozzle Size For a nozzle with a developed pattern, the average drop size can be estimated to fall with rising ΔP (pressure drop) by Eq. (14-196):

$$\frac{D_1}{D_2} = \left(\frac{\Delta P_2}{\Delta P_1} \right)^{1/3} \quad (14-196)$$

For similar nozzles and constant ΔP , the drop size will increase with nozzle size as indicated by Eq. (14-197):

$$\frac{D_1}{D_2} = \left(\frac{\text{orifice diameter}_1}{\text{orifice diameter}_2} \right)^{1/2} \quad (14-197)$$

Once again, these relationships are averages of a number of reported values and are intended as rough guides.

The normal operating regime is well below turbulent breakup velocity. However the data of Kennedy [*J. of Engineering for Gas Turbines and Power*, **108**, 191 (1986)] at very high pressure drop in large nozzles shows a shift to a higher dependence on pressure drop. This data suggests that turbulent droplet breakup can also be governing with hydraulic spray nozzles, although this is unusual.

Spray Angle A shift to a smaller-angle nozzle gives slightly larger drops for a given type of nozzle because of the reduced tendency of the sheet to thin. Dietrich [*Proc. 1st Int. Conf. Liq. Atomization Spray Systems*, Tokyo (1978)] shows the following:

Angle	25°	50°	65°	80°	95°
$D_{cm}, \mu\text{m}$	1459	1226	988	808	771

In calculating the impact point of spray, one should recognize that the spray angle closes in as the spray moves away from the nozzle. This is caused by loss of momentum of the spray to the gas.

TABLE 14-19 Atomizer Summary

Types of atomizer	Design features	Advantages	Disadvantages
Pressure.	Flow $\alpha(\Delta P/\rho)^{1/2}$. Only source of energy is from fluid being atomized.	Simplicity and low cost.	Limited tolerance for solids; uncertain spray with high-viscosity liquids; susceptible to erosion. Need for special designs (e.g., bypass) to achieve turndown. Concentrated spray pattern at cone boundaries.
1. Hollow cone.	Liquid leaves as conical sheet as a result of centrifugal motion of liquid. Air core extends into nozzle. Centrifugal motion developed by tangential inlet in chamber upstream of orifice.	High atomization efficiency.	
<i>a.</i> Whirl chamber (see Fig. 14-87 <i>a</i>).	Centrifugal motion developed by tangential inlet in chamber upstream of orifice.	Minimum opportunity for plugging.	
<i>b.</i> Grooved core.	Centrifugal motion developed by inserts in chamber.	Smaller spray angle than 1 <i>a</i> and ability to handle flows smaller than 1 <i>a</i> .	
2. Solid cone (see Fig. 14-87 <i>b</i>).	Similar to hollow cone but with insert to provide even distribution.	More uniform spatial pattern than hollow cone.	Coarser drops for comparable flows and pressure drops. Failure to yield same pattern with different fluids. Small clearances.
3. Fan (flat) spray.	Liquid leaves as a flat sheet or flattened ellipse.	Flat pattern is useful for coating surfaces and for injection into streams.	
<i>a.</i> Oval or rectangular orifice (see Fig. 14-87 <i>c</i>). Numerous variants on cavity and groove exist.	Combination of cavity and orifice produces two streams that impinge within the nozzle.		
<i>b.</i> Deflector (see Fig. 14-87 <i>d</i>).	Liquid from plain circular orifice impinges on curved deflector.	Minimal plugging.	Coarser drops.
<i>c.</i> Impinging jets (see Fig. 14-87 <i>e</i>).	Two jets collide outside nozzle and produce a sheet perpendicular to their plane.	Different liquids are isolated until they mix outside of orifice. Can produce a flat circular sheet when jets impinge at 180°.	Extreme care needed to align jets.
4. Nozzles with wider range of turndown.			
<i>a.</i> Spill (bypass) (see Fig. 14-87 <i>f</i>).	A portion of the liquid is recirculated after going through the swirl chamber.	Achieves uniform hollow cone atomization pattern with very high turndown (50:1).	Waste of energy in bypass stream. Added piping for spill flow.
<i>b.</i> Poppet (see Fig. 14-87 <i>g</i>).	Conical sheet is developed by flow between orifice and poppet. Increased pressure causes poppet to move out and increase flow area.	Simplest control over broad range.	Difficult to maintain proper clearances.
Two-fluid (see Fig. 14-87 <i>h</i>).	Gas impinges coaxially and supplies energy for breakup.	High velocities can be achieved at lower pressures because the gas is the high-velocity stream. Liquid-flow passages can be large, and hence plugging can be minimized. Similar to two-fluid but with greater tolerance for solids.	Because gas is also accelerated, efficiency is inherently lower than pressure nozzles.
Sonic.	Gas generates an intense sound field into which liquid is directed.		Similar to two-fluid.
Rotary wheels (see Fig. 14-87 <i>i</i>) disks, and cups.	Liquid is fed to a rotating surface and spreads in a uniform film. Flat disks, disks with vanes, and bowl-shaped cups are used. Liquid is thrown out at 90° to the axis.	The velocity that determines drop size is independent of flow. Hence these can handle a wide range of rates. They can also tolerate very viscous materials as well as slurries. Can achieve very high capacity in a single unit; does not require a high-pressure pump.	Mechanical complexity of rotating equipment. Radial discharge.
Ultrasound.	Liquid is fed over a surface vibrating at a frequency > 20 kHz.	Fine atomization, small size, and low injection velocity.	Low flow rate and need for ultrasound generator.

At some low flow, pressure nozzles do not develop their normal pattern but tend to approach solid streams. The required flow to achieve the normal pattern increases with viscosity.

Two-Fluid (Pneumatic) Atomizers This general category includes such diverse applications as venturi atomizers and reactor-effluent quench systems in addition to two-fluid spray nozzles. Depending on the manner in which the two fluids meet, several of the breakup mechanisms may be applicable, but the final one is high-level turbulent rupture.

As shown by Table 14-20, empirical correlations for two-fluid atomization show dependence on high gas velocity to supply atomizing energy, usually to a power dependence close to that for turbulent breakup. In addition, the correlations show a dependence on the ratio of gas to liquid and system dimension.

Further differences from hydraulic nozzles (controlled by sheet and ligament breakup) are the stronger increase in drop size with increasing surface tension and decreasing gas density.

The similarity of these dependencies to Eq. (14-190) led to a reformulation with two added terms that arise naturally from the theory of power dissipation per unit mass. The result is Eq. (14-198) which is labeled *power/mass* in Table 14-20.

$$D_{32} = 0.29 \left(\frac{\sigma}{\rho_G} \right)^{0.6} (1/\text{velocity})^{1.2} \left(1 + \frac{L}{G} \right)^{0.4} (D_{\text{nozzle}})^{0.4} \quad (14-198)$$

where σ = surface tension

ρ_G = gas density

L/G = mass ratio of liquid flow to gas flow

D_{nozzle} = diameter of the air discharge

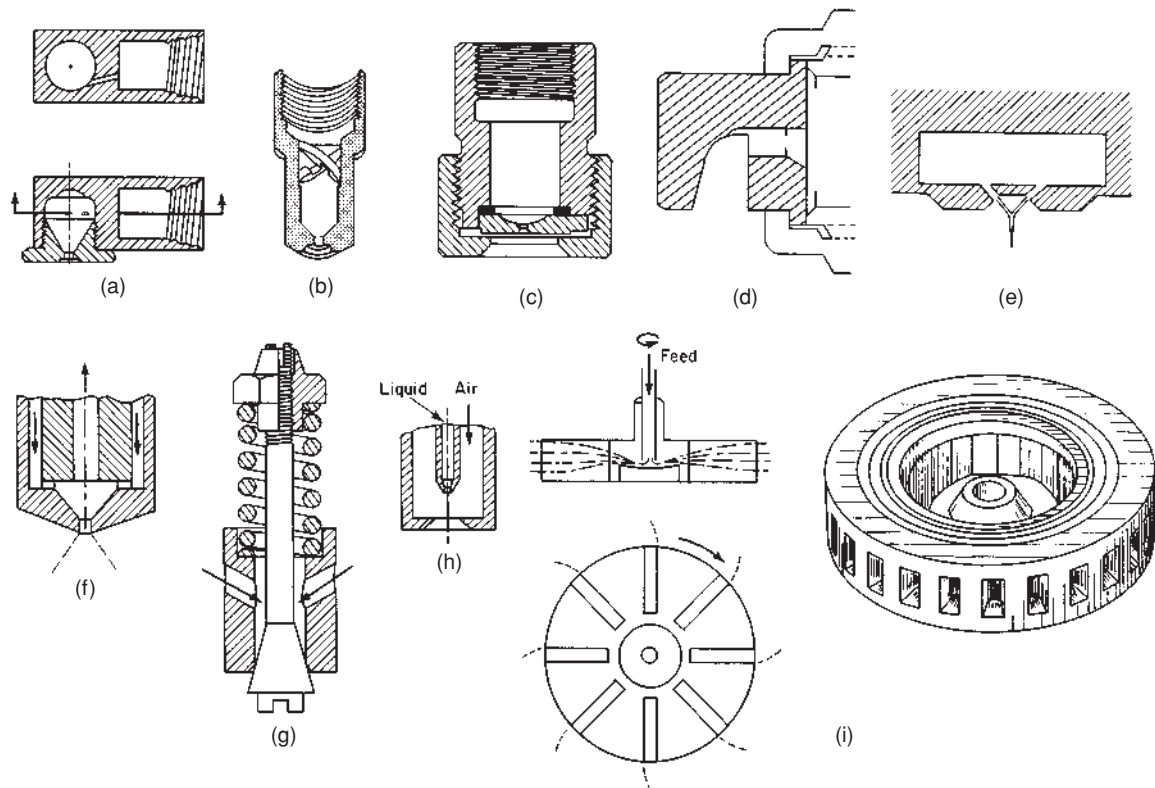


FIG. 14-87 Characteristic spray nozzles. (a) Whirl-chamber hollow cone. (b) Solid cone. (c) Oval-orifice fan. (d) Deflector jet. (e) Impinging jet. (f) Bypass. (g) Poppet. (h) Two-fluid. (i) Vaned rotating wheel.

This is remarkably similar to the empirical two-fluid atomizer relationships of El-Shanawany and Lefebvre [J. Energy, **4**, 184 (1980)] and Jasuja [Trans. Am. Soc. Mech. Engr., **103**, 514 (1981)]. For example, El-Shanawany and Lefebvre give a relationship for a prefilming atomizer:

$$D_{32} = 0.0711(\sigma/\rho_G)^{0.6}(1/\text{velocity})^{1.2}(1 + L/G)(D_{\text{nozzle}})^{0.4}(\rho_L/\rho_G)^{0.1} + 0.015[(\mu_L)^2/(\sigma \times \rho_L)]^{0.5}(D_{\text{nozzle}})^{0.5}(1 + L/G) \quad (14-199)$$

where μ_L is liquid viscosity.

According to Jasuja,

$$D_{32} = 0.17(\sigma/\rho_G)^{0.45}(1/\text{velocity})^{0.9}(1 + L/G)^{0.5}(D_{\text{nozzle}})^{0.55} + \text{viscosity term} \quad (14-200)$$

[Eqs. (14-198), (14-199), and (14-200) are dimensionally consistent; any set of consistent units on the right-hand side yields the droplet size in units of length on the left-hand side.]

The second, additive term carrying the viscosity impact in Eq. (14-199) is small at viscosities around 1 cP but can become controlling as viscosity increases. For example, for air at atmospheric pressure atomizing water, with nozzle conditions

$$\begin{aligned} D_{\text{nozzle}} &= 0.076 \text{ m (3 in)} \\ \text{velocity} &= 100 \text{ m/s (328 ft/s)} \\ L/G &= 1 \end{aligned}$$

El-Shanawany measured 70 μm and his Eq. (14-199) predicted 76 μm . The power/mass correlation [Eq. (14-198)] predicts 102 μm . The agreement between both correlations and the measurement is much better than normally achieved.

Rotary Atomizers For rotating wheels, vaneless disks, and cups, there are three regimes of operation. At low rates, the liquid is shed directly as drops from the rim. At intermediate rates, the liquid leaves the rim as threads; and at the highest rate, the liquid extends from the edge as a thin sheet that breaks down similarly to a fan or hollow-cone spray nozzle. As noted in Table 14-19, rotary devices have many unique advantages such as the ability to handle high viscosity and slurries and produce small droplets without high pressures. The prime applications are in spray drying. See Masters [Spray Drying Handbook, Wiley, New York (1991)] for more details.

Pipeline Contactors The correlation for droplet diameter based on power/mass is similar to that for two-fluid nozzles. The dimensionless correlation is

$$D_{32} = 0.8(\sigma/\rho_G)^{0.6}(1/\text{velocity})^{1.2}(D_{\text{pipe}})^{0.4} \quad (14-201)$$

(The relation is dimensionally consistent; any set of consistent units on the right-hand side yields the droplet size in units of length on the left-hand side.)

The relationship is similar to the empirical correlation of Tatterson, Dallman, and Hanratty [Am. Inst. Chem. Eng. J., **23**(1), 68 (1977)]

$$D_{32} \sim \left(\frac{\sigma}{\rho_G} \right)^{0.5} (1/\text{velocity})^1 (D_{\text{pipe}})^{0.5}$$

Predictions from Eq. (14-201) align well with the Tatterson data. For example, for a velocity of 43 m/s (140 ft/s) in a 0.05-m (1.8-inch) equivalent diameter channel, Eq. (14-201) predicts D_{32} of 490 microns, compared to the measured 460 to 480 microns.

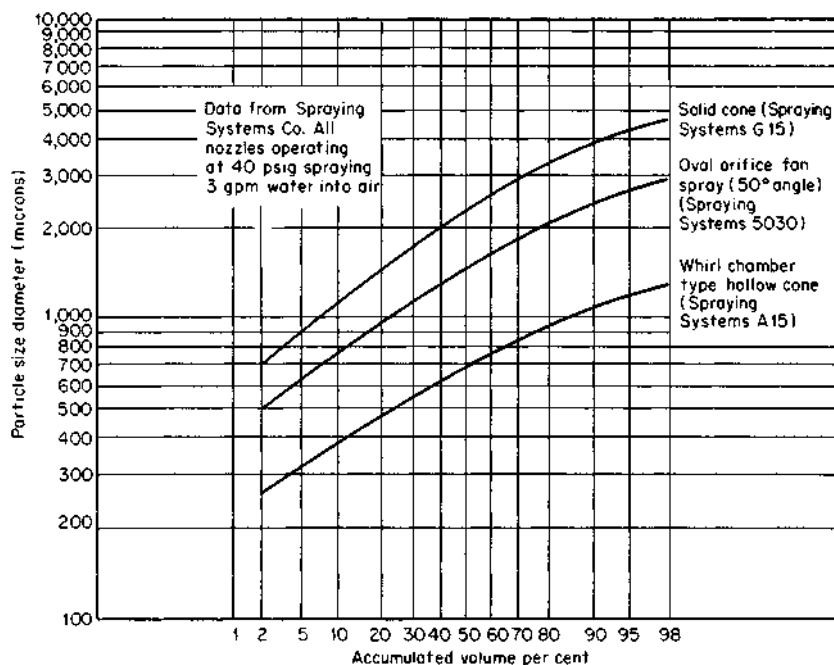


FIG. 14-88 Droplet-size distribution for three different types of nozzles. To convert pounds per square inch gauge to kilopascals, multiply by 6.89; to convert gallons per minute to cubic meters per hour, multiply by 0.227. (Spraying Systems Inc.)

Entrainment due to Gas Bubbling/Jetting through a Liquid

Entrainment generally limits the capacity of distillation trays and is commonly a concern in vaporizers and evaporators. Fortunately, it is readily controllable by simple inertial entrainment capture devices such as wire mesh pads in gravity separators.

In distillation towers, entrainment lowers the tray efficiency, and 1 pound of entrainment per 10 pounds of liquid is sometimes taken as the limit for acceptable performance. However, the impact of entrainment on distillation efficiency depends on the relative volatility of the component being considered. Entrainment has a minor impact on close separations when the difference between vapor and liquid concentration is small, but this factor can be dominant for systems where the liquid concentration is much higher than the vapor in equilibrium with it (i.e., when a component of the liquid has a very low volatility, as in an absorber).

As shown by Fig. 14-90, entrainment droplet sizes span a broad range. The reason for the much larger drop sizes of the upper curve is the short disengaging space. For this curve, over 99 percent of the entrainment has a terminal velocity greater than the vapor velocity. For contrast, in the lower curve the terminal velocity of the largest particle reported is the same as the vapor velocity. For the

settling velocity to limit the maximum drop size entrained, at least 0.8 m (30 in) disengaging space is usually required. Note that even for the lower curve, less than 10 percent of the entrainment is in drops of less than 50 μm . The coarseness results from the relatively low power dissipation per mass on distillation trays. This means that it is relatively easy to remove by a device such as a wire mesh pad. Over 50 percent is typically captured by the underside of the next higher tray or by a turn in the piping leaving an evaporator. Conversely, though small on a mass basis, the smaller drops are extremely numerous. On a number basis, more than one-half of the drops in the lower curve are under 5 μm . These can serve as nuclei for fog condensation in downstream equipment.

Entrainment E is inherent in the bubbling process and can stem from a variety of sources, as shown by Fig. 14-89. However, the biggest practical problem is entrainment generated by the kinetic energy of the flowing vapor rather than the bubbling process. As vapor velocity approaches the flooding limit [Eq. (14-168)], the entrainment rises approximately with (velocity)⁶.

Pinczewski and Fell [*Trans. Inst. Chem Eng.*, **55**, 46 (1977)] show that the velocity at which vapor jets onto the tray sets the droplet size, rather than the superficial tray velocity. The power/mass correlation predicts an average drop size close to that measured by Pinczewski and Fell. Combination of this prediction with the estimated fraction of the droplets entrained gave a relationship for entrainment, Eq. (14-202). The dependence of entrainment with the eighth power of velocity even approximates the observed velocity dependence, as flooding is approached.

$$E = \frac{\text{constant}(\text{velocity})^8(\rho_G)^4}{(A_f)^3(\rho_L - \rho_G)^{2.5}\sigma^{1.5}} \quad (14-202)$$

(Here E is the mass of entrainment per mass of vapor and A_f is the fractional open area on the tray.)

When flooding is defined as the condition that gives E of 1, the flood velocity is estimated by Eq. (14-203).

TABLE 14-20 Exponential Dependence of Drop Size on Different Parameters in Two-Fluid Atomization

	Relative velocity	Surface tension	Gas density	1 + L/G	Atomizer dimension
Jasuja (empirical for small nozzle)	-0.9	0.45	-0.45	0.5	0.55
El-Shanawany and Lefebvre (empirical for small nozzle)	-1.2	0.6	-0.7	1	0.40
Tattersson, Dallman, and Hanratty (pipe flow)	-1	0.5	-0.5		0.5
Power/mass	-1.2	0.6	-0.6	0.4	0.4

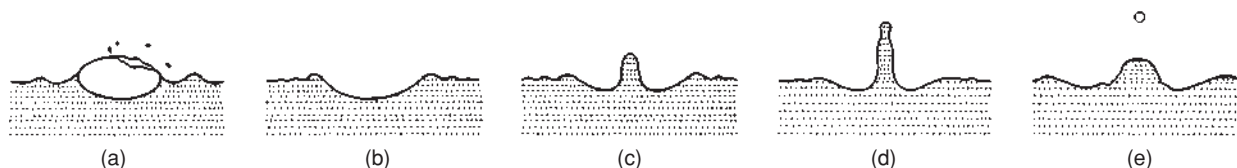


FIG. 14-89 Mechanism of the burst of an air bubble on the surface of water. [Newitt, Dombrowski, and Knellman, Trans. Inst. Chem. Eng., 32, 244 (1954).]

$$U_{\text{flood}} = \frac{1.25(A_f)^{3/8}[\sigma(\rho_L - \rho_G)]^{0.3125} \sigma^{0.1875}}{\rho_G^{0.5}} \quad (14-203)$$

The relationship is dimensionally consistent; any set of consistent units on the right-hand side yields velocity units on the left-hand side. It is similar in form to Eq. (14-168) and provides a conceptual framework for understanding the ultimate distillation column capacity concept.

“Upper Limit” Flooding in Vertical Tubes If, instead of a gas jet being injected into a liquid as in distillation, the liquid runs down the walls and the gas moves up the center of the tube, higher velocities can be achieved than shown by Eq. (14-168) or (14-203). This application is important in the design of vertical condensers.

Maharudraya and Jayanti [AIChE J., 48(2), 212–220 (2002)] show that peak pressure drop in a 25-mm vertical tube occurs at a value close to that predicted by Eq. (14-168) or (14-203). At this velocity about 20 percent of the injected liquid is being entrained out the top of the tube. However, the condition where essentially all liquid was entrained didn't occur until a velocity over twice the value estimated from Eq. (14-168) and Eq. (14-203).

The higher velocities at modest entrainment observed by Maharudraya and Jayanti were obtained with special smooth entry of gas (and exit of liquid) at the bottom of the tube. Hewitt (*Handbook of Heat Exchanger Design*, pp. 2.3.2–23, 1992) suggests that these values should be derated by at least 35 percent for more typical sharp heat exchanger tube entry. Similar to the smooth entry effect, other data suggest that countercurrent capacity can be increased by providing an extension of the tube below the tube sheet, with the bottom of the extension cut on a steep angle (>60°) to the horizontal. The tapered extension facilitates drainage of liquid.

An extensive data bank correlated by Diehl and Koppany [Chem. Eng. Prog. Symp. Ser., 65, 77–83 (1965)] also gave higher allowable entry velocities than Eq. (14-168) or (14-203). Diehl and Koppany's correlation [Eq. (14-204)] is dimensional, and appears to give a much higher dependence on σ than the more recent work. However, for many fluids, $\sigma^{0.5}$ is essentially the same as the combination $\sigma^{0.1875}(\rho_L - \rho_G)^{0.3125}$ that appears in Eq. (14-203). Hence Eq. (14-204) gives a similar physical property dependence.

$$U_f = F_1 F_2 (\sigma / \rho_g)^{0.5} \quad (14-204)$$

where U_f = flooding gas velocity, m/s

$F_1 = 1.22$ when $3.2 d_i / \sigma > 1.0$

$= 1.22 (3.2 d_i / \sigma)^{0.4}$ when $3.2 d_i / \sigma < 1.0$

$F_2 = (G/L)^{0.25}$

G/L = gas-liquid mass ratio

d_i = inside diameter of column, mm

σ = surface tension, mN/m (dyn/cm)

ρ_g = gas density, kg/m³

The primary reason for citing Eq. (14-204) is the large successful experience base in practical applications. Note that the reduction in allowable gas velocity for small diameters given by the F_1 factor is conceptually the same as the effect of using smaller-diameter packing in distillation. Note also that over the range of G/L between 1 and 0.1, the Maharudraya and Jayanti data show a similar reduction in allowable gas rate to the F_2 factor in Eq. (14-204). The phenomenon behind

this is that a thicker liquid film on the tube wall is more easily entrained.

While the limiting phenomenon of upper limit flooding in a vertical pipe is similar to *ultimate capacity* in distillation, there is a distinct difference. *Upper limit* in a vertical pipe applies to a design where a conscious effort should be made to minimize gas-liquid contact. Carried to extremes, it would involve separate tubes for liquid flowing down and vapor going up. In contrast, *ultimate capacity* in a distillation column corresponds to the condition where effective mass transfer disappears due to high entrainment. One could force more vapor up through the contactor, but fractionation would be poor.

Fog Condensation—The Other Way to Make Little Droplets For a variety of reasons, a gas or vapor can become supersaturated with a condensable component. Surface tension and mass transfer impose barriers on immediate condensation, so growth of fog particles lags behind what equilibrium predicts. Droplets formed by fog condensation are usually much finer (0.1 to 10 μm) than those formed by mechanical breakup and hence more difficult to collect. Sometimes fog can be a serious problem, as in the atmospheric discharge of a valuable or a hazardous material. More commonly, fog is a curiosity rather than a dominating element in chemical processing.

Fog particles grow because of excess saturation in the gas. Usually this means that the gas is supersaturated (i.e., it is below its dew point). Sometimes, fog can also grow on soluble foreign nuclei at partial pressures below saturation. Increased saturation can occur through a variety of routes:

1. Mixing of two saturated streams at different temperatures. This is commonly seen in the plume from a stack. Since vapor pressure is an exponential function of temperature, the resultant mixture of two saturated streams will be supersaturated at the mixed temperature. Uneven flow patterns and cooling in heat exchangers make this route to supersaturation difficult to prevent.

2. Increased partial pressure due to reaction. An example is the reaction of SO_3 and H_2O to yield H_2SO_4 , which has much lower vapor pressure than its components.

3. Isoentropic expansion (cooling) of a gas, as in a steam nozzle.

4. Cooling of a gas containing a condensable vapor. Here the problem is that the gas cools faster than condensable vapor can be removed by mass transfer.

These mechanisms can be observed in many common situations. For example, fog via mixing can be seen in the discharge of breath on a cold day. Fog via adiabatic expansion can be seen in the low-pressure area over the wing of an airplane landing on a humid summer day; and fog via condensation can be seen in the exhaust from an automobile air conditioner (if you follow closely enough behind another car to pick up the ions or NO molecules needed for nucleation). All of these occur at a very low supersaturation and appear to be keyed to an abundance of foreign nuclei. All of these fogs also quickly dissipate as heat or unsaturated gas is added.

The supersaturation in condensers arises for two reasons. First, the condensable vapor is generally of higher molecular weight than the noncondensable gas. This means that the molecular diffusivity of the vapor will be much less than the thermal diffusivity of the gas. Restated, the ratio of N_{sc}/N_v is greater than 1. The result is that a condenser yields more heat-transfer units $dT_g/(T_g - T_i)$ than mass-transfer units $dY_g/(Y_g - Y_i)$. Second, both transfer processes derive their driving force from the temperature difference between the gas T_g and the interface T_i . Each incremental decrease in interface temperature

TABLE 14-21 Simulation of Three Heat Exchangers with Varying Foreign Nuclei

	1	2	3
Weight fraction, noncondensable			
Inlet	0.51	0.42	0.02
Outlet	0.80	0.80	0.32
Molecular weight			
Inert	28	29	29
Condensable	86	99	210
Temperature difference between gas and liquid interface, K			
Inlet	14	24	67
Outlet	4	10	4
Percent of liquid that leaves unit as fog			
Nuclei concentration in inlet particles/cm ³			
100	0.05	1.1	2.2
1,000	0.44	5.6	3.9
10,000	3.2	9.8	4.9
100,000	9.6	11.4	5.1
1,000,000	13.3	11.6	
10,000,000	14.7		
∞	14.7	11.8	5.1
Fog particle size based on 10,000 nuclei/cm ³ at inlet, μm	28	25	4

yields the same relative increase in temperature driving force. However, the interface vapor pressure can only approach the limit of zero. Because of this, for equal molecular and thermal diffusivities a saturated mixture will supersaturate when cooled. The tendency to supersaturate generally increases with increased molecular weight of the condensable, increased temperature differences, and reduced initial superheating. To evaluate whether a given condensing step yields fog requires rigorous treatment of the coupled heat-transfer and mass-transfer processes through the entire condensation. Steinmeyer [*Chem. Eng. Progr.*, **68**(7), 64 (1972)] illustrates this, showing the impact of foreign-nuclei concentration on calculated fog formation. See Table 14-21. Note the relatively large particles generated for cases 1 and 2 for 10,000 foreign nuclei per cm³. These are large enough to be fairly easily collected. There have been very few documented problems with industrial condensers despite the fact that most calculate to generate supersaturation along the condensing path. The explanation appears to be a limited supply of foreign nuclei.

Ryan et al. [*Chem. Eng. Progr.*, **90**(8), 83 (1994)] show that separate mass and heat transfer-rate modeling of an HCl absorber predicts 2 percent fog in the vapor. The impact is equivalent to lowering the stage efficiency to 20 percent.

Spontaneous (Homogeneous) Nucleation This process is quite difficult because of the energy barrier associated with creation of the interfacial area. It can be treated as a kinetic process with the rate a very steep function of the supersaturation ratio (S = partial pressure of condensable per vapor pressure at gas temperature). For water, an increase in S from 3.4 to 3.9 causes a 10,000-fold increase in the nucleation rate. As a result, below a critical supersaturation (S_{crit}), homogeneous nucleation is slow enough to be ignored. Generally, S_{crit} is defined as that which limits nucleation to one particle produced per cubic centimeter per second. It can be estimated roughly by traditional theory (*Theory of Fog Condensation*, Israel Program for Scientific Translations, Jerusalem, 1967) using the following equation:

$$S_{crit} = \exp \left[0.56 \frac{M}{\rho_l} \left(\frac{\sigma}{T} \right)^{3/2} \right] \quad (14-205)$$

where σ = surface tension, mN/m (dyn/cm)
 ρ_l = liquid density, g/cm³
 T = temperature, K
 M = molecular weight of condensable

Table 14-22 shows typical experimental values of S_{crit} taken from the work of Russel [*J. Chem. Phys.*, **50**, 1809 (1969)]. Since the critical supersaturation ratio for homogeneous nucleation is typically greater

TABLE 14-22 Experimental Critical Supersaturation Ratios

	Temperature, K	S_{crit}
H ₂ O	264	4.91
C ₂ H ₅ OH	275	2.13
CH ₃ OH	264	3.55
C ₆ H ₆	253	5.32
CCl ₄	247	6.5
CHCl ₃	258	3.73
C ₆ H ₅ Cl	250	9.5

than 3, it is not often reached in process equipment. However, fog formation is typically found in steam turbines. Gyarnathy [*Proc. Inst. Mech. E., Part A: J. Power and Energy* **219**(A6), 511–521 (2005)] reports fog in the range 3.5 to 5 percent of total steam flow, with average fog diameter in the range of 0.1 to 0.2 μm.

Growth on Foreign Nuclei As noted above, foreign nuclei are often present in abundance and permit fog formation at much lower supersaturation. For example,

1. **Solids.** Surveys have shown that air contains thousands of particles per cubic centimeter in the 0.1-μm to 1-μm range suitable for nuclei. The sources range from ocean-generated salt spray to combustion processes. The concentration is highest in large cities and industrial regions. When the foreign nuclei are soluble in the fog, nucleation occurs at S values very close to 1.0. This is the mechanism controlling atmospheric water condensation. Even when not soluble, a foreign particle is an effective nucleus if wet by the liquid. Thus, a 1-μm insoluble particle with zero contact angle requires an S of only 1.001 in order to serve as a condensation site for water.

2. **Ions.** Amelin [*Theory of Fog Condensation*, Israel Program for Scientific Translations, Jerusalem, (1967)] reports that ordinary air contains even higher concentrations of ions. These ions also reduce the required critical supersaturation, but by only about 10 to 20 percent, unless multiple charges are present.

3. **Entrained liquids.** Production of small droplets is inherent in the bubbling process, as shown by Fig. 14-90. Values range from near zero to 10,000/cm³ of vapor, depending on how the vapor breaks through the liquid and on the opportunity for evaporation of the small drops after entrainment.

As a result of these mechanisms, most process streams contain enough foreign nuclei to cause some fogging. While fogging has been reported in only a relatively low percent of process partial condensers, it is rarely looked for and volunteers its presence only when yield losses or pollution is intolerable.

Dropsie Distribution Monodisperse (nearly uniform droplet size) fogs can be grown by providing a long retention time for growth. However, industrial fogs usually show a broad distribution, as in Fig. 14-91. Note also that for this set of data, the sizes are several orders of magnitude smaller than those shown earlier for entrainment and atomizers.

The result, as discussed in a later subsection, is a demand for different removal devices for the small particles.

While generally fog formation is a nuisance, it can occasionally be useful because of the high surface area generated by the fine drops. An example is insecticide application.

GAS-IN-LIQUID DISPERSIONS

GENERAL REFERENCES: Design methods for agitated vessels are presented by Penney in Couper et al., *Chemical Process Equipment, Selection and Design*, Chap. 10, Gulf Professional Publishing, Burlington, Mass., 2005. A comprehensive review of all industrial mixing technology is given by Paul, Atomo-Obeng, and Kresta, *Handbook of Industrial Mixing*, Wiley, Hoboken, N.J., 2004. Comprehensive treatments of bubbles or foams are given by Akers, *Foams: Symposium 1975*, Academic Press, New York, 1973; Bendure, *Tappi*, **58**, 83 (1975); Benfratello, *Energ. Elettr.*, **30**, 80, 486 (1953); Berkman and Egloff, *Emulsions and Foams*, Reinhold, New York, 1941, pp. 112–152; Bikerman, *Foams*, Springer-Verlag, New York, 1975; Kirk-Othmer *Encyclopedia of Chemical Technology*, 4th ed., Wiley, New York, 1993, pp. 82–145; Haberman and Morton, *Report 802*, David W. Taylor Model Basin, Washington, 1953; Levich, *Physicochemical Hydrodynamics*, Prentice-Hall, Englewood Cliffs, NJ, 1962; and Soo, *Fluid Dynamics of Multiphase Systems*, Blaisdell, Waltham, Massachusetts,

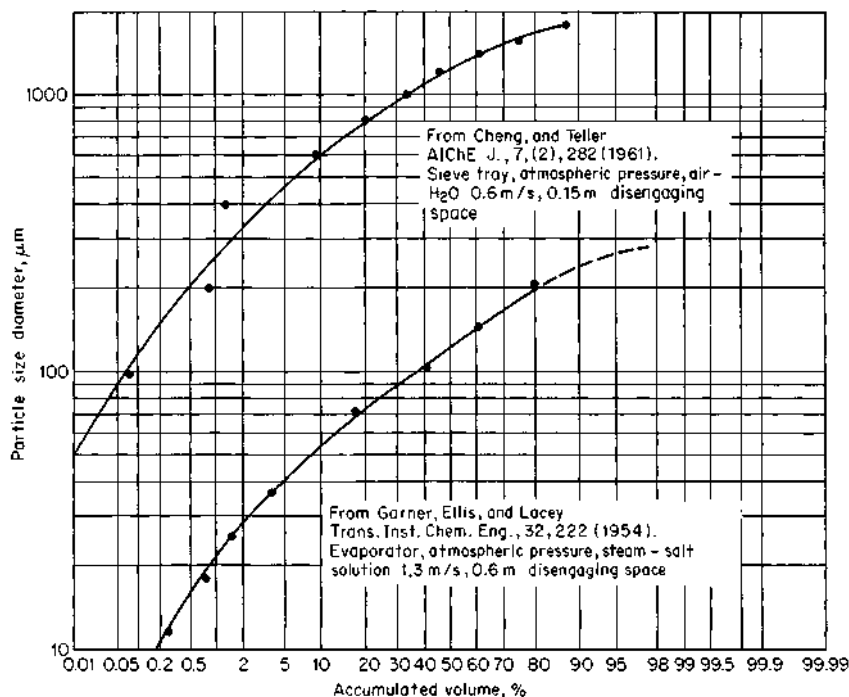


FIG. 14-90 Entrainment droplet-size distribution. To convert meters per second to feet per second, multiply by 3.28, to convert meters to feet multiply by 3.28.

1967. The formation of bubbles is comprehensively treated by Clift, Grace, and Weber, *Bubbles, Drops and Particles*, Academic, New York, 1978; Kumar and Kuloor, *Adv. Chem. Eng.*, **8**, 255-368 (1970); and Wilkinson and Van Dierendonck, *Chem. Eng. Sci.*, **49**, 1429-1438 (1994). Design methods for units operation in bubble columns and stirred vessels are covered by Atika and Yoshida, *Ind. Eng. Chem. Process Des. Dev.*, **13**, 84 (1974); Calderbank, *The Chem. Eng.* (London), CE209 (October, 1967); and Mixing, vol. II, Academic, New York, 1967, pp. 1-111; Fair, *Chem. Eng.*, **74**, 67 (July 3, 1967); Jordan, *Chemical Process Des.*, Interscience, New York, 1968, part 1, pp. 111-175; Mersmann, *Ger. Chem. Eng.*, **1**, 1 (1978); Resnick and Gal-Or, *Adv. Chem. Eng.*, **7**, 295-395 (1968); Valentin, *Absorption in Gas-Liquid Dispersions*, E. & F. N. Spon, London, 1967; Tattersson, *Fluid Mixing and Gas Dispersion in Agitated Tanks*, McGraw-Hill, 1991; and Deckwer and Schumpe, *Chem. Eng. Sci.*, **48**, 889-991 (1993).

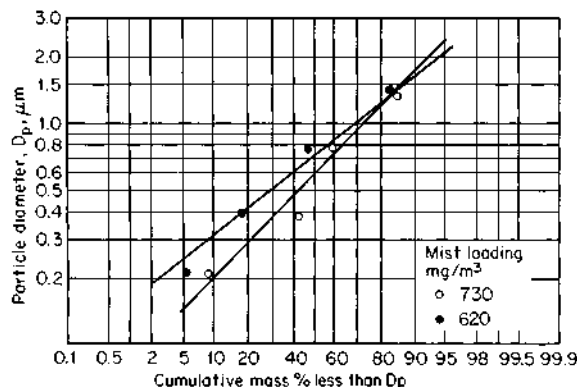


FIG. 14-91 Particle-size distribution and mist loading from absorption tower in a contact H₂SO₄ plant. [Gillespie and Johnstone, *Chem. Eng. Prog.*, **51**(2), 74 (1955).]

Review of foam rheology is given by Herzhaft [*Oil & Gas Sci. & Technol.*, **54**, 587 (1999)] and Heller and Kuntamukkula [*Ind. Eng. Chem. Res.*, **26**, 318 (1987)]. The influence of surface-active agents on bubbles and foams is summarized in selected passages from Schwartz and Perry, *Surface Active Agents*, vol. 1, Interscience, New York, 1949; and from Schwartz, Perry, and Berch, *Surface Active Agents and Detergents*, vol. 2, Interscience, New York, 1958. See also Elenkov, *Theor. Found. Chem. Eng.*, **1**, 1, 117 (1967); and Rubel, *Antifoaming and Defoaming Agents*, Noyes Data Corp., Park Ridge, NJ, 1972.

A review of foam stability also is given by de Vries, Meded, *Rubber Sticht. Delft*, No. 328, 1957. Foam-separation methodology is discussed by Aguiyoy and Lemlich, *Ind. Eng. Chem. Process Des. Dev.*, **13**, 153 (1974); and Lemlich, *Ind. Eng. Chem.*, **60**, 16 (1968). The following reviews of specific applications of gas-to-liquid dispersions are recommended: Industrial fermentations Aiba, Humphrey, and Millis, *Biochemical Engineering*, Academic, New York, 1965. Finn, *Bacteriol. Rev.*, **18**, 254 (1954). Oldshue, "Fermentation Mixing Scale-Up Techniques," in *Biotechnology and Bioengineering*, vol. 8, 1966, pp. 3-24. Aerobic oxidation of wastes: Eckenfelder and McCabe, *Advances in Biological Waste Treatment*, Macmillan, New York, 1963. Eckenfelder and O'Connor, *Biological Waste Treatment*, Pergamon, New York, 1961. McCabe and Eckenfelder, *Biological Treatment of Sewage and Industrial Wastes*, vol. 1, Reinhold, New York, 1955. Proceedings of Industrial Waste Treatment Conference, Purdue University, annually. Zlokarnik, *Adv. Biochem. Eng.*, **11**, 158-180 (1979). *Cellular elastomers*: Fling, *Natural Rubber Latex and Its Applications: The Preparation of Latex Foam Products*, British Rubber Development Board, London, 1954. Gould, in *Symposium on Application of Synthetic Rubbers*, American Society for Testing and Materials, Philadelphia, 1944, pp. 90-103. *Firefighting foams*: Perri, in Bikerman, op. cit., Chap. 12. Ratzer, *Ind. Eng. Chem.*, **48**, 2013 (1956). *Froth-flotation methods and equipment*: Booth, in Bikerman, op. cit., Chap. 13. Gaudin, *Flotation*, McGraw-Hill, New York, 1957. Taggart, *Handbook of Mineral Dressing*, Wiley, New York, 1945, Sec. 12, pp. 52-81. Tattersson, *Fluid Mixing and Gas Dispersion in Agitated Tanks*, McGraw-Hill, New York, 1991.

Objectives of Gas Dispersion The dispersion of gas as bubbles in a liquid or in a plastic mass is effected for one of the following purposes: (1) gas-liquid contacting (to promote absorption or stripping, with or without chemical reaction), (2) agitation of the liquid phase, or

(3) foam or froth production. Gas-in-liquid dispersions also may be produced or encountered inadvertently, sometimes undesirably.

Gas-Liquid Contacting Usually this is accomplished with conventional columns or with spray absorbers (see preceding subsection "Liquid-in-Gas Dispersions"). For systems containing solids or tar likely to plug columns, absorptions accomplished by strongly exothermic reactions, or treatments involving a readily soluble gas or a condensable vapor, however, bubble columns or agitated vessels may be used to advantage.

Agitation Agitation by a stream of gas bubbles (often air) rising through a liquid is often employed when the extra expense of mechanical agitation is not justified. Gas spargers may be used for simple blending operations involving a liquid of low volatility or for applications where agitator shaft sealing is difficult.

Foam Production This is important in froth-flotation separations; in the manufacture of cellular elastomers, plastics, and glass; and in certain special applications (e.g., food products, fire extinguishers). Unwanted foam can occur in process columns, in agitated vessels, and in reactors in which a gaseous product is formed; it must be avoided, destroyed, or controlled. Berkman and Egloff (*Emulsions and Foams*, Reinhold, New York, 1941, pp. 112–152) have mentioned that foam is produced only in systems possessing the proper combination of interfacial tension, viscosity, volatility, and concentration of solute or suspended solids. From the standpoint of gas comminution, foam production requires the creation of small bubbles in a liquid capable of sustaining foam.

Theory of Bubble and Foam Formation A bubble is a globule of gas or vapor surrounded by a mass or thin film of liquid. By extension, globular voids in a solid are sometimes called bubbles. Foam is a group of bubbles separated from one another by thin films, the aggregation having a finite static life. Although nontechnical dictionaries do not distinguish between foam and froth, a technical distinction is often made. A highly concentrated dispersion of bubbles in a liquid is considered a froth even if its static life is substantially nil (i.e., it must be dynamically maintained). Thus, all foams are also froths, whereas the reverse is not true. The term *lather* implies a froth that is worked up on a solid surface by mechanical agitation; it is seldom used in technical discussions. The thin walls of bubbles comprising a foam are called *laminae* or *lamellae*.

Bubbles in a liquid originate from one of three general sources: (1) They may be formed by desupersaturation of a solution of the gas or by the decomposition of a component in the liquid; (2) They may be introduced directly into the liquid by a bubbler or sparger or by mechanical entrainment; and (3) They may result from the disintegration of larger bubbles already in the liquid.

Generation Spontaneous generation of gas bubbles within a homogeneous liquid is theoretically impossible (Bikerman, *Foams: Theory and Industrial Applications*, Reinhold, New York, 1953, p. 10). The appearance of a bubble requires a gas nucleus as a void in the liquid. The nucleus may be in the form of a small bubble or of a solid carrying adsorbed gas, examples of the latter being dust particles, boiling chips, and a solid wall. A void can result from cavitation, mechanically or acoustically induced. Basu, Warriar, and Dhir [*J. Heat Transfer*, **124**, 717 (2002)] have reviewed boiling nucleation, and Blander and Katz [*AIChE J.*, **21**, 833 (1975)] have thoroughly reviewed bubble nucleation in liquids.

Theory permits the approximation of the maximum size of a bubble that can adhere to a submerged horizontal surface if the contact angle between bubble and solid (angle formed by solid-liquid and liquid-gas interfaces) is known [Wark, *J. Phys. Chem.*, **37**, 623 (1933); Jakob, *Mech. Eng.*, **58**, 643 (1936)]. Because the bubbles that actually rise from a surface are always considerably smaller than those so calculated and inasmuch as the contact angle is seldom known, the theory is not directly useful.

Formation at a Single Orifice The formation of bubbles at an orifice or capillary immersed in a liquid has been the subject of much study, both experimental and theoretical. Kulkarni and Joshi [*Ind. Eng. Chem. Res.*, **44**, 5873 (2005)] have reviewed bubble formation and rise. Bikerman (op. cit., Secs. 3 to 7), Valentin (op. cit., Chap. 2), Jackson (op. cit.), Soo (op. cit., Chap. 3), Fair (op. cit.), Kumer et al. (op. cit.), Clift et al. (op. cit.) and Wilkinson and Van Dierendonck

[*Chem. Eng. Sci.*, **49**, 1429 (1994)] have presented reviews and analyses of this subject.

There are three regimes of bubble production (Silberman in *Proceedings of the Fifth Midwestern Conference on Fluid Mechanics*, Univ. of Michigan Press, Ann Arbor, 1957, pp. 263–284): (1) single-bubble, (2) intermediate, and (3) jet.

Single-Bubble Regime Bubbles are produced one at a time, their size being determined primarily by the orifice diameter d_o , the interfacial tension of the gas-liquid film σ , the densities of the liquid ρ_L and gas ρ_G , and the gravitational acceleration g according to the relation

$$d_b/d_o = [6\sigma g d_o^2 (\rho_L - \rho_G)]^{1/3} \quad (14-206)$$

where d_b is the bubble diameter.

$$f = Q/(\pi d_b^3/6) = Qg(\rho_L - \rho_G)/(\pi d_o \sigma) \quad (14-207)$$

where f is the frequency of bubble formation and Q is the volumetric rate of gas flow in consistent units.

Equations (14-206) and (14-207) result from a balance of bubble buoyancy against interfacial tension. They include no inertia or viscosity effects. At low bubbling rates ($<1/s$), these equations are quite satisfactory. Van Krevelen and Hofstijzer [*Chem. Eng. Prog.*, **46**, 29 (1950)], Guyer and Peterhaus [*Helv. Chim. Acta*, **26**, 1099 (1943)] and Wilkinson (op. cit.) report good agreement with Eq. (14-206) for water, transformer oil, ether, and carbon tetrachloride for vertically oriented orifices with $0.004 < D < 0.95$ cm. If the orifice diameter becomes too large, the bubble diameter will be smaller than the orifice diameter, as predicted by Eq. (14-206), and instability results; consequently, stable, stationary bubbles cannot be produced. Kulkarni et al. (op. cit.) have discussed much more detailed models for predicting bubble size; however, the models are very difficult and tedious to use in practice. These more sophisticated models need to be considered if the added accuracy warrants the extra effort.

For bubbles formed in water, the orifice diameter that permits bubbles of about its own size is calculated as 0.66 cm. Davidson and Amick [*AIChE J.*, **2**, 337 (1956)] confirmed this estimate in their observation that stable bubbles in water were formed at a 0.64-cm orifice but could not be formed at a 0.79-cm orifice.

For very thin liquids, Eqs. (14-206) and (14-207) are expected to be valid up to a gas-flow Reynolds number of 200 (Valentin, op. cit., p. 8). For liquid viscosities up to 100 cP, Datta, Napier, and Newitt [*Trans. Inst. Chem. Eng.*, **28**, 14 (1950)] and Siems and Kauffman [*Chem. Eng. Sci.*, **5**, 127 (1956)] have shown that liquid viscosity has very little effect on the bubble volume, but Davidson and Schuler [*Trans. Inst. Chem. Eng.*, **38**, 144 (1960)] and Krishnamurthi et al. [*Ind. Eng. Chem. Fundam.*, **7**, 549 (1968)] have shown that bubble size increases considerably over that predicted by Eq. (14-206) for liquid viscosities above 1000 cP. In fact, Davidson et al. (op. cit.) found that their data agreed very well with a theoretical equation obtained by equating the buoyant force to drag based on Stokes' law and the velocity of the bubble equator at break-off:

$$d_b = \left(\frac{6}{\pi} \right) \left(\frac{4\pi}{3} \right)^{1/4} \left(15 \times \frac{vQ}{2g} \right)^{3/4} \quad (14-208)$$

where v is the liquid kinematic viscosity and Q is the gas volumetric flow rate. This equation is dimensionally consistent. The relative effect of liquid viscosity can be obtained by comparing the bubble diameters calculated from Eqs. (14-206) and (14-208). If liquid viscosity appears significant, one might want to use the long and tedious method developed by Krishnamurthi et al. (op. cit.) and the review by Kulkarni et al. (op. cit.) that considers both surface-tension forces and viscous-drag forces.

Intermediate Regime This regime extends approximately from a Reynolds number of 200 to one of 2100. As the gas flow through a

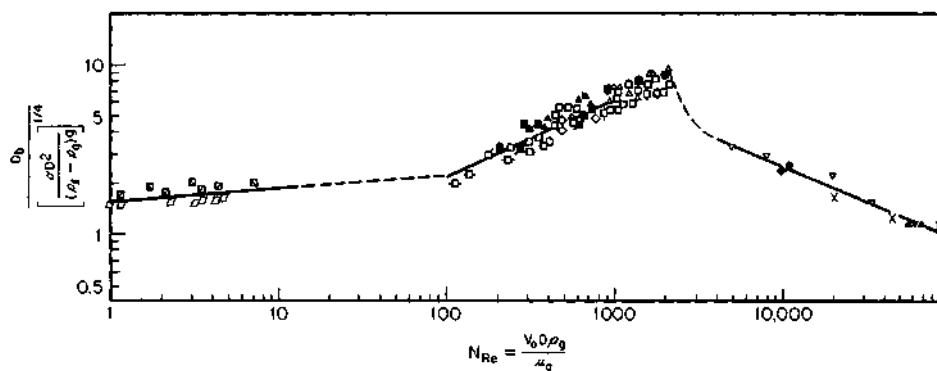


FIG. 14-92 Bubble-diameter correlation for air sparged into relatively inviscid liquids. D_b = bubble diameter, D = orifice diameter, V_o = gas velocity through sparging orifice, P = fluid density, and μ = fluid viscosity. [From Can. J. Chem. Eng., **54**, 503 (1976).]

submerged orifice increases beyond the limit of the single-bubble regime, the frequency of bubble formation increases more slowly, and the bubbles begin to grow in size. Between the two regimes there may indeed be a range of gas rates over which the bubble size decreases with increasing rate, owing to the establishment of liquid currents that nip the bubbles off prematurely. The net result can be the occurrence of a minimum bubble diameter at some particular gas rate [Mater, U.S. Bur. Mines Bull. 260 (1927) and Bikerman, op. cit., p. 4]. At the upper portion of this region, the frequency becomes very nearly constant with respect to gas rate, and the bubble size correspondingly increases with gas rate. The bubble size is affected primarily by (1) orifice diameter, (2) liquid-inertia effects, (3) liquid viscosity, (4) liquid density, and (5) the relationship between the constancy of gas flow and the constancy of pressure at the orifice.

Kumar et al. have done extensive experimental and theoretical work reported in *Ind. Eng. Chem. Fundam.*, **7**, 549 (1968); *Chem. Eng. Sci.*, **24**, part 1, 731; part 2, 749; part 3, 1711 (1969) and summarized in *Adv. Chem. Eng.*, **8**, 255 (1970). They, along with other investigators—Swope [*Can. J. Chem. Eng.*, **44**, 169 (1972)], Tsuge and Hibino [*J. Chem. Eng. Japan*, **11**, 307 (1972)], Pinczewski [*Chem. Eng. Sci.*, **36**, 405 (1981)], Tsuge and Hibino [*Int. Chem. Eng.*, **21**, 66 (1981)], and Takahashi and Miyahara [ibid., p. 224]—have solved the equations resulting from a force balance on the forming bubble, taking into account buoyancy, surface tension, inertia, and viscous-drag forces for both conditions of constant flow through the orifice and constant pressure in the gas chamber. The design method is complex and tedious and involves the solution of algebraic and differential equations. Although Mersmann [*Ger. Chem. Eng.*, **1**, 1 (1978)] claims that the results of Kumar et al. (loc. cit.) well fit experimental data, Lanauze and Harn [*Chem. Eng. Sci.*, **29**, 1663 (1974)] claim differently:

Further, it has been shown that the mathematical formulation of Kumar's model, including the condition of detachment, could not adequately describe the experimental situation—Kumar's model has several fundamental weaknesses, the computational simplicity being achieved at the expense of physical reality.

In lieu of careful independent checks of predictive accuracy, the results of the comprehensive theoretical work will not be presented here. Simpler, more easily understood predictive methods, for certain important limiting cases, will be presented. As a check on the accuracy of these simpler methods, it will perhaps be prudent to calculate the bubble diameter from the graphical representation by Mersmann (loc. cit.) of the results of Kumar et al. (loc. cit.) and the review by Kulkarni et al. (op. cit.)

For conditions approaching constant flow through the orifice, a relationship derived by equating the buoyant force to the inertia force of the liquid [Davidson et al., *Trans. Instn. Chem. Engrs.*, **38**, 335 (1960)] (dimensionally consistent),

$$d_b = 1.378 \times \frac{6Q^{6/5}}{\pi g^{3/5}} \quad (14-209)$$

fits experimental data reasonably well. Surface tension and liquid viscosity tend to increase the bubble size—at a low Reynolds number. The effect of surface tension is greater for large orifice diameters. The magnitude of the diameter increase due to high liquid viscosity can be obtained from Eq. (14-208).

For conditions approaching constant pressure at the orifice entrance, which probably simulates most industrial applications, there is no independently verified predictive method. For air at near atmospheric pressure sparged into relatively inviscid liquids (11 ~ 100 cP), the correlation of Kumar et al. [*Can. J. Chem. Eng.*, **54**, 503 (1976)] fits experimental data well. Their correlation is presented here as Fig. 14-92.

Wilkinson et al. (op. cit.) make the following observation about the effect of gas density on bubble size: "The fact that the bubble size decreases slightly for higher gas densities can be explained on the basis of a force balance."

Jet Regime With further rate increases, turbulence occurs at the orifice, and the gas stream approaches the appearance of a continuous jet that breaks up 7.6 to 10.2 cm above the orifice. Actually, the stream consists of large, closely spaced, irregular bubbles with a rapid swirling motion. These bubbles disintegrate into a cloud of smaller ones of random size distribution between 0.025 cm or smaller and about 1.25 cm, with a mean size for air and water of about 0.4 cm (Leibson et al., loc. cit.). According to Wilkinson et al. (op. cit.), jetting begins when

$$N_{We,g} = \frac{\rho_g d_o U_o^2}{\sigma} \leq 2 \quad (14-210)$$

There are many contradictory reports about the jet regime, and theory, although helpful (see, for example, Siberman, loc. cit.), is as yet unable to describe the phenomena observed. The correlation of Kumar et al. (Fig. 14-92) is recommended for air-liquid systems.

Formation at Multiple Orifices At high velocities, coalescence of bubbles formed at individual orifices occurs; Helsby and Tuson [*Research (London)*, **8**, 270 (1955)], for example, observed the frequent coalescence of bubbles formed in pairs or in quartets at an orifice. Multiple orifices spaced by the order of magnitude of the orifice diameter increase the probability of coalescence, and when the magnitude is small (as in a sintered plate), there is invariably some. The broken lines of Fig. 14-92 presumably represent zones of increased coalescence and relatively less effective dispersion as the gas rate through porous-carbon tubes is increased. Savitskaya [*Kolloidn. Zh.*, **13**, 309 (1951)] found that the average bubble size formed at the

surface of a porous plate was such as to maintain constancy of the product of bubble specific surface and interfacial tension as the latter was varied by addition of a surfactant. König et al. [*Ger. Chem. Eng.*, **1**, 199 (1978)] produced bubble sizes varying from 0.5 to 4 mm by the use of two porous-plate spargers and one perforated-plate sparger with superficial gas velocities from 1 to 8 cm/s. The small bubble sizes were stabilized by adding up to 0.5 percent of various alcohols to water.

At high-flow rates through perforated plates such as those that occur in distillation columns, Calderbank and Rennie [*Trans. Instn. Chem. Engrs.*, **40**, T3 (1962)]; Porter et al. [*ibid.*, **45**, T265 (1967)]; Rennie and Evans [*Br. Chem. Eng.*, **7**, 498 (1962)]; and Valentin (op. cit., Chap. 3) have investigated and discussed the effect of the flow conditions through the multiple orifices on the froths and foams that occur above perforated plates.

Entrainment and Mechanical Disintegration Gas can be entrained into a liquid by a solid or a stream of liquid falling from the gas phase into the liquid, by surface ripples or waves, and by the vertical swirl of a mass of agitated liquid about the axis of a rotating agitator. Small bubbles probably form near the surface of the liquid and are caught into the path of turbulent eddies, whose velocity exceeds the terminal velocity of the bubbles. The disintegration of a submerged mass of gas takes place by the turbulent tearing of smaller bubbles away from the exterior of the larger mass or by the influence of surface tension on the mass when it is attenuated by inertial or shear forces into a cylindrical or disk form. A fluid cylinder that is greater in length than in circumference is unstable and tends to break spontaneously into two or more spheres. These effects account for the action of fluid attrition and of an agitator in the disintegration of suspended gas. Quantitative correlations for gas entrainment by liquid jets and in agitated vessels will be given later.

Foams Two excellent reviews (Shedlovsky, op. cit.; Lemlich, op. cit.; Herzhaft, op. cit.; and Heller et al., op. cit.) covering the literature pertinent to foams have been published. A foam is formed when bubbles rise to the surface of a liquid and persist for a while without coalescence with one another or without rupture into the vapor space. The formation of foam, then, consists simply of the formation, rise, and aggregation of bubbles in a liquid in which foam can exist. The life of foams varies over many magnitudes—from seconds to years—but in general is finite. Maintenance of a foam, therefore, is a dynamic phenomenon.

Gravitational force favors the separation of gas from liquid in a dispersion system, causing the bubbles to rise to the liquid surface and the liquid contained in the bubble walls to drain downward to the main body of the liquid. Interfacial tension favors the coalescence and ultimate disappearance of bubbles; indeed, it is the cause of bubble destruction upon the rupture of the laminae.

The viscosity of the liquid in a film opposes the drainage of the film and its displacement by the approach of coalescing bubbles. The higher the viscosity, the slower will be the film-thinning process; furthermore, if viscosity increases as the film grows thinner, the process becomes self-retarding. The viscosity of films appears to be greater than that of the main body of the parent liquid in many cases. Sometimes this is a simple temperature effect, the film being cooler because of evaporation; sometimes it is a concentration effect, with dissolved or fine suspended solids migrating to the interface and producing classical or anomalous increases in viscosity; at yet other times, the effect seems to occur without explanation.

If the liquid laminae of a foam system can be converted to impermeable solid membranes, the film viscosity can be regarded as having become infinite, and the resulting solid foam will be permanent. Likewise, if the laminae are composed of a gingham plastic or a thixotrope, the foam will be permanently stable for bubbles whose buoyancy does not permit exceeding the yield stress. For other non-newtonian fluids, however, and for all newtonian ones, no matter how viscous, the viscosity can only delay but never prevent foam disappearance. The popular theory, held since the days of Plateau, that foam life is proportional to surface viscosity and inversely proportional to interfacial tension, is not correct, according to Bikerman (op. cit., p. 161), who points out that it is contradicted by experiment.

The idea that foam films drain to a critical thickness at which they spontaneously burst is also rejected by Bikerman. Foam stability, rather, is keyed to the existence of a surface skin of low interfacial tension immediately overlying a solution bulk of higher tension, latent until it is exposed by rupture of the superficial layer [Maragoni, *Nuovo Cimento*, **2** (5-6), 239 (1871)]. Such a phenomenon of surface elasticity, resulting from concentration differences between bulk and surface of the liquid, accounts for the ability of bubbles to be penetrated by missiles without damage. It is conceivable that films below a certain thickness no longer carry any bulk of solution and hence have no capacity to close surface ruptures, thus becoming vulnerable to mechanical damage that will destroy them. The Maragoni phenomenon is consistent also with the observation that neither pure liquids nor saturated solutions will sustain a foam, since neither extreme will allow the necessary differences in concentration between surface and bulk of solution.

The specific ability of certain finely divided, insoluble solids to stabilize foam has long been known [Berkman and Egloff, op. cit., p. 133; and Bikerman, op. cit., Chap. 11]. Bartsch [*Kolloidchem. Beih.*, **20**, 1 (1925)] found that the presence of fine galena greatly extended the life of air foam in aqueous isoamyl alcohol, and the finer the solids, the greater the stability. Particles on the order of 50 μ m length extended the life from 17 seconds to several hours. This behavior is consistent with theory, which indicates that a solid particle of medium contact angle with the liquid will prevent the coalescence of two bubbles with which it is in simultaneous contact. Quantitative observations of this phenomenon are scanty.

Berkman and Egloff explain that some additives increase the flexibility or toughness of bubble walls, rather than their viscosity, to render them more durable. They cite as illustrations the addition of small quantities of soap to saponin solutions or of glycerin to soap solution to yield much more stable foam. The increased stability with ionic additives is probably due to electrostatic repulsion between charged, nearly parallel surfaces of the liquid film, which acts to retard draining and hence rupture.

Characteristics of Dispersion

Properties of Component Phases As discussed in the preceding subsection, dispersions of gases in liquids are affected by the viscosity of the liquid, the density of the liquid and of the gas, and the interfacial tension between the two phases. They also may be affected directly by the composition of the liquid phase. Both the formation of bubbles and their behavior during their lifetime are influenced by these quantities as well as by the mechanical aspects of their environment.

Viscosity and density of the component phases can be measured with confidence by conventional methods, as can the interfacial tension between a pure liquid and a gas. The interfacial tension of a system involving a solution or micellar dispersion becomes less satisfactory, because the interfacial free energy depends on the concentration of solute at the interface. Dynamic methods and even some of the so-called static methods involve the creation of new surfaces. Since the establishment of equilibrium between this surface and the solute in the body of the solution requires a finite amount of time, the value measured will be in error if the measurement is made more rapidly than the solute can diffuse to the fresh surface. Eckenfelder and Barnhart (*Am. Inst. Chem. Engrs.*, 42d national meeting, Repr. 30, Atlanta, 1960) found that measurements of the surface tension of sodium lauryl sulfate solutions by maximum bubble pressure were higher than those by DuNuoy tensiometer by 40 to 90 percent, the larger factor corresponding to a concentration of about 100 ppm, and the smaller to a concentration of 2500 ppm of sulfate.

Even if the interfacial tension is measured accurately, there may be doubt about its applicability to the surface of bubbles being rapidly formed in a solution of a surface-active agent, for the bubble surface may not have time to become equilibrated with the solution. Coppock and Meiklejohn [*Trans. Instn. Chem. Engrs.*, **29**, 75 (1951)] reported that bubbles formed in the single-bubble regime at an orifice in a solution of a commercial detergent had a diameter larger than that calculated in terms of the measured surface tension of the solution [Eq. (14-206)]. The disparity is probably a reflection of unequilibrated bubble laminae.

One concerned with the measurement of gas-liquid interfacial tension should consult the useful reviews of methods prepared by Harkins [in Chap. 9 of Weissberger, *Techniques of Organic Chemistry*, 2d ed., vol. 1, part 2, Interscience, New York, 1949], Schwartz and coauthors [*Surface Active Agents*, vol. 1, Interscience, New York, 1949, pp. 263-271; *Surface Active Agents and Detergents*, vol. 2, Interscience, New York, 1958, pp. 389-391, 417-418], and by Adamson [*Physical Chemistry of Surfaces*, Interscience, New York, 1960].

Dispersion Characteristics The chief characteristics of gas-liquid dispersions, like those of liquid-in-gas suspensions, are heterogeneity and instability. The composition and structure of an unstable dispersion must be observed in the dynamic situation by looking at the mixture, with or without the aid of optical devices, or by photographing it, preferably in nominal steady state; photographs usually are required for quantitative treatment. Stable foams may be examined after the fact of their creation if they are sufficiently robust or if an immobilizing technique such as freezing is employed [Chang et al., *Ind. Eng. Chem.*, **48**, 2035 (1956)].

The rate of rise of bubbles has been discussed by Kulkarni, op. cit.; Benfratello, *Energ. Elettr.*, **30**, 80 (1953); Haberman and Morton, *Report 802: David W. Taylor Model Basin*, Washington, September 1953; Jackson, loc. cit.; Valentin, op. cit., Chap. 2; Soo, op. cit., Chap. 3; Calderbank, loc. cit., p. CE220; and Levich, op. cit., Chap. 8). A comprehensive and apparently accurate predictive method has been published [Jamialahmadi et al., *Trans. ICE*, **72**, part A, 119-122 (1994)]. Small bubbles (below 0.2 mm in diameter) are essentially rigid spheres and rise at terminal velocities that place them clearly in the laminar-flow region; hence their rising velocity may be calculated from Stokes' law. As bubble size increases to about 2 mm, the spherical shape is retained, and the Reynolds number is still sufficiently small (<10) that Stokes' law should be nearly obeyed.

As bubble size increases, two effects set in, however, that alter the velocity. At about $N_{Re} = 100$, a wobble begins that can develop into a helical path if the bubbles are not liberated too closely to one another [Houghton, McLean, and Ritchie, *Chem. Eng. Sci.*, **7**, 40 (1957); and Houghton et al., *ibid.*, p. 111]. Furthermore, for bubbles in the range of 1 mm and larger (until distortion becomes serious) internal circulation can set in [Garner and Hammerton, *Chem. Eng. Sci.*, **3**, (1954); and Haberman and Morton, loc. cit.], and according to theoretical analyses by Hadamard and Rybczynski and given by Levich (op. cit.), the drag coefficient for a low-viscosity dispersed phase and a high-viscosity continuous phase will approach two-thirds of the drag coefficient for rigid spheres, namely $C_D = 16/N_{Re}$. The rise velocity of a circulating bubble or drop will thus be 1.5 times that of a rigid sphere. Redfield and Houghton [*Chem. Eng. Sci.*, **20**, 131 (1965)] have found that CO_2 bubbles rising in pure water agree with the theoretical solution for a circulating drop below $N_{Re} = 1$. Many investigators (see Valentin, op. cit.) have found that extremely small quantities of impurities can retard or stop this internal circulation. In this behavior may lie the explanation of the fact that the addition of long-chain fatty acids to water to produce a concentration of 1.5×10^{-4} molar markedly reduces the rate of rise of bubbles [Stuke, *Naturwissenschaften*, **39**, 325 (1952)].

Above diameters of about 2 mm, bubbles begin to change to ellipsoids, and above 1 cm they become lens-shaped, according to Davies and Taylor [*Proc. Roy. Soc. (London)*, **A200**, 379 (1950)]. The rising velocity in thin liquids for the size range $1 \text{ mm} < D_b < 20 \text{ mm}$ has been reported as 20 to 30 cm/s by Haberman and Morton (op. cit.) and Davenport, Richardson, and Bradshaw [*Chem. Eng. Sci.*, **22**, 1221 (1967)]. Schwerdtfeger [*ibid.*, **23**, 937 (1968)] even found the same for argon bubbles rising in mercury. Surface-active agents have no effect on the rise velocity of bubbles larger than 4 mm in thin liquids (Davenport et al., loc. cit.).

Above a Reynolds number of the order of magnitude of 1000, bubbles assume a helmet shape, with a flat bottom (Eckenfelder and Barnhart, loc. cit.; and Leibson et al., loc. cit.). After bubbles become large enough to depart from Stokes' law at their terminal velocity, behavior is generally complicated and erratic, and the reported data scatter considerably. The rise can be slowed, furthermore, by a wall effect if the diameter of the container is not greater than 10 times the diameter of the bubbles, as shown by Uno and Kintner [*AIChE J.*, **2**, 420 (1956); and Collins, *J. Fluid Mech.*, **28**(1), 97 (1967)]. Work has

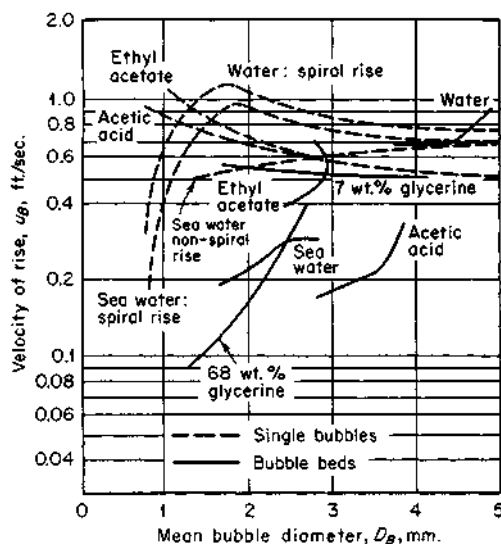


FIG. 14-93 Velocity of rising bubbles, singly and in clouds. To convert feet per second to meters per second, multiply by 0.305. [From *Chem. Eng. Sci.*, **7**, 48 (1957).]

been done to predict the rise velocity of large bubbles [Rippin and Davidson, *Chem. Eng. Sci.*, **22**, 217 (1967); Grace and Harrison, *ibid.*, 1337; Mendelson, *AIChE J.*, **13**, 250 (1967); Cole, *ibid.*; Lehrer, *J. Chem. Eng. Japan*, **9**, 237 (1976); and Lehrer, *AIChE J.*, **26**, 170 (1980)]. The works of Lehrer present correlations that accurately predict rise velocities for a wide range of system properties. Reviews of the technical literature concerning the rise of single bubbles and drops have been published by Kulkarni et al. (op. cit.) and Clift, Grace, and Weber (*Bubbles, Drops and Particles*, Academic, New York, 1978). Mendelson has used a wave theory to predict the terminal velocity, and Cole has checked the theory with additional data. The other authors listed solved some simplified form of the Navier-Stokes equations. Jamialahmadi et al., loc. cit., have developed a single equation predictive method for bubble rise velocity, which covers the entire range of bubble diameters.

When bubbles are produced in clouds, as by a porous disperser, their behavior during rising is further complicated by interaction among themselves. In addition to the tendency for small bubbles to coalesce and large ones to disintegrate, there are two additional opposing influences on the rate of rise of bubbles of any particular size: (1) A "chimney effect" can develop in which a massive current upward appears at the axis of the bubble stream, leading to increased net bubble velocity; and (2) the proximity of the bubbles to one another can result in a hindered-settling condition, leading to reduced average bubble velocity. Figure 14-93 shows the data of Houghton et al. (op. cit.) for clouds of bubbles compared with their single-bubble data for pure water and seawater and of Peebles and Garber [*Chem. Eng. Progr.*, **49**, 88 (1953)] for acetic acid and ethyl acetate. The bubble clouds were produced with a sintered-glass plate of mean pore size (inferred from air wet-permeability data) of 81 μm .

The difference between the curves for pure water and seawater again illustrates the significance of small concentrations of solute with respect to bubble behavior. In commercial bubble columns and agitated vessels coalescence and breakup are so rapid and violent that the rise velocity of a single bubble is meaningless. The average rise velocity can, however, be readily calculated from holdup correlations that will be given later.

The quantitative examination of bubble systems is aided by the use of proper illumination and photography. The formation of bubbles at single sources often is sufficiently periodic to be stopped by stroboscopic light. Clouds of rising bubbles are more difficult to assess and

require careful technique. Satisfactory photographic methods have been developed by Vermeulen, Williams, and Langlois [*Chem. Eng. Progr.*, **51**, 85 (1955)] and by Calderbank [*Trans. Instn. Chem. Engrs.*, **36**, 443 (1958)] and are described by these authors. Calderbank's technique resulted in particularly precise measurements that permitted a good estimation of the surface area of the dispersed bubbles.

Methods of Gas Dispersion The problem of dispersing a gas in a liquid may be attacked in several ways: (1) The gas bubbles of the desired size or which grow to the desired size may be introduced directly into the liquid; (2) a volatile liquid may be vaporized by either decreasing the system pressure or increasing its temperature; (3) a chemical reaction may produce a gas; or (4) a massive bubble or stream of gas is disintegrated by fluid shear and/or turbulence in the liquid.

Spargers: Simple Bubblers The simplest method of dispersing gas in a liquid contained in a tank is to introduce the gas through an open-end standpipe, a horizontal perforated pipe, or a perforated plate at the bottom of the tank. At ordinary gassing rates (corresponding to the jet regime), relatively large bubbles will be produced regardless of the size of the orifices.

Perforated-pipe or -plate spargers usually have orifices 3 to 12 mm in diameter. Effective design methods to minimize maldistribution are presented in the fifth edition of this handbook, p. 5-47, 1973, and by Knaebel [*Chem. Eng.*, 116 (Mar. 9, 1981)]. For turbulent flow conditions into the sparger, the following relationship will allow design of a perforated-pipe sparger for a given degree of maldistribution provided $N_h > 5$ and length/diameter < 300 .

$$d_p = 0.95(N_h C_v)^{1/2} \times \left(\frac{d_h}{\Delta U_h / U_h} \right)^{1/4} \quad (14-211)$$

where d_p = pipe diameter, d_h = sparging hole diameter, N_h = number of holes in sparger, C_v = orifice coefficient for sparger hole (see *Chemical Engineers' Handbook*, 5th ed., pp. 5-13, 5-34), U_h = average velocity through sparger holes, ΔU_h = difference between maximum and minimum velocities through sparger holes, and $\Delta U_h / U_h$ = fractional maldistribution of flow through sparger holes.

Simple spargers are used as agitators for large tanks, principally in the cement and oil industries. Kauffman [*Chem. Metall. Eng.*, **37**, 178-180 (1930)] reported the following air rates for various degrees of agitation in a tank containing 2.7 m (9 ft) of liquid:

Degree of agitation	Air rate, m ³ /(m ² tank cross section, min)
Moderate	0.0033
Complete	0.0066
Violent	0.016

For a liquid depth of 0.9 m (3 ft), Kauffman recommended that the listed rates be doubled.

An air lift consisting of a sparger jetting into a draft tube with ports discharging at several heights has been recommended by Heiser [*Chem. Eng.*, **55**(1), 135 (1948)] for maintaining agitation in a heavy, coarse slurry, the level of which varies widely. The design is illustrated in Fig. 14-94.

The ability of a sparger to blend miscible liquids might be described in terms of a fictitious diffusivity. Siemes did so, reporting that the agitation produced by a stream of bubbles rising in a tube with a superficial velocity of about 8.2 cm/s corresponded to an apparent diffusion coefficient as large as 75 cm²/s [*Chem. Ing. Tech.*, **29**, 727 (1957)]. The blending rate thus is several orders of magnitude higher than it would be by natural diffusive action. These results are typical of subsequent investigations on back mixing, which will be discussed in more detail later.

Lehrer [*Ind. Eng. Chem. Process Des. Dev.*, **7**, 226 (1968)] conducted liquid-blending tests with air sparging in a 0.61-m-diameter by 0.61-m-tall vessel and found that an air volume equal to about one-half of the vessel volume gave thorough blending of inviscid liquids of equal viscosities. Using an analogy to mechanically agitated vessels in which equal tank turnovers give equal blend times, one would expect this criterion to be applicable to other vessel sizes. Liquids of unequal density would require somewhat more air.

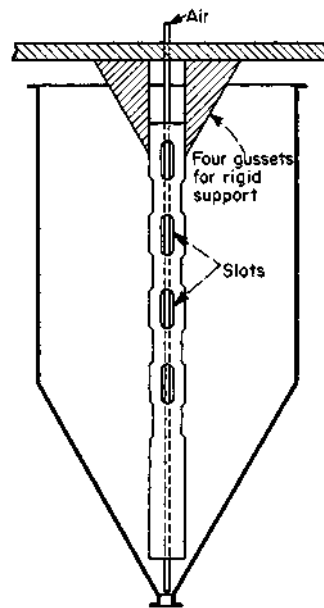


FIG. 14-94 Slotted air lift for agitation of a variable-level charge. [From *Chem. Eng.*, **55**(1), 135 (1948).]

Open-end pipes, perforated plates, and ring- or cross-style perforated-pipe spargers are used without mechanical agitation to promote mass transfer, as in chlorinators and biological sewage treatment. In the "quiescent regime" (superficial gas velocity less than 4.57 to 6.1 cm/s [0.15 to 0.2 ft/s]) the previously mentioned spargers are usually operated at orifice Reynolds numbers in excess of 6000 in order to get small bubbles so as to increase the interfacial area and thus increase mass transfer. In the "turbulent regime" (superficial gas velocity greater than 4.57 to 6.1 cm/s), sparger design is not critical because a balance between coalescence and breakup is established very quickly according to Towell et al. [*AIChE Symp. Ser. No. 10*, 97 (1965)]. However, a reasonably uniform orifice distribution over the column cross section is desirable, and according to Fair [*Chem. Eng.*, **74**, 67 (July 3, 1967); 207 (July 17, 1967)] the orifice velocity should be less than 75 to 90 m/s.

Porous Septa In the quiescent regime porous plates, tubes, disks, or other shapes that are made by bonding or sintering together carefully sized particles of carbon, ceramic, polymer, or metal are frequently used for gas dispersion, particularly in foam fractionators. The resulting septa may be used as spargers to produce much smaller bubbles than will result from a simple bubbler. Figure 14-95 shows a comparison of the bubbles emitted by a perforated-pipe sparger [0.16-cm orifices] and a porous carbon septum (120 μm pores). The gas flux through a porous septum is limited on the lower side by the requirement that, for good performance, the whole sparger surface should bubble more or less uniformly, and on the higher side by the onset of serious coalescence at the surface of the septum, resulting in poor dispersion. In the practical range of fluxes, the size of the bubbles produced depends on both the size of pores in the septum and the pressure drop imposed across it, being a direct function of both.

Table 14-23 lists typical grades of porous carbon, silica, alumina, stainless steel (type 316), and polymers commercially available.

Porous media are also manufactured from porcelain, glass, silicon carbide, and a number of metals: Monel, Inconel, nickel, bronze, Hastelloy C, Stellite L-605, gold, platinum, and many types of stainless steel. Two manufacturers of porous septa are Mott (mottcorp.com) and Pall (pall.com). The air permeabilities of Table 14-23 indicate the relative flow resistances of the various grades to homogeneous fluid but may not be used in designing a disperser for submerged

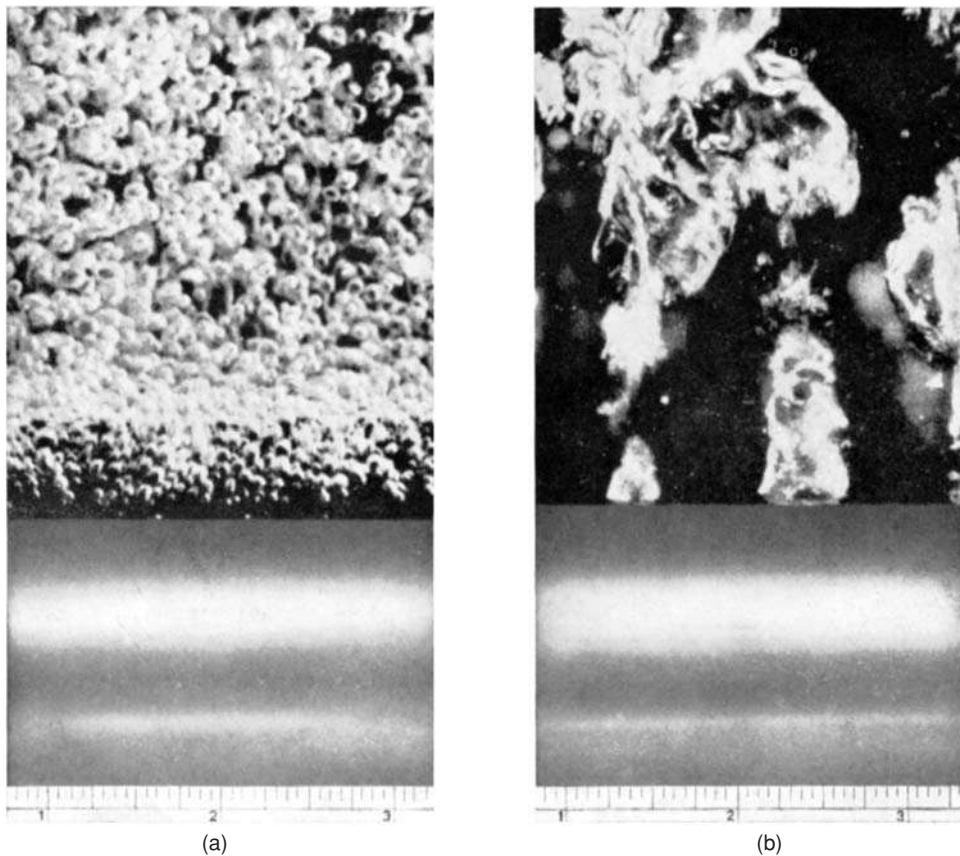


FIG. 14-95 Comparison of bubbles from a porous septum and from a perforated-pipe sparger. Air in water at 70°F. (a) Grade 25 porous-carbon diffuser operating under a pressure differential of 13.7 in. of water. (b) Karbate pipe perforated with 1/16-in holes on 1-in centers. To convert inches to centimeters, multiply by 2.54; °C = $\frac{5}{9}$ (°F - 32). (National Carbon Co.)

operation, for the resistance of a septum to the flow of gas increases when it is wet. The air permeabilities for water-submerged porous carbon of some of the grades listed in the table are shown in Fig. 14-96. The data were determined with septa 0.625 in thick in water at 70°F. Comparable wet-permeability data for 1-in Alundum plates of two grades of fineness are given in Table 14-24.

The gas rate at which coalescence begins to reduce the effectiveness of dispersion appears to depend not only on the pore size and pore structure of the dispersing medium but also on the liquid properties, liquid depth, agitation, and other features of the sparging environment; coalescence is strongly dependent on the concentration of surfactants capable of forming an electrical double layer and thus produce ionic bubbles, long-chain alcohols in water being excellent examples. For porous-carbon media, the manufacturer suggests that the best dispersion performance will result if the broken-line regions of Fig. 14-96 are avoided. For porous stainless-steel spargers, which extend to a lower pore size than carbon, Micro Metallic Division, Pall Corp., recommends (Release 120A, 1959) a working limit of 8 ft/min (0.044 m/s) to avoid serious coalescence. This agrees with the data reported by König et al. (loc. cit.), in which 0.08 m/s was used and bubbles as small as 1 mm were produced from a 5- μ m porous sparger.

Slabs of porous material are installed by grouting or welding together to form a diaphragm, usually horizontal. Tubes are prone to produce coalesced gas at rates high enough to cause bubbling from their lower faces, but they have the advantage of being demountable for cleaning or replacement (U.S. Patent 2,328,655). Roe [Sewage

Works J., **18**, 878 (1945)] claimed that silicon carbide tubes are superior to horizontal plates, principally because of the wiping action of the liquid circulating past the tube. He reported respective maximum capacities of 2.5 and 1.5 cm³/s of gas/cm² of sparger for a horizontal tube and a horizontal plate of the same material (unspecified grade). Mounting a flat-plate porous sparger vertically instead of horizontally seriously reduces the effectiveness of the sparger for three reasons: (1) The gas is distributed over a reduced cross section; (2) at normal rates, the lower portion of the sparger may not operate because of difference in hydrostatic head; and (3) there is a marked tendency for bubbles to coalesce along the sparger surface. Bone (M.S. thesis in chemical engineering, University of Kansas, 1948) found that the oxygen sulfite solution coefficient for a 3.2- by 10-cm rectangular porous carbon sparger was 26 to 41 percent lower for vertical than for horizontal operation of the sparger, the greatest reduction occurring when the long dimension was vertical.

Precipitation and Generation Methods For a thorough understanding of the phenomena involved, bubble nucleation should be considered. A discussion of nucleation phenomena is beyond the scope of this handbook; however, a starting point with recent references is Deng, Lee, and Cheng, *J. Micromech. Microeng.*, **15**, 564 (2005), and Jones, Evans, and Galvin, *Adv. Colloid and Interface Sci.*, **80**, 27 (1999).

Gas Dispersion—Vessel Headspace Boerma and Lankester have measured the surface aeration of a nine-bladed disk-type turbine (NOTE: A well-designed pitched-blade turbine will give equal or better performance). In a fully baffled vessel, the optimum depth to obtain maximum gas dispersion was 15 to 50 percent of the impeller diameter.

TABLE 14-23 Characteristics of Porous Septa

Grade	Avg. % porosity	Avg. pore diam.	Air-permeability data		
			Diaphragm thickness, in	Pressure differential, in water	Air flow, cu ft/(sq ft)(min)
Alundum porous alumina*					
P2220		25	1	2	0.35
P2120	36	60	1	2	2
P260	35	164	1	2	15
P236	34	240	1	2	40
P216		720	1	2	110
National porous carbon†					
60	48	33	1	2	
45	48	58	1	2	2
25	48	120	1	2	13
Filtros porous silica‡					
Extra fine	26.0	55	1.5	2	1-3
Fine	28.8	110	1.5	2	4-8
Medium fine	31.1	130	1.5	2	9-12
Medium	33.7	150	1.5	2	13-20
Medium coarse	33.8	200	1.5	2	21-30
Coarse	34.5	250	1.5	2	31-59
Extra coarse	36.5	300	1.5	2	60-100
Porous plastic§					
Teflon		9	0.125	1.38	5
Kel-F		15	0.125	1.38	13
Micro Metallic porous stainless steel¶, ¶					
H	45	5	0.125	1.38	1.8
G	50	10	0.125	1.38	3
F	50	20	0.125	1.38	5
E	50	35	0.125	1.38	18
D	50	65	0.125	1.38	60
C	55	165	0.125	27.7	990

*Data by courtesy of Norton Co., Worcester, Mass. A number of other grades between the extremes listed are available.

†Data by courtesy of National Carbon Co., Cleveland, Ohio.

‡Data by courtesy of Filtros Inc., East Rochester, N.Y.

§Data by courtesy of Pall Corp., Glen Cove, N.Y.

¶Similar septa made from other metals are available.

In a vessel with baffles extending only halfway to the liquid surface the optimum impeller submergence increased with agitator speed because of the vortex formed. At optimum depth, the following correlation is recommended for larger vessels:

$$Q = 0.00015(N/10)^{2.5}(D/0.1)^{4.5} \quad (14-214)$$

with Q in m^3/s , N in rps, and D in m. Thus, a typical 3-m-diameter plant-size vessel, with four-blade pitched impeller ($D = 1$ m) operating at 2 rps will give gas dispersion from the headspace into the batch of $Q = 0.00015(2/10)^{2.5}(2/0.1)^{4.5} = 1.9 \text{ m}^3/\text{s}$ (4000 ft^3/min).

Borema et al. (op. cit.) recommend headspace gas dispersion with partial baffling for fatty oils hydrogenation in stirred reactors: "The hydrogen in the head-space of the closed reactor can again be brought into contact with the liquid by a stirrer under the liquid level. . . ." And Penney in Paul et al. (loc. cit.) says, "partial baffling can be very effective to produce a vortex, which can effectively drawdown gas from the headspace. . . ." However, Middleton et al. in Paul et al. (op. cit.) say, "The simplest self-inducer for an agitated vessel is an impeller located near the surface, sometimes with the upper part of the baffles removed so as to encourage the formation of a vortex. This is, however, a sensitive and unstable arrangement. It is better, although probably more expensive, to use a self-inducing impeller system in which gas is drawn down a hollow shaft to the low-pressure region behind the blades of a suitable, often shrouded impeller. . . . Various proprietary designs are available, such as the Ekato gasjet, Prasair AGR and the Frings Friborator. . . ." Figure 14-97 illustrates a gas-inducing hollow shaft/hollow impeller agitator. In many hydrogenation reactors, the impeller just

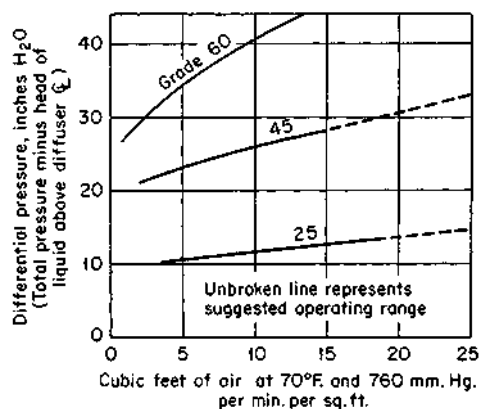


FIG. 14-96 Pressure drop across porous-carbon diffusers submerged in water at 70°F. To convert feet per minute to meters per second, multiply by 0.0051; to convert inches to millimeters, multiply by 25.4; $^{\circ}\text{C} = \frac{5}{9} (^{\circ}\text{F} - 32)$. (National Carbon Co.)

underneath the free surface has, without any doubt, performed admirably; consequently, one must consider this fact very carefully before using a self-inducing impeller.

Gas dispersion through the free surface by mechanical aerators is commonplace in aerobic waste-treatment lagoons. Surface aerators are generally of three types: (1) large-diameter flow-speed turbines operating just below the free surface of the liquid, often pontoon-mounted; (2) small-diameter high-speed (normally motor-speed) propellers operating in draft tubes, the units of which are always pontoon-mounted; and (3) hollow-tube turbines (Fig. 14-97). An example of the turbine type is illustrated in Fig. 14-98 and the propeller type is illustrated in Fig. 14-99. There are several other styles of the turbine type; for instance, Mixing Equipment Co., Inc., (www.light-ninmixers.com), uses an unshrouded 45° axial-flow turbine [see Dykman and Michel, *Chem. Eng.*, **117** (Mar. 10, 1969)], and Infilco (www.infilcoegremont.com) makes a unit that has a large-diameter vaned disk operating just below the free surface with a smaller-diameter submerged-disk turbine for additional solids suspension.

Aeration injectors, like the one shown in Fig. 14-100 by Penberthy [a division of Houdialle Industries (penberthy-online.com/jet1.asp)], are used to provide mass transfer in gas-liquid applications, and simple impingement aerators (Fig. 14-101) are sometimes used for mass-transfer applications.

Equipment Selection Ideally, selection of equipment to produce a gas-in-liquid dispersion should be made on the basis of a complete economic analysis. The design engineer and especially the pilot-plant engineer seldom have sufficient information or time to do a complete economic analysis. In the following discussion, some

TABLE 14-24 Wet Permeability of Alundum Porous Plates 1 in Thick*

Dry permeability at 2 in of water differential, cu ft/(min)(sq ft)	Pressure differential across wet plate, in of water	Air flow through wet plate, cu ft/(min)(sq ft)
4.3	20.67	2.0
	21.77	3.0
	22.86	4.0
	23.90	5.0
55.0	4.02	1.0
	4.14	2.0
	4.22	3.0
	4.27	4.0
	4.30	5.0

*Data by courtesy of Norton Company, Worcester, Mass. To convert inches to centimeters, multiply by 2.54; to convert feet per minute to meters per second, multiply by 0.0051.

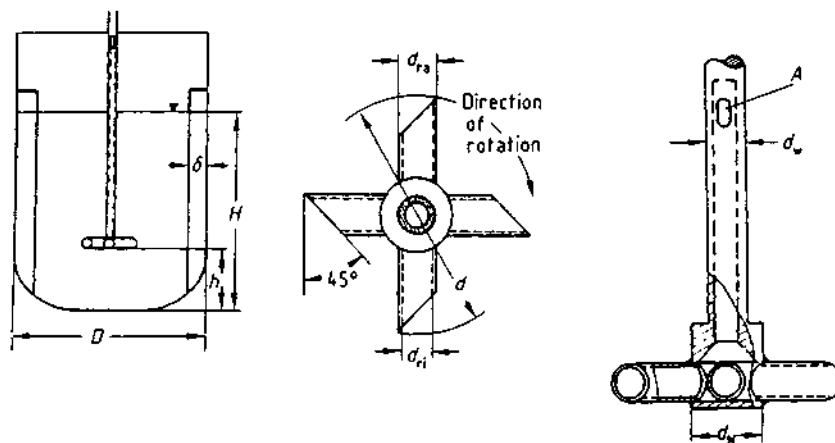


FIG. 14-97 Installation and dimensions of a tube stirrer: $h/d = 1$; $H/D = 1$; $D/\delta = 10$; $A = 1.5 d_w^2$; $D/d_N = 10$; $d/d_N = 3$; $d/d_m = 7.5$; $d/d_m = 6$. [Zlokarnik, Ullman's Encyclopedia of Industrial Chemistry, Sec. 25, VCH, Weinheim, Germany, 1988.]

guidelines are given as to what equipment might be feasible and what equipment might prove most economical.

For producing foam for foam-separation processes, perforated-plate or porous-plate spargers are normally used. Mechanical agitators are often not effective in the light foams needed in foam fractionation. Dissolved-air flotation, based on the release of a pressurized flow in which oxygen was dissolved, has been shown to be effective for particulate removal when sparged air failed because the bubbles formed upon precipitation are smaller—down to $80 \mu\text{m}$ —than bubbles possible with sparging, typically $1000 \mu\text{m}$ [Grieves and Ettelt, *AIChE J.*, **13**, 1167 (1967)]. Mechanically agitated surface aerators such as the Wemco-Fagergren flotation unit (Fig. 14-102) are used extensively for ore flotation.

To produce foam in batch processes, mechanical agitators are used almost exclusively. The gas can either be introduced through the free surface by the entraining action of the impeller or alternatively sparged beneath the impeller. In such batch operation, the liquid level gradually rises as the foam is generated; thus, squatly impellers such as turbines are rapidly covered with foam and must almost always be sparged from below. Tall impellers such as wire whips (Fig. 14-103) are especially well suited to entrain gas from the vapor space. Inter-meshing wire whips are standard kitchen utensils for producing foamed meringues, consisting of air, vegetable oil, and egg whites. For a new application, generally some experimentation with different impellers is necessary in order to get the desired fine final bubble size without getting frothing over initially. For producing foams continually, an aspirating venturi nozzle and restrictions in pipes such as baffles and metal gauzes are generally most economical.

For gas absorption, the equipment possibilities are generally packed columns; plate distillation towers, possibly with mechanical agitation on every plate; deep-bed contactors (bubble columns or sparged lagoons); and mechanically agitated vessels or lagoons. Packed towers and plate distillation columns are discussed elsewhere. Generally these

devices are used when a relatively large number of stages (more than two or three) is required to achieve the desired result practically.

The volumetric mass-transfer coefficients and heights of transfer units for bubble columns and packed towers have been compared for absorption of CO_2 into water by Houghton et al. [*Chem. Eng. Sci.*, **7**, 26 (1957)]. The bubble column will tolerate much higher vapor velocities, and in the overlapping region (superficial gas velocities of 0.9 to 1.8 cm/s), the bubble column has about three times higher mass-transfer coefficient and about 3 times greater height of transfer unit. The liquid in a bubble column is, for practical purposes, quite well mixed; thus, chemical reactions and component separations requiring significant plug flow of the liquid cannot be effected with bubble columns. Bubble columns and agitated vessels are the ideal equipment for processes in which the fraction of gas absorbed need not be great, possibly the gas can be recycled, and the liquid can or should be well mixed. The gas phase in bubble columns is not nearly so well back-mixed as the liquid, and often plug flow of the gas is a logical assumption, but in agitated vessels the gas phase is also well mixed.

The choice of a bubble column or an agitated vessel depends primarily on the solubility of the gas in the liquid, the corrosiveness of the liquid (often a gas compressor can be made of inexpensive material, whereas a mechanical agitator may have to be made of exotic, expensive materials), and the rate of chemical reaction as compared with the mass-transfer rate. Bubble columns and agitated vessels are seldom used for gas absorption except in chemical reactors. As a general rule, if the overall reaction rate is five times greater than the mass-transfer rate in a simple bubble column, a mechanical agitator will be most economical unless the mechanical agitator would have to be made from considerably more expensive material than the gas compressor.

In bubble columns and simply sparged lagoons, selecting the sparger is a very important consideration. In the turbulent regime (superficial gas velocity greater than 4.6 to 6 cm/s), inexpensive

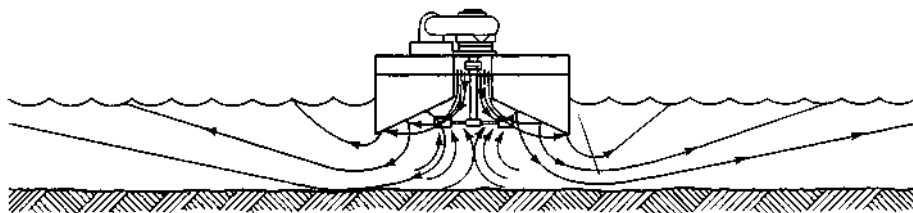


FIG. 14-98 The Cyclox surface aerator. (Cleveland Mixer Co.)

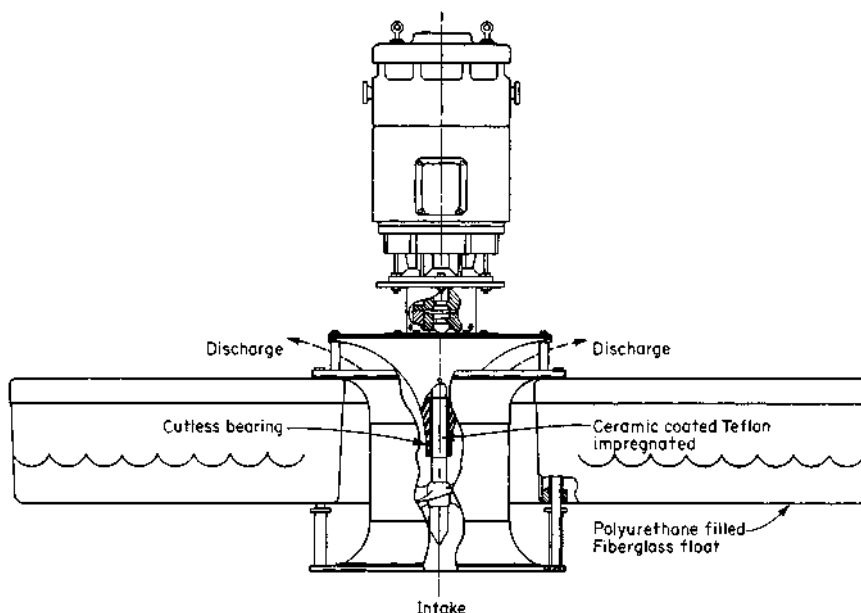


FIG. 14-99 Propeller-type surface aerator. (Ashbrook-Simon-Hartley Corp.)

perforated-pipe spargers should be used. Often the holes must be placed on the pipe bottom in order to make the sparger free-draining during operation. In the quiescent regime, porous septa will often give considerably higher overall mass-transfer coefficients than perforated plates or pipes because of the formation of tiny bubbles that do not coalesce. Chain and coworkers (*First International Symposium on Chemical Microbiology*, World Health Organization, Monograph Ser. 10, Geneva, 1952) claimed that porous disks are about twice as effective as open-pipe and ring spargers for the air oxidation of sodium sulfite. Eckenfelder [*Chem. Eng. Progr.*, **52**(7), 290 (1956)] has compared the oxygen-transfer capabilities of various devices on the basis of the operating power required to absorb a given quantity of O_2 . The installed cost of the various pieces of equipment probably would not vary sufficiently to warrant being included in an economic analysis. Surface mechanical aerators are not included in this comparison. Of the units compared, it appears that porous tubes give the most efficient power usage. Kalinske (*Adv. Biol. Waste Treatment*, 1963, p. 157) has compared submerged sparged aerators with mechanical surface aerators. He has summarized this comparison in *Water Sewage Works*, **33** (January 1968). He indicates that surface aerators are significantly more efficient than subsurface aeration, both for oxygen absorption and for gas-stripping operations.

Zlokarnik and Mann (paper at Mixing Conf., Rindge, New Hampshire, August 1975) have found the opposite of Kalinske, i.e., subsurface diffusers, subsurface sparged turbines, and surface aerators compare approximately as 4:2:1 respectively in terms of O_2 transfer efficiency; however, Zlokarnik [*Adv. Biochem. Eng.*, **11**, 157 (1979)] later indicates that the scale-up correlation used earlier might be somewhat inaccurate. When all available information is considered, it appears that with near-optimum design any of the aeration systems (diffusers, submerged turbines, or surface impellers) should give a transfer efficiency of at least 2.25 kg O_2 /kWh. Thus, the final selection should probably be made primarily on the basis of operational reliability, maintenance, and capital costs.

Mass Transfer Mass transfer in plate and packed gas-liquid contactors has been covered earlier in this subsection. Attention here will be limited to deep-bed contactors (bubble columns and agitated vessels). Theory underlying mass transfer between phases is discussed in Sec. 5 of this handbook.

To design deep-bed contactors for mass-transfer operations, one must have, in general, predictive methods for the following design parameters:

- Flooding (for both columns and agitator impellers)
- Agitator power requirements

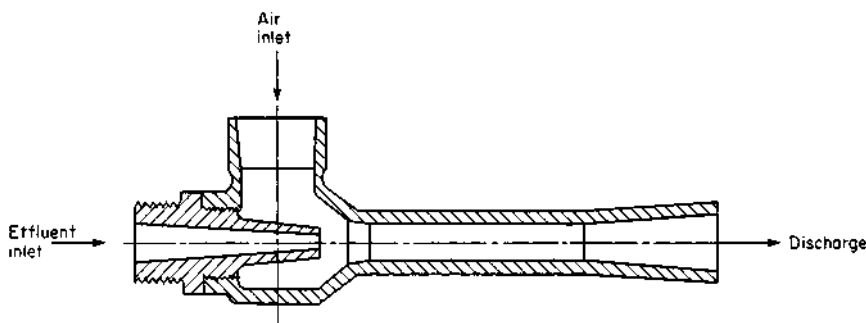


FIG. 14-100 Aeration ejector. (Penberthy, a division of Houdaille Industries, Inc.)

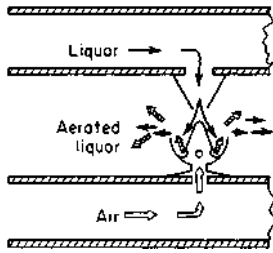


FIG. 14-101 Impingement aerator.

- Gas-phase and liquid-phase mass-transfer coefficients
- Interfacial area
- Interface resistance
- Mean concentration driving force for mass transfer

In most cases, available methods are incomplete or unreliable, and some supporting experimental work is necessary, followed by scale-up. The methods given here should allow theoretical feasibility studies, help minimize experimentation, and permit a measure of optimization in final design.

Flooding of Agitator Impellers Impeller flooding correlations for six-blade disk (6BD) Rushton turbines and six-blade disk Smith turbines (Chemineer designation: CD-6) are presented by Bakker, Myers, and Smith [*Chem. Eng.*, **101**, 98 (Dec. 1994)] and a review of impeller flooding. The Bakker et al. (loc. cit.) correlation is

$$Q/ND^3 = C_{FL} N_{Fr} (D/T)^{3.5} \quad (14-215)$$

where $C_{FL} = 30$ for a 6BD impeller and $C_{FL} = 70$ for a concave blade CD-6 impeller and $N_{Fr} = \text{Froude number} = N^2 D/g$; $Q = \text{gas flow rate at flooding, m}^3/\text{s}$; $N = \text{impeller speed, rps}$; $D = \text{impeller diameter, m}$; and $T = \text{tank diameter, m}$. Note that the CD-6 impeller will handle $70/30 = 2.33$ times the gas a 6BD will handle, without flooding, at the

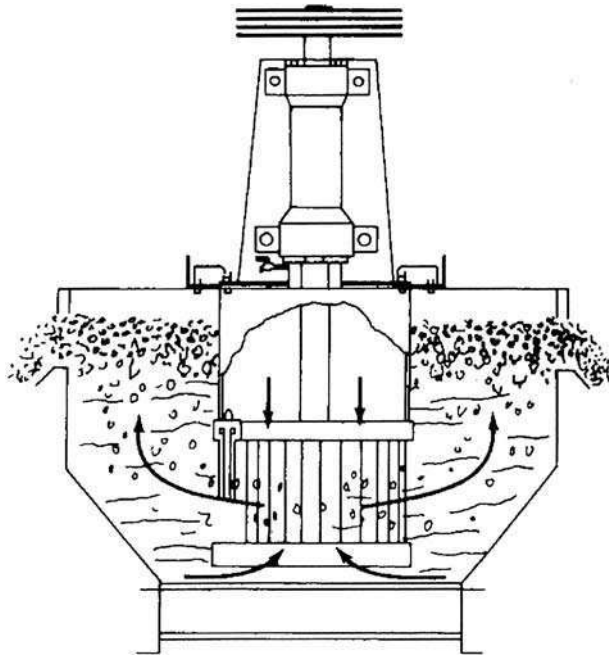
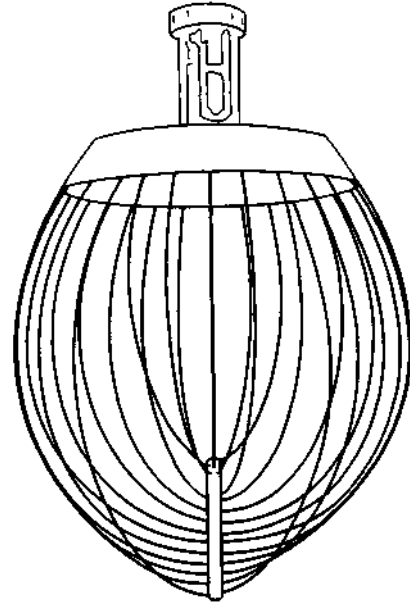

 FIG. 14-102 The Wemco-Fagergren flotation machine. [From [www.tucottbus.de/BTU/Fak4/Aufbtech/pages/pbrr_N7_Sep2-\(elstat-flotat\).pdf](http://www.tucottbus.de/BTU/Fak4/Aufbtech/pages/pbrr_N7_Sep2-(elstat-flotat).pdf)].


FIG. 14-103 Wire whip.

same N and D ; this is the great advantage of the CD-6 along with lower power decrease as the gas flow rate increases.

Gassed Impeller Power Bakker et al. (op. cit.) have given a gassed power correlation for the 6BD and CD-6 impellers.

$$P_g/P_u = [1 - (b - a\mu)] N_{Fr}^d \tanh(cN_A) \quad (14-216a)$$

where $P_g = \text{gassed power, W}$; $P_u = \text{ungassed power, W}$; $N_A = Q/ND^3$; and the constants of Eq. (14-216a) are given in Table 14-25.

As mentioned previously, the CD-6 suffers much less power decrease with increased gassing compared to the 6BD. For example, at $N_A = 0.15$, $P_g/P_u = 0.7$ for the CD-6 and 0.5 for the 6BD.

The ungassed power is calculated by

$$P_u = N_p \rho N^3 D^5 \quad (14-216b)$$

where the impeller power numbers N_p are given Table 14-25.

Bakker et al. (op. cit.) and Sensel et al. (op. cit.) have given correlations for gas holdup in agitated vessels. The Bakker et al. correlation is

$$\epsilon = C_e = (P_g/V)^A v_{sg}^B \quad (14-217)$$

where $C_e = 0.16$, $A = 0.33$, $B = 0.67$; $V = \text{batch volume, m}^3$; $v_{sg} = \text{superficial gas velocity} = Q/[(\pi/4)T^2]$; $T = \text{tank diameter, m}$. Equation (14-217) applies for both 6BD and CD-6.

Interfacial Area This consideration in agitated vessels has been reviewed and summarized by Tattersson (op. cit.). Predictive methods for interfacial area are not presented here because correlations are given for the overall volumetric mass transfer coefficient liquid phase controlling mass transfer.

TABLE 14-25 Constants in Eq. (14-216) and Impeller Power Numbers

Impeller type	a	b	c	d	N_p
6BD	0.72	0.72	24	0.25	3.2
CD-6	0.12	0.44	12	0.37	5.5

Overall Mass-Transfer Coefficient In systems with relatively sparingly soluble gases, where the gas-phase resistance is negligible, the mass-transfer rate can be determined by using the concept of an overall volumetric mass-transfer coefficient $k_L a$ as follows:

$$M_s = k_L a (C_s^* - C_{s,b}) \quad (14-218)$$

where M_s = solute molar mass-transfer rate, kg-mol/s; $k_L a$ = overall mass-transfer coefficient, 1/s; C_s^* = solute concentration in equilibrium with the liquid phase, kg-mol/s; and $C_{s,b}$ = solute concentration in bulk of liquid.

Bakker et al. (op. cit.) have given a correlation for $k_L a$ for aqueous systems in the absence of significant surface active agents.

$$k_L a = C_{kLa} (P_g/V)^n v_{sg}^b \quad (14-219)$$

where $C_{kLa} = 0.015$, 1/s; Eq. (14-219) applies for both 6BD and CD-6.

Interfacial Phenomena These can significantly affect overall mass transfer. Deckwer, *Bubble Column Reactors*, Wiley, Hoboken, N.J., 1992, has covered the effect of surfactants on mass transfer in bubble columns. In fermentation reactors, small quantities of surface-active agents (especially antifoaming agents) can drastically reduce overall oxygen transfer (Aiba et al., op. cit., pp. 153, 154), and in aerobic mechanically aerated waste-treatment lagoons, overall oxygen transfer has been found to be from 0.5 to 3 times that for pure water from tests with typical sewage streams (Eckenfelder et al., op. cit., p. 105).

One cannot quantitatively predict the effect of the various interfacial phenomena; thus, these phenomena will not be covered in detail here. The following literature gives a good general review of the effects of interfacial phenomena on mass transfer: Goodridge and Robb, *Ind. Eng. Chem. Fund.*, **4**, 49 (1965); Calderbank, *Chem. Eng. (London)*, CE 205 (1967); Gal-Or et al., *Ind. Eng. Chem.*, **61**(2), 22 (1969); Kintner, *Adv. Chem. Eng.*, **4** (1963); Resnick and Gal-Or, op. cit., p. 295; Valentin, loc. cit.; and Elenkov, loc. cit., and *Ind. Eng. Chem. Ann. Rev. Mass Transfer*, **60**(1), 67 (1968); **60**(12), 53 (1968); **62**(2), 41 (1970). In the following outline, the effects of the various interfacial phenomena on the factors that influence overall mass transfer are given. Possible effects of interfacial phenomena are tabulated below:

1. Effect on continuous-phase mass-transfer coefficient
 - a. Impurities concentrate at interface. Bubble motion produces circumferential surface-tension gradients that act to retard circulation and vibration, thereby decreasing the mass-transfer coefficient.
 - b. Large concentration gradients and large heat effects (very soluble gases) can cause interfacial turbulence (the Marangoni effect), which increases the mass-transfer coefficient.
2. Effect on interfacial area
 - a. Impurities will lower static surface tension and give smaller bubbles.
 - b. Surfactants can electrically charge the bubble surface (produce ionic bubbles) and retard coalescence (soap stabilization of an oil-water emulsion is an excellent example of this phenomenon), thereby increasing the interfacial area.
 - c. Large concentration gradients and large heat effects can cause bubble breakup.
3. Effect on mean mass-transfer driving force
 - a. Relatively insoluble impurities concentrate at the interface, giving an interfacial resistance. This phenomenon has been used in retarding evaporation from water reservoirs.
 - b. The axial concentration variation can be changed by changes in coalescence. The mean driving force for mass transfer is therefore changed.

Gas Holdup (ϵ) in Bubble Columns With coalescing systems, holdup may be estimated from a correlation by Hughmark [*Ind. Eng. Chem. Process Des. Dev.*, **6**, 218–220 (1967)] reproduced here as Fig. 14-104. For noncoalescing systems, with considerably smaller bubbles, ϵ can be as great as 0.6 at $U_{sg} = 0.05$ m/s, according to Mersmann [*Ger. Chem. Eng.*, **1**, 1 (1978)].

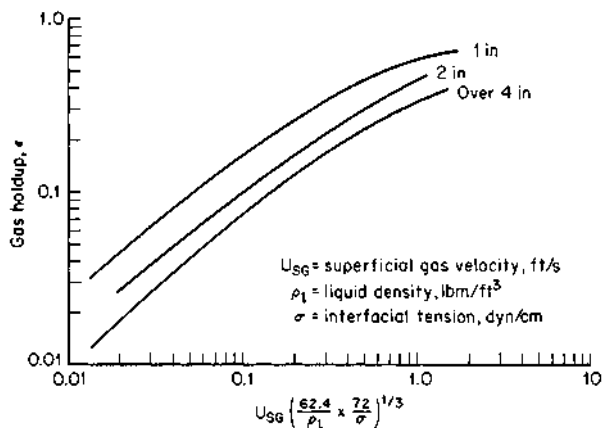


FIG. 14-104 Gas holdup correlation. [*Ind. Eng. Chem. Process Des. Dev.*, **6**, 218 (1967).]

It is often helpful to use the relationship between ϵ and superficial gas velocity (U_{sg}) and the rise velocity of a gas bubble relative to the liquid velocity ($U_r + U_L$, with U_L defined as positive upward):

$$\epsilon = \frac{U_{sg}}{U_r + U_L} \quad (14-220)$$

Rise velocities of bubbles through liquids have been discussed previously.

For a better understanding of the interactions between parameters, it is often helpful to calculate the effective bubble rise velocity U_r from measured values of ϵ ; for example, the data of Mersmann (loc. cit.) indicated $\epsilon = 0.6$ for $U_{sg} = 0.05$ m/s, giving $U_r = 0.083$ m/s, which agrees with the data reported in Fig. 14-43 for the rise velocity of bubble clouds. The rise velocity of single bubbles, for $d_b \sim 2$ mm, is about 0.3 m/s, for liquids with viscosities not too different from water. Using this value in Eq. (14-220) and comparing with Fig. 14-104, one finds that at low values of U_{sg} , the rise velocity of the bubbles is less than the rise velocity of a single bubble, especially for small-diameter tubes, but that the opposite occurs for large values of U_{sg} .

More recent literature regarding generalized correlational efforts for gas holdup is adequately reviewed by Tsuchiya and Nakanishi [*Chem. Eng. Sci.*, **47**(13/14), 3347 (1992)] and Sotelo et al. [*Int. Chem. Eng.*, **34**(1), 82–90 (1994)]. Sotelo et al. (op. cit.) have developed a dimensionless correlation for gas holdup that includes the effect of gas and liquid viscosity and density, interfacial tension, and diffuser pore diameter. For systems that deviate significantly from the waterlike liquids for which Fig. 14-104 is applicable, their correlation (the fourth numbered equation in the paper) should be used to obtain a more accurate estimate of gas holdup. Mersmann (op. cit.) and Deckwer et al. (op. cit.) should also be consulted.

Liquid-phase mass-transfer coefficients in bubble columns have been reviewed by Calderbank ("Mixing," loc. cit.), Fair (*Chem. Eng.*, loc. cit.), Mersmann [*Ger. Chem. Eng.*, **1**, 1 (1978), *Int. Chem. Eng.*, **32**(3) 397–405 (1991)], Deckwer et al. [*Can. J. Chem. Eng.*, **58**, 190 (1980)], Hikita et al. [*Chem. Eng. J.*, **22**, 61 (1981)] and Deckwer and Schumpe [*Chem. Eng. Sci.*, **48**(5), 889–911 (1993)]. The correlation of Ozturk, Schumpe, and Deckwer [*AIChE J.*, **33**, 1473–1480 (1987)] is recommended. Deckwer et al. (op. cit.) have documented the case for using the correlation:

Ozturk et al. (1987) developed a new correlation on the basis of a modification of the Akita-Yoshida correlation suggested by Nakanoh and Yoshida (1980). In addition, the bubble diameter d_b rather than the column diameter was used as the characteristic length as the column diameter has little influence on $k_L a$. The value of d_b was assumed to be approximately constant ($d_b = 0.003$ m). The correlation was obtained by nonlinear regression is as follows:

$$\left(\frac{k_L a d_B}{D_L}\right) = 0.62 \left[\frac{\mu_L}{(\rho_L D_L)} \right]^{0.5} \left(\frac{g \rho_L d_B^2}{\sigma} \right)^{0.33} \left(\frac{g \rho_L^2 d_B}{\mu_L^2} \right)^{0.29} \times \left[\frac{U_G}{(g d_B)^{0.5}} \right]^{0.68} \left(\frac{\rho_G}{\rho_L} \right)^{0.04} \quad (14-221)$$

where $k_L a$ = overall mass-transfer coefficient, d_B = bubble diameter = 0.003 m, D_L = diffusivity of gas in liquid, ρ = density, μ = viscosity, σ = interfacial tension, g = gravitational acceleration.

As mentioned earlier, surfactants and ionic solutions significantly affect mass transfer. Normally, surface effects act to retard coalescence and thus increase the mass transfer. For example, Hikata *et al.* [*Chem. Eng. J.*, **22**, 61–69 (1981)] have studied the effect of KCl on mass transfer in water. As KCl concentration increased, the mass transfer increased up to about 35 percent at an ionic strength of 6 gm/l. Other investigators have found similar increases for liquid mixtures.

Axial Dispersion Backmixing in bubble columns has been extensively studied. Wiemann and Mewes [*Ind. Eng. Chem. Res.*, **44**, 4959 (2005)] and Wild *et al.* [*Int. J. Chemical Reactor Eng.*, **1**, R7 (2003)] give a long list of references pertaining to backmixing in bubble columns. An excellent review article by Shah *et al.* [*AIChE J.*, **24**, 369 (1978)] has summarized the literature prior to 1978. Works by König *et al.* [*Ger. Chem. Eng.*, **1**, 199 (1978)], Lucke *et al.* [*Trans. Inst. Chem. Eng.*, **58**, 228 (1980)], Riquarts [*Ger. Chem. Eng.*, **4**, 18 (1981)], Mersmann (*op. cit.*), Deckwer (*op. cit.*), Yang *et al.* [*Chem. Eng. Sci.*, **47**(9–11), 2859 (1992)], and Garcia-Calvo and Leton [*Chem. Eng. Sci.*, **49**(21), 3643 (1994)] are particularly useful references.

Axial dispersion occurs in both the liquid and the gas phases. The degree of axial dispersion is affected by vessel diameter, vessel internals, gas superficial velocity, and surface-active agents that retard coalescence. For systems with coalescence-retarding surfactants the initial bubble size produced by the gas sparger is also significant. The gas and liquid physical properties have only a slight effect on the degree of axial dispersion, except that liquid viscosity becomes important as the flow regime becomes laminar. With pure liquids, in the absence of coalescence-inhibiting, surface-active agents, the nature of the sparger has little effect on the axial dispersion, and experimental results are reasonably well correlated by the dispersion model. For the liquid phase the correlation recommended by Deckwer *et al.* (*op. cit.*), after the original work by Baird and Rice [*Chem. Eng. J.*, **9**, 171(1975)] is as follows:

$$\frac{E_L}{(DU_G)} = 0.35 \left(\frac{gD}{U_G^2} \right)^{1/3} \quad (14-222)$$

where E_L = liquid-phase axial dispersion coefficient, U_G = superficial velocity of the gas phase, D = vessel diameter, and g = gravitational acceleration.

The recommended correlation for the gas-phase axial-dispersion coefficient is given by Field and Davidson (*loc. cit.*):

$$E_G = 56.4 D^{1.33} \left(\frac{U_G}{\epsilon} \right)^{3.56} \quad (14-223)$$

where E_G = gas-phase axial-dispersion coefficient, m^2/s ; D = vessel diameter, m; U_G = superficial gas velocity, m/s; and ϵ = fractional gas holdup, volume fraction.

The correlations given in the preceding paragraphs are applicable to vertical cylindrical vessels with pure liquids without coalescence inhibitors. For other vessel geometries such as columns of rectangular cross section, packed columns, and coiled tubes, the work of Shah *et al.* (*loc. cit.*) should be consulted. For systems containing coalescence-inhibiting surfactants, axial dispersion can be vastly different from that in systems in which coalescence is negligible. König *et al.* (*loc. cit.*) have well demonstrated the effects of surfactants and sparger type by conducting tests with weak alcohol solutions using three different porous spargers. With pure water, the sparger—and, consequently, initial bubble size—had little effect on back mixing because coalescence produced a dynamic-equilibrium bubble size not far above the sparger. With surfactants, the average bubble size was smaller than the dynamic-equilibrium bubble size. Small bubbles produced minimal back mixing up to $\epsilon \approx 0.40$; however, above $\epsilon \approx 0.40$ backmixing increased very rapidly as U_G increased. The rapid increase in back mixing as ϵ exceeds 0.40 was postulated to occur indirectly because a bubble carries upward with it a volume of liquid equal to about 70 percent of the bubble volume, and, for $\epsilon \approx 0.40$, the bubbles carry so much liquid upward that steady, uniform bubble rise can no longer be maintained and an oscillating, slugging flow develops, which produces fluctuating pressure at the gas distributor and the formation of large eddies. The large eddies greatly increase backmixing. For the air alcohol-water system, the minimum bubble size to prevent unsteady conditions was about 1, 1.5, and 2 mm for $U_G = 1, 3$, and 5 cm/s, respectively. Any smaller bubble size produced increased backmixing. The results of König *et al.* (*loc. cit.*) clearly indicate that the interaction of surfactants and sparger can be very complex; thus, one should proceed very cautiously in designing systems for which surfactants significantly retard coalescence. Caution is particularly important because surfactants can produce either much more or much less backmixing than surfactant-free systems, depending on the bubble size, which, in turn, depends on the sparger utilized.

PHASE SEPARATION

Gases and liquids may be intentionally contacted as in absorption and distillation, or a mixture of phases may occur unintentionally as in vapor condensation from inadvertent cooling or liquid entrainment from a film. Regardless of the origin, it is usually desirable or necessary ultimately to separate gas-liquid dispersions. While separation will usually occur naturally, the rate is often economically intolerable and separation processes are employed to accelerate the step.

GAS-PHASE CONTINUOUS SYSTEMS

Practical separation techniques for liquid particles in gases are discussed. Since gas-borne particulates include both liquid and solid particles, many devices used for dry-dust collection (discussed in Sec. 17 under "Gas-Solids Separation") can be adapted to liquid-particle separation. Also, the basic subject of particle mechanics is covered in Sec. 6. Separation of liquid particulates is frequently desirable in chemical processes such as in countercurrent-stage contacting because liquid entrainment with the gas partially reduces true countercurrency. Sep-

aration before entering another process step may be needed to prevent corrosion, to prevent yield loss, or to prevent equipment damage or malfunction. Separation before the atmospheric release of gases may be necessary to prevent environmental problems and for regulatory compliance.

GENERAL REFERENCES

- G-1. Buonicore and Davis, eds., *Air Pollution Engineering Manual*, Van Nostrand Reinhold, New York, 1992.
- G-2. Calvert and Englund, eds., *Handbook of Air Pollution Technology*, Wiley, New York, 1984.
- G-3. Cheremisinoff, ed., *Encyclopedia of Environmental Control Technology*, vol. 2, Gulf Pub., Houston, 1989.
- G-4. McKetta, *Unit Operations Handbook*, vol. 1–2, Dekker, New York, 1992.
- G-5. Wark and Warner, *Air Pollution: Its Origin and Control*, 2d ed., Harper & Row, New York, 1981.
- G-6. Hesketh, *Air Pollution Control*, 1979; *Fine Particles in Gaseous Media*, Ann Arbor Science Pubs., Ann Arbor, MI, 1977.
- G-7. Stern, *Air Pollution*, 3d ed., vols. 3–5, Academic, Orlando, FL, 1976–77.

- G-8. Strauss, *Industrial Gas Cleaning*, 2d ed., Pergamon, New York, 1975.
 G-9. Theodore and Buonicore, *Air Pollution Control Equipment; Selection, Design, Operation and Maintenance*, Prentice Hall, Englewood Cliffs, NJ, 1982.
 G-10. Wang and Pereira, eds., *Handbook of Environmental Engineering*, vol. 1, Humana, Clifton, NJ 1979.
 G-11. Chermisinoff and Young, *Air Pollution Control and Design Handbook*, parts 1-2, Dekker, New York, 1977.
 G-12. Nonhebel, *Gas Purification Processes for Air Pollution Control*, Newnes-Butterworth, London, 1972.

Sampling

- R-1. *Code of Federal Regulations*, 40 (CFR 40), subchapter C—Air Programs, parts 50–99, Office of the Federal Register, Washington.
 R-2. Ref. G-11, part 1, pp. 65–121.
 R-3. Cooper and Rossano, *Source Testing for Air Pollution Control*, Environmental Science Services, Wilton, Connecticut, 1970.
 R-4. Ref. G-7, vol. 3, pp. 525–587.
 R-5. *Methods of Air Sampling and Analysis*, 2d Ed., American Public Health Assoc., Washington, 1977.
 R-6. Stockham and Fochtman, *Particle Size Analysis*, Ann Arbor Science Pubs., Ann Arbor, Michigan, 1977.
 R-7. Ref. G-2, Ch. 31, pp. 785–832.
 R-8. Ref. G-8, Ch. 2, pp. 39–79.

Specific

- R-9. Calvert, Goldchmid, Leith, and Mehta, NTIS Publ. PB-213016, 213017, 1972.
 R-10. Calvert, *J. Air Pollut. Control Assoc.*, **24**, 929 (1974).
 R-11. Calvert, *Chem. Eng.*, **54**(18), 54 (1977).
 R-12. Calvert, Yung, and Leung, NTIS Publ. PB-248050, 1975.
 R-13. Calvert and Lundgren, *J. Air Pollut. Control Assoc.*, **18**, 677 (1968).
 R-14. Calvert, Lundgren, and Mehta, *J. Air Pollut. Control Assoc.*, **22**, 529 (1972).
 R-15. Yung, Barbarika, and Calvert, *J. Air Pollut. Control Assoc.*, **27**, 348, (1977).
 R-16. Katz, M.S. thesis, Pennsylvania State University, 1958.
 R-17. York and Poppele, *Chem. Eng. Prog.*, **59**(6), 45 (1963).
 R-18. York, *Chem. Eng. Prog.*, **50**, 421 (1954).
 R-19. Ref. G-2, Ch. 10, pp. 215–248.
 References with the notation (R-) are cited in the text.

Definitions: Mist and Spray Little standardization has been adopted in defining gas-borne liquid particles, and this frequently leads to confusion in the selection, design, and operation of collection equipment. Aerosol applies to suspended particulate, either solid or liquid, which is slow to settle by gravity and to particles from the sub-micrometer range up to 10 to 20 μm . Mists are fine suspended liquid dispersions usually resulting from condensation and ranging upward in particle size from around 0.1 μm . Spray refers to entrained liquid droplets. The droplets may be entrained from atomizing processes previously discussed under "Liquid-in-Gas Dispersions" in this section. In such instances, size will range from the finest particles produced up to a particle whose terminal settling velocity is equal to the entraining gas velocity if some settling volume is provided. Process spray is often created unintentionally, such as by the condensation of vapors on cold duct walls and its subsequent reentrainment, or from two-phase flow in pipes, gas bubbling through liquids, and entrainment from boiling liquids. Entrainment size distribution from sieve trays has been given by Cheng and Teller [*Am. Inst. Chem. Eng. J.*, **7**(2), 282 (1961)] and evaporator spray by Garner et al. [*Trans. Inst. Chem. Eng.*, **32**, 222 (1954)]. In general, spray can range downward in particle size from 5000 μm . There can be overlapping in size between the coarsest mist particles and the finest spray particles, but some authorities have found it convenient arbitrarily to set a boundary of 10 μm between the two. Actually, considerable overlap exists in the region of 5 to 40 μm . Table 14-26 lists typical ranges of particle size created by different mechanisms. The sizes actually entrained can be influenced by the local gas velocity. Figure 14-105 compares the approximate size range of liquid particles with other particulate material and the approximate applicable size range of collection devices. Figure 17-34 gives an expanded chart by Lapple for solid particles. Mist and fog formation has been discussed previously.

Gas Sampling The sampling of gases containing mists and sprays may be necessary to obtain data for collection-device design, in which case particle-size distribution, total mass loading, and gas volume,

TABLE 14-26 Particle Sizes Produced by Various Mechanisms

Mechanism or process	Particle-size range, μm
Liquid pressure spray nozzle	100–5000
Gas-atomizing spray nozzle	1–100
Gas bubbling through liquid or boiling liquid	20–1000
Condensation processes with fogging	0.1–30
Annular two-phase flow in pipe or duct	10–2000

temperature, pressure, and composition may all be needed. Other reasons for sampling may be to determine equipment performance, measure yield loss, or determine compliance with regulations.

Location of a sample probe in the process stream is critical especially when larger particles must be sampled. Mass loading in one portion of a duct may be severalfold greater than in another portion as affected by flow patterns. Therefore, the stream should be sampled at a number of points. The U.S. Environmental Protection Agency (R-1) has specified 8 points for ducts between 0.3 and 0.6 m (12 and 24 in) and 12 points for larger ducts, provided there are no flow disturbances for eight pipe diameters upstream and two downstream from the sampling point. When only particles smaller than 3 μm are to be sampled, location and number of sample points are less critical since such particles remain reasonably well dispersed by brownian motion. However, some gravity settling of such particles and even gases of high density have been observed in long horizontal breeching. Isokinetic sampling (velocity at the probe inlet is equal to local duct velocity) is required to get a representative sample of particles larger than 3 μm (error is small for 4- to 5- μm particles). Sampling methods and procedures for mass loading have been developed (R-1 through R-8).

Particle Size Analysis Many particle-size-analysis methods suitable for dry-dust measurement are unsuitable for liquids because of coalescence and drainage after collection. Measurement of particle sizes in the flowing aerosol stream by using a cascade impactor is one

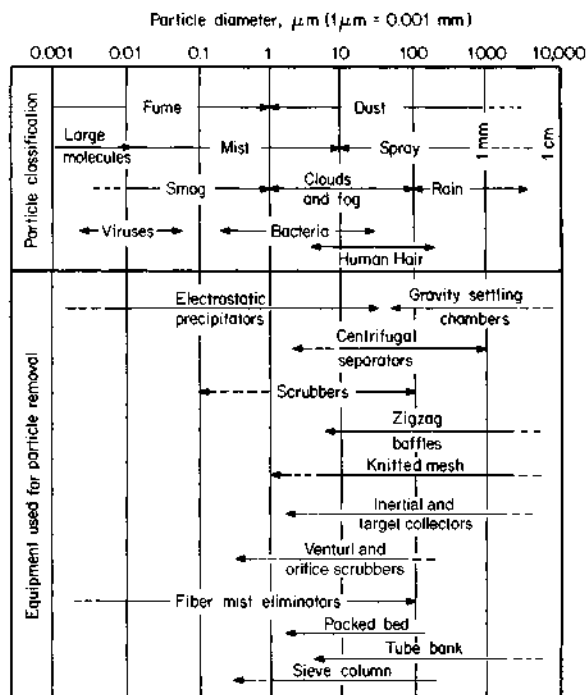


FIG. 14-105 Particle classification and useful collection equipment versus particle size.

of the better means. The impacting principle has been described by Ranz and Wong [Ind. Eng. Chem., **44**, 1371 (1952)] and Gillespie and Johnstone [Chem. Eng. Prog., **51**, 75F (1955)]. The Andersen, Sierra, and University of Washington impactors may be used if the sampling period is kept short so as not to saturate the collection substrate. An impactor designed specifically for collecting liquids has been described by Brink, Kennedy, and Yu [Am. Inst. Chem. Eng. Symp. Ser., **70**(137), 333 (1974)].

Collection Mechanisms Mechanisms which may be used for separating liquid particles from gases are (1) gravity settling, (2) inertial (including centrifugal) impaction, (3) flow-line interception, (4) diffusional (brownian) deposition, (5) electrostatic attraction, (6) thermal precipitation, (7) flux forces (thermophoresis, diffusiophoresis, Stefan flow), and (8) particle agglomeration (nucleation) techniques. Equations and parameters for these mechanisms are given in Table 17-2. Most collection devices rarely operate solely with a single mechanism, although one mechanism may so predominate that it may be referred to, for instance, as an inertial-impaction device.

After collection, liquid particles coalesce and must be drained from the unit, preferably without reentrainment. Calvert (R-12) has studied the mechanism of reentrainment in a number of liquid-particle collectors. Four types of reentrainment were typically observed: (1) transition from separated flow of gas and liquid to a two-phase region of separated-entrained flow, (2) rupture of bubbles, (3) liquid creep on the separator surface, and (4) shattering of liquid droplets and splashing. Generally, reentrainment increased with increasing gas velocity. Unfortunately, in devices collecting primarily by centrifugal and inertial impaction, primary collection efficiency increases with gas velocity; thus overall efficiency may go through a maximum as reentrainment overtakes the incremental increase in efficiency. Prediction of collection efficiency must consider both primary collection and reentrainment.

Procedures for Design and Selection of Collection Devices Calvert and coworkers (R-9 to R-12 and R-19) have suggested useful design and selection procedures for particulate-collection devices in which direct impingement and inertial impaction are the most significant mechanisms. The concept is based on the premise that the mass median aerodynamic particle diameter d_{p50} is a significant measure of the difficulty of collection of the liquid particles and that the collection device cut size d_{pc} (defined as the aerodynamic particle diameter collected with 50 percent efficiency) is a significant measure of the capability of the collection device. The aerodynamic diameter for a particle is the diameter of a spherical particle (with an arbitrarily assigned density of 1 g/cm³) which behaves in an air stream in the same fashion as the actual particle. For real spherical particles of diameter d_p , the equivalent aerodynamic diameter d_{pa} can be obtained from the equation $d_{pa} = dp(\rho_p C')^{1/2}$, where ρ_p is the apparent particle density (mass/volume) and C' is the Stokes-Cunningham correction factor for the particle size, all in consistent units. If particle diameters are expressed in micrometers, ρ_p can be in grams per cubic centimeter and C' can be approximated by $C' = 1 + A_c(2\lambda/d_p)$, where A_c is a constant dependent upon gas composition, temperature, and pressure ($A_c = 0.88$ for atmospheric air at 20°C) and λ is the mean free path of the gas molecules ($\lambda = 0.10 \mu\text{m}$ for 20°C atmospheric air). For other temperatures or pressures, or gases other than air, calculations using these more precise equations may be made: $A_c = 1.257 + 0.4 \exp[-1.1(d_p/2\lambda)]$ and $\lambda = \mu_g/0.499\rho_g \times \mu_m$ (where μ_g is the gas viscosity, kg/m-h; ρ_g is gas density, g/cm³; and μ_m is the mean molecular speed, m/s. $\mu_m = [8RT/\pi M]^{0.5}$, where R_u is the universal gas constant, 8.315 kJ/kg-mol-K; T is the gas absolute temperature, K; and M is the molar mass or equivalent molecular weight of the gas. (π is the usual geometric constant.) For test purposes (air at 25°C and 1 atm), $\rho_g = 1.183 \text{ kg/m}^3$, $\mu_g = 0.0666 \text{ kg/m-h}$, $\lambda = 0.067 \mu\text{m}$, and $u_m = 467 \text{ m/s}$. For airborne liquid particles, the assumption of spherical shape is reasonably accurate, and ρ_p is approximately unity for dilute aqueous particles at ambient temperatures. C' is approximately unity at ambient conditions for such particles larger than 1 to 5 μm , so that often the actual liquid particle diameter and the equivalent aerodynamic diameter are identical.

When a distribution of particle sizes which must be collected is present, the actual size distribution must be converted to a mass dis-

tribution by aerodynamic size. Frequently the distribution can be represented or approximated by a log-normal distribution (a straight line on a log-log plot of cumulative mass percent of particles versus diameter) which can be characterized by the mass median particle diameter d_{p50} and the standard statistical deviation of particles from the median σ_g . σ_g can be obtained from the log-log plot by $\sigma_g = D_{p950}/D_{p50}$ at 15.87 percent = D_{p50} at 84.13 percent/ D_{p50} .

The grade efficiency η of most collectors can be expressed as a function of the aerodynamic particle size in the form of an exponential equation. It is simpler to write the equation in terms of the particle penetration P_i (those particles not collected), where the fractional penetration $P_i = 1 - \eta$, when η is the fractional efficiency. The typical collection equation is

$$P_i = e^{(-A_i D_{p50} B)} \quad (14-224)$$

where A_i and B are functions of the collection device. Calvert (R-12) has determined that for many devices in which the primary collection mechanism is direct interception and inertial impaction, such as packed beds, knitted-mesh collectors, zigzag baffles, target collectors such as tube banks, sieve-plate columns, and venturi scrubbers, the value of B is approximately 2.0. For cyclonic collectors, the value of B is approximately 0.67. The overall integrated penetration $\bar{P}_i$ for a device handling a distribution of particle sizes can be obtained by

$$\bar{P}_i = \int_0^W \left(\frac{dW}{W} \right) P_i \quad (14-225)$$

where (dW/W) is the mass of particles in a given narrow size distribution and P_i is the average penetration for that size range. When the particles to be collected are log-normally distributed and the collection device efficiency can be expressed by Eq. (14-224), the required overall integrated collection efficiency $\bar{P}_i$ can be related to the ratio of the device aerodynamic cut size D_{pc} to the mass median aerodynamic particle size D_{p50} . This required ratio for a given distribution and collection is designated R_{iL} and these relationships are illustrated graphically in Fig. 14-106. For the many devices for which B is approximately 2.0, a simplified plot (Fig. 14-107) is obtained. From these figures, by knowing the desired overall collection efficiency and particle distribution, the value of R_{iL} can be read. Substituting the mass median particle diameter gives the aerodynamic cut size

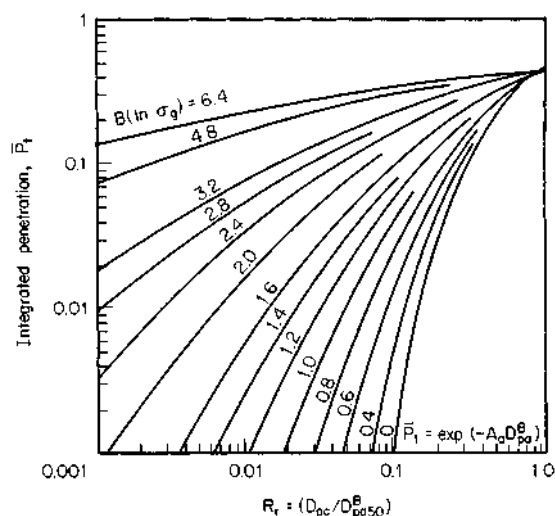


FIG. 14-106 Overall integrated penetration as a function of particle-size distribution and collector parameters. (Calvert, Yung, and Leung, NTIS Publ. PB-248050, 1975.)

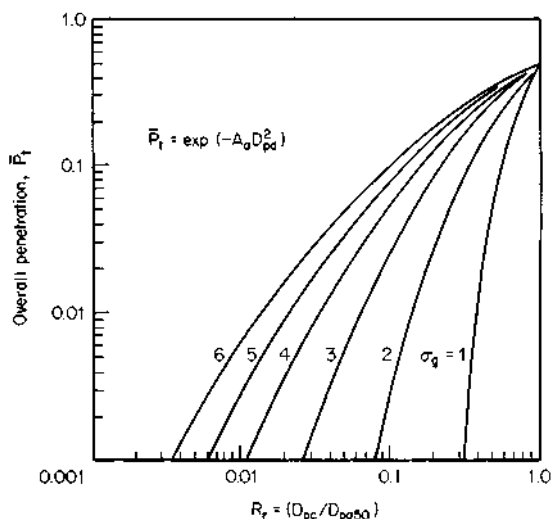


FIG. 14-107 Overall integrated penetration as a function of particle-size distribution and collector cut diameter when $B = 2$ in Eq. (14-224). (Calvert, Goldschmid, Leith, and Mehta, NTIS Publ. PB-213016, 213017, 1972.)

required from the collection device being considered. Therefore, an experimental plot of aerodynamic cut size for each collection device versus operating parameters can be used to determine the device suitability.

Collection Equipment

Gravity Settlers Gravity can act to remove larger droplets. Settling or disengaging space above aerated or boiling liquids in a tank or spray zone in a tower can be very useful. If gas velocity is kept low, all particles with terminal settling velocities (see Sec. 6) above the gas velocity will eventually settle. Increasing vessel cross section in the settling zone is helpful. Terminal velocities for particles smaller than $50\ \mu\text{m}$ are very low and generally not attractive for particle removal. Laminar flow of gas in long horizontal paths between trays or shelves on which the droplets settle is another effective means of employing gravity. Design equations are given in Sec. 17 under "Gas-Solids Separations." Settler pressure drop is very low, usually being limited to entrance and exit losses.

Centrifugal Separation Centrifugal force can be utilized to enhance particle collection to several hundredfold that of gravity. The design of cyclone separators for dust removal is treated in detail in Sec. 17 under "Gas-Solids Separations," and typical cyclone designs are shown in Fig. 17-43. Dimension ratios for one family of cyclones are given in Fig. 17-36. Cyclones, if carefully designed, can be more efficient on liquids than on solids since liquids coalesce on capture and are easy to drain from the unit. However, some precautions not needed for solid cyclones are necessary to prevent reentrainment.

Tests by Calvert (R-12) show high primary collection efficiency on droplets down to $10\ \mu\text{m}$ and in accordance with the efficiency equations of Leith and Licht [*Am. Inst. Chem. Eng. Symp. Ser.*, 68(126), 196-206 (1972)] for the specific cyclone geometry tested if entrainment is avoided. Typical entrainment points are (1) creep along the gas outlet pipe, (2) entrainment by shearing of the liquid film from the walls, and (3) vortex pickup from accumulated liquid in the bottom (Fig. 14-108a). Reentrainment from creep of liquid along the top of the cyclone and down the outlet pipe can be prevented by providing the outlet pipe with a flared conical skirt (Fig. 14-108b), which provides a point from which the liquid can drip without being caught in the outlet gas. The skirt should be slightly shorter than the gas outlet pipe but extend below the bottom of the gas inlet. The cyclone inlet gas should not impinge on this skirt. Often the bottom edge of the skirt is V-notched or serrated.

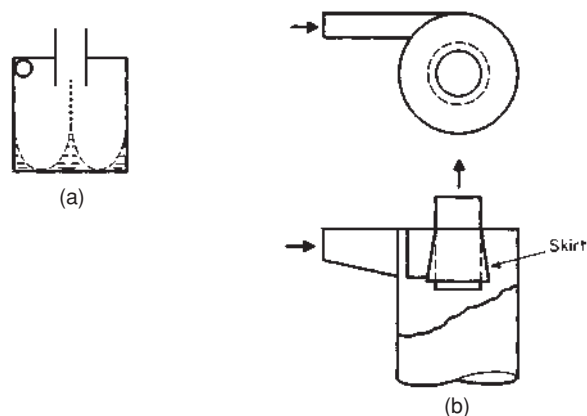


FIG. 14-108 (a) Liquid entrainment from the bottom of a vessel by centrifugal flow. (Rietema and Verver, *Cyclones in Industry*, Elsevier, Amsterdam, 1961.) (b) Gas-outlet skirt for liquid cyclones. (Stern et al., *Cyclone Dust Collectors*, American Petroleum Institute, New York, 1955.)

Reentrainment is generally reduced by lower inlet gas velocities. Calvert (R-12) reviewed the literature on predicting the onset of entrainment and found that of Chien and Ibele (ASME Pap. 62-WA170) to be the most reliable. Calvert applies their correlation to a liquid Reynolds number on the wall of the cyclone, $N_{Re,L} = 4Q_L/h_i v_L$, where Q_L is the volumetric liquid flow rate, cm^3/s ; h_i is the cyclone inlet height, cm ; and v_L is the kinematic liquid viscosity, cm^2/s . He finds that the onset of entrainment occurs at a cyclone inlet gas velocity V_{ci} , m/s , in accordance with the relationship in $V_{ci} = 6.516 - 0.2865 \ln N_{Re,L}$.

Reentrainment from the bottom of the cyclone can be prevented in several ways. If a typical long-cone dry cyclone is used and liquid is kept continually drained, vortex entrainment is unlikely. However, a vortex breaker baffle in the outlet is desirable, and perhaps a flat disk on top extending to within 2 to 5 cm (0.8 to 2 in) of the walls may be beneficial. Often liquid cyclones are built without cones and have dished bottoms. The modifications described earlier are definitely needed in such situations. Stern, Caplan, and Bush (*Cyclone Dust Collectors*, American Petroleum Institute, New York, 1955) and Rietema and Verver (in Tengbergen, *Cyclones in Industry*, Elsevier, Amsterdam, 1961, chap. 7) have discussed liquid-collecting cyclones.

As with dust cyclones, no reliable pressure-drop equations exist (see Sec. 17), although many have been published. A part of the problem is that there is no standard cyclone geometry. Calvert (R-12) experimentally obtained $\Delta P = 0.000513 \rho_g (Q_g/h_i W_i)^2 (2.8 h_i w_i/d_o^3)$, where ΔP is in cm of water; ρ_g is the gas density, g/cm^3 ; Q_g is the gas volumetric flow rate, cm^3/s ; h_i and w_i are cyclone inlet height and width respectively, cm ; and d_o is the gas outlet diameter, cm . This equation is in the same form as that proposed by Shepherd and Lapple [*Ind. Eng. Chem.*, **31**, 1246 (1940)] but gives only 37 percent as much pressure drop.

Liquid cyclone efficiency can be improved somewhat by introducing a coarse spray of liquid in the cyclone inlet. Large droplets which are easily collected collide with finer particles as they sweep the gas stream in their travel to the wall. (See subsection "Wet Scrubbers" regarding optimum spray size.) Cyclones may also be operated wet to improve their operation on dry dust. Efficiency can be improved through reduction in entrainment losses since the dust particles become trapped in the water film. Collision between droplets and dust particles aids collection, and adequate irrigation can eliminate problems of wall buildup and fouling. The most effective operation is obtained by spraying countercurrently to the gas flow in the cyclone inlet duct at liquid rates of 0.7 to 2.0 L/m^3 of gas. There are also many proprietary designs of liquid separators using centrifugal force, some of which are illustrated in Fig. 14-109. Many of these were originally

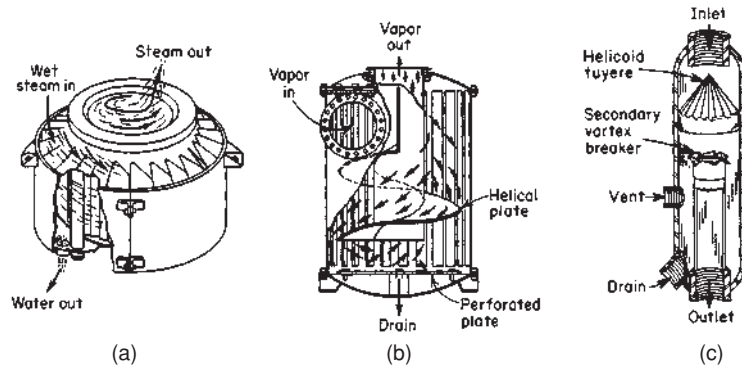


FIG. 14-109 Typical separators using impingement in addition to centrifugal force. (a) Hi-eF purifier. (V. D. Anderson Co.) (b) Flick separator. (Wurster & Sanger, Inc.) (c) Type RA line separator. (Centrifix Corp., Bull. 220.)

developed as steam separators to remove entrained condensate. In some designs, impingement on swirl baffles aids separation.

Impingement Separation Impingement separation employs direct impact and inertial forces between particles, the gas streamlines, and target bodies to provide capture. The mechanism is discussed in Sec. 17 under "Gas-Solids Separations." With liquids, droplet coalescence occurs on the target surface, and provision must be made for drainage without reentrainment. Calvert (R-12) has studied droplet collection by impingement on targets consisting of banks of tubes, zigzag baffles, and packed and mesh beds. Figure 14-110 illustrates some other types of impingement-separator designs.

In its simplest form, an impingement separator may be nothing more than a target placed in front of a flow channel such as a disk at the end of a tube. To improve collection efficiency, the gas velocity may be increased by forming the end into a nozzle (Fig. 14-110a). Particle collection as a function of size may be estimated by using the target-efficiency correlation in Fig. 17-39. Since target efficiency will be low for systems with separation numbers below 5 to 10 (small particles, low gas velocities), the mist will frequently be subjected to a number of targets in series as in Fig. 14-110c, d , and g .

The overall droplet penetration is the product of penetration for each set of targets in series. Obviously, for a distribution of particle sizes, an integration procedure is required to give overall collection efficiency. This target-efficiency method is suitable for predicting efficiency when the design effectively prevents the bypassing or short-circuiting of targets by the gas stream and provides adequate time to accelerate the liquid droplets to gas velocity. Katz (R-16) investigated a jet and target-plate entrainment separator design and found the pressure drop less than would be expected to supply the kinetic energy both for droplet acceleration and gas friction. An estimate based on his results indicates that the liquid particles on the average were being accelerated to only about 60 percent of the gas velocity. The largest droplets, which are the easiest to collect, will be accelerated less than the smaller particles. This factor has a leveling effect on collection efficiency as a function of particle size so that experimental results on such devices may not show as sharp a decrease in efficiency with particle size as predicted by calculation. Such results indicate that in many cases our lack of predicting ability results, not from imperfections in the theoretical treatment, but from our lack of knowledge of velocity distributions within the system.

Katz (R-16) also studied *wave-plate impingement separators* (Fig. 14-110b) made up of 90° formed arcs with an 11.1-mm (0.44-in) radius and a 3.8-mm (0.15-in) clearance between sheets. The pressure drop is a function of system geometry. The pressure drop for Katz's system and collection efficiency for seven waves are shown in Fig. 14-111. Katz used the Souders-Brown expression to define a design velocity for the gas between the waves:

$$U = K \sqrt{(\rho_l - \rho_g)/\rho_g} \quad (14-226)$$

K is 0.12 to give U in ms^{-1} (0.4 for ft/s), and ρ_l and ρ_g are liquid and gas densities in any consistent set of units. Katz found no change in efficiency at gas velocities from one-half to 3 times that given by the equation.

Calvert (R-12) investigated *zigzag baffles* of a design more like Fig. 14-110e. The baffles may have spaces between the changes in direction or be connected as shown. He found close to 100 per collection for water droplets of 10 μm and larger. Some designs had high efficiencies down to 5 or 8 μm . Desirable gas velocities were 2 to 3.5 m/s (6.6 to 11.5 ft/s), with a pressure drop for a six-pass baffle of 2 to 2.5 cm (0.8 to 1.0 in) of water. On the basis of turbulent mixing, an equation was developed for predicting primary collection efficiency as a function of particle size and collector geometry:

$$\eta = 1 - \exp \left[- \frac{u_{te} n W \theta}{57.3 U_g b \tan \theta} \right] \quad (14-227)$$

where η is the fractional primary collection efficiency; u_{te} is the drop terminal centrifugal velocity in the normal direction, cm/s ; U_g is the superficial gas velocity, cm/s ; n is the number of rows of baffles or bends; θ is the angle of inclination of the baffle to the flow path, °; W is the width of the baffle, cm ; and b is the spacing between baffles in the same row, cm . For conditions of low Reynolds number ($N_{Re,D} < 0.1$) where Stokes' law applies, Calvert obtains the value for drop terminal centrifugal velocity of $u_{te} = d_p^2 \rho_g a / 18 \mu_g$, where d_p and ρ_p are the drop particle diameter, cm , and particle density, g/cm^3 , respectively; μ_g is the gas viscosity, P ; and a is the acceleration due to centrifugal force. It is defined by the equation $a = 2U_g^2 \sin \theta / W \cos^3 \theta$. For situations in which Stokes' law does not apply, Calvert recommends substitution in the derivation of Eq. (14-227) for u of drag coefficients from drag-coefficient data of Foust et al. (*Principles of Unit Operations*, Toppan Co., Tokyo, 1959).

Calvert found that reentrainment from the baffles was affected by the gas velocity, the liquid-to-gas ratio, and the orientation of the baffles. Horizontal gas flow past vertical baffles provided the best drainage and lowest reentrainment. Safe operating regions with vertical baffles are shown in Fig. 14-112. Horizontal baffles gave the poorest drainage and the highest reentrainment, with inclined baffles intermediate in performance. Equation (14-228), developed by Calvert, predicts pressure drop across zigzag baffles. The indicated summation must be made over the number of rows of baffles present.

$$\Delta P = \sum_{i=1}^{i=n} 1.02 \times 10^{-3} f_D \rho_g \frac{U_g^2 A_p}{2 A_t} \quad (14-228)$$

ΔP is the pressure drop, cm of water; ρ_g is the gas density, g/cm^3 ; A_p is the total projected area of an entire row of baffles in the direction of inlet gas flow, cm^2 ; and A_t is the duct cross-sectional area, cm^2 . The

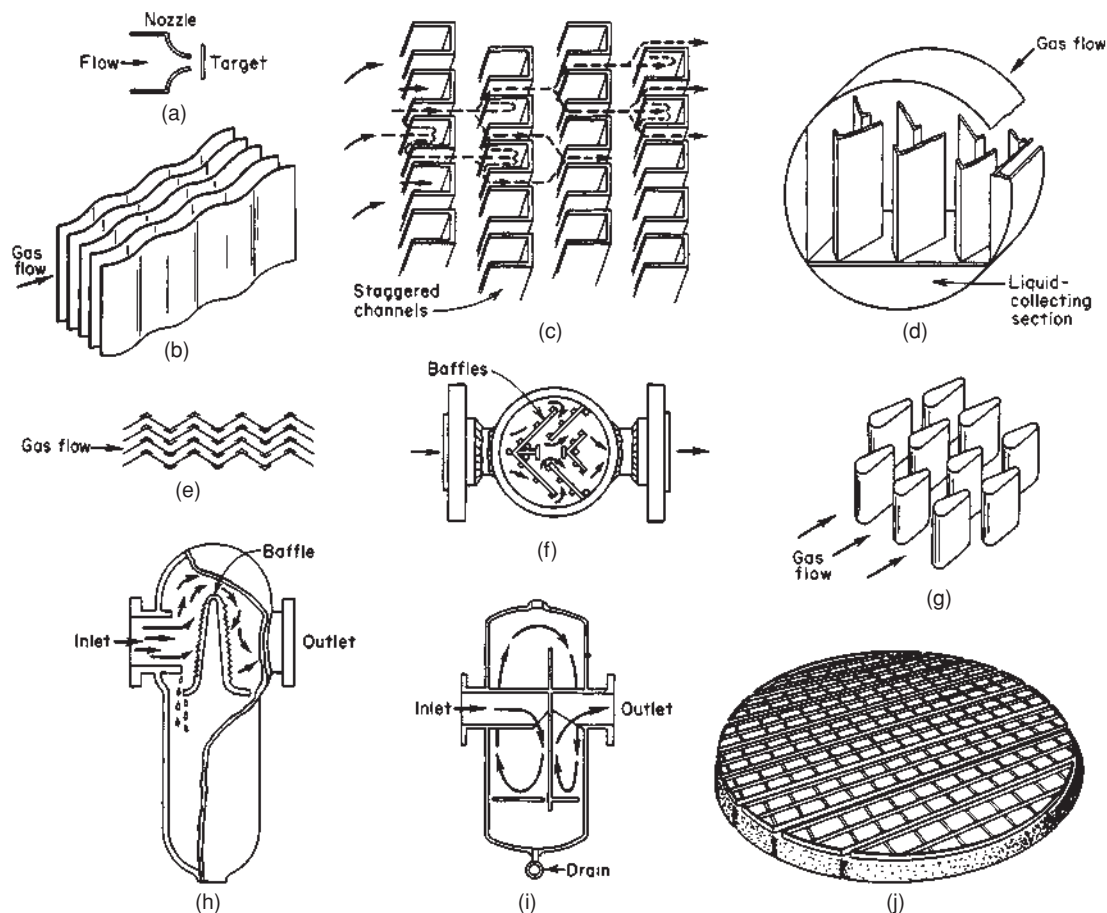


FIG. 14-110 Typical impingement separators. (a) Jet impactor. (b) Wave plate. (c) Staggered channels. (Blaw-Knox Food & Chemical Equipment, Inc.) (d) Vane-type mist extractor. (Maloney-Crawford Tank and Mfg. Co.) (e) Peerless line separator. (Peerless Mfg. Co.) (f) Strong separator. (Strong Carlisle and Hammond.) (g) Karbate line separator. (Union Carbide Corporation) (h) Type E horizontal separator. (Wright-Austin Co.) (i) PL separator. (Ingersoll Rand.) (j) Wire-mesh demister. (Otto H. York Co.)

value f_D is a drag coefficient for gas flow past inclined flat plates taken from Fig. 14-113, while U_g'' is the actual gas velocity, cm/s, which is related to the superficial gas velocity U_g by $U_g'' = U_g / \cos \theta$. It must be noted that the angle of incidence θ for the second and successive rows of baffles is twice the angle of incidence for the first row. Most of

Calvert's work was with 30° baffles, but the method correlates well with other data on 45° baffles.

The Karbate line separator (Fig. 14-110g) is composed of several layers of teardrop-shaped target rods of Karbate. A design flow constant K in Eq. (14-226) of 0.035 m/s (1.0 ft/s) is recommended by the

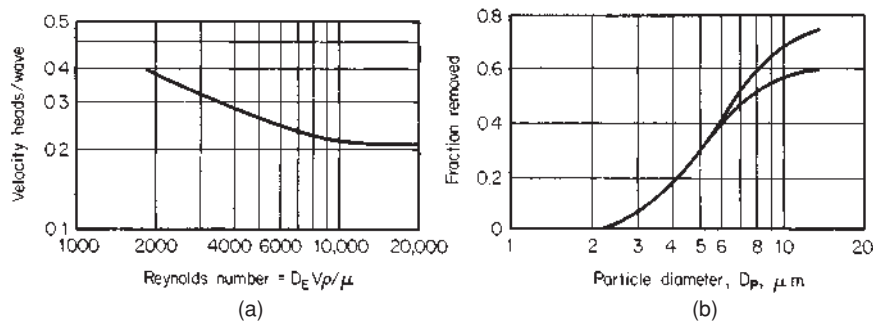


FIG. 14-111 Pressure drop and collection efficiency of a wave-plate separator. (a) Pressure drop. (b) Efficiency D_E = clearance between sheets. (Katz, M.S. thesis, Pennsylvania State University, 1958.)

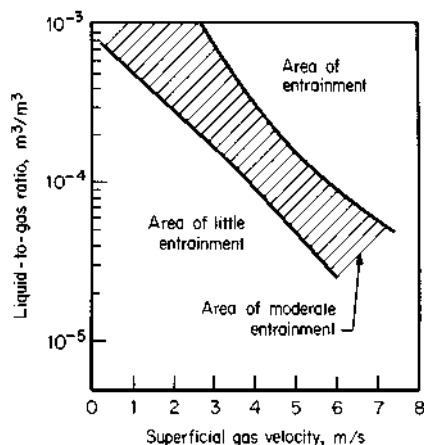


FIG. 14-112 Safe operating region to prevent reentrainment from vertical zigzag baffles with horizontal gas flow. (Calvert, Yung, and Leung, NTIS Publ. PB-248050, 1975.)

manufacturer. Pressure drop is said to be $5\frac{1}{2}$ velocity heads on the basis of the superficial gas velocity. This value would probably increase at high liquid loads. Figure 14-114 gives the manufacturer's reported grade efficiency curve at the design air velocity.

The use of multiple tube banks as a droplet collector has also been studied by Calvert (R-12). He reports that collection efficiency for closely packed tubes follows equations for rectangular jet impaction which can be obtained graphically from Fig. 14-115 by using a dimensional parameter β which is based on the tube geometry: $\beta = 2l_i/b$, where b is the open distance between adjacent tubes in the row (orifice width) and l_i is the impaction length (distance between orifice and impingement plane), or approximately the distance between centerlines of successive tube rows. Note that the impaction parameter K_p is plotted to the one-half power in Fig. 14-115 and that the radius of the droplet is used rather than the diameter. Collection efficiency overall for a given size of particle is predicted for the entire tube bank by

$$\eta = 1 - (1 - \eta_b)^N \quad (14-229)$$

where η_b is the collection efficiency for a given size of particle in one stage of a rectangular jet impactor (Fig. 14-115) and N is the number of stages in the tube bank (equal to one less than the number of rows).

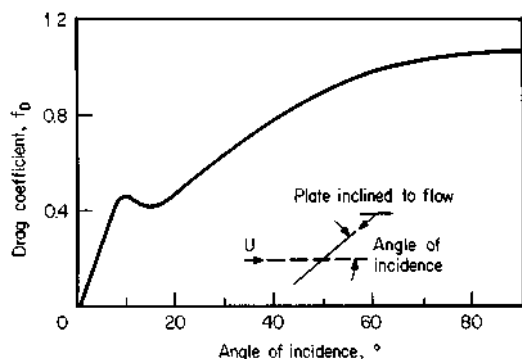


FIG. 14-113 Drag coefficient for flow past inclined flat plates for use in Eq. (14-228). [Calvert, Yung, and Leung, NTIS Publ. PB-248050; based on Fage and Johansen, Proc. R. Soc. (London), 116A, 170 (1927).]

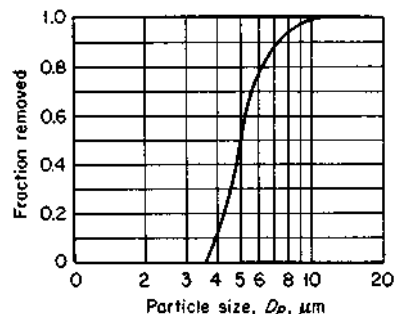


FIG. 14-114 Collection efficiency of Karbate line separator, based on particles with a specific gravity of 1.0 suspended in atmospheric air with a pressure drop of 2.5 cm water gauge. (Union Carbide Corporation Cat. Sec. S-6900, 1960.)

For widely spaced tubes, the target efficiency η_g can be calculated from Fig. 17-39 or from the impaction data of Golovin and Putnam [Ind. Eng. Chem. Fundam., 1, 264 (1962)]. The efficiency of the overall tube banks for a specific particle size can then be calculated from the equation $\eta = 1 - (1 - \eta_g/A)^n$, where A' is the cross-sectional area of all tubes in one row, A is the total flow area, and n is the number of rows of tubes.

Calvert reports pressure drop through tube banks to be largely unaffected by liquid loading and indicates that Grimison's correlations in Sec. 6 ("Tube Banks") for gas flow normal to tube banks or data for gas flow through heat-exchanger bundles can be used. However, the following equation is suggested:

$$\Delta P = 8.48 \times 10^{-3} n \rho_g U_g^2 \quad (14-230)$$

where ΔP is cm of water; n is the number of rows of tubes; ρ_g is the gas density, g/cm³; and U_g is the actual gas velocity between tubes in a row, cm/s. Calvert did find an increase in pressure drop of about 80 to 85 percent above that predicted by Eq. (14-230) in vertical upflow of gas through tube banks due to liquid holdup at gas velocities above 4 m/s.

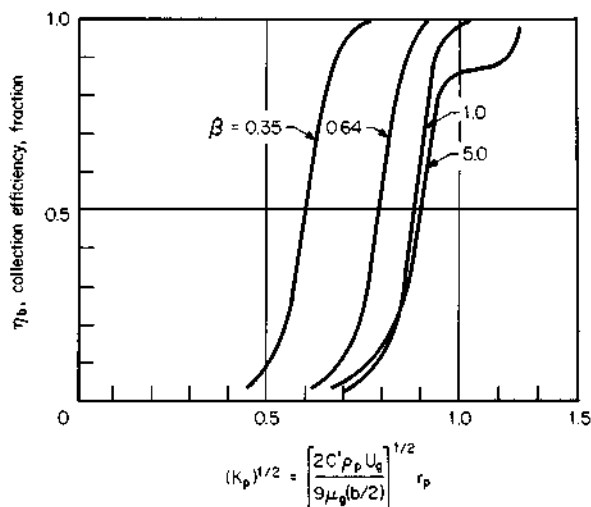


FIG. 14-115 Experimental collection efficiencies of rectangular impactors. C' is the Stokes-Cunningham correction factor; ρ_p , particle density, g/cm³; U_g , superficial gas velocity, approaching the impactor openings, cm/s; and μ_g , gas viscosity, P. [Calvert, Yung, and Leung, NTIS Publ. PB-248050; based on Mercer and Chow, J. Coll. Interface Sci., 27, 75 (1968).]

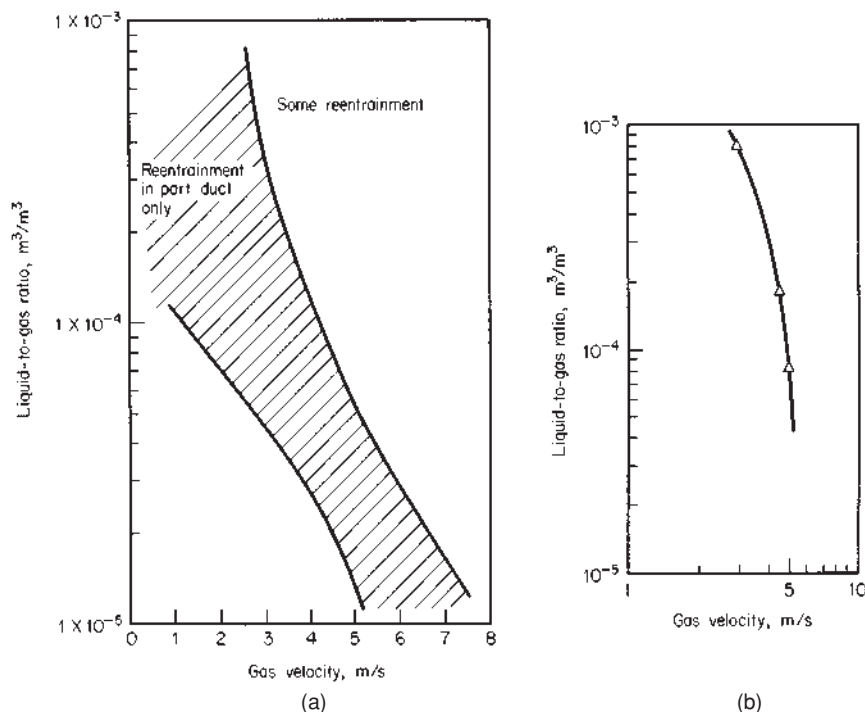


FIG. 14-116 Experimental results showing effect of gas velocity and liquid load on entrainment from (a) vertical tube banks with horizontal gas flow and (b) horizontal tube banks with upflow. To convert meters per second to feet per second, multiply by 3.281. (Calvert, Yung, and Leung, NTIS Publ. PB-248050.)

The onset of liquid reentrainment from tube banks can be predicted from Fig. 14-116. Reentrainment occurred at much lower velocities in vertical upflow than in horizontal gas flow through vertical tube banks. While the top of the cross-hatched line of Fig. 14-116a predicts reentrainment above gas velocities of 3 m/s (9.8 ft/s) at high liquid loading, most of the entrainment settled to the bottom of the duct in 1 to 2 m (3.3 to 6.6 ft), and entrainment did not carry significant distances until the gas velocity exceeded 7 m/s (23 ft/s).

Packed-Bed Collectors Many different materials, including coal, coke, broken solids of various types such as brick, tile, rock, and stone, as well as normal types of tower-packing rings, saddles, and special plastic shapes, have been used over the years in packed beds to remove entrained liquids through impaction and filtration. Separators using natural materials are not available as standard commercial units but are designed for specific applications. Coke boxes were used extensively in the years 1920 to 1940 as sulfuric acid entrainment separators (see *Chemical Engineers' Handbook*, 5th ed., p. 18-87) but have now been largely superseded by more sophisticated and efficient devices.

Jackson and Calvert [*Am. Inst. Chem. Eng. J.*, **12**, 1075 (1966)] studied the collection of fine fuel-oil-mist particles in beds of ½-in glass spheres, Raschig rings, and Berl and Intalox saddles. The mist had a mass median particle diameter of 6 μm and a standard deviation of 2.0. The collection efficiency as a function of particle size and gas velocity in a 355-mm- (14-in-) diameter by 152-mm- (6-in-) thick bed of Intalox saddles is given in Fig. 14-117. This and additional work have been generalized by Calvert (R-12) to predict collection efficiencies of liquid particles in any packed bed. Assumptions in the theoretical development are that the drag force on the drop is given by Stokes' law and that the number of semicircular bends to which the gas is subjected, η_1 , is related to the length of the bed, Z (cm), in the direction of gas flow, the packing diameter, d_c (cm), and the gas-flow channel width, b (cm), such that $\eta_1 = Z/(d_c + b)$. The gas velocity through the channels, U_{gb} (cm/s), is inversely proportional to the bed

free volume for gas flow such that $U_{gb} = U_g [1/(\epsilon - h_b)]$, where U_g is the gas superficial velocity, cm/s, approaching the bed, ϵ is the bed void fraction, and h_b is the fraction of the total bed volume taken up with liquid which can be obtained from data on liquid holdup in packed beds. The width of the semicircular channels b can be expressed as a fraction j of the diameter of the packing elements, such that $b = jd_c$. These assumptions (as modified by G. E. Goltz, personal communica-

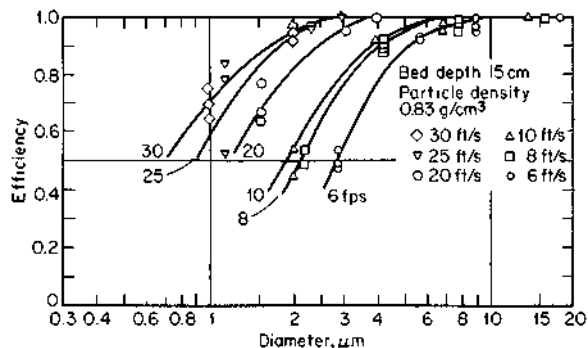


FIG. 14-117 Experimental collection efficiency, ½-in Intalox saddles. To convert feet per second to meters per second, multiply by 0.3048; to convert centimeters to inches, multiply by 0.394; and to convert grams per cubic centimeter to pounds per cubic foot, multiply by 62.43. [Jackson and Calvert, *Am. Inst. Chem. Eng. J.*, **12**, 1975 (1968).]

TABLE 14-27 Experimental Values for j , Channel Width in Packing as a Fraction of Packing Diameter

Packing size		Type of packing	j
cm	in		
1.27	0.5	Berl and Intalox saddles, marbles, Raschig rings	0.192
2.54	1.0	Berl and Intalox saddles, pall rings	0.190
3.8	1.5	Berl and Intalox saddles, pall rings	0.165
7.6–12.7	3–5	Coke	0.03

tion) lead to an equation for predicting the penetration of a given size of liquid particle through a packed bed:

$$P_t = \exp \left[\frac{-\pi}{2(j+j^2)(\epsilon - h_p)} \left(\frac{Z}{d_c} \right) K_p \right] \quad (14-231)$$

where

$$K_p = \frac{\rho_p d_p^2 U_g}{9 \mu_g d_c} \quad (14-232)$$

Values of ρ_p and d_p are droplet density, g/cm^3 , and droplet diameter, cm; μ_g is the gas viscosity, P. All other terms were defined previously. Table 14-27 gives values of j calculated from experimental data of Jackson and Calvert. Values of j for most manufactured packing appear to fall in the range from 0.16 to 0.19. The low value of 0.03 for coke may be due to the porosity of the coke itself.

Calvert (R-12) has tested the correlation in cross-flow packed beds, which tend to give better drainage than countercurrent beds, and has found the effect of gas-flow orientation insignificant. However, the onset of reentrainment was somewhat lower in a bed of 2.5-cm (1.0-in) pall rings with gas upflow [6 m/s (20 ft/s)] than with horizontal cross-flow of gas. The onset of reentrainment was independent of liquid loading (all beds were nonirrigated), and entrainment occurred at values somewhat above the flood point for packed beds as predicted by conventional correlations. In beds with more than 3 cm (1.2 in) of water pressure drop, the experimental drop with both vertical and horizontal gas flow was somewhat less than predicted by generalized packed-bed pressure-drop correlations. However, Calvert recommends these correlations for design as conservative.

Calvert's data indicate that packed beds irrigated only with the collected liquid can have collection efficiencies of 80 to 90 percent on mist particles down to 3 μm but have low efficiency on finer mist particles. Frequently, irrigated packed towers and towers with internals will be used with liquid having a wetting capability for the fine mist which must be collected. Tennessee Valley Authority (TVA) experiments with the collection of 1.0- μm mass median phosphoric acid mist in packed towers have shown that the strength of the circulating phosphoric acid is highly important [see Baskerville, *Am. Inst. Chem. Eng. J.*, **37**, 79 (1941); and p. 18–87, 5th ed. of the *Handbook*]. Hesketh (*J. Air Pollut. Control Assoc.*, **24**, 942 (1974)) has reported up to 50 percent improvement in collection efficiency in venturi scrubbers on fine particles with the addition of only 0.10 percent of a low-foaming nonionic surfactant to the scrubbing liquid, and others have experienced similar results in other gas-liquid-contacting devices. Calvert (R-9 and R-10) has reported on the efficiency of various gas-liquid-contacting devices for fine particles. Figure 14-118 gives the particle aerodynamic cut size for a single-sieve-plate gas scrubber as a function of sieve hole size d_h , cm; hole gas velocity u_h , m/s; and froth or foam density on the plate F , g/cm^3 . This curve is based on standard air and water properties and wettable (hydrophilic) particles. The cut diameter decreases with an increase in froth density, which must be predicted from correlations for sieve-plate behavior (see Fig. 14-32). Equation (14-231) can be used to calculate generalized design curves for collection in packed columns in the same fashion by finding parameters of packing size, bed length, and gas velocity which give collection efficiencies of 50 percent for various size particles. Figure 14-119 illustrates such a plot for three gas velocities and two sizes of packing.

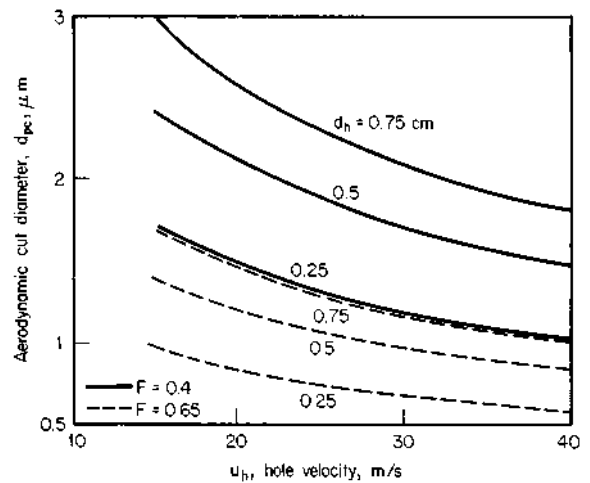


FIG. 14-118 Aerodynamic cut diameter for a single-sieve-plate scrubber as a function of hole size, hole-gas velocity, and froth density, F , g/cm^3 . To convert meters per second to feet per second, multiply by 3.281; to convert grams per cubic centimeter to pounds per cubic foot, multiply by 62.43. [Calvert, J. Air Pollut. Control Assoc., **24**, 929 (1974).]

Wire-Mesh Mist Collectors Knitted mesh of varying density and voidage is widely used for entrainment separators. Its advantage is close to 100 percent removal of drops larger than 5 μm at superficial gas velocities from about 0.2 m/s (0.6 ft/s) to 5 m/s (16.4 ft/s), depending somewhat on the design of the mesh. Pressure drop is usually no more than 2.5 cm (1 in) of water. A major disadvantage is the ease with which tars and insoluble solids plug the mesh. The separator can be made to fit vessels of any shape and can be made of any material which can be drawn into a wire. Stainless-steel and plastic fibers are most common, but other metals are sometimes used. Generally three basic types of mesh are used: (1) layers with a crimp in the same direction (each layer is actually a nested double layer); (2) layers with a crimp in

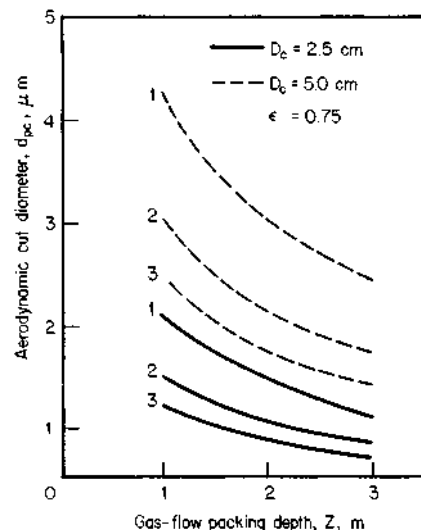


FIG. 14-119 Aerodynamic cut diameter for a typical packed-bed entrainment separator as a function of packing size, bed depth, and three gas velocities: curve 1–1.5 m/s, curve 2–3.0 m/s, and curve 3–4.5 m/s. To convert meters to feet, multiply by 3.281; to convert centimeters to inches, multiply by 0.394. [Calvert, J. Air Pollut. Control Assoc., **24**, 929 (1974).]

alternate directions, which increases voidage, reduces sheltering and increases target efficiency per layer, and gives a lower pressure drop per unit length; and (3) spiral-wound layers which reduce pressure drop by one-third, but fluid creep may lead to higher entrainment. Some small manufacturers of plastic meshes may offer other weaves claimed to be superior. The filament size can vary from about 0.15 mm (0.006 in) for fine-wire pads to 3.8 mm (0.15 in) for some plastic fibers. Typical pad thickness varies from 100 to 150 mm (4 to 6 in), but occasionally pads up to 300 mm (12 in) thick are used. A typical wire diameter for standard stainless mesh is 0.28 mm (0.011 in), with a finished mesh density of 0.15 g/cm³ (9.4 lb/ft³). A lower mesh density may be produced with standard wire to give 10 to 20 percent higher flow rates.

Figure 14-120 presents an early calculated estimate of mesh efficiency as a fraction of mist-particle size. Experiments by Calvert (R-12) confirm the accuracy of the equation of Bradie and Dickson (*Joint Symp. Proc. Inst. Mech. Eng./Yorkshire Br. Inst. Chem. Eng.*, 1969, pp. 24–25) for primary efficiency in mesh separators:

$$\eta = 1 - \exp(-2/3)\pi a/l\eta_i \quad (14-232)$$

where η is the overall collection efficiency for a given-size particle; l is the thickness of the mesh, cm, in the direction of gas flow; a is the surface area of the wires per unit volume of mesh pad, cm²/cm³; and η_i , the target collection efficiency for cylindrical wire, can be calculated from Fig. 17-39 or the impaction data of Golovin and Putnam [*Ind. Eng. Chem.*, **1**, 264 (1962)]. The factor 2/3, introduced by Carpenter and Othmer [*Am. Inst. Chem. Eng. J.*, **1**, 549 (1955)], corrects for the fact that not all the wires are perpendicular to the gas flow and gives the projected perpendicular area. If the specific mesh surface area a is not available, it can be calculated from the mesh void area ϵ and the mesh wire diameter d_w in cm, $a = 4(1 - \epsilon)/d_w$.

York and Poppele (R-17) have stated that factors governing maximum allowable gas velocity through the mesh are (1) gas and liquid density, (2) liquid surface tension, (3) liquid viscosity, (4) specific wire surface area, (5) entering-liquid loading, and (6) suspended-solids content. York (R-18) has proposed application of the Souders-Brown equation [Eq. (14-226)] for correlation of maximum allowable gas velocity with values of K for most cases of 0.1067 m/s to give U in m/s (0.35 for ft/s). When liquid viscosity or inlet loading is high or the liquid is dirty, the value of K must be reduced. Schroeder (M.S. thesis, Newark College of Engineering, 1962) found lower values for K necessary when liquid surface tension is reduced such as by the presence of surfactants in water. Ludwig (*Applied Process Design for Chemical and Petrochemical Plants*, 2d ed., vol. I, Gulf, Houston, 1977, p. 157) recommends reduced K values of (0.061 m/s) under vacuum at an absolute pressure of 6.77 kPa (0.98 lb/in²) and $K = 0.082$ m/s at 54 kPa (7.83 lb/in²) absolute. Most manufacturers suggest setting the design velocity at three-fourths of the maximum velocity to allow for surges in gas flow.

York and Poppele (R-17) have suggested that total pressure drop through the mesh is equal to the sum of the mesh dry pressure drop

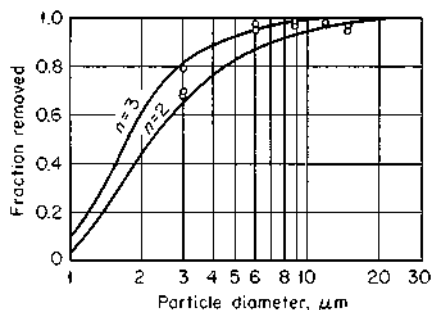


FIG. 14-120 Collection efficiency of wire-mesh separator; 6-in thickness, 98.6 percent free space, 0.006-in-diameter wire used for experiment points. Curves calculated for target area equal to 2 and 3 times the solids volume of packing. To convert inches to millimeters, multiply by 25.4.

plus an increment due to the presence of liquid. They considered the mesh to be equivalent to numerous small circular channels and used the D'Arcy formula with a modified Reynolds number to correlate friction factor (see Fig. 14-121) for Eq. (14-233) giving dry pressure drop.

$$\Delta P_{\text{dry}} = f a \rho_g U_g^2 / 981 \epsilon^3 \quad (14-233)$$

where ΔP is in cm of water; f is from Fig. (14-121); ρ_g is the gas density, g/cm³; U_g is the superficial gas velocity, cm/s; and ϵ is the mesh porosity or void fraction; l and a are as defined in Eq. (14-232). Figure 14-121 gives data of York and Poppele for mesh crimped in the same and alternating directions and also includes the data of Satsangee, of Schuring, and of Bradie and Dickson.

The incremental pressure drop for wet mesh is not available for all operating conditions or for mesh of different styles. The data of York and Poppele for wet-mesh incremental pressure drop, ΔP_L in cm of water, are shown in Fig. 14-122 or parameters of liquid velocity L/A , defined as liquid volumetric flow rate, cm³/min per unit of mesh cross-sectional area in cm²; liquid density ρ_L is in g/cm³.

York generally recommends the installation of the mesh horizontally with upflow of gas as in Fig. 14-110f; Calvert (R-12) tested the mesh horizontally with upflow and vertically with horizontal gas flow. He reports better drainage with the mesh vertical and somewhat higher permissible gas velocities without reentrainment, which is contrary to past practice. With horizontal flow through vertical mesh, he found collection efficiency to follow the predictions of Eq. (14-232) up to 4 m/s (13 ft/s) with air and water. Some reentrainment was encountered at higher velocities, but it did not appear serious until velocities exceeded 6.0 m/s (20 ft/s). With vertical upflow of gas, entrainment was encountered at velocities above and below 4.0 m/s (13 ft/s), depending on inlet liquid quantity (see Fig. 14-123). Figure 14-124 illustrates the onset of entrainment from mesh as a function of liquid loading and gas velocity and the safe operating area recommended by Calvert. Measurements of dry pressure drop by Calvert gave values only about one-third of those predicted from Eq. (14-233). He found the pressure drop to be highly affected by liquid load. The pressure drop of wet mesh could be correlated as a function of $U_g^{1.65}$ and parameters of liquid loading L/A , as shown in Fig. 14-125.

As indicated previously, mesh efficiency drops rapidly as particles decrease in size below 5 μm . An alternative is to use two mesh pads in series. The first mesh is made of fine wires and is operated beyond the

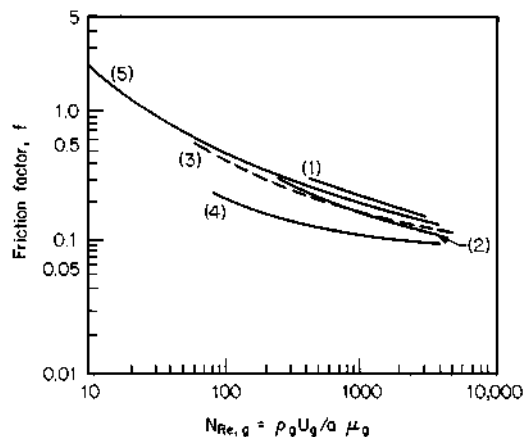


FIG. 14-121 Value of friction factor f for dry knitted mesh for Eq. (14-233). Values of York and Poppele [*Chem. Eng. Prog.*, **50**, 421 (1954)] are given in curve 1 for mesh crimped in the alternating direction and curve 2 for mesh crimped in the same direction. Data of Bradie and Dickson (*Joint Symp. Proc. Inst. Mech. Eng./Yorkshire Br. Inst. Chem. Eng.*, 1969, pp. 24–25) are given in curve 3 for layered mesh and curve 4 for spiral-wound mesh. Curve 5 is data of Satsangee (M.S. thesis, Brooklyn Polytechnic Institute, 1948) and Schurig (D.Ch.E. dissertation, Brooklyn Polytechnic Institute, 1946). (From Calvert, Yung, and Leung, NTIS Publ. PB-248050, 1975.)

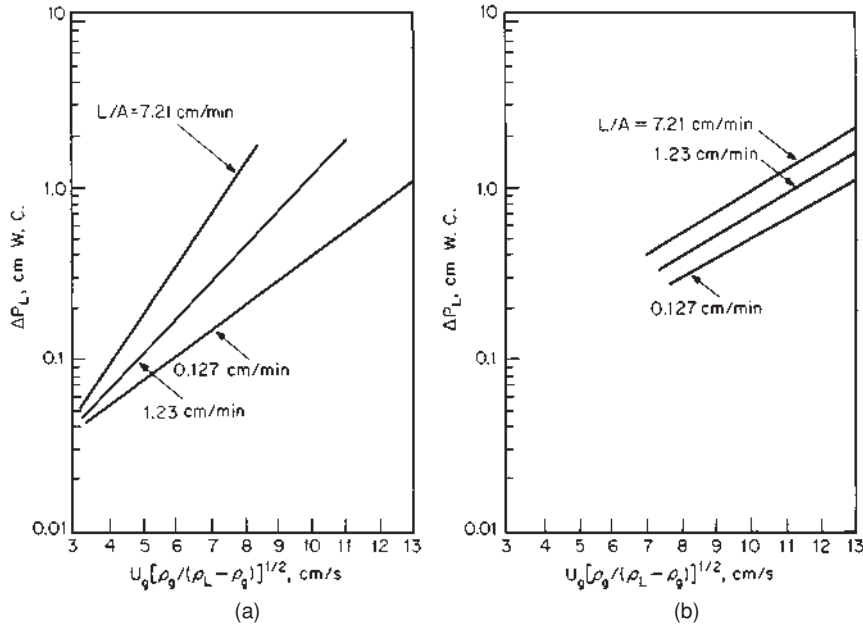


FIG. 14-122 Incremental pressure drop in knitted mesh due to the presence of liquid (a) with the mesh crimps in the same direction and (b) with crimps in the alternating direction, based on the data of York and Poppele [Chem. Eng. Prog., **50**, 421 (1954)]. To convert centimeters per minute to feet per minute, multiply by 0.0328; to convert centimeters per second to feet per second, multiply by 0.0328. (From Calvert, Yung, and Leung, NTIS Publ. PB-248050, 1975.)

flood point. It results in droplet coalescence, and the second mesh, using standard wire and operated below flooding, catches entrainment from the first mesh. Coalescence and flooding in the first mesh may be assisted with water sprays or irrigation. Massey [Chem. Eng. Prog., **53**(5), 114 (1959)] and Coykendall et al. [J. Air Pollut. Control Assoc., **18**, 315 (1968)] have discussed such applications. Calvert (R-12) presents data on the particle size of entrained drops from mesh as a function of gas velocity which can be used for sizing the secondary collector. A major disadvantage of this approach is high pressure drop, which can be in the range from 25 cm (10 in) of water to as high as 85 cm (33 in) of water if the mist is mainly submicrometer.

Wet Scrubbers Scrubbers have not been widely used for the collection of purely liquid particulate, probably because they are generally more complex and expensive than impaction devices of the types previously discussed. Further, scrubbers are no more efficient than

the former devices for the same energy consumption. However, scrubbers of the types discussed in Sec. 17 and illustrated in Figs. 17-48 to 17-54 can be used to capture liquid particles efficiently. Their use is primarily indicated when it is desired to accomplish simultaneously another task such as gas absorption or the collection of solid and liquid particulate mixtures.

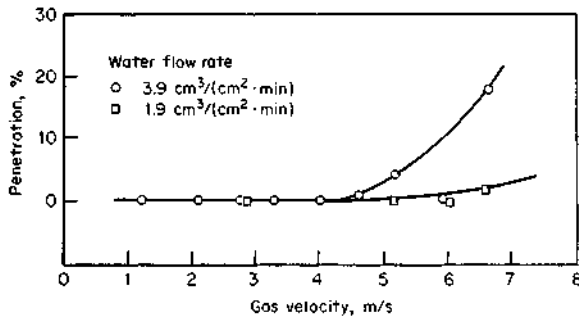


FIG. 14-123 Experimental data of Calvert with air and water in mesh with vertical upflow, showing the effect of liquid loading on efficiency and reentrainment. To convert meters per second to feet per second, multiply by 3.281; to convert cubic centimeters per square centimeter-minute to cubic feet per square foot-minute, multiply by 0.0328. (Calvert, Yung, and Leung, NTIS Publ. PB-248050, 1975.)

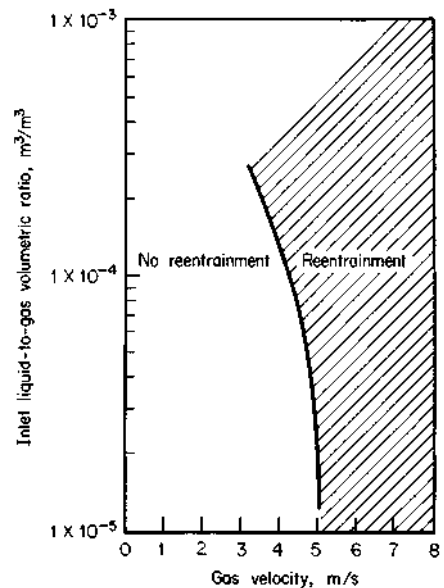


FIG. 14-124 Effect of gas and liquid rates on onset of mesh reentrainment and safe operating regions. To convert meters per second to feet per second, multiply by 3.281. (Calvert, Yung, and Leung, NTIS Publ. PB-248050, 1975.)

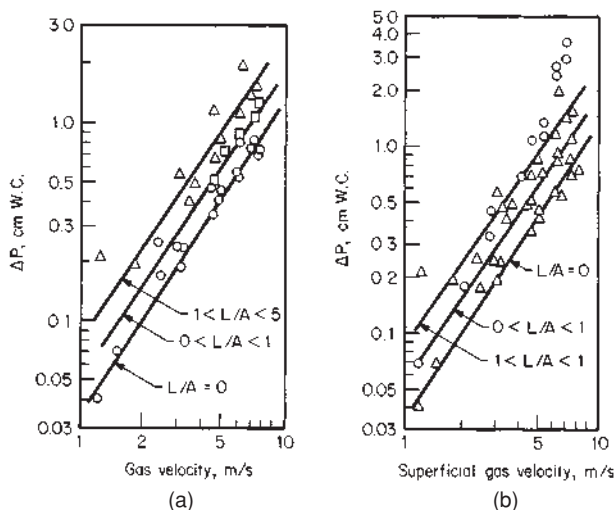


FIG. 14-125 Experimental pressure measured by Calvert as a function of gas velocity and liquid loading for (a) horizontal gas flow through vertical mesh and (b) gas upflow through horizontal mesh. Mesh thickness was 10 cm with 2.8-mm wire and void fraction of 98.2 percent, crimped in alternating directions. To convert meters per second to feet per second, multiply by 3.281; to convert centimeters to inches, multiply by 0.394. (Calvert, Yung, and Leung, NTIS Publ. PB-248050, 1975.)

Table 20-41 [*Chemical Engineers' Handbook*, 5th ed.], showing the minimum size of particles collectible in different types of scrubbers at reasonably high efficiencies, is a good selection guide. Cyclonic spray towers can effectively remove liquid particles down to around 2 to 3 μm . Figures 20-112 and 20-113 (*Chemical Engineers' Handbook*, 5th ed.), giving target efficiency between spray drop size and particle size as calculated by Stairmand or Johnstone and Roberts, should be considered in selecting spray atomization for the most efficient tower operation. Figure 14-126 gives calculated particle cut size as a function of tower height (or length) for vertical countercurrent spray towers and for horizontal-gas-flow, vertical-liquid-flow cross-current spray towers with parameters for liquid

drop size. These curves are based on physical properties of standard air and water and should be used under conditions in which these are reasonable approximations. Lack of uniform liquid distribution or liquid flowing down the walls can affect the performance, requiring empirical correction factors. Calvert (R-10) suggests that a correction factor of 0.2 be used in small-diameter scrubbers to account for the liquid on the walls, i.e., let $Q_L/Q_G = 0.2 (Q_L/Q_G)_{\text{actual}}$. Many more complicated wet scrubbers employ a combination of sprays or liquid atomization, cyclonic action, baffles, and targets. These combinations are not likely to be more efficient than similar devices previously discussed that operate at equivalent pressure drop. The vast majority of wet scrubbers operate at moderate pressure drop [8 to 15 cm (3 to 6 in) of water or 18 to 30 cm (7 to 12 in) of water] and cannot be expected to have high efficiency on particles smaller than 10 μm or 3 to 5 μm respectively. Fine and submicrometer particles can be captured efficiently only in wet scrubbers having high energy input such as venturi scrubbers, two-phase eductor scrubbers, and flux-force-condensation scrubbers.

Venturi Scrubbers One type of venturi scrubber is illustrated in Fig. 17-48. Venturi scrubbers have been used extensively for collecting fine and submicrometer solid particulate, condensing tars and mists, and mixtures of liquids and solids. To a lesser extent, they have also been used for simultaneous gas absorption, although Lundy [*Ind. Eng. Chem.*, **50**, 293 (1958)] indicates that they are generally limited to three transfer units. They have been used to collect submicrometer chemical incinerator fume and mist as well as sulfuric and phosphoric acid mists. The collection efficiency of a venturi scrubber is highly dependent on the throat velocity or pressure drop, the liquid-to-gas ratio, and the chemical nature of wettability of the particulate. Throat velocities may range from 60 to 150 m/s (200 to 500 ft/s). Liquid injection rates are typically 0.67 to 1.4 $\text{m}^3/1000 \text{ m}^3$ of gas. A liquid rate of 1.0 m^3 per 1000 m^3 of gas is usually close to optimum, but liquid rates as high as 2.7 m^3 (95 ft³) have been used. Efficiency improves with increased liquid rate but only at the expense of higher pressure drop and energy consumption. Pressure-drop predictions for a given efficiency are hazardous without determining the nature of the particulate and the liquid-to-gas ratio. In general, particles coarser than 1 μm can be collected efficiently with pressure drops of 25 to 50 cm of water. For appreciable collection of submicrometer particles, pressure drops of 75 to 100 cm (30 to 40 in) of water are usually required. When particles are appreciably finer than 0.5 μm , pressure drops of 175 to 250 cm (70 to 100 in) of water have been used.

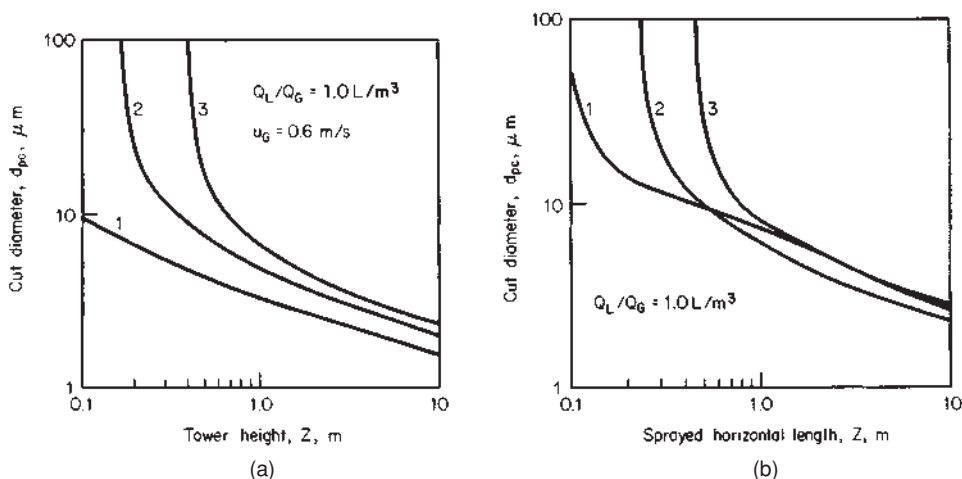


FIG. 14-126 Predicted spray-tower cut diameter as a function of sprayed length and spray droplet size for (a) vertical-countercurrent towers and (b) horizontal-cross-flow towers per Calvert [J. Air Pollut. Control Assoc., **24**, 929 (1974)]. Curve 1 is for 200- μm spray droplets, curve 2 for 500- μm spray, and curve 3 for 1000- μm spray. Q_L/Q_G is the volumetric liquid-to-gas ratio, L liquid/ m^3 gas, and u_G is the superficial gas velocity in the tower. To convert liters per cubic meter to cubic feet per cubic foot, multiply by 10^{-3} .

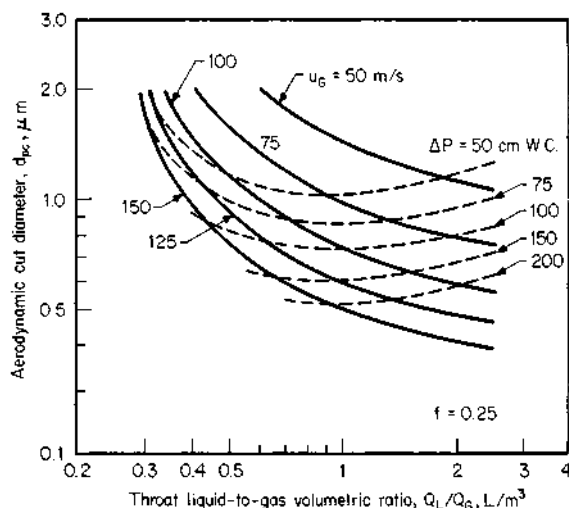


FIG. 14-127 Prediction of venturi-scrubber cut diameter for hydrophobic particles as functions of operating parameters as measured by Calvert [Calvert, Goldshmid, Leith, and Mehta, NTIS Publ. PB-213016, 213017, 1972; and Calvert, J. Air Pollut. Control Assoc., **24**, 929 (1974).] u_g is the superficial throat velocity, and ΔP is the pressure drop from converging to diverging section. To convert meters per second to feet per second, multiply by 3.281; to convert liters per cubic meter to cubic feet per cubic foot, multiply by 10^{-3} ; and to convert centimeters to inches, multiply by 0.394.

One of the problems in predicting efficiency and required pressure drop of a venturi is the chemical nature or wettability of the particulate, which on 0.5- μm -size particles can make up to a threefold difference in required pressure drop for its efficient collection. Calvert (R-9, R-10) has represented this effect by an empirical factor f , which is based on the hydrophobic ($f = 0.25$) or hydrophilic ($f = 0.50$) nature of the particles. Figure 14-127 gives the cut diameter of a venturi scrubber as a function of its operating parameters (throat velocity, pressure drop, and liquid-to-gas ratio) for hydrophobic particles. Figure 14-129 compares cut diameter as a function of pressure drop for an otherwise identically operating venturi on hydrophobic and hydrophilic particles. Calvert (R-9) gives equations which can be used for constructing cut-size curves similar to those of Fig. 14-127 for other values of the empirical factor f . Most real particles are neither completely hydrophobic nor completely hydrophilic but have f values lying between the two extremes. Phosphoric acid mist, on the basis of data of Brink and Contant [Ind. Eng. Chem., **50**, 1157 (1958)] appears to have a value of $f = 0.46$. Unfortunately, no chemical-test methods have yet been devised for determining appropriate f values for a particulate in the laboratory.

Pressure drop in a venturi scrubber is controlled by throat velocity. While some venturis have fixed throats, many are designed with variable louvers to change throat dimensions and control performance for changes in gas flow. Pressure-drop equations have been developed by Calvert (R-13, R-14, R-15), Boll [Ind. Eng. Chem. Fundam., **12**, 40 (1973)], and Hesketh [J. Air Pollut. Control Assoc., **24**, 939 (1974)]. Hollands and Goel [Ind. Eng. Chem. Fundam., **14**, 16 (1975)] have developed a generalized pressure-drop equation.

The Hesketh equation is empirical and is based upon a regression analysis of data from a number of industrial venturi scrubbers:

$$\Delta P = U_{gt}^2 \rho_g A_t^{0.155} L^{0.78} / 1270 \quad (14-234)$$

where ΔP is the pressure drop, in of water; U_{gt} is the gas velocity in the throat, ft/s; ρ_g is the gas density, lb/ft³; A_t is the throat area, ft²; and L is the liquid-to-gas ratio, gal/1000 acf.

Calvert (R-15) critiqued the many pressure-drop equations and suggested the following simplified equation as accurate to ± 10 percent:

$$\Delta P = \frac{2\rho_l U_g^2}{981g_c} \left(\frac{Q_l}{Q_g} \right) [1 - x^2 + \sqrt{(x^4 - x^2)^{0.5}}] \quad (14-235)$$

where

$$x = (3l_i C_{Di} \rho_g / 16d_l \rho_l) + 1 \quad (14-236)$$

ΔP is the pressure drop, cm of water; ρ_l and ρ_g are the density of the scrubbing liquid and gas respectively, g/cm³; U_g is the velocity of the gas at the throat inlet, cm/s; Q_l/Q_g is the volumetric ratio of liquid to gas at the throat inlet, dimensionless; l_i is the length of the throat, cm; C_{Di} is the drag coefficient, dimensionless, for the mean liquid diameter, evaluated at the throat inlet; and d_l is the Sauter mean diameter, cm, for the atomized liquid. The atomized-liquid mean diameter must be evaluated by the Nukiyama and Tanasawa [Trans. Soc. Mech. Eng. (Japan), **4**, **5**, **6** (1937–1940)] equation:

$$d_l = \frac{0.0585}{U_g} \left(\frac{\sigma_l}{\rho_l} \right)^{0.5} + 0.0597 \left[\frac{\mu_l}{(\sigma_l \rho_l)^{0.5}} \right]^{0.45} \left(\frac{Q_l}{Q_g} \right)^{1.5} \quad (14-237)$$

where σ_l is the liquid surface tension, dyn/cm; and μ_l is the liquid viscosity; P . The drag coefficient C_{Di} should be evaluated by the Dickinson and Marshall [Am. Inst. Chem. Eng. J., **14**, 541 (1968)] correlation $C_{Di} = 0.22 + (24/N_{Re})^{0.6} (1 + 0.15 N_{Re}^{0.6})$. The Reynolds number, N_{Re} , is evaluated at the throat inlet considerations as $d_l C_{Di} \rho_g$.

All venturi scrubbers must be followed by an entrainment collector for the liquid spray. These collectors are usually centrifugal and will have an additional pressure drop of several centimeters of water, which must be added to that of the venturi itself.

Other Scrubbers A liquid-ejector venturi (Fig. 17-49), in which high-pressure water from a jet induces the flow of gas, has been used to collect mist particles in the 1- to 2- μm range, but submicrometer particles will generally pass through an eductor. Power costs for liquid pumping are high if appreciable motive force must be imparted to the gas because jet-pump efficiency is usually less than 10 percent. Harris [Chem. Eng. Prog., **42**(4), 55 (1966)] has described their application. Two-phase eductors have been considerably more successful on capture of submicrometer mist particles and could be attractive in situations in which large quantities of waste thermal energy are available. However, the equivalent energy consumption is equal to that required for high-energy venturi scrubbers, and such devices are likely to be no more attractive than venturi scrubbers when the thermal energy is priced at its proper value. Sparks [J. Air Pollut. Control Assoc., **24**, 958 (1974)] has discussed steam ejectors giving 99 percent collection of particles 0.3 to 10 μm . Energy requirements were 311,000 J/m³ (8.25 Btu/scf). Gardener [J. Air Pollut. Control Assoc., **24**, 954 (1974)] operated a liquid eductor with high-pressure (6900- to 27,600-kPa) (1000- to 4000-lbf/in²) hot water heated to 200°C (392°F) which flashed into two phases as it issued from the jet. He obtained 95 to 99 percent collection of submicrometer particulate. Figure 14-128 shows the water-to-gas ratio required as a function of particle size to achieve 99 percent collection.

Effect of Gas Saturation in Scrubbing If hot unsaturated gas is introduced into a wet scrubber, spray particles will evaporate to cool and saturate the gas. The evaporating liquid molecules moving away from the target droplets will repel particles which might collide with them. This results in the forces of diffusio-phoresis opposing particle collection. Semrau and Witham (Air Pollut. Control Assoc. Prepr. 75-30.1) investigated temperature parameters in wet scrubbing and found a definite decrease in the efficiency of evaporative scrubbers and an enhancement of efficiency when a hot saturated gas is scrubbed with cold water rather than recirculated hot water. Little improvement was experienced in cooling a hot saturated gas below a 50°C dew point.

Energy Requirements for Inertial-Impaction Efficiency Semrau [J. Air Pollut. Control Assoc., **13**, 587 (1963)] proposed a "contacting-power" principle which states that the collecting efficiency of a given size of particle is proportional to the power expended and that the smaller the particle, the greater the power required.

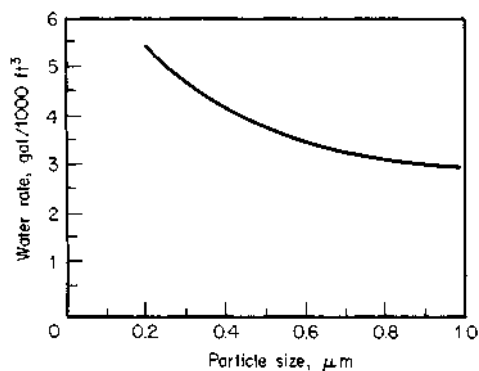


FIG. 14-128 Superheated high-pressure hot-water requirements for 99 percent collection as a function of particle size in a two-phase eductor jet scrubber. To convert gallons per 1000 cubic feet to cubic meters per 1000 cubic meters, multiply by 0.134. [Gardenier, J. Air Pollut. Control Assoc., **24**, 954 (1974).]

Mathematically expressed, $N_T = \propto P_T^\gamma$, where N_T is the number of particulate transfer units achieved and P_T is the total energy expended within the collection device, including gas and liquid pressure drop and thermal and mechanical energy added in atomizers. N_T is further defined as $N_T = \ln [1/(1 - \eta)]$, where η is the overall fractional collection efficiency. This was intended as a universal principle, but the constants $\propto$ and γ have been found to be functions of the chemical nature of the system and the design of the control device. Others have pointed out that the principle is applicable only when the primary collection mechanism is impaction and direct interception. Calvert (R-10, R-12) has found that plotting particle cut size versus pressure drop (or power expended) as in Fig. 14-129 is a more suitable way to develop a generalized energy-requirement curve for impaction

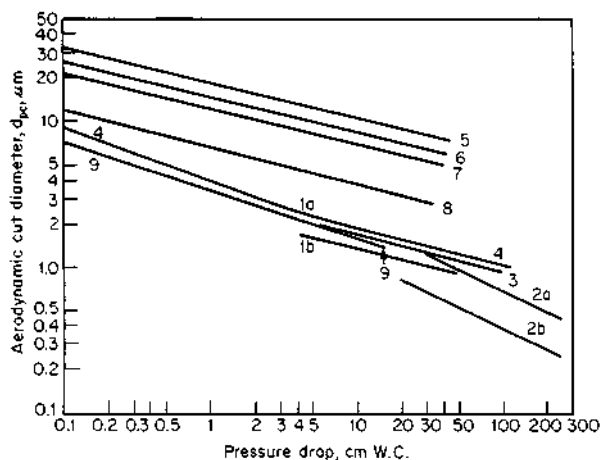


FIG. 14-129 Typical cut diameter as a function of pressure drop for various liquid-particle collectors. Curves 1a and b are single-sieve plates with froth density of 0.4 g/cm^3 ; 1a has sieve holes of 0.5 cm and 1b holes of 0.3 cm. Curves 2a and b are for a venturi scrubber with hydrophobic particles (2a) and hydrophilic particles (2b). Curve 3 is an impingement plate, and curve 4 is a packed column with 2.5-cm-diameter packing. Curve 5 is a zigzag baffle collector with six baffles at $\theta = 30^\circ$. Curve 7 is for six rows of staggered tubes with 1-cm spacing between adjacent tube walls in a row. Curve 8 is similar, except that tube-wall spacing in the row is 0.3 cm. Curve 9 is for wire-mesh pads. To convert grams per cubic centimeter to pounds per cubic foot, multiply by 62.43; to convert centimeters to inches, multiply by 0.394. [Calvert, J. Air Pollut. Control Assoc., **24**, 929 (1974); and Calvert, Yung, and Leung, NTIS Publ. PB-248050, 1975.]

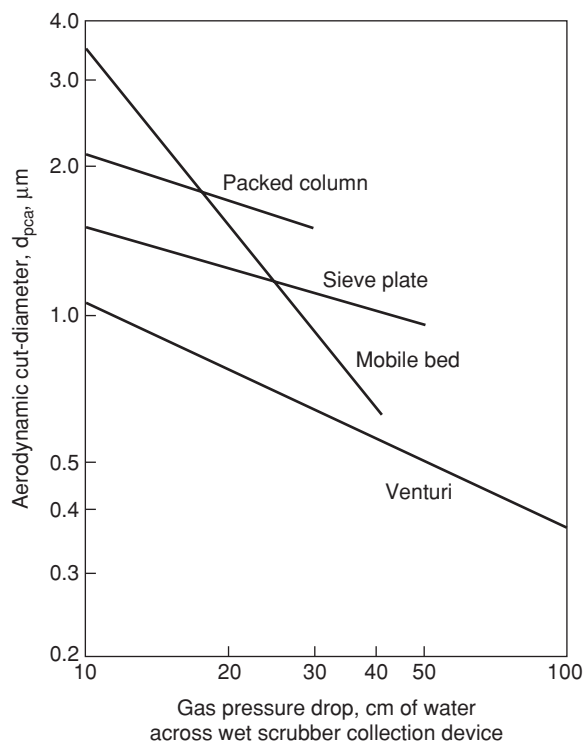


FIG. 14-130 Calvert's refined particle cut-size/power relationship for particle inertial impaction wet collectors. Ref. (R-19) by permission.

devices. The various curves fall close together and outline an imaginary curve that indicates the magnitude of pressure drop required as particle size decreases bound by the two limits of hydrophilic and hydrophobic particles. By calculating the required cut size for a given collection efficiency, Fig. 14-129 can also be used as a guide to deciding between different collection devices.

Subsequently, Calvert (R-19, p. 228) has combined mathematical modeling with performance tests on a variety of industrial scrubbers and has obtained a refinement of the power-input/cut-size relationship as shown in Fig. 14-130. He considers these relationships sufficiently reliable to use this data as a tool for selection of scrubber type and performance prediction. The power input for this figure is based solely on gas pressure drop across the device.

Collection of Fine Mists Inertial-impaction devices previously discussed give high efficiency on particles above $5 \mu\text{m}$ in size and often reasonable efficiency on particles down to $3 \mu\text{m}$ in size at moderate pressure drops. However, this mechanism becomes ineffective for particles smaller than $3 \mu\text{m}$ because of the particle gaslike mobility. Only impaction devices having extremely high energy input such as venturi scrubbers and a flooded mesh pad (the pad interstices really become miniature venturi scrubbers in parallel and in series) can give high collection efficiency on fine particles, defined as 2.5 or $3 \mu\text{m}$ and smaller, including the submicrometer range. Fine particles are subjected to brownian motion in gases, and diffusional deposition can be employed for their collection. Diffusional deposition becomes highly efficient as particles become smaller, especially below 0.2 to $0.3 \mu\text{m}$. Table 14-28 shows typical displacement velocity of particles. Randomly oriented fiber beds having tortuous and narrow gas passages are suitable devices for utilizing this collection mechanism. (The diffusional collection mechanism is discussed in Sec. 17 under "Mechanisms of Dust Collection.") Other collection mechanisms which are efficient for fine particles are electrostatic forces and flux forces such as thermophoresis and diffusiophoresis. Particle growth and nucleation methods are also applicable. Efficient collection of fine particles

TABLE 14-28 Brownian Movement of Particles*

Particle diameter, μm	Brownian displacement of particle, $\mu\text{m/s}$
0.1	29.4
0.25	14.2
0.5	8.92
1.0	5.91
2.5	3.58
5.0	2.49
10.0	1.75

*Brink, *Can. J. Chem. Eng.*, **41**, 134 (1963). Based on spherical water particles in air at 21°C and 1 atm.

is important because particles in the range of 2.0 to around 0.2 μm are the ones which penetrate and are deposited in the lung most efficiently. Hence, particles in this range constitute the largest health hazard.

Fiber Mist Eliminators These devices are produced in various configurations. Generally, randomly oriented glass or polypropylene fibers are densely packed between reinforcing screens, producing fiber beds varying in thickness usually from 25 to 75 mm (1 to 3 in), although thicker beds can be produced. Units with efficiencies as high as 99.9 percent on fine particles have been developed (see *Chemical Engineers' Handbook*, 5th ed., p. 18-88). A combination of mechanisms interacts to provide high overall collection efficiency. Particles larger than 2 to 3 μm are collected on the fibers by inertial impaction and direct interception, while small particles are collected by brownian diffusion. When the device is designed to use this latter mechanism as the primary means, efficiency turndown problems are eliminated as collection efficiency by diffusion increases with residence time. Pressure drop through the beds increases with velocity to the first power since the gas flow is laminar. This leads to design capability trade-offs. As pressure drop is reduced and energy is conserved, capital increases because more filtering area is required for the same efficiency.

Three series of fiber mist eliminators are typically available. A spray-catcher series is designed primarily for essentially 100 percent capture of droplets larger than 3 μm . The high-velocity type is designed to give moderately high efficiency on particles down to 1.0 μm as well. Both of these types are usually produced in the form of flat panels of 25- to 50-mm (1- to 2-in) thickness. The high-efficiency type is illustrated in Fig. 14-131. As mist particles are collected, they coalesce into a liquid film which wets the fibers. Liquid is moved horizontally through the bed by the gas drag force and downward by gravity. It drains down the downstream retaining screen to the bottom of the element and is returned to the process through a liquid seal. Table 14-29 gives typical operating characteristics of the three types of collectors. The application of these devices to sulfuric acid plants and other process gases has been discussed by Brink (see *Chemical Engineers' Handbook*, 5th ed., pp. 18-89, 18-90).

Solid particulates are captured as readily as liquids in fiber beds but can rapidly plug the bed if they are insoluble. Fiber beds have frequently been used for mixtures of liquids and soluble solids and with soluble solids in condensing situations. Sufficient solvent (usually water) is atomized into the gas stream entering the collector to irrigate the fiber elements and dissolve the collected particulate. Such fiber beds have been used to collect fine fumes such as ammonium nitrate and ammonium chloride smokes, and oil mists from compressed air.

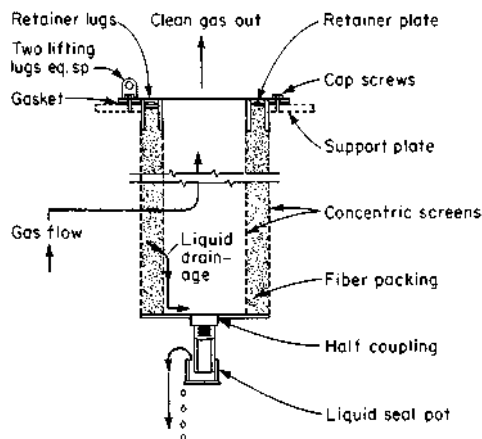


FIG. 14-131 Monsanto high-efficiency fiber-mist-eliminator element. (Monsanto Company.)

Electrostatic Precipitators The principles and operation of electrical precipitators are discussed in Sec. 17 under "Gas-Solids Separations." Precipitators are admirably suited to the collection of fine mists and mixtures of mists and solid particulates. Tube-type precipitators have been used for many years for the collection of acid mists and the removal of tar from coke-oven gas. The first practical installation of a precipitator by Cottrell was made on sulfuric acid mist in 1907. Most older installations of precipitators were tube-type rather than plate-type. However, recently two plate-type wet precipitators employing water sprays or overflowing weirs have been introduced by Mikropul Corporation [Bakke, *J. Air Pollut. Control Assoc.*, **25**, 163 (1975)] and by Fluid Ionics. Such precipitators operate on the principle of making all particles conductive when possible, which increases the particle migration velocity and collection efficiency. Under these conditions, particle dielectric strength becomes a much more important variable, and particles with a low dielectric constant such as condensed hydrocarbon mists become much more difficult to collect than water-wettable particles. Bakke (U.S.-U.S.S.R. Joint Work. Group Symp.: Fine Particle Control, San Francisco, 1974) has developed equations for particle charge and relative collection efficiency in wet precipitators that show the effect of dielectric constant. Wet precipitators can also be used to absorb soluble gases simultaneously by adjusting the pH or the chemical composition of the liquid spray. The presence of the electric field appears to enhance absorption. Wet precipitators have found their greatest usefulness to date in handling mixtures of gaseous pollutants and submicrometer particulate (either liquid or solid, or both) such as fumes from aluminum-pot lines, carbon anode baking, fiber-glass-fume control, coke-oven and metallurgical operations, chemical incineration, and phosphate-fertilizer operations. Two-stage precipitators are used increasingly for moderate-volume gas streams containing nonconductive liquid mists which will drain from the collecting plates. Their application on hydrocarbon mists has been quite successful, but careful attention must be given to fire and explosion hazards.

Electrically Augmented Collectors A new area for enhancing collection efficiency and lowering cost is the combining of electrostatic forces with devices using other collecting mechanisms such as

TABLE 14-29 Operating Characteristics of Various Types of Fiber Mist Eliminators as Used on Sulfuric Acid Plants*

	High efficiency	High velocity	Spray catcher
Controlling mechanism for mist collection	Brownian movement	Impaction	Impaction
Superficial velocity, m/s	0.075-0.20	2.0-2.5	2.0-2.5
Efficiency on particles greater than 3 μm , %	Essentially 100	Essentially 100	Essentially 100
Efficiency on particles 3 μm and smaller, %	95-99+	90-98	15-30
Pressure drop, cm H ₂ O	12-38	15-20	1.0-2.5

*Brink, Burggrabe, and Greenwell, *Chem. Eng. Prog.*, **64**(11), 82 (1968). To convert centimeters to inches, multiply by 0.394.

impaction and diffusion. Cooper (Air Pollut. Control Assoc. Prepr. 75-02.1) evaluated the magnitude of forces operating between charged and uncharged particles and concluded that electrostatic attraction is the strongest collecting force operating on particles finer than 2 μm . Nielsen and Hill [*Ind. Eng. Chem. Fundam.*, **15**, 149 (1976)] have quantified these relationships, and a number of practical devices have been demonstrated. Pilat and Meyer (NTIS Publ. PB-252653, 1976) have demonstrated up to 99 percent collection of fine particles in a two-stage spray tower in which the inlet particles and water spray are charged with opposite polarity. The principle has been applied to retrofitting existing spray towers to enhance collection.

Klugman and Sheppard (Air Pollut. Control Assoc. Prepr. 75-30.3) have developed an ionizing wet scrubber in which the charged mist particles are collected in a grounded, irrigated cross-flow bed of Tellerette packing. Particles smaller than 1 μm have been collected with 98 percent efficiency by using two units in series. Dembinsky and Vicard (Air Pollut. Control Assoc. Prepr. 78-17.6) have used an electrically augmented low-pressure [5 to 10 cm (2 to 4 in) of water] venturi scrubber to give 95 to 98 percent collection efficiency on submicrometer particles.

Particle Growth and Nucleation Fine particles may be subjected to conditions favoring the growth of particles either through condensation or through coalescence. Saturation of a hot gas stream with water, followed by condensation on the particles acting as nuclei when the gas is cooled, can increase particle size and ease of collection. Addition of steam can produce the same results. Scrubbing of the humid gas with a cold liquid can bring diffusiohphoresis into play. The introduction of cold liquid drops causes a reduction in water-vapor pressure at the surface of the cold drop. The resulting vapor-pressure gradient causes a hydrodynamic flow toward the drop known as Stefan flow which enhances the movement of mist particles toward the spray drop. If the molecular mass of the diffusing vapor is different from the carrier gas, this density difference also produces a driving force, and the sum of these forces is known as diffusiohphoresis. A mathematical description of these forces has been presented by Calvert (R-9) and by Sparks and Pilat [*Atmos. Environ.*, **4**, 651 (1970)]. Thermal differences between the carrier gas and the cold scrubbing droplets can further enhance collection through thermophoresis. Calvert and Jhaseri [*J. Air Pollut. Control Assoc.*, **24**, 946 (1974)]; and NTIS Publ. PB-227307, 1973)] have investigated condensation scrubbing in multiple-sieve plate towers.

Submicrometer droplets can be coagulated through brownian diffusion if given ample time. The introduction of particles 50 to 100 times larger in diameter can enhance coagulation, but the addition of a broad range of particle sizes is discouraged. Increasing turbulence will aid coagulation, so fans to stir the gas or narrow, tortuous passages such as those of a packed bed can be beneficial. Sonic energy can also produce coagulation, especially the production of standing waves in the confines of long, narrow tubes. Addition of water and oil mists can sometimes aid sonic coagulation. Sulfuric acid mist [Danser, *Chem. Eng.*, **57**(5), 158 (1950)] and carbon black [Stokes, *Chem. Eng. Prog.*, **46**, 423 (1950)] have been successfully agglomerated with sonic energy. Frequently sonic agglomeration has been unsuccessful because of the high energy requirement. Most sonic generators have very poor energy-transformation efficiency. Wegrzyn et al. (U.S. EPA Publ. EPA-600/7-79-004C, 1979, p. 233) have reviewed acoustic agglomerators. Mednikov (U.S.S.R. Akad. Soc. Moscow, 1963) suggested that the incorporation of sonic agglomeration with electrostatic precipitation could greatly reduce precipitator size.

Other Collectors Tarry particulates and other difficult-to-handle liquids have been collected on a dry, expendable phenol formaldehyde-bonded glass-fiber mat (Goldfield, *J. Air Pollut. Control Assoc.*, **20**, 466 (1970)) in roll form which is advanced intermittently into a filter frame. Superficial gas velocities are 2.5 to 3.5 m/s (8.2 to 11.5 ft/s), and pressure drop is typically 41 to 46 cm (16 to 18 in) of water. Collection efficiencies of 99 percent have been obtained on submicrometer particles. Brady [*Chem. Eng. Prog.*, **73**(8), 45 (1977)] has discussed a cleanable modification of this approach in which the gas is passed through a reticulated foam filter that is slowly rotated and solvent-cleaned.

In collecting very fine (mainly submicron) mists of a hazardous nature where one of the collectors previously discussed has been used

as the primary one (fiber-mist eliminators of the Brownian diffusion type and electrically augmented collectors are primarily recommended), there is the chance that the effluent concentration may still be too high for atmospheric release when residual concentration must be in the range of 1–2 μm . In such situations, secondary treatment may be needed. Probably removal of the residual mist by adsorption will be in order. See "Adsorption," Sec. 16. Another possibility might be treatment of the remaining gas by membrane separation. A separator having a gas-permeable membrane that is essentially nonliquid-permeable could be useful. However, if the gas-flow volumes are appreciable, the device could be expensive. Most membranes have low capacity (requiring high membrane surface area) to handle high gas-permeation capacity. See "Membrane Separation Processes," Sec. 20.

Continuous Phase Uncertain Some situations exist such as in two-phase gas-liquid flow where the volume of the liquid phase may approach being equal to the volume of the vapor phase, and where it may be difficult to be sure which phase is the continuous phase. Svrcek and Monnery [*Chem. Eng. Prog.*, **89**(10), 53–60 (Oct. 1993)] have discussed the design of two-phase separation in a tank with gas-liquid separation in the middle, mist elimination in the top, and entrained gas-bubble removal from the liquid in the bottom. Monnery and Svrcek [*Chem. Eng. Prog.*, **90**(9), 29–40 (Sept. 1994)] have expanded the separation to include multiphase flow, where the components are a vapor and two immiscible liquids and these are also separated in a tank. A design approach for sizing the gas-liquid disengaging space in the vessel is given using a tangential tank inlet nozzle, followed by a wire mesh mist eliminator in the top of the vessel for final separation of entrained mist from the vapor. Design approaches and equations are also given for sizing the lower portion of the vessel for separation of the two immiscible liquid phases by settling and separation of discontinuous liquid droplets from the continuous liquid phase.

LIQUID-PHASE CONTINUOUS SYSTEMS

Practical separation techniques for gases dispersed in liquids are discussed. Processes and methods for dispersing gas in liquid have been discussed earlier in this section, together with information for predicting the bubble size produced. Gas-in-liquid dispersions are also produced in chemical reactions and electrochemical cells in which a gas is liberated. Such dispersions are likely to be much finer than those produced by the dispersion of a gas. Dispersions may also be unintentionally created in the vaporization of a liquid.

GENERAL REFERENCES: Adamson, *Physical Chemistry of Surfaces*, 4th ed., Wiley, New York, 1982. Akers, *Foams*, Academic, New York, 1976. Bikerman, *Foams*, Springer-Verlag, New York, 1973. Bikerman, et al., *Foams: Theory and Industrial Applications*, Reinhold, New York, 1953. Chermisinoff, ed., *Encyclopedia of Fluid Mechanics*, vol. 3, Gulf Publishing, Houston, 1986. Kerner, *Foam Control Agents*, Noyes Data Corp., Park Ridge, NJ, 1976. Rubel, *Antifoaming and Defoaming Agents*, Noyes Data Corp., Park Ridge, NJ, 1972. Rosen, *Surfactants and Interfacial Phenomena*, 2d ed., Wiley, New York, 1989. Sonntag and Streng, *Coagulation and Stability of Disperse Systems*, Halsted-Wiley, New York, 1972. Wilson, ed., *Foams: Physics, Chemistry and Structure*, Springer-Verlag, London, 1989. "Defoamers" and "Foams," *Encyclopedia of Chemical Technology*, 4th ed., vols. 7, 11, Wiley, New York, 1993–1994.

Types of Gas-in-Liquid Dispersions Two types of dispersions exist. In one, gas bubbles produce an unstable dispersion which separates readily under the influence of gravity once the mixture has been removed from the influence of the dispersing force. Gas-liquid contacting means such as bubble towers and gas-dispersing agitators are typical examples of equipment producing such dispersions. More difficulties may result in separation when the gas is dispersed in the form of bubbles only a few micrometers in size. An example is the evolution of gas from a liquid in which it has been dissolved or released through chemical reaction such as electrolysis. Coalescence of the dispersed phase can be helpful in such circumstances.

The second type is a stable dispersion, or foam. Separation can be extremely difficult in some cases. A pure two-component system of gas and liquid cannot produce dispersions of the second type. Stable foams can be produced only when an additional substance is adsorbed

TABLE 14-30 Terminal Velocity of Standard Air Bubbles Rising in Water at 20°C*

Bubble diameter, μm	10	30	50	100	200	300
Terminal velocity, mm/s	0.061	0.488	1.433	5.486	21.95	49.38

*Calculated from Stokes' law. To convert millimeters per second to feet per second, multiply by 0.003281.

at the liquid-surface interface. The substance adsorbed may be in true solution but with a chemical tendency to concentrate in the interface such as that of a surface-active agent, or it may be a finely divided solid which concentrates in the interface because it is only poorly wetted by the liquid. Surfactants and proteins are examples of soluble materials, while dust particles and extraneous dirt including traces of nonmiscible liquids can be examples of poorly wetted materials.

Separation of gases and liquids always involves coalescence, but enhancement of the rate of coalescence may be required only in difficult separations.

Separation of Unstable Systems The buoyancy of bubbles suspended in liquid can frequently be depended upon to cause the bubbles to rise to the surface and separate. This is a special case of gravity settling. The mixture is allowed to stand at rest or is moved along a flow path in laminar flow until the bubbles have surfaced. Table 14-30 shows the calculated rate of rise of air bubbles at atmospheric pressure in water at 20°C (68°F) as a function of diameter. It will be observed that the velocity of rise for 10- μm bubbles is very low, so that long separating times would be required for gas which is more finely dispersed.

For liquids other than water, the rise velocity can be approximated from Table 14-30 by multiplying by the liquid's specific gravity and the reciprocal of its viscosity (in centipoises). For bubbles larger than 100 μm , this procedure is erroneous, but the error is less than 15 percent for bubbles up to 1000 μm . More serious is the underlying assumption of Table 14-30 that the bubbles are rigid spheres. Circulation within the bubble causes notable increases in velocity in the range of 100 μm to 1 mm, and the flattening of bubbles 1 cm and larger appreciably decreases their velocity. However, in this latter size range the velocity is so high as to make separation a trivial problem.

In design of separating chambers, static vessels or continuous-flow tanks may be used. Care must be taken to protect the flow from turbulence, which could cause back mixing of partially separated fluids or which could carry unseparated liquids rapidly to the separated-liquid outlet. Vertical baffles to protect rising bubbles from flow currents are sometimes employed. Unseparated fluids should be distributed to the separating region as uniformly and with as little velocity as possible. When the bubble rise velocity is quite low, shallow tanks or flow channels should be used to minimize the residence time required.

Quite low velocity rise of bubbles due either to small bubble size or to high liquid viscosity can cause difficult situations. With low-viscosity liquids, separation-enhancing possibilities in addition to those previously enumerated are to sparge the liquid with large-diameter gas bubbles or to atomize the mixture as a spray into a tower. Large gas bubbles rising rapidly through the liquid collide with small bubbles and aid their coalescence through capture. Atomizing of the continuous phase reduces the distance that small gas bubbles must travel to reach a gas interface. Evacuation of the spray space can also be beneficial in promoting small-bubble growth and especially in promoting gas evolution when the gas has appreciable liquid solubility. Liquid heating will also reduce solubility.

Surfaces in the settling zone for bubble coalescence such as closely spaced vertical or inclined plates or tubes are beneficial. When clean low-viscosity fluids are involved, passage of the undegassed liquid through a tightly packed pad of mesh or fine fibers at low velocity will result in efficient bubble coalescence. Problems have been experienced in degassing a water-based organic solution that has been passed through an electrolytic cell for chemical reaction in which extremely fine bubbles of hydrogen gas are produced in the liquid within the cell. Near-total removal of hydrogen gas from the liquid is needed for process safety. This is extremely difficult to achieve by gravity settling alone because of the fine bubble size and the need for a coalescing surface. Utilization of a fine fiber media is strongly recommended in such situations. A low-forward liquid flow through the

media is desirable to provide time for the bubbles to attach themselves to the fiber media through Brownian diffusion. Spielman and Goren [*Ind. Eng. Chem.*, **62**(10), (1970)] reviewed the literature on coalescence with porous media and reported their own experimental results [*Ind. Eng. Chem. Fundam.*, **11**(1), 73 (1972)] on the coalescence of oil-water liquid emulsions. The principles are applicable to a gas-in-liquid system. Glass-fiber mats composed of 3.5-, 6-, or 12- μm diameter fibers, varying in thickness from 1.3 to 3.3 mm, successfully coalesced and separated 1- to 7- μm oil droplets at superficial bed velocities of 0.02 to 1.5 cm/s (0.0067 to 0.049 ft/s).

In the deaeration of high-viscosity fluids such as polymers, the material is flowed in thin sheets along solid surfaces. Vacuum is applied to increase bubble size and hasten separation. The Versator (Cornell Machine Co.) degasses viscous liquids by spreading them into a thin film by centrifugal action as the liquids flow through an evacuated rotating bowl.

Separation of Foam Foam is a colloidal system containing relatively large volumes of dispersed gas in a relatively small volume of liquid. Foams are thermodynamically unstable with respect to separation into their components of gas and vapor, and appreciable surface energy is released in the bursting of foam bubbles. Foams are dynamic systems in which a third component produces a surface layer that is different in composition from the bulk of the liquid phase. The stabilizing effect of such components (often present only in trace amounts) can produce foams of troubling persistence in many operations. (Foams which have lasted for years when left undisturbed have been produced.) Bendure [*TAPPI*, **58**(2), 83 (1975)], Keszthelyi [*J. Paint Technol.*, **46**(11), 31 (1974)], Ahmad [*Sep. Sci.*, **10**, 649 (1975)], and Shedlovsky ("Foams," *Encyclopedia of Chemical Technology*, 2d ed., Wiley, New York, 1966) have presented concise articles on the characteristics and properties of foams in addition to the general references cited at the beginning of this subsection.

Foams can be a severe problem in chemical-processing steps involving gas-liquid interaction such as distillation, absorption, evaporation, chemical reaction, and particle separation and settling. It can also be a major problem in pulp and paper manufacture, oil-well drilling fluids, production of water-based paints, utilization of lubricants and hydraulic fluids, dyeing and sizing of textiles, operation of steam boilers, fermentation operations, polymerization, wet-process phosphoric acid concentration, adhesive production, and foam control in products such as detergents, waxes, printing inks, instant coffee, and glycol antifreeze.

Foams, as freshly generated, are gas emulsions with spherical bubbles separated by liquid films up to a few millimeters in thickness. They age rapidly by liquid drainage and form polyhedrals in which three bubbles intersect at corners with angles of approximately 120°. During drainage, the lamellae become increasingly thinner, especially in the center (only a few micrometers thickness), and more brittle. This feature indicates that with some foams if a foam layer can be tolerated, it may be self-limiting, as fresh foam is added to the bottom of the layer with drained foam collapsing on the top. (A quick-breaking foam may reach its maximum life cycle in 6 s. A moderately stable foam can persist for 140 s.) During drainage, gas from small foam bubbles, which is at a high pressure, will diffuse into large bubbles so that foam micelles increase with time. As drainage proceeds, weak areas in the lamella may develop. However, the presence of a higher concentration of surfactants in the surface produces a lower surface tension. As the lamella starts to fail, exposing bulk liquid with higher surface tension, the surface is renewed and healed. This is known as the *Marangoni effect*. If drainage can occur faster than Marangoni healing, a hole may develop in the lamella. The forces involved are such that collapse will occur in milliseconds without concern for rupture propagation. However, in very stable foams, electrostatic surface forces (zeta potential) prevent complete drainage and collapse. In

some cases, stable lamella thicknesses of only two molecules have been measured.

Drainage rate is influenced by surface viscosity, which is very temperature-sensitive. At a critical temperature, which is a function of the system, a temperature change of only a few degrees can change a slow-draining foam to a fast-draining foam. This change in drainage rate can be a factor of 100 or more; thus increasing the temperature of foam can cause its destruction. An increase in temperature may also cause liquid evaporation and lamella thinning. As the lamellae become thinner, they become more brittle and fragile. Thus, mechanical deformation or pressure changes, which cause a change in gas-bubble volume, can also cause rupture.

Bendure indicates 10 ways to increase foam stability: (1) increase bulk liquid viscosity, (2) increase surface viscosity, (3) maintain thick walls (higher liquid-to-gas ratio), (4) reduce liquid surface tension, (5) increase surface elasticity, (6) increase surface concentration, (7) reduce surfactant-adsorption rate, (8) prevent liquid evaporation, (9) avoid mechanical stresses, and (10) eliminate foam inhibitors. Obviously, the reverse of each of these actions, when possible, is a way to control and break foam.

Physical Defoaming Techniques Typical physical defoaming techniques include mechanical methods for producing foam stress, thermal methods involving heating or cooling, and electrical methods. Combinations of these methods may also be employed, or they may be used in conjunction with chemical defoamers. Some methods are only moderately successful when conditions are present to reform the foam such as breaking foam on the surface of boiling liquids. In some cases it may be desirable to draw the foam off and treat it separately. Foam can always be stopped by removing the energy source creating it, but this is often impractical.

Thermal Methods Heating is often a suitable means of destroying foam. As indicated previously, raising the foam above a critical temperature (which must be determined experimentally) can greatly decrease the surface viscosity of the film and change the foam from a slow-draining to a fast-draining foam. Coupling such heating with a mechanical force such as a revolving paddle to cause foam deformation is frequently successful. Other effects of heating are expansion of the gas in the foam bubbles, which increases strain on the lamella walls as well as requiring their movement and flexing. Evaporation of solvent may occur causing thinning of the walls. At sufficiently high temperatures, desorption or decomposition of stabilizing substances may occur. Placing a high-temperature bank of steam coils at the maximum foam level is one control method. As the foam approaches or touches the coil, it collapses. The designer should consider the fact that the coil will frequently become coated with solute.

Application of radiant heat to a foam surface is also practiced. Depending on the situation, the radiant source may be electric lamps, Glowbar units, or gas-fired radiant burners. Hot gases from burners will enhance film drying of the foam. Heat may also be applied by jetting or spraying hot water on the foam. This is a combination of methods since the jetting produces mechanical shear, and the water itself provides dilution and change in foam-film composition. Newer approaches might include foam heating with the application of focused microwaves. This could be coupled with continuous or intermittent pressure fluctuations to stress lamella walls as the foam ages.

Cooling can also destroy foam if it is carried to the point of freezing since the formation of solvent crystals destroys the foam structure. Less drastic cooling such as spraying a hot foam with cold water may be effective. Cooling will reduce the gas pressure in the foam bubbles and may cause them to shrink. This is coupled with the effects of shear and dilution mentioned earlier. In general, moderate cooling will be less effective than heating since the surface viscosity is being modified in the direction of a more stable foam.

Mechanical Methods Static or rotating breaker bars or slowly revolving paddles are sometimes successful. Their application in conjunction with other methods is frequently better. As indicated in the theory of foams, they will work better if installed at a level at which the foam has had some time to age and drain. A rotating breaker works by deforming the foam, which causes rupture of the lamella walls. Rapidly moving slingers will throw the foam against the vessel wall and may cause impact on other foam outside the envelope of the

slinger. In some instances, stationary bars or closely spaced plates will limit the rise of foam. The action here is primarily one of providing surface for coalescence of the foam. Wettability of the surface, whether moving or stationary, is frequently important. Usually a surface not wetted by the liquid is superior, just as is frequently the case of porous media for foam coalescence. However, in both cases there are exceptions for which wettable surfaces are preferred. Shkodin [*Kolloidn. Zh.*, **14**, 213 (1952)] found molasses foam to be destroyed by contact with a wax-coated rod and unaffected by a clean glass rod.

Goldberg and Rubin [*Ind. Eng. Chem. Process Des. Dev.*, **6** 195 (1967)] showed in tests with a disk spinning vertically to the foam layer that most mechanical procedures, whether centrifugation, mixing, or blowing through nozzles, consist basically of the application of shear stress. Subjecting foam to an air-jet impact can also provide a source of drying and evaporation from the film, especially if the air is heated. Other effective means of destroying bubbles are to lower a frame of metal points periodically into the foam or to shower the foam with falling solid particles.

Pressure and Acoustic Vibrations These methods for rupturing foam are really special forms of mechanical treatment. Change in pressure in the vessel containing the foam stresses the lamella walls by expanding or contracting the gas inside the foam bubbles. Oscillation of the vessel pressure subjects the foam to repeated film flexing. Parlow [*Zucker*, **3**, 468 (1950)] controlled foam in sugar-sirup evaporators with high-frequency air pulses. It is by no means certain that high-frequency pulsing is necessary in all cases. Lower frequency and higher amplitude could be equally beneficial. Acoustic vibration is a similar phenomenon causing localized pressure oscillation by using sound waves. Impulses at 6 kHz have been found to break froth from coal flotation [Sun, *Min. Eng.*, **3**, 865 (1958)]. Sonntag and Strenge (*Coagulation and Stability of Disperse Systems*, Halsted-Wiley, New York, 1972, p. 121) report foam suppression with high-intensity sound waves (11 kHz, 150 dB) but indicate that the procedure is too expensive for large-scale application. The Sontrifuge (Teknika Inc., a subsidiary of Chemineer, Inc.) is a commercially available low-speed centrifuge employing sonic energy to break the foam. Walsh [*Chem. Process.*, **29**, 91 (1966)], Carlson [*Pap. Trade J.*, **151**, 38 (1967)], and Thorhildsen and Rich [*TAPPI*, **49**, 95A (1966)] have described the unit.

Electrical Methods As colloids, most foams typically have electrical double layers of charged ions which contribute to foam stability. Accordingly, foams can be broken by the influence of an external electric field. While few commercial applications have been developed, Sonntag and Strenge (op. cit., p. 114) indicate that foams can be broken by passage through devices much like electrostatic precipitators for dusts. Devices similar to two-stage precipitators having closely spaced plates of opposite polarity should be especially useful. Sonntag and Strenge, in experiments with liquid-liquid emulsions, indicate that the colloid structure can be broken at a field strength of the order of 8 to 9×10^5 V/cm.

Chemical Defoaming Techniques Sonntag and Strenge (op. cit., p. 111) indicate two chemical methods for foam breaking. One method is causing the stabilizing substances to be desorbed from the interface, such as by displacement with other more surface-active but nonstabilizing compounds. Heat may also cause desorption. The second method is to carry on chemical changes in the adsorption layer, leading to a new structure. Some defoamers may act purely by mechanical means but will be discussed in this subsection since their action is generally considered to be chemical in nature. Often chemical defoamers act in more than one way.

Chemical Defoamers The addition of chemical foam breakers is the most elegant way to break a foam. Effective defoamers cause very rapid disintegration of the foam and frequently need be present only in parts per million. The great diversity of compounds used for defoamers and the many different systems in which they are applied make a brief and orderly discussion of their selection difficult. Compounds needed to break aqueous foams may be different from those needed for aqueous-free systems. The majority of defoamers are insoluble or nonmiscible in the foam continuous phase, but some work best because of their ready solubility. Lichtman (*Defoamers*, 3d ed., Wiley, New York, 1979) has presented a concise summary of the application and use of defoamers. Rubel (*Antifoaming and Defoaming*

TABLE 14-31 Major Types and Applications of Defoamers

Classification	Examples	Applications
Silicones	Dimethyl silicone, trialkyl and tetraalkyl silanes	Lubricating oils; distillation; fermentation; jam and wine making; food processing
Aliphatic acids or esters	Mostly high-molecular-weight compounds; diethyl phthalate; lauric acid	Papermaking; wood-pulp suspensions; water-based paints; food processing
Alcohols	Moderate- to high-molecular-weight monohydric and polyhydric alcohols; octyl alcohol; C-12 to C-20 alcohols; lauryl alcohol	Distillation; fermentation; papermaking; glues and adhesives
Sulfates or sulfonates	Alkali metal salts of sulfated alcohols, sulfonic acid salts; alkyl-aryl sulfonates; sodium lauryl sulfate	Nonaqueous systems; mixed aqueous and nonaqueous systems; oil-well drilling muds; spent H_2SO_4 recovery; deep-fat frying
Amines or amides	Alkyl amines (undecyloctyl and diamyl methyl amine); polyamides (acyl derivatives of piperazine)	Boiler foam; sewage foam; fermentation; dye baths
Halogenated compounds	Fluochloro hydrocarbons with 5 to 50 C atoms; chlorinated hydrocarbons	Lubrication-oil and grease distillation; vegetable-protein glues
Natural products	Vegetable oils; waxes, mineral oils plus their sulfated derivatives (including those of animal oils and fats)	Sugar extraction; glue manufacture; cutting oils
Fatty-acid soaps	Alkali, alkaline earth, and other metal soaps; sodium stearate; aluminum stearate	Gear oils; paper stock; paper sizing; glue solutions
Inorganic compounds	Monosodium phosphate mixed with boric acid and ethyl carbonate, disodium phosphate; sodium aluminate, bentonite and other solids	Distillation; instant coffee; boiler feedwater; sugar extraction
Phosphates	Alkyl-alkalene diphosphates; tributyl phosphate in isopropanol	Petroleum-oil systems; foam control in soap solutions
Hydrophobic silica	Finely divided silica in polydimethyl siloxane	Aqueous foaming systems
Sulfides or thio derivatives	Metallic derivatives of thio ethers and disulfides, usually mixed with organic phosphite esters; long-chain alkyl thienyl ketones	Lubricating oils; boiler water

Agents, Noyes Data Corp., Park Ridge, N.J., 1972) has reviewed the extensive patent literature on defoamers. Defoamers are also discussed extensively in the general references at the beginning of this subsection.

One useful method of aqueous defoaming is to add a nonfoam stabilizing surfactant which is more surface-active than the stabilizing substance in the foam. Thus a foam stabilized with an ionic surfactant can be broken by the addition of a very surface-active but nonstabilizing silicone oil. The silicone displaces the foam stabilizer from the interface by virtue of its insolubility. However, it does not stabilize the foam because its foam films have poor elasticity and rupture easily.

A major requirement for a defoamer is cost-effectiveness. Accordingly, some useful characteristics are low volatility (to prevent stripping from the system before it is dispersed and does its work), ease of dispersion and strong spreading power, and surface attraction-orientation. Chemical defoamers must also be selected in regard to their possible effect on product quality and their environmental and health suitability. For instance, silicone antifoam agents are effective in textile jet dyeing but reduce the fire retardancy of the fabric. Mineral-oil defoamers in sugar evaporation have been replaced by specifically approved materials. The tendency is no longer to use a single defoamer compound but to use a formulation specially tailored for the application comprising carriers, secondary antifoam agents, emulsifiers, and stabilizing agents in addition to the primary defoamer. Carriers, usually hydrocarbon oils or water, serve as the vehicle to support the release and spread of the primary defoamer. Secondary defoamers may provide a synergistic effect for the primary defoamer or modify its properties such as spreadability or solubility. Emulsifiers may enhance the speed of dispersion, while stabilizing agents may enhance defoamer stability or shelf life.

Hydrophobic silica defoamers work on a basis which may not be chemical at all. They are basically finely divided solid silica particles dispersed in a hydrocarbon or silicone oil which serves as a spreading vehicle. Kulkarni [*Ind. Eng. Chem. Fundam.*, **16**, 472 (1977)] theorizes that this mixture defoams by the penetration of the silica particle into the bubble and the rupture of the wall. Table 14-31 lists major types of defoamers and typical applications.

Other Chemical Methods These methods rely chiefly on destroying the foam stabilizer or neutralizing its effect through methods other than displacement and are applicable when the process will permit changing the chemical environment. Forms stabilized with alkali esters can be broken by acidification since the equivalent free

acids do not stabilize foam. Foams containing sulfated and sulfonated ionic detergents can be broken with the addition of fatty-acid soaps and calcium salts. Several theories have been proposed. One suggests that the surfactant is tied up in the foam as double calcium salts of both the sulfonate and the soap. Another suggests that calcium soaps oriented in the film render it inelastic.

Ionic surfactants adsorb at the foam interface and orient with the charged group immersed in the lamellae and their uncharged tails pointed into the gas stream. As the film drains, the charged groups, which repel each other, tend to be moved more closely together. The repulsive force between like charges hinders drainage and stabilizes the film. Addition of a salt or an electrolyte to the foam screens the repulsive effect, permits additional drainage, and can reduce foam stability.

Foam Prevention Chemical prevention of foam differs from defoaming only in that compounds or mixtures are added to a stream prior to processing to prevent the formation of foam either during processing or during customer use. Such additives, sometimes distinguished as antifoam agents, are usually in the same chemical class of materials as defoamers. However, they are usually specifically formulated for the application. Typical examples of products formulated with antifoam agents are laundry detergents (to control excess foaming), automotive antifreeze, instant coffee, and jet-aircraft fuel. Foaming in some chemical processes such as distillation or evaporation may be due to trace impurities such as surface-active agents. An alternative to antifoam agents is their removal before processing such as by treatment with activated carbon [Pool, *Chem. Process.*, **21**(9), 56 (1958)].

Automatic Foam Control In processing materials when foam can accumulate, it is often desirable to measure the height of the foam layer continuously and to dispense defoamer automatically as required to control the foam. Other corrective action can also be taken automatically. Methods of sensing the foam level have included electrodes in which the electrical circuit is completed when the foam touches the electrode [Nelson, *Ind. Eng. Chem.*, **48**, 2183 (1956); and Browne, U.S. Patent 2,981,693, 1961], floats designed to rise in a foam layer [Carter, U.S. Patent 3,154,577, 1964], and change in power input required to turn a foam-breaking impeller as the foam level rises [Yamashita, U.S. Patent 3,317,435, 1967]. Timers to control the duration of defoamer addition have also been used. Browne has suggested automatic addition of defoamer through a porous wick when the foam level reaches the level of the wick. Foam control has also been discussed by Kroll [*Ind. Eng. Chem.*, **48**, 2190 (1956)].

SECTION 15

Liquid-Liquid
Extraction
and Other
Liquid-Liquid
Operations and
Equipment

PERRY'S CHEMICAL ENGINEERS' HANDBOOK

8TH EDITION



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Liquid-Liquid Extraction and Other Liquid-Liquid Operations and Equipment*

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Nomenclature

A given symbol may represent more than one property. The appropriate meaning should be apparent from the context. The equations given in Sec. 15 reflect the use of the SI or cgs system of units and not ft-lb-s units, unless otherwise noted in the text. The gravitational conversion factor g_c needed to use ft-lb-s units is not included in the equations.

Symbol	Definition	SI units	U.S. Customary System units	Symbol	Definition	SI units	U.S. Customary System units
a	Interfacial area per unit volume	m^2/m^3	ft^2/ft^3	$\mathcal{E}^C$	Extraction factor for case C [Eq. (15-98)]	Dimensionless	Dimensionless
a_p	Specific packing surface area (area per unit volume)	m^2/m^3	ft^2/ft^3	$\mathcal{E}_i$	Extraction factor for component i	Dimensionless	Dimensionless
a_w	Specific wall surface area (area per unit volume)	m^2/m^3	ft^2/ft^3	$\mathcal{E}_s$	Stripping section extraction factor	Dimensionless	Dimensionless
b_{ij}	NRTL model regression parameter (see Table 15-10)	K	K	$\mathcal{E}_w$	Washing section extraction factor	Dimensionless	Dimensionless
A	Envelope-style downcomer area	m^2	ft^2	f_{da}	Fractional downcomer area in Eq. (15-160)	Dimensionless	Dimensionless
A	Area between settled layers in a decanter	m^2	ft^2	f_{ha}	Fractional hole area in Eq. (15-159)	Dimensionless	Dimensionless
A_{col}	Column cross-sectional area	m^2	ft^2	F	Mass or mass flow rate of feed phase	kg or kg/s	lb or lb/h
A_{dow}	Area for flow through a downcomer (or upcomer)	m^2	ft^2	F	Force	N	lb _f
A_{ij}/RT	van Laar binary interaction parameter	Dimensionless	Dimensionless	F'	Feed mass or mass flow rate (feed solvent only)	kg or kg/s	lb or lb/h
A_o	Cross-sectional area of a single hole	m^2	in^2	F_R	Solute reduction factor (ratio of inlet to outlet concentrations)	Dimensionless	Dimensionless
C	Concentration (mass or mol per unit volume)	kg/m^3 or kgmol/m^3 or gmol/L	lb/ft^3 or lbmol/ft^3	g	Gravitational acceleration	9.807 m/s^2	32.17 ft/s^2
C_A^i	Concentration of component A at the interface	kg/m^3 or kgmol/m^3 or gmol/L	lb/ft^3 or lbmol/ft^3	G_{ij}	NRTL model parameter	Dimensionless	Dimensionless
C^o	Concentration at equilibrium	kg/m^3 or kgmol/m^3 or gmol/L	lb/ft^3 or lbmol/ft^3	h	Height of coalesced layer at a sieve tray	m	in
C_D	Drag coefficient	Dimensionless	Dimensionless	h	Head loss due to frictional flow	m	in
C_o	Initial concentration	kg/m^3 or kgmol/m^3 or gmol/L	lb/ft^3 or lbmol/ft^3	h	Height of dispersion band in batch decanter	m	in
C_t	Concentration at time t	kg/m^3 or kgmol/m^3 or gmol/L	lb/ft^3 or lbmol/ft^3	h_i^E	Excess enthalpy of mixing	J/gmol	Btu/lbmol or cal/gmol
d	Drop diameter	m	in	H	Dimensionless group defined by Eq. (15-123)	Dimensionless	Dimensionless
d_C	Critical packing dimension	m	in	H	Dimension of envelope-style downcomer (Fig. 15-39)	m	in or ft
d_i	Diameter of an individual drop	m	in	ΔH	Steady-state dispersion band height in a continuously fed decanter	m	in
d_m	Characteristic diameter of media in a packed bed	m	in	HDU	Height of a dispersion unit	m	in
d_o	Orifice or nozzle diameter	m	in	H_e	Height of a transfer unit due to resistance in extract phase	m	in
d_p	Sauter mean drop diameter	m	in	HETS	Height equivalent to a theoretical stage	m	in
d_{32}	Sauter mean drop diameter	m	in	H_{or}	Height of an overall mass-transfer unit based on raffinate phase	m	in
D_{col}	Column diameter	m	in or ft	H_r	Height of a transfer unit due to resistance in raffinate phase	m	in
D_{eq}	Equivalent diameter giving the same area	m	in	I	Ionic strength in Eq. (15-26)		
D_h	Equivalent hydraulic diameter	m	in	k	Individual mass-transfer coefficient	m/s or cm/s	ft/h
$\mathcal{D}_i$	Distribution ratio for a given chemical species including all its forms (unspecified units)			k	Mass-transfer coefficient (unspecified units)		
D_i	Impeller diameter or characteristic mixer diameter	m	in or ft	k_m	Membrane-side mass-transfer coefficient	m/s or cm/s	ft/h
D_{sm}	Static mixer diameter	m	in or ft	k_o	Overall mass-transfer coefficient	m/s or cm/s	ft/h
D_t	Tank diameter	m	ft	k_c	Continuous-phase mass-transfer coefficient	m/s or cm/s	ft/h
$\mathcal{D}$	Molecular diffusion coefficient (diffusivity)	m^2/s	cm^2/s	k_d	Dispersed-phase mass-transfer coefficient	m/s or cm/s	ft/h
$\mathcal{D}_{AB}$	Mutual diffusion coefficient for components A and B	m^2/s	cm^2/s	k_s	Setschenow constant	L/gmol	L/gmol
E	Mass or mass flow rate of extract phase	kg or kg/s	lb or lb/h	k_s	Shell-side mass-transfer coefficient	m/s or cm/s	ft/h
E'	Solvent mass or mass flow rate (in the extract phase)			k_t	Tube-side mass-transfer coefficient	m/s or cm/s	ft/h
E	Axial mixing coefficient (eddy diffusivity)	m^2/s	cm^2/s	K	Partition ratio (unspecified units)		
				K_s	Stripping section partition ratio (in Bancroft coordinates)	Mass ratio/mass ratio	Mass ratio/mass ratio

Nomenclature (Continued)

Symbol	Definition	SI units	U.S. Customary System units	Symbol	Definition	SI units	U.S. Customary System units
K'_w	Washing section partition ratio (in Bancroft coordinates)	Mass ratio/ mass ratio	Mass ratio/ mass ratio	Re	Reynolds number: for pipe flow, $Vd\rho/\mu$; for an impeller, $\rho_m\omega D_i^2/\mu_m$; for drops, $V_w d_p \rho_c/\mu_c$; for flow in a packed-bed coalescer, $V_d \rho_c/\mu$; for flow through an orifice, $V_o d_o \rho_d/\mu_d$	Dimensionless	Dimensionless
K'	Partition ratio, mass ratio basis (Bancroft coordinates)	Mass ratio/ mass ratio	Mass ratio/ mass ratio	Re _{Stokes}	$\rho_c \Delta p g d_i^2/18\mu_c^2$	Dimensionless	Dimensionless
K''	Partition ratio, mass fraction basis	Mass fraction/ mass fraction	Mass fraction/ mass fraction	S	Mass or mass flow rate of solvent phase	kg or kg/s	lb or lb/h
K^o	Partition ratio, mole fraction basis	Mole fraction/ mole fraction	Mole fraction/ mole fraction	S	Dimension of envelope-style downcomer (Fig. 15-39)	m	ft
K^{vol}	Partition ratio (volumetric concentration basis)	Ratio of kg/m ³ or kgmol/m ³ or gmol/L	Ratio of lb/ft ³ or lbmol/ft ³	S'	Solvent mass or mass flow rate (extraction solvent only)	kg or kg/s	lb or lb/h
L	Downcomer (or upcomer) length	m	in or ft	S'_s	Mass flow rate of extraction solvent within stripping section	kg/s	lb/h
L_{fp}	Length of flow path in Eq. (15-161)	m	in or ft	S'_w	Mass flow rate of extraction solvent within washing section	kg/s	lb/h
m	Local slope of equilibrium line (unspecified concentration units)			S_{ij}	Separation power for separating component i from component j [defined by Eq. (15-105)]	Dimensionless	Dimensionless
m'	Local slope of equilibrium line (in Bancroft coordinates)	Mass ratio/ mass ratio	Mass ratio/ mass ratio	S_{tip}	Impeller tip speed	m/s	ft/s
m_{dc}	Local slope of equilibrium line for dispersed-phase concentration plotted versus continuous-phase concentration			t_b	Batch mixing time	s or h	min or h
m_{er}	Local slope of equilibrium line for extract-phase concentration plotted versus raffinate-phase concentration			T	Temperature (absolute)	K	°R
m^{vol}	Local slope of equilibrium line (volumetric concentration basis)	Ratio of kg/m ³ or kgmol/m ³ or gmol/L	Ratio of lb/ft ³ or lbmol/ft ³ units	u_t	Stokes' law terminal or settling velocity of a drop	m/s or cm/s	ft/s or ft/min
M	Mass or mass flow rate	kg or kg/s	lb or lb/h	u_{∞}	Unhindered settling velocity of a single drop	m/s or cm/s	ft/s or ft/min
MW	Molecular weight	kg/kgmol or g/gmol	lb/lbmol	v	Molar volume	m ³ /kgmol or cm ³ /gmol	ft ³ /lbmol
N	Number of theoretical stages	Dimensionless	Dimensionless	V	Liquid velocity (or volumetric flow per unit area)	m/s	ft/s or ft/min
N_A	Flux of component A (mass or mol/area/unit time)	(kg or kgmol)/(m ² ·s)	(lb or lbmol)/(ft ² ·s)	V'	Volume	m ³	ft ³ or gal
N_{holes}	Number of holes	Dimensionless	Dimensionless	V_{cf}	Continuous-phase flooding velocity	m/s	ft/s or ft/min
N_{or}	Number of overall mass-transfer units based on the raffinate phase	Dimensionless	Dimensionless	V_{cflow}	Cross-flow velocity of continuous phase at sieve tray	m/s	ft/s or ft/min
N_s	Number of theoretical stages in stripping section	Dimensionless	Dimensionless	V_{df}	Dispersed-phase flooding velocity	m/s	ft/s or ft/min
N_w	Number of theoretical stages in washing section	Dimensionless	Dimensionless	V_{drop}	Average velocity of a dispersed drop	m/s	ft/s or ft/min
P	Pressure	bar or Pa	atm or lb _f /in ²	V_{ic}	Interstitial velocity of continuous phase	m/s	ft/s or ft/min
P	Dimensionless group defined by Eq. (15-122)	Dimensionless	Dimensionless	$V_{o,max}$	Maximum velocity through an orifice or nozzle	m/s	ft/s or ft/min
P	Power	W or kW	HP or ft·lb _f /h	V_s	Slip velocity	m/s	ft/s or ft/min
Pe	Péclet number Vb/E , where V is liquid velocity, E is axial mixing coefficient, and b is a characteristic equipment dimension	Dimensionless	Dimensionless	V_{so}	Slip velocity at low dispersed-phase flow rate	m/s	ft/s or ft/min
$P_{i,extract}$	Purity of solute i in extract (in wt %)	wt %	wt %	V_{sm}	Static mixer superficial liquid velocity (entrance velocity)	m/s	ft/s or ft/min
$P_{i,feed}$	Purity of solute i in feed (in wt %)	wt %	wt %	W	Mass or mass flow rate of wash solvent phase	kg or kg/s	lb or lb/h
P_o	Power number $P/(\rho_m \omega^3 D_i^5)$	Dimensionless	Dimensionless	W'_s	Mass flow rate of wash solvent within stripping section	kg/s	lb/h
ΔP_{dow}	Pressure drop for flow through a downcomer (or upcomer)	bar or Pa	atm or lb _f /in ²	W'_w	Mass flow rate of wash solvent within washing section	kg/s	lb/h
ΔP_o	Orifice pressure drop	bar or Pa	atm or lb _f /in ²	We	Weber number: for an impeller, $\rho_c \omega^2 D_i^3/\sigma$; for flow through an orifice or nozzle, $V_o^2 d_o \rho_d/\sigma$; for a static mixer, $V_{sm}^2 D_{sm} \rho_c/\sigma$	Dimensionless	Dimensionless
q	MOSCED induction parameter	Dimensionless	Dimensionless	x	Mole fraction solute in feed or raffinate	Mole fraction	Mole fraction
Q	Volumetric flow rate	m ³ /s	ft ³ /min	X	Concentration of solute in feed or raffinate (unspecified units)		
R	Universal gas constant	8.31 J·K/ kgmol	1.99 Btu·°R/ lbmol	X''	Mass fraction solute in feed or raffinate	Mass fractions	Mass fractions
R	Mass or mass flow rate of raffinate phase	kg or kg/s	lb or lb/h	X'	Mass solute/mass feed solvent in feed or raffinate	Mass ratios	Mass ratios
R_A	Rate of mass-transfer (moles per unit time)	kgmol/s	lbmol/h	X_f^B	Pseudoconcentration of solute in feed for case B [Eq. (15-95)]	Mass ratios	Mass ratios

Nomenclature (Concluded)

Symbol	Definition	SI units	U.S. Customary System units
X_f^C	Pseudoconcentration of solute in feed for case C [Eq. (15-97)]	Mass ratios	Mass ratios
$X_{i,extract}$	Concentration of solute i in extract	Mass fraction	Mass fraction
$X_{i,feed}$	Concentration of solute i in feed	Mass fraction	Mass fraction
X_{ij}	Concentration of component i in the phase richest in j	Mass fraction	Mass fraction
y	Mole fraction solute in solvent or extract	Mole fraction	Mole fraction
Y	Concentration of solute in the solvent or extract (unspecified units)		
Y''	Mass fraction solute in solvent or extract	Mass fraction	Mass fraction
Y'	Mass solute/mass extraction solvent in solvent or extract	Mass ratio	Mass ratio
Y_s^B	Pseudoconcentration of solute in solvent for case B [Eq. (15-96)]	Mass ratio	Mass ratio
z	Dimension or direction of mass transfer	m	in or ft
z	Sieve tray spacing	m	in or ft
z	Point representing feed composition on a tie line		
z_i	Number of electronic charges on an ion	Dimensionless	Dimensionless
Z_t	Total height of extractor	m	ft
Greek Symbols			
α	MOSCED hydrogen-bond acidity parameter	(J/cm ³) ^{1/2}	(cal/cm ³) ^{1/2}
α	Solvatochromic hydrogen-bond acidity parameter	(J/cm ³) ^{1/2}	(cal/cm ³) ^{1/2}
α_{ij}	Separation factor for solute i with respect to solute j	Dimensionless	Dimensionless
α_{ij}	NRTL model parameter	Dimensionless	Dimensionless
β	MOSCED hydrogen-bond basicity parameter	(J/cm ³) ^{1/2}	(cal/cm ³) ^{1/2}
β	Solvatochromic hydrogen-bond basicity parameter	(J/cm ³) ^{1/2}	(cal/cm ³) ^{1/2}
γ_{ij}	Activity coefficient of i dissolved in j	Dimensionless	Dimensionless
γ^∞	Activity coefficient at infinite dilution	Dimensionless	Dimensionless
γ_i^C	Activity coefficient, combinatorial part of UNIFAC	Dimensionless	Dimensionless
γ_i^I	Activity coefficient of component i in phase I	Dimensionless	Dimensionless
γ_i^R	Activity coefficient, residual part of UNIFAC	Dimensionless	Dimensionless
ϵ	Void fraction	Dimensionless	Dimensionless
ϵ	Fractional open area of a perforated plate	Dimensionless	Dimensionless
δ	Solvatochromic polarizability parameter	(J/cm ³) ^{1/2}	(cal/cm ³) ^{1/2}
δ_d	Hansen nonpolar (dispersion) solubility parameter	(J/cm ³) ^{1/2}	(cal/cm ³) ^{1/2}
δ_h	Hansen solubility parameter for hydrogen bonding	(J/cm ³) ^{1/2}	(cal/cm ³) ^{1/2}
δ_p	Hansen polar solubility parameter	(J/cm ³) ^{1/2}	(cal/cm ³) ^{1/2}
Greek Symbols			
δ^i	Solubility parameter for component i	(J/cm ³) ^{1/2}	(cal/cm ³) ^{1/2}
$\bar{\delta}$	Solubility parameter for mixture	(J/cm ³) ^{1/2}	(cal/cm ³) ^{1/2}
ζ	Tortuosity factor defined by Eq. (15-147)	Dimensionless	Dimensionless
θ	Residence time for total liquid	s	s or min
θ_i	Fraction of solute i extracted from feed	Dimensionless	Dimensionless
λ	MOSCED dispersion parameter	(J/cm ³) ^{1/2}	(cal/cm ³) ^{1/2}
λ_m	Membrane thickness	mm	in
μ	Liquid viscosity	Pa·s	cP
μ_i^I	Chemical potential of component i in phase I	J/gmol	Btu/lbmol
μ_m	Mixture mean viscosity defined in Eq. (15-180)	Pa·s	cP
μ_w	Reference viscosity (of water)	Pa·s	cP
ξ_1	MOSCED asymmetry factor	Dimensionless	Dimensionless
ξ_{batch}	Efficiency of a batch experiment [Eq. (15-175)]	Dimensionless	Dimensionless
$\xi_{continuous}$	Efficiency of a continuous process [Eq. (15-176)]	Dimensionless	Dimensionless
ξ_m	Murphree stage efficiency	Dimensionless	Dimensionless
ξ_{md}	Murphree stage efficiency based on dispersed phase	Dimensionless	Dimensionless
ξ_o	Overall stage efficiency	Dimensionless	Dimensionless
π	Solvatochromic polarity parameter	(J/cm ³) ^{1/2}	(cal/cm ³) ^{1/2}
$\Delta\pi$	Osmotic pressure gradient	bar or Pa	atm or lb _f /in ²
ρ	Liquid density	kg/m ³	lb/ft ³
ρ_m	Mixture mean density defined in Eq. (15-178)	kg/m ³	lb/ft ³
σ	Interfacial tension	N/m	dyn/cm
τ	MOSCED polarity parameter	(J/cm ³) ^{1/2}	(cal/cm ³) ^{1/2}
τ_{ij}	NRTL model parameter	Dimensionless	Dimensionless
ϕ	Volume fraction	Dimensionless	Dimensionless
ϕ_d	Volume fraction of dispersed phase (holdup)	Dimensionless	Dimensionless
$\phi_{d,feed}$	Volume fraction of dispersed phase in feed	Dimensionless	Dimensionless
ϕ_o	Initial dispersed-phase holdup in feed to a decanter	Dimensionless	Dimensionless
φ	Volume fraction of voids in a packed bed	Dimensionless	Dimensionless
Φ	Factor governing use of Eqs. (15-148) and (15-149)	Dimensionless	Dimensionless
χ	Parameter in Eq. (15-41) indicating which phase is likely to be dispersed	Dimensionless	Dimensionless
ω	Impeller speed	Rotations/s	Rotations/min
Additional Subscripts			
c	Continuous phase		
d	Dispersed phase		
e	Extract phase		
f	Feed phase or flooding condition (when combined with d or c)		
i	Component i		
j	Component j		
H	Heavy liquid		
L	Light liquid		
max	Maximum value		
min	Minimum value		
o	Orifice or nozzle		
r	Raffinate phase		
s	Solvent		

GENERAL REFERENCES: Wankat, *Separation Process Engineering*, 2d ed. (Prentice-Hall, 2006); Seader and Henley, *Separation Process Principles*, 2d ed. (Wiley, 2006); Seibert, "Extraction and Leaching," Chap. 14 in *Chemical Process Equipment: Selection and Design*, 2d ed., Couper et al., eds. (Elsevier, 2005); Aguilar and Cortina, *Solvent Extraction and Liquid Membranes: Fundamentals and Applications in New Materials* (Dekker, 2005); Glatz and Parker, "Enriching Liquid-Liquid Extraction," *Chem. Eng. Magazine*, **111**(11), pp. 44-48 (2004); *Solvent Extraction Principles and Practice*, 2d ed., Rydberg et al., eds. (Dekker, 2004); *Ion Exchange and Solvent Extraction*, vol. 17, Marcus and SenGupta, eds. (Dekker, 2004), and earlier volumes in the series; Leng and Calabrese, "Immiscible Liquid-Liquid Systems," Chap. 12 in *Handbook of Industrial Mixing: Science and Practice*, Paul, Atiemo-Obeng, and Kresta, eds. (Wiley, 2004); Cheremisinoff, *Industrial Solvents Handbook*, 2d ed. (Dekker, 2003); Van Brunt and Kanel, "Extraction with Reaction," Chap. 3 in *Reactive Separation Processes*, Kulprathipanja, ed. (Taylor & Francis, 2002); Mueller et al., "Liquid-Liquid Extraction" in *Ullmann's Encyclopedia of Industrial Chemistry*, 6th ed. (VCH, 2002); Benitez, *Principles and Modern Applications of Mass Transfer Operations* (Wiley, 2002); Wypych, *Handbook of Solvents* (Chemtec, 2001); Flick, *Industrial Solvents Handbook*, 5th ed. (Noyes, 1998); Robbins, "Liquid-Liquid Extraction," Sec. 1.9 in *Handbook of Separation Techniques for Chemical Engineers*, 3d ed., Schweitzer, ed. (McGraw-Hill, 1997); Lo, "Commercial Liquid-Liquid Extraction Equipment," Sec. 1.10 in *Handbook of Separation Techniques for Chemical Engineers*, 3d ed., Schweitzer, ed. (McGraw-Hill, 1997); Humphrey and Keller, "Extraction," Chap. 3 in *Separation Process Technology* (McGraw-Hill, 1997), pp. 113-151; Cusack and Glatz, "Apply Liquid-Liquid Extraction to Today's Problems," *Chem. Eng. Magazine*, **103**(7), pp. 94-103 (1996); *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994); Zaslavsky, *Aqueous Two-Phase Partitioning* (Dekker, 1994); Strigle, "Liquid-Liquid Extraction," Chap. 11 in *Packed Tower Design and Applications*, 2d ed. (Gulf, 1994); Schügerl, *Solvent Extraction in Biotechnology* (Springer-Verlag, 1994); Schügerl, "Liquid-Liquid Extraction (Small Molecules)," Chap. 21 in *Biotechnology*, 2d ed., vol. 3, Stephanopoulos, ed. (VCH, 1993); Kelley and Hattori, "Protein Purification by Liquid-Liquid Extraction," Chap. 22 in *Biotechnology*, 2d ed., vol. 3, Stephanopoulos, ed. (VCH, 1993); Lo and Baird, "Extraction,

Liquid-Liquid," in *Kirk-Othmer Encyclopedia of Chemical Technology*, 4th ed., vol. 10, Kroschwitz and Howe-Grant, eds. (Wiley, 1993), pp. 125-180; *Science and Practice of Liquid-Liquid Extraction*, vol. 1, *Phase Equilibria; Mass Transfer and Interfacial Phenomena; Extractor Hydrodynamics, Selection, and Design*, and vol. 2, *Process Chemistry and Extraction Operations in the Hydrometallurgical, Nuclear, Pharmaceutical, and Food Industries*, Thornton, ed. (Oxford, 1992); Cusack, Fremaux, and Glatz, "A Fresh Look at Liquid-Liquid Extraction," pt. 1, "Extraction Systems," *Chem. Eng. Magazine*, **98**(2), pp. 66-67 (1991); Cusack and Fremaux, pt. 2, "Inside the Extractor," *Chem. Eng. Magazine*, **98**(3), pp. 132-138 (1991); Cusack and Karr, pt. 3, "Extractor Design and Specification," *Chem. Eng. Magazine*, **98**(4), pp. 112-120 (1991); *Methods in Enzymology*, vol. 182, *Guide to Protein Purification*, Deutscher, ed. (Academic, 1990); Wankat, *Equilibrium Staged Separations* (Prentice Hall, 1988); Blumberg, *Liquid-Liquid Extraction* (Academic, 1988); Skelland and Tedder, "Extraction—Organic Chemicals Processing," Chap. 7 in *Handbook of Separation Process Technology*, Rousseau, ed. (Wiley, 1987); Chapman, "Extraction—Metals Processing," Chap. 8 in *Handbook of Separation Process Technology*, Rousseau, ed. (Wiley, 1987); Novak, Matous, and Pick, *Liquid-Liquid Equilibria*, Studies in Modern Thermodynamics Series, vol. 7 (Elsevier, 1987); Bailes et al., "Extraction, Liquid-Liquid" in *Encyclopedia of Chemical Processing and Design*, vol. 21, McKetta and Cunningham, eds. (Dekker, 1984), pp. 19-166; *Handbook of Solvent Extraction*, Lo, Baird, and Hanson, eds. (Wiley, 1983; Krieger, 1991); Sorenson and Arlt, *Liquid-Liquid Equilibrium Data Collection*, DECHEMA, *Binary Systems*, vol. V, pt. 1, 1979, *Ternary Systems*, vol. V, pt. 2, 1980, *Ternary and Quaternary Systems*, vol. 5, pt. 3, 1980, Macedo and Rasmussen, Suppl. 1, vol. V, pt. 4, 1987; Wisniak and Tamir, *Liquid-Liquid Equilibrium and Extraction, a Literature Source Book*, vols. I and II (Elsevier, 1980-1981), Suppl. 1 (1985); Treybal, *Mass Transfer Operations*, 3d ed. (McGraw-Hill, 1980); King, *Separation Processes*, 2d ed. (McGraw-Hill, 1980); Laddha and Degaleesan, *Transport Phenomena in Liquid Extraction* (McGraw-Hill, 1978); Brian, *Staged Cascades in Chemical Processing* (Prentice-Hall, 1972); Pratt, *Countercurrent Separation Processes* (Elsevier, 1967); Treybal, "Liquid Extractor Performance," *Chem. Eng. Prog.*, **62**(9), pp. 67-75 (1966); Treybal, *Liquid Extraction*, 2d ed. (McGraw-Hill, 1963); Alders, *Liquid-Liquid Extraction*, 2d ed. (Elsevier, 1959).

INTRODUCTION AND OVERVIEW

Liquid-liquid extraction is a process for separating the components of a liquid (the feed) by contact with a second liquid phase (the solvent). The process takes advantage of differences in the chemical properties of the feed components, such as differences in polarity and hydrophobic/hydrophilic character, to separate them. Stated more precisely, the transfer of components from one phase to the other is driven by a deviation from thermodynamic equilibrium, and the equilibrium state depends on the nature of the interactions between the feed components and the solvent phase. The potential for separating the feed components is determined by differences in these interactions.

A liquid-liquid extraction process produces a solvent-rich stream called the extract that contains a portion of the feed and an extract-free stream called the raffinate. A commercial process almost always includes two or more auxiliary operations in addition to the extraction operation itself. These extra operations are needed to treat the extract and raffinate streams for the purposes of isolating a desired product, recovering the solvent for recycle to the extractor, and purging unwanted components from the process. A typical process includes two or more distillation operations in addition to extraction.

Liquid-liquid extraction is used to recover desired components from a crude liquid mixture or to remove unwanted contaminants. In developing a process, the project team must decide what solvent or solvent mixture to use, how to recover solvent from the extract, and how to remove solvent residues from the raffinate. The team must also decide what temperature or range of temperatures should be used for the extraction, what process scheme to employ among many possibilities, and what type of equipment to use for liquid-liquid contacting and phase separation. The variety of commercial equipment options is large and includes stirred tanks and decanters, specialized mixer-settlers, a wide variety of agitated and nonagitated extraction columns or towers, and various types of centrifuges.

Because of the availability of hundreds of commercial solvents and extractants, as well as a wide variety of established process schemes and equipment options, liquid-liquid extraction is a versatile technology with a wide range of commercial applications. It is utilized in the

processing of numerous commodity and specialty chemicals including metals and nuclear fuel (hydrometallurgy), petrochemicals, coal and wood-derived chemicals, and complex organics such as pharmaceuticals and agricultural chemicals. Liquid-liquid extraction also is an important operation in industrial wastewater treatment, food processing, and the recovery of biomolecules from fermentation broth.

HISTORICAL PERSPECTIVE

The art of solvent extraction has been practiced in one form or another since ancient times. It appears that prior to the 19th century solvent extraction was primarily used to isolate desired components such as perfumes and dyes from plant solids and other natural sources [Aftalion, *A History of the International Chemical Industry* (Univ. Penn. Press, 1991); and Taylor, *A History of Industrial Chemistry* (Abelard-Schuman, 1957)]. However, several early applications involving liquid-liquid contacting are described by Blass, Liebel, and Haerberl ["Solvent Extraction—A Historical Review," *International Solvent Extraction Conf. (ISEC) '96 Proceedings* (Univ. of Melbourne, 1996)], including the removal of pigment from oil by using water as the solvent.

The modern practice of liquid-liquid extraction has its roots in the middle to late 19th century when extraction became an important laboratory technique. The partition ratio concept describing how a solute partitions between two liquid phases at equilibrium was introduced by Berthelot and Jungfleisch [*Ann. Chim. Phys.*, **4**, p. 26 (1872)] and further defined by Nernst [*Z. Phys. Chemie*, **8**, p. 110 (1891)]. At about the same time, Gibbs published his theory of phase equilibrium (1876 and 1878). These and other advances were accompanied by a growing chemical industry. An early countercurrent extraction process utilizing ethyl acetate solvent was patented by Goering in 1883 as a method for recovering acetic acid from "pyroigneous acid" produced by pyrolysis of wood [Othmer, p. xiv in *Handbook of Solvent Extraction* (Wiley, 1983; Krieger, 1991)], and Pfeleiderer patented a stirred extraction column in 1898 [Blass, Liebel, and Haerberl, *ISEC '96 Proceedings* (Univ. of Melbourne, 1996)].

With the emergence of the chemical engineering profession in the 1890s and early 20th century, additional attention was given to process fundamentals and development of a more quantitative basis for process design. Many of the advances made in the study of distillation and absorption were readily adapted to liquid-liquid extraction, owing to its similarity as another diffusion-based operation. Examples include application of mass-transfer coefficients [Lewis, *Ind. Eng. Chem.*, **8**(9), pp. 825–833 (1916); and Lewis and Whitman, *Ind. Eng. Chem.*, **16**(12), pp. 1215–1220 (1924)], the use of graphical stagewise design methods [McCabe and Thiele, *Ind. Eng. Chem.*, **17**(6), pp. 605–611 (1925); Evans, *Ind. Eng. Chem.*, **26**(8), pp. 860–864 (1934); and Thiele, *Ind. Eng. Chem.*, **27**(4), pp. 392–396 (1935)], the use of theoretical-stage calculations [Kremser, *National Petroleum News*, **22**(21), pp. 43–49 (1930); and Souders and Brown, *Ind. Eng. Chem.*, **24**(5), pp. 519–522 (1932)], and the transfer unit concept introduced in the late 1930s by Colburn and others [Colburn, *Ind. Eng. Chem.*, **33**(4), pp. 459–467 (1941)]. Additional background is given by Hampe, Hartland, and Slater [Chap. 2 in *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994)].

The number of commercial applications continued to grow, and by the 1930s liquid-liquid extraction had replaced various chemical treatment methods for refining mineral oil and coal tar products [Varteressian and Fenske, *Ind. Eng. Chem.*, **28**(8), pp. 928–933 (1936)]. It was also used to recover acetic acid from waste liquors generated in the production of cellulose acetate, and in various nitration and sulfonation processes [Hunter and Nash, *The Industrial Chemist*, **9**(102–104), pp. 245–248, 263–266, 313–316 (1933)]. The article by Hunter and Nash also describes early mixer-settler equipment, mixing jets, and various extraction columns including the spray column, baffle tray column, sieve tray column, and a packed column filled with Raschig rings or coke breeze, the material left behind when coke is burned.

Much of the liquid-liquid extraction technology in practice today was first introduced to industry during a period of vigorous innovation and growth of the chemical industry as a whole from about 1920 to 1970. The advances of this period include development of fractional extraction schemes including work described by Cornish et al. [*Ind. Eng. Chem.*, **26**(4), pp. 397–406 (1934)] and by Thiele [*Ind. Eng. Chem.*, **27**(4), pp. 392–396 (1935)]. A well-known commercial example involving the use of extract reflux is the Udex process for separating aromatic compounds from hydrocarbon mixtures using diethylene glycol, a process developed jointly by The Dow Chemical Company and Universal Oil Products in the 1940s. This period also saw the introduction of many new equipment designs including specialized mixer-settler equipment, mechanically agitated extraction columns, and centrifugal extractors as well as a great increase in the availability of different types of industrial solvents. A variety of alcohols, ketones, esters, and chlorinated hydrocarbons became available in large quantities beginning in the 1930s, as petroleum refiners and chemical companies found ways to manufacture them inexpensively using the byproducts of petroleum refining operations or natural gas. Later, a number of specialty solvents were introduced including sulfolane (tetrahydrothiophene-1,1-dioxane) and NMP (*N*-methyl-2-pyrrolidone) for improved extraction of aromatics from hydrocarbons. Specialized extractants also were developed including numerous organophosphorous extractants used to recover or purify metals dissolved in aqueous solutions.

The ready availability of numerous solvents and extractants, combined with the tremendous growth of the chemical industry, drove the development and implementation of many new industrial applications. Handbooks of chemical process technology provide a glimpse of some of these [Riegel's *Handbook of Industrial Chemistry*, 10th ed., Kent, ed. (Springer, 2003); *Chemical Processing Handbook*, McKetta, ed. (Dekker, 1993); and Austin, *Shreve's Chemical Process Industries*, 5th ed. (McGraw-Hill, 1984)], but many remain proprietary and are not widely known. The better-known examples include the separation of aromatics from aliphatics, as mentioned above, extraction of phenolic compounds from coal tars and liquors, recovery of ϵ -caprolactam for production of polyamide-6 (nylon-6), recovery of hydrogen peroxide from oxidized anthraquinone solution, plus many processes involving the washing of crude organic streams with alkaline or acidic

solutions and water, and the detoxification of industrial wastewater prior to biotreatment using steam-strippable organic solvents. The pharmaceutical and specialty chemicals industry also began using liquid-liquid extraction in the production of new synthetic drug compounds and other complex organics. In these processes, often involving multiple batch reaction steps, liquid-liquid extraction generally is used for recovery of intermediates or crude products prior to final isolation of a pure product by crystallization. In the inorganic chemical industry, extraction processes were developed for purification of phosphoric acid, purification of copper by removal of arsenic impurities, and recovery of uranium from phosphate-rock leach solutions, among other applications. Extraction processes also were developed for bioprocessing applications, including the recovery of citric acid from broth using trialkylamine extractants, the use of amyl acetate to recover antibiotics from fermentation broth, and the use of water-soluble polymers in aqueous two-phase extraction for purification of proteins.

The use of supercritical or near-supercritical fluids for extraction, a subject area normally set apart from discussions of liquid-liquid extraction, has received a great deal of attention in the R&D community since the 1970s. Some processes were developed many years before then; e.g., the propane deasphalting process used to refine lubricating oils uses propane at near-supercritical conditions, and this technology dates back to the 1930s [McHugh and Krukonsis, *Supercritical Fluid Processing*, 2d ed. (Butterworth-Heinemann, 1993)]. In more recent years the use of supercritical fluids has found a number of commercial applications displacing earlier liquid-liquid extraction methods, particularly for recovery of high-value products meant for human consumption including decaffeinated coffee, flavor components from citrus oils, and vitamins from natural sources.

Significant progress continues to be made toward improving extraction technology, including the introduction of new methods to estimate solvent properties and screen candidate solvents and solvent blends, new methods for overall process conceptualization and optimization, and new methods for equipment design. Progress also is being made by applying the technology developed for a particular application in one industry to improve another application in another industry. For example, much can be learned by comparing equipment and practices used in organic chemical production with those used in the inorganic chemical industry (and vice versa), or by comparing practices used in commodity chemical processing with those used in the specialty chemicals industry. And new concepts offering potential for significant improvements continue to be described in the literature. (See "Emerging Developments.")

USES FOR LIQUID-LIQUID EXTRACTION

For many separation applications, the use of liquid-liquid extraction is an alternative to the various distillation schemes described in Sec. 13, "Distillation." In many of these cases, a distillation process is more economical largely because the extraction process requires extra operations to process the extract and raffinate streams, and these operations usually involve the use of distillation anyway. However, in certain cases the use of liquid-liquid extraction is more cost-effective than using distillation alone because it can be implemented with smaller equipment and/or lower energy consumption. In these cases, differences in chemical or molecular interactions between feed components and the solvent provide a more effective means of accomplishing the desired separation compared to differences in component volatilities.

For example, liquid-liquid extraction may be preferred when the relative volatility of key components is less than 1.3 or so, such that an unusually tall distillation tower is required or the design involves high reflux ratios and high energy consumption. In certain cases, the distillation option may involve addition of a solvent (extractive distillation) or an entrainer (azeotropic distillation) to enhance the relative volatility. Even in these cases, a liquid-liquid extraction process may offer advantages in terms of higher selectivity or lower solvent usage and lower energy consumption, depending upon the application. Extraction may be preferred when the distillation option requires operation at pressures less than about 70 mbar (about 50 mmHg) and an unusually large-diameter distillation tower is required, or when most of the

feed must be taken overhead to isolate a desired bottoms product. Extraction may also be attractive when distillation requires use of high-pressure steam for the reboiler or refrigeration for overheads condensation [Null, *Chem. Eng. Prog.*, **76**(8), pp. 42–49 (August 1980)], or when the desired product is temperature-sensitive and extraction can provide a gentler separation process.

Of course, liquid-liquid extraction also may be a useful option when the components of interest simply cannot be separated by using distillation methods. An example is the use of liquid-liquid extraction employing a steam-strippable solvent to remove nonstrippable, low-volatility contaminants from wastewater [Robbins, *Chem. Eng. Prog.*, **76**(10), pp. 58–61 (1980)]. The same process scheme often provides a cost-effective alternative to direct distillation or stripping of volatile impurities when the relative volatility of the impurity with respect to water is less than about 10 [Robbins, U.S. Patent 4,236,973 (1980); Hwang, Keller, and Olson, *Ind. Eng. Chem. Res.*, **31**, pp. 1753–1759 (1992); and Frank et al., *Ind. Eng. Chem. Res.*, **46**(11), pp. 3774–3786 (2007)].

Liquid-liquid extraction also can be an attractive alternative to separation methods, other than distillation, e.g., as an alternative to crystallization from solution to remove dissolved salts from a crude organic feed, since extraction of the salt content into water eliminates the need to filter solids from the mother liquor, often a difficult or expensive operation. Extraction also may compete with process-scale chromatography, an example being the recovery of hydroxytyrosol (3,4-dihydroxyphenylethanol), an antioxidant food additive, from olive-processing wastewaters [Guzman et al., U.S. Patent 6,849,770 (2005)].

The attractiveness of liquid-liquid extraction for a given application compared to alternative separation technologies often depends upon the concentration of solute in the feed. The recovery of acetic acid from aqueous solutions is a well-known example [Brown, *Chem. Eng. Prog.*, **59**(10), pp. 65–68 (1963)]. In this case, extraction generally is more economical than distillation when handling dilute to moderately concentrated feeds, while distillation is more economical at higher concentrations. In the treatment of water to remove trace amounts of organics, when the concentration of impurities in the feed is greater than about 20 to 50 ppm, liquid-liquid extraction may be more economical than adsorption of the impurities by using carbon beds, because the latter may require frequent and costly replacement of the adsorbent [Robbins, *Chem. Eng. Prog.*, **76**(10), pp. 58–61 (1980)]. At lower concentrations of impurities, adsorption may be the more economical option because the usable lifetime of the carbon bed is longer.

Examples of cost-effective liquid-liquid extraction processes utilizing relatively low-boiling solvents include the recovery of acetic acid from aqueous solutions using ethyl ether or ethyl acetate [King, Chap. 18.5 in *Handbook of Solvent Extraction*, Lo, Baird, and Hanson, eds. (Wiley, 1983; Krieger, 1991)] and the recovery of phenolic compounds from water by using methyl isobutyl ketone [Greminger et al., *Ind. Eng. Chem. Process Des. Dev.*, **21**(1), pp. 51–54 (1982)]. In these processes, the solvent is recovered from the extract by distillation, and dissolved solvent is removed from the raffinate by steam stripping (Fig. 15-1). The solvent circulates through the process in a closed loop.

One of the largest applications of liquid-liquid extraction in terms of total worldwide production volume involves the extraction of aromatic compounds from hydrocarbon mixtures in petrochemical operations using high-boiling polar solvents. A number of processes have been developed to recover benzene, toluene, and xylene (BTX) as feedstock for chemical manufacturing or to refine motor oils. This general technology is described in detail in “Single-Solvent Fractional Extraction with Extract Reflux” under “Calculation Procedures.” A typical flow diagram is shown in Fig. 15-2. Liquid-liquid extraction also may be used to upgrade used motor oil; an extraction process employing a relatively light polar solvent such as *N,N*-dimethylformamide or acetonitrile has been developed to remove polynuclear aromatic and sulfur-containing contaminants [Sherman, Hershberger, and Taylor, U.S. Patent 6,320,090 (2001)]. An alternative process utilizes a blend of methyl ethyl ketone + 2-propanol and small amounts of aqueous KOH [Rincón, Cañizares, and García, *Ind. Eng. Chem. Res.*, **44**(20), pp. 7854–7859 (2005)].

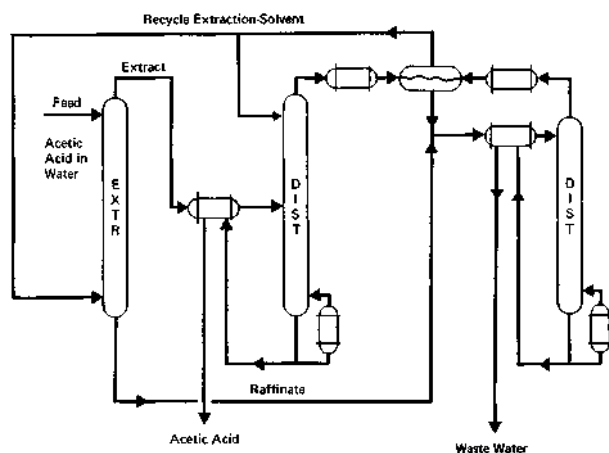


FIG. 15-1 Typical process for extraction of acetic acid from water.

Extraction also is used to remove CO_2 , H_2S , and other acidic contaminants from liquefied petroleum gases (LPGs) generated during operation of fluid catalytic crackers and cokers in petroleum refineries, and from liquefied natural gas (LNG). The acid gases are extracted from the liquefied hydrocarbons (primarily C_1 to C_3) by reversible reaction with various amine extractants. Typical amines are methyldiethanolamine (MDEA), diethanolamine (DEA), and monoethanolamine (MEA). In a typical process (Fig. 15-3), the treated hydrocarbon liquid (the raffinate) is washed with water to remove residual amine, and the loaded amine solution (the extract) is regenerated in a stripping tower for recycle back to the extractor [Nielsen et al., *Hydrocarbon Proc.*, **76**, pp. 49–59 (1997)]. The technology is similar to that used to scrub CO_2 and H_2S from gas streams [Oyenekan and Rochelle, *Ind. Eng. Chem. Res.*, **45**(8), pp. 2465–2472 (2006); and Jassim and Rochelle, *Ind. Eng. Chem. Res.*, **45**(8), pp. 2457–2464 (2006)], except that the process involves liquid-liquid contacting instead of gas-liquid contacting. Because of this, a common stripper often is used to regenerate solvent from a variety of gas absorbers and liquid-liquid extractors operated within a typical refinery. In certain applications, organic acids such as formic acid are present in low concentrations in the hydrocarbon feed. These contaminants will react with the amine extractant to form heat-stable amine salts that accumulate in the solvent loop over time, requiring periodic purging or regeneration of the solvent solution [Price and Burns, *Hydrocarbon Proc.*, **74**, pp. 140–141 (1995)]. The amine-based extraction process is an alternative to washing with caustic or the use of solid adsorbents.

A typical extraction process used in hydrometallurgical applications is outlined in Fig. 15-4. This technology involves transferring the desired element from the ore leachate liquor, an aqueous acid, into an organic solvent phase containing specialty extractants that form a complex with the metal ion. The organic phase is later contacted with an aqueous solution at a different pH and temperature to regenerate the solvent and transfer the metal into a clean solution from which it can be recovered by electrolysis or another method [Cox, Chap. 1 in *Science and Practice of Liquid-Liquid Extraction*, vol. 2, Thornton, ed. (Oxford, 1992)]. Another process technology utilizes metals complexed with various organophosphorus compounds as recyclable homogeneous catalysts; liquid-liquid extraction is used to transfer the metal complex between the reaction phase and a separate liquid phase after reaction. Different ligands having different polarities are chosen to facilitate the use of various extraction and recycle schemes [Kanel et al., U.S. Patents 6,294,700 (2001) and 6,303,829 (2001)].

Another category of useful liquid-liquid extraction applications involves the recovery of antibiotics and other complex organics from fermentation broth by using a variety of oxygenated organic solvents such as acetates and ketones. Although some of these products are unstable at the required extraction conditions (particularly if pH must

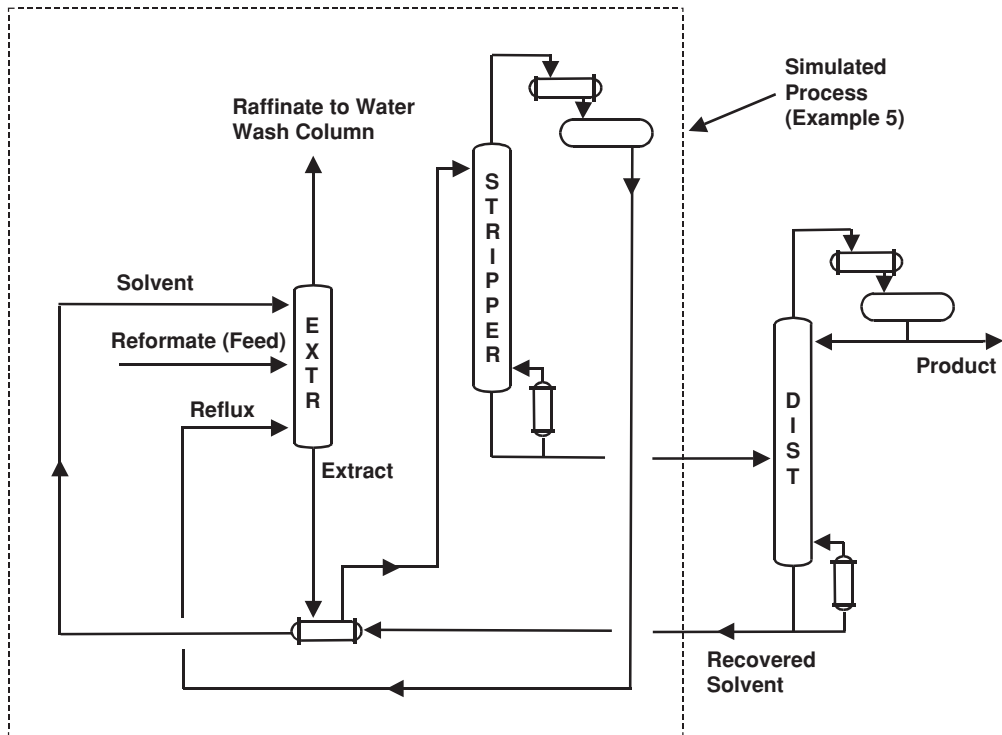


FIG. 15-2 Flow sheet of a simplified aromatic extraction process (see Example 5).

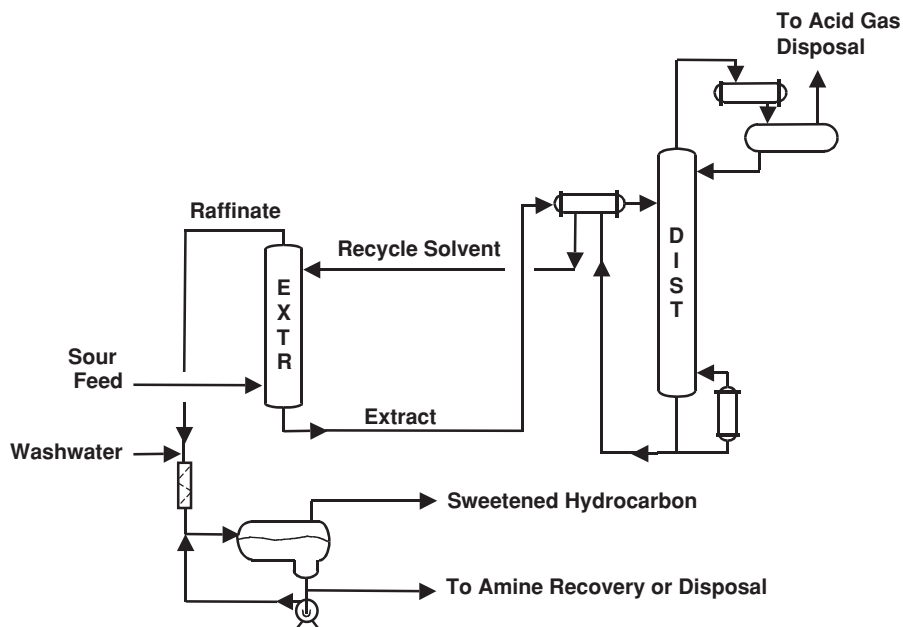


FIG. 15-3 Typical process for extracting acid gases from LPG or LNG.

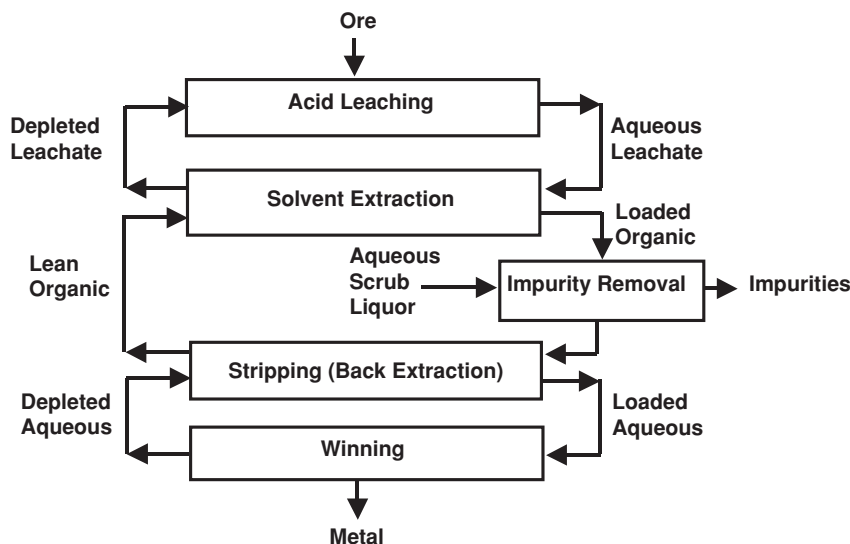


FIG. 15-4 Example process scheme used in hydrometallurgical applications. [Taken from Cox, Chap. 1 in *Science and Practice of Liquid-Liquid Extraction*, vol. 2, Thornton, ed. (Oxford, 1992), with permission. Copyright 1992 Oxford University Press.]

be low for favorable partitioning), short-contact-time centrifugal extractors may be used to minimize exposure. Centrifugal extractors also help overcome problems associated with formation of emulsions between solvent and broth. In a number of applications, the whole broth can be processed without prior removal of solids, a practice that can significantly reduce costs. For detailed information, see “The History of Penicillin Production,” Elder, ed., *Chemical Engineering Progress Symposium Series No. 100*, vol. 66, pp. 37–42 (1970); Queener and Swartz, “Penicillins: Biosynthetic and Semisynthetic,” in *Secondary Products of Metabolism, Economic Microbiology*, vol. 3, Rose, ed. (Academic, 1979); and Chaung et al., *J. Chinese Inst. Chem. Eng.*, **20**(3), pp. 155–161 (1989). Another well-known commercial application of liquid-liquid extraction in bioprocessing is the Baniel process for the recovery of citric acid from fermentation broth with tertiary amine extractants [Baniel, Blumberg, and Hadju, U.S. Patent 4,275,234 (1980)]. This type of process is discussed in “Reaction-Enhanced Extraction” under “Commercial Process Schemes.”

DEFINITIONS

Extraction terms defined by the International Union of Pure and Applied Chemistry (IUPAC) generally are recommended. [See Rice, Irving, and Leonard, *Pure Appl. Chem.* (IUPAC), **65**(11), pp. 2673–2396 (1993); and J. Inczédy, *Pure Appl. Chem.* (IUPAC), **66**(12), pp. 2501–2512 (1994).] **Liquid-liquid extraction** is a process for separating components dissolved in a liquid feed by contact with a second liquid phase. **Solvent extraction** is a broader term that describes a process for separating the components of any matrix by contact with a liquid, and it includes liquid-solid extraction (leaching) as well as liquid-liquid extraction. The **feed** to a liquid-liquid extraction process is the solution that contains the components to be separated. The major liquid component (or components) in the feed can be referred to as the **feed solvent** or the **carrier solvent**. Minor components in solution often are referred to as **solutes**. The **extraction solvent** is the immiscible or partially miscible liquid added to the process to create a second liquid phase for the purpose of extracting one or more solutes from the feed. It is also called the **separating agent** and may be a mixture of several individual solvents (a **mixed solvent** or a **solvent blend**). The extraction solvent also may be a liquid comprised of an **extractant** dissolved in a liquid **diluent**. In this case, the extractant species is primarily responsible for extraction of solute due to a relatively strong attractive

interaction with the desired solute, forming a reversible adduct or molecular complex. The diluent itself does not contribute significantly to the extraction of solute and in this respect is not the same as a true extraction solvent. A **modifier** may be added to the diluent to increase the solubility of the extractant or otherwise enhance the effectiveness of the extractant. The phase leaving a liquid-liquid contactor rich in extraction solvent is called the **extract**. The **raffinate** is the liquid phase left from the feed after it is contacted by the extract phase. The word *raffinate* originally referred to a “refined product”; however, common usage has extended its meaning to describe the feed phase after extraction whether that phase is a product or not.

Industrial liquid-liquid extraction most often involves processing two immiscible or partially miscible liquids in the form of a **dispersion** of droplets of one liquid (the **dispersed phase**) suspended in the other liquid (the **continuous phase**). The dispersion will exhibit a distribution of drop diameters d_i often characterized by the volume to surface area average diameter or **Sauter mean drop diameter**. The term **emulsion** generally refers to a liquid-liquid dispersion with a dispersed-phase mean drop diameter on the order of 1 μm or less.

The tension that exists between two liquid phases is called the **interfacial tension**. It is a measure of the energy or work required to increase the surface area of the liquid-liquid interface, and it affects the size of dispersed drops. Its value, in units of force per unit length or energy per unit area, reflects the compatibility of the two liquids. Systems that have low compatibility (low mutual solubility) exhibit high interfacial tension. Such a system tends to form relatively large dispersed drops and low interfacial area to minimize contact between the phases. Systems that are more compatible (with higher mutual solubility) exhibit lower interfacial tension and more easily form small dispersed droplets.

A **theoretical or equilibrium stage** is a device or combination of devices that accomplishes the effect of intimately mixing two liquid phases until equilibrium concentrations are reached, then physically separating the two phases into clear layers. The **partition ratio K** is commonly defined for a given solute as the solute concentration in the extract phase divided by that in the raffinate phase after equilibrium is attained in a single stage of contacting. A variety of concentration units are used, so it is important to determine how partition ratios have been defined in the literature for a given application. The term *partition ratio* is preferred, but it also is referred to as the **distribution constant**, **distribution coefficient**, or the **K value**. It is a measure of the

thermodynamic potential of a solvent for extracting a given solute and can be a strong function of composition and temperature. In some cases, the partition ratio transitions from a value less than unity to a value greater than unity as a function of solute concentration. A system of this type is called a **solutrope** [Smith, *Ind. Eng. Chem.*, **42**(6), pp. 1206–1209 (1950)]. The term *distribution ratio*, designated by $\mathcal{D}_i$, is used in analytical chemistry to describe the distribution of a species that undergoes chemical reaction or dissociation, in terms of the total concentration of analyte in one phase over that in the other, regardless of its chemical form.

The **extraction factor** $\mathcal{E}$ is a process variable that characterizes the capacity of the extract phase to carry solute relative to the feed phase. Its value largely determines the number of theoretical stages required to transfer solute from the feed to the extract. The extraction factor is analogous to the stripping factor in distillation and is the ratio of the slope of the equilibrium line to the slope of the operating line in a McCabe-Thiele type of stagewise graphical calculation. For a standard extraction process with straight equilibrium and operating lines, $\mathcal{E}$ is constant and equal to the partition ratio for the solute of interest times the ratio of the solvent flow rate to the feed flow rate. The **separation factor** α_{ij} measures the relative enrichment of solute i in the extract phase, compared to solute j , after one theoretical stage of extraction. It is equal to the ratio of K values for components i and j and is used to characterize the **selectivity** a solvent has for a given solute.

A **standard extraction** process is one in which the primary purpose is to transfer solute from the feed phase into the extract phase in a manner analogous to stripping in distillation. **Fractional extraction** refers to a process in which two or more solutes present in the feed are sharply separated from each other, one fraction leaving the extractor in the extract and the other in the raffinate. **Cross-current or cross-flow extraction** (Fig. 15-5) is a series of discrete stages in which the raffinate R from one extraction stage is contacted with additional fresh solvent S in a subsequent stage. **Countercurrent extraction** (Fig. 15-6) is an extraction scheme in which the extraction solvent enters the stage or end of the extraction farthest from where the feed F enters, and the two phases pass each other in countercurrent fashion. The objective is to transfer one or more components from the feed solution F into the extract E . Compared to cross-current operation, countercurrent operation generally allows operation with less solvent. When a **staged contactor** is used, the two phases are mixed with droplets of one phase suspended in the other, but the phases are separated before leaving each stage. A **countercurrent cascade** is a process utilizing multiple staged contactors with countercurrent flow of solvent and feed streams from stage to stage. When a **differential contactor** is used, one of the phases can remain dispersed as drops throughout the contactor as the phases pass each other in countercurrent fashion. The dispersed phase is then allowed to coalesce at the end of the device before being discharged. For these types of processes, **mass-transfer units** (or the related **mass-transfer coefficients**) often are used instead of theoretical stages to characterize separation performance. For a given phase, mass-transfer units are

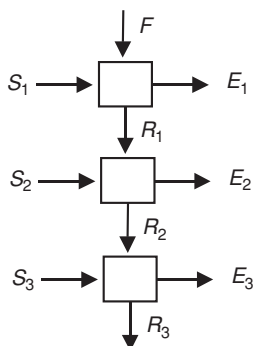


FIG. 15-5 Cross-current extraction.

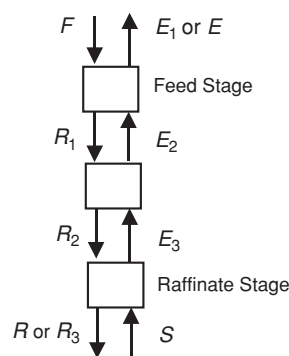


FIG. 15-6 Standard countercurrent extraction.

defined as the integral of the differential change in solute concentration divided by the deviation from equilibrium, between the limits of inlet and outlet solute concentrations. A single transfer unit represents the change in solute concentration equal to that achieved by a single theoretical stage when the extraction factor is equal to 1.0. It differs from a theoretical stage at other values of the extraction factor.

The term **flooding** generally refers to excessive breakthrough or entrainment of one liquid phase into the discharge stream of the other. The flooding characteristics of an extractor limit its hydraulic capacity. Flooding can be caused by excessive flow rates within the equipment, by **phase inversion** due to accumulation and coalescence of dispersed droplets, or by formation of stable dispersions or emulsions due to the presence of surface-active impurities or excessive agitation. The **flood point** typically refers to the specific total volumetric throughput in ($\text{m}^3/\text{h}/\text{m}^2$ or gpm/ft^2 of cross-sectional area (or the equivalent **phase velocity** in m/s or ft/s) at which flooding begins.

DESIRABLE SOLVENT PROPERTIES

Common industrial solvents generally are single-functionality organic solvents such as ketones, esters, alcohols, linear or branched aliphatic hydrocarbons, aromatic hydrocarbons, and so on; or water, which may be acidic or basic or mixed with water-soluble organic solvents. More complex solvents are sometimes used to obtain specific properties needed for a given application. These include compounds with multiple functional groups such as diols or triols, glycol ethers, and alkanol amines as well as heterocyclic compounds such as pine-derived solvents (terpenes), sulfolane (tetrahydrothiophene-1,1-dioxane), and NMP (*N*-methyl-2-pyrrolidinone). Solvent properties have been summarized in a number of handbooks and databases including those by Cheremisinoff, *Industrial Solvents Handbook*, 2d ed. (Dekker, 2003); Wypych, *Handbook of Solvents* (ChemTech, 2001); Wypych, *Solvents Database*, CD-ROM (ChemTec, 2001); Yaws, *Thermodynamic and Physical Property Data*, 2d ed. (Gulf, 1998); and Flick, *Industrial Solvents Handbook*, 5th ed. (Noyes, 1998). Solvents are sometimes blended to obtain specific properties, another approach to achieving a multifunctional solvent with properties tailored for a given application. Examples are discussed by Escudero, Cabezas, and Coca [*Chem. Eng. Comm.*, **173**, pp. 135–146 (1999)] and by Delden et al. [*Chem. Eng. Technol.*, **29**(10), pp. 1221–1226 (2006)]. As discussed earlier, a solvent also may be a liquid containing a dissolved extractant species, the extractant chosen because it forms a specific attractive interaction with the desired solute.

In terms of desirable properties, no single solvent or solvent blend can be best in every respect. The choice of solvent often is a compromise, and the relative weighting given to the various considerations depends on the given situation. Assessments should take into account long-term sustainability and overall cost of ownership. Normally, the factors considered in choosing a solvent include the following.

1. **Loading capacity.** This property refers to the maximum concentration of solute the extract phase can hold before two liquid phases can no longer coexist or solute precipitates as a separate phase.

If a specialized extractant is used, loading capacity may be determined by the point at which all the extractant in solution is completely occupied by solute and extractant solubility limits capacity. If loading capacity is low, a high solvent-to-feed ratio may be needed even if the partition ratio is high.

2. *Partition ratio* $K_i = Y_i/X_i$. Partition ratios on the order of $K_i = 10$ or higher are desired for an economical process because they allow operation with minimal amounts of solvent (more specifically, with a minimal solvent-to-feed ratio) and production of higher solute concentrations in the extract—unless the solute concentration in the feed already is high and a limitation in the solvent's loading capacity determines the required solvent-to-feed ratio. Since high partition ratios generally allow for low solvent use, smaller and less costly extraction equipment may be used and costs for solvent recovery and recycle are lower. In principle, partition ratios less than $K_i = 1.0$ may be accommodated by using a high solvent-to-feed ratio, but usually at much higher cost.

3. *Solute selectivity*. In certain applications, it is important not only to recover a desired solute from the feed, but also to separate it from other solutes present in the feed and thereby achieve a degree of solute purification. The selectivity of a given solvent for solute i compared to solute j is characterized by the separation factor $\alpha_{ij} = K_i/K_j$. Values must be greater than $\alpha_{ij} = 1.0$ to achieve an increase in solute purity (on a solvent-free basis). When solvent blends are used in a commercial process, often it is because the blend provides higher selectivity, and often at the expense of a somewhat lower partition ratio. The degree of purification that can be achieved also depends on the extraction scheme chosen for the process, the amount of extraction solvent, and the number of stages employed.

4. *Mutual solubility*. Low liquid-liquid mutual solubility between feed and solvent phases is desirable because it reduces the separation requirements for removing solvents from the extract and raffinate streams. Low solubility of extraction solvent in the raffinate phase often results in high relative volatility for stripping the residual solvent in a raffinate stripper, allowing low-cost desolventizing of the raffinate [Hwang, Keller, and Olson, *Ind. Eng. Chem. Res.*, **31**(7), pp. 1753–1759 (1992)]. Low solubility of feed solvent in the extract phase reduces separation requirements for recovering solvent for recycle and producing a purified product solute. In some cases, if the solubility of feed solvent in the extract is high, more than one distillation operation will be required to separate the extract phase. If mutual solubility is nil (as for aliphatic hydrocarbons dissolved in water), the need for stripping or another treatment method may be avoided as long as efficient liquid-liquid phase separation can be accomplished without entrainment of solvent droplets into the raffinate. However, very low mutual solubility normally is achieved at the expense of a lower partition ratio for extracting the desired solute—because a solvent that has very little compatibility with the feed solvent is not likely to be a good extractant for something that is dissolved in the feed solvent—and therefore has some compatibility. Mutual solubility also limits the solvent-to-feed ratios that can be used, since a point can be reached where the solvent stream is so large it dissolves the entire feed stream, or the solvent stream is so small it is dissolved by the feed, and these can be real limitations for systems with high mutual solubility.

5. *Stability*. The solvent should have little tendency to react with the product solute and form unwanted by-products, causing a loss in yield. Also it should not react with feed components or degrade to undesirable contaminants that cause development of undesirable odors or color over time, or cause difficulty achieving desired product purity, or accumulate in the process because they are difficult to purge.

6. *Density difference*. As a general rule, a difference in density between solvent and feed phases on the order of 0.1 to 0.3 g/mL is preferred. A value that is too low makes for poor or slow liquid-liquid phase separation and may require use of a centrifuge. A value that is too high makes it difficult to build high dispersed-droplet population density for good mass transfer; i.e., it is difficult to mix the two phases together and maintain high holdup of the dispersed phase within the extractor—but this depends on the viscosity of the continuous phase.

7. *Viscosity*. Low viscosity is preferred since higher viscosity generally increases mass-transfer resistance and liquid-liquid phase

separation difficulty. Sometimes an extraction process is operated at an elevated temperature where viscosity is significantly lower for better mass-transfer performance, even when this results in a lower partition ratio. Low viscosity at ambient temperatures also facilitates transfer of solvent from storage to processing equipment.

8. *Interfacial tension*. Preferred values for interfacial tension between the feed phase and the extraction solvent phase generally are in the range of 5 to 25 dyn/cm (1 dyn/cm is equivalent to 10^{-3} N/m). Systems with lower values easily emulsify. For systems with higher values, dispersed droplets tend to coalesce easily, resulting in low interfacial area and poor mass-transfer performance unless mechanical agitation is used.

9. *Recoverability*. The economical recovery of solvent from the extract and raffinate is critical to commercial success. Solvent physical properties should facilitate low-cost options for solvent recovery, recycle, and storage. For example, the use of relatively low-boiling organic solvents with low heats of vaporization generally allows cost-effective use of distillation and stripping for solvent recovery. Solvent properties also should enable low-cost methods for purging impurities from the overall process (lights and/or heavies) that may accumulate over time. One of the challenges often encountered in utilizing a high-boiling solvent or extractant involves accumulation of heavy impurities in the solvent phase and difficulty in removing them from the process. Another consideration is the ease with which solvent residues can be reduced to low levels in final extract or raffinate products, particularly for food-grade products and pharmaceuticals.

10. *Freezing point*. Solvents that are liquids at all anticipated ambient temperatures are desirable since they avoid the need for freeze protection and/or thawing of frozen solvent prior to use. Sometimes an “antifreeze” compound such as water or an aliphatic hydrocarbon can be added to the solvent, or the solvent is supplied as a mixture of related compounds instead of a single pure component—to suppress the freezing point.

11. *Safety*. Solvents with low potential for fire and reactive chemistry hazards are preferred as inherently safe solvents. In all cases, solvents must be used with a full awareness of potential hazards and in a manner consistent with measures needed to avoid hazards. For information on the safe use of solvents and their potential hazards, see Sec. 23, “Safety and Handling of Hazardous Materials.” Also see Crowl and Louvar, *Chemical Process Safety: Fundamentals with Applications* (Prentice-Hall, 2001); Yaws, *Handbook of Chemical Compound Data for Process Safety* (Elsevier, 1997); Lees, *Loss Prevention in the Process Industries* (Butterworth, 1996); and Bretherick's *Handbook of Reactive Chemical Hazards*, 6th ed., Urben and Pitt, eds. (Butterworth-Heinemann, 1999).

12. *Industrial hygiene*. Solvents with low mammalian toxicity and good warning properties are desired. Low toxicity and low dermal absorption rate reduce the potential for injury through acute exposure. A thorough review of the medical literature must be conducted to ascertain chronic toxicity issues. Measures needed to avoid unsafe exposures must be incorporated into process designs and implemented in operating procedures. See Goetsch, *Occupational Safety and Health for Technologists, Engineers, and Managers* (Prentice-Hall, 2004).

13. *Environmental requirements*. The solvent must have physical or chemical properties that allow effective control of emissions from vents and other discharge streams. Preferred properties include low aquatic toxicity and low potential for fugitive emissions from leaks or spills. It also is desirable for a solvent to have low photoreactivity in the atmosphere and be biodegradable so it does not persist in the environment. Efficient technologies for capturing solvent vapors from vents and condensing them for recycle include activated carbon adsorption with steam regeneration [Smallwood, *Solvent Recovery Handbook* (McGraw-Hill, 1993), pp. 7–14] and vacuum-swing adsorption [Pezolt et al., *Environmental Prog.*, **16**(1), pp. 16–19 (1997)]. The optimization of a process to increase the efficiency of solvent utilization is a key aspect of waste minimization and reduction of environmental impact. An opportunity may exist to reduce solvent use through application of countercurrent processing and other chemical engineering principles aimed at improving processing efficiencies. For a discussion of environmental issues in

process design, see Allen and Shonnard, *Green Engineering: Environmentally Conscious Design of Chemical Processes* (Prentice-Hall, 2002)]. Also see Sec. 22, "Waste Management."

14. **Multiple uses.** It is desirable to use as the extraction solvent a material that can serve a number of purposes in the manufacturing plant. This avoids the cost of storing and handling multiple solvents. It may be possible to use a single solvent for a number of different extraction processes practiced in the same facility, either in different equipment operated at the same time or by using the same equipment in a series of product campaigns. In other cases, the solvent used for extraction may be one of the raw materials for a reaction carried out in the same facility, or a solvent used in another operation such as a crystallization.

15. **Materials of construction.** It is desirable for a solvent to allow the use of common, relatively inexpensive materials of construction at moderate temperatures and pressures. Material compatibility and potential for corrosion are discussed in Sec. 25, "Materials of Construction."

16. **Availability and cost.** The solvent should be readily available at a reasonable cost. Considerations include the initial fill cost, the investment costs associated with maintaining a solvent inventory in the plant (particularly when expensive extractants are used), as well as the cost of makeup solvent.

COMMERCIAL PROCESS SCHEMES

For the purpose of illustrating process concepts, liquid-liquid extraction schemes typically practiced in industry may be categorized into a number of general types, as discussed below.

Standard Extraction Also called simple extraction or single-solvent extraction, standard extraction is by far the most widely practiced type of extraction operation. It can be practiced using single-stage or multistage processing, cross-current or countercurrent flow of solvent, and batch-wise or continuous operation. Figure 15-6 illustrates the contacting stages and liquid streams associated with a typical multistage, countercurrent scheme. Standard extraction is analogous to stripping in distillation because the process involves transferring or stripping components from the feed phase into another phase. Note that the feed (F) enters the process where the extract stream (E) leaves the process, analogous to feeding the top of a stripping tower. And the raffinate (R) leaves where the extraction solvent (S) enters. Standard extraction is used to remove contaminants from a crude liquid feed (product purification) or to recover valuable components from the feed (product recovery). Applications can involve very dilute feeds, such as when purifying a liquid product or detoxifying a wastewater stream, or concentrated feeds, such as when recovering a crude product from a reaction mixture. In either case, standard extraction can be used to transfer a high fraction of solute from the feed phase into the extract. Note, however, that transfer of the desired solute or solutes may be accompanied by transfer of unwanted solutes. Because of this, standard extraction normally cannot achieve satisfactory solute purity in the extract stream unless the separation factor for the desired solute with respect to unwanted solutes is at least $\alpha_{i,j} = K_i/K_j = 20$ and usually much higher. This depends on the crude feed purity and the product purity specification. (See "Potential for Solute Purification Using Standard Extraction" under "Process Fundamentals and Basic Calculation Methods.")

Fractional Extraction Fractional extraction combines solute recovery with cosolute rejection. In principle, the process can achieve high solute recovery and high solute purity even when the solute separation factor is fairly low, as low as $\alpha_{i,j} = 4$ or so (see "Dual-Solvent Fractional Extraction" under "Calculation Procedures"). Dual-solvent fractional extraction utilizes an extraction solvent (S) and a wash solvent (W) and includes a stripping section at the raffinate end of the process (for product-solute recovery) and a washing section at the extract end of the process (for cosolute rejection and product purification) (Fig. 15-7). The feed enters the process at an intermediate stage located between the extract and raffinate ends. In this respect, the process is analogous to a middle-fed fractional distillation, although the analogy is not exact since wash solvent is added to the extract end of the process instead of returning a reflux stream. The

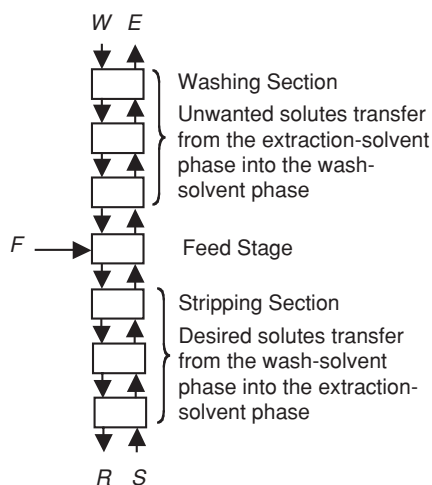


FIG. 15-7 Dual-solvent fractional extraction without reflux.

desired solutes transfer into the extraction solvent (the extract phase) within the stripping section, and unwanted solutes transfer into the wash solvent (the raffinate phase) within the washing section. Typically, the feed stream consists of feed solutes predissolved in wash solvent or extraction solvent; or, if they are liquids, they may be injected directly into the process. To maximize performance, a fractional extraction process may be operated such that the washing and stripping sections are carried out in different equipment and at different temperatures. The stripping section is sometimes called the extraction section, and the washing section is sometimes called the enriching section, the scrubbing section, or the absorbing section. A dual-solvent fractional extraction process involving reflux to the washing section is shown in Fig. 15-8.

In a special case referred to as single-solvent fractional extraction with extract reflux, the wash solvent is comprised of components that

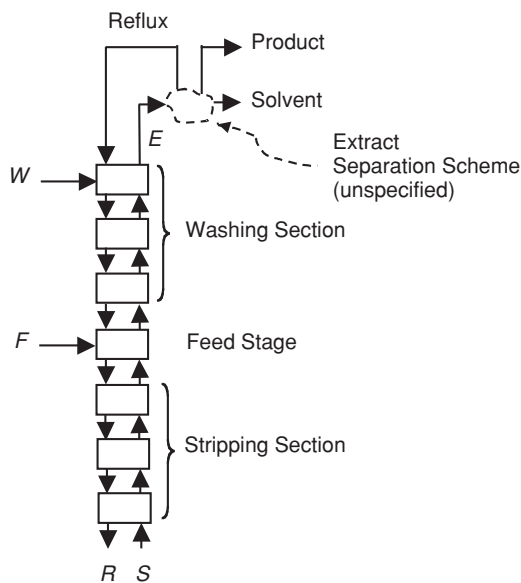


FIG. 15-8 Process concepts for dual-solvent fractional extraction with extract reflux.

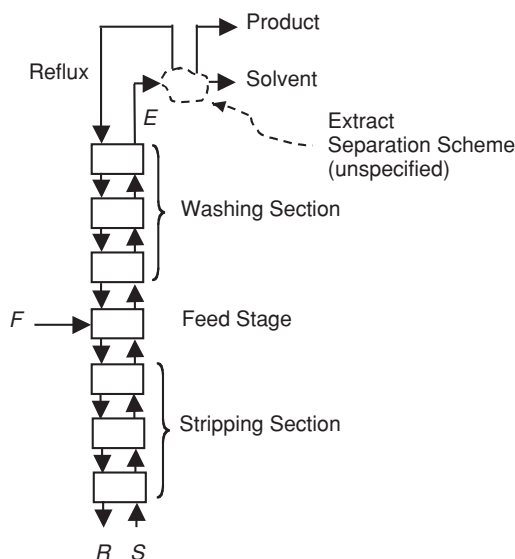


FIG. 15-9 Process concepts for single-solvent fractional extraction with extract reflux. The process flow sheet shown in Fig. 15-2 is an example of this general process scheme.

enter the overall process with the feed and return as reflux (Fig. 15-9). This is the type of extraction scheme commonly used to recover aromatic components from crude hydrocarbon mixtures using high-boiling polar solvents (as in Fig. 15-2). A reflux stream rich in light aromatics including benzene is refluxed to the washing section to serve as wash solvent. This process scheme is very similar in concept to fractional distillation. It is used only in a very limited number of applications [Stevens and Pratt, Chap. 6, in *Science and Practice of Liquid-Liquid Extraction*, vol. 1, Thornton, ed. (Oxford, 1992), pp. 379–395]. More detailed discussion is given in “Single-Solvent Fractional Extraction with Extract Reflux” under “Calculation Procedures.”

In terms of common practice, fractional extraction operations may be classified into several types: (1) standard extraction augmented by addition of a washing section utilizing a relatively small amount of feed solvent as the wash solvent; (2) full fractionation (less common); and (3) full fractionation with solute reflux (much less common). The first two categories are examples of dual-solvent fractional extraction. The third category can be practiced as dual-solvent or single-solvent fractional extraction.

In the first type of operation, a relatively small amount of feed solvent is added to a short washing section as wash solvent. (The word *short* is used here in an extraction column context, but refers in general to a relatively few theoretical stages.) This approach is useful for systems exhibiting a moderate to high solute separation factor ($\alpha_{ij} > 20$ or so) and requiring a boost in product-solute purity. An example involves recovery of an organic solute from a dilute brine feed by using a partially miscible organic solvent. In this case, the inorganic salt present in the aqueous feed stream has some solubility in the organic solvent phase because of water that saturates that phase, and the partition ratio for transfer of salt into the organic phase is small (i.e., the partition ratio for transfer of salt into wash water is high). Adding wash water to the extract end of the process has the effect of washing a portion of the soluble salt content out of the organic extract. The reduction in salt content depends on how much wash water is added and how many washing stages or transfer units are used in the design.

The second type of fractional extraction operation involves the use of stripping and washing sections without reflux (Fig. 15-7) to separate a mixture of feed solutes with close K values. In this case, the solute separation factor is low to moderate. Normally, α_{ij} must be greater than about 4 for a commercially viable process. Scheibel [*Chem. Eng. Prog.*, **44**(9), pp. 681–690 (1948); and **44**(10), pp. 771–782 (1948)] gives several

instructive examples of fractional extraction: (1) separation of ortho and para chloronitrobenzenes using heptane and 85% aqueous methanol as solvents ($\alpha_{\text{para,ortho}} \approx 1.6$ to 1.8); (2) separation of ethanol and isopropanol by using water and xylene ($\alpha_{\text{ethanol,isopropanol}} \approx 2$); and (3) separation of ethanol and methyl ethyl ketone (MEK) by using water and kerosene ($\alpha_{\text{ethanol,MEK}} \approx 10$ to 20). The first two applications demonstrate fractional extraction concepts, but a sharp separation is not achieved because the selectivity of the solvent is too low. In these kinds of applications, fractional extraction might be combined with another separation operation to complete the separation. (See “Hybrid Extraction Processes.”) In Scheibel’s third example, the selectivity is much higher and nearly complete separation is achieved by using a total of about seven theoretical stages. In another example, Venter and Nieuwoudt [*Ind. Eng. Chem. Res.*, **37**(10), pp. 4099–4106 (1998)] describe a dual-solvent extraction process using hexane and aqueous tetraethylene glycol to selectively recover *m*-cresol from coal pyrolysis liquors also containing *o*-toluonitrile. This process has been successfully implemented in industry. The separation factor for *m*-cresol with respect to *o*-toluonitrile varies from 5 to 70 depending upon solvent ratios and the resulting liquid compositions. The authors compare a standard extraction configuration (bringing the feed into the first stage) with a fractional extraction configuration (bringing the feed into the second stage of a seven theoretical-stage process).

Another example of the use of dual-solvent fractional extraction concepts involves the recovery of ϵ -caprolactam monomer (for nylon-6 production) from a two-liquid-phase reaction mixture containing ammonium sulfate plus smaller amounts of other impurities, using water and benzene as solvents [Simons and Haasen, Chap. 18.4 in *Handbook of Solvent Extraction* (Wiley, 1983; Krieger, 1991)]. In this application, the separation factor for caprolactam with respect to ammonium sulfate is high because the salt greatly favors partitioning into water; however, separation factors with respect to the other impurities are smaller. Alessi et al. [*Chem. Eng. Technol.*, **20**, pp. 445–454 (1997)] describe two process schemes used in industry. These are outlined in Fig. 15-10. The simpler scheme (Fig. 15-10a) is a straightforward dual-solvent fractional extraction process that isolates caprolactam (CPL) in a benzene extract stream and ammonium sulfate (AS) in the aqueous raffinate. The feed stage is comprised of mixer M1 and settler S1, and separate extraction columns are used for the washing and stripping sections. In Fig. 15-10a, these are denoted by C1 and C2, respectively. Minor impurity components also present in the feed must exit the process in either the extract or the raffinate. The more complex scheme (Fig. 15-10b) eliminates addition of benzene to the feed stage and adds a back-extraction section at the extract end of the process (denoted by C4) to extract CPL from the benzene phase leaving the washing section. Also, a separate fractional extractor (denoted as C1 in Fig. 15-10b) is added between the original stripping and washing sections to treat the benzene phase leaving the stripping section and recover the CPL content of the CPL-rich aqueous stream leaving the feed stage. In the C1 extractor, the CPL transfers into the benzene stream that ultimately enters the upper washing section, leaving hydrophilic impurities in an aqueous purge stream that exits at the bottom. The resulting process scheme includes two purge streams for rejecting minor impurities: a stream rich in heavy organic impurities leaving the bottom of the benzene distillation tower and the aqueous stream rich in hydrophilic impurities leaving the bottom of the C1 extractor. This sophisticated design separates the feed into four streams instead of just two, allowing separate removal of two impurity fractions to increase the purity of the two main products. The caprolactam is made to transfer into either an aqueous or a benzene-rich stream as desired, by judicious choice of solvent-to-feed ratio at the various sections in the process (perhaps aided by adjustment of temperature).

A dual-solvent fractional extraction process can provide a powerful separation scheme, as indicated by the examples given above, and some authors suggest that fractional extraction is not utilized as much as it could be. In many cases, instead of using full fractional extraction, standard extraction is used to recover solute from a crude feed; and if the solvent-to-feed ratio is less than 1.0, concentrate the solute in a smaller solute-bearing stream. Another operation such as crystallization, adsorption, or process chromatography is then used downstream for solute purification. Perhaps fractional extraction schemes should be evaluated more often as an alternative processing scheme that may have advantages.

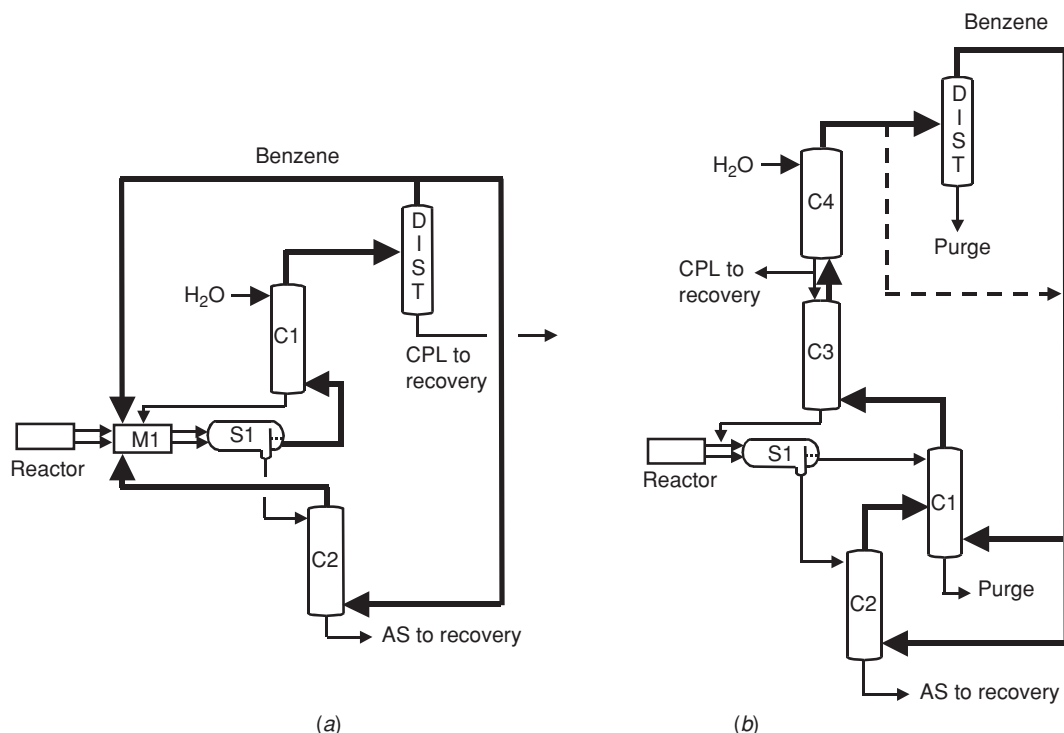


FIG. 15-10 Two industrial extraction processes for separation of caprolactam (CPL) and ammonium sulfate (AS): (a) a simpler fractional extraction scheme; (b) a more complex scheme. Heavy lines denote benzene-rich streams; light lines denote aqueous streams. [Taken from Alessi, Penzo, Slater, and Tessari, *Chem. Eng. Technol.*, **20**(7), pp. 445–454 (1997), with permission. Copyright 1997 Wiley-VCH.]

The third type of fractional extraction operation involves refluxing a portion of the extract stream back to the extract end (washing section) of the process. As mentioned earlier, this process can be practiced as a dual-solvent process (Fig. 15-8) or as a single-solvent process (Figs. 15-2 and 15-9). However, unlike in distillation, the use of reflux is not common. The reflux consists of a portion of the extract stream from which a significant amount of solvent has been removed. Injection of this solvent-lean, concentrated extract back into the washing section increases the total amount of solute and the amount of raffinate phase present in that section of the extractor. This can boost separation performance by allowing the process to operate at a more favorable location within the phase diagram, resulting in a reduction in the number of theoretical stages or transfer units needed within the washing section. This also allows the process to boost the concentration of solute in the extract phase above that in equilibrium with the feed phase. The increased amount of solute present within the process may require use of extra solvent to avoid approaching the plait point at the feed stage (the composition at which only a single liquid phase can exist at equilibrium). Because of this, utilizing reflux normally involves a tradeoff between a reduction in the number of theoretical stages and an increase in the total liquid traffic within the process equipment, requiring larger-capacity equipment and increasing the cost of solvent recovery and recycle. This tradeoff is discussed by Scheibel with regard to extraction column design [*Ind. Eng. Chem.*, **47**(11), pp. 2290–2293 (1955)]. The potential benefit that can be derived from the use of extract reflux is greatest for applications utilizing solvents with a low solute separation factor and low partition ratios (as in the example illustrated in Fig. 15-2). In these cases, reflux serves to reduce the number of required theoretical stages or transfer units to a practical number on the order of 10 or so, or reduce the solvent-to-feed ratio required for the desired separation.

The fractional extraction schemes described above are typical of those practiced in industry. A related kind of process employs a second solvent in a separate extraction operation to wash the raffinate

produced in an upstream extraction operation. This process scheme is particularly useful when the wash solvent is only slightly soluble in the raffinate and can easily be removed. An example is the use of water to remove residual amine solvent from the treated hydrocarbon stream in an acid-gas extraction process (Fig. 15-3).

A potential fourth type of fractional extraction operation involves the use of reflux at both ends of a dual-solvent process, i.e., reflux to the raffinate end of the process (the stripping section) as well as reflux to the extract end of the process (the washing section). The authors are not aware of a commercial application of this kind; however, Scheibel [*Chem. Eng. Prog.*, **62**(9), pp. 76–81 (1966)] discusses such a process scheme in light of several potential flow sheets. In the special case of single-solvent fractional extraction with extract reflux, Skelland [*Ind. Eng. Chem.*, **53**(10), pp. 799–800 (1961)] has pointed out that addition of raffinate reflux is not effective from a strictly thermodynamic point of view as it cannot reduce the required number of theoretical stages in this special case.

Dissociative Extraction This process scheme normally involves partitioning of weak organic acids or bases between water and an organic solvent. Whether the solute partitions mainly into one phase or the other depends upon whether it is in its neutral state or its charged ionic state and the ability of each phase to solvate that form of the solute. In general, water interacts much more strongly with the charged species, and the ionic form will strongly favor partitioning into the aqueous phase. The nonionic form generally will favor partitioning into the organic phase.

The pK_a is the pH at which 50 percent of the solute is in the dissociated (ionized) state. It is a function of solute concentration and normally is reported for dilute conditions. For an organic acid ($RCOOH$) dissolved in aqueous solution, the amount of solute in the dissociated state relative to that in the nondissociated state is $[RCOO^-]/[RCOOH] = 10^{pH-pK_a}$. Extraction of an organic acid out of an organic feed into an aqueous phase is greatly facilitated by operating at a pH

above the acid's pK_a value because the majority of the acid will be deprotonated to yield the dissociated form ($RCOO^-$). On the other hand, partitioning of the organic acid from an aqueous feed into an organic solvent is favored by operating at a pH below its pK_a to ensure most of the acid is in the protonated (nondissociated) form. Another example involves extraction of a weak base, such as a compound with amine functionality (RNH_2), out of an organic phase into water at a pH below the pK_a . This will protonate or neutralize the majority of the base, yielding the ionized form (RNH_3^+) and favoring extraction into water. It follows that extracting an organic base out of an aqueous feed into an organic solvent is favored by operating at a pH above its pK_a since this yields most of the solute in the free base (nonionized) form. For weak bases, $pK_a = 14 - pK_b$, and the relative amount of solute in the dissociated state in the aqueous phase is given by $10^{pK_a - pH}$. In principle, to obtain the maximum partition ratio for an extraction, the pH should be maintained about 2 units from the solute's pK_a value to obtain essentially complete dissociation or nondissociation, as appropriate for the extraction. In a typical continuous application, the pH of the aqueous stream leaving the process is controlled at a constant pH set point by injection of acid or base at the opposite end of the process, and a pH gradient exists within the process. The pH set point may be adjusted to optimize performance. The effect of pH on the partition ratio is discussed in "Effect of pH for Ionizable Organic Solutes" under "Thermodynamic Basis for Liquid-Liquid Extraction." Determination of the optimum pH for extraction of compounds with multiple ionizable groups and thus multiple pK_a values is discussed by Crocker, Wang, and McCauley [*Organic Process Res. Dev.*, **5**(1), pp. 77–79 (2001)].

In fractional dissociative extraction, a sharp separation of feed solutes is achieved by taking advantage of a difference in their pK_a values. If the difference in pK_a is sufficient, controlling pH at a specific value can yield high K values for one solute fraction and very low K values for another fraction, thus allowing a sharp separation. For example, a mixture of two organic bases can be separated by contacting the mixture with an aqueous acid containing less than the stoichiometric amount of acid needed to neutralize (ionize) both bases. The stronger of the two bases reacts with the acid to yield the dissociated form in the aqueous phase, while the other base remains undissociated in a separate organic phase. Buffer compounds may be used to control pH within a desired range for improved separation results [Ma and Jha, *Organic Process Res. Dev.*, **9**(6), pp. 847–852 (2005)]. Buffers are discussed by Perrin and Dempsey [*Buffers for pH and Metal Ion Control* (Chapman and Hall, 1979)]. For additional discussion, see Pratt, Chap. 21 in *Handbook of Solvent Extraction*, Lo, Baird, and Hanson, eds. (Wiley, 1983; Krieger, 1991), and Anwar, Arif, and Pritchard, *Solvent Ext. Ion Exch.*, **16**, p. 931 (1998).

pH-Swing Extraction A pH-swing extraction process utilizes dissociative extraction concepts to recover and purify ionizable organic solutes in a forward- and back-extraction scheme, each extraction operation carried out at a different pH. For example, in the forward extraction, the desired solute may be in its nonionized state so it can be extracted out of a crude aqueous feed into an organic solvent. The extract stream from this operation is then fed to a separate extraction operation where the solute is ionized by readjustment of pH and back-extracted into clean water. This scheme can achieve both high recovery and high purity if the impurity solutes are not ionizable or have pK_a values that differ greatly from those of the desired solute. A pH-swing extraction scheme commonly is used for recovery and purification of antibiotics and other complex organic solutes with some ionizable functionality. The production of high-purity food-grade phosphoric acid from lower-grade acid is another example of a pH-swing process ["Purification of Wet Phosphoric Acid" in *Ullmann's Encyclopedia of Industrial Chemistry*, 6th ed. (VCH, 2002)].

Reaction-Enhanced Extraction Reaction-enhanced extraction involves enhancement of the partition ratio for extraction through the use of a reactive extractant that forms a reversible adduct or molecular complex with the desired solute. Normally, the extractant compound is dissolved in a diluent liquid such as kerosene or another high-boiling hydrocarbon. Because reactive extractants form strong specific interactions with the solute molecule, they can provide much

higher partition ratios and generally are more selective compared to conventional solvents. Also, when used to recover relatively volatile compounds, extractants may allow significant reduction in the energy required to separate the extract phase by distillation. Extractants are successfully used at very large scales to recover metals in hydrometallurgical processing, among other applications. However, it is important to note that the use of high-boiling extractants can present severe difficulties whenever high-boiling impurities are present. A number of commercial processes have failed because there was no economical option for purging high-boiling contaminants that accumulated in the solvent phase over time, so care must be taken to address this possibility when developing a new application. The advantages and disadvantages of using high-boiling solvents or extractants versus low-boiling solvents are discussed by King in the context of acetic acid recovery [Chap. 18.5 in *Handbook of Solvent Extraction*, Lo, Baird, and Hanson, eds. (Wiley, 1983; Krieger, 1991)].

Detailed reviews of reactive extractants are given by Cox [Chap. 1 in *Science and Practice of Liquid-Liquid Extraction*, vol. 2 (Oxford, 1992), (pp. 1–27)] and by King [Chap. 15 in *Handbook of Separation Process Technology*, Rousseau, ed. (Wiley, 1987)]. Also see *Solvent Extraction Principles and Practice*, 2d ed., Rydberg et al., eds. (Dekker, 2004). Cox has classified extractants as either acidic, ion-pair-forming or solvating (nonionic) according to the mechanism of solute-solvent interaction in solution. In hydrometallurgical applications involving recovery or purification of metals dissolved in aqueous feed solutions, commercial extractants include acid chelating agents, alkyl amines, and various organophosphorous compounds including trioctylphosphine oxide (TOPO) and tri-*n*-butyl phosphate, plus quaternary ammonium salts. A well-known example is the use of TOPO to remove arsenic impurities from copper electrolyte solutions produced in copper refining operations. Another well-known class of applications involves formation of ion-pair interactions between a carboxylic acid dissolved in an aqueous feed and alkylamine extractants such as trioctylamine dissolved in a hydrocarbon diluent, as discussed by Wennersten [*J. Chem. Technol. Biotechnol.*, **33B**, pp. 85–94 (1983)], by King and others [*Ind. Eng. Chem. Res.*, **29**(7), pp. 1319–1338 (1990); and *Chemtech*, **22**, p. 285 (1992)], and by Schunk and Maurer [*Ind. Eng. Chem. Res.*, **44**(23), pp. 8837–8851 (2005)]. Extractants also may be used to facilitate extraction of other ionizable organic solutes including certain antibiotics [Pai, Doherty, and Malone, *AIChE J.*, **48**(3), pp. 514–526 (2002)]. Sometimes mixing extractants with promoter compounds (called modifiers) provides synergistic effects that dramatically enhance the partition ratio. An example is discussed by Atanassova and Dukov [*Sep. Purif. Technol.*, **40**, pp. 171–176 (2004)]. Also see the discussion of combined physical (hydrogen-bonding) and reaction-enhanced extraction by Lee [*Biotechnol. Prog.*, **22**(3), pp. 731–736 (2006)].

Extractive Reaction Extractive reaction combines reaction and separation in the same unit operation for the purpose of facilitating a desired reaction. To avoid confusion, the term *extractive reaction* is recommended for this type of process, while the term *reaction-enhanced extraction* is recommended for a process involving formation of reversible solute-extractant interactions and enhanced partition ratios for the purpose of facilitating a desired separation. The term *reactive extraction* is a more general term commonly used for both types of processes.

In general, extractive reaction involves carrying out a reaction in the presence of two liquid phases and taking advantage of the partitioning of reactants, products, and homogeneous catalyst (if used) between the two phases to improve reaction performance. The classes of reactions that can benefit from an extractive reaction scheme include chemical-equilibrium-limited reactions (such as esterifications, transesterifications, and hydrolysis reactions), where it is important to remove a product or by-product from the reaction zone to drive conversion, and consecutive or sequential reactions (such as nitrations, sulfonations, and alkylations), where the goal may be to produce only the mono- or difunctional product and minimize formation of subsequent addition products. For additional discussion, see Gorissen, *Chem Eng. Sci.*, **58**, pp. 809–814 (2003); Van Brunt and Kanel, Chap. 3, in *Reactive Separation Processes*, S. Kulprathipanja, ed. (Taylor & Francis, 2002), pp. 51–92; and Hanson, "Extractive Reaction Processes," Chap. 22 in *Handbook of Solvent*

Extraction, Lo, Baird, and Hanson, eds. (Wiley, 1983; Krieger, 1991), pp. 615–618.

The manufacture of fatty acid methyl esters (FAME) for use as biodiesel fuel, by transesterification of triglyceride oils and greases [Canakci and Van Gerpen, *ASAE Trans.*, **46**(4), pp. 945–954 (2003)], provides an example of a chemical-equilibrium-limited extractive reaction. Low-grade triglycerides are reacted with methanol to produce FAME plus glycerin as a by-product. Because glycerin is only partially miscible with the feed and the FAME product, it transfers from the reaction zone into a separate glycerin-rich liquid phase, driving further conversion of the triglycerides. In another example, Minotti, Doherty, and Malone [*Ind. Eng. Chem. Res.*, **37**(12), pp. 4748–4755 (1998)] studied the esterification of aqueous acetic acid by reaction with butanol in an extractive reaction process involving extraction of the butyl acetate product into a separate butanol-rich phase. The authors concluded that cocurrent processing is preferred over countercurrent processing in this case. Their general conclusions likely apply to other applications involving extraction of a reaction product out of the reaction phase to drive conversion. The cocurrent scheme is equivalent to a series of two-liquid-phase stirred-tank reactors approaching the performance of a plug-flow reactor. Rohde, Marr, and Siebenhofer [Paper no. 232f, AIChE Annual Meeting, Austin, Tex., Nov. 7–12, 2004] studied the esterification of acetic acid with methanol to produce methyl acetate. Their extractive reaction scheme involves selective transfer of methyl acetate into a high-boiling solvent such as *n*-nonane.

An example of a sequential-reaction extractive reaction is the manufacture of 2,4-dinitrotoluene, an important precursor to 2,4-diaminotoluene and toluene diisocyanate (TDI) polyurethanes. The reaction involves nitration of toluene by using concentrated nitric and sulfuric acids which form a separate phase. Toluene transfers into the acid phase where it reacts with nitronium ion, and the reaction product transfers back into the organic phase. Careful control of liquid-liquid contacting conditions is required to obtain high yield of the desired product and minimize formation of unwanted reaction products. A similar reaction involves nitration of benzene to mononitrobenzene, a precursor to aniline used in the manufacture of many products including methylenediphenylisocyanate (MDI) for polyurethanes [Quadros, Reis, and Baptista, *Ind. Eng. Chem. Res.*, **44**(25), pp. 9414–9421 (2005)].

Another category of extractive reaction involves the extraction of a product solute during microbial fermentation (biological reaction) to avoid microbe inhibition effects, allowing an increase in fermenter productivity. An example involving production of ethanol is discussed by Weilhammer and Blass [*Chem. Eng. Technol.*, **17**, pp. 365–373 (1994)], and an example involving production of propionic acid is discussed by Gu, Glatz, and Glatz [*Biotechnol. and Bioeng.*, **57**(4), pp. 454–461 (1998)]. Finally, the scrubbing of reactive components from a feed liquid, by irreversible reaction with a treating solution, also may be considered an extractive reaction. An example is removal of acidic components from petroleum liquids by reaction with aqueous NaOH.

Temperature-Swing Extraction Temperature-swing processes take advantage of a change in *K* value with temperature. An extraction example is the commercial process used to recover citric acid from whole fermentation broth by using trioctylamine (TOA) extractant [Baniel et al., U.S. Patent 4,275,234 (1981); Wennersten, *J. Chem. Biotechnol.*, **33B**, pp. 85–94 (1983); and Pazouki and Panda, *Bioprocess Eng.*, **19**, pp. 435–439 (1998)]. This process involves a forward reaction-enhanced extraction carried out at 20 to 30°C in which citric acid transfers from the aqueous phase into the extract phase. Relatively pure citric acid is subsequently recovered by back extraction into clean water at 80 to 100°C, also liberating the TOA extractant for recycle. This temperature-swing process is feasible because partitioning of citric acid into the organic phase is favored at the lower temperature but not at 80 to 100°C.

Partition ratios can be particularly sensitive to temperature when solute-solvent interactions in one or both phases involve specific attractive interactions such as formation of ion-pair bonds (as in trialkylamine–carboxylic acid interactions) or hydrogen bonds, or when mutual solubility between feed and extraction solvent involves hydrogen bonding. An interesting example is the extraction of citric acid from water with 1-butoxy-2-propanol (common name propylene glycol *n*-butyl ether) as solvent (Fig. 15-11). This example illustrates how important it can be when developing and optimizing an extraction operation to understand how *K* varies with temperature, regardless of whether a temperature-swing process is contemplated. Of course, changes in other properties such as mutual solubility and viscosity also must be considered. For additional discussion, see “Temperature Effect” under “Thermodynamic Basis for Liquid-Liquid Extraction.”

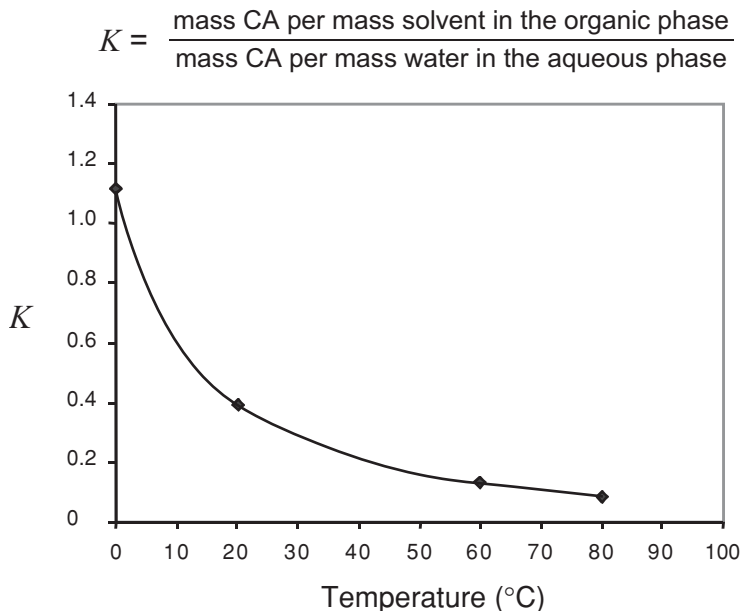


FIG. 15-11 Partition ratio as a function of temperature for recovery of citric acid (CA) from water using 1-butoxy-2-propanol (propylene glycol *n*-butyl ether). (Data generated by The Dow Chemical Company.)

Reversed Micellar Extraction This scheme involves use of microscopic water-in-oil micelles formed by surfactants and suspended within a hydrophobic organic solvent to isolate proteins from an aqueous feed. The micelles essentially are microdroplets of water having dimensions on the order of the protein to be isolated. These stabilized water droplets provide a compatible environment for the protein, allowing its recovery from a crude aqueous feed without significant loss of protein activity [Ayala et al., *Biotechnol. and Bioeng.*, **39**, pp. 806–814 (1992); and Bordier, *J. Biol. Chem.*, **256**(4), pp. 1604–1607 (February 1981)]. Also see the discussion of ultrafiltration membranes for concentrating micelles in “Liquid-Liquid Phase Separation Equipment.”

Aqueous Two-Phase Extraction Also called aqueous biphasic extraction, this technique generally involves use of two incompatible water-miscible polymers [normally polyethylene glycol (PEG) and dextran, a starch-based polymer], or a water-miscible polymer and a salt (such as PEG and Na_2SO_4), to form two immiscible aqueous phases each containing 75+% water. This technology provides mild conditions for recovery of proteins and other biomolecules from broth or other aqueous feeds with minimal loss of activity [Walter and Johansson, eds., *Aqueous Two Phase Systems, Methods in Enzymology*, vol. 228 (Academic, 1994); Zaslavsky, *Aqueous Two-Phase Partitioning* (Dekker, 1994); and Blanch and Clark, Chap. 6 in *Biochemical Engineering* (Dekker, 1997) pp. 474–482]. The effect of salts on the liquid-liquid phase equilibrium of polyethylene glycol + water mixtures has been extensively studied [Salabat, *Fluid Phase Equil.*, **187–188**, pp. 489–498 (2001)]. A typical phase diagram, for PEG 6000 + Na_2SO_4 + water, is shown in Fig. 15-12. The hydraulic characteristics of the aqueous two-phase system PEG 4000 + Na_2SO_4 + water in a countercurrent sieve plate column have been reported by Hamidi et al. [*J. Chem. Technol. Biotechnol.*, **74**, pp. 244–249 (1999)]. Two immiscible aqueous phases also may be formed by using two incompatible salts. An example is the system formed by using the hydrophilic organic salt 1-butyl-3-methylimidazolium chloride and a water-structuring (kosmotropic) salt such as K_3PO_4 [Gutowski et al., *J. Am. Chem. Soc.*, **125**, p. 6632 (2003)].

Hybrid Extraction Processes Hybrid processes employ an extraction operation in close association with another unit operation. In these processes, the individual unit operations may not be able to achieve all the separation goals, or the use of one or the other operation alone may not be as economical as the hybrid process. Common examples include the following.

Extraction-distillation An example involves the use of extraction to break the methanol + dichloromethane azeotrope. The near-azeotropic overheads from a distillation tower can be fed to an extrac-

tor where water is used to extract the methanol content and generate nearly methanol-free dichloromethane (saturated with roughly 2000 ppm water). A related type of extraction-distillation operation involves closely coupling extraction with the distillate or bottoms stream produced by a distillation tower, such that the distillation specification for that stream can be relaxed. For example, this approach has been used to facilitate distillation of aqueous acetic acid to produce acetic acid as a bottoms product, taking a mixture of acetic acid and water overhead [Gualy et al., U.S. Patent 5,492,603 (1996)]. The distillate is sent to an extraction tower to recover the acetic acid content for recycle back to the process. The hybrid process allows operation with lower energy consumption compared to distillation alone, because it allows the distillation tower to operate with a reduced requirement for recovering acetic acid in the bottoms stream, which permits relaxation of the minimum concentration of acetic acid allowed in the distillate. Another type of hybrid process involves combining liquid-liquid extraction with azeotropic or extractive distillation of the extract [Skelland and Tedder, chap. 7, in *Handbook of Separation Process Technology*, Rousseau, ed. (Wiley, 1987), pp. 449–453]. The solvent serves both as the extraction solvent for the upstream liquid-liquid extraction operation and as the entrainer for a subsequent azeotropic distillation or as the distillation solvent for a subsequent extractive distillation. (For a detailed discussion of azeotropic and extraction distillation concepts, see Sec. 13, “Distillation.”) The solvent-to-feed ratio must be optimized with regard to both the liquid-liquid extraction operation and the downstream distillation operation. An example is the use of ethyl acetate to extract acetic acid from an aqueous feed, followed by azeotropic distillation of the extract to produce a dry acetic acid bottoms product and an ethyl acetate + water overheads stream. In this example, ethyl acetate serves as the extraction solvent in the extractor and as the entrainer for removing water overhead in the distillation tower. Examples involving extractive distillation and high-boiling solvents can be seen in the various processes used to recover aromatics from aliphatic hydrocarbons, as described by Mueller et al., in *Ullmann's Encyclopedia of Industrial Chemistry*, 5th ed., vol. B3, Gerhartz, ed. (VCH, 1988), pp. 6-34 to 6-43.

Extraction-crystallization Extraction often is used in association with a crystallization operation. In the pharmaceutical and specialty chemical industries, extraction is used to recover a product compound (or remove impurities) from a crude reaction mixture, with subsequent crystallization of the product from the extract (or from the preextracted reaction mixture). In many of these applications, the product needs to be delivered as a pure crystalline solid, so crystallization is a necessary

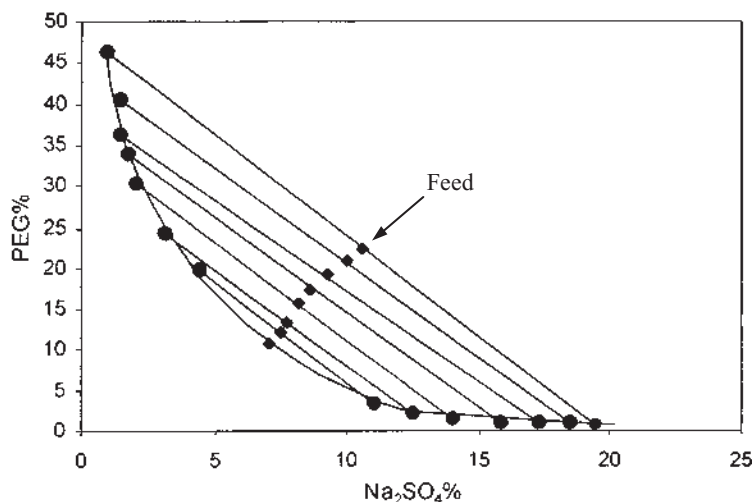


FIG. 15-12 Equilibrium phase diagram for PEG 6000 + Na_2SO_4 + water at 25°C. [Reprinted from Salabat, *Fluid Phase Equil.*, **187–188**, pp. 489–498 (2001), with permission. Copyright 2001 Elsevier B. V.]

operation. (For a detailed discussion of crystallization operations, see Sec. 18, "Liquid-Solid Operations and Equipment.") The desired solute can sometimes be crystallized directly from the reaction mixture with sufficient purity and yield, thus avoiding the cost of the extraction operation; however, direct crystallization generally is more difficult because of higher impurity concentrations. In cases where direct crystallization is feasible, deciding whether to use extraction prior to crystallization or crystallization alone involves consideration of a number of tradeoffs and ultimately depends on the relative robustness and economics of each approach [Anderson, *Organic Process Res. Dev.*, **8**(2), pp. 260–265 (2004)]. A well-known example of extraction-crystallization is the recovery of penicillin from fermentation broth by using a pH-swing forward and back extraction scheme followed by final purification using crystallization [Queener and Swartz, "Penicillins: Biosynthetic and Semisynthetic," in *Secondary Products of Metabolism, Economic Microbiology*, vol. 3, Rose, ed. (Academic, 1979)]. Extraction is used for solute recovery and initial purification, followed by crystallization for final purification and isolation as a crystalline solid. Another category of extraction-crystallization processes involves use of extraction to recover solute from the spent mother liquor leaving a crystallization operation. In yet another example, Maeda et al., [*Ind. Eng. Chem. Res.*, **38**(6), pp. 2428–2433 (1999)] describe a crystallization-extraction hybrid process for separating fatty acids (lauric and myristic acids). In comparing these process options, the potential uses of extraction should include efficient countercurrent processing schemes, since these may significantly reduce solvent usage and cost.

Neutralization-extraction A common example of neutralization-extraction involves neutralization of residual acidity (or basicity) in a crude organic feed by injection of an aqueous base (or aqueous acid) combined with washing the resulting salts into water. The neutralization and washing operations may be combined within a single extraction column as illustrated in Fig. 15-13. Also see the discussion by Koolen [*Design of Simple and Robust Process Plants* (Wiley-VCH, 2001), pp. 159–161].

Reaction-extraction This technique involves chemical modification of solutes in solution in order to more easily extract them in a subsequent extraction operation. Applications generally involve modification of impurity compounds to facilitate purification of a desired product. An example is the oxygenation of sulfur-containing aromatic impurities present in fuel oil by using H_2O_2 and acetic acid, followed by liquid-liquid extraction into an aqueous acetonitrile solution [Shiraishi and Hirai, *Energy and Fuels*, **18**(1), pp. 37–40 (2004); and Shiraishi et al., *Ind. Eng. Chem. Res.*, **41**, pp. 4362–4375 (2002)]. Another example involves esterification of aromatic alcohol impurities to facilitate their separation from apolar hydrocarbons by using an aqueous extractant solution [Kuzmanovid et al., *Ind. Eng. Chem. Res.*, **43**(23), pp. 7572–7580 (2004)].

Reverse osmosis-extraction In certain applications, reverse osmosis (RO) or nanofiltration membranes may be used to reduce the volume of an aqueous stream and increase the solute concentration, in

order to reduce the size of downstream extraction and solvent recovery equipment. Wytcherley, Gentry, and Gualy [U.S. Patents 5,492,625 (1996) and 5,624,566 (1997)] describe such a process for carboxylic acid solutes. Water is forced through the membrane when the operating pressure drop exceeds the natural osmotic pressure difference generated by the concentration gradient:

$$\text{Flux} = \frac{\mathcal{P}}{\lambda_m} (\Delta P - \Delta \pi) \quad (15-1)$$

where $\mathcal{P}$ is a permeability coefficient for water, λ_m is the membrane thickness, ΔP is the operating pressure drop, and $\Delta \pi$ is the osmotic pressure gradient, a function of solute concentration on each side of the membrane. Normally the solute also will permeate the membrane to a small extent. The maximum possible concentration of solute in the concentrate is limited by that corresponding to an osmotic pressure of about 70 bar (about 1000 psig), since this is the maximum pressure rating of commercially available membrane modules (typical). For acetic acid, this maximum concentration is about 25 wt %. Depending upon whether the particular organic permeate of interest can swell or degrade the membrane material, the concentration achieved in practice may need to be reduced below this osmotic-pressure limit to avoid excessive membrane deterioration. In general, a membrane preconcentrator is considered for feeds containing on the order of 3 wt % solute or less. In these cases, a moderate membrane operating pressure may be used, and the preconcentrator can provide a large reduction in the volume of feed entering the extraction process. In these processes, the stream entering the membrane module normally must be carefully prefiltered to avoid fouling the membrane. The general application of RO and nanofiltration membranes is described in Sec. 20, "Alternative Separation Processes." The modeling of mass transfer through RO membranes, with an emphasis on cases involving solute-membrane interactions, is discussed by Mehdizadeh, Molaiee-Nejad, and Chong [*J. Membrane Sci.*, **267**, pp. 27–40 (2005)].

Liquid-Solid Extraction (Leaching) Extraction of solubles from porous solids is a form of solvent extraction that has much in common with liquid-liquid extraction [Prabhudesai, "Leaching," Sec. 5.1 in *Handbook of Separation Techniques for Chemical Engineers*, Schweitzer, ed., pp. 5-3 to 5-31 (McGraw-Hill, 1997)]. The main differences come from the need to handle solids and the fact that mass transfer of soluble components out of porous solids generally is much slower than mass transfer between liquids. Because of this, different types of contacting equipment operating at longer residence times often are required. Washing of nonporous solids is a related operation that generally exhibits faster mass-transfer rates compared to leaching. On the other hand, purification of nonporous solids or crystals by removal of impurities that reside within the bulk solid phase often is not economical or even feasible by using these methods, because the rate of mass transfer of impurities through the bulk solid is extremely slow. Liquid-solid extraction is covered in Sec. 18, "Liquid-Solid Operations and Equipment."

Liquid-Liquid Partitioning of Fine Solids This process involves separation of small-particle solids suspended in a feed liquid, by contact with a second liquid phase. Robbins describes such a process for removing ash from pulverized coal [U.S. Patent 4,575,418 (1986)]. The process involves slurring pulverized coal fines into a hydrocarbon liquid and contacting the resulting slurry with water. The coal slurry is cleaned by preferential transfer of ash particles into the aqueous phase. The process takes advantage of differences in surface-wetting properties to separate the different types of solid particles present in the feed.

Supercritical Fluid Extraction This process generally involves the use of CO_2 or light hydrocarbons to extract components from liquids or porous solids [Brunner, *Gas Extraction: An Introduction to Fundamentals of Supercritical Fluids and the Application to Separation Processes* (Springer-Verlag, 1995); Brunner, ed., *Supercritical Fluids as Solvents and Reaction Media* (Elsevier, 2004); and McHugh and Krukonis, *Supercritical Fluid Extraction*, 2d ed. (Butterworth-Heinemann, 1993)]. Supercritical fluid extraction differs from liquid-liquid or liquid-solid extraction in that the operation is carried out at high-pressure, supercritical (or near-supercritical) conditions where the extraction fluid exhibits

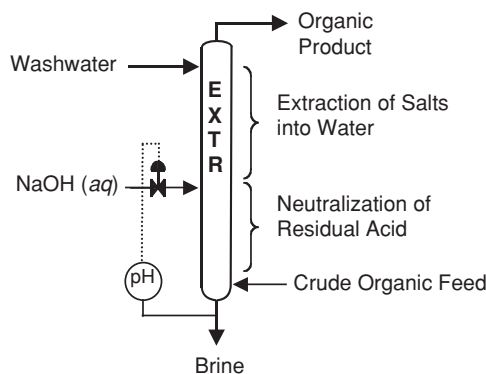


FIG. 15-13 Example of neutralization-extraction hybrid process implemented in an extraction column.

physical and transport properties that are in between those of liquid and vapor phases (intermediate density, viscosity, and solute diffusivity). Most applications involve the use of CO₂ (critical pressure = 73.8 bar at 31°C) or propane (critical pressure = 42.5 bar at 97°C). Other supercritical fluids and their critical-point properties are discussed by Poling, Prausnitz, and O'Connell [*The Properties of Gas and Liquids*, 5th ed. (McGraw-Hill, 2001)].

Supercritical CO₂ extraction often is considered for extracting high-value soluble components from natural materials or for purifying low-volume specialty chemicals. For products derived from natural materials, this can involve initial processing of solids followed by further processing of the crude liquid extract. Applications include decaffeination of coffee and recovery of active ingredients from plant- and animal-derived feeds including recovery of flavor components and vitamins from natural oils. An example is the use of supercritical CO₂ fractional extraction to remove terpenes from cold-pressed bergamot oil [Kondo et al., *Ind. Eng. Chem. Res.*, **39**(12), pp. 4745–4748 (2000)]. A nonfood example involves the removal of unreacted dodecanol from nonionic surfactant mixtures and fractionation of the surfactant mixture based on polymer chain length [Eckert et al., *Ind. Eng. Chem. Res.*, **31**(4), pp. 1105–1110 (1992)]. In these applications, process advantages may be obtained because solvent residues are easily removed or are nontoxic, the process can be operated at mild temperatures that avoid product degradation, the product is easily recovered from the extract fluid, or the solute separation factor and product purity can be adjusted by making small changes in the operating temperature and pressure. Although the loading capacity of supercritical CO₂ typically is low, addition of cosolvents such as methanol, ethanol, or tributylphosphate can dramatically boost capacity and enhance selectivity [Brennecke and Eckert, *AIChE J.*, **35**(9), pp. 1409–1427 (1989)].

For processing liquid feeds, some supercritical fluid extraction processes utilize packed columns, in which the liquid feed phase wets the packing and flows through the column in film flow, with the supercritical fluid forming the continuous phase. In other applications, sieve trays give improved performance [Seibert and Moosberg, *Sep. Sci. Technol.*, **23**, p. 2049 (1988)]. In a number of these applications, concentrated solute is added back to the column as reflux to boost separation power (a form of single-solvent fractional extraction). Supercritical fluid extraction requires high-pressure equipment and may involve a high-pressure compressor. These requirements add considerable capital and operating costs. In certain cases, pumps can be used instead of compressors, to bring down the cost. The separators are run slightly below the critical point at slightly elevated pressure and reduced temperature to ensure the material is in the liquid state so it can be pumped. As a rule, supercritical fluid extraction is considerably more expensive than liquid-liquid extraction, so when the required separation can be accomplished by using a liquid solvent, liquid-liquid extraction often is more cost-effective.

Although most commercial applications of supercritical fluid extraction involve processing of high-value, low-volume products, a notable exception is the propane deasphalting process used to refine lubricating oils. This is a large-scale, commodity chemical process dating back to the 1930s. In this process and more recent versions, lube oils are extracted into propane at near-supercritical conditions. The extract phase is depressurized or cooled in stages to isolate various fractions. Compared to operation at lower pressures, operation at near-supercritical conditions minimizes the required pressure or temperature change—so the process is more efficient. For further discussion of supercritical fluid separation processes, see Sec. 20, “Alternative Separation Processes,” Gironi and Maschietti, *Chem. Eng. Sci.*, **61**, pp. 5114–5126 (2006), and Fernandes et al., *AIChE J.*, **53**(4), pp. 825–837 (2007).

KEY CONSIDERATIONS IN THE DESIGN OF AN EXTRACTION OPERATION

Successful approaches to designing an extraction process begin with an appreciation of the fundamentals (basic phase equilibrium and mass-transfer principles) and generally rely on both experimental studies and mathematical models or simulations to define the commercial technology. Small-scale experiments using representative feed usually are needed to accurately quantify physical properties and phase equilibrium. Additionally, it is common practice in industry to perform

miniplant or pilot-plant tests to accurately characterize the mass-transfer capabilities of the required equipment as a function of throughput [Robbins, *Chem. Eng. Prog.*, **75**(9), pp. 45–48 (1979)]. In many cases, mass-transfer resistance changes with increasing scale of operation, so an ability to accurately scale up the data also is needed. The required scale-up know-how often comes from experience operating commercial equipment of various sizes or from running pilot-scale equipment of sufficient size to develop and validate a scale-up correlation. Mathematical models are used as a framework for planning and analyzing the experiments, for correlating the data, and for estimating performance at untested conditions by extrapolation. Increasingly, designers and researchers are utilizing computational fluid dynamics (CFD) software or other simulation tools as an aid to scale-up.

Typical steps in the work process for designing and implementing an extraction operation include the following:

1. Outline the design basis including specification of feed composition, required solute recovery or removal, product purity, and production rate.
2. Search the published literature (including patents) for information relevant to the application.
3. For dilute feeds, consider options for preconcentrating the feed to reduce the volumes of feed and solvent that must be handled by the extraction operation. Consider evaporation or distillation of a high-volatility feed solvent or the use of reverse osmosis membranes to concentrate aqueous feeds. (See “Hybrid Extraction Processes” under “Commercial Process Schemes.”)
4. Generate a list of candidate solvents based on chemical knowledge and experience. Consider solvents similar to those used in analogous applications. Use one or more of the methods described in “Solvent Screening Methods” to identify additional candidates. Include consideration of solvent blends and extractants.
5. Estimate key physical properties and review desirable solvent properties. Give careful consideration to safety, industrial hygiene, and environmental requirements. Use this preliminary information to trim the list of candidate solvents to a manageable size. (See “Desirable Solvent Properties.”)
6. Measure partition ratios for selected solvents at representative conditions.
7. Evaluate the potential for trace chemistry under extraction and solvent recovery conditions to determine whether solutes and candidate solvents are likely to degrade or react to produce unwanted impurities. For example, it is well known that penicillin G easily degrades at commercial extraction conditions, and short contact time is required for good results. Also under certain conditions acetate solvents may hydrolyze to form alcohols, certain alcohols and ethers can form peroxides, sulfur-containing solvents may degrade at elevated regeneration temperatures to form acids, chlorinated solvents may hydrolyze at elevated temperatures to form trace HCl with severe corrosion implications, and so on. In other cases, leakage of air into the process may cause formation of trace oxidation products. Understanding the potential for trace chemistry, the fate of potential impurities (i.e., where they go in the process), their possible effects on the process (including impact on product purity and interfacial tension) and devising means to avoid or successfully deal with impurities often are critical to a successful process design. Laboratory tests designed to probe the stability of feed and solvent mixtures may be needed.
8. Characterize mass-transfer difficulty in terms of the required number of theoretical stages or transfer units as a function of the solvent-to-feed ratio. Keep in mind that there will be a limit to the number of theoretical stages that can be achieved. For most cost-effective extraction operations, this limit will be in the range of 3 to 10 theoretical stages, although some can achieve more, depending upon the chemical system, type of equipment, and flow rate (throughput).
9. Estimate the cost of the proposed extraction operation relative to alternative separation technologies, such as extractive distillation, adsorption, and crystallization. Explore other options if they appear less expensive or offer other advantages.
10. If technical and economic feasibility looks good, determine accurate values of physical properties and phase equilibria, particularly liquid densities, mutual solubilities (miscibility), viscosities, interfacial tension, and *K* values (at feed, extract, and raffinate ends of the

proposed process), as well as data needed to evaluate solvent recycle options. Search available literature and databases. Assess data quality and generate additional data as needed. Develop the appropriate data correlations. Finalize the choice of solvent.

11. Outline an overall process flow sheet and material balance including solvent recovery and recycle. This should be done with the aid of process simulation software. [See Seider, Seader, and Lewin, *Product and Process Design Principles: Synthesis, Analysis, and Evaluation*, 2d ed. (Wiley, 2004); and Turton et al., *Analysis, Synthesis, and Design of Chemical Processes*, 2d ed. (Prentice-Hall, 2002)]. In the flow sheet include methods needed for controlling emissions and managing wastes. Carefully consider the possibility that impurities may accumulate in the recycled solvent, and devise methods for purging these impurities, if needed.

12. In some cases, especially with multiple solutes and complex phase equilibria, it may be useful to perform laboratory batch experiments to simulate a continuous, countercurrent, multistage process. These experiments can be used to test/verify calculation results and determine the correct distribution of components. For additional information, see Treybal, Chap. 9 in *Liquid Extraction*, 2d ed. (McGraw-Hill, 1963), pp. 359–393, and Baird and Lo, Chap. 17.1 in *Handbook of Solvent Extraction* (Wiley, 1983; Krieger, 1991).

13. Identify useful equipment options for liquid-liquid contacting and liquid-liquid phase separation, estimate approximate equipment size, and outline preliminary design specifications. (See “Extractor Selection” under “Liquid-Liquid Extraction Equipment.”) Where appropriate, consult with equipment vendors. Using small-scale experiments, determine whether sludgelike materials are likely to accumulate at the liquid-liquid interface (called formation of a rag layer). If so, it will be important to identify equipment options that can tolerate accumulation of a rag layer and allow the rag to be drained or otherwise purged periodically.

14. For the most promising equipment option, run miniplant or pilot-plant tests over a range of operating conditions. Utilize representative feed including all anticipated impurities, since even small concentrations of surface-active components can dramatically affect interfacial behavior. Whenever possible, the miniplant tests should be conducted by using actual material from the manufacturing plant, and should include solvent recycle to evaluate the effects of impurity accumulation or possible solvent degradation. Run the miniplant long enough that the solvent encounters numerous cycles so that recycle effects can be seen. If difficulties arise, consider alternative solvents.

15. Analyze miniplant data and update the preliminary design. Carefully evaluate loss of solvent to the raffinate, and devise methods to minimize losses as needed. Consult equipment vendors or other specialists regarding recommended scale-up methods.

16. Specify the final material balance for the overall process and carry out detailed equipment design calculations. Try to add some flexibility (depending on the cost) to allow for some adjustment of the process equipment during operation—to compensate for uncertainties in the design.

17. Install and start up the equipment in the manufacturing plant.

18. Troubleshoot and improve the operation as needed. Once a unit is operational, carefully measure the material balance and characterize mass-transfer performance. If performance does not meet expectations, look for defects in the equipment installation. If none are found, revisit the scale-up methodology and its assumptions.

LABORATORY PRACTICES

An equilibrium or theoretical stage in liquid-liquid extraction, as defined earlier, is routinely utilized in laboratory procedures. A feed solution is contacted with a solvent to remove one or more of the solutes from the feed. This can be carried out in a separating funnel or, preferably, in an agitated vessel that can produce droplets about 1 mm in diameter. After agitation has stopped and the phases separate, the two clear liquid layers are isolated by decantation. The partition ratio can then be determined directly by measuring the concentration of solute in the extract and raffinate layers. (Additional discussion is given in “Liquid-Liquid Equilibrium Experimental Meth-

ods” under “Thermodynamic Basis for Liquid-Liquid Extraction.”) When an appropriate analytical method is available only for the feed phase, the partition ratio can be determined by measuring the solute concentration in the feed and raffinate phases and calculating the partition ratio from the material balance. When the initial concentration of solute in the extraction solvent is zero (before extraction), the partition ratio expressed in terms of mass fractions is given by

$$K'' = \frac{Y''_e}{X''_r} = \frac{M_r}{M_e} \left(\frac{M_f}{M_r} \frac{X''_f}{X''_r} - 1 \right) \quad (15-2)$$

where K'' = mass fraction solute in extract divided by that in raffinate

M_f = total mass of feed added to vial

M_s = total mass of extraction solvent before extraction

M_r = mass of raffinate phase after extraction

M_e = mass of extract phase after extraction

X''_f = mass fraction solute in feed prior to extraction

X''_r = mass fraction solute in raffinate, at equilibrium

Y''_e = mass fraction solute in extract, at equilibrium

For systems with low mutual solubility between phases, $K'' \approx (M_f/M_s)(X''_f/X''_r - 1)$. An actual analysis of solute concentration in the extract and raffinate is preferred in order to understand how well the material balance closes (a check of solute accountability).

After a single stage of liquid-liquid contact, the phase remaining from the feed solution (the raffinate) can be contacted with another quantity of fresh extraction solvent. This cross-current (or cross-flow) extraction scheme is an excellent laboratory procedure because the extract and raffinate phases can be analyzed after each stage to generate equilibrium data for a range of solute concentrations. Also, the feasibility of solute removal to low levels can be demonstrated (or shown to be problematic because of the presence of “extractable” and “non-extractable” forms of a given species). The number of cross-current treatments needed for a given separation, assuming a constant K value, can be estimated from

$$N = \frac{\ln \left(\frac{X_{in} - Y_{in}/K}{X_{out} - Y_{in}/K} \right)}{\ln(KS^*/F + 1)} \quad (15-3)$$

where F is the amount of feed, the feed and solvent are presaturated, and equal amounts of solvent (denoted by S^*) are used for each treatment [Treybal, *Liquid Extraction*, 2d ed. (McGraw-Hill, 1963), pp. 209–216]. The total amount of solvent is $N \times S^*$. The variable Y_{in} is the concentration of solute in the fresh solvent, normally equal to zero. Equation (15-3) is written in a general form without specifying the units, since any consistent system of units may be used. (See “Process Fundamentals and Basic Calculation Methods.”)

A cross-current scheme, although convenient for laboratory practice, is not generally economically attractive for large commercial processes because solvent usage is high and the solute concentration in the combined extract is low. A number of batchwise countercurrent laboratory techniques have been developed and can be used to demonstrate countercurrent performance. (See item 12 in the previous subsection, “Key Considerations in the Design of an Extraction Operation.”) Several equipment vendors also make available continuously fed laboratory-scale extraction equipment. Examples include small-scale mixer-settler extraction batteries offered by Rousselet-Robatel, Normag, MEAB, and Schott/QVF. Small-diameter extraction columns also may be used, such as the $\frac{3}{8}$ -in. (16-mm-) diameter reciprocating-plate agitated column offered by Koch Modular Process Systems, and a 60-mm-diameter rotary-impeller agitated column offered by Kühni. Static mixers also may be useful for mixer-settler studies in the laboratory [Benz et al., *Chem. Eng. Technol.*, **24**(1), pp. 11–17 (2001)].

For additional discussion of laboratory techniques, see “Liquid-Liquid Equilibrium Experimental Methods” as well as “High-Throughput Experimental Methods” under “Solvent-Screening Methods.”

THERMODYNAMIC BASIS FOR LIQUID-LIQUID EXTRACTION

GENERAL REFERENCES: See Sec. 4, "Thermodynamics," as well as Sandler, *Chemical, Biochemical, and Engineering Thermodynamics* (Wiley, 2006); Solvent Extraction Principles and Practice, 2d ed., Rydberg et al., eds. (Dekker, 2004); Smith, Abbott, and Van Ness, *Introduction to Chemical Engineering Thermodynamics*, 7th ed. (McGraw-Hill, 2004); Schwarzenbach, Gschwend, and Imboden, *Environmental Organic Chemistry*, 2d ed. (Wiley-VCH, 2002); Elliot and Lira, *Introduction to Chemical Engineering Thermodynamics* (Prentice-Hall, 1999); Prausnitz, Lichtenthaler, and Gomez de Azevedo, *Molecular Thermodynamics of Fluid-Phase Equilibria*, 3d ed. (Prentice-Hall, 1999); Seader and Henley, Chap. 2 in *Separation Process Principles* (Wiley, 1998); Bolz et al., *Pure Appl. Chem. (IUPAC)*, **70**, pp. 2233–2257 (1998); Grant and Higuchi, *Solubility Behavior of Organic Compounds*, Techniques of Chemistry Series, vol. 21 (Wiley, 1990); Abbott and Prausnitz, "Phase Equilibria," in *Handbook of Separation Process Technology*, Rousseau, ed. (Wiley, 1987), pp. 3–59; Novak, Matous, and Pick, *Liquid-Liquid Equilibria*, Studies in Modern Thermodynamics Series, vol. 7 (Elsevier, 1987); Walas, *Phase Equilibria in Chemical Engineering* (Butterworth-Heinemann, 1985); and Rowlinson and Swinton, *Liquids and Liquid Mixtures*, 3d ed. (Butterworths, 1982).

ACTIVITY COEFFICIENTS AND THE PARTITION RATIO

Two phases are at equilibrium when the total Gibbs energy for the system is at a minimum. This criterion can be restated as follows: Two nonreacting phases are at equilibrium when the chemical potential of each distributed component is the same in each phase; i.e., for equilibrium between two phases I and II containing n components

$$\mu_i^I = \mu_i^{II} \quad i = 1, 2, \dots, n \quad (15-4)$$

For two phases at the same temperature and pressure, Eq. (15-4) can be expressed in terms of mole fractions and activity coefficients, giving

$$y_i \gamma_i^I = x_i \gamma_i^{II} \quad i = 1, 2, \dots, n \quad (15-5)$$

where y_i and x_i represent mole fractions of component i in phases I and II, respectively. The equilibrium partition ratio, in units of mole fraction, is then given by

$$K_i^o = \frac{y_i}{x_i} = \frac{\gamma_i^{\text{raffinate}}}{\gamma_i^{\text{extract}}} \quad (15-6)$$

where y_i is the mole fraction in the extract phase and x_i is the mole fraction in the raffinate. Note that, in general, activity coefficients and K_i^o are functions of temperature and composition. For ionic compounds that dissociate in solution, the species that form and the extent of dissociation in each phase also must be taken into account. Similarly, for extractions involving adduct formation or other chemical reactions, the reaction stoichiometry is an important factor. For discussion of these special cases, see Choppin, Chap. 3, and Rydberg et al., Chap. 4, in *Solvent Extraction Principles and Practice*, 2d ed., Rydberg et al., eds. (Dekker, 2004).

The activity coefficient for a given solute is a measure of the non-ideality of solute-solvent interactions in solution. In this context, the solvent is either the feed solvent or the extraction solvent depending on which phase is considered, and the composition of the "solvent" includes all components present in that phase. For an ideal solution, activity coefficients are unity. For solute-solvent interactions that are repulsive relative to solvent-solvent interactions, γ_i is greater than 1. This is said to correspond to a positive deviation from ideal solution behavior. For attractive interactions, γ_i is less than 1.0, corresponding to a negative deviation. Activity coefficients often are reported for binary pairs in the limit of very dilute conditions (infinite dilution) since this represents the interaction of solute completely surrounded by solvent molecules, and this normally gives the largest value of the activity coefficient (denoted as γ_i^∞). Normally, useful approximations of the activity coefficients at more concentrated conditions can be obtained by extrapolation from infinite dilution using an appropriate activity coefficient correlation equation. (See Sec. 4, "Thermodynamics.") Extrapolation in the reverse direction, i.e., from finite concentration to infinite dilution, often does not provide reliable results.

In units of mass fraction, the partition ratio for a nonreacting/nondissociating solute is given by

$$K_i'' \text{ (mass frac. basis)} = \frac{Y_i''}{X_i''} = K_i^o \text{ (mole frac. basis)} \times \left[\frac{y_i(MW_i - MW_{\text{raffinate}}) + MW_{\text{raffinate}}}{x_i(MW_i - MW_{\text{extract}}) + MW_{\text{extract}}} \right] \quad (15-7)$$

Here, the notation MW refers to the molecular weight of solute i and the effective average molecular weights of the extract and raffinate phases, as indicated by the subscripts. For dilute systems, $K_i'' \approx K_i^o$ ($MW_{\text{raffinate}}/MW_{\text{extract}}$). For theoretical stage or transfer unit calculations, often it is useful to express the partition ratio in terms of mass ratio coordinates introduced by Bancroft [*Phys. Rev.*, **3**(1), pp. 21–33; **3**(2), pp. 114–136; and **3**(3), pp. 193–209 (1895)]:

$$K_i' = \frac{Y_i'}{X_i'} = \frac{M_{\text{solute}}/M_{\text{extraction solvent}} \text{ in extract phase}}{M_{\text{solute}}/M_{\text{feed solvent}} \text{ in raffinate phase}} \quad (15-8)$$

Partition ratios also may be expressed on a volumetric basis. In that case,

$$K_i^{\text{vol}} \text{ (mass/vol. basis)} = K_i'' \frac{\rho_{\text{extract}}}{\rho_{\text{raffinate}}} \quad (15-9)$$

$$K_i^{\text{vol}} \text{ (mole/vol. basis)} = K_i^o \left(\frac{\rho_{\text{extract}}}{\rho_{\text{raffinate}}} \right) \left(\frac{MW_{\text{raffinate}}}{MW_{\text{extract}}} \right) \quad (15-10)$$

Extraction Factor The extraction factor is defined by

$$\mathcal{E}_i = m_i \frac{S}{F} \quad (15-11)$$

where $m_i = dY_i/dX_i$, the slope of the equilibrium line, and F and S are the flow rates of the feed phase and the extraction-solvent phase, respectively. On a McCabe-Thiele type of diagram, $\mathcal{E}$ is the slope of the equilibrium line divided by the slope of the operating line F/S . (See "McCabe-Thiele Type of Graphical Method" under "Process Fundamentals and Basic Calculation Methods.") For dilute systems with straight equilibrium lines, the slope of the equilibrium line is equal to the partition ratio $m_i = K_i$.

To illustrate the significance of the extraction factor, consider an application where K_i , S , and F are constant (or nearly so) and the extraction solvent entering the process contains no solute. When $\mathcal{E}_i = 1$, the extract stream has just enough capacity to carry all the solute present in the feed:

$$SY_{i,\text{extract}} = FX_{i,\text{feed}} \quad \text{at } \mathcal{E}_i = 1 \text{ and equilibrium conditions} \quad (15-12)$$

At $\mathcal{E}_i < 1.0$, the extract's capacity to carry solute is less than this amount, and the maximum fraction that can be extracted θ_i is numerically equal to the extraction factor:

$$(\theta_i)_{\text{max}} = \mathcal{E}_i \quad \text{when } \mathcal{E}_i < 1.0 \quad (15-13)$$

At $\mathcal{E}_i > 1.0$, the extract phase has more than sufficient carrying capacity (in principle), and the actual amount extracted depends on the extraction scheme, number of contacting stages, and mass-transfer resistance. Even a solute for which $m_i < 1.0$ (or $K_i < 1.0$) can, in principle, be extracted to a very high degree—by adjusting S/F so that $\mathcal{E}_i > 1$.

Thus, the extraction factor characterizes the relative capacity of the extract phase to carry solute present in the feed phase. Its value is a major factor determining the required number of theoretical stages or transfer units. (For further discussion, see "The Extraction Factor and

General Performance Trends.”) In general, the value of the extraction factor can vary at each point along the equilibrium curve, although in many cases it is nearly constant. Many commercial extraction processes are designed to operate with an average or overall extraction factor in the range of 1.3 to 5. Exceptions include applications where the partition ratio is very large and the solvent-to-feed ratio is set by hydraulic considerations.

Because the extraction factor is a dimensionless variable, its value should be independent of the units used in Eq. (15-11), as long as they are consistently applied. Engineering calculations often are carried out by using mole fraction, mass fraction, or mass ratio units (Bancroft coordinates). The flow rates S and F then need to be expressed in terms of total molar flow rates, total mass flow rates, or solute-free mass flow rates, respectively. In the design of extraction equipment, volume-based units often are used. Then the appropriate concentration units are mass or mole per unit volume, and flow rates are expressed in terms of the volumetric flow rate of each phase.

Separation Factor The separation factor in extraction is analogous to relative volatility in distillation. It is a dimensionless factor that measures the relative enrichment of a given component in the extract phase after one theoretical stage of extraction. For cosolutes i and j ,

$$\alpha_{ij} = \frac{(Y_i/Y_j)_{\text{extract}}}{(X_i/X_j)_{\text{raffinate}}} = \frac{(Y_i)_{\text{extract}}/(X_i)_{\text{raffinate}}}{(Y_j)_{\text{extract}}/(X_j)_{\text{raffinate}}} = \frac{K_i}{K_j} \quad (15-14)$$

The enrichment of solute i with respect to solute j can be further increased with the use of multiple contacting stages. The solute separation factor α_{ij} is used to characterize the selectivity a solvent has for extracting a desired solute from a feed containing other solutes. It can be calculated by using any consistent units. As in distillation, α_{ij} must be greater than 1.0 to achieve an increase in product-solute purity (on a solvent-free basis). In practice, if solute purity is an important requirement of a given application, α_{ij} must be greater than 20 for standard extraction (at least) and greater than about 4 for fractional extraction, in order to have sufficient separation power. (See “Potential for Solute Purification Using Standard Extraction” in “Process Fundamentals and Basic Calculation Methods” and “Dual-Solvent Fractional Extraction” in “Calculation Procedures.”)

The separation factor also can be evaluated for solute i with respect to the feed solvent denoted as component f . The value of α_{if} must be greater than 1.0 if the proposed separation is to be feasible, i.e., in order to be able to enrich solute i in a separate extract phase. Note that the feed may still be separated if $\alpha_{if} < 1.0$, but this would have to involve concentrating solute i in the feed phase by preferential transfer of component f into the extract phase. Although $\alpha_{if} > 1.0$ represents a minimum theoretical requirement for enriching solute i in a separate extract phase, most commercial extraction processes operate with values of α_{if} on the order of 20 or higher. There are exceptions to this rule, such as the Udex process and similar processes involving extraction of aromatics from aliphatic hydrocarbons. In these applications, α_{if} can be as low as 10 and sometimes even lower. Applications such as these involve particularly difficult design challenges because of low solute partition ratios and high mutual solubility between phases. (For more detailed discussion of these kinds of systems, see “Single Solvent Fractional Extraction with Extract Reflux” in “Fractional Extraction Calculations.”)

Minimum and Maximum Solvent-to-Feed Ratios Normally, it is possible to quickly estimate the physical constraints on solvent usage for a standard extraction application in terms of minimum and maximum solvent-to-feed ratios. As discussed above, the minimum theoretical amount of solvent needed to transfer a high fraction of solute i is the amount corresponding to $\mathcal{E}_i = 1$. In practice, the minimum practical extraction factor is about 1.3, because at lower values the required number of theoretical stages increases dramatically. This gives a minimum solvent-to-feed ratio for a practical process equal to

$$\left(\frac{S}{F}\right)_{\min} \approx \frac{1.3}{K_i} \quad (15-15)$$

Note that this minimum is achievable only if a sufficient number of contacting stages or transfer units can be used. (For additional discussion,

see “The Extraction Factor and General Performance Trends.”) It is also achievable only if the amount of solvent added to the feed is greater than the solubility limit in the feed phase (including solute); otherwise, only one liquid phase can exist. In certain cases involving fairly high mutual solubilities, this can be an important consideration when running a process using minimal solvent—because if the process operates close to the solubility limit, an upset in the solvent-to-feed ratio may cause the solvent phase to disappear.

The maximum possible solvent-to-feed ratio is obtained when the amount of extraction solvent is so large that it dissolves the feed phase. Assuming the feed entering the process does not contain extraction solvent,

$$\left(\frac{S}{F}\right)_{\max} = \frac{1}{1 - Y_s^{\text{SAT}}} \quad (15-16)$$

where Y_s^{SAT} denotes the concentration of extraction solvent in the extract phase at equilibrium after contact with the feed phase. The denominator in Eq. (15-16) represents the solubility limit on the solvent-rich side of the miscibility envelope, including the effect of the presence of solute on solubility. Normally, the solubility limits are easily measured in small-scale experiments by adding solvent until the solvent phase appears (representing the feed-rich side of the miscibility envelope) and continuing to add solvent until the feed phase disappears (the solvent-rich side). For dilute feeds containing less than about 1% solute, reasonable estimates often can be obtained by using mutual solubility data for the feed solvent + extraction solvent binary pair.

If an application proves to be technically feasible, the choice of solvent-to-feed ratio is determined by identifying the most cost-effective ratio between the minimum and maximum limits. For most applications, the maximum solvent-to-feed ratio will be much larger than the ratio chosen for the commercial process; however, the maximum ratio can be a real constraint when dealing with applications exhibiting high mutual solubility, especially for systems that involve high solute concentrations. Additional discussion is given by Seader and Henley [Chap. 8 in *Separations Process Principles* (Wiley, 1998)]. Solvent ratios are further constrained for a fractional extraction scheme, as discussed in “Fractional Extraction Calculations.”

Temperature Effect The effect of temperature on the value of the partition ratio can vary greatly from one system to another. This depends on how the activity coefficients of the components in each phase are affected by changes in temperature, including any effects due to changes in mutual solubility with temperature. For a given phase, the Gibbs-Helmholtz equation indicates that

$$\left[\frac{\partial \ln \gamma_i^\infty}{\partial (1/T)}\right]_{p,x} = \frac{h_i^{E,\infty}}{R} \quad (15-17)$$

where γ_i^∞ is the activity coefficient for solute i at infinite dilution and $h_i^{E,\infty}$ is the partial molar excess enthalpy of mixing relative to ideal solution behavior [Atik et al., *J. Chem. Eng. Data*, **49**(5), pp. 1429–1432 (2004); and Sherman et al., *J. Phys. Chem.*, **99**, pp. 11239–11247 (1995)].

Systems with specific interactions between solute and solvent, such as hydrogen bonds or ion-pair bonds, often are particularly sensitive to changes in temperature because the specific interactions are strongly temperature-dependent. In general, hydrogen bonding and ion-pair formation are disrupted by increasing temperature (increasing molecular motion), and this can dominate the overall temperature dependence of the partition ratio. An example of a temperature-sensitive hydrogen bonding system is toluene + diethylamine + water [Morello and Beckmann, *Ind. Eng. Chem.*, **42**, pp. 1079–1087 (1950)]. The partition ratio for transfer of diethylamine from water into toluene increases with increasing temperature (on a weight percent basis, $K = 0.7$ at 20°C and $K = 2.8$ at 58°C). For further discussion of the temperature dependence of K for this type of system, see Frank et al., *Ind. Eng. Chem. Res.*, **46**(11), pp. 3774–3786 (2007). An example of a temperature-sensitive system involving ion-pair formation is the commercial process used to recover citric acid from fermentation broth using trioctylamine (TOA) extractant [Pazouki and Panda, *Bioprocess*

Engineering, **19**, pp. 435–439 (1998)]. In this case, the partition ratio for transfer of citric acid into the TOA phase decreases with increasing temperature. Temperature-sensitive ion-pair interactions in the extract phase are disrupted with increasing temperature, and this appears to dominate the temperature sensitivity of the partition ratio, not the interactions between citric acid and water in the aqueous raffinate phase [Canari and Eyal, *Ind. Eng. Chem. Res.*, **43**, pp. 7608–7617 (2004)]. Also see the discussion of “Temperature-Swing Extraction” in “Commercial Extraction Schemes.”

Salting-out and Salting-in Effects for Nonionic Solutes It is well known that the presence of an inorganic salt can significantly affect the solubility of a nonionic (nonelectrolyte) organic solute dissolved in water. In most cases the inorganic salt reduces the organic solute's solubility (salting-out effect). Here, the salt increases the organic solute's activity coefficient in the aqueous solution. As a result, certain solutes that are not easily extracted from water may be quite easily extracted from brine, depending upon the type of solute and the salt. In principle, the deliberate addition of a salt to an aqueous feed is an option for enhancing partition ratios and reducing the mutual solubility of the two liquid phases; however, this approach complicates the overall process and normally is not cost-effective. Difficulties include the added complexity and costs associated with recovery and recycle of the salt in the overall process, or disposal of the brine after extraction and the need to purchase makeup salt. The potential use of NaCl to enhance the extraction of ethanol from fermentation broth is discussed by Gomis et al. [*Ind. Eng. Chem. Res.*, **37**(2), pp. 599–603 (1998)].

When an aqueous feed contains a salt, the effect of the dissolved salt on the partition ratio for a given organic solute may be estimated by using an expression introduced by Setschenow [*Z. Phys. Chem.*, **4**, pp. 117–128 (1889)] and commonly written in the form

$$\log \frac{\gamma_{i,\text{brine}}}{\gamma_{i,\text{water}}} = k_s C_{\text{salt}} \quad (15-18)$$

where C_{salt} is the concentration of salt in the aqueous phase in units of gmol/L and k_s is the Setschenow constant. Equation (15-18) generally is valid for dilute organic solute concentrations and low to moderate salt concentrations. In many cases, the salt has no appreciable effect on the activity coefficient in the organic phase since the salt solubility in that phase is low or negligible. Then

$$\log \frac{K_{i,\text{brine}}}{K_{i,\text{water}}} \approx \log \frac{\gamma_{i,\text{brine}}}{\gamma_{i,\text{water}}} = k_s C_{\text{salt}} \quad (15-19)$$

for extraction from the aqueous phase into an organic phase. For aromatic solutes dissolved in NaCl brine at room temperature, typical values of k_s fall within the range of 0.2 to 0.3 L/gmol. In general, k_s is found to vary with salt composition (i.e., with the type of salt) and increase with increasing organic-solute molar volume. Kojima and Davis [*Int. J. Pharm.*, **20**(1–2), pp. 203–207 (1984)] showed that partition ratio data for extraction of phenol dissolved in NaCl brine (at low concentration) using CCl_4 solvent is well fit by a Setschenow equation for salt concentrations up to 4 gmol/L (about 20 wt % NaCl). Experimental values and methods for estimating Setschenow constants are discussed by Ni and Yalkowski [*Int. J. Pharm.*, **254**(2), pp. 167–172 (2003)] and by Xie, Shiu, and MacKay [*Marine Environ. Res.*, **44**, pp. 429–444 (1997)].

In special cases, salts with large ions (such as tetramethylammonium chloride and sodium toluene sulfonate) may cause a “salting in” or “hydrotropic” effect where by the salt increases the solubility of an organic solute in water, apparently by disordering the structure of associated water molecules in solution [Sugunan and Thomas, *J. Chem. Eng. Data.*, **38**(4), pp. 520–521 (1993)]. Agrawal and Gaikar [*Sep. Technol.*, **2**, pp. 79–84 (1992)] discuss the use of hydrotropic salts to facilitate extraction processes. For additional discussion, see Ruckenstein and Shulgin, *Ind. Eng. Chem. Res.*, **41**(18), pp. 4674–4680 (2002); and Akia and Feyzi, *AIChE J.*, **52**(1), pp. 333–341 (2006).

Effect of pH for Ionizable Organic Solutes The distribution of weak acids and bases between organic and aqueous phases is dramatically affected by the pH of the aqueous phase relative to the $\text{p}K_a$ of the solute. As discussed earlier, the $\text{p}K_a$ is the pH at which 50 percent of the solute is in the ionized state. (See “Dissociative Extraction” in “Commercial Extraction Schemes.”) For a weak organic acid (RCOOH) that dissociates into RCOO^- and H^+ , the overall partition ratio for extraction into an organic phase depends upon the extent of dissociation such that

$$K_{\text{weak acid}} = K_{\text{nonionized}} \div \left(1 + \frac{[\text{RCOO}^-]_{\text{aq}}}{[\text{RCOOH}]_{\text{aq}}} \right) \quad (15-20)$$

where $K_{\text{weak acid}} = [\text{RCOOH}]_{\text{org}} / ([\text{RCOO}^-]_{\text{aq}} + [\text{RCOOH}]_{\text{aq}})$ is the partition ratio for both ionized and nonionized forms of the acid, and $K_{\text{nonionized}} = [\text{RCOOH}]_{\text{org}} / [\text{RCOOH}]_{\text{aq}}$ is the partition ratio for the nonionized form alone [Treybal, *Liquid Extraction*, 2d ed. (McGraw-Hill, 1963), pp. 38–40]. Equation (15-20) can be rewritten in terms of the $\text{p}K_a$ for a weak acid or weak base:

$$K_{\text{weak acid}} = K_{\text{nonionized}} \div (1 + 10^{\text{pH} - \text{p}K_a}) \quad (15-21)$$

and

$$K_{\text{weak base}} = K_{\text{nonionized}} \div (1 + 10^{\text{p}K_b - \text{pH}}) \quad (15-22)$$

For weak bases, $\text{p}K_a = 14 - \text{p}K_b$. Appropriate values for $K_{\text{nonionized}}$ may be obtained by measuring the partition ratio at sufficiently low pH (for acids) or high pH (for bases) to ensure the solute is in its nonionized form (normally at a pH at least 2 units from the $\text{p}K_a$ value). In Eqs. (15-21) and (15-22), it is assumed that concentrations are dilute, that dissociation occurs only in the aqueous phase, and that the acid does not associate (dimerize) in the organic phase. The effect of pH on the partition ratio for extraction of penicillin G, a complex organic containing a carboxylic acid group, is illustrated in Fig. 15-14. For a discussion of the effect of pH on the extraction of carboxylic acids with tertiary amines, see Yang, White, and Hsu, *Ind. Eng. Chem. Res.*, **30**(6), pp. 1335–1342 (1991). Another example is discussed by Greminger et al., [*Ind. Eng. Chem. Process Des. Dev.*, **21**(1), pp. 51–54 (1982)]; they present partition ratio data for various phenolic compounds as a function of pH.

For compounds with multiple ionizable groups, such as amino acids, the effect of pH on partitioning behavior is more complex. Amino acids are zwitterionic (dipolar) molecules with two or three ionizable groups; the $\text{p}K_a$ values corresponding to RCOOH acid groups generally are between 2 and 3, and $\text{p}K_a$ values for RNH_3^+ amino groups generally are between 9 and 10. Amino acid partitioning is discussed by Schügerl [*Solvent Extraction in Biotechnology* (Springer-Verlag, 1994); Chap. 21 in *Biotechnology*, 2d ed., vol. 3, Stephanopoulos, ed. (VCH, 1993)]; and by Gude, Meuwissen, van der Wielen, and Luyben [*Ind. Eng. Chem. Res.*, **35**, pp. 4700–4712

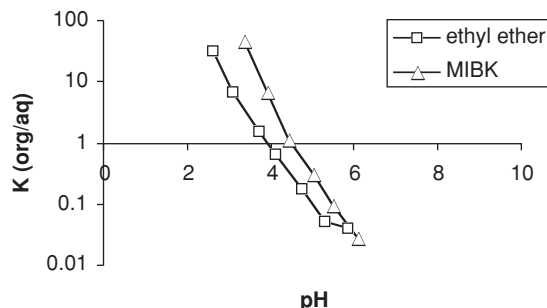


FIG. 15-14 The effect of pH on the partition ratio for extraction of penicillin G ($\text{p}K_a = 2.75$) from broth using an oxygenated organic solvent. The partition ratio is expressed in units of grams per liter in the organic phase over that in the aqueous phase. [Data from R. L. Feder, M.S. thesis (Polytechnic Institute of Brooklyn, 1947).]

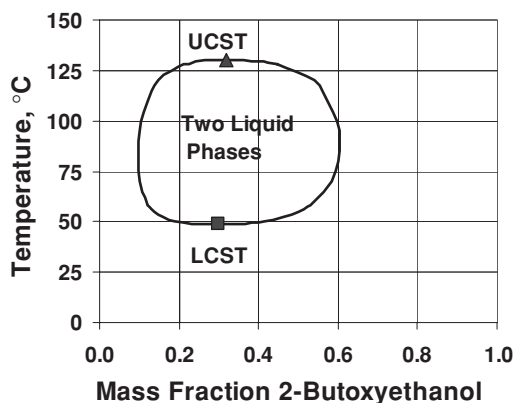


FIG. 15-15 Temperature-composition diagram for water + 2-butoxyethanol (ethylene glycol *n*-butyl ether). [Reprinted from Christensen, Donat, Frank, LaTulip, and Wilson, *J. Chem. Eng. Data*, **50**(3), pp. 869–877 (2005), with permission. Copyright 2005 American Chemical Society.]

(1996)]. The aqueous solubility of amino acids as a function of pH is discussed by Fuchs et al., *Ind. Eng. Chem. Res.*, **45**(19), pp. 6578–6584 (2006). Solution pH also has a strong effect on the solubility of proteins (complex polyaminoacids) in aqueous solution; solubility is lowest at the pH corresponding to the protein's isoelectric point (the pH at which all negative charges are balanced by all positive charges and the protein has zero net charge) [van Holde, Johnson, and Ho, *Principles of Physical Biochemistry* (Prentice-Hall, 1998)]. Partition ratios for partitioning of proteins in two-aqueous-phase systems depend upon many factors and are difficult to predict [Zaslavsky, *Aqueous Two-Phase Partitioning* (Dekker, 1994); and Kelley and Hatton, Chap. 22, "Protein Purification by Liquid-Liquid Extraction," in *Biotechnology*, 2d ed., vol. 3, Stephanopoulos, ed. (VCH, 1993)].

For general discussions of organic acid and base ionic equilibria, see Butler, *Ionic Equilibrium: Solubility and pH Calculations* (Wiley,

1998); and March, *Advanced Organic Chemistry: Reactions, Mechanisms, and Structure*, 5th ed., Chap. 8 (Wiley, 2000). The dissociation of inorganic salts is discussed in the book edited by Perrin [*Ionization Constants of Inorganic Acids and Bases in Aqueous Solution*, vol. 29 (Franklin, 1982)]. Compilations of pK_a values are given in several handbooks [Jencks and Regenstein, "Ionization Constants of Acids and Bases," in *Handbook of Biochemistry and Molecular Biology: Physical and Chemical Data*, vol. 1, 3d ed., Fasman, ed. (CRC Press, 1976), pp. 305–351; and *CRC Handbook of Chemistry and Physics*, 84th ed., Lide, ed. (CRC Press, 2003–2004)]. Also see Perrin, Dempsey, and Serjeant, *pK_a Prediction for Organic Acids and Bases* (Chapman and Hall, 1981).

PHASE DIAGRAMS

Phase diagrams are used to display liquid-liquid equilibrium data across a wide composition range. Consider the binary system of water + 2-butoxyethanol (common name ethylene glycol *n*-butyl ether) plotted in Fig. 15-15. This system exhibits both an upper critical solution temperature (UCST), also called the upper consolute temperature, and a lower critical solution temperature (LCST), or lower consolute temperature. The mixture is only partially miscible at temperatures between 48°C (the LCST) and 130°C (the UCST). Most mixtures tend to become more soluble in each other as the temperature increases; i.e., they exhibit UCST behavior. The presence of a LCST in the phase diagram is less common. Mixtures that exhibit LCST behavior include hydrogen-bonding mixtures such as an amine, a ketone, or an etheric alcohol plus water. Numerous water + glycol ether mixtures behave in this way [Christensen et al., *J. Chem. Eng. Data*, **50**(3), pp. 869–877 (2005)]. For these systems, hydrogen bonding leads to complete miscibility below the LCST. As temperature increases, hydrogen bonding is disrupted by increasing thermal (kinetic) energy, and hydrophobic interactions begin to dominate, leading to partial miscibility at temperatures above the LCST. The ethylene glycol + triethylamine system shown in Fig. 15-16 is another example.

Most of the ternary or pseudoternary systems used in extraction are of two types: one binary pair has limited miscibility (termed a type I system), or two binary pairs have limited miscibility (a type II system). The water + acetic acid + methyl isobutyl ketone (MIBK) system

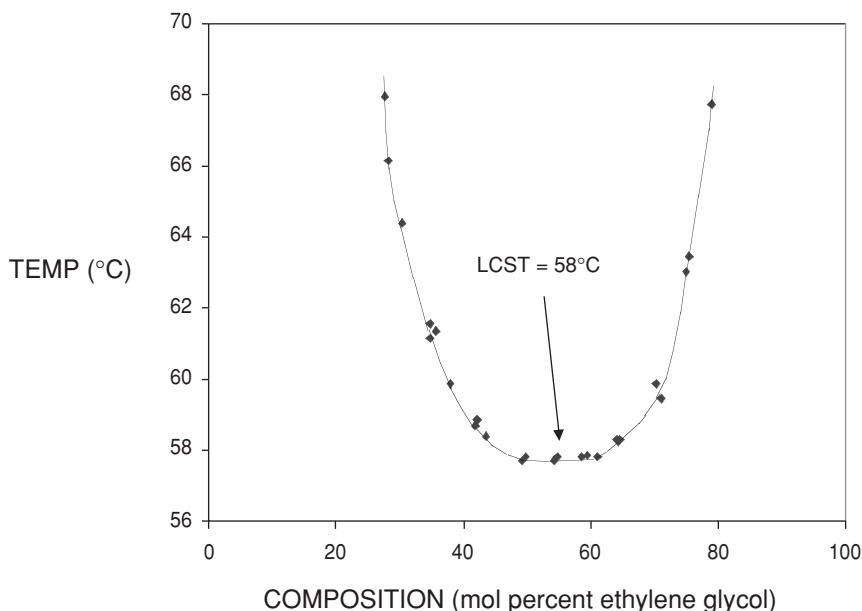


FIG. 15-16 Temperature-composition diagram for ethylene glycol + triethylamine. [Data taken from Sorenson and Arlt, *Liquid-Liquid Equilibrium Data Collection, DECHEMA, Binary Systems*, vol. V, pt. 1, 1979.]

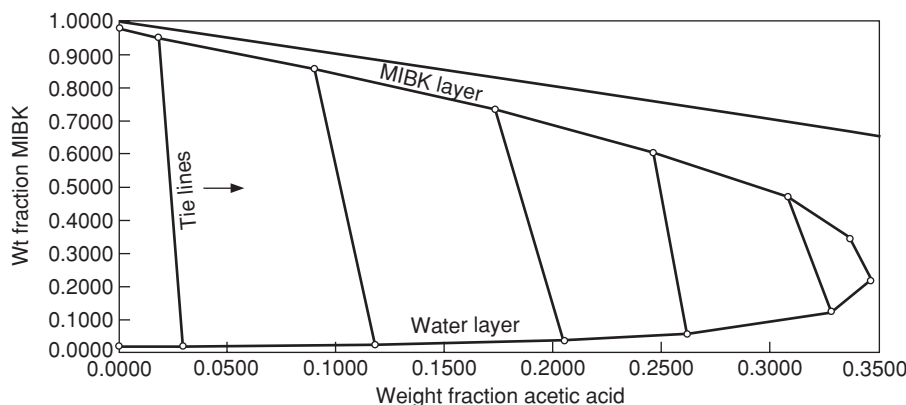


FIG. 15-17 Water + acetic acid + methyl isobutyl ketone at 25°C, a type I system.

shown in Fig. 15-17 is a type I system where only one of the binary pairs, water + MIBK, exhibits partial miscibility. The heptane + toluene + sulfolane system is another example of a type I system. In this case, only the heptane + sulfolane binary is partially miscible (Fig. 15-18). An example of a type II system is MIBK + phenol + water (Fig. 15-19), where MIBK + water and phenol + water are only partially miscible. Some systems form more complicated phase diagrams. For example, the system water + dodecane + 2-butoxyethanol can form three liquid phases in equilibrium at 25°C [Lin and Chen, *J. Chem. Eng. Data*, **47**(4), pp. 992–996 (2002)]. Complex systems such as this rarely are encountered in extraction applications; however, Shen, Chang, and Liu [*Sep. Purif. Technol.*, **49**(3), pp. 217–222 (2006)] describe a single-stage, three-liquid-phase extraction process for transferring phenol and *p*-nitrophenol from wastewater in separate phases. In this process, the three-phase system consists of ethylene oxide–propylene oxide copolymer + ammonium sulfate + water + an oxygenated organic solvent such as butyl acetate or 2-octanol.

For ternary systems, a three-dimensional plot is required to represent the effects of both composition and temperature on the phase behavior. Normally, ternary phase data are plotted on isothermal, two-dimensional triangular diagrams. These can be right-triangle plots, as in Fig. 15-17, or equilateral-triangle plots, as in Figs. 15-18 and 15-19. In Fig. 15-18, the line delineating the region where two liquid phases form is called the binodal locus. The lines connecting equilibrium compositions for each phase are called tie lines, as illustrated by lines *ab* and *cd*. The tie lines converge on the plait point, the point on the binodal locus where both liquid phases attain the same composition

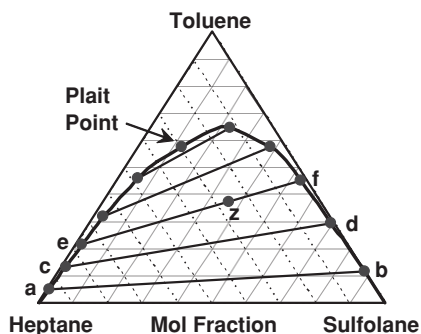


FIG. 15-18 Heptane + toluene + sulfolane at 25°C, a type I system. [Data taken from De Fre and Verhoeve, *J. Appl. Chem. Biotechnol.*, **26**, pp. 1-19 (1976).]

and the tie line length goes to zero. To calculate the relative amounts of the liquid phases, the lever rule is used. For the total feed composition z , the fraction of phase 1 with the composition e is equal to the ratio of the lengths of the line segments given by fz/ez in Fig. 15-18. Data often are plotted on a mass fraction basis when differences in the molecular weights of the components are large, since plotting the phase diagram on a mole basis tends to compress the data into a small region and details are hidden by the scale. This often is the case for systems involving water, for example.

An extraction application normally involves more than three components, including the key solute, the feed solvent, and extraction solvent (or solvent blend), plus impurity solutes. Usually, the minor impurity components do not have a major impact on the phase equilibrium. Phase equilibrium data for multicomponent systems may be represented by using an appropriate activity coefficient correlation. (See “Data Correlation Equations.”) However, for many dilute and moderately concentrated feeds, process design calculations are carried out as if the system were a ternary system comprised only of a single solute plus the feed solvent and extraction solvent (a pseudoternary). Partition ratios are determined for major and minor solutes by using a representative feed, and solute transfer calculations are carried out using solute K values as if they were completely independent of one another. This approach often is satisfactory, but its validity should be checked with a few key experiments. For industrial mixtures containing numerous impurities, a mass fraction or mass ratio basis often is used to avoid

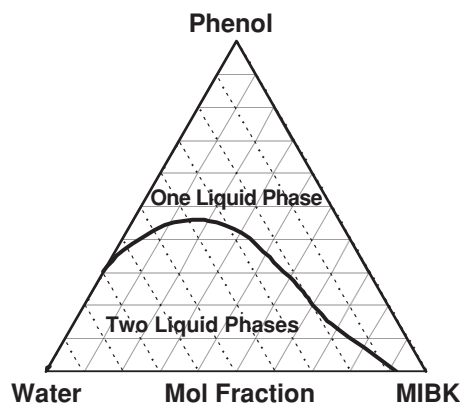


FIG. 15-19 Methyl isobutyl ketone + phenol + water at 30°C, a type II system. [Data taken from Narashinhan, Reddy, and Chari, *J. Chem. Eng. Data*, **7**, p. 457 (1962).]

difficulties accounting for impurities of unknown structure and molecular weight.

LIQUID-LIQUID EQUILIBRIUM EXPERIMENTAL METHODS

GENERAL REFERENCES: Raal, Chap. 3, "Liquid-Liquid Equilibrium Measurements," in *Vapor-Liquid Equilibria Measurements and Calculations* (Taylor & Francis, 1998); Newsham, Chap. 1 in *Science and Practice of Liquid-Liquid Extraction*, vol. 1, Thornton, ed. (Oxford, 1992); and Novak, Matous, and Pick, *Liquid-Liquid Equilibria, Studies in Modern Thermodynamics Series*, vol. 7, pp. 266–282 (Elsevier, 1987).

Three general types of experimental methods commonly are used to generate liquid-liquid equilibrium data: (1) titration with visual observation of liquid clarity or turbidity; (2) visual observation of clarity or turbidity for known compositions as a function of temperature; and (3) direct analysis of equilibrated liquids typically using GC or LC methods. In the titration method, one compound is slowly titrated into a known mass of the second compound during mixing. The titration is terminated when the mixture becomes cloudy, indicating that a second liquid phase has formed. A tie line may be determined by titrating the second compound into the first at the same temperature. This method is reasonably accurate for binary systems composed of pure materials. It also may be applied to ternaries by titrating the third component into a solution of the first and second components, at least to some extent. This method also requires the least time to perform. Since the method is visual, a trace impurity in the "titrant" that is less soluble in the second compound may cause cloudiness at a lower concentration than if pure materials were used. This method has poor precision for sparingly soluble systems. Normally, it is used at ambient temperature and pressure for systems that do not pose a significant health risk to the operator.

In the second method, several mixtures of known composition are formulated and placed in glass vials or ampoules. These are placed in a bath or oven and heated or cooled until two phases become one, or vice versa. In this way, the phase boundaries of a binary system may be determined. Again, impurities in the starting materials may affect the results, and this method does not work well for sparingly soluble systems or for systems that develop significant pressure.

To obtain tie-line data for systems that involve three or more significant components, or for systems that cannot be handled in open containers, both phases must be sampled and analyzed. This generally requires the greatest effort but gives the most accurate results and can be used over the widest range of solubilities, temperatures, and pressures. This method also may be used on multicomponent systems, which are more likely to be encountered in an industrial process. For this method, an appropriate glass vessel or autoclave is selected, based on the temperature, pressure, and compounds in the mixture. It is best to either place the vessel in an oven or submerge it in a bath to ensure there are no cold or hot spots. The mixture is introduced, thermostatted, and thoroughly mixed, and the phases are allowed to separate fully. Samples are then carefully withdrawn through lines that have the minimum dead volume feasible. The sampling should be done isothermally; otherwise the collected sample may not be exactly the same as what was in the equilibrated vessel. Adding a carefully chosen, nonreactive diluent to the sample container will prevent phase splitting, and this can be an important step to ensure accuracy in the subsequent sample workup and analysis. Take sufficient purges and at least three samples from each phase. Use the appropriate analytical method and analyze a calibration standard along with the samples. Try to minimize the time between sampling and analysis.

Rydberg and others describe automated equipment for generating tie line data, including an apparatus called AKUFVE offered by MEAB [Rydberg et al., Chap. 4 in *Solvent Extraction Principles and Practice*, 2d ed., Rydberg et al., eds. (Dekker, 2004), pp. 193–197]. The AKUFVE apparatus employs a stirred cell, a centrifuge for phase separation, and online instrumentation for rapid generation of data. As an alternative, Kuzmanović et al. [*J. Chem. Eng. Data*, **48**, pp. 1237–1244 (2003)] describe a fully automated workstation for rapid measurement of liquid-liquid equilibrium using robotics for automated sampling.

DATA CORRELATION EQUATIONS

Tie Line Correlations Useful correlations of ternary data may be obtained by using the methods of Hand [*J. Phys. Chem.*, **34**(9), pp. 1961–2000 (1930)] and Othmer and Tobias [*Ind. Eng. Chem.*, **34**(6), pp. 693–696 (1942)]. Hand showed that plotting the equilibrium line in terms of mass ratio units on a log-log scale often gave a straight line. This relationship commonly is expressed as

$$\log \frac{X_{23}}{X_{33}} = a + b \log \frac{X_{21}}{X_{11}} \quad (15-23)$$

where X_{ij} represents the mass fraction of component i dissolved in the phase richest in component j , and a and b are empirical constants. Subscript 2 denotes the solute, while subscripts 1 and 3 denote feed solvent and extraction solvent, respectively. An equivalent expression can be written by using the Bancroft coordinate notation introduced earlier: $Y' = cX'^b$, where $c = 10^a$. Othmer and Tobias proposed a similar correlation:

$$\log \frac{1 - X_{33}}{X_{33}} = d + e \log \frac{1 - X_{11}}{X_{11}} \quad (15-24)$$

where d and e are constants. Equations (15-23) and (15-24) may be used to check the consistency of tie line data, as discussed by Awwad et al. [*J. Chem. Eng. Data*, **50**(3), pp. 788–791 (2005)] and by Kirbaslar et al. [*Braz. J. Chem. Eng.*, **17**(2), pp. 191–197 (2000)].

A particularly useful diagram is obtained by plotting the solute equilibrium line on log-log scales as X_{23}/X_{33} versus X_{21}/X_{11} [from Eq. (15-23)] along with a second plot consisting of X_{23}/X_{33} versus X_{23}/X_{13} and X_{21}/X_{31} versus X_{21}/X_{11} . This second plot is termed the limiting solubility curve. The plait point may easily be found from the intersection of the solute equilibrium line with this curve, as shown by Treybal, Weber, and Daley [*Ind. Eng. Chem.*, **38**(8), pp. 817–821 (1946)]. This type of diagram also is helpful for interpolation and limited extrapolation when equilibrium data are scarce. An example diagram is shown in Fig. 15-20 for the water + acetic acid + methyl isobutyl ketone (MIBK) system. For additional discussion of various

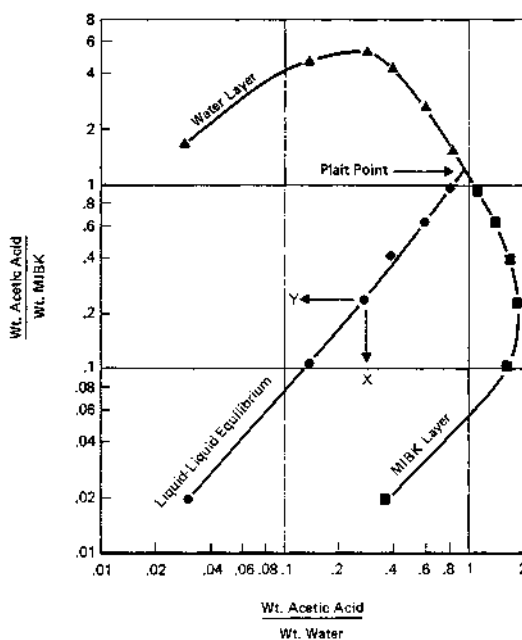


FIG. 15-20 Hand-type ternary diagram for water + acetic acid + MIBK at 25°C.

correlation methods, see Laddha and Degaleesan, *Transport Phenomena in Liquid Extraction* (McGraw-Hill, 1978), Chap. 2.

Thermodynamic Models The thermodynamic theories and equations used to model phase equilibria are reviewed in Sec. 4, "Thermodynamics." These equations provide a framework for data that can help minimize the required number of experiments. An accurate liquid-liquid equilibrium (LLE) model is particularly useful for applications involving concentrated feeds where partition ratios and mutual solubility between phases are significant functions of solute concentration. Sometimes it is difficult to model LLE behavior across the entire composition range with a high degree of accuracy, depending upon the chemical system. In that case, it is best to focus on the composition range specific to the particular application at hand—to ensure the model accurately represents the data in that region of the phase diagram for accurate design calculations. Such a model can be a powerful tool for extractor design or when used with process simulation software to conceptualize, evaluate, and optimize process options. However, whether a complete LLE model is needed will depend upon the application. For dilute applications where partition ratios do not vary much with composition, it may be satisfactory to characterize equilibrium in terms of a simple Hand-type correlation or in terms of partition ratios measured over the range of anticipated feed and raffinate compositions and fit to an empirical equation. Also, when partition ratios are always very large, on the order of 100 or larger, as can occur when washing salts from an organic phase into water, a continuous extractor is likely to operate far from equilibrium. In this case, a precise equilibrium model may not be needed because the extraction factor always is very large and solute diffusion rates dominate performance. (See "Rate-Based Calculations" under "Process Fundamentals and Basic Calculation Methods.")

LLE models for nonionic systems generally are developed by using either the NRTL or UNIQUAC correlation equations. These equations can be used to predict or correlate multicomponent mixtures using only binary parameters. The NRTL equations [Renon and Prausnitz, *AIChE J.*, **14**(1), pp. 135–144 (1968)] have the form

$$\ln \gamma_i = \frac{\sum_j \tau_{ji} G_{ji} x_j}{\sum_j G_{ji} x_j} + \sum_k \frac{G_{ki} x_k}{\sum_k G_{ki} x_k} \left(\tau_{ij} - \frac{\sum_k \tau_{kj} G_{kj} x_k}{\sum_k G_{kj} x_k} \right) \quad (15-25)$$

where τ_{ij} and $G_{ij} = \exp(-\alpha_{ij} \tau_{ij})$ are model parameters. The UNIQUAC equations [Abrams and Prausnitz, *AIChE J.*, **21**(1), pp. 116–128 (1975)] are somewhat more complex. (See Sec. 4, "Thermodynamics.") Most commercial simulation software packages include these models and allow regression of data to determine model parameters. One should refer to the process simulator's operating manual for specific details. Not all simulation software will use exactly the same equation format and parameter definitions, so parameters reported in the literature may not be appropriate for direct input to the program but need to be converted to the appropriate form. In theory, activity coefficient data from binary or ternary vapor-liquid equilibria can be used for calculating liquid-liquid equilibria. While this may provide a reasonable starting point, in practice at least some of the binary parameters will need to be determined from liquid-liquid tie line data to obtain an accurate model [Lafayatis et al., *Ind. Eng. Chem. Res.*, **28**(5), pp. 585–590 (1989)]. Detailed discussion of the application and use of NRTL and UNIQUAC is given by Walas [*Phase Equilibria in Chemical Engineering* (Butterworth-Heinemann, 1985)]. The application of NRTL in the design of a liquid-liquid extraction process is discussed by van Grieken et al. [*Ind. Eng. Chem. Res.*, **44**(21), pp. 8106–8112 (2005)], by Venter and Nieuwoudt [*Ind. Eng. Chem. Res.*, **37**(10), pp. 4099–4106 (1998)], and by Coto et al. [*Chem. Eng. Sci.*, **61**, pp. 8028–8039 (2006)]. The use of the NRTL model also is discussed in Example 5 under "Single-Solvent Fractional Extraction with Extract Reflux" in "Calculation Procedures." The application of UNIQUAC is discussed by Anderson and Prausnitz [*Ind. Eng. Chem. Process Des. Dev.*, **17**(4), pp. 561–567 (1978)].

Although the NRTL or UNIQUAC equations generally are recommended for nonionic systems, a number of alternative approaches have been introduced. Some include explicit terms for association of

molecules in solution, and these may have advantages depending upon the application. An example is the statistical associating fluid theory (SAFT) equation of state introduced by Chapman et al. [*Ind. Eng. Chem. Res.*, **29**(8), pp. 1709–1721 (1990)]. SAFT approximates molecules as chains of spheres and uses statistical mechanics to calculate the energy of the mixture [Müller and Gubbins, *Ind. Eng. Chem. Res.*, **40**(10), pp. 2193–2211 (2001)]. Yu and Chen discuss the application of SAFT to correlate data for 41 binary and 8 ternary liquid-liquid systems [*Fluid Phase Equilibria*, **94**, pp. 149–165 (1994)]. Note that at present not all commercial simulation software packages include SAFT as an option; or if it is included, the association term may be left out. The SAFT equation often is used to correlate LLE data for polymer-solvent systems [Jog et al., *Ind. Eng. Chem. Res.*, **41**(5), pp. 887–891 (2002)]. In another approach, Asprion, Hasse, and Maurer [*Fluid Phase Equil.*, **205**, pp. 195–214 (2003)] discuss the addition of chemical theory association terms to the UNIQUAC model and other phase equilibrium models in general. With this approach, molecular association is treated as a reversible chemical reaction, and parameter values for the association terms may be determined from spectroscopic data. Another activity coefficient correlation called COSMO-SPACE is presented as an alternative to UNIQUAC [Klamt, Krooshof, and Taylor, *AIChE J.*, **48**(10), pp. 2332–2349 (2002)].

Other methods are used to describe the behavior of ionic species (electrolytes). The activity coefficient of an ion in solution may be expressed in terms of modified Debye-Hückel theory. A common expression suitable for low concentrations has the form

$$\log \gamma_i = \frac{-a z_i^2 I^{1/2}}{1 + I^{1/2}} + b z_i^2 I \quad (15-26)$$

where I is ionic strength, z_i is the number of electronic charges, and a and b are parameters that depend upon temperature. Ionic strength is defined in terms of the ion molal concentration. Equation (15-26) represents the activity coefficient for a single ion. For a compound MX that dissociates into M^+ and X^- in solution, the mean ionic activity coefficient is given by $\gamma^\pm = (\gamma^+ \gamma^-)^{1/2}$. Activity coefficients for most electrolytes dissolved in water are less than unity because of the strong attractive interaction between water and a charged species, but this can vary depending upon the organic character of the ion and its concentration. For more detailed discussions focusing on extraction, see Marcus, Chap. 2, and Grenthe and Wanner, Chap. 6, in *Solvent Extraction Principles and Practice*, 2d ed., Rydberg et al., eds. (Dekker, 2004). For general discussions, see *Activity Coefficients in Electrolyte Solutions*, 2d ed., Pitzer, ed. (CRC Press, 1991); Zemaitis et al., *Handbook of Aqueous Electrolyte Thermodynamics* (DIPPR, AIChE, 1986); and Robinson and Stokes, *Electrolyte Solutions* (Butterworths, 1959). The concepts of molecular association have been applied to modeling electrolyte solutions with good success [Stokes and Robinson, *J. Soln. Chem.*, **2**, p. 173 (1973)].

Modeling phase equilibria for mixed-solvent electrolyte systems including nonionic organic compounds is discussed by Polka, Li, and Gmehling [*Fluid Phase Equil.*, **94**, pp. 115–127 (1994)]; Li, Lin, and Gmehling [*Ind. Eng. Chem. Res.*, **44**(5), pp. 1602–1609 (2005)]; and Wang et al. [*Fluid Phase Equil.*, **222–223**, pp. 11–17 (2004)]. Another computer program is discussed by Baes et al. [*Sep. Sci. Technol.*, **25**, p. 1675 (1990)]. Ahlem, Abdeslam-Hassen, and Mossaab [*Chem. Eng. Technol.*, **24**(12), pp. 1273–1280 (2001)] discuss two approaches to modeling metal ion extraction for purification of phosphoric acid.

Data Quality Normally, it is not possible to evaluate LLE data for thermodynamic consistency [Sorenson and Arlt, *Liquid-Liquid Equilibrium Data Collection*, Binary Systems, vol. V, pt. 1 (DECHEMA, 1979), p. 12]. The thermodynamic consistency test for VLE data involves calculating an independently measured variable from the others (usually the vapor composition from the temperature, pressure, and liquid composition) and comparing the measurement with the calculated value. Since LLE data are only very weakly affected by change in pressure, this method is not feasible for LLE. However, if the data were produced by equilibration and analysis of both phases, then at least the data can be checked to determine how well the material balance closes. This can be done by plotting the total

TABLE 15-1 Selected Partition Ratio Data

Partition ratios are listed in units of weight percent solute in the extract divided by weight percent solute in the raffinate, generally for the lowest solute concentrations given in the cited reference. The partition ratio tends to be greatest at low solute concentrations. Consult the original references for more information about a specific system.

Solute	Feed solvent	Extraction solvent	Temp. (°C)	K (wt % basis)	Reference
Ethanol	Cyclohexane	Ethanolamine	25	2.79	1
Acetone	Ethylene glycol	Amyl acetate	31	1.84	2
Acetone	Ethylene glycol	Ethyl acetate	31	1.85	2
Acetone	Ethylene glycol	Butyl acetate	31	1.94	2
Trilinolein	Furfural	Heptane	30	47.5	3
<i>o</i> -Xylene	Heptane	Tetraethylene glycol	20	0.15	4
<i>o</i> -Xylene	Heptane	Tetraethylene glycol	30	0.15	4
<i>o</i> -Xylene	Heptane	Tetraethylene glycol	40	0.16	4
Toluene	Heptane	Sulfolane	25	0.34	5
Toluene	Heptane	Sulfolane	50	0.36	5
Toluene	Heptane	Sulfolane	75	0.31	5
Toluene	Heptane	Sulfolane	100	0.33	5
Toluene	Hexane	Sulfolane	25	0.34	6
Xylene	Hexane	Sulfolane	25	0.30	6
Toluene	<i>n</i> -Hexane	Sulfolane	25	0.34	6
Xylene	<i>n</i> -Hexane	Sulfolane	25	0.30	6
Toluene	<i>n</i> -Octane	Sulfolane	25	0.35	6
Xylene	<i>n</i> -Octane	Sulfolane	25	0.25	6
Toluene	Octane	Sulfolane	25	0.35	6
Xylene	Octane	Sulfolane	25	0.25	6
1,2-Dimethoxyethane	Water	Dodecane	25	0.46	7
1,4-Dioxane	Water	Ethyl acetate	30	1.29	8
1-Butanol	Water	Benzonitrile	25	3.01	9
1-Butanol	Water	Ethyl acetate	40	5.48	10
1-Butanol	Water	Methyl <i>t</i> -butyl ether	25	7.95	11
1-Heptene	Water	1-Propanol	25	3.95	12
1-Octanol	Water	Methyl <i>t</i> -butyl ether	25	10.9	13
1-Propanol	Water	1-Heptene	25	1.36	12
1-Propanol	Water	Butyraldehyde	25	4.14	14
1-Propanol	Water	Cyclohexane	25	0.34	15
1-Propanol	Water	Di-isobutyl ketone	25	0.93	14
1-Propanol	Water	Methyl <i>tert</i> -butyl ether	25	3.79	11
2,3-Butanediol	Water	2,4-Dimethylphenol	40	1.89	16
2,3-Butanediol	Water	2-Butoxyethanol	70	1.79	17
2,3-Dichloropropene	Water	Epichlorohydrin	20	181	18
2,3-Dichloropropene	Water	Epichlorohydrin	77	69.5	18
2-Butoxyethanol	Water	Decane	22	0.45	19
2-Methoxyethanol	Water	Cyclohexanone	70	0.54	20
2-Methyl-1-propanol	Water	Benzene	25	1.18	21
2-Methyl-1-propanol	Water	Toluene	25	0.88	21
2-Propanol	Water	1-Methylcyclohexanol	20	3.66	22
2-Propanol	Water	2,2,4-Trimethylpentane	20	0.045	23
2-Propanol	Water	Carbon tetrachloride	20	1.41	24
2-Propanol	Water	Dichloromethane	20	3.56	22
2-Propanol	Water	Di-isopropyl ether	25	0.41	25
2-Propanol	Water	Di-isopropyl ether	25	0.98	26
3-Cyanopyridine	Water	Benzene	30	1.55	27
Acetaldehyde	Water	Furfural	16	0.97	28
Acetaldehyde	Water	1-Pentanol	18	1.43	28
Acetic acid	Water	1-Butanol	27	1.61	29
Acetic acid	Water	1-Hexene	25	0.0073	30
Acetic acid	Water	1-Octanol	20	0.56	31
Acetic acid	Water	20 vol % Trioctylamine + 20 vol % 1-Decanol + 60 vol % dodecane	20	0.61	32
Acetic acid	Water	2-Butanone	25	1.20	33
Acetic acid	Water	2-Ethyl-1-hexanol	20	0.58	34
Acetic acid	Water	2-Pentanol	25	1.35	35
Acetic acid	Water	2-Pentanone	25	1.00	30
Acetic acid	Water	4-Heptanone	25	0.30	30
Acetic acid	Water	70 vol % Tributylphosphate + 30 vol % dodecane	20	0.31	36
Acetic acid	Water	Cyclohexanol	27	1.33	29
Acetic acid	Water	Diethyl phthalate	20	0.22	37
Acetic acid	Water	Di-isopropyl carbinol	25	0.80	38
Acetic acid	Water	Dimethyl phthalate	20	0.34	37
Acetic acid	Water	Di- <i>n</i> -butyl ketone	25	0.38	39
Acetic acid	Water	Ethyl acetate	30	0.91	40
Acetic acid	Water	Isopropyl ether	20	0.25	41
Acetic acid	Water	Methyl cyclohexanone	25	0.93	38
Acetic acid	Water	Methylisobutyl ketone	25	0.66	42
Acetic acid	Water	Methylisobutyl ketone	25	0.76	38
Acetic acid	Water	Toluene	25	0.06	43
Acetone	Water	1-Octanol	25	0.81	44
Acetone	Water	1-Pentanol	25	4.11	44

15-30 LIQUID-LIQUID EXTRACTION AND OTHER LIQUID-LIQUID OPERATIONS AND EQUIPMENT

TABLE 15-1 Selected Partition Ratio Data (Continued)

Partition ratios are listed in units of weight percent solute in the extract divided by weight percent solute in the raffinate, generally for the lowest solute concentrations given in the cited reference. The partition ratio tends to be greatest at low solute concentrations. Consult the original references for more information about a specific system.

Solute	Feed solvent	Extraction solvent	Temp. (°C)	K (wt % basis)	Reference
Acetone	Water	1-Pentanol	30	1.14	44
Acetone	Water	2-Octanol	30	0.66	44
Acetone	Water	Chloroform	25	1.83	45
Acetone	Water	Chloroform	25	1.72	46
Acetone	Water	Dibutyl ether	25	1.94	38
Acetone	Water	Diethyl ether	30	1.00	47
Acetone	Water	Ethyl acetate	30	1.50	48
Acetone	Water	Ethyl butyrate	30	1.28	48
Acetone	Water	Methyl acetate	30	1.15	48
Acetone	Water	Methylisobutyl ketone	25	1.91	38
Acetone	Water	Hexane	25	0.34	49
Acetone	Water	Toluene	25	0.84	38
Acrylic acid	Water	89.6 wt % Kerosene/10.4 wt % trialkylphosphine oxide (C7–C9)	25	6.50	50
Aniline	Water	Methylcyclohexane	25	2.05	51
Aniline	Water	Methylcyclohexane	50	3.41	51
Aniline	Water	Heptane	25	1.43	51
Aniline	Water	Heptane	50	2.20	51
Aniline	Water	Toluene	25	12.9	52
Benzoic acid	Water	87.4 wt % Kerosene/12.6 wt % tributylphosphate	25	36.0	53
Benzoic acid	Water	89.6 wt % Kerosene/10.4 wt % trialkylphosphine oxide (C7–C9)	25	1.30	50
Butyric acid	Water	20 vol % Trioctylamine + 20 vol % 1-decanol + 60 vol % dodecane	20	6.16	36
Butyric acid	Water	70 vol % Tributylphosphate + 30 vol % dodecane	20	2.51	36
Butyric acid	Water	Methyl butyrate	30	6.75	54
Citric acid	Water	25 wt % Tri-isooctylamine + 75 wt % Chloroform	25	14.1	55
Citric acid	Water	26 wt % Tri-isooctylamine + 75 wt % 1-Octanol	25	41.5	55
Epichlorohydrin	Water	2,3-Dichloropropene	20	11.4	56
Epichlorohydrin	Water	2,3-Dichloropropene	77	13.4	56
Ethanol	Water	1-Octanol	25	0.66	57
Ethanol	Water	1-Octene	25	0.036	58
Ethanol	Water	2,2,4-Trimethylpentane	5	0.027	59
Ethanol	Water	2,2,4-Trimethylpentane	40	0.041	59
Ethanol	Water	3-Heptanol	25	0.78	60
Ethanol	Water	1-Butanol	20	3.00	61
Ethanol	Water	Di- <i>n</i> -propyl ketone	25	0.59	38
Ethanol	Water	1-Hexanol	28	1.00	62
Ethanol	Water	2-Octanol	28	0.83	62
Ethyl acetate	Water	1-Butanol	40	11.1	10
Ethylene glycol	Water	Furfural	25	0.32	64
Formic acid	Water	20 vol % Trioctylamine + 20 vol % 1-decanol + 60 vol % dodecane	20	1.77	36
Formic acid	Water	70 vol % Tributylphosphate + 30 vol % dodecane	20	0.37	36
Formic acid	Water	Methylisobutyl carbinol	30	1.22	65
Furfural	Water	Toluene	25	5.64	66
Glycolic acid	Water	89.6 wt % Kerosene/10.4 wt % trialkylphosphine oxide (C7–C9)	25	0.29	67
Glyoxylic acid	Water	89.6 wt % Kerosene/10.4 wt % trialkylphosphine oxide (C7–C9)	25	0.067	67
Lactic acid	Water	20 vol % Trioctylamine + 20 vol % 1-decanol + 60 vol % dodecane	20	0.65	36
Lactic acid	Water	25 wt % Tri-isooctylamine + 75 wt % chloroform	25	19.2	55
Lactic acid	Water	26 wt % Tri-isooctylamine + 75 wt % 1-octanol	25	25.9	55
Lactic acid	Water	70 vol % Tributylphosphate + 30 vol % dodecane	20	0.14	36
Lactic acid	Water	<i>iso</i> -Amyl alcohol	25	0.35	68
Malic acid	Water	25 wt % Tri-isooctylamine + 75 wt % chloroform	25	30.7	55
Malic acid	Water	25 wt % Tri-isooctylamine + 75 wt % 1-octanol	25	59.0	55
Methanol	Water	1-Octanol	25	0.28	57
Methanol	Water	Ethyl acetate	0	0.059	69
Methanol	Water	Ethyl acetate	20	0.24	69
Methanol	Water	1-Butanol	0	0.60	70
Methanol	Water	1-Hexanol	28	0.57	71
Methanol	Water	<i>p</i> -Cresol	35	0.31	72

TABLE 15-1 Selected Partition Ratio Data (Concluded)

Partition ratios are listed in units of weight percent solute in the extract divided by weight percent solute in the raffinate, generally for the lowest solute concentrations given in the cited reference. The partition ratio tends to be greatest at low solute concentrations. Consult the original references for more information about a specific system.

Solute	Feed solvent	Extraction solvent	Temp. (°C)	K (wt % basis)	Reference
Methanol	Water	Phenol	25	1.33	72
Methyl <i>t</i> -butyl ether	Water	1-Octanol	25	2.61	13
Methyl <i>t</i> -amyl ether	Water	2,2,4-Trimethylpentane	25	131	73
Methylethyl ketone	Water	1,1,2-Trichloroethane	25	3.44	74
Methylethyl ketone	Water	Hexane	25	1.78	75
1-Propanol	Water	Ethyl acetate	20	1.54	69
1-Propanol	Water	Heptane	38	0.54	76
<i>p</i> -Cresol	Water	Methylnaphthalene	35	9.89	72
Phenol	Water	Ethyl acetate	25	0.048	77
Phenol	Water	Isoamyl acetate	25	0.046	77
Phenol	Water	Isopropyl acetate	25	0.040	77
Phenol	Water	Methyl isobutyl ketone	30	39.8	78
Phenol	Water	Methylnaphthalene	25	7.06	79
Phosphoric acid	Water	4-Methyl-2-pentanone	25	0.0012	80
Propionic acid	Water	20 vol % Trioctylamine + 20 vol % 1-decanol + 60 vol % dodecane	20	2.13	36
Propionic acid	Water	70 vol % Tributylphosphate + 30 vol % dodecane	20	1.02	36
Propionic acid	Water	Ethyl acetate	30	2.77	81
Propionic acid	Water	Toluene	31	0.52	82
Pyridine	Water	Chlorobenzene	25	2.10	83
Pyridine	Water	Toluene	25	1.90	84
Pyridine	Water	Xylene	25	1.26	84
<i>t</i> -Butanol	Water	Ethyl acetate	20	1.74	69
Tetrahydrofuran	Water	1-Octanol	20	3.31	85

References:

- Harris et al., *J. Chem. Eng. Data*, **47**, pp. 781–787 (2002).
- Garner, Ellis, and Roy, *Chem. Eng. Sci.*, **2**, p. 14 (1953).
- Beech and Glasstone, *J. Chem. Soc.*, p. 67 (1938).
- Darwish et al., *J. Chem. Eng. Data*, **48**, pp. 1614–1619 (2003).
- De Fre, Thesis, University of Gent, 1976.
- Barbaudy, *Compt. Rend.*, **182**, p. 1279 (1926).
- Burgdorf, Thesis, Technische University, Berlin, 1995.
- Komatsu and Yamamoto, *Kagaku Kogaku Ronbunshu*, **22**(2), pp. 378–384 (1996).
- Grande et al., *J. Chem. Eng. Data*, **41**(4), pp. 926–928 (1996).
- De Andrade and D'Avila, Private communication to DDB, pp. 1–7 (1991).
- Letcher, Ravindran, and Radloff, *Fluid Phase Equil.*, **69**, pp. 251–260 (1991).
- Letcher et al., *J. Chem. Eng. Data*, **39**(2), pp. 320–323 (1994).
- Arce et al., *J. Chem. Thermodyn.*, **28**, pp. 3–6 (1996).
- Letcher et al., *J. Chem. Eng. Data*, **41**(4), pp. 707–712 (1996).
- Plackov and Stern, *Fluid Phase Equil.*, **71**, pp. 189–209 (1992).
- Escudero, Cabezas, and Coca, *J. Chem. Eng. Data*, **39**(4), pp. 834–839 (1994).
- Escudero, Cabezas, and Coca, *J. Chem. Eng. Data*, **41**(6), pp. 1383–1387 (1996).
- Zhang and Liu, *J. Chem. Ind. Eng. (China)*, **46**(3), pp. 365–369 (1995).
- Sazonov, *Zh. Obshch. Khim.*, **61**(1), pp. 59–64 (1991).
- Hauschild and Knapp, *J. Solution Chem.*, **23**(3), pp. 363–377 (1994).
- Stephenson, *J. Chem. Eng. Data*, **37**(1), pp. 80–95 (1992).
- Sayar, *J. Chem. Eng. Data*, **36**(1), pp. 61–65 (1991).
- Arda and Sayar, *Fluid Phase Equil.*, **73**, pp. 129–138 (1992).
- Blumberg, Cejtlin, and Fuchs, *J. Appl. Chem.*, **10**, p. 407 (1960).
- Chang and Moulton, *Ind. Eng. Chem.*, **45**, p. 2350 (1953).
- Letcher, Ravindran, and Radloff, *Fluid Phase Equil.*, **71**, pp. 177–188 (1992).
- Hu, Shi, and Yun, *Shiyou Huagong*, **21**, pp. 38–42 (1992).
- Elgin and Browning, *Trans. Am. Inst. Chem. Engrs.*, **31**, p. 639 (1935).
- Griswold, Chu, and Winsauer, *Ind. Eng. Chem.*, **41**, p. 2352 (1949).
- Nakahara, Masamoto, and Arai, *Kagaku Kogaku Ronbunshu*, **19**(4), pp. 663–668 (1993).
- Ratkovics et al., *J. Chem. Thermodyn.*, **23**, pp. 859–865 (1991).
- Morales et al., *J. Chem. Eng. Data*, **48**, pp. 874–886 (2003).
- Gomis et al., *Fluid Phase Equil.*, **106**, pp. 203–211 (1995).
- Ratkovics et al., *J. Chem. Thermodyn.*, **23**, pp. 859–865 (1991).
- Al-Muhtaseb and Fahim, *Fluid Phase Equil.*, **123**, pp. 189–203 (1996).
- Morales et al., *J. Chem. Eng. Data*, **48**, pp. 874–886 (2003).
- Dramur and Tatli, *J. Chem. Eng. Data*, **38**(1), pp. 23–25 (1993).
- Fairburn, Cheney, and Chernovsky, *Chem. Eng. Progr.*, **43**, p. 280 (1947).
- Fairburn, Cheney, and Chernovsky, *Chem. Eng. Progr.*, **43**, p. 280 (1947).
- Briggs and Comings, *Ind. Eng. Chem.*, **35**, p. 411 (1943).
- Buchanan, *Ind. Eng. Chem.*, **44**, p. 2449 (1952).
- Griswold, Chew, and Klecka, *Ind. Eng. Chem.*, **42**, p. 1246 (1950).
- Johnson and Bliss, *Trans. Am. Inst. Chem. Engrs.*, **42**, p. 331 (1946).
- Tiryaki, Guruz, and Orbey, *Fluid Phase Equil.*, **94**, pp. 267–280 (1994).
- Church and Briggs, *J. Chem. Eng. Data*, **9**, p. 207 (1964).
- Baker, *Phys. Chem.*, **59**, p. 1182 (1955).
- Conway and Phillips, *Ind. Eng. Chem.*, **46**, p. 1474 (1954).
- Hixon and Bockelmann, *Trans. Am. Inst. Chem. Engrs.*, **38**, p. 891 (1942).
- Hirata and Hirose, *Kagaku Kogaku*, **27**, p. 407 (1963).
- Li et al., *J. Chem. Eng. Data*, **48**, pp. 621–624 (2003).
- Charles and Morton, *J. Appl. Chem.*, **7**, p. 39 (1957).
- Hand, *J. Phys. Chem.*, **34**, p. 1961 (1930).
- Mei, Qin, and Dai, *J. Chem. Eng. Data*, **47**, pp. 941–943 (2002).
- Durand et al., *Rev. Inst. Franc. Petrole*, **11**, p. 811 (1956).
- Davison, Smith, and Hood, *J. Chem. Eng. Data*, **11**, p. 304 (1966).
- Zhang and Liu, *J. Chem. Ind. Eng. (China)*, **46**(3), pp. 365–369 (1995).
- Arce et al., *J. Chem. Eng. Data*, **39**(2), pp. 378–380 (1994).
- Purvanto et al., *J. Chem. Eng. Data*, **41**(6), pp. 1414–1417 (1996).
- Wagner and Sandler, *J. Chem. Eng. Data*, **40**(5), pp. 1119–1123 (1995).
- Forbes and Coolidge, *J. Am. Chem. Soc.*, **41**, p. 150 (1919).
- Boobar et al., *Ind. Eng. Chem.*, **43**, p. 2922 (1951).
- Crook and Van Winkle, *Ind. Eng. Chem.*, **46**, p. 1474 (1954).
- De Andrade and D'Avila, private communication to DDB, pp. 1–7 (1991).
- Berg, Manders, and Switzer, *Chem. Eng. Prog.*, **47**, p. 11 (1951).
- Fritzsche and Stockton, *Ind. Eng. Chem.*, **38**, p. 737 (1946).
- Conway and Norton, *Ind. Eng. Chem.*, **43**, p. 1433 (1951).
- Li et al., *J. Chem. Eng. Data*, **48**, pp. 621–624 (2003).
- Jeffreys, *J. Chem. Eng. Data*, **8**, p. 320 (1963).
- Bancroft and Hubard, *J. Am. Chem. Soc.*, **64**, p. 347 (1942).
- Durand et al., *Rev. Inst. Franc. Petrole*, **9**, p. 296 (1954).
- Coull and Hope, *J. Phys. Chem.*, **39**, 967 (1935).
- Frere, *Ind. Eng. Chem.*, **41**, p. 2365, (1949).
- Peschke and Sandler, *J. Chem. Eng. Data*, **40**(1), pp. 315–320 (1995).
- Eaglesfield, Kelly, and Short, *Ind. Chemist*, **29**, pp. 147, 243 (1953).
- Henty, McManamey, and Price, *J. Appl. Chem.*, **14**, p. 148 (1964).
- Denzler, *J. Phys. Chem.*, **49**, p. 358 (1945).
- Alberty and Washburn, *J. Phys. Chem.*, **49**, p. 4 (1945).
- Narashimhan, Reddy, and Chari, *J. Chem. Eng. Data*, **7**, p. 457 (1962).
- Pruett, Wlash, and Desai, *Ind. Eng. Chem.*, **42**, p. 1210 (1950).
- Feki et al., *Can. J. Chem. Eng.*, **72**, pp. 939–944 (1994).
- Gladel and Lablaude, *Rev. Inst. Franc. Petrole*, **12**, p. 1236 (1957).
- Fuoss, *J. Am. Chem. Soc.*, **62**, p. 3183 (1940).
- Fowler and Noble, *J. Appl. Chem.*, **4**, p. 546 (1954).
- Hunter and Brown, *Ind. Eng. Chem.*, **39**, p. 1343 (1947).
- Senol, Alptekin, and Sayar, *J. Chem. Thermodyn.*, **27**, pp. 525–529 (1995).

feed composition used in the experiments along with the measured tie line compositions on a ternary diagram. The feed composition should lie on the tie line. For very low solute concentrations, this plot may be unrevealing. Alternatively, a plot of Y_i''/Z_i'' versus X_i''/Z_i'' (where Y_i'' is the mass fraction of component i in the extract phase, X_i'' is the mass fraction of component i in the raffinate phase, and Z_i'' is the mass fraction of component i in the total feed) should give a straight line that passes through the point (1, 1). The tie line data also may be checked for consistency by plotting the data in the form of a Hand plot or Othmer-Tobias plot, as described in "Tie Line Correlations," and looking for outliers. Another approach is to plot the partition ratio as a function of solute concentration and look for data points that deviate significantly from otherwise smooth trends. If the NRTL equation is used, refit all the binary data sets by using the same value for model parameter α . A value of 0.3 is recommended by Walas [*Phase Equilibria in Chemical Engineering* (Butterworth-Heinemann, 1985), p. 203] for nonaqueous systems, and a higher value of 0.4 is recommended for aqueous systems. Sorensen and Arlt [*Chemistry Data Series: Liquid-Liquid Equilibrium Data Collection*, Vol. V, pt. 1 (DECHEMA, 1979), p. 14] use a value of 0.2 for all their work. The particular value chosen appears to be less important than using the same value for all binaries of the same type (aqueous or nonaqueous). Try for a reasonable fit of the overall data, but be sure to focus on achieving a good fit of the data in the region most relevant to the application at hand.

TABLE OF SELECTED PARTITION RATIO DATA

Table 15-1 summarizes typical partition ratio data for selected systems.

PHASE EQUILIBRIUM DATA SOURCES

A comprehensive collection of phase equilibrium data (including vapor-liquid, liquid-liquid, and solid-liquid data) is maintained by a group headed by Prof. Juergen Gmehling at the University of Oldenburg, Germany. This collection, known as the Dortmund Data Bank, includes LLE measurements as well as NRTL and UNIQUAC fitted parameters. The data bank also includes a compilation of infinite-dilution activity coefficients. The LLE collection is available as a series of

books [Sorensen and Arlt, *Chemistry Data Series: Liquid-Liquid Equilibrium Data Collection, Binary Systems*, vol. V, pts. 1-4 (DECHEMA, 1979-1980)], as a proprietary database including retrieval and modeling software, and online by subscription. There also is a new online database offered by FIZ-Berlin Infotherm. Other sources of thermodynamic data include the IUPAC Solubility Data Series published by Oxford University Press, and compilations prepared by the Thermodynamics Research Center (TRC) in Boulder, Colo., a part of the Physical and Chemical Properties Division of the National Institute of Standards and Technology. An older but still useful data collection is that of Stephens and Stephens [*Solubilities of Inorganic and Organic Compounds*, vol. 1, pts. 1 and 2 (Pergamon, 1960)]. Also, a database of activity coefficients is included in the supporting information submitted with the article by Lazzaroni et al. [*Ind. Eng. Chem. Res.*, **44**(11), pp. 4075-4083 (2005)] and available from the publisher. A listing of the original sources is included. Additional sources of data are discussed by Skrzecz [*Pure Appl. Chem. (IUPAC)*, **69**(5), pp. 943-950 (1997)].

RECOMMENDED MODEL SYSTEMS

To facilitate the study and comparison of various types of extraction equipment, Bart et al. [Chap. 3 in Godfrey and Slater, *Liquid-Liquid Extraction Equipment* (Wiley, 1994)] recommend several model systems. These include (1) water + acetone + toluene (high interfacial tension); (2) water + acetone + butyl acetate (moderate interfacial tension); and (3) water + succinic acid + *n*-butanol (low interfacial tension). All have solute partition ratios near $K = 1.0$. Misek, Berger, and Schröter [*Standard Test Systems for Liquid Extraction* (The Instn. of Chemical Engineers, 1985)] summarize phase equilibrium, viscosities, densities, diffusion coefficients, and interfacial tensions for these systems. Note that methyl isobutyl ketone + acetic acid + water was replaced with the water + acetone + butyl acetate system because of concerns over acetic acid dimerization and Marangoni instabilities. (See "Liquid-Liquid Dispersion Fundamentals.") For test systems with a partition ratio near $K = 10$, Bart et al. recommend (1) water + methyl isopropyl ketone + toluene (high interfacial tension) and (2) water + methyl isopropyl ketone + butyl acetate (medium interfacial tension) and give references to data sources. Bart et al. also recommend a number of systems involving reactive extractants.

SOLVENT SCREENING METHODS

A variety of methods may be used to estimate solvent properties as an aid to identifying useful solvents for a new application. Many of these methods focus on thermodynamic properties; a favorable partition ratio and low mutual solubility often are necessary for an economical extraction process, so ranking candidates according to thermodynamic properties provides a useful initial screen of the more promising candidates. Keep in mind, however, that other factors also must be taken into account when selecting a solvent, as discussed in "Desirable Solvent Properties" under "Introduction and Overview." When using the following methods, also note that the level of uncertainty may be fairly high. The uncertainty depends upon how closely the chemical system of interest resembles the systems used to develop the method.

USE OF ACTIVITY COEFFICIENTS AND RELATED DATA

Compilations of infinite-dilution activity coefficients, when available for the solute of interest, may be used to rank candidate solvents. Partition ratios at finite concentrations can be estimated from these data by extrapolation from infinite dilution using a suitable correlation equation such as NRTL [Eq. (15-25)]. Examples of these kinds of calculations are given by Walas [*Phase Equilibria in Chemical Engineering* (Butterworth-Heinemann, 1985)]. Most activity coefficients available in the literature are for small organic molecules and are derived from vapor-liquid equilibrium measurements or azeotropic composition data.

Partition ratios at infinite dilution can be calculated directly from the ratio of infinite-dilution activity coefficients for solute dissolved in the extraction solvent and in the feed solution, often providing a reasonable estimate of the partition ratio for dilute concentrations. Infinite-dilution

activity coefficients often are reported in terms of a van Laar binary interaction parameter [Smallwood, *Solvent Recovery Handbook* (McGraw-Hill, 1993)] such that

$$\ln \gamma_{ij}^{\infty} = \frac{A_{ij}}{RT} \quad (15-27)$$

$$K_i^0 = \frac{\gamma_i^{\infty}}{\gamma_i^{\infty,*}} = \frac{\exp(A_{ij}^*/RT)}{\exp(A^{oij}/RT)} \quad (15-28)$$

where $*$ denotes the extraction solvent phase. For example, the partition ratio for transferring acetone from water into benzene at 25°C and dilute conditions may be estimated as follows: For acetone dissolved in benzene $A_{ij}/RT = 2.47$, and for acetone dissolved in water $A_{ij}/RT = 2.29$. Then $K_i^0 = e^{2.29}/e^{0.47} = 9.87/1.6 = 6.17$ (mol/mol) ≈ 1.4 (wt/wt). Briggs and Comings [*Ind. Eng. Chem.*, **35**(4), pp. 411-417 (1943)] report experimental values that range between 1.06 and 1.39 (wt/wt).

For screening candidate solvents, comparing the magnitude of the activity coefficient for the solute of interest dissolved in the solvent phase often is a good way to rank solvents, since a smaller value of $\gamma_{i,\text{solvent}}$ indicates a higher K value. Solubility data available for a given solute dissolved in a range of solvents also can be used to rank solvents, since higher solubility in a candidate solvent indicates a more attractive interaction (a lower activity coefficient) and therefore a higher partition ratio.

ROBBINS' CHART OF SOLUTE-SOLVENT INTERACTIONS

When available data are not sufficient (the most common situation), Robbins' chart of functional group interactions (Table 15-2) is a useful

TABLE 15-2 Robbins' Chart of Solute-Solvent Interactions*

Solute class		Solvent class											
		1	2	3	4	5	6	7	8	9	10	11	12
H donor groups													
1	Phenol	0	0	–	0	–	–	–	–	–	–	+	+
2	Acid, thiol	0	0	–	0	–	–	0	0	0	0	+	+
3	Alcohol, water	–	–	0	+	+	0	–	–	+	+	+	+
4	Active H on multihalogen paraffin	0	0	+	0	–	–	–	–	–	–	0	+
H acceptor groups													
5	Ketone, amide with no H on N, sulfone, phosphine oxide	–	–	+	–	0	+	+	+	+	+	+	+
6	Tertiary amine	–	–	0	–	+	0	+	+	0	+	0	0
7	Secondary amine	–	0	–	–	+	+	0	0	0	0	0	+
8	Primary amine, ammonia, amide with 2H on N	–	0	–	–	+	+	0	0	+	+	+	+
9	Ether, oxide, sulfoxide	–	0	+	–	+	0	0	+	0	+	0	+
10	Ester, aldehyde, carbonate, phosphate, nitrate, nitrite, nitrile, intramolecular bonding, e.g., <i>o</i> -nitrophenol	–	0	+	–	+	+	0	+	+	0	+	+
11	Aromatic, olefin, halogen aromatic, multihalogen paraffin without active H, monohalogen paraffin	+	+	+	0	+	0	0	+	0	+	0	0
Non-H-bonding groups													
12	Paraffin, carbon disulfide	+	+	+	+	+	0	+	+	+	+	0	0

*From Robbins, *Chem. Eng. Prog.*, **76**(10), pp. 58–61 (1980), by permission. Copyright 1980 AIChE.

guide to ranking general classes of solvents. It is based on an evaluation of hydrogen bonding and electron donor-acceptor interactions for 900 binary systems [Robbins, *Chem. Eng. Prog.*, **76** (10), pp. 58–61 (1980)]. The chart includes 12 general classes of functional groups, divided into three main types: hydrogen-bond donors, hydrogen-bond acceptors, and non-hydrogen-bonding groups. Compounds representative of each class include (1) phenol, (2) acetic acid, (3) pentanol, (4) dichloromethane, (5) methyl isobutyl ketone, (6) triethylamine, (7) diethylamine, (8) *n*-propylamine, (9) ethyl ether, (10) ethyl acetate, (11) toluene, and (12) hexane. Robbins' chart is applicable to any process where liquid-phase activity coefficients are important, including liquid-liquid extraction, extractive distillation, azeotropic distillation, and crystallization from solution. The activity coefficient in the liquid phase is common to all these separation processes.

Robbins' chart predicts positive, negative, or zero deviations from ideal behavior for functional group interactions. For example, consider an application involving extraction of acetone from water into chloroform solvent. Acetone contains a ketone carbonyl group which is a hydrogen acceptor and a member of solute class 5 according to Table 15-2. Chloroform contains a hydrogen donor group (solvent class 4). The solute class 5 and solvent class 4 interaction in Table 15-2 is shown to give a negative deviation from ideal behavior. This indicates an attractive interaction which enhances the liquid-liquid partition ratio. Other classes of solvents shown in Table 15-2 that yield a negative deviation with a ketone (class 5) are classes 1 and 2 (phenolics and acids). Other ketones (solvent class 5) are shown to be compatible with acetone (solute class 5) and tend to give activity coefficients near 1.0, that is, nearly ideal behavior. The solvent classes 6 through 12 tend to provide repulsive interactions between these groups and acetone, and so they are not likely to exhibit partition ratios for ketones as high as the other solvent groups do.

Most of the classes in Table 15-2 are self-explanatory, but some can use additional definition. Class 4 includes halogenated solvents that have highly active hydrogens as described by Ewell, Harrison, and Berg [*Ind. Eng. Chem.*, **36**(10), pp. 871–875 (1944)]. These are molecules that have two or three halogen atoms on the same carbon as a hydrogen atom, such as dichloromethane, trichloromethane, 1,1-dichloroethane, and 1,1,2,2-tetrachloroethane. Class 4 also includes molecules that have one halogen on the same carbon atom as a hydrogen atom and one or more halogen atoms on an adjacent carbon atom, such as 1,2-dichloroethane and 1,1,2-trichloroethane.

Apparently, the halogens interact intramolecularly to leave the hydrogen atom highly active. Monohalogen paraffins such as methyl chloride and ethyl chloride are in class 11 along with multihalogen paraffins and olefins without active hydrogen, such as carbon tetrachloride and perchloroethylene. Chlorinated benzenes are also in class 11 because they do not have halogens on the same carbon as a hydrogen atom. Intramolecular bonding on aromatics is another fascinating interaction which gives a net result that behaves much as does an ester group, class 10. Examples of this include *o*-nitrophenol and *o*-hydroxybenzaldehyde (salicylaldehyde). The intramolecular hydrogen bonding is so strong between the hydrogen donor group (phenol) and the hydrogen acceptor group (nitrate or aldehyde) that the molecule acts as an ester. One result is its low solubility in hot water. By contrast, the *para* derivative is highly soluble in hot water.

ACTIVITY COEFFICIENT PREDICTION METHODS

Robbins' chart provides a useful qualitative indication of interactions between classes of molecules but does not give quantitative differences within each class. For this, a number of methods are available. Many have been implemented in commercial and university-supported software packages. Perhaps the most widely used of these is the UNIFAC group contribution method [Fredenslund et al., *Ind. Eng. Chem. Proc. Des. Dev.*, **16**(4), pp. 450–462 (1977); and Wittig et al., *Ind. Eng. Chem. Res.*, **42**(1), pp. 183–188 (2003). Also see Jakob et al., *Ind. Eng. Chem. Res.*, **45**, pp. 7924–7933 (2006)]. The use of UNIFAC for estimating LLE is discussed by Gupte and Danner [*Ind. Eng. Chem. Res.*, **26**(10), pp. 2036–2042 (1987)] and by Hooper, Michel, and Prausnitz [*Ind. Eng. Chem. Res.*, **27**(11), pp. 2182–2187 (1988)]. Vakili-Nezhad, Modarress, and Mansoori [*Chem. Eng. Technol.*, **22**(10), pp. 847–852 (1999)] discuss its use for representing a complex feed containing a large number of components for which available LLE data are incomplete.

UNIFAC calculates activity coefficients in two parts:

$$\ln \gamma_i = \ln \gamma_i^C + \ln \gamma_i^R \quad (15-29)$$

The combinatorial part $\ln \gamma_i^C$ is calculated from pure-component properties. The residual part $\ln \gamma_i^R$ is calculated by using binary interaction parameters for solute-solvent group pairs determined by fitting phase equilibrium data. Both parts are based on the UNIQUAC set

of equations. With this approach, a molecule is treated as an assembly of various groups of atoms. Compounds for which phase equilibrium already has been measured are used to regress constants for these different groups. These constants are then used in a correlation to predict properties for a new molecule. There are several UNIFAC parameter sets available. It is important to use a consistent set of parameters since the different parameter databases are not necessarily compatible.

A number of methods based on regular solution theory also are available. Only pure-component parameters are needed to make estimates, so they may be applied when UNIFAC group-interaction parameters are not available. The Hansen solubility parameter model divides the Hildebrand solubility parameter into three parts to obtain parameters δ_d , δ_p , and δ_h accounting for nonpolar (dispersion), polar, and hydrogen-bonding effects [Hansen, *J. Paint Technol.*, **39**, pp. 104–117 (1967)]. An activity coefficient may be estimated by using an equation of the form

$$\ln \gamma_i = \frac{v_i}{RT} \left\{ \left(\bar{\delta}_i - \delta_i^i \right)^2 + 0.25 \left[\left(\bar{\delta}_p - \delta_p^i \right)^2 + \left(\bar{\delta}_h - \delta_h^i \right)^2 \right] \right\} \quad (15-30)$$

where $\bar{\delta}$ is the solubility parameter for the mixture, δ^i is the solubility parameter for component i , v is molar volume, R is the universal gas constant, and T is absolute temperature [Frank, Downey, and Gupta, *Chem. Eng. Prog.*, **95**(12), pp. 41–61 (1999)]. The Hansen model has been used for many years to screen solvents and facilitate development of product formulations. Hansen parameters have been determined for more than 500 solvents [Hansen, *Hansen Solubility Parameters: A User's Handbook* (CRC, 2000); and *CRC Handbook of Solubility Parameters and Other Cohesion Parameters*, 2d ed., Barton, ed. (CRC, 1991)].

MOSCED, another modified regular solution model, utilizes two parameters to represent hydrogen bonding: one for proton donor capability (acidity) and one for proton acceptor capability (basicity) [Thomas and Eckert, *Ind. Eng. Chem. Proc. Des. Dev.*, **23**(2), pp. 194–209 (1984)]. This provides a more realistic representation of hydrogen bonding that allows more accurate modeling of a wider range of solvents, and unlike the Hansen model, MOSCED can predict negative deviations from ideal solution (activity coefficients less than 1.0). MOSCED calculates infinite-dilution activity coefficients by using an equation of the general form

$$\ln \gamma_{21}^\infty = \frac{v_2}{RT} \left[\frac{(\lambda_1 - \lambda_2)^2}{\psi_1} + \frac{q_1^2 q_2^2 (\tau_1 - \tau_2)^2}{\xi_1} + \frac{(\alpha_1 - \alpha_2)(\beta_1 - \beta_2)}{\xi_1} \right] \quad (15-31)$$

There are five adjustable parameters per molecule: λ , the dispersion parameter; q , the induction parameter; τ , the polarity parameter; α , the hydrogen-bond acidity parameter; and β , the hydrogen-bond basicity parameter. The induction parameter q often is set to a value of 1.0, yielding a four-parameter model. The terms ψ_1 and ξ_1 are asymmetry factors calculated from the other parameters. A database of parameter values for 150 compounds, determined by regression of phase equilibrium data, is given by Lazzaroni et al. [*Ind. Eng. Chem. Res.*, **44**(11), pp. 4075–4083 (2005)]. An application of MOSCED in the study of liquid-liquid extraction is described by Escudero, Cabezas, and Coca [*Chem. Eng. Comm.*, **173**, pp. 135–146 (1999)]. Also see Frank et al., *Ind. Eng. Chem. Res.*, **46**, pp. 4621–4625 (2007).

Another method for estimating activity coefficients is described by Chen and Song [*Ind. Eng. Chem. Res.*, **43**(26), pp. 8354–8362 (2004); **44**(23), pp. 8909–8921 (2005)]. This method involves regression of a small data set in a manner similar to the way the Hansen and MOSCED models typically are used. The model is based on a modified NRTL framework called NRTL-SAC (for segment activity coefficient) that utilizes only pure-component parameters to represent polar, hydrophobic, and hydrophilic segments of a molecule. An electrolyte parameter may be added to characterize ion-ion and ion-molecule interactions attributed to ionized segments of species in solution. The resulting model may be used to estimate activity coefficients and related properties for mixtures of non-ionic organics plus electrolytes in aqueous and nonaqueous solvents.

A method developed by Meyer and Maurer [*Ind. Eng. Chem. Res.*, **34**(1), pp. 373–381 (1995)] uses the linear solvation energy relationships (LSER) model [Taft et al., *Nature*, **313**, p. 384 (1985); and Taft et al.,

J. Pharma Sci., **74**, pp. 807–814 (1985)] to estimate infinite-dilution partition ratios for solute distributed between water and an organic solvent. The model uses 36 generalized parameters and four solvatochromic parameters to characterize a given solute. The solvatochromic parameters are α (acidity), β (basicity), π (polarity), and δ (polarizability). Another method utilizing LSER concepts is the SPACE model for estimating infinite-dilution activity coefficients [Hait et al., *Ind. Eng. Chem. Res.*, **32**(11), pp. 2905–2914 (1993)]. Also see Abraham, Ibrahim, and Zissimos, *J. Chromatography*, **1037**, pp. 29–47 (2004).

The thermodynamic methods described above glean information from available data to make estimates for other systems. As an alternative approach, quantum chemistry calculations and molecular simulation methods are finding more and more use in engineering applications [Gupta and Olson, *Ind. Eng. Chem. Res.*, **42**(25), pp. 6359–6374 (2003); and Chen and Mathias, *AIChE J.*, **48**(2), pp. 194–200 (2002)]. These methods minimize the need for data; however, the computational effort and specialized expertise required to use them generally are higher, and the accuracy of the results may not be known. An important method gaining increasing application in the chemical industry is the conductorlike screening model (COSMO) introduced by Klamt and colleagues [Klamt, *J. Phys. Chem.*, **99**, p. 2224 (1995); Klamt and Eckert, *Fluid Phase Equil.*, **172**, pp. 43–72 (2000); Eckert and Klamt, *AIChE J.*, **48**(2), pp. 369–385 (2002); and Klamt, *From Quantum Chemistry to Fluid Phase Thermodynamics and Drug Design* (Elsevier, 2005)]. Also see Grensemann and Gmehling, *Ind. Eng. Chem. Res.*, **44**(5), pp. 1610–1624 (2005). This method utilizes computational quantum mechanics to calculate a two-dimensional electron density profile to characterize a given molecule. This profile is then used to estimate phase equilibrium using statistical mechanics and solvation theory. The Klamt model is called COSMO-RS (for realistic solvation). A similar model is COSMO-SAC (segment activity coefficient) published by Lin and Sandler [*Ind. Eng. Chem. Res.*, **41**(5), pp. 899–913, 2332 (2002)]. Databases of electron density profiles (sigma profiles) are available from a number of vendors and universities. For example, a sigma-profile database of more than 1000 molecules is available from the Virginia Polytechnic Institute and State University [Mullins et al., *Ind. Eng. Chem. Res.*, **45**(12), pp. 4389–4415 (2006)]. Once determined, the profiles allow convenient calculation of phase equilibria using available software. An application of COSMO-RS to predict liquid-liquid equilibria is discussed by Banerjee et al. [*Ind. Eng. Chem. Res.*, **46**(4), pp. 1292–1304 (2007)].

METHODS USED TO ASSESS LIQUID-LIQUID MISCIBILITY

In evaluating potential solvents, it is important to determine whether a given candidate will exhibit sufficiently limited miscibility with the feed liquid. Mutual solubility data for organic-solvent + water mixtures often are listed somewhere in the literature and can be obtained through a literature search. (See “Phase Equilibrium Data Sources” under “Thermodynamic Basis for Liquid-Liquid Extraction.”) However, data often are not available for pairs of organic solvents and for multicomponent mixtures showing the effect of dissolved solutes. In these cases, estimates can provide useful guidance. Note, however, that the available estimation methods normally provide limited accuracy, so it is best to measure these properties for the more promising candidates.

Phase splitting behavior can be inferred from activity coefficients. In general, partial miscibility will not occur whenever the infinite-dilution activity coefficients of the components in solution are less than 7. This is a reliable rule but it depends upon the quality of the activity coefficient data or estimates. If γ_i^∞ for any one of the components is greater than 7, then partial miscibility may occur at some finite composition. The criterion $\gamma_i^\infty > 7$ often is cited as a general rule indicating a partially miscible system, but there are many exceptions. For detailed discussion, see Prausnitz, Lichtenthaler, and Gomez de Azevedo, *Molecular Thermodynamics of Fluid-Phase Equilibria*, 3d ed. (Prentice-Hall, 1999). Solubility parameters also can be used to assess miscibility [*Handbook of Solubility Parameters and Other Cohesion Parameters*, 2d ed., Barton, ed. (CRC, 1991)].

As a complementary alternative, Godfrey's data-based method [*CHEMTECH*, **2**(6), pp. 359–363 (1972)] provides a quick way of qualitatively assessing whether an organic-solvent pair of interest is likely to

TABLE 15-3 Godfrey Miscibility Numbers

Acetal	23	Chloroform	19
Acetic acid	14	1-Chloronaphthalene	22
Acetic anhydride	12, 19	3-Chlorophenetole	15, 20
Acetol	8	2-Chlorophenol	16
Acetol acetate	10	2-Chloropropane	23
Acetol formate	9, 17	2-Chlorotoluene	20
Acetone	15, 17	Coconut oil	29
Acetonitrile	11, 17	<i>p</i> -Cresol	14
Acetophenone	15, 18	4-Cyano-2,2-dimethylbutyraldehyde	11, 18
<i>N</i> -Acetylmorpholine	11	Cyclohexane	28
Acrylonitrile	14, 18	Cyclohexanecarboxylic acid	16
Adiponitrile	8, 19	Cyclohexanol	16
Allyl alcohol	14	Cyclohexanone	17
Allyl ether	22	Cyclohexene	26
2-Allyloxyethanol	13	Cyclooctane	29
2-Aminoethanol	2	Cyclooctene	27
2-(2-Aminoethoxy) ethanol	2	<i>p</i> -Cymene	25
Aminoethylethanolamine	5	Decalin	29
1-(2-Aminoethyl) piperazine	12	Decane	29
1-Amino-2-propanol	6	1-Decanol	18
Aniline	12	1-Decene	29
Anisole	20	Diacetone alcohol	14
Benzaldehyde	15, 19	Diallyl adipate	21
Benzene	21	1,2-Dibromobutane	22
Benzonitrile	15, 19	1,4-Dibromobutane	21
Benzyl alcohol	13	Dibromoethane	19
Benzyl benzoate	15, 21	1,2-Dibromopropane	21
Bicyclohexyl	29	1,2-Dibutoxyethane	25
Bis(2-butoxyethyl) ether	23	<i>N,N</i> -Dibutylacetamide	17
Bis(2-chloroethyl) ether	20	Dibutyl ether	26
Bis(2-chloroisopropyl) ether	20	Dibutyl maleate	22
Bis(2-ethoxyethyl) ether	15	Dibutyl phthalate	22
Bis(2-hydroxyethyl) thiodipropionate	5	1,3-Dichloro-2-propanol	12
Bis(2-hydroxypropyl) maleate	6	Dichloroacetic acid	13
Bis(2-methoxyethyl) ether	15, 17	1,2-Dichlorobenzene	21
Bis(2-methoxyethyl) phthalate	11, 19	1,4-Dichlorobutane	20
Bromobenzene	21	1,1-Dichloroethane	20
1-Bromobutane	23	1,2-Dichloroethane	20
Bromocyclohexane	25	<i>cis</i> -1,2-Dichloroethylene	20
1-Bromodecane	27	<i>trans</i> -1,2-Dichloroethylene	21
1-Bromododecane	27	Dichloromethane	20
Bromoethane	21	1,2-Dichloropropane	20
1-Bromohexane	24	1,3-Dichloropropane	20
1-Bromo-3-methylbutane	24	Dicyclopentadiene	26
1-Bromooctane	26	Didecyl phthalate	26
2-Bromooctane	26	Diethanolamine	1
1-Bromotetradecane	29	Diethoxydimethylsilane	26
1,2-Butanediol	6	<i>N,N</i> -Diethylacetamide	14
1,3-Butanediol	4	Diethyl adipate	19
1,4-Butanediol	3	Diethyl carbonate	21
2,3-Butanediol	12, 17	Diethyl ketone	18
1-Butanol	15	Diethyl oxalate	14, 20
2-Butanol	16	Diethyl phthalate	13, 20
<i>t</i> -Butanol	16	Diethyl sulfate	12, 21
2-Buten-1-ol	15	Diethylene glycol	5
2-Buten-1,4-diol	3	Diethylene glycol diacetate	12, 19
2-Butoxyethanol	16	Diethylenetriamine	9
2-(2-Butoxyethoxy) ethanol	15	Diethyl ether	23
Butyl acetate	22	2,5-Dihydrofuran	17
Butyl formate	19	Di-isobutyl ketone	23
Butyl methacrylate	23	Di-isopropyl ketone	23
Butyl oleate	26	Di-isopropylbenzene	25
Butyl sulfide	26	1,2-Dimethoxyethane	17
Butylaldoxime	15	<i>N,N</i> -Dimethylacetamide	13
Butyric acid	16	<i>N,N</i> -Dimethylacetoacetamide	10
Butyric anhydride	21	2-Dimethylaminoethanol	14
Butyrolactone	10	Dimethyl carbonate	14, 19
Butyronitrile	14, 19	Dimethylformamide	12
Carbon disulfide	26	Dimethyl maleate	12, 19
Carbon tetrachloride	24	Dimethyl malonate	11, 19
Castor oil	25	Dimethyl phthalate	12, 19
1-Chlorobutane	23	1,4-Dimethylpiperazine	16
2-Chloroethanol	11	2,5-Dimethylpyrazine	16
3-Chloro-1,2-propanediol	4	Dimethyl sebacate	22
1-Chloro-2-propanol	14	2,4-Dimethylsulfonate	12, 17
Chlorobenzene	21	Dimethyl sulfoxide	9
1-Chlorobutane	23	Dioctyl phthalate	24
1-Chlorodecane	27	1,4-Dioxane	17

TABLE 15-3 Godfrey Miscibility Numbers (Continued)

1,4-Dioxene	15, 19	Isobromobutane	23
Dipentene	26	2-Isobutoxyethanol	15, 17
Dipentyl ether	26	Isobutyl acetate	21
Diphenyl ether	22	Isobutyl isobutyrate	23
Diphenyl methane	21	Isobutanol	15
Dipropyl sulfone	12, 17	Isophorone	18
Dipropylene glycol	11	Isoprene	25
Dodecane	29	Isopropenyl acetate	19
1-Dodecanol	18	Isopropyl acetate	19
1-Dodecene	29	Isopropyl ether	26
Epichlorohydrin	14, 19	Isopropylbenzene	24
Epoxyethylbenzene	15, 19	Kerosene	30
Ethanesulfonic acid	5	2-Mercaptoethanol	9
Ethanol	14	Mesityl oxide	18
2-Ethoxyethanol	14	Mesitylene	24
2-(2-Ethoxy) ethanol	13	Methacrylonitrile	15, 19
2-Ethoxyethylacetate	15, 19	Methanesulfonic acid	4
Ethyl acetate	19	Methanol	12
Ethyl acetoacetate	13, 19	5-Methoxazolidinone	7
Ethyl benzene	24	Methoxyacetic acid	8
Ethyl benzoate	21	Methoxyacetoneitrile acetamide	11, 19
2-Ethylbutanol	17	3-Methoxybutanol	14
Ethyl butyrate	22	2-Methoxyethanol	13
Ethylene carbonate	6, 17	2-(2-Methoxyethoxy) ethanol	12
Ethylenediamine	9	2-Methoxyethyl acetate	14, 17
Ethylene glycol	2	2-Methoxyethyl methoxyacetate	15
Ethylene glycol bis(methoxyacetate)	9, 17	1-[2-Methoxy-1-methylethoxy]-2-propanol	15
Ethylene glycol diacetate	12, 19	3-Methoxy-1,2-propanediol	5
Ethylene glycol diformate	8, 17	1-Methoxy-2-propanol	15
Ethylene monobiscarbonate	10, 19	3-Methoxypropionitrile	11, 17
Ethylformamide	9	3-Methoxypropylamine	15
Ethyl formate	15, 19	3-Methoxypropylformamide	10
2-Ethyl-1,3-hexanediol	14, 17	Methyl acetate	15, 17
2-Ethylhexanol	17	Methylal	19
Ethyl hexoate	23	2-Methylaminoethanol	11
Ethyl lactate	14	2-Methyl-1-butene	27
N-Ethylmorpholine	16	2-Methyl-2-butene	26
Ethyl orthoformate	23	Methylchloroacetate	13, 19
Ethyl propionate	21	Methylcyanoacetate	8, 17
2-Ethylthioethanol	13	Methylcyclohexane	29
Ethyl trichloroacetate	21	1-Methylcyclohexene	27
Fluorobenzene	20	Methylcyclopentane	28
1-Fluoronaphthalene	21	Methyl ethyl ketone	17
Formamide	3	Methyl formate	14, 19
Formic acid	5	2,2'-Methyliminodiethanol	8
N-Formylmorpholine	10	Methyl isoamyl ketone	19
Furan	20	Methyl isobutyl ketone	19
Furfural	11, 17	Methyl methacrylate	20
Furfuryl alcohol	11	Methyl methoxyacetate	13
Glycerol (glycerin)	1	N-Methylmorpholine	16
Glycerol carbonate	3	1-Methylnaphthalene	22
Glycidyl phenyl ether	13, 19	Methyl oleate	26
Heptane	29	2-Methylpentane	29
1-Heptanol	17	3-Methylpentane	29
3-Heptanone	22	4-Methyl-2-pentanol	17
4-Heptanone	23	2-Methyl-2,4-pentanediol	14
1-Heptene	28	4-Methyl-1-pentene	28
Hexachlorobutadiene	26	cis-4-Methyl-2-pentene	27
Hexadecane	30	N-Methyl-2-pyrrolidinone	13
1-Hexadecene	29	Methyl stearate	26
Hexamethylphosphoramide	15	α -Methylstyrene	23
Hexane	29	3-Methylsulfolane	10, 17
2,5-Hexanediol	5	Mineral spirits	29
2,5-Hexanedione	12, 17	Morpholine	14
1,2,6-Hexanetriol	2	Nitrobenzene	14, 20
1-Hexanol	17	Nitroethane	13, 20
Hexanoic acid	17	Nitromethane	10, 19
1-Hexene	27	2-Nitropropane	15, 20
2-Hydroxyethyl carbamate	2	1-Nonanol	17
2-Hydroxyethylformamide	1	Nonylphenol	17
2-Hydroxyethylmethacrylate	12	1-Octadecene	30
1-(2-Hydroxyethoxy)-2-propanol	8	1,7-Octadiene	27
2-Hydroxypropyl carbamate	3	Octane	29
Hydroxypropyl methacrylate	14, 17	1-Octanethiol	26
Iodobenzene	22	1-Octanol	17
Iodoethane	22	2-Octanol	17
Iodomethane	21	2-Octanone	22
Isoamylbenzene	25	1-Octene	28

TABLE 15-3 Godfrey Miscibility Numbers (Concluded)

<i>cis</i> -2-Octene	27	Tetrahydrofuran	17
<i>trans</i> -2-Octene	28	Tetrahydrofurfuryl alcohol	13
3,3'-Oxydipropionitrile	6	Tetrahydrothiophene	21
Paraldehyde	15, 19	Tetralin	24
Polyethylene glycol PEG-200	7	Tetramethylsilane	29
Polyethylene glycol PEG-300	8	Tetramethylurea	15
Polyethylene glycol PEG-600	8	Tetrapropylene	29
1,3-Pentadiene	25	1,1-Thiodi-2-propanol	8
Pentaethylene glycol	7	2,2'-Thiodiethanol	4
Pentaethylenhexamine	9	3,3'-Thiodipropionitrile	6, 19
Pentafluoroethanol	9	Thiophene	20
1,5-Pentanediol	3	Toluene	23
2,4-Pentanedione	12, 18	Triacetin	11, 19
1-Pentanol	17	Tributylphosphate	18
<i>t</i> -Pentanol	16	Tributylamine	28
Petrolatum (C ₁₄ -C ₁₆ alkanes)	31	1,2,4-Trichlorobenzene	24
Phenetole	20	1,1,1-Trichloroethane	22
2-Phenoxyethanol	12	1,1,2-Trichloroethane	19
1-Phenoxy-2-propanol	13, 17	Trichloroethylene	20
Phenyl acetate	23	1,1,2-Trichloro-2,2,2-trifluoroethane	27
Phenylacetoneitrile	12, 19	1,2,3-Trichloropropane	20
<i>N</i> -Phenylethanamine	10	Tricresyl phosphate	21
2-Picoline	16	Triethanolamine	2
Polypropyleneglycol PPG-1000	14, 23	Triethyl phosphate	14
Polypropyleneglycol PPG-400	14	Triethylamine	26
Propanediamine	11, 11	Triethylbenzene	25
1,2-Propanediol	4	Triethylene glycol	6
1,3-Propanediol	3	Triethylene glycol monobutyl ether	14
Propanesulfone	7, 19	Triethylene glycol monomethyl ether	13
1-Propanol	15	Triethylenetetramine	9
2-Propanol	15	Triisobutylene	29
Propionic acid	15	Trimethyl borate	16
Propionitrile	13, 17	Trimethyl nitrilotripropionate	12
Propyl acetate	19	Trimethyl phosphate	10
Propylene carbonate	9, 17	2,4,4'-Trimethyl-1-pentane	27
Propylene oxide	17	2,4,4'-Trimethyl-2-pentane	27
Pyridine	16	Trimethylboroxin	12, 17
2-Pyrrolidinone	10	2,2,4-Trimethylpentene	29
Styrene	22	Tripropylamine	26
Sulfolane	9, 17	Tripropylene glycol	12
1,1,2,2-Tetrabromoethane	11, 19	Vinyl acetate	20
1,1,2,2-Tetrachloroethane	19	Vinyl butyrate	22
Tetrachloroethylene	25	4-Vinylcyclohexene	26
Tetradecane	30	Naphtha	29
1-Tetradecene	29	<i>m</i> -Xylene	23
Tetraethyl orthosilicate	23	<i>o</i> -Xylene	23
Tetraethylene glycol	7	<i>p</i> -Xylene	24
Tetraethylenepentamine	9		

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exhibit partial miscibility at near-ambient temperatures. Godfrey assigned miscibility numbers to approximately 400 organic solvents (Table 15-3) by observing their miscibility in a series of 31 standard solvents (Table 15-4). He then showed that the general miscibility behavior of a given solvent pair can be predicted by comparing their miscibility numbers. Godfrey's rules, slightly modified, are summarized below:

1. If $\Delta \leq 12$, where Δ is the difference in miscibility numbers, the solvents are likely to be miscible in all proportions at 25°C.

2. If $13 \leq \Delta \leq 15$, the solvents may be only partially miscible with an upper critical solution temperature (UCST) between 25 and 50°C. This is a borderline case. If the binary mixture is miscible, then adding a relatively small amount of water likely will induce phase splitting.

3. If $\Delta = 16$, the solvents are likely to exhibit a UCST between 25 and 75°C.

4. If $\Delta \geq 17$, the solvents are likely to exhibit a UCST above 75°C.

About 15 percent of the solvents in Table 15-3 have dual miscibility numbers *A* and *B* because the appropriate difference in miscibility numbers depends upon which end of the hydrophobic-lipophilic scale is being considered. If one of the solvents has dual miscibility numbers *A* and *B* and the other has a single miscibility number *C*, then Δ should be calculated as follows:

5. If $C > B$, then the solvent having miscibility number *C* is somewhat more lipophilic than the solvent having numbers *A* and *B*. At

this end of the lipophilicity scale, the number *A* characterizes the solvent's miscibility behavior. Apply rules 1 through 3 above, using $\Delta = C - A$.

6. If $C < A$, then the solvent having miscibility number *C* is somewhat less lipophilic than the solvent with numbers *A* and *B*. At this end of the lipophilicity scale, the number *B* characterizes the solvent's miscibility behavior. Apply rules 1 through 3, using $\Delta = B - C$.

7. If $A \leq C \leq B$, then evaluate $\Delta = C - A$ and $\Delta = B - C$ and use the larger of the Δ values in applying rules 1 through 3. Such a mixture is likely to be miscible in all proportions at 25°C.

8. If both members of a solvent pair have dual miscibility numbers, then the pair is likely to be miscible in all proportions at 25°C.

If a compound of interest is not listed in Table 15-3 or 15-4, a compound of the same type or class may help to gauge its miscibility behavior. In cases where Godfrey's rules indicate that partial miscibility is likely, whether phase splitting actually occurs depends upon the composition of the mixture and the temperature. The composition may be close to but still outside the two-liquid-phase region on a temperature-composition diagram.

Godfrey's method is a useful guide for compounds that exhibit behavior similar to the 31 standard solvents used to define miscibility numbers. The method deals with the common situation in which a mixture exhibits a UCST; i.e. solubility tends to increase with

TABLE 15-4 Godfrey Standard Solvents

Miscibility Number	Solvent	
1	Glycerol ("glycerin")	Hydrophilic end of scale
2	1,2-Ethanediol ("ethylene glycol")	
3	1,4-Butanediol	
4	2,2'-Thiodiethanol	
5	Diethylene glycol	
6	Triethylene glycol	
7	Tetraethylene glycol	(decreasing hydrophilicity)
8	Methoxyacetic acid	(increasing lipophilicity)
9	Dimethylsulfoxide	
10	N-Formylmorpholine	
11	Furfuryl alcohol	
12	2-(2-Methoxyethoxy) ethanol ("diethylene glycol methyl ether")	
13	2-Methoxyethanol ("ethylene glycol methyl ether")	
14	2-Ethoxyethanol ("ethylene glycol ethyl ether")	
15	2-(2-Butoxyethoxy) ethanol ("diethylene glycol n-butyl ether")	
16	2-Butoxyethanol ("ethylene glycol n-butyl ether")	
17	1,4-Dioxane	
18	3-Pentanone	
19	1,1,2,2-Tetrachloroethane	
20	1,2-Dichloroethane	
21	Chlorobenzene	
22	1,2-Dibromobutane	
23	1-Bromobutane	
24	1-Bromo-3-methylbutane	
25	sec-Amylbenzene	
26	4-Vinylcyclohexene	
27	1-Methylcyclohexene	
28	Cyclohexane	
29	Heptane	
30	Tetradecane	
31	Petrolatum (C ₁₄ -C ₁₆ alkanes)	Lipophilic end of scale

Reprinted from Godfrey, *CHEMTECH*, **2**(6), pp. 359-363 (1972), with permission. Published 1972 by the American Chemical Society.

increasing temperature. Exceptions to Godfrey's rules include binary mixtures that form unusually strong hydrogen-bonding interactions. Normally, mixtures of this type are completely miscible, or they exhibit a lower critical solution temperature (LCST). Examples include ethylene glycol + triethylamine (Fig. 15-16) and glycerin + ethylbenzylamine (UCST = 280°C and LCST = 49°C) [Sorenson and Arlt, *Liquid-Liquid Equilibrium Data Collection*, vol. V, pt. 1 (DECHEMA, 1979)]. As mentioned earlier, it is not unusual for mixtures of water and amines or water and glycol ethers to exhibit LCST behavior. (See "Phase Diagrams" under "Thermodynamic Basis for Liquid-Liquid Extraction.") This is a reason why Godfrey's method does not include water.

Sometimes the mutual solubility of a solvent pair of interest can easily be decreased by adding a third component. For example, it is common practice to add water to a solvent system containing a water-miscible organic solvent (the polar phase) and a hydrophobic organic solvent (the nonpolar phase). A typical example is the solvent system (methanol + water) + dichloromethane. An anhydrous mixture of methanol and dichloromethane is completely miscible, but adding water causes phase splitting. Adjusting the amount of water added to the polar phase also may be used to alter the *K* values for the extraction, density difference, and interfacial tension. Table 15-5 lists some common examples of solvent systems of this type. These systems are common candidates for fractional extractions.

COMPUTER-AIDED MOLECULAR DESIGN

Many specialized computer programs have been written specifically to identify candidate solvents with properties that best match those needed for a particular application—by weighing various considerations of the kind outlined in “Desired Solvent Properties” in addition to the partition ratio. These Computer-Aided Molecular Design (CAMD) programs generally utilize a group contribution method such as UNIFAC, or a group contribution Hansen parameter model, as the means for estimating phase equilibrium, plus methods for estimating

physical properties and other relevant factors. The goal is to determine the optimal solvent structure that best meets the specified set of performance factors [Brignole, Botini, and Gani, *Fluid Phase Equil.*, **29**, pp. 125–132 (1986); and Joback and Stephanopoulos, *Proc. FOCAPD*, **11**, p. 631 (1989)]. Recent studies that include reviews of previous work are given by Papadopoulos and Linke [*AIChE J.*, **52**(3), pp. 1057–1070 (2006)]; Karunanithi, Achenie, and Gani [*Ind. Eng. Chem. Res.*, **44**(13), pp. 4785–4797 (2005)]; Cismondi and Brignole [*Ind. Eng. Chem. Res.*, **43**(3), pp. 784–790 (2004)]; and Giovanoglou et al. [*AIChE J.*, **49**(12), pp. 3095–3109 (2003)]. A variety of creative search strategies have been employed including use of stochastic algorithms to account for uncertainty [Kim and Diwekar, *Ind. Eng. Chem. Res.*, **41**(5), pp. 1285–1296 (2002)], the use of quantum chemistry methods for property estimation [Lehmann and Maranas, *Ind. Eng. Chem. Res.*, **43**(13), pp. 3419–3432 (2004)], and the application of a genetic theory of evolution (survival of the fittest) [Nieuwoudt, Paper No.

TABLE 15-5 Common Solvent Systems Involving a Water-Miscible Organic Solvent and Addition of Water to Control Properties

Polar component	Nonpolar component
Methanol	<i>n</i> -Hexane, <i>n</i> -heptane, other alkanes, dichloromethane
Acetonitrile	<i>n</i> -Hexane, <i>n</i> -heptane, other alkanes, dichloromethane
Ethylene glycol, diethylene glycol, triethylene glycol, tetraethylene glycol, and propylene glycol analogs	<i>n</i> -Hexane, <i>n</i> -heptane, other alkanes, dichloromethane, amyl acetate, toluene, xylene
Ethylene glycol mono methyl ether and other glycol ethers	<i>n</i> -Hexane, <i>n</i> -heptane, other alkanes, and dichloromethane

233a, AIChE National Meeting, Austin, Tex. (2004); and Van Dyk and Nieuwoudt, *Ind. Eng. Chem. Res.*, **39**(5), pp. 1423–1429 (2000)]. Similar programs have been written to facilitate identification of alternative solvents or solvent blends as replacements for a given solvent, by attempting to identify compounds that match the physical properties of the solvent the user wishes to replace. An example is the PARIS II program developed by the U.S. Environmental Protection Agency [Cabezas, Harten, and Green, *Chem. Eng. Magazine*, pp. 107–109 (March 2000)].

HIGH-THROUGHPUT EXPERIMENTAL METHODS

In addition to the methods described above, it may be useful to devise a rapid experimental method for screening solvents and extraction conditions. High-throughput methods are designed to measure a key property and automatically carry out tens or hundreds

of experiments in a short time. An example involves automation of liquid-liquid extraction using a 96-well sample plate and a robotic liquid-handling workstation in conjunction with automated liquid chromatography for analysis [Peng et al., *Anal. Chem.*, **72**(2), pp. 261–266 (2000)]. The authors developed this method to purify libraries of compounds for accelerated discovery of active compounds (such as new pharmaceuticals); however, the same approach may prove useful for screening solvents for a particular extraction application. Another paper describes a high throughput screening method for rapid optimization of aqueous two-phase extraction applications [Bensch et al., *Chem. Eng. Sci.*, **62**, pp. 2011–2021 (2007)]. For a review of high-throughput methods in general, see Murray, *Principles and Practice of High Throughput Screening* (Blackwell, 2005). The automated methods described in “Liquid-Liquid Equilibrium Experimental Methods” under “Thermodynamic Basis for Liquid-Liquid Extraction” also may be useful for screening solvents.

LIQUID DENSITY, VISCOSITY, AND INTERFACIAL TENSION

GENERAL REFERENCES: See Sec. 2, “Prediction and Correlation of Physical Properties,” and Rosen, *Surfactants and Interfacial Phenomena*, 3d ed. (Wiley, 2004); Hartland, *Surface and Interfacial Phenomena* (Dekker, 2004); and Poling, Prausnitz, and O’Connell, *The Properties of Gases and Liquids*, 5th ed. (McGraw-Hill, 2000).

The utility of liquid-liquid extraction as a separation tool depends upon both phase equilibria and transport properties. The most important physical properties that influence transport properties are liquid-liquid interfacial tension, liquid density, and viscosity. These properties influence solute diffusion and the formation and coalescence of drops, and so are critical factors affecting the performance of liquid-liquid contactors and phase separators.

DENSITY AND VISCOSITY

Many handbooks, including this one, contain an extensive compilation of liquid density data. These same sources often include liquid viscosity data, although fewer experimental data may be available for a particular compound. Available data compilations include those by Wypych, *Handbook of Solvents* (ChemTech, 2001); Wypych, *Solvents Database*, CD-ROM (ChemTec, 2001); Yaws, *Thermodynamic and Physical Property Data*, 2d ed. (Gulf, 1998); and Flick, *Industrial Solvents Handbook*, 5th ed. (Noyes, 1998). In addition, viscosity data for C₁–C₂₈ organic compounds have been compiled by Yaws in *Handbook of Viscosity*, vols. 1–3 (Elsevier, 1994). Density and viscosity data also are available from the Thermodynamics Research Center at the National Institute of Standards and Technology (Boulder, Colo.) and from the DIPPR physical property databank of AIChE.

Methods for estimating density and viscosity are reviewed in Sec. 2, “Prediction and Correlation of Physical Properties,” and in the book by Poling, Prausnitz, and O’Connell, *The Properties of Gases and Liquids*, 5th ed. (McGraw-Hill, 2000). However, it is best to measure density and viscosity in the laboratory whenever possible. The methods used to measure viscosity are described in numerous books including *Measurement of Transport Properties of Fluids*, vol. 3, Wakeham, Nagashima, and Sengers, eds. (Blackwell, 1991); and Leblanc, Secco, and Kostic, “Viscosity Measurement,” Chap. 30 in *Measurement, Instrumentation, and Sensors Handbook*, Webster, ed. (CRC Press, 1999). A new instrument introduced by the Anton Paar Company utilizes Stabinger’s methods for simultaneous measurement of viscosity and density [American Society for Testing and Materials, ASTM D7042-04 (2005)].

INTERFACIAL TENSION

Typical values of interfacial tension are listed in Tables 15-6 and 15-7. Refer to the references listed in these tables for the full data sets and for data on other mixtures. Table 15-6 shows typical values for organic + water binary mixtures. Table 15-7 shows the strong effect of the addition of a third component. Also, Treybal’s classic plot of interfacial tension versus mutual solubility is given in Fig. 15-21. This information can be helpful in assessing whether interfacial tension is likely to be low, moderate, or high for a new application. However, for design purposes, interfacial tension should be measured by using representative feed and solvent because even small amounts of surface-active impurities can significantly impact the result.

Methods used to measure interfacial tension are reviewed by Drelich, Fang, and White [“Measurement of Interfacial Tension in Fluid-Fluid Systems,” in *Encyclopedia of Surface and Colloid Science* (Dekker, 2003), pp. 3152–3156]. Also see Megias-Alguacil, Fischer, and Windhab, *Chem. Eng. Sci.*, **61**, pp. 1386–1394 (2006). One class of methods derives interfacial tension values from measurement of the shape, contact angle, or volume of a drop suspended in a second liquid. These methods include the pendant drop method (a drop of heavy liquid hangs from a vertically mounted capillary tube immersed

TABLE 15-6 Typical Interfacial Tensions for Different Classes of Organic + Water Binary Mixtures at 20 to 25°C

Class of organic compounds	Interfacial tension, dyn/cm
Alkanes (C ₅ –C ₁₂)	45–53
Halogenated alkanes (C ₁ –C ₄)	30–40
Halogenated aromatics (single ring)	35–40
Aromatics (single ring)	30–40
Mononitro aromatics (single ring)	25–28
Ethers (C ₄ –C ₆)	10–30
Esters (C ₄ –C ₆)	10–20
Ketones (C ₄ –C ₈)	5–15
Organic acids (C ₅ –C ₁₂)	3–15
Aniline	6–7
Alcohols (C ₄ –C ₈)	2–8

References:

1. Demond and Lindner, *Environ. Sci. Technol.*, **27**(12), pp. 2318–2331 (1993).
2. Fu, Li, and Wang, *Chem. Eng. Sci.*, **41**(10), pp. 2673–2679 (1986).
3. Backes et al., *Chem. Eng. Sci.*, **45**(1), pp. 275–286 (1990).

TABLE 15-7 Example Interfacial-Tension Data for Selected Ternary Mixtures

Component 1	Component 2 in phase 1, wt %	Component 3 in phase 1, wt %	Component 2 in phase 2, wt %	Component 3 in phase 2, wt %	Interfacial tension, dyn/cm
Water	Benzene	Ethanol	Benzene	Ethanol	At 25°C
	0.2	10.8	98.6	1.2	17.2
	3.6	43.7	91.3	7.9	1.99
	21.2	52.0	79.3	18.0	0.04
Water	Benzene	Acetone	Benzene	Acetone	At 30°C
	0.1	1.9	98.1	1.8	25.9
	0.2	10.3	91.2	8.6	16.1
	0.6	23.6	81.9	17.8	9.5
	2.7	45.5	68.2	30.9	3.8
Water	Benzene	Acetic acid	Benzene	Acetic acid	At 25°C
	0.3	17.2	98.6	1.3	17.3
	1.1	45.1	92.2	7.5	7.0
	7.9	64.7	77.0	21.9	2.0
Water	Hexane	Ethanol	Hexane	Ethanol	At 20°C
	0.1	32.5	99.5	0.5	9.82
	8.2	73.0	93.9	6.0	1.5
	30.0	64.0	86.2	13.2	0.096
Hexane	Methyl ethyl ketone	Water	Methyl ethyl ketone	Water	At 25°C
	0.4	99.6	0.59	0.01	40.1
	11.7	88.3	35.56	0.09	9.0
	24.5	75.5	89.88	9.97	1.1

References:

1. Sada, Kito, and Yamashita, *J. Chem. Eng. Data*, **20**(4), pp. 376–377 (1975).
2. Pliskin and Treybal, *J. Chem. Eng. Data*, **11**(1), pp. 49–52 (1966).
3. Paul and de Chazal, *J. Chem. Eng. Data*, **12**(1), pp. 105–107 (1967).
4. Ross and Patterson, *J. Chem. Eng. Data*, **24**(2), pp. 111–115 (1979).
5. Backes et al., *Chem. Eng. Sci.*, **45**(1), pp. 275–286 (1990).

in the light liquid), the sessile drop method (a drop of heavy liquid lies on a plate immersed in the light liquid), and the spinning drop method (a drop of one liquid is suspended in a rotating tube filled with the second liquid). The sessile drop method is particularly useful for following the change in interfacial tension when surfactants or macromolecules accumulate at the surface of the drop. The spinning drop method is well suited to measuring low interfacial tensions. Another class of methods derives interfacial tension values from measurement of the force required to detach a ring of wire (Du Noüy's method), or a plate of glass or platinum foil (the Wilhelmy method), from the liquid-liquid interface. The ring or plate must be extremely clean. For the commonly used ring-pull method, the wire is usually flamed before the experiment and must be kept very horizontal and located exactly at the interface of the two liquids.

For an initial assessment, an approximate value for the interfacial tension may be obtained, at least in principle, from knowledge of the maximum size of drops that can persist in a dispersion at equilibrium and without agitation. For example, if it is possible to determine drop size from a photograph of the dispersion of interest at quiescent conditions, then an estimate of interfacial tension may be obtained from the balance between interfacial tension and buoyancy forces

$$\sigma \approx d_{\max}^2 \Delta \rho g \quad (15-32)$$

where $d_{\max}$ is the maximum drop diameter. Antonov's rule also may be used to obtain an approximate value. This rule states that interfacial tension between two liquids is approximately equal to the difference in their liquid-air surface tensions measured at the same conditions. For an organic + water system,

$$\sigma \approx |\sigma_{w(o)} - \sigma_{o(w)}| \quad (15-33)$$

where $\sigma_{w(o)}$ represents the surface tension of the water saturated with the organic and $\sigma_{o(w)}$ represents the surface tension of organic saturated with water.

Measurements of interfacial tension are not always feasible, and calculation methods are sometimes used. The results are least reliable for interfacial tensions below about 10 dyn/cm (10^{-2} N/m). A commonly used empirical correlation of interfacial tension and mutual solubilities is given by Donahue and Bartell [*J. Phys. Chem.*, **56**, pp. 480–484 (1952)]:

$$\sigma = -3.33 - 7.21 \ln (x_1'' + x_2') \quad (15-34)$$

where σ = interfacial tension, dyn/cm (10^{-3} N/m)
 x_1'' = mole fraction solubility of organic in aqueous phase
 x_2' = mole fraction solubility of water in organic phase

Treybal [*Liquid Extraction*, 2d ed. (McGraw-Hill, 1963)] modified Eq. (15-36) to expand its application to ternary systems:

$$\sigma = -5.0 - 7.355 \ln [x_1'' + x_2' + 0.5(x_3' + x_3'')] \quad (15-35)$$

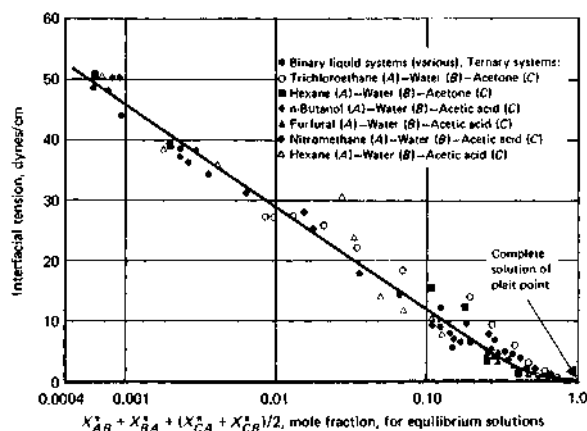


FIG. 15-21 Correlation of interfacial tension with mutual solubility for binary and ternary two-liquid-phase mixtures. [Reprinted from Treybal, *Liquid Extraction*, 2d ed. (McGraw-Hill, 1963). Copyright 1963 McGraw-Hill, Inc.]

where σ = interfacial tension, dyn/cm (10^{-3} N/m)
 x_3'' = mole fraction solute in aqueous phase
 x_3' = mole fraction solute in organic phase

The results are plotted in Fig. 15-21. More recently, Fu, Li, and Wang [*Chem. Eng. Sci.*, **41**(10), pp. 2673–2679 (1986)] derived a relationship for ternary mixtures:

$$\sigma = \frac{0.914RT\chi}{(A_o \exp \chi)(x_1''q_1 + x_2'q_2 + x_{3r}q_3)} \quad (15-36)$$

$$\chi = -\ln(x_1'' + x_2' + x_{3r}) \quad (15-37)$$

where σ = interfacial tension, dyn/cm (10^{-3} N/m)
 R = ideal gas law constant

T = absolute temperature
 x_1'' = solubility of extract phase in raffinate phase (mole fraction)
 x_2' = solubility of raffinate phase in extract phase (mole fraction)
 x_{3r} = mole fraction of solute 3 in bulk phase richest in solute 3
 A_o = van der Waals area of standard segment (2.5×10^9 cm²/mol)
 q_i = van der Waals surface area ratio, usually calculated from UNIQUAC

For additional discussion, see Suarez, Torres-Marchal, and Rasmussen, *Chem. Eng. Sci.*, **44**(3), pp. 782–786 (1989); Wu and Zhu, *Chem. Eng. Sci.*, **54**, pp. 433–440 (1990); and Li and Fu, *Fluid Phase Equil.*, **81**, pp. 129–152 (1992).

LIQUID-LIQUID DISPERSION FUNDAMENTALS

GENERAL REFERENCES: Leng and Calabrese, Chap. 12 in *Handbook of Industrial Mixing*, Paul, Atiemo-Obeng, and Kresta, eds. (Wiley, 2004); Becher, *Emulsions: Theory and Practice*, 3d ed. (American Chemical Society, 2001); Binks, *Modern Aspects of Emulsion Science* (Royal Chemical Society, 1998); Adamson and Gast, *Physical Chemistry of Surfaces*, 6th ed. (Wiley, 1997); *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994); *Encyclopedia of Emulsion Technology*, vols. 1–4, Becher, ed. (Decker, 1983–); and Laddha and Degaleesan, Chap. 4 in *Handbook of Solvent Extraction*, Lo, Hanson, and Baird, eds. (Wiley, 1983; Krieger, 1991).

HOLDUP, SAUTER MEAN DIAMETER, AND INTERFACIAL AREA

Most liquid-liquid extractors are designed to generate drops of one liquid suspended in the other rather than liquid films. The volume fraction of the dispersed phase (or holdup) within the extractor is defined as

$$\phi_d = \frac{\text{volume of dispersed phase}}{\text{total contacting volume}} \quad (15-38)$$

where the total contacting volume is the volume within the extractor minus the volume of any internals such as impellers, packing, or trays. A distribution of drop sizes will be present. The Sauter mean drop diameter d_{32} represents a volume to surface-area average diameter

$$d_{32} = \frac{\sum_{i=1}^n N_i d_i^3}{\sum_{i=1}^n N_i d_i^2} \quad (15-39)$$

where N_i is the number of drops with diameter d_i . The Sauter mean diameter often is used in the analysis and modeling of extractor performance because it is directly related to holdup and interfacial area (assuming spherical drops). It is calculated from the total dispersed volume divided by total interfacial area, and often it is expressed in the form

$$d_{32} = \frac{6\epsilon\phi_d}{a} \quad (15-40)$$

where a is interfacial area per unit volume and ϵ is the void fraction within the extractor, i.e., the fraction of internal volume not occupied by any packing, trays, and so on. In the remainder of Sec. 15, the Sauter mean diameter is denoted simply by d_p .

Much less is known about the actual distribution of drop sizes existing within liquid-liquid extractors, particularly at high holdup and as a function of agitation intensity (if agitation is used) and location within

the extractor. For a review, see Kumar and Hartland, Chap. 17 in *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994). Experimental methods used to measure drop size distribution include the use of a high-speed video camera [Ribeiro, et al., *Chem. Eng. J.*, **97**, pp. 173–182 (2004)], real-time optical measurements [Ritter and Kraume, *Chem. Eng. Technol.*, **23**(7), pp. 579–581 (2000)], and phase-Doppler anemometry [Lohner, Bauckhage, and Schombacher, *Chem. Eng. Technol.*, **21**(4), pp. 337–341 (1998); and Willie, Langer, and Werner, *Chem. Eng. Technol.*, **24**(5), pp. 475–479 (2001)].

FACTORS AFFECTING WHICH PHASE IS DISPERSED

Consider mixing a batch of two liquid phases in a stirred tank. The minority phase generally will be the dispersed phase whenever the ratio of minority to majority volume fractions, or phase ratio, is less than about 0.5 (equivalent to a dispersed-phase volume fraction or holdup less than 0.33). For phase ratios between 0.5 and about 2, a region called the ambivalent range, the phase that becomes dispersed is determined in large part by the protocol used to create the dispersion. For example, pouring liquid A into a stirred tank already containing liquid B will tend to create a dispersion of A suspended in B, as long as agitation is maintained. When more of the dispersed-phase material is added to the system, the population density of dispersed drops will increase and eventually reach a point where the drops are so close together that they rapidly coalesce and the phases become inverted, i.e., the formerly dispersed phase becomes the continuous phase. In the ambivalent range, a sudden increase in the agitation intensity also can trigger phase inversion by increasing the number of drop-to-drop collisions. Once phase inversion occurs, it is not easily reversed because the new condition corresponds to a more stable configuration.

This phase behavior may be roughly correlated in terms of light and heavy phase properties including relative density and viscosity as follows:

$$\chi = \frac{\phi_L}{\phi_H} \left(\frac{\rho_L \mu_H}{\rho_H \mu_L} \right)^{0.3} = \frac{\phi_L}{1 - \phi_L} \left(\frac{\rho_L \mu_H}{\rho_H \mu_L} \right)^{0.3} \quad (15-41)$$

where $\chi < 0.3$ light phase always dispersed
 $\chi = 0.3 - 0.5$ light phase probably dispersed
 $\chi = 0.5 - 2.0$ either phase can be dispersed, and phase inversion may occur
 $\chi = 2.0 - 3.3$ heavy phase probably dispersed
 $\chi > 3.3$ heavy phase always dispersed

The symbol ϕ denotes the volume fraction of light (L) and heavy (H) phases existing within the vessel. Equation (15-41) is taken from the expression recommended by Hooper [Sec. 1.11 in *Handbook of*

Separation Techniques for Chemical Engineers, 3d ed., Schweitzer, ed. (McGraw-Hill, 1997)] and Jacobs and Penney [Chap. 3 in *Handbook of Separation Process Technology*, Rousseau, ed. (Wiley, 1987)] for design of continuous decanters. It is based on the dispersed-phase data of Selker and Slicher [Can. J. Chem. Eng., **43**, pp. 298–301 (1965)].

Equation (15-41) should apply to continuously fed extraction columns and other continuous extractors as well as batch vessels. The equation is expressed here in terms of volume fractions ϕ_L/ϕ_H existing within the vessel, not volumetric flow rates of each phase entering the vessel Q_L/Q_H . The ratio of volume fractions within a continuously fed vessel can be very different from Q_L/Q_H —primarily because buoyancy allows the dispersed-phase drops to travel rapidly through the continuous phase relative to the dispersed-phase superficial velocity. For example, a continuously fed extraction column can be designed to operate with either phase being the dispersed phase, with the main liquid-liquid interface controlled at the top of the column (for a light-phase dispersed system) or at the bottom (for a heavy-phase dispersed system). As the dispersed-to-continuous phase ratio within the column is increased, through either changes in operating variables or changes in the design of the internals, a point may be reached where the population density or holdup of dispersed drops is too large and phase inversion occurs. In the absence of stabilizing surfactants, the point of phase inversion should correspond roughly to the same general phase-ratio rules given in Eq. (15-41), with the exact conditions at which phase inversion occurs depending upon agitation intensity (if used) and the geometry of any internals (baffles, packing, trays, and so on). Certain extractors such as sieve-tray columns often are designed to disperse the majority flowing phase. In extreme cases, the ratio Q_d/Q_c (where d and c represent dispersed and continuous phases) may be as high as 50, and the continuous phase may be nearly stagnant with a superficial velocity as low as 0.02 cm/s; yet the phase ratio within the extractor can be controlled within the guidelines needed to avoid phase inversion [approximated by Eq. (15-41)].

The stability of a dispersion also can be affected by the presence of fine solids or gas bubbles as well as surfactants. For additional discussion of factors affecting which phase is dispersed, see Norato, Tsouris, and Tavarides, *Can. J. Chem. Eng.*, **76**, pp. 486–494 (1998); and Pacek et al., *AIChE J.*, **40**(12), pp. 1940–1949 (1994). For a given application, the precise conditions that lead to phase inversion must be determined by experiment. For organic + water dispersions, experimental determination may be facilitated by measuring the conductivity of the mixture, since conductivity normally will be significantly higher when water is in the continuous phase [Gilchrist, et al., *Chem. Eng. Sci.*, **44**(10), pp. 2381–2384 (1989)]. Another method involves monitoring the dynamics of phase inversion by using a stereo microscope and video camera [Pacek et al., *AIChE J.*, **40**(12), pp. 1940–1949 (1994)].

SIZE OF DISPERSED DROPS

In nonagitated (static) extractors, drops are formed by flow through small holes in sieve plates or inlet distributor pipes. The maximum size of drops issuing from the holes is determined not by the hole size but primarily by the balance between buoyancy and interfacial tension forces acting on the stream or jet emerging from the hole. Neglecting any viscosity effects (i.e., assuming low dispersed-phase viscosity), the maximum drop size is proportional to the square root of interfacial tension σ divided by density difference $\Delta\rho$:

$$d_{\max} = \text{const} \sqrt{\frac{\sigma}{\Delta\rho g}} \quad \text{for static extractors} \quad (15-42)$$

The proportionality constant typically is close to unity [Seibert and Fair, *Ind. Eng. Chem. Res.*, **27**(3), pp. 470–481 (1988)]. Note that Eq. (15-42) indicates the maximum stable drop diameter and not the Sauter mean diameter (although the two are proportionally related and may be close in value). Smaller drops may be formed at the distributor due to jetting of the inlet liquid through the distributor holes or by mechanical pulsation of the liquid inside the distributor [Koch and Vogelpohl, *Chem. Eng. Technol.*, **24**(12), pp. 1245–1248 (2001)]. In static extractors, hydrodynamic stresses within the main body of the

extractor away from the distributor are small and normally not sufficient to cause significant drop breakage as drops flow through the extractor, although small drops may collide and coalesce into larger drops. Some authors report a small amount of drop breakage in packed columns due to collisions with packing materials [Mao, Godfrey, and Slater, *Chem. Eng. Technol.*, **18**, pp. 33–40 (1995)]. Additional discussion is given in “Static Extraction Columns” under “Liquid-Liquid Extraction Equipment.”

In agitated extractors, drop size is determined by the equilibrium established between drop breakage and coalescence rates occurring within the extractor. Breakage is due to turbulent stresses caused by the agitator, so it is mainly confined to the vicinity of the agitator. Drop coalescence, however, can happen anywhere in the vessel where drops can come into close proximity with one another. Dispersed drops will begin to break into smaller droplets when turbulent stresses exceed the stabilizing forces of interfacial tension and liquid viscosity. Kolmogorov [*Dokl. Akad. Nauk*, **66**, pp. 825–828 (1949)] and Hinze [*AIChE J.*, **1**(3), pp. 289–295 (1955)] developed expressions for the maximum size of drops in an agitated liquid-liquid dispersion. Their results can be expressed as follows:

$$d_{\max} = (\text{const}) \sigma^{3/5} \rho_c^{-1/5} \left(\frac{P}{V} \right)^{-2/5} \quad \text{for agitated extractors} \quad (15-43)$$

where P/V is the rate of mechanical energy dissipation (or power P) input to the dispersion per unit volume V . Equation (15-43) assumes dispersed-phase holdup is low. It also assumes viscous forces that resist breakage can be neglected, a valid assumption for water and typical low- to moderate-viscosity organic solvents. Wang and Calabrese discuss how to determine when viscous resistance to breakage becomes important and show that this depends upon interfacial tension as well as dispersed-phase viscosity [Wang and Calabrese, *AIChE J.*, **32**(4), pp. 667–676 (1986)]. Equation (15-43) can be restated as

$$\frac{d_{\max}}{D_i} \propto We^{-3/5} \quad (15-44)$$

where We is a dimensionless Weber number (disruptive shear stress/cohesive interfacial tension) and D_i is a characteristic diameter. For applications involving the use of rotating impellers, D_i is the impeller diameter and the appropriate Weber number is $We = \rho \omega^2 D_i^3 / \sigma$ where ω is the impeller speed (in rotations per unit time). For static mixers, $D_i = D_{sm}$ and $We = \rho V_{sm}^2 D_{sm} / \sigma$, where D_{sm} is the static mixer pipe diameter and V_{sm} is the superficial liquid velocity (entrance velocity). A variety of drop size models derived for various mixers and operating conditions have been tabulated by Leng and Calabrese [Chap. 12 in *Handbook of Industrial Mixing*, Paul, Atiemo-Obeng, and Kresta, eds. (Wiley, 2004), pp. 669–675]. Also see Naseef, Soultan, and Stamatoudis, *Chem. Eng. Technol.*, **29**(5), pp. 583–587 (2006).

Equation (15-44) represents a limiting operating regime where the rate of drop breakage dominates performance and the coalescence rate can be neglected. Drop coalescence requires that two drops collide, and the coalescence rate increases with increasing holdup since there is greater opportunity for drop-drop collisions. For agitated systems with fast coalescence at high holdup, i.e., when drop coalescence dominates, drop size appears best correlated by an expression of the form $d_p/D \propto We^{-n}$, where n varies between 0.35 and 0.45 [Pacek, Man, and Nienow, *Chem. Eng. Sci.*, **53**(11), pp. 2005–2011 (1998); and Kraume, Gabler, and Schulze, *Chem. Eng. Technol.*, **27**(3), pp. 330–334 (2004)]. This is similar to the theoretical expression derived by Shinnar [*J. Fluid Mech.*, **10**, p. 259 (1961)].

When two drops first come into contact in the process of coalescing, a film of continuous phase becomes trapped between them. The film is compressed at the point of encounter until it drains away and the two drops can merge. Decreasing the viscosity of the continuous phase, by heating or by addition of a low-viscosity diluent, may promote drop coalescence by increasing the rate of film drainage. Surface-active impurities or surfactants, when present, also can affect the coalescence rate, by accumulating at the surface of the drop. Surfactants tend to stabilize the film and reduce coalescence rates. Fine

solid particles that are wetted by the continuous phase tend to slow film drainage, also reducing the rate of drop coalescence.

A number of semiempirical drop size data correlations have been developed for different types of extractors (static and agitated) including a term for holdup. See Kumar and Hartland, *Ind. Eng. Chem. Res.*, **35**(8), pp. 2682–2695 (1996); and Kumar and Hartland, Chap. 17 in *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994). These equations predict a characteristic drop size. They do not provide information about the drop size distribution or the minimum drop size. For discussion of minimum drop size, see Zhou and Kresta, *Chem. Eng. Sci.*, **53**(11), pp. 2063–2079 (1998).

STABILITY OF LIQUID-LIQUID DISPERSIONS

In designing a liquid-liquid extraction process, normally the goal is to generate an unstable dispersion that provides reasonably high interfacial area for good mass transfer during extraction and yet is easily broken to allow rapid liquid-liquid phase separation after extraction. Given enough time, most dispersions will break on standing. Often this process occurs in two distinct periods. The first is a relatively short initial period or primary break during which an interface forms between two liquid layers, one or both of which remain cloudy or turbid. This is followed by a longer period or secondary break during which the liquid layers become clarified. During the primary break, the larger drops migrate to the interface where they accumulate and begin to coalesce. If the coalescence rate is relatively slow compared to the rate at which drops rise or fall to the interface, then a layer of coalescing drops or *dispersion band* will form at the interface. The initial interface can form within a few minutes or less for drop sizes on the order of 100 to 1000 μm (0.1 to 1 mm), as in a water + toluene system, for example. When the drop size distribution in the feed dispersion is wide, smaller droplets remain suspended in one or both phases. Longer residence times are then required to break this secondary dispersion. In extreme cases, the secondary dispersion can take days or even longer to break.

When a dispersion requires a long time to break, the presence of surfactantlike impurities may be a contributing factor. Surfactants are molecules with a hydrophobic end (such as a long hydrocarbon chain) and a hydrophilic end (such as an ionic group or oxygen-containing short chain). Surfactants stabilize droplets by forming an adsorbed film at the interface and by introducing electrical repulsions between drops [Tcholakova, Denkov, and Danner, *Langmuir*, **20**(18), pp. 7444–7458 (2004)]. Both effects can interfere with drop coalescence. Surfactants also decrease the interfacial tension of the system. As more surfactant is introduced into a solution, the concentration of free surfactant molecules in the bulk liquid increases and reaches a plateau called the critical micelle concentration. At this point, any excess molecules begin forming aggregates with other surfactant molecules at the interface of the two liquids to minimize interaction with the continuous phase. The dispersed phase is then trapped inside the micelles. As more surfactant is added to the mixture, more micelles can form and in most cases the droplets become smaller to maximize interfacial area. In theory, the maximum volume fraction of the dispersed phase should be limited to 0.74 due to the close packing density of spheres; but in practice much higher values are possible when the micelles change to other structures of different geometries such as a mix of small drops among larger ones and nonspherical shapes.

Emulsions are broken by changing conditions to promote drop coalescence, either by disrupting the film formed at the interface between adjacent drops or by interfering with the electrical forces that stabilize the drops. Water droplets are usually positively charged while oil droplets are negatively charged. Physical techniques used to break emulsions include heating (including application of microwave radiation), freezing and thawing, adsorption of surface-active compounds, filtration of fine particles that stabilize films between drops, and application of an electric field. Heating can be particularly effective for nonionic surfactants, since heating disrupts hydrogen bonding interactions that contribute to micelle stability. Chemical techniques include adding a salt to alter the charges around drops, changing the

pH of the system, and adding a deemulsifier compound (or even another type of surfactant) to interact with and alter the surfactant layer. Ionic surfactants are particularly sensitive to change in pH. Additives include bases and acids, aluminum or ferric salts, chelating agents, charged polymers (polyamines or polyacrylates), polyalcohols, silicone oils, various fatty acid esters and fatty alcohols, as well as adsorbents such as clay and lime. For further discussion, see Rajaković and Skala, *Sep. Purif. Technol.*, **49**(2), pp. 192–196 (2006); and Alther, *Chem. Eng. Magazine*, **104**(3) pp. 82–88 (1998). Chemical additives need to be used in sufficiently small concentrations so as not to interfere with other operations in the overall process or product quality. General information is available in Schramm, *Emulsions, Foams, and Suspensions* (Wiley-VCH, 2005); Becher, *Emulsions: Theory and Practice*, 3d ed. (American Chemical Society, 2001); and Binks, *Modern Aspects of Emulsion Science* (Royal Society of Chemical 1998).

EFFECT OF SOLID-SURFACE WETTABILITY

The stability of a dispersion also may depend upon the surface properties of the container or equipment used to process the dispersion, since the walls of the vessel, or more importantly, the surfaces of any internal structures, may promote drop coalescence. In a liquid-liquid extractor or a liquid-liquid phase separator, the wetting of a solid surface by a liquid is a function of the interfacial tensions of both the liquid-solid and the liquid-liquid interfaces. For dispersed drops with low liquid-solid interfacial tension, the drops tend to spread out into films when in contact with the solid surface. In general, an aqueous liquid will tend to wet a metal or ceramic surface better than an organic liquid will, and an organic liquid will tend to wet a polymer surface better than an aqueous liquid will. However, there are many exceptions. Strigle [*Packed Tower Design and Applications*, 2d ed., Chap. 11 (Gulf, 1994)] indicates that for packed extractors, metal packings may be wetted by either an aqueous or an organic solvent depending upon the initial exposure of the metal surface (whether the unit is started up filled with the aqueous phase or the organic phase). In general, however, metals tend to be preferentially wetted by an aqueous phase. Also, it is not uncommon for materials of construction to acquire different surface properties after aging in service, since the solid surface can change due to adsorption of impurities, corrosion, or fouling. This aging effect often is observed for polymer materials. Small-scale lab tests are recommended to determine these wetting effects. For detailed discussion of wettability and its characterization, see *Contact Angle, Wettability, and Adhesion*, vols. 1–3, Mittal, ed. (VSP, 1993–); or *Wettability*, Berg, ed. (Dekker, 1993).

In liquid-liquid extraction equipment, the internals generally should be preferentially wetted by the continuous phase—in order to maintain dispersed-phase drops with a high population density (high holdup). If the dispersed phase preferentially wets the internals, then drops may coalesce on contact with these surfaces, and this can result in loss of interfacial area for mass transfer and even in the formation of rivulets that flow along the internals. In an agitated extractor, this tendency may be mitigated somewhat, if needed, by increasing the agitation intensity.

MARANGONI INSTABILITIES

Numerous studies have shown that mass transfer of solute from one phase to the other can alter the behavior of a liquid-liquid dispersion—because of interfacial tension gradients that form along the surface of a dispersed drop. For example, see Sawistowski and Goltz, *Trans. Inst. Chem. Engrs.*, **41**, p. 174 (1963); Bakker, van Buytenen, and Beek, *Chem. Eng. Sci.*, **21**(11), pp. 1039–1046 (1966); Ruckenstein and Berbente, *Chem. Eng. Sci.*, **25**(3), pp. 475–482 (1970); Lode and Heideger, *Chem. Eng. Sci.*, **25**(6), pp. 1081–1090 (1970); and Takeuchi and Numata, *Int. Chem. Eng.*, **17**(3), p. 468 (1977). These interfacial tension gradients can induce interfacial turbulence and circulation within drops. These effects, known as Marangoni instabilities, have been shown to enhance mass-transfer rates in certain cases.

The direction of mass transfer also can have a significant effect upon drop-drop coalescence and the resulting drop size. Seibert and

Fair [*Ind. Eng. Chem. Res.*, **27**(3), pp. 470–481 (1988)] showed that mass transfer out of the drop will promote coalescence. Larger drop sizes were observed when transferring solute into the continuous phase (interfacial tension was increasing as the drop traveled through the extractor). Kumar and Hartland [*Ind. Eng. Chem. Res.*, **35**(8), pp. 2682–2695 (1996)] suggest that transfer of solute from the dispersed to the continuous phase ($d \rightarrow c$) tends to produce larger drops because the concentration of transferring solute in the draining film between two approaching drops is higher than that in the surrounding continuous liquid. This accelerates drainage, thus promoting drop coalescence. For mass transfer in the opposite direction

($c \rightarrow d$), smaller drops tend to form because the solute concentration in the draining film between drops is relatively low. The magnitude of these effects depends upon system properties, the surface activity of the transferring solute, and the degree of mass transfer. Unless the solute is unusually surface-active, the effect will be small. For more information, see Gourdon, Casamatta, and Muratet, Chap. 7 in *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994); Perez de Ortiz, Chap. 3, "Marangoni Phenomena," in *Science and Practice of Liquid-Liquid Extraction*, vol. 1, Thornton, ed. (Oxford, 1992); and Grahn, *Chem. Eng. Sci.*, **61**, pp. 3586–3592 (2006).

PROCESS FUNDAMENTALS AND BASIC CALCULATION METHODS

GENERAL REFERENCES: See Sec. 5, "Mass Transfer," as well as Wankat, *Separation Process Engineering*, 2d ed. (Prentice-Hall, 2006); Seader and Henley, *Separation Process Principles* (Wiley, 1998); Godfrey and Slater, *Liquid-Liquid Extraction Equipment* (Wiley, 1994); Thornton, ed., *Science and Practice of Liquid-Liquid Extraction*, vol. 1 (Oxford, 1992); Wankat, *Equilibrium Staged Separations* (Prentice-Hall, 1988); Kirwin, Chap. 2 in *Handbook of Separation Process Technology*, Rousseau, ed. (Wiley, 1987); Skelland and Tedder, Chap. 7 in *Handbook of Separation Process Technology*, Rousseau, ed. (Wiley, 1987); Lo, Baird, and Hanson, eds., *Handbook of Solvent Extraction* (Wiley, 1983; Krieger, 1991); King, *Separation Processes*, 2d ed. (McGraw-Hill, 1980); Brian, *Staged Cascades in Chemical Processing* (Prentice-Hall, 1972); Geankoplis, *Mass Transport Phenomena* (Holt, Rinehart and Winston, 1972); and Treybal, *Liquid Extraction*, 2d ed. (McGraw-Hill, 1963).

The fundamental mechanisms for solute mass transfer in liquid-liquid extraction involve molecular diffusion driven by a deviation from equilibrium. When a liquid feed is contacted with a liquid solvent, solute transfers from the interior of the feed phase across a liquid-liquid interface into the interior of the solvent phase. Transfer of solute will continue until the solute's chemical potential is the same in both phases and equilibrium is achieved.

The calculation methods used to quantify extraction processes generally involve either the calculation of theoretical stages, with application of an operating efficiency to reflect mass-transfer resistance, or calculations based on consideration of mass-transfer rates using expressions related in some way to molecular diffusion. Theoretical-stage calculations commonly are used to characterize separation difficulty regardless of the type of extractor to be used. They are also used for extractor design purposes, although for this purpose they generally should be reserved for single-stage contactors or mixer-settler cascades involving discrete stages, or for other equipment where discrete contacting zones exist, such as in a sieve-tray column. The appropriate stage efficiency reflects how closely an actual contacting stage approaches equilibrium, and is a function of operating variables that affect drop size, population density, and contact time.

The development and application of rate-based models for analysis and design of extraction processes are becoming more common. For example, Jain, Sen, and Chopra [*ISEC '02 Proceedings*, **2**, pp. 1265–1270 (2002)] recently described a rate-based model for a lube oil extraction process. Rate-based models most often are applied to differential-type contactors that lack discrete contacting stages, to staged contactors with low stage efficiencies, or to processes with extraction factors greater than about 3, indicating a mass-transfer-limited operating regime. Differential-type contactors operating at extraction factors less than 3 also can be adequately modeled with theoretical stages since these contactors operate reasonably close to equilibrium. With either approach, appropriate values for model parameters typically are determined by fitting data generated by using laboratory or pilot-plant experiments, or by analysis of the performance of large-scale commercial units. In certain cases, parameter values have been correlated as a function of physical properties and operating conditions for specific types of equipment using model systems. The reliability of the resulting correlations is generally limited to applications very similar to those used to develop the correlations. Also, most calculation methods have been developed for

continuous steady-state operation. The dynamic modeling of extraction processes is discussed elsewhere [Mohanty, *Rev. Chem. Eng.*, **16**(3), p. 199 (2000); Weinstein, Semiat, and Lewin, *Chem. Eng. Sci.*, **53**(2), pp. 325–339 (1998); and Steiner and Hartland, Chap. 7 in *Handbook of Solvent Extraction*, Lo, Baird, and Hanson, eds. (Wiley, 1983; Krieger, 1991)].

The calculation methods used for designing extraction operations are analogous in many respects to methods used to design absorbers and strippers in vapor-liquid and gas-liquid contacting such as those described by Ortiz-Del Castillo, et al. [*Ind. Eng. Chem. Res.*, **39**(3), pp. 731–739 (2000)] and by Kohl ["Absorption and Stripping," Chap. 6 in *Handbook of Separation Process Technology* (Wiley-Interscience, 1987)]. Unlike in stripping and absorption, however, liquid-liquid extraction always deals with highly nonideal systems; otherwise, only one liquid phase would exist. This nonideality contributes to difficulties in modeling and predicting phase equilibrium, liquid-liquid phase behavior (hydraulics), and thus mass transfer. Also, the mass-transfer efficiency of an extractor generally is much less than that observed in distillation, stripping, or absorption equipment. For example, an overall sieve tray efficiency of 70 percent is common in distillation, but it is rare when a sieve tray extractor achieves an overall efficiency greater than 30 percent. The difference arises in part because generation of interfacial area, normally by dispersing drops of one phase in the other, generally is more difficult in liquid-liquid contactors. Unlike in distillation, formation of liquid films often is purposely avoided; generation of dispersed droplets provides greater interfacial area for mass transfer per unit volume of extractor. (Film formation may be important in extraction applications involving centrifugal contactors or baffle tray extractors, but this is not generally the case.) In certain cases, mass-transfer rates also may be slower compared to those of gas-liquid contactors because the second phase is a liquid instead of a gas, and transport properties in that phase are less favorable. Although mass-transfer efficiency generally is lower, the specific throughput of liquid-liquid extraction equipment (in kilograms of feed processed per hour per unit volume) can be higher than is typical of vapor-liquid contactors, simply because liquids are much denser than vapors.

THEORETICAL (EQUILIBRIUM) STAGE CALCULATIONS

Calculating the number of theoretical stages is a convenient method used by process designers to evaluate separation difficulty and assess the compromise between the required equipment size (column height or the number of actual stages) and the ratio of solvent rate to feed rate required to achieve the desired separation. In any mass-transfer process, there can be an infinite number of combinations of flow rates, number of stages, and degrees of solute transfer. The optimum is governed by economic considerations. The cost of using a high solvent rate with relatively few stages should be carefully compared with the cost of using taller extraction equipment (or more equipment) capable of achieving more theoretical stages at a reduced solvent rate and operating cost. While the operating cost of an extractor is generally quite low, the operating cost for a solvent recovery distillation tower can be quite high. Another common objective for calculating the number of countercurrent theoretical stages is to evaluate

the performance of liquid-liquid extraction test equipment in a pilot plant or to evaluate production equipment in an industrial plant. As mentioned earlier, most liquid-liquid extraction equipment in common use can be designed to achieve the equivalent of 1 to 8 theoretical countercurrent stages, with some designed to achieve 10 to 12 stages.

McCabe-Thiele Type of Graphical Method Graphical methods may be used to determine theoretical stages for a ternary system (solute plus feed solvent and extraction solvent) or for a pseudo-ternary system with the focus placed on a key solute of interest. Although developed long ago, graphical methods are still valuable today because they help visualize the problem, clearly illustrating pinch points and other design issues not readily apparent by using other techniques. Even with computer simulations, often it is useful to plot the results for a key solute as an aid to analyzing the design. This section briefly reviews the commonly used McCabe-Thiele type of graphical method. More detailed discussions of this and other graphical methods are available elsewhere. For example, see Seibert, "Extraction and Leaching," Chap. 14 in *Chemical Process Equipment: Selection and Design*, 2d ed., Couperet et al., eds. (Elsevier, 2005); Wankat, *Separation Process Engineering* (Prentice-Hall, 2006); and King, *Separation Processes*, 2d ed. (McGraw-Hill, 1980), among others.

In distillation calculations, the McCabe-Thiele graphical method assumes constant molar vapor and liquid flow rates and allows convenient stepwise calculation with straight operating lines and a curved equilibrium line. A similar concept can be achieved in liquid-liquid extraction by using Bancroft coordinates and expressing flow rates on a solute-free basis, i.e., a constant flow rate of feed solvent F' and a constant flow rate of extraction solvent S' through the extractor [Evans, *Ind. Eng. Chem.*, **26**(8), pp. 860–864 (1934)]. The solute concentrations are then given as the mass ratio of solute to feed solvent X' and the mass ratio of solute to extraction solvent Y' . These concentrations and coordinates give a straight operating line on an X' - Y' diagram for stages 2 through $r-1$ in Fig. 15-22. The ratio of solute-free extraction solvent to solute-free feed solvent will be constant within the extractor except at the outer stages where unsaturated feed and extraction solvent enter the process. Equilibrium data using these mass ratios have been shown to follow straight-line segments on a log-log plot (see Fig. 15-20), and they will be approximately linear over some composition range on an X' - Y' plot. When expressed in terms of Bancroft coordinates, the equilibrium line typically will curve upward at high solute concentrations, as shown in Fig. 15-23.

To illustrate the McCabe-Thiele method, consider the simplified case where feed and extraction solvents are immiscible; i.e., mutual solubility is nil. Then the rate of feed solvent alone in the feed stream

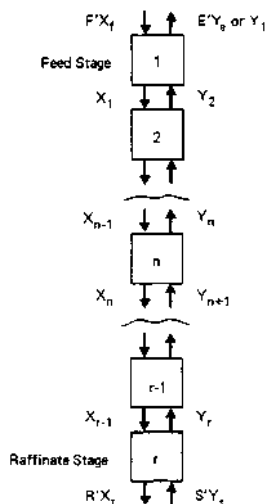


FIG. 15-22 Countercurrent extraction cascade.

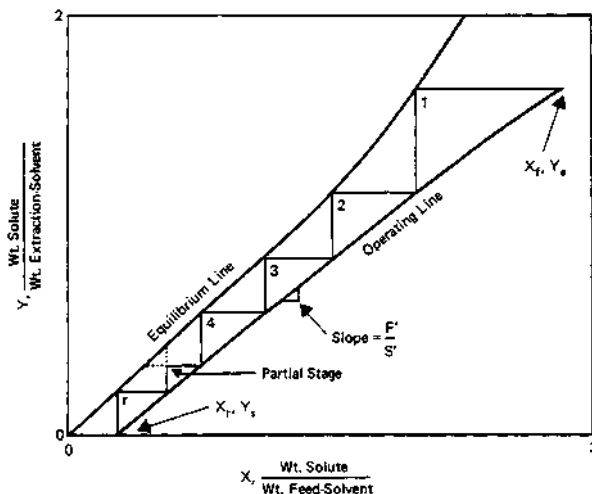


FIG. 15-23 McCabe-Thiele type of graphical stage calculation using Bancroft coordinates.

F' is the same as the rate of feed solvent alone in the raffinate stream R' . In like manner, the rate of extraction solvent alone is the same in the entering stream S' as in the leaving extract stream E' . The ratio of extraction-solvent to feed-solvent flow rates is therefore $S'/F' = E'/R'$. A material balance can be written around the feed end of the extractor down to any stage n (as shown in Fig. 15-22) and then rearranged to a McCabe-Thiele type of operating line with a slope of F'/S' :

$$Y'_{n+1} = \frac{F'}{S'} X'_n + \frac{E'Y'_e - F'X'_f}{S'} \quad (15-45)$$

Similarly, the same operating line can be derived from a material balance around the raffinate end of the extractor up to stage n :

$$Y'_n = \frac{F'}{S'} X'_{n-1} + \frac{S'Y'_s - R'X'_r}{S'} \quad (15-46)$$

The overall extractor material balance is given by

$$Y'_e = \frac{F'X'_f + S'Y'_s - R'X'_r}{E'} \quad (15-47)$$

The endpoints of the operating line on an X' - Y' plot (Fig. 15-23) are the points (X'_f, Y'_e) and (X'_r, Y'_s) where X' and Y' are the mass ratios for solute in the feed phase and extract phase, respectively, and subscripts f, r, s , and e denote the feed, raffinate, entering extraction solvent, and leaving extract streams. The number of theoretical stages can then be stepped off graphically as illustrated in Fig. 15-23.

Kremser-Souders-Brown Theoretical Stage Equation The Kremser-Souders-Brown (KSB) equation [Kremser, *Natl. Petrol. News*, **22**(21), pp. 43–49 (1930); and Souders and Brown, *Ind. Eng. Chem.*, **24**(5), pp. 519–522 (1932)] provides a way of calculating performance equivalent to that of a McCabe-Thiele type of graphical calculation with straight equilibrium and operating lines. In terms of Bancroft coordinates, the KSB equation may be written

$$N = \frac{\ln \left[\left(\frac{X'_f - Y'_s/m'}{X'_r - Y'_s/m'} \right) \left(1 - \frac{1}{E} \right) + \frac{1}{E} \right]}{\ln E} \quad E = m' \frac{S'}{F'}, E \neq 1 \quad (15-48)$$

where N = number of theoretical stages

X'_f = mass ratio solute to feed solvent in feed entering process (Bancroft coordinates)

- X'_r = mass ratio solute to feed solvent in raffinate leaving process
 Y'_s = mass ratio solute to extraction solvent in extraction solvent entering process
 $\mathcal{E}$ = extraction factor
 $m' = dY'/dX'$, local slope of equilibrium line in Bancroft coordinates
 S' = mass flow rate of extraction solvent (solute-free basis)
 F' = mass flow rate of feed solvent (solute-free units)

Solutions to Eq. (15-48) are shown graphically in Fig. 15-24. The concentration of solute in the extract leaving the process Y'_e is determined from the material balance, as in Eq. (15-47). (Note that other systems of units also may be used in these equations, as long as they are consistently applied.)

Rearranging Eq. (15-48) yields another common form of the KSB equation:

$$\frac{X'_f - Y'_e/m'}{X'_r - Y'_e/m'} = \frac{\mathcal{E}^N - 1/\mathcal{E}}{1 - 1/\mathcal{E}} \quad \mathcal{E} \neq 1 \quad (15-49)$$

Equations (15-48) and (15-49) can be used whenever $\mathcal{E} > 1$ or $\mathcal{E} < 1$. They cannot be used when $\mathcal{E}$ is exactly equal to unity because this would involve division by zero. When $\mathcal{E} = 1$, the number of theoretical stages is given by

$$N = \frac{X'_f - Y'_e/m'}{X'_r - Y'_e/m'} - 1 \quad \text{for } \mathcal{E} = 1 \quad (15-50)$$

Equation (15-50) may be rewritten

$$\frac{X'_f - Y'_e/m'}{X'_r - Y'_e/m'} = N + 1 \quad \text{for } \mathcal{E} = 1 \quad (15-51)$$

In the special case where $\mathcal{E} < 1$, the *maximum* performance potential is represented by

$$\left(\frac{X'_f - Y'_e/m'}{X'_r - Y'_e/m'} \right)_{\max} \approx \frac{1}{1 - \mathcal{E}} \quad \text{for } \mathcal{E} < 1 \text{ and large } N \quad (15-52)$$

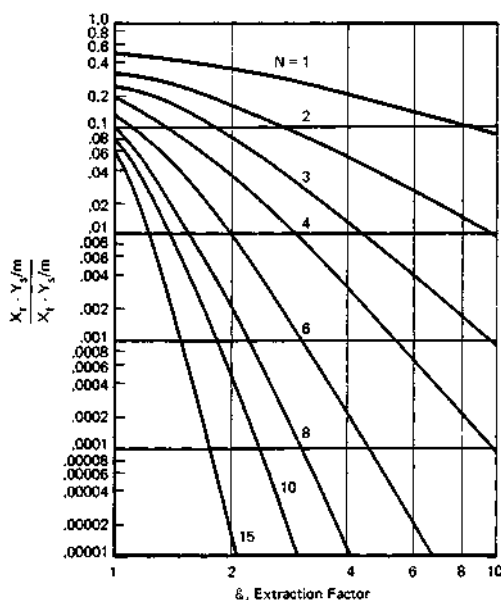


FIG. 15-24 Graphical solutions to the KSB equation [(Eq. 15-48)].

Equation (15-52) reflects the fact that the carrying capacity of the extract stream limits performance at $\mathcal{E} < 1$, as noted in earlier discussions.

In general, Eqs. (15-48) through (15-52) (and Fig. 15-24) are valid for any concentration range in which equilibrium can be represented by a linear relationship $Y = mX + b$ (written here in general form for any system of units). For applications that involve dilute feeds, the section of the equilibrium line of interest is a straight line that extends through the origin where $Y_i = 0$ at $X_i = 0$. In this case, $b = 0$ and the slope of the equilibrium line is equal to the partition ratio ($m = K$). The KSB equation also may be used to represent a linear segment of the equilibrium curve at higher solute concentrations. In this case, the linear segment is represented by a straight line that does not extend through the origin, and m is the local slope of the equilibrium line, so $b \neq 0$ and $m \neq K$. Furthermore, a series of KSB equations may be used to model a highly curved equilibrium line by dividing the analysis into linear segments and matching concentrations where the segments meet. For equilibrium lines with moderate curvature, an approximate average slope of the equilibrium line may be obtained from the geometric mean of the slopes at low and high solute concentrations:

$$m_{\text{average}} \approx m_{\text{geometric mean}} = \sqrt{m_{\text{low}} m_{\text{high}}} \quad (15-53)$$

As noted above, other systems of units such as mass fraction and total mass flow rates or mole fraction and total molar flow rates also may be used with the KSB equation; however, Bancroft coordinates and solute-free mass flow rates are recommended because then the operating line must be linear, and this normally extends the concentration range over which the KSB analysis may be used. It is important to check whether equilibrium can be adequately represented by a straight line over the concentration range of interest. The application of the KSB equation is discussed in "Shortcut Calculations" under "Calculation Procedures." Additional discussion is given by Wankat [*Equilibrium Staged Separations* (Prentice-Hall, 1988)] and by King [*Separation Processes*, 2d ed. (McGraw-Hill, 1980)]. To facilitate use of the KSB equation in computer calculations where the singularity around $\mathcal{E} = 1$ can present difficulties, Shenoy and Fraser have proposed an alternative form of the equation [*Chem. Eng. Sci.*, **58**(22) pp. 5121-5124 (2003)].

Stage Efficiency For a multistage process, the overall stage efficiency is simply the number of theoretical stages divided by the number of actual stages times 100:

$$\xi_o (\%) = \frac{\text{theoretical stages}}{\text{actual stages}} \times 100 \quad (15-54)$$

The fundamental stage efficiency is referred to as the Murphree stage efficiency ξ_m . The Murphree efficiency based on the dispersed phase is defined as

$$\xi_{md} = \frac{C_{d,n+1} - C_{d,n}}{C_{d,n+1} - C_d^*} \quad (15-55)$$

where $C_{d,n+1}$ = concentration of solute i in dispersed phase at stage $n + 1$

$C_{d,n}$ = concentration of solute i in dispersed phase at stage n

C_d^* = concentration of solute i in dispersed phase, at equilibrium

The overall stage efficiency is related to the Murphree stage efficiency and the extraction factor ($\mathcal{E}$):

$$\xi_o (\%) = \frac{\ln [1 + \xi_{md}(\mathcal{E} - 1)]}{\ln \mathcal{E}} \times 100 \quad (15-56)$$

For applications involving extraction of multiple solutes, sometimes the extraction rate and mass-transfer efficiency for each solute are significantly different. In these cases, individual efficiencies will need to be determined for each solute.

Stage efficiencies normally are determined by running miniplant tests to measure performance as a function of process variables such as feed rates, operating temperature, physical properties, impurities,

and agitation (if used). A number of data correlations have been developed for various types of mixing equipment. In principle, these can be used in the estimation of mass-transfer rates and stage efficiencies, but in practice reliable design generally requires generation of mini-plant data and application of mixing scale-up methods. (See "Mixer-Settler Equipment" under "Liquid-Liquid Extraction Equipment.")

The overall efficiency of an extraction column also can be expressed as the height equivalent to a theoretical stage (HETS). This is simply the total contacting height Z_t divided by the number of theoretical stages achieved.

$$\text{HETS} = \frac{Z_t}{N} \quad (15-57)$$

The HETS often is used to compare staged contactors with differential contactors.

RATE-BASED CALCULATIONS

This section reviews the basics of the mass-transfer coefficient and mass-transfer unit approaches to modeling extraction performance. These methods have been used for many years and continue to provide a useful basis for the design of extractors and extraction processes. Additional discussions of these and other rate-based methods are given in the books edited by Godfrey and Slater [*Liquid-Liquid Extraction Equipment* (Wiley, 1994)] and by Thornton [*Science and Practice of Liquid-Liquid Extraction*, vol. 1 (Oxford, 1992)]. For discussions of more mechanistic methods that include characterization of drop breakage and coalescence rates, drop size distributions, and drop population balances, see Leng and Calabrese, Chap. 12 in *Handbook of Industrial Mixing*, Paul, Atiemo-Obeng, and Kresta, eds. (Wiley, 2004); Goodson and Kraft, *Chem. Eng. Sci.*, **59**, pp. 3865–3881 (2004); Attarakih, Bart, and Faqir, *Chem. Eng. Sci.*, **61**, pp. 113–123 (2006); and Schmidt et al., *Chem. Eng. Sci.*, **61**, pp. 246–256 (2006). These methods are the subject of current research. Also see the discussion of general approaches to analyzing dispersed-phase systems given by Ramkrishna, Sathiyagal, and Narsimhan [*AIChE J.*, **41**(1), pp. 35–44 (1995)]. For discussions of the effect of contaminants on mass-transfer rates, see Saïen et al., *Ind. Eng. Chem. Res.*, **45**(4), pp. 1434–1440 (2006); and Dehkordi et al., *Ind. Eng. Chem. Res.*, **46**(5), pp. 1563–1571 (2007).

Solute Diffusion and Mass-Transfer Coefficients For a binary system consisting of components A and B, the overall rate of mass transfer of component A with respect to a fixed coordinate is the sum of the rates due to diffusion and bulk flow:

$$N_A = -\mathcal{D}_{AB} \frac{\partial C_A}{\partial z} + N_A \frac{C_A}{C} \quad (15-58)$$

where N_A = flux for component A (moles per unit area per unit time)
 $\mathcal{D}_{AB}$ = mutual diffusion coefficient of A into B (area/unit time)
 z = dimension or direction of mass transfer (length)
 C = total concentration of A and B (mass or mole per unit volume)
 C_A = concentration of A (mass or mole per unit volume)

Equation (15-58) is written for steady-state unidirectional diffusion in a quiescent liquid, assuming that the net transfer of component B is negligible. For transfer of component A across an interface or film between two liquids, it may be rewritten in the form

$$N_A = \frac{\mathcal{D}_{AB}}{\Delta z(1 - x_A)_m} (C_A - C_A^i) \quad (15-59)$$

where $(1 - x_A)_m$ = mean mole fraction of component B
 C_A^i = concentration of component A at interface
 C_A = concentration of component A in bulk

For steady-state counter diffusion where $N_A + N_B = 0$, the flux equation simplifies to

$$N_A = \frac{\mathcal{D}_{AB}}{\Delta z} (C_A - C_A^i) \quad (15-60)$$

The flux also may be written in terms of an individual mass-transfer coefficient k

$$N_A = k(C_A - C_A^i) \quad (15-61)$$

$$\text{where } k = \frac{\mathcal{D}_{AB}}{\Delta z(1 - x_A)_m} \quad (15-62)$$

In Eqs. (15-58) to (15-62), the flux is expressed in terms of mass or moles per unit area per unit time, and the concentration driving force is defined in terms of mass or moles per unit volume. The units of the mass-transfer coefficients are then length per unit time. Other definitions of the flux and resulting mass-transfer coefficients also are used. When mass-transfer coefficients are used, it is important to understand their definition and how they were determined; they need to be used in the same way in any subsequent calculations. Additional discussion of mass-transfer coefficients and mass-transfer rate is given in Sec. 5. Also see Laddha and Degaleesan, *Transport Phenomena in Liquid Extraction* (McGraw-Hill, 1978), Chap. 3; Skelland, *Diffusional Mass Transfer* (Krieger, 1985); Skelland and Tedder, Chap. 7 in *Handbook of Separation Process Technology*, Rousseau, ed. (Wiley, 1987); Curtiss and Bird, *Ind. Eng. Chem. Res.*, **38**(7), pp. 2515–2522 (1999); and Bird, Stewart, and Lightfoot, *Transport Phenomena*, 2d ed. (Wiley, 2002). Available correlations of molecular diffusion coefficients (diffusivities) are discussed in Sec. 5 and in Poling, Prausnitz, and O'Connell, *The Properties of Gases and Liquids*, 5th ed. (McGraw-Hill, 2000). The prediction of diffusion coefficients is discussed by Bosse and Bart, *Ind. Eng. Chem. Res.*, **45**(5), pp. 1822–1828 (2006).

Mass-Transfer Rate and Overall Mass-Transfer Coefficients

In transferring from one phase to the other, a solute must overcome certain resistances: (1) movement from the bulk of the raffinate phase to the interface; (2) movement across the interface; and (3) movement from the interface to the bulk of the extract phase, as illustrated in Fig. 15-25. The two-film theory first used to model this process [Lewis and Whitman, *Ind. Eng. Chem.*, **16**, pp. 1215–1220 (1924)] assumes that motion in the two phases is negligible near the interface such that the entire resistance to transfer is contained within two laminar films on each side of the interface, and mass transfer occurs by molecular diffusion through these films. The theory further invokes the following simplifying assumptions: (1) The rate of mass transfer within each phase is proportional to the difference in concentration in the bulk liquid and the interface; (2) mass-transfer resistance across the interface itself is negligible, and the phases are in equilibrium at the interface; and (3) steady-state diffusion occurs with negligible holdup of diffusing solute at the interface. Within a liquid-liquid extractor, the rate of steady-state mass transfer between the dispersed phase and the continuous phase (mass or moles per unit time per unit volume of extractor) is then expressed as

$$R_A = \frac{dC}{dt} = k_d a (C_{d,i} - C_d) = k_c a (C_c - C_{c,i}) \quad (15-63)$$

where C_i = concentration at interface (mass or moles per unit volume)
 C = concentration in bulk liquid (mass or moles per unit volume)
 k_c = continuous-phase mass-transfer coefficient (length per unit time)
 k_d = dispersed-phase film mass-transfer coefficient (length per unit time)
 a = interfacial area for mass transfer per unit volume of extractor (length⁻¹)

Subscripts d and c denote the dispersed and continuous phases. The concentrations at the interface normally are not known, so the rate expression is written in terms of equilibrium concentrations assuming that the rate is proportional to the deviation from equilibrium:

$$R_A = \frac{dC}{dt} = k_{oc} a (C_d^* - C_d) = k_{oc} a (C_c - C_c^*) \quad (15-64)$$

where the superscript $*$ denotes equilibrium, and k_{oc} is an overall mass-transfer coefficient given by

$$\frac{1}{k_{oc}} = \underbrace{\frac{1}{k_c}}_{\text{Continuous phase resistance}} + \underbrace{\frac{1}{m_{dc}^* k_d}}_{\text{Dispersed phase resistance}} \quad (15-65)$$

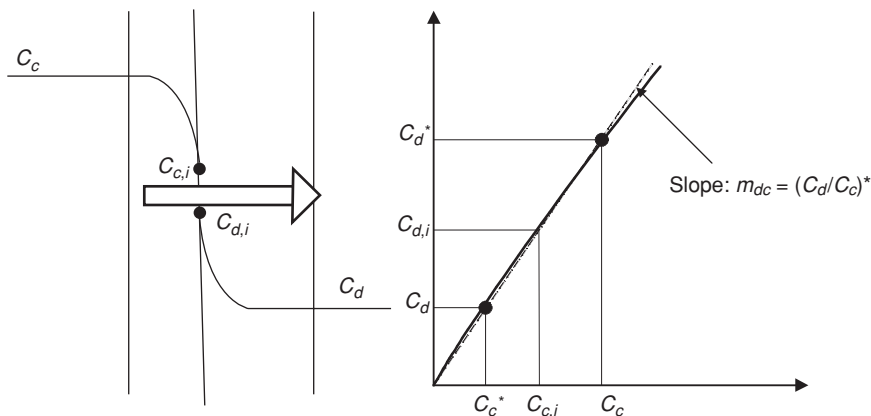


FIG. 15-25 Two-film mass transfer.

Similarly, the overall mass-transfer coefficient based on the dispersed phase is given by

$$\frac{1}{k_{od}} = \underbrace{\frac{1}{k_d}}_{\text{Dispersed phase resistance}} + \underbrace{\frac{m_{dc}^{\text{vol}}}{k_c}}_{\text{Continuous phase resistance}} \quad (15-66)$$

Assuming mass-transfer coefficients are constant over the range of conditions of interest, Eq. (15-64) may be integrated to give

$$\frac{C_c - C_c^*}{C_{c,\text{initial}} - C_c^*} = \exp(-k_{oc}a\theta) \approx \frac{C_c}{C_{c,\text{initial}}} \quad (15-67)$$

where θ is the contact time.

In Eqs. (15-65) and (15-66), $m_{dc}^{\text{vol}} = dC_d/dC_c$ is the local slope of the equilibrium line, with the equilibrium concentration of solute in the dispersed phase plotted on the ordinate (y axis), and the equilibrium concentration of solute in the continuous phase plotted on the abscissa (x axis). Note that m_{dc}^{vol} is expressed on a volumetric basis (denoted by superscript *vol*), i.e., in terms of mass or mole per unit volume, because of the way the mass-transfer coefficients are defined. The mass-transfer coefficients will not necessarily be the same for each solute being extracted, so depending upon the application, mass-transfer coefficients may need to be determined for a range of different solutes. As noted earlier, other systems of units also may be used as long as they are consistently applied.

The mass-transfer coefficient in each film is expected to depend upon molecular diffusivity, and this behavior often is represented by a power-law function $k \propto \mathcal{D}^n$. For two-film theory, $n = 1$ as discussed above [(Eq. (15-62)]. Subsequent theories introduced by Higbie [*Trans. AIChE*, **31**, p. 365 (1935)] and by Dankwerts [*Ind. Eng. Chem.*, **43**, pp. 1460–1467 (1951)] allow for surface renewal or penetration of the stagnant film. These theories indicate a 0.5 power-law relationship. Numerous models have been developed since then where $0.5 < n < 1.0$; the results depend upon such things as whether the dispersed drop is treated as a rigid sphere, as a sphere with internal circulation, or as oscillating drops. These theories are discussed by Skelland [“Interphase Mass Transfer,” Chap. 2 in *Science and Practice of Liquid-Liquid Extraction*, vol. 1, Thornton, ed. (Oxford, 1992)].

In the design of extraction equipment with complex flows, mass-transfer coefficients are determined by experiment and then correlated as a function of molecular diffusivity and system properties. The available theories provide an approximate framework for the data. The correlation constants vary depending upon the type of equipment and operating conditions. In most cases, the dominant mass-transfer resistance resides in the feed (raffinate) phase, since

the slope of the equilibrium line usually is greater than unity. In that case, the overall mass-transfer coefficient based on the raffinate phase may be written

$$\frac{1}{k_{or}} = \frac{1}{k_r} + \frac{1}{m_{er}^{\text{vol}} k_c} \approx \frac{1}{k_r} \quad \text{for large } m_{er}^{\text{vol}} \quad (15-68)$$

where m_{er}^{vol} is defined by the usual convention in terms of concentration in the extract phase over that in the raffinate phase, $m_{er}^{\text{vol}} = dC_{i,\text{extract}}/dC_{i,\text{raffinate}}$. This approximation is particularly useful when the extraction solvent is significantly less viscous than the feed liquid, so the solute diffusivity and mass-transfer coefficient in the extract phase are relatively large.

Mass-Transfer Units The mass-transfer unit concept follows directly from mass-transfer coefficients. The choice of one or the other as a basis for analyzing a given application often is one of preference. Colburn [*Ind. Eng. Chem.*, **33**(4), pp. 450–467 (1941)] provides an early review of the relationship between the height of a transfer unit and volumetric mass-transfer coefficients ($k_{or}a$). From a differential material balance and application of the flux equations, the required contacting height of an extraction column is related to the height of a transfer unit and the number of transfer units

$$Z_t = \left(\frac{V_r}{k_{or}a} \right) \int_{X_{\text{out}}}^{X_{\text{in}}} \frac{dX}{X - X^*} = H_{or} \times N_{or} \quad (15-69)$$

where V_r is the velocity of the raffinate phase, a is the interfacial area per unit volume, and the superscript $*$ denotes the equilibrium concentration. The transfer unit model has proved to be a convenient framework for characterizing mass-transfer performance.

Thus, mass-transfer units are defined as the integral of the differential change in solute concentration divided by the deviation from equilibrium, between the limits of inlet and outlet solute concentrations:

$$N_{or} = \int_{X_{\text{out}}}^{X_{\text{in}}} \frac{dX}{X - X^*} \quad (15-70)$$

When the equilibrium and operating lines are linear, the solution to Eq. (15-70) can be expressed as

$$N_{or} = \frac{\ln \left[\left(\frac{X'_f - Y'_s/m'}{X'_r - Y'_s/m'} \right) \left(1 - \frac{1}{E} \right) + \frac{1}{E} \right]}{1 - \frac{1}{E}} \quad E = m' \frac{S'}{F'}, E \neq 1 \quad (15-71)$$

where N_{or} is the number of overall mass-transfer units based on the raffinate phase. The units are the same as those used previously for the KSB equation [Eq. (15-48)]. Rearranging Eq. (15-71) gives

$$\frac{X'_f - Y'_f/m'}{X'_r - Y'_r/m'} = \frac{\exp[N_{or}(1 - 1/\mathcal{E})] - 1/\mathcal{E}}{1 - 1/\mathcal{E}} \quad (15-72)$$

Note that Eq. (15-71) is the same as the KSB equation except in the denominator. Comparing these equations shows that the number of overall raffinate phase transfer units is related to the number of theoretical stages by

$$N_{or} = N \times \frac{\ln \mathcal{E}}{1 - 1/\mathcal{E}} \quad (15-73)$$

The difference becomes pronounced when values of the extraction factor are high. When $\mathcal{E} = 1$, the number of mass-transfer units and number of theoretical stages are the same:

$$N_{or} = N = \frac{X'_f - Y'_f/m'}{X'_r - Y'_r/m'} - 1 \quad \text{for } \mathcal{E} = 1 \quad (15-74)$$

As with the KSB equation, in the special case where $\mathcal{E} < 1$, the maximum performance potential is represented by

$$\left(\frac{X'_f - Y'_f/m'}{X'_r - Y'_r/m'} \right)_{\max} \approx \frac{1}{1 - \mathcal{E}} \quad \text{for } \mathcal{E} < 1 \text{ and large } N_{or} \quad (15-75)$$

Equation (15-71) often is referred to as the Colburn equation. Although commonly used to represent the performance of a differential contactor, it models any steady-state, diffusion-controlled processes with straight equilibrium and operating lines. As with the KSB equation, the operating line is straight even when solute concentration changes significantly as long as Bancroft coordinates are used, and both the KSB and Colburn equations can be used to model applications involving a highly curved equilibrium line by dividing the analysis into linear segments. With these approaches, these equations often can be used for applications involving high concentrations of solute.

Solutions to the Colburn equation are shown graphically in Fig. 15-26. Note the contrast to the KSB equation solutions shown in Fig. 15-24. The KSB equations are best used to model countercurrent contact devices where the separation is primarily governed by equilibrium limitations, such as extractors involving discrete stages with high stage efficiencies. The Colburn equation, on the other hand, better represents the performance of a diffusion rate-controlled contactor because performance approaches a definite limit as the extraction factor increases beyond $\mathcal{E} = 10$ or so, corresponding to a diffusion rate limitation where addition of extra solvent has little or no effect. Note that in Eq. (15-71) the extraction factor always appears as $1/\mathcal{E}$, and this is how a finite diffusion rate is taken into account. The KSB equation can be misleading in this regard because it predicts continued improvement as the extraction factor increases without limit. Rate-based models most often are utilized for applications with no discrete stages; however, even staged equipment may be modeled best by the number of mass-transfer units when the extraction factor is higher than about 3, especially when stage efficiencies are low.

The height of an overall mass-transfer unit based on raffinate phase compositions H_{or} is the total contacting height Z_t divided by the number of transfer units achieved by the column.

$$H_{or} = \frac{Z_t}{N_{or}} \quad (15-76)$$

The value of H_{or} is the sum of contributions from the resistance to mass transfer in the raffinate phase (H_r) plus resistance to mass transfer in the extract phase (H_e) divided by the extraction factor $\mathcal{E}$:

$$H_{or} = H_r + \frac{H_e}{\mathcal{E}} \quad (15-77)$$

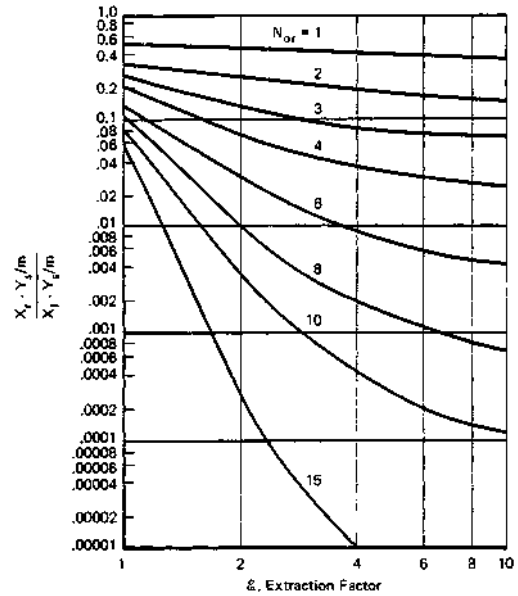


FIG. 15-26 Graphical solutions to the Colburn equation [Eq. (15-71)].

The individual transfer unit heights are given by

$$H_r = \frac{Q_r}{A_{col} k_r a} \quad (15-78)$$

$$H_e = \frac{Q_e}{A_{col} k_e a} \quad (15-79)$$

where Q = volumetric flow rate

A_{col} = column cross-sectional area

k = film mass-transfer coefficient (length per unit time)

a = interfacial mass-transfer area per unit volume of extractor

and subscripts r and e denote the raffinate and extract phases, respectively. As discussed earlier, the main resistance to mass transfer generally resides in the feed (raffinate) phase.

The lumped parameter H_{or} often is employed for design of extraction columns. Its value reflects the efficiency of the differential contactor; higher contacting efficiency is reflected in a lower value of H_{or} . It deals directly with the ultimate design criterion, the height of the column, and reliable values often can be obtained from miniplant experiments and experience with commercial units. For processes with discrete contacting stages, mass-transfer efficiency may be expressed as the number of transfer units achieved per actual stage. For applications involving transfer of multiple solutes, the value of H_{or} or N_{or} per actual stage may differ for each solute, as discussed earlier with regard to stage efficiencies and mass-transfer coefficients.

EXTRACTION FACTOR AND GENERAL PERFORMANCE TRENDS

Because of their simplicity, the KSB equation [Eq. (15-48)] and Colburn equation [Eq. (15-71)] are useful for illustrating a number of general trends in mass-transfer performance, in particular, helping to show how the extraction factor is related to process performance for different process configurations. For illustration, consider a dilute system involving immiscible liquids and zero solute concentration in the entering extraction solvent. The resulting expressions that follow are written in a general form without regard to a specific set of units.

15-50 LIQUID-LIQUID EXTRACTION AND OTHER LIQUID-LIQUID OPERATIONS AND EQUIPMENT

For a single-stage batch process or a continuous extraction process that achieves one theoretical stage, the solute reduction factor is given by

$$F_R = \frac{X_{in}}{X_{out}} = \frac{\mathcal{E} - 1/\mathcal{E}}{1 - 1/\mathcal{E}} \quad \text{for } N = 1 \quad (15-80)$$

The required solvent-to-feed ratio is then approximated by

$$\frac{S}{F} = \frac{F_R - 1}{K} \quad \text{for } N = 1 \quad (15-81)$$

After extraction, the concentration of solute in the extract, no matter what the extraction configuration, is given by

$$Y_{out} = \frac{X_{in}}{S/F} \left(1 - \frac{1}{F_R} \right) \quad \text{for } Y_{in} = 0 \quad (15-82)$$

Equation (15-82) follows from Eq. (15-47).

If the performance of a single-stage extraction is not adequate, repeated cross-current extractions can be carried out to increase solute recovery or removal. For this configuration, the reduction factor is given by

$$F_R = \left(1 + \frac{\mathcal{E}}{N} \right)^{\xi_o N} \quad \text{for cross-current operation} \quad (15-83)$$

where N is the number of repeated extractions or stages employing equal amounts of solvent, ξ_o is overall stage efficiency, and the extraction factor is expressed in terms of the total amount of solvent used by the process. Although high solute recoveries can be obtained by using cross-current processing, the required solvent usage will be high, as indicated by

$$\frac{S}{F} = \frac{N}{K} (F_R^{1/\xi_o N} - 1) \quad \text{for cross-current-operation} \quad (15-84)$$

where S is the total amount of solvent. The concentration of solute in the combined extract will be low, as calculated by using Eq. (15-82). Comparing the results of Eqs. (15-80) and (15-81) with Eqs. (15-83) and (15-84) will show that multistage cross-current extraction yields improved performance relative to using single-stage extraction with the same total amount of solvent, but at the cost of additional contacting steps.

Compared to single-stage or cross-current processing, multistage, countercurrent processing allows a significant reduction in solvent use or an increase in separation performance. For this type of process, the reduction factor is approximated by

$$F_R = \frac{\mathcal{E}^{\xi_o N} - 1/\mathcal{E}}{1 - 1/\mathcal{E}} \quad \text{for countercurrent operation} \quad (15-85)$$

Inspection of Eqs. (15-80) and (15-85) will show how the addition of countercurrent stages magnifies the effect of the extraction factor on performance. Note that Eq. (15-85) predicts that performance will continue to improve as the value of $\mathcal{E}$ increases, approaching $F_R = \mathcal{E}^{\xi_o N}$ at high values of $\mathcal{E}$. However, stage efficiency must remain high, and this likely will require a change in some operating variable such as residence time per stage.

Multistage countercurrent processing may be practiced batchwise as well as in a continuous cascade. A batchwise countercurrent operation involves first treating a batch with extract solution as the extract leaves the process, and the last treatment is carried out by using fresh solvent as it enters the process (as in Figs. 15-6 and 15-22). A multistage, countercurrent process with discrete contacting stages (practiced either batchwise or using a continuous cascade) is well suited to applications with fairly slow rates of mass transfer because liquid-

liquid contacting is carried out stagewise in separate vessels or compartments, and long residence times can be designed into each stage.

For a countercurrent extraction column with no discrete stages (or for processes operated within a diffusion-controlled regime far from equilibrium), performance is well modeled by the Colburn equation, where

$$F_R = \frac{\exp [N_{or}(1 - 1/\mathcal{E})] - 1/\mathcal{E}}{1 - 1/\mathcal{E}} \quad \text{for countercurrent operation} \quad (15-86)$$

and

$$Z_t = N_{or} \times H_{or} \quad (15-87)$$

Extraction columns are most attractive for applications with fairly fast mass transfer because residence time in the column is limited. Performance becomes mass-transfer-limited at high values of $\mathcal{E}$, approaching $F_R = \exp N_{or}$. At this point, a significant increase in performance can be achieved only by adding transfer units (column height).

With countercurrent processing, carried out using either a multistage cascade or an extraction column, the required solvent-to-feed ratio generally can be reduced by adding more and more stages or transfer units. As discussed in "Minimum and Maximum Solvent-to-Feed Ratios," the minimum practical solvent-to-feed ratio is approximated by

$$\left(\frac{S}{F} \right)_{\min} \approx \frac{1.3}{K} \quad \text{for countercurrent processing} \quad (15-88)$$

Below this value, the required number of stages or transfer units increases rapidly. At $\mathcal{E} = 1$, the number of theoretical stages and number of transfer units are equal, and

$$F_R = N + 1 = N_{or} + 1 \quad \text{for } \mathcal{E} = 1 \quad (15-89)$$

For $\mathcal{E} < 1$, the fraction of solute removed from the feed θ_i will approach a value equal to the extraction factor. In this case,

$$(F_R)_{\max} = \frac{1}{1 - \mathcal{E}} \quad \text{for } \mathcal{E} < 1 \quad (15-90)$$

POTENTIAL FOR SOLUTE PURIFICATION USING STANDARD EXTRACTION

As noted earlier, the ability of a standard extraction process to isolate a desired solute from other solutes is limited. This can be illustrated by using the KSB equation [Eq. (15-48)] to calculate solute transfer for a dilute feed containing a desired solute i and an impurity solute j . On a solvent-free basis, the purity of solute i in the feed is given by

$$P_{i,\text{feed}} (\text{in units of wt \%}) = 100 \left(\frac{X''_{i,\text{feed}}}{X''_{i,\text{feed}} + X''_{j,\text{feed}}} \right) \quad (15-91)$$

Similarly, the purity of solute i in the extract is given by

$$P_{i,\text{extract}} (\text{wt \%}) = 100 \left(\frac{\theta_i X''_{i,\text{feed}}}{\theta_i X''_{i,\text{feed}} + \theta_j X''_{j,\text{feed}}} \right) \quad (15-92)$$

where θ_i is the fraction of solute extracted from the feed into the extract. By using the KSB equation to estimate θ for solutes i and j , the following expression is derived:

$$\begin{aligned} P_{i,\text{extract}} (\text{wt \%}) &= \frac{100}{1 + \left(\frac{\theta_j}{\theta_i} \right) \left(\frac{X''_{j,\text{feed}}}{X''_{i,\text{feed}}} \right)} \\ &= \frac{100}{1 + \left(\frac{\mathcal{E}_j^N - 1}{\mathcal{E}_i^N - 1} \right) \left(\frac{\mathcal{E}_i^N - 1/\mathcal{E}_i}{\mathcal{E}_j^N - 1/\mathcal{E}_j} \right) \left(\frac{X''_{j,\text{feed}}}{X''_{i,\text{feed}}} \right)} \quad \text{for } \mathcal{E} \neq 1.0 \end{aligned} \quad (15-93)$$

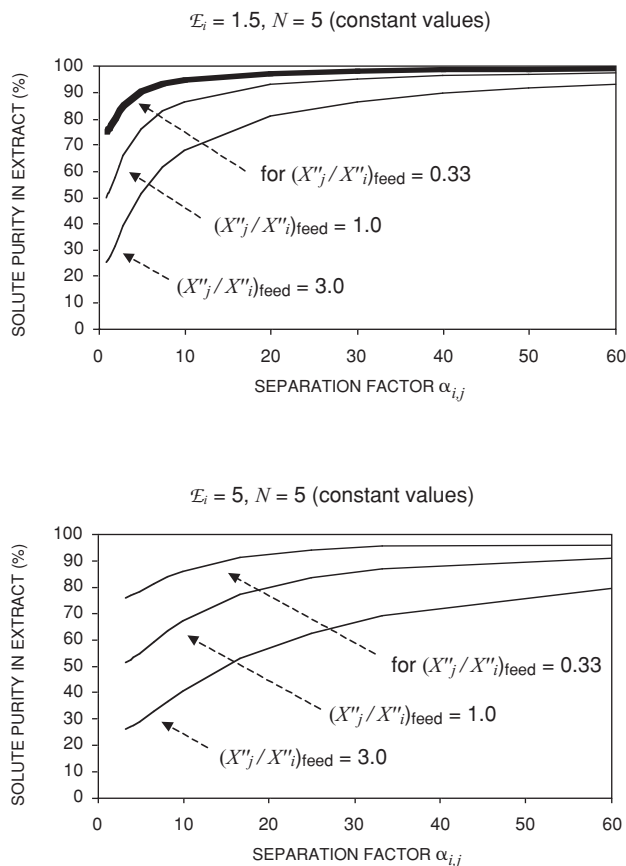


FIG. 15-27 Approximate purity of solute i in the extract ($P_{i,\text{extract}}$) versus separation factor α_{ij} for standard extraction involving dilute feeds containing solutes i and j . Results obtained by using Eq. (15-93). Concentrations are in mass fraction (X'').

Equation (15-93) assumes that no solute enters the process with the extraction solvent and that $\mathcal{E}_i$ and $\mathcal{E}_j$ are constant. An alternative expression can be written in terms of transfer units; however, the calculated results are essentially the same as a function of the number of stages or the number of transfer units—because the models assume that both solute i and solute j experience the same mass-transfer resistance. Example results obtained by using Eq. (15-93) are shown in Fig. 15-27. Note that performance is not uniquely determined by a given value of $\alpha_{ij} = K_i/K_j = \mathcal{E}_i/\mathcal{E}_j$, but depends upon the absolute value of $\mathcal{E}_i$, as well. In principle, the purity of solute i in the extract will approach a maximum value as the number of stages or transfer units

approaches infinity:

$$\text{Maximum } P_{i,\text{extract}} (\%) = 100 \div \left[1 + \frac{1}{\alpha_{ij}} \left(\frac{X''_{j,\text{feed}}}{X''_{i,\text{feed}}} \right) \right] \quad \text{in limit as } N \rightarrow \infty \quad (15-94)$$

Of course, this theoretical maximum can never be attained in practice. Equation (15-94) follows from Eq. (15-93), noting that $\theta_i/\theta_j = 1/\alpha_{ij}$ for $N \rightarrow \infty$ as discussed by Brian [*Staged Cascades in Chemical Processing* (Prentice-Hall, 1972), p. 50]. As noted earlier, the ability to purify a desired solute is greatly enhanced by using fractional extraction (see “Fractional Extraction Calculations”).

CALCULATION PROCEDURES

SHORTCUT CALCULATIONS

Shortcut calculations can be quite useful to the process designer or run-plant engineer; they may be used to outline process requirements (stream and equipment sizes) early in a design project, to check the output of a process simulation program for reasonableness, to help analyze or troubleshoot a unit operating in the manufacturing plant or pilot plant, or to help explain performance trends and relationships between key process variables. In some applications involving dilute

or even moderately concentrated feeds, they also may be used to specify the final design of an extraction process. In carrying out such calculations, Robbins [Sec. 1.9 in *Handbook of Separation Techniques for Chemical Engineers*, Schweitzer, ed. (McGraw-Hill, 1997)] indicates that most liquid-liquid extraction systems can be treated as having immiscible solvents (case A), partially miscible solvents with a low solute concentration in the extract (case B), or partially miscible solvents with a high solute concentration in the extract (case C). These cases are illustrated in Examples 1 through 3 below.

Example 1: Shortcut Calculation, Case A Consider a 100-kg/h feed stream containing 20 wt % acetic acid in water that is to be extracted with 200 kg/h of recycle MIBK that contains 0.1 wt % acetic acid and 0.01 wt % water. The aqueous raffinate is to be extracted down to 1% acetic acid. How many theoretical stages will be required and what will the extract composition be? The equilibrium data for this system are listed in Table 15-8 (in units of weight percent). The corresponding Hand plot is shown in Fig. 15-20. The Hand correlation (in mass ratio units) can be expressed as $Y' = 0.930(X')^{1.10}$, for X' between 0.03 and 0.25.

Assuming immiscible solvents, we have

$$F' = 100(1 - 0.2) = 80 \text{ kg water/h}$$

$$X'_f = \frac{0.2}{0.8} = 0.25 \text{ kg acetic acid/kg water}$$

$$X'_r = \frac{0.01}{0.99} = 0.01 \text{ kg acetic acid/kg water}$$

$$S' = 200(1 - 0.001) = 199.8 \text{ kg MIBK/h}$$

$$Y'_e = \frac{0.2}{199.8} = 0.001 \text{ kg acetic acid/kg MIBK}$$

If we assume $R' = F'$ and $E' = S'$, we can calculate Y'_e from Eq. (15-47):

$$Y'_e = \frac{80(0.25) + 199.8(0.001) - 80(0.01)}{199.8} = 0.097 \frac{\text{kg acetic acid}}{\text{kg MIBK}}$$

Calculate $X'_1 = (0.097/0.930)^{1/1.10} = 0.128$. Then

$$m' = \frac{dY'}{dX'} = (0.930)(1.10)(X')^{0.1} \quad \text{for } X' \text{ between 0.03 and 0.25}$$

$$m'_1 = 0.833 \quad \text{at } X' = 0.128$$

$$m'_r = \frac{dY'}{dX'} = K' = 0.656 \quad \text{for } X' \text{ below 0.03}$$

$$K'_e = 0.656 \quad \text{at } Y'_e = 0.001$$

$$\mathcal{E} = \sqrt{m'_1 m'_r} \frac{S'}{F'} = \frac{0.739(199.8)}{80} = 1.85$$

And N is determined from Fig. 15-24 and Eq. (15-48).

$$N = \frac{\ln \left[\left(\frac{0.25 - 0.001/0.656}{0.01 - 0.001/0.656} \right) \left(1 - \frac{1}{1.85} \right) + \frac{1}{1.85} \right]}{\ln 1.85} = 4.3 \text{ theoretical stages}$$

This result is very close to that obtained by using a McCabe-Thiele diagram (Fig. 15-23). From solubility data at $Y' = 0.1039$ kg acetic acid/kg MIBK (given in Table 15-8), the extract layer contains $5.4/85.7 = 0.0630$ kg water/kg MIBK, and $Y'_e = (0.097)/(1 + 0.097 + 0.063) = 0.084$ mass fraction acetic acid in the extract.

For cases B and C , Robbins developed the concept of pseudosolute concentrations for the feed and solvent streams entering the extractor that will allow the KSB equations to be used. In case B the solvents are partially miscible, and the miscibility is nearly constant through the extractor. This frequently occurs when all solute concentrations are relatively low. The feed stream is assumed to dissolve extraction solvent only in the feed stage and to retain the same amount throughout the extractor. Likewise, the extraction solvent is assumed to dissolve feed solvent only in the raffinate stage. With these assumptions the primary extraction solvent rate moving through the extractor is

assumed to be S' , and the primary feed solvent rate is assumed to be F' . The extract rate E' is less than S' , and the raffinate rate R' is less than F' because of solvent mutual solubilities.

The slope of the operating line is F'/S' , just as in Eqs. (15-45) and (15-46), but only stages 2 through $r - 1$ will fall directly on the operating line. And X'_1 must be on the equilibrium line in equilibrium with Y'_e by definition. One can also calculate a pseudofeed concentration X'_f that will fall on the operating line at $Y'_{n+1} = Y'_e$ as follows:

$$X'_f = X'_f + \frac{S' - E'}{F'} Y'_e \quad (15-95)$$

Likewise, one knows that Y'_r will be on the equilibrium line with X'_r . One can therefore calculate a pseudoconcentration of solute in the inlet extraction solvent Y'_s that will fall on the operating line where $X'_{n-1} = X'_r$, as follows:

$$Y'_s = Y'_s + \frac{F' - R'}{S'} X'_r \quad (15-96)$$

For case B , the pseudo inlet concentration X'_f can be used in the KSB equation with the actual value of X'_r and $\mathcal{E} = m'S'/F'$ to calculate rapidly the number of theoretical stages required. The graphical stepwise method illustrated in Fig. 15-23 also can be used. The operating line will go through points (X'_r, Y'_s) and (X'_f, Y'_e) with a slope of F'/S' .

Example 2: Shortcut Calculation, Case B Let us solve the problem in Example 1 by assuming case B . The solute (acetic acid) concentration is low enough in the extract that we may assume that the mutual solubilities of the solvents remain nearly constant. The material balance can be calculated by an iterative method.

From equilibrium data (Table 15-8) the extraction solvent (MIBK) loss in the raffinate will be about $0.016/0.984 = 0.0163$ kg MIBK/kg water, and the feed solvent (water) loss in the extract will be about $5.4/85.7 = 0.0630$ kg water/kg MIBK.

First iteration: Assume $R' = F' = 80$ kg water/h. Then extraction solvent in raffinate $= (0.0163)(80) = 1.30$ kg MIBK/h. Estimate $E' = 199.8 - 1.3 = 198.5$ kg MIBK/h. Then feed solvent in extract $= (0.063)(198.5) = 12.5$ kg water/h.

Second iteration: Calculate $R' = 80 - (0.063)(198.7) = 67.5$ kg water/h. And $E' = 199.8 - (0.0163)(67.5) = 198.7$ kg MIBK/h.

Third iteration: Converge $R' = 80 - (0.063)(198.7) = 67.5$ kg water/h. And Y'_e is calculated from the overall extractor material balance [(Eq. (15-47))]:

$$Y'_e = \frac{(80)(0.25) + (199.8)(0.001) - (67.5)(0.01)}{198.7} = 0.0983 \frac{\text{kg acetic acid}}{\text{kg MIBK}}$$

$$Y_e = \frac{0.0983}{1 + 0.0983 + 0.0630} = 0.0846 \text{ mass fraction acetic acid in extract}$$

From the Hand correlation of equilibrium data,

$$Y'_e = 0.930(X')^{1.10} \quad \text{for } X' \text{ between 0.03 and 0.25}$$

The raffinate composition leaving the feed (first stage) is

$$X'_1 = \left(\frac{0.0983}{0.930} \right)^{1/1.10} = 0.130$$

$$m'_1 = \frac{dY'}{dX'} = (0.930)(1.10)(X')^{0.1}$$

TABLE 15-8 Water + Acetic Acid + Methyl Isobutyl Ketone Equilibrium Data at 25°C

Weight percent in raffinate			X'	Weight percent in extract			Y'
Water	Acetic acid	MIBK	Acetic acid	Water	Acetic acid	MIBK	Acetic acid
98.45	0	1.55	0	2.12	0	97.88	0
95.46	2.85	1.7	0.0299	2.80	1.87	95.33	0.0196
85.8	11.7	2.5	0.1364	5.4	8.9	85.7	0.1039
75.7	20.5	3.8	0.2708	9.2	17.3	73.5	0.2354
67.8	26.2	6.0	0.3864	14.5	24.6	60.9	0.4039
55.0	32.8	12.2	0.5964	22.0	30.8	47.2	0.6525
42.9	34.6	22.5	0.8065	31.0	33.6	35.4	0.9492

SOURCE: Sherwood, Evans, and Longcor, *Ind. Eng. Chem.*, **31**(9), pp. 1144-1150 (1939).

$$\begin{aligned} m'_r &= \frac{dY}{dX} = K' = 0.656 \\ m'_1 &= 0.834 \quad \text{at } X'_1 = 0.13 \\ m'_r &= 0.656 \quad \text{at } X'_r = 0.01 \\ K'_s &= 0.656 \quad \text{at } Y'_s = 0.001 \\ E &= \sqrt{m'_1 m'_r} \frac{S'}{F'} = \frac{(0.740)(199.8)}{80} = 1.85 \end{aligned}$$

And X_f^B is calculated from Eq. (15-95)

$$X_f^B = 0.25 + \frac{(199.8 - 198.7)(0.0983)}{80} = 0.251$$

and Y_s^B from Eq. (15-96):

$$Y_s^B = 0.001 + \frac{(80 - 67.5)(0.01)}{199.8} = 0.0016$$

Now N is determined from Fig. 15-24, Eq. (15-48), or the McCabe-Thiele type of plot (Fig. 15-23). For case B ,

$$N = \frac{\ln \left[\left(\frac{0.251 - 0.0016/0.656}{0.01 - 0.0016/0.656} \right) \left(1 - \frac{1}{1.85} \right) + \frac{1}{1.85} \right]}{\ln 1.85} = 4.5 \text{ theoretical stages}$$

A less frequent situation, case C , can occur when the solute concentration in the extract is so high that a large amount of feed solvent is dissolved in the extract stream in the "feed stage" but a relatively small amount of feed solvent (say one-tenth as much) is dissolved by the extract stream in the "raffinate stage." The feed stream is assumed to dissolve the extraction solvent only in the feed stage just as in case B . But the extract stream is assumed to dissolve a large amount of feed solvent leaving the feed stage and a negligible amount leaving the raffinate stage. With these assumptions the primary feed solvent rate is assumed to be R' , so the slope of the operating line for case C is R'/S' . Again the extract rate E' is less than S' , and the raffinate rate R' is less than F' .

The pseudofeed concentration for case C , X_f^C , can be calculated from

$$X_f^C = \frac{F'}{R'} X'_f + \frac{S' - E'}{R'} Y'_e \quad (15-97)$$

For case C , the value of Y'_s will fall on the operating line, and the extraction factor is given by

$$E^C = \frac{m' S'}{R'} \quad (15-98)$$

On an X' - Y' diagram for case C , the operating line will go through points (X'_r, Y'_r) and (X_f^C, Y_e') with a slope of R'/S' similar to Fig. 15-23. When the KSB equation is used for case C , use the pseudofeed concentration X_f^C from Eq. (15-97) and the extraction factor E^C from Eq. (15-98). The raffinate concentration X'_r and inlet solvent concentration Y'_s are used without modification. For more detailed discussion, see Robbins, Sec. 1.9 in *Handbook of Separation Techniques for Chemical Engineers*, Schweitzer, ed. (McGraw-Hill, 1997).

Example 3: Number of Transfer Units Let us calculate the number of transfer units required to achieve the separation in Example 1. The solution to the problem is the same as in Example 1 except that the denominator is changed. From Eq. (15-73):

$$N_{or} = 4.5 \frac{\ln 1.85}{1 - 1/1.85} = 6.0 \text{ transfers units}$$

COMPUTER-AIDED CALCULATIONS (SIMULATIONS)

A number of process simulation programs such as Aspen Plus® from Aspen Technology, HYSYS® from Honeywell, ChemCAD® from Chemstations, and PRO/II® from SimSci E&S, among others, can

facilitate rigorous calculation of the number of theoretical stages required by a given application, provided an accurate liquid-liquid equilibrium model is employed. At the time of this writing, commercially available simulation packages do not include rate-based programs specifically designed for extraction process simulation; however, the equivalent number of transfer units at each stage can be calculated from knowledge of the extraction factor by using Eq. (15-73). Process simulation programs are particularly useful for concentrated systems that exhibit highly nonlinear equilibrium and operating lines, significant change in extract and raffinate flow rates within the process due to transfer of solute from one phase to the other, significant changes in the mutual solubility of the two phases as solute concentration changes, or nonisothermal operation. They also facilitate convenient calculation for complex extraction configurations such as fractional extraction with extract reflux as well as calculations involving more than three components (more than one solute). They can also facilitate process optimization by allowing rapid evaluation of numerous design cases. These programs do not provide information about mass-transfer performance in terms of stage efficiencies or extraction column height requirements, or information about the throughput and flooding characteristics of the equipment; these factors must be determined separately by using other methods. The use of simulation software to analyze extraction processes is illustrated in Examples 4 and 5.

In using simulation software, it is important to keep in mind that the quality of the results is highly dependent upon the quality of the liquid-liquid equilibrium (LLE) model programmed into the simulation. In most cases, an experimentally validated model will be needed because UNIFAC and other estimation methods are not sufficiently accurate. It also is important to recognize, as mentioned in earlier discussions, that binary interaction parameters determined by regression of vapor-liquid equilibrium (VLE) data cannot be relied upon to accurately model the LLE behavior for the same system. On the other hand, a set of binary interaction parameters that model LLE behavior properly often will provide a reasonable VLE fit for the same system—because pure-component vapor pressures often dominate the calculation of VLE.

Commercially available simulation programs often are used in a fashion similar to the classic graphical methods. When separation of specific solutes is important, the design of a new process generally focuses on determining the optimum solvent rates and number of theoretical stages needed to comply with the separation specifications according to relative K values for solutes of interest. Calculations often are made by focusing on a "soluble" key solute with a relatively high K value, and an "insoluble" key solute, expressing the design specification in terms of the maximum concentration of soluble key left in the raffinate and the maximum concentration of insoluble key contaminating the extract (analogous to light and heavy key components in distillation design). Then solutes with K values higher than that of the soluble key will go out with the extract to a greater extent, and solutes with K values less than that of the insoluble key will go out with the raffinate. If the desired separation is not feasible using a standard extraction scheme, then fractional extraction schemes should be evaluated.

For rating an existing extractor, the designer must make an estimate of the number of theoretical stages the unit can deliver and then determine the concentrations of key solutes in extract and raffinate streams as a function of the solvent-to-feed ratio, keeping in mind the fact that the number of theoretical stages a unit can deliver can vary depending upon operating conditions.

The use of process simulation software for process design is discussed by Seider, Seader, and Lewin [*Product and Process Design Principles: Synthesis, Analysis, and Evaluation*, 2d ed. (Wiley, 2004)] and by Turton et al. [*Analysis, Synthesis, and Design of Chemical Processes*, 2d ed. (Prentice-Hall, 2002)]. Various computational procedures for extraction simulation are discussed by Steiner [Chap. 6 in *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994)]. In addition, a number of authors have developed specialized methods of analysis. For example, Sanpui, Singh, and Khanna [*AIChE J.*, **50**(2), pp. 368–381 (2004)] outline a computer-based approach to rate-based, nonisothermal modeling of extraction processes. Harjo,

Ng, and Wibowo [*Ind. Eng. Chem. Res.*, **43**(14), pp. 3566–3576 (2004)] describe methods for visualization of high-dimensional liquid-liquid equilibrium phase diagrams as an aid to process conceptualization. Since in general it is not economically feasible to generate precise phase equilibrium data for the entire multicomponent phase diagram, this methodology can help focus the design effort by identifying specific composition regions where the design analysis will be particularly sensitive to uncertainties in the equilibrium behavior. The method of Minotti, Doherty, and Malone [*Ind. Eng. Chem. Res.*, **35**(8), pp. 2672–2681 (1996)] facilitates a feasibility analysis of potential solvents and process options by locating fixed points or pinches in the composition profiles determined by equilibrium and operating constraints. Marcilla et al. [*Ind. Eng. Chem., Res.*, **38**(8), pp. 3083–3095 (1999)] developed a method involving correlation of tie lines to calculate equilibrium compositions at each stage without iterations. To optimize the design and operating parameters of an extraction cascade, Reyes-Labarta and Grossmann [*AIChE J.*, **47**(10), pp. 2243–2252 (2001)] have proposed a calculation framework that employs nonlinear programming techniques to systematically evaluate a wide range of potential process configurations and interconnections. Focusing on another aspect of process design, Ravi and Rao [*Ind. Eng. Chem. Res.*, **44**(26), pp. 10016–10020 (2005)] provide an analysis of the phase rule (number of degrees of freedom) for liquid-liquid extraction processes. For discussion of reactive extraction process conceptualization methods, see Samant and Ng, *AIChE J.*, **44**(12), pp. 2689–2702 (1998); and Gorissen, *Chem. Eng. Sci.*, **58**, pp. 809–814 (2003).

Example 4: Extraction of Phenol from Wastewater The amount of 350 gpm (79.5 m³/h) of wastewater from a coke oven plant contains an average of 700 ppm phenol by weight that needs to be reduced to 1 ppm or less to meet environmental requirements [Karr and Ramanujam, *St. Louis AIChE Symp.* (March 19, 1987)]. The wastewater comes from the bottom of an ammonia stripping tower at 105°C and is to be extracted at 1.7 atm with recycle methylisobutyl ketone (MIBK) containing 5 ppm phenol. The extraction will be carried out by using a reciprocating-plate extractor (Karr column). How many theoretical stages will be required in the extractor at a solvent-to-feed ratio of 1:15, and what is the resulting extract composition?

The Aspen Plus® process simulation program is used in this example, but it should be recognized that any of a number of process simulation programs such as mentioned above may be used for this purpose. In Aspen Plus, the EXTRACT liquid-liquid extraction unit-operation block is used to model the phenol wastewater extraction. As is typical in process simulation programs, the EXTRACT block is fundamentally a rating calculation rather than a design calculation, so the determination of the required number of stages for the separation cannot be made directly. In addition, since the EXTRACT block can only handle integral numbers of theoretical stages, the fractional number of required theoretical stages must be determined by an interpolation method.

The partition ratio for transfer of phenol from water into MIBK at 105°C is $K'' = 34$ on a mass fraction basis [Greminger et al., *Ind. Eng. Chem. Process Des. Dev.*, **21**(1), pp. 51–54 (1982)]. Because the partition ratio is so high, a fairly low solvent-to-feed ratio of 1:15 can be used and still give an extraction factor of about 2. In the EXTRACT block, a property option is available that allows the user to specify liquid-liquid K value correlations (designated as “KLL Correlation” in Aspen Plus) for the components involved in the extraction rather than a complete set of binary interaction parameters to define the liquid-liquid equilibria. In this example, it is time-consuming to regress a set of liquid-liquid binary interaction parameters that results in representative partition ratios, so the option of simply specifying K values directly is highly recommended.

Because phenol will be relatively dilute in both the raffinate and extract phases, appropriate liquid-liquid K values for distribution of water and MIBK between phases at 105°C can be estimated from water-MIBK liquid-liquid equilibrium data [Rehak et al., *Collect. Czech Chem. Commun.*, **65**, pp. 1471–1486 (2000)] to yield $K''_{\text{water}} = 0.0532$ and $K''_{\text{MIBK}} = 53.8$ (mass fraction basis). It is important in Aspen Plus to specify K values for all the components in the extractor in order to properly model the liquid-liquid equilibria with this approach.

The temperatures and compositions of the wastewater and solvent feed streams, as well as the wastewater feed flow rate, are specified in the problem statement. The solvent flow rate is specified as one-fifteenth of the wastewater flow rate as described above. In the EXTRACT block, the number of stages will be manually varied from 2 to 10 to observe the effect on the raffinate and extract concentrations, and it will be specified as operating adiabatically at 1.7 atm. Water is specified as the key component in the first liquid phase, and MIBK is specified as the key component in the second liquid phase. The rest of the block parameters (convergence, report, and miscellaneous block options) are allowed to remain at their default values.

The raffinate and extract concentrations resulting from successive simulation runs for 2 through 10 theoretical stages are given in Table 15-9, and the raffinate phenol concentrations are presented graphically in Fig. 15-28. Examining the results, we can see that the number of theoretical stages required to achieve the 1 ppm phenol discharge limitation falls somewhere between 7 and 8. In addition, we can see from Fig. 15-28 that the dependence of raffinate phenol concentration on number of stages yields nearly a straight line on a semilog plot. As a result, performing a linear interpolation of the log of the raffinate concentration between 7 and 8 stages yields the number of stages required to achieve 1 ppm phenol in the raffinate:

$$N = 7 + (8 - 7) \left(\frac{\log 1.47 - \log 1}{\log 1.47 - \log 0.707} \right) = 7.53 \text{ theoretical stages}$$

From examining the extract phenol concentrations in Table 15-9, it is clear that for 5 or more stages, they varied little with number of stages, as is expected since nearly all the phenol contained in the wastewater feed was extracted in stages 1 through 4. As a result, the extract will contain 1.3 wt % phenol, 5.2% water, and 93.5% MIBK.

The simulation results can be checked by using a shortcut calculation—to provide confidence that the simulation is delivering a reasonable result. The KSB equation [Eq. (15-48)] can be used for this purpose with values taken from the problem specification and estimates of the phenol K' value (in Bancroft coordinates). Since phenol is always quite dilute in both the extract and raffinate phases, its K' value can be calculated from the component mass fraction K'' values according to the following approximation:

$$K'_{\text{PhOH}} \approx K''_{\text{PhOH}} \left[\frac{K_{\text{MIBK}} - 1}{K_{\text{MIBK}}(1 - K_{\text{H}_2\text{O}})} \right] = 34 \left[\frac{53.8 - 1}{53.8(1 - 0.0532)} \right] = 35.24$$

This value compares favorably with the value of 35.28 calculated directly from phenol mass ratios taken from extractor internal profile data in the simulation output. The extraction factor [Eq. (15-11)] is then calculated with the dilute system approximation that $m_{\text{PhOH}} \approx K'_{\text{PhOH}}$ and solute-free water and MIBK feed rates of 159,841 and 10,668 lb/h taken from the simulation output:

$$E_{\text{PhOH}} = m_{\text{PhOH}} \frac{S}{F} \approx K'_{\text{PhOH}} \frac{S''}{F''} = K'_{\text{PhOH}} \frac{S'}{F'} = 35.24 \times \frac{10,668}{159,841} = 2.35$$

It is interesting to note that this value of the extraction factor, 2.35, is the same as those calculated on mole fraction, mass fraction, and Bancroft coordinate bases from extractor internal profile data in the simulation, a confirmation that the extraction factor is indeed independent of units as long as consistent values of m , S , and F are used. By substituting the above values into Eq. (15-48) along

TABLE 15-9 Simulation Results for Extraction of Phenol from Wastewater Using MIBK (Example 4)

N	Raffinate compositions			Extract compositions		
	X''_{PhOH} , ppm	$X''_{\text{H}_2\text{O}}$, mass fraction	X''_{MIBK} , mass fraction	Y''_{PhOH} , mass fraction	$Y''_{\text{H}_2\text{O}}$, mass fraction	Y''_{MIBK} , mass fraction
2	101	0.98235	0.01755	0.01146	0.05223	0.93631
3	41.8	0.98237	0.01759	0.01260	0.05223	0.93517
4	17.7	0.98238	0.01761	0.01306	0.05223	0.93471
5	7.55	0.98238	0.01761	0.01326	0.05223	0.93451
6	3.28	0.98238	0.01762	0.01334	0.05223	0.93443
7	1.47	0.98238	0.01762	0.01337	0.05223	0.93440
8	0.707	0.98238	0.01762	0.01339	0.05223	0.93438
9	0.381	0.98238	0.01762	0.01340	0.05223	0.93437
10	0.242	0.98238	0.01762	0.01340	0.05223	0.93437

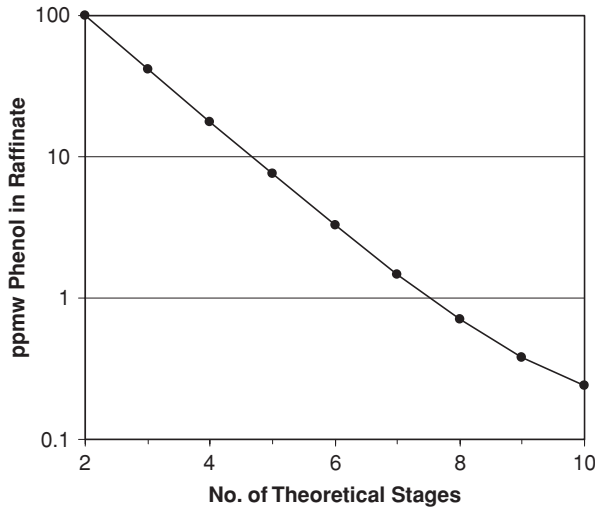


FIG. 15-28 Simulation results showing phenol concentration in the raffinate versus number of theoretical stages (Example 4).

with concentrations taken from the problem statement and Table 15-9, the required number of stages is estimated as

$$N \approx \frac{\ln \left[\frac{0.0007/0.9993 - (0.000005)/(0.999995)/35.24}{0.000001/0.9824 - (0.000005)/(0.999995)/35.24} (1 - 1/2.35) + 1/2.35 \right]}{\ln 2.35} = 7.18 \text{ theoretical stages}$$

The simulation result of 7.53 theoretical stages is close to this shortcut estimate, indicating that the simulation is indeed delivering reasonable results.

FRACTIONAL EXTRACTION CALCULATIONS

Dual-Solvent Fractional Extraction As discussed in “Commercial Process Schemes,” under “Introduction and Overview,” fractional extraction often may be viewed as combining product purification with product recovery by adding a washing section to the stripping section of a standard extraction process. In the stripping section, the mass transfer we focus on is the transfer of the product solute from the wash solvent into the extraction solvent. If we assume dilute conditions and use shortcut calculations for illustration, the extraction factor is given by

$$\mathcal{E}_s = K'_s \frac{S'_s}{W'_s} \quad (15-99)$$

where $\mathcal{E}_s$ = stripping section extraction factor (dimensionless)

K'_s = stripping section partition ratio, defined as equilibrium concentration of product solute in extraction solvent divided by that in wash solvent (Bancroft coordinates)

S'_s = mass flow rate of extraction solvent within stripping section (solute-free basis)

W'_s = mass flow rate of wash solvent in stripping section (solute-free basis)

The change in the concentration of product dissolved in the wash solvent, within the stripping section, can be calculated by using the KSB equation

$$\left(\frac{X'_{out}}{X'_{in}} \right)_{\text{product}} \approx \frac{1 - 1/\mathcal{E}_s}{(\mathcal{E}_s)^N - 1/\mathcal{E}_s} \quad (15-100)$$

where N_s = number of theoretical stages in stripping section

X'_{in} = concentration of product solute in wash solvent at inlet to stripping section (feed stage)

X'_{out} = concentration of product solute in wash solvent at outlet from stripping section (raffinate end of overall process)

In the washing section, we focus on transfer of impurity solute from the extraction solvent into the wash solvent. A washing extraction factor can be defined as

$$\mathcal{E}_w = \frac{1}{K'_w} \frac{W'_w}{S'_w} \quad (15-101)$$

where $\mathcal{E}_w$ = washing section extraction factor (dimensionless)

K'_w = washing section partition ratio (equilibrium concentration of impurity solute in extraction solvent divided by that in wash solvent, in Bancroft coordinates)

S'_w = mass flow rate of extraction solvent within washing section (solute-free basis)

W'_w = mass flow rate of wash solvent in washing section (solute-free basis)

Then the change in the concentration of impurity solute dissolved in the extraction solvent, within the washing section, is given by

$$\left(\frac{Y'_{out}}{Y'_{in}} \right)_{\text{impurities}} \approx \frac{1 - 1/\mathcal{E}_w}{(\mathcal{E}_w)^{N_w} - 1/\mathcal{E}_w} \quad (15-102)$$

where N_w = number of theoretical stages in washing section

Y'_{in} = concentration of impurity solute in extraction solvent at inlet to washing section (feed stage)

Y'_{out} = concentration of impurity solute in extraction solvent at outlet from washing section (extract end of overall process)

The ratio of extraction solvent to wash solvent in each section will be different if either solvent enters the process with the feed. Note that both K'_s and K'_w are defined as the ratio of the appropriate solute concentration in the *extraction* solvent to that in the *wash* solvent.

The shortcut calculations outlined above illustrate the general considerations involved in analyzing a fractional extraction process. The analysis requires locating the feed stage and matching the calculations for each section with the material balance at the feed stage, an iterative procedure. Buford and Brinkley [AIChE J., 6(3), pp. 446–450 (1960)] discuss application of the KSB equation to fractional extraction calculations including the use of reflux. Transfer unit calculations also may be used. When equilibrium and operating lines are not linear, more sophisticated calculations will be needed to take this into account. Commercially available simulation software or other computer programs often are used to carry out this procedure (see “Computer-Aided Calculations”). Note that with dual-solvent fractional extraction, solute concentrations always are highest at the feed stage. This can lead to undesired behavior such as tendencies toward emulsion formation or even formation of a single liquid phase at the plait point. The minimum amounts of solvent needed to avoid these effects can be determined in laboratory tests.

Early in a project, it may be useful to consider a simplified case in which the ratio of extraction solvent to wash solvent is constant and the same in the stripping and washing sections (i.e., the amount of solvent entering with the feed is negligible) and the extraction factors for each section are equal. For this special case, termed a *symmetric separation*, the extraction factors are

$$\mathcal{E}_s = \mathcal{E}_w = \sqrt{\alpha_{ij}} \quad (15-103)$$

and the ratio of extraction solvent to wash solvent is given by

$$\frac{S}{W} \approx \frac{1}{\sqrt{K_s K_w}} \approx \frac{1}{\sqrt{\alpha_{ij} K_w}} = \frac{\sqrt{\alpha_{ij}}}{K_s} \quad (15-104)$$

Using these relationships, we find the number of stages required for the stripping and washing sections will be about the same and the total number of stages required likely will be close to the minimum number—assuming symmetric separation requirements. The effects of the separation factor and the number of stages on the separation performance can be estimated by using expressions given by Brian [*Staged Cascades in Chemical Processing* (Prentice-Hall, 1972)]. For a process containing two solutes i and j , with the feed entering at the middle stage, it follows from Brian's analysis that

$$S_{ij} = \frac{Y_i}{X_i} \frac{1 - X_i}{1 - Y_i} \approx \alpha_{ij}^{(N+1)/2} \quad (15-105)$$

where S_{ij} is termed the *separation power* of the process. Equation (15-105) is derived by assuming that the ratio of extract phase to raffinate phase within the process is constant, and that α_{ij} is constant. Interestingly, Eq. (15-105) is very similar in its general form to the equation obtained by using the Fenske equation to calculate fractional distillation performance for a binary feed, assuming that the required number of theoretical stages is twice the minimum number obtained at total reflux. (See Sec. 13, "Distillation.")

For a proposed symmetric separation, Eqs. (15-104) and (15-105) can be used to gauge the required flow rates, number of theoretical stages, and separation factor. For example, consider a hypothetical application with the goal of transferring 99 percent of a key solute i into the extract and 99 percent of an impurity solute j into the raffinate. For illustration, let $K_i = 2.0$ and $K_j = 0.5$, so $\alpha_{ij} = 4$. From Eq. (15-104), the extraction solvent to wash solvent ratio should be about $S/W = 1.0$ for a symmetric separation. The number of theoretical stages is estimated by using Eq. (15-105): $S_{ij} = 99 \times 99 = 9801$ gives $N \approx 12$ total stages for $\alpha_{ij} = 4$. When one is evaluating candidate solvent pairs for a proposed fractional extraction process, a useful first step is to measure the equilibrium K values for product and impurity solutes and then assess process feasibility by using Eqs. (15-104) and (15-105). This can provide a quick way of assessing whether the measured separation factor is sufficiently large to achieve the separation goals, using a reasonable number of stages.

Single-Solvent Fractional Extraction with Extract Reflux As discussed earlier, single-solvent fractional extraction with extract reflux is widely practiced in the petrochemical industry to separate aromatics from crude hydrocarbon feeds. For example, a variety of extraction processes utilizing different high-boiling, polar solvents are used to separate benzene, toluene, and xylene (BTX) from aliphatic hydrocarbons and naphthenes (cycloalkanes), although processes involving extractive distillation are displacing some of the older extraction processes, depending upon the application. A typical hydrocarbon feed is a distillation cut containing mostly C_5 to C_9 components. Commercial extraction processes include the Udex process (employing diethylene and/or triethylene glycol), the AROSOLVAN process (employing *N*-methyl-2-pyrrolidone), and the Sulfolane process (employing tetrahydrothiophene-1,1-dioxane), among others. Although the flow diagrams for these processes differ, they all involve use of a liquid-liquid extractor followed by a top-fed extract stripper or extractive distillation tower. A number of different processing schemes are used to isolate the aromatics and recycle the heavy solvent. For detailed discussion, see Chaps. 18.1 to 18.3 in *Handbook of Solvent Extraction*, Lo, Baird, and Hanson, eds. (Wiley, 1983; Krieger, 1991); Mueller et al., *Ullmann's Encyclopedia of Industrial Chemistry*, 5th ed., vol. B3, Gerhartz, ed. (VCH, 1988), pp. 6-34 to 6-43; Gaile et al., *Chem. Technol. Fuels Oils*, **40**(3), pp. 131-136, and **40**(4), pp. 215-221 (2004); and Schneider, *Chem. Eng. Prog.*, **100**(7), pp. 34-39 (2004).

Consider a process scheme involving a liquid-liquid extractor followed by a top-fed extract stripper (as illustrated in Fig. 15-2). In the extractor, the feed is contacted with the polar solvent to transfer aromatics into the solvent phase. Some nonaromatics (NAs) also transfer into the solvent. In the stripper, low-boiling NAs plus some aromatics are stripped out of the extract. The overheads stream also contains some high-boiling NAs because their low solubility in the polar solvent boosts their relative volatility in the stripper. In this respect, the

stripper may be thought of as an extractive distillation tower with the high-boiling polar solvent serving as the extractive distillation solvent. The stripper overheads are then condensed and returned to the bottom of the extractor as extract reflux. As the backwash of extract reflux passes up through the extractor, the aromatics and a portion of the low-boiling NAs transfer back into the solvent phase, preferentially displacing high-boiling NAs from the extract phase because of their lower solubilities in the polar solvent. Without extract reflux, the concentration of higher-boiling NAs in the extract phase would be significantly higher, and they would be difficult to completely remove in the stripper in spite of their low solubilities in the polar solvent. In this manner, low-boiling aromatics and NAs tend to build up in the extract reflux loop to provide a sort of barrier that minimizes entry of higher-boiling NAs into the extract phase.

The use of simulation software to analyze this type of process is illustrated in Example 5, which considers a simplified ternary system for illustration. The simulation of an actual aromatics extraction process is more complex and can exhibit considerable difficulty converging on a solution; however, Example 5 illustrates the basic considerations involved in carrying out the calculations. For more detailed discussion of process simulation and optimization methods, see Seider, Seader, and Lewin, *Product and Process Design Principles: Synthesis, Analysis, and Evaluation*, 2d ed. (Wiley, 2004); and Turton et al., *Analysis, Synthesis, and Design of Chemical Processes*, 2d ed. (Prentice-Hall, 2002).

Example 5: Simplified Sulfolane Process—Extraction of Toluene from *n*-Heptane The amount of 40 metric tons (t) per hour (t/h) of distilled catalytic reformate from petroleum refining, containing 50% by weight aromatics, is to be extracted with recovered sulfolane containing 0.4 vol % aromatics in a 10-stage column contactor operating nearly adiabatically at 3 bar (gauge pressure). The extract will be fed to a 10-stage top-fed extract/paraffin stripper operating at 1 bar gauge to recover 98 percent of the aromatics with no more than 500 ppm by weight of nonaromatics. The catalytic reformate at 90°C is fed into the extractor at three stages up from the bottom, and the recovered sulfolane leaving the bottom of a solvent recovery tower at 185°C is cross-exchanged with the extract stream leaving the bottom of the extractor before being fed to the top of the extractor at 105°C. Extract reflux is returned from the paraffin stripper's condenser to the bottom of the extractor with subcooling to 105°C.

1. What solvent flow and stripper reboiler duty are required to achieve the performance specifications, and what are the extract reflux rate and composition?

2. If the required aromatics recovery is increased to 99 percent, what is the effect on solvent flow and stripper reboiler duty?

In real-world commercial catalytic reformate streams, a wide range of aromatic and nonaromatic hydrocarbons must be considered, and the liquid-liquid extraction and distillation simulation becomes quite complicated. In addition, real-world applications of sulfolane extraction normally add a few percent of water to the sulfolane to reduce its pure-component freezing point of 27 to 28°C during shipping and storage [Kosters, Chap. 18.2.3 in *Handbook of Solvent Extraction*, Lo, Baird, and Hanson, eds. (Wiley, 1983; Krieger, 1991)]. Also, in many processes, steam is injected into the bottom of the solvent recovery tower to help strip the aromatics (i.e., the tower is both steam-stripped and reboiled). This also allows operation of the recovery tower at higher pressures without incurring (excessive) solvent thermal degradation. In a real-world process, water also may be used to wash the raffinate to recover solvent. To simplify the problem for this example, however, we model the aromatics as toluene and the NAs as *n*-heptane, consider only sulfolane as the extraction solvent, and do not include water in the calculations—to reduce the problem to a simple ternary system for illustration.

As in Example 4, the EXTRACT block in the Aspen Plus process simulation program (version 12.1) is used to model this problem, but any of a number of process simulation programs such as mentioned earlier may be used for this purpose. The first task is to obtain an accurate fit of the liquid-liquid equilibrium (LLE) data with an appropriate model, realizing that liquid-liquid extraction simulations are very sensitive to the quality of the LLE data fit. The NRTL liquid activity-coefficient model [Eq. (15-27)] is utilized for this purpose since it can represent a wide range of LLE systems accurately. The regression of the NRTL binary interaction parameters is performed with the Aspen Plus Data Regression System (DRS) to ensure that the resulting parameters are consistent with the form of the NRTL model equations used within Aspen Plus.

Since the extractor operates nearly isothermally only slightly above and below 100°C, the 100°C data of De Fre and Verhoeve [*J. Appl. Chem. Biotechnol.*, **26**, pp. 1-19 (1976)] are used as the basis for the toluene + *n*-heptane + sulfolane LLE. Because of the liquid-liquid miscibility gap for the *n*-heptane + sulfolane binary, the NRTL α_{ij} parameter for this pair is given a value of 0.2. The NRTL

TABLE 15-10 NRTL Binary Interaction Parameters for Example 5

Component <i>i</i>	Component <i>j</i>	<i>b_{ij}</i> , K
<i>n</i> -Heptane	Toluene	23.2040
Toluene	<i>n</i> -Heptane	-34.3180
Toluene	Sulfolane	238.952
Sulfolane	Toluene	203.243
<i>n</i> -Heptane	Sulfolane	1476.41
Sulfolane	<i>n</i> -Heptane	719.006

$\tau_{ij} = b_{ij}/T$ (K); $\alpha_{ij} = 0.2$ for *n*-heptane + sulfolane; $\alpha_{ij} = 0.3$ for toluene + sulfolane and for *n*-heptane + sulfolane. Aspen Plus regression parameters a_{ij} , d_{ij} , e_{ij} , and f_{ij} are set to zero; $c_{ij} = \alpha_{ij}$; $\tau_{ii} = 0$; and $G_{ii} = 1$.

α_{ij} parameters for toluene + sulfolane and *n*-heptane + toluene are allowed to remain at the default value of 0.3 because of their low levels of nonideality. The temperature dependence of α_{ij} is set to zero (Aspen Plus parameter $d_{ij} = 0$). In Aspen Plus, the τ_{ij} parameter may be regressed as a function of temperature by using the expression $\tau_{ij} = a_{ij} + b_{ij}/T + e_{ij} \ln T + f_{ij}/T$. In this example, all the regression parameters are set to zero except b_{ij} . The component activity coefficients are chosen as the objective function for the regression to obtain a fit that models the liquid-liquid *K* values closely, generally found to be within 5 to 10 percent in this case. The resulting b_{ij} binary parameters given in Table 15-10 are then entered into the properties section of the Aspen Plus flow sheet simulation. Pure-component properties were taken from the standard Aspen Plus pure-component databases supplied with the program.

The major unit operations in the sulfolane process usually include an extractor, paraffin stripper, solvent recovery tower, raffinate wash tower, solvent regenerator, and numerous heat exchangers; but for the purposes of this example, the simulation includes only the extractor, paraffin stripper, and extract/recovered solvent cross-exchanger—the portion of the flow sheet shown in Fig. 15-2 outlined by dotted lines. It should be recognized that the exclusion of the solvent recovery tower ignores the highly interactive behavior of the extractor, stripper, and recovery tower; but this is done here to simplify the analysis for the purposes of illustration. Note that the stripper's condenser is modeled as a separate Aspen Plus HEATER block rather than being included in the stripper block, because the Aspen Plus RADFRAC multistage distillation block used to model the stripper requires some distillate reflux if a condenser is included within the block, and generally none is required for the top-fed stripper in the sulfolane process. As a result, the stripper RADFRAC block is specified with no condenser. Also note that in the sulfolane process, the sulfolane solvent enters the top of the extractor since it is denser than the catalytic reformate feed stream.

The 40,000 kg/h of catalytic reformate fed to stage 7 (counting from the top according to the convention in the EXTRACT block) is modeled as 50/50 *n*-heptane/toluene on a mass basis, and the residual aromatic content of the recovered sulfolane fed to the top of the extractor is 0.4 vol % toluene as given

in the problem statement. As an initial guess, the sulfolane rate to the extractor was set at 120,000 kg/h or a solvent-to-feed ratio of 3.0 since depending on the feedstock, solvent-to-feed ratios can range from about 2.0 to 4.0 (Huggins, "Sulfolane Extraction of Aromatics," Paper 67C, AIChE, Spring National Meeting, Houston, March 1997). In the EXTRACT block, sulfolane must be specified as the key component in the first liquid phase, and *n*-heptane must be specified as the key component in the second liquid phase, since the EXTRACT block requires that the first liquid be the one exiting the bottom of the extractor. A constant-temperature profile of 105°C in the extractor is entered as an initial estimate. The rest of the block parameters (convergence, report, and miscellaneous block options) are allowed to remain at their default values.

The paraffin stripper RADFRAC block is specified with feed to the first of 10 stages, a reboiler but no condenser, a 1-bar gauge top pressure, no internal pressure drop, and a molar boil-up ratio (boil-up rate/bottoms rate) of 0.2 as an initial guess. An internal RADFRAC design specification is entered to vary the boil-up ratio from 0.10 to 0.30 to achieve a mass purity of 500 ppm *n*-heptane in the stripper bottoms on a sulfolane solvent-free basis. To aid RADFRAC convergence, the standard algorithm was changed to Petroleum/Wide-boiling (Sum-Rates) because of the large volatility difference between the hydrocarbons and the sulfolane solvent.

A separate flow sheet Design Spec block (termed a *controller* block in some other simulators) is entered to vary the solvent feed rate to the extractor to achieve the required 98 percent toluene recovery. In addition, the extract reflux stream is called out as the flow sheet tear stream in a Wegstein convergence block to provide proper block sequencing in the simulation. (This is a numerical technique used to accelerate convergence to a solution.) Since the EXTRACT block will not execute with a zero extract reflux flow to the bottom of the extractor, an initial guess is required for that stream: 10,000 kg/h of 50/50 by weight *n*-heptane/toluene at 100°C is chosen.

During simulation execution, we found that reflux tear stream convergence with the default Wegstein parameters is very oscillatory, with no convergence even with maximum iterations raised to 200. As a result, significant damping needs to be provided in the convergence block. We raised the bounds of the Wegstein *q* acceleration parameter to be between 0.75 and 1.0 for nearly full damping, after which flow sheet convergence was achieved in less than 50 iterations of every reflux tear stream loop. We also found that good initial guesses and bounds on variables needed to be set to keep the simulation from converging to an aberrant solution that was not physically valid.

With these modifications, the result is that 125,500 kg/h of sulfolane feed to the extractor is required to recover 98 percent of the toluene in the simplified reformate feed. The stage-by-stage mass fraction profile in the extractor is given in Table 15-11, from which we can see that there is very little change in concentration in either phase from the feed stage downward. This is so because in our simplified example we have only a single NA hydrocarbon component (*n*-heptane) to deal with, so the benefit of a backwash section in the extractor below the feed is not apparent. In a real-world profile, however, concentrations of higher-boiling NAs would decrease from the feed point to the bottom of the extractor. Also given in Table 15-11 are stage-by-stage *K''* values, the separation factor (toluene with respect to *n*-heptane), and the extraction factor profiles in

TABLE 15-11 Stage Profiles for 98 Percent Recovery (Example 5)

Stage	Liquid 1 profile (extract) (mass fractions)			Liquid 2 profile (raffinate) (mass fractions)		
	<i>n</i> -Heptane	Toluene	Sulfolane	<i>n</i> -Heptane	Toluene	Sulfolane
1	0.02630	0.00624	0.96746	0.97239	0.01945	0.00815
2	0.02683	0.01145	0.96171	0.95565	0.03521	0.00914
3	0.02777	0.02031	0.95192	0.92796	0.06106	0.01098
4	0.02940	0.03519	0.93542	0.88354	0.10196	0.01450
5	0.03225	0.05955	0.90821	0.81568	0.16281	0.02151
6	0.03721	0.09769	0.86510	0.71952	0.24481	0.03568
7	0.04568	0.15309	0.80123	0.59750	0.33926	0.06324
8	0.04570	0.15323	0.80107	0.59680	0.33963	0.06357
9	0.04590	0.15419	0.79991	0.59524	0.34081	0.06395
10	0.04729	0.16016	0.79255	0.58397	0.34838	0.06766
Stage	<i>K''</i> values (mass fraction basis)			α_{ij}		$\mathcal{E}$
	<i>n</i> -Heptane	Toluene	Sulfolane	Toluene/ <i>n</i> -heptane	Toluene	
1	0.0270	0.321	118.7	11.87	2.02	
2	0.0281	0.325	105.2	11.58	1.73	
3	0.0299	0.333	86.7	11.12	1.73	
4	0.0333	0.345	64.5	10.37	1.73	
5	0.0395	0.366	42.2	9.25	1.73	
6	0.0517	0.399	24.2	7.72	1.73	
7	0.0765	0.451	12.7	5.90	1.66	
8	0.0766	0.451	12.6	5.89	5.90	
9	0.0771	0.452	12.5	5.87	5.91	
10	0.0810	0.460	11.7	5.68	5.88	

TABLE 15-12 Stream Compositions and Conditions (Example 5)

	Extract	Raffinate	Reflux	Stripper bottoms
98% Recovery—125,500 kg/h required solvent flow				
Wt. fraction <i>n</i> -heptane	0.04729	0.97239	0.57717	69 ppm
Wt. fraction toluene	0.16016	0.01945	0.41273	0.13765
Wt. fraction sulfolane	0.79255	0.00815	0.01010	0.86228
Total flow, kg/h	157,817	205,578	12,914	144,903
Temperature, °C	96.4	103.7	105.0	194.1
99% Recovery—212,800 kg/h required solvent flow				
Wt. fraction <i>n</i> -heptane	0.03821	0.98258	0.62353	44 ppm
Wt. fraction toluene	0.10444	0.00982	0.36100	0.08771
Wt. fraction sulfolane	0.85736	0.00759	0.01547	0.91225
Total flow, kg/h	247,592	20,344	15,155	232,437
Temperature, °C	98.9	103.9	105.0	215.9

the extractor. From these we can see that the separation factor for toluene with respect to *n*-heptane varies from about 6 at the bottom of the extractor to 12 at the top, and that the extraction factor is about 2 above the feed and about 6 below the feed. These separation factors are somewhat higher than the value of 4 or so normally seen in real-world aromatic extraction cases; this, too, is an artifact of the simplified ternary system used to model the sulfolane process.

Another result of the simulation is that a molar boil-up ratio of 0.180 is required in the stripper to achieve the bottoms mass purity of 500 ppm *n*-heptane considering only the hydrocarbons (solvent-free basis). This boil-up ratio corresponds to a reboiler duty of 3695 kW, or roughly 6700 kg/h of 12-bar gauge steam, and results in 12,914 kg/h of extract reflux for an extractor reflux-to-feed ratio of 0.323. Compositions and rates of the extract, raffinate, reflux, and stripper bottoms streams are given in Table 15-12.

To determine the solvent flow and other conditions required to achieve 99 percent toluene recovery, we merely need to change the specification of the recovery Design Spec block from 98 to 99 percent and reconverge the simula-

tion. With this change and an additional 180 total reflux tear stream iterations, the result is that 212,800 kg/h of sulfolane feed to the extractor is required, 1.7 times the amount needed for 98 percent toluene recovery. A molar boil-up ratio of 0.158 is required in the stripper to maintain the bottoms mass purity of 500 ppm *n*-heptane on a solvent-free basis, even lower than that for the 98 percent recovery case. Likewise, only a slightly higher extract reflux rate is required, 15,155 kg/h, for an extractor reflux-to-feed ratio of 0.379. However, this boil-up ratio corresponds to a reboiler duty of 7191 kW, or roughly 13,100 kg/h of 12-bar gauge steam, about 95 percent higher than for the 98 percent recovery case. The much higher stripper reboiler duty required for 99 percent recovery results from the significantly greater sulfolane feed rate, indicating that the sizes of the extractor and stripper as well as the energy consumption would need to be significantly greater for that increased recovery, probably making it uneconomical in most applications with a 10-stage extractor and stripper. Compositions and rates of the extract, raffinate, reflux, and stripper bottoms streams for the 99 percent recovery case are also given in Table 15-12.

LIQUID-LIQUID EXTRACTION EQUIPMENT

GENERAL REFERENCES: Seibert, "Extraction and Leaching," Chap. 14 in *Chemical Process Equipment: Selection and Design*, 2d ed., Couper et al., eds. (Elsevier, 2005); Robbins, Sec. 1.9 in *Handbook of Separation Techniques for Chemical Engineers*, 3d ed., Schweitzer, ed. (McGraw-Hill, 1997); Lo, Sec. 1.10 in *Handbook of Separation Techniques for Chemical Engineers*, 3d ed., Schweitzer, ed. (McGraw-Hill, 1997); *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994); *Science and Practice of Liquid-Liquid Extraction*, vol. 1, Thornton, ed. (Oxford, 1992); *Handbook of Solvent Extraction*, Lo, Baird, and Hanson, eds. (Wiley, 1983; Krieger, 1991); Laddha and Degaleesan, *Transport Phenomena in Liquid Extraction* (McGraw-Hill, 1978); and Treybal, *Liquid Extraction*, 2d ed. (McGraw-Hill, 1963).

EXTRACTOR SELECTION

The common types of commercially available extraction equipment and their general features are outlined in Table 15-13. The choice of extractor type depends upon many factors including the required number of theoretical stages or transfer units, required residence time (due to slow or fast extraction kinetics or limited solute stability), required production rate, tolerance to fouling, ease of cleaning, availability of the required materials of construction, as well as the ability to handle high or low interfacial tension, high or low density difference, and high or low viscosities. Other factors that influence the choice of extractor include familiarity and tradition (the preferences among designers and operating companies often differ), confidence in scale-up, height constraints, and, of course, the relative capital and operating costs. The flexibility of the extractor to adjust to changes in feed properties also can be an important consideration. For example, compared to a static extractor, a mechanically agitated extractor typically provides a greater turndown ratio (ability to handle a wider range of flow rates), and agitation intensity can be adjusted in the field as needed to accommodate changes in the feed over time. Other factors

that may be important include the ability to operate under pressure, to handle corrosive, highly toxic, or flammable materials, and to meet maintenance requirements, among many other possible considerations. Experience with applications similar to the current application and the use of pilot-plant testing play important roles in equipment selection. Pilot testing can address critical issues including demonstration of separation capabilities and equipment scale-up. The simplest extractor design that can meet the process requirements generally will be selected over other competing designs.

Figure 15-29 outlines the decision process recommended by Robbins [Sec. 1.9 in *Handbook of Separation Techniques for Chemical Engineers*, 3d ed., Schweitzer, ed. (McGraw-Hill, 1997)]. As an aid to decision making, Robbins recommends characterizing the feed by measuring a flooding curve using a 1-in.-diameter reciprocating-plate (Karr column) miniplant extractor. This is a plot of maximum specific throughput (very close to flooding) versus agitation intensity in the Karr column. The position of the resulting curve may be used to identify the type of extractor best suited for commercial development, as illustrated in Fig. 15-30. The flooding curve results reflect the liquid-liquid dispersion behavior of the system, and so they can point to options most in line with those properties. The test typically requires 40 to 200 L of feed materials (10 to 50 gal).

A number of equipment selection guides have been published. Pratt and Hanson [Chap. 16 in *Handbook of Solvent Extraction*, Lo, Baird, and Hanson, eds. (Wiley, 1983; Krieger, 1991)] provide a detailed comparison chart for 20 equipment types considering 14 characteristics. Pratt and Stevens [Chap. 8 in *Science and Practice of Liquid-Liquid Extraction*, vol. 1, Thornton, ed. (Oxford, 1992)] modified the Pratt and Hanson selection guide to include solvent volatility and flammability design parameters. Stichlmair [*Chem. Ing. Tech.*, **52**(3) pp. 253–255 (1980)] and Holmes, Karr, and Cusack (AIChE

TABLE 15-13 Common Liquid-Liquid Extraction Equipment and Applications

Type of extractor	General features	Fields of industrial application
Static extraction columns Spray column Baffle column Packed column (random and structured packing) Sieve tray column	Deliver low to medium mass-transfer efficiency, simple construction (no internal moving parts), low capital cost, low operating and maintenance costs, best suited to systems with low to moderate interfacial tension, can handle high production rates	Petrochemical Chemical Food
Mixer-settlers Stirred-vessels with integral or external settling zones	Can deliver high stage efficiencies with long residence time, can handle high-viscosity liquids, can be adjusted in the field (good flexibility), with proper mixer-settler design can handle systems with low to high interfacial tension, can handle very high production rates	Petrochemical Nuclear Fertilizer Metallurgical
Rotary-agitated columns Rotary disc contactor (RDC) Asymmetric rotating disc (ARD) contactor Oldshue-Rushton column Scheibel column Kühni column	Can deliver moderate to high efficiency (many theoretical stages possible in a single column), moderate capital cost, low operating cost, can be adjusted in the field (good flexibility), suited to low to moderate viscosity (up to several hundred centipoise), well suited to systems with moderate to high interfacial tension, can handle moderate production rates	Petrochemical Chemical Pharmaceutical Metallurgical Fertilizer Food
Reciprocating-plate column Karr column	Can deliver moderate to high efficiency (many theoretical stages possible in a single column), moderate capital cost, low operating cost, can be adjusted in the field (good flexibility), well suited to systems with low to moderate interfacial tension including mixtures with emulsifying tendencies, can handle moderate production rates	Petrochemical Chemical Pharmaceutical Metallurgical Food
Pulsed columns Packed column Sieve tray column	No internal moving parts, can deliver moderate to high efficiency, can handle moderate production rates, well suited to highly corrosive or toxic feeds requiring a hermetically sealed system	Nuclear Petrochemical Metallurgical
Centrifugal extractors	Allow short contact time for unstable solutes, minimal space requirements (minimal footprint and height), can handle systems with low density difference or tendency to easily emulsify	Petrochemical Chemical Pharmaceutical Nuclear

Summer National Meeting, August 1987) compared performance characteristics of various equipment designs in the form of a Stichlmair plot. This is a plot of typical mass-transfer efficiency versus characteristic specific throughput (for combined feed and solvent flows) for various types of extractors. Figure 15-31 represents typical performance data generated by using various small-diameter (2- to 6-in, equal to 5- to 15-cm) extractors. This type of plot is intended for use in comparing the relative performance of different extractor types and can be very helpful in this regard. It should not be used for design purposes.

Volumetric efficiency is another characteristic used to compare the different types of extractors. It can be expressed as the product of specific throughput (including feed and extraction solvent) in total volumetric flow rate per unit area (or a characteristic liquid velocity) times the number of theoretical stages achieved per unit length of extractor. It has the units of stages per unit time, or simply reciprocal time (h^{-1}). Thus, volumetric efficiency is inversely proportional to the volume of the column needed to perform a given separation. The Karr reciprocating-plate extractor provides relatively high volumetric efficiency, as it has both a high capacity per unit area and a high number of stages per meter. The Scheibel rotary-impeller column also can provide a high number of stages per meter, but the column throughput typically is less than that of a Karr column, so volumetric efficiency is less. Thus, for a given separation a Scheibel column might be somewhat shorter than a Karr column, but it will need to have a larger diameter to process the same flow rate of feed and extraction solvent. The sieve plate extractor generally exhibits moderate to high throughput, but the number of stages per meter typically is low. The Graesser raining-bucket contactor exhibits low to moderate throughput, but is reported to have a high separating capability in certain applications.

The ability of an extractor to tolerate the presence of surface-active impurities also may be an important factor in choosing the most

appropriate design. Karr, Holmes, and Cusack [*Solvent Extraction and Ion Exchange*, 8(30), pp. 515-528 (1990)] investigated the performance of small-diameter agitated columns and found that the performance of a rotating-disk contactor (RDC) declined faster on addition of trace surface-active impurities compared to the Karr or Scheibel column. The test results indicate that care should be taken when comparing pilot tests of different types of extractors when the data were generated by using high-purity materials. The presence of surface-active impurities can lower column capacity by 20+ percent and efficiency by as much as 60 percent.

Production capacity also may be a deciding factor, since some extractors are available only in small to moderate sizes suitable for low to moderate production rates, as in specialty chemical manufacturing, while others are available in very large sizes designed to handle the very high production rates needed in the petroleum and petrochemical industries. An estimate of relative production rates (feed plus solvent) for selected extractors is given in Table 15-14. Note that the numbers are intended to represent approximate maximum values for a rough comparison. The actual values likely will vary depending upon the particular application. Keep in mind that the relative mass-transfer performance of the various designs is not represented in Table 15-14, and that very large-diameter columns are limited as to how tall they can be built.

HYDRODYNAMICS OF COLUMN EXTRACTORS

Flooding Phenomena The hydraulic capacity of a countercurrent extractor is constrained by breakthrough of one liquid phase into the discharge stream of the other, a condition called *flooding*. The point at which an extractor floods is a function of the design of the internals (as this affects the pressure drop and holdup characteristics of the extractor), the solvent-to-feed ratio and physical properties (as

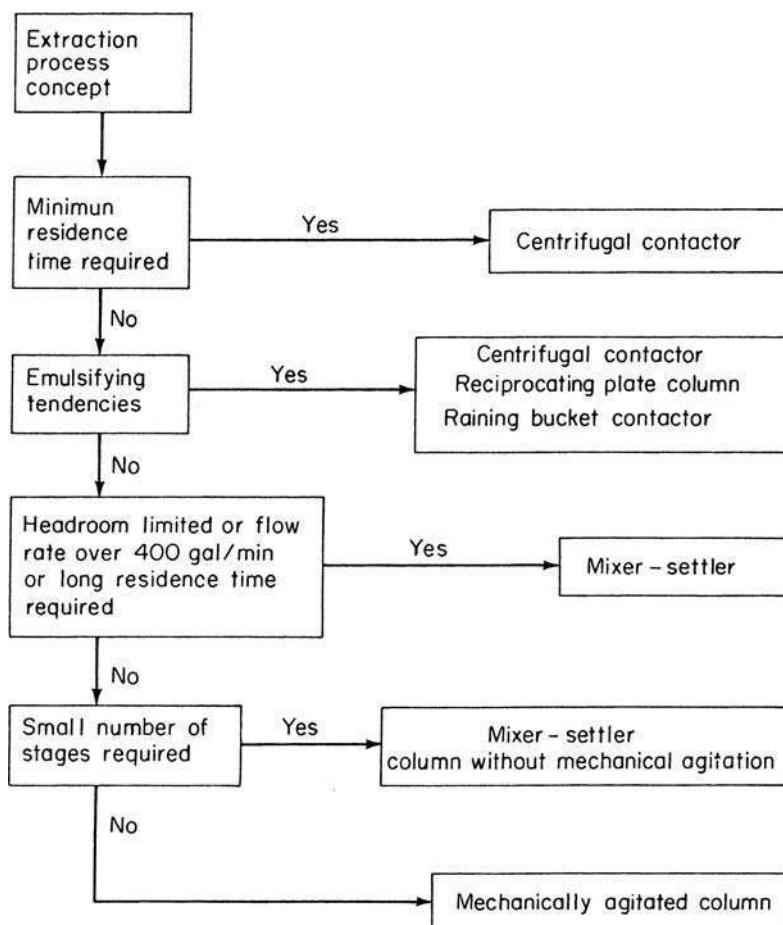


FIG. 15-29 Decision guide for extractor selection. [Reprinted from Robbins, Sec. 1.9 in Handbook of Separation Techniques for Chemical Engineers, 3d ed., Schweitzer, ed. (McGraw-Hill, 1997), with permission. Copyright 1997 McGraw-Hill, Inc.]

this affects the liquid-liquid dispersion behavior), the agitation intensity (if agitation is used), and the specific throughput. The latter often is expressed in terms of the volumetric flow rate per cross-sectional area; or, equivalently, in terms of liquid velocity. A plot of the maximum throughput that can be sustained just prior to flooding versus a key operating variable is called a flooding curve. Ideally, extractors are designed to operate near flooding to maximize productivity. In practice, however, many new column extractors are designed to operate at 40 to 60 percent of the predicted flood point because of uncertainties in the design, process impurity uncertainties, and to allow for future capacity increases. This practice varies from one type of extractor to another and one designer to another. In a static extraction column, countercurrent flow of the two liquid phases is maintained by virtue of the difference in their densities and the pressure drop through the equipment. Only one of the liquids may be pumped through the equipment at any desired flow rate or velocity; the maximum velocity of the other phase is then fixed by the flood point. If an attempt is made to exceed this hydraulic limit, the extractor will flood.

In extraction equipment, flooding may occur through a variety of mechanisms [Seibert, Bravo, and Fair, *ISEC '02 Proc.*, **2**, pp. 1328–1333 (2002)]:

1. Excessive flow rates of either dispersed-phase or continuous-phase, or high agitation intensity, cause dispersed-phase holdup or population density to exceed the volumetric capacity of the equipment.

2. Excessively high continuous-phase flow rate causes excessive entrainment of dispersed phase into the continuous-phase outlet.

3. Inadequate drop coalescence causes formation of dispersion bands or layers of uncoalesced drops that entrap continuous phase between them. The continuous phase can then be entrained into the wrong outlet.

4. Operation at a high ratio of dispersed phase to continuous phase results in phase inversion. (See "Liquid-Liquid Dispersion Fundamentals.")

5. Operating too close to the liquid-liquid phase boundary causes complete miscibility during an upset. A slight change in solvent or feed rates or an increase in solute concentration in the feed can potentially cause formation of a single phase.

6. In sieve tray columns, excessive orifice and/or downcomer pressure drop within the extractor causes formation of large coalesced layers that back up and overflow the trays.

7. Poor interface control allows the main liquid-liquid interface to leave the extractor. This may result from inadequate size of interface flow control valves, or operation with internals that provide inverse control responses such as those observed with sieve tray extractors. (See "Process-Control Considerations.")

8. Mechanical problems such as plugging of internals or outlet flow control valves can develop.

Accounting for Axial Mixing Differential-type column extractors are subject to axial (longitudinal) mixing, also called axial dispersion

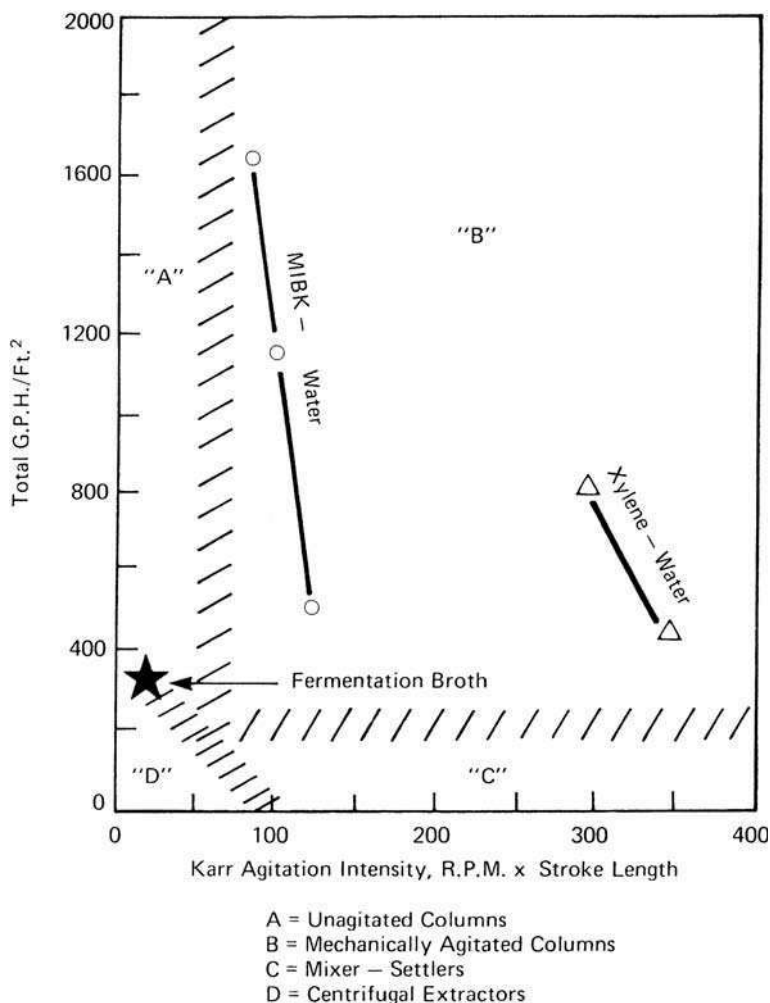


FIG. 15-30 Typical Karr column flooding characteristics. Example flooding data are shown for two applications involving MIBK + water and xylene + water (flooding occurs to the right of the indicated flooding curve). A data point for extraction of a fermentation broth is indicated by the star. Results will vary depending upon process variables including solute concentration, the presence of other solutes, and temperature. [Reprinted from Robbins, Sec. 1.9 in *Handbook of Separation Techniques for Chemical Engineers*, 3d ed., Schweitzer, ed. (McGraw-Hill, 1997), with permission. Copyright 1997 McGraw-Hill, Inc.]

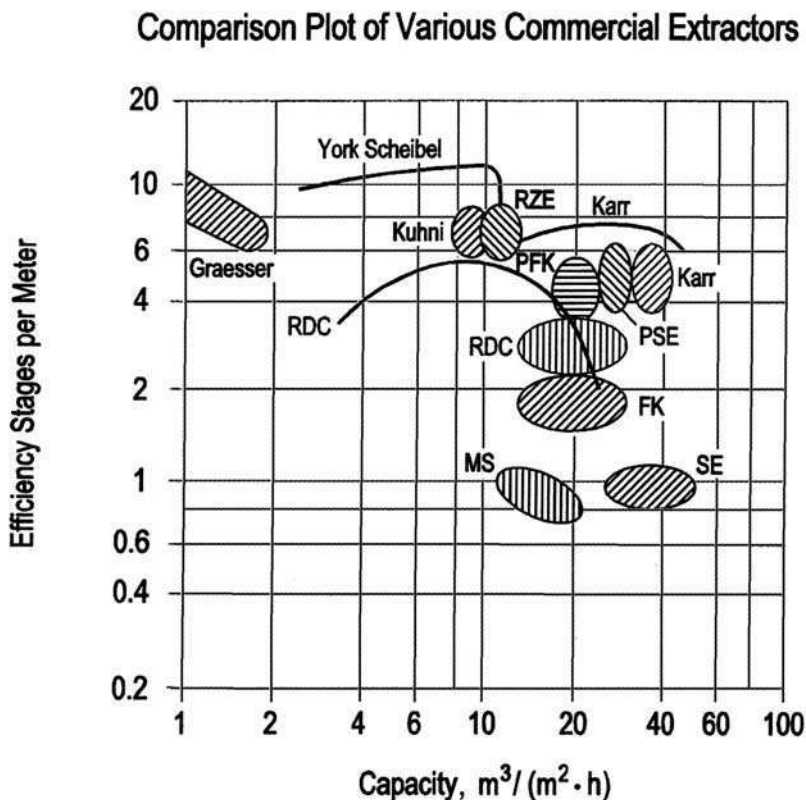
and generally referred to as backmixing. This condition refers to a departure from uniform plug flow of the swarm of dispersed drops as drops rise or fall in the column, as well as any departure from plug flow of continuous phase in the opposite direction. As a result of axial mixing, the elements of the dispersed phase and the continuous phase exhibit a distribution of residence times within the equipment, and this decreases the effective or overall concentration driving force in the contactor. Because of this effect, the actual column must be taller than simple application of an ideal, plug flow model would indicate. When one is approaching the design of a contactor, factors that may contribute to axial mixing should be considered so that measures might be taken to reduce their effects. This may involve design of baffles to help direct the liquid traffic within the column. Also, if the transfer of solute occurs such that the continuous phase is significantly denser at the top of an extraction column than at the bottom, this may encourage circulation of continuous phase, and it may be advisable to switch the phase that is dispersed. For more information on this effect, see Holmes, Karr, and Baird, *AIChE J.*, **37**(3), pp. 360-366

(1991); and Aravamudan and Baird, *AIChE J.*, **42**(8), pp. 2128-2140 (1996).

Axial mixing effects commonly are taken into account by using a diffusion analogy and an axial mixing coefficient E , also called the longitudinal dispersion coefficient or eddy diffusivity, to account for the spreading of the concentration profiles. At steady state, the conservation equation has the general form

$$E \frac{\partial^2 C}{\partial z^2} + V \frac{\partial C}{\partial z} + k_o a (C - C^*) = 0 \quad (15-106)$$

where V is phase velocity, k_o is an overall mass-transfer coefficient, C is solute concentration (mass or moles per unit volume), and the superscript asterisk denotes equilibrium. By using Eq. (15-106) as a foundation, the required height of extractor may be calculated from a simplified plug flow model plus application of a correction factor expressed as a function of E or a Péclet number $Pe = Vb/E$, where b is a characteristic equipment



Key:

Graesser	= Raining Bucket Extractor	RDC	= Rotating Disc Contactor
MS	= Mixer Settler	RZE	= Agitated Cell Extractor
SE	= Sieve Plate Extractor	Karr	= Karr Reciprocating Plate Column
FK	= Random Packed Column	Kuhni	= Kuhni Extractor
PFK	= Pulsed Packed Column	York-Scheibel	= Scheibel Column
PSE	= Pulsed Sieve Plate Column		

Source: Shaded Area = 
Curves = 

(1) J. Stichlmair, University of Essen, Wiesbaden, Germany, "Performance and Cost Comparison of Various Equipment Designs for Liquid-Liquid Extraction", *Chemie-Ingenieur-Technik* 52(3), 1980

(2) T. Holmes, R. Cusack, A. Karr, The Otto H. York Company, AIChE Summer National Meeting, August 17, 1987

FIG. 15-31 Modified Stichlmair chart. (Courtesy of Koch Modular Process Systems.)

dimension. The required values of E must be determined by experiment. A variety of models and data correlations have been developed for various types of column extractors. For detailed discussion, see Sleicher, *AIChE J.*, **5**(2), pp. 145–149 (1959); Vermeulen et al., *Chem. Eng. Prog.*, **62**(9), pp. 95–102 (1966); and Li and Zeigler, *Ind. Eng. Chem.*, **59**(3), pp. 30–36 (1967). Also see the detailed discussions in Laddha and Degaleesan, *Transport Phenomena in Liquid Extraction* (McGraw-Hill, 1978); Pratt and Baird, Chap. 6 in *Handbook of Solvent Extraction*, Lo, Baird, and Hanson, eds. (Wiley, 1983;

Krieger, 1991); and *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994). The method used by Becker [*Chem. Eng. Technol.*, **26**(1), pp. 35–41 (2003)] is discussed in "Static Extraction Columns."

Computational fluid dynamics (CFD) simulations are beginning to be developed for certain types of extractors to better understand flow patterns in column extractors. The simulation of two-liquid-phase flows around complex internals is an active research area. For an example of this approach, see the discussion of CFD calculations for a

TABLE 15-14 Estimated Maximum Production Rate for Selected Extractors

Extractor type	Maximum ^a specific throughput		Maximum ^b diameter (typical)	Estimated maximum production rate	
	m ³ /h/m ²	gal/h/ft ²	m	m ³ /h	gal/min
Mixer-settler ^c	30	750	10	~2,400	10,000
Baffle tray column	60	1,500	5	~1,200	5,200
Sieve plate column	50	1,200	5	~1,000	4,300
Packed column	50	1,200	5	~1,000	4,300
Spray column ^d	70	1,700	4	~900	4,000
Rotating disk contactor	35	850	4	~450	1,900
Kühni rotating-impeller column ^e	40	1,000	3	~280	1,200
Karr reciprocating-plate column	40	1,000	3	~280	1,200
Scheibel rotating-impeller column	25	600	3	~200	800
Graesser raining-bucket contactor	10	250	3	~70	300

^aTypical maximum value for dispersed + continuous phase flow rates. The actual value for a given application will depend upon physical properties and may be much lower.

^bTypical value. Larger diameters may be possible.

^cThroughput and equivalent diameter are based on mixer-settler footprint.

^dLarger diameters possible but not recommended due to severe backmixing.

^eHigher throughput may be achieved by increasing the column open area.

rotating-disk contactor by Modes and Bart [*Chem. Eng. Technol.*, **24**(12), pp. 1242–1244 (2001)].

Liquid Distributors and Dispersers It should be recognized that the performance of a column extractor can be significantly affected by how uniformly the feed and solvent inlet streams are distributed to the cross section of the column. The requirements for distribution and redistribution vary depending upon the type of column internals (packing, trays, agitators, or baffles) and the impact of the internals on the flow of dispersed and continuous phases within the column. Important considerations in specifying a distributor include the number of holes and the hole pattern (geometric layout), hole size, number of downcomers or upcomers (if used) and their placement, the maximum to minimum flow rates the design can handle (turndown ratio), and resistance to fouling. Various types of liquid distributors are available, including sieve tray dispersers and ladder-type pipe distributors designed to give uniform distribution of drops across the column cross section. (See “Packed Columns” and “Sieve Tray Columns” under “Liquid-Liquid Extraction Equipment” for more information about these. The height of the coalesced layer on a disperser plate may be calculated by using the method described in “Sieve Tray Columns.”) Ring-type distributors also are used, primarily for agitated extractors. Equipment vendors should be consulted for additional information.

Typical hole sizes for distributors and dispersers are between 0.05 in (1.3 mm) and 0.25 in (6.4 mm). Small holes should be avoided in applications where the potential for plugging or fouling of the holes is a concern. For plate dispersers, the holes should be spaced no closer than about 3 hole diameters to avoid coalescence of drops emerging from adjacent holes. Design velocities for liquid exiting the holes generally are in the range of 0.5 to 1.0 ft/s (15 to 30 cm/s). Several methods have been proposed for more precisely specifying the design velocities. For detailed discussion, see Kumar and Hartland, Chap. 17 in *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994), pp. 631–635; Ruff, *Chem. Ing. Tech.*, **50**(6), pp. 441–443 (1978); and Laddha and Degaleesan, *Transport Phenomena in Liquid Extraction* (McGraw-Hill, 1978), Chap. 11, pp. 307–310. These methods are relevant for the design of distributors/dispersers used in all types of column extractors. The liquid should issue from the hole as a jet that breaks up into drops. The jet should yield a drop size distribution that provides good interfacial area, with an average drop size smaller than the maximum given by $d_{\max} = [\sigma/(\Delta\rho g)]^{0.5}$, but without creating small secondary drops that cause entrainment problems or formation of an emulsion. (See “Size of Dispersed Drops” in “Liquid-

Liquid Dispersion Fundamentals.”) As a general guideline, the maximum recommended design velocity corresponds to a Weber number of about 12:

$$V_{o,\max} \approx \left(\frac{12\sigma}{d_o \rho_d} \right)^{1/2} \quad (15-107)$$

The minimum Weber number that ensures jetting in all the holes is about 2. It is common practice to specify a Weber number between 8 and 12 for a new design. For a detailed discussion of fundamentals, see Homma et al., *Chem. Eng. Sci.*, **61**, pp. 3986–3996 (2006).

It is well established that the dispersed phase must issue cleanly from the holes. This requires that the material of the pipe or disperser plate be preferentially wetted by the continuous phase (requiring the use of plastics or plastic-coated trays in some instances), or that the dispersed phase issue from nozzles projecting beyond the surface. For plate dispersers, these may be formed by punching the holes and leaving the burr in place [Mayfield and Church, *Ind. Eng. Chem.*, **44**(9), pp. 2253–2260 (1952)]. Once the design velocity is set, the number of holes is given by

$$N_{\text{holes}} = \frac{Q_d}{A_o V_o} \quad (15-108)$$

where Q_d is the total volumetric flow rate of dispersed phase and A_o is the cross-sectional area of a single hole.

STATIC EXTRACTION COLUMNS

Common Features and Design Concepts Static extractors include spray-type, packed, and trayed columns often used in the petrochemical industries (Fig. 15-32). They offer the advantages of (1) availability in large diameters for very high production rates, (2) simple operation with no moving parts and associated seals, (3) requirement for control of only one operating interface, and (4) relatively small required footprint compared to mixer-settler equipment. Their primary disadvantage is low mass-transfer efficiency compared to that of mechanically agitated extractors. This usually limits applications to those involving low viscosities (less than about 5 cP), low to moderate interfacial tensions (typically 3 to 20 dyn/cm equal to 0.003 to 0.02 N/m), and no more than three to five equilibrium stages. Although the spray column is the least efficient static extractor in terms of mass-transfer performance, due to considerable backmixing effects, it finds

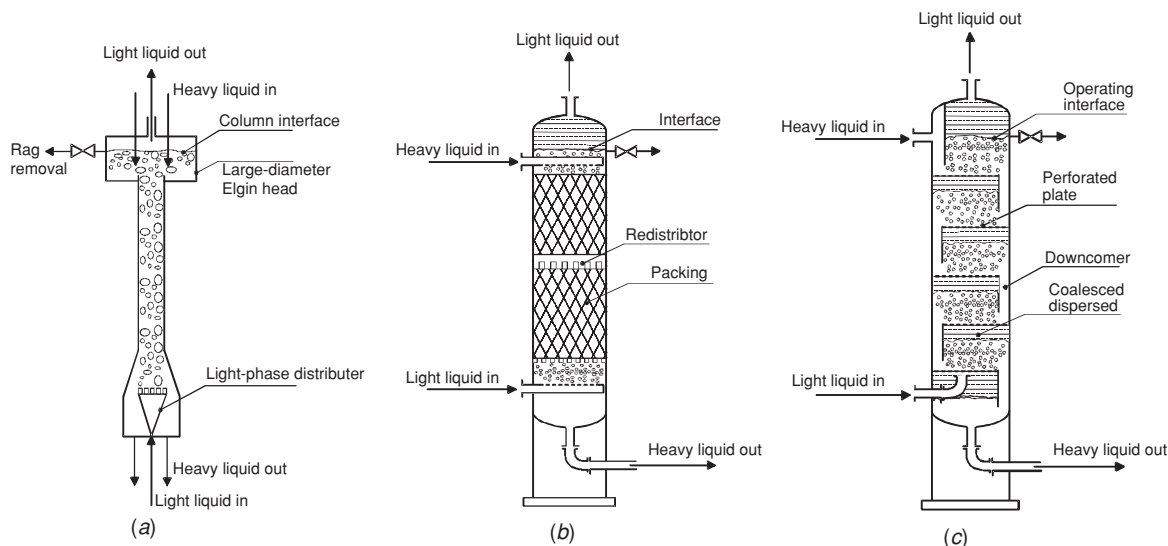


FIG. 15-32 Schematic of common static extractors. (a) Spray column. (b) Packed column. (c) Sieve tray column.

use in processing feeds that would easily foul other equipment. Packed and trayed column designs provide improved mass-transfer performance by limiting backmixing.

An understanding of the general hydraulics of a static contactor is necessary for estimating the diameter and height of the column, as this affects both capacity and mass-transfer efficiency. Accurate evaluations of characteristic drop diameter, dispersed-phase holdup, slip velocity, and flooding velocities usually are necessary. Fortunately, the relative simplicity of these devices facilitates their analysis and the approaches taken to modeling performance.

Choice of Dispersed Phase In general, formation of dispersed drops is preferred over formation of films or rivulets in order to maximize contact area and mass transfer. Static extractors generally are designed with the majority phase dispersed in order to maximize interfacial area needed for mass transfer; i.e., the phase with the greatest flow rate entering the column generally is dispersed. The choice of dispersed phase also depends upon the relative viscosity of the two phases. If one phase is particularly viscous, it may be necessary to disperse that phase.

Drop Size and Dispersed-Phase Holdup Various models used to estimate the size of dispersed drops in static extractors are listed in Table 15-15. Also see "Size of Dispersed Drops" under "Liquid-Liquid Dispersion Fundamentals." Measurements of dispersed-phase holdup within a column-type extractor often are made by stopping all flows in and out of the extractor and measuring the change in the main interface level. This technique can be prone to significant experimental error as a result of end effects, static holdup present in small laboratory packings, inaccurate measurement of the baseline interface level, and holdup variations within a column as flooding conditions are approached. Examples of models for prediction of holdup are provided in Table 15-16. Additional models are given in *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994). In general, an implicit calculation of the dispersed-phase holdup is usually encountered. One must be very careful in evaluating the roots of these equations, especially in the region of high dispersed-phase holdup ($\phi_d > 0.2$).

Interfacial Area The mass-transfer efficiency of most extraction devices is proportional to the area available for mass transfer (neglecting any axial mixing effects). As discussed in "Liquid-Liquid Dispersion Fundamentals," for the general case where the dispersed phase travels through the column as drops, an average liquid-liquid interfacial area can be calculated from the Sauter mean drop diameter and dispersed-phase holdup:

$$a = \frac{6\epsilon\phi_d}{d_p} \quad (15-109)$$

In most cases, the drop size distribution is not known.

Drop Velocity and Slip Velocity The hydraulic characteristics of a static extractor depend upon drop diameter, liquid velocities, and physical properties. The average velocity of a dispersed-phase drop (V_{drop}) and the interstitial velocity of the continuous phase V_{ic} are given by

$$V_{drop} = \frac{V_d}{\epsilon\phi_d} \quad (15-110)$$

$$V_{ic} = \frac{V_c}{\epsilon(1 - \phi_d)} \quad (15-111)$$

where V_d = superficial velocity of dispersed phase
 V_c = superficial velocity of continuous phase
 ϕ_d = fraction of void volume occupied by dispersed phase
 ϵ = void fraction of column ($\epsilon = 1.0$ for sprays and sieve trays)

The relative velocity between the counterflowing phases is referred to as the slip velocity and defined by

$$V_s = V_{drop} + V_{ic} = \frac{V_d}{\epsilon\phi_d} + \frac{V_c}{\epsilon(1 - \phi_d)} \quad (15-112)$$

The slip velocity of a dispersed-phase drop of diameter d_p can be estimated from a balance of gravitational, buoyancy, and frictional forces:

$$F_{buoyancy} - F_{gravity} - F_{drag} = 0 \quad (15-113)$$

$$F_{buoyancy} = \rho_c \left(\frac{\pi}{6} d_p^3 \right) g \quad (15-114)$$

$$F_{gravity} = \rho_d \left(\frac{\pi}{6} d_p^3 \right) g \quad (15-115)$$

$$F_{drag} = \frac{1}{2} C_D \rho_c \left(\frac{\pi}{4} d_p^2 \right) V_{so}^2 \quad (15-116)$$

TABLE 15-15 Example Drop Diameter Models for Static Extractors

Example	Eq.	Comments	Ref.
$d_p = 1.15\eta \sqrt{\frac{\sigma}{\Delta\rho g}}$, $\eta = 1.0$ for no mass transfer and $c \rightarrow d$, $\eta = 1.4$ for $d \rightarrow c$	1	Spray, packing, and sieve tray	1
$d_p = 0.12D_h We_c^{-0.5} Re_c^{0.15}$, developed with no mass transfer, We_c and Re_c are calculated based on slip velocity	2	SMV structured packing	2
$d_p = C_p \sqrt{\frac{\sigma}{\Delta\rho g}}$, $C_p = 1$ for $\rho_d < \rho_c$, $C_p = 0.8$ for $\rho_d > \rho_c$, developed with no mass transfer	3	Packing	3
$d_p = 1.09 \sqrt{\frac{\sigma}{\Delta\rho g}} \left(1 + 700 \frac{V_{sl}}{\sigma} \right)$, developed with no mass transfer	4	Packing	4
$d_p = 0.74C_\psi \sqrt{\frac{\sigma}{\Delta\rho g}} \left(\frac{\Delta\rho \rho_d \sigma}{\rho_w^2 \sigma_w} \right)^{-0.12}$, $C_\psi = 1$ for no mass transfer, $C_\psi = 0.84$ for $c \rightarrow d$, $C_\psi = 1.23$ for $d \rightarrow c$	5	Packing	5
$d_p = \frac{C_\psi}{\frac{1}{(6d_o\sigma/\Delta\rho g)^{1/3}} + \frac{1}{2.04(12\sigma/\rho_d V_o^2)}}$, $C_\psi = 1.0$ for $c \rightarrow d$ and no mass transfer, $C_\psi = 1.06$ for $d \rightarrow c$	6	Spray nozzles	5
$d_p = d_o E\ddot{o}_o^{-0.35} \left[0.80 + \exp \left(-2.73 \times 10^{-2} \frac{We_o}{E\ddot{o}_o} \right) \right]$ $E\ddot{o}_o = \frac{d_o^2 \Delta\rho g}{\sigma}$, $We_o = \frac{\rho_d d_o V_o^2}{\sigma}$	7	Perforated plate	6

References:

- Seibert and Fair, *Ind. Eng. Chem. Res.*, **27**(3), pp. 470–481 (1988).
- Streiff and Jancic, *Ger. Chem. Eng.*, **7**, pp. 178–183 (1984).
- Billet, Mackowiak, and Pajak, *Chem. Eng. Process.*, **19**, pp. 39–47 (1985).
- Lewis, Jones, and Pratt, *Trans. Instn. Chem. Engrs.*, **29**, pp. 126–148 (1951).
- Kumar and Hartland, *Ind. Eng. Chem. Res.*, **35**(8), pp. 2682–2695 (1996).
- Kumar and Hartland, Chap. 17 in *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994), pp. 625–735.

Refer to the original articles for details.

where V_{so} is defined as the characteristic slip velocity obtained at low dispersed-phase flow rate. Rearranging Eqs. (15-113) to (15-116) gives

$$V_{so} = \sqrt{\frac{4\Delta\rho g d_p}{3\rho_c C_D}} \quad (15-117)$$

The slip velocity at higher holdup often is estimated from $V_s \approx V_{so}(1 - \phi_d)$.

Equation (15-117) provides the basis for various methods used to predict the characteristic slip velocity. For additional discussion, see Mišek, Chap. 5 in *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994). Equation (15-117) can be difficult to use for design because of difficulty estimating the drag coefficient C_D and difficulty accounting for packing resistance or drop-drop interactions. The drag coefficient can be affected by internal circulation within the drop. For good mass transfer, it is most desirable to have circulating drops traveling through a relatively nonviscous continuous phase. Particular care should be taken in utilizing models developed primarily from studies involving small laboratory packings, because the packing resistance is particularly significant in that case. Also many studies do not include low-interfacial-tension systems, even though most applications of static extractors involve low to moderate interfacial tension. Also note that surface-active impurities can reduce the characteristic

drop velocity [Garner and Skelland, *Ind. Eng. Chem.*, **48**(1), pp. 51–58 (1956); and Skelland and Caenepeel, *AIChE J.*, **18**(6), pp. 1154–1163 (1972)], which is another reason to approach these models with care.

The following method is recommended for calculating slip velocity in static extractors at low dispersed-phase holdup:

$$\text{If } Re_{\text{Stokes}} = \frac{\rho_c \Delta\rho g d_p^3}{18\mu_c^2} < 2, \quad \text{then } V_{so} = \frac{\Delta\rho g d_p^2}{18\mu_c} \quad (\text{Stokes' law}) \quad (15-118)$$

For $Re_{\text{Stokes}} > 2$, Seibert and coworkers [Seibert and Fair, *Ind. Eng. Chem. Res.*, **27**(3), pp. 470–481 (1988); and Seibert, Reeves, and Fair [*Ind. Eng. Chem. Res.*, **29**(9), pp. 1901–1907 (1990)] recommend the model of Grace, Wairegi, and Nguyen [*Trans. Inst. Chem. Eng.*, **54**, p. 167 (1976)]. In this case, the characteristic slip velocity may be calculated from

$$V_{so} = \frac{Re_{\mu_c}}{d_p \rho_c} \quad (15-119)$$

TABLE 15-16 Example Hold-up Models for Static Extractors

Example	Eq.	Comments	Ref.
$\phi_d = \frac{V_d [\cos(\pi\zeta/4)]^2}{\varepsilon [V_{so} \exp(-6\phi_d/\pi) - V_c/\varepsilon(1 - \phi_d)]}$, $\zeta = \frac{a_p d_p}{2}$	1	Spray, packing, and sieve tray V_{so} is calculated by the method of Grace et al. (1976), Eqs. (15-118) to (15-123).	1
$\frac{V_d}{\phi_d} + \frac{V_c}{1 - \phi_d} = \frac{4c_p}{3} \sqrt{\frac{d_p \Delta \rho g}{C_D \rho_c}}$	2	SMV structured packing. Drag coefficient, C_D is calculated by assuming a drop is a rigid sphere. Parameter c_p depends upon drop-drop and drop-packing interactions.	2
$\frac{V_d}{\phi_d} + \frac{V_c}{1 - \phi_d} = \varepsilon C \left(\frac{4g \Delta \rho \sigma}{\rho_c^2} \right)^{0.25} \exp(-b\phi_d)$	3	Packing. Constants C and b differ for different packings. Drag coefficient = 1.	3
$\left(\frac{V_d}{\phi_d} + \frac{V_c}{1 - \phi_d} \right) \left(\frac{a_p \rho_c}{\varepsilon^3 g \Delta \rho} \right)^{0.5} = 0.683 \phi_d (1 - \phi_d)$	4	Packing	4
$\phi_d = A \left\{ 0.27 + \left[\frac{\varepsilon}{g} \left(\frac{\rho_c}{g \sigma} \right)^{0.25} \right]^{0.78} \right\} \left[V_d \left(\frac{\rho_c}{g \sigma} \right)^{0.25} \right]^{0.57} \exp(B)$ $B = 3.34 V_c \left(\frac{\rho_c}{g \sigma} \right)^{0.25}$ $A = \left(\frac{\Delta \rho}{\rho_c} \right)^{-0.58} \left(\frac{\mu_d}{\mu_w} \right)^{0.18} C \varepsilon^n \left[l \left(\frac{\rho_c g}{\sigma} \right)^{0.5} \right]^{-0.39}$	5	Unified model for packing, spray, Karr, pulsed perforated plate, Kühni, rotating disk. Constants C , n , and l depend upon type of contactor.	5

References:

1. Seibert and Fair, *Ind. Eng. Chem. Res.*, **27**(3), pp.470–481 (1988).
2. Streiff and Jancic, *Ger. Chem. Eng.*, **7**, pp. 178–183 (1984).
3. Billet, Mackowiak, and Pajak, *Chem. Eng. Process.*, **19**, pp. 39–47 (1985).
4. Sitaramayya and Laddha, *Chem. Eng. Sci.*, **13**, p. 263 (1961).
5. Kumar and Hartland *Ind. Eng. Chem. Res.*, **34**, pp. 3925–3940 (1995).

Refer to the original articles for details.

where Re is obtained from the correlation:

$$\frac{Re}{P^{0.149}} = 0.94H^{0.757} - 0.857 \quad H \leq 59.3 \quad (15-120)$$

$$\frac{Re}{P^{0.149}} = 3.42H^{0.441} - 0.857 \quad H > 59.3 \quad (15-121)$$

And P and H are dimensionless groups defined by

$$P = \frac{\rho_c^2 \sigma^3}{\mu_c^2 g \Delta \rho} \quad (15-122)$$

$$H = \left(\frac{4d_p^2 g \Delta \rho}{3\sigma} \right) \left(\frac{\mu_w}{\mu_c} \right)^{0.14} P^{0.149} \quad (15-123)$$

and μ_w is a reference viscosity equal to 0.9 cP (9×10^{-4} Pa·s). For discussion of methods to correct slip velocity to account for the effect of high dispersed-phase holdup, see Augier, Masbernat, and Guiraud, *AIChE J.*, **49**(9), pp. 2300–2316 (2003).

Flooding Velocity Maximum flow through a countercurrent extractor is limited by the flooding velocity. See “Hydrodynamics of Column Extractors” for a general discussion of flooding mechanisms. Because of the many possible causes of flooding, published data and

models should be viewed with some caution. In addition, models developed from laboratory data can lead to problems when used for design of commercial-scale columns. For example, in packed columns a column diameter/packing diameter ratio of at least 8 is recommended to avoid channeling due to wall effects. This means that laboratory studies must utilize small packings with high specific packing surface areas (packing area/contacting volume). The high packing area will provide significant resistance to drop flow, greater than that encountered in large columns containing large commercial packings. In addition, many of the published laboratory data on flooding velocities were generated by using moderate to high-interfacial-tension systems. In this case, the packing surface area resistance controls the flooding mechanism.

Several correlations of flooding velocity have the general form

$$V_{fj} \propto C_1 \frac{C_2}{a_p^n} \quad 0 < n < 1 \quad (15-124)$$

where V_{fj} is the continuous-phase velocity at which flooding occurs, a_p is the specific packing surface area, and C_1 and C_2 are empirical constants that depend upon the specific type of packing, fluid physical properties, and flow ratio. While these types of models have excellent reported fits of data, they were primarily developed by using laboratory-scale packings. Furthermore, in the limit as the packing surface area approaches zero, the predicted flooding velocity becomes infinite, an unrealistic result. Care should be taken when extrapolating such models to a larger packing size.

Seibert, Reeves, and Fair [*Ind. Eng. Chem. Res.*, **29**(9), pp. 1901–1907 (1990)] proposed a more mechanistically consistent flooding model that is derived by assuming a tightly packed arrangement of drops at flooding and yields

$$\frac{1}{V_{cf}} = C_1 + \frac{C_2}{C_3 \cos^2(C_4 a_p)} \quad (15-125)$$

where parameters C_1 , C_2 , C_3 , and C_4 are functions of the system properties and flow ratio. An advantage of this flooding model is that as the packing surface area approaches zero, a finite flooding velocity is calculated since the $\cos 0 = 1$. In the absence of packing, Eq. (15-125) can be rewritten to predict flooding in a spray column and the ultimate capacity of a tray column. Examples of published flooding models for static extractors are given in Table 15-17. Unfortunately, very

few flooding data are available for columns greater than 30 cm (12 in) in diameter. Also, many of the available flooding data have been obtained in the absence of mass transfer. With this in mind, for new designs it is recommended that flow velocities be limited to no more than 50 percent of the calculated flooding values. The final design should be refined in miniplant or pilot-plant tests using actual feed materials.

Drop Coalescence Rate The rate of drop coalescence often is assumed to be rapid (not rate-limiting) in the design of static extractors. However, this is not necessarily the case, particularly during operation at high dispersed-phase holdup and high flow ratios of dispersed phase to continuous phase. Under these conditions, a large number of drops flow through a nearly stagnant continuous phase, and these drops must coalesce at the main operating interface located at the top or bottom of the column. Seibert, Bravo, and Fair [*ISEC '02 Proc.* **2**, pp. 1328–1333 (2002)] report that problems with coalescence are most likely when the

TABLE 15-17 Example Flooding Models for Static Extractors

Example	Eq.	Comments	Ref.
$V_{cf} = \frac{0.178 \epsilon V_{so}}{1 + 0.925(V_{df}/V_{cf}) \{1/[\cos(\pi\zeta/4)]^2\}} \quad \zeta = \frac{a_p d_p}{2}$	1	Spray, packing, and ultimate capacity of sieve tray V_{so} is calculated by the method of Grace et al. (1976), Eqs. 15-118 to 15-123.	1
$V_{cf} = \sqrt{\frac{L_{dc} - A}{B(V_{df}/V_{cf})^2 + C}} \quad A = \frac{6\sigma}{d_p \Delta \rho g} \quad B = \frac{1.1 \rho_d}{g \Delta \rho f_{fa}^2} \quad C = \frac{2.7 \rho_c}{2 g \Delta \rho f_{da}^2}$	2	Sieve tray capacity limited by coalesced layer flood	2
$V_{cf} = \epsilon C \left(\frac{4g \Delta \rho \sigma}{\rho_c^2} \right)^{1/4} (1 - \phi_{df})^2 [\exp(-b\phi_{df})] (1 - b\phi_{df})$	3	Packing	3
$\frac{V_d}{V_c} = \frac{\phi_{d,f}^2}{(1 - \phi_{d,f})^2} \frac{1 + b(1 - \phi_{d,f})}{1 - b\phi_{d,f}}$	4	Constants C and b depend on packing. Drag coefficient = 1.	
$V_{cf} = \left\{ \frac{C[(a_p/g\epsilon^3)(\rho_c/\Delta \rho)\sigma^{0.25}]^{-0.25}}{1 + 0.835(\rho_d/\rho_c)^{0.25}(V_{df}/V_{cf})^{0.5}} \right\}^2$	5	Packing C is a constant for each packing.	4
$V_{cf} = \left[\left[1 + \left(\frac{V_{df}}{V_{cf}} \right)^{0.5} \right]^2 \sqrt{\frac{a_p}{g}} \right]^{-1} \alpha C_1 \epsilon^{1.54} \left(\frac{\Delta \rho}{\rho_d} \right)^{0.41} \left[\frac{1}{a_p} \left(\frac{\Delta \rho^2 g}{\mu_c^2} \right)^{1/3} \right]^{0.3} \left(\frac{\mu_c}{\sqrt{\Delta \rho \sigma a_p}} \right)^{0.15}$ $\alpha = 1$ for continuous-phase wetting, $\alpha = 1.29$ for dispersed-phase wetting	6	Packing C_1 is a constant that depends upon the type of packing.	5
$\frac{(V_{df} + V_{cf})d_c \rho_c}{\mu_c} = 0.30137 \left(\frac{A_{col}}{\pi d_c^2 N_c/4} \right)^{0.0948} A \left(\frac{\Delta \rho}{\rho_c} \right)^{0.1397} B \left(\frac{d_d^3 \rho_c^2 g}{\mu_c^2} \right)^{0.3875}$ $A = \left(\frac{V_{df}}{V_{cf}} \right)^{0.0593} \quad B = \left(\frac{\sigma \rho_d d_o}{\mu_d^2} \right)^{0.0127} \quad A_{col} = \frac{1}{1 - \frac{A_{dow}}{A_{col}}} \frac{Q_d}{f V_{dfn}}$ f = fraction of flood	7	Sieve tray	6

References:

- Seibert and Fair, *Ind. Eng. Chem. Res.*, **27**(3), pp. 470–481 (1988).
 - Seibert and Fair, *Ind. Eng. Chem. Res.*, **32**(10), pp. 2213–2219 (1993).
 - Billet, Mackowiak, and Pajak, *Chem. Eng. Process.*, **19**, pp. 39–47 (1985).
 - Dell and Pratt, *Trans. Inst. Chem. Eng.*, **29**, p. 89 (1951).
 - Kumar and Hartland, *Trans. Inst. Chem. Eng.*, **72**(Pt. A), pp. 89–104 (1994).
 - Rocha et al., *Ind. Eng. Chem. Res.*, **28**(12), pp. 1873–1878 (1989).
- Refer to the original articles for details.

superficial dispersed-phase velocity V_{df} is greater than about 12 percent of the characteristic slip velocity given by Eqs. (15-118) to (15-123). For these systems, miniplant tests normally are needed to understand the rate of coalescence. If coalescence is slow, design rates will need to be reduced below those predicted by assuming rapid coalescence.

For slowly coalescing systems, placement of coalescing material within the column at the main interface may significantly improve performance. The height of the uncoalesced layer located at the main operating interface may be reduced by adding a high-surface-area mesh type of coalescer that is wetted by the dispersed phase. If plugging is a concern, a more open (lower-surface-area) structured packing may be preferred. It also may be useful to add a separate liquid-liquid phase separator outside the extractor to clarify the extract or raffinate streams. See "Liquid-Liquid Phase Separation Equipment."

Mass-Transfer Coefficients As described in "Rate-Based Calculations," the overall mass-transfer coefficient may be written as

$$\frac{1}{k_{od}} = \frac{1}{k_d} + \frac{m_{dc}^{\text{vol}}}{k_c} \quad (15-126)$$

where the slope of the equilibrium line m_{dc}^{vol} is expressed in volumetric concentration units. The dispersed-phase and continuous-phase film coefficients k_d and k_c generally are functions of convection and turbulence effects, as well as molecular diffusion coefficients and the thicknesses of stagnant films at the interface between drops and the continuous phase. Examples of mass-transfer coefficient models for static extractors are given in Table 15-18. For additional discussion of

TABLE 15-18 Example Mass-Transfer Coefficient Models

Example	Eq.	Comments	Ref.
$k_c = 0.698 \left(\frac{\mathcal{D}_c}{d_p} \right) \left(\frac{d_p V_s \rho_c}{\mu_c} \right)^{0.5} \left(\frac{\mu_c}{\rho_c \mathcal{D}_c} \right)^{0.4} (1 - \phi_d)$	1	For nonrigid drops. Spray, packing, and sieve trays.	1
$k_d = 0.023 V_s \sqrt{\frac{\rho_c \mathcal{D}_c}{\mu_c}}$	2	Model of Laddha and Degaleesan. For nonrigid drops. Spray, packing, and sieve trays.	2
$k_d = \frac{0.00375 V_s}{1 + \mu_d/\mu_c}$	3	Model of Handlos and Baron. Approximate solution to series model. Independent of molecular diffusion. Use for large drops.	3
$\Phi = \frac{\sqrt{\mu_c \rho_c \mathcal{D}_c}}{1 + \mu_d/\mu_c}$	4	Spray, packing, and sieve trays. Use Eq. (3) if $\Phi < 6$.	1
$k_d = \frac{17.9 \mathcal{D}_d}{d_p}$	5	Laminar circulation within drops. Recommended for long contact times. For $Re < 50$.	4
$k_d = 1.14 \left(\frac{\mathcal{D}_d}{d_p} \right) \left(\frac{\rho_d d_p^2 \bar{\omega}}{\mu_d} \right)^{0.56} \left(\frac{\mu_d}{\rho_d \mathcal{D}_d} \right)^{0.5}$	6	For oscillating drops. Simplified version for assumption of $\theta = 0.2$.	6
$\bar{\omega} = \frac{1}{2\pi} \left[\frac{192 \sigma b}{d_p^3 (3\rho_d + 2\rho_c)} \right]^{0.5} \quad b = 0.805 d_p^{0.225}, d_p \text{ in cm}$	7		
$k_c = \frac{\mathcal{D}_c}{d_p} \left[2 + 0.95 \left(\frac{\rho_c V_s d_p}{\mu_c} \right)^{0.5} \left(\frac{\mu_c}{\rho_c \mathcal{D}_c} \right)^{0.33} \right]$	8	For rigid drops.	7
$k_c = 0.725 \left(\frac{\mathcal{D}_c}{d_p} \right) \left(\frac{\rho_c V_s d_p}{\mu_c} \right)^{0.57} \left(\frac{\mu_c}{\rho_c \mathcal{D}_c} \right)^{0.42} (1 - \phi_d)$	9	For circulating drops. Developed from correlation of spray column data.	8
$k_c = 1.4 \left(\frac{\mathcal{D}_d}{d_p} \right) \left(\frac{\rho_c d_p^2 \bar{\omega}}{\mu_c} \right)^{0.5} \left(\frac{\mu_c}{\rho_c \mathcal{D}_c} \right)^{0.5}$	10	For oscillating drops.	9

References:

- Seibert and Fair, *Ind. Eng. Chem. Res.*, **27**(3), pp. 470–481 (1988).
 - Laddha and Degaleesan, *Transport Phenomena in Liquid Extraction* (McGraw-Hill, 1978).
 - Handlos and Baron, *AIChE J.*, **3**, p. 127 (1957).
 - Kronig and Brink, *Appl. Sci. Res.*, **A2**, p. 142 (1950).
 - Johnson and Hamielec, *AIChE J.*, **6**, p. 145 (1960).
 - Yamaguchi, Fujimoto, and Katayama, *J. Chem. Japan*, **8**, p. 361 (1975).
 - Garner and Suckling, *AIChE J.*, **4**, p. 114 (1958).
 - Treybal, *Liquid Extraction* (McGraw-Hill, 1963).
 - Yamaguchi, Watanabe, and Katayama, *J. Chem. Japan*, **8**, p. 415 (1975).
- Refer to the original articles for details.

film coefficient models, see *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994).

Axial Mixing See “Accounting for Axial Mixing” under “Liquid-Liquid Extraction Equipment.” Many approaches have been developed. Becker recommends the concept of the height of a dispersion unit (HDU) to correct the height of a transfer unit for axial mixing in a static contactor [Becker, *Chem. Eng. Technol.*, **26**(1), pp. 35–41 (2003); *Chem. Ing. Tech.*, **74**, pp. 59–66 (2002); and Becker and Seibert, *Chem. Ing. Tech.*, **72**, pp. 359–364 (2000)]:

$$H_{or|axial} = H_{or|plug} + HDU_o \quad (15-127)$$

where
$$HDU_o = \left(p_0 + \frac{0.8}{Z_t} \frac{\mathcal{E} \ln \mathcal{E}}{\mathcal{E} - 1} \right)^{-1} \quad (15-128)$$

$$p_0 = \frac{0.1Z_t/H_{or}^* + 1}{0.1Z_t/H_{or}^* + p_1/p_2} \cdot p_1 \quad (15-129)$$

$$\frac{1}{p_1} = \frac{1}{\mathcal{E}} HDU_r + HDU_e \quad \frac{1}{p_2} = HDU_r + \frac{1}{\mathcal{E}} HDU_e \quad (15-130)$$

For a given phase,
$$HDU = \frac{\mathcal{E}}{V} \quad (15-131)$$

In these equations, the superscript* denotes the plug flow overall height of a transfer unit, subscript *r* denotes the raffinate phase, subscript *e* denotes the extract phase, and *Z_t* is the contacting height. For $\mathcal{E} = 1$, the equations reduce to

$$HDU_o = \left(\frac{1}{HDU_r + HDU_e} + \frac{0.8}{Z_t} \right)^{-1} \quad (15-132)$$

The axial mixing coefficient is correlated by

$$\frac{E_c \rho_c}{\mu_c} = (C_1 Re_c^a + C_2 Re_d^b) \left(\frac{D_{col}}{100} \right)^c \quad (15-133)$$

D_{col} = column diameter, cm

where
$$Re_c = \frac{V_{sc} \rho_c}{\mu_c (a_p + a_w)} \quad (15-134)$$

$$a_w = \frac{4}{D_{col}} \quad (15-135)$$

$$Re_d = \frac{V_d d_p \rho_c}{\mu_c} \quad (15-136)$$

In Eq. (15-135), *a_w* is the specific wall surface (cm²/cm³) and *a_p* is the specific packing surface (cm²/cm³). This term is dropped for a spray column (*C₁* = 0). The model coefficients are summarized in Table 15-19. Most of the axial mixing data available in the literature are for the continuous phase; dispersed-phase axial mixing data are rare. Becker recommends assuming *HDU_d* = *HDU_e* when dispersed-phase data are not available. Becker presents a parity plot (Fig. 15-33) based on small- and large-scale data for packed and spray columns.

Spray Columns The spray column is one of the simplest and oldest types of equipment used to contact two liquid phases in counter-current flow. Normally it consists of an empty vertical vessel with a distributor located at one end. The distributor disperses one of the liquids into drops. These drops then rise or fall against the flow of the continuous phase, collecting at the other end of the column and finally coalescing to form a layer of clear liquid that is withdrawn from the column. Because spray columns often are used when solids are present, phases often are dispersed through pipe distributors with large holes oriented in the direction of flow. In cases where the ratio of volumetric flow rates entering the column is far from unity, the liquid entering the extractor at the smaller rate generally should be dispersed to avoid excessive backmixing. Sometimes liquid distributors are used at each end to disperse both phases, with the main liquid-liquid interface located in the middle of the column (Fig. 15-34). See “Liquid Distributors and Dispersers” under “Liquid-Liquid Extraction Equipment.”

Spray columns are inexpensive and easy to operate and provide high volumetric throughput. However, because the continuous phase flows freely through the column, backmixing effects generally are severe. As a result, spray columns rarely achieve more than one theoretical stage. Spray columns may be used when only one theoretical stage is required or when solid precipitation is prevalent and no other contacting device can be used because of plugging. Spray columns also are used for direct heat transfer between large immiscible liquid streams.

Drop Size, Holdup, and Interfacial Area Drop size is estimated by using one of the models listed in Table 15-15, and holdup is estimated from expressions given in Table 15-16. Interfacial area is then calculated by using Eq. (15-109).

Flooding Several empirical and mechanistic flooding models have been reported. These have been reviewed by Kumar and Hartland [Chap. 17 in *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994), pp. 680–686]. Seibert, Reeves, and Fair [*Ind. Eng. Chem. Res.*, **29**(9), pp. 1901–1907 (1990)] propose an alternative model:

$$V_{cf} = \frac{0.178 V_{so}}{1 + 0.925 (V_{df}/V_{cf})} \quad (15-137)$$

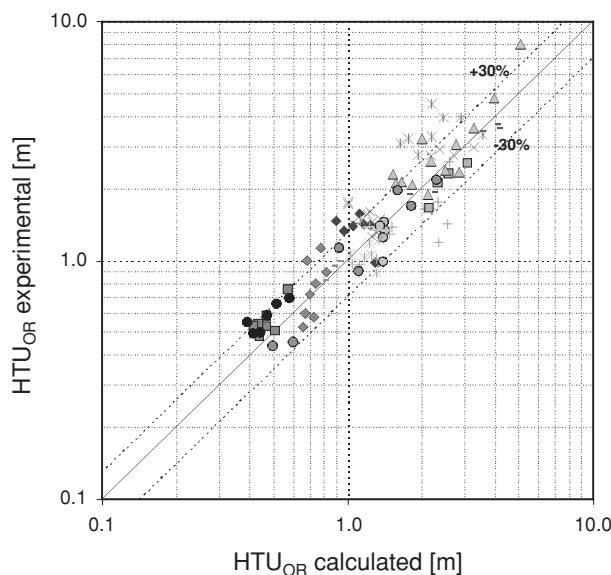
where *V_{so}* is the characteristic slip velocity determined by using Eqs. (15-118) to (15-123).

Mass-Transfer Efficiency As mentioned earlier, spray columns rarely develop more than one theoretical stage due to axial mixing within the column. Nevertheless, it is necessary to determine the column height that will give this theoretical stage. Cavers [Chap. 10 in

TABLE 15-19 Correlation Constants for the Becker Axial Mixing Model*

	No. of data points	<i>C₁</i>	<i>a</i>	<i>C₂</i>	<i>B</i>	<i>c</i>	Average relative error, %
Spray column	197	0	0	45.6	1.058	0.917	24.8
Structured packed columns and IMTP random packing	118	405.1	0.798	27.7	0.914	1.178	32.0
Structured packed columns with dual flow plates	57	284.5	0.494	35.0	0.406	0.847	18.7

*Becker, *Chem. Eng. Technol.*, **24**(12), pp. 1242–1244 (2001).



- ◆ SMVP, Hexane/Methanol/Water, 42 cm, d-c
- SMV, Hexane/Methanol/Water, 10 cm, d-c
- ◊ Spray, Hexane/Methanol/Water, 42 cm, d-c
- × Pall Rings, Hexane/Methanol/Water, 42 cm, d-c
- * IMTP®40 Random Packing, Hexane/MeOH/Water, 42 cm d-c
- SMVP, Hexane/Methanol/Water, 10 cm, d-c
- + SMV, Toluene/Acetone/Water, 10 cm, d-c
- Spray C., Toluene/Acetone/Water, 10 cm, d-c
- SMV, Water/MIBK/BuAc, 5 cm, d-c
- ◆ BX Water/Ethanol/CO₂, 6, 7 cm, d-c
- INTALOX®2T, Toluene/Acetone/Water, 42 cm, c-d
- ▲ Spray, Toluene/Acetone/Water, 42 cm, c-d
- × IMTP®40 Random Packing, Tol./Ac./Water, 42 cm, c-d
- * IMTP®25 Random Packing, Tol./Ac./Water, 42 cm, c-d
- ▲ IMTP®25 Random Packing, Tol./Ac./Water, 42 cm, c-d (Redistr.)
- ◊ SMV, Toluene/Acetone/Water, 10 cm, c-d
- SMV, Water/MIBK/BuAc, 5 cm, c-d

FIG. 15-33 Parity plot comparing spray and packed column results incorporating axial mixing model. [Reprinted from Becker, *Chemie Ing. Technik*, **74**(1–2), pp. 59–66 (2002). Copyright 2002 Wiley-VCH.]

Handbook of Solvent Extraction, Lo, Baird, and Hanson, eds. (Wiley, 1983; Krieger, 1991)] recommends the following equation from Laddha and Degaleesan [*Transport Phenomena in Liquid Extraction* (McGraw-Hill, 1978), p. 233] to estimate the overall volumetric mass-transfer coefficient:

$$k_{oc}a = m_{dc}^{vol} k_{od}a = 0.08 \times \frac{\phi_d(1 - \phi_d)(g^3 \Delta \rho^3 / \sigma \rho_c^2)^{1/4}}{(\mu_c / \rho_c \mathcal{D}_c)^{1/2} + (1/m_{dc})(\mu_d / \rho_d \mathcal{D}_d)^{1/2}} \quad (15-138)$$

Here $\mathcal{D}_c$ and $\mathcal{D}_d$ are the solute diffusivity coefficients in the continuous and dispersed phases, respectively. The height of a transfer unit can then be estimated from

$$H_{oc} = \frac{V_c}{k_{oc}a} \quad (15-139)$$

where H_{oc} is the height of an overall transfer unit based on the continuous phase and V_c is the superficial velocity of the continuous phase. Equation (15-138) provides only a rough approximation.

Packed Columns Packing is used in a column extractor to reduce axial mixing (backmixing). Packing also affects interfacial area and mass transfer through its impact on the holdup and flow path of dispersed drops. For reviews of packed-column extractor design, see Strigle, *Packed Tower Design and Applications*, 2d ed., Chap. 11 (Gulf, 1994); and Stevens, Chap. 8 in *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994).

The packings used for liquid-liquid extraction are essentially the same as those used in distillation and absorption service, although the distributors and dispersers and many of the associated internals are not the same. Various commercially available packings offered by Koch-Glitsch and Sulzer Chemtech for liquid-liquid extraction service are listed in Table 15-20. Other manufacturers of packings include Raschig, Montz, Lantec, and Jaeger Products. It is a good idea to consult a variety of vendors before making a selection. Illustrations of various types of packings are given in Sec. 14, "Equipment for Distillation, Gas Absorption, Phase Dispersion, and Phase Separation."

Packings are classified as either random or structured. Random packings may be wet-loaded into a column by filling the column

with liquid and slowly adding the packing at the liquid surface so the packing pieces gently fall to the surface of the forming bed (typical of ceramic packings); or they may be dry-loaded by transferring them into an empty column through a chute or fabric sock (typical of metal or plastic packings). The familiar ring and saddle packings such as Raschig rings, Berl saddles, Intalox saddles, and Lessing rings are examples of ceramic packings. The more modern metal and plastic random packings such as Pall rings, Hy-Pak®, and IMTP® packings are ring or saddle shapes with internal fingers and slots in the wall. These packings are more open and provide greater access to the interior surfaces for improved capacity and mass-transfer performance. Structured packings are modular assemblies placed inside the column in a specific arrangement. Many are in the form of woven wire mesh or crimped sheets arranged in layers at specific angles. For packing made from sheets, it is not clear whether surface treatments such as perforations and embossing are important in liquid-liquid extraction, so a number of smooth-surface structured packings are marketed for extraction applications. For best performance, the packing should be preferentially wetted by the continuous phase. (See "Effect of Solid-Surface Wettability" under "Liquid-Liquid Dispersion Fundamentals.") Many older packed extractors are being refurbished with newer packings and internals to achieve higher throughput and better separation performance. As with any packing and the associated internals, installation procedures recommended by the packing vendor need to be carefully followed to ensure the packing performs as designed. In addition to mass-transfer performance and throughput, another important consideration when choosing metal packing is the packing material and wall thickness relative to corrosion rates. The packing should have sufficient wall thickness for a reasonable service life.

Liquid Distribution Good initial distribution of the dispersed phase is very important for good performance. Strigle [*Packed Tower Design and Applications*, 2d ed., Chap. 11 (Gulf, 1994)] describes typical packed-column internals for liquid-liquid contacting. When the light phase is dispersed, a combination liquid disperser/packing support is preferred because a separate support plate can adversely affect the flow of dispersed drops. An example of a disperser plate is shown in Fig. 15-35. A ladder-type pipe distributor commonly is used to distribute the dispersed-phase feed to the initial disperser plate. Other

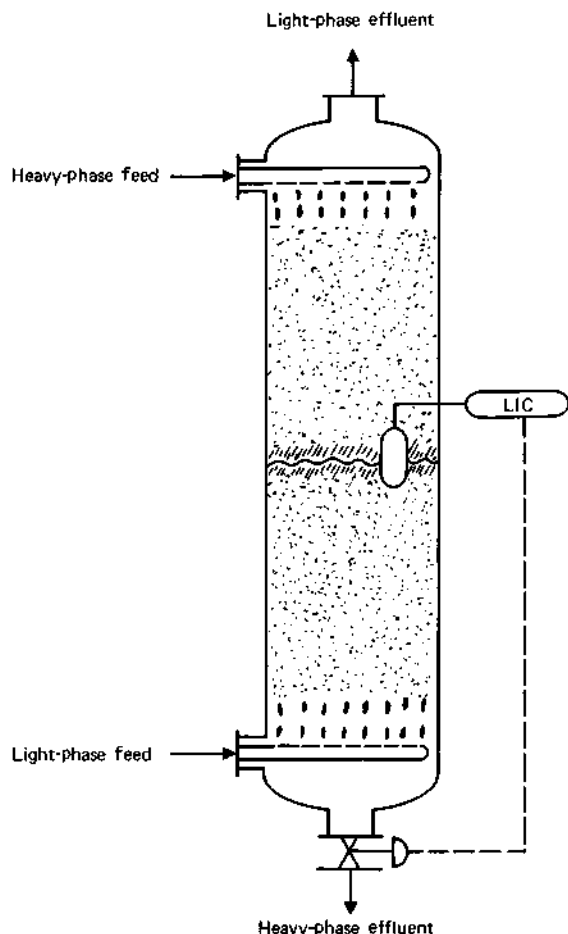


FIG. 15-34 Spray column with both phases dispersed.

distributor designs also are available. Koch and Vogelpohl [*Chem. Eng. Technol.*, **24**(7), pp. 695–698 (2001); and *Chem. Eng. Technol.*, **24**(8), pp. 795–798 (2001)] discuss a sieve plate distributor design that includes a predistributor plate. Many of the concepts concerning geometric uniformity for liquid distribution in packed gas-liquid contactors [Perry, Nutter, and Hale, *Chem. Eng. Prog.*, **86**(1), pp. 30–35 (1990)] are relevant to liquid-liquid contactors as well. See “Liquid Distributors and Dispersers” under “Liquid-Liquid Extraction Equipment.”

Redistribution Seibert, Reeves, and Fair [*Ind. Eng. Chem. Res.*, **29**(9), pp. 1901–1907 (1990)] and Nemunaitis et al. [*Chem. Eng. Prog.*, **67**(11), p. 60 (1971)] report data showing little benefit from a packed height greater than 10 ft (3 m) and recommend redistributing the dispersed phase about every 5 to 10 ft (1.5 to 3 m) to generate new droplets and constrain backmixing. A random packed column often is designed with a redistributor placed between two or more packed sections. Structured packings sometimes are installed with a dual-flow perforated plate (with no downcomer) between elements, without coalescence of dispersed drops.

Minimum Packing Size and Drop Size For a given application there will be a minimum packing size or dimension below which random packing is too small for good extraction performance [Lewis, Jones, and Pratt, *Trans. Inst. Chem. Eng.*, **29**, pp. 126–148 (1951); Gayler and Pratt, *Trans. Inst. Chem. Eng.*, **31**, pp. 69–77 (1953); and Laddha and Degaleesan, *Transport Phenomena in Liquid Extraction*

TABLE 15-20 Random and Structured Packings Used in Packed Extractors

Packing	Surface area a_p , m^2/m^3	Void fraction ^a , ϵ
Metal random packing		
Koch-Glitsch IMTP [®] 25	224	0.964
Koch-Glitsch IMTP [®] 40	151	0.980
Koch-Glitsch IMTP [®] 50	102	0.979
Koch-Glitsch IMTP [®] 60	84	0.983
Sulzer I-Ring #25	224	0.964
Sulzer I-Ring #40	151	0.980
Sulzer I-Ring #50	102	0.979
Nutter Ring [®] NR 0.7	226	0.977
Nutter Ring [®] NR 1	168	0.977
Nutter Ring [®] NR 1.5	124	0.976
Nutter Ring [®] NR 2	96	0.982
Nutter Ring [®] NR 2.5	83	0.984
HY-PAK [®] #1	172	0.965
HY-PAK [®] #1½	118	0.976
HY-PAK [®] #2	84	0.979
FLEXIRING [®] 1 in	200	0.959
FLEXIRING [®] 1½ in	128	0.974
FLEXIRING [®] 2 in	97	0.975
CMR [™] 1	246	0.973
CMR [™] 2	157	0.970
CMR [™] 3	102	0.980
BETA RING [®] #1	186	0.963
BETA RING [®] #2	136	0.973
FLEXIMAX [®] 200	189	0.973
FLEXIMAX [®] 300	148	0.979
FLEXIMAX [®] 400	92	0.983
Plastic random packing		
Super INTALOX [®] Saddles #1	204	0.896
Super INTALOX [®] Saddles #2	105	0.934
BETA RING [®] #1 in	167	0.942
BETA RING [®] #2	114	0.940
SNOWFLAKE [®]	93	0.949
FLEXIRING [®] 1 in	205	0.922
FLEXIRING [®] 1½ in	119	0.925
FLEXIRING [®] 2 in	99	0.932
Ceramic random packing		
INTALOX [®] Saddles 1 in	256	0.730
INTALOX [®] Saddles 1½ in	195	0.750
INTALOX [®] Saddles 2 in	118	0.760
Ceramic structured packing		
FLEXERAMIC [®] 28	282	0.720
FLEXERAMIC [®] 48	157	0.770
FLEXERAMIC [®] 88	102	0.850
Metal structured packing†		
Koch-Glitsch SMV-8	417	0.978
Koch-Glitsch SMV-10	292	0.985
Koch-Glitsch SMV-16	223	0.989
Koch-Glitsch SMV-32	112	0.989
Sulzer SMV 2Y	205	0.990
Sulzer SMV 250Y	256	0.988
Sulzer SMV 350Y	353	0.983
INTALOX [®] 2T	214	0.989
INTALOX [®] 3T	170	0.989
INTALOX [®] 4T	133	0.987
Plastic structured packing†		
Koch Glitsch SMV-8	330	0.802
Koch-Glitsch SMV-16	209	0.875
Koch-Glitsch SMV-32	93	0.944
Sulzer SMV 250Y	256	0.875

^aTypical value for standard wall thickness. Values will vary depending upon thickness.

†SMV structured packings also are available with horizontal dual-flow perforated plates installed between elements (typically designated SMVP packing). These plates generally reduce backmixing and improve mass-transfer performance at the expense of a reduction in the open cross-sectional area and somewhat reduced capacity.

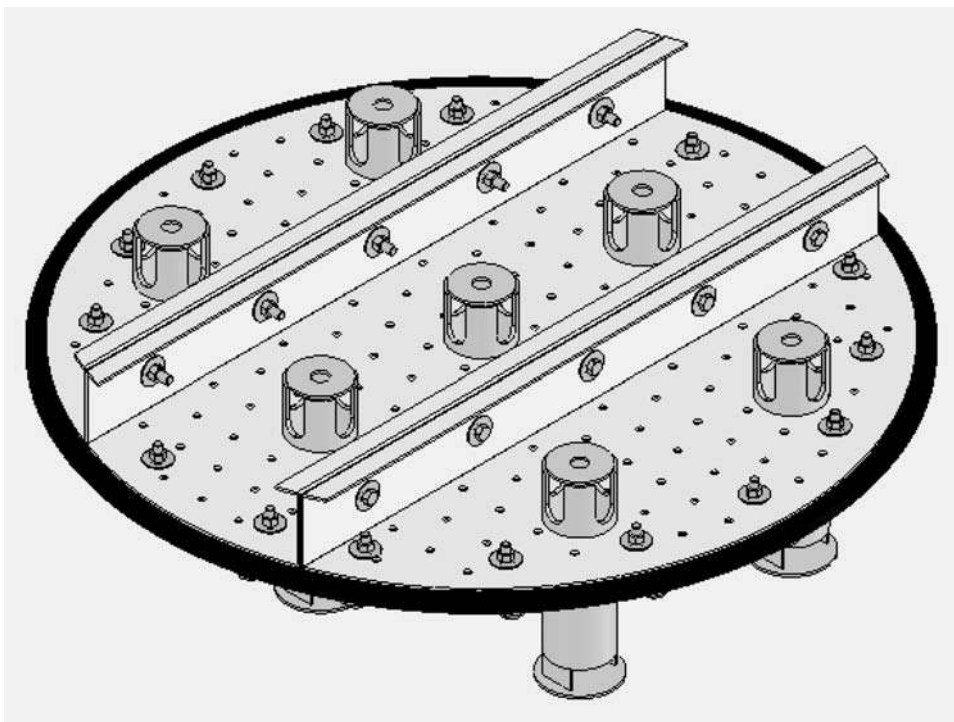


FIG. 15-35 Example of disperser plate (Sulzer Chemtech model VSX). (Courtesy of Sulzer Chemtech.)

(McGraw-Hill, 1978), Chap. 10, pp. 288–289]. The critical packing dimension is given by

$$d_c = 2.4 \sqrt{\frac{\sigma}{\Delta \rho g}} \quad (15-140)$$

In many cases, the minimum random packing size is about 0.5 in (1.3 cm). A similar effect may be seen with short-crimp-height structured sheet packings that might act as a parallel-plate type of coalescer. For packings smaller than the critical size, the packing acts to promote growth of dispersed drops somewhat as a packed-bed coalescer as drops flow through the spaces between the packing elements. (For a discussion of packed-bed coalescers, see “Coalescers” under “Liquid-Liquid Phase Separation Equipment.”) For packing sizes larger than d_c , the characteristic drop diameter is independent of packing size and may be estimated by using the models listed in Table 15-15. The choice of packing size above d_c generally involves a tradeoff; throughput increases with increasing packing size, while mass-transfer performance may decrease with increasing packing size due to an increase in backmixing effects. Typical random packings for commercial-scale columns are in the range of $\frac{3}{4}$ to 2 in (or about 2 to 5 cm). For small columns, the packing should be no larger than one-eighth the column diameter to avoid channeling at the wall. This effectively restricts the size of laboratory extractors packed with random packings to no less than 4 in (10 cm) in diameter if they are intended to generate directly scalable data.

Holdup and Interfacial Area The dispersed-phase holdup in a packed-column extractor may be placed into two categories: (1) a small portion that is held in the column for extended periods (essentially permanent) and (2) a larger portion that is free to move through the packing. This is the portion that participates in transfer of solute between phases. The total is ϕ_d which here refers to the volume of dispersed phase expressed as a fraction of the void space in the packed section. Pratt and coworkers [Trans. Inst. Chem. Eng.,

29, pp. 89–109, 110–125, 126–148 (1951); 31, pp. 57–68, 69–77, 78–93 (1953)] developed relationships between dispersed-phase velocity and holdup for packed columns. For standard commercial packings of 0.5 in (1.27 cm) and larger, they found that ϕ_d varies linearly with V_d up to values of $\phi_d = 0.10$ (for low values of V_d). With further increase of V_d , ϕ_d increases sharply up to a “lower transition point” resembling loading in gas-liquid contact. At still higher values of V_d an upper transition point occurs, the drops of dispersed phase tend to coalesce, and V_d can increase without a corresponding increase in ϕ_d . This regime ends in flooding. Below the upper transition point, Pratt and coworkers calculated dispersed-phase holdup from the expression

$$\frac{V_d}{\phi_d} + \frac{V_c}{1 - \phi_d} = \epsilon V_{so} (1 - \phi_d) \quad (15-141)$$

where V_{so} is the characteristic slip velocity at low dispersed-phase flow rate. The slip velocity may be estimated by using Eqs. (15-118) to (15-123) or alternative methods listed in Table 15-16. (See the related discussion in “Common Features and Design Concepts.”) Interfacial area is calculated from Eq. (15-109).

Flooding Numerous methods have been proposed for correlating flooding velocities in packed extractors as a function of the packing specific surface area and void volume. Most were developed by using the older-style packings such as Raschig rings and Berl saddles. For example, the well-known flooding correlation $(\sigma/\rho_c)^{0.2}(\mu_c/\Delta\rho)(a_p/\epsilon)^{1.5}$ versus $(V_c^{1/2} + V_d^{1/2})^2 \rho_c/(a_p \mu_c)$, developed by Crawford and Wilke [Chem. Eng. Prog., 47(8), pp. 423–431 (1951)], is plotted in Fig. 15-36. This is a dimensional correlation developed by using U.S. Customary System units, so the following units must be used: viscosity in lb/ft·h (equal to 2.42 times the value in cP), density in lb/ft³, interfacial tension in dyn/cm, specific packing surface area in ft²/ft³, and velocities in ft/h based on total column cross section. Nemunaitis et al. [Chem. Eng. Prog., 67(11),

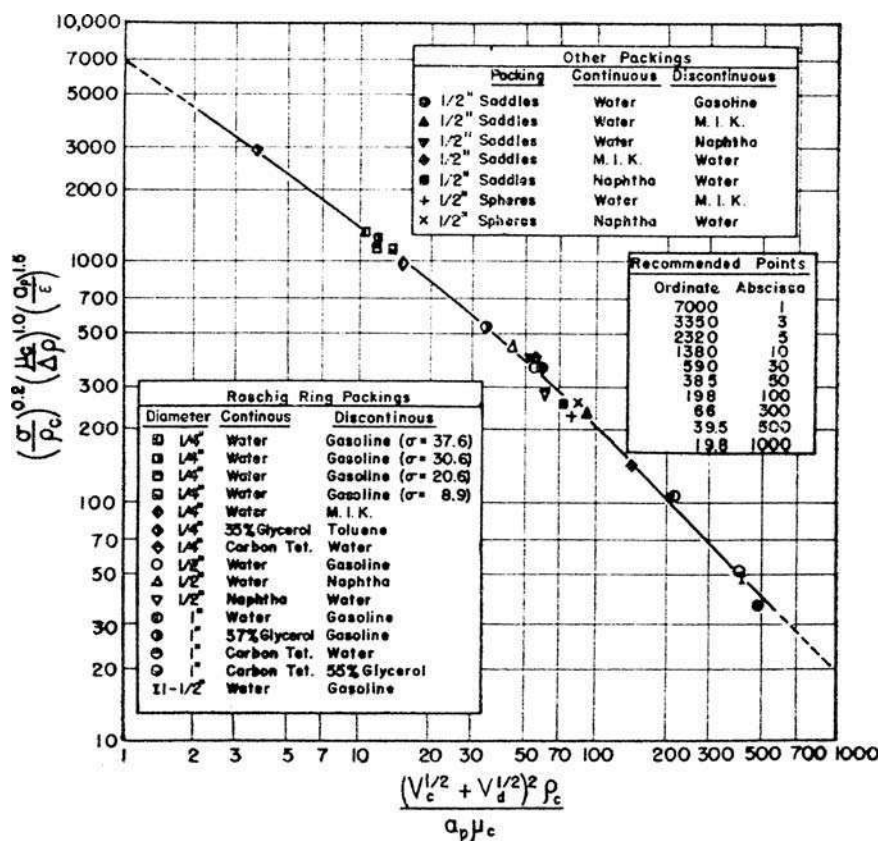


FIG. 15-36 Crawford-Wilke correlation for flooding in packed columns. Use only the units given in the text. [Reprinted from Crawford and Wilke, *Chem. Eng. Prog.*, 47(8), pp. 423-431 (1951), with permission.]

pp. 60-67 (1971)] modified the Crawford-Wilke correlation to include packing factors for specific types of packings (including Raschig rings, Intalox® saddles, and Pall rings). Another correlation that uses packing factors is given by Sakiadis and Johnson [*Ind. Eng. Chem.*, 46(6), pp. 1229-1239 (1954)]:

$$1 + 0.835 \left(\frac{\rho_c}{\rho_d} \right)^{1/4} \left(\frac{V_{cf}}{V_{df}} \right)^{1/2} = C_p \left[\frac{V_{cf}^2 a_p}{g \epsilon^3} \left(\frac{\rho_c}{\Delta \rho} \right) \sigma^{1/4} \mu_c \right]^{-1/4} \quad (15-142)$$

$$\text{where } C_p = 0.87 \frac{\epsilon^{0.0068}}{a_p^{0.048}} \quad \text{for nonribbed Raschig rings} \quad (15-143)$$

$$C_p = 1.2 \frac{\epsilon^{0.78}}{a_p^{0.0351}} \quad \text{for Berl saddles} \quad (15-144)$$

$$C_p = 1.02 \frac{\epsilon^{0.0068}}{a_p^{0.048}} \quad \text{for Lessing rings} \quad (15-145)$$

In Eqs. (15-142) to (15-145), the units are as follows: viscosity in cP, interfacial tension in dyn/cm, and specific packing surface area in ft²/ft³. Other correlation methods are listed in Table 15-17. The generalized flooding model of Seibert, Reeves, and Fair [*Ind. Eng. Chem. Res.*, 29(9), pp. 1901-1907 (1990)] was developed by using data for several types of packing and a range of operating scales, including data from a larger-scale column (42.5-cm inner diameter) using more modern packings: No. 25 IMTP® and No. 40 IMTP®

random packings and Intalox® Structured Packing 2T. It has the form

$$V_{cf} = \frac{0.178 \epsilon V_{so}}{1 + 0.925 (V_{df}/V_{cf}) \{1/[\cos(\pi \zeta/4)]^2\}} \quad (15-146)$$

$$\zeta = \frac{a_p d_p}{2} \quad (15-147)$$

where ζ is a dimensionless tortuosity factor. The quantity V_{so} is calculated by using Eqs. (15-118) to (15-123).

These correlations may be used to estimate extractor capacity for various types and sizes of packings; however, the results must be used with care due to considerable uncertainties in the calculation. This is particularly true when data for the packing of interest were not included in the data used to develop the correlation equation, and this is generally the case for the more modern packings. Nemunaitis et al. [*Chem. Eng. Prog.*, 67(11), pp. 60-67 (1971)] recommend designing for only 20 percent of the flood point calculated by using the Crawford-Wilke correlation (or their modified version). Because of this level of uncertainty, it is recommended that some experimental data be generated for a new design. In this regard, the flooding correlations may be used to scale up the pilot plant data to a larger packing size needed for the commercial-scale unit—by calculating the expected percentage change in capacity. This extrapolation approach also may be taken to estimate the improvement that might be achieved by retrofitting an existing commercial unit with a new packing. But again, the results should be used with caution, and consultation with packing vendors is recommended.

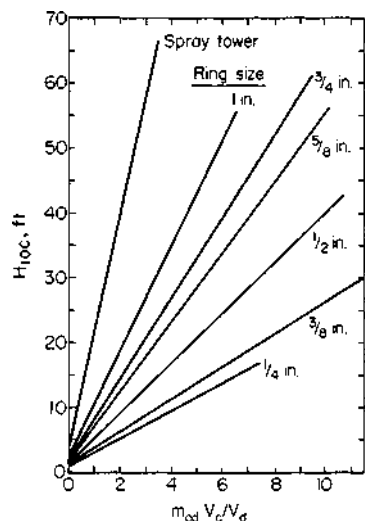


FIG. 15-37 Extraction of diethylamine from water into toluene (dispersed) in columns packed with unglazed porcelain Raschig rings. To convert feet to meters, multiply by 0.3048; to convert inches to centimeters, multiply by 2.54. [Reprinted from Leibson and Beckman, *Chem. Eng. Prog.*, **49**, p. 405 (1953), with permission.]

Pressure Drop In general, the pressure drop through a packed extractor is due to the hydrostatic head pressure. The resistance to flow caused by the packing itself normally is negligible because typical packings are large, and flooding velocities are much lower than those that would be needed to develop significant ΔP from resistance to flow between the packing elements. In some applications, solids may accumulate in the region of the packing support over time, and this may cause added pressure drop and premature flooding. For additional discussion, see Laddha and Degaleesan, *Transport Phenomena in Liquid Extraction* (McGraw-Hill, 1978), Chap. 10, pp. 271–273.

Mass Transfer Figure 15-37 plots the height of an overall transfer unit based on the raffinate phase H_{or} versus the extraction factor $\mathcal{E}$ for a series of Raschig rings of different sizes. The data are for transfer of diethylamine from water into toluene, where toluene is the dispersed phase. The data are typical in that mass-transfer performance is shown to improve (H_{or} decreases) as the size of the packing decreases. At the same time, the pressure drop must increase and hydraulic capacity decrease, so the design problem involves finding the economic optimum for the given production rate. The system water + ethylenamine + toluene is a high-interfacial-tension system, so the H_{or} data in Fig. 15-37 are expected to be somewhat high compared to those in systems with lower interfacial tension due to larger drop size in a nonagitated extractor. Note that most extractor designs will involve extraction factors in the range of $\mathcal{E} = 1.3$ to 2.

Table 15-21 lists typical mass-transfer performance of various packing sizes, as given by Strigle [*Packed Tower Design and Applications*, 2d ed., Chap. 11 (Gulf, 1994)]. Strigle's table is based on experience

TABLE 15-21 Typical Packed Extractor Performance According to Strigle*

Transfer units per bed	Required bed depth for modern random packings, ft (m)		
	Nominal packing size of 1 in (2.5 cm)	Nominal packing size of 1.5 in (3.8 cm)	Nominal packing size of 2 in (5 cm)
1.5	4.4 (1.3)	5.3 (1.6)	6.2 (1.9)
2.0	7.2 (2.2)	8.6 (2.6)	10.1 (3.1)
2.5	9.9 (3.0)	11.9 (3.6)	14.0 (4.3)

*Taken from Strigle, Chap. 11 in *Packed Tower Design and Applications*, 2d ed. (Gulf, 1994), with permission. Copyright 1994 Gulf Publishing Company. The numbers represent typical performance achieved with good liquid distribution.

with organic aqueous systems and the use of metal slotted-ring or ceramic saddle packings, using high-performance dispersion plates for liquid distribution and redistribution between packed sections.

Figure 15-37 and Table 15-21 provide only general guidelines. To estimate mass-transfer rates in packed towers, the calculation procedure outlined by Seibert, Reeves, and Fair [*Ind. Eng. Chem. Res.*, **29**(9), pp. 1901–1907 (1990)] and corrected for axial mixing [as in Eqs. (15-127) to (15-136)] is recommended. The overall mass-transfer coefficient is obtained by using Eq. (15-126). The predictive method of Handlos and Baron [*AIChE J.*, **3**(1), pp. 127–136 (1957)] allows calculation of the dispersed-phase coefficient:

$$k_d = \frac{0.00375V_s}{1 + \mu_d/\mu_c} \quad \text{when } \Phi = \frac{(\mu_d/\rho_d D_d)^{1/2}}{1 + \mu_d/\mu_c} < 6 \quad (15-148)$$

For $\Phi > 6$, the method given by Laddha and Degaleesan [*Transport Phenomena in Liquid Extraction* (McGraw-Hill, 1978)] is recommended:

$$k_d = 0.023V_s \left(\frac{\mu_d}{\rho_d D_d} \right)^{-1/2} \quad (15-149)$$

The continuous-phase coefficient may be calculated from

$$\frac{k_c d_p}{D_c} = 0.698 \left(\frac{\mu_c}{\rho_c D_c} \right)^{2/5} \left(\frac{d_p V_s \rho_c}{\mu_c} \right)^{1/2} (1 - \phi_d) \quad (15-150)$$

where V_s is the slip velocity of the dispersed drop [Seibert and Fair, *Ind. Eng. Chem. Res.*, **27**(3), pp. 470–481 (1988)]. While this calculation procedure can provide useful estimates, it does not replace the need for good pilot tests for any new design. Table 15-22 lists selected sources of data for mass transfer in packed columns.

Sieve Tray Columns A schematic diagram of the most common design of sieve tray column (also called a sieve plate or perforated-plate column) is shown in Fig. 15-32c. The light liquid is shown as the dispersed phase. The liquid flows up through the perforations of each tray and is thereby dispersed into drops that rise up through the continuous phase. The continuous liquid flows horizontally across each tray and passes to the tray beneath through the downcomer. For dispersing the heavy phase, the same design may be used, but turned upside down. The trays serve to eliminate (or at least greatly reduce) the vertical recirculation of continuous phase. Mass-transfer rates may be enhanced by the repeated coalescence and redispersion into droplets of the dispersed phase at each tray, although in general the overall efficiency of a sieve tray is fairly low, on the order of 15 to 30 percent. The higher efficiencies generally are achieved for systems with low to moderate interfacial tension. As discussed earlier, the liquid entering the column at the larger volumetric flow rate generally should be dispersed to obtain satisfactory interfacial area for mass transfer. Example mass-transfer data are plotted in Fig. 15-38 for low values of $\mathcal{E}$. The advantage gained by dispersing the liquid flowing at the larger rate, which results in lower values on the x axis of Fig. 15-38 and consequently lower transfer unit heights, is clear.

Liquid Distribution Good initial distribution is not as essential in a sieve tray extractor as it is in a packed extractor, since the trays provide redistribution. While the same distributors used in packed columns are applicable, simpler devices also are used. Capped pipes with holes drilled uniformly have been found to be adequate in many cases.

Drop Size, Holdup, and Interfacial Area Drop size is estimated by using one of the models listed in Table 15-15, and holdup is estimated from expressions given in Table 15-16. Interfacial area is then calculated by using Eq. (15-109).

Sieve Tray Design Perforations usually are in the range of 0.125 to 0.25 in (0.32 to 0.64 cm) in diameter, set 0.5 to 0.75 in (1.27 to 1.81 cm) apart, on square or triangular pitch. There appears to be relatively little effect of hole size on the mass-transfer rate, except that with systems of high interfacial tension, smaller holes will produce somewhat better mass transfer. The entire hole area is normally set at 15 to 25 percent of the column cross section, although adjustments may be needed. The velocity through the holes should be such that drops do not form slowly at the holes, but rather the

TABLE 15-22 Mass-Transfer Data for Packed Columns

System	Column diameter, in	Packing	Ref.
Water-acetic acid-ethyl acetate, cyclohexane, methylcyclohexane, ethyl acetate + benzene	1	0.25-in saddles	3
Water-acetic acid-methyl isobutyl ketone	1.95	0.23-in rings	9
	3	0.375-in plastic spheres	12
		0.375-in plastic, ceramic rings	14
		0.5-in plastic, ceramic saddles	14
Water-acetic acid-toluene	6	Montz B1-300 1-in stacked Bialecki rings	2
Water-acetone-hydrocarbon	1.88	0.25- and 0.375-in rings, 6-mm beads	16
Water-acetone-toluene	2-4	0.5- and 0.75-in rings	1
	4	0.5-in rings, $\frac{5}{8}$ -in Pall rings, IMTP® 15, SMV structured, spray	18
	16.8	IMTP® 25, IMTP® 40, Intalox® 2T structured, spray	19
	6	Montz B1-300 1-in stacked Bialecki rings	2
Water-adipic acid-ethyl ether	4	SMV	22
	6	0.5- and 0.75-in rings, 0.375-in spheres	7
Water-benzoic acid-carbon tetrachloride	1.95	0.25-in rings	8
Water-benzoic acid-toluene	8.7	0.5-in saddles, 0.5-in rings	17
Water-diethylamine-toluene	3, 4, 6	0.25- to 1-in rings	11
	3	0.375-in rings	20
Water-ethyl acetate	4	0.5-in rings	5
Water-isopar(m)	16.8	IMTP® 25, IMTP® 40, Intalox® 2T structured, spray	19
Water-kerosene	4	SMV	22
Water-methyl ethyl ketone-kerosene	18	1-in rings, 1-in saddles, 1-in Pall rings, spray	13, 4
Water-methylisobutyl-carbinol	4	0.5-in rings	21
Water-methyl ethyl ketone	4	0.5-in rings	21
Water-propionic acid-methyl isobutyl ketone	1.88	0.25- and 0.375-in rings, 6-mm beads	16
Water-propionic acid-carbon tetrachloride	4	SMV	22
Water-succinic acid-1-butanol	4	0.5-in rings, $\frac{5}{8}$ -in Pall rings, IMTP® 15, SMV structured, spray	18
	4	SMV	22
Water-toluene	6	Montz B1-300 1-in stacked Bialecki rings	2
Acetone (aq)-soybean oil, linseed oil	2	0.25-in saddles, 0.5-in rings	23
Petroleum-furfural	2	0.25-in rings	6
	1.2	0.16-in rings	15
Toluene-heptane-diethylene glycol	1.4, 2.25	Glass and brass rings	10

NOTE: To convert inches to centimeters, multiply by 2.54.

References:

1. Degaleesan and Laddha, *Chem. Eng. Sci.*, **21**, p. 199 (1966); *Indian Chem. Eng.*, **8**(1), p. 6 (1966).
2. Billet and Mackowiak, *Fette-Seifen-Anstrichmittel*, **87**, pp. 205-208 (1985).
3. Eaglesfield, Kelly, and Short, *Ind. Chem.*, **29**, pp. 147, 243 (1953).
4. Eckert, *Hydrocarbon Processing*, **55**(3), pp. 117-124 (1976).
5. Gaylor and Pratt, *Trans. Inst. Chem. Eng. (London)*, **31**, p. 78 (1953).
6. Garwin and Barber, *Pet. Refiner*, **32**(1), p. 144 (1953).
7. Gier and Hougen, *Ind. Eng. Chem.*, **45**, p. 1362 (1953).
8. Guyer, Guyer, and Mauli, *Helv. Chim. Acta*, **38**, p. 790 (1955).
9. Guyer, Guyer, and Mauli, *Helv. Chim. Acta*, **38**, p. 955 (1955).
10. Kishinevskii and Mochalova, *Zh. Prikl. Khim.*, **33**, p. 2344 (1960).
11. Liebson and Beckmann, *Chem. Eng. Prog.*, **49**, pp. 405-416 (1953).
12. Moorhead and Himmelblau, *Ind. Eng. Chem. Fundam.*, **1**, p. 68 (1962).
13. Nemunaitis, Eckert, Foote and Rollison, *Chem. Eng. Prog.*, **67**(11), pp. 60-67 (1971).
14. Osmon and Himmelblau, *J. Chem. Eng. Data*, **6**, p. 551 (1961).
15. Sef and Moretu, *Nafta (Zagreb)*, **5**, p. 125 (1954).
16. Rao and Rao, *J. Chem. Eng. Data*, **6**, p. 200 (1961).
17. Row, Koffolt, and Withrow, *Trans. Am. Inst. Chem.*, **46**, p. 1229 (1954).
18. Seibert and Fair, *Ind. Chem. Eng. Res.*, **27**(3), p. 470 (1988).
19. Seibert, Reeves, and Fair, *Ind. Chem. Eng. Res.*, **29**(9), p. 1901 (1990).
20. Shih and Kraybill, *Ind. Eng. Chem. Process. Des. Dev.*, **5**, p. 260 (1966).
21. Smith and Beckmann, *Am. Inst. Chem. Eng. J.*, **4**, p. 180 (1958).
22. Streiff and Jancic, *Ger. Chem. Eng.*, **7**, pp. 178-183 (1984).
23. Young and Sullans, *J. Am. Oil Chem. Soc.*, **32**, p. 397 (1955).

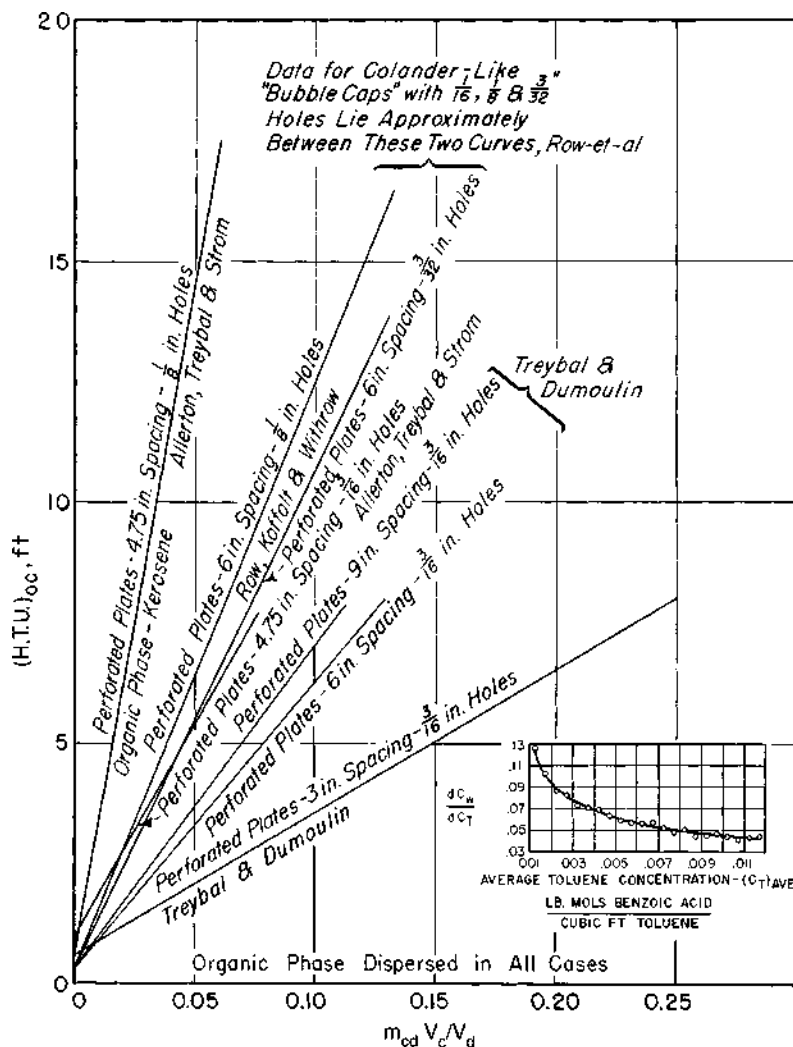


FIG. 15-38 Mass-transfer data for sieve plate and modified bubble plate columns. System: benzoic acid + water + toluene, except where noted. To convert feet to meters, multiply by 0.3048; to convert inches to centimeters, multiply by 2.54. [Data taken from Allerton, Strom, and Treybal, *Trans. AIChE*, **39**, p. 361 (1943); Row, Koffelt, and Withrow, *Trans. AIChE*, **37**, p. 559 (1941); and Treybal and Dumoulin, *Ind. Eng. Chem.*, **34**, p. 709 (1942).]

dispersed phase streams through the openings as a jet that breaks up into drops at a slight distance from the tray. It is common practice to set the velocity of liquid exiting the holes to correspond to a Weber number between 8 and 12. This normally gives velocities in the range of 0.5 to 1.0 ft/s (15 to 30 cm/s). The same general guidelines used to specify hole size and velocities for plate dispersers apply to sieve tray design. See Eqs. (15-107) and (15-108) and the related discussions in "Liquid Distributors and Dispersers" under "Liquid-Liquid Extraction Equipment."

The velocity of the continuous phase in the downcomer (or upcomer) V_{down} , which sets the downcomer cross-sectional area, should be set at a value lower than the terminal velocity of some arbitrarily small droplet of dispersed phase, say, $\frac{1}{32}$ or $\frac{1}{16}$ in (0.08 or 0.16 cm) in diameter; otherwise, recirculation of entrained dispersed phase around a tray will result in flooding. The terminal velocity of these small drops can be calculated by using Stokes' law: $u_t = (gd_p^2 \Delta \rho) / 18\mu_c$.

Downcomer area typically is in the range of 5 to 20 percent of the total cross-sectional area, depending upon the ratio of continuous- to dispersed-phase volumetric flow rates. The downcomers should extend beyond the accumulated layer of dispersed phase on the tray, and the tray area directly opposite downcomers should be kept free of perforations.

The spacing between trays should be sufficient that (1) the "streamers" of dispersed liquid from the holes break up into drops before coalescing into the layer of liquid on the next tray; (2) the cross-flow velocity of continuous-phase liquid does not cause excessive entrainment of the dispersed phase; and (3) the column may be entered through handholes or manholes in the sides for inspection and cleaning. For systems that accumulate an interface rag, provision may be made for periodic withdrawal of the rag through the side of the column between trays. For large columns, tray spacing between 18 and 24 in (45 and 60 cm) is generally recommended.

The height of the coalesced layer at each tray is given by

$$h = \frac{\Delta P_o + \Delta P_{\text{down}} - \phi_d g \Delta \rho L}{(1 - \phi_d) g \Delta \rho} \quad (15-151)$$

where L is the downcomer length. Equation (15-151) is a slightly simplified form of the expression given by Mewes and Kunkel [Ger. Chem. Eng., **1**, pp. 111–115 (1978)]. In most cases holdup is low, and Eq. (15-151) reduces to $h = (\Delta P_o + \Delta P_{\text{down}})/(g \Delta \rho)$. The orifice pressure drop ΔP_o may be calculated by using the model of Pilhofer and Goedl [Chem. Eng. Tech., **49**, p. 431 (1977)]:

$$\Delta P_o = \frac{1}{2} \left(1 - \frac{0.71}{\log \text{Re}} \right)^{-2} \rho_d V_o^2 + 3.2 \left(\frac{d_o^2 g \Delta \rho}{\sigma} \right)^{0.2} \frac{\sigma}{d_o} \quad (15-152)$$

where V_o is the velocity through the orifice, d_o is the orifice diameter, and $\text{Re} = V_o d_o \rho_d / \mu_d$. The pressure drop through the downcomer ΔP_{down} includes losses due to (1) friction in the downcomer, which should be negligible; (2) contraction and expansion upon entering and leaving the downcomer; and (3) two abrupt changes in direction. These losses total 4.5 velocity heads:

$$\Delta P_{\text{down}} = \frac{4.5 V_{\text{down}}^2 \rho_c}{2} \quad (15-153)$$

For large columns, the design should be specified such that the height of the coalesced layer is at least 1 in (2.5 cm) to ensure all the holes are adequately covered, and one should allow for the trays to be slightly out of level. On the other hand, the height of the coalesced layer should not be too large, since this is unproductive column height that unnecessarily increases the total column height requirement. A typical design value is about 2 in (5 cm).

Envelope-style segmental downcomers (Fig. 15-39) often are used in commercial-scale sieve tray extractors instead of circular or pipe-style downcomers. The area of an envelope downcomer is given by

$$A = \frac{H}{6S} (3H^2 + 4S^2) \quad (15-154)$$

The distance S is determined from the column diameter. The distance H is obtained from

$$S = \left[8H \left(\frac{D_{\text{col}}}{2} - \frac{H}{2} \right) \right]^{1/2} \quad (15-155)$$

The diameter of a circular downcomer with equivalent area is given by

$$D_{\text{eq}} = \sqrt{\frac{4}{\pi} A} \quad (15-156)$$

Sieve Tray Capacity at Flooding The capacity of a sieve tray is determined by hydraulic mechanisms involved in flooding and is not

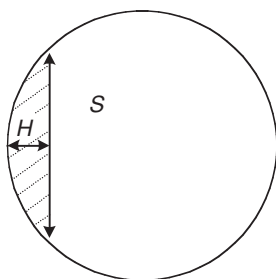


FIG. 15-39 Dimensions of an envelope-style segmental downcomer or upcomer (shaded area).

completely understood, especially for larger-diameter columns. Three studies using larger equipment have been reported by Oloidi, Jeffreys, and Mumford [Inst. Chem. Eng. Symp. Ser., **103**, pp. 117–132 (1987)]; Seibert and Fair [Ind. Eng. Chem., **32**, pp. 2213–2219 (1993)]; and Eldridge and Fair [Ind. Eng. Chem., **38**, pp. 218–222 (1999)]. An example of sieve tray flooding data is illustrated in Fig. 15-40.

The sieve tray capacity and efficiency are strongly influenced by the height of the coalesced layer. If the height of this layer grows to the outlet of the downcomer, a sharp reduction in efficiency will result since the mass-transfer height will be significantly reduced. In this case, the downcomer area and/or total perforated area should be increased. A flooding model based on the height of the coalesced layer is given by Seibert and Fair [Ind. Eng. Chem. Res., **32**(10), pp. 2213–2219 (1993)]

$$V_{cf} = \left[\frac{L - A}{B(V_{df}/V_{cf})^2 + C} \right]^{0.5} \quad (15-157)$$

$$A = \frac{6\sigma}{d_{cs} \Delta \rho g} \quad (15-158)$$

$$B = \frac{1.11 \rho_d}{g \Delta \rho f_{ha}^2} \quad (15-159)$$

$$C = \frac{2.7 \rho_c}{2g \Delta \rho f_{da}} \quad (15-160)$$

where L is the downcomer length, f_{ha} is the fractional hole area, and f_{da} is the fractional downcomer area.

High Cross-flow of the Continuous Phase Miniplant tests of sieve tray extractors are often performed prior to the final design of a commercial-scale column. The design often is scaled up based on superficial velocities of the dispersed and continuous phases calculated from the volumetric flow rates and the column cross-sectional area. However, in scaling up one must be careful about the cross-flow velocity (V_{cflow}) of the continuous phase. A value may be estimated from

$$V_{\text{cflow}} \approx \frac{L_{fp}}{z - h} V_c \quad (15-161)$$

where L_{fp} is the length of flow path, z is the tray spacing, h is the height of coalesced layer, and V_c is the superficial continuous-phase velocity. The magnitude of the cross-flow velocity of the continuous phase can be much greater than that studied in the miniplant. Multiple downcomers or upcomers reduce the flow path length and can be utilized in new designs to reduce cross-flow velocity. Large-diameter multiple downcomer (or upcomer) trays have been reported to provide 10 to 15 percent greater capacity relative to the single-pass tray. Seibert, Bravo, and Fair [ISEC '02 Proc., **2**, pp. 1328–1333 (2002)] propose a model for correcting the sieve tray capacity for high cross-flow velocity.

Mass-Transfer Data Mass-transfer data are available from the sources listed in Table 15-23. Mass-transfer performance can be expressed in terms of the number of transfer units per actual tray, or in terms of overall heights of transfer units for a given column configuration, as in Fig. 15-38. The system of Fig. 15-38 is one of high interfacial tension, so the heights of transfer units are expected to be relatively large. For systems of low interfacial tension, mass-transfer performance is likely to be much improved. Since sieve trays resemble and basically behave in the manner of stages, performance also can be expressed in terms of a stage efficiency, either as an overall ξ_o for the entire tower or, more satisfactorily, as a Murphree efficiency for each tray.

Tray Efficiency The overall efficiencies of sieve trays typically are between 10 and 30 percent. One of the earliest models for predicting the overall tray efficiency was an empirical one reported by Treybal [Liquid Extraction, 2d ed. (McGraw-Hill, 1963)]. Krishna, Murty, and Rao [Ind. Eng. Chem. Process Des. Dev., **7**(2),

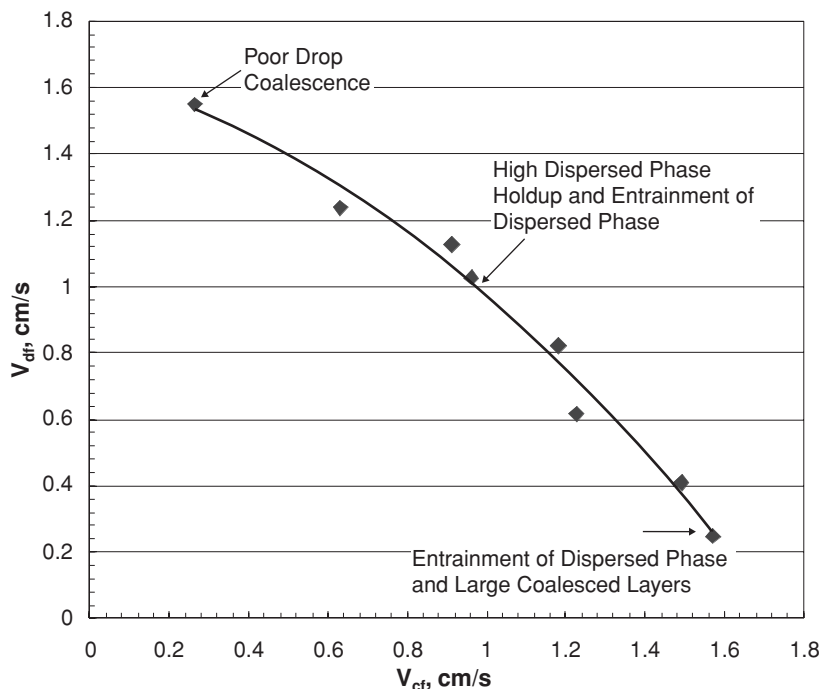


FIG. 15-40 Sieve tray flooding data. System: toluene (dispersed) + water (continuous). Tray spacing = 30.5 cm. Column diameter = 42.8 cm. [Taken from Seibert, Bravo, and Fair, ISEC '02 Proc., 2, pp. 1328–1333 (2002), with permission. Copyright 2002 South African Institute of Mining and Metallurgy.]

pp. 166–172 (1968)] modified the Treybal model to account for hole diameter:

$$\xi_o = 0.21 \left(\frac{z^{0.5}}{\sigma d_o^{0.35}} \right) \left(\frac{V_d}{V_c} \right)^{0.42} \quad (15-162)$$

where z is the tray spacing, cm; d_o is the hole diameter, cm; and σ is interfacial tension, dyn/cm. Seibert and Fair [Ind. Eng. Chem., 32(10), pp. 2213–2219 (1993)] recommend calculating the local Murphree stage efficiency based on the dispersed phase, assuming a log mean driving force and negligible mass-transfer contribution from drop formation:

$$\xi_{md} = 1 - \exp \left[- \frac{6k_{od} \phi_d (z - h)}{d_p V_d} \right] \quad (15-163)$$

The overall tray efficiency may then be estimated by using

$$\xi_o = \frac{\ln [1 + \xi_{md} (E - 1)]}{\ln E} \quad (15-164)$$

$$E = m_{dc}^{vol} \frac{V_d}{V_c} \quad (15-165)$$

Equation (15-163) assumes plug flow of the rising or falling drop population and complete mixing of the continuous phase on the tray. Also see Eldridge and Fair, Ind. Eng. Chem. Res., 38, pp. 218–222 (1999); Rocha et al., Ind. Eng. Chem. Res., 28(12), pp. 1873–1878 (1989); and Rocha, Cárdenas, and García, Ind. Eng. Chem. Res., 28(12), pp. 1879–1883 (1989).

Baffle Tray Columns Baffle tray columns are similar to spray columns except that baffles are added to reduce backmixing. The

baffles usually are slightly sloped to drain any solids that might settle out in the column and are designed to provide a high open area. Lemieux [Hydrocarbon Proc., 62(9), pp. 106–111 (1983)] and Fair [Hydrocarbon Proc., 72(5), pp. 75–79 (1993)] report on the performance and design of these columns for gas-liquid contacting. Treybal [Liquid Extraction, 2d ed. (McGraw-Hill, 1963)] provides a brief but valuable description of a baffle tray extractor. Although no design equations or performance data are provided, Treybal indicates that commercial tray spacings should be in the range of 10 to 15 cm (4 to 6 in). Treybal also provides an interesting illustration of a baffle tray extractor in operation (Fig. 15-41). This figure shows multiple trays with a very short spacing, with the dispersed light phase moving as a layer of liquid under each tray.

Because baffle tray performance data are not widely available, the results of a pilot-scale study (Seibert, Lewis, and Fair, Paper No. 112a, AIChE National Meeting, Indianapolis, 2002) are summarized in Figs. 15-42 to 15-47. The study was carried out using a 4.0-in- (10.2-cm-) diameter column set up with 5 to 30 trays. The trays were arranged in a side-to-side horizontal arrangement, as indicated in Fig. 15-41a. The data were generated by using the toluene (dispersed) + acetone + water (continuous) and butanol (dispersed) + succinic acid + water (continuous) systems. The effects of changes in baffle spacing and tray overlap (expressed as the percentage of total tray area covered by the next tray above or below) were measured for transfer of solute from the organic to the aqueous phase.

Hydraulic Capacity The capacity of the baffle trays at flooding was found to depend strongly on system properties, as shown in Fig. 15-42. The butanol system with its lower interfacial tension provided a much lower capacity relative to the toluene system with its higher interfacial tension. The capacity was found to be independent of tray spacing, as shown in Fig. 15-43. However, capacity was strongly affected by the degree of tray overlap, as shown in Figs. 15-44 and 15-45. See Seibert, Lewis, and Fair (Paper No. 112a, AIChE National Meeting, Indianapolis, 2002) for discussion of a proposed flooding model.

TABLE 15-23 Mass-Transfer Data for Sieve Tray Columns

System	Column diameter, in	Tray spacing, in	Ref.
Benzene-acetic acid-water	1.97	3.9-6.3	25
	1.97	3.2-6.3	24
	2.2	2.8-6.3	23
	1.6 × 3.2	5.9	20
Benzene-acetone-water	3	4, 8	13
Benzene-benzoic acid-water	3	4	13
Benzene-monochloroacetic acid-water	1.97	3.9-6.3	25
Benzene-propionic acid-water	1.97	3.2-6.3	24
Carbon tetrachloride-propionic acid-water	1.97	3.9-6.3	25
Clairsol-water	17.7	13-15	14
Ethyl acetate-acetic acid-water	2	8-24	10
Ethyl ether-acetic acid-water	8.63	4-7.2	15
Gasoline-methyl ethyl ketone-water	3.75	4.5, 6	11
Isopar(M)-water	16.8	12	21
Kerosene-acetone-water	3	4, 8	13
Kerosene-benzoic acid-water	3.63	4.75	1
Kerosene-benzoic acid-water	6	6, 12	9
Isopar-H-benzyl alcohol, methyl benzyl alcohol, acetophenone-water	2 × 12	24	2
Methylisobutylcarbinol-acetic acid-water	3	6	12
Methyl isobutyl ketone-adipic acid-water	4.18	6	5
Methyl isobutyl ketone-butyric acid-water	4.8	6-23	8
Methyl isobutyl ketone-acetic acid-water	4	6-12	17
Pegasol-propionic acid-water	9.7	8-24	18, 19
	4.8	6-11	7
	8.75	6	16
	3.63	4.75	1
Toluene-benzoic acid-water	3.56	3-9	22
	3	6	12
	2.72	9	6
	2	24	10
Toluene-diethylamine-water	4.18	6	3, 4
Toluene-water	16.8	12	21
	9.7	8-24	18
	16.8	12	21
	9.7	8-24	19
Toluene-acetone-water	4	6-12	17
	3.75	4.5, 6	11
2,2,4-Trimethylpentane-methyl ethyl ketone-water	3.75	4.5, 6	11

NOTE: To convert inches to centimeters, multiply by 2.54.

References:

1. Allerton, Strom, and Treybal, *Trans. Am. Inst. Chem. Eng.*, **39**, p. 361 (1943).
2. Angelo and Lightfoot, *Am. Inst. Chem. Eng. J.*, **14**, p. 531 (1968).
3. Garner, Ellis, and Fosbury, *Trans. Inst. Chem. Eng. (London)*, **31**, p. 348 (1953).
4. Garner, Ellis, and Hill, *Am. Inst. Chem. Eng. J.*, **1**, p. 185 (1955).
5. Garner, Ellis, and Hill, *Trans. Inst. Chem. Eng. (London)*, **34**, p. 223 (1956).
6. Goldberger and Benenati, *Ind. Eng. Chem.*, **51**, p. 641 (1959).
7. Krishnamurty and Rao, *Indian J. Technol.*, **5**, p. 205 (1967).
8. Krishnamurty and Rao, *Ind. Eng. Chem. Process Des. Dev.*, **7**, p. 166 (1968).
9. Lodh and Rao, *Indian J. Technol.*, **4**, p. 163 (1966).
10. Mayfield and Church, *Ind. Eng. Chem.*, **44**, p. 2253 (1952).
11. Moulton and Walkey, *Trans. Am. Inst. Chem. Eng.*, **40**, p. 695 (1944).
12. Murali and Rao, *J. Chem. Eng. Data*, **7**, p. 468 (1962).
13. Nandi and Ghosh, *J. Indian Chem. Soc., Ind. News Ed.*, **13**, pp. 93, 103, 108 (1950).
14. Ololdi and Mumford, *ISEC Proc.* (Munich, 1986).
15. Pyle, Duffey, and Colburn, *Ind. Eng. Chem.*, **42**, p. 1042 (1950).
16. Row, Koffolt, and Withrow, *Trans. Am. Inst. Chem. Eng.*, **37**, p. 559 (1941).
17. Rocha, Humphrey, and Fair, *Ind. Eng. Chem. Process Des.*, **25**, p. 862 (1986).
18. Rocha et al., *Ind. Eng. Chem. Res.*, **28**(12), pp. 1873-1878 (1989).
19. Rocha, Cardenas, and Garcia, *Ind. Eng. Chem. Res.*, **37**(12), pp. 1879-1883 (1989).
20. Shiotsuka and Murakami, *Kagaku Kogaku*, **30**, p. 727 (1966).
21. Seibert and Fair, *Ind. Eng. Chem. Res.*, **32**(10), pp. 2213-2219 (1993).
22. Treybal and Dumoulin, *Ind. Eng. Chem.*, **34**, p. 709 (1942).
23. Ueyama and Koboyashi, *Bull. Univ. Osaka Prefect.*, **A7**, p. 113 (1959).
24. Zheliznyak, *Zh. Prikl. Khim.*, **40**, p. 689 (1967).
25. Zheliznyak and Brounshtein, *Zh. Prikl. Khim.*, **40**, p. 584 (1967).

Baffle Tray Efficiency Baffle tray mass-transfer efficiency was observed to depend strongly on the tray spacing and system properties, as shown in Figs. 15-46 and 15-47. In these studies, a tray spacing of about 10 cm provided a minimum HETS. The data indicate that the performance of baffle trays relative to sieve trays depends upon the interfacial tension of the system. For the high-interfacial-tension system (Fig. 15-46), the baffle tray performance (in terms of capacity and mass transfer) is relatively low compared to that of a sieve tray. However, for the low-interfacial-tension system (Fig. 15-47), performance was somewhat better using 62 percent tray overlap.

AGITATED EXTRACTION COLUMNS

In certain applications, the mass-transfer efficiency of a static extraction column is quite low, especially for systems with moderate to high interfacial tension. In these cases, efficiency may be improved by

mechanically agitating the liquid-liquid dispersion within the column to better control drop size and population density (dispersed-phase holdup). Many different types of mechanically agitated extraction columns have been proposed. The more common types include various rotary-impeller columns, the reciprocating-plate column, and the rotating-disk contactor (RDC). The following is a brief review. For more detailed discussion, see *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994); *Science and Practice of Liquid-Liquid Extraction*, vol. 1, Thornton, ed. (Oxford, 1992); and *Handbook of Solvent Extraction*, Lo, Baird, and Hanson, eds. (Wiley, 1983; Krieger, 1991).

Rotating-Impeller Columns A number of different rotating-impeller column extractors have been proposed and built over the years. Only the Scheibel and Kühni designs are reviewed here. For information about the Oldshue-Rushton design, see the previous edition of this handbook. Also see Oldshue, Chap. 13.4 in *Handbook of*

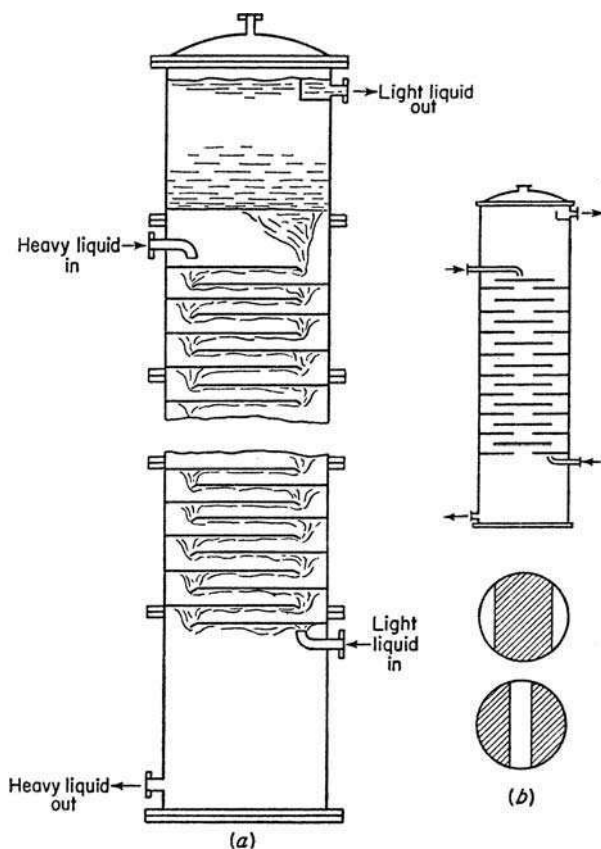


FIG. 15-41 Baffle towers. (a) Side-to-side flow at each tray. (b) Center-to-center flow (disk-and-doughnut style). (c) Center-to-side flow. [Reprinted from Treybal, *Liquid Extraction* (McGraw-Hill, 1963), with permission. Copyright 1963 McGraw-Hill, Inc.]

Solvent Extraction, Lo, Baird, and Hanson, eds. (Wiley, 1983; Krieger, 1991).

Scheibel Extraction Column The original Scheibel column design consisted of a series of knitted-wire-mesh packed sections placed within a vertical column, with a centrally located impeller between each section and no baffles [Scheibel and Karr, *Ind. Eng. Chem.*, **42**(6) pp. 1048–1057 (1950)]. A second-generation Scheibel design [AICHE J., **2**(1), pp. 74–78 (1956); U.S. Patent 2,850,362 (1958)] added flat partitions or baffles to the ends of each packed section, and the impellers were surrounded by stationary shroud baffles to direct the flow of droplets discharged from the impeller tips. The new baffling arrangement improved efficiency, allowing design of larger-diameter columns with less power input and decreased height per theoretical stage. A third design by Scheibel [U.S. Patent 3,389,970 (1968)] eliminated the wire-mesh packing and retained the use of baffles and shrouded impellers (Fig. 15-48). The packed sections were replaced by agitated sections. This design was developed because the wire-mesh packed sections were prone to fouling (plugging) and difficult to clean. A Scheibel extractor of this type is very well suited to handling mixtures with high interfacial tension and can be designed with a large number of stages. It is not as well suited for systems that tend to emulsify easily owing to the high shear rate generated by a rotating impeller. Because of its internal baffling, which controls the mixing patterns on the stages, the Scheibel column has proved to be one of the more efficient extractors in terms of height of a theoretical stage; this makes it well suited to applications that

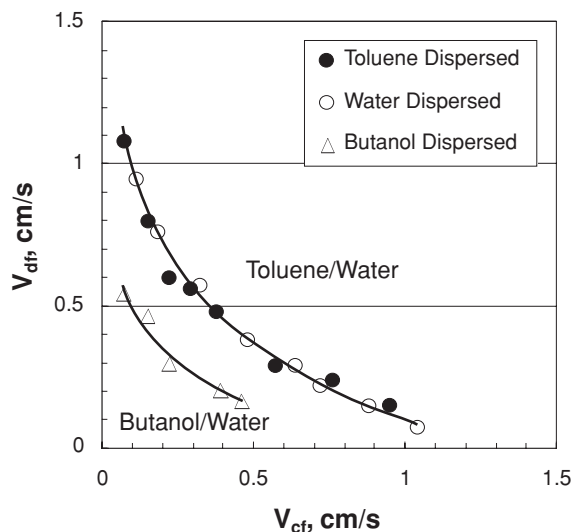


FIG. 15-42 Capacity characteristics of a baffle tray extractor. Tray overlap = 62 percent. Column diameter = 10.2 cm. [Taken from Seibert, Lewis, and Fair, Paper No. 112a, AIChE National Meeting, Indianapolis (November 2002), with permission. Copyright 2002 AIChE.]

require a large number of stages or are located indoors with headroom restrictions. Holmes, Karr, and Cusack [*Solvent Extraction and Ion Exchange*, **8**(3), pp. 515–528 (1990)] have published results comparing the efficiency of the Scheibel column to that of other extractors using the system toluene + acetone + water. For additional discussion, see Scheibel, Chap. 13.3 in *Handbook of Solvent Extraction*, Lo, Baird, and Hansen, eds. (Wiley, 1983; Kreiger, 1991). A related column design called the AP column consists of alternating sections of Scheibel-type agitators and structured packing [Cusack, Glatz, and Holmes, *Proc. ESEC'99, Soc. Chem. Ind.*, p. 427 (2001)]. The high open area of the packing allows for higher capacity while the agitation provides increased efficiency.

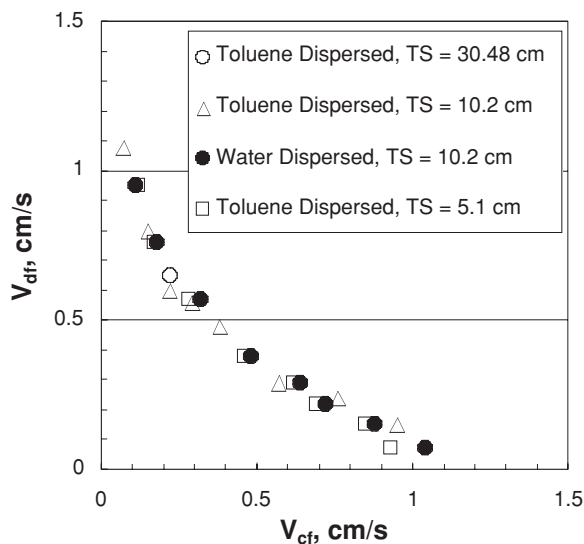


FIG. 15-43 Effect of tray spacing on baffle tray capacity. [Taken from Seibert, Lewis, and Fair, Paper No. 112a, AIChE National Meeting, Indianapolis (November 2002), with permission. Copyright 2002 AIChE.]

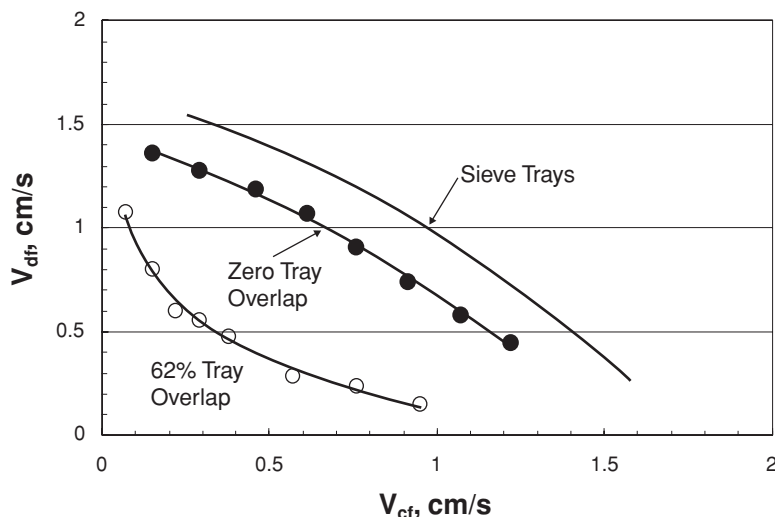


FIG. 15-44 Effect of tray overlap on baffle tray capacity. System: toluene (*d*) + acetone + water (*c*). [Taken from Seibert, Lewis, and Fair, Paper No. 112a, AIChE National Meeting, Indianapolis (November 2002), with permission. Copyright 2002 AIChE.]

As with most agitated extractors, the final design of a Scheibel column typically involves scale-up of data generated in a miniplant or pilot-plant test. The column vendor should be consulted for specific information. The key scale-up guidelines are as follows: (1) $D_i(2)/D_i(1) = [Q(2)/Q(1)]^{0.4}$; (2) $Z_i(2)/Z_i(1) = [D_i(2)/D_i(1)]^{0.70}$; (3) stage efficiency is the same for the pilot and full scale; and (4) power per unit volume is the same for each scale [Cusack and Karr, *Chem. Eng. Magazine*, pp. 112–119 (1991)]. Industrial columns up to 10 ft (3 m) in diameter and containing 90 actual stages have been designed using the following general procedures and a 3-in (75-mm) pilot column:

1. Pilot tests usually are conducted in 3-in (75-mm-) diameter columns. The column should contain a sufficient number of stages to complete the extraction. This may require several iterations on column height.

2. The column is run over a range of throughputs $V_d + V_c$ and agitation speeds. At each condition, the concentrations of solute in extract and raffinate streams are measured after steady-state operation has been achieved (usually after 3 to 5 turnovers of column volume). At each throughput, the flood point is determined by increasing the agitation until flooding is induced. A minimum of three throughput ranges are examined in this manner. Mass-transfer performance is measured at several agitation speeds up to the flood point.

3. From the above mass-transfer and flooding data, the combination of specific throughput and agitation speed that gives the optimum economic performance for the required separation can be determined. This information is used to specify the specific throughput value [$\text{gal}/(\text{h} \cdot \text{ft}^3)$ or $\text{m}^3/(\text{h} \cdot \text{m}^3)$] and agitation speed (rpm) for the commercial design. However, unlike the RDC and Karr columns, for

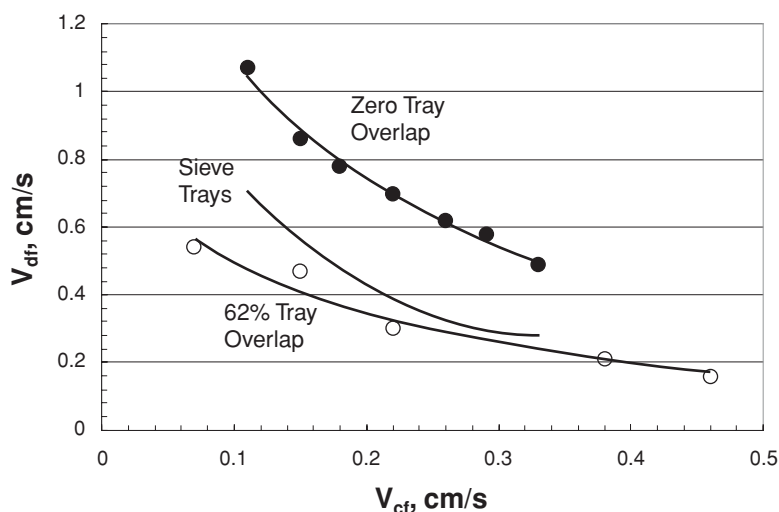


FIG. 15-45 Effect of tray overlap on baffle tray capacity. System: *n*-Butanol (*d*) + succinic acid + water (*c*). [Taken from Seibert, Lewis, and Fair, Paper No. 112a, AIChE National Meeting, Indianapolis (November 2002), with permission. Copyright 2002 AIChE.]

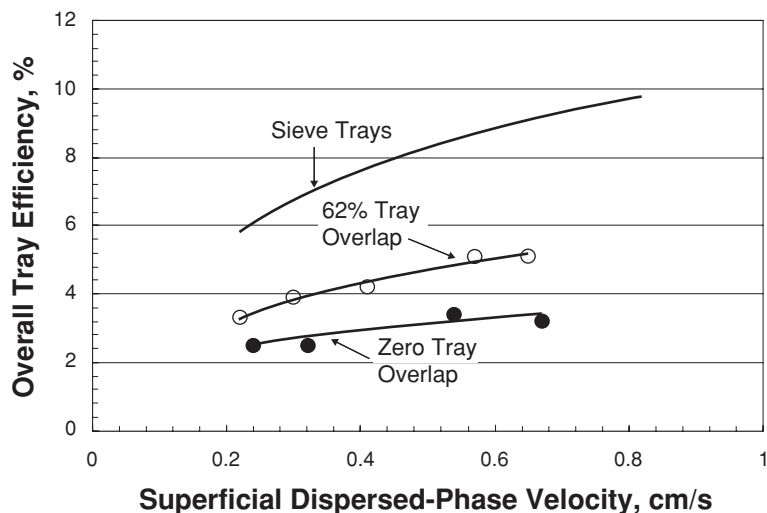


FIG. 15-46 Effect of tray overlap on baffle tray efficiency. System: toluene (*d*) + acetone + water (*c*). Tray spacing = 10.2 cm. [Taken from Seibert, Lewis, and Fair, Paper No. 112a, AIChE National Meeting, Indianapolis (November 2002), with permission. Copyright 2002 AIChE.]

which the specific throughput of the scaled-up version is the same as that of the pilot column, it is a characteristic of the Scheibel column that the throughput of the scaled-up column is on the order of 3 to 5 times greater than that achieved on the 3-in.-diameter pilot column. The limited throughput of the 3-in. column is due to its restrictive geometry; these restrictions are removed in the scaled-up columns.

4. Once the column diameter is determined, the stage geometry can be fixed. The geometry of a stage is a complex function of the column diameter. In the 3-in. pilot column, the stage height-to-diameter ratio is on the order of 1:3. On a 10-ft- (3-m-) diameter column, it is on the order of 1:8. The recommended ratio of height to diameter is $Z_i(2)/Z_i(1) = [D_i(2)/D_i(1)]^{10.70}$.

5. The principle of the Scheibel column scale-up procedure is to maintain the same stage efficiency. Therefore, the scaled-up column

will have the same number of actual stages as the pilot column. The only difference is that the stages will be taller, to take into account the effect of axial mixing. With the agitator dimensions determined, the speed is then calculated to give the same power input per unit of throughput. Scheibel found that power input can be correlated by

$$P = 1.85 \rho \omega^3 D_i^5 \quad (15-166)$$

where P is the power input per mixing stage, D_i is the impeller diameter, ρ is the average liquid density, and ω is the impeller speed (rotations per unit time).

Kühni Column Like the Scheibel column, the Kühni column uses shrouded (closed) turbine impellers as mixing elements on a central shaft (Fig. 15-49). Perforated partitions or stator plates extend

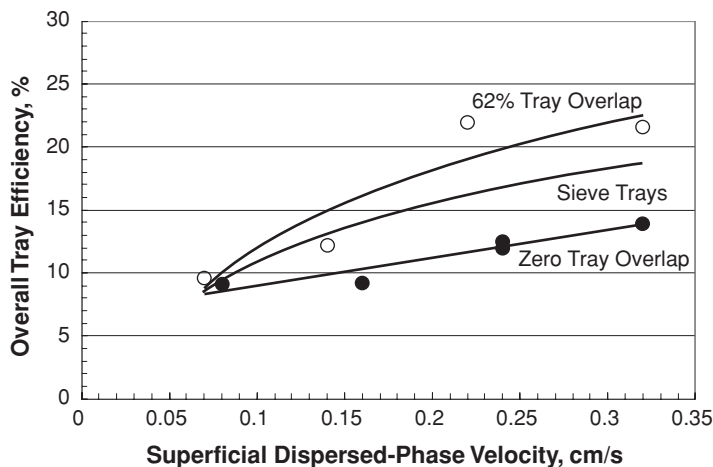


FIG. 15-47 Effect of tray overlap on baffle tray efficiency. System: *n*-butanol + succinic acid + water. Tray spacing = 10.2 cm. [Taken from Seibert, Lewis, and Fair, Paper No. 112a, AIChE National Meeting, Indianapolis (November 2002), with permission. Copyright 2002 AIChE.]

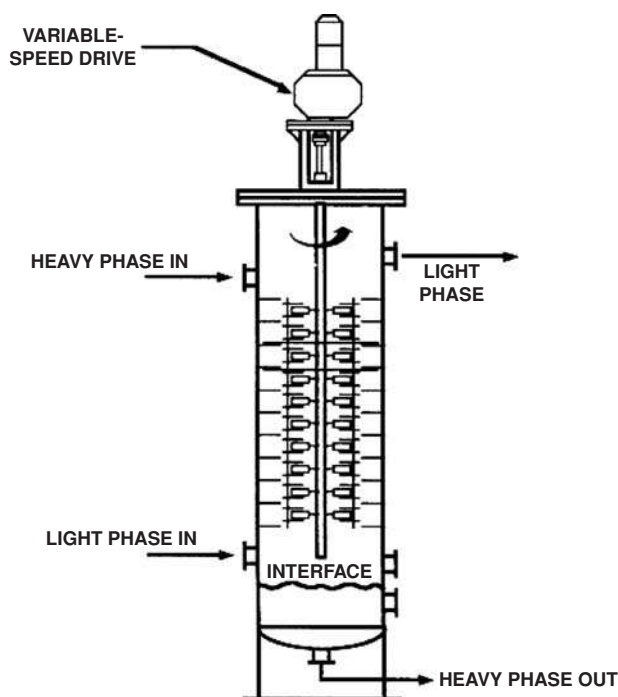


FIG. 15-48 Scheibel column extractor (third-generation design). (Courtesy of Koch Modular Process Systems.)

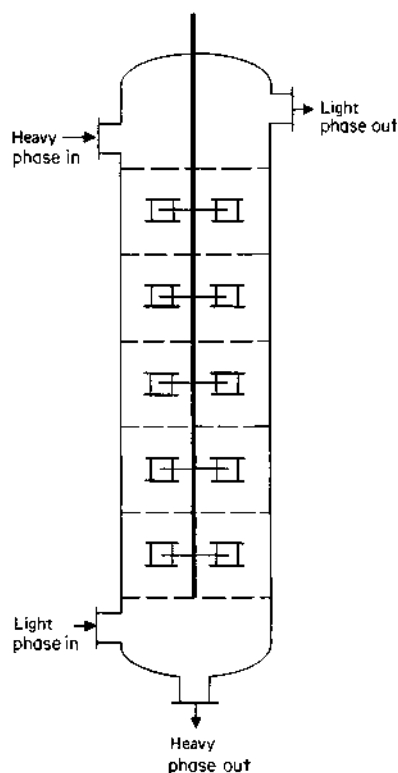


FIG. 15-49 Kühni column extractor.

over the vessel cross section to separate the extraction stages and reduce backmixing between stages. The fractional free-flow area between compartments can be adjusted by changing the free area around the rotor shaft and/or the perforations in the stator plate. As the free-flow area increases, throughput increases at the expense of increased axial mixing of the continuous phase and reduced mass-transfer performance. Throughput typically varies from $30 \text{ m}^3/(\text{h} \cdot \text{m}^2)$ [$750 \text{ gal}/(\text{h} \cdot \text{ft}^2)$] to significantly higher values depending upon the specific design factors chosen to meet the requirements of a given application.

Mögli and Bühlmann [Chap. 13.5 in *Handbook of Solvent Extraction*, Lo, Baird, and Hanson, eds. (Wiley, 1983; Krieger, 1991)] outline general considerations for specifying a commercial design from pilot data. The column vendor should be consulted for specific information. The scale-up procedures are based upon hydrodynamic and geometric similarity between the pilot-scale and plant-scale designs. Individual stage geometry (impeller size and free area of the stator plate) may be tailored for each stage, especially in cases where physical properties vary significantly along the column length. Mögli and Bühlmann suggest design options to maintain a somewhat uniform interfacial area along the column to minimize the impacts of axial mixing. Pratt and Stevens [*Science and Practice of Liquid-Liquid Extraction*, vol. 1, Thornton, ed. (Oxford, 1992), [Chap. 8, p. 541] provide recommended scale-up factors for a Kühni column as follows: $D_i/D_t = 0.33$ to 0.5 , compartment height $= 0.2$ to $0.3D_t$, and the fractional free area of the stator plates $= 0.2$ to 0.4 . The minimum recommended diameter for the pilot column is 60 mm (2.4 in) for specifying columns up to 1 m in diameter and 150 mm (6 in) for specifying larger-diameter columns.

A stagewise computational procedure is proposed by Kumar and Hartland [*Ind. Eng. Chem. Res.*, **38**(3), pp. 1040–1056 (1999)] for design of a Kühni column. The procedure considers backflow of the continuous phase, with an attempt to estimate average drop size, drop size distribution, dispersed-phase holdup, flooding velocities, mass-transfer coefficients, and axial mixing. A design example for extraction of aniline from water is presented. This approach to design can be very useful for initial estimates, but as with all agitated extractors, some pilot testing is recommended for a final commercial design. Also see the discussion by Gomes et al. [*Ind. Eng. Chem. Res.*, **43**(4), pp. 1061–1070 (2004)].

Reciprocating-Plate Columns Another approach to agitating a dispersion within an extraction column is the use of reciprocating plates. This generally results in a more uniform drop size distribution because the shear forces are more evenly distributed over the entire cross section of the column. Reciprocating-plate extractors have a wide turndown range and are well suited to systems with moderate interfacial tension. They often can handle systems exhibiting a tendency to emulsify, and because of their high open-area design, they can handle slurries of solids, some containing as much as 30 percent solids by weight. Several types of reciprocating-plate extractors have been designed; design differences generally involve differences in the plate open area and plate spacing as well as the inclusion or omission of static baffles or downcomers. For detailed discussion of these designs, see Lo and Procházka, Chap. 12 in *Handbook of Solvent Extraction*, Lo, Baird, and Hanson, eds. (Wiley, 1983; Krieger, 1991); and Baird et al., Chap. 11 in *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994).

The Karr reciprocating-plate column (Fig. 15-50) is a popular example. It uses dual-flow plates with 50 to 60 percent open area and has no downcomers [Karr, *AIChE J.*, **5**(4), pp. 446–452 (1959); Karr and Lo, *Chem. Eng. Prog.*, **72**(11), pp. 68–70 (1976); and Karr, *AIChE J.*, **31**(4), pp. 690–692 (1985)]. Because of the high open area, a Karr column may be operated with relatively high throughput compared to other types of agitated columns, up to about $1000 \text{ gal}/(\text{h} \cdot \text{ft}^2)$ [$40 \text{ m}^3/(\text{h} \cdot \text{m}^2)$] depending upon the application. The plates are mounted on a central shaft that moves up and down through a stroke length of up to 2 in (5 cm). As the diameter of the column increases, the HETS achieved by the column tends to increase due to axial mixing effects. For columns with a diameter greater than 1 ft (0.3 m), doughnut-shaped baffle plates may be added every 5 plates (typically) within the plate stack to minimize axial mixing. A Karr column also is well suited

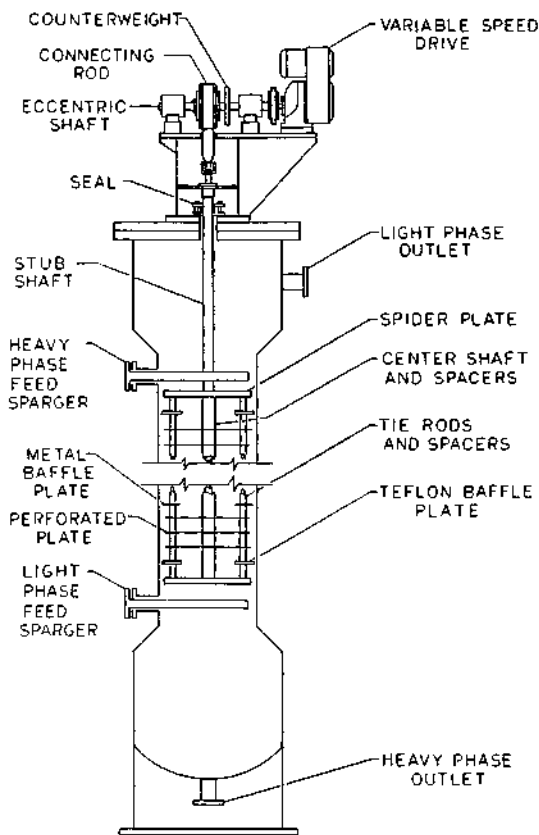


FIG. 15-50 Karr reciprocating-plate extraction column.

for corrosive systems since the plates can be fabricated from non-metallic materials. Pratt and Stevens [Chap. 8 in *Science and Practice of Liquid-Liquid Extraction*, vol. 1, Thornton, ed. (Oxford, 1992), p. 556] provide recommended geometric design and operating conditions for a Karr column as follows: reciprocation amplitude = 1 to 2 in (2.5 to 5 cm) with a 1-in amplitude being most common; reciprocation speed = 10 to 400 complete strokes (up and down) per minute; plate spacing = 2 to 6 in (5 to 15 cm); hole pitch = 0.625 to 0.75 in (1.6 to 1.9 cm); hole diameter = 0.50 to 0.625 in (1.3 to 1.6 cm); plate wall clearance = 1.25 to 2.5 in (3.2 to 6.4 cm). The plate spacing may be graduated to produce uniform drop size and population density along the length of the column, particularly for systems with high solute concentrations and depending upon how physical properties change along the column length [Karr, U.S. Patent 4,200,525 (1980)].

Baird et al. [Chap. 11 in *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994)] discuss and summarize correlations for predicting holdup and flooding, mean drop diameter, axial mixing, mass transfer, and reciprocating-plate column performance. Kumar and Hartland [*Ind. Eng. Chem. Res.*, **38**(3), pp. 1040–1056 (1999)] present a correlation-based computational procedure for design of a Karr reciprocating-plate column, and they give an example for separation of acetone from water by using toluene. A backmixing model is described by Stella et al. [*Ind. Eng. Chem. Res.*, **45**(19), pp. 6555–6562 (2006)].

As with other agitated extractors, the final design of a commercial-scale Karr column is based on pilot test data. The column vendor should be consulted for specific information. The following general procedure is recommended:

1. For specifying commercial columns up to 6.5 ft (2 m) in diameter, testing in a pilot column of 1-in (25-mm) diameter is sufficient. If the anticipated scaled-up diameter is greater than 6.5 ft, then the pilot

tests should be conducted in a 2-in- (50-mm-) diameter column. The column should be tall enough to accomplish the complete extraction. This may require several iterations on column height.

2. The column is first optimized with regard to plate spacing. The plate spacing is adjusted along the length of the column to obtain the same tendency to flood everywhere in the column. If one particular section appears to flood early, limiting the throughput, then the plate spacing should be increased in this section. This will decrease the power input into that section. Similarly, in sections that appear to be undermixed because the population of drops is low, the plate spacing should be decreased.

3. Once the plate spacing is optimized, the column is run over a range of total throughputs ($V_d + V_c$) and agitation speeds. There should be a minimum of three throughput levels and at each throughput three agitation speeds. After steady state is attained at each condition (usually 3 to 5 turnovers of column volume), samples are taken and the separation is measured. At each condition the flood point also is determined. In small-scale tests, the data used for scale-up should be collected at a point very close to flooding, say, 95 percent of flooding. Scaling these data typically results in a commercial-scale unit that operates at roughly 80 or 85 percent of flooding.

4. From the data, plots are made of volumetric efficiency and agitation speed at each throughput level. From these plots the condition that gives the maximum volumetric efficiency is selected for scale-up. For additional discussion, see Lo and Prochazka, Chap. 12 in *Handbook of Solvent Extraction*, Lo, Baird, and Hanson, eds. (Wiley, 1983; Krieger, 1991).

5. For scale-up, the following parameters are kept constant: total throughput per unit area, plate spacing, and stroke length. The height and agitation speed of the scaled-up column are then calculated from the following relationships:

$$\frac{Z_i(2)}{Z_i(1)} = \left(\frac{D_{col}(2)}{D_{col}(1)} \right)^{0.38} \quad (15-167)$$

$$\frac{SPM(2)}{SPM(1)} = \left(\frac{D_{col}(1)}{D_{col}(2)} \right)^{0.14} \quad (15-168)$$

Here Z_i is the plate stack height, D_{col} is the column diameter, SPM is the reciprocating speed (complete strokes per minute), and 1 and 2 denote the pilot column and the scaled-up column, respectively.

Karr and Ramanujam [St. Louis AIChE Symposium (March 19, 1987)] propose a power per unit volume normalization factor for scale-up of the reciprocation speed if the pilot column plates have a different open area than the industrial scale plates, as follows:

$$\frac{SPM(2)}{SPM(1)} = \left(\frac{D_{col}(1)}{D_{col}(2)} \right)^{0.14} \left(\frac{\epsilon(2)^2}{1 - \epsilon(2)^2} \right) \left(\frac{1 - \epsilon(1)^2}{\epsilon(1)^2} \right) \quad (15-169)$$

where ϵ is the fractional open area of the perforated plate.

Rotating-Disk Contactor The rotary-disk contactor (RDC) is a vertical column containing an assembly of rotating disks and stationary baffles or stators. A typical design is illustrated in Fig. 15-51. The column is formed into compartments by horizontal doughnut-shaped or annular baffles, and within each compartment agitation is provided by a rotating, centrally located, horizontal disk. The rotating disk is smooth and flat and has a diameter less than that of the opening in the stationary baffles. The RDC extractor has been widely used because of its simplicity of construction, availability in relatively large diameters for high production rates, and low power consumption. For detailed reviews, see Chaps. 9 and 17 in *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994); and Chaps. 13.1 and 13.2 in *Handbook of Solvent Extraction*, Lo, Baird, and Hanson, eds. (Wiley, 1983; Krieger, 1991). Also see Al-Rahawi, *Chem. Eng. Technol.*, **30**(2), pp. 184–192 (2007); Drumm and Bart, *Chem. Eng. Technol.*, **29**(11), pp. 1297–1302 (2006).

The RDC has a moderate throughput typically in the range of 20 to 35 m³/(h·m²) [500 to 850 gal/(h·ft²)], and it can be turned down to 20 to 35 percent of the design rate. However, the relatively open arrangement leads to some backmixing and results in only moderate mass-transfer performance. As a consequence, some RDC columns are being replaced by more efficient extractor designs. The RDC can be

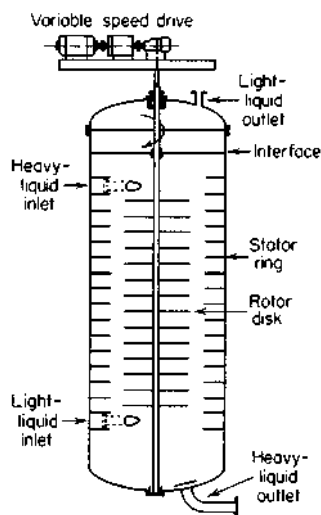


FIG. 15-51 Typical rotating-disk contactor.

used for systems with moderate viscosities up to about 100 cP and can be used for systems that tend to foul easily. The RDC also is suitable for systems with slow mass-transfer rates requiring only a few theoretical stages. An RDC can have difficulty handling feeds with emulsion formation tendencies, so it may not be suitable for some systems with low interfacial tension and low density difference.

Pulsed-Liquid Columns These are packed or tray column extractors in which a rapid reciprocating motion of relatively short amplitude is applied to the liquid contents to give improved rates of extraction (Fig. 15-52). Liquid pulsing improves the mass-transfer performance at a cost of somewhat reduced throughput. For detailed reviews of this technology, see Logsdail and Slater, Chap. 11.2 in *Handbook of Solvent Extraction*, Lo, Baird, and Hanson, eds. (Wiley, 1983; Krieger, 1991); Pratt and Stevens, Chap. 8 in *Science and Practice of Liquid-Liquid Extraction*, vol. 1, Thorton, ed. (Oxford, 1992); and Haverland and Slater, Chap. 10 in *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994). Also see Bujalski et al., *Chem. Eng. Sci.*, **61**, pp. 2930-2938 (2006), for discussion of a disk and doughnut type of column extractor operated with pulsed liquid. Externally pulsing the liquid to impart mechanical agitation allows for a sealed agitated extraction column with

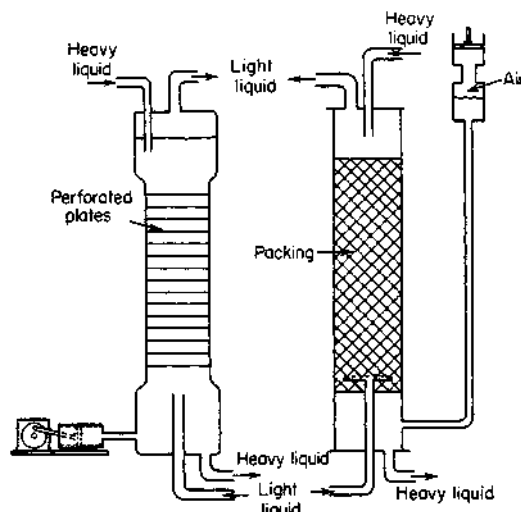


FIG. 15-52 Pulsed-liquid columns. (a) Sieve tray column with pump-type pulse generator. (b) Packed column with air pulser.

no moving parts. This feature is important for special applications involving highly corrosive or dangerously radioactive liquids, and it is the main reason why pulsed columns commonly are applied in the extraction and separation of metals from solutions in atomic energy operations. Pulsed-liquid contactors are similar to reciprocating-plate extractors in their basic operation. However, considerably more energy generally is required to move the entire column of liquid than to move the plates. For this reason, a reciprocating-plate or other type of mechanically agitated column design generally is preferred, unless special conditions require a sealed extraction column.

Raining-Bucket Contactor (a Horizontal Column) The "raining-bucket" contactor, originally developed by the Graesser Company in the United Kingdom, consists of a horizontal column or shell, as illustrated in Fig. 15-53. The shell slowly rotates about a central axis, and during operation a main liquid-liquid interface is maintained near the centerline. The light phase is continuous in the upper half of the shell, and the heavy phase is continuous in the lower half. Buckets mounted within the shell pick up continuous phase in one half and discharge it as dispersed droplets into the other half. As a result, each phase is dispersed. The raining-bucket design is intended for systems

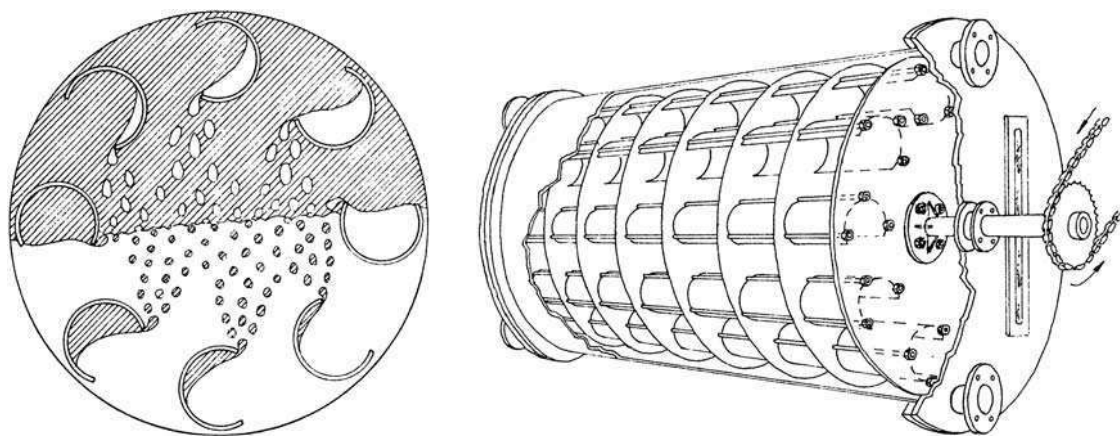


FIG. 15-53 Schematic views of a Graesser raining-bucket contactor. [Reprinted from Coleby, Chap. 13.6 in *Handbook of Solvent Extraction*, Lo, Baird, and Hanson, eds. (Wiley, 1983; Krieger, 1991), with permission.]

with low density difference and low interfacial tension, i.e., systems that tend to emulsify easily. It was originally developed for handling difficult settling systems in the coal-tar industry. A detailed review is given by Coleby [Chap. 13.6 in *Handbook of Solvent Extraction*, Lo, Baird, and Hanson, eds. (Wiley, 1983; Kreiger, 1991)]. Units currently are available through the Biotechna Company.

The rotor assembly of a raining-bucket contactor is made of a series of disks that divide the shell into a series of compartments. Each compartment contains an assembly of buckets. A small gap is maintained between the edge of the disks and the interior wall of the shell to allow for flow between compartments. The gap needs to be small to minimize backmixing. During operation, the phases are fed and removed from opposite ends of the column to produce a countercurrent flow. Throughput generally is low compared to that of other mechanically agitated extractors owing to the limited cross-sectional area available for flow. Rotational speeds are in the range of 0.25 to 40 rpm depending upon the contactor diameter and physical properties of the phases. Coleby [Chap. 13.6 in *Handbook of Solvent Extraction*, Lo, Baird, and Hanson, eds. (Wiley, 1983; Kreiger, 1991)] indicates that raining-bucket contactors can achieve up to 0.3 theoretical stage per compartment depending upon the application. Applications should not involve too high a viscosity in either phase, since dispersing drops in a high-viscosity continuous phase can result in slow liquid-liquid phase separation, and this can severely limit mass-transfer performance and the throughput of the extractor. Experience indicates that careful attention to this possibility is needed if viscosity is on the order of 30 cP or greater. A theoretical approach to estimating axial mixing and efficiency in a raining-bucket extractor is presented by Dente and Bozzano [*Ind. Eng. Chem. Res.*, **43**(16), pp. 4761–4767 (2004)]. A biotechnology application is described by Jarndillokikul, Paulsen, and Stuckey [*Biotechnol. Prog.*, **16**(6), pp. 1071–1078 (2000)].

MIXER-SETTLER EQUIPMENT

Mixer-settlers are used in hydrometallurgical processing for recovery of metals from aqueous acid solutions, and in multistep batchwise production of specialty chemicals including pharmaceuticals and agricultural chemicals, among other applications. In principle, any mixer may be coupled with any settler to obtain a complete stage. The function of a single stage within the cascade is to contact the liquids so that equilibrium is closely approached (achieving a high stage efficiency), and then to separate the liquids so they can be routed to the next stage. The design must strike a balance between contacting and settling requirements; i.e., the liquids should be mixed with sufficient intensity to suspend drops and facilitate good mass transfer, but not so intensely that drop sizes are too small and settling of the resulting dispersion is problematic.

A mixer-settler operation may be carried out batchwise or with a continuous feed. If batchwise operation is chosen, the same vessel used for mixing often is used for settling. Batchwise extraction in a stirred tank is a common operation in multistep, batchwise manufacture of complex organics. Such equipment allows flexibility to accommodate batch-to-batch variability, can ensure a single batch remains isolated from other batches throughout the manufacturing process (sometimes a regulatory requirement for pharmaceuticals), and is suitable for multipurpose plants producing a variety of products in campaigns. A batchwise process may be implemented in cocurrent, cross-current, or countercurrent multistage arrangements. A countercurrent operation is carried out as in Figs. 15-6 and 15-22, by initially treating the feed batch with extract solution as the extract leaves the process. The final treatment is carried out using fresh solvent as it enters the process. A two-stage batchwise countercurrent process scheme is common practice.

Continuously operated devices may place the mixing and settling functions in separate vessels or combine them into a single, specially designed vessel with compartments for mixing and settling. Continuous mixer-settlers are particularly attractive for applications requiring several equilibrium stages and long residence times due to slow extraction kinetics, especially for applications involving the use of reactive extractants or viscous fluids. Mixing commonly is done using rotating impellers. Impeller type, shape, size, tip speed, and position

within the mixing vessel may be adjusted to optimize the overall design. A static mixer may be a feasible alternative, but only if the required mass transfer can be accomplished in the short contacting time these devices allow, without generating a difficult-to-separate dispersion. Mixer-settlers may offer other advantages including easy start-up and operation, the ability to handle very high production rates and suspended solids, and the ability to achieve high stage efficiency with proper design. For systems that accumulate rag layers (sludges) between settled liquid layers, the rag material may easily be removed at each settler. As a potential disadvantage, difficult-to-break emulsions may be formed from the shear due to mixing and pumping liquids between tanks. Mixer-settlers also generally require large floor space, and the relatively long residence time in a mixer-settler can be a disadvantage if the desired solute is degraded over time at the required extraction conditions.

Mass-Transfer Models Because the mass-transfer coefficient and interfacial area for mass transfer of solute are complex functions of fluid properties and the operational and geometric variables of a stirred-tank extractor or mixer, the approach to design normally involves scale-up of miniplant data. The mass-transfer coefficient and interfacial area are influenced by numerous factors that are difficult to precisely quantify. These include drop coalescence and breakage rates as well as complex flow patterns that exist within the vessel (a function of impeller type, vessel geometry, and power input). Nevertheless, it is instructive to review available mass-transfer coefficient and interfacial area models for the insights they can offer.

The correlation of Skelland and Moeti [*Ind. Eng. Chem. Res.*, **29**(11), pp. 2258–2267 (1990)] for estimating individual continuous-phase mass-transfer coefficients is given by

$$\frac{k_d d_p}{D_c} = 1.237 \times 10^{-5} \left(\frac{\mu_c}{\rho_c D_c} \right)^{1/3} \left(\frac{D_i \omega^2}{g} \right)^{5/12} \left(\frac{D_i}{d_p} \right)^2 \times \left(\frac{d_p}{D_i} \right)^{1/2} \left(\frac{\rho_d D_{p,c}^2 g}{\sigma} \right)^{5/4} \phi_d^{-1/2} \quad (15-170)$$

where ω is impeller speed (rotations per unit time), D_i is impeller diameter, D_t is tank diameter, and D_c is the solute diffusion coefficient in the continuous phase. Equation (15-170) is restricted to dispersed-phase holdup values less than $\phi_d = 0.06$. Other studies are described by Schindler and Treybal [*AIChE J.*, **14**(5), pp. 790–798 (1968)] and by Keey and Glen [*AIChE J.*, **15**(6), pp. 942–947 (1969)]. Equation (15-170) normally is used to estimate performance for applications in which the feed phase is the continuous phase and the partition ratio for transfer of solute into the raffinate phase is large. In this case, the overall resistance to mass transfer is dominated by the continuous-phase resistance. Relatively little information is available about individual dispersed-phase mass-transfer coefficients. Skelland and Xien [*Ind. Eng. Chem. Res.*, **29**(3), pp. 415–420 (1990)] offer a correlation of k_d values for batchwise extraction of solute from the dispersed phase into the continuous phase.

To use these correlation equations, it is necessary to identify which phase will be dispersed and to estimate the dispersed drop size and holdup as a function of throughput near flooding conditions. For relevant discussions, see “Factors Affecting Which Phase Is Dispersed” and “Size of Dispersed Drops” under “Liquid-Liquid Dispersion Fundamentals.” Holdup is a complex function of flow rates, impeller type, vessel geometry, and power input, as well as physical properties. For most impeller types, correlations for estimating holdup are not available. However, Weinstein and Treybal [*AIChE J.*, **19**(2), pp. 304–312; **19**(4), pp. 851–852 (1973)] offer the following correlations for estimating holdup in a vessel agitated using a six-blade disk-style flat-blade turbine (Rushton): For a baffled vessel with a gas-liquid surface:

$$\frac{\phi_d}{\phi_{d,\text{feed}}} = 0.764 \left(\frac{P Q_d \mu_c^2}{V_i \sigma^3} \right)^{0.300} \left(\frac{\mu_c^2}{Q_d \rho_c^2 \sigma} \right)^{0.178} \left(\frac{\rho_c}{\Delta \rho} \right)^{0.0741} \times \left(\frac{\sigma^2 \rho_c}{\mu_c^2 g} \right)^{0.276} \left(\frac{\mu_d}{\mu_c} \right)^{0.136} \quad (15-171)$$

For a liquid-full vessel without baffles:

$$\frac{\phi_d}{\phi_{d,\text{feed}}} = 3.39 \left(\frac{P Q_d \mu_c^2}{V_t \sigma^3} \right)^{0.247} \left(\frac{\mu_c^3}{Q_d \rho_c^2 \sigma} \right)^{0.427} \left(\frac{\rho_c}{\Delta \rho} \right)^{0.430} \times \left(\frac{\sigma^3 \rho_c}{\mu_c^4 g} \right)^{0.401} \left(\frac{\mu_d}{\mu_c} \right)^{0.0987} \quad (15-172)$$

Baffles are not needed if the vessel is operated full of liquid with no head space. In Eqs. (15-171) and (15-172), $\phi_{d,\text{feed}}$ is the volume fraction of the phase that ultimately becomes the dispersed phase, for the combined streams entering the vessel: $\phi_{d,\text{feed}} = Q_d / (Q_d + Q_c)$. If $\phi_d / \phi_{d,\text{feed}}$ is calculated to be greater than 1.0, it should be taken as 1.0. These equations are not applicable to other types of impellers.

When an estimate of ϕ_d is available, then $a \approx 6\epsilon \phi_d / d_p$ [Eq. (15-109)]. If the individual mass-transfer coefficients can be estimated with reasonable accuracy, a value for the overall coefficient k_{or} can be calculated from the individual coefficients as in Eq. (15-68). The stage efficiency for a continuous process can then be estimated from

$$\xi_{mr} = 1 - \exp \left(-\frac{k_{or} a \theta}{\phi_d} \right) \quad (15-173)$$

where ξ_{mr} is the Murphree raffinate-based stage efficiency and θ is the residence time for total liquid in the vessel [Treybal, "Liquid Extractor Performance," *Chem. Eng. Prog.*, **62**(9), pp. 67–75 (1966); and Laddha and Degaleesan, *Transport Phenomena in Liquid Extraction* (McGraw-Hill, 1978), p. 418]. Also see the discussion by Skelland and Kanel [*Ind. Eng. Chem. Res.*, **31**(3), pp. 908–920 (1992)]. These authors describe an extraction model framework that includes terms representing drop breakage and coalescence effects.

Miniplant Tests As mentioned earlier, for most liquid-liquid extraction applications involving mixer-settlers, the requirements for satisfactory performance with respect to mixing and settling are determined by using small miniplant or pilot-plant tests. For mixer design, the usual procedure is to run continuous experiments for a specific mixer geometry and type of impeller, generating performance data over a range of residence times and agitation intensities. The experimental program typically involves testing a variety of impellers and impeller locations until satisfactory results are obtained, with the ultimate goal of scaling up the miniplant design to achieve the same performance at the commercial scale. The design of settlers is discussed in the section "Liquid-Liquid Separation Equipment." With careful design, most extractions require residence times in the range of 1 to 3 min. However, for reaction-enhanced extractions having relatively slow chemical kinetics compared to mass transfer, longer times in the range of 10 to 15 min are not unusual. As noted earlier, it is important to consider the time required to settle the dispersion after mixing and to determine the optimum mixing intensity that provides good mass transfer with reasonable ease of settling.

In these tests, extraction efficiency may be expressed in terms of a Murphree efficiency as

$$\xi = \frac{C_o - C_t}{C_o - C^*} \quad (15-174)$$

where C_o is the initial concentration of solute in the feed, C_t is the concentration in the outlet for a given residence time or at time t for a batch process, and C^* is the concentration at equilibrium. Normally, the extraction efficiency is determined from continuous experiments. If batch extraction data are available for the same solvent-to-feed ratio, the efficiency of a continuous process may be estimated by fitting the batch data to a first-order rate expression

$$\xi_{\text{batch}} = 1 - \exp(-kt_b) \quad (15-175)$$

where ξ_{batch} for the batch experiment is measured as a function of t_b , the batch mixing time [Godfrey, Chap. 12 in *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994)]. The efficiency of the continuous process is calculated from the expression

$$\xi_{\text{continuous}} = \frac{k\theta}{1 + k\theta} \quad (15-176)$$

where θ is the total liquid residence time for the continuous process. This approach is valid for most diffusion rate controlled processes, but may not be valid for reaction-enhanced processes in which the chemical reaction rate may be rate-limiting and not necessarily first-order.

When the ratio of phases entering a mixer-settler stage is far from unity, recycling a portion of the minority phase from the settler back to the mixer sometimes improves the settling of the dispersion by boosting the phase ratio in the settler. (See "Gravity Decanters (Settlers)" under "Liquid-Liquid Phase Separation Equipment.") The stage efficiency also may be enhanced. For example, when the extract (solvent) is the minority phase (because K is greater than unity) and mass-transfer rates are poor, recycling the settled extract phase can boost the mass-transfer efficiency [Treybal, *Ind. Eng. Chem. Fundam.*, **3**(3), pp. 185–188 (1964)].

Liquid-Liquid Mixer Design Many different types of impellers are used for liquid-liquid extraction, including flat-blade and pitched-blade turbines, marine-type propellers, and special pump-mix impellers. With pump-mix designs, the impeller serves not only to mix the fluids, but also to move the fluids through the extraction stages of a mixer-settler cascade. The agitated vessel should be baffled if the vessel is operated with a gas-liquid surface, to avoid forming a vortex. As noted earlier in reference to Eq. (15-172), baffles are not needed if the vessel is operated with the liquid full [Weinstein and Treybal, *AIChE J.*, **19**(2), pp. 304–312 (1973)].

The design of a liquid-liquid mixer includes specification of impeller type and rotational speed (or tip speed), the number of impellers required, the ratio of impeller diameter to vessel diameter D_i/D_v , and the location of impeller(s) and any baffles within the vessel. A single impeller generally can be used for vessels with a height-to-diameter ratio less than 1.2 and liquid density ratios within the range of $0.9 < \rho_d/\rho_c < 1.1$. Multiple impeller designs are used to improve circulation and power distribution in tall vessels. For detailed discussions of liquid-liquid mixer design, see Leng and Calabrese, Chap. 12 in *Handbook of Industrial Mixing, Science and Practice*, Paul, Atiemo-Obeng, and Kresta, eds. (Wiley, 2004); and Edwards and Baker, Chap. 7, and Edwards, Baker, and Godfrey, Chap. 8, in *Mixing in the Process Industries*, 2d ed., Hamby, Edwards, and Nienow, eds. (Butterworth-Heinemann, 1992). Also see Daglas and Stamatoudis, *Chem. Eng. Technol.*, **23**(5), pp. 437–440 (2000), for discussion of the effect of impeller vertical position on drop size; and Willie, Langer, and Werner, *Chem. Eng. Technol.*, **24**(5), pp. 475–479 (2001), for discussion of the influence of power input on drop size distribution for a variety of impeller types.

The mixing power per unit volume P/V is a function of impeller rotational speed ω , impeller diameter D_i , and the Power number (P_o) for the type of impeller and vessel geometry:

$$\frac{P}{V} = P_o \left(\frac{\rho_m \omega^3 D_i^5}{V_{\text{tank}}} \right) \quad (15-177)$$

In Eq. (15-177), the mixture mean density is given by

$$\rho_m = \phi_d \rho_d + (1 - \phi_d) \rho_c \quad (15-178)$$

Power numbers for different impeller types depend upon the impeller Reynolds number. Representative relationships of Power number versus Reynolds number for several types of impellers are given in Fig. 15-54. For additional information on a variety of impellers, see Sec. 6 and Hemrajani and Tattersson, Chap. 6 in *Handbook of Industrial Mixing, Science and Practice*, Paul, Atiemo-Obeng, and Kresta, eds. (Wiley, 2004).

The power P in Eq. (15-177) does not include losses associated with the motor and drive unit. These losses can contribute as much as 30 to 40 percent to the overall power requirement. The drive supplier should be consulted for specific information. For pump-mix impellers, knowledge of the power characteristics for pumping is required in addition to that for mixing. For a discussion of these special cases, see Godfrey,

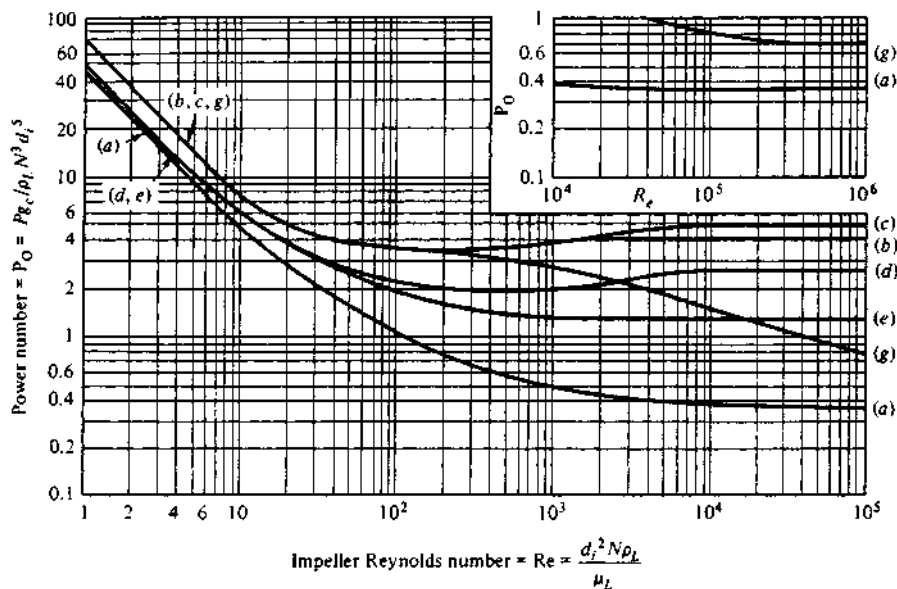


FIG. 15-54 Power for agitation impellers immersed in single-phase liquids, baffled vessels with a gas-liquid surface (except curves *c* and *g*). Curves correspond to (a) marine impellers; (b) flat-blade turbines, width = $D_i/5$; (c) disk flat-blade turbines (Rushton) with or without a gas-liquid surface; (d) curved blade turbines; (e) pitched blade turbines; (g) flat-blade turbines, no baffles, no gas-liquid interface, no vortex.

Notes on Fig. 15-54:

1. All the curves are for axial impeller shafts, with liquid depth equal to the tank diameter D_t .
2. Curves *a* to *e* are for open vessels, with a gas-liquid surface, fitted with four baffles, baffle width = $D_i/10$ to $D_i/12$. The impeller is set at a distance $C = D_i$ or greater from the bottom of the vessel.
3. Curve *a* is for marine propellers, $D_i/D_t \approx \frac{1}{3}$. The effect of changing D_i/D_t is apparently felt only at very high Reynolds numbers.
4. Curves *b* to *e* are for turbines. For disk flat-blade (Rushton) turbines, curve *c*, the effect of changing D_i/D_t is negligible in the range $0.15 < D_i/D_t < 0.50$. For open types (without the disk), curve *b*, the effect of D_i/D_t may be strong.
5. Curve *g* is for disk flat-blade turbines operated in unbaffled vessels filled with liquid and covered, so that no vortex forms. If baffles are present, the power characteristics at high Reynolds numbers are essentially the same as curve *b* for baffled open vessels, with only a slight increase in power.

6. For very deep tanks, two impellers normally are mounted on the same shaft, one above the other. For all flat-blade turbines, at a spacing of $1.5D_i$ or greater, the combined power for both will approximate that for a single turbine.

SOURCE: Treybal, *Mass-Transfer Operations* (McGraw-Hill, 1980), p. 152. For more detailed information, consult *Handbook of Industrial Mixing*, Paul, Atiemo-Obeng, and Kresta, eds. (Wiley, 2004).

Chap. 12 in *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994); and Singh et al., *Ind. Eng. Chem. Res.*, **46**(7), pp. 2180–2190 (2007).

Skelland and Ramsay [*Ind. Eng. Chem. Res.*, **26**(1), pp. 77–81 (1987)] correlated the minimum impeller speed needed to completely disperse one liquid in another in an agitated vessel with standard baffles as follows:

$$\frac{\omega_{\min}^2 \rho_m D_i}{g \Delta \rho} = C^2 \left(\frac{D_t}{D_i} \right)^{2\alpha} \phi^{0.106} \left(\frac{\mu_m^2 \sigma}{D_i^5 \rho_m g^2 \Delta \rho^2} \right)^{0.084} \quad (15-179)$$

The mixture mean density is given by Eq. (15-178), and the mixture mean viscosity is given by

$$\mu_m = \frac{\mu_c}{1 - \phi_d} \left(1 + \frac{1.5 \mu_d \phi_d}{\mu_d + \mu_c} \right) \quad (15-180)$$

The authors determined correlation constants C and α for five common types of impellers (two axial-flow impellers and three radial-flow impellers) and four impeller locations within a standard tank configuration. The specific power requirement can then be estimated by using Eq. (15-177). The power required to disperse one liquid phase

into another typically is in the range of 0.2 to 0.8 kW/m³ (1 to 4 hp/1000 gal) [Edwards, Baker, and Godfrey, Chap. 8 in *Mixing in the Process Industries*, 2d ed., Harnby, Edwards, and Nienow, eds. (Butterworth-Heinemann, 1992), p. 144].

Scale-up Criteria It is common practice to scale up a miniplant design on the basis of equal residence time, constant power per unit volume, and geometric similarity such that the ratio D_i/D_t is held constant and the same types of impeller, tank geometry, and baffling are used. Treybal [*Chem. Eng. Prog.*, **62**(9), pp. 67–75 (1966)] indicates that in using this criterion, stage efficiency for liquid-liquid extraction is likely to increase on scale-up, so it is expected to yield a conservative design. With this approach, P/D_i^3 is constant and proportional to $P_o \omega^3 D_i^3 / D_i^3 = P_o \omega^3 D_i^3$. Assuming that the Power number is independent of scale, this yields the relationship

$$\frac{\omega(2)}{\omega(1)} = \left(\frac{D_i(1)}{D_i(2)} \right)^{2/3} = \left(\frac{D_t(1)}{D_t(2)} \right)^{2/3} \quad (15-181)$$

Skelland and Ramsay [*Ind. Eng. Chem. Res.*, **26**(1), pp. 77–81 (1987)] indicate that Eq. (15-181) is somewhat conservative, in general agreement with Treybal. Based on an analysis of mixing data generated at low holdup, they indicate that the exponent $\frac{2}{3}$ may be replaced with 0.71 as

a scale-up rule. Skelland and Ramsay also discuss the criteria for scale-up to a tank design involving a different ratio of D_2/D_1 at the large scale.

Leng and Calabrese [Chap. 12 in *Handbook of Industrial Mixing: Science and Practice*, Paul, Atiemo-Obeng, and Kresta, eds. (Wiley, 2004), p. 732] show that constant power per unit volume also yields the following relationship if a change in drop size is desired (again, for applications with low holdup):

$$\frac{d_{\max}(1)}{d_{\max}(2)} \approx \frac{\omega(2)^{6/5} D_1(2)^{4/5}}{\omega(1)^{6/5} D_1(1)^{4/5}} \quad \text{for } \text{Re} = \frac{\rho_m \omega D_i^2}{\mu_m} > 10^4 \quad (15-182)$$

Equation (15-182) reduces to Eq. (15-181) when $d_{\max}(1)$ is set equal to $d_{\max}(2)$.

The constant power per unit volume scale-up criterion is equivalent to scaling the impeller tip speed ($S_{\text{tip}} = \pi D_i \omega$) by the ratio $S_{\text{tip}}(2)/S_{\text{tip}}(1) = [D(2)/D(1)]^{1/3}$. It follows that when the tank diameter is doubled, the impeller tip speed must increase by a factor of 1.26 to maintain constant power per unit volume. If the Skelland and Ramsay exponent of 0.71 is applied in Eq. (15-181) instead of $\frac{2}{3}$, then tip speed scales as $S_{\text{tip}}(2)/S_{\text{tip}}(1) = [D(2)/D(1)]^{0.29}$ and doubling the tank diameter involves increasing the tip speed by a factor of 1.22.

Podgórska and Bałdyga [*Chem. Eng. Sci.*, **56**, pp. 741–746 (2001)] present a model of drop breakage and coalescence and compare four scale-up criteria for agitated liquid-liquid dispersions:

- I. Equal power per unit volume and geometric similarity
- II. Equal average circulation time and geometric similarity
- III. Equal power per unit volume and equal average circulation time ($D_2/D_1 \neq \text{constant}$)
- IV. Equal tip speed and geometric similarity

For slow-coalescing systems and systems at low holdup, the rate of drop breakage dominates. In this case, according to the analysis of Podgórska

and Bałdyga, criteria I and II yield smaller drops on scale-up, and criteria III and IV yield larger drops. For fast-coalescing systems, the rate of drop coalescence begins to dominate breakage. In this case, the authors indicate that I and III yield almost constant drop size with scale-up, II yields much smaller drops, and IV yields larger drops. Podgórska and Bałdyga recommend III for fast-coalescing systems, although they point out a limitation in terms of the maximum size of tank that this criterion will allow. See Leng and Calabrese, Chap. 12 in *Handbook of Industrial Mixing: Science and Practice*, Paul, Atiemo-Obeng, and Kresta, eds. (Wiley, 2004), pp. 682–687, for detailed discussion of factors influencing coalescence and their impact on scale-up difficulty.

Based on the analyses described above, taken together, it appears that scaling according to constant power per unit volume and geometric similarity generally will give satisfactory results, although the resulting design may not be optimal. For a new design, generally it is advisable to specify a variable-speed drive that can operate within a range of tip speeds. This provides flexibility for further adjustment and optimization of the process in the plant, and it also allows flexibility to accommodate variability in feed composition (a likely scenario in an industrial process).

Specialized Mixer-Settler Equipment As mentioned earlier, any mixer and settler can be combined to produce a stage, and the stages in turn arranged in a multistage cascade. A great many specialized designs have been developed in an effort to reduce costs, e.g., by minimizing or eliminating interstage pumping or by combining the various stages into a single vessel. With proper design, these devices generally can achieve overall stage efficiencies in excess of 80 percent, with many providing 90 to 95 percent stage efficiency. Only a few of the more commonly used types are mentioned here. For more detailed discussions, see Chaps. 9.1 to 9.5 in *Handbook of Solvent Extraction*, Lo, Baird, and Hanson, eds. (Wiley, 1983; Krieger, 1991).

Several pump-mix combinations have been developed by industry to simplify overall plant layout and minimize the number of pumps, at the expense of more expensive mixer design or complexity. The IMI axial pump-mix and draft tube (Fig. 15-55a) has the pumping

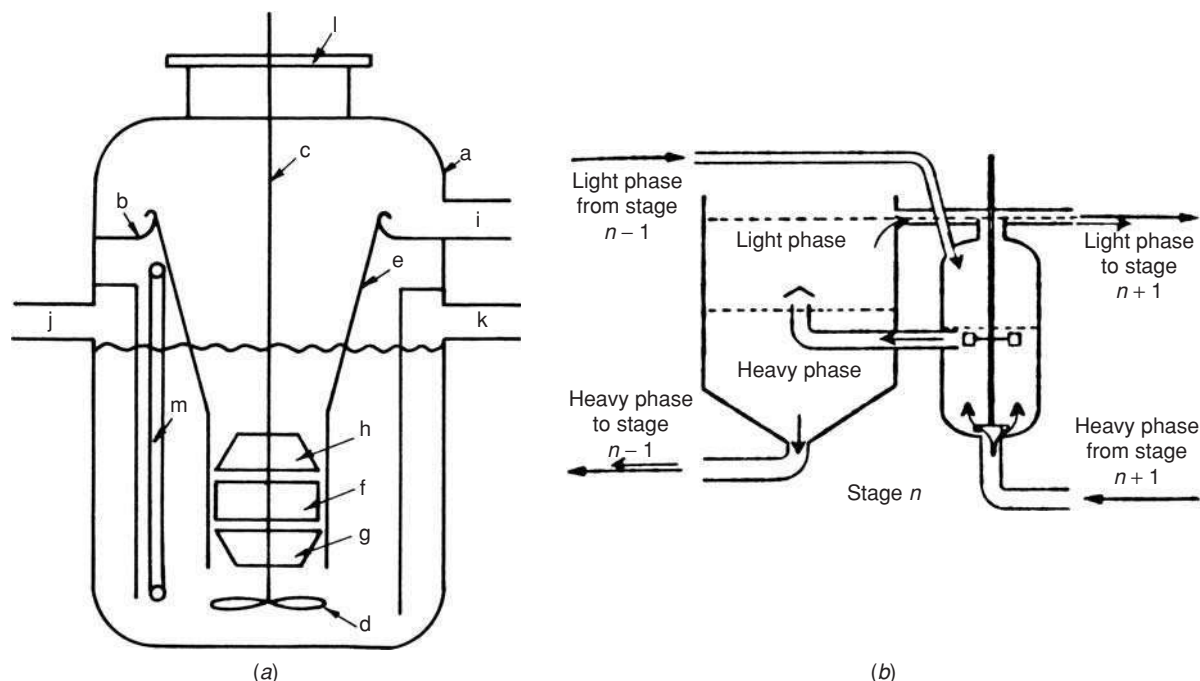


FIG. 15-55 Types of pump-mix arrangements for mixer-settler extractors. (a) IMI pump mix with mixing and pumping impellers (a, vessel; b, internal deck; c, shaft; d, mixing impeller; e, draft tube; f, pumping impeller; g and h, guide vanes; i, dispersion discharge; j, light-phase feed; k, heavy-phase feed; l, mounting flange; m, sight glass). (b) Kemira mixer-settler. [Figure 15-55a taken from *Handbook of Solvent Extraction*, Lo, Baird, and Hansen, eds. (Wiley, 1983; Krieger, 1991), with permission. Figure 15-55b taken from Mattila, ISEC '74 Proc., London, 1974, with permission.]

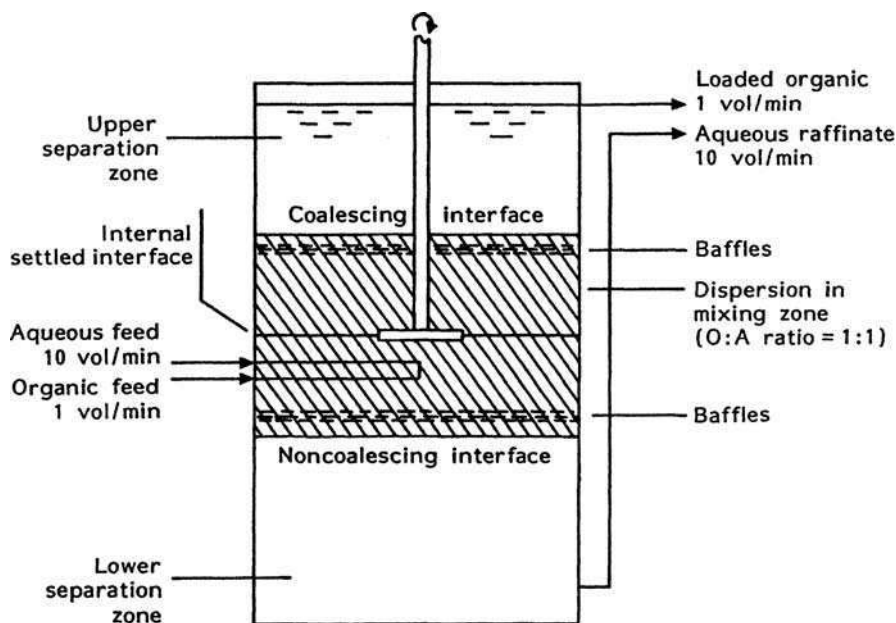


FIG. 15-56 Davy CMS extractor with pump-mix impeller and phase separation zones. [Reprinted from *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994), with permission. Copyright 1994 John Wiley & Sons Ltd.]

and mixing impellers on the same shaft. The upper part of the tank contains the draft tube and the mixing-impeller. The pumping-impeller for transferring the dispersion to the settler is in the lower part of the tank. There is a potential disadvantage of forming smaller and hard to separate drops when pumping a dispersion versus pumping a single phase. The Kemira design (Fig. 15-55*b*) uses a pumping-impeller located near the bottom of the tank along with a mixing-impeller located near the central zone of the tank. The draft tube is eliminated and a dispersion is not pumped in this design. The Davy CMS design (Fig. 15-56) uses a pump-mix impeller in a large tank that provides both mixing and settling capability over a wide range of phase flow ratios. The dispersion occurs in the central section of the tank, and the separation occurs in the upper and lower separation zones.

A compact alternating arrangement of mixers and settlers has been adopted in many of the "box-type" extractors developed originally for processing radioactive solutions. These designs are used for many other processes, with literally dozens of modifications. An example is the pump-mix mixer-settler (Fig. 15-57), in which adjacent stages have common walls [Coplan, Davidson, and Zebroski, *Chem. Eng. Prog.*, **50**(8), pp. 403–408 (1954)]. In this case, the impellers pump as well as mix by drawing the heavy liquid upward through the hollow impeller shaft and discharging it at a higher level through the hollow impeller. Rectangular tanks are not ideal for good mixing; however, the compromise in mixing and settling performance is offset by the compact and economical design.

Vertical arrangement of the stages is desirable, since then a single drive may be used for agitators and the floor space requirement of a cascade is reduced to that of a single stage. The Lurgi extractor configuration has the mixer and settlers in separate vertical shells interconnected with piping [Guccione, *Chem. Eng. Magazine*, **73**(4), pp. 78–80 (1966)]. A great many other designs are known. For example, the Fenske and Long extractor [Fenske and Long, *Chem. Eng. Prog.*, **51**(4), pp. 194–198 (1955); Long and Fenske, *Ind. Eng. Chem.*, **53**(10), pp. 791–798 (1961); Long, *Ind. Eng. Chem. Fundam.*, **1**, p. 152 (1962)] is a vertical stack of mixer-settler stages. This

design employs a reciprocating plate at each stage to mix the two phases.

Suspended-Fiber Contactor The Merichem Fiber-Film® contactor is used in petroleum refining operations to wash hydrocarbon streams with caustic or other treating solutions [Suarez, U.S. Patent 5,997,731 (1999)]. The hydrocarbon feed and wash fluid are brought together within a vertical pipe or wash column containing fibers suspended from the top, as shown in Fig. 15-58. The two liquids flow concurrently down the column through the bed of fibers. The fibers are attached at the top of the column but not at the bottom. Liquid-liquid contacting is facilitated through capillary and surface-wetting effects. This arrangement avoids (or minimizes) formation of small dispersed

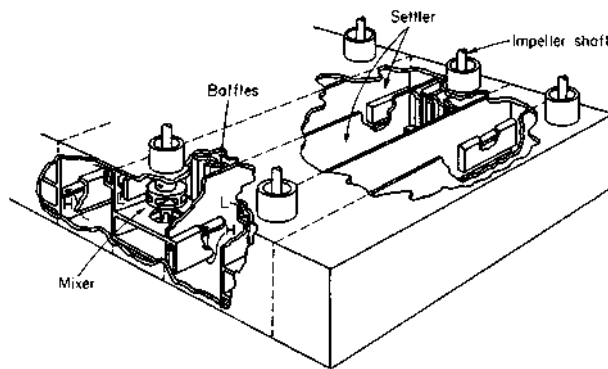


FIG. 15-57 Pump-mix box-type mixer-settler. [Taken from Coplan, Davidson, and Zebroski, *Chem. Eng. Prog.*, **50**, p. 403 (1954), with permission.]

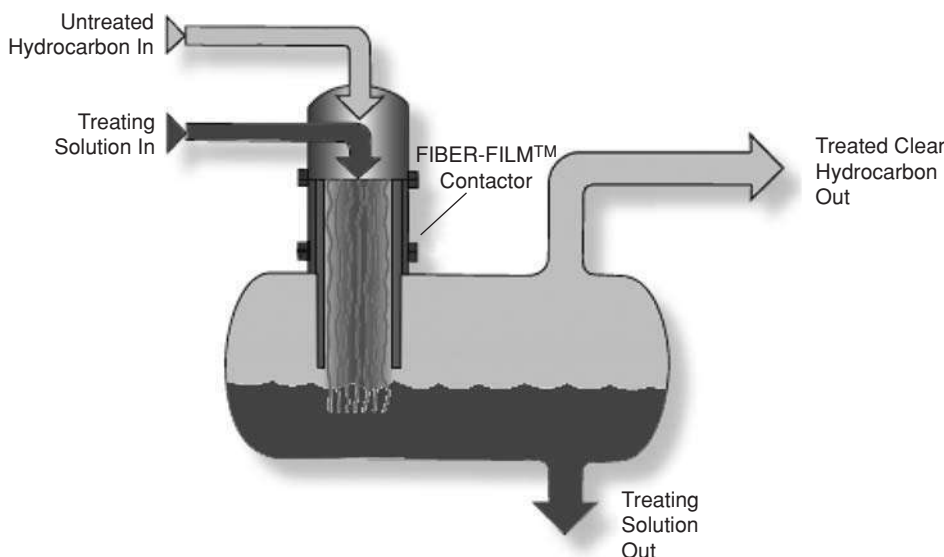


FIG. 15-58 Merichem Fiber-Film™ contactor. (Courtesy of Merichem Chemicals and Refinery Services, LLC.)

drops, and this helps to minimize entrainment of aqueous phase into the hydrocarbon outlet. Little information about the mass-transfer performance and design requirements for this type of contactor has been published.

CENTRIFUGAL EXTRACTORS

A centrifugal extractor multiplies the force of gravity acting on two liquid phases. Centrifugal extractors can facilitate a liquid-liquid extraction process by reducing diffusion path lengths and increasing the driving force for liquid-liquid phase separation. They can achieve very high specific throughput with very low liquid residence time. A wide variety of machine types are available, ranging from relatively simple devices used primarily for phase separation or for single-stage liquid-liquid contacting with separation to more complex machines designed to provide the equivalent of multistage liquid-liquid contacting within a single unit. Some machines are designed to handle feeds containing solids such as whole fermentation broth. This section provides a brief overview with a description of several machines for illustration. More detailed descriptions of centrifuge design and performance are available from equipment vendors. For additional discussion, see Janoske and Piesche, *Chem. Eng. Technol.*, **22**(3), pp. 213–216 (1999); Leonard, Chamberlain, and Conner, *Sep. Sci. Tech.*, **32**(1–4), pp. 193–210 (1997); Blass, Chap. 14 in *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994); Schügerl, *Solvent Extraction in Biotechnology* (Springer-Verlag, 1994); Ollinger and Blass, “Mass Transfer in Centrifugal Extractors,” *Chem. Eng. Technol.*, **11**, pp. 312–320 (1988); and Hafez, Chap. 15 in *Handbook of Solvent Extraction*, Lo, Baird, and Hanson, eds. (Wiley, 1983; Krieger, 1991).

Centrifugal extractors can be beneficial when the liquid density difference is small, when short contact time is needed to avoid product degradation, when feed and solvent easily emulsify, or in cases where high specific throughput is needed due to limitations in available floor space or ceiling height. Centrifugal extractors also can provide flexibility in operation in cases where feed variability is high, by allowing adjustment of feed rate and rotational speed as needed to obtain satisfactory performance. Potential disadvantages generally derive from difficulties associated with maintaining high-speed rotating machinery, relatively high purchase prices compared to those of some other types of extractors, and limitations as to the number of theoretical stages that can be achieved per machine (generally < 1 or up to 5 or 6

theoretical stages depending upon throughput and the type of machine). Another consideration for some machines with close internal clearances is the potential for plugging if any solids are present in the feed; however, as noted above, some machines are specifically designed to handle and discharge solids.

Commercial-scale centrifuges almost always are continuously fed machines, unless the scale of the operation is very low, as in some low-volume bioprocessing operations where very-high-*g* operation and long processing times are needed. A continuously fed centrifugal extractor can deliver high multiples of *g*, but at much lower residence time (given by holdup volume of the feed phase divided by volumetric feed rate) compared to a batch process. The maximum hydraulic capacity (or nominal capacity) of a continuously operated machine often is not realized in commercial applications, because the feed rate needs to be turned down in order to have sufficient residence time for good extraction and phase separation performance.

In evaluating options, it generally is not possible to accurately predict performance because of the complexity of the hydrodynamics within a centrifuge. While high-*g* operation can promote good performance, in certain cases the extremely rapid acceleration generated within the machine also can promote backmixing or emulsification. Miniplant tests using small units generally are needed, and vendors often offer testing services.

Single-Stage Centrifugal Extractors The types of centrifuges used in extraction operations are quite varied. Differences include vertical versus horizontal configuration, fluid-filled versus operation with an air core, pressurized or unpressurized operation, generation of low to extremely high multiples of gravitational acceleration (500 up to 20,000 $\times g$ or higher), as well as differences in the liquid holdup volume, design of internals, internal clearances, and purchase price. The simpler machines, such as the CINC separator from CINC Processing Equipment, Inc. (Fig. 15-59) and the Rousselet-Robatel model BXP, have relatively large internal clearances. An air core is maintained within the machine, and liquid layers decant over internal weirs. Flow restrictions in the overflow piping need to be minimized to avoid any pressure imbalance between light- and heavy-liquid overflow lines, since this can affect the location of the liquid-liquid interface and the liquid overflow/underflow split. These machines often are used for washing operations and other extraction applications with high *K* values requiring few theoretical stages. They often serve as the separator in a mixer-settler stage, such that solvent and

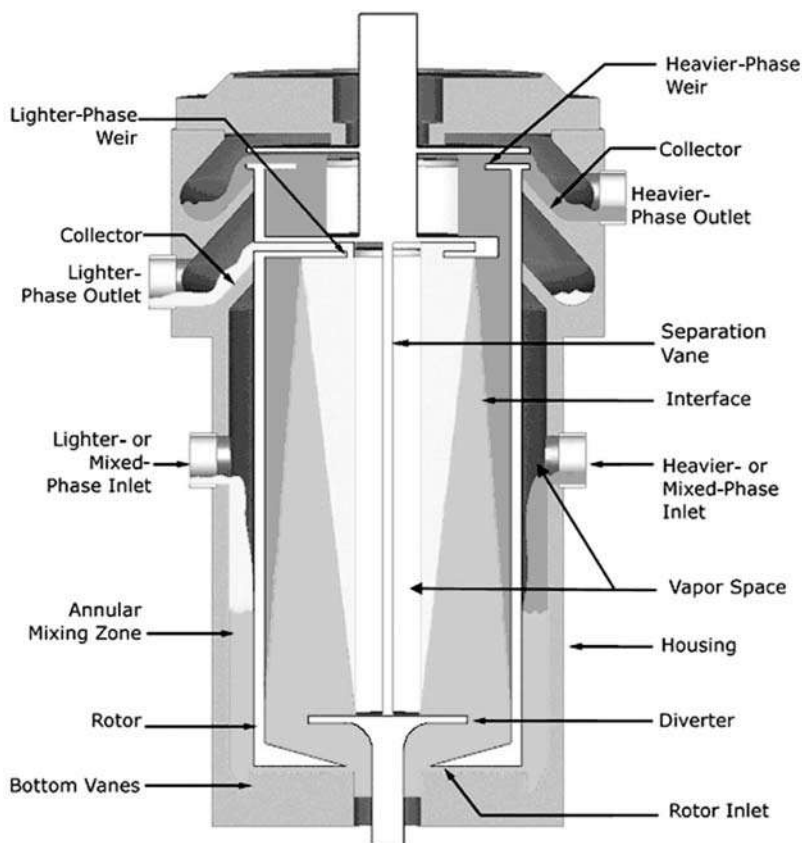


FIG. 15-59 CINC centrifugal separator. (Courtesy of CINC Processing Equipment, Inc.)

feed are first mixed in a static mixer or a separate vessel before being fed to the centrifuge. Some mixing occurs within the centrifuge itself; so if the extraction is sufficiently fast, solvent and feed might be fed directly to the centrifuge to accomplish both mixing and phase separation. Multiple units can be connected in a countercurrent mixer-settler cascade if needed. Processes with 5 to 7 units are typical, while processes with as many as 50 units have been reported. Multiple-unit mixer-settler processes utilizing centrifuges at each stage generally involve production of high-value, low-volume products. Stacked-disk types of machines also are available from numerous vendors and may be used in a similar extraction scheme (generally requiring some type of mixer in the feed line). These machines contain an internal stack of conical disks with a small gap between disks on the order of millimeters [Janoske and Piesche, *Chem. Eng. Technol.*, **22**(3), pp. 213–216 (1999); and Mannweiler and Hoare, *Bioproc. Biosystems Eng.*, **8**(1–2), pp. 19–25 (1992)]. Stacked-disk machines can be thought of as inclined-plate or lamella-type decanters operating in a centrifugal field (see “Liquid-Liquid Phase Separation Equipment”). They magnify the separation power by greatly reducing the distance the dispersed phase must travel before coalescing at a surface, at the expense of somewhat higher complexity and closer internal clearances.

Figure 15-59 shows a cutaway drawing of a CINC separator showing an outer annular space where solvent and feed mix before entering the interior of a rotating drum. Although this type of machine is not designed to separate solids from feeds, a clean-in-place option is offered to facilitate periodic removal of solids that accumulate in the internals. In applications in which one or more of the feed liquids is somewhat viscous, special consideration must be given to the design

of the centrifuge internals such that pressure drop through the machine is not excessive. In certain applications, feed with viscosities as high as several hundred centipoise may be handled; however, special modifications to the internals are needed, and throughput must be reduced compared to that in typical operation. Maximum or nominal volumetric flow capacities for CINC machines range from 110 L/h to 136 m³/h (0.5 to 600 gal/min) depending upon the size of the unit. The Rousselet-Robatel design is somewhat similar. These machines range in size from 50 L/h up to 80 m³/h (0.2 to 350 gal/min). They are designed to generate only moderate centrifugal force and are generally limited to applications requiring no more than about 25,000*g*·s (maximum *g* acceleration times the liquid residence time based on total volumetric flow rate and liquid holdup in the machine).

The CENTREK single-stage extractor from MEAB consists of a funnel-shaped centrifugal-bowl centrifuge mounted above a mixing tank containing a submerged stirrer. An internal “hydrolock” is used to control the position of the liquid-liquid interface in the bowl. According to the manufacturer, this is especially important for multistage, cascade operation. The unit can tolerate some amount of solids in the feed and is available in nominal capacities of 20 L/h to 20 m³/h (0.1 to 90 gal/min).

Centrifugal Extractors Designed for Multistage Performance At the other end of the spectrum are the more complex machines designed to provide multistage or differential liquid-liquid contacting and separation within a single unit. Some machines promote formation of very thin films for efficient liquid-liquid contacting and separation. Others provide multiple zones for mixing and separating the phases. All are designed with complex internals and close clearances. These machines typically achieve 2 to 5 theoretical stages

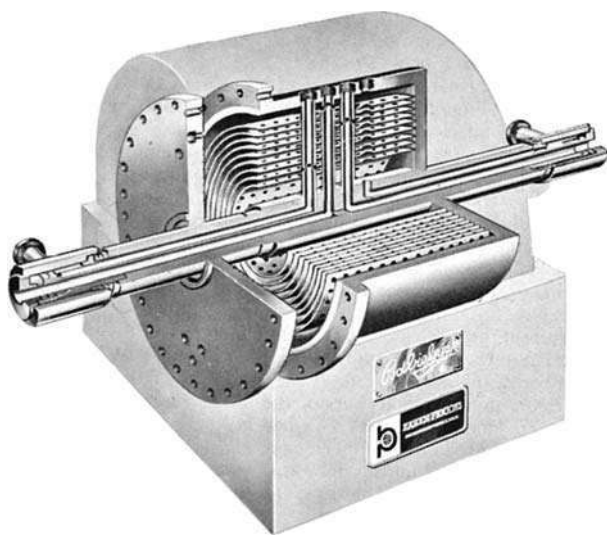


FIG. 15-60 Podbielniak centrifugal extractor. (Courtesy of Baker Perkins, Inc.)

depending upon operating conditions, with some authors claiming as many as 7 or 8 stages.

The classic machine of this type is the Podbielniak extractor available from Baker-Perkins (Fig. 15-60). The body of the extractor is a horizontal cylindrical drum containing concentric perforated cylinders. The liquids are introduced through the horizontal rotating shaft with the help of special mechanical seals; the light liquid is fed internally to the drum periphery and the heavy liquid to the axis of the drum. Rapid rotation (up to several thousand revolutions per minute, depending on size) causes radial counterflow of the liquids, which then flow out through the shaft. Materials of construction include steel, stainless steel, Hastelloy, and other corrosion-resistant alloys. The Podbielniak design provides extremely low holdup of liquid per stage, and this led to its extensive use in the extraction of antibiotics, such as penicillin and the like, for which multistage extraction and phase separation must be done rapidly to avoid chemical destruction of the product under conditions of extraction

[Podbielniak, Kaiser, and Ziegenhorn, Chap. VI in *Chemical Engineering Progress Symposium Series No. 100*, vol. 66, pp. 43-50 (1970)]. Podbielniak extractors have been used in all phases of pharmaceutical manufacturing, in petroleum processing (both solvent refining and acid treating), in extraction of uranium from ore leach liquors, and for clarification and phase separation work. Jacobsen and Beyer [*AIChE J.*, 2(3), pp. 283-289 (1956)] describe operating characteristics and the number of theoretical stages achieved for a specific application.

The Quadronics (Liquid Dynamics) extractor is a horizontally rotated device, a variant of the Podbielniak extractor, in which either fixed or adjustable orifices may be inserted radially as a package. These permit control of the mixing intensity as the liquids pass radially through the extractor. Flow capacities, depending on machine size, range from 0.34 to 340 m³/h (1.5 to 1500 gal/min).

The Luwesta (Centriwesta) extractor is a development from Coutour [Eisenlohr, *Ind. Chem.*, 27, p. 271 (1951)]. This centrifuge revolves about a vertical axis and contains three actual stages. It operates at 3800 rotations per minute and handles approximately 5 m³/h (1300 gal/h) total liquid flow at 12-kW power requirement. Provision is made in the machine for the accumulation of solids separated from the liquids, for periodic removal. It is used, more extensively in Europe than in the United States, for the extraction of acetic acid, pharmaceuticals, and similar products.

The de Laval extractor contains a number of perforated cylinders revolving about a vertical shaft [Palmqvist and Beskow, U.S. Patent 3,108,953 (1959)]. The liquids follow a spiral path about 25 m (82 ft) long, in countercurrent fashion radially, and mix when passing through the perforations. There are no published performance data.

The Rousselet-Robatel LX multistage centrifugal extractor is designed with up to 7 internal mixing/separation stages. Each stage consists of a mixing chamber where the two phases are mixed by means of a stationary agitation disk mounted on a central drum. The high relative speed between the stationary disk and the rotating walls of the mixing chamber creates a liquid-liquid dispersion with high interfacial area to facilitate rapid mass transfer. The agitation disk and the mixing chamber's inlet and outlet channels form a pump which draws the two phases from the adjacent stages and transfers the dispersion to a settling chamber, where it is separated by centrifugal force. The manufacturer claims that high stage efficiencies can be achieved. Extract and raffinate phases are removed from the machine by gravity discharge, or an internal centripetal pump can be employed to discharge these streams under pressure. Nominal flow rates range from 25 L/h up to 80 m³/h.

PROCESS CONTROL CONSIDERATIONS

GENERAL REFERENCES: Wilkinson and Ingham, Chap. 27.2, and S. Plonsky, Chap. 27.3, in *Handbook of Solvent Extraction*, Lo, Baird, and Hanson, eds. (Wiley, 1983; Krieger, 1991).

STEADY-STATE PROCESS CONTROL

Control of a continuous liquid-liquid extraction process generally refers to maintaining satisfactory dispersion of one phase in another for good mass-transfer performance while also maintaining the required production rate. This must be done without entering a flooding condition. It is common practice to set up a continuously fed extractor to handle a range of feed rates while maintaining other operating variables at constant preset values. These include the solvent flow rate, temperatures, and mechanical variables (if agitation or centrifugation is employed). For extraction processes that experience large swings in feed flow rate, the solvent flow rate may be manipulated to maintain a constant solvent-to-feed ratio, in order to reduce the volume of extract that needs to be processed. In this case, the extractor must be able to operate within a fairly wide range of volumetric throughput.

A common cause of upsets in operation is contamination of the feed by trace amounts of impurities that affect interfacial tension, so it is

important to control upstream operations to avoid contamination. Upsets or deviations from desired performance also can be caused by changes in the purity of solvent entering from solvent recovery equipment, so adequate control of closely coupled auxiliary operations is needed to ensure good extractor performance. Periodic monitoring of the interfacial tension of light and heavy phases at the feed location (where interfacial tension is likely to be lowest due to higher solute concentration) may be useful for understanding the range of values that can be tolerated, and trends in the data may provide warning of an impending flooding or coalescence problem.

Steady-state control of a continuously fed extraction column requires maintenance of the location of the liquid-liquid interface at one end of the column. The main interface will appear at the top of the column when the light phase is dispersed and at the bottom of the column when the heavy phase is dispersed. If needed, extraction columns can be designed with an expanded-diameter settling zone to facilitate liquid-liquid phase separation by reducing liquid velocities. If sufficient clarification of the phases cannot be achieved, then it may be necessary to add an external device such as a gravity decanter or centrifuge. (See "Liquid-Liquid Phase Separation Equipment.") Sometimes a column is built with expanded ends at

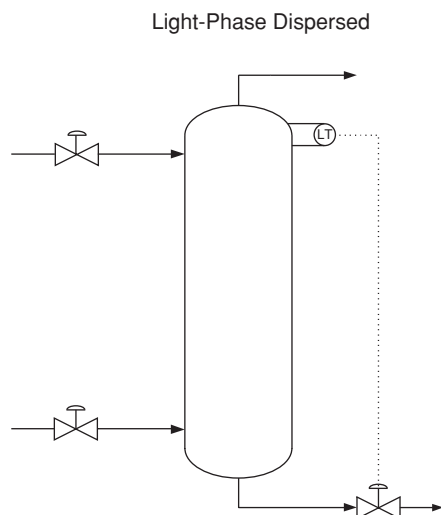


FIG. 15-61 Typical interface control for a light-phase dispersed process (with the main interface located at the top of the column). The same basic arrangement can be used for the heavy-phase dispersed case, but the level transmitter would be located differently to reflect the location of the main interface at the bottom of the column.

both top and bottom to allow the option of operating with either phase dispersed.

The position of the main operating interface in an extraction column, whether located at the top or the bottom, generally is controlled by adjusting the outlet flow of the heavy phase; the heavy-phase outlet valve opens to lower the interface and closes to raise the interface, and the light phase is allowed to overflow the top of the column. The location of the interface often can be maintained at a set position by measuring the differential pressure (if density difference is sufficiently large) or the capacitance of the liquid across the settling zone (for aqueous/organic systems) and manipulating the control valve in the bottom outlet stream to control a set point. Another technique uses a float that rests at the position of the interface. The general concept is illustrated in Fig. 15-61. Weinstein, Semiat, and Lewin [*Chem. Eng. Sci.*, **53**(2), pp. 325–339 (1998)] studied the light-phase dispersed case (with the main interface maintained at the top of the column) and recommend controlling the main interface level by manipulating the continuous-phase feed flow rate instead of the continuous-phase outlet flow rate. The authors developed a dynamic model of the hydrodynamics and mass transfer in a countercurrent liquid-liquid extraction column, and the simulation results indicate faster dynamic response using their alternative scheme.

When a continuous extraction column begins to flood, often one of the first indications is the appearance of an interface at the wrong end of the column; so adding instrumentation that can detect such an interface (such as one or more conductivity probes when phase inversion involves formation of a continuous aqueous phase) may help identify a flooding condition in time to take corrective action. Sometimes a rag layer will accumulate at the liquid-liquid interface, and it is necessary to provide a means for periodically draining the rag to avoid entrainment into the extract or raffinate. It may be useful to add instrumentation that can detect the rag at high positions to warn an operator before breakthrough occurs; however, often the approach taken is to drain the interface region on a predetermined schedule. Installing sensors to detect a rag layer can be problematic because they are easily fouled.

For a continuous extraction column, it is important to control the holdup of each phase within the column to obtain high interfacial area for good mass transfer. For nonagitated extraction columns, this is set by proper design of the internals and maintaining flow rates during operation within a fairly narrow range of values needed for good performance. Agitated columns allow greater flexibility in this regard,

because agitation intensity can be adjusted in the plant to maintain good performance over a wider range of flow rates and as the properties of the feed change. In industrial practice, agitation intensity normally is set at a constant rate or manually adjusted at infrequent intervals in response to a significant change in feed characteristics. Model-based control schemes offer potential for automatic adjustment of agitation intensity and other variables for faster response [Mjalli, *Chem. Eng. Sci.*, **60**(1), pp. 239–253 (2005); and Mjalli, Abdel-Jabbar, and Fletcher, *Chem. Eng. Processing*, **44**, pp. 531–542 and 543–555 (2005)]. Careful programming will be needed to avoid inappropriate control actions when sensors are out of calibration. Real-time measurement of dispersed-phase holdup also may be helpful; Chen et al. [*Ind. Eng. Chem. Res.*, **41**(7), pp. 1868–1872 (2002)] report a method for a pulsed-liquid column. They studied a system consisting of 30% trialkyl(C_{6-s}) phosphine oxide in kerosene + nitric acid solution, with the acid phase dispersed.

For some extraction operations, particularly fractional extractions, it may be useful to control a temperature profile across the process. In extraction columns, this is normally done by controlling the temperature of entering feed and solvent streams. Heating jackets generally are not effective because of insufficient heat-transfer area. Internal heating or cooling coils are problematic because they are difficult and expensive to install and can interfere with other column internals and liquid-liquid traffic within the column. For fractional extraction, the stripping and washing operations may be carried out in separate equipment with external heating or cooling of the streams entering the equipment.

For startup of column extractors, it generally is best to start from dilute-solute conditions to avoid unstable operation. For example, when starting a column in which the feed is the continuous phase, first fill the column with solute-lean feed liquid before starting the flow of solvent and actual feed. This way, the solvent quickly becomes dispersed and mass transfer approaches steady state from dilute conditions, promoting faster and more stable startup.

SIEVE TRAY COLUMN INTERFACE CONTROL

Control of the main liquid-liquid interface for a sieve tray column can be counterintuitive because of complexity caused by the presence of multiple interfaces within the column. For example, if the interface level is too high, the usual control response is to allow the heavy phase to flow out the bottom of the column for a time until the desired level is reached (using the scheme outlined in Fig. 15-61). Ideally, this should lower the interface level, as shown in Fig. 15-62a. This is a typical response for most differential contactors such as packed or spray columns. However, for the sieve tray column the initial response can actually be a rise in the interface level for a short time, as shown in Fig. 15-62b. In some cases, this can result in entrainment of heavy phase out the top of the tower.

The inverse response is caused by changes in the coalesced layer heights at each tray. Neglecting any correction for dispersed-phase holdup, the height of the coalesced layer is affected by the pressure drop through the sieve holes and downcomer:

$$h \approx \frac{\Delta P_o + \Delta P_{\text{dow}}}{\Delta \rho g} = \frac{C_1 V_o^2 + C_2 V_{\text{dow}}^2}{\Delta \rho g} \quad (15-183)$$

where h is the coalesced layer height, ΔP_o is the pressure drop through perforations, ΔP_{dow} is the pressure drop through the downcomer, V_o is the average velocity through a perforation (orifice), V_{dow} is the average velocity through the downcomer, and C_1 and C_2 are constants related to tray geometry and physical properties. Tray designs often vary as to which contribution, orifice or downcomer pressure drop, controls the height of the coalesced layer. The inverse response can cause significant control problems if the downcomer pressure drop is much greater than the orifice pressure drop, and this issue should be addressed during design.

CONTROLLED-CYCLING MODE OF OPERATION

Extraction columns usually are operated in a steady-state continuous-flow mode of operation with one liquid dispersed in the other. Mass transfer is then promoted by using various fixed or moving elements (various types of packings, trays, or agitators). These elements are

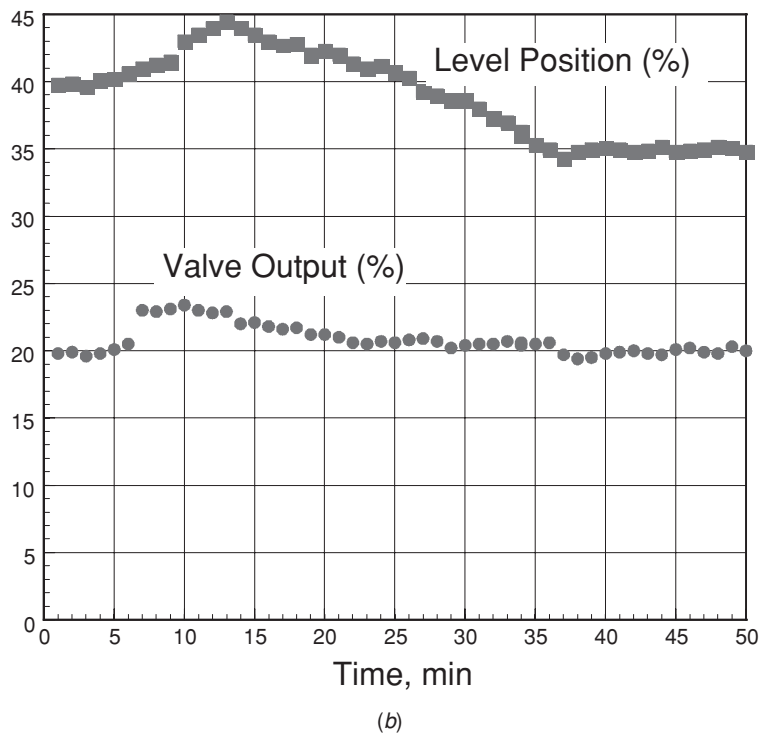
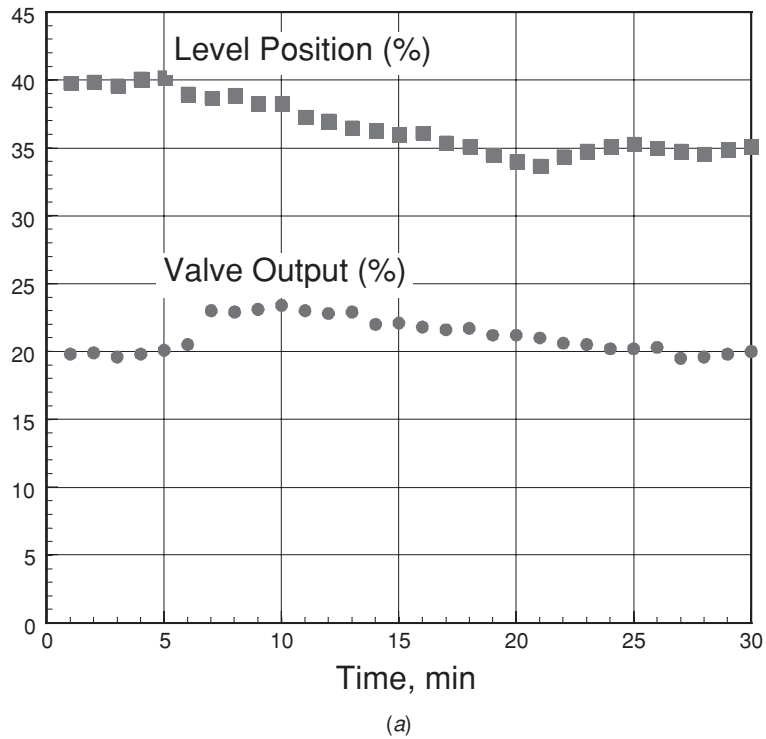


FIG. 15-62 Dynamic response to a change in heavy-phase flow rate. (a) Normal dynamic response to increasing outlet heavy-phase flow (packing). (b) Dynamic response to increasing outlet heavy-phase flow rate (sieve trays).

designed to strike a balance between throughput capacity and mass-transfer efficiency. An alternative mode of operation is the controlled-cycling mode in which light and heavy phases are alternately dispersed and coalesced. Flow is stopped periodically so the phases can switch roles (dispersed versus continuous phase) for the next portion of the cycle. While these coalescing periods reduce the net throughput, the overall mass-transfer effectiveness can be enhanced.

The concept of controlled cycling of phase contactors in general was introduced in the early 1950s by Cannon [*Oil Gas J.*, **51**(12), p. 268 (1952); *Oil Gas J.*, **55**(38), p. 68 (1956); and *Ind. Eng. Chem.*, **53**(8), p. 629 (1961)]. When applied to extraction, it normally involves the use of perforated tray columns, where both phases can flow through the same openings. Since only one phase flows at a time, downcomers are not necessary, and dual-flow trays generally are used. A cycle is completed by the following sequence of events: (1) A light-phase flow period, during which the heavy phase does not flow; (2) a coalescing period, during which neither phase flows; (3) a heavy-phase flow period, during which the light phase does not flow; and (4) a repeat of the coalescing period. The net result can be an increase in overall stage efficiency, roughly doubling the number of theoretical stages the column can achieve, provided the total holdup of each phase is dis-

placed during each cycle. Robinson and Engel [*Ind. Eng. Chem.*, **59**(3), pp. 22–29 (1967)] provide a theoretical analysis for describing the advantages of controlled cycling, and Lövsland [*Ind. Eng. Chem. Proc. Des. Dev.*, **7**(1), pp. 65–67 (1968)] discussed a graphical method for determining the number of theoretical stages.

Belter and Speaker [*Ind. Eng. Chem. Proc. Des. Dev.*, **6**(1), pp. 36–42 (1967)] reported studies using a 6-in-diameter column and the system cyclohexane + ethyl acetate + ethanol + water, a low-interfacial-tension system (1.2 dyn/cm, equal to 1.2×10^{-3} N/m). Excellent stage efficiencies were reported in the range of 50 to 75 percent. Darsi and Feick [*Can. J. Chem. Eng.*, **49**(2), p. 95 (1971)] determined the effects of hole size, direction of solute transfer, and throughput using a 4-in-diameter extractor and a MIBK + acetic acid + water test mixture. They reported that smaller holes and transfer from the organic phase enhanced mass transfer. Stage efficiencies ranged up to 50 percent. Seibert, Humphrey, and Fair [*Solvent Extraction and Ion Exchange*, **4**(5), p. 1049 (1986)] observed that the volume of phase transferred within a cycle should be less than the total holdup volume per stage to minimize backmixing. They also showed that the capacity of a controlled cyclic extractor, while lower than that of a conventional sieve tray extractor, could be higher than that of a pulsed sieve tray extractor.

LIQUID-LIQUID PHASE SEPARATION EQUIPMENT

GENERAL REFERENCES: Sinnott, Coulson and Richardson's *Chemical Engineering*, vol. 6, 4th ed. (Butterworth-Heinemann, 2005); Mueller et al., "Liquid-Liquid Extraction," in *Ullmann's Encyclopedia of Industrial Chemistry*, 6th ed. (VCH, 2002); Hooper, Sec. 1.11 in *Handbook of Separation Techniques for Chemical Engineers*, 3d ed., Schweitzer, ed. (McGraw-Hill, 1997); Hartland and Jeelani, Chap. 13 in *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994); Monnery and Srcek, *Chem. Eng. Prog.*, **90**(9), pp. 29–40 (1994); and Jacobs and Penney, Chap. 3 in *Handbook of Separation Process Technology*, Rousseau, ed. (Wiley, 1987).

OVERALL PROCESS CONSIDERATIONS

The ability to separate a mixture of two liquid phases is critical to the successful operation of many chemical and petrochemical processes. Besides its obvious importance to liquid-liquid extraction and washing operations, liquid-liquid phase separation can be a critical factor in other operations including two-liquid-phase reaction, azeotropic distillation, and industrial wastewater treatment. Sometimes the required phase separation can be accomplished within the main process equipment, such as in using an extraction column or a batch-wise, stirred-tank reactor; but in many cases a stand-alone separator is used. These include many types of gravity decanters, filter-type coalescers, coalescers filled with granular media, centrifuges, and hydrocyclones.

The path that a liquid-liquid mixture takes through a chemical process on its way to the separator often has a dramatic impact on separation difficulty once the mixture arrives. For this reason, the first steps toward designing a decanter or other type of liquid-liquid phase separator should include a study of the overall process flow sheet to determine whether changes in upstream processing conditions can make for an easier and more robust separation. For example, if the main stream entering the separator is produced by mixing a number of smaller streams, look for opportunities to remove fine solids that contaminate the main stream by filtering solids from one or more small streams before they enter the larger stream. Also, standard centrifugal pumps are notorious for producing stable dispersions. If this type of pump is used, determine whether the turbulence caused by the pump is contributing to phase separation difficulty; and if so, consider using gravity flow (if possible) or replacing a high-shear pump and piping system with a lower-shear design. If a dispersion proves to be particularly difficult to separate, it may be due to the presence of some contaminant acting as a surfactant. Contaminants may be oxidation products produced in trace amounts owing to leakage of air into

the process, or they may be the products of corrosion of upstream equipment. They also may be materials that are intentionally added upstream to solve a problem there, such as cleaning agents and antifouling agents, but their presence, even in very small concentration, may cause unintended phase separation difficulties downstream.

FEED CHARACTERISTICS

Traditionally, the guidelines for selection and design of a gravity decanter or other type of separator focus on the size of dispersed drops. However, drop diameter often cannot be accurately predicted during the design of a new process, especially the size of the smaller drops in the distribution of drop sizes, and often this information is not available for an existing process because of sampling difficulties. Furthermore, knowledge of drop size alone is not sufficient because it says nothing about the rate of drop coalescence. In light of this, it is recommended instead to characterize the feed material in terms of the results of simple shake tests, as indicated in Table 15-24. This basic information can be very helpful in identifying an appropriate separator.

In Table 15-24, feed materials are classified into four main types according to the results of a shake test. Typical values of interfacial tension, density difference, and viscosity also are listed. The shake test can be as simple as vigorously shaking a representative feed by hand in a sealed graduated cylinder (about an inch in diameter) for 30 s or 1 min. The graduated cylinder is then placed on the bench, the time is recorded, and the progress of the separation is observed. For systems with drops that coalesce quickly, a sharp interface will quickly form between two settling liquid layers, and the rate at which drops fall or rise to the interface will determine the rate of phase separation or clarification of the layers. For many other systems, however, drops will accumulate at the interface forming a dispersion band, i.e., a layer of slowly coalescing drops, and the rate at which the drops coalesce determines the rate of phase separation. Whether a system is fast-coalescing or slow-coalescing is an important question that is easily answered by performing a simple shake test. Figure 15-63 illustrates the details of a batch settling profile. Once the dispersion band has disappeared, one or both of the phases may remain cloudy. If so, this typically indicates the presence of droplets on the order of 100 μm in diameter or smaller. For additional discussion of dispersion properties, see "Liquid-Liquid Dispersion Fundamentals."

TABLE 15-24 Shake Test Characterizations

Type	Shake test observations	Interfacial tension*	Density difference*	Viscosity of each phase*	Presence of fine solids or surfactants*
I	Dispersion band collapses within 5 min with crystal-clear liquids on top and bottom	Moderate to high, 10 dyn/cm or higher	$\Delta\rho > 0.1 \text{ g/cm}^3$	$\mu < 5 \text{ cP}$	Negligible
II	Dispersion band collapses within 10 to 20 min with clear liquids on top and bottom	Moderate, ~10 dyn/cm	$\Delta\rho > 0.1 \text{ g/cm}^3$	$\mu < 20 \text{ cP}$	Negligible
III	Dispersion band collapses within 20 min but one or more phases remain cloudy	Low to moderate, 3–10 dyn/cm	$\Delta\rho > 0.05 \text{ g/cm}^3$	$\mu < 100 \text{ cP}$	Might be present in low concentration
IVa	Stable dispersion is formed (dispersion band does not collapse within an hour or longer)—high viscosity	Low to high	$\Delta\rho > 0.1 \text{ g/cm}^3$	$\mu > 100 \text{ cP}$ in one of the phases	Negligible
IVb	Stable dispersion is formed—low interfacial tension	$< 3 \text{ dyn/cm}$	$\Delta\rho > 0.1 \text{ g/cm}^3$	$\mu < 100 \text{ cP}$	Negligible
IVc	Stable dispersion is formed—low density difference	Low to high	$\Delta\rho < 0.05 \text{ g/cm}^3$	$\mu < 100 \text{ cP}$	Negligible
IVd	Stable dispersion is formed—stabilized by surface-active components or solids	Low	$\Delta\rho > 0.1 \text{ g/cm}^3$	$\mu < 100 \text{ cP}$	Enough surfactant/solids to keep emulsion stable

*Typical physical properties. Behavior also depends upon the shear history of the fluid. For this test, a sample is characterized by the results of the shake test (second column), not its physical properties. Physical properties are listed only as typical values.

GRAVITY DECANTERS (SETTLERS)

Gravity decanters or settlers are simple vessels designed to allow time for two liquid phases to settle into separate layers (Fig. 15-64). Ideally, clear top and bottom layers form above and below a sharp interface or dispersion band. The top and bottom layers serve as clarifying zones. The height of the dispersion band, if present, generally remains constant during steady-state operation, although it may vary with position. The choice of where to locate the phase boundary within the vessel depends on whether more or less height is needed in the upper or lower clarification zones to obtain the desired clarity in the discharge streams. It can also depend on whether the inventory of one particular layer within the vessel should be minimized, as when handling reactive fluids such as monomers. Gravity decanters are well suited for separating type I feeds defined in Table 15-24 and, in most cases, type II feeds as well. It is common for coalescence to be the limiting factor in the separation of type II mixtures, so the design and sizing of the decanter will differ from those of the fast-coalescing systems.

Design Considerations Gravity decanters normally are specified as horizontal vessels with a length-to-diameter ratio greater than 2 (and often greater than 4) to maximize the phase boundary (cross-sectional area) between the two settled layers. This provides more effective utilization of the vessel volume compared to vertical decanters, although vertical decanters may be more practical for low-flow applications or when space requirements limit the footprint of the vessel.

The volume fraction of the minority phase is an important parameter in the operation of a decanter. Vessels handling less than 10 to 20 percent dispersed phase typically contain a wider distribution of droplet diameters with a long tail in the small size range [Barnea and Mizrahi, *Trans. Instn. Chem. Engrs.*, **53**, pp. 61–69 (1975)]. These decanters have a smaller capacity than when they contain more-concentrated dispersions. If one of the phases has a concentration lower than 20 percent in the feed mixture, it might be worthwhile to recycle the low-concentration phase to the feed point to boost the phase ratio within the separator vessel. Also, in certain cases increasing the operating temperature increases the drop coalescence rate. The result is a reduction in the dispersion band height for a given throughput, allowing an increase in the capacity of the settler. This behavior often can be attributed to a reduction in the continuous-phase viscosity.

Numerous methods are used to control the location of the interface inside the decanter. A boot or sump sometimes is included in the design to increase the path traveled by the heavy phase before exiting the vessel, to maximize the clarification zone for the light phase, or to minimize the inventory of heavy phase within the vessel. The interface can even be located inside the boot for one of these reasons. When a rag layer forms at the interface between settled layers, adding one or more nozzles in the vicinity of the interface will allow periodic draining of the rag (Fig. 15-65). Instruments such as differential pressure cells, conductance probes, or density meters are commonly used to control the location of the interface in a decanter. These instruments can be prone to fouling, and their operation can be compromised by the presence of a dispersion band or a rag layer. In that case, an alternative is to use an overflow leg or seal loop as illustrated in Figs. 15-64 and 15-65. The following expression can be used to specify the loop dimensions [Bocangel, *Chem. Eng. Magazine*, **93**(2), pp. 133–135 (1986); and Aerstlin and Street, *Applied Chemical Process Design* (Plenum, 1982)]:

$$Z_2 = \frac{(h_L + Z_1 - Z_3)\rho_L}{\rho_H} + Z_3 - h_H \quad (15-184)$$

where Z_1 , Z_2 , and Z_3 are the heights shown in Fig. 15-65 and h_L and h_H are the head losses in the light- and heavy-liquid discharge piping. An overflow leg can work reasonably well, provided that the densities of the two phases and the height of the dispersion band do not change significantly in operation (as in an upset). The light phase also may be removed through a takeoff tube entering the vessel from the bottom. This design provides added flexibility by allowing adjustment of the pipe length in the field without altering the vessel itself. Care should be taken to avoid the possibility of inducing a swirling motion as liquid enters the top of the weir. Swirling motions may be avoided or minimized by adding vanes or slots at the entrance.

To allow the phases to settle and remain calm, any form of turbulence or vortexing inside the decanter should be avoided. Introduction of the feed stream into the decanter should be located close to the interface to facilitate phase separation. Turbulence can arise from the inlet liquid entering the vessel at too high a velocity, forming a jet that disturbs the liquid layers. To counter these flow patterns, the feed into the gravity settler should enter the vessel at a velocity of less than

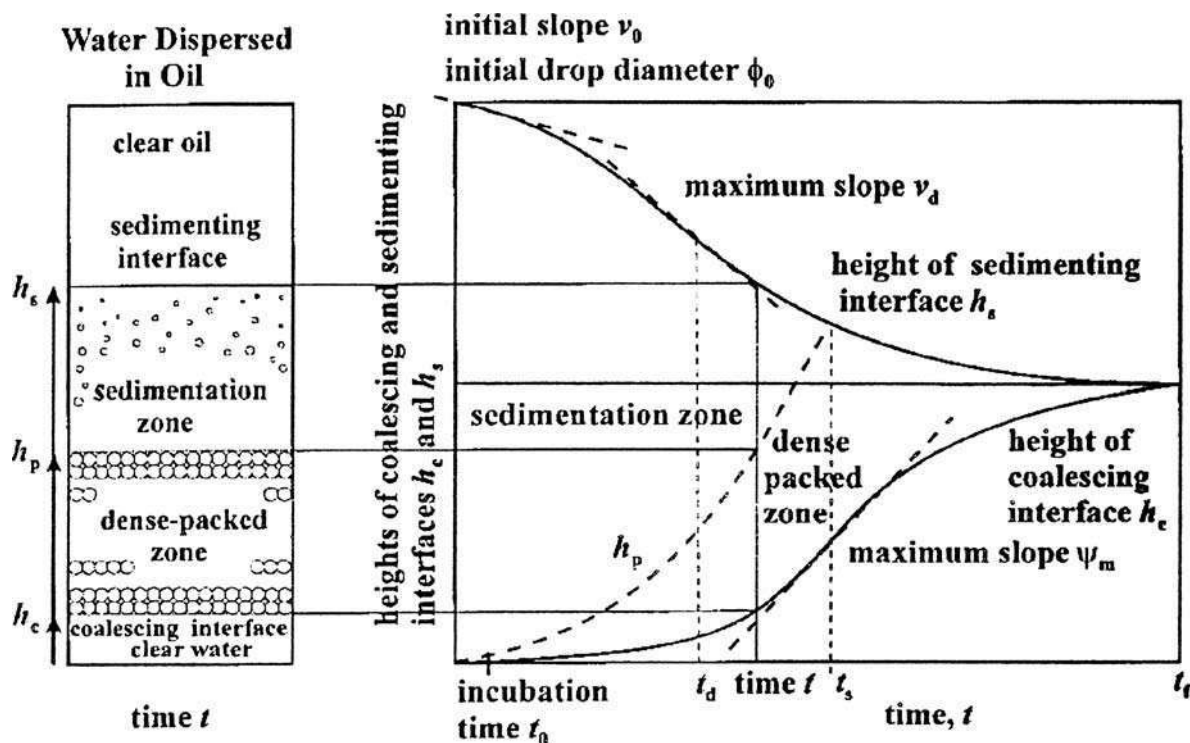


FIG. 15-63 Batch settling profile showing four regions: a top clarified phase, a sedimentation zone, a dense-packed dispersion zone, and a bottom clarified phase. [Reprinted from Jeelani, Panoussopoulos, and Hartland, *Ind. Eng. Chem. Res.*, **38**(2), pp. 493–501 (1999), with permission. Copyright 1999 American Chemical Society.] Consult the original article for a detailed description.

about 1 m/s (3 ft/s) as a general rule. This can be achieved by enlarging the feed line in the last 1 to 2 m (3 to 6 ft) leading to the vessel, to slow down the feed velocity at the inlet nozzle. In addition, a quiet feed zone may be created by installing a baffle plate in front of the feed pipe or a cap at the end of the feed line, with slots machined into the side of the pipe. Some designers are now using computational fluid dynamics (CFD) methods to analyze general flow patterns as an aid to specifying decanter designs.

Vented Decanters When the liquid-liquid stream to be decanted also contains a gas or vapor, provisions for venting the decanter must be included. This often is the case when decanting overheads condensate from an azeotropic distillation tower operating under vacuum, since

some amount of air leakage is virtually unavoidable, or when decanting liquids from an extractor operating at a higher pressure. A common design used for this service when the amount of gas is low is shown in Fig. 15-66. The feed enters the vessel at a point below the liquid level, so any gas must flow up through the liquid before disengaging in the vapor head space. An alternative design is illustrated in Fig. 15-67. With this design, the feed is introduced to the top of the vessel in the vapor headspace so that gases can be freely discharged and disengaged with no back-pressure. One drawback to this approach is that the feed liquids are dropped onto the light liquid surface, and significant quantities of heavy liquid may be carried over to the light liquid draw-off nozzle owing to the resulting turbulence. To mitigate this effect, a quiescent zone may be

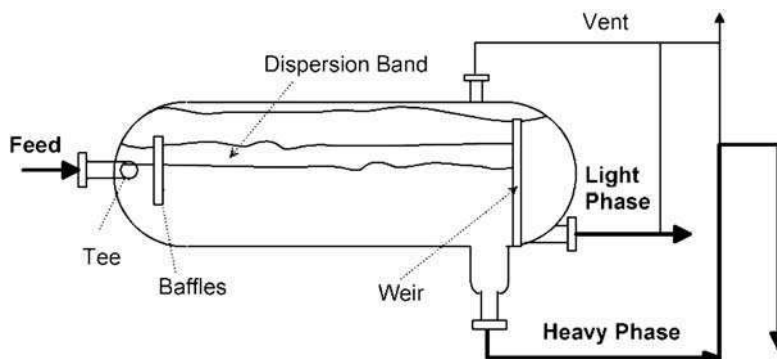


FIG. 15-64 Typical horizontal gravity decanter design.

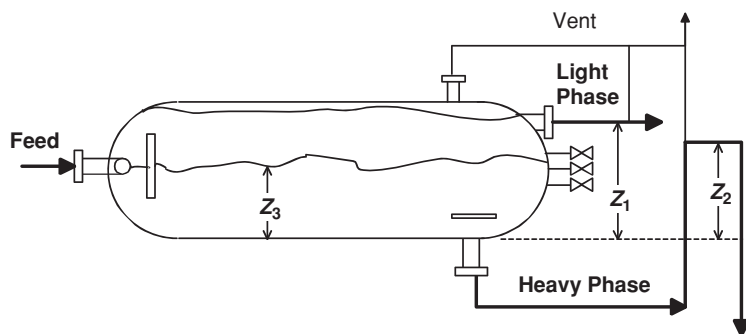


FIG. 15-65 Overflow loop for the control of the main interface in a decanter.

provided immediately below the top feed nozzle by means of a perforated baffle, as shown in Fig. 15-67. The baffle separates the disturbance caused by the entering feed from a calm separation zone where the two liquid phases can coalesce and disengage prior to draw-off.

Decanters with Coalescing Internals Adding coalescing internals may improve decanter performance by promoting the growth of drops and may reduce the size of vessel required to handle dispersions with slow coalescence (as in type II systems in Table 15-24). A wide variety of internals have been used including wire mesh, knitted wire or fibers, and flat or corrugated plates. When plates are used, the coalescer is sometimes referred to as a lamella-type coalescer. Plates typically are arranged in packets installed at a slight angle with respect to horizontal. The plates shorten the distance that drops must rise or fall to a coalescing surface and guide the flow of the resulting coalesced film [Menon, Rommel, and Blass, *Chem. Eng. Sci.*, **48**(1), pp. 159–168 (1993); and Menon and Blass, *Chem. Eng. Technol.*, **14**, pp. 11–19 (1991)]. Arranging the plates in packets of opposite slopes promotes flow reversal, and this may lead to more frequent drop-drop collisions [Berger, *Int. Chem. Eng.*, **29**(3), pp. 377–387 (1989)]. The Merichem Fiber-Film® contactor described earlier in “Suspended-Fiber Contactor” under “Mixer-Settler Equipment” also may be used

to promote growth of dispersed drops in a stream feeding a gravity decanter. In any case, the dispersed phase normally must preferentially wet the coalescence media for the media to be effective. If the feed contains solids, the potential for plugging the internals should be carefully evaluated. In certain cases, it may be necessary to allow access to the vessel internals for thorough cleaning. For more information, see Mueller et al., “Liquid-Liquid Extraction,” *Ullmann's Encyclopedia of Industrial Chemistry*, 6th ed. (Wiley-VCH, 2002).

Sizing Methods Sizing a decanter involves quantifying the relationship between the velocity of liquid to the phase boundary between settled layers and the average height of a dispersion band formed at the boundary. For fast-coalescing systems, the height of the dispersion band is negligible. Performance is determined solely by the rate of droplet rise or fall to the interface compared with the rate of flow through the decanter. In this case, design methods based on Stokes' law may be used to size the decanter, and residence time in the vessel becomes a key parameter. In many cases, however, coalescence is slow and the shake tests show a coalescence band that requires a fair amount of time to disappear. Then performance is determined by the volumetric flow rate of liquid to the boundary between the two settled layers, the boundary area available for coalescence, and the steady-state height of the dispersion band. For these systems, residence time is not a useful parameter for characterizing performance requirements.

Stokes' Law Design Method This method is described by Hooper [Sec. 1.11 in *Handbook of Separation Techniques for Chemical Engineers*, 3d ed., Schweitzer, ed. (McGraw-Hill, 1997)]; and by Jacobs and Penney [Chap. 3 in *Handbook of Separation Process Technology*, Rousseau, ed. (Wiley, 1987)]. It assumes that the drop coalescence rate is rapid and relies on knowledge of drop size. The terminal settling velocity of a drop is computed by using Stokes' law

$$u_t = \frac{gd^2\Delta\rho}{18\mu_c} \quad (15-185)$$

where d is a characteristic minimum drop diameter. (See Sec. 6 for detailed discussion of terminal settling velocity.) Note that which phase is continuous and which is dispersed can make a significant difference, since only the continuous-phase viscosity appears in Eq. (15-185). The decanter size is then specified such that

$$\frac{Q_c}{A} < u_t \quad (15-186)$$

where Q_c is the volumetric flow rate of the continuous phase and A is the cross-sectional area between the settled layers. This analysis assumes no effect of swirling or other deviation from quiescent flow, so a safety factor of 20 percent often is applied. Hooper and Jacobs indicate that designing for a Reynolds number $Re = VD_h\rho_c/\mu_c$ less than 5000 or so should provide sufficiently quiescent conditions, where V is the continuous-phase cross-flow velocity and D_h is the

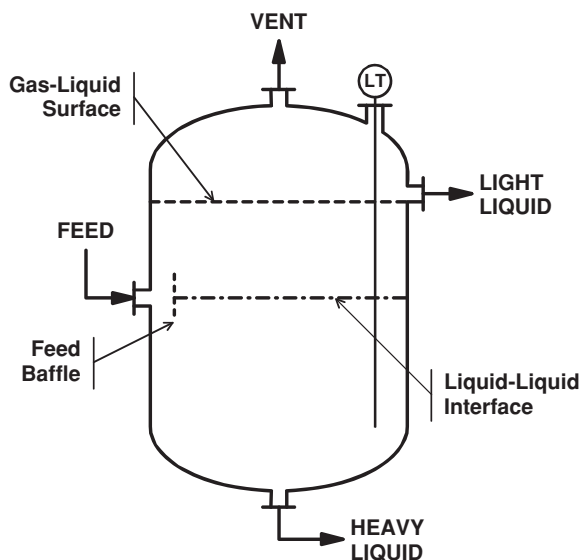


FIG. 15-66 Vertical decanter with submerged feed.

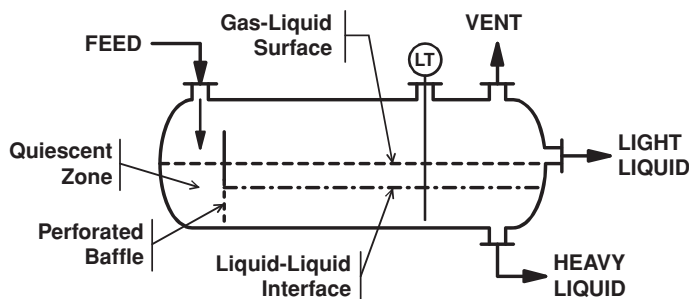


FIG. 15-67 Horizontal decanter with feed entering from the top and a baffled quiescent zone.

hydraulic diameter of the continuous-phase layer (given by 4 times the flow area divided by the perimeter of the flow channel including the interface). Decanter design methods based on Stokes' law generally assume a minimum droplet size of 150 μm , and this appears to be a reasonably conservative value for many chemical process applications. For separating secondary dispersions, it is common to assume a drop size in the range 70 to 100 μm . For more detailed discussion, see Hartland and Jeelani, Chap. 13, pp. 509–516, in *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994).

The method described above neglects any reduction in settling velocity due to the presence of neighboring drops at high population density (hindered settling). For best results, experimental data showing the relationship between settling velocity and initial dispersed-phase holdup should be generated. A simplified expression that neglects any drop coalescence during settling may be suitable for approximate design purposes

$$u_i \approx u_{t\infty}(1 - \phi_0) \frac{\mu_c}{\mu_d} \quad (15-187)$$

where u_i is an average settling velocity used to specify the decanter design, $u_{t\infty}$ is the velocity of an isolated drop calculated from Eq. (15-185), and ϕ_0 is the initial holdup. For more detailed discussion, see Ishii and Zuber, *AIChE J.*, **25**, pp. 843–855 (1979); and Das, *Chem. Eng. Technol.*, **20**, pp. 475–477 (1997).

Design Methods for Systems with Slow Coalescence For slow-coalescing systems, simple Stokes' law calculations will not provide a reliable design. Instead, it is necessary to understand the height of the dispersion band as a function of throughput. Jeelani and Hartland [*AIChE J.*, **31**, pp. 711–720 (1985)] recommend correlating decanter performance by using an expression of the form

$$\frac{1}{Q/A} = \frac{1}{k_1 \Delta H} + \frac{1}{k_2} \quad (15-188)$$

where ΔH is an average steady-state dispersion band height, Q is total volumetric throughput, and k_1 and k_2 are empirical constants. The general relationship between ΔH and Q/A also may be expressed in terms of a power law equation of the form

$$\Delta H \propto \left(\frac{Q}{A}\right)^a \propto \left(\frac{Q_c}{A}\right)^a \propto \left(\frac{Q_d}{A}\right)^a \quad (15-189)$$

Equations (15-188) and (15-189) represent decanter performance for a given feed with constant properties, i.e., a constant composition and phase ratio. Note that the analysis can be done in terms of total flow Q or the flow of continuous phase Q_c or dispersed phase Q_d . Typically, the value of the exponent a is greater than 2.5 [Barnea and Mizrahi, *Trans. Inst. Chem. Eng.*, **53**, pp. 61–91 (1975); and Golob and Modic,

Trans. Inst. Chem. Eng., **55**, pp. 207–211 (1977)]. The required size of a commercial-scale decanter may be determined by operating a small miniplant decanter to obtain values for the constants in Eqs. (15-188) and (15-189), since scale-up to the larger size generally follows the same relationship as long as the phase ratio and other operating variables are maintained constant. A commercial-scale decanter normally is designed for a throughput Q/A that yields a value of ΔH no larger than 15 percent of the total decanter height. Designs specifying taller dispersion bands are avoided because a sudden change in feed rate can trigger a dramatic increase in the height of the dispersion band that quickly floods the vessel. The dynamic response of ΔH has been studied by Jeelani and Hartland [*AIChE J.*, **34**(2), pp. 335–340 (1988)].

In certain cases, batch experiments may be used to size a continuous decanter [Jeelani and Hartland, *AIChE J.*, **31**, pp. 711–720 (1985)]. In a batch experiment similar to the simple shake test described earlier, the change in the height of the dispersion band with time may follow a relationship given by

$$\frac{1}{-dh/dt} = \frac{1}{k_1 h} + \frac{1}{k_2} \quad (15-190)$$

where h is the height of the batch dispersion band varying with time t . The constants k_1 and k_2 in Eq. (15-190) are the same as those used in the steady-state equation [Eq. (15-188)], assuming the batch test conditions (phase ratio and turbulence) are the same. Jeelani and Hartland have derived a number of models for systems with different coalescence behaviors [Jeelani and Hartland, *Chem. Eng. Sci.*, **42**(8), pp. 1927–1938 (1987)]. The most appropriate coalescence model is determined in batch tests and then is used to estimate ΔH versus throughput Q/A for a continuous decanter. For additional information, see Hartland and Jeelani, Chap. 13 in *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994); Nadiv and Semiat, *Ind. Eng. Chem. Res.*, **34**(7), pp. 2427–2435 (1995); Jeelani and Hartland, *Ind. Eng. Chem. Res.*, **37**(2), pp. 547–554 (1998); Jeelani, Panoussopoulos, and Hartland, *Ind. Eng. Chem. Res.*, **38**(2), pp. 493–501 (1999); and Yu and Mao, *Chem. Eng. Technol.*, **27**(4), pp. 407–413 (2004). Development of design methods for specifying continuous decanters with coalescing internals using batch test data is a current area of research [Hülswitt and Pfennig, *ISEC '05*, Biejing, China (September 2005)].

Several authors have derived correlations relating the height of the dispersion band to the density of each phase, the density difference, the viscosities, and the interfacial tension of aqueous/organic or aqueous/organic two-phase systems [Golob and Modic, *Trans. Inst. Chem. Eng.*, **55**, pp. 207–211 (1977); and Asenjo et al., *Biotech. and Bioeng.*, **79**(2), pp. 217–223 (2002)]. These correlations can provide useful estimates, but the results are generally valid only for the systems used to develop the correlations and should be used with caution. For new applications, some experimental work will be needed for reliable design.

OTHER TYPES OF SEPARATORS

Coalescers As noted earlier, adding coalescing internals to a decanter can improve decanter performance by promoting growth of small drops. The same concept can be applied in a separate coalescer vessel to treat the stream feeding the decanter. Systems of type III or type IV (Table 15-24) in particular may benefit, i.e., applications involving a need to break a secondary dispersion. Coalescers typically are packed with a granular material, a mesh made of metal wire or polymer filaments (or both), or fine fibers in woven or nonwoven composite sheets. The typical flow configuration is upflow if the light phase is dispersed and downflow if the heavy phase is dispersed. Coalescers containing fairly large media such as beds of granules or wire mesh may be able to tolerate a feed containing some fine solids. Coalescers containing fine granules or fine fibers require that the feed be free of solids to avoid plugging, so prefiltration may be necessary. For more detailed information, see Li and Gu, *Sep. and Purif. Tech.*, **42**, pp. 1-13 (2005); Shin and Chase, *AIChE J.*, **50**(2), pp. 343-350 (2004); Wines and Brown, *Chem. Eng. Magazine*, **104**(12), pp. 104-109 (1997); Hennessey et al., *Hydrocarbon Proc.*, **74**, pp. 107-124 (1995); Madia et al., *Env. Sci. Technol.*, **10**(10), pp. 1044-1046 (1976); Davies, Jeffreys, and Azfal, *Brit. Chem. Eng. Proc. Tech.*, **17**(9), pp. 709-712 (1972); and Hazlett, *Ind. Eng. Chem. Fund.*, **8**(4), pp. 625-632 (1969).

In most applications, the packing material should be wetted by the dispersed phase to some degree for best performance; however, this will depend on the size of dispersed droplets. For very fine droplets on the order of 10 μm or smaller, surface wetting is not the primary coalescence mechanism [Davies and Jeffreys, *Filtration and Separation*, pp. 349-354 (July/August 1969)]. In these cases, the packing promotes coalescence by providing a tortuous path that holds dispersed drops in close contact, facilitating drop-drop collisions. In other cases involving larger drops, a drop interception and wettability mechanism becomes important; i.e., the media provide a target for drop-solid surface collisions, and the surface becomes wetted with drops that merge together and leave the media as larger drops. In this case, an intermediate (optimum) wettability may be needed to most effectively promote the growth and dislodging of drops from the media [Shin and Chase, *AIChE J.*, **50**(2), pp. 343-350 (2004)]. In general, the degree to which flow path/collision mechanisms and/or surface wettability are important for good performance depends on the drop size distribution and dispersed-phase holdup in the feed, as well as system physical properties and whether surfactants or fine particulates are present. (See "Stability of Liquid-Liquid Dispersions" under "Liquid-Liquid Dispersion Fundamentals.") All this affects the choice of media, media size and porosity, and coalescer dimensions as a function of throughput. For a given application, some experimental work generally will be needed to sort this out and identify an effective and reliable design.

In cases where wettability is important, various types of sand, zeolites, glass fibers, and other inorganic materials may be used to facilitate coalescence of aqueous drops dispersed in organic feeds. Carbon granules, polymer beads, or polymer fibers may be useful in coalescing organic drops dispersed in water. The packing material should resist disarming by impurities, meaning that impurities should not become adsorbed and degrade the surface wettability characteristics over time. This can happen with charged or surfactantlike impurities; Paria and Yuet [*Ind. Eng. Chem. Res.*, **45**(2), pp. 712-718 (2006)] describe the adsorption of cationic surfactants at sand-water interfaces, a phenomenon that can alter surface wettability. In a few cases, the packing needs to age in service to develop its most effective surface properties.

Madia et al. [*Env. Sci. Technol.*, **10**(10), pp. 1044-1046 (1976)] describe a chromatography method for screening potential media with regard to surface wettability. The method involves measuring the retention times of water and heptane (or other components of interest) by using columns filled with the packing materials of interest (reduced in size if needed); the longer the relative retention time, the greater is the wettability of the packing for that component. The authors used gas chromatography of water and heptane to characterize coalescence for an oil-in-water dispersion; but it should be possible to characterize other systems by using this approach, and liquid chromatography methods might be used for components with low volatility.

For granular bed coalescers, typical granule sizes include 12 $\times$ 16 Tyler screen mesh (between 1.4 and 1 mm) and 24 $\times$ 48 Tyler mesh (0.7 to 0.3 mm). Smaller sizes sometimes are used as well. Typical bed heights range from 8 in to 4 ft (0.2 to 1.2 m), with the taller beds used with the larger granules. Layered beds may be used. For example, the front of the coalescer may contain a thin layer of fine media with low porosity and high tortuosity characteristics to facilitate drop-drop collisions of very small droplets, followed by a layer of coarser media having the wetting characteristics needed to further grow and shed larger drops.

For fine-fiber coalescers, the coalescing media normally are arranged in the form of a filter cartridge. Wines and Brown [*Chem. Eng. Magazine*, **104**(12), pp. 104-109 (1997)] describe a coalescing mechanism in which a drop (on the order of 0.2 to 50 μm) becomes adsorbed onto a fiber and then moves along the fiber with the bulk liquid flow until colliding with another adsorbed drop at the intersection where two fibers cross. Fiber diameter and wettability are important properties as they affect porosity (tortuous path) and wettable surface area. Like a packed-bed coalescer, a filter-type coalescer may be constructed in layers: an initial prefilter zone to remove particulates and minimize fouling, a primary coalescence zone where small droplets grow to larger ones, and a secondary coalescence zone with greater porosity and having surface-wetting characteristics optimized to grow the larger drops.

Pressure drop, an important consideration in the design of any coalescer, depends upon media size and shape, bed height or filter thickness, and throughput. Methods for calculating pressure drop through packed beds and porous media are described in Sec. 6. For approximately spherical media, the pressure drop due to frictional losses, assuming incompressible media, may be estimated from

$$\frac{\Delta P}{L} = \frac{150(1-\phi)^2 \mu V}{d_m^2 \phi^3} + \frac{1.75 \rho_c V^2}{d_m \phi^3} \quad \text{Re}_{\text{particle}} = \frac{V \rho_c d_m}{\mu} \leq 10 \quad (15-191)$$

where L is the length of the packed section, V is the superficial velocity of the total liquid flow, d_m is an equivalent spherical diameter of the media particles (given by 6 times the mean ratio of particle volume to particle surface area), and ϕ is the volume fraction of voids (flow channels) within the bed [Ergun, *Chem. Eng. Prog.*, **48**(2), pp. 89-94 (1952)]. Also see Leva, *Chem. Eng. Magazine*, **56**(5), pp. 115-117 (1949), or Leva, *Fluidization* (McGraw-Hill, 1959). The minimum value of ϕ for a tightly ordered bed of uniform spherical particles is 0.26, but of course for real media this will vary depending upon the particle size distribution and particle shape. The second term in Eq. (15-191) often is neglected at $\text{Re}_{\text{particle}} \leq 1$. For fiber media, d_m can be thought of as a characteristic fiber dimension. For discussion of pressure drop through fiber beds, see Shin and Chase, *AIChE J.*, **50**(2), pp. 343-350 (2004); and Li and Gu, *Sep. and Purif. Tech.*, **42**, pp. 1-13 (2005). In practice, pressure drop data may be correlated by using an equation of the same form as Eq. (15-191), $\Delta P/L = aV + bV^2$, where a and b are empirically determined constants. Media and equipment suppliers generally will have some experimental data showing $\Delta P/L$ versus flow rate.

Centrifuges A stacked-disk centrifuge or other type of centrifuge may be a cost-effective option for liquid-liquid phase separation whenever use of a gravity decanter/coalescer proves to be impractical because rates of drop settling or coalescence are too low. This may be the case for type III and type IV systems (Table 15-24) in particular. Factors involved in specifying a centrifuge are discussed in "Centrifugal Extractors" under "Liquid-Liquid Extraction Equipment."

Hydrocyclones Liquid-liquid hydrocyclones, like centrifuges, utilize centrifugal force to facilitate the separation of two liquid phases [*Hydrocyclones: Analysis and Applications*, Svarovsky and Thew, eds. (Kluwer, 1992); and Bradley, *The Hydrocyclone* (Pergamon, 1965)]. Instead of using rotating internals, as in a centrifuge, a hydrocyclone

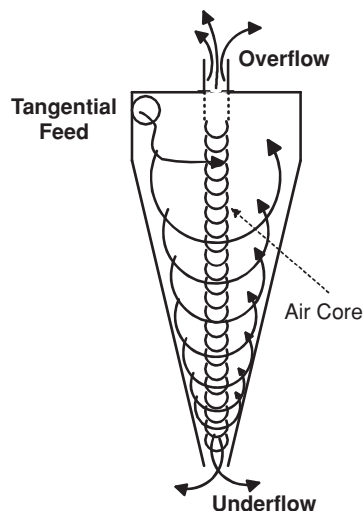


FIG. 15-68 Flow patterns in a hydrocyclone.

generates centrifugal force through fluid pressure to create rotational fluid motion (Fig. 15-68). Feed enters the hydrocyclone through a tangential-entry nozzle. A primary vortex rich in the heavy phase forms along the inner wall, and a secondary vortex rich in the light phase forms near the centerline. The underflow stream (heavy phase) exits the cyclone through the apex of the cone (underflow nozzle). The overflow stream (light phase) exits through the vortex finder, a tube extending from the cylinder roof into the interior. The feed split can be adjusted by changing the relative diameters of the vortex finder and underflow nozzle. A hydrocyclone is not completely filled with liquid; an air core exists at the centerline. A commercial-scale hydrocyclone multiplies the force of gravity by a factor of 100 to 1000 or so, depending on the diameter and operating pressure. Hydrocyclones traditionally have been used for liquid-solid separations, but by adjusting their design (cone angle and length, vortex finder length, and so on) they can be applied to liquid-liquid separations [Mozley, *Filtration and Sep.*, pp. 474–477 (Nov/Dec. 1983)].

Since the fluid flow is turbulent at the top of the unit and the rotation of liquid within the device produces a high shear field, mixtures with low interfacial tension tend to emulsify or create foam within a hydrocyclone. However, hydrocyclones may be well suited for type I or possibly type II mixtures containing some solids, especially if only a rough cut is needed. The flow pattern established within a hydrocyclone normally requires that a considerable part of the feed leave in the overflow outlet. For this reason, hydrocyclones are generally more efficient for feeds containing only a small fraction of heavy phase, although some authors indicate they can be effective for feeds with a small fraction of light phase through careful specification of hydrocyclone geometry.

The main operating variables for a hydrocyclone are the feed pressure, the feed flow rate, and the split ratio, i.e., the relative amounts of fluid exiting top and bottom. The split ratio may be adjusted by specifying the size of the underflow and overflow nozzles. Choosing a material of construction wetted by the heavy phase for the cone may improve the effectiveness of the device. Experimental work is needed to determine the efficiency of the separation as a function of the split ratio for a series of flow rates and hydrocyclone geometries [Sheng, *Sep. and Purif. Methods*, **6**(1), pp. 89–127 (1977); and Colman and Thew, *Chem. Eng. Res. Des.*, **61**(7), pp. 233–240 (1983)]. If testing indicates satisfactory performance, hydrocyclones can be relatively inexpensive and simple-to-operate units (no moving parts). Because sufficient centrifugal force cannot be generated in large-diameter units, scale-up consists of connecting multiple small units in parallel.

Units are sometimes placed in series to provide multiple stages of separation. Hydrocyclones are used on ships and drilling platforms for removing oil from water [Bednarski and Listewnik, *Filtration and Sep.*, pp. 92–97 (March/April 1988)]. Numerical simulations of hydrocyclone performance and flow profiles are described by Bai and Wang [Chem. Eng. Technol., **29**(10), pp. 1161–1166 (2006)] and by Murphy et al. [Chem. Eng. Sci., **62**, pp. 1619–1635 (2007)].

Ultrafiltration Membranes These are microporous membranes with pore sizes in the range of 0.1 and 0.001 μm [Porter, "Ultrafiltration," in *Handbook of Industrial Membrane Technology* (Noyes, 1990)]. In this size range, the pores may be used to "filter out" and concentrate micelles from a liquid feed without disrupting (breaking) the micellar structure. Such a membrane may also be used to remove micrometer size droplets from a dilute dispersion. However, if the dispersed-phase content is too high, the membrane may become fouled owing to deposition of a coalesced layer that obstructs the pores. This can be a particular problem when removing oil droplets for an oil-in-water dispersion using a polymeric membrane.

The feed solution is fed to the membrane module under pressure (normally less than 6 bar). The majority of the continuous phase flows through the pores of the membranes by pressure difference and collects on the permeate side as a clarified solution. The micelles or microdroplets are rejected and flow with the remaining continuous phase, tangentially along the membrane surface, to the retentate outlet of the membrane module [Voges, Wu, and Dalan, *Chem. Processing*, pp. 40–43 (April 2001)]. The shear at the surface of the membrane should be high enough to stop the micelles from aggregating on the polymeric surface of the membrane, but low enough to avoid breaking the colloidal particles.

Ultrafiltration membranes can be very efficient at removing colloidal particles of an emulsion but normally will not stop dissolved oil from permeating. Since most membranes are polymeric, they are more stable in the presence of water, so they are best suited for aqueous systems. Since they produce only one well-clarified phase (the permeate), they should be applied to processes with stable micelles where clear continuous phase is required and where losses of continuous phase with the micellar phase can be tolerated. The use of ultrafiltration membranes in an extractive ultrafiltration process for recovery of carboxylic acids is discussed by Rodríguez et al. [*J. Membrane Sci.*, **274**(1–2), pp. 209–218 (2006)].

Selecting the membrane best suited for a given application is best accomplished experimentally. The membrane material must be compatible with the feed, and the module should exhibit high permeation flow while maintaining good micelle rejection. The pore size and the molecular weight cutoff reported by the manufacturer are good indications of membrane performance; but since other factors such as membrane/solute interaction and fouling impact the separation, this information is only a starting point. Key operating parameters include temperature, feed flow rate, and permeate-to-feed ratio. Scale-up consists of adding membrane modules to handle the required production rate [Eykamp and Steen, Chap. 18 in *Handbook of Separation Process Technology*, Rousseau, ed. (Wiley, 1987)].

Electrotreaters In an electrostatic coalescer, an electric field is applied to a dispersion to induce dipoles or net charges on the suspended drops. The drops are then attracted to one another, facilitating their coalescence [Waterman, *Chem. Eng. Prog.*, **61**(10), pp. 51–57 (1965); and Yamaguchi, Chap. 16 in *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994)]. This technology is applicable only to a nonconductive continuous phase and an aqueous dispersed phase. Once the water drops are sufficiently large, they settle to the bottom of the vessel while the clarified oil phase migrates to the top. The top and bottom zones are kept quiet and out of the electric field. In cases where inlet salt content is high, a multistage, countercurrent desalting system can be used. Units with ac or dc voltage are available.

Electrostatic separators are high-voltage electrostatic devices that can arc under certain conditions. For this reason, a careful review of safety considerations is needed, especially for applications involving flammable liquids. Evaluating feasibility and generating design data normally involve close consultation with the equipment vendors. This technology is applied on a very large scale in the petroleum industry for crude oil desalting.

EMERGING DEVELOPMENTS

MEMBRANE-BASED PROCESSES

Polymer Membranes Extraction processes employing polymer membranes are sometimes referred to as nondispersive or pertraction operations. The use of membranes in extraction offers a number of potential advantages including (1) constant well-defined mass-transfer area; (2) the ability to operate at very low solvent-to-feed ratios independent of other operating variables; (3) very low holdup of solvent and product within the extractor, thus providing low residence time similar to a centrifugal extractor; (4) dispersion-free liquid-liquid contacting that eliminates the need for liquid-liquid interface control and phase separation; (5) no requirement for a difference in density between liquid phases; and (6) linear scale-up by addition of extra modules, so performance at large scale can be determined directly from small-scale tests using a single module. This last point suggests, however, that the economy of scale may not be as large as it is for extractors that are scaled up as a single larger unit.

The most important advantages that membranes can offer to the process designer are those that overcome an inherent limitation of another type of extractor, as in the ability to handle liquids with close or even equal densities and the ability to operate at extremely low solvent-to-feed ratios. Thus, the types of applications where membranes are likely to be most attractive include applications with close densities and/or a K value greater than 50 or so. In principle, $K > 50$ would allow operation using a solvent-to-feed ratio of 1 : 25 or less (for an extraction factor of 2), something that can be difficult to accomplish by using conventional extractors. To take full advantage, the feed would have to be sufficiently dilute that the loading capacity of the solvent is not exceeded. The primary disadvantages of membrane-based extractors are the added mass-transfer resistance across the membrane, limited fiber-side or tube-side throughput, and concerns about fouling and limited membrane life in industrial service. Applications are limited to feeds that are free of solid particles (or can be cost effectively prefiltered); otherwise, the membranes are easily fouled. The useful life of a membrane module also is a critical factor since the frequency with which membrane modules must be replaced has a dramatic impact on overall cost.

The use of nonporous polymer membranes for liquid-liquid extraction suffers from very slow permeation of solute through the membrane, although this approach has been developed for a special case

involving reaction-enhanced extraction of an aromatic acid from wastewater through a nonporous silicone membrane into a caustic solution [Ferreira et al., *Desalination*, **148**(1–3), pp. 267–273 (2002)]. For most liquid-liquid extraction applications, however, a porous membrane is used and extraction involves transfer through a liquid-liquid meniscus maintained within the pores. One of the most promising contactors for this type of extraction is the microporous hollow-fiber (MHF) contactor (Fig. 15-69). The MHF contactor resembles a shell-and-tube heat exchanger in which the tube walls are porous and are capable of immobilizing a liquid-liquid interface within the pores. For a hydrophobic polymeric membrane, the aqueous phase normally is fed to the interior of the fiber (the fiber-bore side), while the organic phase is fed to the shell side. In this configuration, the aqueous fluid is maintained at a higher pressure relative to the organic phase, to immobilize the liquid-liquid interface within each pore. Care must be taken to avoid too high an aqueous pressure, or else breakthrough of the aqueous phase can occur. This breakthrough pressure is a function of the interfacial tension and pore size. Earlier versions of MHF contactors provided a parallel-flow design, but this design suffered from shell-side bypassing [Seibert et al., *Sep. Sci. Technol.*, **28**(1–3), p. 343 (1993)]. An improved design that incorporates a central baffle and uniform fiber spacing is currently available (Fig. 15-69). The dimensions are listed in Table 15-25.

In the baffled design, the shell-side fluid is fed through a central perforated distributor. It flows radially through the fiber bundle, around a baffle located in the middle of the module, and leaves the module through the central distributor. As in conventional extraction, the mass transfer of solute occurs across a liquid-liquid interface. However, unlike in conventional extraction, the interface is maintained at micrometer-size pores, and three mass-transfer resistances are present: tube-side (k_t), shell-side (k_s), and pore or membrane-side (k_m). The overall mass-transfer coefficient based on the tube-side liquid k_{ot} is given by

$$\frac{1}{k_{ot}} = \frac{1}{k_t} + \frac{m^{vol}}{k_s} + \frac{1}{k_m} \quad (15-192)$$

where m^{vol} is the local slope of the equilibrium line for the solute of interest, with the equilibrium concentration of solute in the tube-side

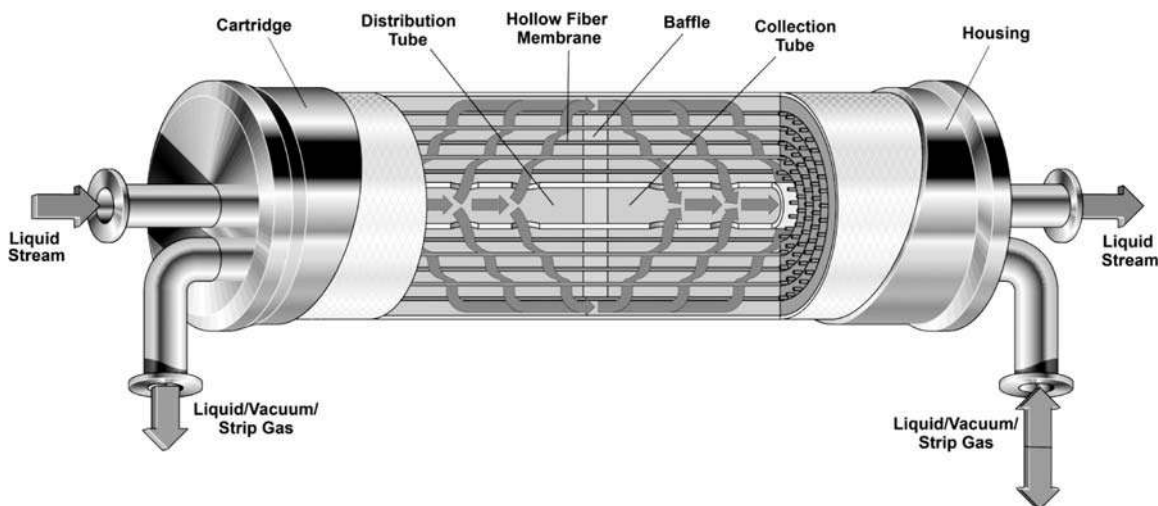


FIG. 15-69 Schematic of the Liqui-Cel Membrane Contactor. (Courtesy of Membrana-Charlotte. Liqui-Cel is a registered trademark of Membrana-Charlotte, a division of Celgard, LLC.)

TABLE 15-25 Baffled MHF Contactor Geometric Characteristics

Baffles per module	1
Module diameter, cm	9.8
Module length, cm	71
Effective fiber length, cm	63.5
Fiber outside diameter, μm	300
Fiber inside diameter, μm	240
Porosity of fiber	0.3
Number of fibers per module	30,000
Contact area per module, cm^2	81,830
Interfacial area, cm^2/cm^3	27
Tortuosity	2.6

Reprinted from Seibert and Fair, *Sep. Sci. Technol.*, **32**(1-4), pp. 573-583 (1997), with permission. Copyright 1997 Taylor & Francis.

liquid plotted on the y axis and the equilibrium concentration of solute in the shell-side liquid plotted on the x axis. Equation (15-192) assumes the tube-side fluid wets the pores.

The mass-transfer efficiencies of various MHF contactors have been studied by many researchers. Dahuron and Cussler [*AIChE J.*, **34**(1), pp. 130-136 (1988)] developed a membrane mass-transfer coefficient model (k_m); Yang and Cussler [*AIChE J.*, **32**(11), pp. 1910-1916 (1986)] developed a shell-side mass-transfer coefficient model (k_s) for flow directed radially into the fibers; and Prasad and Sirkar [*AIChE J.*, **34**(2), pp. 177-188 (1988)] developed a tube-side mass-transfer coefficient model (k_t). Additional studies have been published by Prasad and Sirkar ["Membrane-Based Solvent Extraction," in *Membrane Handbook*, Ho and Sirkar, eds. (Chapman & Hall, 1992)]; by Reed, Semmens, and Cussler ["Membrane Contactors," *Membrane Separations Technology: Principles and Applications*, Noble and Stern, eds. (Elsevier, 1995)]; by Qin and Cabral [*AIChE J.*, **43**(8), pp. 1975-1988 (1997)]; by Baudot, Flourey, and Smorenburg [*AIChE J.*, **47**(8), pp. 1780-1793 (2001)]; by González-Muñoz et al. [*J. Membrane Sci.*, **213**(1-2), pp. 181-193 (2003) and *J. Membrane Sci.*, **255**(1-2), pp. 133-140 (2005)]; by Saikia, Dutta, and Dass [*J. Membrane Sci.*, **225**(1-2), pp. 1-13 (2003)]; by Bocquet et al. [*AIChE J.*, **51**(4), pp. 1067-1079 (2005)]; and by Schlosser, Kertesz, and Martak [*Sep. Purif. Technol.*, **41**, p. 237 (2005)]. A review of mass-transfer correlations for hollow-fiber membrane modules is given by Liang and Long [*Ind. Eng. Chem. Res.*, **44**(20), pp. 7835-7843 (2005)]. Eksangri, Habaki, and Kawasaki [*Sep. Purif. Technol.*, **46**, pp. 63-71 (2005)] discuss the effect of hydrophobic versus hydrophilic membranes for a specific application involving transfer of solute from an aqueous feed to an organic solvent. Karabelas and Asimakopoulou [*J. Membrane Sci.*, **272**(1-2), pp. 78-92 (2006)] discuss process and equipment design considerations.

In general, researchers have treated MHF contactors as differential contacting devices. However, Seibert and Fair [*Sep. Sci. Technol.*, **32**(1-4), pp. 573-583 (1997)] and Seibert et al. [*ISEC '96 Proc.*, **2**, p. 1137 (1996)] suggest that the baffled MHF contactor can be treated as a staged countercurrent contactor. Their recommendations are based on studies using a commercial-scale skid-mounted extraction system. Their semi-work-scale study demonstrated the performance advantages of the MHF contactor relative to a column filled with structured packing for a system with a high partition ratio. Seibert et al. [*ISEC '96 Proc.*, **2**, p. 1137 (1996)] also provide limited economic data for the extraction of *n*-hexanol from water by using *n*-octanol. Also see the discussion by Yeh [*J. Membrane Sci.*, **269**(1-2), pp. 133-141 (2006)] regarding the use of internal reflux in a cross-flow membrane configuration to boost liquid velocities for enhanced performance.

Liquid Membranes Emulsion liquid-membrane (ELM) extraction involves intentional formation of an emulsion between two immiscible liquid phases followed by suspension of the emulsion in a third liquid that forms an outer continuous phase. The encapsulated liquid and the continuous phase are miscible. The liquid-membrane phase is immiscible with the other phases and normally must be stabilized by using surfactants.

If the continuous phase is aqueous, the suspended phase is a water-in-oil emulsion. If the continuous phase is organic, the emulsion is the oil-in-water type. This technology differs from traditional liquid-liquid extraction processes in that it allows transfer of solute between miscible liquids by introducing an immiscible liquid membrane between them. A typical process involves first forming a stable emulsion and contacting it with the continuous phase to transfer solute between the encapsulated phase and the continuous phase, followed by steps for separating the emulsion and continuous phases and breaking the emulsion. The emulsion must be sufficiently stable to remain intact during processing, but not so stable that it cannot be broken after processing, and this may present a challenge for commercial implementation. The technology is described by Frankendorf and Li [Chap. 19 in *Handbook of Separation Process Technology*, Rousseau, ed. (Wiley, 1987)].

Potential applications of ELM extraction include separation of aromatic and aliphatic hydrocarbons [Chakraborty and Bart, *Chem. Eng. Technol.*, **28**(12), pp. 1518-1524 (2005)], separation and concentration of amino acids [Thien, Hatton, and Wang, *Biotech. and Bioeng.*, **32**(5), pp. 604-615 (1988)], and recovery of penicillin G from fermentation broth [Lee, Lee, and Lee, *J. Chem. Technol. Biotechnol.*, **59**(4), pp. 365-370, 371-376 (1994)]; Lee et al. [*J. Membrane Sci.*, **124**, pp. 43-51 (1997)]; and Lee and Yeo, *J. Ind. Eng. Chem.*, **8**(2), p. 114 (2002)]. The latter application involves transfer of the penicillin G solute ($\text{p}K_a = 2.7$) from the continuous phase (consisting of a filtered broth adjusted to a pH of about 3) into the membrane phase (typically *n*-lauryltrialkylmethyl amine extractant dissolved in kerosene) and then into the interior aqueous phase (clean water at a pH of about 8). Lee et al. [*J. Membrane Sci.*, **124**, pp. 43-51 (1997)] show that the operation can be carried out in a continuous countercurrent extraction column. The product is later obtained by separating the emulsion droplets from the continuous phase by using filtration, and this is followed by breaking the emulsion and isolating the interior aqueous phase from the amine extractant phase. A polyamine surfactant is used to stabilize the emulsion during extraction.

Supported liquid-membrane (SLM) processes involve introduction of a microporous solid membrane to serve as a support for the liquid-membrane phase. The microporous membrane provides well-defined interfacial area and eliminates the need for a surfactant. As in the penicillin ELM application described above, SLM applications often employ an extractant solution as the liquid-membrane phase to enable a facilitated transport mechanism. The extractant species interacts with the desired solute at the feed side and then carries the solute across the membrane to the other side, where solute transfers into a stripping solution. Such a process, whether using a surfactant-stabilized emulsion or a supported liquid membrane, allows forward and back extraction (or stripping) in a single operation. Ho and Wang [*Ind. Eng. Chem. Res.*, **41**(3), pp. 381-388 (2002)] discuss the application of SLM technology to remove radioactive strontium, Sr-90, from contaminated waters. Other examples involve extraction of metal ions from water [Canet and Seta, *Pure Appl. Chem. (IUPAC)*, **73**(12), pp. 2039-2046 (2001)] and recovery of aromatic acids or bases from wastewater [Dastgir et al., *Ind. Eng. Chem. Res.*, **44**(20), pp. 7659-7667 (2005)]. One of the challenges encountered in using supported liquid membranes is the difficulty in controlling trans-membrane pressure drop and maintaining the liquid membrane on the support; it may become dislodged and entrained into the flowing phases. Various approaches to stabilizing the supported liquid have been proposed. These are discussed by Dastgir et al. [*Ind. Eng. Chem. Res.*, **44**(20), pp. 7659-7667 (2005)].

ELECTRICALLY ENHANCED EXTRACTION

An electric field may be used to enhance the performance of an aqueous-organic liquid-liquid contactor, by promoting either drop breakup or drop coalescence, depending upon the operating conditions and how the field is applied. The technology normally involves dispersing an electrically conductive phase (the aqueous phase) within a continuous nonconductive phase, applying a high-voltage electric field (either ac or dc) across the continuous phase, and taking advantage of the effect of the electric field

on the shape, size, and motion of the dispersed drops. The potential advantages of this technology include more precise control of drop size and motion for improved control of mass transfer and phase separation within an extractor. Potential disadvantages include the requirement for more complex equipment, difficulties in scaling up the technology to handle large production rates, and safety hazards involved in processing flammable liquids in high-voltage equipment.

A number of different equipment configurations and operating concepts have been proposed. Yamaguchi [Chap. 16 in *Liquid-Liquid Extraction Equipment*, Godfrey and Slater, eds. (Wiley, 1994)] classifies the proposed equipment into three general types: perforated-plate and spray columns, mixed contactors, and liquid-film contactors. For example, Yamaguchi and Kanno [*AIChE J.*, **42**(9), pp. 2683–2686 (1996)] describe an apparatus in which a dc voltage is applied between two electrodes in the presence of a nitrogen gas interface. Aqueous drops form in the presence of the electric field, and they are first attracted to the gas-liquid interface. Once the drops contact the interface, the charge on the drops is reversed, and the drops fall back to coalesce at the bottom of the vessel. Bailes and Stitt [U.S. Patent 4,747,921 (1988)] describe a rotating-impeller extraction column containing alternating zones of high voltage (to promote dispersed drop coalescence) and high-intensity mixing (to promote redispersion of drops). In this design, the electric field serves to promote drop coalescence so that dispersed drops experience alternating drop breakup and growth as they move through the agitated column. Scott and Wham [*Ind. Eng. Chem. Res.*, **28**(1), pp. 94–97 (1989)] and Scott, DePaoli, and Sisson [*Ind. Eng. Chem. Res.*, **33**(5), pp. 1237–1244 (1994)] describe a nonagitated apparatus called an emulsion-phase contactor. This device employs an electric field to induce formation of a stable emulsion or dispersion band, with clear organic and aqueous layers above and below. The aqueous phase is fed to the middle or top of the dispersion band; it flows down through the band and is removed from a clarified aqueous zone maintained at the bottom. The lighter organic phase is fed to the bottom; it moves up through the dispersion band and is removed from the top. The net result is countercurrent contacting with very high interfacial area and significantly improved mass transfer in terms of the number of transfer units achieved for a given contactor height.

Another approach involves electrostatically spraying aqueous solutions into a continuous organic phase to create dispersed drops within a spray column contactor [Weatherley et al., *J. Chem. Technol. Biotechnol.*, **48**(4), pp. 427–438 (1990)]. A high voltage is applied between electrodes, one connected to a nozzle where dispersed drops are formed and the other placed within the continuous organic phase. Petera et al. [*Chem. Eng. Sci.*, **60**, pp. 135–149 (2005)] discuss the modeling of drop size and motion within such a device. For additional discussion, see Tsouris et al. [*Ind. Eng. Chem. Res.*, **34**(4), pp. 1394–1403 (1995)], Tsouris et al. [*AIChE J.*, **40**(11), pp. 1920–1923 (1994)], Gneist and Bart [*Chem. Eng. Technol.*, **25**(2), pp. 129–133 (2002)], Gneist and Bart [*Chem. Eng. Technol.*, **25**(9), pp. 899–904 (2002)], and Elperin and Fominykh [*Chem. Eng. Technol.*, **29**(4), pp. 507–511 (2006)].

PHASE TRANSITION EXTRACTION AND TUNABLE SOLVENTS

Phase transition extraction (PTE) involves transitioning between single-liquid-phase and two-liquid-phase states to facilitate a desired separation. Ullmann, Ludmer, and Shinnar [*AIChE J.*, **41**(3), pp. 488–500 (1995)] showed that extraction of an antibiotic from fermentation broth into an organic solvent could be improved by transitioning across a UCST phase boundary using heating and cooling. The results showed much higher stage efficiency compared to a standard extraction technique without phase transition and much faster phase separation. The phase transition may be induced by a change in temperature or a change in composition through addition and/or removal of organic solvents or antisolvents [Gupta, Mauri, and Shinnar, *Ind. Eng. Chem. Res.*, **35**(7), pp. 2360–2368 (1996)]. Alizadeh and Ashtari describe a temperature-induced phase transition process for extracting silver(I) from aqueous solution using dinitrile solvents [*Sep. Purification*

Technol., **44**, pp. 79–84 (2005)]. Another process that exploits a phase transition to facilitate separation and recycle of solvent after extraction utilizes ethylene oxide-propylene oxide copolymers in aqueous two-phase extraction of proteins [Persson et al., *J. Chem. Technol. Biotechnol.*, **74**, pp. 238–243 (1999)]. After extraction, the polymer-rich extract phase is heated above its LCST to form two layers: an aqueous layer containing the majority of protein and a polymer-rich layer that can be decanted and recycled to the extraction.

Another approach utilizes pressurized CO₂ to control phase splitting and tune partition ratios in organic-water mixtures. Addition of pressurized CO₂ yields an organic phase rich in CO₂ (the gas-expanded phase) and an aqueous phase containing little CO₂. Adrian, Freitag, and Maurer [*Chem. Eng. Technol.*, **23**(10), pp. 857–860 (2000)] report data demonstrating the ability to induce phase splitting in the completely miscible 1-propanol + water system by pressurization with CO₂ at near-critical pressures above 74 bar (about 1100 psia). The authors also show that the partition ratio for transfer of methyl anthranilate from the aqueous phase to the organic phase can be varied between 1 and about 13 by adjusting pressure and temperature. Jie Lu et al. [*Ind. Eng. Chem. Res.*, **43**(7), pp. 1586–1590 (2004)] demonstrate a reduction in the lower critical solution temperature for the partially miscible THF + water system by addition of CO₂ at more moderate pressures (on the order of 10 bar, or about 145 psia). The authors show that the partition ratio for transfer of a water-soluble dye from the organic phase to the aqueous phase can be increased dramatically by increasing CO₂ pressure. For more detailed discussion of gas-expanded-liquid techniques used to facilitate various reaction and extraction processes, see Eckert et al., *J. Phys. Chem. B*, **108**(47), pp. 18108–18118 (2004).

IONIC LIQUIDS

The potential use of ionic liquids for liquid-liquid extraction is gaining considerable attention [Parkinson, *Chem. Eng. Prog.*, **100**(9), pp. 7–9 (2004)]. Ionic liquids are low-melting organic salts that form highly polar liquids at or near ambient temperature [Rogers and Seddon, *Science*, **302**, p. 792 (2003)]. The potential use of ionic liquids to extract metal ions from aqueous solution is discussed by Visser et al. [*Sep. Sci. Technol.*, **36**(5–6), pp. 785–804 (2001)] and by Nakashima et al. [*Ind. Eng. Chem. Res.*, **44**(12), pp. 4368–4372 (2005)]. In another example, phenolic impurities are extracted from an organic reaction mixture using an acidic ionic liquid such as methylimidazolium chloride [BASF promotional literature (2005)]. After extraction, the extract phase is separated by evaporation of the phenolic content, and the raffinate containing the desired product is washed with water to remove small amounts of ionic liquid that saturate that phase. Other potential applications are described in *Ionic Liquids IIIB: Fundamentals, Challenges, and Opportunities*, Rogers and Seddon, eds. (Oxford, 2005). The possibility of switching a solvent system from ionic to nonionic states also is being investigated [Jessop et al., *Nature*, **436**, p. 1102 (2005)]. The authors report that a 50/50 blend of 1-hexanol and 1,8-diazabicyclo[5.4.0]undec-7-ene (DBU) becomes ionic when CO₂ is bubbled through the solution. The CO₂ reacts to form a mixture of 1-hexylcarbonate anion and DBUH⁺ cation, a viscous ionic liquid. The reaction can be reversed by using N₂ to strip the weakly bound CO₂ from solution. This returns the solution to its less viscous, nonionic state and provides a basis for a switchable solvent system.

The challenges involved in using ionic liquids for extraction appear similar to those encountered using nonvolatile extractants dissolved in a diluent, including difficulty dealing with buildup of heavy impurities in the solvent phase over time. Additionally, solvent stability and recovery need to be very high for the process to be economical due to the high cost of makeup solvent. Potential advantages include the possibility of obtaining higher *K* values, allowing use of lower solvent-to-feed ratios, and simplification of extract and raffinate separation requirements. For example, volatile components may easily be removed from the ionic liquid by using evaporation under vacuum instead of multistage distillation; and, in certain cases, the solubility of ionic liquid in the raffinate may be very low.

SECTION 16

Adsorption and
Ion Exchange

PERRY'S CHEMICAL ENGINEERS' HANDBOOK

8TH EDITION



M. DOUGLAS LEVAN
GIORGIO CARTA

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Adsorption and Ion Exchange*

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GENERAL REFERENCES

1. Adamson, *Physical Chemistry of Surfaces*, Wiley, New York, 1990.
2. Barrer, *Zeolites and Clay Minerals as Adsorbents and Molecular Sieves*, Academic Press, New York, 1978.
3. Breck, D. W., *Zeolite Molecular Sieves*, Wiley, New York, 1974.
4. Cheremisinoff and Ellerbusch, *Carbon Adsorption Handbook*, Ann Arbor Science, Ann Arbor, 1978.
5. Cooney, *Adsorption Design for Wastewater Treatment*, CRC Press, Boca Raton, Fla., 1998.
6. Do, *Adsorption Analysis: Equilibria and Kinetics*, Imperial College, London, 1998.
7. Dorfner (ed.), *Ion Exchangers*, W. deGruyter, Berlin, 1991.
8. Dyer, *An Introduction to Zeolite Molecular Sieves*, Wiley, New York, 1988.
9. EPA, *Process Design Manual for Carbon Adsorption*, U.S. Envir. Protect. Agency., Cincinnati, 1973.
10. Gembicki, Oroskar, and Johnson, "Adsorption, Liquid Separation" in *Kirk-Othmer Encyclopedia of Chemical Technology*, 4th ed., Wiley, 1991.
11. Guiochon, Felinger-Shirazi, and Katti, *Fundamentals of Preparative and Nonlinear Chromatography*, Elsevier, 2006.
12. Gregg and Sing, *Adsorption, Surface Area and Porosity*, Academic Press, New York, 1982.
13. Helfferich, *Ion Exchange*, McGraw-Hill, New York, 1962; reprinted by University Microfilms International, Ann Arbor, Michigan.
14. Helfferich and Klein, *Multicomponent Chromatography*, Marcel Dekker, New York, 1970.
15. Jaroniec and Madey, *Physical Adsorption on Heterogeneous Solids*, Elsevier, New York, 1988.
16. Kärger and Ruthven, *Diffusion in Zeolites and Other Microporous Solids*, Wiley, New York, 1992.
17. Keller, Anderson, and Yon, "Adsorption" in Rousseau (ed.), *Handbook of Separation Process Technology*, Wiley-Interscience, New York, 1987.
18. Keller and Staudt, *Gas Adsorption Equilibria: Experimental Methods and Adsorption Isotherms*, Springer, New York, 2005.
19. Ladisch, *Bioseparations Engineering: Principles, Practice, and Economics*, Wiley, New York, 2001.
20. Rhee, Aris, and Amundson, *First-Order Partial Differential Equations: Volume 1. Theory and Application of Single Equations; Volume 2. Theory and Application of Hyperbolic Systems of Quasi-Linear Equations*, Prentice Hall, Englewood Cliffs, New Jersey, 1986, 1989.
21. Rodrigues, LeVan, and Tondeur (eds.), *Adsorption: Science and Technology*, Kluwer Academic Publishers, Dordrecht, The Netherlands, 1989.
22. Rudzinski and Everett, *Adsorption of Gases on Heterogeneous Surfaces*, Academic Press, San Diego, 1992.
23. Ruthven, *Principles of Adsorption and Adsorption Processes*, Wiley, New York, 1984.
24. Ruthven, Farooq, and Knaebel, *Pressure Swing Adsorption*, VCH Publishers, New York, 1994.
25. Seader and Henley, *Separation Process Principles*, 2d ed., Wiley, New York, 2006.
26. Sherman and Yon, "Adsorption, Gas Separation" in *Kirk-Othmer Encyclopedia of Chemical Technology*, 4th ed., Wiley, 1991.
27. Streat and Cloete, "Ion Exchange," in Rousseau (ed.), *Handbook of Separation Process Technology*, Wiley, New York, 1987.
28. Suzuki, *Adsorption Engineering*, Elsevier, Amsterdam, 1990.
29. Thomas and Crittenden, *Adsorption Technology and Design*, Butterworth-Heinemann, Oxford, U.K., 1998.
30. Tien, *Adsorption Calculations and Modeling*, Butterworth-Heinemann, Newton, Massachusetts, 1994.
31. Valenzuela and Myers, *Adsorption Equilibrium Data Handbook*, Prentice Hall, Englewood Cliffs, New Jersey, 1989.
32. Vermeulen, LeVan, Hiester, and Klein, "Adsorption and Ion Exchange" in Perry, R. H. and Green, D. W. (eds.), *Perry's Chemical Engineers' Handbook* (6th ed.), McGraw-Hill, New York, 1984.
33. Wankat, *Large-Scale Adsorption and Chromatography*, CRC Press, Boca Raton, Florida, 1986.
34. Yang, *Adsorbents: Fundamentals and Applications*, Wiley, Hoboken, N.J., 2003.
35. Yang, *Gas Separation by Adsorption Processes*, Butterworth, Stoneham, Mass., 1987.
36. Young and Crowell, *Physical Adsorption of Gases*, Butterworths, London, 1962.

DESIGN CONCEPTS

INTRODUCTION

Adsorption and ion exchange share so many common features in regard to application in batch and fixed-bed processes that they can be grouped together as sorption for a unified treatment. These processes involve the transfer and resulting distribution of one or more solutes between a fluid phase and particles. The partitioning of a single solute between fluid and sorbed phases or the selectivity of a sorbent toward multiple solutes makes it possible to separate solutes from a bulk fluid phase or from one another.

This section treats batch and fixed-bed operations and reviews process cycles and equipment. As the processes indicate, fixed-bed operation with the sorbent in granule, bead, or pellet form is the predominant way of conducting sorption separations and purifications. Although the fixed-bed mode is highly useful, its analysis is complex. Therefore, fixed beds including chromatographic separations are given primary attention here with respect to both interpretation and prediction.

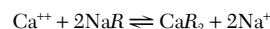
Adsorption involves, in general, the accumulation (or depletion) of solute molecules at an interface (including gas-liquid interfaces, as in foam fractionation, and liquid-liquid interfaces, as in detergency). Here we consider only gas-solid and liquid-solid interfaces, with solute distributed selectively between the fluid and solid phases. The accumulation per unit surface area is small; thus, highly porous solids with very large internal area per unit volume are preferred. Adsorbent surfaces are often physically and/or chemically heterogeneous, and bonding energies may vary widely from one site to another. We seek to promote physical adsorption or *physisorption*, which involves van der Waals forces (as in vapor condensation), and retard chemical adsorption or *chemisorption*, which involves chemical bonding (and often dissociation, as in catalysis). The former is well suited for a regenerable process, while the latter generally destroys the capacity of the adsorbent.

Adsorbents are natural or synthetic materials of amorphous or microcrystalline structure. Those used on a large scale, in order of sales volume, are activated carbon, molecular sieves, silica gel, and activated alumina [Keller et al., gen. refs.].

Ion exchange usually occurs throughout a polymeric solid, the solid being of gel-type, which dissolves some fluid-phase solvent, or truly porous. In ion exchange, ions of positive charge in some cases (cations) and negative charge in others (anions) from the fluid (usually an aqueous solution) replace dissimilar ions of the same charge initially in the solid. The ion exchanger contains permanently bound functional groups of opposite charge-type (or, in special cases, notably weak-base exchangers act as if they do). Cation-exchange resins generally contain bound sulfonic acid groups; less commonly, these groups are carboxylic, phosphonic, phosphinic, and so on. Anionic resins involve quaternary ammonium groups (strongly basic) or other amino groups (weakly basic).

Most ion exchangers in large-scale use are based on synthetic resins—either preformed and then chemically reacted, as for polystyrene, or formed from active monomers (olefinic acids, amines, or phenols). Natural zeolites were the first ion exchangers, and both natural and synthetic zeolites are in use today.

Ion exchange may be thought of as a reversible reaction involving chemically equivalent quantities. A common example for cation exchange is the familiar water-softening reaction



where *R* represents a stationary univalent anionic site in the polyelectrolyte network of the exchanger phase.

Table 16-1 classifies sorption operations by the type of interaction and the basis for the separation. In addition to the normal sorption operations of adsorption and ion exchange, some other similar separations are included. Applications are discussed in this section in "Process Cycles."

TABLE 16-1 Classification of Sorptive Separations

Type of interaction	Basis for separation	Examples
Adsorption	Equilibrium	Numerous purification and recovery processes for gases and liquids Activated carbon-based applications Desiccation using silica gels, aluminas, and zeolites Oxygen from air by PSA using LiX and 5A zeolites Nitrogen from air by PSA using carbon molecular sieve Nitrogen and methane using titanosilicate ETS-4 Separation on <i>n</i> - and <i>iso</i> -paraffins using 5A zeolite Separation of xylenes using zeolite
	Rate	
	Molecular sieving	
Ion exchange (electrostatic)	Equilibrium	Deionization Water softening Rare earth separations Recovery and separation of pharmaceuticals (e.g., amino acids, proteins)
Ligand exchange	Equilibrium	Chromatographic separation of glucose-fructose mixtures with Ca-form resins Removal of heavy metals with chelating resins Affinity chromatography
Solubility	Equilibrium	Partition chromatography
None (purely steric)	Equilibrium partitioning in pores	Size exclusion or gel permeation chromatography

Example 1: Surface Area and Pore Volume of Adsorbent A simple example will show the extent of internal area in a typical granular adsorbent. A fixed bed is packed with particles of a porous adsorbent material. The bulk density of the packing is 500 kg/m^3 , and the interparticle void fraction is 0.40. The intraparticle porosity is 0.50, with two-thirds of this in cylindrical pores of diameter 1.4 nm and the rest in much larger pores. Find the surface area of the adsorbent and, if solute has formed a complete monomolecular layer 0.3 nm thick inside the pores, determine the percent of the particle volume and the percent of the total bed volume filled with adsorbate.

From surface area to volume ratio considerations, the internal area is practically all in the small pores. One gram of the adsorbent occupies 2 cm^3 as packed and has 0.4 cm^3 in small pores, which gives a surface area of $1150 \text{ m}^2/\text{g}$ (or about 1 mi^2 per 5 lb or $6.3 \text{ mi}^2/\text{ft}^3$ of packing). Based on the area of the annular region filled with adsorbate, the solute occupies 22.5 percent of the internal pore volume and 13.5 percent of the total packed-bed volume.

DESIGN STRATEGY

The design of sorption systems is based on a few underlying principles. First, knowledge of *sorption equilibrium* is required. This equilibrium, between solutes in the fluid phase and the solute-enriched phase of the solid, supplants what in most chemical engineering separations is a fluid-fluid equilibrium. The selection of the sorbent material with an understanding of its equilibrium properties (i.e., capacity and selectivity as a function of temperature and component concentrations) is of primary importance. Second, because sorption operations take place in batch, in fixed beds, or in simulated moving beds, the processes have *dynamical character*. Such operations generally do not run at steady state, although such operation may be approached in a simulated moving bed. Fixed-bed processes often approach a periodic condition called a periodic state or cyclic steady state, with several different feed steps constituting a cycle. Thus, some knowledge of how transitions travel through a bed is required. This introduces both time and space into the analysis, in contrast to many chemical engineering operations that can be analyzed at steady state with only a spatial dependence. For good design, it is crucial to understand fixed-bed performance in relation to adsorption equilibrium and rate behavior. Finally, many *practical aspects* must be included in design so that a process starts up and continues to perform well, and that it is not so overdesigned that it is wasteful. While these aspects are process-specific, they include an understanding of dispersive phenomena at the bed scale and, for regenerative processes, knowledge of aging characteristics of the sorbent material, with consequent changes in sorption equilibrium.

Characterization of Equilibria Phase equilibrium between fluid and sorbed phases for one or many components in adsorption or two or more species in ion exchange is usually the single most

important factor affecting process performance. In most processes, it is much more important than mass and heat transfer rates; a doubling of the stoichiometric capacity of a sorbent or a significant change in the shape of an isotherm would almost always have a greater impact on process performance than a doubling of transfer rates.

A difference between adsorption and ion exchange with completely ionized resins is indicated in the *variance* of the systems. In adsorption, part of the solid surface or pore volume is vacant. This diminishes as the fluid-phase concentration of solute increases. In contrast, for ion exchange the sorbent has a fixed total capacity and merely exchanges solutes while conserving charge. Variance is defined as the number of independent variables in a sorption system at equilibrium—that is, variables that one can change separately and thereby control the values of all others. Thus, it also equals the difference between the total number of variables and the number of independent relations connecting them. Numerous cases arise in which ion exchange is accompanied by chemical reaction (neutralization or precipitation, in particular), or adsorption is accompanied by evolution of sensible heat. The concept of variance helps greatly to assure correct interpretations and predictions.

The working capacity of a sorbent depends on fluid concentrations and temperatures. Graphical depiction of sorption equilibrium for single component adsorption or binary ion exchange (monovariance) is usually in the form of isotherms [$n_i = n_i(c_i)$ or $n_i(p_i)$ at constant T] or isosteres [$p_i = p_i(T)$ at constant n_i]. Representative forms are shown in Fig. 16-1. An important dimensionless group dependent on adsorption equilibrium is the **partition ratio** [see Eq. (16-125)], which is a measure of the relative affinities of the sorbed and fluid phases for solute.

Historically, isotherms have been classified as *favorable* (concave downward) or *unfavorable* (concave upward). These terms refer to the spreading tendencies of transitions in fixed beds. A favorable isotherm gives a compact transition, whereas an unfavorable isotherm leads to a broad one.

Example 2: Calculation of Variance In mixed-bed deionization of a solution of a single salt, there are 8 concentration variables: 2 each for cation, anion, hydrogen, and hydroxide. There are 6 connecting relations: 2 for ion exchange and 1 for neutralization equilibrium, and 2 ion-exchanger and 1 solution electroneutrality relations. The variance is therefore $8 - 6 = 2$.

Adsorbent/Ion Exchanger Selection Guidelines for sorbent selection are different for regenerative and nonregenerative systems. For a nonregenerative system, one generally wants a high capacity and a strongly favorable isotherm for a purification and additionally high selectivity for a separation. For a regenerative

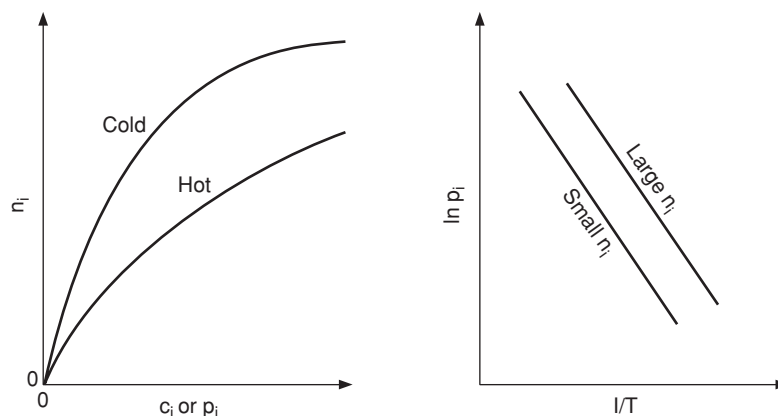


FIG. 16-1 Isotherms (left) and isosteres (right). Isosteres plotted using these coordinates are nearly straight parallel lines, with deviations caused by the dependence of the isosteric heat of adsorption on temperature and loading.

system, high overall capacity and selectivity are again desired, but needs for cost-effective regeneration leading to a reasonable working capacity influence what is sought after in terms of isotherm shape. For separations by pressure swing adsorption (or vacuum pressure swing adsorption), generally one wants a linear to slightly favorable isotherm (although purifications can operate economically with more strongly favorable isotherms). Temperature-swing adsorption usually operates with moderately to strongly favorable isotherms, in part because one is typically dealing with heavier solutes and these are adsorbed fairly strongly (e.g., organic solvents on activated carbon and water vapor on zeolites). Exceptions exist, however; for example, water is adsorbed on silica gel and activated alumina only moderately favorably, with some isotherms showing unfavorable sections. Equilibria for ion exchange separations generally vary from moderately favorable to moderately unfavorable; depending on feed concentrations, the alternates often exist for the different steps of a regenerative cycle. Other factors in sorbent selection are mechanical and chemical stability, mass transfer characteristics, and cost.

Fixed-Bed Behavior The number of transitions occurring in a fixed bed of initially uniform composition before it becomes saturated by a constant composition feed stream is generally equal to the variance of the system. This introductory discussion will be limited to single transition systems.

Methods for analysis of fixed-bed transitions are shown in Table 16-2. Local equilibrium theory is based solely of stoichiometric concerns and system nonlinearities. A transition becomes a “simple wave” (a gradual transition), a “shock” (an abrupt transition), or a combination of the two. In other methods, mass-transfer resistances are incorporated.

The asymptotic behavior of transitions under the influence of mass-transfer resistances in long, “deep” beds is important. The three basic asymptotic forms are shown in Fig. 16-2. With an unfavorable isotherm, the breadth of the transition becomes proportional to the depth of bed it has passed through. For the linear isotherm, the breadth becomes proportional to the square root of the depth. For the favorable isotherm, the transition approaches a constant breadth called a *constant pattern*.

Design of nonregenerative sorption systems and many regenerative ones often relies on the concept of the mass-transfer zone or MTZ, which closely resembles the constant pattern [Collins, *Chem. Eng. Prog. Symp. Ser. No. 74*, **63**, 31 (1974); Keller et al., gen. refs.]. The length of this zone (depicted in Fig. 16-3) together with stoichiometry can be used to predict accurately how long a bed can be utilized prior to breakthrough. Upstream of the mass-transfer zone, the adsorbent is in equilibrium with the feed. Downstream, the adsorbent is in its initial state. Within the mass-transfer zone, the fluid-phase concentration drops from the feed value to the initial, presaturation state.

TABLE 16-2 Methods of Analysis of Fixed-Bed Transitions

Method	Purpose	Approximations
Local equilibrium theory	Shows wave character—simple waves and shocks Usually indicates best possible performance Better understanding	Mass and heat transfer very rapid Dispersion usually neglected If nonisothermal, then adiabatic
Mass-transfer zone	Design based on stoichiometry and experience	Isothermal MTZ length largely empirical Regeneration often empirical
Constant pattern and related analyses	Gives asymptotic transition shapes and upper bound on MTZ	Deep bed with fully developed transition
Full rate modeling	Accurate description of transitions Appropriate for shallow beds, with incomplete wave development General numerical solutions by finite difference or collocation methods	Various to few

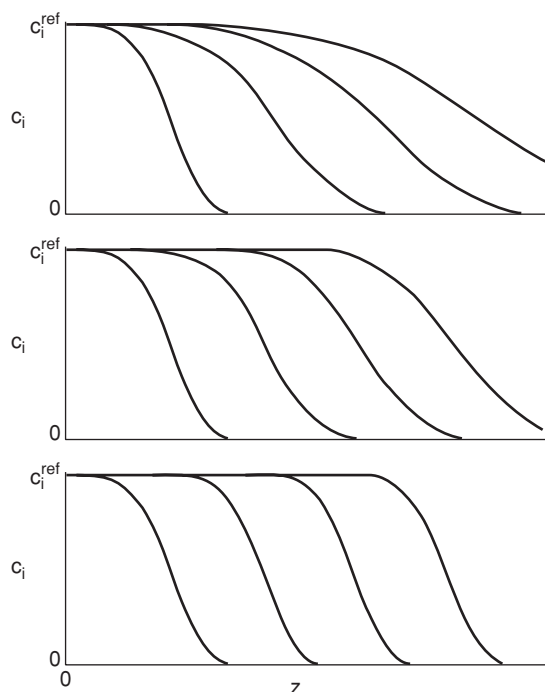


FIG. 16-2 Limiting fixed-bed behavior: simple wave for unfavorable isotherm (top), square-root spreading for linear isotherm (middle), and constant pattern for favorable isotherm (bottom). [From LeVan in Rodrigues *et al.* (eds.), *Adsorption: Science and Technology*, Kluwer Academic Publishers, Dordrecht, The Netherlands, 1989; reprinted with permission.]

Equilibrium with the feed is not attained in this region. As a result, because an adsorption bed must typically be removed from service shortly after breakthrough begins, the full capacity of the bed is not utilized. Obviously, the broader that the mass-transfer zone is, the greater will be the extent of unused capacity. Also shown in the figure is the length of the equivalent equilibrium section (LES) and the length of equivalent unused bed (LUB). The length of the MTZ is divided between these two.

Adsorption with strongly favorable isotherms and ion exchange between strong electrolytes can usually be carried out until most of the stoichiometric capacity of the sorbent has been utilized, corresponding to a thin MTZ. Consequently, the total capacity of the bed is practically constant regardless of the composition of the solution being treated.

The effluent concentration history is the breakthrough curve, also shown in Fig. 16-3. The effluent concentration stays at or near zero or a low residual concentration until the transition reaches the column outlet. The effluent concentration then rises until it becomes unacceptable, this time being called the breakthrough time. The feed step must stop and, for a regenerative system, the regeneration step begins.

Two dimensionless variables play key roles in the analysis of single transition systems (and some multiple transition systems). These are the **throughput parameter** [see Eq. (16-129)] and the **number of transfer units** (see Table 16-13). The former is time made dimensionless so that it is equal to unity at the stoichiometric center of a breakthrough curve. The latter is, as in packed tower calculations, a measure of mass-transfer resistance.

Cycles Design methods for cycles rely on mathematical modeling (or empiricism) and often extensive pilot plant experiments. Many

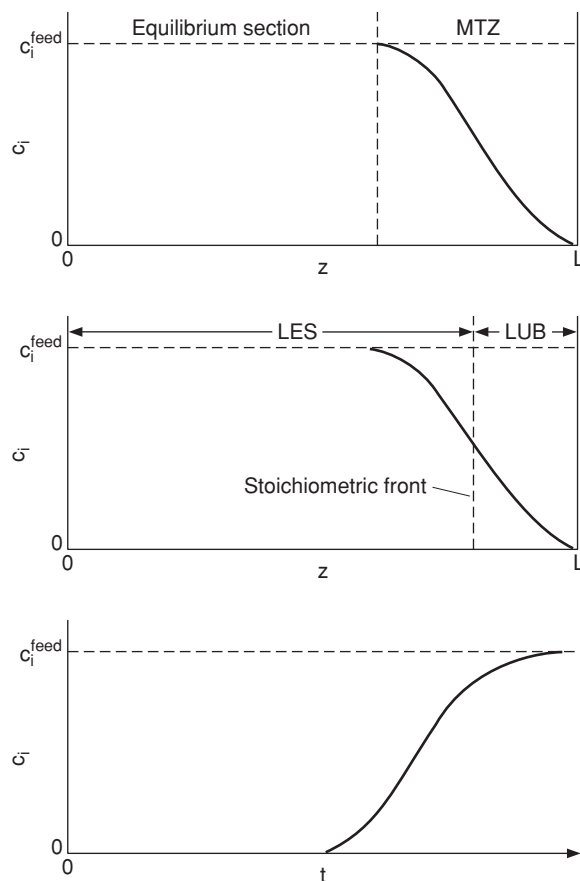


FIG. 16-3 Bed profiles (top and middle) and breakthrough curve (bottom). The bed profiles show the mass-transfer zone (MTZ) and equilibrium section at breakthrough. The stoichiometric front divides the MTZ into two parts with contributions to the length of equivalent equilibrium section (LES) and the length of equivalent unused bed (LUB).

cycles can be easily analyzed using the methods described above applied to the collection of steps. In some cycles, however, especially those operated with short cycle times or in shallow beds, transitions may not be very fully developed, even at a periodic state, and the complexity may be compounded by multiple sorbates.

A wide variety of complex process cycles have been developed. Systems with many beds incorporating multiple sorbents, possibly in layered beds, are in use. Mathematical models constructed to analyze such cycles can be complex. With a large number of variables and nonlinear equilibria involved, it is usually not beneficial to make all variables in such models dimensionless; doing so does not help appreciably in making comparisons with other largely dissimilar systems. If dimensionless variables are used, these usually begin with a dimensionless bed length and a dimensionless time, which is often different from the throughput parameter.

Practical Aspects There are a number of process-specific concerns that are accounted for in good design. In regenerable systems, sorbents age, losing capacity because of fouling by heavy contaminants, loss of surface area or crystallinity, oxidation, and the like. Mass-transfer resistances may increase over time. Because of particle shape, size distribution, or column packing method,

dispersion may be more pronounced than would normally be expected. The humidity of an entering stream will usually impact a solvent recovery application. Safety, including the possibility of a fire, may be a concern. For gas-phase adsorption, scale-up from an isothermal laboratory column to a nonisothermal pilot plant col-

umn to a largely adiabatic process column requires careful judgment. If the MTZ concept is utilized, the length of the MTZ cannot be reliably determined solely from knowledge on other systems. Experience plays the key role in accounting for these and other such factors.

ADSORBENTS AND ION EXCHANGERS

CLASSIFICATIONS AND CHARACTERIZATIONS

Adsorbents Table 16-3 classifies common adsorbents by structure type and water adsorption characteristics. Structured adsorbents take advantage of their crystalline structure (zeolites and silicalite) and/or their molecular sieving properties. The hydrophobic (nonpolar surface) or hydrophilic (polar surface) character may vary depending on the competing adsorbate. A large number of zeolites have been identified, and these include both synthetic and naturally occurring (e.g., mordenite and chabazite) varieties.

TABLE 16-3 Classification of Common Adsorbents

	Amorphous	Structured
Hydrophobic	Activated carbon Polymers	Carbon molecular sieves Silicalite
Hydrophilic	Silica gel Activated alumina	Common zeolites: 3A (KA), 4A (NaA), 5A (CaA), 13X (NaX), Mordenite, Chabazite, etc.

The classifications in Table 16-3 are intended only as a rough guide. For example, a carbon molecular sieve is truly amorphous but has been manufactured to have certain structural, rate-selective properties. Similarly, the extent of hydrophobicity of an activated carbon will depend on its ash content and its level of surface oxidation.

Zeolites are crystalline aluminosilicates. Zeolitic adsorbents have had their water of hydration removed by calcination to create a structure with well-defined openings into crystalline cages. The molecular sieving properties of zeolites are based on the size of these openings. Two crystal types are common: type A (with openings formed by 4 sodalite cages) and type X or Y (with openings formed by 6 sodalite cages). Cations balancing charge and their locations determine the size of opening into a crystal unit cell. Nominal opening sizes for the most common synthetic zeolites are 0.3 nm for KA, 0.4 nm for NaA, 0.5 nm for CaA, and 1.0 nm for NaX. Further details, including effective molecular diameters, are widely available [Barrer; Breck; Ruthven; Yang 1987, gen. refs.].

Many adsorbents, particularly the amorphous adsorbents, are characterized by their pore size distribution. The distribution of small pores is usually determined by analysis, using one of several available methods, of a cryogenic nitrogen adsorption isotherm, although other probe molecules are also used. The distribution of large pores is usually determined by mercury porosimetry [Gregg and Sing, gen. refs.].

Table 16-4 shows the IUPAC classification of pores by size. Micropores are small enough that a molecule is attracted to both of the opposing walls forming the pore. The potential energy functions for these walls superimpose to create a deep well, and strong adsorption results. Hysteresis is generally not observed. (However, water vapor adsorbed in the micropores of activated carbon shows a large hysteresis loop, and the desorption branch is sometimes used with the Kelvin equation to determine the pore size distribution.) Capillary condensation occurs in mesopores and a hysteresis loop is typically found. Macropores form important paths for molecules to diffuse into a par-

ticle; for gas-phase adsorption, they do not fill with adsorbate until the gas phase becomes saturated.

New adsorbents and ion-exchange materials are constantly under development. The class of microporous crystalline materials with molecular sieving properties is extensive and includes many metal silicates and phosphates, with the zeolites (aluminosilicates) being the most broadly useful. The titanosilicate ETS-4 is currently being implemented for separation of CH₄ and N₂. Metal organic framework (MOF) materials are currently in the research stage but appear to have great potential for adsorbing light gases and specific adsorbates. Among other classes of materials, the recently developed π -complexation sorbents are particularly promising; these use weak chemical bonds instead of purely physical interaction as the basis for separation (Yang, 2003, gen. refs.). Two other adsorbent materials of considerable research interest, but which have not been implemented for commercial separations, are carbon nanotubes and the mesoporous silica MCM-41.

Ion Exchangers Ion exchangers are classified according to (1) their functionality and (2) the physical properties of the support matrix. Cation and anion exchangers are classified in terms of their ability to exchange positively or negatively charged species. Strongly acidic and strongly basic ion exchangers are ionized and thus are effective at nearly all pH values (pH 0–14). Weakly acidic exchangers are typically effective in the range of pH 5–14. Weakly basic resins are effective in the range of pH 0–9. Weakly acidic and weakly basic exchangers are often easier to regenerate, but leakage due to incomplete exchange may occur. Chelating resins containing iminodiacetic acid form specific metal complexes with metal ions with complex stability constants that follow the same order as those for EDTA. However, depending on pH, they also function as weak cation exchangers. The achievable ion-exchange capacity depends on the concentration of ionogenic groups and their availability as an exchange site. The latter is a function of the support matrix.

Polymer-based, synthetic ion-exchangers known as resins are available commercially in gel type or truly porous forms. Gel-type resins are not porous in the usual sense of the word, since their structure depends upon swelling in the solvent in which they are immersed. Removal of the solvent usually results in a collapse of the three-dimensional structure, and no significant surface area or pore diameter can be defined by the ordinary techniques available for truly porous materials. In their swollen state, gel-type resins approximate a true molecular-scale solution. Thus, we can identify an internal porosity ϵ_p only in terms of the equilibrium uptake of water or other liquid. When crosslinked polymers are used as the support matrix, the internal porosity so defined varies in inverse proportion to the degree of crosslinking, with swelling and therefore

TABLE 16-4 Classification of Pore Sizes

Type	Slit width* (w)	Characteristic
Micropore†	$w < 2$ nm	Superimposed wall potentials
Mesopore	$2 \text{ nm} < w < 50$ nm	Capillary condensation
Macropore	$w > 50$ nm	Effectively flat walled until $p \rightarrow P^*$

*Or pore diameter.

†Further subdivided into ultramicropores and supermicropores (Gregg and Sing, gen. refs.).

porosity typically being more pronounced in solvents with a high dielectric constant. The ion held by the exchanger also influences the resin swelling. Thus, the size of the resin particles changes during the ion-exchange process as the resin is changed from one form to another, and this effect is more dramatic for resins with a lower degree of crosslinking. The choice of degree of crosslinking is dependent on several factors including: the extent of swelling, the exchange capacity, the intraparticle diffusivity, the ease of regeneration, and the physical and chemical stability of the exchanger under chosen operating conditions. The concentration of ionogenic groups determines the capacity of the resin. Although the capacity per unit mass of dry resin is insensitive to the degree of crosslinking, except for very highly crosslinked resins, the exchange capacity per unit volume of swollen resin increases significantly with degree of crosslinking, so long as the mesh size of the polymer network allows the ions free access to functional groups within the interior of the resin. The degree of crosslinking also affects the rate of ion exchange. The intraparticle diffusivity decreases nearly exponentially with the mesh size of the matrix. As a result, resins with a lower degree of crosslinking are normally required for the exchange of bulky species, such as organic ions with molecular weight in excess of 100. The regeneration efficiency is typically greater for resins with a lower degree of crosslinking. Finally, the degree of crosslinking also affects the long-term stability of the resin. Strongly acidic and strongly basic resins are subject to irreversible oxidative degradation of the polymer and thermal and oxidative decomposition of functional groups. Generally, more highly crosslinked resins are less prone to irreversible chemical degradation but they may be subject to osmotic breakage caused by volume changes that occur during cyclic operations. In general, experience shows that an intermediate degree of crosslinking is often preferred. However, readers are referred to manufacturers' specifications for resin stability data at different operating conditions.

Truly porous, synthetic ion exchangers are also available. These materials retain their porosity even after removal of the solvent and have measurable surface areas and pore size. The term macroporous is commonly used for resins prepared from a phase separation technique, where the polymer matrix is prepared with the addition of a liquid that is a good solvent for the monomers, but in which the polymer is insoluble. Matrices prepared in this way usually have the appearance of a conglomerate of gel-type microspheres held together to form an interconnected porous network. Macroporous resins possessing a more continuous cellular structure interlaced with a pore network have also been obtained with different techniques and are commercially available. Since higher degrees of crosslinking are typically used to produce truly porous ion-exchange resins, these materials tend to be more stable under highly oxidative conditions, more attrition-resistant, and more resistant to breakage due to osmotic shock than their gel-type counterparts. Moreover, since their porosity does not depend entirely on swelling, they can be used in solvents with low dielectric constant where gel-type resins can be ineffective. In general, compared to gel-type resins, truly porous resins typically have somewhat lower capacities and can be more expensive. Thus, for ordinary ion-exchange applications involving small ions under nonharsh conditions, gel-type resins are usually preferred.

Adsorbents for biomacromolecules such as proteins have special properties. First, they need to have large pore sizes. A ratio of pore radius to molecule radius larger than 5 is desirable to prevent excessive diffusional hindrance (see "Intraparticle Mass Transfer" in this section). Thus, for typical proteins, pore radii need to be in excess of 10–15 nm. Second, functional groups for interactions with the protein are usually attached to the adsorbent backbone via a spacer arm to provide accessibility. Third, adsorbents based on hydrophilic structures are preferred to limit nonspecific interactions with the adsorbent backbone and prevent global unfolding or denaturation of the protein. Thus, if hydrophobic supports are used, their surfaces are usually rendered hydrophilic by incorporating hydrophilic coatings such as dextran or polyvinyl alcohol. Finally, materials stable in sodium hydroxide solutions (used for clean-in-place) are

preferred. Support matrices for protein adsorption can be classified into the following four broad groups: hydrophilic gels, including cross-linked agarose, cross-linked dextran, cross-linked polyacrylamide, and cellulose; rigid macroporous media, including silica/ceramic, alumina, polystyrene-DVB, and polymethacrylate; composite media including agarose-dextran and ceramic-polyacrylamide; and monoliths, including polystyrene-DVB, polymethacrylate, and silica-based materials. Monoliths for biomacromolecule adsorption are generally synthesized in place in a column, resulting in a continuous random structure of linked, generally nonporous microparticles intercalated by a network of flow-through pores. The continuous structure allows independent control of microparticle size and flow-through pore sizes resulting in rapid adsorption and moderate pressure drop. Although the adsorption capacity for proteins is smaller than for porous-particle-based adsorbents, they offer significant advantages for rapid high-resolution separations and for adsorption of very large bioparticles such as plasmid DNA and viruses [see Josic et al., *J. Chrom. B.*, **752**, 191 (2001)].

Functional groups or ligands incorporated in materials for biomacromolecule adsorption can be classified in the following four broad groups: cation exchange, including sulfopropyl (SP), methyl sulfonate (S), and carboxymethyl (CM); anion exchange, including quaternary ammonium ion (Q), quaternary aminoethyl (QAE), and diethylamino ethyl (DEAE); hydrophobic interaction, including ether, isopropyl, butyl, octyl, and phenyl ligands in order of increasing hydrophobicity; and biospecific interaction, including amino acids, dyes, and proteins covalently bound to the adsorbent through coupling ligands and transition metals supported via complexing ligands. Polystyrene-based particles and silica particles functionalized with hydrocarbon chains from C2 to C18 in length are also used for reversed-phase chromatographic separations. However, because of the limited stability of biomacromolecules, their process applications are limited. Inert, hydrophilic porous matrices and gels are used extensively for size exclusion chromatography (SEC). A useful general reference is Janson and Ryden, *Protein Purification: Principles, High-Resolution Methods, and Applications*, Wiley-Liss, 1998.

PHYSICAL PROPERTIES

Selected data on some commercially available adsorbents and ion exchangers are given in Tables 16-5 and 16-6. The purpose of the tables is twofold: to assist the engineer or scientist in identifying materials suitable for a needed application, and to supply typical physical property values.

In addition to the particulate adsorbents listed in Table 16-5, some adsorbents are available in structured form for specific applications. Monoliths, papers, and paint formulations have been developed for zeolites, with these driven by the development of wheels (Fig. 16-60), adsorptive refrigeration, etc. Carbon monoliths are also available as activated carbon fibers, created from polymeric materials, and sold in the forms of fabrics, mats, felts, and papers for use in various applications including in pleated form in filters. Zeolitic and carbon membranes are also available, with the latter developed for separation by "selective surface flow" [Rao and Sircar, *J. Membrane Sci.*, **85**, 253 (1993)].

Excellent sources of information on the characteristics of adsorbents or ion exchange products for specific applications are the manufacturers themselves. Additional information on adsorbents and ion exchangers is available in many of the general references and in several articles in *Kirk-Othmer Encyclopedia of Chemical Technology*. A comprehensive summary of commercial ion-exchangers, including manufacturing methods, properties, and applications, is given by Dorfner (*Ion Exchangers*, de Gruyter, New York, 1991).

Several densities and void fractions are commonly used. For adsorbents, usually the **bulk density** ρ_b , the weight of clean material per unit bulk volume as packed in a column, is reported. The dry **particle density** ρ_p is related to the (external) **void fraction of**

16-10 ADSORPTION AND ION EXCHANGE

TABLE 16-5 Physical Properties of Adsorbents

Material and uses	Shape* of particles	Size range, U.S. standard mesh†	Internal porosity, %	Bulk dry density, kg/L	Average pore diameter, nm	Surface area, km ² /kg	Sorptive capacity, kg/kg (dry)
Aluminas							
Low-porosity (fluoride sorbent)	G, S	8–14, etc.	40	0.70	~7	0.32	0.20
High-porosity (drying, separations)	G	Various	57	0.85	4–14	0.25–0.36	0.25–0.33
Desiccant, CaCl ₂ -coated	G	3–8, etc.	30	0.91	4.5	0.2	0.22
Activated bauxite	G	8–20, etc.	35	0.85	5		0.1–0.2
Chromatographic alumina	G, P, S	80–200, etc.	30	0.93			~0.14
Silicates and aluminosilicates							
Molecular sieves	S, C, P	Various					
Type 3A (dehydration)			~30	0.62–0.68	0.3	~0.7	0.21–0.23
Type 4A (dehydration)			~32	0.61–0.67	0.4	~0.7	0.22–0.26
Type 5A (separations)			~34	0.60–0.66	0.5	~0.7	0.23–0.28
Type 13X (purification)			~38	0.58–0.64	1.0	~0.6	0.25–0.36
Silicalite (hydrocarbons)	S, C, P	Various		0.64–0.70	0.6	~0.4	0.12–0.16
Dealuminated Y (hydrocarbons)	S, C, P	Various		0.48–0.53	0.8	0.5–0.8	0.28–0.42
Mordenite (acid drying)				0.88	0.3–0.8		0.12
Chabazite (acid drying)				0.72	0.4–0.5		0.20
LiX (air separation)	S	14–40		0.61	1.0		
Titanosilicate ETS-4 (N ₂ /CH ₄)	S	Various	25–30	1.0	0.37		
Silica gel (drying, separations)	G, P	Various	38–48	0.70–0.82	2–5	0.6–0.8	0.35–0.50
Magnesium silicate (decolorizing)	G, P	Various	~33	~0.50		0.18–0.30	
Calcium silicate (fatty-acid removal)	P		75–80	~0.20		~0.1	
Clay, acid-treated (refining of petroleum, food products)	G	4–8		0.85			
Fuller's earth (same)	G, P	<200		0.80			
Diatomaceous earth	G	Various		0.44–0.50		~0.002	
Carbons							
Shell-based	G	Various	60	0.45–0.55	2	0.8–1.6	0.40
Wood-based	G	Various	~80	0.25–0.30		0.8–1.8	~0.70
Petroleum-based	G, C	Various	~80	0.45–0.55	2	0.9–1.3	0.3–0.4
Peat-based	G, C, P	Various	~55	0.30–0.50	1–4	0.8–1.6	0.5
Lignite-based	G, P	Various	70–85	0.40–0.70	3	0.4–0.7	0.3
Bituminous-coal-based	G, P	8–30, 12–40	60–80	0.40–0.60	2–4	0.9–1.2	0.4
Synthetic-polymer-based (pyrolyzed)	S	20–100	40–70	0.49–0.60		0.1–1.1	
Carbon molecular sieve (air separation)	C	Various	35–50	0.5–0.7	0.3–0.6		0.5–0.20
Organic polymers							
Polystyrene (removal of organics, e.g., phenol; antibiotics recovery)	S	20–60	40–60	0.64	4–20	0.3–0.7	
Polyacrylic ester (purification of pulping wastewaters; antibiotics recovery)	G, S	20–60	50–55	0.65–0.70	10–25	0.15–0.4	
Phenolic (also phenolic amine) resin (decolorizing and deodorizing of solutions)	G	16–50	45	0.42		0.08–0.12	0.45–0.55

*Shapes: C, cylindrical pellets; F, fibrous flakes; G, granules; P, powder; S, spheres.

†U.S. Standard sieve sizes (given in parentheses) correspond to the following diameters in millimeters: (3) 6.73, (4) 4.76, (8) 2.98, (12) 1.68, (14) 1.41, (16) 1.19, (20) 0.841, (30) 0.595, (40) 0.420, (50) 0.297, (60) 0.250, (80) 0.177, (200) 0.074.

packing ϵ by

$$\rho_p = \frac{\rho_b}{1 - \epsilon} \quad (16-1)$$

The **skeletal density** ρ_s of a particle (or crystalline density for a pure chemical compound) is given in terms of **internal porosity** ϵ_p by

$$\rho_s = \frac{\rho_p}{1 - \epsilon_p} \quad (16-2)$$

For an adsorbent or ion exchanger, the **wet density** ρ_w of a particle is related to these factors and to the liquid density ρ_f by

$$\rho_w = \rho_p + \epsilon_p \rho_f \quad (16-3)$$

The **total voidage** ϵ_b in a packed bed (outside and inside particles) is

$$\epsilon_b = \epsilon + (1 - \epsilon)\epsilon_p \quad (16-4)$$

TABLE 16-6 Physical Properties of Ion-Exchange Materials

Material	Shape ^a of particles	Bulk wet density (drained), kg/L	Moisture content (drained), % by weight	Swelling due to exchange, %	Maximum operating temperature, † °C	Operating pH range	Exchange capacity	
							Dry, equivalent/kg	Wet, equivalent/L
Cation exchangers: strongly acidic								
Polystyrene sulfonate								
Homogeneous (gel) resin	S				120–150	0–14		
4% cross-linked		0.75–0.85	64–70	10–12			5.0–5.5	1.2–1.6
6% cross-linked		0.76–0.86	58–65	8–10			4.8–5.4	1.3–1.8
8–10% cross-linked		0.77–0.87	48–60	6–8			4.6–5.2	1.4–1.9
12% cross-linked		0.78–0.88	44–48	5			4.4–4.9	1.5–2.0
16% cross-linked		0.79–0.89	42–46	4			4.2–4.6	1.7–2.1
20% cross-linked		0.80–0.90	40–45	3			3.9–4.2	1.8–2.0
Porous structure								
12–20% cross-linked	S	0.81	50–55	4–6	120–150	0–14	4.5–5.0	1.5–1.9
Cation exchangers: weakly acidic								
Acrylic (pK 5) or methacrylic (pK 6)								
Homogeneous (gel) resin	S	0.70–0.75	45–50	20–80	120	4–14	8.3–10	3.3–4.0
Macroporous	S	0.67–0.74	50–55	10–100	120		~8.0	2.5–3.5
Polystyrene phosphonate	G, S	0.74	50–70	<40	120	3–14	6.6	3.0
Polystyrene iminodiacetate	S	0.75	68–75	<100	75	3–14	2.9	0.7
Polystyrene amidoxime	S	~0.75	58	10	50	1–11	2.8	0.8–0.9
Zeolite (Al silicate)	G	0.85–0.95	40–45	0	60	6–8	1.4	0.75
Zirconium tungstate	G	1.15–1.25	~5	0	>150	2–10	1.2	1.0
Anion exchangers: strongly basic								
Polystyrene-based								
Trimethyl ammonium (type I)								
Homogeneous, 8% CL	S	0.70	46–50	~20	60–80	0–14	3.4–3.8	1.3–1.5
Porous, 11% CL	S	0.67	57–60	15–20	60–80	0–14	3.4	1.0
Dimethyl hydroxyethyl ammonium (type II)								
Homogeneous, 8% CL	S	0.71	~42	15–20	40–80	0–14	3.8–4.0	1.2
Porous, 10% CL	S	0.67	~55	12–15	40–80	0–14	3.8	1.1
Acrylic-based								
Homogeneous (gel)	S	0.72	~70	~15	40–80	0–14	~5.0	1.0–1.2
Porous	S	0.67	~60	~12	40–80	0–14	3.0–3.3	0.8–0.9
Anion exchangers: intermediately basic (pK 11)								
Polystyrene-based	S	0.75	~50	15–25	65	0–10	4.8	1.8
Epoxy-polyamine	S	0.72	~64	8–10	75	0–7	6.5	1.7
Anion exchangers: weakly basic (pK 9)								
Aminopolystyrene								
Homogeneous (gel)	S	0.67	~45	8–12	100	0–7	5.5	1.8
Porous	S	0.61	55–60	~25	100	0–9	4.9	1.2
Acrylic-based amine								
Homogeneous (gel)	S	0.72	~63	8–10	80	0–7	6.5	1.7
Porous	S	0.72	~68	12–15	60	0–9	5.0	1.1

^aShapes: C, cylindrical pellets; G, granules; P, powder; S, spheres.

†When two temperatures are shown, the first applies to H form for cation, or OH form for anion, exchanger; the second, to salt ion.

SORPTION EQUILIBRIUM

The quantity of a solute adsorbed can be given conveniently in terms of moles or volume (for adsorption) or ion-equivalents (for ion exchange) per unit mass or volume (dry or wet) of sorbent. Common units for adsorption are mol/(m³ of fluid) for the fluid-phase concentration c_i and mol/(kg of clean adsorbent) for adsorbed-phase concentration n_i . For gases, partial pressure may replace concentration.

Many models have been proposed for adsorption and ion exchange equilibria. The most important factor in selecting a model from an engineering standpoint is to have an accurate mathematical description over the entire range of process conditions. It is usually fairly easy to obtain correct capacities at selected points, but isotherm shape over the entire range is often a critical concern for a regenerable process.

GENERAL CONSIDERATIONS

Forces Molecules are attracted to surfaces as the result of two types of forces: dispersion-repulsion forces (also called London or van der Waals forces) such as described by the Lennard-Jones potential for molecule-molecule interactions; and electrostatic forces, which exist as the result of a molecule or surface group having a permanent electric dipole or quadrupole moment or net electric charge.

Dispersion forces are always present and in the absence of any stronger force will determine equilibrium behavior, as with adsorption of molecules with no dipole or quadrupole moment on nonoxidized carbons and silicalite.

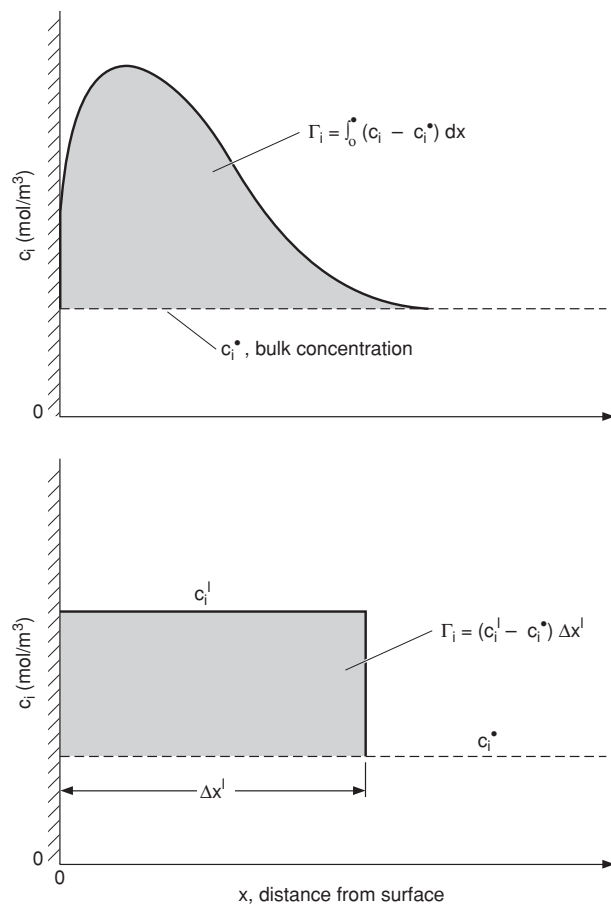


FIG. 16-4 Depictions of surface excess Γ_i . *Top*: The force field of the solid concentrates component i near the surface; the concentration c_i is low at the surface because of short-range repulsive forces between adsorbate and surface. *Bottom*: Surface excess for an imagined homogeneous surface layer of thickness $\Delta x'$.

If a surface is polar, its resulting electric field will induce a dipole moment in a molecule with no permanent dipole and, through this polarization, increase the extent of adsorption. Similarly, a molecule with a permanent dipole moment will polarize an otherwise nonpolar surface, thereby increasing the attraction.

For a polar surface and molecules with permanent dipole moments, attraction is strong, as for water adsorption on a hydrophilic adsorbent. Similarly, for a polar surface, a molecule with a permanent quadrupole moment will be attracted more strongly than a similar molecule with a weaker moment; for example, nitrogen is adsorbed more strongly than oxygen on zeolites (Sherman and Yon, gen. refs.).

Surface Excess With a Gibbs dividing surface placed at the surface of the solid, the surface excess of component i , Γ_i (mol/m²), is the amount per unit area of solid contained in the region near the surface, above that contained at the fluid-phase concentration far from the surface. This is depicted in two ways in Fig. 16-4. The quantity adsorbed per unit mass of adsorbent is

$$n_i = \Gamma_i A \quad (16-5)$$

where A (m²/kg) is the surface area of the solid.

For a porous adsorbent, the amount adsorbed in the pore structure per unit mass of adsorbent, based on surface excess, is obtained by the difference

$$n_i = n_i^{\text{tot}} - V_p c_i^{\infty} \quad (16-6)$$

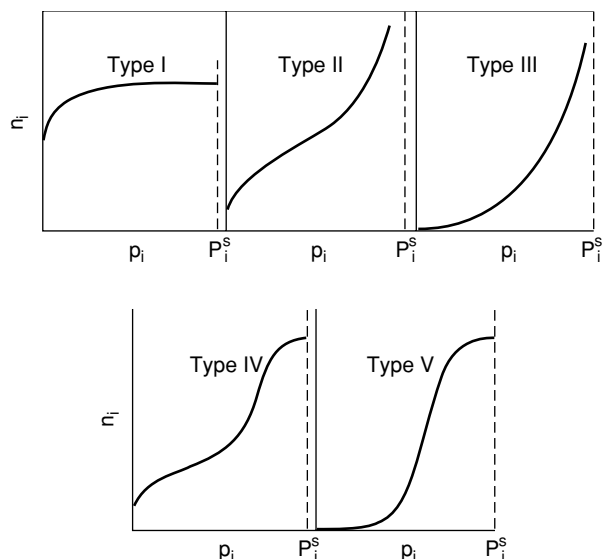


FIG. 16-5 Representative isotherm types. p_i and P_i^s are pressure and vapor pressure of the solute. [Brunauer, J. Am. Chem. Soc., **62**, 1723 (1940); reprinted with permission.]

where n_i^{tot} (mol/kg) is the total amount of component i contained within the particle's pore volume V_p (m³/kg), and c_i^{∞} is the concentration outside of the particle. If n_i differs significantly from n_i^{tot} (as it will for weakly adsorbed species), then it is important to consider adsorbed-phase quantities in terms of surface excesses.

Classification of Isotherms by Shape Representative isotherms are shown in Fig. 16-5, as classified by Brunauer and coworkers. Curves that are concave downward throughout (type I) have historically been designated as "favorable," while those that are concave upward throughout (type III) are "unfavorable." Other isotherms (types II, IV, and V) have one or more inflection points. The designations "favorable" and "unfavorable" refer to fixed-bed behavior for the uptake step, with a favorable isotherm maintaining a compact wave shape. A favorable isotherm for uptake is unfavorable for the discharge step. This becomes particularly important for a regenerative process, in which a favorable isotherm may be too favorable for regeneration to occur effectively.

Categorization of Equilibrium Models Historically, sorption equilibrium has been approached from different viewpoints. For adsorption, many models for flat surfaces have been used to develop explicit equations and equations of state for pure components and mixtures, and many of the resulting equations are routinely applied to porous materials. Explicit equations for pore filling have also been proposed, generally based on the Polanyi potential theory. Ion exchange adds to these approaches concepts of absorption or dissolution (absorption) and exchange reactions. Statistical mechanics and molecular dynamics contribute to our understanding of all of these approaches (Steele, *The Interaction of Gases with Solid Surfaces*, Pergamon, Oxford, 1974; Nicholson and Parsonage, *Computer Simulation and the Statistical Mechanics of Adsorption*, Academic Press, New York, 1982). Mixture models are often based on the adsorbed solution theory, which uses thermodynamic equations from vapor-liquid equilibria with volume replaced by surface area and pressure replaced by a two-dimensional spreading pressure. Other approaches include lattice theories and mass-action exchange equilibrium.

Heterogeneity Adsorbents and ion exchangers can be physically and chemically heterogeneous. Although exceptions exist, solutes generally compete for the same sites. Models for adsorbent heterogeneity have been developed for both discrete and continuous distributions of energies [Ross and Olivier, *On Physical Adsorption*, Interscience, New York, 1964; Jaroniec and Madey, Rudzinski and Everett, gen. refs.].

Isosteric Heat of Adsorption The most useful heat of adsorption for fixed-bed calculations is the *isosteric heat of adsorption*, which

is given by the Clausius-Clapeyron type relation

$$\lambda_i = \mathcal{R}T^2 \left. \frac{\partial \ln p_i}{\partial T} \right|_{n_i, n_j} \quad (16-7)$$

where the n_j can be dropped for single-component adsorption. λ_i is positive by convention. If isosteres are straight lines when plotted as $\ln p_i$ versus T^{-1} (see Fig. 16-1), then Eq. (16-7) can be integrated to give

$$\ln p_i = f(n_i) - \frac{\lambda_i}{\mathcal{R}T} \quad (16-8)$$

where $f(n_i)$ is an arbitrary function dependent only on n_i . Many other heats of adsorption have been defined and their utility depends on the application (Ross and Olivier; Young and Crowell, gen. refs.).

From Eq. (16-8), if a single isotherm is known in the pressure explicit form $p_i = p_i(n_i, T)$ and if λ_i is known at least approximately, then equilibria can be estimated over a narrow temperature range using

$$\ln \frac{p_i}{p_i^{\text{ref}}} = \frac{\lambda_i}{\mathcal{R}} \left(\frac{1}{T^{\text{ref}}} - \frac{1}{T} \right) \quad (\text{const } n_i) \quad (16-9)$$

Similarly, Eq. (16-9) is used to calculate the isosteric heat of adsorption from two isotherms.

Experiments Sorption equilibria are measured using apparatuses and methods classified as volumetric, gravimetric, flow-through (frontal analysis), and chromatographic. Apparatuses are discussed elsewhere [Yang 1987; Keller and Staudt (gen. refs.)]. Heats of adsorption can be determined from isotherms measured at different temperatures or measured independently by calorimetric methods.

Dimensionless Concentration Variables Where appropriate, isotherms will be written here using the dimensionless system variables

$$c_i^{\circ} = \frac{c_i}{c_i^{\text{ref}}} \quad n_i^{\circ} = \frac{n_i}{n_i^{\text{ref}}} \quad (16-10)$$

where the best choice of reference values depends on the operation.

In some cases, to allow for some preloading of the adsorbent, it will be more convenient to use the dimensionless transition variables

$$c_i^{\circ} = \frac{c_i - c_i'}{c_i'' - c_i'} \quad n_i^{\circ} = \frac{n_i - n_i'}{n_i'' - n_i'} \quad (16-11)$$

where single and double primes indicate initial and final concentrations, respectively. Figure 16-6 shows n° plotted versus c° for a sample system. Superimposed are an upward transition (loading) and a downward transition (unloading), shown by the respective positions of (c_i', n_i') and (c_i'', n_i'') .

SINGLE COMPONENT OR EXCHANGE

The simplest relationship between solid-phase and fluid-phase concentrations is the **linear isotherm**

$$n_i = K_i c_i \quad \text{or} \quad n_i = K_i' p_i \quad (16-12)$$

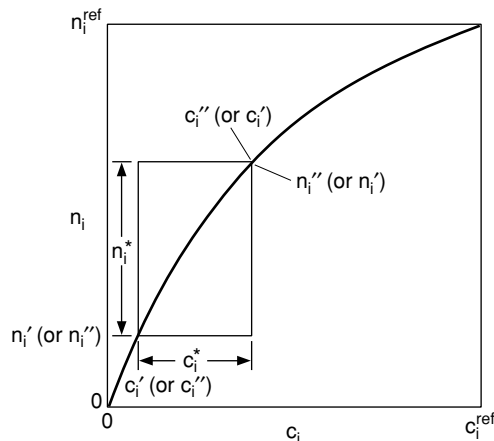


FIG. 16-6 Isotherm showing concentration variables for a transition from (c_i', n_i') to (c_i'', n_i'') .

Thermodynamics requires that a linear limit be approached in the Henry's law region for all isotherm equations.

Flat-Surface Isotherm Equations The classification of isotherm equations into two broad categories for flat surfaces and pore filling reflects their origin. It does not restrict equations developed for flat surfaces from being applied successfully to describe data for porous adsorbents.

The classical isotherm for a homogeneous flat surface, and most popular of all nonlinear isotherms, is the **Langmuir isotherm**

$$n_i = \frac{n_i^s K_i c_i}{1 + K_i c_i} \quad (16-13)$$

where n_i^s is the monolayer capacity approached at large concentrations and K_i is an equilibrium constant. These parameters are often determined by plotting $1/n_i$ versus $1/c_i$. The derivation of the isotherm assumes negligible interaction between adsorbed molecules. For temperature-dependent adsorption equilibrium and a constant isosteric heat of adsorption, K_i is given by

$$K_i = K_i^0 \exp(\lambda_i / \mathcal{R}T) \quad (16-14)$$

where K_i^0 is a constant. A variation on Eq. (16-13) is the multisite Langmuir isotherm, which is [Nitta et al., *J. Chem. Eng. Japan*, **17**, 39 (1984)]

$$K_i p_i = \frac{n_i / n_i^s}{(1 - n_i / n_i^s)^{a_i}} \quad (16-15)$$

where a_i is a constant.

The classical isotherm for multilayer adsorption on a homogeneous, flat surface is the **BET isotherm** [Brunauer, Emmett, and Teller, *J. Am. Chem. Soc.*, **60**, 309 (1938)]

$$n_i = \frac{n_i^s K_i p_i}{[1 + K_i p_i - (p_i / P_i^s)][1 - (p_i / P_i^s)]} \quad (16-16)$$

where p_i is the pressure of the adsorbable component and P_i^s is its vapor pressure. It is useful for gas-solid systems in which condensation is approached, fitting type-II behavior.

For a heterogeneous flat surface, a classical isotherm is the **Freundlich isotherm**

$$n_i = K_i c_i^{m_i} \quad (16-17)$$

where m_i is positive and generally not an integer. The isotherm corresponds approximately to an exponential distribution of heats of adsorption. Although it lacks the required linear behavior in the Henry's law region, it can often be used to correlate data on heterogeneous adsorbents over wide ranges of concentration.

Several isotherms combine aspects of both the Langmuir and Freundlich equations. One that has been shown to be effective in describing data mathematically for heterogeneous adsorbents is the **Toth isotherm** [*Acta Chim. Acad. Sci. Hung.*, **69**, 311 (1971)]

$$n_i = \frac{n_i^s p_i}{[(1/K_i) + p_i^{m_i}]^{1/m_i}} \quad (16-18)$$

This three-parameter equation behaves linearly in the Henry's law region and reduces to the Langmuir isotherm for $m_i = 1$. Other well-known isotherms include the **Langmuir-Freundlich isotherm** or **Sips isotherm** [Sips, *J. Chem. Phys.*, **16**, 490 (1948); Koble and Corrigan, *Ind. Eng. Chem.*, **44**, 383 (1952)] or **loading ratio correlation** with prescribed temperature dependence [Yon and Turnock, *AIChE Symp. Ser.*, **67**(117), 75 (1971)]

$$n_i = \frac{n_i^s (K_i p_i)^m}{1 + (K_i p_i)^m} \quad (16-19)$$

Equations of state are also used. Given such an equation written in terms of the two-dimensional spreading pressure π , the corresponding isotherm is easily determined, as described later for mixtures [see Eq. (16-43)]. The two-dimensional equivalent of an ideal gas is an ideal surface gas, which is described by

$$\pi A = n_i \mathcal{R}T \quad (16-20)$$

which readily gives the linear isotherm, Eq. (16-12). Many more complicated equations of state are available, including two-dimensional analogs of the virial equation and equations of van der Waals, Redlich-Kwong, Peng-Robinson, etc. [Adamson, gen. refs.; Patrykiewicz et al., *Chem. Eng. J.*, **15**, 147 (1978); Haydel and Kobayashi, *Ind. Eng. Chem. Fundam.*, **6**, 546 (1967)].

Pore-Filling Isotherm Equations Most pore-filling models are grounded in the **Polanyi potential theory**. In Polanyi's model, an attracting potential energy field is assumed to exist adjacent to the surface of the adsorbent and concentrates vapors there. Adsorption takes place wherever the strength of the field, independent of temperature, is great enough to compress the solute to a partial pressure greater than its vapor pressure. The last molecules to adsorb form an equipotential surface containing the adsorbed volume. The strength of this field, called the adsorption potential ϵ (J/mol), was defined by Polanyi to be equal to the work required to compress the solute from its partial pressure to its vapor pressure

$$\epsilon = RT \ln \left(\frac{P_i}{p_i} \right) \quad (16-21)$$

The same result is obtained by considering the change in chemical potential. In the basic theory, W (m³/kg), the volume adsorbed as saturated liquid at the adsorption temperature, is plotted versus ϵ to give a characteristic curve. Data measured at different temperatures for the same solute-adsorbent pair should fall on this curve. Using the method, it is possible to use data measured at a single temperature to predict isotherms at other temperatures. Data for additional, homologous solutes can be collapsed into a single "correlation curve" by defining a scaling factor β , the most useful of which has been V/V^{ref} , the adsorbate molar volume as saturated liquid at the adsorption temperature divided by that for a reference compound. Thus, by plotting W versus ϵ/β or ϵ/V for data measured at various temperatures for various similar solutes and a single adsorbent, a single curve should be obtained. Variations of the theory are often used to evaluate properties for components near or above their critical points [Grant and Manes, *Ind. Eng. Chem. Fund.*, **5**, 490 (1966)].

The most popular equations used to describe the shape of a characteristic curve or a correlation curve are the two-parameter **Dubinin-Radushkevich** (DR) equation

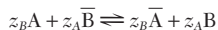
$$\frac{W}{W_0} = \exp \left[-k \left(\frac{\epsilon}{\beta} \right)^2 \right] \quad (16-22)$$

and the three-parameter **Dubinin-Astakhov** (DA) equation

$$\frac{W}{W_0} = \exp \left[-k \left(\frac{\epsilon}{\beta} \right)^m \right] \quad (16-23)$$

where W_0 is micropore volume and m is related to the pore size distribution [Gregg and Sing, gen. refs.]. Neither of these equations has correct limiting behavior in the Henry's law regime.

Ion Exchange A useful tool is provided by the **mass action law** for describing the general exchange equilibrium in fully ionized exchanger systems as



where overbars indicate the ionic species in the ion exchanger phase and z_A and z_B are the valences of ions A and B. The associated equilibrium relation is of the form

$$K_{A,B} = K_{A,B} \left(\frac{\gamma_A}{\gamma_B} \right)^{z_B} \left(\frac{\gamma_B}{\gamma_A} \right)^{z_A} = \left(\frac{n_A}{c_A} \right)^{z_B} \left(\frac{c_B}{n_B} \right)^{z_A} \quad (16-24)$$

where $K'_{A,B}$ is the apparent equilibrium constant or molar selectivity coefficient, $K_{A,B}$ is the thermodynamic equilibrium constant based on activities, and the γ s are activity coefficients. Often it is desirable to represent concentrations in terms of equivalent ionic fractions based on solution normality c_{tot} and fixed exchanger capacity n^* as $c_i^* = z_i c_i / c_{\text{tot}}$ and $n_i^* = z_i n_i / n_{\text{tot}}$, where $c_{\text{tot}} = \sum z_i c_i$ and $n_{\text{tot}} = \sum z_i n_i = n^*$ with the summations extended to all counter-ion species. A rational selectivity coefficient is then defined as

$$K'_{A,B} = K_{A,B} \left(\frac{n_{\text{tot}}}{c_{\text{tot}}} \right)^{z_A - z_B} = \left(\frac{n_A^*}{c_A^*} \right)^{z_B} \left(\frac{c_B^*}{n_B^*} \right)^{z_A} \quad (16-25)$$

For the exchange of ions of equal valence ($z_A = z_B$), $K'_{A,B}$ and $K_{A,B}$ are

coincident and, to a first approximation, independent of concentration. When $z_A > z_B$, $K'_{A,B}$ decreases with solution normality. This reflects the fact that ion exchangers exhibit an increasing affinity for ions of lower valence as the solution normality increases.

An alternate form of Eq. (16-25) is

$$n_A^* = K_{A,B}^c \left(\frac{n_{\text{tot}}}{c_{\text{tot}}} \right)^{(z_A - z_B)/z_B} \left(\frac{1 - n_A^*}{1 - c_A^*} \right)^{z_A/z_B} c_A^* \quad (16-26)$$

When B is in excess ($c_A^*, n_A^* \rightarrow 0$), this reduces to

$$n_A^* = K_{A,B}^c \left(\frac{n_{\text{tot}}}{c_{\text{tot}}} \right)^{(z_A - z_B)/z_B} c_A^* = K_A c_A^* \quad (16-27)$$

where the linear equilibrium constant $K_A = K_{A,B}^c (n_{\text{tot}}/c_{\text{tot}})^{(z_A/z_B)-1}$ decreases with solution normality c_{tot} when $z_A > z_B$ or increases when $z_A < z_B$.

Table 16-7 gives equilibrium constants for crosslinked cation and anion exchangers for a variety of counterions. The values given for cation exchangers are based on ion A replacing Li⁺, and those given for anion exchangers are based on ion A replacing Cl⁻. The selectivity for a particular ion generally increases with decreasing hydrated ion size and increasing degree of crosslinking. The selectivity coefficient for any two ions A and D can be obtained from Table 16-7 from values of $K_{A,B}$ and $K_{D,B}$ as

$$K_{A,D} = K_{A,B}/K_{D,B} \quad (16-28)$$

The values given in this table are only approximate, but they are adequate for process screening purposes with Eqs. (16-24) and (16-25). Rigorous calculations generally require that activity coefficients be accounted for. However, for the exchange between ions of the same valence at solution concentrations of 0.1 N or less, or between any ions at 0.01 N or less, the solution-phase activity coefficients will be similar enough that they can be omitted.

Models for ion exchange equilibria based on the mass-action law taking into account solution and exchanger-phase nonidealities with equations similar to those for liquid mixtures have been developed by several authors [see Smith and Woodburn, *AIChE J.*, **24**, 577 (1978); Mehlabia et al., *Chem. Eng. Sci.*, **49**, 2277 (1994)]. Thermodynamics-based approaches are also available [Soldatov in Dorfner, gen. refs.; Novosad and Myers, *Can J. Chem. Eng.*, **60**, 500 (1982); Myers and Byington in Rodrigues, ed., *Ion Exchange Science and Technology*, NATO ASI Series, No. 107, Nijhoff, Dordrecht, 1986, pp. 119–145] as well as approaches for the exchange of macromolecules, taking into account steric-hindrance effects [Brooks and Cramer, *AIChE J.*, **38**, 12 (1992)].

Example 3: Calculation of Useful Ion-Exchange Capacity An 8 percent crosslinked sulfonated resin is used to remove calcium from a solution containing 0.0007 mol/l Ca²⁺ and 0.01 mol/l Na⁺. The resin capacity is 2.0 equiv/l. Estimate the resin capacity for calcium removal.

From Table 16-7, we obtain $K_{Ca,Na} = 5.16/1.98 = 2.6$. Since $n_{\text{tot}} = 2$ equiv/l and $c_{\text{tot}} = 2 \times 0.0007 + 0.01 = 0.011$ equiv/l, $K'_{Ca,Na} = 2.6 \times 2/0.011 = 470$. Thus, with $c_{Ca}^* = 2 \times 0.0007/0.011 = 0.13$, Eq. (16-25) gives $470 = (n_{Ca}^*/0.13)/[(1 - 0.13)/(1 - n_{Na}^*)]^2$ or $n_{Ca}^* = 0.9$. The available capacity for calcium is $n_{Ca} = 0.9 \times 2/2 = 0.9$ mol/l.

Donnan Uptake The uptake of an electrolyte as a neutral ion pair of a salt is called Donnan uptake. It is generally negligible at low ionic concentrations. Above 0.5 g-equiv/l with strongly ionized exchangers (or at lower concentrations with those more weakly ionized), the resin's fixed ion-exchange capacity is measurably exceeded as a result of electrolyte invasion. With only one coion species Y (matching the charge sign of the fixed groups in the resin), its uptake n_Y equals the total excess uptake of the counterion. Equilibrium is described by the mass-action law. For the case of a resin in A-form in equilibrium with a salt AY, the excess counterion uptake is given by [Helfferich, gen. refs., pp. 133–147]

$$z_Y n_Y = \left[\frac{n^2}{4} + (z_Y c_Y)^2 \left(\frac{\gamma_{AY}}{\gamma_Y} \right)^2 \left(\frac{a_w}{a_w} \right)^{z_A/z_Y} \right]^{1/2} - \frac{n^*}{2} \quad (16-29)$$

where γ s are activity coefficients, a s water activities, and v s partial molar volumes. For dilute conditions, Eq. (16-29) predicts a squared dependence of n_Y on c_Y . Thus, the electrolyte sorption isotherm has a strong positive curvature. Donnan uptake is more pronounced for resins of lower degree of crosslinking and for counterions of low valence.

Separation Factor By analogy with the mass-action case and appropriate for both adsorption and ion exchange, a **separation factor** r

TABLE 16-7 Equilibrium Constants for Polystyrene DVB Cation and Anion Exchangers

Strong acid sulfonated cation exchangers (Li ⁺ reference ion)							
Degree of crosslinking				Degree of crosslinking			
Counter-ion	4% DVB	8% DVB	16% DVB	Counter-ion	4% DVB	8% DVB	16% DVB
Li ⁺	1.00	1.00	1.00	Mg ⁺⁺	2.95	3.29	3.51
H ⁺	1.32	1.27	1.47	Zn ⁺⁺	3.13	3.47	3.78
Na ⁺	1.58	1.98	2.37	Co ⁺⁺	3.23	3.74	3.81
NH ₄ ⁺	1.90	2.55	3.34	Cu ⁺⁺	3.29	3.85	4.46
K ⁺	2.27	2.90	4.50	Cd ⁺⁺	3.37	3.88	4.95
Rb ⁺	2.46	3.16	4.62	Ni ⁺⁺	3.45	3.93	4.06
Cs ⁺	2.67	3.25	4.66	Ca ⁺⁺	4.15	5.16	7.27
Ag ⁺	4.73	8.51	22.9	Pb ⁺⁺	6.56	9.91	18.0
Tl ⁺	6.71	12.4	28.5	Ba ⁺⁺	7.47	11.5	20.8

Strong base anion exchangers, 8% DVB (Cl ⁻ reference ion)					
Counterion	Type I resin*	Type II resin†	Counterion	Type I resin*	Type II resin†
Salicylate	32	28	Cyanide	1.6	1.3
Iodide	8.7	7.3	Chloride	1.0	1.0
Phenoxide	5.2	8.7	Hydroxide	0.05-0.07	0.65
Nitrate	3.8	3.3	Bicarbonate	0.3	0.5
Bromide	2.8	2.3	Formate	0.2	0.2
Nitrite	1.2	1.3	Acetate	0.2	0.2
Bisulfite	1.3	1.3	Fluoride	0.09	0.1
Cyanide	1.6	1.3	Sulfate	0.15	0.15

*Trimethylamine.

†Dimethyl-hydroxyethylamine.

Data from Bonner and Smith, *J. Phys. Chem.*, **61**, 326 (1957) and Wheaton and Bauman, *Ind. Eng. Chem.*, **45**, 1088 (1951).

can be defined based on dimensionless system variables [Eq. (16-10)] by

$$r = \frac{c_i^o(1 - n_i^o)}{n_i^o(1 - c_i^o)} \quad \text{or} \quad r = \frac{n_B^o/c_B^o}{n_A^o/c_A^o} \quad (16-30)$$

This term is analogous to relative volatility or its reciprocal (or to an equilibrium selectivity). Similarly, the assumption of a constant separation factor is a useful assumption in many sorptive operations. [It is constant for the Langmuir isotherm, as described below, and for mass-action equilibrium with $z_a = z_b$ in Eq. (16-24).] This gives the constant separation factor isotherm

$$n_i^o = \frac{c_i^o}{r + (1 - r)c_i^o} \quad (16-31)$$

The separation factor r identifies the equilibrium increase in n_i^o from 0 to 1, which accompanies an increase in c_i^o from 0 to 1. For a concentration change over only part of the isotherm, a separation factor R can be defined for the dimensionless transition variables [Eq. (16-11)]. This separation factor is

$$R = \frac{n_i''/c_i''}{n_i'/c_i'} = \frac{c_i^o(1 - n_i^o)}{n_i^o(1 - c_i^o)} \quad (16-32)$$

and gives an equation identical to Eq. (16-31) with R replacing r .

Figure 16-7 shows constant separation factor isotherms for a range of r (or R) values. The isotherm is linear for $r = 1$, favorable for $r < 1$, rectangular (or irreversible) for $r = 0$, and unfavorable for $r > 1$. As a result of symmetry properties, if r is defined for adsorption of component i or exchange of ion A for B, then the reverse process is described by $1/r$.

The Langmuir isotherm, Eq. (16-13), corresponds to the constant separation factor isotherm with

$$r = 1/(1 + K_i c_i^{\text{ref}}) \quad (16-33)$$

for system variables Eq. (16-10) or

$$R = \frac{r + (1 - r)(c_i'/c_i^{\text{ref}})}{r + (1 - r)(c_i''/c_i^{\text{ref}})} \quad (16-34)$$

for transition variables [Eq. (16-11)]. Vermeulen et al. [gen. refs.] give additional properties of constant separation factor isotherms.

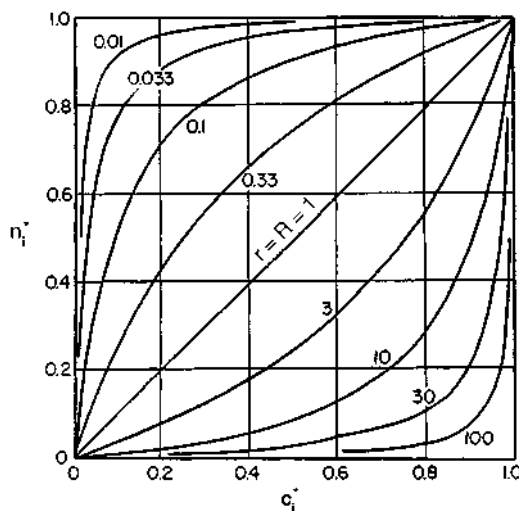


FIG. 16-7 Constant separation factor isotherm as a function of the separation factor r (or interchangeably R). Each isotherm is symmetric about the perpendicular line connecting (0,1) and (1,0). Isotherms for r and $1/r$ are symmetric about the 45° line.

Example 4: Application of Isotherms Thomas [*Ann. N.Y. Acad. Sci.*, **49**, 161 (1948)] provides the following Langmuir isotherm for the adsorption of anthracene from cyclohexane onto alumina:

$$n_i = \frac{22c_i}{1 + 375c_i}$$

with n_i in mol anthracene/kg alumina and c_i in mol anthracene/l liquid.

a. What are the values of K_i and n_i^s according to Eq. (16-13)?

$$K_i = 375 \text{ l/mol}$$

$$n_i^s = \frac{22}{K_i} = 0.0587 \text{ mol/kg}$$

16-16 ADSORPTION AND ION EXCHANGE

b. For a feed concentration of 8.11×10^{-4} mol/l, what is the value of r ?

$$r = \frac{1}{1 + K_i(8.11 \times 10^{-4})} = 0.766 \quad [\text{from Eq. (16-33)}]$$

c. If the alumina is presaturated with liquid containing 2.35×10^{-4} mol/l and the feed concentration is 8.11×10^{-4} mol/l, what is the value of R ?

$$R = 0.834 \quad [\text{from Eq. (16-32) or (16-34)}]$$

MULTIPLE COMPONENTS OR EXCHANGES

When more than one adsorbed species or more than two ion-exchanged species interact in some manner, equilibrium becomes more complicated. Usually, thermodynamics provides a sound basis for prediction.

Adsorbed-Solution Theory The common thermodynamic approach to multicomponent adsorption treats adsorption equilibrium in a way analogous to fluid-fluid equilibrium. The theory has as its basis the Gibbs adsorption isotherm [Young and Crowell, gen. refs.], which is

$$Ad\pi = \sum_i n_i d\mu_i \quad (\text{const } T) \quad (16-35)$$

where μ is chemical potential. For an ideal gas ($d\mu = \mathcal{R}T d \ln p_i$), if it is assumed that an adsorbed solution is defined with a pure-component standard state (as is common for a liquid solution), then Eq. (16-35) can be integrated to give [Rudisill and LeVan, *Chem. Eng. Sci.*, **47**, 1239 (1992)]

$$p_i = \gamma_i x_i P_i^{\text{ref}}(T, \pi) \quad (16-36)$$

where γ_i and x_i are the adsorbed-phase activity coefficient and mole fraction of component i and P_i^{ref} is the standard state, specified to be at the temperature and spreading pressure of the mixture.

Equation (16-36) with $\gamma_i = 1$ provides the basis for the ideal adsorbed-solution theory [Myers and Prausnitz, *AIChE J.*, **11**, 121 (1965)]. The spreading pressure for a pure component is determined by integrating Eq. (16-35) for a pure component to obtain

$$\frac{\pi A}{\mathcal{R}T} = \int_0^{P_i^{\text{ref}}} \frac{n_i}{p_i} dp_i \quad (16-37)$$

where n_i is given by the pure-component isotherm. Also, since $\sum x_i = 1$, Eq. (16-36) with $\gamma_i = 1$ gives $\sum (p_i/P_i^{\text{ref}}) = 1$. With no area change on mixing for the ideal solution, the total number of moles adsorbed per unit weight of adsorbent is determined using a two-dimensional form of Amagat's law:

$$\frac{1}{n_{\text{tot}}} = \sum \frac{x_i}{n_i^{\text{ref}}} \quad (16-38)$$

where $n_{\text{tot}} = \sum n_i$ and n_i^{ref} is given by the pure-component isotherm at P_i^{ref} . Adsorbed-phase concentrations are calculated using $n_i = x_i n_{\text{tot}}$. Generally, different values of π [or $\pi A/(\mathcal{R}T)$] must be tried until the one is found that satisfies Eq. (16-37) and $\sum x_i = 1$.

Example 5: Application of Ideal Adsorbed-Solution Theory

Consider a binary adsorbed mixture for which each pure component obeys the Langmuir equation, Eq. (16-13). Let $n_1^s = 4$ mol/kg, $n_2^s = 3$ mol/kg, $K_1 p_1 = K_2 p_2 = 1$. Use the ideal adsorbed-solution theory to determine n_1 and n_2 .

Substituting the pure component Langmuir isotherm

$$n_i = \frac{n_i^s K_i p_i}{1 + K_i p_i}$$

into Eq. (16-37) and integrating gives

$$\frac{\pi A}{\mathcal{R}T} = n_1^s \ln(1 + K_1 P_i^{\text{ref}})$$

which can be solved explicitly for $K_i P_i^{\text{ref}}$. Values are guessed for $\pi A/(\mathcal{R}T)$, values of $K_i P_i^{\text{ref}}$ are calculated from the equation above, and $\sum x_i = \sum K_i p_i / (K_i P_i^{\text{ref}}) = 1$ is checked to see if it is satisfied. Trial and error gives $\pi A/(\mathcal{R}T) = 3.8530$ mol/kg, $K_1 P_1^{\text{ref}} = 1.6202$, $K_2 P_2^{\text{ref}} = 2.6123$, and $x_1 = 0.61720$. Evaluating the pure-component isotherms at the reference pressures and using Eq. (16-38) gives $n_{\text{tot}} = 2.3475$ mol/kg, and finally $n_i = x_i n_{\text{tot}}$ gives $n_1 = 1.4489$ mol/kg and $n_2 = 0.8986$ mol/kg.

Other approaches to account for various effects have been developed. Negative deviations from Raoult's law (i.e., $\gamma_i < 1$) are frequently found due to adsorbent heterogeneity [e.g., Myers, *AIChE J.*, **29**, 691

(1983)]. Thus, contributions include accounting for adsorbent heterogeneity [Valenzuela et al., *AIChE J.*, **34**, 397 (1988)] and excluded pore-volume effects [Myers, in Rodrigues et al., gen. refs.]. Several activity coefficient models have been developed to account for non-ideal adsorbate-adsorbate interactions including a spreading pressure-dependent activity coefficient model [e.g., Talu and Zwiebel, *AIChE J.*, **32**, 1263 (1986)] and a vacancy solution theory [Suwanayuen and Danner, *AIChE J.*, **26**, 68, 76 (1980)].

Langmuir-Type Relations For systems composed of solutes that individually follow Langmuir isotherms, the traditional multicomponent Langmuir equation, obtained via a kinetic derivation, is

$$n_i = \frac{n_i^s K_i p_i}{1 + \sum_j K_j p_j} \quad (16-39)$$

where j is summed over all components. This equation has been criticized on thermodynamic grounds because it does not satisfy the Gibbs adsorption isotherm unless all monolayer capacities n_i^s are equal.

To satisfy the Gibbs adsorption isotherm for unequal monolayer capacities, explicit isotherms can be obtained in the form of a series expansion [LeVan and Vermeulen, *J. Phys. Chem.*, **85**, 3247 (1981)]. A two-term form is

$$n_1 = \frac{(n_1^s + n_2^s) K_1 p_1}{2(1 + K_1 p_1 + K_2 p_2)} + \frac{(n_1^s - n_2^s) K_1 p_1 K_2 p_2}{(K_1 p_1 + K_2 p_2)^2} \ln(1 + K_1 p_1 + K_2 p_2) \quad (16-40)$$

where the subscripts may be interchanged. Multicomponent forms are also available [Frey and Rodrigues, *AIChE J.*, **40**, 182 (1994)].

A variation on the multicomponent Langmuir isotherm for application to heterogeneous surfaces with sites of various energies involves summing terms given by the right-hand side of Eq. (16-39) to give the multipatch Langmuir isotherm

$$n_i = \sum_k \frac{n_{i,k}^s K_{i,k} p_i}{1 + \sum_j K_{j,k} p_j} \quad (16-41)$$

where k is summed over all patches.

Example 6: Comparison of Binary Langmuir Isotherms Use the numerical values in Example 5 to evaluate the binary Langmuir isotherms given by Eqs. (16-39) and (16-40) and compare results with the exact answers given in Example 5.

Equation (16-39) gives $n_1 = 1.3333$ mol/kg and $n_2 = 1.0000$ mol/kg for an average deviation from the exact values of approximately 10 percent. Equation (16-40) gives $n_1 = 1.4413$ mol/kg and $n_2 = 0.8990$ mol/kg for an average deviation of about 0.6 percent.

Freundlich-Type Relations A binary Freundlich isotherm, obtained from the ideal adsorbed solution theory in loading-explicit closed form [Crittenden et al., *Environ. Sci. Technol.*, **19**, 1037 (1985)], is

$$c_i = \frac{n_i}{\sum_{j=1}^N n_j} \left(\frac{\sum_{j=1}^N (n_j/m_j)}{K_i/m_i} \right)^{1/m_i} \quad (16-42)$$

Equations of State If an equation of state is specified for a multicomponent adsorbed phase of the form $\pi A/(\mathcal{R}T) = f(n_1, n_2, \dots)$, then the isotherms are determined using [Van Ness, *Ind. Eng. Chem. Fundam.*, **8**, 464-473 (1969)]

$$\ln \left(\frac{K_i p_i}{n_i/M_s} \right) = \int_A^\infty \left[\frac{\partial [\pi A/(\mathcal{R}T)]}{\partial n_i} \right]_{T,A,n_j} - 1 \Bigg] \frac{dA}{A} \quad (16-43)$$

where, because integration is over A , M_s is mass of adsorbent, units for n and A are mol and m^2 (rather than mol/kg and m^2/kg), and $K_i p_i / (n_i/M_s) = 1$ is the linear lower limit approached for the ideal surface gas [see Eqs. (16-12) and (16-20)].

Ion Exchange—Stoichiometry In most applications, except for some weak-electrolyte and some concentrated-solution cases, the following summations apply:

$$\sum_i z_i c_i = c_{\text{tot}} = \text{const} \quad \sum_i z_i n_i = n_{\text{tot}} = \text{const} \quad (16-44a)$$

In equivalent-fraction terms, the sums become

$$\sum_i c_i^* = 1 \quad \sum_i n_i^* = 1 \quad (16-44b)$$

Mass Action Here the equilibrium relations, consistent with Eq. (16-25), are

$$K'_{ij} = \left(\frac{n_i^*}{c_i^*} \right)^{z_j} \left(\frac{c_j^*}{n_j^*} \right)^{z_i} \quad (16-45)$$

For an N -species system, with N c^* 's (or n^* 's) known, the N n^* 's (or c^* 's) can be found by simultaneous solution of the $N - 1$ independent i, j combinations for Eq. (16-45) using Eq. (16-44a); one n_j^*/c_j^* is assumed, the other values can be calculated using Eq. (16-45), and the sum of the trial n^* 's (or c^* 's) is compared with Eq. (16-44b).

Because an N -component system has $N - 1$ independent concentrations, a three-component equilibrium can be plotted in a plane and a four-component equilibrium in a three-dimensional space. Figure 16-8 shows a triangular plot of c^* contours in equilibrium with the corresponding n^* coordinates.

Improved models for ion-exchange equilibria are in Smith and Woodburn [*AIChE J.*, **24**, 577 (1978)]; Mehablia et al. [*Chem. Eng. Sci.*, **49**, 2277 (1994)]; Soldatov [in Dorfner, gen. refs.]; Novosad and Myers, [*Can J. Chem. Eng.*, **60**, 500 (1982)]; and Myers and Byington [in Rodrigues, ed., *Ion Exchange: Science and Technology*, NATO ASI Series, No. 107, Nijhoff, Dordrecht, 1986, pp. 119–145].

Constant Separation-Factor Treatment If the valences of all species are equal, the separation factor α_{ij} applies, where

$$\alpha_{ij} = K_{ij} = \frac{n_i^* c_j^*}{c_i^* n_j^*} \quad (16-46)$$

For a binary system, $r = \alpha_{BA} = 1/\alpha_{AB}$. The symbol r applies primarily to the process, while α is oriented toward interactions between pairs of solute species. For each binary pair, $r_{ij} = \alpha_{ij} = 1/\alpha_{ji}$.

Equilibrium then is given explicitly by

$$n_i^* = \frac{c_i^*}{\sum_j \alpha_{ji} c_j^*} = \frac{\alpha_{iN} c_i^*}{\sum_j \alpha_{ji} n_j^*} \quad (16-47)$$

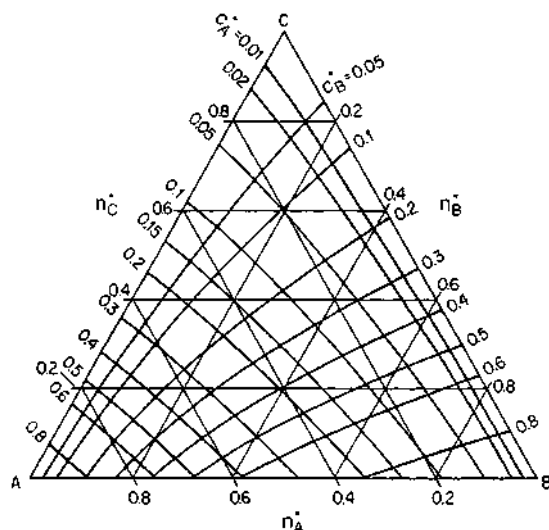


FIG. 16-8 Ideal mass-action equilibrium for three-component ion exchange with unequal valences. $K'_{A/C} = 8.06$; $K'_{B/C} = 3.87$. Duolite C-20 polystyrenesulfonate resin, with Ca as A, Mg as B, and Na as C. [Klein et al., *Ind. Eng. Chem. Fund.*, **6**, 339 (1967); reprinted with permission.]

$$\text{and} \quad c_i^* = \frac{n_i^*}{\sum_j \alpha_{ji} n_j^*} = \frac{\alpha_{Ni} n_i^*}{\sum_j \alpha_{Nj} n_j^*} \quad (16-48)$$

For the constant separation factor case, the c^* contours in a plot like Fig. 16-8 are linear.

CONSERVATION EQUATIONS

Material balances, often an energy balance, and occasionally a momentum balance are needed to describe an adsorption process. These are written in various forms depending on the specific application and desire for simplicity or rigor. Reasonably general material balances for various processes are given below. An energy balance is developed for a fixed bed for gas-phase application and simplified for liquid-phase application. Momentum balances for pressure drop in packed beds are given in Sec. 6.

MATERIAL BALANCES

At a microscale, a sorbable component exists at three locations—in a sorbed phase, in pore fluid, and in fluid outside particles. As a consequence, in material balances time derivatives must be included of terms involving n_p , c_{pi} (the pore concentration), and c_i (the extraparticle concentration). Let $\bar{n}_i$ represent n_i averaged over particle volume, and let $\bar{c}_{pi}$ represent c_{pi} averaged over pore fluid volume.

For batch or stirred tank processes, in terms of the mass of adsorbent M_s (kg), extraparticle volume of fluid V_f (m^3), and volumetric flow rates F_v (m^3/s) in and out of a tank, the material balance on component i is

$$M_s \frac{d\bar{n}_i}{dt} + \frac{d(V_f c_i)}{dt} = F_{v,in} c_{i,in} - F_{v,out} c_i \quad (16-49)$$

$$\text{with} \quad \hat{n}_i = \bar{n}_i + (\epsilon_p/\rho_p) \bar{c}_{pi} = \bar{n}_i + [(1 - \epsilon)\epsilon_p/\rho_b] \bar{c}_{pi} \quad (16-50)$$

where ρ_p and ρ_b are particle and bulk densities, and ϵ and ϵ_p are void fraction (extra particle volume fraction) and particle porosity, respectively.

For a fixed-bed process, the material balance for component i is

$$\rho_b \frac{\partial \hat{n}_i}{\partial t} + \epsilon \frac{\partial c_i}{\partial t} + \epsilon \frac{\partial (v c_i)}{\partial z} = \epsilon D_L \frac{\partial}{\partial z} \left(c \frac{\partial y_i}{\partial z} \right) \quad (16-51)$$

where v is interstitial fluid velocity, D_L is a Fickian axial dispersion coefficient, and $y_i = c_i/c$ is the fluid-phase mole fraction of component i . In certain applications such as adsorption of biological molecules in gels, where the density of the adsorbent material is not known, the group $\rho_b \hat{n}_i$ in the first term of Eq. (16-51) can be replaced by $(1 - \epsilon) q_i$, where q_i has units of $\text{mol}/(\text{m}^3 \text{ of adsorbent})$. An alternative form, grouping together fluid-phase concentrations rather than intraparticle concentrations, is

$$\rho_b \frac{\partial \bar{n}_i}{\partial t} + \epsilon_b \frac{\partial \hat{c}_i}{\partial t} + \epsilon \frac{\partial (v c_i)}{\partial z} = \epsilon D_L \frac{\partial}{\partial z} \left(c \frac{\partial y_i}{\partial z} \right) \quad (16-52)$$

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where, noting Eq. (16-4), $\hat{c}_i$ is defined by

$$\varepsilon_b \hat{c}_i = \varepsilon c_i + (1 - \varepsilon) \varepsilon_{pf} \bar{c}_{pi} \quad (16-53)$$

For moving-bed processes, we add a term to Eq. (16-51) to obtain

$$\rho_b \left[\frac{\partial \hat{n}_i}{\partial t} + v_s \frac{\partial \hat{n}_i}{\partial z} \right] + \varepsilon \left[\frac{\partial c_i}{\partial t} + \frac{\partial (v c_i)}{\partial z} - D_L \frac{\partial}{\partial z} \left(c \frac{\partial y_i}{\partial z} \right) \right] = 0 \quad (16-54)$$

where v_s is the solid-phase velocity (opposite in sign to v for a countercurrent process).

ENERGY BALANCE

Many different forms of the energy balance have been used in fixed-bed adsorption studies. The form chosen for a particular study depends on the process considered (e.g., temperature swing adsorption or pressure swing adsorption) and on the degree of approximation that is appropriate [see Walton and LeVan, *Ind. Eng. Chem. Research*, **42**, 6938 (2003); **44**, 7474 (2005)].

An energy balance for a fixed-bed process, ignoring dispersion, is

$$\rho_b \frac{\partial u_s}{\partial t} + \frac{\partial (\varepsilon_b \hat{c} u_f)}{\partial t} + \frac{\partial (\varepsilon v c h_f)}{\partial z} = - \frac{2 h_w (T - T_w)}{r_c} \quad (16-55)$$

where h_w is a heat transfer coefficient for energy transfer with the column wall and r_c is the radius of the column. The second term of Eq. (16-55) combines contributions from both pore and extraparticle fluid.

Thermodynamic paths are necessary to evaluate the enthalpy (or internal energy) of the fluid phase and the internal energy of the stationary phase. For gas-phase processes at low and modest pressures, the enthalpy departure function for pressure changes can be ignored and a reference state for each pure component chosen to be ideal gas at temperature T^{ref} , and a reference state for the stationary phase (adsorbent plus adsorbate) chosen to be adsorbate-free solid at T^{ref} . Thus, for the gas phase we have

$$h_f = \sum_i y_i h_{fi} = \sum_i y_i \left[h_{fi}^{\text{ref}} + \int_{T^{\text{ref}}}^T C_{pfi}^{\circ} dT \right] \quad (16-56)$$

$$u_f = h_f - P/c \quad (16-57)$$

and for the stationary phase

$$u_s = u_{\text{sol}} + n u_a \approx u_{\text{sol}} + n h_a \quad (16-58)$$

$$u_{\text{sol}} = u_{\text{sol}}^{\text{ref}} + \int_{T^{\text{ref}}}^T C_s dT \quad (16-59)$$

The enthalpy of the adsorbed phase h_a is evaluated along a path for which the gas-phase components undergo temperature change from T^{ref} to T and then are adsorbed isothermally, giving

$$h_a = \sum_i x_i h_{fi} - \left(\frac{1}{n} \right) \sum_i \int_0^{n_i} q_i^{\text{st}}(n_i, n_j, T) dn_i \quad (16-60)$$

The isosteric heat of adsorption q_i^{st} is composition-dependent, and the sum of integrals in Eq. (16-60) is difficult to evaluate for multicomponent adsorption if the isosteric heats indeed depend on loading. Because each isosteric heat depends on the loadings of all components, the sum must be evaluated for a path beginning with clean adsorbent and ending with the proper loadings of all components. If the isosteric heat of adsorption is constant, as is commonly assumed, then the energy balance [Eq. (16-55)] becomes

$$\left[\rho_b \left(C_s + \sum_i n_i C_{pfi}^{\circ} \right) + \varepsilon_b c C_{pff}^{\circ} \right] \frac{\partial T}{\partial t} - \rho_b \sum_i q_i^{\text{st}} \frac{\partial n_i}{\partial t} - \frac{\partial (\varepsilon_b P)}{\partial t} + \varepsilon v c C_{pff}^{\circ} \frac{\partial T}{\partial z} = - \frac{2 h_w (T - T_w)}{r_c} \quad (16-61)$$

where Eq. (16-52) with $D_L = 0$ has been used. Equation (16-61) is a popular form of the energy balance for fixed-bed adsorption calculations. Often the first summation on the left-hand side, which involves gas-phase heat capacities, is neglected, or the gas-phase heat capacities are replaced by adsorbed-phase heat capacities. It is also possible to estimate the adsorbed-phase heat capacities from thermodynamic paths given gas-phase heat capacities and temperature-dependent adsorption equilibrium [Walton and LeVan, *Ind. Eng. Chem. Research*, **44**, 178 (2005)].

Nonisothermal liquid-phase processes may be driven by changes in feed temperature or heat addition or withdrawal through a column wall. For these, heats of adsorption and pressure effects are generally of less concern. For this case a suitable energy balance is

$$(\rho_b C_s + \varepsilon_b c C_{pff}^{\circ}) \frac{\partial T}{\partial t} + \varepsilon v c C_{pff}^{\circ} \frac{\partial T}{\partial z} = - \frac{2 h_w (T - T_w)}{r_c} \quad (16-62)$$

RATE AND DISPERSION FACTORS

The performance of adsorption processes results in general from the combined effects of thermodynamic and rate factors. It is convenient to consider first thermodynamic factors. These determine the process performance in a limit where the system behaves ideally; i.e. without mass transfer and kinetic limitations and with the fluid phase in perfect piston flow. **Rate factors** determine the efficiency of the real process in relation to the ideal process performance. Rate factors include heat- and mass-transfer limitations, reaction kinetic limitations, and hydrodynamic dispersion resulting from the velocity distribution across the bed and from mixing and diffusion in the interparticle void space.

TRANSPORT AND DISPERSION MECHANISMS

Figure 16-9 depicts porous adsorbent particles in an adsorption bed with sufficient generality to illustrate the nature and location of individual transport and dispersion mechanisms. Each mechanism involves a different driving force and, in general, gives rise to a different form of mathematical result.

Intraparticle Transport Mechanisms Intraparticle transport may be limited by *pore diffusion*, *solid diffusion*, *reaction kinetics* at phase boundaries, or two or more of these mechanisms together.

1. *Pore diffusion in fluid-filled pores.* These pores are sufficiently large that the adsorbing molecule escapes the force field of the adsorbent surface. Thus, this process is often referred to as *macropore diffusion*. The driving force for such a diffusion process can be approximated by the gradient in mole fraction or, if the molar concentration is constant, by the gradient in concentration of the diffusing species within the pores.

2. *Solid diffusion in the adsorbed phase.* Diffusion in pores sufficiently small that the diffusing molecule never escapes the force field of the adsorbent surface. In this case, transport may occur by an activated process involving jumps between adsorption sites. Such a process is often called *surface diffusion* or, in the case of zeolites, *micropore* or *intracrystalline* diffusion. The driving force for the process can thus be approximated by the gradient in concentration of the species in its adsorbed state. Phenomenologically, the process is not distinguishable from that of homogeneous diffusion that occurs inside a sorbent gel or in a pore-filling fluid that is immiscible with the external fluid. The generic term *solid diffusion* is used here to encompass the general traits of these physically different systems.

3. *Reaction kinetics at phase boundaries.* Rates of adsorption and desorption in porous adsorbents are generally controlled by mass

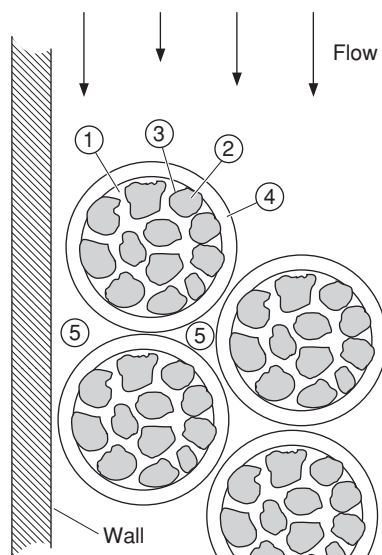


FIG. 16-9 General scheme of adsorbent particles in a packed bed showing the locations of mass transfer and dispersive mechanisms. Numerals correspond to numbered paragraphs in the text: 1, pore diffusion; 2, solid diffusion; 3, reaction kinetics at phase boundary; 4, external mass transfer; 5, fluid mixing.

transfer within the pore network rather than by the kinetics of sorption at the surface. Exceptions are the cases of chemisorption and affinity-adsorption systems used for biological separations, where the kinetics of bond formation can be exceedingly slow.

Intraparticle convection can also occur in packed beds when the adsorbent particles have very large and well-connected pores. Although, in general, bulk flow through the pores of the adsorbent particles is only a small fraction of the total flow, intraparticle convection can affect the transport of very slowly diffusing species such as macromolecules. The driving force for convection, in this case, is the pressure drop across each particle that is generated by the frictional resistance to flow experienced by the fluid as this flows through the packed bed [Rodrigues et al., *Chem. Eng. Sci.*, **46**, 2765 (1991); Carta et al., *Sep. Technol.*, **2**, 62 (1992); Frey et al., *Biotechnol. Progr.*, **9**, 273 (1993); Liapis and McCoy, *J. Chromatogr.*, **599**, 87 (1992)]. Intraparticle convection can also be significant when there is a total pressure difference between the center of the particle and the outside, such as is experienced in pressurization and depressurization steps of pressure swing adsorption or when more gas is drawn into an adsorbent to equalize pressure as adsorption occurs from the gas phase within a porous particle [Lu et al., *AIChE J.*, **38**, 857 (1992); Lu et al., *Gas Sep. Purif.*, **6**, 89 (1992); Taqvi et al., *Adsorption*, **3**, 127 (1997)].

Extraparticle Transport and Dispersion Mechanisms Extraparticle mechanisms are affected by the design of the contacting device and depend on the hydrodynamic conditions outside the particles.

4. *External mass transfer* between the external surfaces of the adsorbent particles and the surrounding fluid phase. The driving force is the concentration difference across the boundary layer that surrounds each particle, and the latter is affected by the hydrodynamic conditions outside the particles.

5. *Mixing*, or lack of mixing, between different parts of the contacting equipment. This may occur through the existence of a velocity distribution or dead zones in a packed bed or through inefficient mixing in an agitated contactor. In packed-bed adsorbers, mixing is often described in terms of an axial dispersion coefficient whereby all mechanisms contributing to axial mixing are lumped together in a single effective coefficient.

Heat Transfer Since adsorption is generally accompanied by the evolution of heat, the rate of heat transfer between the adsorbent

particles and the fluid phase may be important. In addition, heat transfer can occur across the column wall in small diameter beds and is important in energy applications of adsorption. In gas adsorption systems, even with highly porous particles, the controlling heat transfer resistance is generally associated with extraparticle transport [Lee and Ruthven, *Can. J. Chem. Eng.*, **57**, 65 (1979)], so that the temperature within the particles is essentially uniform. In liquid-phase adsorption, intraparticle and extraparticle heat transfer resistances are generally comparable. However, in this case the heat capacity of the fluid phase is sufficiently high that temperature effects may be negligible except in extreme cases. General discussions of heat-transfer effects in adsorbents and adsorption beds are found in Suzuki (gen. refs., pp. 187–208 and pp. 275–290) and in Ruthven (gen. refs., pp. 189–198 and pp. 215–219).

INTRAPARTICLE MASS TRANSFER

The phenomenological aspects of diffusional mass transfer in adsorption systems can be described in terms of Fick's law:

$$J_i = -D_i(c_i) \frac{\partial c_i}{\partial x} \quad (16-63)$$

This expression can be used to describe both pore and solid diffusion so long as the driving force is expressed in terms of the appropriate concentrations. Although the driving force should be more correctly expressed in terms of chemical potentials, Eq. (16-63) provides a qualitatively and quantitatively correct representation of adsorption systems so long as the diffusivity is allowed to be a function of the adsorbate concentration. The diffusivity will be constant only for a thermodynamically ideal system, which is an adequate approximation for only a limited number of adsorption systems.

Pore Diffusion When fluid transport through a network of fluid-filled pores inside the particles provides access for solute adsorption sites, the diffusion flux can be expressed in terms of a pore diffusion coefficient D_{pi} as:

$$J_i = -\epsilon_p D_{pi} \frac{\partial c_{pi}}{\partial r} \quad (16-64)$$

D_{pi} is smaller than the diffusivity in a straight cylindrical pore as a result of the random orientation of the pores, which gives a longer diffusion path, and the variation in the pore diameter. Both effects are commonly accounted for by a tortuosity factor τ_p such that $D_{pi} = D_i/\tau_p$. In principle, predictions of the tortuosity factor can be made if the pore structure, pore size, and shape distributions are known (see Dullien, *Porous Media: Fluid Transport and Pore Structure*, Academic Press, NY, 1979). In some cases, approximate prediction can be obtained from the following equations.

Mackie and Meares, *Proc. Roy. Soc.*, **A232**, 498 (1955):

$$\tau_p = \frac{(2 - \epsilon_p)^2}{\epsilon_p} \quad (16-65a)$$

Wakao and Smith, *Chem. Eng. Sci.*, **17**, 825 (1962):

$$\tau_p = \frac{1}{\epsilon_p} \quad (16-65b)$$

Suzuki and Smith, *Chem. Eng. J.*, **3**, 256 (1972):

$$\tau_p = \epsilon_p + 1.5(1 - \epsilon_p) \quad (16-65c)$$

In practice, however, the predictive value of these equations is rather uncertain, and vastly different results are obtained from each. All of them, on the other hand, predict that τ_p increases as the porosity decreases.

For catalyst particles, Satterfield (*Heterogeneous Catalysis in Practice*, McGraw-Hill, 1980) recommends the use of a value of $\tau_p = 4$ when no other information is available, and this can be used for many adsorbents. In general, however, it is more reliable to treat the tortuosity as an empirical constant that is determined experimentally for any particular adsorbent.

For adsorbent materials, experimental tortuosity factors generally fall in the range 2–6 and generally decrease as the particle porosity is

increased. Higher apparent values may be obtained when the experimental measurements are affected by other resistances, while values much lower than 2 generally indicate that surface or solid diffusion occurs in parallel to pore diffusion.

Ruthven (gen. refs.) summarizes methods for the measurement of effective pore diffusivities that can be used to obtain tortuosity factors by comparison with the estimated pore diffusion coefficient of the adsorbate. Molecular diffusivities can be estimated with the methods in Sec. 6.

For gas-phase diffusion in small pores at low pressure, the molecular mean free path may be larger than the pore diameter, giving rise to Knudsen diffusion. Satterfield (*Mass Transfer in Heterogeneous Catalysis*, MIT, Cambridge, MA, 1970, p. 43) gives the following expression for the pore diffusivity:

$$D_{pi} = \frac{1}{\tau_p} \left[\frac{3}{4r_{pore}} \left(\frac{\pi M_{fi}}{2RT} \right)^{1/2} + \frac{1}{D_i} \right]^{-1} \quad (16-66)$$

where r_{pore} is the average pore radius, T the absolute temperature, and M_{fi} the molecular weight.

For liquid-phase diffusion of large adsorbate molecules, when the ratio $\lambda_m = r_m/r_{pore}$ of the molecule radius r_m to the pore radius is significantly greater than zero, the pore diffusivity is reduced by steric interactions with the pore wall and hydrodynamic resistance. When $\lambda_m < 0.2$, the following expressions derived by Brenner and Gaydos [*J. Coll. Int. Sci.*, **58**, 312 (1977)] for a hard sphere molecule (a particle) diffusing in a long cylindrical pore, can be used

$$D_{pi} = \frac{D_i}{\tau_p} (1 - \lambda_m)^{-2} \left[1 + \frac{9}{8} \lambda_m \ln \lambda_m - 1.539 \lambda_m \right] \quad (16-67)$$

r_m is the Stokes-Einstein radius of the solute that can be determined from the free diffusivity as

$$r_m = \frac{\kappa T}{6\pi\mu D_i} \quad (16-68)$$

where κ is the Boltzmann constant. When $\lambda_m > 0.2$, the centerline approximation [Anderson and Quinn, *Biophys. J.*, **14**, 130, (1974)] can be used instead of Eq. (16-67)

$$D_{pi} = \frac{D_i}{\tau_p} (1 - 2.1044\lambda_m + 2.089\lambda_m^3 - 0.984\lambda_m^5) \times 0.865 \quad (16-69)$$

The 0.865 factor is used to match this equation to the Brenner and Gaydos expression for $\lambda_m = 0.2$. In these cases, the pore concentration C_{pi} is related to the external concentration C_i by the partition ratio $(1 - \lambda_m)^2$.

Solid Diffusion In the case of pore diffusion discussed above, transport occurs within the fluid phase contained inside the particle; here the solute concentration is generally similar in magnitude to the external fluid concentration. A solute molecule transported by pore diffusion may attach to the sorbent and detach many times along its path. In other cases, attachment can be essentially permanent, but in both cases, only detached molecules undergo transport. In contrast, the following four instances illustrate cases where diffusion of adsorbate molecules occurs in their adsorbed state within phases that are distinct from the pore fluid:

1. Movement of mobile adsorbed solute molecules along pore surfaces, without detaching
2. Transport in a homogeneously dissolved state, as for a neutral molecule inside a sorbent gel or in a pore filled with a liquid which is immiscible with the external fluid
3. Ion transport in charged ion-exchange resins
4. Advance of an adsorbate molecule from one cage to another within a zeolite crystal

In these cases, the diffusion flux may be written in terms of the adsorbed solute concentration as

$$J_i = -\rho_p D_{si} \frac{\partial n_i}{\partial r} \quad (16-70)$$

The diffusion coefficient in these phases D_{si} is usually considerably smaller than that in fluid-filled pores; however, the adsorbate concentration is often much larger. Thus, the diffusion rate can be smaller or

larger than can be expected for pore diffusion, depending on the magnitude of the fluid/solid partition coefficient.

Numerical values for solid diffusivities D_{si} in adsorbents are sparse and vary greatly. Moreover, they may be strongly dependent on the adsorbed phase concentration of solute. Hence, locally conducted experiments and interpretation must be used to a great extent. Summaries of available data for surface diffusivities in activated carbon and other adsorbent materials and for micropore diffusivities in zeolites are given in Ruthven, Yang (1987), Suzuki, and Karger and Ruthven (gen. refs.).

Surface diffusivities are generally strongly dependent on the fractional surface coverage and increase rapidly at surface coverage greater than 80 percent [see, for example, Yang et al., *AIChE J.*, **19**, 1052 (1973)]. For estimation purposes, the correlation of Sladek et al. [*Ind. Eng. Chem. Fundam.*, **13**, 100 (1974)] can be used to predict surface diffusivities for gas-phase adsorption on a variety of adsorbents.

Zeolite crystallite diffusivities for sorbed gases range from 10^{-7} to 10^{-14} cm²/s. These diffusivities generally show a strong increase with the adsorbate concentration that is accounted for by the Darken thermodynamic correction factor

$$D_{si} = D_{oi} \frac{d \ln a_i}{d \ln n_i} \quad (16-71)$$

where D_{oi} is the corrected diffusivity and a_i the thermodynamic activity of the species in the adsorbed phase. Corrected diffusivities D_{oi} calculated according to this equation are often found to be essentially independent of concentration. If the adsorption equilibrium isotherm obeys the Langmuir equation [Eq. (16-13)], Eq. (16-71) yields:

$$D_{si} = D_{oi} \left(1 - \frac{n_i}{n_i^s} \right)^{-1} \quad (16-72)$$

The effect of temperature on diffusivities in zeolite crystals can be expressed in terms of the Eyring equation (see Ruthven, gen. refs.).

In ion-exchange resins, diffusion is further complicated by electrical coupling effects. In a system with M counterions, diffusion rates are described by the Nernst-Planck equations (Helfferich, gen. refs.). Assuming complete Donnan exclusion, these equations can be written as:

$$J_i = -\rho_p \frac{1}{z_i} \sum_{j=1}^{M-1} \bar{D}_{i,j} \frac{\partial z_j n_j}{\partial r} \quad (16-73)$$

$$\bar{D}_{i,j} = -\frac{\bar{D}_i(\bar{D}_j - \bar{D}_M)z_i^2 n_i}{\sum_{k=1}^M \bar{D}_k z_k^2 n_k} \quad (16-74a)$$

$$\bar{D}_{i,i} = \bar{D}_i - \frac{\bar{D}_i(\bar{D}_i - \bar{D}_M)z_i^2 n_i}{\sum_{k=1}^M \bar{D}_k z_k^2 n_k} \quad (16-74b)$$

which are dependent on the **ionic self diffusivities** $\bar{D}_i$ of the individual species. As a qualitative rule, ionic diffusivities of inorganic species in crosslinked polystyrene-DVB ion-exchange resins compared with those in water are 1:10 for monovalent ions, 1:100 for divalent ions, and 1:1000 for trivalent ions. Table 16-8 shows typical ionic diffusivities of inorganic ions in cation and anion exchange resins; larger organic ions, however, can have ionic diffusivities much smaller than inorganic ions of the same valence [see, for example, Jones and Carta, *Ind. Eng. Chem. Research*, **32**, 117 (1993)].

For mixtures of unlike ions (the usual case), the apparent diffusivity will be intermediate between these values because of the electrical coupling effect. For a system with two counterions A and B, with charge z_A and z_B , Eqs. (16-73) and (16-74) reduce to:

$$J_A = -\rho_p \bar{D}_{A,B} \frac{\partial n_A}{\partial r} = -\rho_p \frac{\bar{D}_A \bar{D}_B [z_A^2 n_A + z_B^2 n_B]}{z_A^2 \bar{D}_A n_A + z_B^2 \bar{D}_B n_B} \frac{\partial n_A}{\partial r} \quad (16-75)$$

which shows that the apparent diffusivity $\bar{D}_{A,B}$ varies between $\bar{D}_A$ when the ionic fraction of species A in the resin is very small and $\bar{D}_B$ when the ionic fraction of A in the resin approaches unity, indicating that the ion present in smaller concentration has the stronger effect on the local interdiffusion rate.

TABLE 16-8 Self Diffusion Coefficients in Polystyrene-Divinylbenzene Ion Exchangers
(Units of $10^{-7} \text{ cm}^2/\text{s}$)^a

Temperature	0.3 °C					25 °C				
Crosslinking, %	4	6	8	12	16	4	6	8	12	16
Cation exchangers (sulfonated): Dowex 50										
Na ⁺	6.7		3.4	1.15	0.66	14.1		9.44		2.40
Cs ⁺			6.6		1.11			13.7		3.10
Ag ⁺			2.62		1.00			6.42		2.75
Zn ²⁺			0.21		0.03			0.63		0.14
La ³⁺	0.30		0.03		0.002			0.092		0.005
Anion exchangers (dimethyl hydroxyethylamine): Dowex 2										
Cl ⁻		1.25					3.54			
Br ⁻	(1.8)	1.50	0.63		0.06	(4.3)	3.87	2.04		0.26
I ⁻		0.35					1.33			
BrO ₃ ⁻		1.76					4.55			
WO ₄ ²⁻		0.60					1.80			
PO ₄ ³⁻		0.16					0.57			

^aData from Boyd and Soldano, *J. Am. Chem. Soc.*, **75**, 6091 (1953).

Combined Pore and Solid Diffusion In porous adsorbents and ion-exchange resins, intraparticle transport can occur with pore and solid diffusion in parallel. The dominant transport process is the faster one, and this depends on the relative diffusivities and concentrations in the pore fluid and in the adsorbed phase. Often, equilibrium between the pore fluid and the solid phase can be assumed to exist locally at each point within a particle. In this case, the mass-transfer flux is expressed by:

$$J_i = -\left[\epsilon_p D_{pi} + \rho_p D_{si} \frac{dn_i^e}{dc_i}\right] \frac{\partial c_{pi}}{\partial r} = -D_{ei}(c_{pi}) \frac{\partial c_{pi}}{\partial r} \quad (16-76)$$

where dn_i^e/dc_i is the derivative of the adsorption isotherm and it has been assumed that at equilibrium $c_{pi} = c_i$. This equation suggests that in such an adsorbent, pore and solid diffusivities can be obtained by determining the apparent diffusivity D_{ei} for conditions of no adsorption ($dn_i^e/dc_i = 0$) and for conditions of strong adsorption, where dn_i^e/dc_i is large. If the adsorption isotherm is linear over the range of experimental measurement:

$$D_{ei} = \epsilon_p D_{pi} + \rho_p K_i D_{si} \quad (16-77)$$

Thus, a plot of the apparent diffusivity versus the linear adsorption equilibrium constant should be linear so long as D_{pi} and D_{si} remain constant.

In a particle having a **bidispersed pore structure** comprising spherical adsorptive subparticles of radius r_s forming a macroporous aggregate, separate flux equations can be written for the macroporous network in terms of Eq. (16-64) and for the subparticles themselves in terms of Eq. (16-70) if solid diffusion occurs.

EXTERNAL MASS TRANSFER

Because of the complexities encountered with a rigorous treatment of the hydrodynamics around particles in industrial contactors, mass transfer to and from the adsorbent is described in terms of a mass-transfer coefficient k_f . The flux at the particle surface is:

$$N_i = k_f(c_i - c_i^s) \quad (16-78)$$

where c_i and c_i^s are the solute concentrations in the bulk fluid and at the particle surface, respectively. k_f can be estimated from available correlations in terms of the Sherwood number $Sh = k_f d_p / D_i$ and the Schmidt number $Sc = \nu / D_i$. For packed-bed operations, the correlations in Table 16-9 are recommended. A plot of these equations is given in Fig. 16-10 for representative ranges of Re and Sc with $\epsilon = 0.4$.

External mass-transfer coefficients for particles suspended in agitated contactors can be estimated from equations in Levins and Glastonbury [*Trans. Instn. Chem. Eng.*, **50**, 132 (1972)] and Armenante and Kirwan [*Chem. Eng. Sci.*, **44**, 2871 (1989)].

AXIAL DISPERSION IN PACKED BEDS

The axial dispersion coefficient [cf. Eq. (16-51)] lumps together all mechanisms leading to axial mixing in packed beds. Thus, the axial dispersion coefficient must account not only for molecular diffusion and convective mixing but also for nonuniformities in the fluid velocity across the packed bed. As such, the axial dispersion coefficient is best determined experimentally for each specific contactor.

The effects of **flow nonuniformities**, in particular, can be severe in gas systems when the ratio of bed-to-particle diameters is small; in liquid systems when viscous fingering occurs as a result of large

TABLE 16-9 Recommended Correlations for External Mass Transfer Coefficients in Adsorption Beds ($Re = \epsilon \nu d_p / \nu$, $Sc = \nu / D$)

Equation	Re	Phase	Ref.
$Sh = 1.15 \left(\frac{Re}{\epsilon} \right)^{0.5} Sc^{0.33}$	$Re > 1$	Gas/liquid	Carberry, <i>AIChE J.</i> , 6 , 460 (1960)
$Sh = 2.0 + 1.1 Re^{0.6} Sc^{0.33}$	$3 < Re < 10^4$	Gas/liquid	Wakao and Funazkri, <i>Chem. Eng. Sci.</i> , 33 , 1375 (1978)
$Sh = 1.85 \left(\frac{1 - \epsilon}{\epsilon} \right)^{0.33} Re^{0.33} Sc^{0.33}$	$Re < 40$	Liquid	Kataoka et al., <i>J. Chem. Eng. Japan</i> , 5 , 132 (1972)
$Sh = \frac{1.09}{\epsilon} Re^{0.33} Sc^{0.33}$	$0.0015 < Re < 55$	Liquid	Wilson and Geankoplis, <i>Ind. Eng. Chem. Fundam.</i> , 5 , 9 (1966)
$Sh = \frac{0.25}{\epsilon} Re^{0.69} Sc^{0.33}$	$55 < Re < 1050$	Liquid	Wilson and Geankoplis, <i>Ind. Eng. Chem. Fundam.</i> , 5 , 9 (1966)

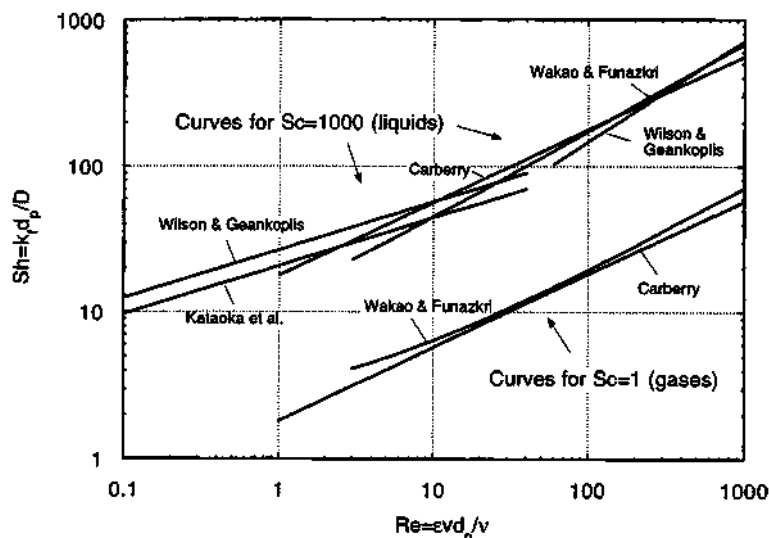


FIG. 16-10 Sherwood number correlations for external mass-transfer coefficients in packed beds for $\epsilon = 0.4$. (Adapted from Suzuki, *gen. refs.*)

viscosity gradients in the adsorption bed; when very small particles ($<50 \mu\text{m}$) are used, such as in high performance liquid chromatography systems; and in large-diameter beds. A lower bound of the axial dispersion coefficient can be estimated for well-packed beds from correlations that follow.

Neglecting flow nonuniformities, the contributions of molecular diffusion and turbulent mixing arising from stream splitting and recombination around the sorbent particles can be considered additive [Langer et al., *Int. J. Heat and Mass Transfer*, **21**, 751 (1978)]; thus, the axial dispersion coefficient D_L is given by:

$$\frac{D_L}{D_i} = \gamma_1 + \gamma_2 \frac{d_p v}{D_i} = \gamma_1 + \gamma_2 \frac{(Re)(Sc)}{\epsilon} \quad (16-79)$$

or, in terms of a particle-based Peclet number ($Pe = d_p v/D_L$), by:

$$\frac{1}{Pe} = \frac{\gamma_1 \epsilon}{(Re)(Sc)} + \gamma_2 \quad (16-80)$$

The first term in Eqs. (16-79) and (16-80) accounts for molecular diffusion, and the second term accounts for mixing. For the first term, Wicke [Ber. Bunsenges, **77**, 160 (1973)] has suggested:

$$\gamma_1 = 0.45 + 0.55\epsilon \quad (16-81)$$

which, for typical void fractions, $\epsilon = 0.35 - 0.5$ gives $\gamma_1 = 0.64 - 0.73$ (Ruthven, *gen. refs.*). Expressions for the axial mixing term, γ_2 in Eq. (16-79) are given in Table 16-10. The expression of Wakao and Funazkri includes an axial diffusion term, γ_1 , that varies from 0.7 for nonporous particles to $20/\epsilon$, depending on the intraparticle mass-transfer mechanism. For strongly adsorbed species, Wakao and Funazkri suggest that the effective axial dispersion coefficient is much larger than that predicted on the basis of nonporous, nonadsorbing particles. The Gunn expression includes a term σ_v^2 accounting for deviations from plug flow. σ_v^2 is defined as the dimensionless variance of the distribution of the ratio of velocity to average velocity over the cross section of the bed. The parameter values included in this equation are valid for spherical particles. Values for nonspherical particles can be found in the original reference.

Figure 16-11 compares predicted values of D_L/D_i for $\sigma_v = 0$ and $\epsilon = 0.4$ with $Sc = 1$ (gases at low pressure), and $Sc = 1000$ (liquids), based on the equations in Table 16-10.

Correlations for axial dispersion in beds packed with very small particles ($<50 \mu\text{m}$) that take into account the holdup of liquid in a bed are discussed by Horvath and Lin [*J. Chromatogr.*, **126**, 401 (1976)].

RATE EQUATIONS

Rate equations are used to describe interphase mass transfer in batch systems, packed beds, and other contacting devices for sorptive processes and are formulated in terms of fundamental transport properties of adsorbent and adsorbate.

General Component Balance For a spherical adsorbent particle:

$$\epsilon_p \frac{\partial c_{pi}}{\partial t} + \rho_p \frac{\partial n_i}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} (-r^2 N_i) \quad (16-82)$$

For particles that have no macropores, such as gel-type ion-exchange resins, or when the solute holdup in the pore fluid is small, ϵ_p may be taken as zero. Ignoring bulk flow terms, the fluxes N_i and J_i are equal. In this case, coupling the component balance with the flux expressions previously introduced gives the rate equations in Table 16-11. Typical boundary conditions are also included in this table. Generally, these equations apply to particles that can be approximated as spherical and of a uniform size and properties. An appropriately chosen mean particle size must be used in these equations when dealing with adsorbents having a broad particle size distribution. The appropriate average depends on the controlling mass-transfer mechanism. For intraparticle mass-transfer mechanisms, the volume or mass-average particle size usually provides the best prediction.

Linear Driving Force Approximation Simplified expressions can also be used for an approximate description of adsorption in terms of rate coefficients for both extraparticle and intraparticle mass transfer controlling. As an approximation, the rate of adsorption on a particle can be written as:

$$\frac{\partial \hat{n}_i}{\partial t} = k f(n_i, c_i) \quad (16-83)$$

where k is a rate coefficient, and the function $f(n_i, c_i)$ is a driving force relationship. The variables k_c and k_n are used to denote rate coefficients based on fluid-phase and adsorbed-phase concentration driving forces, respectively.

TABLE 16-10 Coefficients for Axial Dispersion Correlations in Packed Beds Based on Eq. (16-79)

γ_1	γ_2	Ref.
0.73	$0.5 \left(1 + \frac{13\gamma_1\epsilon}{ReSc} \right)^{-1}$	Edwards and Richardson, <i>Chem. Eng. Sci.</i> , 23 , 109 (1968)
Nonporous particles: 0.7 Porous particles: $\leq 20/\epsilon$	0.5	Wakao and Funazkri, <i>Chem. Eng. Sci.</i> , 33 , 1375 (1978)
1	$\frac{3}{4}\epsilon + \frac{\pi^2\epsilon(1-\epsilon)}{6} \ln(ReSc)$	Koch and Brady, <i>J. Fluid Mech.</i> , 154 , 399 (1985)
0.714	$\frac{\sigma_v^2}{2} + (1 + \sigma_v^2) \left\{ \gamma(1-p)^2 + \gamma^2 p(1-p)^3 \left[e^{-\frac{1}{\gamma p(1-p)}} - 1 \right] \right\}$ with $\gamma = 0.043 ReSc/(1-\epsilon)$ $p = 0.33 \exp(-24/Re) + 0.17$	Gunn, <i>Chem. Eng. Sci.</i> , 2 , 363 (1987)

Commonly used forms of this rate equation are given in Table 16-12, where $\Lambda = \rho_p n_i^{ref}/c_i^{ref}$ is the partition ratio based on the feed concentration as a reference [cf. Eq. (16-125)]. For adsorption bed calculations with constant separation factor systems, somewhat improved predictions are obtained using correction factors ψ_s and ψ_p defined in Table 16-12.

The linear driving force (LDF) approximation is obtained when the driving force is expressed as a concentration difference. It was originally developed to describe packed-bed dynamics under linear equilibrium conditions [Glueckauf, *Trans. Far. Soc.*, **51**, 1540 (1955)]. This form is exact for a nonlinear isotherm only when external mass transfer is controlling. However, it can also be used for nonlinear systems with pore or solid diffusion mechanisms as an approximation, since it provides qualitatively correct results.

Alternate driving force approximations, item 2B in Table 16-12 for solid diffusion, and item 3B in Table 16-12 for pore diffusion, provide somewhat more accurate results in constant pattern packed-bed calculations with pore or solid diffusion controlling for constant separation factor systems.

The reaction kinetics approximation is mechanistically correct for systems where the reaction step at pore surfaces or other fluid-solid interfaces is controlling. This may occur in the case of chemisorption on porous catalysts and in affinity adsorbents that involve very slow binding steps. In these cases, the mass-transfer parameter k is replaced

by a second-order reaction rate constant k_a . The driving force is written for a constant separation factor isotherm (column 4 in Table 16-12). When diffusion steps control the process, it is still possible to describe the system by its apparent second-order kinetic behavior, since it usually provides a good approximation to a more complex exact form for single transition systems (see "Fixed Bed Transitions").

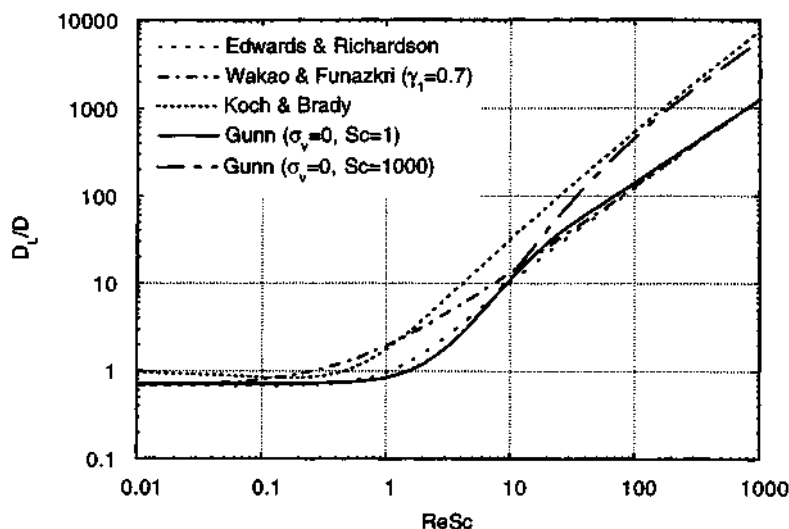
Combined Intraparticle Resistances When solid diffusion and pore diffusion operate in parallel, the effective rate is the sum of these two rates. When solid diffusion predominates, mass transfer can be represented approximately in terms of the LDF approximation, replacing k_n in column 2 of Table 16-12 with

$$k_n^c = \frac{15\psi_s D_{sl}}{r_p^2} + \frac{15(1-\epsilon)\psi_p \epsilon_p D_{pi}}{\Lambda r_p^2} \quad (16-84)$$

When pore diffusion predominates, use of column 3 in Table 16-12 is preferable, with k_n^c replacing k_n .

For particles with a **bidispersed pore structure**, the mass-transfer parameter k_n in the LDF approximation (column 2 in Table 16-12) can be approximated by the series-combination of resistances as:

$$\frac{1}{k_n} = \frac{1}{b_s} \left[\frac{\Lambda r_p^2}{15(1-\epsilon)\psi_p \epsilon_p D_{pi}} + \frac{r_s^2}{15\psi_s D_{sl}} \right] \quad (16-85)$$

FIG. 16-11 Axial dispersion coefficient correlations for well-packed beds for $\epsilon = 0.4$.

16-24 ADSORPTION AND ION EXCHANGE

TABLE 16-11 Rate Equations for Description of Mass Transfer in Spherical Adsorbent Particles

Mechanism	Flux equation	Rate equation
A. Pore diffusion	16-64	$\epsilon_p \frac{\partial c_{pi}}{\partial t} + \rho_p \frac{\partial n_i}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(\epsilon_p D_{pi} r^2 \frac{\partial c_{pi}}{\partial r} \right)$ $(\partial c_{pi} / \partial r)_{r=0} = 0, (\epsilon_p D_{pi} \partial c_{pi} / \partial r)_{r=r_p} = k_f (c_i - c_{pi _{r=r_p}})$ or $(c_{pi})_{r=r_p} = c_i$ for no external resistance
B. Solid diffusion	16-70	$\frac{\partial n_i}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(D_{si} r^2 \frac{\partial n_i}{\partial r} \right)$ $(\partial n_i / \partial r)_{r=0} = 0, (\rho_p D_{si} \partial n_i / \partial r)_{r=r_p} = k_f (c_i - c_i^e)$ or $(n_i)_{r=r_p} = n_i^e (c_i)$ for no external resistance
C. Parallel pore and solid diffusion (local equilibrium between pore and adsorbed phase)	16-76	$\left(\epsilon_p + \rho_p \frac{dn_i^e}{dc_i} \right) \frac{\partial c_{pi}}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left[r^2 \left(\epsilon_p D_{pi} + \rho_p D_{si} \frac{dn_i^e}{dc_i} \right) \frac{\partial c_{pi}}{\partial r} \right]$ $(\partial c_{pi} / \partial r)_{r=0} = 0, [(\epsilon_p D_{pi} + \rho_p D_{si} dn_i^e / dc_i) \partial c_{pi} / \partial r]_{r=r_p} = k_f (c_i - c_{pi _{r=r_p}})$ or $(c_{pi})_{r=r_p} = c_i$ for no external resistance
D. Diffusion in bidispersed particles (no external resistance)	16-64 and 16-70	$\frac{\partial n_i}{\partial t} = \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left(D_{si} \rho^2 \frac{\partial n_i}{\partial \rho} \right), (\partial n_i / \partial \rho)_{\rho=0} = 0, (n_i)_{\rho=r_p} = n_i^e (c_{pi})$ $\bar{n}_i(r, t) = \frac{3}{r_s^3} \int_0^{r_s} \rho^2 n_i d\rho$ $\epsilon_p \frac{\partial c_{pi}}{\partial t} + \rho_p \frac{\partial \bar{n}_i}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(\epsilon_p D_{pi} r^2 \frac{\partial c_{pi}}{\partial r} \right), (\partial c_{pi} / \partial r)_{r=0} = 0, (c_{pi})_{r=r_p} = c_i$

where b_s is a correction to the driving force that is described below. In the limiting cases where the controlling resistance is diffusion through the particle pores or diffusion within the subparticles, the rate coefficients $k_n = 15(1 - \epsilon)\psi_p \epsilon_p D_{pi} / \Lambda r_p^2$ and $k_n = 15\psi_s D_{si} / r_s^2$ are obtained.

Overall Resistance With a linear isotherm ($R = 1$), the overall mass transfer resistance is the sum of intraparticle and extraparticle resistances. Thus, the overall LDF coefficient for use with a particle-side

driving force (column 2 in Table 16-12) is:

$$\frac{1}{k_n^o} = \frac{\Lambda r_p}{3(1 - \epsilon)k_f} + \frac{1}{k_n^c} \quad (16-86)$$

or

$$\frac{1}{k_n^o} = \frac{\rho_b r_p}{3(1 - \epsilon)k_f} + \frac{\rho_b}{\Lambda k_n^c} \quad (16-87)$$

for use with a fluid-phase driving force (column 1 in Table 16-12).

TABLE 16-12 Expressions for Rate Coefficient k and Driving Force Relationships for Eq. (16-83)

Mechanism	1. External film	2. Solid diffusion	3. Pore diffusion	4. Reaction kinetics
Expression for rate coefficient, k	$k_c = \frac{k_f a}{\rho_b} = \frac{3(1 - \epsilon)k_f}{\rho_b r_p}$	$k_n = \frac{15\psi_s D_{si}}{r_p^2}$	$k_n = \frac{15\psi_p (1 - \epsilon)\epsilon_p D_{pi}}{\Lambda r_p^2}$	k_a
A. Linear driving force (LDF)	$c - c_i^e$	$n_i^e - \bar{n}_i$	$n_i^e - \bar{n}_i$	—
LDF for constant R	$c_i - \frac{R c_i^{\text{ref}} \bar{n}_i / n_i^{\text{ref}}}{1 - (R - 1)\bar{n}_i / n_i^{\text{ref}}}$	$\frac{n_i^{\text{ref}} c_i / c_i^{\text{ref}}}{R + (R - 1)c_i / c_i^{\text{ref}}} - \bar{n}_i$	$\frac{n_i^{\text{ref}} c_i / c_i^{\text{ref}}}{R + (R - 1)c_i / c_i^{\text{ref}}} - \bar{n}_i$	—
Correction factors ψ for constant R	—	$\frac{0.894}{1 - 0.106R^{0.5}}$	$\frac{0.775}{1 - 0.225R^{0.5}}$	—
B. Alternate driving force for constant R	—	$\frac{n_i^{e2} - \bar{n}_i^2}{2\bar{n}_i}$	$\frac{n_i^e - \bar{n}_i}{[1 - (R - 1)\bar{n}_i / n_i^{\text{ref}}]^{0.5}}$	—
Correction factors ψ for alternate driving force	—	$\frac{0.590}{1 - 0.410R^{0.5}}$	$\frac{0.548}{1 - 0.452R^{0.5}}$	—
C. Reaction kinetics for constant R	—	—	—	$\frac{c_i(n_i^{\text{ref}} - \bar{n}_i) - R\bar{n}_i(c_i^{\text{ref}} - c_i)}{1 - R}$

References: 1A. Beaton and Furnas, *Ind. Eng. Chem.*, **33**, 1500 (1941); Michaels, *Ind. Eng. Chem.*, **44**, 1922 (1952)
2A,3A. Glueckauf and Coates, *J. Chem. Soc.*, 1315 (1947); *Trans. Faraday Soc.*, **51**, 1540 (1955); Hall et al., *Ind. Eng. Chem. Fundam.*, **5**, 212 (1966)
2B. Vermeulen, *Ind. Eng. Chem.*, **45**, 1664 (1953)
3B. Vermeulen and Quilici, *Ind. Eng. Chem. Fundam.*, **9**, 179 (1970)
4C. Hiester and Vermeulen, *Chem. Eng. Progr.*, **48**, 505 (1952)

In either equation, k_n^c is given by Eq. (16-84) for parallel pore and surface diffusion or by Eq. (16-85) for a bidispersed particle. For nearly linear isotherms ($0.7 < R < 1.5$), the same linear addition of resistance can be used as a good approximation to predict the adsorption behavior of packed beds, since solutions for all mechanisms are nearly identical. With a highly favorable isotherm ($R \rightarrow 0$), however, the rate at each point is controlled by the resistance that is locally greater, and the principle of additivity of resistances breaks down. For approximate calculations with intermediate values of R , an overall transport parameter for use with the LDF approximation can be calculated from the following relationship for solid diffusion and film resistance in series:

$$\frac{\Lambda r_p}{3(1-\varepsilon)k_f} + \frac{r_p^2}{15\Psi_s D_{si}} = \frac{b_s}{k_n^c} = \frac{b_f \Lambda}{\rho_b k_c^o} \quad (16-88)$$

b_s and b_f are correction factors that are given by Fig. 16-12 as a function of the separation factor R and the mechanism parameter

$$\Psi = \frac{10\Psi_s D_{si} \Lambda}{D_i} \frac{1}{Sh} \quad (16-89)$$

Axial Dispersion Effects In adsorption bed calculations, axial dispersion effects are typically accounted for by the axial diffusion-like term in the bed conservation equations [Eqs. (16-51) and (16-52)]. For nearly linear isotherms ($0.5 < R < 1.5$), the combined effects of axial dispersion and mass-transfer resistances on the adsorption behavior of packed beds can be expressed approximately in terms of an apparent rate coefficient k_c for use with a fluid-phase driving force (column 1, Table 16-12):

$$\frac{1}{k_c} = \frac{\rho_b}{\varepsilon} \frac{D_L}{v^2} + \frac{1}{k_c^o} \quad (16-90)$$

which extends the linear addition principle to combined axial dispersion and mass-transfer resistances. Even for a highly nonlinear isotherm ($R = 0.33$), the linear addition principle expressed by this equation provides a useful approximation except in the extreme case

of low mass-transfer resistance and large axial dispersion, when $D_L \rho_b k_c^o / v^2 \varepsilon \gg 5$ [Garg and Ruthven, *Chem. Eng. Sci.*, **30**, 1192 (1975)]. However, when the isotherm is irreversible ($R \rightarrow 0$), the linear addition principle breaks down and axial dispersion has to be taken into account by explicit models (see "Fixed Bed Transitions").

Rapid Adsorption-Desorption Cycles For rapid cycles with particle diffusion controlling, when the cycle time t_c is much smaller than the time constant for intraparticle transport, the LDF approximation becomes inaccurate. The generalized expression

$$\frac{\partial \hat{n}_i}{\partial t} = \Omega k_n (n_i^e - \bar{n}_i) \quad (16-91)$$

can be used for packed-bed calculations when the parameter Ω is defined to be a function of the cycle time such that the amount of solute adsorbed and desorbed during a cycle is equal to that obtained by solution of the complete particle diffusion equations. Graphical and analytical expressions for Ω in the case of a single particle, usable for very short beds, are given by Nakao and Suzuki [*J. Chem. Eng. Japan*, **16**, 114 (1983)] and Carta [*Chem. Eng. Sci.*, **48**, 622 (1993)]. With equal adsorption and desorption times, $t_a = t_d = t_c/2$, Ω approaches the value $\pi^2/15$ for long cycle times and the asymptote $\Omega = 1.877/\sqrt{t_c k_n}$ for short cycle times [Alpay and Scott, *Chem. Eng. Sci.*, **47**, 499 (1992)]. However, other results by Raghavan et al. [*Chem. Eng. Sci.*, **41**, 2787 (1986)] indicate that a limiting constant value of Ω (larger than 1) is obtained for very short cycles, when calculations are carried out for beds of finite length.

Determination of Controlling Rate Factor The most important physical variables determining the controlling dispersion factor are particle size and structure, flow rate, fluid- and solid-phase diffusivities, partition ratio, and fluid viscosity. When multiple resistances and axial dispersion can potentially affect the rate, the spreading of a concentration wave in a fixed bed can be represented approximately in terms of the single rate parameter k . In customary separation-process calculations, the height of an adsorption bed can be calculated approximately as the product of the number of transfer units times the height of one fluid-phase transfer unit (HTU). The HTU is related to the LDF rate parameters k_c and k_n by:

$$HTU = \frac{\varepsilon v}{\rho_b k_c} = \frac{\varepsilon v}{\Lambda k_n} \quad (16-92)$$

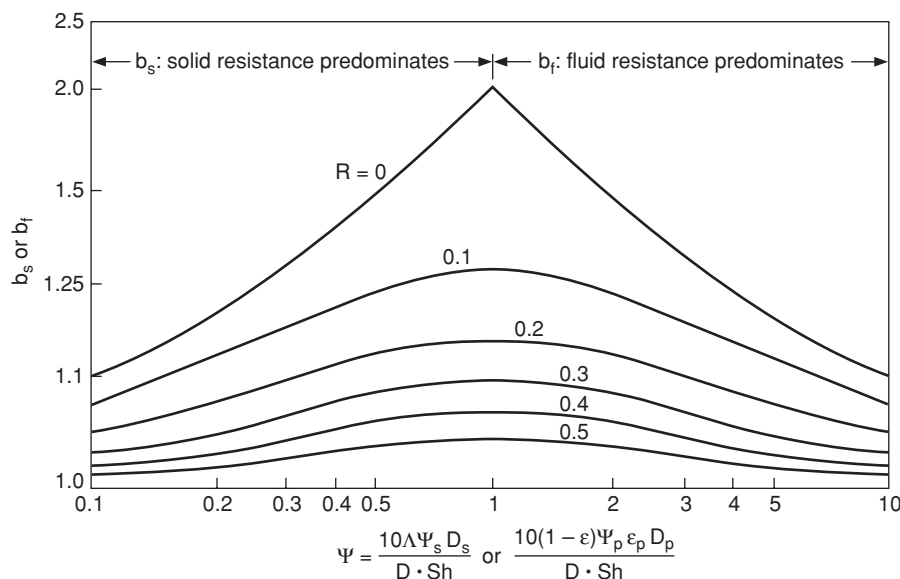


FIG. 16-12 Correction factors for addition of mass-transfer resistances, relative to effective overall solid phase or fluid phase rates, as a function of the mechanism parameter. Each curve corresponds to both b_s and b_f over its entire range.

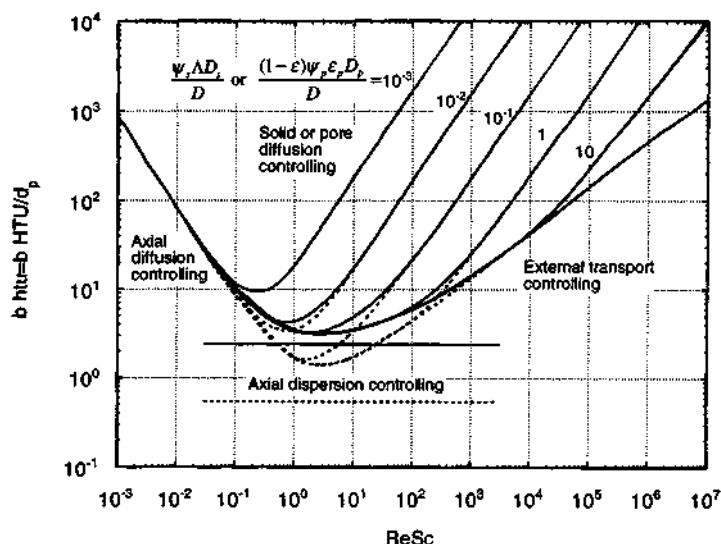


FIG. 16-13 Effect of $ReSc$ group, distribution ratio, and diffusivity ratio on height of a transfer unit. Dotted lines for gas and solid lines for liquid-phase systems.

Figure 16-13 is a plot of the dimensionless HTU ($htu = HTU/d_p$) multiplied times the correction factor b_f (between 1 and 2) as a function of the dimensionless velocity $(Re)(Sc) = \epsilon v d_p / D$ and a ratio of the controlling diffusivity to the fluid-phase diffusivity, generated on the basis of results of Vermeulen et al. (gen. ref.) using typical values of the individual physical factors likely to be found in adsorption beds. This figure can be used to determine the controlling rate factor from a knowledge of individual physical parameters. If fluid-side effects control, the dimensionless HTU is given by the bottom curve (dotted for gas and solid for liquid-phase systems). If particle-side diffusivities control, the dimensionless HTU is given by a point above the lower envelope on the appropriate diffusional contour (through the ψ_p , the contour value depends slightly on the separation factor R). If pore and solid diffusion occur in parallel, the reciprocal of the HTU is the sum of the reciprocals of the HTU values for the two mechanisms. Near the intersections of the diffusional contours with the envelope, the dimensionless HTU is the sum of the HTU values for fluid-side and particle-side resistances.

Example 7: Estimation of Rate Coefficient for Gas Adsorption

An adsorption bed is used to remove methane from a methane-hydrogen mixture at 10 atm (abs.) (10.1 bar) and 25°C (298 K), containing 10 mol % methane. Activated carbon particles having a mean diameter $d_p = 0.17$ cm, a surface area $A = 1.1 \times 10^7$ cm²/g, a bulk density $\rho_b = 0.509$ g/cm³, a particle density $\rho_p = 0.777$ g/cm³, and a skeletal density $\rho_s = 2.178$ g/cm³ is used as the adsorbent. Based on data of Grant et al. [AIChE J., 8, 403 (1962)], adsorption equilibrium is represented by $n = 2.0 \times 10^{-3} K_A p_A / (1 + K_A p_A)$ mol/g adsorbent, with $K_A = 0.346$ atm⁻¹. Estimate the rate coefficient and determine the controlling rate factor for a superficial velocity of 30 cm/s.

1. The intraparticle void fraction is $\epsilon_p = (0.777^{-1} - 2.178^{-1}) / (0.777^{-1}) = 0.643$ and the extraparticle void fraction is $\epsilon = (0.509^{-1} - 0.777^{-1}) / (0.509^{-1}) = 0.345$. The pore radius is estimated from $r_p = 2\epsilon_p / (A \rho_p) = 1.5 \times 10^{-7}$ cm.

2. The fluid phase diffusivity is $D = 0.0742$ cm²/s. The pore diffusivity is estimated from Eq. (16-66) with a tortuosity factor $\tau_p = 4$; $D_p = 1.45 \times 10^{-3}$ cm²/s.

3. The fluid-side mass transfer coefficient is estimated from Fig. 16-10. For these conditions, $v = 0.108$ cm/s, $Re = 30 \times 0.17/0.108 = 47$, and $Sc = 0.108/0.0742 = 1.5$. From Fig. 16-10 or equations in Table 16-9, $Sh = 13$.

4. The isotherm parameters based on the feed concentration are $R = 1 / (1 + K_A p_A) = 0.4$ and $\Lambda = 0.509 \times 7.68 \times 10^{-3} / 4.09 \times 10^{-5} = 9.56$. For pore diffusion, $\psi_p = 0.961$ from item 3A in Table 16-12. Thus, $(1 - \epsilon) \psi_p \epsilon_p D_p / D = (1 - 0.345) \times 0.961 \times 0.643 \times 1.45 \times 10^{-3} / 0.0742 = 7.9 \times 10^{-3}$. From Fig. 16-13 at $ReSc = 69$, b htu ~ 150 . b is found from Fig. 16-12. However, since the mechanism parameter Ψ is very small, $b \sim 1$. Thus, $k_a = \epsilon v / (htu d_p \Lambda) = 0.12$ s⁻¹. This value applies to the driving force $n_i^* - \bar{n}_i$. Since pore diffusion is dominant, this

value is very close to the value $k_a = 15(1 - \epsilon) \psi_p \epsilon_p D_p / (r_p^2 \Lambda) = 0.13$ s⁻¹ obtained directly from Table 16-12. It should be noted that surface diffusion is neglected in this estimation. Its occurrence could significantly affect the overall mass transfer rate (see Suzuki, gen. refs., pp. 70–85).

Example 8: Estimation of Rate Coefficient for Ion Exchange

Estimate the rate coefficient for flow of a 0.01-M water solution of NaCl through a bed of cation exchange particles in hydrogen form with $\epsilon = 0.4$. The superficial velocity is 0.2 cm/s and the temperature is 25°C. The particles are 600 μ m in diameter, and the diffusion coefficient of sodium ion is 1.2×10^{-5} cm²/s in solution and 9.4×10^{-7} cm²/s inside the particles (cf. Table 16-8). The bulk density is 0.7 g dry resin/cm³ of bed, and the capacity of the resin is 4.9 mequiv/g dry resin. The mass action equilibrium constant is 1.5.

1. Estimate the fluid-side mass transfer coefficient; $Re = \epsilon v d_p / \nu = 0.2 \times 0.06/0.00913 = 1.3$, $Sc = \nu/D = 0.00913/1.2 \times 10^{-5} = 761$. From Fig. 16-10 or Table 16-9, $Sh = 23$. Thus, $k_f = D Sh/d_p = 4.5 \times 10^{-3}$ cm/s.

2. From the equilibrium constant, $R = 1/K_{Na,H} = 0.67$. Thus, from Table 16-12, item 2A, $\psi_p = 0.979$. Using $n^{ref} = 4.9$ mequiv/g and $c^{ref} = 0.01$ mmole/cm³, $\Lambda = \rho_p n^{ref}/c^{ref} = 343$. Thus, $\psi_p D_p \Lambda/D = 26$ and the external mass-transfer resistance is controlling (cf. Fig. 16-13).

3. The rate coefficient for use with a fluid-phase driving force is $k_c = 3(1 - \epsilon)k_f/(\rho_p r_p) = 0.39$ cm³/g(s).

Example 9: Estimation of Rate Coefficient for Protein Adsorption

Estimate the rate coefficient for the adsorption of an antibody ($M_r = 150,000$) from a 1 mg/cm³ aqueous solution ($\mu = 1$ cP) in a column with $\epsilon = 0.4$ operated at 0.05 cm/s and 25°C. The adsorbent particles have diameter $d_p = 100$ μ m, porosity $\epsilon_p = 0.5$, and pore radius $r_{pore} = 30$ nm. The adsorption isotherm is $n = 500c / (1 + 9.0c)$, where n is in mg/cm³ particle volume and c in mg/cm³.

1. Estimate k_f . $Re = \epsilon v d_p / \nu = 0.05 \times 100 \times 10^{-4} / 0.01 = 0.05$. For a 150,000 molecular mass globular protein in water $D = 4 \times 10^{-7}$ cm²/s [Tyn and Gusek, *Biotechnol. Bioeng.*, 35, 327 (1990)], $Sc = \nu/D = 25,000$. From Table 16-9, $Sh = (1.09/0.4)(0.05)^{0.33}(25,000)^{0.33} = 29$. Thus, $k_f = D Sh/d_p = 1.1 \times 10^{-3}$ cm/s.

2. Determine the controlling resistance. From the isotherm, $n^{ref} = (500 \times 1) / (1 + 9.0 \times 1) = 50$ mg/cm³ particle. Thus, $\Lambda = (1 - \epsilon)n^{ref}/c^{ref} = (1 - 0.4) \times 50/1 = 30$ and $R = 1/(1 + 9.0 \times 1) = 0.1$ [cf. Eqs. (16-33) and (16-34)]. From Eq. (16-68), $r_m = \kappa T / 6\pi\eta D = (1.38 \times 10^{-16} \times 298) / (6\pi \times 0.01 \times 4 \times 10^{-7}) = 5.5$ nm. Thus, $\lambda_{m0} = r_m / r_{pore} = 5.5/30 = 0.18$. From Eq. (16-67) with $\tau_p = 4$ obtain $D_p = 5.8 \times 10^{-8}$ cm²/s. From Table 16-12, item 3A, $\psi_p = 0.83$, giving $10(1 - \epsilon)\psi_p \epsilon_p D_p / (D \cdot Sh) = 0.012$. Thus, from Fig. 16-13, the intraparticle pore diffusion resistance is dominant. For these conditions, from Table 16-12 we have $k_a = 15\psi_p(1 - \epsilon)\epsilon_p D_p / (\Lambda r_p^2) = 2.8 \times 10^{-4}$ s⁻¹. A similar approximate result is obtained directly from Fig. 16-13 with $(Re)(Sc) = 0.05 \times 25,000 = 1250$ and $(1 - \epsilon)\psi_p \epsilon_p D_p / D \sim 0.04$, giving $HTU/d_p \sim 1,000$. $k_a \sim 2 \times 10^{-4}$ s⁻¹ is obtained by using the HTU value in Eq. (16-92). These calculations assume that intraparticle transport is by pore diffusion alone, typically limiting for protein adsorption in porous adsorbents.

BATCH ADSORPTION

In this section, we consider the transient adsorption of a solute from a dilute solution in a constant-volume, well-mixed batch system or, equivalently, adsorption of a pure gas. The solutions provided can approximate the response of a stirred vessel containing suspended adsorbent particles, or that of a very short adsorption bed. Uniform, spherical particles of radius r_p are assumed. These particles, initially of uniform adsorbate concentration, are assumed to be exposed to a step change in concentration of the external fluid.

In general, solutions are obtained by coupling the basic conservation equation for the batch system, Eq. (16-49) with the appropriate rate equation. Rate equations are summarized in Tables 16-11 and 16-12 for different controlling mechanisms.

Solutions are provided for external mass-transfer control, intraparticle diffusion control, and mixed resistances for the case of constant V_f and $F_{v, in} = F_{v, out} = 0$. The results are in terms of the fractional approach to equilibrium $F = (\hat{n}_i - \hat{n}_i^0)/(\hat{n}_i^\infty - \hat{n}_i^0)$, where $\hat{n}_i^0$ and $\hat{n}_i^\infty$ are the initial and ultimate solute concentrations in the adsorbent. The solution concentration is related to the amount adsorbed by the material balance $c_i = c_i^0 - (\hat{n}_i - \hat{n}_i^0)M_s/V_f$.

Two general cases are considered: (1) adsorption under conditions of constant or nearly constant external solution concentration (equivalent to infinite fluid volume); and (2) adsorption in a batch with finite volume. In the latter case, the fluid concentration varies from c_i^0 to c_i^∞ when equilibrium is eventually attained. $\Lambda^\infty = (c_i^0 - c_i^\infty)/c_i^0 = M_s(\hat{n}_i^\infty - \hat{n}_i^0)/(V_f c_i^0)$ is a partition ratio that represents the fraction of adsorbate that is ultimately adsorbed. It determines which general case should be considered in the analysis of experimental systems. Generally, when $\Lambda^\infty \geq 0.1$, solutions for the second case are required.

EXTERNAL MASS-TRANSFER CONTROL

The intraparticle concentration is uniform, and the rate equation is given by column 1 in Table 16-12.

For a **Langmuir isotherm** with negligible solute accumulation in the particle pores, the solution for an infinite fluid volume:

$$(1-R)(1 - n_i^0/n_i^0)F - R \ln(1-F) = (3k_f t/r_p)(c_i^0/\rho_p n_i^0) \quad (16-93)$$

where $n_i^0 = n_i^\infty = n_i^0 K_i c_i^0/(1 + K_i c_i^0)$ is the adsorbate concentration in the particle at equilibrium with the fluid concentration. The predicted behavior is shown in Fig. 16-14 for $n_i^0 = 0$. In the **irreversible limit**

($R = 0$), F increases linearly with time; and in the **linear limit** ($R = 1$), $1 - F$ decreases exponentially with time.

For a finite fluid volume ($\Lambda^\infty > 0$), the fractional approach to equilibrium is given by:

$$\left[1 - \frac{b'(1-R^\infty)}{2c'}\right] \frac{1}{q'} \ln \frac{(2c'F - b' - q')(-b' + q')}{(2c'F - b' + q')(-b' - q')} - \frac{1-R^\infty}{2c'} \ln(1 - b'F + c'F^2) = \frac{3k_f t}{r_p} \frac{c_i^0}{\rho_p n_i^\infty} \quad (16-94)$$

where

$$b' = \frac{1-R^\infty}{1-R^0} + \Lambda^\infty \quad (16-95a)$$

$$c' = \Lambda^\infty(1-R^\infty) \quad (16-95b)$$

$$q' = (b'^2 - 4c')^{0.5} \quad (16-95c)$$

$$R^0 = \frac{1}{1 + K_i c_i^0} \quad (16-95d)$$

$$R^\infty = \frac{1}{1 + K_i c_i^\infty} \quad (16-95e)$$

The predicted behavior is shown in Fig. 16-15 for $R^0 = 0.5$ with different values of Λ^∞ .

SOLID DIFFUSION CONTROL

For a constant diffusivity and an infinite fluid volume the solution is:

$$F = 1 - \frac{6}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \exp\left(-\frac{n^2 \pi^2 D_{st} t}{r_p^2}\right) \quad (16-96)$$

For short times, this equation does not converge rapidly. The following approximations can be used instead (Helfferich and Hwang, in Dorfner, gen. refs., pp. 1277-1309):

$$F = \frac{6}{r_p} \left(\frac{D_{st} t}{\pi}\right)^{0.5} \quad F < 0.2 \quad (16-97)$$

$$F = \frac{6}{r_p} \left(\frac{D_{st} t}{\pi}\right)^{0.5} - \frac{3D_{st} t}{r_p^2} \quad F < 0.8 \quad (16-98)$$

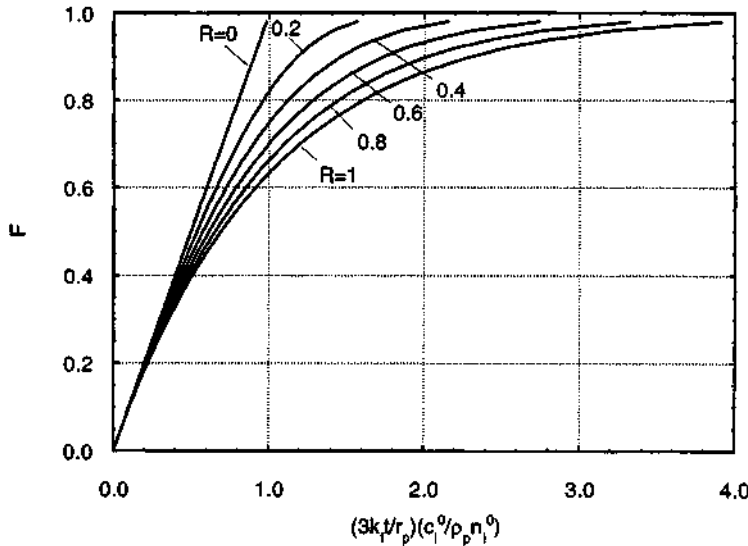


FIG. 16-14 Constant separation factor batch adsorption curves for external mass-transfer control with an infinite fluid volume and $n_i^0 = 0$.

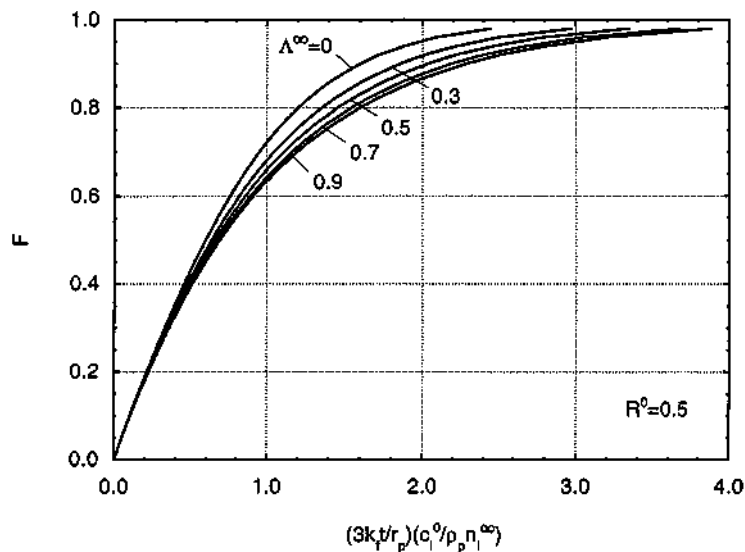


FIG. 16-15 Constant separation factor batch adsorption curves for external mass-transfer control with a finite fluid volume, $n_1^0 = 0$ and $R^2 = 0.5$.

For values of $F > 0.8$, the first term ($n = 1$) in Eq. (16-96) is generally sufficient. If the controlling resistance is diffusion in the subparticles of a bidispersed adsorbent, Eq. (16-96) applies with r_s replacing r_p .

For a finite fluid volume the solution is:

$$F = 1 - 6 \sum_{n=1}^{\infty} \frac{\exp(-p_n^2 D_{st} t / r_p^2)}{9\Lambda^\infty / (1 - \Lambda^\infty) + (1 - \Lambda^\infty) p_n^2} \quad (16-99)$$

where the p_n s are the positive roots of

$$\frac{\tan p_n}{p_n} = \frac{3}{3 + (1/\Lambda^\infty - 1)p_n^2} \quad (16-100)$$

The predicted behavior is shown in Fig. 16-16. F is calculated from Eq. (16-96) for $\Lambda^\infty = 0$ and from Eq. (16-99) for $\Lambda^\infty > 0$. Significant deviations from the $\Lambda^\infty = 0$ curve exist for $\Lambda^\infty > 0.1$.

For nonconstant diffusivity, a numerical solution of the conservation equations is generally required. In molecular sieve zeolites, when equilibrium is described by the Langmuir isotherm, the concentration dependence of the intracrystalline diffusivity can often be approximated by Eq. (16-72). The relevant rate equation is:

$$\frac{\partial n_i}{\partial t} = \frac{D_{0i}}{r^2} \frac{\partial}{\partial r} \left(\frac{r^2}{1 - n_i/n_i^\infty} \frac{\partial n_i}{\partial r} \right) \quad (16-101)$$

A numerical solution of this equation for a constant surface concentration (infinite fluid volume) is given by Garg and Ruthven [*Chem.*

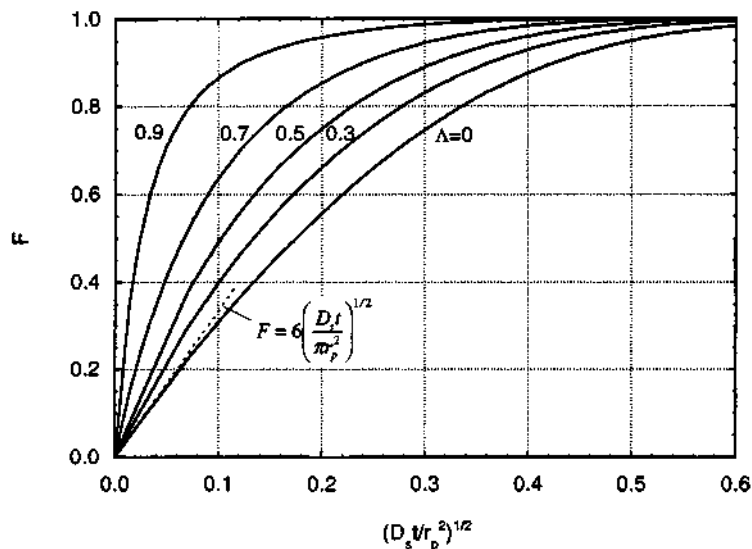


FIG. 16-16 Batch adsorption curves for solid diffusion control. The curve for $\Lambda^\infty = 0$ corresponds to an infinite fluid volume. (Adapted from Ruthven, *gen. refs.*, with permission.)

Eng. Sci., **27**, 417 (1972)]. The solution depends on the value of $\lambda = (n_i^0 - n_i^s)/(n_i^s - n_i^0)$. Because of the effect of adsorbate concentration on the effective diffusivity, for large concentration steps adsorption is faster than desorption, while for small concentration steps, when D_s can be taken to be essentially constant, adsorption and desorption curves are mirror images of each other as predicted by Eq. (16-96); see Ruthven, gen. refs., p. 175.

In binary ion-exchange, intraparticle mass transfer is described by Eq. (16-75) and is dependent on the ionic self diffusivities of the exchanging counterions. A numerical solution of the corresponding conservation equation for spherical particles with an infinite fluid volume is given by Helfferich and Plesset [*J. Chem. Phys.*, **66**, 28, 418 (1958)]. The numerical results for the case of two counterions of equal valence where a resin bead, initially partially saturated with A, is completely converted to the B form, is expressed by:

$$F = \{1 - \exp[\pi^2(f_1(\alpha')\tau_D + f_2(\alpha')\tau_D^2 + f_3(\alpha')\tau_D^3)]\}^{1/2} \quad (16-102)$$

$$\text{with } f_1(\alpha') = -(0.570 + 0.430\alpha'^{0.775})^{-1} \quad (16-103a)$$

$$f_2(\alpha') = (0.260 + 0.782\alpha')^{-1} \quad (16-103b)$$

$$f_3(\alpha') = -(0.165 + 0.177\alpha')^{-1} \quad (16-103c)$$

where $\tau_D = \bar{D}_A t / r_p^2$ and $\alpha' = 1 + (\bar{D}_A / \bar{D}_B - 1)n_A^0 / n^s$ for $0.1 \leq \alpha' \leq 10$. The predicted behavior is shown in Fig. 16-17. When $\alpha' = 1$ (equal ion diffusivities or $n_A^0 \sim 0$), the solution coincides with Eq. (16-96). For $\alpha' \neq 1$, the exchange rate is faster or slower depending on which counterion is initially present in the ion exchanger and on the initial level of saturation.

For an initially fully saturated particle, the exchange rate is faster when the faster counterion is initially in the resin, with the difference in rate becoming more important as conversion from one form to the other progresses. Helfferich (gen. refs., pp. 270–271) gives explicit expressions for the exchange of ions of unequal valence.

PORE DIFFUSION CONTROL

The rate equation is given by item A in Table 16-11. With pore fluid and adsorbent at equilibrium at each point within the particle and for a constant diffusivity, the rate equation can be written as:

$$\frac{\partial c_{pi}}{\partial t} = \frac{\epsilon_p D_{pi}}{\epsilon_p + \rho_p dn_i^s/dc_i} \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c_{pi}}{\partial r} \right) \quad (16-104)$$

For a **linear isotherm** ($n_i = K_i c_i$), this equation is identical to the conservation equation for solid diffusion, except that the solid diffusivity D_{si} is replaced by the equivalent diffusivity $D_{ei} = \epsilon_p D_{pi} / (\epsilon_p + \rho_p K_i)$. Thus, Eqs. (16-96) and (16-99) can be used for pore diffusion control with infinite and finite fluid volumes simply by replacing D_{si} with D_{ei} .

When the adsorption isotherm is nonlinear, a numerical solution is generally required. For a **Langmuir system** with negligible solute holdup in the pore fluid, item A in Table 16-11 gives:

$$\frac{\partial n_i}{\partial t} = \frac{\epsilon_p D_{pi}}{\rho_p n_i^s K_i} \frac{1}{r^2} \frac{\partial}{\partial r} \left[\frac{r^2}{(1 - n_i/n_i^s)^2} \frac{\partial n_i}{\partial r} \right] \quad (16-105)$$

This equation has the same form as that obtained for solid diffusion control with D_{si} replaced by the equivalent concentration-dependent diffusivity $D_{ei} = \epsilon_p D_{pi} / [\rho_p n_i^s K_i (1 - n_i/n_i^s)^2]$. Numerical results for the case of adsorption on an initially clean particle are given in Fig. 16-18 for different values of $\lambda = n_i^0/n_i^s = 1 - R$. The uptake curves become increasingly steeper, as the nonlinearity of the isotherm, measured by the parameter λ , increases. The desorption curve shown for a particle with $n_i^0/n_i^s = 0.9$ shows that for the same step in concentration, adsorption occurs much more quickly than desorption. This difference, however, becomes smaller as the value of λ is reduced, and in the linear region of the adsorption isotherm ($\lambda \rightarrow 0$), adsorption and desorption curves are mirror images. The solution in Fig. 16-18 is applicable to a nonzero initial adsorbent loading by redefining λ as $(n_i^0 - n_i^s)/(n_i^s - n_i^0)$ and the dimensionless time variable as $[\epsilon_p D_{pi} t / \rho_p (1 - n_i^0/n_i^s)^2 n_i^s K_i r_p^2]^{1/2}$ (Ruthven, gen. refs.).

In the **irreversible limit** ($R < 0.1$), the adsorption front within the particle approaches a shock transition separating an inner core into which the adsorbate has not yet penetrated from an outer layer in which the adsorbent phase concentration is uniform at the saturation value. The dynamics of this process is described approximately by the shrinking-core model [Yagi and Kunii, *Chem. Eng. (Japan)*, **19**, 500 (1955)]. For an infinite fluid volume, the solution is:

$$\frac{\epsilon_p D_{pi} t}{r_p^2} \frac{c_i^0}{\rho_p n_i^s} = \frac{1}{2} - \frac{1}{3} F - \frac{1}{2} (1 - F)^{2/3} \quad (16-106)$$

or, in explicit form [Brauch and Schlunder, *Chem. Eng. Sci.*, **30**, 540 (1975)]:

$$F = 1 - \left\{ \frac{1}{2} + \cos \left[\frac{\pi}{3} + \frac{1}{3} \cos^{-1} \left(1 - \frac{12 \epsilon_p D_{pi} t}{r_p^2} \frac{c_i^0}{\rho_p n_i^s} \right) \right] \right\}^3 \quad (16-107)$$

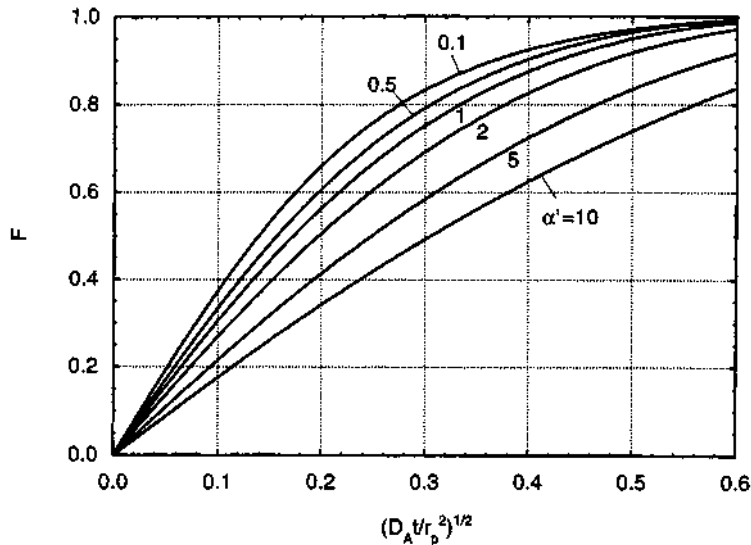


FIG. 16-17 Batch ion exchange for two equal-valence counterions. The exchanger is initially uniformly loaded with ion A in concentration n_A^0 and is completely converted to the B form. $\alpha' = 1 + (\bar{D}_A / \bar{D}_B - 1)n_A^0 / n^s$.

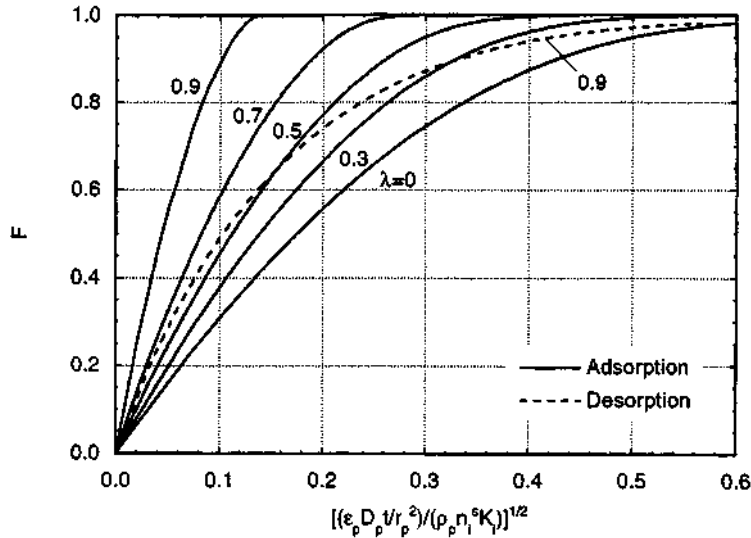


FIG. 16-18 Constant separation factor batch adsorption curves for pore diffusion control with an infinite fluid volume. λ is defined in the text.

For a finite fluid volume with $0 < \Lambda^\infty \leq 1$, the solution is [Teo and Ruthven, *Ind. Eng. Chem. Process Des. Dev.*, **25**, 17 (1986)]:

$$\frac{\epsilon_p D_{pi} t}{r_p^2} \frac{c_i^0}{\rho_p n_i^0} = I_2 - I_1 \quad (16-108)$$

where

$$I_1 = \frac{1}{\lambda' \Lambda^\infty \sqrt{3}} \left[\tan^{-1} \frac{2\eta - \lambda'}{\lambda' \sqrt{3}} - \tan^{-1} \frac{2 - \lambda'}{\lambda' \sqrt{3}} \right] + \frac{1}{6\lambda' \Lambda^\infty} \ln \left[\frac{\lambda'^3 + \eta^3}{\lambda'^3 + 1} \left(\frac{\lambda' + 1}{\lambda' + \eta} \right)^3 \right] \quad (16-109a)$$

$$I_2 = \frac{1}{3\Lambda^\infty} \ln \frac{\lambda'^3 + \eta^3}{\lambda'^3 + 1} \quad (16-109b)$$

$$\eta = (1 - F)^{1/3} \quad (16-109c)$$

$$\lambda' = \left(\frac{1}{\Lambda^\infty} - 1 \right)^{1/3} \quad (16-109d)$$

COMBINED RESISTANCES

In general, exact analytic solutions are available only for the linear ($R = 1$) and irreversible limits ($R \rightarrow 0$). Intermediate cases require numerical solution or use of approximate driving force expressions (see "Rate and Dispersion Factors").

Parallel Pore and Solid Diffusion Control With a **linear isotherm**, assuming equilibrium between the pore fluid and the solid adsorbent, batch adsorption can be represented in terms of an equivalent solid diffusivity $\bar{D}_{ei} = (\epsilon_p D_{pi} + \rho_p D_{si}) / (\epsilon_p + \rho_p K_i)$. Thus, Eqs. (16-96) and (16-99) can be used for this case with D_{si} replaced by $\bar{D}_{ei}$.

External Mass Transfer and Intraparticle Diffusion Control With a **linear isotherm**, the solution for combined external mass transfer and pore diffusion control with an infinite fluid volume is (Crank, *Mathematics of Diffusion*, 2d ed., Clarendon Press, 1975):

$$F = 1 - \sum_{n=1}^{\infty} \frac{6Bi^2 \exp[-(p_n^2 \epsilon_p D_{pi} t / r_p^2) / (\epsilon_p + \rho_p K_i)]}{p_n^2 [p_n^2 + Bi(Bi - 1)]} \quad (16-110)$$

where $Bi = k_f r_p / \epsilon_p D_{pi}$ is the **Biot number** and the p_n s are the positive roots of

$$p_n \cot p_n = 1 - Bi \quad (16-111)$$

For a finite fluid volume the solution is:

$$F = 1 - 6 \sum_{n=1}^{\infty} \frac{\exp[-(p_n^2 \epsilon_p D_{pi} t / r_p^2) / (\epsilon_p + \rho_p K_i)]}{\frac{9\Lambda^\infty}{1 - \Lambda^\infty} + (1 - \Lambda^\infty)p_n^2 - (5\Lambda^\infty + 1) \frac{p_n^2}{Bi} + (1 - \Lambda^\infty) \frac{p_n^4}{Bi^2}} \quad (16-112)$$

where the p_n s are the positive roots of

$$\tan p_n = \frac{3 - \frac{1 - \Lambda^\infty}{\Lambda^\infty} \frac{p_n^2}{Bi}}{3 + \frac{1 - \Lambda^\infty}{\Lambda^\infty} \frac{(Bi - 1)p_n^2}{Bi}} \quad (16-113)$$

These expressions can also be used for the case of external mass transfer and solid diffusion control by substituting D_{si} for $\epsilon_p D_{pi} / (\epsilon_p + \rho_p K_i)$ and $k_f r_p / (\rho_p K_i D_{si})$ for the Biot number.

In the **irreversible limit**, the solution for combined external resistance and pore diffusion with infinite fluid volume is (Yagi and Kunii):

$$\frac{\epsilon_p D_{pi} t}{r_p^2} \frac{c_i^0}{\rho_p n_i^0} = \frac{1}{2} - \frac{1}{3} \left(1 - \frac{1}{Bi} \right) F - \frac{1}{2} (1 - F)^{2/3} \quad (16-114)$$

For a finite fluid volume the solution is (Teo and Ruthven):

$$\frac{\epsilon_p D_{pi} t}{r_p^2} \frac{c_i^0}{\rho_p n_i^0} = \left(1 - \frac{1}{Bi} \right) I_2 - I_1 \quad (16-115)$$

where I_1 and I_2 are given by Eqs. (16-109a) and (16-109b).

Bidispersed Particles For particles of radius r_p comprising adsorptive subparticles of radius r_s that define a macropore network, conservation equations are needed to describe transport both within the macropores and within the subparticles and are given in Table 16-11, item D. Detailed equations and solutions for a linear isotherm are given in Ruthven (gen. refs., p. 183) and Ruckenstein et al. [*Chem. Eng. Sci.*, **26**, 1306 (1971)]. The solution for a **linear isotherm** with no external resistance and an infinite fluid volume is:

$$F = 1 - \frac{18}{\beta + 3\alpha} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \left(\frac{n^2 \pi^2}{p_{n,m}^4} \right) \times \frac{\exp(-p_{n,m}^2 D_{si} t / r_s^2)}{\alpha + \frac{\beta}{2} \left[1 + \frac{\cot p_{n,m}}{p_{n,m}} (p_{n,m} \cot p_{n,m} - 1) \right]} \quad (16-116)$$

where the $p_{n,m}$ values are the roots of the equation

$$\alpha p_{n,m}^2 - n^2 \pi^2 = \beta (p_{n,m} \cot p_{n,m} - 1) \quad (16-117)$$

and

$$\alpha = \frac{D_{si}/r_p^2}{D_{pi}/r_p^2} \quad (16-118a)$$

$$\beta = \frac{3\alpha p_p K_i}{\epsilon_p} \quad (16-118b)$$

In these equations, D_{si} is the diffusivity in the subparticles, and D_{pi} is the diffusivity in the pore network formed by the subparticles.

For large K_i values, the uptake curve depends only upon the value of the parameter β representing the ratio of characteristic time constants for diffusion in the pores and in the subparticles. For small β values, diffusion in the subparticles is controlling and the solution coincides with Eq. (16-96) with r_s replacing r_p . For large β values, pore diffusion is controlling, and the solution coincides with Eq. (16-96) with $\epsilon_p D_{pi}/(\epsilon_p + \rho_p K_i)$ replacing D_{si} .

Lee [AIChE J., 24, 531 (1978)] gives the solution for batch adsorption with bidispersed particles for the case of a finite fluid volume.

FIXED-BED TRANSITIONS

As discussed in "Design Concepts," a large fraction of adsorption and ion-exchange processes takes place in fixed beds. Two classical methods for analyzing fixed-bed transitions are described here. First, local equilibrium theory is presented. In this, all mass-transfer resistances are ignored to focus on the often dominating role of isotherm shape. Second, results of constant pattern analysis are presented. This gives the maximum breadth to which a mass-transfer zone will spread for various rate mechanisms. It is therefore conservative for design purposes. Both of these methods pertain to behavior in deep beds. For shallow beds, the equations given below must be solved for the particular case of interest.

DIMENSIONLESS SYSTEM

For the methods, we consider Eq. (16-52), the material balance for a fixed bed, written in the form

$$\rho_b \frac{\partial \bar{n}_i}{\partial t} + \epsilon_b \frac{\partial c_i}{\partial t} + \epsilon \frac{\partial (v c_i)}{\partial z} = \epsilon D_L \frac{\partial}{\partial z} \left(c \frac{\partial y_i}{\partial z} \right) \quad (16-119)$$

where it has been assumed that D_L is constant and that $\hat{c}_i \approx c_i$ (or that the second term in the balance is small compared to the first—usually a good assumption).

Dimensionless variables can be defined for time, the axial coordinate, and velocity:

$$\tau = \frac{\epsilon v^{\text{ref}} t}{L} \quad (16-120)$$

$$\zeta = \frac{z}{L} \quad (16-121)$$

$$v^* = \frac{v}{v^{\text{ref}}} \quad (16-122)$$

where L is bed length, v^{ref} is the interstitial velocity at the bed inlet, and τ is equal to the number of empty bed volumes of feed passed into the bed. The material balance becomes

$$\rho_b \frac{\partial \bar{n}_i}{\partial \tau} + \epsilon_b \frac{\partial c_i}{\partial \tau} + \frac{\partial (v^* c_i)}{\partial \zeta} = \frac{1}{N_{Pe}} \frac{\partial}{\partial \zeta} \left(c \frac{\partial y_i}{\partial \zeta} \right) \quad (16-123)$$

where $N_{Pe} = v^{\text{ref}} L / D_L$ is a Peclet number for the bed or a **number of dispersion units**. Equation (16-123) or a similar equation is often the material balance used in nonisothermal problems, in problems involving adsorption of nontrace components, and in calculations of cycles.

For a trace, isothermal system, we have $v^* = 1$, and using the dimensionless system variables for concentrations [Eq. (16-10)], Eq. (16-123) becomes

$$\Lambda \frac{\partial \bar{n}_i^*}{\partial \tau} + \epsilon_b \frac{\partial c_i^*}{\partial \tau} + \frac{\partial c_i^*}{\partial \zeta} = \frac{1}{N_{Pe}} \frac{\partial^2 c_i^*}{\partial \zeta^2} \quad (16-124)$$

where Λ is the **partition ratio** defined by

$$\Lambda = \frac{\rho_b n_i^{\text{ref}}}{c_i^{\text{ref}}} \quad (16-125)$$

This important dimensionless group is the volumetric capacity of the bed for the sorbable component divided by the concentration of the sorbable component in the feed. The stoichiometric capacity of the bed for solute is exactly equal to Λ empty bed volumes of feed (to saturate the sorbent at the feed concentration) plus a fraction of a bed volume of feed to fill the voids outside and inside the particles. Alternatively, we also obtain Eq. (16-124) using the dimensionless transition variables for concentrations [Eq. (16-11)], but now the partition ratio in the first term of Eq. (16-124) pertains to the transition and is given by

$$\Lambda = \rho_b \frac{n_i'' - n_i'}{c_i'' - c_i'} \quad (16-126)$$

Equation (16-124) is a commonly used form of material balance for a fixed-bed adsorber.

If the system under consideration involves use of the sorbent for only a single feed step or reuse after uniform regeneration, as in many applications with activated carbons and ion exchangers, then one of two paths is often followed at this point to simplify Eq. (16-124) further. The second term on the left-hand side of the equation is often assumed to be negligibly small (usually a good assumption), and time is redefined as

$$\tau_1 = \frac{\tau}{\Lambda} \quad (16-127)$$

$$\text{to give} \quad \frac{\partial \bar{n}_i^*}{\partial \tau_1} + \frac{\partial c_i^*}{\partial \zeta} = \frac{1}{N_{Pe}} \frac{\partial^2 c_i^*}{\partial \zeta^2} \quad (16-128)$$

Alternatively, in the absence of axial dispersion, a variable of the form

$$\tau_1 = \frac{\tau - \epsilon_b \zeta}{\Lambda} \quad (16-129)$$

can be defined to reduce Eq. (16-124) directly to

$$\frac{\partial \bar{n}_i^*}{\partial \tau_1} + \frac{\partial c_i^*}{\partial \zeta} = 0 \quad (16-130)$$

The variable τ_1 defined by Eq. (16-127) or (16-129) is a **throughput parameter**, equal to unity (hence, the "1" subscript) at the time when the stoichiometric center of the concentration wave leaves the bed. This important group, in essence a dimensionless time variable, determines the location of the stoichiometric center of the transition in the bed at any time.

LOCAL EQUILIBRIUM THEORY

In local equilibrium theory, fluid and sorbed phases are assumed to be in local equilibrium with one another at every axial position in the bed. Thus, because of uniform concentrations, the overbar on n_i^* is not necessary and we have $\hat{c}_i \approx c_i$ [note Eqs. (16-52) and (16-119)].

Single Transition System For a system described by a single material balance, Eq. (16-130) gives

$$\frac{d\tau_1}{d\zeta} = -\frac{\partial c_i^*/\partial \zeta}{\partial c_i^*/\partial \tau_1} = \frac{dn_i^*}{dc_i^*} \quad (16-131)$$

where $d\tau_1/d\zeta$ is the reciprocal of a concentration velocity. Equation (16-131) is the equation for a **simple wave** (or gradual transition or proportionate pattern). If a bed is initially uniformly saturated, then $d\tau_1/d\zeta = \tau_1/\zeta$. Thus, for the dimensionless system, the reciprocal of the velocity of a concentration is equal to the slope of the isotherm at that concentration. Furthermore, from Eq. (16-131), the depth of penetration of a given concentration into the bed is directly proportional to time, so the breadth of a simple wave increases in direct proportion to the depth of its penetration into the bed (or to time). Thus, for the simple wave, the length of the MTZ is proportional to the depth of the bed through which the wave has passed. Consideration of isotherm shape indicates that a simple wave occurs for an unfavorable dimensionless isotherm ($d^2n_i^*/dc_i^{*2} > 0$), for which low concentrations will go faster than high concentrations. Equation (16-131) also pertains to a linear isotherm, in which case the wave is called a **contact discontinuity**, because it has neither a tendency to spread nor sharpen. If a mass-transfer resistance is added to the consideration of wave character for unfavorable isotherms, the wave will still asymptotically approach the simple wave result given by Eq. (16-131).

For a favorable isotherm ($d^2n_i^*/dc_i^{*2} < 0$), Eq. (16-131) gives the impossible result that three concentrations can coexist at one point in the bed (see example below). The correct solution is a **shock** (or abrupt transition) and not a simple wave. Mathematical theory has been developed for this case to give "weak solutions" to conservation laws. The form of the solution is

$$\text{Shock speed} = \frac{\text{change in flux}}{\text{change in accumulated quantity}}$$

where the changes are jump discontinuities across the shock. The reciprocal of this equation, using Eq. (16-130), is

$$\frac{d\tau_1}{d\zeta} = \frac{\Delta n_i^*}{\Delta c_i^*} \quad (16-132)$$

where the differences are taken across the shock.

It is also possible to have a **combined wave**, which has both gradual and abrupt parts. The general rule for an isothermal, trace system is that in passing from the initial condition to the feed point in the isotherm plane, the slope of the path must not decrease. If it does, then a shock chord is taken for part of the path. Referring to Fig. 16-19, for

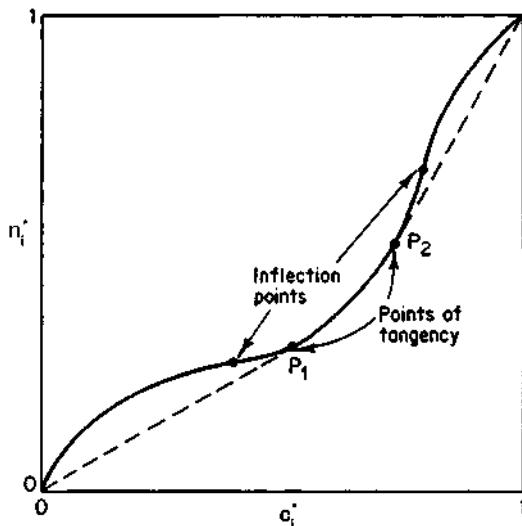


FIG. 16-19 Path in isotherm plane for a combined wave. (After Tudge.)

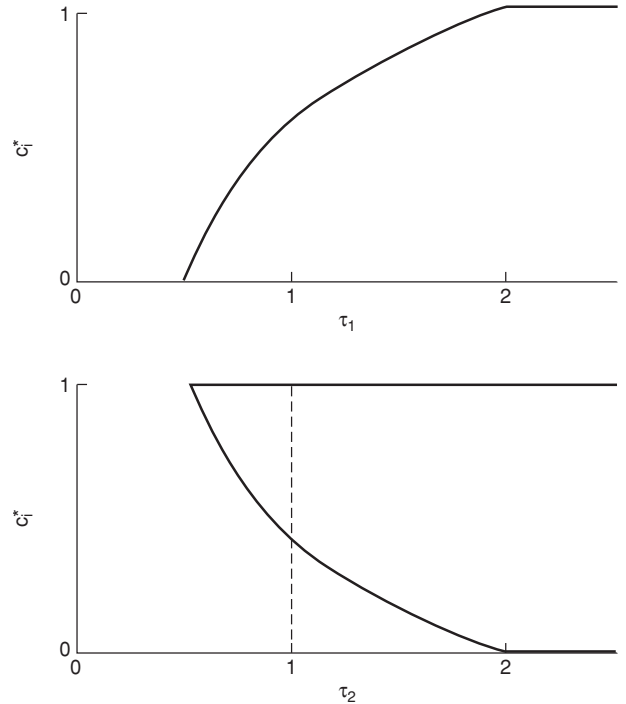


FIG. 16-20 Breakthrough curves for $r=2$ (top) and $r=0.5$ (bottom) for Example 9.

a transition from (0,0) to (1,1), the dashes indicate shock parts, which are connected by a simple wave part between points P_1 and P_2 .

Example 10: Transition Types For the constant separation-factor isotherm given by Eq. (16-31), determine breakthrough curves for $r=2$ and $r=0.5$ for transitions from $c_i^*=0$ to $c_i^*=1$.

Using Eq. (16-131), we obtain

$$\frac{\tau_1}{\zeta} = \frac{r}{[r + (1-r)c_i^*]^2}$$

This equation, evaluated at $\zeta=1$, is plotted for $r=2$ and $r=0.5$ in Fig. 16-20. Clearly, the solution for $r=0.5$ is not physically correct. Equation (16-132), with $d\tau_1/d\zeta = \tau_1/\zeta$, is applied to this case to give the shock indicated by the dashed line. Alternatively, we could have obtained bed profiles by evaluating equations at $\tau_1 = \text{const}$.

Multiple Transition System Local equilibrium theory for multiple transitions begins with some combination of material and energy balances, written

$$\rho_b \frac{\partial n_i}{\partial \tau} + \epsilon_b \frac{\partial c_i}{\partial \tau} + \frac{\partial(v^*c_i)}{\partial \zeta} = 0 \quad (i = 1, 2, \dots) \quad (16-133)$$

$$\rho_b \frac{\partial u_s}{\partial \tau} + \epsilon_b \frac{\partial(cu_f)}{\partial \tau} + \frac{\partial(v^*ch_f)}{\partial \zeta} = 0 \quad (16-134)$$

which are Eq. (16-123) written with no dispersion and Eq. (16-55) written for an adiabatic bed.

For a simple wave, application of the method of characteristics (hodograph transformation) gives

$$\frac{d\tau}{d\zeta} = \frac{d(\rho_b n_1 + \epsilon_b c_1)}{d(v^*c_1)} = \frac{d(\rho_b n_2 + \epsilon_b c_2)}{d(v^*c_2)} = \dots = \frac{d(\rho_b u_s + \epsilon_b cu_f)}{d(v^*ch_f)} \quad (16-135)$$

where the derivatives are taken along the path of a transition (i.e., directional derivatives).

If a simple wave is not possible on physical grounds, then it (or part of it) is replaced by a shock, given by

$$\frac{d\tau}{d\zeta} = \frac{\Delta(\rho_b n_1 + \epsilon_b c_1)}{\Delta(v^* c_1)} = \frac{\Delta(\rho_b n_2 + \epsilon_b c_2)}{\Delta(v^* c_2)} = \dots = \frac{\Delta(\rho_b u_s + \epsilon_b c u_f)}{\Delta(v^* c u_f)} \quad (16-136)$$

Extensions When more than two conservation equations are to be solved simultaneously, matrix methods for eigenvalues and left eigenvectors are efficient [Jeffrey and Taniuti, *Nonlinear Wave Propagation*, Academic Press, New York, 1964; Jacob and Tondeur, *Chem. Eng. J.*, **22**, 187 (1981), **26**, 41 (1983); Davis and LeVan, *AIChE J.*, **33**, 470 (1987); Rhee et al., gen. refs.].

Nontrace isothermal systems give the “adsorption effect” (i.e., significant change in fluid velocity because of loss or gain of solute). Criteria for the existence of simple waves, contact discontinuities, and shocks are changed somewhat [Peterson and Helfferich, *J. Phys. Chem.*, **69**, 1283 (1965); LeVan et al., *AIChE J.*, **34**, 996 (1988); Frey, *AIChE J.*, **38**, 1649 (1992)].

Local equilibrium theory also pertains to adsorption with axial dispersion, since this mechanism does not disallow existence of equilibrium between stationary and fluid phases across the cross section of the bed [Rhee et al., *Chem. Eng. Sci.*, **26**, 1571 (1971)]. It is discussed below in further detail from the standpoint of the constant pattern.

Example 11: Two-Component Isothermal Adsorption Two components present at low mole fractions are adsorbed isothermally from an inert fluid in an initially clean bed. The system is described by $\rho_b = 500 \text{ kg/m}^3$, $\epsilon_b = 0.7$, and the binary Langmuir isotherm

$$n_i = \frac{n_i^s K_i c_i}{1 + K_i c_i + K_2 c_2} \quad (i = 1, 2)$$

with $n_1^s = n_2^s = 6 \text{ mol/kg}$, $K_1 = 40 \text{ m}^3/\text{mol}$, and $K_2 = 20 \text{ m}^3/\text{mol}$. The feed is $c_1 = c_2 = 0.5 \text{ mol/m}^3$. Find the bed profile.

Using the isotherm to calculate loadings in equilibrium with the feed gives $n_1 = 3.87 \text{ mol/kg}$ and $n_2 = 1.94 \text{ mol/kg}$. An attempt to find a simple wave solution for this problem fails because of the favorable isotherms (see the next example for the general solution method). To obtain the two shocks, Eq. (16-136) is written

$$\frac{d\tau}{d\zeta} = \frac{\Delta(\rho_b n_1 + \epsilon_b c_1)}{\Delta c_1} = \frac{\Delta(\rho_b n_2 + \epsilon_b c_2)}{\Delta c_2}$$

The concentration of one of the components will drop to zero in the shock nearest the bed inlet. If it is component 1, then using feed values and the equation above, that shock would be at

$$\frac{\tau}{\zeta} = \frac{\rho_b n_1 + \epsilon_b c_1}{c_1} \approx \frac{\rho_b n_1}{c_1} = 3870$$

Similarly, if the second component were to disappear in the first shock, we would have $\tau/\zeta = 1940$. Material balance considerations require that we accept the shorter distance, so component 1 disappears in the first shock.

The concentrations of component 2 on the plateau downstream of the first shock are then calculated from

$$\frac{\tau}{\zeta} = \frac{\Delta(\rho_b n_2 + \epsilon_b c_2)}{\Delta c_2} \approx \frac{\rho_b \Delta n_2}{\Delta c_2} = 3870$$

and its pure component isotherm, giving $c_2 = 0.987 \text{ mol/m}^3$ and $n_2 = 5.71 \text{ mol/kg}$. The location of this shock is determined using these concentrations and

$$\frac{\tau}{\zeta} = \frac{\rho_b n_2 + \epsilon_b c_2}{c_2} \approx \frac{\rho_b n_2}{c_2}$$

which gives $\tau/\zeta = 2890$. The bed profile is plotted in Fig. 16-21 using ζ/τ as the abscissa. (This example can also be worked with the h -transformation described in this section under chromatography.)

Example 12: Adiabatic Adsorption and Thermal Regeneration

An initially clean activated carbon bed at 320 K is fed a vapor of benzene in nitrogen at a total pressure of 1 MPa. The concentration of benzene in the feed is 6 mol/m^3 . After the bed is uniformly saturated with feed, it is regenerated using benzene-free nitrogen at 400 K and 1 MPa. Solve for both steps. For simplicity, neglect fluid-phase accumulation terms and assume constant mean heat capacities for stationary and fluid phases and a constant velocity. The system is described by

$$\rho_b n_1 = \frac{\rho_b n_1^s K_1 c_1}{1 + K_1 c_1}$$

$$\rho_b n_1^s = 2750 \text{ mol/m}^3$$

$$K_1 = 3.88 \times 10^{-4} \sqrt{T} \exp [q_1^s / (RT)] \text{ m}^3/(\text{mol K}^{1/2})$$

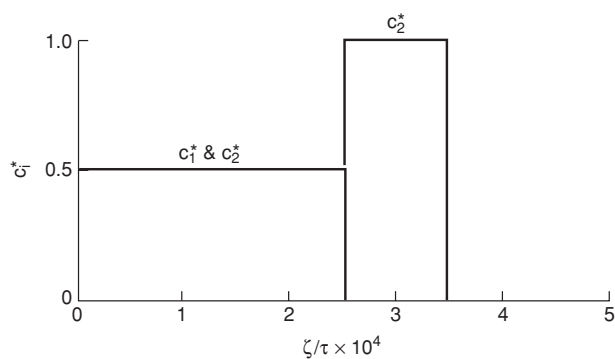


FIG. 16-21 Bed profiles for two-component isothermal adsorption, Example 10.

$$\rho_b C_{sm} = 850 \text{ kJ}/(\text{m}^3 \text{ K})$$

$$c C_{pfm} = 11.3 \text{ kJ}/(\text{m}^3 \text{ K})$$

$$q_1^s = 43.5 \text{ kJ/mol}$$

$$T^{\text{ref}} = 320 \text{ K}$$

Extensive analysis has been made of this system [Rhee et al., *Chem. Eng. J.*, **1**, 241 (1970) and gen. refs.; LeVan, in Rodrigues et al., gen. refs.].

To obtain the concentration and temperature profiles, the two transitions are first assumed to be gradual. Equation (16-135) is written in the form

$$\frac{\tau}{\zeta} = \rho_b \frac{dn_1}{dc_1} = \rho_b \frac{du_s}{d(ch_f)}$$

where, from Eqs. (16-56) to (16-60),

$$\begin{aligned} \rho_b u_s &= \rho_b C_{sm}(T - T^{\text{ref}}) - \rho_b n_1 q_1^s \\ ch_f &= c C_{pfm}(T - T^{\text{ref}}) \end{aligned}$$

This equation is expanded in terms of c_1 and T to obtain

$$\frac{\tau}{\zeta} = \rho_b \left(\frac{\partial n_1}{\partial c_1} + \frac{\partial n_1}{\partial T} \frac{dT}{dc_1} \right) = \frac{\rho_b C_{sm}}{c C_{pfm}} - \frac{\rho_b q_1^s}{c C_{pfm}} \left(\frac{\partial n_1}{\partial T} + \frac{\partial n_1}{\partial c_1} \frac{dc_1}{dT} \right)$$

Solving the rightmost equality for the directional derivative using the quadratic formula gives

$$\frac{dc_1}{dT} = \frac{-b \pm \sqrt{b^2 - 4ad}}{2a}$$

with

$$a = \left[\frac{\rho_b q_1^s}{c C_{pfm}} \right] \frac{\partial n_1}{\partial c_1}$$

$$b = \left[\frac{\rho_b q_1^s}{c C_{pfm}} \right] \frac{\partial n_1}{\partial T} - \frac{\rho_b C_{sm}}{c C_{pfm}} + \frac{\rho_b \partial n_1}{\partial c_1}$$

$$d = \frac{\rho_b \partial n_1}{\partial T}$$

The plus sign corresponds to the downstream transition and the minus sign to the upstream one. This equation is solved along each path beginning at the respective end points—the initial condition of the bed for the downstream transition and the feed condition for the upstream transition. If either path fails to evolve continuously in the expected direction, the difference form, from Eq. (16-136),

$$\frac{d\tau}{d\zeta} = \rho_b \frac{\Delta n_1}{\Delta c_1} = \rho_b \frac{\Delta u_s}{\Delta(ch_f)}$$

is used for that path (or part thereof, if appropriate); two solutions of this equation pass through each composition point, and care must be taken to ensure that the correct path is taken. The two correct paths found intersect to give the composition and temperature of an intermediate plateau region.

Letting $c_1^{\text{ref}} = 6 \text{ mol/m}^3$, the isotherm gives $\rho_b n_1^{\text{ref}} = 2700 \text{ mol/m}^3$, and the partition ratio is $\Lambda = 450$.

In the figures, $\Gamma(k)$ and $\Sigma(k)$ symbolize simple waves and shocks, respectively, with $k = 1$ downstream and $k = 2$ upstream.

Adiabatic Adsorption The construction is shown in Figs. 16-22 and 16-23. The first path begins at the initial condition, point A ($T = 320 \text{ K}$, $c_1 = 0 \text{ mol/m}^3$). Since $\partial n_1 / \partial T = 0$ there, we obtain $dc_1 / dT = 0$ and $\tau/\zeta = \rho_b C_{sm} / (c C_{pfm}) = 75.2$ (or

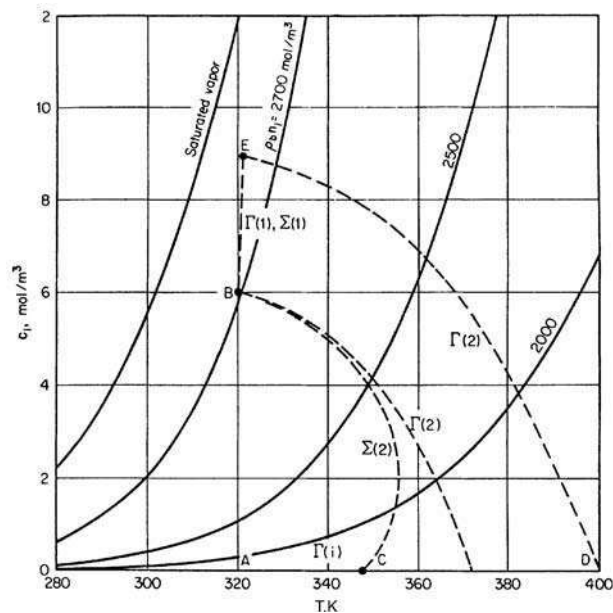


FIG. 16-22 Transition paths in c_1 , T plane for adiabatic adsorption and thermal regeneration.

$\tau/\Lambda = 0.167$ at $\zeta = 1$), corresponding to a pure thermal wave along the $c_1 = 0$ axis of Fig. 16-22. The second path begins at the feed condition, point B ($T = 320$ K, $c_1 = 6$ mol/m³). A $\Gamma(2)$ fails, so a $\Sigma(2)$ is calculated and plotted in the two figures. The $\Gamma(1)$ and $\Sigma(2)$ intersect at point C ($T = 348$ K, $c_1 = 0$ mol/m³). Breakthrough curves of temperature and concentration are shown in Fig. 16-24.

Thermal Regeneration The bed, initially at point B, is fed with pure nitrogen. The feed, at point D ($T = 400$ K, $c_1 = 0$ mol/m³), provides a successful $\Gamma(2)$. The $\Gamma(1)$ from point B fails, so a $\Sigma(1)$, which differs imperceptibly from $\Gamma(1)$, is determined. The $\Sigma(1)$ and $\Gamma(2)$ meet at point E ($T = 320.8$ K, $c_1 = 8.93$ mol/m³),

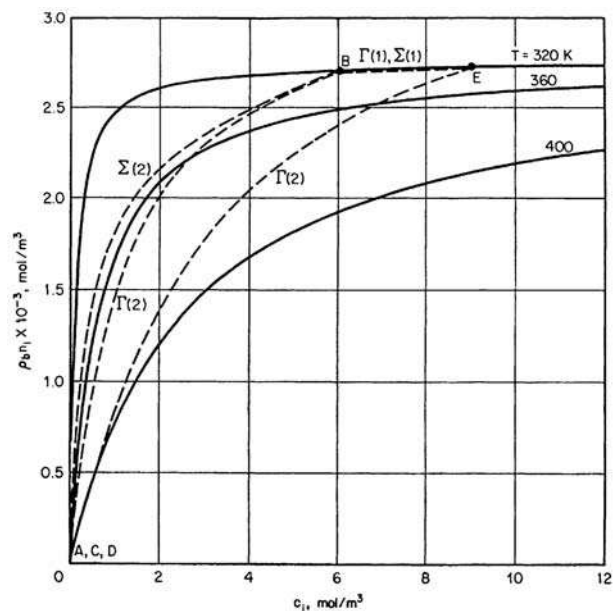


FIG. 16-23 Transition paths in isotherm plane for adiabatic adsorption and thermal regeneration.

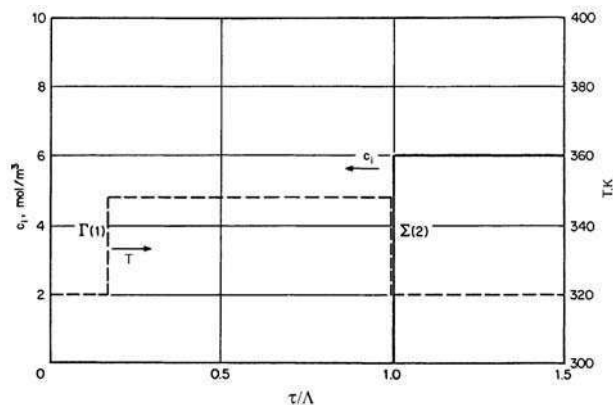


FIG. 16-24 Breakthrough curves for adiabatic adsorption.

where $p_0n_1 = 2716$ mol/m³, which is greater than the initial loading, indicating “roll-up,” a term attributed to Basmadjian et al. [*Ind. Eng. Chem. Process Des. Dev.*, **14**, 328 (1975)], of the adsorbed-phase concentration. Breakthrough curves are shown in Fig. 16-25. The $\Sigma(1)$ leaves the bed at $\tau/\Lambda = 0.011$. The $\Gamma(2)$ begins to emerge at $\tau/\Lambda = 0.169$, but regeneration is not complete until $\tau/\Lambda = 2.38$.

It is possible (with lower initial bed temperature, higher initial loading, or higher regeneration temperature or pressure) for the transition paths to contact the saturated vapor curve in Fig. 16-22 rather than intersect beneath it. For this case, liquid benzene condenses in the bed, and the effluent vapor is saturated during part of regeneration [Friday and LeVan, *AIChE J.*, **30**, 679 (1984)].

CONSTANT PATTERN BEHAVIOR FOR FAVORABLE ISOTHERMS

With a favorable isotherm and a mass-transfer resistance or axial dispersion, a transition approaches a **constant pattern**, which is an asymptotic shape beyond which the wave will not spread. The wave is said to be “self-sharpening.” (If a wave is initially broader than the constant pattern, it will sharpen to approach the constant pattern.) Thus, for an initially uniformly loaded bed, the constant pattern gives the maximum breadth of the MTZ. As bed length is increased, the constant pattern will occupy an increasingly smaller fraction of the bed. (Square-root spreading for a linear isotherm gives this same qualitative result.)

The treatment here is restricted to the Langmuir or constant separation factor isotherm, single-component adsorption, dilute systems, isothermal behavior, and mass-transfer resistances acting alone. References to extensions are given below. Different isotherms have been considered, and the theory is well understood for general isotherms.

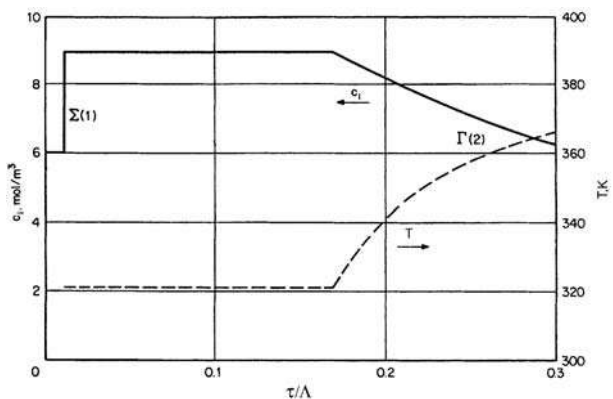


FIG. 16-25 Breakthrough curves for thermal regeneration.

TABLE 16-13 Constant Pattern Solutions for Constant Separation Factor Isotherm ($R < 1$)

Mechanism	N	Dimensionless rate equation ¹	Constant pattern	Refs.
Pore diffusion	$\frac{15(1-\epsilon_p)\epsilon_p D_p L}{\epsilon v^{\text{ref}} r_p^2}$	$\frac{\partial n_i^*}{\partial \tau_1} = \frac{N}{15} \frac{1}{\xi^2} \frac{\partial}{\partial \xi} \left(\xi^2 \frac{\partial c_i^*}{\partial \xi} \right)$	Numerical for $0 < R < 1$ Analytical for $R = 0$	A B
Solid diffusion	$\frac{15\Delta D_s L}{\epsilon v^{\text{ref}} r_p^2}$	$\frac{\partial n_i^*}{\partial \tau_1} = \frac{N}{15} \frac{1}{\xi^2} \frac{\partial}{\partial \xi} \left(\xi^2 \frac{\partial n_i^*}{\partial \xi} \right)$	Numerical for $0 < R < 1$ Analytical for $R = 0$	C D
External mass transfer	$\frac{k_f a L}{\epsilon v^{\text{ref}}}$	$\frac{\partial n_i^*}{\partial \tau_1} = N (c_i^* - c_i^e)$	$\frac{1}{1-R} \ln \left[\frac{(1-c_i^e)^R}{c_i^e} \right] - 1 = N (\zeta - \tau_1)$	E
Linear driving force	$\frac{k_n \Delta L}{\epsilon v^{\text{ref}}}$	$\frac{\partial n_i^*}{\partial \tau_1} = N (n_i^e - \bar{n}_i^*)$	$\frac{1}{1-R} \ln \left[\frac{1-c_i^e}{c_i^{eR}} \right] + 1 = N (\zeta - \tau_1)$	F
Reaction kinetics	$\frac{k_r c^{\text{ref}} \Delta L}{(1-R)\epsilon v^{\text{ref}}}$	$\frac{\partial \bar{n}_i^*}{\partial \tau_1} = N [(1-\bar{n}_i^*)c_i^* - R\bar{n}_i^*(1-c_i^*)]$	$\frac{1}{1-R} \ln \left[\frac{1-c_i^*}{c_i^*} \right] = N (\zeta - \tau_1)$	G
Axial dispersion	$\frac{v^{\text{ref}} L}{D_L}$	Eq. (16-128)	$\frac{1}{1-R} \ln \left[\frac{1-c_i^*}{c_i^{eR}} \right] = N (\zeta - \tau_1)$	H

1: Dimensional rate equations are given in Tables 16-11 and 16-12.

A: Hall et al., *Ind. Eng. Chem. Fundam.*, **5**, 212 (1966).

B: Hall et al., *Ind. Eng. Chem. Fundam.*, **5**, 212 (1966); Cooper and Liberman, *Ind. Eng. Chem. Fundam.*, **9**, 620 (1970). Analytical solution for $R = 0$ is

$$\frac{15}{2} \ln [1 + (1 - c_i^*)^{1/3} + (1 - c_i^*)^{2/3}] - \frac{15}{\sqrt{3}} \tan^{-1} \left[\frac{2(1 - c_i^*)^{1/3} + 1}{\sqrt{3}} \right] + \frac{5\pi}{2\sqrt{3}} - \frac{5}{2} = N(\zeta - \tau_1)$$

C: Hall et al., *Ind. Eng. Chem. Fundam.*, **5**, 212 (1966); Garg and Ruthven, *Chem. Eng. Sci.*, **28**, 791, 799 (1973).

D: Hall et al., *Ind. Eng. Chem. Fundam.*, **5**, 212 (1966) [see also Cooper, *Ind. Eng. Chem. Fundam.*, **4**, 308 (1965); Ruthven (gen. refs.)]. Analytical solution for $R = 0$ is

$$c_i^* = 1 - \frac{6}{\pi^2} \sum_{n=0}^{\infty} \frac{1}{n^2} \exp \left\{ -n^2 \left[\left(\frac{\pi^2}{15} \right) N(\tau_1 - \zeta) + 0.64 \right] \right\}$$

E: Michaels, *Ind. Eng. Chem.*, **44**, 1922 (1952); Miura and Hashimoto, *J. Chem. Eng. Japan*, **10**, 490 (1977).

F: Glueckauf Coates, *J. Chem. Soc.*, **1947**, 1315 (1947); Vermeulen, *Advances in Chemical Engineering*, **2**, 147 (1958); Hall et al., *Ind. Eng. Chem. Fundam.*, **5**, 212 (1966); Miura and Hashimoto, *J. Chem. Eng. Japan*, **10**, 490 (1977).

G: Walter, *J. Chem. Phys.*, **13**, 229 (1945); Hiester and Vermeulen, *Chem. Eng. Progress*, **48**, 505 (1952).

H: Acrivos, *Chem. Eng. Sci.*, **13**, 1 (1960); Coppola and LeVan, *Chem. Eng. Sci.*, **36**, 967 (1981).

Asymptotic Solution Rate equations for the various mass-transfer mechanisms are written in dimensionless form in Table 16-13 in terms of a number of transfer units, $N = L/\text{HTU}$, for particle-scale mass-transfer resistances, a number of reaction units for the reaction kinetics mechanism, and a number of dispersion units, N_{pe} , for axial dispersion. For pore and solid diffusion, $\xi = r/r_p$ is a dimensionless radial coordinate, where r_p is the radius of the particle. If a particle is bidisperse, then r_p can be replaced by r_s , the radius of a subparticle. For preliminary calculations, Fig. 16-13 can be used to estimate N for use with the LDF approximation when more than one resistance is important.

In constant pattern analysis, equations are transformed into a new coordinate system that moves with the wave. Variables are changed from (ζ, τ_1) to $(\zeta - \tau_1, \tau_1)$. The new variable $\zeta - \tau_1$ is equal to zero at the stoichiometric center of the wave. Equation (16-130) for a bed with no axial dispersion, when transformed to the $(\zeta - \tau_1, \tau_1)$ coordinate system, becomes

$$-\frac{\partial \bar{n}_i^*}{\partial (\zeta - \tau_1)} + \frac{\partial \bar{n}_i^*}{\partial \tau_1} + \frac{\partial c_i^*}{\partial (\zeta - \tau_1)} = 0 \quad (16-137)$$

The constant pattern is approached as the τ_1 dependence in this equation disappears. Thus, discarding the derivative with respect to τ_1 and integrating, using the condition that $\bar{n}_i^*$ and c_i^* approach zero as $N(\zeta - \tau_1) \rightarrow \infty$ [or approach unity as $N(\zeta - \tau_1) \rightarrow -\infty$], gives simply

$$\bar{n}_i^* = c_i^* \quad (16-138)$$

For adsorption with axial dispersion, the material balance transforms to

$$-\frac{\partial n_i^*}{\partial (\zeta - \tau_1)} + \frac{\partial n_i^*}{\partial \tau_1} + \frac{\partial c_i^*}{\partial (\zeta - \tau_1)} = \frac{1}{N_{pe}} \frac{\partial^2 c_i^*}{\partial (\zeta - \tau_1)^2} \quad (16-139)$$

The partial derivative with respect to τ_1 is discarded and the resulting equation integrated once to give

$$-n_i^* + c_i^* = \frac{1}{N_{pe}} \frac{dc_i^*}{d(\zeta - \tau_1)} \quad (16-140)$$

After eliminating n_i^* or c_i^* using the adsorption isotherm, Eq. (16-140) can be integrated directly to obtain the constant pattern.

For other mechanisms, the particle-scale equation must be integrated. Equation (16-138) is used to advantage. For example, for external mass transfer acting alone, the dimensionless rate equation in Table 16-13 would be transformed into the $(\zeta - \tau_1, \tau_1)$ coordinate system and derivatives with respect to τ_1 discarded. Equation (16-138) is then used to replace c_i^* with $\bar{n}_i^*$ in the transformed equation. Furthermore, for this case there are assumed to be no gradients within the particles, so we have $\bar{n}_i^* = n_i^*$. After making this substitution, the transformed equation can be rearranged to

$$-\frac{dn_i^*}{n_i^* - c_i^*} = d(\zeta - \tau_1) \quad (16-141)$$

Since n_i^* and c_i^* are related by the adsorption isotherm, Eq. (16-141) can be integrated.

The integration of Eq. (16-140) or (16-141) as an indefinite integral will give an integration constant that must be evaluated to center the transition properly. The material balance depicted in Fig. 16-26 is used. The two shaded regions must be of equal area if the stoichiometric center of the transition is located where the throughput parameter is unity. Thus, we have

$$\int_{-\infty}^0 (1 - n_i^*) d(\zeta - \tau_1) = \int_0^{\infty} n_i^* d(\zeta - \tau_1) \quad (16-142)$$

Integrating Eq. (16-142) by parts gives

$$\int_0^1 (\zeta - \tau_1) dn_i^* = 0 \quad (16-143)$$

For all mechanisms except axial dispersion, the transition can be centered just as well using c_i^* , because of Eq. (16-138). For axial dispersion, the transition should be centered using n_i^* , provided the fluid-phase accumulation term in the material balance, Eq. (16-124), can be neglected. If fluid-phase accumulation is important, then the

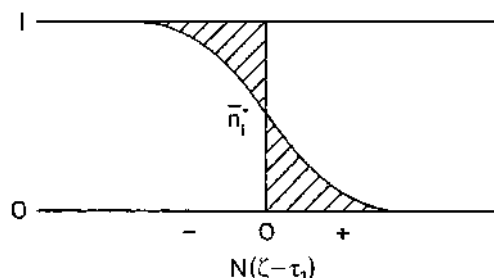


FIG. 16-26 Material balance for centering profile. [From LeVan in Rodrigues *et al.* (eds), *Adsorption: Science and Technology*, Kluwer Academic Publishers, Dordrecht, The Netherlands, 1989; reprinted with permission.]

transition for axial dispersion can be centered by taking into account the relative quantities of solute held in the fluid and adsorbed phases.

Constant pattern solutions for the individual mechanisms and constant separation factor isotherm are given in Table 16-13. The solutions all have the expected dependence on R —the more favorable the isotherm, the sharper the profile.

Figure 16-27 compares the various constant pattern solutions for $R = 0.5$. The curves are of a similar shape. The solution for reaction kinetics is perfectly symmetrical. The curves for the axial dispersion fluid-phase concentration profile and the linear driving force approximation are identical except that the latter occurs one transfer unit further down the bed. The curve for external mass transfer is exactly that for the linear driving force approximation turned upside down [i.e., rotated 180° about $c_i^* = n_i^* = 0.5$, $N(\zeta - \tau_1) = 0$]. The linear driving force

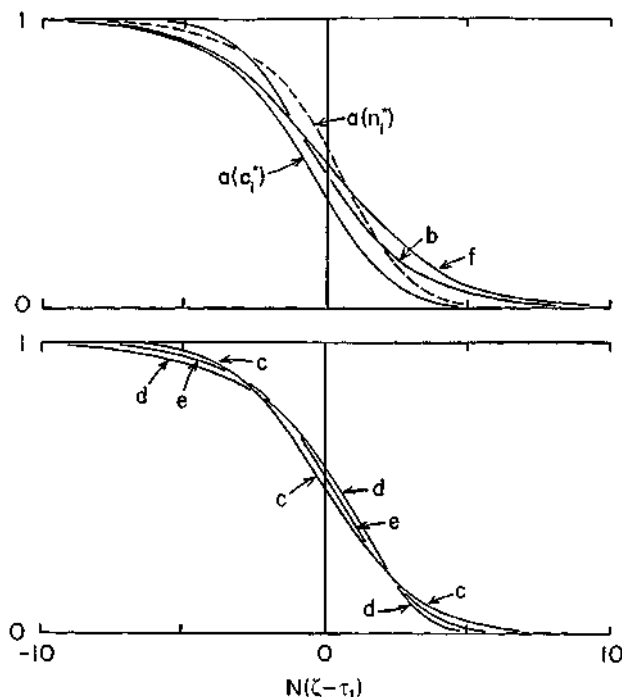


FIG. 16-27 Constant pattern solutions for $R = 0.5$. Ordinate is c_i^* or n_i^* except for axial dispersion for which individual curves are labeled: a, axial dispersion; b, external mass transfer; c, pore diffusion (spherical particles); d, surface diffusion (spherical particles); e, linear driving force approximation; f, reaction kinetics. [From LeVan in Rodrigues *et al.* (eds.), *Adsorption: Science and Technology*, Kluwer Academic Publishers, Dordrecht, The Netherlands, 1989; reprinted with permission.]

approximation provides a good approximation for both pore diffusion and surface diffusion.

Because of the close similarity in shape of the profiles shown in Fig. 16-27 (as well as likely variations in parameters; e.g., concentration-dependent surface diffusion coefficient), a controlling mechanism cannot be reliably determined from transition shape. If reliable correlations are not available and rate parameters cannot be measured in independent experiments, then particle diameters, velocities, and other factors should be varied and the observed impact considered in relation to the definitions of the numbers of transfer units.

Example 13: Estimation of Breakthrough Time With reference to Example 9, determine the 10 percent breakthrough time and the column dynamic binding capacity if the column is 20 cm long.

1. From Example 9, $\Lambda = 30$, $R = 0.1$, and $D_p = 5.8 \times 10^{-8}$ cm²/s. For pore diffusion control (Table 16-13, item A), $N = 15(1 - \epsilon)\epsilon_p D_p L / (\epsilon V \cdot r_p^2) = 4.2$. Based on the numerical solution of Hall *et al.*, $N(1 - \tau_1) \sim 1.3$ with $c_i^* = 0.1$, giving $\tau_1 = 0.69$. The corresponding time is $t = (L/\epsilon v)(\epsilon + \Lambda\tau_1) = (20/0.05)(0.4 + 30 \times 0.69) \sim 8,400$ s. The column dynamic binding capacity (DBC_{10%}) is the amount of adsorbate retained in the column per unit column volume when the outlet concentration reaches 10 percent of the feed. As a fraction of equilibrium capacity, DBC_{10%} is equal to τ_1 , neglecting the adsorbate held in the pore volume and the leakage prior to 10 percent of breakthrough.

2. From Example 9, the rate coefficient for the linear driving-force approximation (Table 16-13, item E) is $k_n = 2.8 \times 10^{-4}$ s⁻¹. Thus, $N = k_n \Lambda L / \epsilon v = 2.8 \times 10^{-4} \times 30 \times 20 / 0.05 = 3.4$ and

$$N(1 - \tau_1) = \frac{1}{1 - 0.1} \ln \left(\frac{1 - 0.1}{0.1} \right) + 1 = 1.1$$

with $c_i^* = 0.1$, giving $\tau_1 = 0.67$. The corresponding breakthrough time is $t = 8,200$ s.

3. If the isotherm is approximated as irreversible ($R = 0$), from Table 16-13, item B, we obtain $N(1 - \tau_1) \sim 1.03$ with $c_i^* = 0.1$, giving $\tau_1 = 0.75$ with a corresponding breakthrough time $t = 9,100$ s.

Breakthrough Behavior for Axial Dispersion Breakthrough behavior for adsorption with axial dispersion in a deep bed is not adequately described by the constant pattern profile for this mechanism. Equation (16-128), the partial differential equation of the second order Fickian model, requires two boundary conditions for its solution. The constant pattern pertains to a bed of infinite depth—in obtaining the solution we apply the downstream boundary condition $c_i^* \rightarrow 0$ as $N_{Pe}\zeta \rightarrow \infty$. Breakthrough behavior presumes the existence of a bed outlet, and a boundary condition must be applied there.

The full mathematical model for this problem is Eq. (16-128) with boundary conditions

$$c_i^* - \frac{1}{N_{Pe}} \frac{\partial c_i^*}{\partial \zeta} = 1 \quad \text{at} \quad \zeta = 0 \quad (16-144)$$

$$\frac{\partial c_i^*}{\partial \zeta} = 0 \quad \text{at} \quad \zeta = 1 \quad (16-145)$$

and an initial condition. Equation (16-144) specifies a constant flux at the bed inlet, and Eq. (16-145), the Danckwerts-type boundary condition at the bed outlet, is appropriate for fixed-bed adsorption, provided that the partition ratio is large.

The solution to this model for a deep bed indicates an increase in velocity of the fluid-phase concentration wave during breakthrough. This is most dramatic for the rectangular isotherm—the instant the bed becomes saturated, the fluid-phase profile jumps in velocity from that of the adsorption transition to that of the fluid, and a near shock-like breakthrough curve is observed [Coppola and LeVan, *Chem. Eng. Sci.*, **36**, 967 (1981)].

Extensions Existence, uniqueness, and stability criteria have been developed for the constant pattern [Cooney and Lightfoot, *Ind. Eng. Chem. Fundam.*, **4**, 233 (1965); Rhee *et al.*, *Chem. Eng. Sci.*, **26**, 1571 (1971); Rhee and Amundson, *Chem. Eng. Sci.*, **26**, 1571 (1971), **27**, 199 (1972), **29**, 2049 (1974)].

The rectangular isotherm has received special attention. For this, many of the constant patterns are developed fully at the bed inlet, as shown for external mass transfer [Klotz, *Chem. Revs.*, **39**, 241 (1946)], pore diffusion [Vermeulen, *Adv. Chem. Eng.*, **2**, 147 (1958); Hall *et al.*, *Ind. Eng. Chem. Fundam.*, **5**, 212 (1966)], pore diffusion with film resistance in series [Weber and Chakraborti, *AIChE J.*, **20**, 228 (1974)], the

linear driving force approximation [Cooper, *Ind. Eng. Chem. Fundam.*, **4**, 308 (1965)], reaction kinetics [Hiester and Vermeulen, *Chem. Eng. Progress*, **48**, 505 (1952); Bohart and Adams, *J. Amer. Chem. Soc.*, **42**, 523 (1920)], and axial dispersion [Coppola and LeVan, *Chem. Eng. Sci.*, **38**, 991 (1983)].

Multiple mass-transfer resistances have been considered in many studies [Vermeulen, *Adv. in Chem. Eng.*, **2**, 147 (1958); Vermeulen et al., Ruthven, gen. refs.; Fleck et al., *Ind. Eng. Chem. Fundam.*, **12**, 95 (1973); Yoshida et al., *Chem. Eng. Sci.*, **39**, 1489 (1984)].

Treatments of constant pattern behavior have been carried out for multicomponent adsorption [Vermeulen, *Adv. in Chem. Eng.*, **2**, 147 (1958); Vermeulen et al., Ruthven, gen. refs.; Rhee and Amundson, *Chem. Eng. Sci.*, **29**, 2049 (1974); Cooney and Lightfoot, *Ind. Eng. Chem. Fundam.*, **5**, 25 (1966); Cooney and Strusi, *Ind. Eng. Chem. Fundam.*, **11**, 123 (1972); Bradley and Sweed, *AIChE Symp. Ser. No. 152*, **71**, 59 (1975)]. The behavior is such that coexisting compositions advance through the bed together at a uniform rate; this is the *coherence* concept of Helfferich and coworkers [gen. refs.].

Nontrace systems have been considered [Sircar and Kumar, *Ind. Eng. Chem. Proc. Des. Dev.*, **22**, 271 (1983)].

Constant patterns have been developed for adiabatic adsorption [Pan and Basmadjian, *Chem. Eng. Sci.*, **22**, 285 (1967); Ruthven et al., *Chem. Eng. Sci.*, **30**, 803 (1975); Kaguei et al., *Chem. Eng. Sci.*, **42**, 2964 (1987)].

The constant pattern concept has also been extended to circumstances with nonplug flows, with various degrees of rigor, including flow profiles in tubes [Sartory, *Ind. Eng. Chem. Fundam.*, **17**, 97 (1978); Tereck et al., *Ind. Eng. Chem. Res.*, **26**, 1222 (1987)], wall effects [Vortmeyer and Michael, *Chem. Eng. Sci.*, **40**, 2135 (1985)], channeling [LeVan and Vermeulen in Myers and Belfort (eds.), *Fundamentals of Adsorption*, Engineering Foundation, New York (1984), pp. 305–314, *AIChE Symp. Ser. No. 233*, **80**, 34 (1984)], networks [Avilés and LeVan, *Chem. Eng. Sci.*, **46**, 1935 (1991)], and general structures of constant cross section [Rudisill and LeVan, *Ind. Eng. Chem. Res.*, **29**, 1054 (1991)].

SQUARE ROOT SPREADING FOR LINEAR ISOTHERMS

The simplest isotherm is $n_i^* = c_i^*$ corresponding to $R = 1$. For this isotherm, the rate equation for external mass transfer, the linear driving force approximation, or reaction kinetics, can be combined with Eq. (16-130) to obtain

$$\frac{\partial \bar{n}_i^*}{\partial \tau_1} = - \frac{\partial c_i^*}{\partial \zeta} = N(c_i^* - \bar{n}_i^*) \quad (16-146)$$

The solution to this equation, with initial condition $n_i^* = 0$ at $\tau_1 = 0$ and boundary condition $c_i^* = 1$ at $\zeta = 0$, originally obtained for an analogous heat transfer case [Anzelius, *Z. Angew. Math. Mech.*, **6**, 291 (1926); Schumann, *J. Franklin Inst.*, **208**, 405 (1929)], is

$$c_i^* = J(N\zeta, N\tau_1) \quad n_i^* = 1 - J(N\tau_1, N\zeta) \quad (16-147)$$

where the J function is [Hiester and Vermeulen, *Chem. Eng. Prog.*, **48**, 505 (1952)]

$$J(s, t) = 1 - \int_0^s e^{-t-\xi} I_0(2\sqrt{t\xi}) d\xi \quad (16-148)$$

where I_0 is the modified Bessel function of the first kind of order zero. This linear isotherm result can be generalized to remove the assumption that $\bar{c}_i^* \approx c_i^*$ if the throughput parameter is redefined as

$$\tau_1 = \frac{\tau - \varepsilon \zeta}{(1 - \varepsilon)(\rho_p K_i + \varepsilon_p)}$$

The J function is plotted in Fig. 16-28 and tables are available (e.g., Sherwood et al., *Mass Transfer*, McGraw-Hill, New York, 1975). Vermeulen et al. (gen. refs.) discuss several approximations of the J function; see also Eq. (16-185). For large arguments J approaches

$$J(s, t) = \frac{1}{2} \operatorname{erfc} [\sqrt{s} - \sqrt{t}] \quad (16-149)$$

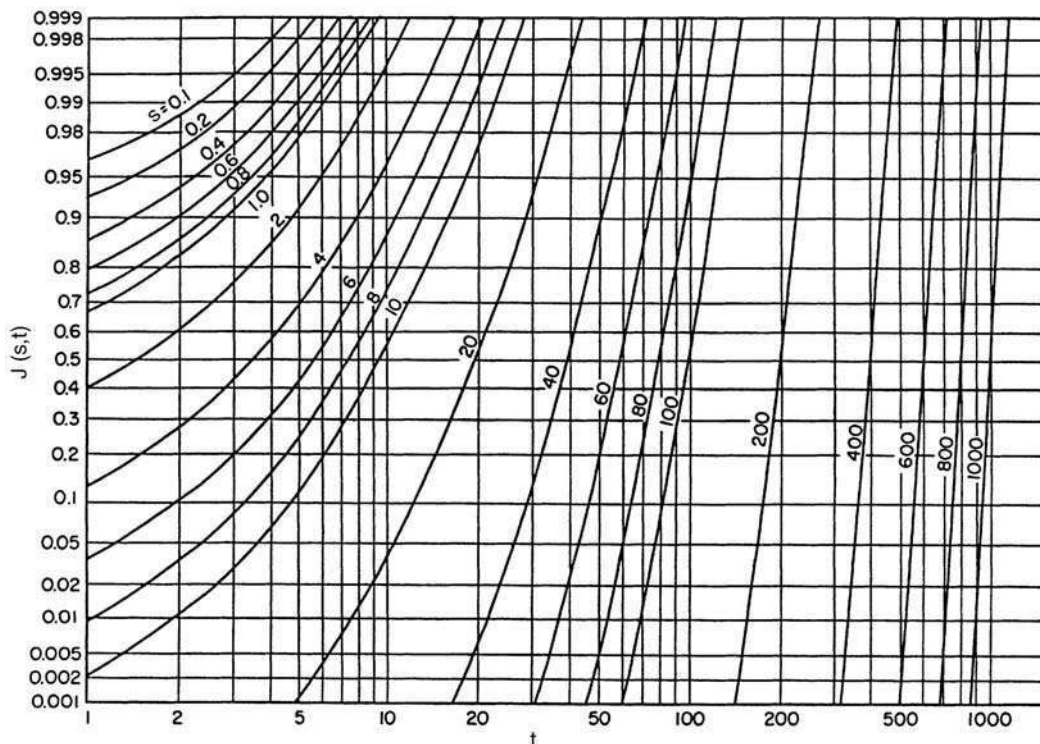


FIG. 16-28 Plot of the function $J(s, t)$ defined by Eq. (16-148).

A derivation for particle-phase diffusion accompanied by fluid-side mass transfer has been carried out by Rosen [*J. Chem. Phys.*, **18**, 1587 (1950); *ibid.*, **20**, 387 (1952); *Ind. Eng. Chem.*, **46**, 1590 (1954)] with a limiting form at $N > 50$ of

$$c_i^* = \frac{1}{2} \operatorname{erfc} \left[\frac{\sqrt{N}}{2} (\zeta - \tau_1) \right] \quad (16-150)$$

For axial dispersion in a semi-infinite bed with a linear isotherm, the complete solution has been obtained for a constant flux inlet boundary condition [Lapidus and Amundson, *J. Phys. Chem.*, **56**, 984 (1952); Brenner, *Chem. Eng. Sci.*, **17**, 229 (1962); Coates and Smith, *Soc. Petrol. Engrs. J.*, **4**, 73 (1964)]. For large N , the leading term is

$$c_i^* = \frac{1}{2} \operatorname{erfc} \left[\frac{\sqrt{N}}{2\sqrt{\tau_1}} (\zeta - \tau_1) \right] \quad (16-151)$$

All of these solutions are very similar and show, for large N , a wave with breadth proportional to the square root of the bed depth through which it has passed.

COMPLETE SOLUTION FOR REACTION KINETICS

In general, full time-dependent analytical solutions to differential equation-based models of the above mechanisms have not been obtained for nonlinear isotherms. Only for reaction kinetics with the constant separation factor isotherm has a full solution been found [Thomas, *J. Amer. Chem. Soc.*, **66**, 1664 (1944)]. Referred to as the *Thomas solution*, it has been extensively studied [Amundson, *J. Phys. Colloid Chem.*, **54**, 812 (1950); Hiester and Vermeulen, *Chem. Eng. Progress*, **48**, 505 (1952); Gilliland and Baddour, *Ind. Eng. Chem.*, **45**, 330 (1953); Vermeulen, *Adv. in Chem. Eng.*, **2**, 147 (1958)]. The solution to Eq. (16-130) for item 4C in Table 16-12 for the same initial and boundary conditions as Eq. (16-146) is

$$\begin{aligned} c_i^* &= \frac{J(RN\zeta, N\tau_1)}{J(RN\zeta, N\tau_1) + e^{-(R-1)N(\zeta-\tau_1)}[1 - J(N\zeta, RN\tau_1)]} \\ n_i^* &= \frac{1 - J(N\tau_1, RN\zeta)}{J(RN\zeta, N\tau_1) + e^{-(R-1)N(\zeta-\tau_1)}[1 - J(N\zeta, RN\tau_1)]} \end{aligned} \quad (16-152)$$

The solution gives all of the expected asymptotic behaviors for large N —the proportionate pattern spreading of the simple wave if $R > 1$, the constant pattern if $R < 1$, and square root spreading for $R = 1$.

NUMERICAL METHODS AND CHARACTERIZATION OF WAVE SHAPE

For the solution of sophisticated mathematical models of adsorption cycles incorporating complex multicomponent equilibrium and rate expressions, two numerical methods are popular, including implementations in commercially available software packages. These are finite difference and weighted residual methods. The former vary in the manner in which distance variables are discretized, ranging from simple backward difference stage models (akin to the plate theory of chromatography) to more involved schemes exhibiting little numerical dispersion. The weighted residual method of orthogonal collocation is often thought to be faster computationally, but oscillations in the polynomial trial function can be a problem. The choice of best method is often the preference of the user.

For both the finite difference and weighted residual methods a set of coupled ordinary differential equations results which are integrated forward in time using the method of lines. Various software packages implementing Gear's method are popular.

The development of mathematical models is described in several of the general references [Guiochon et al., Rhee et al., Ruthven, Ruthven et al., Suzuki, Tien, Wankat, and Yang]. See also Finlayson [*Numerical Methods for Problems with Moving Fronts*, Ravenna Park, Washington, 1992; Holland and Liapis, *Computer Methods for Solving Dynamic Separation Problems*, McGraw-Hill, New York, 1982; Villadsen and Michelsen, *Solution of Differential Equation Models by Polynomial Approximation*, Prentice Hall, Englewood Cliffs, N.J., 1978].

For the characterization of wave shape and breakthrough curves, three methods are popular. The MTZ method [Michaels, *Ind. Eng. Chem.*, **44**, 1922 (1952)] measures the breadth of a wave between two chosen concentrations (e.g., $c_i^* = 0.05$ and 0.95 or $c_i^* = 0.01$ and 0.99). Outside of a laboratory, the measurement of full breakthrough curves is rare, so the breadth of the MTZ is often estimated from an independently determined stoichiometric capacity and a measured small concentration in the "toe" of the breakthrough curve. A second method for characterizing wave shape is by the slope of the breakthrough curve at its midheight (i.e., $c_i^* = 0.5$ [Vermeulen et al., *gen. refs.*]). The use of moments of the slope of breakthrough curves is a third means for characterization. They can often be used to extract numerical values for rate coefficients for linear systems [Ruthven, Suzuki, *gen. refs.*; Nauman and Buffham, *Mixing in Continuous Flow Systems*, Wiley-Interscience, New York, 1983]. The method of moments is discussed further in the following part of this section.

CHROMATOGRAPHY

CLASSIFICATION

Chromatography is a sorptive separation process where a portion of a solute mixture (feed) is introduced at the inlet of a column containing a selective adsorbent (stationary phase) and separated over the length of the column by the action of a carrier fluid (mobile phase) that is continually supplied to the column following introduction of the feed. The mobile phase is generally free of the feed components, but may contain various other species introduced to modulate the chromatographic separation. The separation occurs as a result of the different partitioning of the feed solutes between the stationary phase; the separated solutes are recovered at different times in the effluent from the column. Chromatography is used both in analysis of mixtures and in preparative and process-scale applications. It can be used for both trace-level and for bulk separations both in the gas and the liquid phase.

Modes of Operation The classical modes of operation of chromatography as enunciated by Tiselius [*Kolloid Z.*, **105**, 101 (1943)] are: elution chromatography, frontal analysis, and displacement development. Basic features of these techniques are illustrated in Fig. 16-29. Often, each of the different modes can be implemented with the same equipment and stationary phase. The results are, however, quite different in the three cases.

Elution Chromatography The components of the mobile phase supplied to the column after feed introduction have less affinity for the stationary phase than any of the feed solutes. Under trace conditions, the feed solutes travel through the column as bands or zones at different velocities that depend only on the composition of the mobile phase and the operating temperature and that exit from the column at different times.

Two variations of the technique exists: **isocratic elution**, when the mobile phase composition is kept constant, and **gradient elution**, when the mobile phase composition is varied during the separation. Isocratic elution is often the method of choice for analysis and in process applications when the retention characteristics of the solutes to be separated are similar and not dramatically sensitive to very small changes in operating conditions. Isocratic elution is also generally practical for systems where the equilibrium isotherm is linear or nearly linear. In all cases, isocratic elution results in a dilution of the separated products.

In **gradient elution**, the eluting strength of the mobile phase is gradually increased after supplying the feed to the column. In liquid chromatography, this is accomplished by changing the mobile phase composition. The gradient in eluting strength of the mobile phase that is generated in the column is used to modulate the separation allowing control of the retention time of weakly and strongly retained

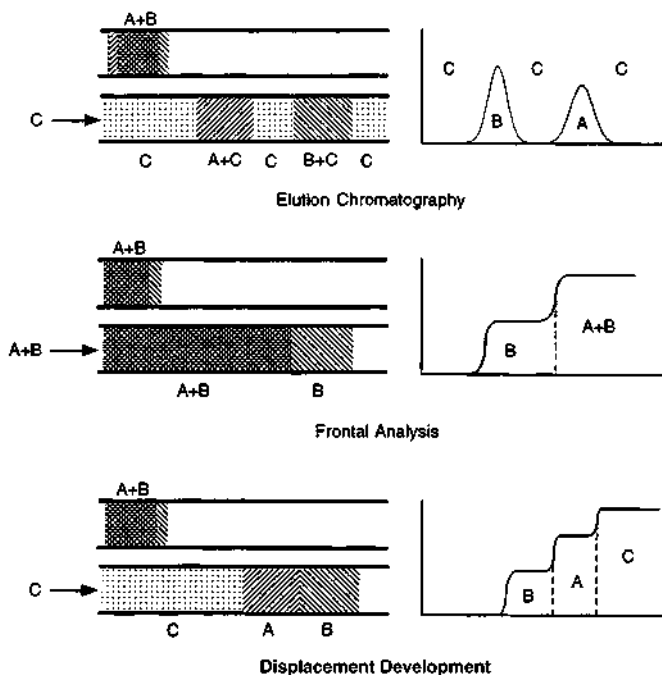


FIG. 16-29 Modes of operation of chromatography for the separation of a mixture of two components A and B. Figures on the left represent a schematic of the column with sample passing through it. Top diagrams show column at end of feed step. Figures on the right show the corresponding effluent concentrations as might be seen by a non-specific detector. C is either the eluent or the displacer. (Adapted from Ettre, 1980.)

components. A similar effect can be obtained in gas chromatography by modulating the column temperature. In either case, the column has to be brought back to the initial conditions before the next cycle is commenced.

Generally, gradient elution is best suited for the separation of complex mixtures that contain both species that interact weakly with the stationary phase and species that interact strongly. Since the eluting strength of the mobile phase is adjusted continuously, weakly retained components of a mixture are separated in the initial phase when the relative elution strength of the mobile phase is low, while strongly retained components are separated later in the gradient when the elution strength is high. In addition, the technique is used to obtain reproducible chromatographic separations when the solute retention characteristics are extremely sensitive to the operating conditions, as in the case of the chromatography of biopolymers, such as proteins. These molecules are often found to be very strongly retained in an extremely small range of mobile phase compositions and completely unretained elsewhere, making it practically impossible to obtain reproducible isocratic separations.

Frontal Analysis The feed mixture to be separated is continuously supplied to the column where the mixture components are competitively retained by the stationary phase. These solutes are partially separated in a series of fronts, preceded downstream by the least strongly retained species forming a pure component band, and upstream by the feed mixture. The technique is best suited for the removal of strongly adsorbed impurities present in trace amounts from an unretained or weakly retained product of interest. In this case, a large amount of feed can be processed before the impurities begin to break through. When this point is reached, the bed is washed to remove any desired product from the interstitial voids, and the adsorbent is regenerated. The method can only provide a single component in pure form, but avoids product dilution completely. Multicomponent

separations require a series of processing steps; either a series of frontal analysis separations, or a combination of elution and displacement separations. Example 11 illustrates bed concentration profiles for the frontal analysis separation of two components.

Displacement Development The column is partially loaded with the feed mixture as in frontal analysis, usually for conditions where all solutes of interest are strongly and competitively retained by the stationary phase. The feed supply is then stopped and a mobile phase containing the **displacer**, a component that has an affinity for the stationary phase stronger than any of the feed components, is fed to the column. The advancement of the displacer front through the column causes desorption of the feed components and their competitive readsorption downstream of the displacer front. As in frontal chromatography, the less strongly retained species tends to migrate faster down the column concentrating in a band farthest from the displacer front, while the most strongly adsorbed solute tends to move more slowly concentrating in a band adjacent to the displacer front. If the column is sufficiently long, all feed components eventually become distributed into a pattern of adjacent pure component bands where each upstream component acts as a displacer for each downstream species located in the band immediately downstream. When this occurs, all bands in the displacement train move at the same velocity which is equal to the velocity of the displacer front and the bed concentration profile is called an **isotachic pattern**.

The various operational steps of a displacement development separation are shown in Fig. 16-30. Ideally, the separated species exit the column as adjacent rectangular bands in order of increasing affinity for the stationary phase as shown in this figure. In practice, dispersion effects result in a partial mixing of adjacent bands requiring recycling of portions of the effluent that do not meet purity requirements. Following separation, the displacer has to be removed from the column with a suitable regenerant and the initial conditions of the column restored

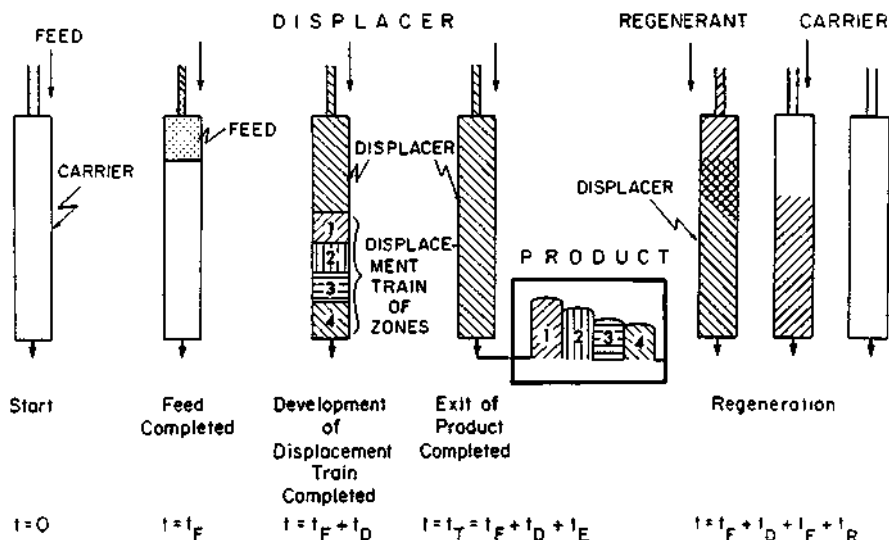


FIG. 16-30 Operational steps in displacement chromatography. The column, initially equilibrated with a carrier solvent at time 0, is loaded with feed until time t_F and supplied with displacer for a time $t_D + t_E$. Development of the displacement train occurs during the time t_D and elution of the separated products ends at time t_E . t_R is the time required to remove the displacer from the column and restore the initial conditions. Components are numbered in order of decreasing affinity for the stationary phase. [Reference: Horvath et al., *J. Chromatogr.*, **218**, 365 (1981). Reprinted with permission of *J. Chromatogr.*]

before the next cycle. Column regeneration may consume a significant portion of the cycle, when removal of the displacer is difficult.

Displacement chromatography is suitable for the separation of multicomponent bulk mixtures. For dilute multicomponent mixtures it allows a simultaneous separation and concentration. Thus, it permits the separation of compounds with extremely low separation factors without the excessive dilution that would be obtained in elution techniques.

Other modes of operation, including recycle and flow reversal schemes and continuous chromatography, are discussed in Ganetsos and Barker (*Preparative and Production Scale Chromatography*, Marcel Dekker, New York, 1993).

CHARACTERIZATION OF EXPERIMENTAL CHROMATOGRAMS

Method of Moments The first step in the analysis of chromatographic systems is often a characterization of the column response to small pulse injections of a solute under trace conditions in the Henry's law limit. For such conditions, the statistical moments of the response peak are used to characterize the chromatographic behavior. Such an approach is generally preferable to other descriptions of peak properties which are specific to Gaussian behavior, since the statistical moments are directly correlated to equilibrium and dispersion parameters. Useful references are Schneider and Smith [*AIChE J.*, **14**, 762 (1968)], Suzuki and Smith [*Chem. Eng. Sci.*, **26**, 221 (1971)], and Carbonell et al. [*Chem. Eng. Sci.*, **9**, 115 (1975); **16**, 221 (1978)].

The relevant moments are:

$$\mu_0 = \int_0^\infty c_i dt \quad (16-153)$$

$$\mu_1 = \frac{1}{\mu_0} \int_0^\infty c_i t dt \quad (16-154)$$

$$\sigma^2 = \frac{1}{\mu_0} \int_0^\infty c_i (t - \mu_1)^2 dt \quad (16-155)$$

where c_i is the peak profile. μ_0 represents the area, μ_1 the mean residence time, and σ^2 the variance of the response peak. Moments can be calculated by numerical integration of experimental profiles.

The **retention factor** is defined as a dimensionless peak locator as:

$$k'_i = \frac{\mu_i - \mu_1^0}{\mu_1^0} \quad (16-156)$$

where μ_1^0 is the first moment obtained for an inert tracer which is excluded from the stationary phase. k'_i is a partition ratio representing the equilibrium ratio of the amount of solute in the stationary phase (including any pores) and the amount in the external mobile phase. Another commonly used definition of the retention factor uses as a reference the first moment of an inert that has access to all the pores.

The number of plates, N_p , and the height equivalent to a theoretical plate, HETP, are defined as measures of dispersion effects as:

$$N_p = \frac{L}{\text{HETP}} = \frac{\mu_1^2}{\sigma^2} \quad (16-157)$$

$$\text{HETP} = \frac{\sigma^2 L}{\mu_1^2} \quad (16-158)$$

A high number of plates and a low HETP indicate a high column efficiency.

Higher moments can also be computed and used to define the skewness of the response peak. However, difficulties often arise in such computations as a result of drifting of the detection system.

In practice, experimental peaks can be affected by extracolumn retention and dispersion factors associated with the injector, connections, and any detector. For linear chromatography conditions, the apparent response parameters are related to their corresponding true column value by

$$\mu_1^{app} = \mu_1 + \mu_1^{inj} + \mu_1^{conn} + \mu_1^{det} \quad (16-159)$$

$$\sigma^{2app} = \sigma^2 + \sigma^{2inj} + \sigma^{2conn} + \sigma^{2det} \quad (16-160)$$

Approximate Methods For certain conditions, symmetrical, Gaussian-like peaks are obtained experimentally. Such peaks may be

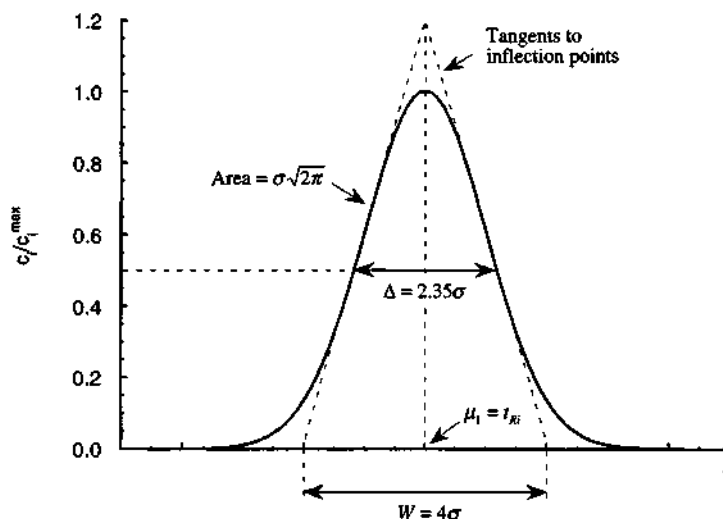


FIG. 16-31 Properties of a Gaussian peak. $c_i^{\max}$ is the peak height; t_{Ri} , the peak apex time; σ , the standard deviation; Δ , the peak width at midheight; and W , the distance between the baseline intercepts of the tangents to the peak.

empirically described by:

$$c_i = \frac{Q_i/F_v}{\sigma\sqrt{2\pi}} \exp\left[-\left(\frac{t-t_{Ri}}{\sigma\sqrt{2}}\right)^2\right] = \frac{Q_i/F_v}{t_{Ri}} \sqrt{\frac{N_p}{2\pi}} \exp\left[-\frac{N_p}{2}\left(\frac{t}{t_{Ri}}-1\right)^2\right] \quad (16-161)$$

where Q_i is the amount of solute injected, F_v is the volumetric flow rate, and t_{Ri} is the peak apex time. The relationships between the moments and other properties of such peaks are shown in Fig. 16-31. For such peaks, approximate calculations of the number of plates can be done with the following equations:

$$N_p = 5.54 \left(\frac{t_{Ri}}{\Delta}\right)^2 \quad (16-162)$$

$$N_p = 16 \left(\frac{t_{Ri}}{W}\right)^2 \quad (16-163)$$

$$N_p = 2\pi \left(\frac{c_i^{\max} t_{Ri}}{\mu_0}\right)^2 \quad (16-164)$$

where Δ is the peak width at half peak height and W is the distance between the baseline intercepts of the tangents to the inflection points of the peak.

In general, Gaussian behavior can be tested by plotting the cumulative fractional recovery $\int_0^t c_i dt/\mu_0$ versus time on probability-linear coordinates; if the plot is linear, Gaussian behavior is confirmed. For nearly Gaussian peaks, calculations of N_p based on Eqs. (16-162) to (16-164) provide results close to those obtained with a rigorous calculation of moments. When deviations from Gaussian behavior are significant, however, large errors can be obtained using these expressions.

Tailing Peaks Tailing peaks can be obtained experimentally when the column efficiency is very low, when there are large extracolumn dispersion effects, when the stationary phase is heterogeneous (in the sense that it contains different adsorption sites), or when the adsorption equilibrium deviates from the Henry's law limit. Asymmetrical tailing peaks can sometimes be described empirically by an exponentially modified Gaussian (EMG) defined as the convolute integral of a Gaussian constituent with mean time t_C and standard deviation σ_C and an exponential decay with time constant τ_C [Grushka, *Anal. Chem.*, **44**, 1733 (1972)]:

$$c_i = \frac{Q_i/F_v}{\sigma_C \tau_C \sqrt{2\pi}} \int_0^\infty \exp\left[-\left(\frac{t-t_C-t'}{\sigma_C\sqrt{2}}\right)^2 - \frac{t'}{\tau_C}\right] dt' \quad (16-165)$$

Although a numerical integration is required to compute the peak profile, the moments are calculated directly as $\mu_1 = t_C + \tau_C$ and $\sigma^2 = \sigma_C^2 + \tau_C^2$ and the **peak skew** as:

$$\text{Peak skew} = \frac{1}{\mu_0 \sigma^3} \int_0^\infty c_i (t - \mu_1)^3 dt \sim \frac{2(\tau_C/\sigma_C)^3}{[1 + (\tau_C/\sigma_C)^2]^{3/2}} \quad (16-166)$$

For EMG peaks, peak skew increases with the ratio τ_C/σ_C . Figure 16-32 illustrates the characteristics of such a peak calculated for $\tau_C/\sigma_C = 1.5$. In general, with $\tau_C/\sigma_C > 1$ (peak skew > 0.7), a direct calculation of the moments is required to obtain a good approximation of the true value of N_p , since other methods give a large error (Yau et al., *Moderns Size-Exclusion Liquid Chromatography*, Wiley, New York, 1979). Alternatively, Eq. (16-165) can be fitted to experimental peaks to determine the optimum values of t_C , σ_C , and τ_C .

In practice, the calculation of peak skew for highly tailing peaks is rendered difficult by baseline errors in the calculation of third moments. The **peak asymmetry factor**, $A_s = b/a$, at 10 percent of peak height (see Fig. 16-32) is thus frequently used. An approximate relationship between peak skew and A_s for tailing peaks, based on data in Yau et al. is: Peak skew = $[0.51 + 0.19/(A_s - 1)]^{-1}$. Values of $A_s < 1.25$ (corresponding to peak skew < 0.7) are generally desirable for an efficient chromatographic separation.

Resolution The chromatographic separation of two components, A and B, under trace conditions with small feed injections can be characterized in terms of the resolution, R_s . For nearly Gaussian peaks:

$$R_s = \frac{2(t_{R,A} - t_{R,B})}{W_A + W_B} \sim \frac{\Delta t_R}{4\sigma_{AB}} \quad (16-167)$$

where Δt_R is the difference in retention time of the two peaks and $\sigma_{AB} = (\sigma_A + \sigma_B)/2$ is the average of their standard deviations. When Eq. (16-161) is applicable, the resolution for two closely spaced peaks is approximated by:

$$R_s = \frac{1}{2} \frac{\alpha - 1}{\alpha + 1} \frac{\bar{k}'_i}{1 + \bar{k}'_i} \sqrt{N_p} \sim \frac{\alpha - 1}{4} \frac{k'_A}{1 + k'_A} \sqrt{N_p} \quad \text{for } \alpha \sim 1 \quad (16-168)$$

where $\alpha = k'_A/k'_B$ and $\bar{k}' = (k'_A + k'_B)/2$.

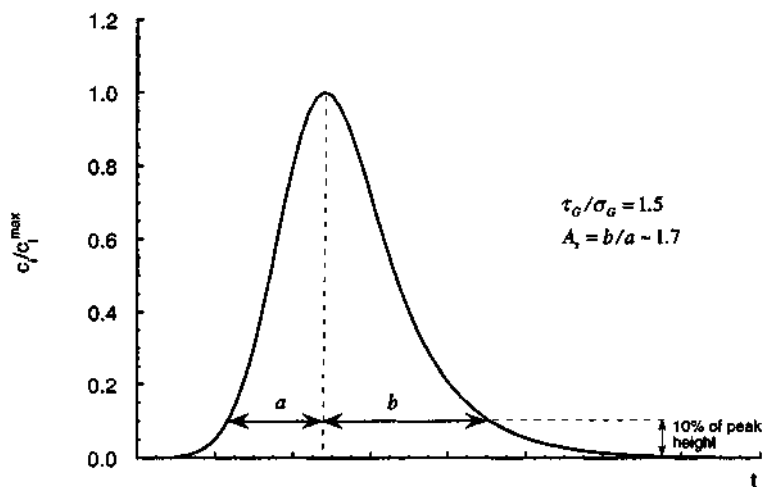


FIG. 16-32 Exponentially modified Gaussian peak with $\tau_c/\sigma_c = 1.5$. The graph also shows the definition of the peak asymmetry factor A_s at 10 percent of peak height.

Equation (16-168) shows that the resolution is the result of independent effects of the separation selectivity (α), column efficiency (N_p), and capacity ($\bar{k}'$). Generally, peaks are essentially completely resolved when $R_s = 1.5$ (>99.5 percent separation). In practice, values of $R_s \sim 1$, corresponding to 98 percent separation, are often considered adequate.

The preceding equations are accurate to within about 10 percent for feed injections that do not exceed 40 percent of the final peak width. For large, rectangular feed injections, the baseline width of the response peak is approximated by:

$$W \sim 4\sigma + t_F \quad (16-169)$$

where 4σ is the baseline width obtained with a small pulse injection and t_F is the duration of feed injection. In this case, the resolution is defined as (see Ruthven, gen. refs., pp. 324–331):

$$R_s = \frac{t_{R,A} - t_{R,B} - t_F}{4\sigma_{AB}} \quad (16-170)$$

For strongly retained components ($\bar{k}' \gg 1$), the number of plates required to obtain a given resolution with a finite feed injection is approximated by:

$$N_p = 4R_s^2 \left(\frac{\alpha + 1}{\alpha - 1} \right)^2 \left(1 + \frac{t_F}{4\sigma_{AB}R_s} \right)^2 \quad (16-171)$$

PREDICTION OF CHROMATOGRAPHIC BEHAVIOR

The conservation equations and the rate models described in the "Rate and Dispersion Factors" can normally be used for a quantitative description of chromatographic separations. Alternatively, **plate models** can be used for an approximate prediction, lumping together all dispersion contributions into a single parameter, the HETP or the number of plates [Sherwood et al., *Mass Transfer*, McGraw-Hill, New York, 1975, p. 576; Dondi and Guiochon, *Theoretical Advancements in Chromatography and Related Techniques*, NATO-ASI, Series C: Mathematical and Physical Sciences, vol. 383, Kluwer, Dordrecht, 1992, pp. 1–61]. Exact analytic solutions are generally available for linear isocratic elution under trace conditions (see Dondi and Guiochon, *ibid.*, Ruthven [gen. refs., pp. 324–335], Suzuki [gen. refs., pp. 224–243]). Other cases generally require numerical solution (see Guiochon et al., gen. refs.) or approximate treatments with simplified rate models.

Isocratic Elution In the simplest case, feed with concentration c_i^F is applied to the column for a time t_F followed by the pure carrier fluid. Under trace conditions, for a linear isotherm with external mass-transfer control, the linear driving force approximation or reaction kinetics (see Table 16-12), solution of Eq. (16-146) gives the following expression for the dimensionless solute concentration at the column outlet:

$$c_i^* = J(N, N\tau_1) - J(N, N\tau_1') \quad (16-172)$$

where N is the number of transfer units given in Table 16-13 and $\tau_1 = (\epsilon v t_F / L - \epsilon) / [(1 - \epsilon)(\rho_p K_i + \epsilon_p)]$ the throughput parameter (see "Square Root Spreading for Linear Isotherms" in "Fixed Bed Transitions"). τ_1' represents the value of τ_1 with time measured from the end of the feed step. Thus, the column effluent profile is obtained as the difference between a breakthrough profile for a feed started at $t = 0$ and another for a feed started at $t = t_F$.

The behavior predicted by this equation is illustrated in Fig. 16-33 with $N = 80$. $\tau_F = (\epsilon v t_F / L) / [(1 - \epsilon)(\rho_p K_i + \epsilon_p)]$ is the dimensionless duration of the feed step and is equal to the amount of solute fed to the column divided by the sorption capacity. Thus, at $\tau_F = 1$, the column has been supplied with an amount of solute equal to the stationary phase capacity. The graph shows the transition from a case where complete saturation of the bed occurs before elution ($\tau_F = 1$) to incomplete saturation as τ_F is progressively reduced. The lower curves with $\tau_F \leq 0.4$ are seen to be nearly Gaussian and centered at a dimensionless time $\tau_m \sim (1 - \tau_F/2)$. Thus, as $\tau_F \rightarrow 0$, the response curve approaches a Gaussian centered at $\tau_1 = 1$.

When τ_F is small ($\ll 0.4$), the solution for a feed pulse represented by a Dirac's delta function at $\zeta = 0$ can be used in lieu of Eq. (16-172). In terms of the dimensionless concentration $c_i^* = c_i/c_i^F$ (Sherwood et al., *ibid.*, pp. 571–577):

$$c_i^* = \frac{\tau_F N}{\sqrt{\tau_1}} e^{-N} e^{-N\tau_1} I_1(2N\sqrt{\tau_1}) \quad (16-173)$$

where I_1 is the Bessel function of imaginary argument. When N is larger than 5, this equation is approximated by:

$$c_i^* = \frac{\tau_F}{2\sqrt{\pi}} \sqrt{\frac{N}{\tau_1}} \frac{\exp[-N(\sqrt{\tau_1} - 1)^2]}{(\tau_1)^{1/4}} \quad (16-174)$$

The behavior predicted by Eq. (16-174) is shown in Fig. 16-34 as $c_i^*/(\tau_F\sqrt{N})$ versus τ_1 for different values of N . For $N > 50$, the response peak is symmetrical, the peak apex occurs at $\tau_1 = 1$, and the

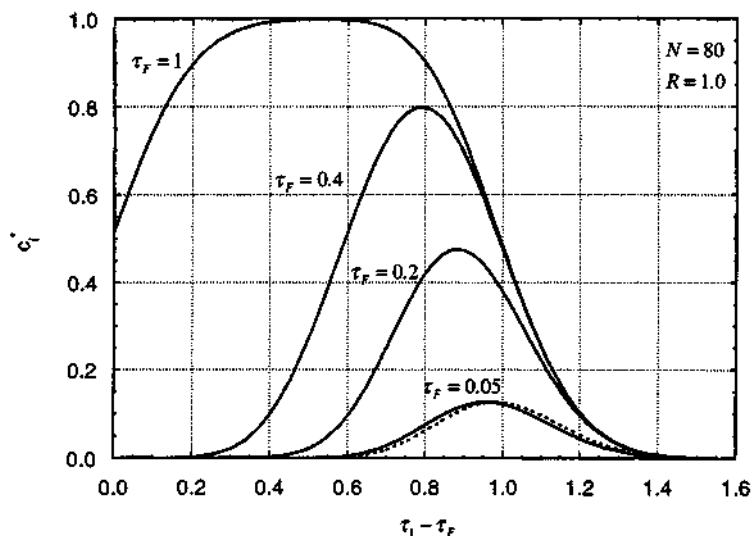


FIG. 16-33 Elution curves under trace linear equilibrium conditions for different feed loading periods and $N = 80$. Solid lines, Eq. (16-172); dashed line, Eq. (16-174) for $\tau_F = 0.05$.

dimensionless peak height is $c_i^{*max} = \tau_F \sqrt{N/4\pi}$. A comparison of this equation with Eq. (16-172) with $N = 80$ is shown in Fig. 16-33 for $\tau_F = 0.05$.

The moments of the response peak predicted by Eq. (16-173) are

$$\mu_1 = \frac{L}{v} \left[1 + \frac{1-\epsilon}{\epsilon} (\epsilon_p + \rho_p K_i) \right] = \frac{L}{v} (1 + k'_i) \quad (16-175)$$

$$\sigma^2 = \frac{2\mu_1^2}{N} \left(\frac{k'_i}{1 + k'_i} \right)^2 \quad (16-176)$$

where $k'_i = (1 - \epsilon)(\epsilon_p + \rho_p K_i)/\epsilon$. Correspondingly, the number of plates and the HETP are:

$$N_p = \frac{N}{2} \left(\frac{1 + k'_i}{k'_i} \right)^2 \quad (16-177)$$

$$\text{HETP} = \frac{2L}{N} \left(\frac{k'_i}{1 + k'_i} \right)^2 \quad (16-178)$$

Since the term $(1 + k'_i)/k'_i$ approaches unity for large k'_i -value, the number of plates is equal to one half the number of transfer units for a strongly retained component. For these conditions, when $N_p = N/2$, Eq. (16-174) and Eq. (16-161) produce the same peak retention time, peak spreading, and predict essentially the same peak profile.

In the general case of axially dispersed plug flow with bidispersed particles, the first and second moment of the pulse response are [Haynes and Sarma, *AIChE J.*, **19**, 1043 (1973)]:

$$\mu_1 = \frac{L}{v} \left[1 + \frac{1-\epsilon}{\epsilon} (\epsilon_p + \rho_p K_i) \right] = \frac{L}{v} (1 + k'_i) \quad (16-179)$$

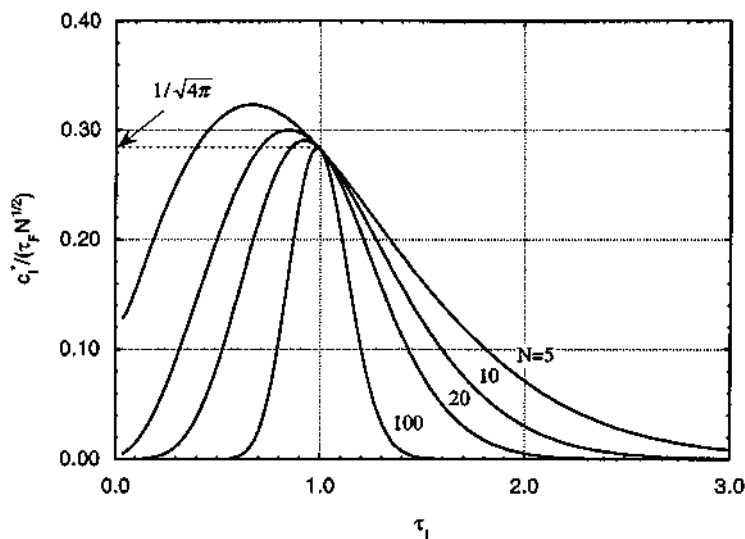


FIG. 16-34 Elution curves under trace linear equilibrium conditions with a pulse feed from Eq. (16-174).

$$\sigma^2 = \frac{2LD_L}{v^3} (1 + k'_i)^2 + 2 \frac{L}{v} \frac{\epsilon k'_i{}^2}{1 - \epsilon} \left[\frac{r_p}{3k_f} + \frac{r_p^2}{15\epsilon_p D_{pi}} + \frac{\rho_p K_i}{(\epsilon_p + \rho_p K_i)^2} \frac{r_s^2}{15D_{si}} \right] \quad (16-180)$$

Correspondingly, the number of plates and the plate height are:

$$N_p = \left\{ \frac{2D_L}{vL} + \frac{2\epsilon}{1 - \epsilon} \frac{v}{L} \left(\frac{k'_i}{1 + k'_i} \right)^2 \times \left[\frac{r_p}{3k_f} + \frac{r_p^2}{15\epsilon_p D_{pi}} + \frac{\rho_p K_i}{(\epsilon_p + \rho_p K_i)^2} \frac{r_s^2}{15D_{si}} \right] \right\}^{-1} \quad (16-181)$$

$$\text{HETP} = \frac{2D_L}{v} + \frac{2\epsilon v}{1 - \epsilon} \left(\frac{k'_i}{1 + k'_i} \right)^2 \times \left[\frac{r_p}{3k_f} + \frac{r_p^2}{15\epsilon_p D_{pi}} + \frac{\rho_p K_i}{(\epsilon_p + \rho_p K_i)^2} \frac{r_s^2}{15D_{si}} \right] \quad (16-182)$$

In dimensionless form, a **reduced HETP**, $h = \text{HETP}/d_p$, analogous to the reduced HTU (cf. Fig. 16-13), is obtained as a function of the dimensionless velocity $ReSc$:

$$h = \frac{b}{ReSc} + a + cReSc \quad (16-183)$$

where $b = 2\epsilon\gamma_1$ (16-184a)

$a = 2\gamma_2$ (16-184b)

$$c = \frac{1}{30} \frac{1}{1 - \epsilon} \left(\frac{k'_i}{1 + k'_i} \right)^2 \left[\frac{10}{Sh} + \frac{\tau_p}{\epsilon_p} + \frac{\rho_p K_i}{(\epsilon_p + \rho_p K_i)^2} \frac{r_s^2}{D_{si}} \frac{D_{pi}}{r_p^2} \right] \quad (16-184c)$$

Equation (16-183) is qualitatively the same as the **van Deemter equation** [van Deemter and Zuidervog, *Chem. Eng. Sci.*, **5**, 271 (1956)] and is equivalent to other empirical reduced HETP expressions such as the **Knox equation** [Knox, *J. Chromatogr. Sci.*, **15**, 352 (1977)].

The Sherwood number, Sh , is estimated from Table 16-9, and the dispersion parameters γ_1 and γ_2 from Table 16-10 for well-packed columns. Typical values are $a = 1-4$ and $b = 0.5-1$. Since $\text{HETP} \sim 2\text{HTU}$, Fig. 16-13 can also be used for approximate calculations.

Concentration Profiles In the general case but with a linear isotherm, the concentration profile can be found by numerical inversion of the Laplace-domain solution of Haynes and Sarma [see Lenhoff, *J. Chromatogr.*, **384**, 285 (1987)] or by direct numerical solution of the conservation and rate equations. For the special case of no axial dispersion, an explicit time-domain solution is also available in the cyclic steady state for repeated injections of arbitrary duration t_F followed by an elution period t_E with cycle time $t_C = t_F + t_E$ [Carta, *Chem. Eng. Sci.*, **43**, 2877 (1988)]. For the linear driving force mechanism, the solution is

$$c_i^* = \phi + \frac{2}{\pi} \sum_{j=1}^{\infty} \left\{ \frac{1}{j^2} \exp \left(- \frac{j^2 N}{j^2 + \omega^2} \right) \sin(j\pi\phi) \times \cos \left[\frac{2j\pi}{t_C} \left(t - \frac{t_F}{2} - \frac{L}{v} \right) - \frac{j\omega N}{j^2 + \omega^2} \right] \right\} \quad (16-185)$$

where

$$\phi = \frac{t_F}{t_C} \quad (16-185a)$$

$$\omega = \frac{N\epsilon v_c/L}{2\pi(1 - \epsilon)(\rho_p K_i + \epsilon_p)} \quad (16-185b)$$

The average effluent concentration in a product cut between times t_1 and t_2 can be calculated directly from

$$\bar{c}_i^* = \frac{\int_{t_1}^{t_2} c_i^* dt}{t_2 - t_1} \phi + \frac{t_C}{t_2 - t_1} \frac{2}{\pi^2}$$

$$\times \sum_{j=1}^{\infty} \left\{ \frac{1}{j^2} \exp \left(- \frac{j^2 N}{j^2 + \omega^2} \right) \sin(j\pi\phi) \sin \left[j\pi \frac{t_2 - t_1}{t_C} \right] \times \cos \left[\frac{2j\pi}{t_C} \left(\frac{t_1 + t_2 - t_F}{2} - \frac{L}{v} \right) - \frac{j\omega N}{j^2 + \omega^2} \right] \right\} \quad (16-186)$$

The same equations can also be used as a good approximation for other rate mechanisms with N calculated according to Table 16-13. When t_C is large, Eq. (16-185) describes a single feed injection and approaches the results of Eqs. (16-173) and (16-148) for small and large values of ϕ , respectively. See Seader and Henley (gen. refs.), and Carta, op. cit., for sample calculations.

When the adsorption equilibrium is nonlinear, skewed peaks are obtained, even when N is large. For a constant separation-factor isotherm with $R < 1$ (favorable), the leading edge of the chromatographic peak is steeper than the trailing edge. When $R > 1$ (unfavorable), the opposite is true.

Figure 16-35 portrays numerically calculated chromatographic peaks for a constant separation factor system showing the effect of feed loading on the elution profile with $R = 0.5$ ($\tau_1 - \tau_F = [\epsilon v(t - t_F)/L - \epsilon]/\Lambda$). When the dimensionless feed time $\tau_F = 1$, the elution curve comprises a sharp leading profile reaching the feed concentration followed by a gradual decline to zero. As τ_F is reduced, breakthrough of the leading edge occurs at later times, while the trailing edge, past the peak apex, continues to follow the same profile. As the amount of feed injected approaches zero, mass-transfer resistance reduces the solute concentration to values that fall in the Henry's law limit of the isotherm, and the peak retention time gradually approaches the value predicted in the infinite dilution limit for a linear isotherm.

For high feed loads, the shape of the diffuse trailing profile and the location of the leading front can be predicted from local equilibrium theory (see "Fixed Bed Transitions"). This is illustrated in Fig. 16-35 for $\tau_F = 0.4$. For the diffuse profile (a "simple wave"), Eq. (16-131) gives:

$$c_i^* = \frac{R}{1 - R} \left[\left(\frac{1}{R(\tau_1 - \tau_F)} \right)^{1/2} - 1 \right] \quad (16-187)$$

Thus, the effluent concentration becomes zero at $\tau_1 - \tau_F = 1/R$. The position of the leading edge (a "shock front") is determined from Eq. (16-132):

$$\tau_{1s} = \tau_F + \frac{1}{R} [1 - \sqrt{(1 - R)\tau_F}]^2 \quad (16-188)$$

and the peak highest concentration by [Golshan-Shirazi and Guiochon, *Anal. Chem.*, **60**, 2364 (1988)]:

$$c_i^{* \text{ max}} = \frac{R}{1 - R} \frac{\sqrt{(1 - R)\tau_F}}{1 - \sqrt{(1 - R)\tau_F}} \quad (16-189)$$

The local equilibrium curve is in approximate agreement with the numerically calculated profiles except at very low concentrations when the isotherm becomes linear and near the peak apex. This occurs because band-spreading, in this case, is dominated by adsorption equilibrium, even if the number of transfer units is not very high. A similar treatment based on local equilibrium for a two-component mixture is given by Golshan-Shirazi and Guiochon [*J. Phys. Chem.*, **93**, 4143 (1989)].

Prediction of multicomponent nonlinear chromatography accounting for rate factors requires numerical solution (see Guiochon et al., gen. refs., and "Numerical Methods and Characterization of Wave Shape" in "Fixed Bed Transitions").

Linear Gradient Elution Analytical solutions are available for special cases under trace conditions with a linear isotherm. Other situations normally require a numerical solution. General references are Snyder [in Horvath (ed.), *High Performance Liquid Chromatography: Advances and Perspectives*, vol. 1, Academic Press, 1980, p. 208], Antia and Horvath [*J. Chromatogr.*, **484**, 1 (1989)], Yamamoto et al. [*Ion Exchange Chromatography of Proteins*, Marcel Dekker, 1988], and Guiochon et al. (gen. refs.).

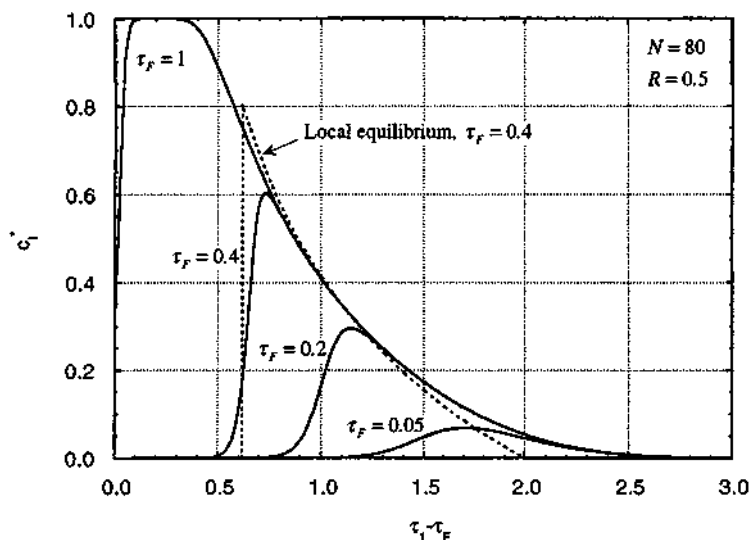


FIG. 16-35 Elution curves under trace conditions with a constant separation factor isotherm for different feed loadings and $N = 80$. Solid lines, rate model; dashed line, local equilibrium theory for $\tau_F = 0.4$.

The most commonly used gradients are linear gradients where the starting solvent is gradually mixed with a second gradient-forming solvent at the column entrance to yield a volume fraction ϕ of the mobile phase modulator that increases linearly with time:

$$\phi = \phi_0 + \beta t \quad (16-190)$$

When the mobile phase modulator is a dilute solute in a solvent, as when the modulator is a salt, ϕ indicates molar concentration.

Under *trace conditions*, the retention of the modulator in the column is independent of the presence of any solutes. The modulator concentration at the column exit is approximated by

$$\phi = \phi_0 + \beta \left[t - \frac{L}{v} (1 + k'_M) \right] \quad (16-191)$$

where k'_M is the retention factor of the modulator.

For a small feed injection, the modulator concentration ϕ_R at which a feed solute is eluted from the column is obtained from the following integral relationship

$$G(\phi_R) = \frac{\beta L}{v} = \int_{\phi_0}^{\phi_R} \frac{d\phi}{k'_M(\phi) - k'_M} \quad (16-192)$$

where $k'_M(\phi)$ is the solute retention factor as a function of the mobile phase modulator concentration. Note that the solute retention time in the column is affected by the steepness of the gradient at the column entrance.

$k'_M(\phi)$ can be obtained experimentally from isocratic elution experiments at different ϕ values, or from linear gradient elution experiments where the ratio $G = \beta L/v$ is varied. In the latter case, the retention factor is obtained by differentiation of Eq. (16-192) from $k'_M(\phi_R) = k'_M + (dG/d\phi_R)^{-1}$.

Table 16-14 gives explicit expressions for chromatographic peak properties in isocratic elution and linear gradient elution for two cases.

In **reversed-phase chromatography** (RPC), the mobile phase modulator is typically a water-miscible organic solvent, and the stationary phase is a hydrophobic adsorbent. In this case, the logarithm of solute retention factor is commonly found to be linearly related to the volume fraction of the organic solvent.

In **ion-exchange chromatography** (IEC), the mobile phase modulator is typically a salt in aqueous solution, and the stationary phase is an ion-exchanger. For dilute conditions, the solute retention factor is commonly found to be a power-law function of the salt normality [cf. Eq. (16-27) for ion-exchange equilibrium].

Band broadening is also affected by the gradient steepness. This effect is expressed in Table 16-14 by a **band compression factor** C , which is a function of the gradient steepness and of equilibrium parameters. Since $C < 1$, gradient elution yields peaks that are sharper than those that would be obtained in isocratic elution at $\phi = \phi_R$.

Other cases, involving an arbitrary relationship between the solute retention factor and the modulator concentration can be handled analytically using the approaches of Frey [*Biotechnol. Bioeng.*, **35**, 1055 (1990)] and Carta and Stringfield [*J. Chromatogr.*, **605**, 151 (1992)].

Displacement Development A complete prediction of displacement chromatography accounting for rate factors requires a numerical solution since the adsorption equilibrium is nonlinear and intrinsically competitive. When the column efficiency is high, however, useful predictions can be obtained with the local equilibrium theory (see "Fixed Bed Transitions").

For constant-separation factor systems, the ***h*-transformation** of Helfferich and Klein (gen. refs.) or the method of Rhee et al. [*AIChE J.*, **28**, 423 (1982)] can be used [see also Helfferich, *Chem. Eng. Sci.*, **46**, 3320 (1991)]. The equations that follow are adapted from Frenz and Horvath [*AIChE J.*, **31**, 400 (1985)] and are based on the *h*-transformation. They refer to the separation of a mixture of $M - 1$ components with a displacer (component 1) that is more strongly adsorbed than any of the feed solutes. The **multicomponent Langmuir isotherm** [Eq. (16-39)] is assumed valid with equal monolayer capacities, and components are ranked numerically in order of decreasing affinity for the stationary phase (i.e., $K_1 > K_2 > \dots > K_M$).

The development of component bands is predicted by mapping the trajectories of the ***h*-function roots** h_i , which are obtained from the solution of

$$\sum_{i=1}^M \frac{K_i c_i}{h \alpha_{i,1} - 1} = 1 \quad (16-193)$$

where $\alpha_{i,j} = K_i/K_j$.

Trivial roots also exist for each component with zero concentration. In displacement chromatography, for each of the transitions at the column entrance, a new set of M roots is generated. These roots are given in Table 16-15. The change from the initial solvent (carrier) to feed changes $M - 1$ roots, generating $M - 1$ boundaries or transitions that move through the column. These boundaries are all self-sharpening, characterized by upstream h'_i values larger than the corresponding downstream h'_i values.

TABLE 16-14 Expressions for Predictions of Chromatographic Peak Properties in Linear Gradient Elution Chromatography under Trace Conditions with a Small Feed Injection and Inlet Gradient Described by $\varphi = \varphi_o + \beta t$ (Adapted from Refs. A and B)

Parameter	Isocratic	Gradient elution—RPC (Ref. A)	Gradient elution—IEC (Ref. B)
Dependence of retention factor on modulator concentration	—	$k' - k'_M = \alpha_o e^{-S\varphi}$	$k' - k'_M = \alpha_o \varphi^{-Z}$
Retention time	$t_R = \frac{L}{v} (1 + k'_o)$	$t_R = \frac{L}{v} (1 + k'_M) + \frac{1}{S\beta} \ln \left[\frac{S\beta L}{v} (k'_o - k'_M) + 1 \right]$	$t_R = \frac{L}{v} (1 + k'_M) + \frac{1}{\beta} \left[\frac{\alpha_Z \beta L}{v} (Z + 1) + \varphi_o^{Z+1} \right]^{1/(Z+1)} - \frac{\varphi_o}{\beta}$
Mobile phase composition at column exit	$\varphi_R = \varphi_o$	$\varphi_R = \varphi_o + \beta \left[t_R - \frac{L}{v} (1 + k'_M) \right]$	$\varphi_R = \varphi_o + \beta \left[t_R - \frac{L}{v} (1 + k'_M) \right]$
Retention factor at peak elution	$k'_R = k'_o$	$k'_R = \frac{k'_o + (k'_o - k'_M) S\beta L/v}{1 + (k'_o - k'_M) S\beta L/v}$	$k'_R = \alpha_Z \left[\frac{\alpha_Z \beta L}{v} (Z + 1) + \varphi_o^{Z+1} \right]^{-Z/(Z+1)} + k'_M$
Peak standard deviation	$\sigma = \frac{L/v}{\sqrt{N_P}} (1 + k'_o)$	$\sigma = C \frac{L/v}{\sqrt{N_{PR}}} (1 + k'_R)$	$\sigma = C \frac{L/v}{\sqrt{N_{PR}}} (1 + k'_R)$
Band compression factor	—	$C = \frac{(1 + p + p^2/3)^{1/2}}{1 + p}$ $p = \frac{(k'_o - k'_M)(1 + k'_M) S\beta L/v}{1 + k'_o}$	$C = \begin{cases} \sqrt{M'} & \text{for } M' < 0.25 \\ \frac{3.22M'}{1 + 3.13M'} & \text{for } 0.25 < M' < 120.25 \\ 1 & \text{for } M' > 120.25 \end{cases}$ $M' = \frac{1}{2} \frac{1 + k'_R}{1 + k'_M} \frac{Z + 1}{Z}$

References: A. Snyder in Horvath (ed.), *High Performance Liquid Chromatography: Advances and Perspectives*, vol. 1, Academic Press, 1980, p. 208.

B. Yamamoto, *Biotechnol. Bioeng.*, **48**, 444 (1995).

Solute equilibrium parameters: α_o, S for RPC and α_Z, Z for IEC

Solute retention factor for initial mobile phase: k'_o

Retention factor of mobile phase modulator: k'_M

Plate number obtained for $k' = k'_R$: N_{PR}

The switch from feed to displacer changes M roots, generating M boundaries. All of these boundaries are self-sharpening, except the boundary associated with the transition from h_{MF} to $\alpha_{1,M+1}$ that can be self-sharpening or diffuse depending on the relative value of these roots. Note that only one root changes value across a particular boundary.

The **adjusted propagation velocity** of each self-sharpening boundary is:

$$u_{si} = \frac{d\zeta}{d\tau_1} = h'_i h''_i P_i \quad (16-194)$$

$$P_i = \prod_{j=1}^{i-1} h'_j \prod_{j=i+1}^M h''_j \prod_{j=1}^{M+1} \alpha_{j,i} \quad (16-195)$$

and that of each diffuse boundary:

$$u_{hi} = h_i^2 P_i \quad (16-196)$$

where $\zeta = z/L$ and $\tau_1 = (\epsilon v t/L - \epsilon_b \zeta)/\Lambda$ with $\Lambda = \rho_b n_1^D / c_1^D$ equal to the partition ratio for the displacer.

The root trajectories are mapped as follows on the ζ - τ_1 plane:

1. The starting points are the beginning and end of the feed step. From these points trajectories are traced according to Eqs. (16-194) and (16-196).

2. At the points of intersection of trajectories, a change in boundary velocity occurs: When two boundaries associated with a change in the same root number intersect, they combine into a single boundary with velocity given by Eq. (16-194) or (16-196), intermediate in value between those of the intersecting trajectories. When two trajectories associated with different root numbers intersect, they both continue beyond the point of intersection with new velocities calculated from Eqs. (16-194) and (16-196), reflecting any change in root values.

3. After constructing the trajectories, the following equation is used to calculate the band profiles

$$c_j = \frac{\prod_{i=1}^M (h_i \alpha_{i,j} - 1)}{K_j \prod_{i=1}^M (\alpha_{i,j} - 1)} \quad (16-197)$$

Example 14: Calculation of Band Profiles in Displacement Chromatography An equimolar mixture of two components (concentrations $c_2^f = c_3^f = 1$ arbitrary unit) is separated with a displacer with concentration $c_1^f = 2$. The equilibrium isotherm is:

$$n_i = \frac{n' K_i c_i}{1 + K_1 c_1 + K_2 c_2 + K_3 c_3} \quad (i = 1, 2, 3)$$

with $K_1 = 2$, $K_2 = 1$, and $K_3 = 0.667$. The dimensionless feed time is $\tau_f = 0.2$. The separation factors used for calculating the trivial roots are $\alpha_{1,1} = 1$, $\alpha_{1,2} = 2$, $\alpha_{1,3} = 3$, $\alpha_{1,4} = 1 + K_1 c_1^f = 5$. The remaining h -function roots are calculated by solving Eq. (16-193) for the carrier, feed, and displacer concentrations. The results are summarized in Table 16-16.

TABLE 16-15 Concentrations and h -Function Roots for Displacement Chromatography of a Mixture of $M - 1$ Components Numbered in Order of Decreasing Affinity for the Stationary Phase (Adapted from Frenz and Horvath, 1985)

Solution	$c_i = 0$		$c_i > 0$	
	i	h_i	i	h_i
Carrier	1, 2, ..., M	$\alpha_{1,1}, \dots, \alpha_{1,M}$	—	—
Feed	1	$\alpha_{1,1}$	2, ..., M	$h_{2F}, \dots, h_{MF}^o$
Displacer	2, 3, ..., M	$\alpha_{1,2}, \dots, \alpha_{1,M}$	1	$\alpha_{1,M+1} = 1 + K_1 c_1^D$

*Roots calculated from the solution of Eq. (16-193).

TABLE 16-16 Values of h -Roots for Example 14

State	h_1	h_2	h_3
Carrier	1	2	3
Feed	1	2.44	6.56
Displacer	2	3	5

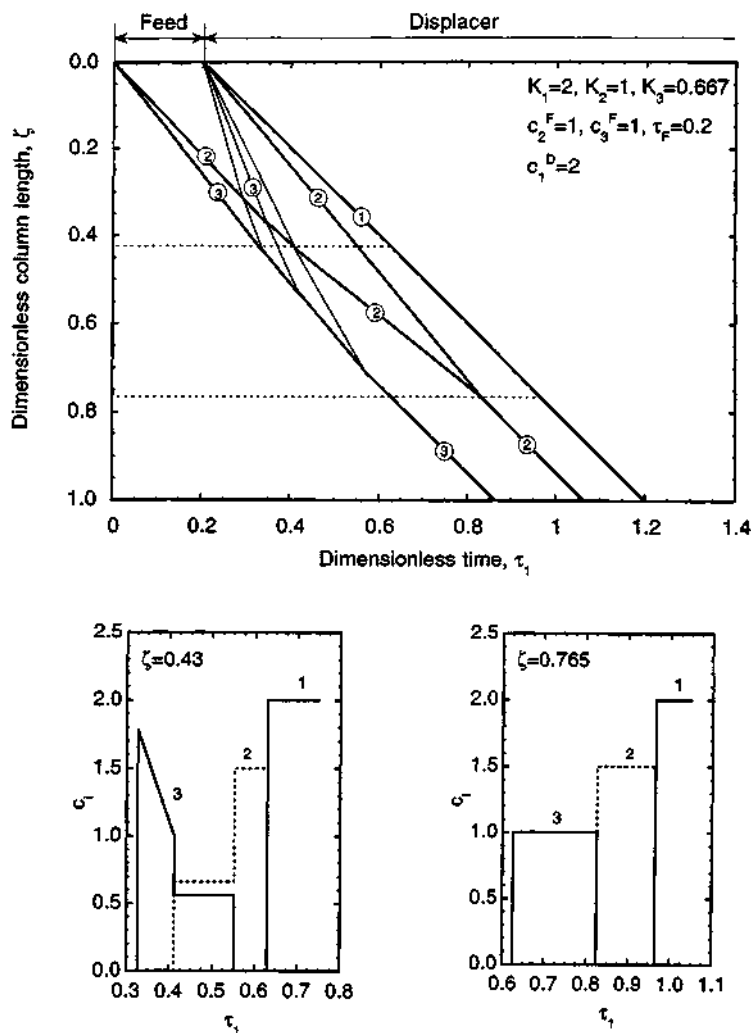


FIG. 16-36 Dimensionless time-distance plot for the displacement chromatography of a binary mixture. The darker lines indicate self-sharpening boundaries and the thinner lines diffuse boundaries. Circled numerals indicate the root number. Concentration profiles are shown at intermediate dimensionless column lengths $\zeta = 0.43$ and $\zeta = 0.765$. The profiles remain unchanged for longer column lengths.

The switch from carrier to feed generates self-sharpening boundaries, since the h'_s (upstream) are larger than the h'_s (downstream). The switch from feed to displacer generates two self-sharpening boundaries (corresponding to h_1 and h_2) and a diffuse boundary (corresponding to h_3) since $h_3'' < h'_s$. The velocity of each boundary is calculated from Eq. (16-194) for self-sharpening boundaries and from Eq. (16-196) for diffuse boundaries. The root trajectories are shown in Fig. 16-36 (top). From one boundary to the next, the root values change one at a time from the values corresponding to the carrier to the values corresponding to the displacer. Concentration profiles calculated from Eq. (16-197) are shown in Fig. 16-36 (bottom) for two different values of ζ . The concentrations in the z and t coordinates are easily reconstructed from the definitions of τ_1 and ζ .

In general, for a constant separation factor system, complete separation is obtained when:

$$c_1^D > \frac{1}{K_1} (\alpha_{1,M} - 1) \quad (16-198)$$

Then, band profiles eventually develop into the **isotachic pattern** of pure component bands moving at the velocity of the displacer front. In Example 16-14, this occurs at $\zeta = 0.765$.

The **isotachic concentrations** c_j^I in the fully developed train are calculated directly from the single component isotherms using:

$$\frac{n_1^I}{c_1^I} = \frac{n_2^I}{c_2^I} = \dots = \frac{n_M^I}{c_M^I} \quad (16-199)$$

$$\text{or} \quad c_j^I = c_1^D - \frac{1}{K_j} (1 - \alpha_{j,1}) \quad (16-200)$$

The graphical interpretation of Eq. (16-199) is shown in Fig. 16-37 for the conditions of Example 16-14. An operating line is drawn from the origin to the point of the pure displacer isotherm at $c_1 = c_1^D$. For displacement to occur, the operating line must cross the pure component isotherms of the feed solutes. The product concentrations in the isotachic train are found where the operating line crosses the isotherms.

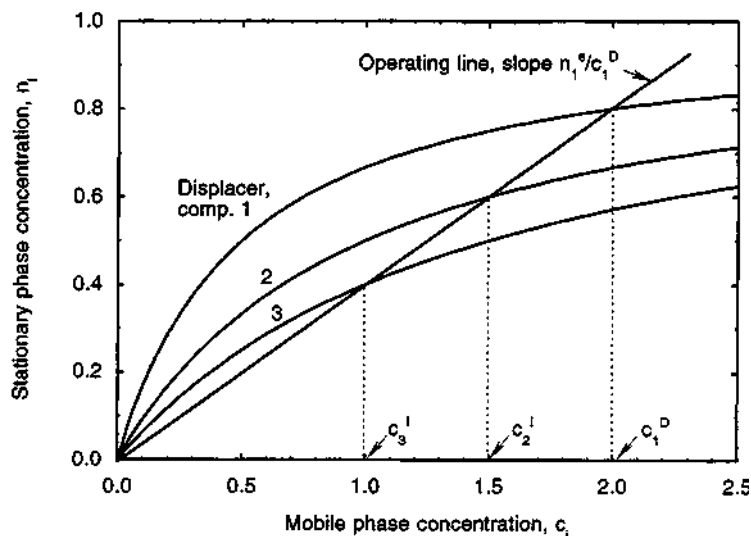


FIG. 16-37 Schematic showing the intersection of the operating line with the pure-component isotherms in displacement chromatography. Conditions are the same as in Fig. 16-36.

When this condition is met, the feed concentrations do not affect the final product concentrations.

Analyses of displacement chromatography by the method of characteristics with non-Langmuirian systems is discussed by Antia and Horvath [*J. Chromatogr.*, **556**, 199 (1991)] and Carta and Dinerman [*AIChE J.*, **40**, 1618 (1994)]. Optimization studies and analyses by computer simulations are discussed by Jen and Pinto [*J. Chromatogr.*, **590**, 47 (1992)], Katti and Guiochon [*J. Chromatogr.*, **449**, 24 (1988)], and Phillips et al. [*J. Chromatogr.*, **54**, 1 (1988)].

DESIGN FOR TRACE SOLUTE SEPARATIONS

The design objectives of the analyst and the production line engineer are generally quite different. For analysis the primary concern is typically resolution. Hence operating conditions near the minimum value of the HETP or the HTU are desirable (see Fig. 16-13).

In preparative chromatography, however, it is generally desirable to reduce capital costs by maximizing the **productivity**, or the amount of feed processed per unit column volume, subject to specified purity requirements. This reduction, however, must be balanced against operating costs that are determined mainly by the mobile phase flow rate and the pressure drop. In practice, preparative chromatography is often carried out under **overload conditions**, i.e., in the nonlinear region of the adsorption isotherm. Optimization under these conditions is discussed in Guiochon et al. (gen. refs.). General guidelines for trace-level, isocratic binary separations, in the Henry's law limit of the isotherm are:

1. The stationary phase is selected to provide the maximum selectivity. Where possible, the retention factor is adjusted (by varying the mobile phase composition, temperature, or pressure) to an optimum value that generally falls between 2 and 10. Resolution is adversely affected when $k' \ll 2$, while product dilution and separation time increase greatly when $k' \gg 10$. When this is not possible for all feed components and large differences exist among the k' -values of the different solutes, gradient elution should be considered.

2. The average feed mixture charging rate, molar or volumetric, is fixed by the raw material supply or the demand for finished product.

3. The value of N_p required to achieve a desired resolution is determined by Eq. (16-168) or (16-171). Since $N = L/HTU \sim 2N_p =$

$2L/HETP$, Fig. 16-13 or Eq. (16-183) can be used to determine the range of the dimensionless velocity $ReSc$ that maximizes N_p for a given particle diameter and column length.

4. The allowable pressure drop influences the choice of the particle size and helps determine the column length. Equations for estimating the pressure drop in packed beds are given in Section 6.

5. When only a few solutes are separated, they may occupy only a small portion of the total column volume at any given instant. In such cases, the productivity is improved by cyclic feed injections, timed so that the most strongly retained component from an injection elutes just before the least strongly retained component from the following injection (see Fig. 16-57). For a mixture of two components with $k' > 1$, when the same resolution is maintained between bands of the same injections and bands of successive injections, the cycle time t_C and the plate number requirement are:

$$t_C = 2(t_{R,A} - t_{R,B}) = \frac{2L}{v} (k'_A - k'_B) \quad (16-201)$$

$$N_p = 4R_s^2 \left(\frac{\alpha + 1}{\alpha - 1} \right)^2 (1 - 2\phi)^{-2} \quad (16-202)$$

where $\phi = t_F/t_C$ is the fraction of the cycle time during which feed is supplied to the column. The productivity, P = volume of feed/(time $\times$ bed volume), is:

$$\begin{aligned} P &= \frac{1}{4R_s^2} \left(\frac{\alpha - 1}{\alpha + 1} \right)^2 \frac{\varepsilon v}{HETP} \phi(1 - 2\phi)^2 \\ &= \frac{1}{4R_s^2} \left(\frac{\alpha - 1}{\alpha + 1} \right)^2 \frac{D_i}{d_p^2} \left(\frac{ReSc}{b/ReSc + a + cReSc} \right) \phi(1 - 2\phi)^2 \end{aligned} \quad (16-203)$$

For a given resolution, P is maximized when $\phi = 1/6$ (i.e., feed is supplied for one sixth of the cycle time), and by the use of small particle sizes. The function $ReSc/(b/ReSc + a + cReSc)$ generally increases with $ReSc$, so that productivity generally increases with the mobile phase velocity. For typical columns, however, this function is within about 10 percent of its maximum value ($\sim 1/c$) when $ReSc$ is in the range 30–100. Thus, increasing the velocity above this range must be balanced against the costs associated with the higher pressure drop. Similar results are predicted with Eq. (16-185).

PROCESS CYCLES

GENERAL CONCEPTS

Some applications of adsorption and ion exchange can be achieved by sorbent-fluid contact in batch equipment. Batch methods are well adapted to laboratory use and have also been applied on a larger scale in several specific instances. In a batch run for either adsorption or ion exchange, a sorbent is added to a fluid, completely mixed, and subsequently separated. Batch treatment is adopted when the capacity and equilibrium of the sorbent are large enough to give nearly complete sorption in a single step, as in purifying and decoloration of laboratory preparations with carbons and clays. Batch runs are useful in the measurement of equilibrium isotherms and also of adsorptive diffusion rate. Batch tests or measurements are often conducted in portions of adsorbent or ion-exchange material intended for larger-scale use. For example, either the equilibrium sorption of a solid and/or its ultimate sorption capacity can be determined in this way. The solid is first equilibrated with a fluid at the concentration of interest; after separating the phases, gravimetric and chemical analyses can be used to determine the sorbed amount and composition.

Like the laboratory batch use, some commercial applications use the adsorbent on a throwaway basis. Reasons for using sorption non-regeneratively are usually: (1) low cost of the sorbent, (2) high value of the product, (3) very low dosage (sorbent-to-fluid ratio), and (4) difficulty in desorbing the sorbates. Magnesium perchlorate and barium oxide are used for drying, iron sponge (hydrated iron oxide on wood chips) is used to remove hydrogen sulfide, and sodium or potassium hydroxide is applied to removal of sulfur compounds or carbon dioxide. In wastewater treatment, powdered activated carbon (PAC) is added to enhance biological treatment but is not regenerated; instead, it remains with the sludge. Silica gel is used as a desiccant in packaging. Activated carbon is used in packaging and storage to adsorb other chemicals for preventing the tarnishing of silver, retarding the ripening or spoiling of fruits, "gettering" (scavenging) out-gassed solvents from electronic components, and removing odors. Synthetic zeolites, or blends of zeolites with silica gel, are used in dual-pane windows to adsorb water during initial dry-down and any in-leakage and to adsorb organic solvents emitted from the sealants during their cure; this prevents fogging between the sealed panes that could result from the condensation of water or the solvents [Ausikaitis, *Glass Digest*, **61**, 69-78 (1982)]. Activated carbon is used to treat recirculated air in office buildings, apartments, and manufacturing plants using thin

filter-like frames to treat the large volumes of air with low pressure drop. On a smaller scale, activated carbon filters are in kitchen hoods, air conditioners, and electronic air purifiers. On the smallest scale, gas masks containing carbon or carbon impregnated with promoters are used to protect individual wearers from industrial odors, toxic chemicals, and gas-warfare chemicals. Activated carbon fibers have been formed into fabrics for clothing to protect against vesicant and percutaneous chemical vapors [Macnair and Arons in Cheremisinoff and Ellerbusch (eds.), *gen. refs.*].

Ion exchangers are sometimes used on a throwaway basis also. In the laboratory, ion exchangers are used to produce deionized water, purify reagents, and prepare inorganic sols. In medicine, they are used as antacid, for sodium reduction, for the sustained release of drugs, in skin-care preparations, and in toxin removal.

Although there are many practical applications for which the sorbent is discarded after one use, most applications of interest to chemical engineers involve the removal of adsorbates from the sorbent (i.e., regeneration). This allows the adsorbent to be reused and the adsorbates to be recovered.

The maximum efficiency that a cyclic adsorption process can approach for any given set of operating conditions is defined by the adsorptive loading in equilibrium with the feed fluid. There are several factors that reduce the practical (or "operating") adsorption: mass-transfer resistance (see above), deactivation (see above), and incomplete regeneration (or *desorption*). The severity of regeneration influences how closely the dynamic capacity of an adsorbent resembles that of fresh, virgin material. Regeneration, or reversal of the adsorption process, requires a reduction in the driving force for adsorption. This is accomplished by increasing the equilibrium driving force for the adsorbed species to desorb from the solid to the surrounding fluid.

TEMPERATURE SWING ADSORPTION

A temperature-swing, or thermal-swing, adsorption (TSA) process cycle is one in which desorption takes place at a temperature much higher than adsorption. The elevation of temperature is used to shift the adsorption equilibrium and affect regeneration of the adsorbent. Figure 16-38 depicts a simplified (and ideal) TSA cycle. The feed fluid containing an adsorbate at a partial pressure of p_1 is passed through an

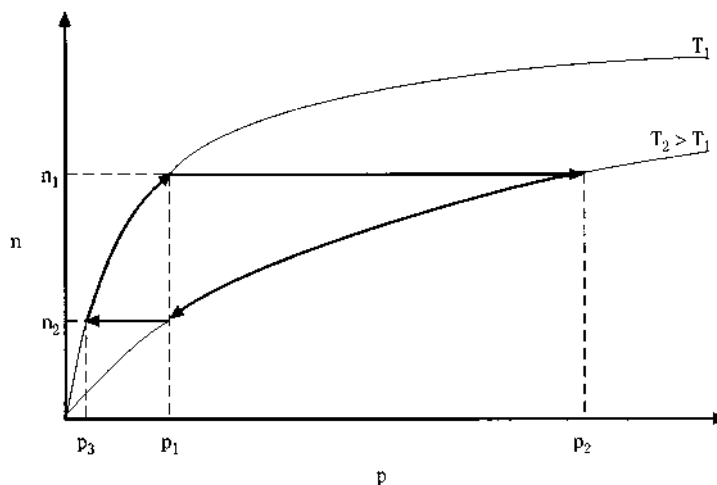


FIG. 16-38 Ideal temperature swing cycle. (Reprinted with permission of UOP.)

adsorbent at temperature T_1 . This adsorption step continues until the equilibrium loading n_1 is achieved with p_1 . Next the adsorbent temperature is raised to T_2 (heating step) so that the partial pressure in equilibrium with n_1 is increased to p_2 , creating a partial pressure driving force for desorption into any fluid containing less than p_2 of the adsorbate. By means of passing a purge fluid across the adsorbent, adsorbate is swept away, and the equilibrium proceeds down the isotherm to some point such as p_1, n_2 . (This point need not coincide with the feed partial pressure; it is selected for illustrative purposes. Also, in some applications, roll-up of the adsorbed-phase concentration can occur during heating such that in some regions of the bed p_2 reaches the condensation pressure of the component, causing a condensed liquid phase to form temporarily in particles [Friday and LeVan, *AIChE J.*, **31**, 1322 (1985)].) During a cooling step, the adsorbent temperature is returned to T_1 . The new equilibrium p_3, n_2 represents the best-quality product that can be produced from the adsorbent at a regenerated loading of n_2 in the simplest cycle. The adsorption step is now repeated. The differential loading, $n_1 - n_2$, is the maximum loading that can be achieved for a TSA cycle operating between a feed containing p_1 at temperature T_1 , regeneration at T_2 , and a product containing a partial pressure p_3 of the adsorbate. The regeneration fluid will contain an average partial pressure between p_2 and p_1 and will therefore have accomplished a concentration of the adsorbate in the regenerant fluid. For liquid-phase adsorption, the partial pressure can be replaced by the fugacity of the adsorbate. Then, the entire discussion above is applicable whether the regeneration is by a fluid in the gas or liquid phase.

In a TSA cycle, the heating step must provide the thermal energy necessary to raise the adsorbate, adsorbent, and adsorber temperatures, to desorb the adsorbate, and to make up for heat losses. TSA regeneration is classified as (1) heating-limited (or stoichiometric-limited) when transfer of energy to the system is limiting, or (2) stripping-limited (or equilibrium-limited) when transferring adsorbate away is limiting. Heating is accomplished by either direct contact of the adsorbent by the heating medium (external heat exchange to a purge gas) or indirect means (heating elements, coils, or panels inside the adsorber). Direct heating is the most commonly used, especially for stripping-limited heating. Indirect heating can be considered for stripping-limited heating, but the complexity of indirect heating limits its practicality to heating-limited regeneration where purge gas is in short supply. Microwave fields [Benchanaa, Lallemand, Simonet-Grange, and Bertrand, *Thermochim. Acta*, **152**, 43–51 (1989)] and dielectric fields [Burkholder, Fanslow, and Bluhm, *Ind. Eng. Chem. Fundam.*, **25**, 414–416 (1986)] are also used to supply indirect heating.

Because high temperatures can be used, resulting in thorough desorption, TSA cycles are characterized by low residual loadings and thus high operating loadings. These high capacities at low concentrations allow long cycle times for reasonably sized adsorbents (hours to days). Long cycle time is needed because particles of adsorbent respond slowly to changes in gas temperature. Most applications of TSA are for systems in which adsorbates are present at low concentration (purification), such as drying, and in which species are strongly adsorbed, such as sweetening, CO_2 removal, and pollution control.

Other Cycle Steps Besides the necessary adsorption and heating steps, TSA cycles may employ additional steps. A purge or sweep gas removes the thermally desorbed components from the adsorbent, and cooling returns it to adsorption temperature. Although the cooling is normally accomplished as a separate step after the heating, sometimes adsorption is started on a hot bed. If certain criteria are met [Basmadjian, *Can. J. Chem. Eng.*, **53**, 234–238 (1975)], the dynamic adsorption efficiency is not significantly affected by the lack of a cooling step.

For liquid-phase adsorption cycles when the unit is to treat a product of significant value, there must be a step to remove the liquid from the adsorbent and one to displace any liquid regenerant thoroughly before filling with the valuable fluid. Because adsorbents and ion exchangers are porous, some retention of the product is unavoidable but needs to be minimized to maximize recovery. When regeneration is by a gas, removal and recovery is accomplished by a drain (or pressure-assisted drain) step using the gas to help displace liquid from the sorbent before heating. When the regenerant is another liquid, the

feed or product can be displaced out of the adsorbent. When heating and cooling are complete, liquid feed must be introduced again to the adsorbent with a corresponding displacement of the gas or liquid regenerant. In ion exchange, these steps for draining and filling are commonly referred to as “sweetening off” and “sweetening on,” respectively.

Applications Drying is the most common gas-phase application for TSA. The natural gas, chemical, and cryogenics industries all use adsorbents to dry streams. Zeolites, activated alumina, and silica gel are used for drying pipeline natural gas. Alumina and silica gel are used because they have higher equilibrium capacity and are more easily regenerated with waste-level heat [Crittenden, *Chem. Engr.*, **452**, 21–24 (1988); Goodboy and Fleming, *Chem. Eng. Progr.*, **80**, 63–68 (1984); Ruthven, *Chem. Eng. Progr.*, **84**, 42–50 (1988)]. The low dew-point that can be achieved with zeolites is especially important when drying cryogenic-process feed-streams to prevent freeze-up. Zeolites dry natural gas before liquefaction to liquefied natural gas (LNG) and before ethane recovery utilizing the cryogenic turboexpander process [Anderson in Katzer (ed.), *Molecular Sieves—II*, *Am. Chem. Soc. Symp. Ser.*, **40**, 637–649 (1977); Brooking and Walton, *The Chem. Engr.*, **257**, 13–17 (1972)]. The feed air to be cryogenically separated into N_2 , O_2 , and argon is purified of both water and CO_2 with 13X zeolites using TSA cycles. Zeolites, silica gel, and activated alumina are used to dry synthesis gas, inert gas, hydrocracker gas, rare gases, and reformer recycle H_2 . Because 3A and pore-closed 4A zeolites size-selectively adsorb water but exclude hydrocarbons, they are used extensively to dry reactive streams such as cracked gas in order to prevent coke formation on the adsorbent. This molecular sieving increases the recovery of hydrocarbons by reducing the coadsorption that would otherwise cause them to be desorbed and lost with the water.

Another area of application for TSA processes is in sweetening. H_2S , mercaptans, organic sulfides and disulfides, and carbonyl sulfide all must be removed from natural gas, H_2 , biogas, and refinery streams in order to prevent corrosion and catalyst poisoning. Natural gas feed to steam methane reforming is sweetened in order to protect the sulfur-sensitive, low-temperature shift catalyst. Well-head natural gas is treated by TSA to prevent pipeline corrosion using 4A zeolites to remove sulfur compounds without the coadsorption of CO_2 that would cause shrinkage. Sweetening and drying of refinery hydrogen streams are needed to prevent poisoning of reformer catalysts. Adsorption can be used to dry and sweeten these in the same unit operation.

TSA processes are applied to the removal of many inorganic pollutants. CO_2 is removed from base-load and peak-shaving natural-gas liquefaction facilities using 4A zeolite in a TSA cycle. The Sulfacid and Hitachi fixed-bed processes, the Sumitomo and BF moving-bed processes, and the Westvaco fluidized-bed process all use activated carbon adsorbents to remove SO_2 from flue gases and sulfuric acid plant tail gases [Juentgen, *Carbon*, **15**, 273–283 (1977)]. Activated carbon with a catalyst is used by the Unitaka process to remove NO , by reacting with ammonia, and activated carbon has been used to convert NO to NO_2 , which is removed by scrubbing. Mercury vapor from air and other gas streams is removed and recovered by activated carbon impregnated with elemental sulfur; the Hg is then recovered by thermal oxidation in a retort [Lovett and Cuniff, *Chem. Eng. Progr.*, **70**, 43–47 (1974)]. Applications for HCl removal from Cl_2 , chlorinated hydrocarbons, and reformer catalyst gas streams use TSA with mordenite and clinoptilolite zeolites [Dyer, gen. refs., pp. 102–105]. Activated aluminas are also used for HCl adsorption as well as fluorine and boron-fluorine compounds from alkylation processes [Crittenden, *ibid.*].

PRESSURE-SWING ADSORPTION

A pressure-swing adsorption (PSA) process cycle is one in which desorption takes place at a pressure much lower than adsorption. Reduction of pressure is used to shift the adsorption equilibrium and affect regeneration of the adsorbent. Figure 16-39 depicts a simplified pressure-swing cycle. Feed fluid containing an adsorbate at a molar concentration of $y_1 = p_1/P_1$ is passed through an adsorbent at conditions

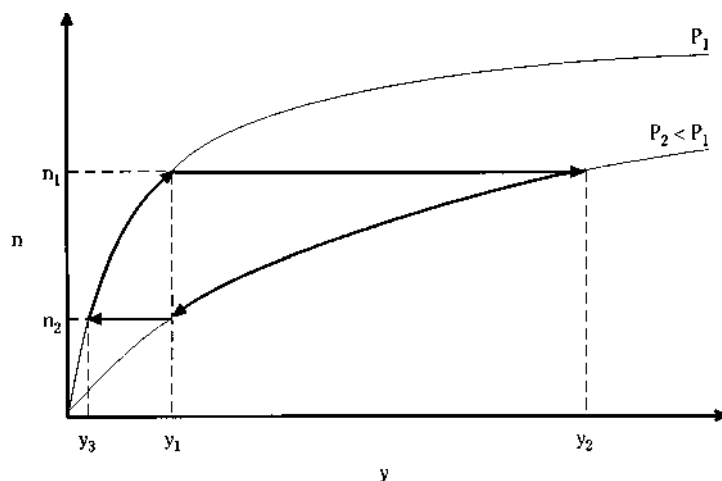


FIG. 16-39 Ideal pressure swing cycle. (Reprinted with permission of UOP.)

T_1 , P_1 , and the adsorption step continues until the equilibrium loading n_1 is achieved with y_1 . Next, the total pressure is reduced to P_2 during the depressurization (or blowdown) step. Now, although the partial pressure in equilibrium with n_1 is still p_1 , there is a concentration driving force of $y_2 = p_1/P_2 > y_1$ for desorption into any fluid containing less than y_2 . By passing a fluid across the adsorbent in a purge step, adsorbate is swept away, and the equilibrium proceeds down the isotherm to some point such as y_1, n_2 . (The choice of y_1 is arbitrary and need not coincide with feed composition.) At this time, the adsorbent is repressurized to P_1 . The new equilibrium y_3, n_2 represents the best quality product that can be produced from the adsorbent at a regenerated loading of n_2 . The adsorption step is now repeated. The differential loading, $n_1 - n_2$, is the maximum loading that can be achieved for a pressure-swing cycle operating between a feed containing y_1 and a product containing a molar concentration y_3 of the adsorbate. The regeneration fluid will contain an average concentration between y_2 and y_1 and will therefore have accomplished a concentration of the adsorbate in the regenerant gas. There is no analog for a liquid-phase PSA process cycle.

Thus, in a PSA process cycle, regeneration is achieved by a depressurization that must reduce the partial pressure of the adsorbates to allow desorption. These cycles operate at constant temperature, requiring no heating or cooling steps. Rather, they use the exothermic heat of adsorption remaining in the adsorbent to supply the endothermic heat needed for desorption. Pressure-swing cycles are classified as: (1) PSA, which, although used broadly, usually swings between a high superatmospheric and a low superatmospheric pressure; (2) VSA (vacuum-swing adsorption), which swings from a superatmospheric pressure to a subatmospheric pressure; and (3) PSPP (pressure-swing parametric pumping) and RPSA (rapid pressure-swing adsorption), which operate at very fast cycle times such that significant pressure gradients develop in the adsorbent bed (see the subsection on parametric pumping). Otherwise, the broad principles remain the same.

Low pressure is not as effective in totally reversing adsorption as is temperature elevation unless very high feed to purge pressure ratios are applied (e.g., deep vacuum). Therefore, most PSA cycles are characterized by high residual loadings and thus low operating loadings. These low capacities at high concentrations require that cycle times be short for reasonably sized beds (seconds to minutes). These short cycle times are attainable because particles of adsorbent respond quickly to changes in pressure. Major uses for PSA processes include purification as well as applications where contaminants are present at high concentration (bulk separations).

Other Cycle Steps A PSA cycle may have several other steps in addition to the basic adsorption, depressurization, and repressurization.

Cocurrent depressurization, purge, and pressure-equalization steps are normally added to increase efficiency of separation and recovery of product. At the end of the adsorption step, the more weakly adsorbed species have been recovered as product, but there is still a significant amount held up in the bed in the inter- and intraparticle void spaces. A cocurrent depressurization step can be added before the blowdown step, which is countercurrent to adsorption. This increases the amount of product produced each cycle. In some applications, the purity of the more strongly adsorbed components has also been shown to be heavily dependent on the cocurrent depressurization step [Cen and Yang, *Ind. Eng. Chem. Fundam.*, **25**, 758–767 (1986)]. This cocurrent blowdown is optional because there is always a countercurrent one. Skarstrom developed criteria to determine when the use of both is justified [Skarstrom in Li, *Recent Developments in Separation Science*, vol. II, CRC Press, Boca Raton, pp. 95–106 (1975)].

Additional stripping of the adsorbates from the adsorbent and purging of them from the voids can be accomplished by the addition of a purge step. The purge can begin toward the end of the depressurization or immediately afterward. Purging is accomplished with a flow of product countercurrent to adsorption to provide a lower residual at the product effluent end of the bed.

The repressurization step that returns the adsorber to feed pressure and completes the steps of a PSA cycle should be completed with pressure equalization steps to conserve gas and compression energy. Portions of the effluent gas during depressurization, blowdown, and enrichment purge can be used for repressurization to reduce the quantity of feed or product gas needed to pressurize the beds. The most efficient cycle is one that most closely matches available pressures and adsorbate concentration to the appropriate portion of the bed at the proper point in the cycle.

A dual reflux PSA cycle with both stripping and enriching has been proposed to obtain high enrichment and recovery. The separation is much better than that obtainable by normal PSA and suggests that it is possible to obtain two nearly pure products from a dilute feed stream [see McIntire et al., *Ind. Eng. Chem. Res.*, **41**, 3499 (2002)].

Applications PSA cycles are used primarily for purification of wet gases and of hydrogen. One of the earliest applications was the original Skarstrom two-bed cycle (adsorption, countercurrent blowdown, countercurrent purge, and cocurrent repressurization) to dry air stream to less than 1 ppm H_2O [Skarstrom, *ibid.*]. Instrument-air dryers still use a PSA cycle similar to Skarstrom's with activated alumina or silica gel [Armond, in Townsend, *The Properties and Applications of Zeolites*, The Chemical Society, London, pp. 92–102 (1980)].

The hydrocarbon exclusion by small-pore zeolites allows PSA to achieve a 10 to 30 K dewpoint depression in air-brake compressors,

even at high discharge air temperatures in the presence of compressor oil [Ausikaitis, in Katzer, *Molecular Sieves—II*, Am. Chem. Soc. Symp. Ser., **40**, pp. 681–695 (1977)]. The high-purity hydrogen employed in processes such as hydrogenation, hydrocracking, and ammonia and methanol production is produced by PSA cycles with adsorbent beds compounded of activated carbon, zeolites, and carbon molecular sieves [Martin, Gotzmann, Notaro, and Stewart, *Adv. Cryog. Eng.*, **31**, 1071–1086 (1986)]. The impurities to be removed include ammonia, carbon oxides, nitrogen, oxygen, methane, and heavier hydrocarbons. In order to be able to produce purities as high as 99.9999 percent, systems such as a UOP™ Polybed™ separation unit use six to ten adsorbents with complex cycles (see below) [Fuderer and Rudelstorfer, U.S. Patent 3,986,849 (1976)].

Air separation, methane enrichment, and iso/normal separations are the major bulk separations utilizing PSA; recovery of CO and CO₂ are also growing uses. PSA process cycles are used to produce oxygen and/or nitrogen from air. Synthetic zeolites, clinoptilolite, mordenite, and carbon molecular sieves are all used in various PSA, VSA, and RPSA cycles. The 85 to 95 percent purity oxygen produced is employed for electric furnace steel, waste water treating, solid waste combustion, and kilns [Martin et al., *ibid.*]. Small PSA oxygen units are used for patients requiring inhalation therapy in the hospital and at home [Cassidy and Holmes, *AIChE Symp. Ser.*, **80** (1984), pp. 68–75] and for pilots on board aircraft [Tedor, Horch, and Dangieri, *SAFE J.*, **12**, 4–9 (1982)]. Lower purity oxygen (25 to 55 percent) can be produced to enhance combustion, chemical reactions, and ozone production [Sircar, in Rodrigues et al., *gen. refs.*, pp. 285–321]. High purity nitrogen (up to 99.99 percent) for inert blanketing is produced in PSA and VSA processes using zeolites and carbon molecular sieves [Kawai and Kaneko, *Gas Sep. & Purif.*, **3**, 2–6 (1989); Ruthven, *ibid.*]. Methane is upgraded to natural gas pipeline quality by another PSA process. The methane is recovered from fermentation gases of landfills and wastewater purification plants and from poor-quality natural gas wells and tertiary oil recovery. Carbon dioxide is the major bulk contaminant but the gases contain water and other “garbage” components such as sulfur and halogen compounds, alkanes, and aromatics [Kumar and VanSloun, *Chem. Eng. Progr.*, **85**, 34–40 (1989)]. These impurities are removed by TSA using activated carbon or carbon molecular sieves and then the CO₂ is adsorbed using a PSA cycle. The cycle can use zeolites or silica gel in an equilibrium-selective separation or carbon molecular sieve in a rate-selective separation [Kapoor and Yang, *Chem. Eng. Sci.*, **44**, 1723–1733 (1989); Richter, Erdoel Kohle, Erdgas, Petrochem., **40**, 432–438 (1987)]. The pore-size selectivity of zeolite 5A is employed to adsorb straight-chain molecules while excluding

branched and cyclic species in the UOP IsoSivSM process. This PSA process separates C₅ to C₉ range hydrocarbons into a normal-hydrocarbon fraction of better than 95 percent purity, and a higher-octane isomer fraction with less than 2 percent normals [Cassidy and Holmes, *ibid.*].

PURGE/CONCENTRATION SWING ADSORPTION

A purge-swing adsorption cycle is usually considered to be one in which desorption takes place at the same temperature and total pressure as adsorption. Desorption is accomplished either by partial-pressure reduction using an inert gas purge or by adsorbate displacement with another adsorbable component. Purge cycles operate adiabatically at nearly constant inlet temperature and require no heating or cooling steps. As with PSA, they utilize the heat of adsorption remaining in the adsorbent to supply the heat of desorption. Purge processes are classified as (1) inert or (2) displacement.

Inert Purge In inert-purge desorption cycles, *inert* refers to the fact that the purge is not adsorbable at the cycle conditions. Inert purging desorbs the adsorbate solely by partial pressure reduction. Regeneration of the adsorbent can be achieved while maintaining the same temperature and pressure by the introduction of an inert purge fluid. Figure 16-40 depicts a simplified inert-purge swing cycle utilizing a nonadsorbing purge fluid. As before, the feed stream containing an adsorbate at a partial pressure of p_1 is passed through an adsorbent at temperature T_1 , and the adsorption step continues until equilibrium n_1 is achieved. Next the nonadsorbing fluid is introduced to reduce the partial pressure below p_1 by dilution. Therefore, there is a partial pressure driving force for desorption into the purge fluid, and the equilibrium proceeds down the isotherm to the point p_2, n_2 , where p_2 represents the best quality product that can be produced from the adsorbent at a regenerated loading of n_2 . The adsorption step is now repeated, and the differential loading is $n_1 - n_2$. The regeneration fluid will contain an average partial pressure between p_2 and p_1 , and the cycle will therefore not have accomplished a concentration of the adsorbate in the regenerant fluid. But it will have transferred the adsorbates to a fluid from which it may be more easily separated by means such as distillation.

Like PSA cycles, inert-purge processes are characterized by high residual loadings, low operating loadings, and short cycle times (minutes). Bulk separations of contaminants not easily separable at high concentration and of weakly adsorbed components are especially suited to inert-purge-swing adsorption. Another version of UOP's

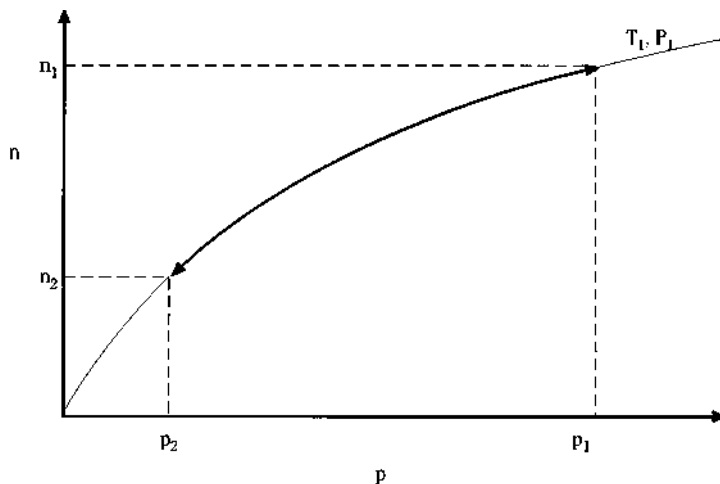


FIG. 16-40 Ideal inert-purge swing cycle. (Reprinted with permission of UOP.)

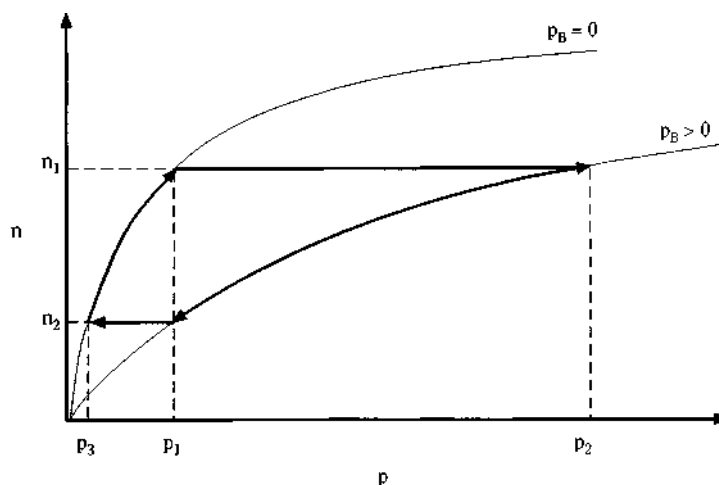


FIG. 16-41 Ideal displacement-purge swing cycle. (Reprinted with permission of UOP.)

IsoSiv process employs H_2 in an inert-purge cycle for separating C_3 to C_9 naphtha by adsorbing straight-chain molecules and excluding branched and cyclic species on size selective 5A zeolite [Cassidy and Holmes, *ibid.*]. Automobiles made in the United States have canisters of activated carbon to adsorb gasoline vapors lost from the carburetor or the gas tank during running, from the tank during diurnal cycling, and from carburetor hot-soak losses; the vapors are desorbed by an inert purge of air that is drawn into the carburetor as fuel when the engine is running [Clarke, Gerrard, Skarstrom, Vardi and Wade, *S.A.E. Trans.*, **76**, 824–842 (1968)]. UOP's Adsorptive Heat Recovery drying system has been commercialized for drying azeotropic ethanol to be blended with gasoline into gasohol; the process uses a closed loop of N_2 as the inert purge to desorb the water [Garg and Yon, *Chem. Eng. Progr.*, **82**, 54–60 (1986)].

Displacement Purge Isothermal, isobaric regeneration of the adsorbent can also be accomplished by using a purge fluid that can adsorb. In displacement-purge stripping, *displacement* refers to the displacing action of the purge fluid caused by its ability to adsorb at the cycle conditions. Figure 16-41 depicts a simplified displacement-purge cycle utilizing an adsorbable purge. Again, the feed stream containing an adsorbate at a partial pressure of p_1 is passed through an adsorbent at temperature T_1 , and the adsorption step continues until equilibrium n_1 is achieved. Next the displacement fluid, B, is introduced. The presence of another adsorbable species reduces the adsorptivity of the key adsorbate, A. Therefore, there exists a partial pressure driving force for desorption into the purge fluid, and the equilibrium proceeds down the isotherm to some point such as p_1, n_2 (again arbitrary.) Next the adsorbent is recharged with a fluid that contains no component B, shifting the effective isotherm to that where the equilibrium of component A is p_3 . The new equilibrium p_3, n_2 represents the best quality product that can be produced from the adsorbent at a regenerated loading of n_2 . The adsorption step is now repeated. The differential loading ($n_1 - n_2$) is the maximum loading that can be achieved for a pressure-swing cycle operating between a feed containing y_1 and a product containing a partial pressure p_3 of the adsorbate. The regeneration fluid will contain an average partial pressure between p_2 and p_1 and will therefore have accomplished a concentration of the adsorbate in the regenerant gas. Displacement-purge cycles are not as dependent on the heat of adsorption remaining on the adsorbent, because the adsorption of purge releases most or all of the energy needed to desorb the adsorbate. It is best if the adsorbate is more selectively adsorbed than the displacement purge, so that the adsorbates can easily desorb the purge fluid during adsorption. The displacement purge must be carefully selected, because it

contaminates both the product stream and the recovered adsorbate, and requires separation for recovery (e.g., distillation).

Displacement-purge processes are more efficient for less selective adsorbate/adsorbent systems, while systems with high equilibrium loading of adsorbate will require more purging [Sircar and Kumar, *Ind. Eng. Chem. Proc. Des. Dev.*, **24**, 358–364 (1985)]. Several displacement-purge-swing processes have been commercialized for the separation of branched and cyclic C_{10} – C_{18} from normals using the molecular-size selectivity of 5A zeolite: Exxon's Ensorb, UOP's IsoSiv, Texaco Selective Finishing (TSF), Leuna Werke's Parex, and the Shell Process [Ruthven, *ibid.*]. All use a purge of normal paraffin or light naphtha with a carbon number of two to four less than the feed stream except for Ensorb, which uses ammonia [Yang, *gen. refs.*]. UOP has also developed a similar process, OlefinSiv, which separates isobutylene from normal butenes with displacement purge and a size-selective zeolite [Adler and Johnson, *Chem. Eng. Progr.*, **75**, 77–79 (1979)]. Solvent extraction to regenerate activated carbon is another example of a displacement-purge cycle; the adsorbent is then usually steamed to remove the purge fluid [Martin and Ng, *Water Res.*, **18**, 59–73 (1984)]. The best use of solvent regeneration is for water phase adsorption where the separation of water from carbon would use too much steam and where purge and water are easily separated; and for vapor-phase where the adsorbate is highly nonvolatile but soluble. Air Products has developed a process for separating ethanol/water on activated carbon using acetone as a displacement agent and adding a water rinse to improve the recovery of two products [Sircar, U.S. Patent 5,026,482 (1991)].

Displacement-purge forms the basis for most simulated continuous countercurrent systems (see hereafter) such as the UOP SorbexSM processes. UOP has licensed close to one hundred Sorbex units for its family of processes: ParexSM to separate *p*-xylene from C_8 aromatics, MolexSM for *n*-paraffin from branched and cyclic hydrocarbons, OlexSM for olefins from paraffin, SarexSM for fructose from dextrose plus polysaccharides, CymexSM for *p*- or *m*-cymene from cymene isomers, and CresexSM for *p*- or *m*-cresol from cresol isomers. Toray Industries' AromaxSM process is another for the production of *p*-xylene [Otani, *Chem. Eng.*, **80**(9), 106–107, (1973)]. Illinois Water Treatment [*Making Waves in Liquid Processing*, Illinois Water Treatment Company, IWT Adsep System, Rockford, IL, **6**(1), (1984)] and Mitsubishi [Ishikawa, Tanabe, and Usui, U.S. Patent 4,182,633 (1980)] have also commercialized displacement-purge processes for the separation of fructose from dextrose.

Chromatography Chromatography is a sorptive separation technique that allows multicomponent separations in both gas and liquid

phase. As a preparative tool, it is often used as a displacement-purge process, although many applications employ an inert-displacement mode, especially for use in analysis. General characteristics and operating modes are discussed in a separate part of this section.

ION EXCHANGE

Except in very small-scale applications, ion-exchangers are used in cyclic operations involving sorption and desorption steps. A typical ion-exchange cycle used in water-treatment applications involves (a) *backwash*—used to remove accumulated solids obtained by an upflow of water to expand (50–80 percent expansion is typical) and fluidize the exchanger bed; (b) *regeneration*—a regenerant is passed slowly through the used to restore the original ionic form of the exchanger; (c) *rinse*—water is passed through the bed to remove regenerant from the void volume and, in the case of porous exchangers, from the resin pores; (d) *loading*—the fresh solution to be treated is passed through the bed until leakage begins to occur. Water softening is practiced in this way with a cation exchange column in sodium form. At the low ionic strength used in the loading step, calcium and magnesium are strongly preferred over sodium, allowing nearly complete removal. Since the selectivity for divalent cations decreases sharply with ionic concentration, regeneration is carried out effectively with a concentrated sodium chloride solution. Removal of sulfates from boiler feed water is done by similar means with anion exchangers in chloride form.

Many ion-exchange columns operate downflow and are regenerated in the same direction (Fig. 16-42a). However, a better regeneration and lower leakage during loading can be achieved by passing the regenerant countercurrently to the loading flow. Specialized equipment is available to perform countercurrent regeneration (see "Equipment" in this section). One approach (Fig. 16-42b) is to apply a vacuum to remove the regenerant at the top of the bed.

Complete deionization with ion-exchange columns is the classical method of producing ultrapure water for boiler feed, in electronics manufacture, and for other general uses in the chemical and allied industries. Deionization requires use of two exchangers with opposite functionality to remove both cations and anions. These can be in separate columns, packed in adjacent layers in the same column, or, more frequently, in a mixed bed. In the latter case, the two exchangers are intimately mixed during the loading step. For regeneration, backwashing separates the usually lighter anion exchanger from the usually denser cation exchanger. The column typically has a screened distributor at the interface between the two exchangers, so that they may be separately regenerated without removing them from the column. The most common cycle (Fig. 16-43) permits sequential regeneration of the two exchangers, first with alkali flowing downward through the anion exchanger to the interface distributor and then acid flowing

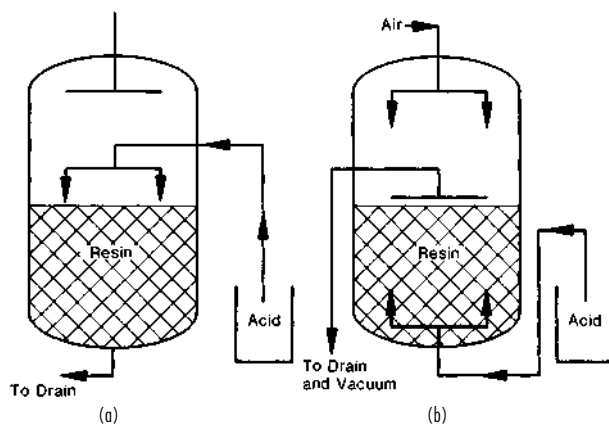


FIG. 16-42 Ion-exchanger regeneration. (a) Conventional. Acid is passed downflow through the cation-exchange resin bed. (b) Counterflow. Regenerant solution is introduced upflow with the resin bed held in place by a dry layer of resin.

downward from the interface distributor through the cation exchanger. After regeneration and rinsing, the exchangers are remixed by compressed air. To alleviate the problem of intermixing of the two different exchangers and chemical penetration through the wrong one, an inert material of intermediate density can be used to provide a buffer zone between layers of cation and anion exchangers.

When recovery of the sorbed solute is of interest, the cycle is modified to include a displacement step. In the manufacture of pharmaceuticals, ion-exchangers are used extensively in recovery and separation. Many of these compounds are amphoteric and are positively or negatively charged depending on the solution pH. Thus using, for example, a cation exchanger, loading can be carried out at a low pH and displacement at a high pH. Differences in selectivity for different species can be used to carry out separations during the displacement [Carta et al., *AIChE Symp. Ser.*, **84**, 54-61 (1988)]. Multibed cycles are also used to facilitate integration with other chemical process operations. Figure 16-44 shows a two-bed ion-exchange system using both cation and anion exchangers to treat and recover chromate from rinse water in plating operations. The cation exchanger removes trivalent chromium, while the anion exchanger removes hexavalent chromium as an anion. Regeneration of the cation exchanger with sulfuric acid produces a concentrated solution of trivalent chromium as the sulfate salt. The hexavalent chromium is eluted from the anion exchanger with sodium hydroxide in a concentrated solution. This solution is recycled

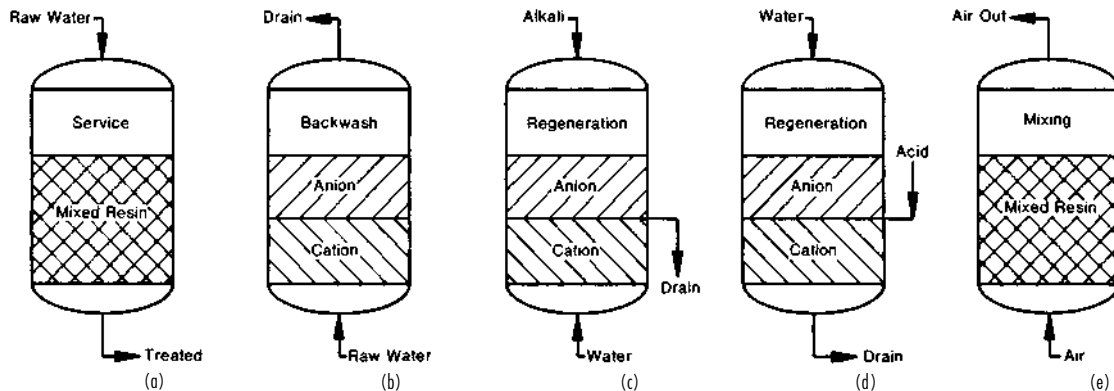


FIG. 16-43 Principles of mixed-bed ion exchange. (a) Service period (loading). (b) Backwash period. (c) Caustic regeneration. (d) Acid regeneration. (e) Resin mixing.

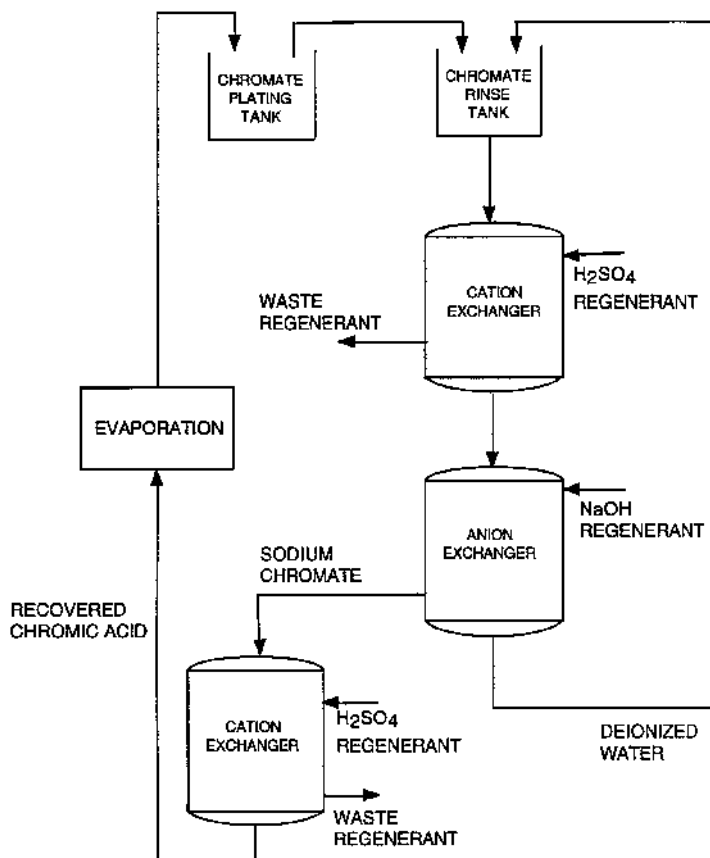


FIG. 16-44 Multicomponent ion-exchange process for chromate recovery from plating rinse water. (Adapted from Rhom & Haas with permission.)

to the plating tank by passing it through a second cation exchange column in hydrogen form to convert the sodium chromate to a dilute chromic acid solution that is concentrated by evaporation.

PARAMETRIC PUMPING

The term *parametric pumping* was coined by Wilhelm et al. [Wilhelm, Rice, and Bendelius, *Ind. Eng. Chem. Fundam.*, **5**, 141–144 (1966)] to describe a liquid-phase adsorption process in which separation is achieved by periodically reversing not only flow but also an intensive thermodynamic property such as temperature, which influences adsorptivity. Moreover, they considered the concurrent cycling of pressure, pH, and electrical and magnetic fields. A lot of research and development has been conducted on thermal, pressure, and pH driven cycles, but to date only gas-phase pressure-swing parametric pumping has found much commercial acceptance.

Temperature Two modes of temperature parametric-pumping cycles have been defined—direct and recuperative. In direct mode, an adsorbent column is heated and cooled while the fluid feed is pumped forward and backward through the bed from reservoirs at each end. When the feed is a binary fluid, one component will concentrate in one reservoir and one in the other. In recuperative mode, the heating and cooling takes place outside the adsorbent column. Parametric pumping, thermal and pH modes, have been widely studied for separation of liquid mixtures. However, the primary success for separating gas mixtures in thermal mode has been the separation of propane/ethane on activated carbon [Jencziewski and Myers, *Ind. Eng. Chem. Fundam.*, **9**, 216–221 (1970)] and of air/SO₂ on silica gel

[Patrick, Schrod, and Kermode, *Sep. Sci.*, **7**, 331–343 (1972)]. The difficulty with applying the thermal mode to gas separation is that in a fixed volume gas pressure increases during the hot step, which defeats the desorption purpose of this step. No thermal parametric-pumping cycle has yet been practiced commercially.

Pressure Another approach to parametric pumping is accomplished by pressure cycling of an adsorbent. An adsorbent bed is alternately pressurized with forward flow and depressurized with backward flow through the column from reservoirs at each end. Like TSA parametric pumping, one component concentrates in one reservoir and one in the other. The pressure mode of parametric pumping has been called pressure-swing parametric pumping (PSPP) and rapid pressure swing adsorption (RPSA). It was developed to minimize process complexity and investment at the expense of product recovery. RPSA is practiced in single-bed [Keller and Jones in Flank, *Adsorption and Ion Exchange with Synthetic Zeolites*, **135** (1980), pp. 275–286] and multiple-bed [Earls and Long, U.S. Patent number 4,194,892, 1980] implementations. Adsorbents are short (about 0.3 to 1.3 m), and particle sizes are very small (about 150 to 400 μm). The total cycle time including adsorption, dead time, countercurrent purge, and sometimes a second dead time, ranges from a few to about 30 seconds. The feature of RPSA that differentiates it from traditional PSA is the existence of axial pressure profiles throughout the cycle much as temperature gradients are present in TSA parametric pumping. Whereas PSA processes have essentially constant pressure through the bed at any given time, the flow resistance of the very small adsorbent particles produce substantial pressure drops in the bed. These pressure dynamics are key to the attainment of separation

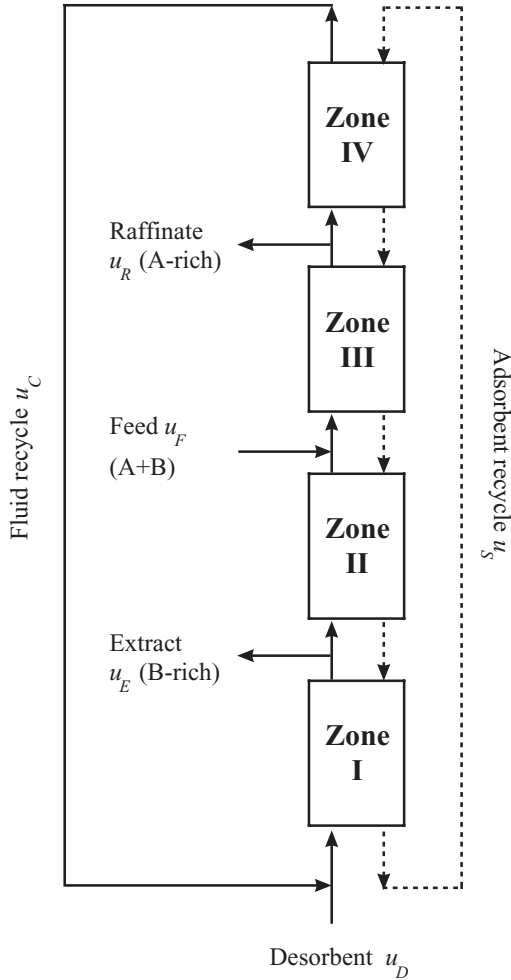


FIG. 16-46 General scheme of a true moving bed (TMB) adsorption system for binary separations. A is less strongly retained than B.

imperfect separation even within the shaded region. As a result, operating away from the vertex and closer to the 45° line is usually needed at the expense of lower productivity. By introducing a safety margin $\beta \geq 1$ (Seader and Henley, gen. refs.), Eqs. (16-206) are transformed to

$$m^I = K_B \beta \quad (16-208a)$$

$$m^{II} = K_A \beta \quad (16-208b)$$

$$m^{III} = K_B / \beta \quad (16-208c)$$

$$m^{IV} = K_A / \beta \quad (16-208d)$$

$\beta = 1$ yields maximum productivity under ideal conditions, while larger values provide a more robust design up to the maximum $\beta = \sqrt{K_B/K_A}$. For a given β , external and internal flow velocities are calculated from

$$u_S = u_F / (K_B / \beta - K_A \beta) \quad (16-209a)$$

$$u_E = u_S (K_B - K_A) \beta \quad (16-209b)$$

$$u_R = u_S (K_B - K_A) / \beta \quad (16-209c)$$

$$u_D = u_E + u_R - u_F \quad (16-209d)$$

$$u^I = u_S (K_B \beta + \epsilon_P / \rho_P) \quad (16-209e)$$

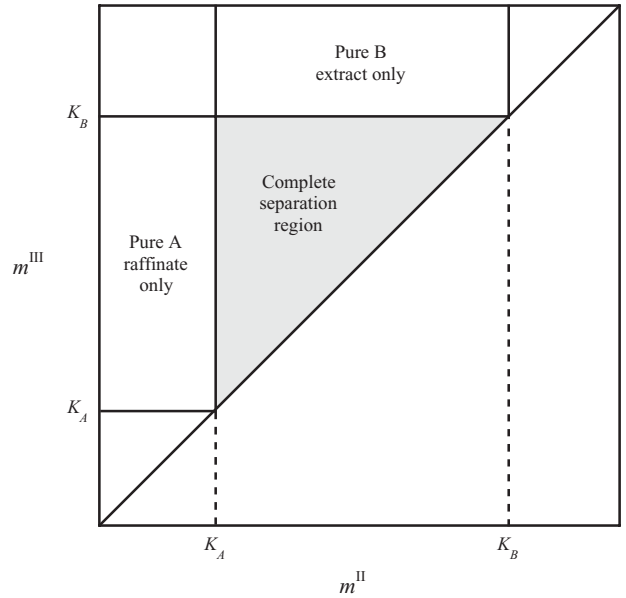


FIG. 16-47 m^{III} - m^{II} plane showing regions of complete and incomplete separation. Area below the 45° line is invalid.

$$u^{II} = u^I - u_E \quad (16-209f)$$

$$u^{III} = u^{II} + u_F \quad (16-209g)$$

$$u^{IV} = u^{III} - u_R \quad (16-209h)$$

$$u_C = u^I - u_D \quad (16-209i)$$

Analogous relationships are derived for SMB systems where each zone comprises a number of fixed beds operated in a merry-go-round sequence, as shown in Fig. 16-48. External flow velocities are calculated from Eqs. (16-209a) to (16-209d), replacing u_S with

$$u_{S,SMB} = \frac{\rho_b L}{p} \quad (16-210)$$

where L is the length of a single bed and p is the switching period. Internal flow velocities equivalent to the TMB operation are increased from the values calculated from Eqs. (16-209e) to (16-209i) to compensate for the extraparticle fluid carried along in each bed at each switch according to

$$u_{SMB}^j = u^j + \frac{\epsilon L}{p} \quad (16-211)$$

In practice, a small number of beds in series in each zone provide a close approach to the performance of ideal, true countercurrent system. Industrial SMB systems normally use 2 to 3 beds per zone.

Complete Design and Extensions Complete design of SMB systems requires a full description of equilibrium and rate factors. For an existing SMB unit, initial stream flow rates and switching period can be selected based on Eqs. (16-209) to (16-211) so that the operating point lies within the desired separation region. Column length design requires a dynamic adsorption model including a description of mass-transfer rates, adsorption kinetics, and axial dispersion. Operation with a nonlinear isotherm is analyzed in a similar manner. In this case, the right triangle defining the complete separation region in Fig. 16-47 is distorted, acquiring one or more curved sides and further restricting the range of conditions leading to complete separation. As is evident from the analysis above, only binary separations are achievable with a four-zone system. Multicomponent separations require multiple SMB units or integrated units comprising more than four zones. Useful references covering SMB design for linear and nonlinear isotherms are

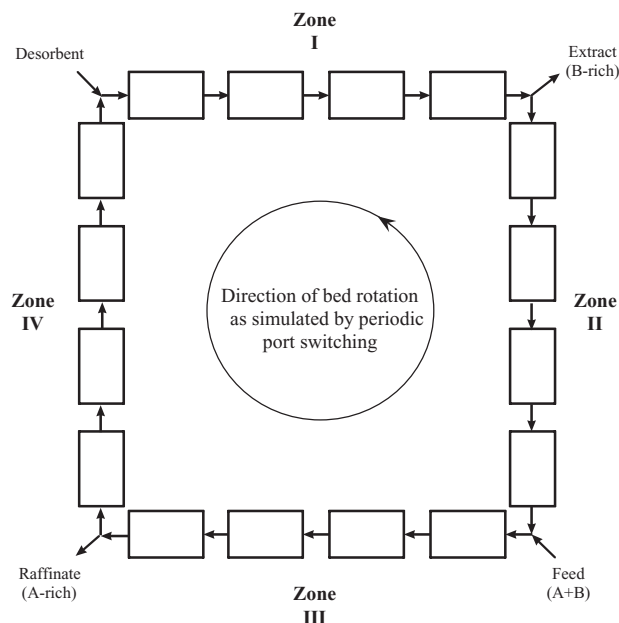


FIG. 16-48 General scheme of a simulated moving bed (SMB) adsorption system. Bed rotation is simulated by periodic switching of ports in the direction of fluid flow. A is less strongly retained than B.

by Ruthven and Ching, *Chem. Eng. Sci.*, **44**, 1011 (1989); Storti et al., *Chem. Eng. Sci.*, **44**, 1329 (1989); Zhong and Guiochon, *Chem. Eng. Sci.*, **51**, 4307 (1996); Mazzotti et al., *J. Chromatogr. A*, **769**, 3 (1997); Mazzotti et al., *AIChE J.*, **40**, 1825 (1994); and Minceva and Rodrigues, *Ind. Eng. Chem. Res.*, **41**, 3454 (2002).

OTHER ADSORPTION CYCLES

Hybrid Recycle Systems Liquid chromatography has been used commercially to separate glucose from fructose and other sugar isomers, for recovery of nucleic acids, and for other uses. A patent to Sanmatsu Kogyo Co., Ltd. (Yoritomi, Kezuka, and Moriya, U.S. Patent number 4,267,054, 1981) presents an improved chromatographic process that is simpler to build and operate than simulated moving-bed processes. Figure 16-49 (Keller et al., gen. refs.) diagrams its use for a binary separation. It is a displacement-purge cycle where pure component cuts are recovered, while cuts that contain both components are recycled to the feed end of the column.

The UOP CyclosorbSM is another adsorptive separation process with semicontinuous recycle. It utilizes a series of chromatographic columns to separate fructose from glucose. A series of internal recycle streams of impure and dilute portions of the chromatograph are used to improve the efficiency (Gerhold, U.S. Patent numbers 4,402,832, 1983; and 4,478,721, 1984). A schematic diagram of a six-vessel UOP Cyclosorb process is shown in Fig. 16-50 (Gembicki et al., gen. refs., p. 595). The process has four external streams and four internal recycles: Dilute raffinate and impure extract are like displacement steps; and impure raffinate and dilute extract are recycled from the bottom of an adsorber to its top. Feed and desorbent are fed to the top of each column in a predetermined sequence. The switching of the feed and desorbent are accomplished by the same rotary valve used for Sorbex switching (see hereafter). A chromatographic profile is established in each column that is moving from top to bottom, and all portions of an adsorber are performing a useful function at any time.

Steam Regeneration When steam is used for regeneration of activated carbon, it is desorbing by a combination thermal swing and displacement purge (described earlier in this section). The exothermic heat released when the steam is adsorbed supplies the thermal energy much more efficiently than is possible with heated gas purging. Slightly superheated steam at about 130°C is introduced into the bed counter-

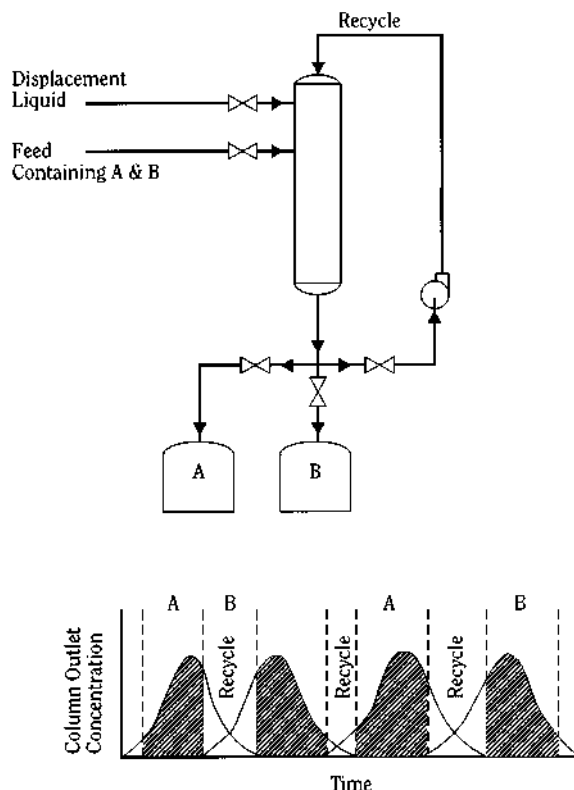


FIG. 16-49 Sanmatsu Kogyo chromatographic process. (Reprinted with permission of Wiley. Reference: Keller, Anderson, and Yon, Chap. 12 in Rousseau, Handbook of Separation Process Technology, John Wiley & Sons, Inc., New York, 1987.)

current to adsorption; for adsorbates with high boiling points, the steam temperature must be higher. Adsorbates are desorbed and purged out of the bed with the steam. Steam and desorbates then go to a condenser for subsequent separation. The water phase can be further cleaned by air stripping, and the sorbate-laden air from that stripper can be recycled with the feed to the adsorption bed.

Steam regeneration is most commonly applied to activated carbon that has been used in the removal and/or recovery of solvents from gases. At volatile organic compound (VOC) concentration levels from 500 to 15,000 ppm, recovery of the VOC from the stream used for regeneration is economically justified. Below about 500 ppm, recovery is not economically justifiable, but environmental concerns often dictate adsorption followed by destruction. While activated carbon is also used to remove similar chemicals from water and wastewater, regeneration by steam is not usual. The reason is that the water-treatment carbon contains 1 to 5 kg of water per kg of adsorbent that must be removed by drying before regeneration or an excessive amount of superheated steam will be needed. In water treatment, there can also be significant amounts of nonvolatile compounds that do not desorb during steam regeneration and that residual will reduce the adsorption working capacity. There is a growing use of reticulated styrene-type polymeric resins for VOC removal from air [Beckett, Wood, and Dixon, *Environ. Technol.*, **13**, 1129–1140 (1992); Heinegaard, *Chem.-Eng.-Tech.*, **60**, 907–908 (1988)]. LeVan and Schweiger [in Mersmann and Scholl (eds.), *Fundamentals of Adsorption*, United Engineering Trustees, New York (1991), pp. 487–496] tabulate reported steam utilizations (kg steam/kg adsorbate recovered) for a number of processes.

Energy Applications Desiccant cooling is a means for more efficiently providing air conditioning for enclosures such as supermarkets, ice rinks, hotels, and hospitals. Adsorbents are integrated with evaporative and electric vapor compression cooling equipment into an overall

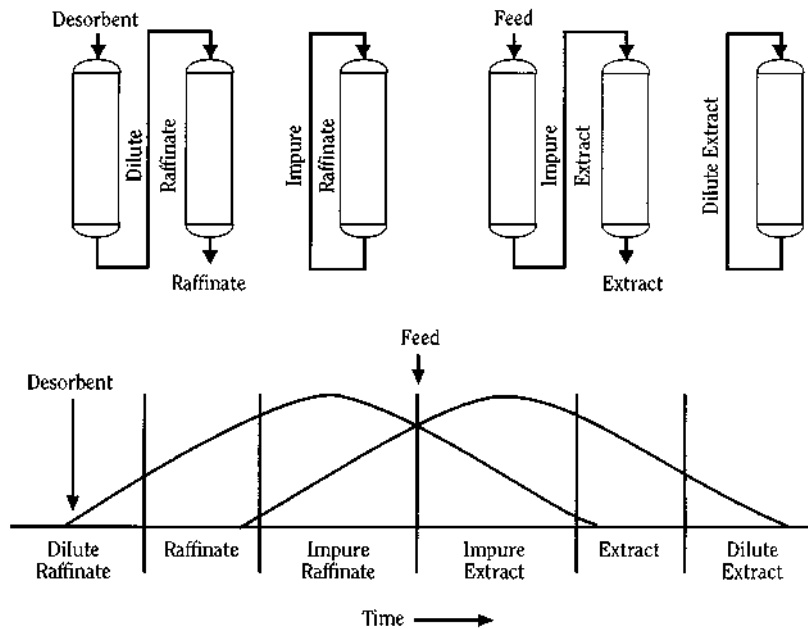


FIG. 16-50 UOP Cyclesorb process. (Reprinted with permission of John Wiley & Sons, Inc. Reference: Gembicki, Orskar, and Johnson, "Adsorption, Liquid Separation," in Kirk-Othmer Encyclopedia of Chemical Technology, 4th ed., Wiley, New York, 1991.)

air handling system. Air conditioning is comprised of two cooling loads, latent heat for water removal and sensible heat for temperature reduction. The energy savings derive from shifting the latent heat load from expensive compression cooling (chilling) to cooling tower load. Early desiccant cooling used adsorption wheels (see hereafter) impregnated with the hygroscopic salt, LiCl. More recently, these wheels are being fabricated with zeolite and/or silica gel. They are then incorporated into a system such as the example shown in Fig. 16-51 [Collier, Cohen, and Slosberg, in Harriman, *Desiccant Cooling and Dehumidification*,

ASHRAE, Atlanta (1992)]. Process air stream 6, to be conditioned, passes through the adsorbent wheel, where it is dried. This is a non-isothermal process due to the release of heat of adsorption and transfer of heat from a wheel that may be above ambient temperature. The dry but heated air (7) is cooled in a heat exchanger that can be a thermal wheel. This stream (8) is further cooled, and the humidity adjusted back up to a comfort range by direct contact evaporative cooling to provide supply air. Regeneration air stream 1, which can be ambient air or exhausted air, is evaporatively cooled to provide a heat sink for the hot,

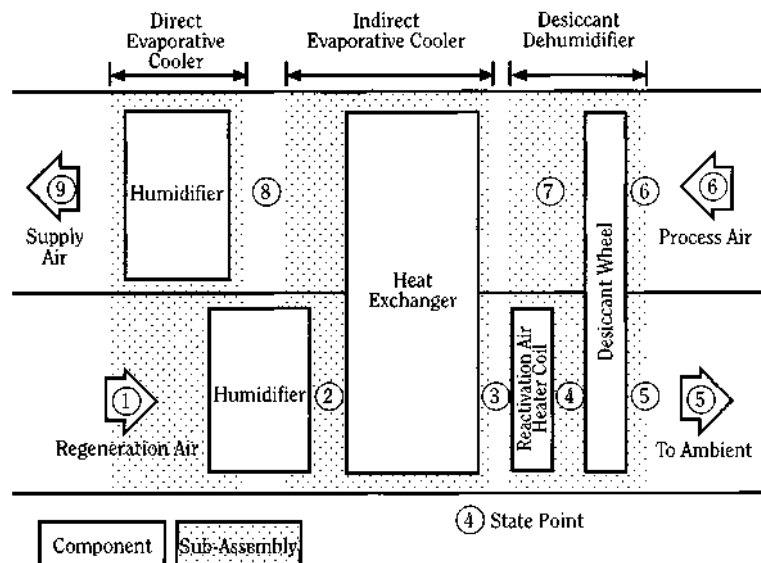


FIG. 16-51 Flow diagram of desiccant cooling cycle. [Reprinted with permission of American Society of Heating, Refrigeration and Air Conditioning Engineers, Inc. (ASHRAE). Reference: Collier, Cohen, and Slosberg in Harriman, *Desiccant Cooling and Dehumidification*, ASHRAE, Atlanta, 1992.]

dry air. This warmed air (3) is heated to the desired temperature for regeneration of the adsorbent wheel and exhausted to the atmosphere. Many other combinations of drying and cooling are used to accomplish desiccant cooling [Belding, in *Proceedings of AFEAS Refrigeration and Air Conditioning Workshop*, Breckenridge, CO (June 23–25, 1993)].

Heat pumps are another developing application of adsorbents. Zeolite/water systems have been evaluated as a means for transferring heat from a low temperature to a higher, more valuable level. Both natural (chabazite and clinoptilolite) and synthetic (NaX and high silica NaY) zeolites have favorable properties for sorption heat pumps. Data have demonstrated that hydrothermally stable Na-mordenite and dealuminated NaY can be used with water in chemical heat pumps to upgrade 100°C heat sources by 50 to 80°C using a 20°C heat sink [Fujiwara, Suzuki, Shin, and Sato, *J. Chem. Eng. Japan*, **23**, 738–743 (1990)]. Other work has shown that integration of two adsorber beds can achieve heating coefficients of performance of 1.56 for the system NaX/water, upgrading 150°C heat to 200°C with a 50°C sink [Douss, Meunier, and Sun, *Ind. Eng. Chem. Res.*, **27**, 310–316 (1988)].

Gas storage of onboard vehicular fuel is important to provide alternatives to gasoline and diesel fuel. Natural gas is cleaner-burning, and hydrogen would burn essentially pollution-free. Onboard storage of natural gas is typically as a high-pressure compressed gas. Adsorbed natural gas systems are a desirable solution because they could operate at lower pressures while maintaining the same capacities. The major problem currently impeding commercialization is the development of adsorbent materials with desirable isotherm capacities and shapes. Also, the exothermic nature of physical adsorption has a negative impact on charge and discharge in a gas storage cycle. Heat released during adsorption will increase the temperature of the adsorbent, thereby lowering the total amount of gas that can be stored. The vessel will cool during the discharge step, decreasing the amount of gas that can be delivered. Technological solutions are being developed and should appear in coming years [Chang and Talu, *Appl. Therm. Eng.*, **16**, 359 (1996); Mota, *AIChE J.*, **45**, 986 (1999)].

Energy Conservation Techniques The major use of energy in an adsorption cycle is associated with the regeneration step, whether it is thermal energy for TSA or compression energy for PSA. Since the regeneration energy per pound of adsorbent tends to be constant, the first step in minimizing consumption is to maximize the operating loading. When the mass-transfer zone (MTZ) is a large portion of the adsorber relative to the equilibrium section, the fraction of the bed being fully utilized is small. Most fixed-bed adsorption systems have two adsorbers so that one is on stream while the other is being regenerated. One means of improving adsorbent utilization is to use a lead/trim (or cascade, or merry-go-round) cycle. Two (or more) adsorbent beds in series treat the feed. The feed enters the lead bed first and then the trim bed. The trim bed is the one that has most recently been regenerated. The MTZ is allowed to proceed through the lead beds but not to break through the trim bed. In this way the lead bed can be almost totally utilized before being regenerated. When a lead bed is taken out of service, the trim bed is placed in lead position, and a regenerated bed is placed in the trim position.

TABLE 16-17 Process Descriptors

Number	Statement
1	Feed is a vaporized liquid or a gas.
2	Feed is a liquid that can be fully vaporized at less than about 200°C.
3	Feed is a liquid that cannot be fully vaporized at 200°C.
4	Adsorbate concentration in the feed is less than about 3 wt %.
5	Adsorbate concentration in the feed is between about 3 and 10 wt %.
6	Adsorbate concentration in the feed is greater than about 10 wt %.
7	Adsorbate must be recovered in high purity (> than 90–99% rejection of nonadsorbed material).
8	Adsorbate can be desorbed by thermal regeneration.
9	Practical purge or displacement agents cannot be easily separated from the adsorbate.

Keller, Anderson, and Yon in Rousseau (ed.), *Handbook of Separation Process Technology*, John Wiley & Sons, Inc., New York, 1987; reprinted with permission.

A thermal pulse cycle is a means of conserving thermal energy in heating-limited desorption. A process cycle that is heat-limited needs only a very small time (dwell) at temperature to achieve satisfactory desorption. If the entire bed is heated before the cooling is begun, every part of the bed will dwell at temperature for the entire time it takes the cooling front to traverse the bed. Thus, much of the heat in the bed at the start of cooling would be swept from the bed. Instead, cooling is begun before any heat front has exited the bed, creating a thermal pulse that moves through the bed. The pulse expends its thermal energy for desorption so that only a small temperature peak remains at the end of the regeneration and no excess heat has been wasted. If the heating step is stripping-limited, a thermal pulse is not applicable.

A series cool/heat cycle is another way in which the heat that is purged from the bed during cooling can be conserved. Sometimes the outlet fluid is passed to a heat sink where energy is stored to be reused to preheat heating fluid, or cross exchanged against the purge fluid to recover energy. However, there is also a process cycle that accomplishes the same effect. Three adsorbers are used, with one on adsorption, one on heating, and one on cooling. The regeneration fluid flows in series, first to cool the bed just heated and then to heat the bed to be desorbed. Thus all of the heat swept from the adsorber during heating can be reused to reduce the heating requirement. Unlike thermal pulse, this cycle is applicable to both heat- and stripping-limited heating.

Process Selection The preceding sections present many process cycles and their variations. It is important to have some guidelines for design engineers to narrow their choice of cycles to the most economical for a particular separation. Keller and coworkers [Keller et al., gen. refs.] have presented a method for choosing appropriate adsorption processes. Their procedure considers the economics of capital, energy, labor, and other costs. Although these costs can vary from site to site, the procedure is robust enough to include most scenarios. In Table 16-17, nine statements are made about the character of the separation being considered. The numbers of the statements that are true (i.e., applicable) are used in the matrix in Table 16-18. A “no” for any true statement under a given

TABLE 16-18 Process Selection Matrix

Statement number, Table 16-17	Gas- or vapor-phase processes					Liquid-phase processes		
	Temperature swing	Inert purge	Displacement purge	Pressure swing	Chromatography	Temperature swing*	Simulated moving bed	Chromatography
1	Yes	Yes	Yes	Yes	Yes	No	No	No
2	Not likely	Yes	Yes	Yes	Yes	Yes	Yes	Yes
3	No	No	No	No	No	Yes	Yes	Yes
4	Yes	Yes	Not likely	Not likely	Not likely	Yes	Not likely	Maybe
5	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes
6	No	Yes	Yes	Yes	Yes	No	Yes	Yes
7	Yes	Yes	Yes	Maybe†	Yes	Yes	Yes	Yes
8	Yes	No	No	No	No	Yes‡	No	No
9	Maybe§	Not likely	Not likely	N/A	Not likely	Maybe§	Not likely	Not likely

*Includes powdered, fixed-bed, and moving-bed processes.

†Very high ratio of feed to desorption pressure (>10:1) will be required. Vacuum desorption will probably be necessary.

‡If adsorbate concentration in the feed is very low, it may be practical to discard the loaded adsorbent or reprocess off-site.

§If it is not necessary to recover the adsorbate, these processes are satisfactory.

Keller, Anderson, and Yon in Rousseau (ed.), *Handbook of Separation Process Technology*, John Wiley & Sons, Inc., New York, 1987; reprinted with permission.

process should remove that process from further consideration. Any process having all "yes" answers for true statements deserves strong

consideration. Entries other than "yes" or "no" provide a means of prioritizing processes when more than one cycle is satisfactory.

EQUIPMENT

ADSORPTION

General Design Adsorbents are used in adsorbers with fixed inventory, with intermittent solids flow, or with continuous-moving solids flow. The most common are fixed beds operating as batch units or as beds of adsorbent through which the feed fluid passes, with periodic interruption for regeneration. Total systems consist of pressure vessels or open tanks along with the associated piping, valves, controls, and auxiliary equipment needed to accomplish regeneration of the adsorbent. Gas treating equipment includes blowers or compressors with a multiplicity of paths to prevent dead-heading. Liquid treating equipment includes pumps with surge vessels as needed to assure continuous flow.

Adsorber Vessel The most frequently used method of fluid-solid contact for adsorption operations is in cylindrical, vertical vessels, with the adsorbent particles in a fixed and closely but randomly packed arrangement. The adsorbers must be designed with consideration for pressure drop and must contain a means of supporting the adsorbent and a means of assuring that the incoming fluid is evenly distributed to the face of the bed. There are additional design considerations for adsorbers when the streams are liquid and for high-performance separation applications using very small particles (<0.05 mm) such as in HPLC.

For most large-scale processes, adsorbent particle size varies from 0.06 to 6 mm (0.0025 to 0.25 in), but the adsorbent packed in a fixed bed will have a fairly narrow particle size range. Pressure drop in adsorbers can be changed by changing the diameter to bed depth ratio and by changing the particle size (see Sec. 5). Adsorbent size also determines separation performance of adsorbent columns—increasing efficiency with decreasing particle size. In liquid-phase processing, total cost of the adsorption step can sometimes be reduced by designing for overall pressure drops as large as 300 to 600 kPa (45 to 90 psi) because pumping is not the major utility cost. In special, high-resolution applications (HPLC), pressure drops as high as 5000–25,000 kPa (800–4000 psi) are sometimes used requiring special pumping and column hardware [Colin in Ganetsos and Barker, *Preparative and Production Scale Chromatography*, Marcel Dekker, New York, 1993, pp. 11–45]. However, the cost of compressing gases is significant. Since blowers are limited to about 5 kPa (20 in wc) of lift, atmospheric gas applications are typically designed with adsorbent pressure drops of 1 to 4 kPa (4 to 16 in wc). To keep compression ratio low, compressed gas adsorption pressure drops are 5 to 100 kPa (0.7 to 15 psi) depending on the pressure level.

Besides influencing how much pressure drop is allowable, the operating pressure determines other design features. When adsorption and/or regeneration is to be performed at pressures above atmospheric, the adsorber vessels are designed like process pressure vessels (Fig. 16-52 [EPA, gen. refs.]). Their flow distributors can consume more gas momentum at higher pressure. On the other hand, for applications near atmospheric pressure, any pressure drop can be costly, and most design choices are made in the direction of minimizing head loss. Beds have large face areas and shallow depth. Many times, the choice is to fabricate a horizontal (horizontal axis) vessel where flows are radial rather than axial as in conventional vertical beds. Figure 16-53 [Leatherdale in Cheremisinoff and Ellerbusch, gen. refs.] depicts how a rectangular, shallow adsorber bed is oriented in a horizontal vessel. Flow distributors, especially for large units, are often elaborate in order to evenly divide the flow rather than consume precious head.

There are two types of support systems used for fixed beds of adsorbent. The first is a progressive series of grid- and screen-layers. In this system, each higher layer screen has successively smaller openings,

with the last small enough to prevent particles from passing through. Each lower layer has greater strength. A series of I-beams can be used to support a layer of subway grating that, in turn, supports several layers of screening. In other cases, special support grills such as Johnson™ screens may rest on the I-beams or on clips at the vessel wall and thus directly support the adsorbent. The topmost screen must retain the original size particles and some reasonable size of broken particles. The second type of support is a graded system of particles such as ceramic balls or gravel that rests directly on the bottom of the adsorber. A typical system might consist of 100 mm (4 in) of 50 mm (2 in) diameter material, covered by succeeding layers of 25, 12, and 6 mm (1, ½, ¼ in) of support material for a 3 mm (¼ in) adsorbent. In water treatment, the support may actually start with filter blocks and have an upper layer of sand (see Fig. 16-54 [EPA, gen. refs.]).

If flow is not evenly distributed throughout the bed of adsorbent, there will be less than maximum utilization of the adsorbent during adsorption and of the desorption fluid during regeneration. Incoming fluids from the nozzles is at a much higher velocity than the average through the bed and may have asymmetric momentum components due to the piping manifold. The simplest means of allowing flow to redistribute across the face of the bed is to employ ample plenum space above and below the fixed bed. A much more cost-effective method is to install simple baffle plates with symmetrically placed inlet and outlet nozzles. The solid, or perforated, baffles are designed to break the momentum of the incoming fluid and redistribute it to prevent direct impingement on the adsorbent. When graded bed support is installed at the bottom, the baffles should be covered by screening to restrain the particles. An alternative to screened baffles is slotted metal or Johnson™ screen distributors. Shallow horizontal beds often have such a large flow area that multiple inlet and outlet nozzles are required. These nozzle headers must be carefully designed to assure balanced flow to each nozzle. In liquid systems, a single inlet may enter the vessel and branch into several pipes that are often perforated along their length (Fig. 16-52). Such "spiders" and "Christmas trees" often have holes that are not uniformly spaced and sized but are distributed to provide equal flow per bed area.

Although allowable pressure loss with liquids is not a restricting factor, there are special considerations for liquid treating systems. Activated carbon adsorbers used in water and wastewater treatment are designed and constructed using the same considerations used for turbidity removal by sand or multilayer filters. A typical carbon bed is shown in Fig. 16-54. Such contactors for liquids at ambient pressure are often nothing more than open tanks or concrete basins with flow distribution simply an overflow weir. In liquid treating, the adsorbers must be designed with a means for liquid draining and filling occasionally or during every cycle when a gas is used for regeneration. Draining is by gravity, sometimes assisted by a 70–140 kPa (10–20 psig) pressure pad. Even with time to drain, there will be significant liquid holdup to recover. As much as 40 cc of liquid per 100 g of adsorbent is retained in the micro- and macropores and bridged between particles. When drain is concurrent to adsorption, the drained liquid can be recovered as treated product. When drain is countercurrent, drained fluid must be returned to feed surge. Minimizing other holdup in dead volume is especially important for liquid separation processes such as chromatography, because it adversely affects product recovery and regeneration efficiency. In filling an adsorber, there must be sufficient time for any gas trapped in the pores to escape. The fill step is preferably upflow to sweep the vapor out and to prevent gas pockets that could cause product contamination, bed lifting, or flow maldistribution. In liquid upflow, the buoyancy force of the liquid plus the pressure drop must not exceed the gravitational forces if bed lifting is to be prevented. Because there is very little increase in pressure

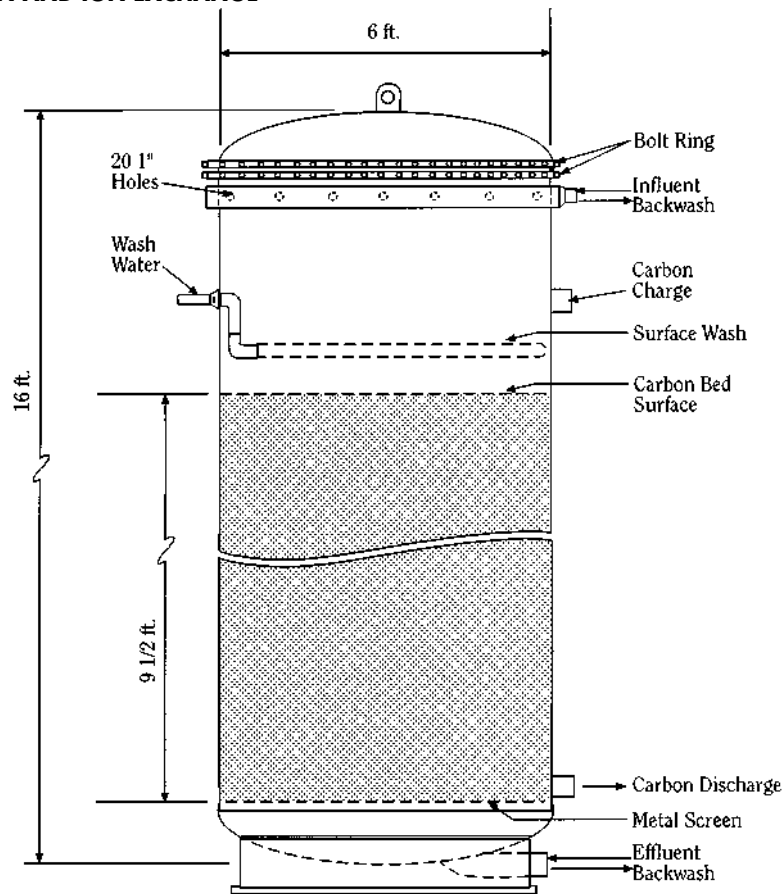


FIG. 16-52 Pressurized adsorber vessel. (Reprinted with permission of EPA. Reference: EPA, Process Design Manual for Carbon Adsorption, U.S. Envir. Protect. Agency., Cincinnati, 1973.)

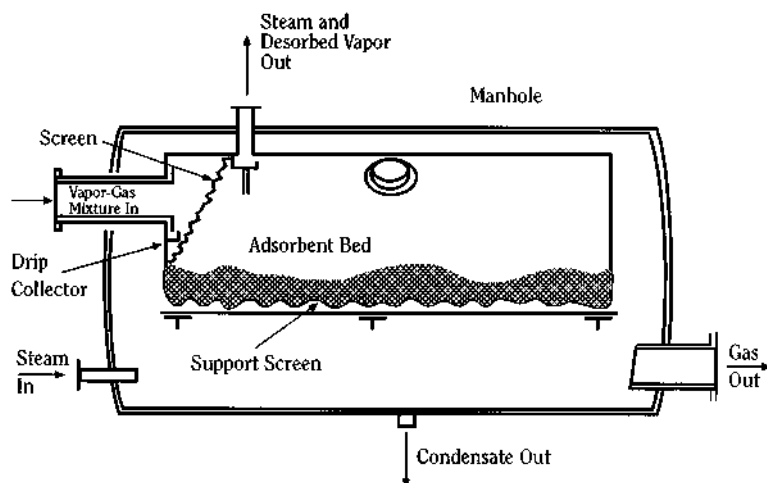


FIG. 16-53 Ambient pressure adsorber vessel. (Reprinted with permission of Ann Arbor Science. Reference: Leatherdale in Cheremisinoff and Ellerbusch, Carbon Adsorption Handbook, Ann Arbor Science, Ann Arbor, 1978.)

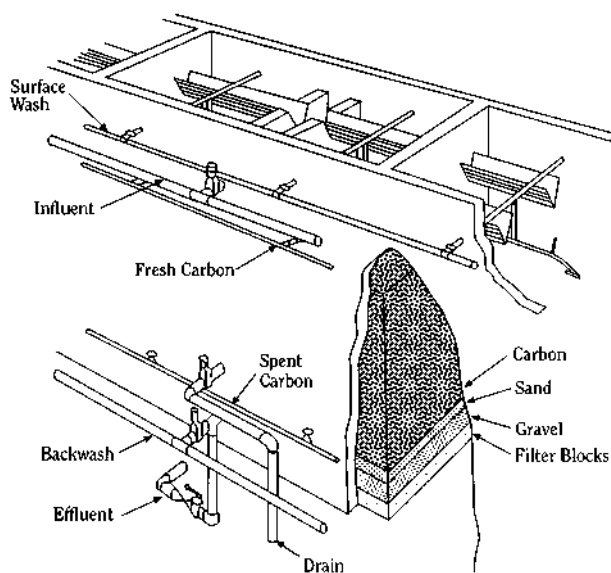


FIG. 16-54 Adsorber vessel with graded support system. (Reprinted with permission of EPA. Reference: EPA, Process Design Manual for Carbon Adsorption, U.S. Envir. Protect. Agency., Cincinnati, 1973.)

drop beyond the lifting (or fluidization) velocity, some liquid systems are designed with bed expansion to limit pressure drop. Upflow-adsorption expanded-beds are also preferred when the liquid contains any suspended solids, so that the bed does not act as a filter and become plugged. Since increased expansion causes the adsorbent to become increasingly well mixed, with accompanying drop in removal efficiency, expansion is usually limited to about 10 percent. Higher velocities also tend to cause too much particle turbulence, abrasion, attrition, and erosion.

Regeneration Equipment Sometimes it is economically justified to remove the adsorbent from the adsorber when it is exhausted and have an outside contractor regenerate it rather than install on-site regeneration equipment. This is feasible only if the adsorbent can treat feed for weeks or months rather than only hours or days. In other cases, the process conditions during regeneration are so much more severe than those for adsorption that a single regenerator with materials of construction capable of handling the conditions is more cost-effective than to make all adsorbers the expensive material. This is true for most water and wastewater treatment with thermally reactivated carbon. Otherwise, desorption is conducted in situ with any additional equipment connected to the adsorbers.

Figure 16-55 [Engineering Data Book, 10th ed., Gas Processors Suppliers Association, Tulsa, 1988, Sec. 20, p. 22] depicts the flow scheme for a typical two-bed TSA dryer system showing the auxiliary equipment associated with regeneration. Some of the dry product gas is externally heated and used countercurrently to heat and desorb water from the adsorber not currently drying feed. The wet, spent regeneration gas is cooled; the water is condensed out; and the gas is recycled to feed for recovery.

The thermal reactivation of spent activated carbon may require the same high temperatures and reaction conditions used for its manufacture. Although the exact conditions to be used for reactivation depend on the type of carbon and the adsorbates to be removed, the objective is to remove the adsorbed material without altering the carbon structure. This occurs in four stages: (a) drying, (b) desorption, (c) pyrolysis and carbonization, and (d) burnoff. Each of these steps is associated with a particular temperature range and is carried out in a multiple hearth furnace such as that depicted in Fig. 16-56. This six-hearth system is a typical configuration for water treatment carbons. The gas temperature ranges from 100°C on the top hearth to 950°C at the bottom. The rotating rabble arms rake the carbon from hearth to hearth. Wet, spent carbon is fed to the top, where it is dried by the top two hearths. Pyrolysis (desorption and decomposition) is accomplished on the next hearth. The bottom four hearths are for reactivation. Varying amounts of air, steam, and/or fuel are added to the different hearths to maintain the conditions established for the particular reactivation. A hearth area requirement of 0.1 m²/kg/hr (0.5 ft²/lb/hr) is adequate for cost estimating, but there are also detailed design procedures available [von Dreusche in Cheremisinoff and Ellerbush, gen. refs.].

In addition to the multiple-hearth furnace, the reactivation system is comprised of additional equipment to transport, store, dewater, and quench the carbon.

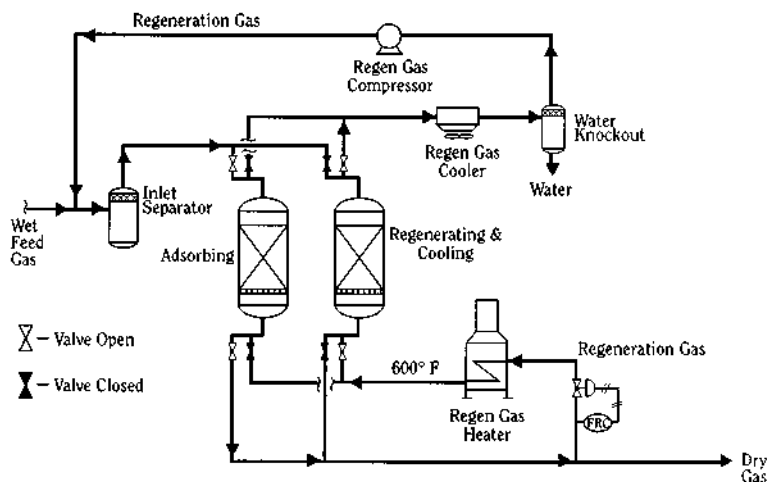


FIG. 16-55 Two-bed TSA system with regeneration equipment. (Reprinted with permission of GPSA. Reference: Engineering Data Book, 10th ed., Gas Processors Suppliers Association, Tulsa, 1988, Sec. 20, p. 22.)

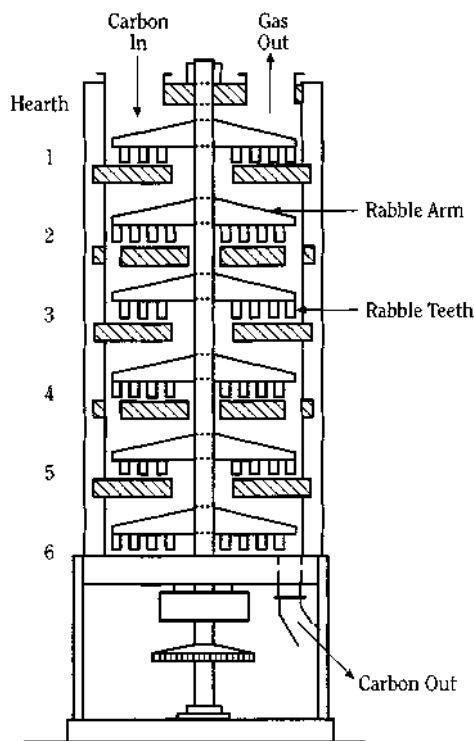


FIG. 16-56 Multiple hearth furnace for carbon reactivation. (Reprinted with permission of EPA. Reference: EPA, Process Design Manual for Carbon Adsorption, U.S. Envir. Protect. Agency, Cincinnati, 1973.)

A typical two-bed PSA dryer process flow scheme is very similar to Fig. 16-55 without the regeneration gas heater. A portion of the dry product gas is used countercurrently to purge the water from the adsorber not currently drying feed. Again, the wet, spent purge gas is cooled, and the water is condensed out and recycled for feed for recovery. If the cycle is VSA, then instead of the regeneration gas compressor, there is a vacuum compressor at the purge outlet upstream of the regeneration cooler. Some PSA cycles such as in air-brake dryers operate with one adsorber, a surge volume, and a three-way valve to switch between pressurizing and countercurrent blowdown. The other extreme of complexity is demonstrated by the UOP nine-bed Polybed PSA H_2 unit shown in Fig. 16-57 [Fuderer and Rudelstorfer, U.S. Patent number 3,986,849, 1976]. This cycle requires nine adsorbers sequenced by 55 valves to maximize purity and recovery.

Cycle Control Valves are the heart of cycle control for cyclic adsorption systems. These on/off valves switch flows among beds so that external to the system it appears as if operation is continuous. In general, one valve is needed for each bed at each end for each step that is performed (e.g., for a two-bed system with an adsorption step plus heating and cooling step [carried out in the same direction and during the same step], only $2 \times 2 \times 2 = 8$ valves would be needed [see Fig. 16-55]). In some cycles such as pressure-swing systems, it may be possible to use valves for more than one function (e.g., repressurization with feed gas using the same manifold as adsorption feed). Without multiple use, the cycle in Fig. 16-57 would need $9 \times 2 \times 5 = 90$ valves instead of 55 to accommodate the five steps of adsorption, purge to product, blowdown, purge, and feed repressurization (even without the equalization). For some applications, 3- and 4-way valves can replace two and four valves, respectively. The ultimate integration of switching valves is the UOP rotary valve discussed below. For long step times (8 hours or more) it is possible for the valves to be manually switched by operators. For most systems, it is

advantageous for the opening and closing of the valves be controlled by automatic timers. The same controller can be responsible for maintaining flows and pressure, logic for proceeding only on completion of events, and safety bypass or shutdown. Automatic control can provide for a period of parallel flow paths to assure transitions. In some applications, process analyzers can interface with the controller to initiate bed-switching when adsorbate is detected breaking through into the effluent.

Continuous Countercurrent Systems Most adsorption systems use fixed-bed adsorbers. However, if the fluid to be separated and that used for desorption can be countercurrently contacted by a moving bed of the adsorbent, there are significant efficiencies to be realized. Because the adsorbent leaves the adsorption section essentially in equilibrium with the feed composition, the inefficiency of the mass-transfer zone is eliminated. The adsorption section only needs to contain a MTZ length of adsorbent compared to a MTZ plus an equilibrium section in a fixed bed. Likewise, countercurrent regeneration is more efficient. Since the adsorbent is moved from an adsorption chamber to another chamber for regeneration, only the regeneration section is designed for the often more severe conditions. Countercurrent adsorption can take advantage of an exceptionally favorable equilibrium in water softening; and the regeneration step can be made favorable by the use of relatively concentrated eluent. Continuous units generally require more headroom but much less footprint. The foremost problems to be overcome in the design and operation of continuous countercurrent sorption operations are the mechanical complexity of equipment and the attrition of the sorbent.

An early example of a commercial countercurrent adsorption processes is the hypersorption process developed by Union Oil Company for the recovery of propane and heavier components from natural gas [Berg, *Chem. Eng. Progr.*, **47**, 585-590 (1951)]. Hypersorption used an activated carbon adsorbent moving as a dense bed continuously downward through a rising gas stream (Fig. 16-58 [Berg, *ibid.*]). However, this process proved to be less economical than cryogenic distillation and had excessive carbon losses resulting from attrition. The commercialization by Kureha Chemical Co. of Japan of a new, highly attrition-resistant, activated-carbon adsorbent, allowed development of a process employing fluidized-bed adsorption and moving bed desorption for removal of VOC compounds from air. The process has been marketed as GASTAKSM in Japan and as PuraSivSM HR (Fig. 16-59 [Anon., *Chem. Eng.*, **84**(18), 39-40 (1977)]) in the United States, and now as SOLDACS by Daikin Industries, Ltd. A similar process using beaded polymeric resin is offered by Chematur [Heinegaard, *Chem.-Ing.-Tech.*, **60**, 907-908 (1988)]. The recent discovery [Acharya and BeVier, U.S. Patent number 4,526,877, 1985] that the graphite coating of zeolites can dramatically improve attrition resistance without significantly impairing adsorption performance could allow the extension of moving-bed technology to other separations. A good review of continuous ion-exchange applications is presented by Wankat [gen. refs.].

Cross-Flow Systems There are at least three implementations of moving-bed adsorption that are cross flow rather than fixed beds or countercurrent flow: (1) panel beds, (2) adsorbent wheels, and (3) rotating annular beds. By *cross flow* is meant that the adsorbent is moving in a direction perpendicular to the fluid flow. All of these employ moving adsorbent—the first, a down-flowing solid; and the others, a constrained solid. Panel beds of activated carbon have been applied to odor control [Lovett and Cuniff, *Chem. Eng. Progr.*, **70**(5), 43-47 (1974)] and to the desulfurization of waste gas [Richter, Knoblauch, and Juntgen, *Verfahrenstechnik (Mainz)*, **14**, 338-342 (1980)]. The spent solid falls from the bottom panel into a load-out bin, and fresh regenerated carbon is added to the top; gas flows across the panel.

The heart of an adsorbent wheel system is a rotating cylinder containing the adsorbent. Figure 16-60 illustrates two types: horizontal and vertical. In some adsorbent wheels, the adsorbent particles are placed in basket segments (a multitude of fixed beds) to form a horizontal wheel that rotates around a vertical axis. In other instances, the adsorbent is integral to the monolithic wheel or coated onto a metal, paper, or ceramic honeycomb substrate. These monolithic or

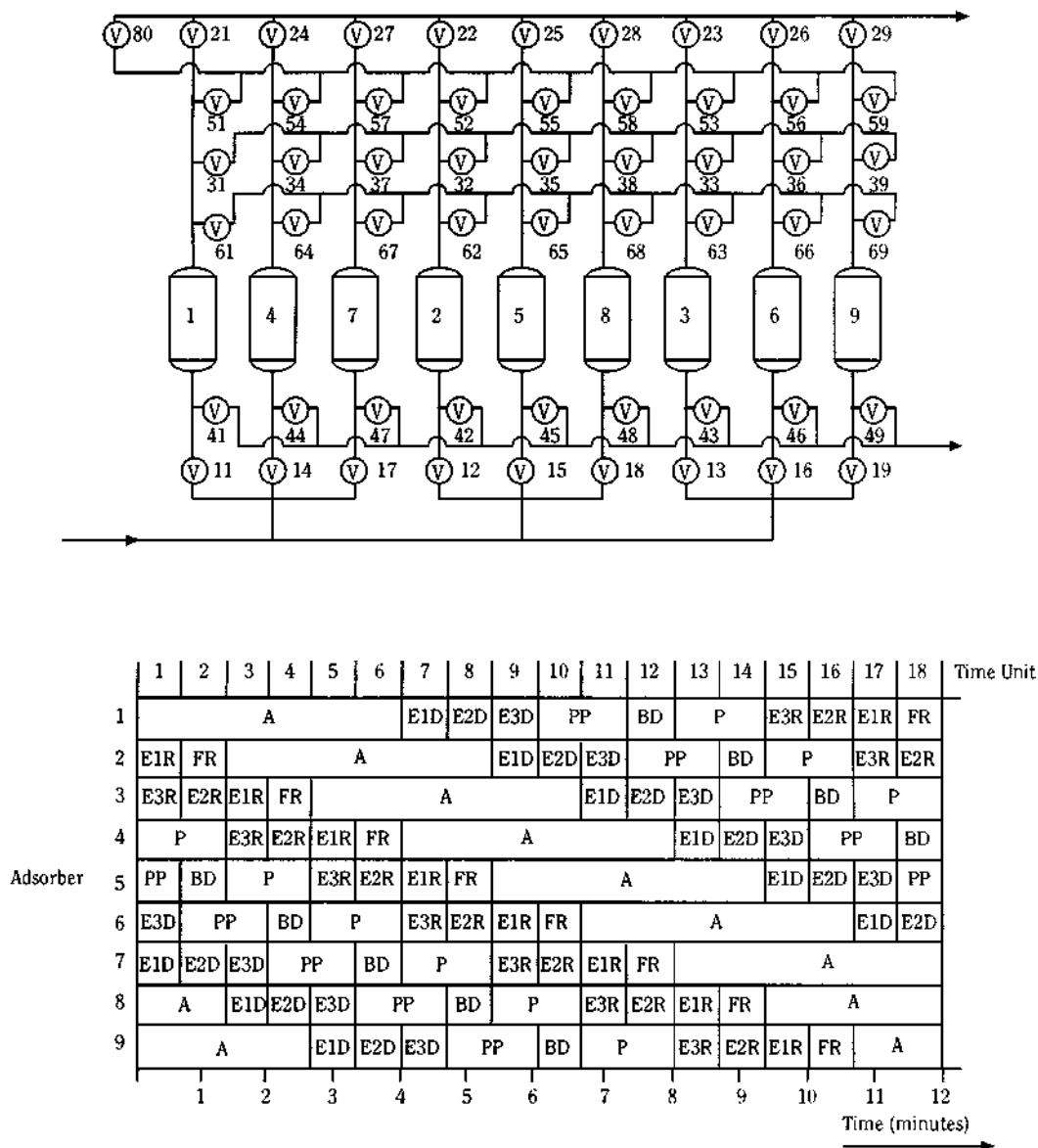


FIG. 16-57 UOP nine-bed polybed PSA H_2 unit: (a) flow scheme; (b) cycle diagram. (Reference: Fudrer and Rudelstorfer, U.S. Patent number 3,986,849, 1976.)

honeycomb structures rotate around either a vertical or a horizontal axis. The gas to be treated usually flows through the wheel parallel to the axis of rotation, although some implementations use radial flow configurations. Most of the wheel is removing adsorbates. The remaining (smaller) portion of the wheel is undergoing thermal regeneration—usually countercurrently. The wheel constantly rotates to provide a continuous treated stream and a steady concentrated stream. Adsorbent wheels are most often used to treat ambient air because they have very low pressure drop. One application of wheels is the removal of VOC where the regeneration stream is usually sent to an incinerator for destruction of VOC. Another use is in desiccant cooling (see previously). They do suffer from low efficiency due to the short contact time, mechanical leakage at seals,

and the tendency to allow the wheel to exceed breakthrough in order to get better adsorbent utilization. Some adsorbent wheels are operated in an intermittent manner such that the wheel periodically indexes to a new position; this is particularly true of radial flow wheels.

The rotating annular bed system for liquid chromatographic separation (two-dimensional chromatography) is analogous to the horizontal adsorbent wheel for gases. Feed to be separated flows to a portion of the top face of an annular bed of sorbent. A displacement purge in the form of a solvent or carrier gas flows to the rest of the annulus. The less strongly adsorbed components travel downward through the sorbent at a higher rate. Thus, they will exit at the bottom of the annulus at a smaller angular distance from the feed sector. The more strongly adsorbed species will exit at a greater angle. The mechanical and

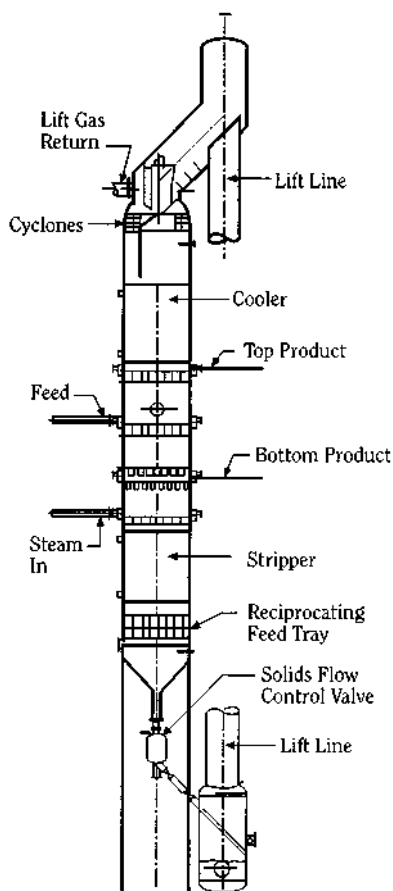


FIG. 16-58 Hypersorption adsorber vessel. [Reprinted with permission of AIChE. Reference: Berg, Chem. Eng. Progr., 47, 585-590 (1951).]

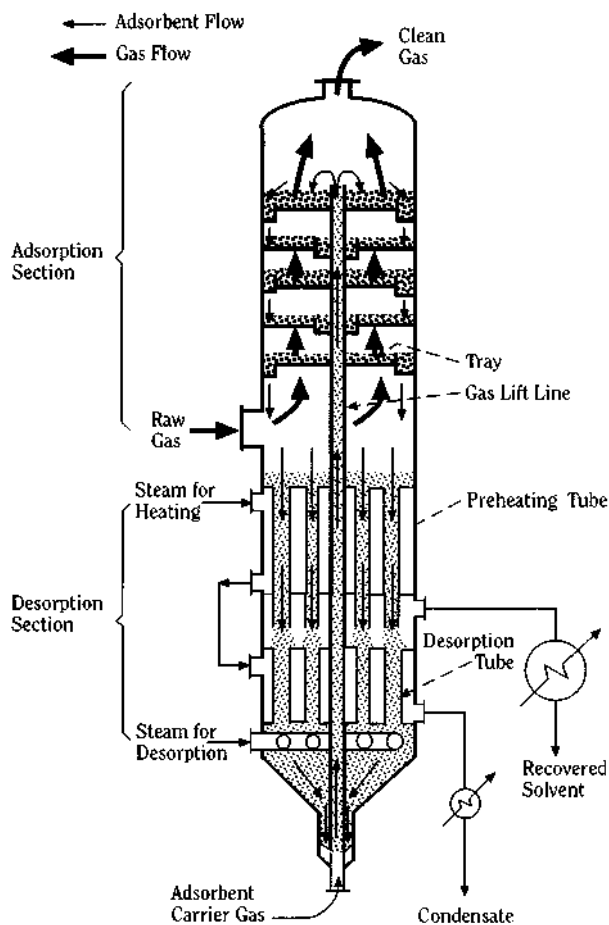


FIG. 16-59 PuraSiv HR adsorber vessel. [Reprinted with permission of Chemical Engineering. Reference: Anon., Chem. Eng., 84(18), 39-40 (1977).]

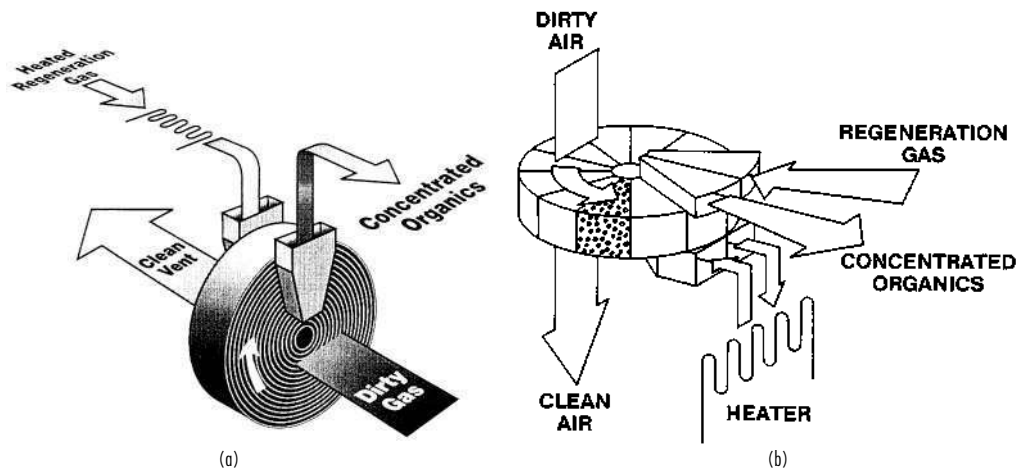


FIG. 16-60 Adsorbent wheels for gas separation: (a) horizontal with fixed beds; (b) vertical monolith. (Reprinted with permission of UOP.)

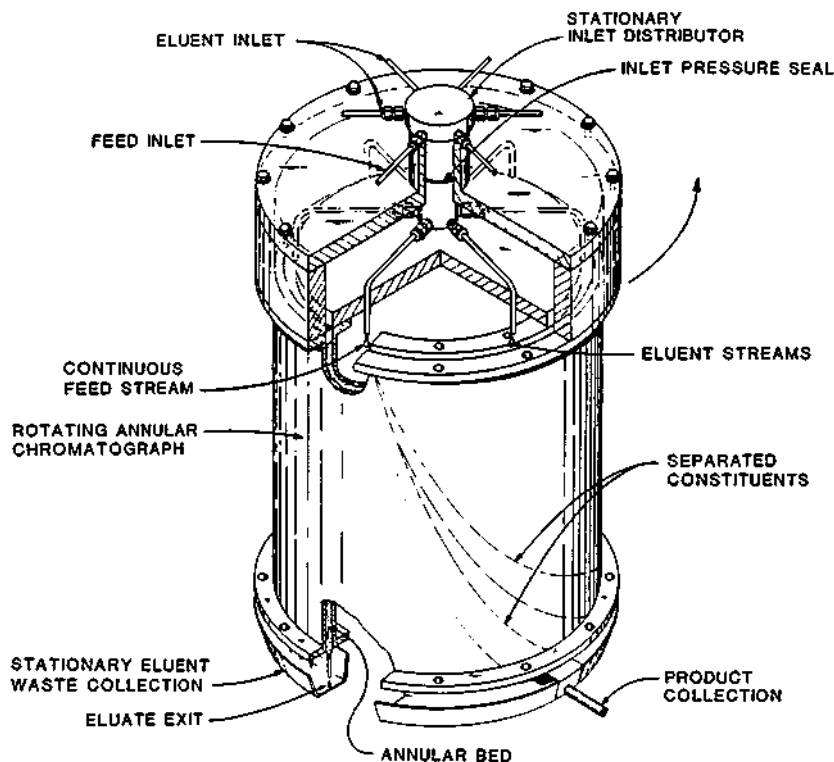


FIG. 16-61 Annular bed for liquid separation. [Reprinted with permission of Marcel Dekker. Reference: Canon, Begovich, and Sisson, *Sep. Sci. Technol.*, **15**, 655-678 (1980).]

packing complexities of such an apparatus have been overcome for a pressurized system by workers at the Oak Ridge National Laboratory shown in Fig. 16-61 [Canon, Begovich, and Sisson, *Sep. Sci. Technol.*, **15**, 655-678 (1980)]. Several potential applications are reviewed by Carta and Byers [*Chromatographic and Membrane Processes in Biotechnology*, NATO ASI Proceeding, Kluwer, 1991].

ION EXCHANGE

A typical fixed-bed ion exchanger consists of a vertical cylindrical vessel of lined steel or stainless steel. Linings are usually of natural or synthetic rubber. Spargers are provided at the top and bottom, and frequently a separate distributor is used for the regenerant solution. The resin bed, consisting of a meter or more of ion-exchange particles, is supported by the screen of the bottom distributor. Externally, the unit is provided with a valve manifold to permit downflow loading, upflow backwash, regeneration, and rinsing. For deionization, a two-exchanger assembly comprising a cation and an anion exchanger, is a common configuration (Fig. 16-62).

Column hardware designed to allow countercurrent, upflow regeneration of ion-exchange resins is available. An example is given in Fig. 16-63. During upflow of the regenerant, bed expansion is prevented by withdrawing the effluent through the application of vacuum. A layer of drained particles is formed at the top of the bed while the rest of the column functions in the usual way.

Typical design data for fixed-bed ion-exchange columns are given in Table 16-19. These should be used for preliminary evaluation purposes only. Characteristic design calculations are presented and illustrated by Applebaum [*Demineralization by Ion Exchange*, Academic, New York, 1968]. Large amounts of data are available in published literature and in bulletins of manufacturers of ion exchangers. In general, however, laboratory testing and pilot-plant work is advisable

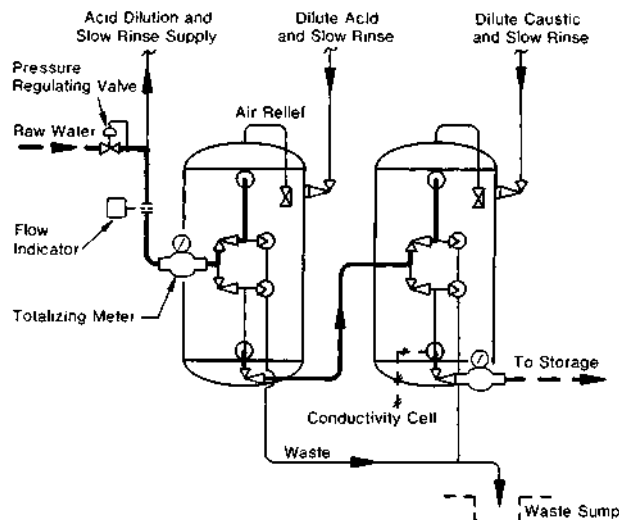


FIG. 16-62 Typical two-bed deionizing system. (Infilco Degremont Inc.)

TABLE 16-19 Design Data for Fixed-Bed Ion Exchanger*

Type of resin	Maximum and minimum flow, m/h [gal/(min-ft ²)]	Minimum bed depth, m (in)	Maximum operating temperatures, °C (°F)	Usable capacity, g-equivalent/L†	Regenerant, g/L resin‡
Weak acid cation	20 max. (8) 3 min. (1)	0.6 (24)	120 (248)	0.5–2.0	110% theoretical (HCl or H ₂ SO ₄)
Strong acid cation	30 max. (12) 3 min. (1)	0.6 (24)	120 (248)	0.8–1.5 0.5–1.0 0.7–1.4	80–250 NaCl 35–200 66° Bé. H ₂ SO ₄ 80–500 20° Bé. HCl
Weak and intermediate base anions	17 max. (7) 3 min. (1)	0.75 (30)	40 (104)	0.8–1.4	35–70 NaOH
Strong-base anions	17 max. (7) 3 min. (1)	0.75 (30)	50 (122)	0.35–0.7	70–140 NaOH
Mixed cation and strong-base anion (chemical-equivalent mixture)	40 max. (16)	1.2 (47)	50 (122)	0.2–0.35 (based on mixture)	Same as cation and anion individually

*These figures represent the usual ranges of design for water-treatment applications. For chemical-process applications, allowable flow rates are generally somewhat lower than the maximums shown, and bed depths are usually somewhat greater.

†To convert to capacity in terms of kilograins of CaCO₃ per cubic foot of resin, multiply by 21.8.

‡To convert to pounds of regenerant per cubic foot of resin, multiply by 0.0625.

to determine usable exchanger capacities, regenerant quantities, exchanger life, and quality of product for any application not already thoroughly proven in other plants for similar conditions. Firms that manufacture ion exchangers and ion-exchange equipment will often cooperate in such tests.

For larger-scale applications, a number of continuous or semicontinuous ion-exchange units are also available. The Higgins contactor (Fig. 16-64) was originally developed to recover uranium from leach slurries at the Oak Ridge National Laboratory. More recently, it has been adapted to a wide variety of applications, including large-volume water softening.

The Asahi process (Fig. 16-65) is used principally for high-volume water treatment. The liquid to be treated is passed upward through a resin bed in the adsorption tank. The upward flow at 30–40 m/h [12–16 gal/(min ft²)] keeps the bed packed against the top. After a preset time, 10 to 60 min, the flow is interrupted for about 30 s,

allowing the entire bed to drop. A small portion (10 percent or less) of the ion-exchange resin is removed from the bottom of the adsorption tank and transferred hydraulically to the hopper feeding the regeneration tank. The process is then resumed. Meanwhile, regeneration is occurring by a similar flow system in the regeneration tank, from which the regenerated ion exchanger is transferred periodically to the hopper above the water-rinse tank. In the latter, the resin particles are fluidized to flush away fines and accumulated foreign matter before the resin is returned to the adsorption tank.

Another continuous ion-exchange system is described by Himsley and Farkas (see Fig. 16-66). Such a system is used to treat 1590 m³/h (7000 gal/min) of uranium-bearing copper leach liquor using fiberglass-construction columns 3.7 m (12 ft) in diameter. The adsorption column is divided vertically into stages. Continuous transfer of batches of resin occurs from stage to stage without any interruption of flow. This is accomplished by pumping solution from one stage (A) to the stage (B) immediately above by means of external pumping in such a manner that the net flow through stage B is downward, carrying with it all of the resin in that stage B. When transfer of the ion exchanger is complete, the resin in stage C above is transferred downward in a similar manner. The process

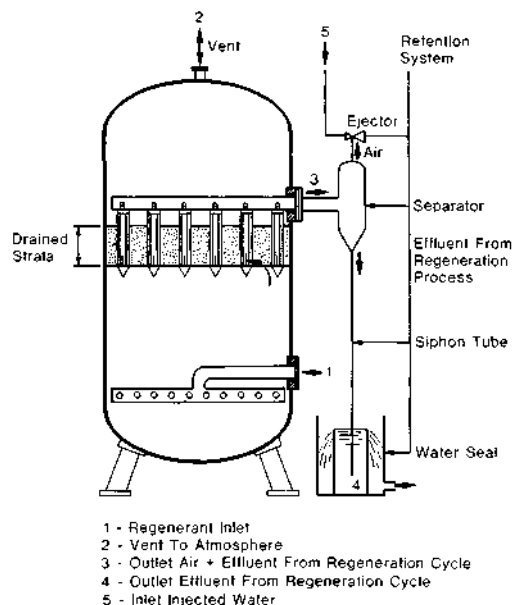


FIG. 16-63 Internals of an upflow regenerated unit. (Inflico Degremont Inc.)

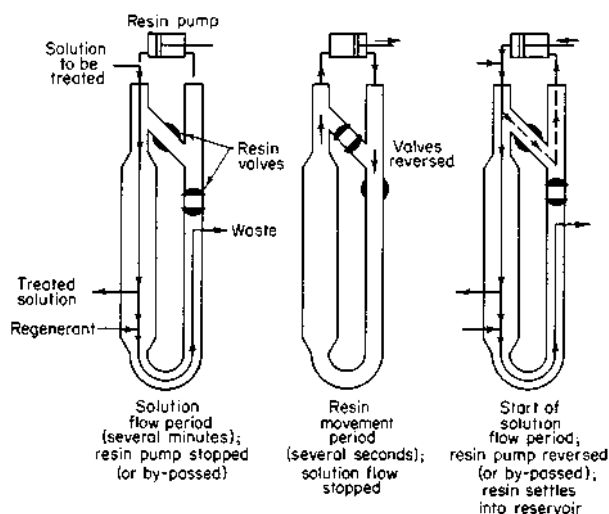


FIG. 16-64 Mode of operation of the Higgins contactor. (ORNL-LR-Dwg. 27857R.)

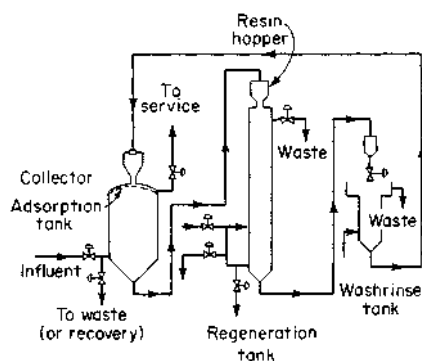


FIG. 16-65 Asahi countercurrent ion-exchange process. [Giltwood, *Chem. Eng.*, **74**(26), 86 (1967); copyright 1967 by McGraw-Hill, Inc., New York. Excerpted with special permission of McGraw-Hill.]

continues until the last stage (F) is empty. The regenerated resin is then transferred from the elution column to the empty stage (F). Elution of the sorbed product is carried out in the elution column in a moving packed bed mode. The countercurrent contact achieved in this column yields a concentrated eluate with a minimum consumption of regenerant.

Disadvantages of these continuous countercurrent systems are associated primarily with the complexity of the equipment required and with the attrition resulting from the transport of the ion exchanger. An effective alternative for intermediate scale processes is the use of merry-go-round systems and SMB units employing only packed-beds with no movement of the ion-exchanger.

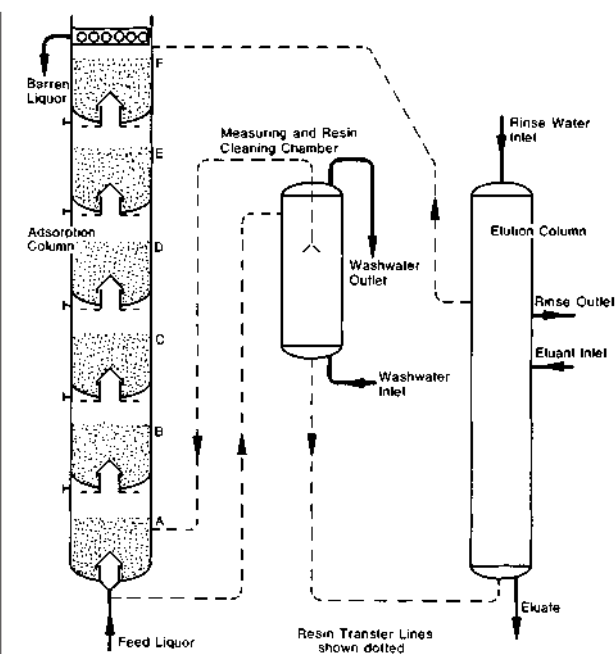


FIG. 16-66 Himsley continuous ion-exchange system. (Himsley and Farkas, "Operating and Design Details of a Truly Continuous Ion Exchange System," *Soc. Chem. Ind. Conf.*, Cambridge, England, July 1976. Used by permission of the Society of Chemical Industry.)

SECTION 17

Gas-Solid
Operations and
Equipment

PERRY'S CHEMICAL ENGINEERS' HANDBOOK

8TH EDITION



MEL PELL, JAMES B. DUNSON
TED M. KNOWLTON

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Gas-Solid Operations and Equipment

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FLUIDIZED-BED SYSTEMS

Consider a bed of particles in a column that is supported by a distributor plate with small holes in it. If gas is passed through the plate so that the gas is evenly distributed across the column, the drag force on the particles produced by the gas flowing through the particles increases as the gas flow through the bed is increased. When the gas flow through the bed causes the drag forces on the particles to equal the weight of the particles in the bed, the particles are fully supported and the bed is said to be fluidized. Further increases in gas flow through the bed cause bubbles to form in the bed, much as in a fluid, and early researchers noted that this resembled a fluid and called this a fluidized state.

When fluidized, the particles are suspended in the gas, and the fluidized mass (called a fluidized bed) has many properties of a liquid. Like a liquid, the fluidized particles seek their own level and assume the shape of the containing vessel. Large, heavy objects sink when added to the bed, and light particles float.

Fluidized beds are used successfully in many processes, both catalytic and noncatalytic. Among the catalytic processes are fluid catalytic cracking and reforming, oxidation of naphthalene to phthalic anhydride, the production of polyethylene and ammoxidation of propylene to acrylonitrile. Some of the noncatalytic uses of fluidized beds are in the roasting of sulfide ores, coking of petroleum residues, calcination of ores, combustion of coal, incineration of sewage sludge, and drying and classification.

Although it is possible to fluidize particles as small as about 1 μm and as large as 4 cm, the range of the average size of solid particles which are more commonly fluidized is about 30 μm to over 2 cm. Particle size affects the operation of a fluidized bed more than particle density or particle shape. Particles with an average particle size of about 40 to 150 μm fluidize smoothly because bubble sizes are relatively small in this size range. Larger particles (150 μm and larger) produce larger bubbles when fluidized. The larger bubbles result in a less homogeneous fluidized bed, which can manifest itself in large pressure fluctuations. If the bubble size in a bed approaches approximately one-half to two-thirds the diameter of the bed, the bed will slug. A slugging bed is characterized by large pressure fluctuations that can result in instability and severe vibrations in the system. Small particles (smaller than 30 μm in diameter) have large interparticle forces (generally van der Waals forces) that cause the particles to stick together, as flour particles do. These type of solids fluidize poorly because of the agglomerations caused by the cohesion. At velocities that would normally fluidize larger particles, channels, or spouts, form in the bed of these small particles, resulting in severe gas bypassing. To fluidize these small particles, it is generally necessary to operate at very high gas velocities so that the shear forces are larger than the cohesive forces of the particles. Adding finer-sized particles to a coarse bed, or coarser-sized particles to a bed of cohesive material (i.e., increasing the particle size range of a material), usually results in better (smoother) fluidization.

Gas velocities in fluidized beds generally range from 0.1 to 3 m/s (0.33 to 9.9 ft/s). The gas velocities referred to in fluidized beds are superficial gas velocities—the volumetric flow through the bed divided by the bed area. More detailed discussions of fluidized beds can be found in Kunii and Levenspiel, *Fluidization Engineering*, 2d ed., Butterworth Heinemann, Boston, 1991; Pell, *Gas Fluidization*, Elsevier, New York, 1990; Geldart (ed.), *Gas Fluidization Technology*, Wiley, New York, 1986; Yang (ed.), *Handbook of Fluidization and Fluid Particle Systems*, Marcel Dekker, New York, 2003; and papers published in periodicals, transcripts of symposia, and the American Institute of Chemical Engineers symposium series.

GAS-SOLID SYSTEMS

Researchers in the fluidization field have long recognized that particles of different size behave differently in fluidized beds, and several have tried to define these differences. Some of these characterizations are described below.

Types of Solids Perhaps the most widely used categorization of particles is that of Geldart [*Powder Technol.* 7: 285–292 (1973)].

Geldart categorized solids into four different groups (groups A, B, C, and D) that exhibited different properties when fluidized with a gas. He classified the four groups in his famous plot, shown in Fig. 17-1. This plot defines the four groups as a function of average particle size d_{av} , μm , and density difference $\rho_s - \rho_f$, g/cm^3 , where ρ_s = particle density, ρ_f = fluid density, and d_{av} = surface volume diameter of the particles. Generally d_{av} is the preferred average particle size for fluid-bed applications, because it is based on the surface area of the particle. The drag force used to generate the pressure drop used to fluidize the bed is proportional to the surface area of the particles. Another widely used average particle is the median particle size $d_{p,50}$.

When the gas velocity through a bed of group A, B, C, or D particles increases, the pressure drop through the bed also increases. The pressure drop increases until it equals the weight of the bed divided by the cross-sectional area of the column. The gas velocity at which this occurs is called the minimum fluidizing velocity U_{mf} . After minimum fluidization is achieved, increases in velocity for a bed of group A (generally in the particle size range between 30 and 100 μm) particles will result in a uniform expansion of the particles without bubbling until at some higher gas velocity the gas bubbles form at a velocity called the minimum bubbling velocity U_{mb} . For Geldart group B (between 100 and about 1000 μm) and group D (1000 μm and larger) particles, bubbles start to form immediately after U_{mf} is achieved, so that U_{mf} and U_{mb} are essentially equal for these two Geldart groups. Group C (generally smaller than 30 μm) particles are termed cohesive particles and clump together in particle agglomerates because of interparticle forces (generally van der Waals forces). When gas is passed through beds of cohesive solids, the gas tends to channel or “rathole” through the bed. Instead of fluidizing the particles, the gas opens channels that extend from the gas distributor to the surface of the bed. At higher gas velocities where the shear forces are great enough to overcome the interparticle forces, or with mechanical agitation or vibration, cohesive particles will fluidize but with larger clumps or clusters of particles formed in the bed.

Two-Phase Theory of Fluidization The two-phase theory of fluidization assumes that all gas in excess of the minimum bubbling velocity passes through the bed as bubbles [Toomey and Johnstone, *Chem. Eng. Prog.* 48: 220 (1952)]. In this view of the fluidized bed, the gas flowing through the emulsion phase in the bed is at the minimum bubbling velocity, while the gas flow above U_{mb} is in the bubble phase. This view of the bed is an approximation, but it is a helpful way

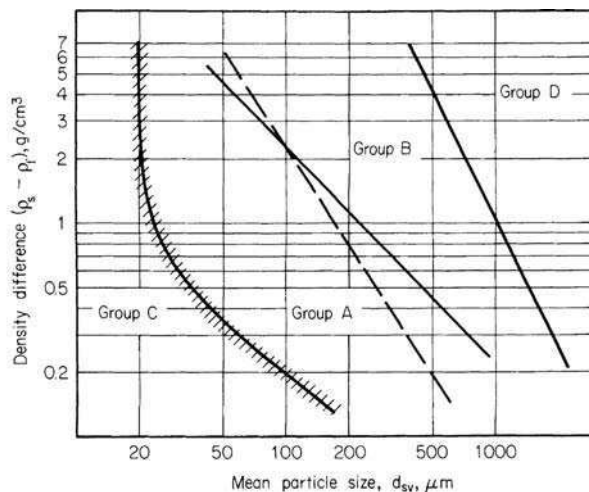


FIG. 17-1 Powder-classification diagram for fluidization by air (ambient conditions). [From Geldart, *Powder Technol.*, 7, 285–292 (1973).]

of understanding what happens as the gas velocity is increased through a fluidized bed. As the gas velocity is increased above U_{mb} , more and larger bubbles are formed in the bed. As more bubbles are produced in the bed, the bed expands and the bed density decreases.

For all Geldart groups (A, B, C, and D), as the gas velocity is increased, the fluidized-bed density is decreased and the turbulence of the bed is increased. In smaller-diameter beds, especially with group B and D powders, slugging will occur as the bubbles increase in size to greater than one-half to two-thirds of the bed diameter. Bubbles grow by vertical and lateral merging and increase in size as the gas velocity is increased [Whitehead, in Davidson and Harrison (eds.), *Fluidization*, Academic, London and New York, 1971]. As the gas velocity is increased further, the stable bubbles break down into unstable voids. When unstable voids characterize the gas phase in fluidized beds, the bed is not in the bubbling regime anymore, but is said to be in the turbulent regime. The turbulent regime is characterized by higher heat- and mass-transfer rates than bubbling fluidized beds, and the pressure fluctuations in the bed are reduced relative to bubbling beds. As the gas velocity is increased above the turbulent fluidized regime, the turbulent bed gradually changes into the pneumatic conveying regime.

Phase Diagram (Zenz and Othmer) As shown in Fig. 17-2, Zenz and Othmer, (*Fluidization and Fluid Particle Systems*, Reinhold, New York, 1960) developed a gas-solid phase diagram for systems in which gas flows upward, as a function of pressure drop per unit length versus gas velocity with solids mass flux as a parameter. Line OAB in Fig. 17-2 is the pressure drop versus gas velocity curve for a packed bed, and line BD is the curve for a fluidized bed with no net solids flow through it. Zenz indicated that there was an instability between points D and H because with no solids flow, all the particles will be

entrained from the bed. However, if solids are added to replace those entrained, system IJ (generally known as the pneumatic conveying region) prevails. The area DHIJ will be discussed in greater detail later.

Phase Diagram (Grace) Grace [Can. J. Chem. Eng., 64: 353-363 (1986); Fig. 17-3] has correlated the various types of gas-solid systems in which the gas is flowing vertically upward in a status graph using the parameters of the Archimedes number Ar for the particle size and a nondimensional velocity U^* for the gas effects. By means of this plot, the fluidization regime for various operating systems can be approximated. This plot is a good guide to estimate the fluidization regime for various particle sizes and operating conditions. However, it should not be substituted for more exact methods of determining the actual fluidization operating regime.

Regime Diagram (Grace) Grace [Can. J. Chem. Eng., 64, 353-363 (1986)] approximated the appearance of the different regimes of fluidization in the schematic drawing of Fig. 17-4. This drawing shows the fluidization regimes that occur as superficial gas velocity is increased from the low-velocity packed bed regime to the pneumatic conveying transport regime. As the gas velocity is increased from the moving packed bed regime, the velocity increases to a value U_{mf} such that the drag forces on the particles equal the weight of the bed particles, and the bed is fluidized. If the particles are group A particles, then a "bubbleless" particulate fluidization regime is formed. At a higher gas velocity U_{mb} , bubbles start to form in the bed. For Geldart group B and D particles, the particulate fluidization regime does not form, but the bed passes directly from a packed bed to a bubbling fluidized bed. As the gas velocity is increased above U_{mb} , the bubbles in the bed grow in size. In small laboratory beds, if the bubble size grows to a value equal to approximately one-half to two-thirds the diameter

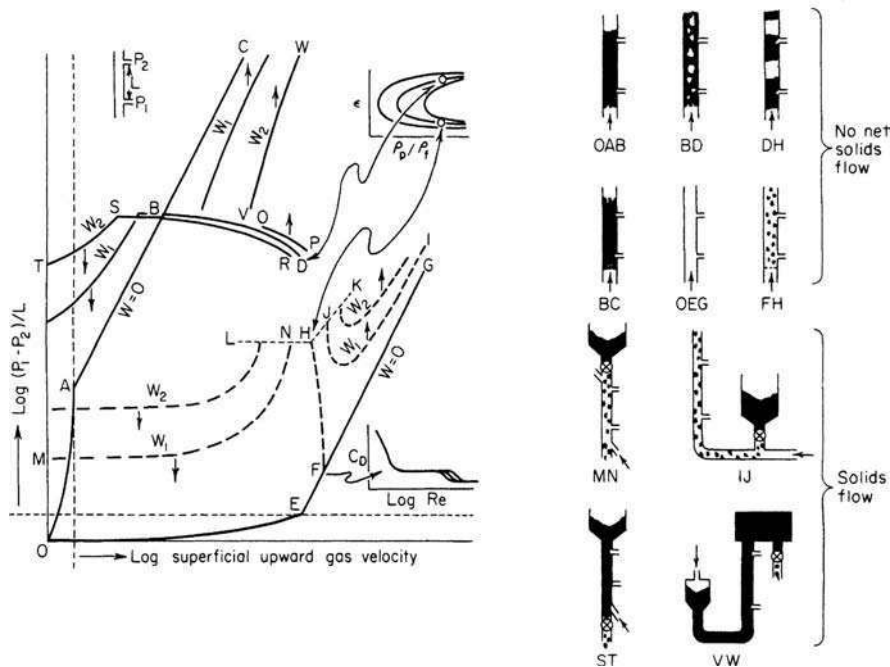


FIG. 17-2 Schematic phase diagram in the region of upward gas flow. W = mass flow solids, $\text{lb}/(\text{h} \cdot \text{ft}^2)$; ϵ = fraction voids; ρ_p = particle density, lb/ft^3 ; ρ_f = fluid density, lb/ft^3 ; C_D = drag coefficient; Re = modified Reynolds number. (Zenz and Othmer, *Fluidization and Fluid Particle Systems*, Reinhold, New York, 1960.)

Key:

OAB = packed bed
BD = fluidized bed
DH = slugging bed

IJ = cocurrent flow
(dilute phase)
ST = countercurrent flow
(dense phase)

AC = packed bed
(restrained at top)
OEG = fluid only
(no solids)

FH = dilute phase
MN = countercurrent flow
(dilute phase)
VW = cocurrent flow
(dense phase)

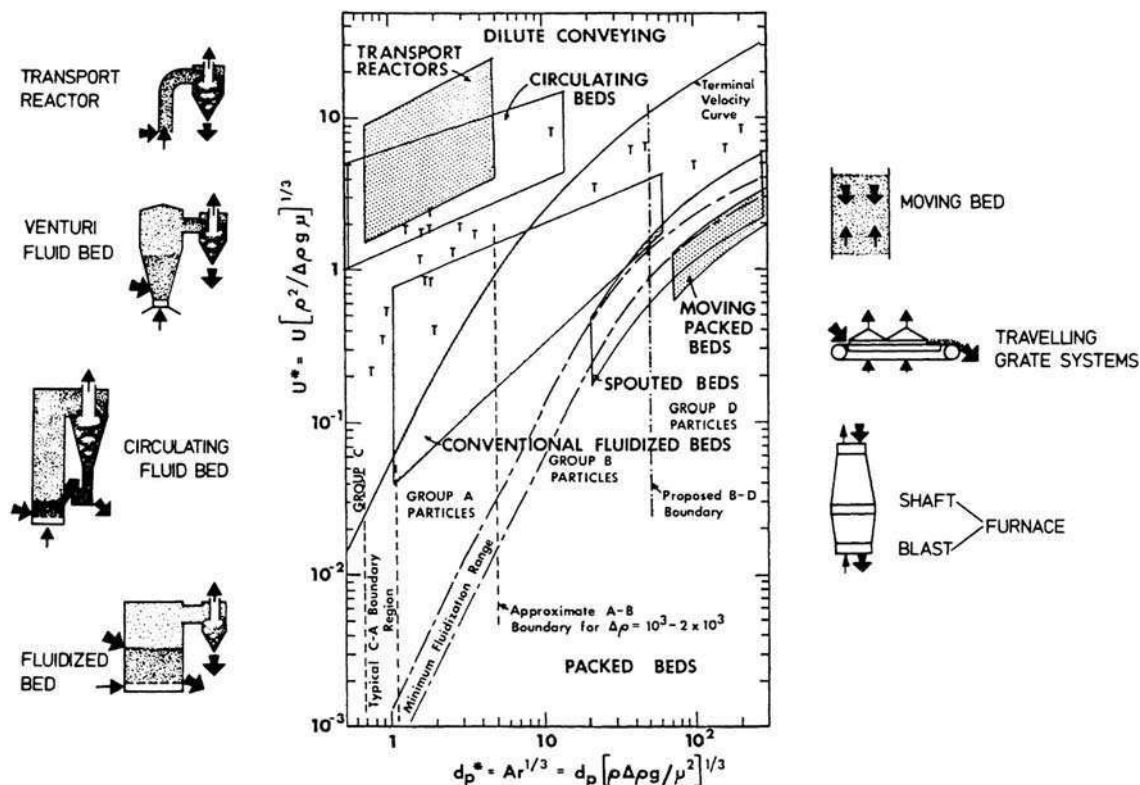


FIG. 17-3 Simplified fluid-bed status graph. [From Grace, Can. J. Chem. Eng., **64**, 353-363 (1986); sketches from Reh, Ger. Chem. Eng., **1**, 319-329 (1978).]

of the fluidization column, the bed will slug. The slugging fluidized bed is characterized by severe pressure fluctuations and limited solids mixing. It only occurs with small-diameter fluidization columns. Commercial fluidized beds are too large for bubbles to grow to the size where slugging will occur.

At high gas velocities in the bed, the stable bubbles break down into unstable voids that continuously disintegrate and reform. This type of bed is said to be operating in the turbulent fluidized-bed regime, and is characterized by higher heat- and mass-transfer rates than in the bubbling bed. As the gas velocity is increased further, the

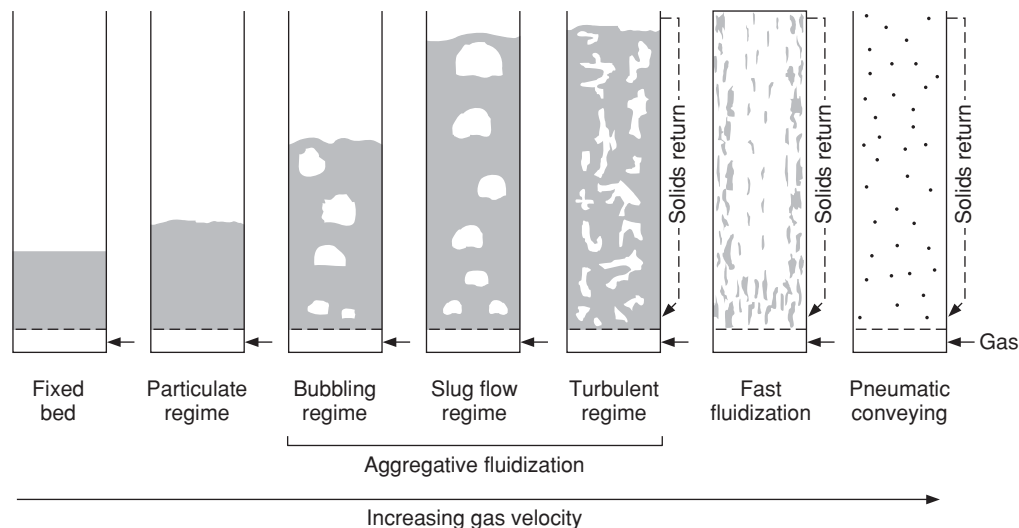


FIG. 17-4 Fluidization regimes. [Adapted from Grace, Can. J. Chem. Eng., **64**, 353-363 (1986).]

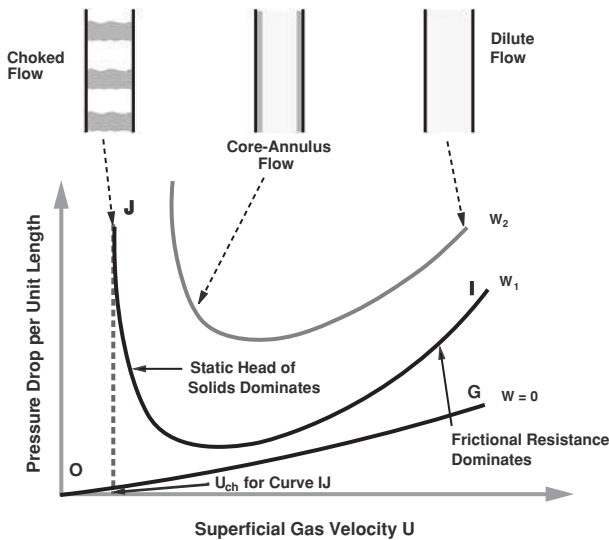


FIG. 17-5 Total transport regime. (Courtesy of PSRI, Chicago, Ill.)

bed transitions from the turbulent bed into the dilute-phase transport regime. This pneumatic conveying regime is composed of two basic regions: the lower-velocity fast fluidized-bed regime and the higher-velocity transport regime (often called the pneumatic conveying regime). The total transport regime is a very important regime, and is defined by the line *IJ* for the constant solids flow rate W_1 in Fig. 17-2. A more detailed drawing of this regime is shown in Fig. 17-5. In this figure, it can be seen that as the gas velocity is decreased from point *J*, the pressure drop per unit length begins to decrease. This occurs because the total pressure drop in the transport regime is composed of two types of terms—a term composed of frictional pressure drops (gas/wall friction, solid/wall friction, and gas/solids friction) and a term required to support the solids in the vertical line (the static head of solids term). At high gas velocities the frictional terms dominate; and as the gas velocity is decreased from point *J*, the frictional terms begin to decrease in magnitude. As this occurs, the concentration of solids in the line starts to increase. At some gas velocity, the static head of solids term and the frictional pressure drop term are equal (the minimum point on the curve). As the gas velocity is decreased below the minimum point, the static head of solids term begins to dominate as the concentration of solids in the line increases. This pressure drop increases until it is no longer possible for the gas to fully support the solids in the line. The gas velocity at which the solids cannot be supported at solids flow rate W_1 is known as the choking velocity for solids flow rate W_1 . Because beds in the turbulent and the transport regimes operate above the terminal velocity of some of or all the particles, a solids collection and return system is necessary to maintain a stable fluidized bed with these regimes.

Solids Concentration versus Height From the foregoing it is apparent that there are several regimes of fluidization. These are, in order of increasing gas velocity, particulate fluidization (Geldart group A), bubbling (aggregative), turbulent, fast, and transport. Each of these regimes has a characteristic solids concentration profile as shown in Fig. 17-6.

Equipment Types Fluidized-bed systems take many forms. Figure 17-7 shows some of the more prevalent concepts with approximate ranges of gas velocities.

Minimum Fluidizing Velocity U_{mf} , the minimum fluidizing velocity, is frequently used in fluid-bed calculations and in quantifying one of the particle properties. This parameter is best measured in small-scale equipment at ambient conditions. The correlation by Wen and Yu [A.I.Ch.E.J., 610–612 (1966)] given below can then be used to back calculate d_p . This gives a particle size that takes into account

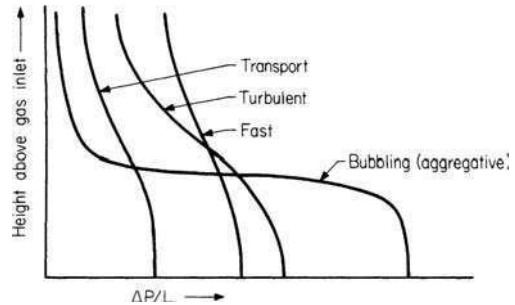


FIG. 17-6 Solids concentration versus height above distributor for regimes of fluidization.

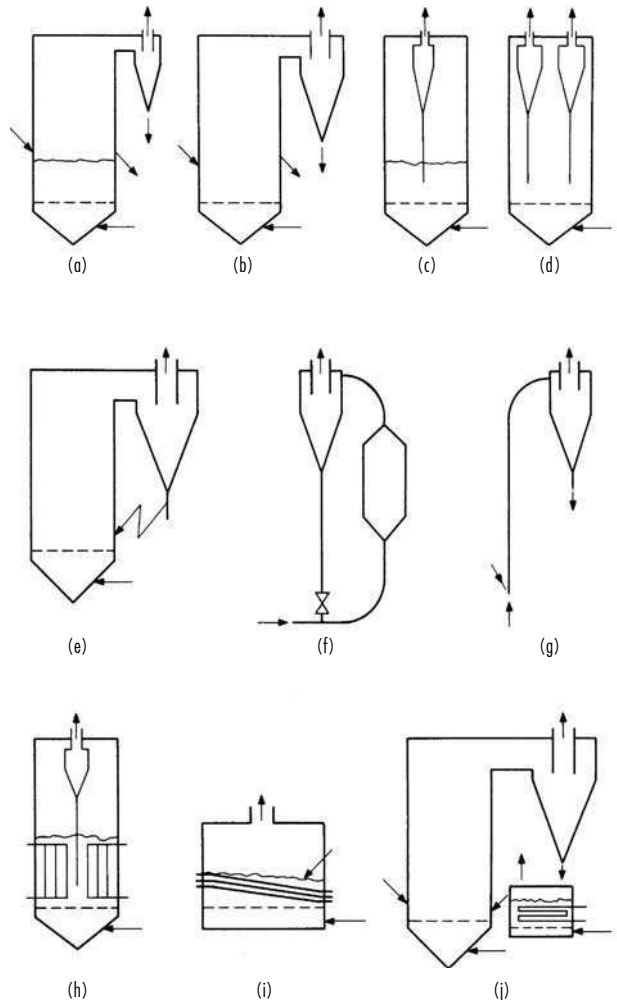


FIG. 17-7 Fluidized-bed systems. (a) Bubbling bed, external cyclone, $U < 20 \times U_{mf}$. (b) Turbulent bed, external cyclone, $20 \times U_{mf} < U < 200 \times U_{mf}$. (c) Bubbling bed, internal cyclones, $U < 20 \times U_{mf}$. (d) Turbulent bed, internal cyclones, $20 \times U_{mf} < U < 200 \times U_{mf}$. (e) Circulating (fast) bed, external cyclones, $U > 200 \times U_{mf}$. (f) Circulating bed, $U > 200 \times U_{mf}$. (g) Transport, $U > U_T$. (h) Bubbling or turbulent bed with internal heat transfer, $2 \times U_{mf} < U < 200 \times U_{mf}$. (i) Bubbling or turbulent bed with internal heat transfer, $2 \times U_{mf} < U < 100 \times U_{mf}$. (j) Circulating bed with external heat transfer, $U > 200 \times U_{mf}$.

17-6 GAS-SOLID OPERATIONS AND EQUIPMENT

effects of size distribution and particle shape, or sphericity. The correlation can then be used to estimate U_{mf} at process conditions. If U_{mf} cannot be determined experimentally, use the expression below directly.

$$Re_{mf} = (1135.7 + 0.0408Ar)^{0.5} - 33.7$$

where $Re_{mf} = d_{sv} \rho_f U_{mf} / \mu$
 $Ar = d_{sv} \rho_f (\rho_s - \rho_f) g / \mu^2$
 $d_{sv} = 1 / \sum (x_i / d_{pi})$

The flow required to maintain a complete homogeneous bed of solids in which coarse or heavy particles will not segregate from the fluidized portion is very different from the minimum fluidizing velocity. See Nienow and Chiba, *Fluidization*, 2d ed., Wiley, 1985, pp. 357–382, for a discussion of segregation or mixing mechanism as well as the means of predicting this flow; also see Baeyens and Geldart, *Gas Fluidization Technology*, Wiley, 1986, 97–122.

Particulate Fluidization Fluid beds of Geldart group A powders that are operated at gas velocities above the minimum fluidizing velocity (U_{mf}) but below the minimum bubbling velocity (U_{mb}) are said to be particulate fluidized. As the gas velocity is increased above U_{mf} , the bed further expands. Decreasing $(\rho_s - \rho_f)$, d_p , and/or increasing μ_f increases the spread between U_{mf} and U_{mb} . Richardson and Zaki [Trans. Inst. Chem. Eng., **32**, 35 (1954)] showed that $U/U_i = \epsilon^n$, where n is a function of system properties, ϵ = void fraction, U = superficial fluid velocity, and U_i = theoretical superficial velocity from the Richardson and Zaki plot when $\epsilon = 1$.

Vibrofluidization It is possible to fluidize a bed mechanically by imposing vibration to throw the particles upward cyclically. This enables the bed to operate with either no gas upward velocity or reduced gas flow. Entrainment can also be greatly reduced compared to unaided fluidization. The technique is used commercially in drying and other applications [Mujumdar and Erdesz, *Drying Tech.*, **6**, 255–274 (1988)], and chemical reaction applications are possible. See Sec. 12 for more on drying applications of vibrofluidization.

DESIGN OF FLUIDIZED-BED SYSTEMS

The use of the fluidization technique requires in almost all cases the employment of a fluidized-bed system rather than an isolated piece of equipment. Figure 17-8 illustrates the arrangement of components of a system.

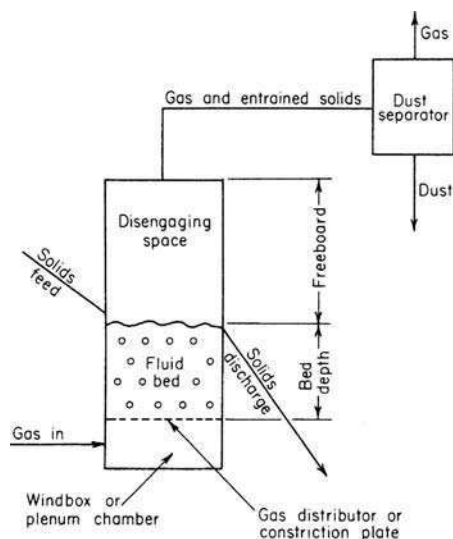


FIG. 17-8 Noncatalytic fluidized-bed system.

The major parts of a fluidized-bed system can be listed as follows:

1. Fluidization vessel
 - a. Fluidized-bed portion
 - b. Disengaging space or freeboard
 - c. Gas distributor
2. Solids feeder or flow control
3. Solids discharge
4. Dust separator for the exit gases
5. Instrumentation
6. Gas supply

Fluidization Vessel The most common shape is a vertical cylinder. Just as for a vessel designed for boiling a liquid, space must be provided for vertical expansion of the solids and for disengaging splashed and entrained material. The volume above the bed is called the disengaging space. The cross-sectional area is determined by the volumetric flow of gas and the allowable or required fluidizing velocity of the gas at operating conditions. In some cases the lowest permissible velocity of gas is used, and in others the greatest permissible velocity is used. The maximum flow is generally determined by the carry-over or entrainment of solids, and this is related to the dimensions of the disengaging space (cross-sectional area and height).

Bed Bed height is determined by a number of factors, either individually or collectively, such as:

1. Gas-contact time
2. L/D ratio required to provide staging
3. Space required for internal heat exchangers
4. Solids-retention time

Generally, bed heights are not less than 0.3 m (12 in) or more than 16 m (50 ft).

Although the reactor is usually a vertical cylinder, generally there is no real limitation on shape. The specific design features vary with operating conditions, available space, and use. The lack of moving parts lends toward simple, clean design.

Many fluidized-bed units operate at elevated temperatures. For this use, refractory-lined steel is the most economical design. The refractory serves two main purposes: (1) it insulates the metal shell from the elevated temperatures, and (2) it protects the metal shell from abrasion by the bed and particularly the splashing solids at the top of the bed resulting from bursting bubbles. Depending on specific conditions, several different refractory linings are used [Van Dyck, *Chem. Eng. Prog.*, 46–51 (December 1979)]. Generally, for the moderate temperatures encountered in catalytic cracking of petroleum, a reinforced-gunite lining has been found to be satisfactory. This also permits the construction of larger units than would be permissible if self-supporting ceramic domes were to be used for the roof of the reactor.

When heavier refractories are required because of operating conditions, insulating brick is installed next to the shell and firebrick is installed to protect the insulating brick. Industrial experience in many fields of application has demonstrated that such a lining will successfully withstand the abrasive conditions in the bed for many years without replacement. Most serious refractory wear occurs with coarse particles at high gas velocities and is usually most pronounced near the operating level of the fluidized bed.

Gas leakage behind the refractory has plagued a number of units. Care should be taken in the design and installation of the refractory to reduce the possibility of the formation of "chimneys" in the refractories. A small flow of solids and gas can quickly erode large passages in soft insulating brick or even in dense refractory. Gas stops are frequently attached to the shell and project into the refractory lining. Care in design and installation of openings in shell and lining is also required.

In many cases, cold spots on the reactor shell will result in condensation and high corrosion rates. Sufficient insulation to maintain the shell and appurtenances above the dew point of the reaction gases is necessary. Hot spots can occur where refractory cracks allow heat to permeate to the shell. These can sometimes be repaired by pumping castable refractory into the hot area from the outside.

The violent motion of a fluidized bed requires an ample foundation and a sturdy supporting structure for the reactor. Even a relatively small differential movement of the reactor shell with the lining will materially shorten refractory life. The lining and shell must be designed as a unit. Structural steel should not be supported from a vessel that is subject to severe vibration.

Freeboard and Entrainment The freeboard or disengaging height is the distance between the top of the fluid bed and the gas-exit nozzle in bubbling- or turbulent-bed units. The distinction between bed and freeboard is difficult to determine in fast and transport units (see Fig. 17-6).

At least two actions can take place in the freeboard: classification of solids and reaction of solids and gases.

As a bubble reaches the upper surface of a fluidized bed, the bubble breaks through the thin upper envelope composed of solid particles entraining some of these particles. The crater-shaped void formed is rapidly filled by flowing solids. When these solids meet at the center of the void, solids are geysered upward. The downward pull of gravity and the upward pull of the drag force of the upward-flowing gas act on the particles. The larger and denser particles return to the top of the bed, and the finer and lighter particles are carried upward. The distance above the bed at which the entrainment becomes constant is the transport disengaging height, TDH. Cyclones and vessel gas outlets are usually located above TDH. Figure 17-9 graphically estimates TDH as a function of velocity and bed size.

The higher the concentration of an entrainable component in the bed, the greater its rate of entrainment. Finer particles have a greater rate of entrainment than coarse ones. These principles are embodied in the method of Geldart (*Gas Fluidization Tech.*, Wiley, 1986, pp. 123–153) via the equation, $E(i) = K^*(i)x(i)$, where $E(i)$ = entrainment rate for size i , $\text{kg/m}^2 \text{ s}$; $K^*(i)$ = entrainment rate constant for particle size i ; and $x(i)$ = weight fraction for particle size i . K^* is a function of operating conditions given by $K^*(i)/(P_j u) = 23.7 \exp [-5.4 U_r(i)/U]$. The composition and the total entrainment are calculated by summing over the entrainable fractions. An alternative is to use the method of Zenz as reproduced by Pell (*Gas Fluidization*, Elsevier, 1990, pp. 69–72).

In batch classification, the removal of fines (particles less than any arbitrary size) can be correlated by treating as a second-order reaction $K = (F/\theta)[1/x(x - F)]$, where K = rate constant, F = fines removed in time θ , and x = original concentration of fines.

Gas Distributor The gas distributor (also often called the grid of a fluidized bed) has a considerable effect on proper operation of the

fluidized bed. For good fluidized-bed operation, it is absolutely necessary to have a properly designed gas distributor. Gas distributors can be used both when the gas is clean and when the gas contains solids. The primary purpose of the gas distributor is to cause uniform gas distribution across the entire bed cross-section. It should operate for years without plugging or breaking, minimize sifting of solids back into the gas inlet to the distributor, and minimize the attrition of the bed material. When the gas is clean, the gas distributor is often designed to prevent backflow of solids during normal operation, and in many cases it is designed to prevent backflow during shutdown. To provide good gas distribution, it is necessary to have a sufficient pressure drop across the grid. This pressure drop should be at least one-third the pressure drop across the fluidized bed for gas upflow distributors, and one-tenth to one-fifth the pressure drop across the fluidized bed for downflow gas distributors. If the pressure drop across the bed is not sufficient, gas maldistribution can result, with the bed being fluidized in one area and not fluidized in another. In units with shallow beds such as dryers or where gas distribution is less crucial, lower gas distributor pressure drops can be used.

When both solids and gas pass through the distributor, such as in some catalytic cracking units, a number of different gas distributor designs have been used. Because the inlet gas contains solids, it is much more erosive than gas alone, and care has to be taken to minimize the erosion of the grid openings as the solids flow through them. Generally, this is done by decreasing the inlet gas/solids velocity so that erosion of the grid openings is low. Some examples of grids that have been used with both solids and gases in the inlet gas are concentric rings in the same plane, with the annuli open (Fig. 17-10a); concentric rings in the form of a cone (Fig. 17-10b); grids of T bars or other structural shapes (Fig. 17-10c); flat metal perforated plates supported or reinforced with structural members (Fig. 17-10d); dished and perforated plates concave both upward and downward (Fig. 17-10e and f). Figure 17-10d, e, and f also uses no solids in the gas to the distributor. The curved distributors of Fig. 17-10d and e are often used because they minimize thermal expansion effects.

There are three basic types of clean inlet gas distributors: (1) a perforated plate distributor, (2) a bubble cap type of distributor, and (3) a sparger or pipe-grid type of gas distributor. The perforated plate distributor (Fig. 17-10d) is the simplest type of gas distributor and consists of a flat or curved plate containing a series of vertical holes. The gas flows upward into the bed from a chamber below the bed called a plenum. This type of distributor is easy and economical to construct. However, when the gas is shut off, the solids can sift downward into

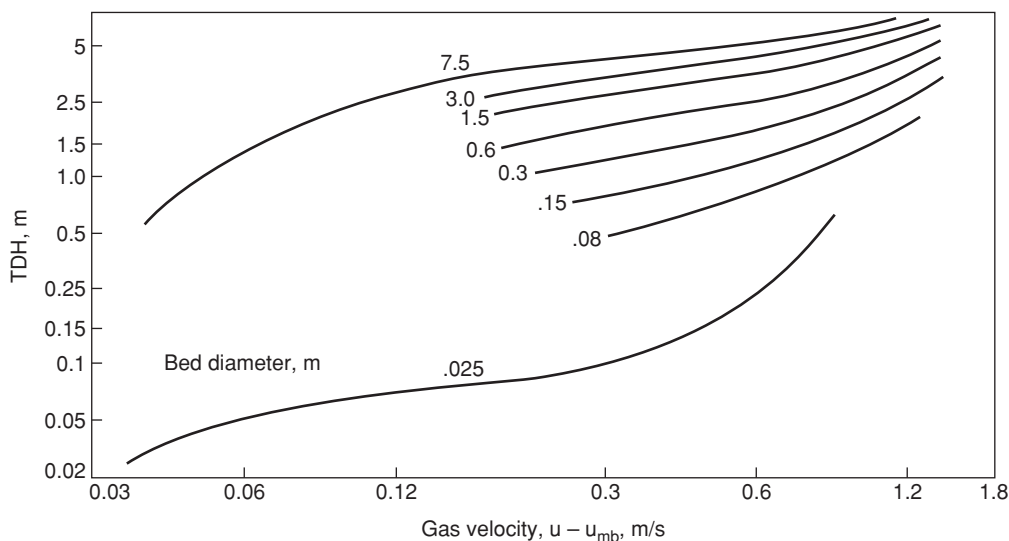


FIG. 17-9 Estimating transport disengaging height (TDH).

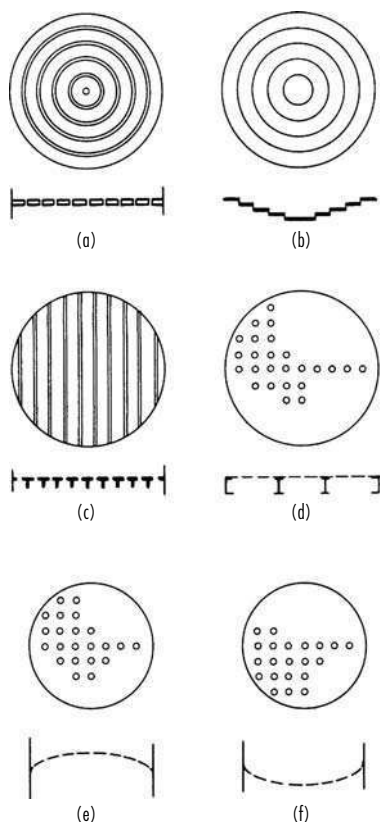


FIG. 17-10 Gas distributors for gases containing solids.

the plenum and may cause erosion of the holes when the bed is started up again. The bubble cap type of distributor is designed to prevent backflow of solids into the plenum chamber or inlet line of the gas distributor on start-up. The cap or tuyere type of distributor generally consists of a vertical pipe containing several small horizontal holes or holes angled downward from 30° to 45° from the horizontal (Fig. 17-11*a* and *b*). It is difficult for the solids to flow back through such a configuration when the fluidizing gas is shut off.

The pipe distributor (often called a sparger) differs from the other two distributor types because it consists of pipes with distribution holes in them that are inserted into the bed. This type of distributor will have solids below it that are not fluidized. If this is not acceptable for a process, then this type of distributor cannot be used. However, the pipe distributor has certain advantages. It does not require a large plenum, the holes in the pipe can be positioned at any angle, and it can be used in cases when multiple gas injections are required in a process. A common type of pipe distributor is the multiple-pipe (manifold sparger) grid shown in Fig. 17-12.

To generate a sufficient pressure drop for good gas distribution, a high velocity through the grid openings may be required. It is best to limit this velocity to less than 60 m/s to minimize attrition of the bed material. The maximum hole velocity allowable may be even lower for very soft materials that attrite easily. The pressure drop and the gas velocity through the hole in the gas distributor are related by the equation

$$\Delta P = \frac{u^2 \rho_f}{2c^2 g_c} \quad \text{for fps units}$$

$$\Delta P = \frac{u^2 \rho_f}{2c^2} \quad \text{for SI units}$$

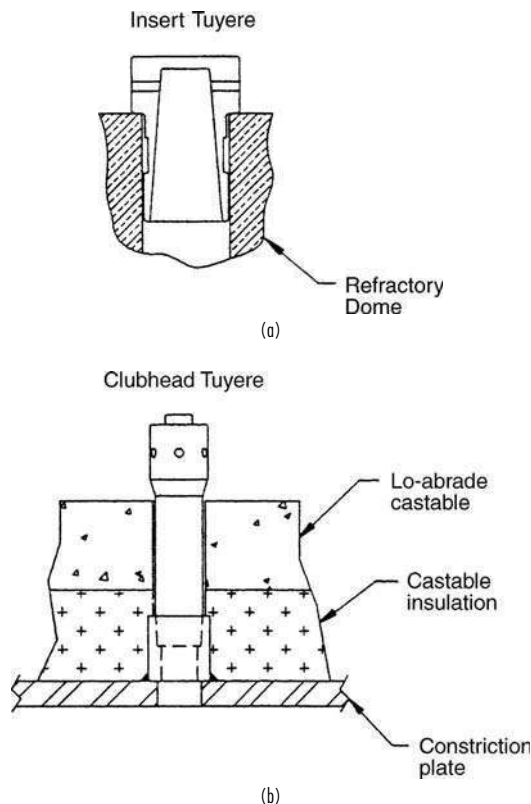


FIG. 17-11 Gas inlets designed to prevent backflow of solids. (a) Insert tuyere; (b) clubhead tuyere. (Dorr-Oliver, Inc.)

where u = velocity in hole at inlet conditions
 ρ_f = fluid density in hole at conditions in inlet to hole
 ΔP = pressure drop in consistent units, kPa or lb/ft²
 c = orifice constant, dimensionless (typically 0.8 for gas distributors)
 g_c = gravitational conversion constant, ft·lbm/(s²·lbf)

Due to the pressure drop requirements across the gas distributor for good gas distribution, the velocity through the grid hole may be higher than desired in order to minimize or limit particle attrition. Therefore, it is common industrial practice to place a length of pipe (called a

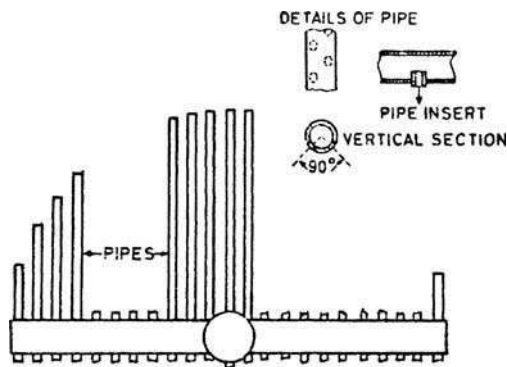


FIG. 17-12 Multiple-pipe gas distributor. [From Stermerding, de Groot, and Kuypers, Soc. Chem. Ind. J. Symp. Fluidization Proc., 35-46, London (1963).]

shroud) over the gas distributor hole such that the diameter of the pipe is larger than the diameter of the distributor hole. This technique effectively allows a smaller hole to give the required pressure drop, but the larger hole diameter of the shroud reduces the exit gas velocity into the bed so that particle attrition at the grid will be minimized. This technique is applied to both plate and pipe spargers.

Experience has shown that a concave-downward (Fig. 17-10f) gas distributor is a better arrangement than a concave-upward (Fig. 17-10e) gas distributor, as it tends to increase the flow of gases in the outer portion of the bed. This counteracts the normal tendency of the gas to flow into the center of the bed after it exits the gas distributor. In addition, the concave-downward type of gas distributor tends to assist the general solids flow pattern in the bed, which is up in the center and down near the walls. The concave-upward gas distributor tends to have a slow-moving region at the bottom near the wall. If solids are large (or if they are slightly cohesive), they can build up in this region.

Structurally, distributors must withstand the differential pressure across the restriction during normal and abnormal flow. In addition, during a shutdown, all or a portion of the bed will be supported by the distributor until sufficient backflow of the solids has occurred into the plenum to reduce the weight of solids above the distributor and to support some of this remaining weight by transmitting the force to the walls and bottom of the reactor. During start-up, a considerable upward thrust can be exerted against the distributor as the settled solids under the distributor are carried up into the normal reactor bed.

When the feed gas is devoid of or contains only small quantities of fine solids, more sophisticated designs of gas distributors can be used to effect economies in initial cost and maintenance. This is most pronounced when the inlet gas is cold and noncorrosive. When this is the case, the plenum chamber gas distributor and distributor supports can be fabricated of mild steel by using normal temperature design factors. The first commercial fluidized-bed ore roaster [Mathews, *Trans. Can. Inst. Min. Metall.* **L11:97** (1949)], supplied by the Dorr Co. (now Dorr-Oliver Inc.) in 1947 to Cochenour-Willans, Red Lake, Ontario, was designed with a mild-steel constriction plate covered with castable refractory to insulate the plate from the calcine and to provide cones in which refractory balls were placed to act as ball checks. The balls eroded unevenly, and the castable cracked. However, when the unit was shut down by closing the air control valve, the runback of solids was negligible because of bridging. If, however, the unit were shut down by deenergizing the centrifugal blower motor, the higher pressure in the reactor would relieve through the blower and fluidizing gas plus solids would run back through the constriction plate. Figure 17-11 illustrates two designs of gas inlets which have been successfully used to prevent flowback of solids. For best results, irrespective of the design, the gas flow should be stopped and the pressure relieved from the bottom upward through the bed. Some units have been built and successfully operated with simple slot-type distributors made of heat-resistant steel. This requires a heat-resistant plenum chamber but eliminates the frequently encountered problem of corrosion caused by condensation of acids and water vapor on the cold metal of the distributor. When the inlet gas is hot, such as in dryers or in the upper distributors of multibed units, ceramic arches or heat-resistant metal grates are generally used. Self-supporting ceramic domes have been in successful use for many years as gas distributors when temperatures range up to 1100°C. Some of these domes are fitted with alloy-steel orifices to regulate air distribution. However, the ceramic arch presents the same problem as the dished head positioned concave upward. Either the holes in the center must be smaller, so that the sum of the pressure drops through the distributor plus the bed is constant across the entire cross section, or the top of the arch must be flattened so that the bed depths in the center and outside are equal. This is especially important when shallow beds are used.

It is important to consider thermal effects in the design of the grid-to-shell seal. Bypassing of the grid at the seal point is a common problem caused by situations such as uneven expansion of metal and ceramic parts, a cold plenum and hot solids in contact with the grid plate at the same time, and start-up and shutdown scenarios. When the atmosphere in the bed is sufficiently benign, a sparger-type

distributor may be used (Fig. 17-12). In some cases, it is impractical to use a plenum chamber under the constriction plate. This condition arises when a flammable or explosive mixture of gases is being introduced to the reactor. One solution is to pipe the gases to a multitude of individual gas inlets in the floor of the reactor. In this way it may be possible to maintain the gas velocities in the pipes above the flame velocity or to reduce the volume of gas in each pipe to the point at which an explosion can be safely contained. Another solution is to provide separate inlets for the different gases and to rely on the rapid axial mixing of the fluidized bed. The inlets should be fairly close to one another, as lateral gas mixing in fluidized beds is poor.

Much attention has been paid to the effect of gas distribution on bubble growth in the bed and the effect of this on catalyst utilization, space-time yield, etc., in catalytic systems. It would appear that the best gas distributor would be a porous membrane because of its even distribution. However, this type of distributor is seldom practical for commercial units because of structural limitations and the fact that it requires absolutely clean gas. Practically, the limitations on hole spacing in a gas distributor are dependent on the particle size of the solids, materials of construction, and type of distributor. If easily worked metals are used, then punching, drilling, and welding are not expensive operations and permit the use of large numbers of holes. The use of tuyeres or bubble caps permits horizontal distribution of the gas so that a smaller number of gas inlet ports can still achieve good gas distribution. If a ceramic arch is used, generally only one hole per brick is permissible and brick dimensions must be reasonable.

Scale-up

Bubbling or Turbulent Beds Scale-up of noncatalytic fluidized beds when the reaction is fast, as in roasting or calcination, is straightforward and is usually carried out on an area basis. Small-scale tests are made to determine physical limitations such as sintering, agglomeration, solids-holdup time required, etc. Slower ($k < 1/s$) catalytic or more complex reactions in which several gas interchanges are required are usually scaled up in several steps, from laboratory to commercial size. The hydrodynamics of gas-solids flow and contacting is quite different in small-diameter high- L/D fluid beds as compared with large-diameter moderate- L/D beds. In small-diameter beds, bubbles tend to be small and cannot grow larger than the vessel diameter. In larger, deeper units, bubbles can grow very large. The large bubbles have less surface for mass transfer to the solids than the same volume of small bubbles. The large bubbles also rise through the bed more quickly.

The size of a bubble as a function of height was given by Darton et al. [*Trans. Inst. Chem. Eng.*, **55**, 274-280 (1977)] as

$$d_b = \frac{0.54(u - u_{mb})^{0.4}(h + 4\sqrt{A_t/N_o})^{0.8}}{g^{0.2}}$$

where d_b = bubble diameter, m
 h = height above the grid, m
 A_t/N_o = grid area per hole

Bubble growth in fluidized beds will be limited by the diameter of the containing vessel and bubble hydrodynamic stability. Bubbles in group B systems can grow to over 1 m in diameter if the gas velocity and the bed height are both high enough. Bubbles in group A materials with high percentage of fines (material less than 44 μ m in size) may reach a maximum stable bubble size in a range of about 5 to 15 cm. Furthermore, solids and gas backmixing are much less in high- L/D beds (whether they are slugging or bubbling) compared with low- L/D beds. Thus, the conversion or yield in large, unstaged reactors is sometimes considerably lower than in small high- L/D units. To overcome some of the problems of scale-up, staged units are used (Fig. 17-13). It is generally concluded that an unstaged 1-m (40-in-) diameter unit will achieve about the same conversion as a large industrial unit. The validity of this conclusion is dependent on many variables, including bed depth, particle size, size distribution, temperature, and system pressure. A brief history of fluidization, fluidized-bed scale-up, and modeling will illustrate the problems.

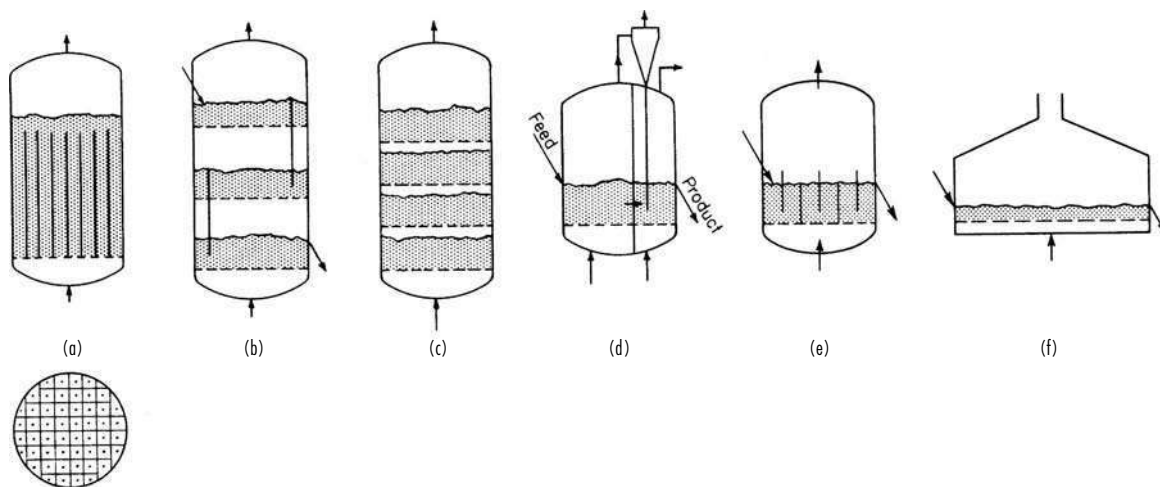


FIG. 17-13 Methods of providing staging in fluidized beds.

Fluidized beds were used in Europe in the 1920s to gasify coal. Scale-up problems either were insignificant or were not publicized. During World War II, catalytic cracking of oil to produce gasoline was successfully commercialized by scaling up from pilot-plant size (a few centimeters in diameter) to commercial size (several meters in diameter). It is fortunate that the kinetics of the cracking reactions are fast, that the ratio of crude oil to catalyst is determined by thermal balance and the required catalyst circulation rates, and that the crude feed point was in the plug-flow riser. The first experience of problems with scale-up was associated with the production of gasoline from natural gas by using the Fischer-Tropsch process. Some 0.10-m- (4-in-), 0.20-m- (8-in-), and 0.30-m- (12-in-) diameter pilot-plant results were scaled to a 7-m-diameter commercial unit, where the yield was only about 50 percent of that achieved in the pilot units. The Fischer-Tropsch synthesis is a relatively slow reaction; therefore, gas-solid contacting is very important. Since this unfortunate experience or perhaps because of it, much effort has been given to the scale-up of fluidized beds. Many models have been developed; these basically are of two types, the two-phase model [May, *Chem. Eng. Prog.*, **55**, 12, 5, 49–55 (1959); and Van Deemter, *Chem. Eng. Sci.*, **13**, 143–154 (1961)] and the bubble model (Kunii and Levenspiel, *Fluidization Engineering*, Wiley, New York, 1969). The two-phase model according to May and Van Deemter is shown in Fig. 17-14. In these models all or most of the gas passes through the bed in plug flow in the bubbles which do not contain solids (catalyst). The solids form a dense

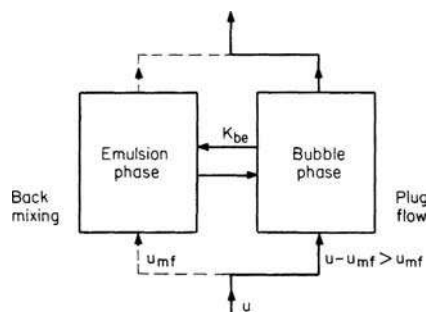


FIG. 17-14 Two-phase model according to May [*Chem. Eng. Prog.*, **55**, 12, 5, 49–55 (1959)] and Van Deemter [*Chem. Eng. Sci.*, **13**, 143–154 (1961)]. U = superficial velocity, U_{mf} = minimum fluidizing velocity, E = axial dispersion coefficient, and K_{bc} = mass-transfer coefficient.

suspension-emulsion phase in which gas and solids mix according to an axial dispersion coefficient (E). Cross flow between the two phases is predicted by a mass-transfer coefficient.

Conversion of a gaseous reactant can be given by $C/C_0 = \exp [-Na \times Nr/(Na + Nr)]$ where C = the exit concentration, C_0 = the inlet concentration, Na = diffusional driving force and Nr = reaction driving force. Conversion is determined by both reaction and diffusional terms. It is possible for reaction to dominate in a lab unit with small bubbles and for diffusion to dominate in a plant size unit. It is this change of limiting regime that makes scale-up so difficult. Refinements of the basic model and predictions of mass-transfer and axial-dispersion coefficients are the subject of many papers [Van Deemter, *Proc. Symp. Fluidization*, Eindhoven (1967); de Groot, *ibid.*; Van Swaaij and Zuidweg, *Proc. 5th Eur. Symp. React. Eng.*, Amsterdam, B9–25 (1972); DeVries, Van Swaaij, Mantovani, and Heijkoop, *ibid.*, B9–59 (1972); Werther, *Ger. Chem. Eng.*, **1**, 243–251 (1978); and Pell, *Gas Fluidization*, Elsevier, 75–81 (1990)].

The bubble model (Kunii and Levenspiel, *Fluidization Engineering*, Wiley, New York, 1969; Fig. 17-15) assumes constant-sized bubbles (effective bubble size d_b) rising through the suspension phase. Gas is transferred from the bubble void to the cloud and wake at mass-transfer coefficient K_{bc} and from the mantle and wake to the emulsion

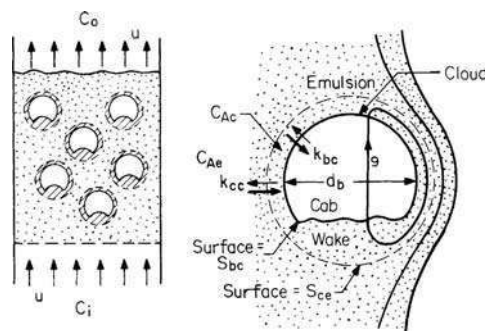


FIG. 17-15 Bubbling-bed model of Kunii and Levenspiel. d_b = effective bubble diameter, C_{Ab} = concentration of A in bubble, C_{Ac} = concentration of A in cloud, C_{Ae} = concentration of A in emulsion, q = volumetric gas flow into or out of bubble, k_{bc} = mass-transfer coefficient between bubble and cloud, and k_{ce} = mass-transfer coefficient between cloud and emulsion. (From Kunii and Levenspiel, *Fluidization Engineering*, Wiley, New York, 1969, and Krieger, Malabar, Fla., 1977.)

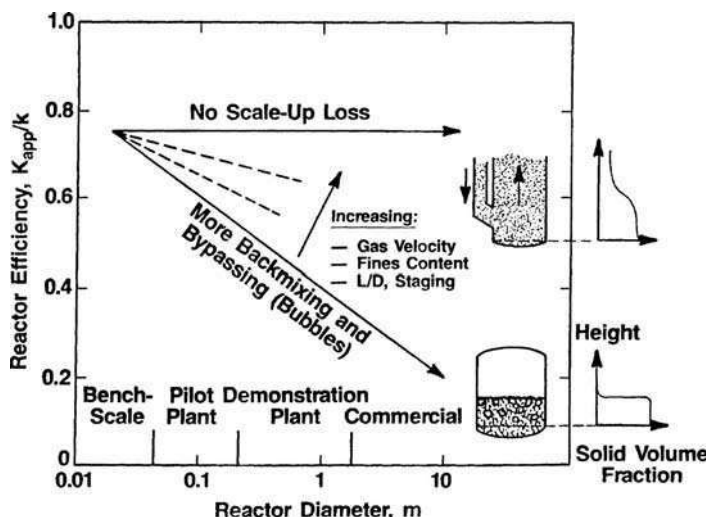


FIG. 17-16 Reducing scale-up loss. (From Krambeck, Avidan, Lee, and Lo, *A.I.Ch.E.J.*, 1727-1734, 1987.)

phase at mass-transfer coefficient K_a . Experimental results have been fitted to theory by means of adjusting the effective bubble size. As mentioned previously, bubble size changes from the bottom to the top of the bed, and thus this model is not realistic though of considerable use in evaluating reactor performance. Several bubble models using bubbles of increasing size from the distributor to the top of the bed and gas interchange between the bubbles and the emulsion phase according to Kunii and Levenspiel have been proposed [Kato and Wen, *Chem. Eng. Sci.*, **24**, 1351-1369 (1969); and Fryer and Potter, in Keairns (ed.), *Fluidization Technology*, vol. I, Hemisphere, Washington, 1975, pp. 171-178].

There are several methods available to reduce scale-up loss. These are summarized in Fig. 17-16. The efficiency of a fluid bed reactor usually decreases as the size of the reactor increases. This can be minimized by the use of high velocity, fine solids, staging methods, and a high L/D . High velocity maintains the reactor in the turbulent mode, where bubble breakup is frequent and backmixing is infrequent. A fine catalyst leads to smaller maximum bubble sizes by promoting instability of large bubbles. Maintaining high L/D minimizes backmixing, as does the use of baffles in the reactor. By these techniques, Mobil was able to scale up its methanol to gasoline technology with little difficulty [Krambeck, Avidan, Lee, and Lo, *A.I.Ch.E.J.*, 1727-1734 (1987)].

Another way to examine scale-up of hydrodynamics is to build a cold or hot scale model of the commercial design. Validated scaling criteria have been developed and are particularly effective for group B and D materials [Glicksman, Hyre, and Woloshun, *Powder Tech.*, 177-199 (1993)].

Circulating or Fast Fluidized Beds The circulating or fast fluidized bed is actually a misnomer in that it is not an extension of the turbulent bed, but is actually a part of the transport regime, as discussed above. However, the fast fluidized bed operates in that part of the transport regime that is dominated by the static head of solids pressure drop term (the part of the regime where the solids concentration is the highest). The solids may constitute up to 10 percent of the volume of the system in this regime. There are no bubbles, mass-transfer rates are high, and there is little gas backmixing in the system. The high velocity in the system results in a high gas throughput which minimizes reactor cost. Because there are no bubbles, scale-up is also less of a problem than with bubbling beds.

Many circulating systems are characterized by an external cyclone return system that usually has as large a footprint as the reactor itself. The axial solids density profile is relatively flat, as indicated in Fig. 17-6. There is a parabolic radial solids density profile that is termed core

annular flow. In the center of the reactor, the gas velocity and the solids velocity may be double the average. The solids in the center of the column (often termed a riser) are in dilute flow, traveling at their expected slip velocity $U_g - U_t$. Near the wall in the annulus, the solids are close to their fluidized-bed density. The solids at the wall can flow either upward or downward. Whether they do so is determined primarily by the velocity used in the system. In circulating fluidized-bed combustor systems, the gas velocity in the rectangular riser is generally in the range of 4 to 6 m/s, and the solids flow down at the wall. In fluid catalytic cracking, the velocity in the riser is typically in the range of 12 to 20 m/s, and the solids flow upward at the wall. Engineering methods for evaluating the hydrodynamics of the circulating bed are given by Kunii and Levenspiel (*Fluidization Engineering*, 2d ed., Butterworth, 1991, pp. 195-209), Werther (*Circulating Fluid Bed Technology IV*, 1994), and Avidan, Grace, and Knowlton (eds.), (*Circulating Fluidized Beds*, Blackie Academic, New York, 1997).

Pneumatic Conveying Pneumatic conveying systems can generally be scaled up on the principles of dilute-phase transport. Mass and heat transfer can be predicted on both the slip velocity during acceleration and the slip velocity at full acceleration. The slip velocity increases as the solids concentration is increased.

Heat Transfer Heat-exchange surfaces have been used to provide the means of removing or adding heat to fluidized beds. Usually, these surfaces are provided in the form of vertical or horizontal tubes manifolded at the tops and bottom or in a trombone shape manifolded exterior to the vessel. Horizontal tubes are extremely common as heat-transfer tubes. In any such installation, adequate provision must be made for abrasion of the exchanger surface by the bed. The prediction of the heat-transfer coefficient for fluidized beds is covered in Secs. 5 and 11.

Normally, the heat-transfer rate is between 5 and 25 times that for the gas alone. Bed-to-surface-heat transfer coefficients vary according to the type of solids in the bed. Group A solids have bed-to-surface heat-transfer coefficients of approximately 300 J/(m²·s·K) [150 Btu/(h·ft²·°F)]. Group B solids have bed-to-surface heat-transfer coefficients of approximately 100 J/(m²·s·K) [50 Btu/(h·ft²·°F)], while group D solids have bed-to-surface heat-transfer coefficients of 60 J/(m²·s·K) [30 Btu/(h·ft²·°F)].

The large area of the solids per cubic foot of bed, 5000 m²/m³ (15,000 ft²/ft³) for 60-μm particles of about 600 kg/m³ (40 lb/ft³) bulk density, results in the rapid approach of gas and solids temperatures near the bottom of the bed. Equalization of gas and solids temperatures generally occurs within 2 to 10 cm (1 to 4 in) of the top of the distributor.

Bed thermal conductivities in the vertical direction have been measured in the laboratory in the range of 40 to 60 kJ/(m²·s·K) [20,000 to 30,000 Btu/(h·ft²·°F·ft)]. Horizontal conductivities for 3-mm (0.12-in) particles in the range of 2 kJ/(m²·s·K) [1000 Btu/(h·ft²·°F·ft)] have been measured in large-scale experiments. Except for extreme *L/D* ratios, the temperature in the fluidized bed is uniform—with the temperature at any point in the bed generally being within 5 K (10°F) of any other point.

Temperature Control Because of the rapid equalization of temperatures in fluidized beds, temperature control can be accomplished in a number of ways.

1. *Adiabatic.* Control gas flow and/or solids feed rate so that the heat of reaction is removed as sensible heat in off gases and solids or heat supplied by gases or solids.

2. *Solids circulation.* Remove or add heat by circulating solids.

3. *Gas circulation.* Recycle gas through heat exchangers to cool or heat.

4. *Liquid injection.* Add volatile liquid so that the latent heat of vaporization equals excess energy.

5. *Cooling or heating surfaces in bed.*

Solids Mixing Solids are mixed in fluidized beds by means of solids entrained in the lower portion of bubbles, and the shedding of these solids from the wake of the bubble (Rowe and Patridge, "Particle Movement Caused by Bubbles in a Fluidized Bed," Third Congress of European Federation of Chemical Engineering, London, 1962). Thus, no mixing will occur at incipient fluidization, and mixing increases as the gas rate is increased. Naturally, particles brought to the top of the bed must displace particles toward the bottom of the bed. Generally, solids upflow is upward in the center of the bed and downward at the wall.

At high ratios of fluidizing velocity to minimum fluidizing velocity, tremendous solids circulation from top to bottom of the bed assures rapid mixing of the solids. For all practical purposes, beds with *L/D* ratios of from 4 to 0.1 can be considered to be completely mixed continuous-reaction vessels insofar as the solids are concerned.

Batch mixing using fluidization has been successfully employed in many industries. In this case there is practically no limitation to vessel dimensions.

All the foregoing pertains to solids of approximately the same physical characteristics. There is evidence that solids of widely different characteristics will classify one from the other at certain gas flow rates [Geldart, Baeyens, Pope, and van de Wijer, *Powder Technol.*, **30**(2), 195 (1981)]. Two fluidized beds, one on top of the other, may be formed, or a lower static bed with a fluidized bed above may result. The latter frequently occurs when agglomeration takes place because of either fusion in the bed or poor dispersion of sticky feed solids. Increased gas flows sometimes overcome the problem; however, improved feeding techniques or a change in operating conditions may be required. Another solution is to remove agglomerates either continuously or periodically from the bottom of the bed.

Gas Mixing The mixing of gases as they pass vertically up through the bed has never been considered a problem. However, horizontal mixing is very poor and requires effective distributors if two gases are to be mixed in the fluidized bed.

In bubbling beds operated at velocities of less than about 5 to 11 times U_{mf} the gases will flow upward in both the emulsion and the bubble phases. At velocities greater than about 5 to 11 times U_{mf} the downward velocity of the emulsion phase is sufficient to carry the contained gas downward. The back mixing of gases increases as U/U_{mf} is increased until the circulating or fast regime is reached where the back mixing decreases as the velocity is further increased.

Size Enlargement Under proper conditions, solid particles can be caused to increase in size in the bed. This can be advantageous or disadvantageous. Particle growth is usually associated with the melting or softening of some portion of the bed material (i.e., addition of soda ash to calcium carbonate feed in lime reburning, tars in fluidized-bed coking, or lead or zinc roasting causes agglomeration of dry particles in much the same way as binders act in rotary pelletizers). The motion of the particles, one against the other, in the bed results in spherical pellets. If the size of these particles is not controlled, rapid agglomeration and segregation of the large particles from the bed will

occur. Control of agglomeration can be achieved by crushing a portion of the bed product and recycling it to form nuclei for new growth. Often, liquids or slurries are fed via a spray nozzle into the bed to cause particles to grow. In drying solutions or slurries of solutions, the location of the feed injection nozzle (spray nozzle) has a great effect on the size of particle that is formed in the bed. Also of importance are the operating temperature, relative humidity of the off-gas, and gas velocity in the bed. Particle growth can occur as agglomeration (two or more particles sticking together) or by the particle growing in layers, often called *onion skinning*.

Size Reduction *Attrition* is the term describing particle reduction in the fluidized bed. Three major attrition mechanisms occur in the fluidized bed: particle fragmentation, particle fracture, and particle thermal decrepitation. Particle fragmentation occurs when the protruding edges on individual particles are broken off in the bed. These particle fragments are very small—usually on the order of 2 to 10 μm . Particle fracture occurs when particle interaction is severe enough to cause the particles to break up into large individual pieces.

Because of the random motion of the solids, some abrasion of the surface occurs in the bed. However, this abrasion is very small relative to the particle breakup caused by the high-velocity jets at the distributor. Typically, particle abrasion (fragmentation) will amount to about 0.25 to 1 percent of the solids per day. In the area of high gas velocities at the distributor, greater rates of attrition will occur because of fracture of the particles by impact. As mentioned above, particle fracture of the grid is reduced by adding shrouds to the gas distributor.

Generally, particle attrition is unwanted. However, at times controlled attrition is desirable. For example, in coking units where agglomeration due to wet particles is frequent, jets are used to attrit particles to control particle size [Dunlop, Griffin, and Moser, *J. Chem. Eng. Prog.*, **54**:39–43 (1958)].

Thermal decrepitation occurs frequently when crystals are rearranged because of transition from one form to another, or when new compounds are formed (i.e., calcination of limestone). Sometimes the stresses on particles in cases such as this are sufficient to reduce the particle to the basic crystal size. All these mechanisms will cause completion of fractures that were started before the introduction of the solids into the fluidized bed.

Standpipes, Solids Feeders, and Solids Flow Control In a fluid catalytic cracking (FCC) unit, hot catalyst is added to aspirated crude oil feed in a riser to crack the feed oil into gasoline and other light and heavy hydrocarbons. The catalyst activity is reduced by this contact as carbon is deposited on the catalyst. The catalyst is then passed through a steam stripper to remove the gas product in the interstices of the catalyst and is transported to a regenerator. The carbon on the catalyst is burned off in the fluidized-bed regenerator, and then the regenerated, hot catalyst is transported back to the bottom of the riser to crack the feed oil. Large FCC units have to control solids flow rates from 10 to 80 tons/min. The units require makeup catalyst to be added to replace solids losses due to attrition, etc. The amount of catalyst makeup is small, and need not be continuous. Therefore, the makeup catalyst is fed into the commercial unit from pressurized hoppers into one of the conveying lines. However, the primary solids flow control problem in this FCC unit is to maintain the correct temperature in the riser reactor by controlling the flow of hot regenerated catalyst around the test unit. This is done by using large, 1.2-m (4-ft)-diameter slide valves (also known as knife-gate valves) located in standpipes to control the flow rates of catalyst.

In the FCC process, the solids are transferred out of the fluidized-bed regenerator into the bottom of the riser via a standpipe. The purpose of a standpipe is to transfer solids from a low-pressure region to a high-pressure region. The point of removal of the solids from the regenerator bed is at a lower pressure than the point of feed introduction into the riser. Therefore, the transfer of solids from the regenerator bed to the bottom of the riser is accomplished with a standpipe. The standpipes in FCC units can be as large as 1.5 m (5 ft) in diameter and as long as about 30 m (100 ft). They can be either vertical or angled (generally approximately 60° from the horizontal). The pressure is higher at the bottom of a standpipe due to the relative flow of gas counter to the solids flow. The gas in the standpipe may be flowing either downward relative to the pipe wall but more slowly than the

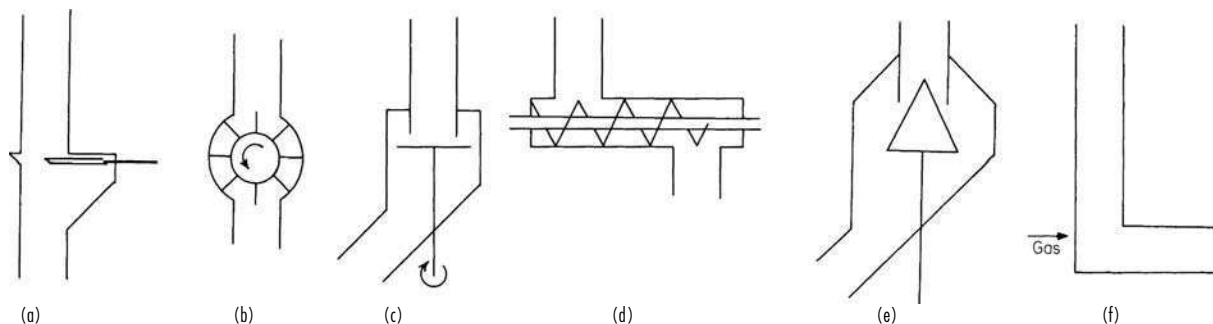


FIG. 17-17 Solids flow control devices. (a) Slide valve. (b) Rotary valve. (c) Table feeder. (d) Screw feeder. (e) Cone valve. (f) L Valve.

solids (the most common occurrence) or upward. The standpipe may be fluidized, or the solids may be in moving packed bed flow.

Fluidized standpipes can accommodate a much higher solids flow rate than moving packed bed standpipes because the friction of the solids flow on the wall of the standpipe is much less in fluidized standpipes. In longer standpipes, the pressure gain over the length of the standpipe is so great that it compresses the gas relative to the conditions at the standpipe inlet. This gas "shrinkage" can cause the gas in and around the particles to compress, which can result in defluidization of the solids in the standpipe unless aeration gas is added to the standpipe to replace the gas volume lost via compression. If the solids defluidize, the flow regime will revert to a moving packed bed with a lower pressure gain across the standpipe. In standpipes operating with group B solids, aeration is added to replace the compressed volume approximately every 1.5 m along the standpipe. In standpipes operating with group A solids, it has been found that aeration is only required at the bottom of the standpipe. Typically, the pressure drop across the solids control valve in the standpipe should be designed for a minimum of approximately 2 psi (14 kPa) for good control. A maximum of no more than 10 to 12 psi (70 to 84 kPa) is recommended to prevent excessive erosion of the valve at high pressure drops [Zenz, *Powder Technol.* pp. 105-113 (1986)].

Several designs of valves for solids flow control are used. These should be chosen with care to suit the specific conditions. Usually, block valves are used in conjunction with the control valves. Figure 17-17 shows schematically some of the devices used for solids flow control. Not shown in Fig. 17-17 is the flow-control arrangement used in the Exxon Research & Engineering Co. model IV catalytic-cracking units. This device consists of a U bend. A variable portion of regenerating air is injected into the riser leg. Changes in air-injection rate change the fluid density in the riser and thereby achieve control of the solids flow rate. Catalyst circulation rates of 1200 kg/s (70 tons/min) have been reported.

When the solid is one of the reactants, such as in ore roasting, the flow must be continuous and precise in order to maintain constant conditions in the reactor. Feeding of free-flowing granular solids into a fluidized bed is not difficult. Standard commercially available solids-weighing and -conveying equipment can be used to control the rate and deliver the solids to the feeder. Screw conveyors, dip pipes, seal legs, and injectors are used to introduce the solids into the reactor proper (Fig. 17-17). Difficulties arise and special techniques must be used when the solids are not free-flowing, such as is the case with most filter cakes. One solution to this problem was developed at Cochenour-Willans. After much difficulty in attempting to feed a wet and sometimes frozen filter cake into the reactor by means of a screw feeder, experimental feeding of a water slurry of flotation concentrates was attempted. This trial was successful, and this method has been used in almost all cases in which the heat balance, particle size of solids, and other considerations have permitted. Gilfillan et al. (*J. Chem. Metall. Min. Soc. S. Afr.*, May 1954) and Soloman and Beal (*Uranium in South Africa*, 1946-56) present complete details on the use of this system for feeding.

When slurry feeding is impractical, recycling of solids product to mix with the feed, both to dry and to achieve a better-handling

material, has been used successfully. Also, the use of a rotary table feeder mounted on top of the reactor, discharging through a mechanical disintegrator, has been successful. The wet solids generally must be broken up into discrete particles of very fine agglomerates either by mechanical action before entering the bed or by rapidly vaporizing water. If lumps of dry or semidry solids are fed, the agglomerates do not break up but tend to fuse together. As the size of the agglomerate is many times the size of the largest individual particle, these agglomerates will segregate out of the bed, and in time the whole of the fluidized bed may be replaced with a static bed of agglomerates.

Solids Discharge The type of discharge mechanism utilized is dependent upon the necessity of sealing the atmosphere inside the fluidized-bed reactor and the subsequent treatment of the solids. The simplest solids discharge is an overflow weir. This can be used only when the escape of fluidizing gas does not present any hazards due to nature or dust content or when the leakage of gas into the fluidized-bed chamber from the atmosphere into which the bed is discharged is permitted. Solids will overflow from a fluidized bed through a port even though the pressure above the bed is maintained at a slightly lower pressure than the exterior pressure. When it is necessary to restrict the flow of gas through the opening, a simple flap-per valve is frequently used. Overflow to combination seal and quench tanks (Fig. 17-18) is used when it is permissible to wet the solids and when disposal or subsequent treatment of the solids in slurry form is desirable. The FluoSeal is a simple and effective way of sealing and purging gas from the solids when an overflow-type discharge is used (Fig. 17-19).

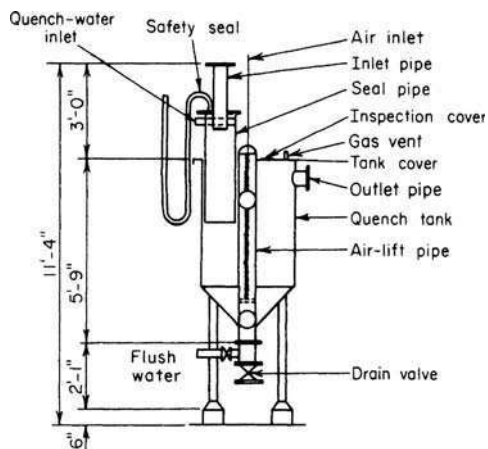


FIG. 17-18 Quench tank for overflow or cyclone solids discharge. [Gilfillan et al., "The FluoSolids Reactor as a Source of Sulphur Dioxide," *J. Chem. Metall. Min. Soc. S. Afr.* (May 1954).]

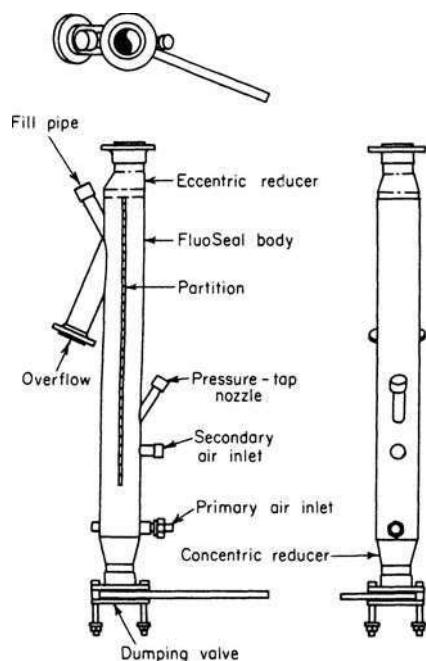


FIG. 17-19 Dorco FluoSeal, type UA. (Dorr-Oliver Inc.)

Either trickle (flapper) or star (rotary) valves are effective sealing devices for solids discharge. Each functions with a head of solids above it. Bottom of the bed discharge is also acceptable via a slide valve with a head of solids.

Seal legs are frequently used in conjunction with solids-flow-control valves to equalize pressures and to strip trapped or adsorbed gases from the solids. The operation of a seal leg is shown schematically in Fig. 17-20. The solids settle by gravity from the fluidized bed into the seal leg or standpipe. Seal and/or stripping gas is introduced near the bottom of the leg. This gas flows both upward and downward. Pressures indicated in the illustration have no absolute value but are only relative. The legs are designed for either fluidized or settled solids.

The L valve is shown schematically in Fig. 17-21. It can act as a seal and as a solids-flow control valve. However, control of solids rate is only practical for solids that deaerate quickly (Geldart *B* and *D* solids). The height at which aeration is added in Fig. 17-21 is usually one exit pipe diameter above the centerline of the exit pipe. For L-valve design equations, see Yang and Knowlton [*Powder Tech.*, **77**, 49–54 (1993)].

In the sealing mode, the leg is usually fluidized. Gas introduced below the normal solids level and above the discharge port will flow upward and downward. The relative flow in each direction is self-adjusting, depending upon the differential pressure between the point of solids feed and discharge and the level of solids in the leg. The length and diameter of the discharge spout are selected so that the undisturbed angle of repose of the solids will prevent discharge of the solids. As solids are fed into the leg, height H of solids increases. This in turn reduces the flow of gas in an upward direction and increases the flow of gas in a downward direction. When the flow of gas downward and through the solids-discharge port reaches a given rate, the angle of repose of the solids is upset and solids discharge commences. Usually, the level of solids above the point of gas introduction will float. When used as a flow controller, the vertical leg is best run in the packed bed mode. The solids flow rate is controlled by varying the aeration gas flow.

In most catalytic-reactor systems, no solids removal is necessary as the catalyst is retained in the system and solids loss is in the form of fines that are not collected by the dust-recovery system.

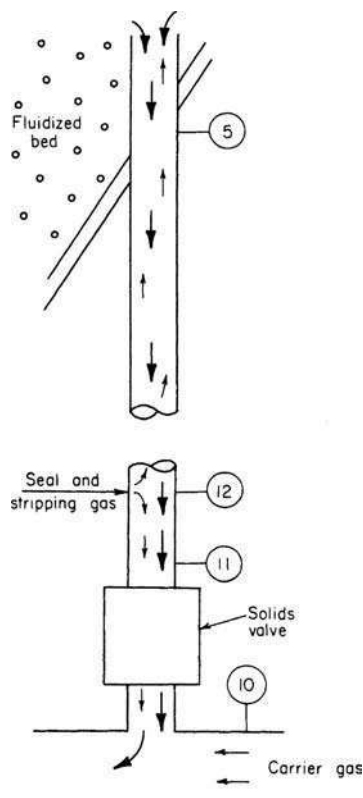


FIG. 17-20 Fluidized-bed seal leg.

Dust Separation It is usually necessary to recover the solids carried by the gas leaving the disengaging space or freeboard of the fluidized bed. Generally, cyclones are used to remove the major portion of these solids (see "Gas-Solids Separation"). However, in a few cases, usually on small-scale units, filters are employed without the use of cyclones to reduce the loading of solids in the gas. For high-temperature usage, either porous ceramic or sintered metal filters have been employed. Multiple units must be provided so that one unit can be blown back with clean gas while one or more are filtering.

Cyclones are arranged generally in any one of the arrangements shown in Fig. 17-22. The effect of cyclone arrangement on the height of the vessel and the overall height of the system is apparent. Details regarding cyclone design and collection efficiencies are to be found in another part of this section.

Discharging of the cyclone into the fluidized bed requires some care. It is necessary to seal the bottom of the cyclone so that the collection efficiency of the cyclone will not be impaired by the passage of

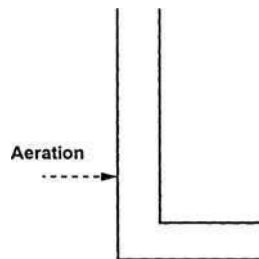


FIG. 17-21 L valve.

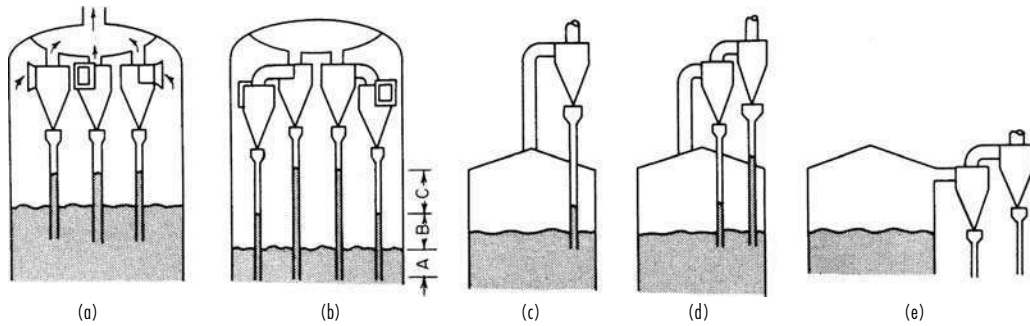


FIG. 17-22 Fluidized-bed cyclone arrangements. (a) Single-stage internal cyclone. (b) Two-stage internal cyclone. (c) Single-stage external cyclone; dust returned to bed. (d) Two-stage external cyclone; dust returned to bed. (e) Two-stage external cyclone; dust collected externally.

appreciable quantities of gas up through the bottom of the dipleg. This is usually done by (1) sealing the dipleg in the fluid bed, or (2) adding a trickle or flapper valve to the bottom of the dipleg if the dipleg is terminated in the freeboard of the fluidized bed. Experience has shown, particularly in the case of deep beds, that the bottom of the dipleg pipe must be protected from the action of large gas bubbles which, if allowed to pass up the leg, would carry quantities of fine solids up into the cyclone and cause momentarily high losses. This is accomplished by attaching a horizontal plate larger in diameter than the pipe to the bottom of the dipleg (see Fig. 17-23e). Care must be taken to ensure that the horizontal plate is located far enough away from the dipleg outlet that the solids discharge from the dipleg is not affected.

Example 1: Length of Seal Leg The length of the seal leg can be estimated as shown.

Given: Fluid density of bed at 0.3-m/s (1-ft/s) superficial gas velocity = 1100 kg/m³ (70 lb/ft³).

Fluid density of cyclone product at 0.15 m/s (0.5 ft/s) = 650 kg/m³ (40 lb/ft³).

Settled bed depth = 1.8 m (6 ft)

Fluidized-bed depth = 2.4 m (8 ft)

Pressure drop through cyclone = 1.4 kPa (0.2 lbf/in²)

In order to assure seal at start-up, the bottom of the seal leg is 1.5 m (5 ft) above the constriction plate or submerged 0.9 m (3 ft) in the fluidized bed.

The pressure at the solids outlet of a gas cyclone is usually about 0.7 kPa (0.1 lbf/in²) lower than the pressure at the discharge of the leg. Total pressure to be balanced by the fluid leg in the cyclone dipleg is

$$(0.9 \times 1100 \times 9.81)/1000 + 1.4 + 0.7 = 11.8 \text{ kPa}$$

$$[(3 \times 70)/144 + 0.2 + 0.1 = 1.7 \text{ lb/in}^2]$$

Height of solids in dipleg = $(11.8 \times 1000)/(650 \times 9.81) = 1.9 \text{ m}$ $[(1.7 \times 144)/40 = 6.1 \text{ ft}]$; therefore, the bottom of the separator pot on the cyclone must be at least 1.9 + 1.5 or 3.4 m (6.1 + 5 or 11.1 ft) above the gas distributor. To allow for upsets, changes in size distribution, etc., use 4.6 m (15 ft).

In addition to the open dipleg, various other devices have been used to seal cyclone solids returns, especially for second-stage cyclones. A number of these are shown in Fig. 17-23. One of the most frequently used is the trickle valve (Fig. 17-23a). There is no general agreement as to whether this valve should discharge below the bed level or in the freeboard. In any event, the legs must be large enough to carry momentarily high rates of solids and must provide seals to overcome cyclone pressure drops as well as to allow for differences in fluid density of bed and cyclone products. It has been reported that, in the case of catalytic-cracking catalysts, the fluid density of the solids collected by the primary cyclone is essentially the same as that in the fluidized bed because the particles in the bed are so small, nearly all are entrained. However, as a general rule the fluidized density of solids collected by the second-stage cyclone is less than the fluidized density of the bed. Each succeeding cyclone collects finer and less dense solids.

As cyclones are less effective as the particle size decreases, secondary collection units are frequently required, i.e., filters, electrostatic precipitators, and scrubbers. When dry collection is not required, elimination of cyclones is possible if allowance is made for heavy solids loads in the scrubber (see "Gas-Solids Separations"; see also Sec. 14).

Instrumentation

Temperature Measurement This is usually simple, and standard temperature-sensing elements are adequate for continuous use. Because of the high abrasion wear on horizontal protection tubes, vertical installations are frequently used. In highly corrosive atmospheres in which metallic protection tubes cannot be used, short, heavy ceramic tubes have been used successfully.

Pressure Measurement Although successful pressure measurement probes or taps have been fabricated by using porous materials, the most universally accepted pressure tap consists of a purged tube

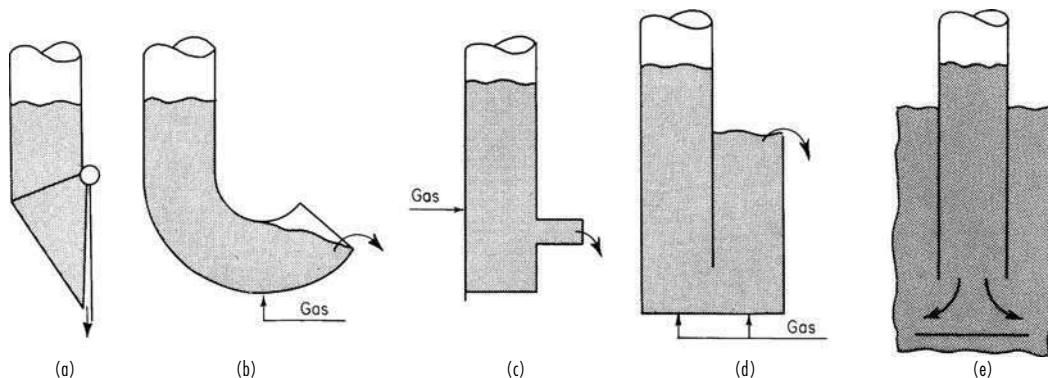


FIG. 17-23 Cyclone solids-return seals. (a) Trickle valve (Ducon Co., Inc.). (b) J valve. (c) L valve. (d) Fluid-seal pot. (e) "Dollar" plate. a, b, c, and d may be used above the bed; a and e are used below the bed.

projecting into the bed. Minimum internal diameters of the tube are 0.6 to 1.2 cm (0.25 to 0.5 in). A purge rate of at least 1.5 m/s (5 ft/s) is usually required to prevent solids from plugging the signal lines. Bed density is determined directly from $\Delta P/L$, the pressure drop inside the bed itself ($\Delta P/L$ in units of weight/area $\times 1/L$). The overall bed weight is obtained from ΔP taken between a point just above the gas distributor and a point in the freeboard. Nominal bed height is determined by dividing the ΔP across the entire bed and dividing it by the $\Delta P/L$ over a section of the bed length. Splashing of the solids by bubbles bursting at the bed surface will eject solids well above the nominal bed height in most cases. The pressure drop signal from fluidized beds fluctuates due to bubble effects and the generally statistical nature of fluid-bed flow parameters. A fast Fourier transform of the pressure drop signal transforms the perturbations to a frequency versus amplitude plot with a maximum at about 3 to 5 Hz and frequencies generally tailing off above 20 Hz. Changes in frequency and amplitude are associated with changes in the quality of the fluidization. Experienced operators of fluidized beds can frequently predict what is happening in the bed from changes in the ΔP signal.

Flow Measurements Measurement of flow rates of clean gases presents no problem. Flow measurement of gas streams containing solids is almost always avoided. The flow of solids is usually controlled but not measured except solids flows added to or taken from the system. Solids flows in the system are usually adjusted on an inferential basis (temperature, pressure level, catalyst activity, gas analysis, heat balance, etc.). In many roasting operations, the color of the calcine discharge material indicates whether the solids feed rate is too high or too low.

USES OF FLUIDIZED BEDS

There are many uses of fluidized beds. A number of applications have become commercial successes; others are in the pilot-plant stage, and others in bench-scale stage. Generally, the fluidized bed is used for gas-solids contacting; however, in some instances the presence of the gas or solid is used only to provide a fluidized bed to accomplish the end result. Uses or special characteristics follow:

- I. Chemical reactions
 - A. Catalytic
 1. Homogeneous
 2. Heterogeneous
 - B. Noncatalytic
 1. Homogeneous
 2. Heterogeneous
- II. Physical contacting
 - A. Heat transfer
 1. To and from fluidized bed
 2. Between gases and solids
 3. Temperature control
 4. Between points in bed
 - B. Solids mixing
 - C. Gas mixing
 - D. Drying
 1. Solids
 2. Gases
 - E. Size enlargement
 - F. Size reduction
 - G. Classification
 1. Removal of fines from solids
 2. Removal of fines from gas
 - H. Adsorption-desorption
 - I. Heat treatment
 - J. Coating

Chemical Reactions

Catalytic Reactions This use has provided the greatest impetus for use, development, and research in the field of fluidized solids. Some of the details pertaining to this use are to be found in the preceding pages of this section. Reference should also be made to Sec. 21.

Cracking The evolution of fluidized catalytic cracking since the early 1940s has resulted in several fluidized-bed process configurations.

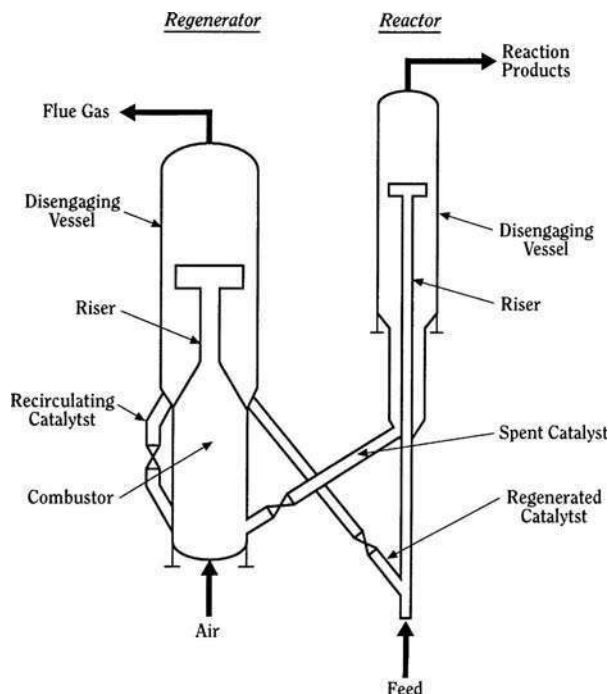


FIG. 17-24 UOP fluid cracking unit. (Reprinted with permission of UOP.)

The high rate of solids transfer between the fluidized-bed regenerator and the riser reactor in this process permits a balancing of the exothermic burning of carbon and tars in the regenerator and the endothermic cracking of petroleum in the reactor. Therefore, the temperature in both units can usually be controlled without resorting to auxiliary heat control mechanisms. The high rate of catalyst circulation also permits the maintenance of the catalyst at a constantly high activity. The original fluidized-bed regenerators were considered to be completely backmixed units. Newer systems have staged regenerators to improve conversion (see Fig. 17-24). The use of the riser reactor operating in the fast fluid-bed mode results in much lower gas and solids backmixing due to the more plug-flow nature of the riser.

The first fluid catalytic cracking unit (called Model I) was placed in operation in Baytown, Texas, in 1942. This was a low-pressure, 14- to 21-kPa (2- to 3-psig) unit operating in what is now called the turbulent fluidized-bed mode with a gas velocity of 1.2 to 1.8 m/s (4 to 6 ft/s). Before the start-up of the Model I cracker, it was realized that by lowering the gas velocity in the bed, a dense, bubbling or turbulent fluidized bed, with a bed density of 300 to 400 kg/m³ (20 to 25 lb/ft³), would be formed. The increased gas/solids contacting time in the denser bed allowed completion of the cracking reaction and catalyst regeneration. System pressure was eventually increased to 140 to 210 kPa (20 to 30 psig).

In the 1970s more-active zeolite catalysts were developed so that the cracking reaction could be conducted in the transport riser. Recently, heavier crude feedstocks have resulted in higher coke production in the cracker. The extra coke causes higher temperatures in the regenerator than are desired. This has resulted in the addition of catalyst cooling to the regeneration step, as shown in Fig. 17-25.

Many companies have participated in the development of the fluid catalytic cracker, including ExxonMobil Research & Engineering Co., UOP, Kellogg Brown and Root, ChevronTexaco, Gulf Research Development Co., and Shell Oil Company. Many of the companies provide designs and/or licenses to operate to others. For further details, see Luckenbach et al., "Cracking, Catalytic," in McKetta (ed.), *Encyclopedia of Chemical Processing and Design*, vol. 13, Marcel Dekker, New York, 1981, pp. 1-132.

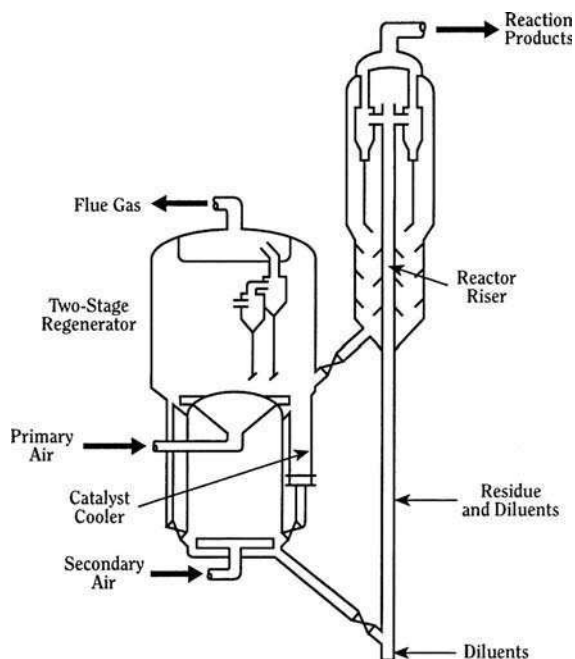


FIG. 17-25 Modern FCC unit configured for high-efficiency regeneration and extra catalyst cooling. (Reprinted with permission of UOP. RCC is a service mark of Ashland Oil Inc.)

Alkyl chloride Olefins are chlorinated to alkyl chlorides in a single fluidized bed. In this process, HCl reacts with O_2 over a copper chloride catalyst to form chlorine. The chlorine reacts with the olefin to form alkyl chloride. The process was developed by Shell Development Co. and uses a recycle of catalyst fines in aqueous HCl to control the temperature [Chem. Proc. 16:42 (1953)].

Phthalic anhydride Naphthalene is oxidized by air to phthalic anhydride in a bubbling fluidized reactor. Even though the naphthalene feed is in liquid form, the reaction is highly exothermic. Temperature control is achieved by removing heat through vertical tubes in the bed to raise steam [Graham and Way, Chem. Eng. Prog. 58:96 (January 1962)].

Acrylonitrile Acrylonitrile is produced by reacting propylene, ammonia, and oxygen (air) in a single fluidized bed of a complex catalyst. Known as the SOHIO process, this process was first operated commercially in 1960. In addition to acrylonitrile, significant quantities of HCN and acetonitrile are produced. This process is also exothermic, and temperature control is achieved by raising steam inside vertical tubes immersed in the bed [Veatch, Hydrocarbon Process. Pet. Refiner 41:18 (November 1962)].

Fischer-Tropsch synthesis The early scale-up of a bubbling bed reactor to produce gasoline from CO and H_2 was unsuccessful (see "Design of Fluidized-Bed Systems: Scale-up"). However, Kellogg Co. later developed a successful Fischer-Tropsch synthesis reactor based on a dilute-phase transport-reactor concept. Kellogg, in its design, prevented gas bypassing by using the transport reactor and maintained temperature control of the exothermic reaction by inserting heat exchangers in the transport line. This process has been very successful and repeatedly improved upon at the South African Synthetic Oil Limited (SASOL) plant in the Republic of South Africa, where politics and economics favor the conversion of coal to gasoline and other hydrocarbons. Refer to Jewell and Johnson, U.S. Patent 2,543,974, Mar. 6, 1951. Recently, the process has been modified to a simpler, less expensive turbulent bed catalytic reactor system (Silverman et al., Fluidization V, Engineering Foundation, 1986, pp. 441-448).

Polyethylene The first commercial fluidized-bed polyethylene plant was constructed by Union Carbide in 1968. Modern units operate at a temperature of approximately $100^\circ C$ and a pressure of

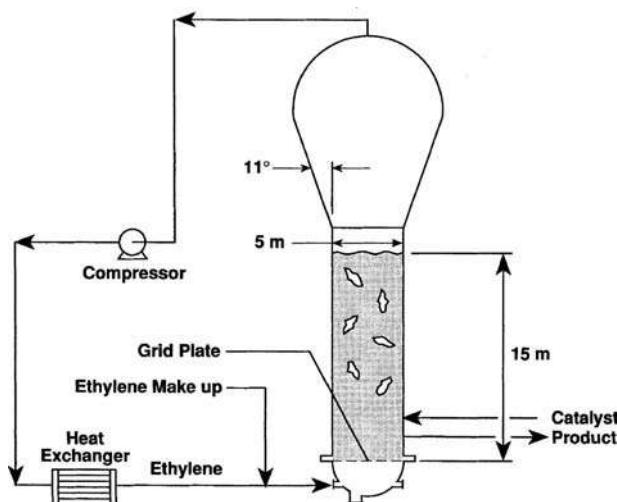


FIG. 17-26 High-pressure polyethylene reactor.

2100 kPa (300 psig). The bed is fluidized with ethylene at about 0.5 to 0.7 m/s (1.65 to 2.3 ft/s) and operates in the turbulent fluidization regime. Small catalyst is added to the bed, and the ethylene polymerizes on the catalyst to form polyethylene particles of approximately 600- to 1000- μm average size, depending on the type of polyethylene product being produced. The excellent mixing provided by the fluidized bed is necessary to prevent hot spots, since the unit is operated near the melting point of the product. A model of the reactor (Fig. 17-26) that couples kinetics to the hydrodynamics was given by Choi and Ray, Chem. Eng. Sci. 40: 2261 (1985).

Additional catalytic processes Nitrobenzene is hydrogenated to aniline (U.S. Patent 2,891,094). Melamine and isophthalonitrile are produced in catalytic fluidized-bed reactors. Badger developed a process to produce maleic anhydride by the partial oxidation of butane (Schaffel, Chen, and Graham, "Fluidized Bed Catalytic Oxidation of Butane to Maleic Anhydride," presented at Chemical Engineering World Congress, Montreal, Canada, 1981). Dupont developed a circulating bed process for production of maleic anhydride (Contractor, Circulating Fluidized Bed Tech. II, Pergamon, 1988, pp. 467-474). Mobil developed a commercial process to convert methanol to gasoline (Grimmer et al., Methane Conversion, Elsevier, 1988, pp. 273-291).

Noncatalytic Reactions

Homogeneous reactions Homogeneous noncatalytic reactions are normally carried out in a fluidized bed to achieve mixing of the gases and temperature control. The solids of the bed act as a heat sink or source and facilitate heat transfer from or to the gas or from or to heat-exchange surfaces. Reactions of this type include chlorination of hydrocarbons or oxidation of gaseous fuels.

Heterogeneous reactions This category covers the greatest commercial use of fluidized beds other than fluid catalytic cracking. Roasting of ores in fluidized beds is very common. Roasting of sulfide, arsenical, and/or antimonial ores to facilitate the release of gold or silver values; the roasting of pyrite, pyrrhotite, or naturally occurring sulfur ores to provide SO_2 for sulfuric acid manufacture; and the roasting of copper, cobalt, and zinc sulfide ores to solubilize the metals are the major metallurgical uses. Figure 17-27 shows the basic items in the roasting process.

Thermally efficient calcination of lime, dolomite, and clay can be carried out in a multicompartiment fluidized bed (Fig. 17-28). Fuels are burned in a fluidized bed of the product to produce the required heat. Bunker C oil, natural gas, and coal are used in commercial units as the fuel. Temperature control is accurate enough to permit production of lime of very high quality with close control of slaking characteristics. Also, half calcination of dolomite is an accepted practice in fluidized

17-18 GAS-SOLID OPERATIONS AND EQUIPMENT

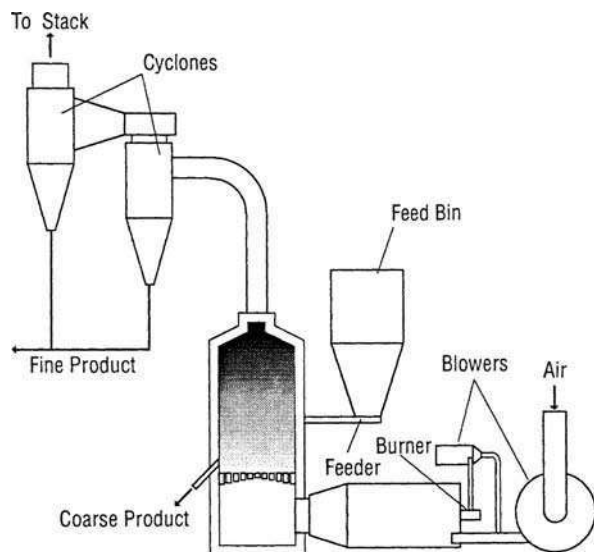


FIG. 17-27 Single-stage FluoSolids roaster or dryer. (Dorr-Oliver, Inc.)

beds. The requirement of large crystal size for the limestone limits application. Small crystals in the limestone result in low yields due to high dust losses from the fluidized bed.

Phosphate rock is calcined to remove carbonaceous material before being digested with sulfuric acid. Several different fluidized-bed processes have been commercialized for the direct reduction of hematite to high-iron, low-oxide products. Foundry sand is also calcined to remove organic binders and release fines. The calcination of $\text{Al}(\text{OH})_3$ to Al_2O_3 in a circulating fluidized process produces a high-grade product. The process combines the use of circulating, bubbling, and transport beds to achieve high thermal efficiency. See Fig. 17-29.

An interesting feature of these high-temperature-calcination applications is the direct injection of heavy oil, natural gas, or fine coal into the fluidized bed. Combustion takes place at well below flame temperatures without atomization. Considerable care in the design of the fuel and air supply system is necessary to take full advantage of the fluidized bed, which serves to mix the air and fuel.

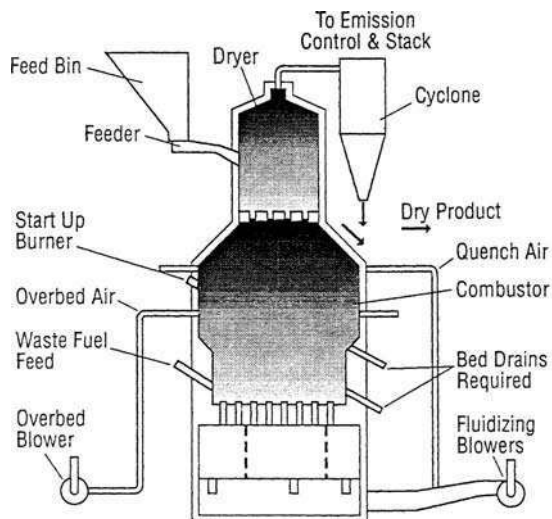


FIG. 17-28 FluoSolids multicompartiment fluidized bed. (Dorr-Oliver, Inc.)

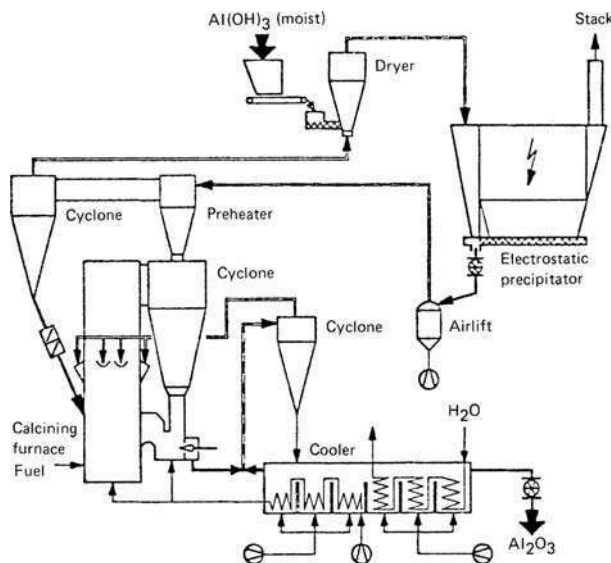


FIG. 17-29 Circulating fluid-bed calciner. (Lurgi Corp.)

Coal can be burned in fluidized beds in an environmentally acceptable manner by adding limestone or dolomite to the bed to react with the SO_2 to form CaSO_4 . Because of moderate combustion temperatures, about 800 to 900°C, NO_x formation, which results from the oxidation of nitrogen compounds contained in the coal, is kept at a low level. NO_x is increased by higher temperatures and higher excess oxygen contents. Two-stage air addition reduces NO_x . Several concepts of fluidized-bed combustion have been or are being developed. Atmospheric fluidized-bed combustion (AFBC), in which most of the heat-exchange tubes are located in the bed, is illustrated in Fig. 17-30.

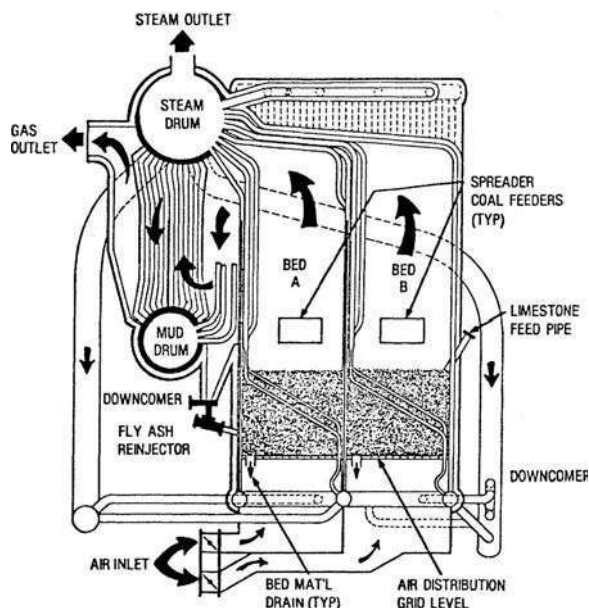


FIG. 17-30 Fluidized-bed steam generator at Georgetown University; 12.6-kg/s (100,000-lb/h) steam at 4.75-MPa (675-psig) pressure. (From Georgetown Univ. Q. Tech. Prog. Rep. METC/DOE/10381/135, July–September 1980.)

This type of unit is most commonly used for industrial applications up to about 50 t/h of steam generation. Larger units are generally of the circulating bed type, as shown in Fig. 17-30. **Circulating fluidized-bed combustors** have many advantages. The gas velocity is significantly higher than in bubbling or turbulent beds, which results in greater throughput. Since all the solids are recycled, fine limestone and coal can be fed to the combustor, which gives better limestone utilization and greater latitude in specifying coal sizing. Because of erosion due to high-velocity coarse solids, heat-transfer surface is usually not designed into the bottom of the combustion zone.

Pressurized fluidized-bed combustion (PFBC) is, as the name implies, operated at above atmospheric pressures. The beds and

heat-transfer surface are stacked to conserve space and to reduce the size of the pressure vessel. This type of unit is usually conceived as a cogeneration unit. Steam raised in the boilers is employed to drive turbines or for other uses. The hot pressurized gases after cleaning are let down through an expander coupled to a compressor to supply the compressed combustion air and/or electric generator. A 71-MWe PFBC unit is shown in Fig. 17-31. Also see Sec. 24, "Energy Resources, Conversion, and Utilization."

Incineration The majority of over 400 units in operation are used for the incineration of biological sludges. These units can be designed to operate autogenously with wet sludges containing as little as 6 MJ/kg (2600 Btu/lb) heating value (Fig. 17-32). Depending

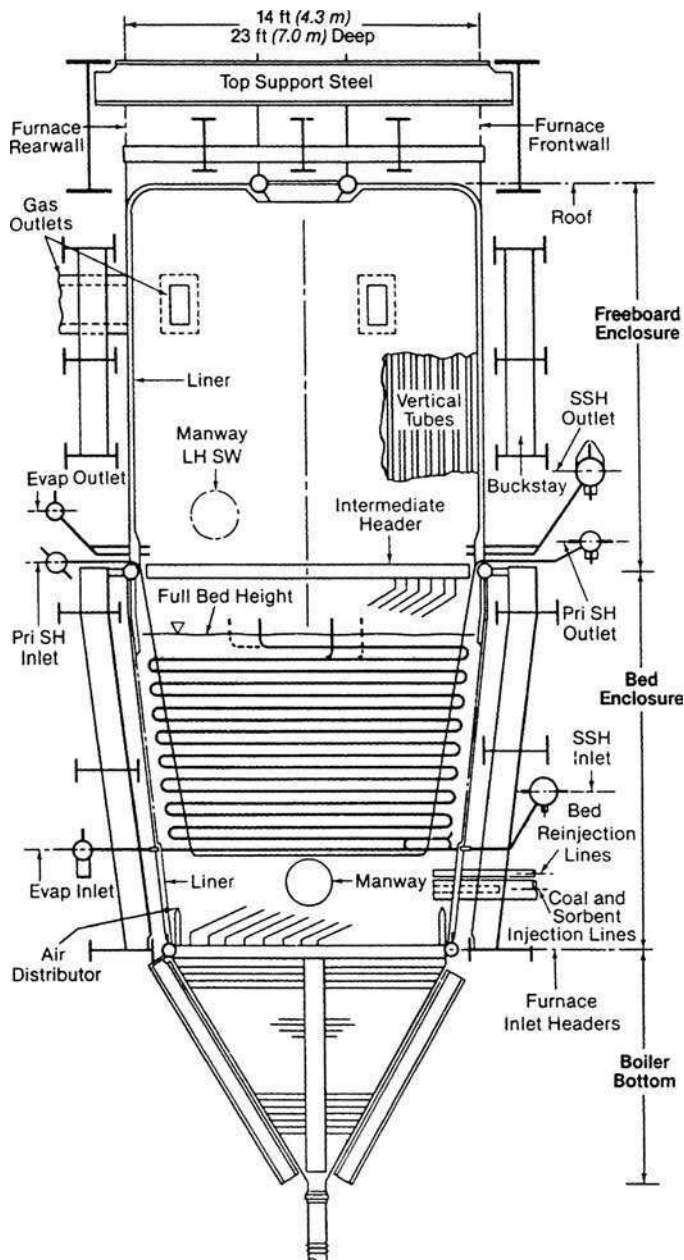


FIG. 17-31 71 MWe PFBC unit. (From *Steam*, 40th ed., 29-9, Babcock & Wilcox, 1992).

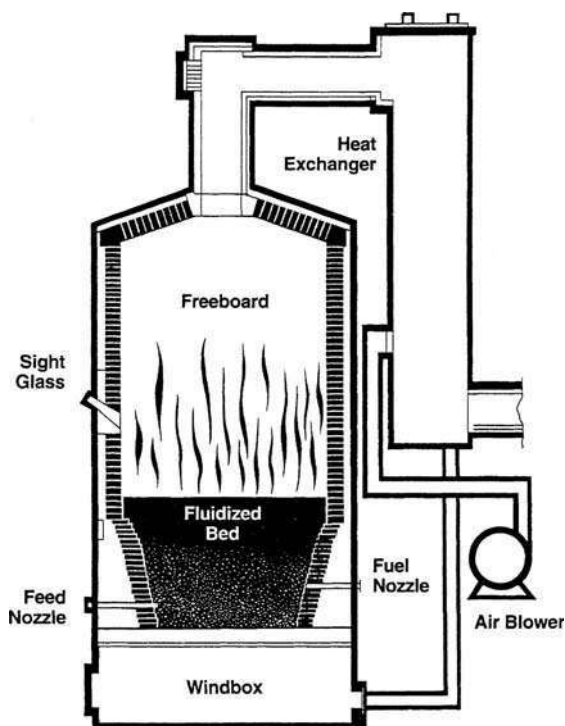


FIG. 17-32 Hot windbox incinerator/reactor with air preheating. (Dorr-Oliver, Inc.)

on the calorific value of the feed, heat can be recovered as steam either by means of waste heat boilers or by a combination of waste heat boilers and the heat-exchange surface in the fluid bed. Several units are used for sulfite papermill waste liquor disposal. Several units are used for oil refinery wastes, which sometimes include a mixture of liquid sludges, emulsions, and caustic waste [Flood and Kernel, *Chem. Proc.* (Sept. 8, 1973)]. Miscellaneous uses include the incineration of sawdust, carbon-black waste, pharmaceutical waste, grease from domestic sewage, spent coffee grounds, and domestic garbage.

Toxic or hazardous wastes can be disposed of in fluidized beds by either chemical capture or complete destruction. In the former case, bed material, such as limestone, will react with halides, sulfides, metals, etc., to form stable compounds which can be landfilled. Contact times of up to 5 or 10 s at 1200 K (900°C) to 1300 K (1000°C) ensure complete destruction of most compounds.

Physical Contacting

Drying Fluidized-bed units for drying solids, particularly coal, cement, rock, and limestone, are in wide use. Economic considerations make these units particularly attractive when large tonnages of solids are to be handled. Fuel requirements are 3.3 to 4.2 MJ/kg (1500 to 1900 Btu/lb of water removed), and total power for blowers, feeders, etc., is about 0.08 kWh/kg of water removed. The maximum feed size is approximately 6 cm (2.4 in) $\times$ 0 coal. One of the major advantages of this type of dryer is the close control of conditions so that a predetermined amount of free moisture may be left with the solids to prevent dusting of the product during subsequent material handling operations. The fluidized-bed dryer is also used as a classifier so that drying and classification operations are accomplished simultaneously. Wall and Ash [*Ind. Eng. Chem.* **41**: 1247 (1949)] state that in drying 4.8-mm (-4-mesh) dolomite with combustion gases at a superficial

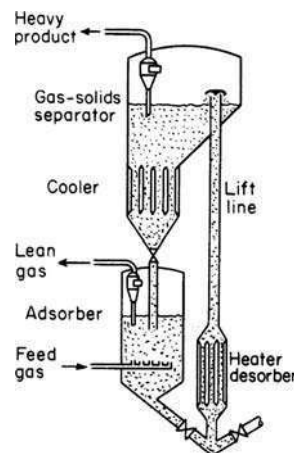


FIG. 17-33 Fluidized bed for gas fractionation. [Sittig, *Chem. Eng.* (May 1953).]

velocity of 1.2 m/s (4 ft/s), the following removals of fines were achieved:

Particle size	% removed
-65 + 100 mesh	60
-100 + 150 mesh	79
-150 + 200 mesh	85
-200 + 325 mesh	89
-325 mesh	89

Classification The separation of fine particles from coarse can be effected by use of a fluidized bed (see "Drying"). However, for economic reasons (i.e., initial cost, power requirements for compression of fluidizing gas, etc.), it is doubtful except in special cases if a fluidized-bed classifier would be built for this purpose alone.

It has been proposed that fluidized beds be used to remove fine solids from a gas stream. This is possible under special conditions.

Adsorption-Desorption An arrangement for gas fractionation is shown in Fig. 17-33.

The effects of adsorption and desorption on the performance of fluidized beds are discussed elsewhere. Adsorption of carbon disulfide vapors from air streams as great as 300 m³/s (540,000 ft³/min) in a 17-m- (53-ft-) diameter unit has been reported by Avery and Tracey ("The Application of Fluidized Beds of Activated Carbon to Recover Solvent from Air or Gas Streams," Tripartate Chemical Engineering Conference, Montreal, Sept. 24, 1968).

Heat Treatment Heat treatment can be divided into two types, treatment of fluidizable solids and treatment of large, usually metallic objects in a fluid bed. The former is generally accomplished in multi-compartment units to conserve heat (Fig. 17-28). The heat treatment of large metallic objects is accomplished in long, narrow heated beds. The objects are conveyed through the beds by an overhead conveyor system. Fluid beds are used because of the high heat-transfer rate and uniform temperature. See Reindl, "Fluid Bed Technology," *American Society for Metals*, Cincinnati, Sept. 23, 1981; Fennell, *Ind. Heat.*, **48**, 9, 36 (September 1981).

Coating Fluidized beds of thermoplastic resins have been used to facilitate the coating of metallic parts. A properly prepared, heated metal part is dipped into the fluidized bed, which permits complete immersion in the dry solids. The heated metal fuses the thermoplastic, forming a continuous uniform coating.

GAS-SOLIDS SEPARATIONS

This subsection is concerned with the application of particle mechanics (see Sec. 5, "Fluid and Particle Mechanics") to the design and application of dust-collection systems. It includes wet collectors, or

scrubbers, for particle collection. Scrubbers designed for purposes of mass transfer are discussed in Secs. 14 and 18. Equipment for removing entrained liquid mist from gases is described in Sec. 18.

Nomenclature

Except where otherwise noted here or in the text, either consistent system of units (SI or U.S. customary) may be used. Only SI units may be used for electrical quantities, since no comparable electrical units exist in the U.S. customary system. When special units are used, they are noted at the point of use.

Symbols	Definition	SI units	U.S. customary units	Special units
B_c	Width of rectangular cyclone inlet duct	m	ft	
B_e	Spacing between wire and plate, or between rod and curtain, or between parallel plates in electrical precipitators	m	ft	
B_s	Width of gravity settling chamber	m	ft	
C°	Dry scrubber pollutant gas equilibrium concentration over sorbent			
C_1	Dry scrubber pollutant gas inlet concentration			
C_2	Dry scrubber pollutant gas outlet concentration			
c_d	Dust concentration in gas stream	g/m^3		grains/ft ³
c_h	Specific heat of gas	$\text{J}/(\text{kg}\cdot\text{K})$	$\text{Btu}/(\text{lbm}\cdot^\circ\text{F})$	
c_{hb}	Specific heat of collecting body	$\text{J}/(\text{kg}\cdot\text{K})$	$\text{Btu}/(\text{lbm}\cdot^\circ\text{F})$	
c_{hp}	Specific heat of particle	$\text{J}/(\text{kg}\cdot\text{K})$	$\text{Btu}/(\text{lbm}\cdot^\circ\text{F})$	
D_b	Diameter or other representative dimension of collector body or device	m	ft	
D_{b1}, D_{b2}	Other characteristic dimensions of collector body or device	m	ft	
D_c	Cyclone diameter	m	ft	
D_d	Outside diameter of wire or discharge electrode of concentric-cylinder type of electrical precipitator	m	ft	
D_e	Diameter of cyclone gas exit duct	m	ft	
D_o	Volume/surface-mean-drop diameter			μm
D_p	Diameter of particle	m	ft	μm
D_{pth}	Cut diameter, diameter of particles of which 50% of those present are collected	m	ft	μm
d_p	Particle diameter of fraction number c'	m	ft	
D_i	Inside diameter of collecting tube of concentric-cylinder type of electrical precipitator	m	ft	
D_c	Diffusion coefficient for particle	m^2/s	ft^2/s	
DI	Decontamination index = $\log_{10}[1/(1 - \eta)]$	Dimensionless	Dimensionless	
e	Natural (napierian) logarithmic base	2.718 . . .	2.718 . . .	
E	Potential difference	V		
E_c	Potential difference required for corona discharge to commence	V		
E_d	Voltage across dust layer	V		
E_s	Potential difference required for sparking to commence	V		
E_L	Cyclone collection efficiency at actual loading			
E_O	Cyclone collection efficiency at low loading			
F_E	Effective friction loss across wetted equipment in scrubber	kPa		in water
F_k	Packed bed friction loss			
g_c	Conversion factor		32.17 (lbm/lbf)/(ft/s ²)	
g_L	Local acceleration due to gravity	m/s^2	ft/s^2	
H_c	Height of rectangular cyclone inlet duct	m	ft	
H_s	Height of gravity settling chamber	m	ft	
I	Electrical current per unit of electrode length	A/m		
j	Corona current density at dust layer	A/m^2		
k_p	Density of gas relative to its density at 0°C, 1 atm	Dimensionless	Dimensionless	Dimensionless
k_t	Thermal conductivity of gas	$\text{W}/(\text{m}\cdot\text{K})$	$\text{Btu}/(\text{s}\cdot\text{ft}\cdot^\circ\text{F})$	
k_{tb}	Thermal conductivity of collecting body	$\text{W}/(\text{m}\cdot\text{K})$	$\text{Btu}/(\text{s}\cdot\text{ft}\cdot^\circ\text{F})$	
k_{tp}	Thermal conductivity of particle	$\text{W}/(\text{m}\cdot\text{K})$	$\text{Btu}/(\text{s}\cdot\text{ft}\cdot^\circ\text{F})$	
K	Empirical proportionality constant for cyclone pressure drop or friction loss	Dimensionless	Dimensionless	
K_1	Resistance coefficient of "conditioned" filter fabric	$\text{kPa}/(\text{m}/\text{min})$	in water/(ft/min)	
K_2	Resistance coefficient of dust cake on filter fabric	$\frac{\text{kPa}}{(\text{m}/\text{min})(\text{g}/\text{m}^2)}$	$\frac{\text{in water}}{(\text{ft}/\text{min})(\text{lbm}/\text{ft}^2)}$	

17-22 GAS-SOLID OPERATIONS AND EQUIPMENT

Nomenclature (Concluded)

Symbols	Definition	SI units	U.S. customary units	Special units
K_a	Proportionality constant, for target efficiency of a single fiber in a bed of fibers	Dimensionless	Dimensionless	
K_c	Resistance coefficient for "conditioned" filter fabric			$\frac{\text{in water}}{(\text{ft}/\text{min})(\text{cP})}$
K_d	Resistance coefficient for dust cake on filter fabric		$\frac{\text{in water}}{(\text{ft}/\text{min})(\text{gr}/\text{ft}^2)(\text{cP})}$	
K_e	Electrical-precipitator constant	s/m	s/ft	
K_F	Resistance coefficient for clean filter cloth			$\frac{\text{in water}}{(\text{ft}/\text{min})(\text{cP})}$
K_v	"Energy-distance" constant for electrical discharge in gases	m		
K_{cu}	Stokes-Cunningham correction factor	Dimensionless	Dimensionless	Dimensionless
L	Thickness of fibrous filter or of dust layer on surface filter	m	ft	
L_c	Length of collecting electrode in direction of gas flow	m	ft	
L_s	Length of gravity settling chamber in direction of gas flow	m	ft	
$\ln$	Natural logarithm (logarithm to the base e)	Dimensionless	Dimensionless	Dimensionless
M	Molecular weight	kg/mol	lbm/mol	
n	Exponent	Dimensionless	Dimensionless	Dimensionless
N_{Kn}	Knudsen number = λ_m/D_b	Dimensionless	Dimensionless	
N_{Ma}	Mach number	Dimensionless	Dimensionless	
N_o	Number of elementary electrical charges acquired by a particle	Dimensionless	Dimensionless	
N_{Re}	Reynolds number = $(D_p \rho V_c / \mu)$ or $(D_p \rho u_c / \mu)$	Dimensionless	Dimensionless	
N_{sc}	Interaction number = $18 \mu / K_m \rho_p D_c$	Dimensionless	Dimensionless	
N_{sd}	Diffusional separation number	Dimensionless	Dimensionless	
N_{sec}	Electrostatic-attraction separation number	Dimensionless	Dimensionless	
N_{sei}	Electrostatic-induction separation number	Dimensionless	Dimensionless	
N_{sf}	Flow-line separation number	Dimensionless	Dimensionless	
N_{sg}	Gravitational separation number	Dimensionless	Dimensionless	
N_{si}	Inertial separation number	Dimensionless	Dimensionless	
N_{st}	Thermal separation number	Dimensionless	Dimensionless	
N_t	Number of transfer units = $\ln [1/(1 - \eta)]$	Dimensionless	Dimensionless	
N_t	Number of turns made by gas stream in a cyclone separator	Dimensionless	Dimensionless	
Δp	Gas pressure drop	kPa	lbf/ft ²	$\frac{\text{in water}}{\text{in water}}$
Δp_i	Gas pressure drop in cyclone or filter			$\frac{\text{in water}}{\text{lbf}/\text{in}^2}$
p_F	Gauge pressure of water fed to scrubber	kPa		
P_G	Gas-phase contacting power	MJ/1000 m ³		hp/(1000 ft ³ /min)
P_L	Liquid-phase contacting power	MJ/1000 m ³		hp/(1000 ft ³ /min)
P_M	Mechanical contacting power	MJ/1000 m ³		hp/(1000 ft ³ /min)
P_T	Total contacting power	MJ/1000 m ³		hp/(1000 ft ³ /min)
q	Gas flow rate	m ³ /s	ft ³ /s	
Q_G	Gas flow rate		ft ³ /s	ft ³ /min
Q_L	Liquid flow rate		ft ³ /s	gal/min
Q_p	Electrical charge on particle	C		
r	Radius; distance from centerline of cyclone separator; distance from centerline of concentric-cylinder electrical precipitator	m	ft	
t_m	Time			min
T	Absolute gas temperature	K	°R	
T_b	Absolute temperature of collecting body	K	°R	
u_s	Velocity of migration of particle toward collecting electrode	m/s	ft/s	
u_t	Terminal settling velocity of particle under action of gravity	m/s	ft/s	ft/s
v_m	Average cyclone inlet velocity, based on area A_c	m/s	ft/s	ft/s
v_p	Actual particle velocity	m/s	ft/s	
V_f	Filtration velocity (superficial gas velocity through filter)	m/min		ft/min
V_o	Gas velocity	m/s	ft/s	
V_s	Average gas velocity in gravity settling	m/s	ft/s	
V_{ct}	Tangential component of gas velocity in cyclone	m/s	ft/s	
w	Loading of collected dust on filter	g/m ²	lbm/ft ²	gr/ft ²

Nomenclature (Concluded)

Symbols	Definition	SI units	U.S. customary units	Special units
Greek symbols				
α	Empirical constant in equation of scrubber performance curve	$\left[\frac{\text{MJ}}{1000 \text{ m}^3} \right]^{-\gamma}$		$\left[\frac{\text{hp}}{100 \text{ ft}^3/\text{min}} \right]^{-\gamma}$
γ	Empirical constant in equation of scrubber performance curve	Dimensionless		Dimensionless
δ	Dielectric constant	Dimensionless		
δ_g	Dielectric constant at 0°C, 1 atm	Dimensionless		
δ_o	Permittivity of free space	F/m		
δ_b	Dielectric constant of collecting body	Dimensionless		
δ_p	Dielectric constant of particle	Dimensionless		
Δ	Fractional free area (for screens, perforated plates, grids)	Dimensionless	Dimensionless	
ϵ	Elementary electrical charge	$1.60210 \times 10^{-19} \text{ C}$		
ϵ_b	Characteristics potential gradient at collecting surface	V/m		
ϵ_c	Fraction voids in bed of solids	Dimensionless	Dimensionless	Dimensionless
ζ	$= 1 + 2 \frac{\delta - 1}{\delta + 2}$ ranges from a value of 1 for materials with a dielectric constant of 1 to 3 for conductors	Dimensionless		
η	Collection efficiency, weight fraction of entering dispersoid collected	Dimensionless	Dimensionless	Dimensionless
η_o	Target efficiency of an isolated collecting body, fraction of dispersoid in swept volume collected on body	Dimensionless	Dimensionless	Dimensionless
η_r	Target efficiency of a single collecting body in an array of collecting bodies, fraction of dispersoid in swept volume collected on body	Dimensionless	Dimensionless	Dimensionless
λ_i	Ionic mobility of gas	(m/s)/(V/m)		
λ_p	Particle mobility = $u_e/\mathcal{E}$	(m/s)/(V/m)		
μ	Gas viscosity	Pa·s	lbm/(s·ft)	cP
μ_L	Liquid viscosity			cP
ρ	Gas density	g/m ³	lb/ft ³	
ρ_d	Resistivity of dust layer	$\Omega\cdot\text{m}$		
ρ_L	Liquid density		lbm/ft ³	lbm/ft ³
ρ_s	True (not bulk) density of solids or liquid drops	kg/m ³	lbm/ft ³	lbm/ft ³
ρ'	Density of gas relative to its density at 25°C, 1 atm	Dimensionless	Dimensionless	Dimensionless
σ	Ion density	Number/m ³		
σ_{avg}	Average ion density	Number/m ³		
σ_L	Liquid surface tension			dyn/cm
Φ_s	Particle shape factor = (surface of sphere)/(surface of particle of same volume)	Dimensionless	Dimensionless	Dimensionless
Script symbols				
$\mathcal{E}$	Potential gradient	V/m		
$\mathcal{E}_c$	Potential gradient required for corona discharge to commence	V/m		
$\mathcal{E}_i$	Average potential gradient in ionization stage	V/m		
$\mathcal{E}_o$	Electrical breakdown constant for gas	V/m		
$\mathcal{E}_p$	Average potential gradient in collection stage	V/m		
$\mathcal{E}_s$	Potential gradient required for sparking to commence	V/m		

GENERAL REFERENCES: Burchsted, Kahn, and Fuller, *Nuclear Air Cleaning Handbook*, ERDA 76-21, Oak Ridge, Tenn., 1976. Cadle, *The Measurement of Airborne Particles*, Wiley, New York, 1975. Davies, *Aerosol Science*, Academic, New York, 1966. Davies, *Air Filtration*, Academic, New York, 1973. Dennis, *Handbook on Aerosols*, ERDA TID-26608, Oak Ridge, Tenn., 1976. Drinker and Hatch, *Industrial Dust*, 2d ed., McGraw-Hill, New York, 1954. Friedlander, *Smoke, Dust, and Haze*, Wiley, New York, 1977. Fuchs, *The Mechanics of Aerosols*, Pergamon, Oxford, 1964. Green and Lane, *Particulate Clouds: Dusts, Smokes, and Mists*, Van Nostrand, New York, 1964. Lapple, *Fluid and Particle Mechanics*, University of Delaware, Newark, 1951. Licht, *Air Pollution Control Engineering—Basic Calculations for Particle Collection*, Marcel Dekker, New York, 1980. Liu, *Fine Particles—Aerosol Generation, Measurement, Sampling, and Analysis*, Academic, New York, 1976. Lunde and Lapple, *Chem. Eng. Prog.*,

53, 385 (1957). Lundgren et al., *Aerosol Measurement*, University of Florida, Gainesville, 1979. Mercer, *Aerosol Technology in Hazard Evaluation*, Academic, New York, 1973. Nonhebel, *Processes for Air Pollution Control*, CRC Press, Cleveland, 1972. Shaw, *Fundamentals of Aerosol Science*, Wiley, New York, 1978. Stern, *Air Pollution: A Comprehensive Treatise*, vols. 3 and 4, Academic, New York, 1977. Strauss, *Industrial Gas Cleaning*, 2d ed., Pergamon, New York, 1975. Theodore and Buonicore, *Air Pollution Control Equipment: Selection, Design, Operation, and Maintenance*, Prentice-Hall, Englewood Cliffs, N.J., 1982. White, *Industrial Electrostatic Precipitation*, Addison-Wesley, Reading, Mass., 1963. White and Smith, *High-Efficiency Air Filtration*, Butterworth, Washington, 1964. ASME Research Committee on Industrial and Municipal Wastes, *Combustion Fundamentals for Waste Incineration*, American Society of Mechanical Engineers, 1974. Buonicore and Davis (eds.), *Air Pollution Engineering*

Manual, Air & Waste Management Association, Van Nostrand Reinhold, 1992. Burchsted, Fuller, and Kahn, *Nuclear Air Cleaning Handbook*, ORNL for the U.S. Energy Research and Development Administration, NTIS Report ERDA 76-21, 1976. Dennis (ed.), *Handbook on Aerosols*, GCA for the U.S. Energy Research and Development Administration, NTIS Report TID-26608, 1976. Stern, *Air Pollution*, 3d ed., Academic Press, 1977 (supplement 1986).

PURPOSE OF DUST COLLECTION

Dust collection is concerned with the removal or collection of solid dispersoids in gases for purposes of:

1. Air-pollution control, as in fly-ash removal from power-plant flue gases
2. Equipment-maintenance reduction, as in filtration of engine-intake air or pyrites furnace-gas treatment prior to its entry to a contact sulfuric acid plant
3. Safety- or health-hazard elimination, as in collection of siliceous and metallic dusts around grinding and drilling equipment and in some metallurgical operations and flour dusts from milling or bagging operations
4. Product-quality improvement, as in air cleaning in the production of pharmaceutical products and photographic film
5. Recovery of a valuable product, as in collection of dusts from dryers and smelters
6. Powdered-product collection, as in pneumatic conveying; the spray drying of milk, eggs, and soap; and the manufacture of high-purity zinc oxide and carbon black

PROPERTIES OF PARTICLE DISPERSOIDS

An understanding of the fundamental properties and characteristics of gas dispersoids is essential to the design of industrial dust-control equipment. Figure 17-34 shows characteristics of dispersoids and other particles together with the types of gas-cleaning equipment that are applicable to their control. Two types of solid dispersoids are shown: (1) dust, which is composed of particles larger than 1 μm ; and (2) fume, which consists of particles generally smaller than 1 μm . Dusts usually result from mechanical disintegration of matter. They may be redispersed from the settled, or bulk, condition by an air blast. Fumes are submicrometer dispersoids formed by processes such as combustion, sublimation, and condensation. Once collected, they cannot be redispersed from the settled condition to their original state of dispersion by air blasts or mechanical dispersion equipment.

The primary distinguishing characteristic of gas dispersoids is particle size. The generally accepted unit of particle size is the micrometer, μm . (Prior to the adoption of the SI system, the same unit was known as the micron and was designated by μ .) The particle size of a gas dispersoid is usually taken as the diameter of a sphere having the same mass and density as the particle in question. Another common method is to designate the screen mesh that has an aperture corresponding to the particle diameter; the screen scale used must also be specified to avoid confusion.

From the standpoint of collector design and performance, the most important size-related property of a dust particle is its dynamic behavior. Particles larger than 100 μm are readily collectible by simple inertial or gravitational methods. For particles under 100 μm , the range of principal difficulty in dust collection, the resistance to motion in a gas is viscous (see Sec. 6, "Fluid and Particle Dynamics"), and for such particles, the most useful size specification is commonly the Stokes settling diameter, which is the diameter of the spherical particle of the same density that has the same terminal velocity in viscous flow as the particle in question. It is yet more convenient in many circumstances to use the "aerodynamic diameter," which is the diameter of the particle of unit density (1 g/cm^3) that has the same terminal settling velocity. Use of the aerodynamic diameter permits direct comparisons of the dynamic behavior of particles that are actually of different sizes, shapes, and densities [Raabe, *J. Air Pollut. Control Assoc.*, **26**, 856 (1976)].

When the size of a particle approaches the same order of magnitude as the mean free path of the gas molecules, the settling velocity is greater than predicted by Stokes' law because of molecular slip. The slip-flow correction is appreciable for particles smaller than 1 μm and is allowed for by the Cunningham correction for Stokes' law [Lapple, *op. cit.*; Licht, *op. cit.*]. The Cunningham correction is applied in

calculations of the aerodynamic diameters of particles that are in the appropriate size range.

Although solid fume particles may range in size down to perhaps 0.001 μm , fine particles effectively smaller than about 0.1 μm are not of much significance in industrial dust and fume sources because their aggregate mass is only a very small fraction of the total mass emission. At the concentrations present in such sources (e.g., production of carbon black) the coagulation, or flocculation, rate of the ultrafine particles is extremely high, and the particles speedily grow to sizes of 0.1 μm or greater. The most difficult collection problems are thus concerned with particles in the range of about 0.1 to 2 μm , in which forces for deposition by inertia are small. For collection of particles under 0.1 μm , diffusional deposition becomes increasingly important as the particle size decreases.

In a gas stream carrying dust or fume, some degree of particle flocculation will exist, so that both discrete particles and clusters of adhering particles will be present. The discrete particles composing the clusters may be only loosely attached to each other, as by van der Waals forces [Lapple, *Chem. Eng.*, **75**(11), 149 (1968)]. Flocculation tends to increase with increases in particle concentration and may strongly influence collector performance.

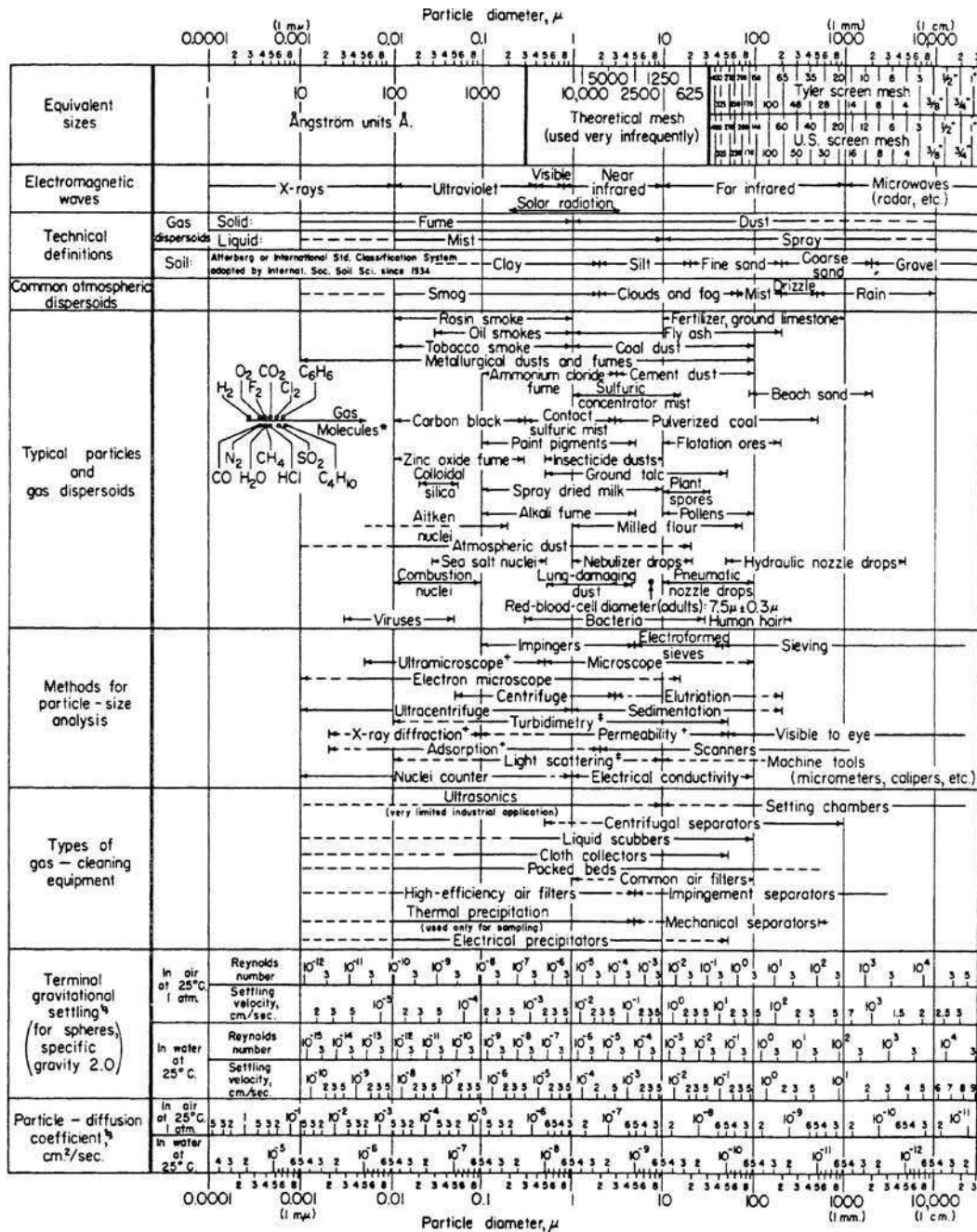
PARTICLE MEASUREMENTS

Measurements of the concentrations and characteristics of dust dispersed in air or other gases may be necessary (1) to determine the need for control measures, (2) to establish compliance with legal requirements, (3) to obtain information for collector design, and (4) to determine collector performance.

Atmospheric-Pollution Measurements The dust-fall measurement is one of the common methods for obtaining a relative long-period evaluation of particulate air pollution. Stack-smoke densities are often graded visually by means of the Ringelmann chart. Plume opacity may be continuously monitored and recorded by a photoelectric device which measures the amount of light transmitted through a stack plume. Equipment for local atmospheric-dust-concentration measurements fall into five general types: (1) the impinger, (2) the hot-wire or thermal precipitator, (3) the electrostatic precipitator, (4) the filter, and (5) impactors and cyclones. The filter is the most widely used, in the form of either a continuous tape, or a number of filter disks arranged in an automatic sequencing device, or a single, short-term, high-volume sampler. Samplers such as these are commonly used to obtain mass emission and particle-size distribution. Impactors and small cyclones are commonly used as size-discriminating samplers and are usually followed by filters for the determination of the finest fraction of the dust (Lundgren et al., *Aerosol Measurement*, University of Florida, Gainesville, 1979; and Dennis, *Handbook on Aerosols*, U.S. ERDA TID-26608, Oak Ridge, Tenn., 1976).

Process-Gas Sampling In sampling process gases either to determine dust concentration or to obtain a representative dust sample, it is necessary to take special precautions to avoid inertial segregation of the particles. To prevent such classification, a traverse of the duct may be required, and at each point the sampling nozzle must face directly into the gas stream with the velocity in the mouth of the nozzle equal to the local gas velocity at that point. This is called "isokinetic sampling." If the sampling velocity is too high, the dust sample will contain a lower concentration of dust than the mainstream, with a greater percentage of fine particles; if the sampling velocity is too low, the dust sample will contain a higher concentration of dust with a greater percentage of coarse particles [Lapple, *Heat. Piping Air Cond.*, **16**, 578 (1944); *Manual of Disposal of Refinery Wastes*, vol. V, American Petroleum Institute, New York, 1954; and Dennis, *op. cit.*].

Particle-Size Analysis Methods for particle-size analysis are shown in Fig. 17-34, and examples of size-analysis methods are given in Table 17-1. More detailed information may be found in Lapple, *Chem. Eng.*, **75**(11), 140 (1968); Lapple, "Particle-Size Analysis," in *Encyclopedia of Science and Technology*, 5th ed., McGraw-Hill, New York, 1982; Cadle, *The Measurement of Airborne Particles*, Wiley, New York, 1975; Lowell, *Introduction to Powder Surface Area*, 2d ed., Wiley, New York, 1993; and Allen, *Particle Size Measurement*, 4th ed., Chapman and Hall, London, 1990. Particle-size distribution may be presented on either a frequency or a cumulative basis; the various methods are discussed in



- * Molecular diameters calculated from viscosity data at 0°C.
 † Furnishes average particle diameter but no size distribution.
 ‡ Size distribution may be obtained by special calibration.
 § Stokes-Cunningham factor included in values given for air but not included for water.

FIG. 17-34 Characteristics of particles and particle dispersoids. (Courtesy of the Stanford Research Institute; prepared by C. E. Lapple.)

the references just cited. The most common method presents a plot of particle size versus the cumulative weight percent of material larger or smaller than the indicated size, on logarithmic-probability graph paper.

For determination of the aerodynamic diameters of particles, the most commonly applicable methods for particle-size analysis are those

based on inertia: aerosol centrifuges, cyclones, and inertial impactors (Lundgren et al., *Aerosol Measurement*, University of Florida, Gainesville, 1979; and Liu, *Fine Particles—Aerosol Generation, Measurement, Sampling, and Analysis*, Academic, New York, 1976). Impactors are the most commonly used. Nevertheless, impactor

TABLE 17-1 Particle Size Analysis Methods and Equipment

Method	Brand names	Size range	Sample size, g
Quantitative image analysis	American Innovation Videometric, Analytical Measuring Systems Quickstep & Optomax, Artec Omnicon Automatrix, Boeckeler, Buehler Omnimat, Compix Imaging Systems, Data Translation—Global Lab, Image, Hamamatsu C-1000, Hitech Olympus Cue-3, Joyce-Loebl Magiscan, Leco AMF System, Leico Quantimet, LeMont Oasys, Millipore_MC, Nachet 1500, Nikon Microphot, Oncor Instrument System, Optomax V, Outokumpu Imagist, Shakespeare Juliet, Tracor Northern, Carl Zeiss Videoplan	1–1000 μm	0.001
Sieves:		4–100 mm	500
Punched Plate	Many		
Woven wire	Many	20 μm –125 mm	25–200
Micromesh	Buckbee Mears, Veco, Endecott	5–500+ μm	1–5
Sieving machines (Air jet, sonic wet and dry)	Alpine, ATM, Gilson, Gradex, Hosokawa, Retsch, Seishin		
Sedimentation	Pipette—Gilson, <i>Photosedimentometer</i> —gravitational, Paar; centrifugal, Joyce Loeb, Brookhaven, Horiba, Seishin, Shimadzu; <i>x-ray absorption</i> —gravitational, Quantachrome, Micromeritics; centrifugal, Brookhaven	Gravitational 1–100 μm Centrifugal 0.05–5 mm	0.1–5+ g
Classification	Air-Bahco, Water-Warman Cyclosizer	5–500 μm	5–100 g
Field scanning (light)	Cilas, Coulter, Insitex, Fritsch, Horiba, Leeds & Northrup (Microtrac), Malvern, Nitto, Seishin, Shimadzu, Sympatec	0.04–3500 μm	<1 g (wet) >20 g (on-line) On-line
Field scanning (ultrasonics)	Sympatec OPUS, Pen-Kem		
Stream scanning	Brinkmann, Climet, Coulter, Dantec, Erdco, Faley, Flowvision, Hiac/Royco, Kowa, Lasentec, Malvern, Met One, Particle Measuring Systems, Polytec, Proceadyne, Rion, Spectrex	0.2–10,000 μm	0.1–10 g (also on-line)
Zeta potential Photon correlation Spectroscopy	Zeta Plus, Micromeritics, Zeta sizer Malvern, Nicomp, Brookhaven, Coulter, Photol	0.001–30 μm	0.1–1 μm

NOTE: This table was compiled with the assistance of T. Allen, DuPont Particle Science and Technology, and is not intended to be comprehensive. Many other fine suppliers of particle analysis equipment are available.

measurements are subject to numerous errors [Rao and Whitby, *Am. Ind. Hyg. Assoc. J.*, **38**, 174 (1977); Marple and Willeke, "Inertial Impactors," in Lundgren et al., *Aerosol Measurement*; and Fuchs, "Aerosol Impactors," in Shaw, *Fundamentals of Aerosol Science*, Wiley, New York, 1978]. Reentrainment due to particle bouncing and blowoff of deposited particles makes a dust appear finer than it actually is, as does the breakup of flocculated particles. Processing cascade-impactor data also presents possibilities for substantial errors (Fuchs, *The Mechanics of Aerosols*, Pergamon, Oxford, 1964) and is laborious as well. Lawless (Rep. No. EPA-600/7-78-189, U.S. EPA, 1978) discusses problems in analyzing and fitting cascade-impactor data to obtain dust-collector efficiencies for discrete particle sizes.

The measured diameters of particles should as nearly as possible represent the effective particle size of a dust as it exists in the gas stream. When significant flocculation exists, it is sometimes possible to use measurement methods based on gravity settling.

For dust-control work, it is recommended that a preliminary qualitative examination of the dust first be made without a detailed particle count. A visual estimate of particle-size distribution will often provide sufficient guidance for a preliminary assessment of requirements for collection equipment.

MECHANISMS OF DUST COLLECTION

The basic operations in dust collection by any device are (1) separation of the gas-borne particles from the gas stream by deposition on a collecting surface; (2) retention of the deposit on the surface; and (3) removal of the deposit from the surface for recovery or disposal. The separation step requires (1) application of a force that produces a differential motion of a particle relative to the gas and (2) a gas retention time sufficient for the particle to migrate to the collecting surface. The principal mechanisms of aerosol deposition that are applied in dust collectors are (1) gravitational deposition, (2) flow-line interception, (3) inertial deposition, (4) diffusional deposition, and (5) electrostatic deposition.

Thermal deposition is only a minor factor in practical dust-collection equipment because the thermophoretic force is small. Table 17-2 lists these six mechanisms and presents the characteristic parameters of their operation [Lunde and Lapple, *Chem. Eng. Prog.*, **53**, 385 (1957)]. The actions of the inertial-deposition, flow-line-interception, and diffusional-deposition mechanisms are illustrated in Fig. 17-35 for the case of a collecting body immersed in a particle-laden gas stream.

Two other deposition mechanisms, in addition to the six listed, may be in operation under particular circumstances. Some dust particles may be collected on filters by sieving when the pore diameter is less than the particle diameter. Except in small membrane filters, the sieving mechanism is probably limited to surface-type filters, in which a layer of collected dust is itself the principal filter medium.

The other mechanism appears in scrubbers. When water vapor diffuses from a gas stream to a cold surface and condenses, there is a net hydrodynamic flow of the noncondensable gas directed toward the surface. This flow, termed the Stefan flow, carries aerosol particles to the condensing surface (Goldsmith and May, in Davies, *Aerosol Science*, Academic, New York, 1966) and can substantially improve the performance of a scrubber. However, there is a corresponding Stefan flow directed away from a surface at which water is evaporating, and this will tend to repel aerosol particles from the surface.

In addition to the deposition mechanisms themselves, methods for preliminary conditioning of aerosols may be used to increase the effectiveness of the deposition mechanisms subsequently applied. One such conditioning method consists of imposing on the gas high-intensity acoustic vibrations to cause collisions and flocculation of the aerosol particles, producing large particles that can be separated by simple inertial devices such as cyclones. This process, termed "sonic (or acoustic) agglomeration," has attained only limited commercial acceptance.

Another conditioning method, adaptable to scrubber systems, consists of inducing condensation of water vapor on the aerosol particles as nuclei, increasing the size of the particles and making them more susceptible to collection by inertial deposition.

TABLE 17-2 Summary of Mechanisms and Parameters in Aerosol Deposition

Deposition	Origin of force field	Deposition mechanism measureable in terms of		System parameters
		Basic parameter	Specific modifying parameters	
Flow-line interception ^o	Physical gradient ^o	$N_{sf} = \left(\frac{D_p}{D_b} \right)$	$N_{sc} = \left(\frac{N_{sf}^2}{N_{st} N_{sd}} \right)$ $= \left(\frac{18\mu}{K_m \rho_p D_v} \right)^\dagger$	Geometry: $(D_{b1}/D_b), (D_{b2}/D_b), \text{ etc.}$ ϵ_p Δ
Inertial deposition	Velocity gradient	$N_{st} = \left(\frac{K_m \rho_s D_p^2 V_o}{18\mu D_b} \right)$		
Diffusional deposition	Concentration gradient	$N_{sd} = \left(\frac{D_v}{V_o D_b} \right)$		
Gravity settling	Elevation gradient	$N_{sg} = \left(\frac{u_t}{V_o} \right)$		Flow pattern: $N_{Re}^\dagger$ N_{Ma} N_{Kn}
Electrostatic precipitation	Electric-field gradient§ a. Attraction b. Induction	$N_{sec} = \left(\frac{K_m Q_p \epsilon_b}{\mu D_p V_o} \right)$ $N_{sei} = \left(\frac{\delta_p - 1}{\delta_p + 2} \right) \left(\frac{K_m D_p^2 \delta_o \epsilon_b 2}{\mu D_b V_o} \right)$	$\delta_p, \delta_b \P$	Surface accommodation
Thermal precipitation	Temperature gradient	$N_{st} =$ $\left(\frac{T - T_b}{T} \right) \left(\frac{\mu}{K_m \rho D_b V_o} \right) \left(\frac{k_t}{2k_t + k_{tp}} \right)$	$(T_b/T), (T_p/T), \dagger (N_{Pr}),$ $(k_{tp}/k_t), (k_{tb}/k_t), \P (c_{tp}/c_b), (c_{tb}/c_b) \P$	

SOURCE: Lunde and Lapple, *Chem. Eng. Prog.*, **53**, 385 (1957).

^oThis has also commonly been termed "direct interception" and in conventional analysis would constitute a physical boundary condition imposed upon the particle path induced by action of other forces. By itself it reflects deposition that might result with a hypothetical particle having finite size but no mass or elasticity.

[†]This parameter is an alternative to N_{sf} , N_{st} , or N_{sd} and is useful as a measure of the interactive effect of one of these on the other two. It is comparable with the Schmidt number.

[‡]When applied to the inertial deposition mechanism, a convenient alternative is $(K_m \rho_s / 18\mu) = N_{st} / (N_{sf}^2 N_{Re})$.

[§]In cases in which the body charge distribution is fixed and known, ϵ_b may be replaced with Q_{bs}/δ_o .

[¶]Not likely to be significant contributions.

Most forms of dust-collection equipment use more than one of the collection mechanisms, and in some instances the controlling mechanism may change when the collector is operated over a wide range of conditions. Consequently, collectors are most conveniently classified by type rather than according to the underlying mechanisms that may be operating.

PERFORMANCE OF DUST COLLECTORS

The performance of a dust collector is most commonly expressed as the collection efficiency η , the weight ratio of the dust collected to the dust entering the apparatus. However, the collection efficiency is usually related exponentially to the properties of the dust and gas and the operating conditions of most types of collectors and hence is an insensitive function of the collector operating conditions as its value approaches 1.0. Performance in the high-efficiency range is better expressed by the penetration $1 - \eta$, the weight ratio of the dust escaping to the dust entering. Particularly in reference to collection of radioactive aerosols, it is common to express performance in terms of the reciprocal of the penetration $1/(1 - \eta)$, which is termed the *decontamination index* (DI). The number of transfer units N_t , which is equal to $\ln [1/(1 - \eta)]$ in the case of dust collection, was first proposed for use by Lapple (Wright, Stasny, and Lapple, "High Velocity Air Filters," WADC Tech. Rep. 55-457, ASTIA No. AD-142075, October 1957) and is more commonly used than the DI. Because of the exponential form of the relationship between efficiency and process variables for most dust collectors, the use of N_t (or DI) is particularly suitable for correlating collector performance data.

In comparing alternative collectors for a given service, a figure of merit is desirable for ranking the different devices. Since power consumption is one of the most important characteristics of a collector, the ratio of N_t to power consumption is a useful criterion. Another is the ratio of N_t to capital investment.

DUST-COLLECTOR DESIGN

In dust-collection equipment, most or all of the collection mechanisms may be operating simultaneously, their relative importance being determined by the particle and gas characteristics, the geometry of the equipment, and the fluid-flow pattern. Although the general case is exceedingly complex, it is usually possible in specific instances to determine which mechanism or mechanisms may be controlling. Nevertheless, the difficulty of theoretical treatment of dust-collection phenomena has made necessary simplifying assumptions, with the introduction of corresponding uncertainties. Theoretical studies have been hampered by a lack of adequate experimental techniques for verification of predictions. Although theoretical treatment of collector performance has been greatly expanded in the period since 1960, few of the resulting performance models have received adequate experimental confirmation because of experimental limitations.

The best-established models of collector performance are those for fibrous filters and fixed-bed granular filters, in which the structures and fluid-flow patterns are reasonably well defined. These devices are also adapted to small-scale testing under controlled laboratory conditions. Realistic modeling of full-scale electrostatic precipitators and scrubbers is incomparably more difficult. Confirmation of the models has been further limited by a lack of monodisperse aerosols that can be generated on a scale suitable for testing equipment of substantial sizes. When a polydisperse test dust is used, the particle-size distributions of the dust both entering and leaving a collector must be determined with extreme precision to avoid serious errors in the determination of the collection efficiency for a given particle size.

The design of industrial-scale collectors still rests essentially on empirical or semiempirical methods, although it is increasingly guided by concepts derived from theory. Existing theoretical models frequently embody constants that must be evaluated by experiment and that may actually compensate for deficiencies in the models.

DUST-COLLECTION EQUIPMENT

Gravity Settling Chambers The gravity settling chamber is probably the simplest and earliest type of dust-collection equipment, consisting of a chamber in which the gas velocity is reduced to enable dust to settle out by the action of gravity. Its simplicity lends it to almost any type of construction. Practically, however, its industrial utility is limited to removing particles larger than 325 mesh (43- μ m diameter). For removing smaller particles, the required chamber size is generally excessive.

Gravity collectors are generally built in the form of long, empty, horizontal, rectangular chambers with an inlet at one end and an outlet at the side or top of the other end. By assuming a low degree of turbulence relative to the settling velocity of the dust particle in question, the performance of a gravity settling chamber is given by

$$\eta = \frac{u_t L_s}{H V_s} = \frac{u_t B_s L_s}{q} \quad (\text{for } \eta \leq 1.0) \quad (17-1)$$

where V_s = average gas velocity. Expressing u_t in terms of particle size (equivalent spherical diameter), the smallest particle that can be completely separated out corresponds to $\eta = 1.0$ and, assuming Stokes' law, is given by

$$D_{p,\min} = \sqrt{\frac{18\mu H V_s}{g_L L_s (\rho_s - \rho)}} \\ = \sqrt{\frac{18\mu q}{g_L B_s L_s (\rho_s - \rho)}} \quad (17-2)$$

where ρ = gas density and ρ_s = particle density. For a given volumetric air-flow rate, collection efficiency depends on the total plan cross section of the chamber and is independent of the height. The height need be made only large enough so that the gas velocity V_s in the chamber is not so high as to cause reentrainment of separated dust. Generally V_s should not exceed about 3 m/s (10 ft/s).

Horizontal plates arranged as shelves within the chamber will give a marked improvement in collection. This arrangement is known as the Howard dust chamber (Fume Arrestor, U.S. Patent 896,111, 1908). The disadvantage of the unit is the difficulty of cleaning owing to the close shelf spacing and warpage at elevated temperatures.

The pressure drop through a settling chamber is small, consisting primarily of entrance and exit losses. Because low gas velocities are used, the chamber is not subject to abrasion and may therefore be used as a precleaner to remove very coarse particles and thus minimize abrasion on subsequent equipment.

Impingement Separators Impingement separators are a class of inertial separators in which particles are separated from the gas by inertial impingement on collecting bodies arrayed across the path of the gas stream, as shown on Fig. 17-35. Fibrous-pad inertial impingement separators for the collection of wet particles are the main application in current technology, as is described in Sec. 14, "Impingement Separation." With the growing need for very high performance dust collectors, there is little application anymore for dry impingement collectors.

Cyclone Separators The most widely used type of dust-collection equipment is the cyclone, in which dust-laden gas enters a cylindrical or conical chamber tangentially at one or more points and

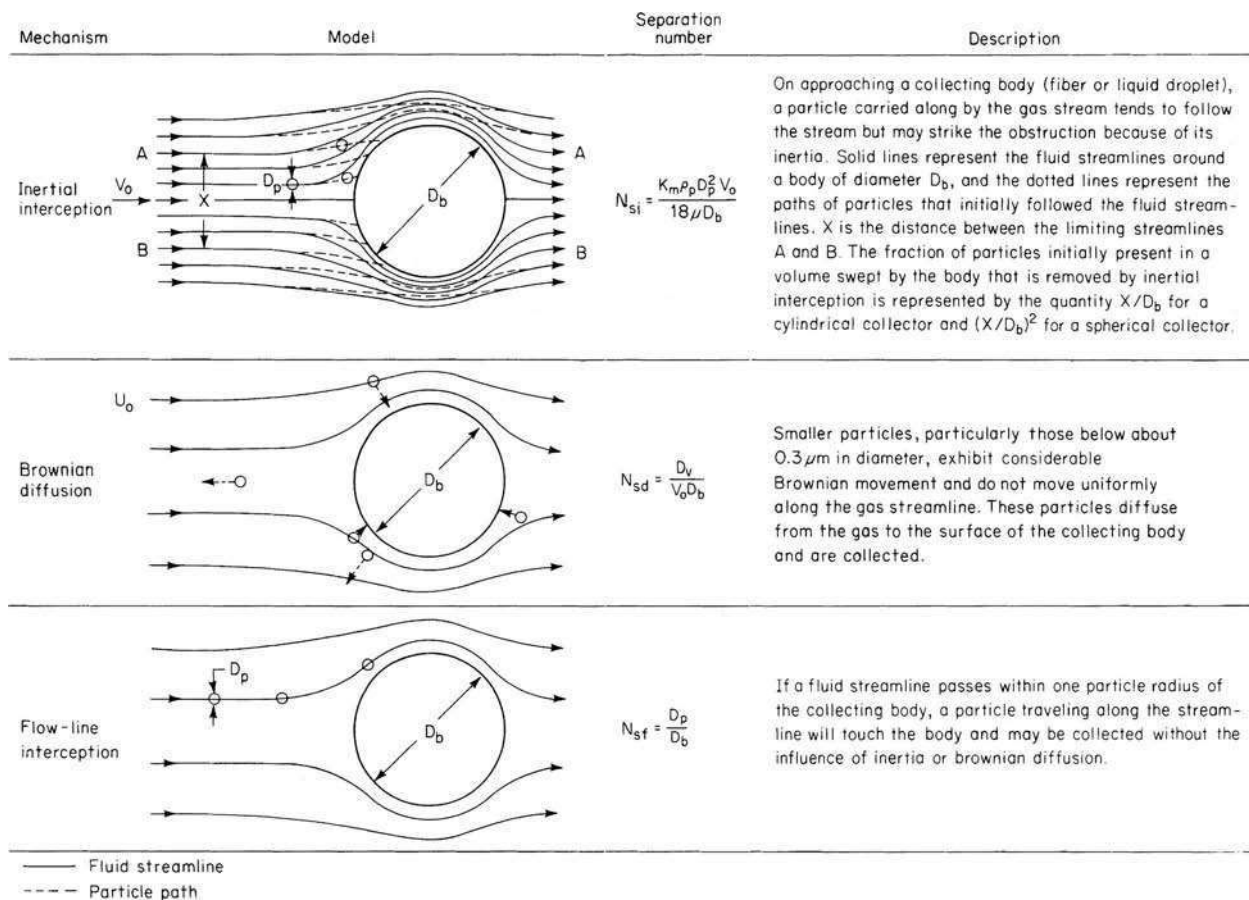


FIG. 17-35 Particle deposition on collector bodies.

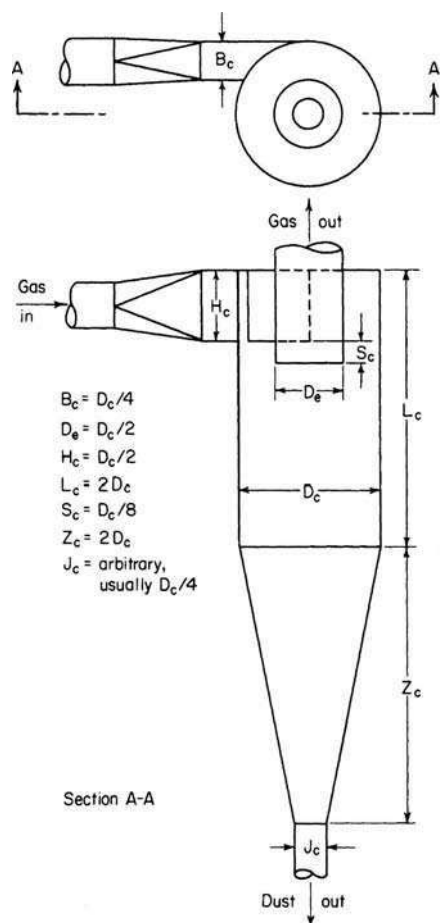


FIG. 17-36 Cyclone-separator proportions.

leaves through a central opening (Fig. 17-36). The dust particles, by virtue of their inertia, will tend to move toward the outside separator wall, from which they are led into a receiver. A cyclone is essentially a settling chamber in which gravitational acceleration is replaced by centrifugal acceleration. At operating conditions commonly employed, the centrifugal separating force or acceleration may range from 5 times gravity in very large diameter, low-resistance cyclones, to 2500 times gravity in very small, high-resistance units. The immediate entrance to a cyclone is usually rectangular.

Fields of Application Within the range of their performance capabilities, cyclone collectors offer one of the least expensive means of dust collection from the standpoint of both investment and operation. Their major limitation is that their efficiency is low for collection of particles smaller than 5 to 10 μm . Although cyclones may be used to collect particles larger than 200 μm , gravity settling chambers or simple inertial separators (such as gas-reversal chambers) are usually satisfactory for this size of particle and are less subject to abrasion. In special cases in which the dust is highly agglomerated or in high dust concentrations (over 230 g/m^3 , or 100 gr/ft^3) are encountered, cyclones will remove dusts having small particle sizes. In certain instances, efficiencies as high as 98 percent have been attained on dusts having ultimate particle sizes of 0.1 to 2.0 μm because of the predominant effect of particle agglomeration due to high interparticle forces. Cyclones are used to remove both solids and liquids from gases and have been operated at temperatures as high as 1200°C and pressures as high as 50,700 kPa (500 atm).

Cyclones can be very small or very large. The smallest cyclones range from approximately 1 to 2 cm in diameter and the largest up to about 10 m in diameter. The number of cyclones used for a single fluidized bed can vary from 1 to up to 22 sets of first-stage and second-stage cyclones (44 cyclones total).

Flow Pattern In a cyclone, the gas moves in a double vortex with the gas initially spiraling downward at the outside after it enters the inlet; then, the gas flows upward in the center of the cyclone before it exits. When the gas enters the cyclone, its velocity undergoes a redistribution so that the tangential component of velocity increases with decreasing radius, as expressed by $V_{ct} \sim r^{-n}$. The tangential velocity in a cyclone may reach a value several times the average inlet gas velocity. Theoretical considerations indicate that n should be equal to 1.0 in the absence of wall friction. Actual measurements [Shepherd and Lapple, *Ind. Eng. Chem.* **31**: 972 (1939); **32**: 1246 (1940)], however, indicate that n may range from 0.5 to 0.7 over a large portion of the cyclone radius. Ter Linden [*Inst. Mech. Eng. J.* **160**: 235 (1949)] found n to be 0.52 for tangential velocities measured in the cylindrical portion of the cyclone at positions ranging from the radius of the gas outlet pipe to the radius of the outer wall. Although the velocity approaches zero at the wall, the boundary layer is sufficiently thin that pitot-tube measurements show relatively high tangential velocities there, as shown in Fig. 17-37. The radial velocity V_r is directed toward the center throughout most of the cyclone, except at the center, where it is directed outward. Superimposed on the "double spiral," there may be a "double eddy" [Van Tongran, *Mech. Eng.* **57**: 753 (1935); and Wellmann, *Feuerungstechnik* **26**: 137 (1938)] similar to that encountered in pipe coils. Measurements on cyclones of the type shown in Fig. 17-36 indicate, however, that such double-eddy velocities are small compared with the tangential velocity (Shepherd and Lapple, op. cit.). Recent analyses of flow patterns can be found in Hoffman et al., *Powder Technol.* **70**: 83 (1992); and Trefz and Muschelknautz, *Chem. Eng. Technol.* **16**: 153 (1993).

The inner vortex (often called the core of the vortex) rotates at a much higher velocity than the outer vortex. In the absence of solids, the radius of this inner vortex has been measured to be 0.4 to 0.8 r . With axial inlet cyclones, the inner core vortex is aligned with the axis of the gas outlet tube. With tangential or volute cyclone inlets, however, the vortex is not exactly aligned with the axis. The non-symmetric entry of the tangential or volute inlet causes the axis of the vortex to be slightly eccentric from the axis of the cyclone. This means that the bottom of the vortex is displaced some distance from the axis and can "pluck off" and reentrain dust from the solids

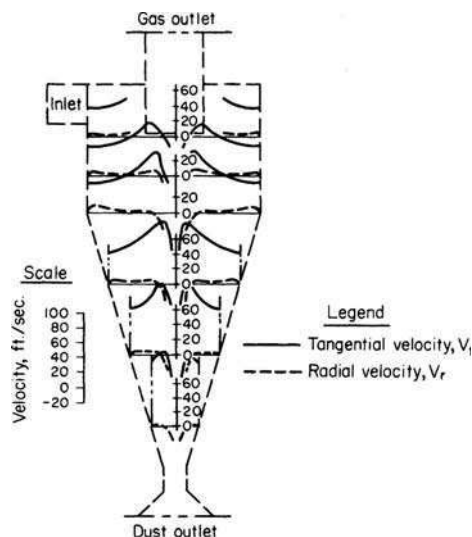


FIG. 17-37 Variation of tangential velocity and radial velocity at different points in a cyclone. [Ter Linden, *Inst. Mech. Eng. J.*, **160**, 235 (1949).]

sliding down the cyclone cone if the vortex gets too close to the wall of the cyclone cone.

At the bottom of the vortex, there is substantial turbulence as the gas flow reverses and flows up the middle of the cyclone into the gas outlet tube. As indicated above, if this region is too close to the wall of the cone, substantial reentrainment of the separated solids can occur. Therefore, it is very important that cyclone design take this into account.

The vortex of a cyclone will precess (or wobble) about the center axis of the cyclone. This motion can bring the vortex into close proximity to the wall of the cone of the cyclone and "pluck" off and reentrain the collected solids flowing down along the wall of the cone. The vortex may also cause erosion of the cone if it touches the cone wall. Sometimes an inverted cone or a similar device is added to the bottom of the cyclone in the vicinity of the cone and dipleg to stabilize and "fix" the vortex. If it is placed correctly, the vortex will attach to the cone and the vortex movement will be stabilized, thus minimizing the efficiency loss due to plucking the solids off the wall and erosion of the cyclone cone.

Hugi and Reh [*Chem. Eng. Technol.* **21**(9): 716–719 (1998)] have reported that (at high solids loadings) enhanced cyclone efficiency occurs when the solids form a coherent, stable strand at the entrance to a cyclone. The formation of such a strand is dependent upon several factors. They reported a higher cyclone efficiency for smaller ($d_{p,50} = 40 \mu\text{m}$) solids than for larger solids ($d_{p,50} = 125 \mu\text{m}$). This is not what theory would predict. However, they also found that the smaller particles formed coherent, stable strands more readily than the larger particles, which explained the reason for the apparent discrepancy.

Cyclone Efficiency The methods described below for pressure drop and efficiency calculations were given by Zenz in *Manual on Disposal of Refinery Wastes—Atmospheric Emissions*, chap. 11 (1975), American Petroleum Institute Publ. 931 and improved by Particulate Solid Research Inc. (PSRI), Chicago. Cyclones work by using centrifugal force to increase the gravity field experienced by the solids. They then move to the wall under the influence of their effectively increased weight. Movement to the wall is improved as the path the solids traverse under centrifugal flow is increased. This path is equated with the number of spirals the solids make in the cyclone barrel. Figure 17-38 gives the number of spirals N_s as a function of the maximum velocity in the cyclone. The maximum velocity may be either the inlet or the outlet velocity depending on the design. The equation for D_{pth} , the theoretical size particle removed by the cyclone at 50 percent collection efficiency, is

$$D_{pth} = \sqrt{\frac{9\mu_g B_c}{\pi N_s v_{max} (\rho_p - \rho_g)}}$$

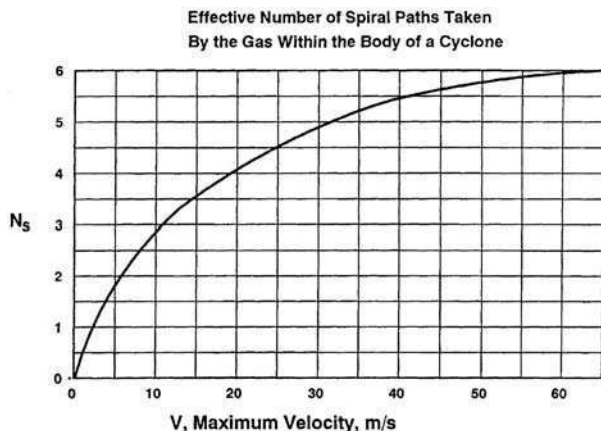


FIG. 17-38 N_s versus velocity—where the larger of either the inlet or outlet velocity is used.

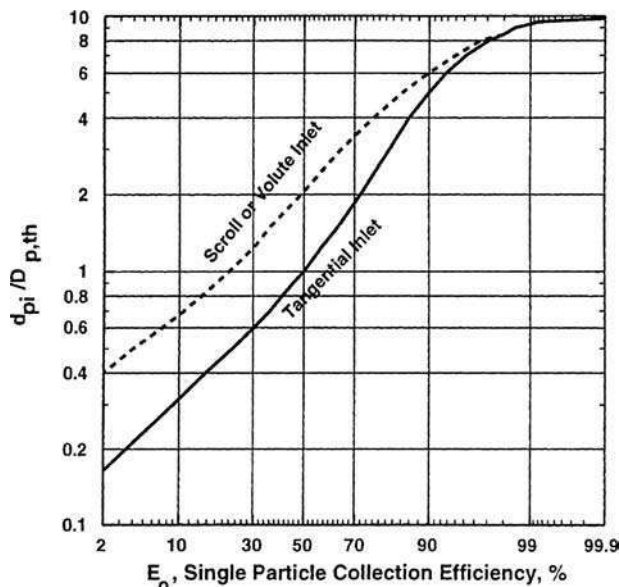


FIG. 17-39 Single particle collection efficiency curve. (Courtesy of PSRI, Chicago.)

This equation is a result of the residence time theory of particle collection. In this theory, the time that it takes for a particle to reach the wall is balanced by the time that a particle spends in the cyclone. The particle size that makes it to the wall by the time that it exits the cyclone is the particle size collected at 50 percent collection efficiency, D_{pth} .

When consistent units are used, the particle size calculated by the above equation will be in either meters or feet. The equation contains effects of cyclone size, gas velocity, gas viscosity, gas density, and particle density of the solids. In practice, a design curve such as given in Fig. 17-39 uses D_{pth} as the size at which 50 percent of solids of a given size are collected by the cyclone. The material entering the cyclone is divided into fractional sizes, and the collection efficiency for each size is determined. The total efficiency of collection is the sum of the collection efficiencies of the cuts.

The above applies for very dilute systems, usually on the order of 1 gr/ft³, or 2.3 g/m³ where 1 gr = (1/7000) lb. When denser flows of solids are present in the inlet gas, cyclone efficiency increases dramatically. This is thought to be due to the coarse particles colliding with fines as they move to the wall, which carry a large percentage of the finer particles along with them. Other explanations are that the solids have a lower drag coefficient or tend to agglomerate in multiparticle environments, thus effectively becoming larger particles. At very high inlet solids loadings, it is believed the gas simply cannot hold that much solid material in suspension at high centrifugal forces, and the bulk of the solids simply "condense" out of the gas stream.

The phenomenon of increasing efficiency with increasing loading is represented by Figs. 17-40 and 17-41 for Geldart group A and B solids, respectively (see beginning of Sec. 17). The initial efficiency of a particle size cut is found on the chart, and the parametric line is followed to the proper overall solids loading. The efficiency for that cut size is then read from the graph.

A single cyclone can sometimes give sufficient gas-solids separation for a particular process or application. However, solids collection efficiency can usually be enhanced by placing cyclones in series. Cyclones in series are typically necessary for most processes to minimize particulate emissions or to minimize the loss of expensive solid reactant or catalyst. Two cyclones in series are most common, but sometimes three cyclones in series are used. Series cyclones can be very efficient. In fluidized catalytic cracking regenerators, two stages of cyclones can give efficiencies of up to and even greater than 99.999 percent.

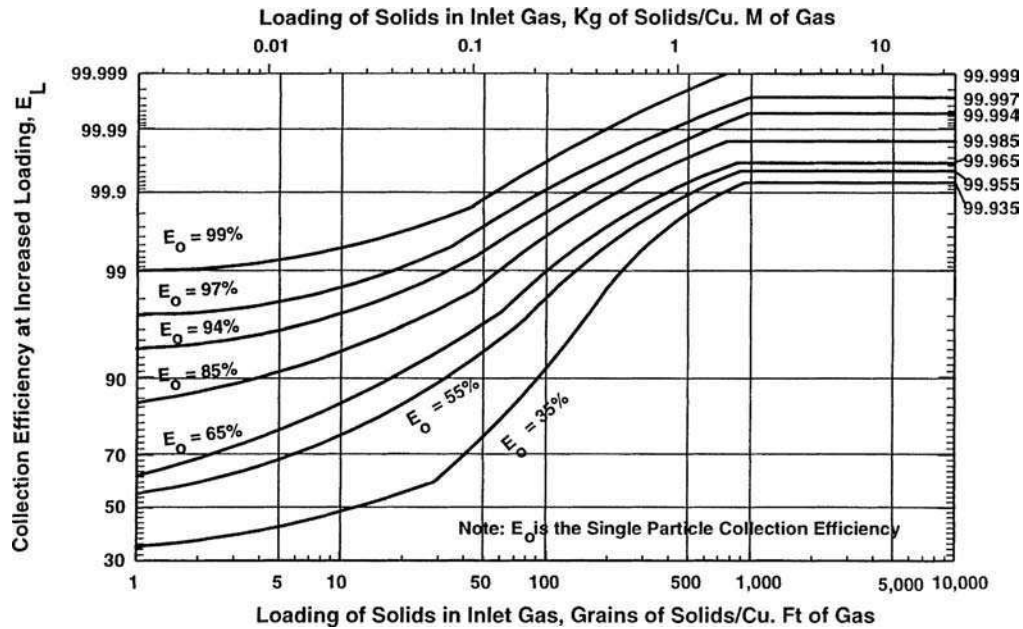


FIG. 17-40 Effect of inlet loading on collection efficiency for Geldart group A and group C particles. (Courtesy of PSRI, Chicago.)

Typically, first-stage cyclones will have an inlet gas velocity less than that of second-stage cyclones. The lower inlet velocity of first-stage cyclones results in lower particle attrition rates and lower wall erosion rates. After most of the solids are collected in the first stage, a higher velocity is generally used in second-stage cyclones to increase the centrifugal force on the solids and increase collection efficiency. Inlet erosion rates are generally low in the second stage because of the vastly reduced flux of solids into the second-stage cyclone. However, cone erosion rates in second-stage cyclones are much greater than in first-stage cyclones.

Pressure Drop Pressure drop is first determined by summing five pressure drop components associated with the cyclone.

1. *Inlet contraction*

$$\Delta P = 0.5\rho_g(v_{in}^2 - v_{vessel}^2 + Kv_{in}^2)$$

where K is taken from Table 17-3. Using SI units gives the pressure drop in Pa. In English units, the factor of 32.2 for g must be included. This loss is primarily associated with cyclones located in the freeboard of a fluidized bed. If the cyclone is located external to a vessel and the

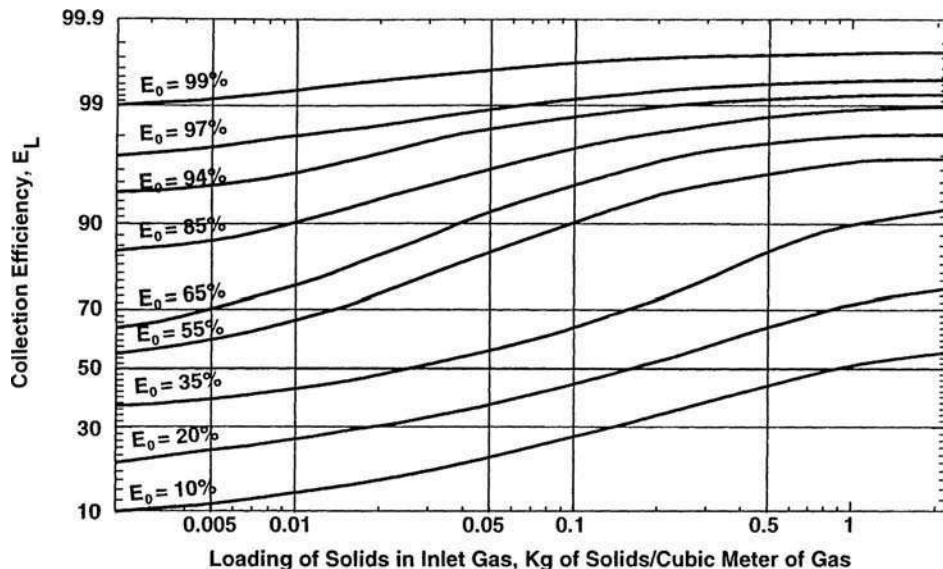


FIG. 17-41 Effect of inlet loading on collection efficiency (Geldart group B and group D) particles. (Courtesy of PSRI, Chicago.)

TABLE 17-3 *K* versus Area Ratio

Area ratio	<i>K</i>
0	0.50
0.1	0.47
0.2	0.43
0.3	0.395
0.4	0.35

high-pressure tap used to measure the cyclone pressure drop is in the inlet pipe before the cyclone, the measured pressure drop will not include this pressure loss, and this term should not be used to calculate total cyclone pressure drop. However, if the high-pressure tap to measure the cyclone pressure drop is located in the freeboard of the bed, this component will be included in the measured pressure drop, and it should be included in the calculation of the total cyclone pressure drop.

2. Particle acceleration

$$\Delta P = L v_{in} (v_{pin} - v_{pvessel})$$

For small particles, the velocity is taken as equal to the gas velocity and L is the solids loading, kg/m^3 .

3. *Barrel friction* The inlet diameter d_{in} is taken as $4 \times (\text{inlet area}) / \text{inlet perimeter}$. Then

$$\Delta P = \frac{2f \rho_g v_{in}^2 \pi D_c N_s}{d_{in}}$$

where the Reynolds number for determining the friction factor f is based on the inlet area.

4. Gas flow reversal

$$\Delta P = \frac{\rho_g v_{in}^2}{2}$$

5. Exit contraction

$$\Delta P = 0.5 \rho_g (v_{exit}^2 - v_c^2 + K v_{exit}^2)$$

where K is determined from Table 17-3 based on the area ratio of barrel and exit tube of the cyclone.

The total pressure drop is the sum of the five individual pressure drops.

However, the actual pressure drop observed turns out to be a function of the solids loading. The pressure drop is high when the gas is free of solids and then *decreases* as the solids loading increases up to about 3 kg/m^3 (0.2 lb/ft^3). The cyclone ΔP then begins to increase with loading. The cause of the initial decline is that the presence of solids decreases the tangential velocity of the gas [Yuu, *Chem. Eng. Sci.*, **33**, 1573 (1978)]. Figure 17-42 gives the actual pressure drop based on the loading. When solids are absent, the observed pressure drop can be 2.5 times the calculated pressure drop with solids present.

Cyclone Design Factors Cyclones are sometimes designed to meet specified pressure drop limitations. For ordinary installations, operating at approximately atmospheric pressure, fan limitations generally dictate a maximum allowable pressure drop corresponding to a cyclone inlet velocity in the range of 8 to 30 m/s (25 to 100 ft/s). Consequently, cyclones are usually designed for an inlet velocity of 15 to 20 m/s (50 to 65 ft/s), although this need not be strictly adhered to.

Because of the relatively high gas velocities at the inlet of cyclones, particle attrition in fluidized-bed systems is generally dominated by the attrition produced in the cyclone. In some catalytic systems with very expensive catalysts, the economics of the process can be dependent on low attrition losses. In such cases, reducing the inlet velocity of the cyclone will significantly reduce the attrition losses in the process. To compensate for the reduction in inlet velocity, the exit gas velocity will generally be increased (by reducing the diameter of the outlet tube) in order to maintain high cyclone efficiencies. Reducing the outlet tube diameter increases the outlet gas velocity and increases the velocity in the vortex of the cyclone—increasing collection efficiency. However, as the vortex velocity is increased, its length is also increased. Therefore, care must be taken to ensure that the cyclone is long enough to contain the increased vortex length. If it is not, the vortex can extend far into the cone and can entrain solids flowing on the sides of the cone as it comes near them.

Cyclone Roughness Large weld beads, etc., can also reduce cyclone efficiency. If the solids flow along the wall of a cyclone encounters a large protuberance such as a weld bead, the weld bead acts as a type of “ski jump” and causes the solids to be deflected farther into the center of the cyclone, where they can be thrown into the vortex and carried out of the cyclone. In small pilot or research cyclones, this is especially common, because the distance between the wall of the cyclone and the vortex tube is very small. Because of their

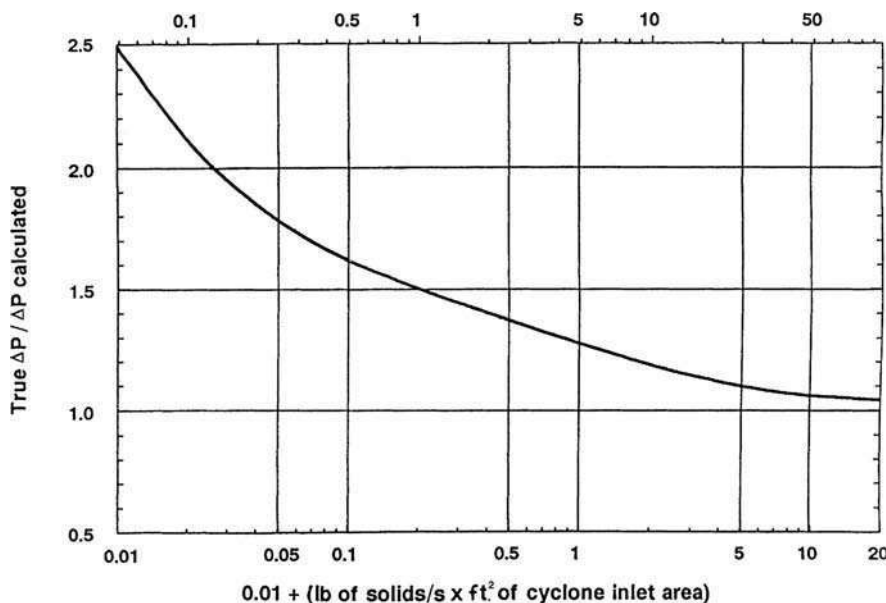


FIG. 17-42 Effect of cyclone inlet loading on pressure drop. (Courtesy of PSRI, Chicago.)

detrimental effect on cyclone efficiency, weld beads should be ground off to make the cyclone inner wall smooth.

In high-temperature processes, cyclones are often lined with refractory to both minimize heat loss and protect the metal surfaces from abrasion. These refractory surfaces are not as smooth as metal, but after a few days of operation, the refractory becomes smoother because of the abrasive action of the solids.

With very small laboratory or pilot cyclones, some solids (large polymer beads, spherical particles, etc.) can sometimes bounce off the cyclone wall immediately across from the cyclone inlet and be deflected into the vortex. Very large particles can be found in the gas outlet stream of the cyclone with these very small cyclones and with particles that bounce. To increase cyclone efficiency with these types of solids, the cyclone barrel diameter can be increased. This increases the distance between the cyclone vortex and the wall and prevents most of the solids from bouncing back into the vortex.

Theoretically, a primary design factor that can be utilized to control collection efficiency is the cyclone diameter. A smaller-diameter unit operating at a fixed pressure drop has a higher efficiency than a larger-diameter cyclone [Anderson, *Chem. Metall.* **40**: 525 (1933); Drijver, *Wärme* **60**: 333 (1937); and Whiton, *Power* **75**: 344 (1932); *Chem. Metall.* **39**: 150 (1932)]. In addition, smaller-diameter cyclones have a much smaller overall length. Small-diameter cyclones, however, will require multiple units in parallel to give the same capacity as a large cyclone. In such cases, the smaller cyclones generally discharge the dust into a common receiving hopper [Whiton, *Trans. Am. Soc. Mech. Eng.* **63**: 213 (1941)]. However, when cyclones discharge into a common hopper, there is a tendency of the gas to produce "cross-talk." This occurs when the gas exiting from one small cyclone passes up the exit of an adjoining cyclone, thus reducing efficiency. Various types of mechanical devices are generally added to the bottom of these small cyclones in parallel to reduce the cross-talk. The final cyclone design involves a compromise between collection efficiency and the complexity of equipment. It is customary to design systems for a single cyclone for a given capacity, resorting to multiple parallel units only if the predicted collection efficiency is inadequate for a single unit or single units in series.

Reducing the gas outlet diameter should increase both collection efficiency and pressure drop. To exit the cyclone, gas must enter the cyclonic flow associated with the outlet tube. If the outlet diameter is reduced, the outlet vortex increases in length to compensate. Therefore, when the outlet area is less than the inlet area, the length of the cyclone must increase. Too short a cyclone is associated with erosion of the cone and reentrainment of solids into the exit flow. Table 17-4 gives the required increase in cyclone length as a function of outlet-to-inlet area. The cyclone length is measured centrally along a cylinder 10 cm larger than the inner diameter of the outlet tube to prevent interference with the cone. If the cone interferes with this "extended vortex," the barrel must be lengthened.

As discussed above, theoretically a smaller-diameter cyclone should be able to collect smaller particles because it can develop a higher centrifugal force. However, using smaller cyclones generally means that many have to be used in parallel to accommodate large gas flows. The problem with parallel cyclones (as indicated above) is that it is difficult to get even distribution of solids into all the cyclones. If maldistribution occurs, this can cause inefficiencies that can negate the natural advantage of the smaller cyclones.

Cyclone diameters can be very large. Perhaps the largest cyclones are those used in circulating fluidized-bed combustors, where cyclone diameters approach 10 m. Large-diameter cyclones also result in very long cyclones, and so these large-diameter, long-length cyclones are

really not feasible as internal cyclones in fluidized beds (they make the vessel too tall).

The minimum cone angle of the cyclone should be 60°. It is generally greater, with steeper cone angles appropriate to materials that are more cohesive. The cyclone inlet is usually rectangular (more efficient at getting material to the wall), but in some cases has been circular. In either case, projection of the inlet flow path should never cause interference with the outlet tube. This generally means that the inlet width of a cyclone should always be less than the distance between the wall and the outside diameter of the outlet tube. If a very heavy solids loading is anticipated, the barrel diameter should be increased slightly to minimize interference with the outlet gas tube.

Collection efficiency is normally increased by increasing the gas throughput (Drijver, op. cit.). However, if the entering dust is agglomerated, high gas velocities may cause breakup of the agglomerated solids in the cyclone, so that efficiency remains the same or actually decreases. Also, variations in design proportions that result in increased collection efficiency with dispersed dusts may be detrimental with agglomerated dusts. Kalen and Zenz [*Am. Inst. Chem. Eng. Symp. Ser.* **70**(137): 388 (1974)] report that collection efficiency increases with increasing gas inlet velocity up to a minimum tangential velocity at which dust is either reentrained or not deposited because of saltation. Koch and Licht [*Chem. Eng.* **84**(24): 80 (1977)] estimate that for typical cyclones the saltation velocity is consistent with cyclone inlet velocities in the range of 15 to 27 m/s (50 to 90 ft/s). Lapple (private communication) reports that in cyclone tests with talc dust, collection efficiency increased steadily as the inlet velocity was increased up to a maximum of 52 m/s (170 ft/s). With ilmenite dust, which was much more strongly flocculated, efficiency decreased over the same inlet velocity range. In later experiments with well-dispersed talc dust, collection efficiency continued to increase at inlet velocities up to the maximum used, 82 m/s (270 ft/s).

Another effect of increasing the cyclone inlet gas velocity is that friable materials may disintegrate (or attrit) as they hit the cyclone wall at high velocity. Thus, the increase in efficiency associated with increased velocity may be more than lost due to generation of fine attrited material that the cyclone cannot contain.

Cyclones can be either placed in the freeboard above the fluidized bed or located outside of the fluidized-bed vessel. There are advantages and disadvantages to each type of placement, and the optimum type of placement depends on what is best for a particular process.

Internal cyclones have the advantages that they require no inlet piping (their inlets can be open to the freeboard) and no high-pressure shell, and they have straight cyclone diplegs. Internal cyclones are generally smaller in diameter than external cyclones because their size is limited by the headspace available in the freeboard above the fluidized bed. These size limitations result in using several smaller cyclones in parallel instead of one large cyclone. In addition, it is difficult to aerate second-stage cyclone diplegs (generally an advantageous technique) when internal cyclones are used. Aerating secondary cyclone diplegs can improve the operation of the diplegs significantly.

The advantages of external cyclones are that (1) they can be much larger than internal cyclones, (2) they are more accessible than internal cyclones, and (3) their diplegs can be aerated more easily. The disadvantages of external cyclones are that (1) they require a pressure shell and (2) external cyclone diplegs generally require a section with an angled or a horizontal pipe to return the solids to the bed. The angled or horizontal dipleg sections can result in poor dipleg operation, if not designed correctly.

Cyclones in series may be justified under the following circumstances:

1. The dust has a broad size distribution, including particles under 10 to 15 μm as well as larger and possibly abrasive particles. A large low-velocity cyclone may be used to remove the coarse particles ahead of a unit with small-diameter multiple tubes.
2. The dust is composed of fine particles but is highly flocculated or tends to flocculate in preceding equipment and in the cyclones themselves. Efficiencies predicted on the basis of ultimate particle size will be highly conservative.
3. The dust is relatively uniform, and the efficiency of the second-stage cyclone is not greatly lower than that of the first stage.

TABLE 17-4 Required Cyclone Length as a Function of Area Ratio

$A_{\text{out}}/A_{\text{in}}$	Length below outlet tube/ D_c
>1.0	2.0
0.8	2.2
0.6	2.6
0.4	3.2

4. Dependable operation is critical. Second-stage or even third-stage cyclones may be used as backup.

Cyclone Inlets The design of the cyclone inlet can greatly affect cyclone performance. It is generally desired to have the width of the inlet B_c as narrow as possible so that the entering solids will be as close as possible to the cyclone wall where they can be collected. However, narrow inlet widths require that the height of the inlet H be very long in order to give an inlet area required for the desired inlet gas velocities. Therefore, a balance between narrow inlet widths and the length of the inlet height has to be struck. Typically, low-loading cyclones (cyclones with inlet loadings less than approximately 2 to 5 kg/m³) have height/width ratios H/B_c of between 2.5 and 3.0. For high-loading cyclones, this inlet aspect ratio can be increased to as high as 7 or so with the correct design. Such high inlet aspect ratios require that the cyclone barrel length increase.

A common cyclone inlet is a rectangular tangential inlet with a constant area along its length. This type of inlet is satisfactory for many cyclones, especially those operating at low solids loadings. However, a better type of inlet is one in which the inner wall of the inlet is angled toward the outer cyclone wall at the cyclone inlet. This induces solids momentum toward the outer wall of the cyclone. The bottom wall of the inlet is angled downward so that the area decrease along the inlet flow path is not too rapid and acceleration is controlled. In addition, the entire inlet can be angled slightly downward to give enhanced efficiencies. This type of inlet is superior to the constant-area tangential inlet, especially for higher solids loadings (greater than 2 to 5 kg/m³).

Hugi and Reh [*Chem. Eng. Technol.* **21**(9):716–719 (1998)] report that continuous acceleration of the solids throughout the inlet is desired for improved efficiency and that the angled inlet described above achieves this. If the momentum of the solids is sufficient and the solids are continuously accelerating along the length of the inlet, the stable, coherent strand important for high collection efficiencies is produced.

The best inlet for high solids loadings is the volute cyclone inlet. At high inlet loadings (above approximately 2 to 3 kg/m³) in a tangential cyclone inlet, the gas-solids stream expands rapidly from its minimum width at the point of contact. This rapid expansion disturbs the laminar gas flow around the gas outlet tube and causes flow separation around the tube. At some loadings, the inlet stream can expand to such an extent that the solids can impact the gas outlet tube. Both effects result in lowered cyclone efficiency. However, when a volute inlet is used, the expanding solids stream is farther from the gas outlet tube and enters at an angle so that the solids do not induce as much flow separation or asymmetric flow around the gas outlet tube. Therefore, cyclone efficiency is not affected to as great a degree. If a tangential cyclone is used at high solids loadings, an extra distance between the gas outlet tube and the cyclone wall should be designed into the cyclone to prevent the solids from impacting on the gas outlet tube. At low solid loadings, the impacting on the gas outlet tube does not occur. Because tangential cyclone inlets are less expensive than volute inlets, the tangential cyclone is typically utilized for low loadings—and the volute inlet cyclone is used for high loadings.

The nature of the gas solids flow in the inlet ducting to the cyclone can affect cyclone efficiency significantly. If the solids in the inlet salt out on the bottom and result in dune formation and the resulting unsteady or pulsing flow, cyclone efficiency is adversely affected. To minimize the possibility of this occurring, it is recommended that the inlet line to the cyclone operate above the saltation velocity [Gauthier et al., in *Circulating Fluidized Bed Technology III*, Basu, Horio, and Hasatani (eds.), 1990, pp. 639–644], which will prevent the solids from operating in the dune or pulsing flow regime. If this is not possible, then the inlet line can be angled downward (approximately 15° to 20°) to let gravity assist in the flow of the solids. Keeping the inlet line as short as possible can also minimize any pulsing of the solids flow.

A cyclone will operate equally well on the suction or pressure side of a fan if the dust receiver is airtight. Probably the greatest single cause of poor cyclone performance, however, is the leakage of air into the dust outlet of the cyclone. A slight air leak at this point can result in a tremendous drop in collection efficiency, particularly with fine dusts. For a cyclone operating under pressure, air leakage at this point is objectionable primarily because of the local dust nuisance created.

For batch operation, an airtight hopper or receiver may be used. For continuous withdrawal of collected dust, a rotary star valve, a double-lock valve, or a screw conveyor may be used, the latter only with fine dusts. A collapsible open-ended rubber tube can be used for cyclones operating under slight negative pressure. Mechanical trickle and flap valves at the end of cyclone diplegs can also be used for continuous withdrawal into fluidized beds or into the freeboard of fluidized beds. Open diplegs simply immersed in a fluidized bed can be used in cases where start-up losses are not excessive and are the simplest type of discharge system returning solids to a fluidized bed (see “Fluidized-Bed Systems: Solids Discharge”). Special pneumatic unloading devices can also be used with dusts. In any case it is essential that sufficient unloading and receiving capacity be provided to prevent collected material from accumulating in the cyclone.

Solids Loading Cyclones can collect solids over a wide range of loadings. Traditionally, solids loadings have been reported as either kilograms of solids per cubic meter of gas (kg/m³), or as kilograms of solids per kilogram of gas (kg/kg_g). However, loading based on mass is probably not the best way to report solids loadings for cyclones. This is so because the volume of solids processed by a cyclone at the same mass loading can vary greatly, depending on the density of the solids. For example, many polymers have a bulk density of approximately 400 kg/m³, and iron ore has a bulk density of approximately 2400 kg/m³. This is a factor of 6. Therefore, a cyclone operating with polymer would have to process 6 times the volume of solids that a cyclone operating with iron ore would process at the same mass loading. If the cyclone operating with the polymer were designed to operate at high loadings on a mass basis, it would probably plug. In addition, the diplegs below the cyclone operating with the polymer may experience operational problems because of the high volumetric loading.

At ambient conditions, cyclones have been operated at solids loadings as low as 0.02 kg/m³ (0.0125 kg/kg_g) and as high as 64 kg/m³ (50 kg/kg_g) or more. This is a factor of 3200. In general, cyclone efficiency increases with increasing solids loading. This is so because at higher loadings, very fine particles are trapped in the interstices of the larger particles, and this entrapment increases the collection efficiency of the small particles. Even though collection efficiencies are increased with increased loading, cyclone loss rates are also increased as loading is increased. This is so because the cyclone efficiency increase is almost always less than the increase in the solids loading.

Generally cone-and-disk baffles, helical guide vanes, etc., placed inside a cyclone, will have a detrimental effect on performance. However, a few of these devices do have some merit under special circumstances. Although an inlet vane will reduce pressure drop (and may result in significant erosion), it generally causes a correspondingly greater reduction in collection efficiency. Its use is recommended only when collection efficiency is normally so high as to be a secondary consideration, and when it is desired to decrease the resistance of an existing cyclone system for purposes of increased air handling capacity or when floorspace or headroom requirements are controlling factors. If an inlet vane is used, it is advantageous to increase the gas exit duct length inside the cyclone chamber.

A disk or cone baffle located beneath the gas outlet duct may be beneficial if air in-leakage at the dust outlet cannot be avoided. A heavy chain suspended from the gas outlet duct has been found beneficial to minimize dust buildup on the cyclone walls in certain circumstances. Such a chain should be suspended from a swivel so that it is free to rotate without twisting. Substantially all devices that have been reported to reduce pressure drop do so by reducing spiral velocities in the cyclone chamber and consequently result in reduced collection efficiency.

At low dust loadings, the pressure in the dust receiver of a single cyclone will generally be lower than in the gas outlet duct. Increased dust loadings will increase the pressure in the dust receiver. Such devices as cones, disks, and inlet vanes will generally cause the pressure in the dust receiver to exceed that in the gas outlet duct. A cyclone will operate as well in a horizontal position as in a vertical position. However, departure from the normal vertical position results in an increasing tendency to plug the dust outlet. If the dust outlet becomes plugged, collection efficiency will, of course, be low. If the

cyclone exit duct must be reduced to tie in with proposed duct sizes, the transition should be made at least five diameters downstream from the cyclone and preferably after a bend. In the event that the transition must be made closer to the cyclone, a Greek cross should be installed in the transition piece to avoid excessive pressure drop.

Cyclone Length As described above, the cyclone length should be great enough to contain the vortex below the gas outlet tube. It is generally advisable to have the cyclone somewhat longer than required so that modifications to the gas outlet tube can be made if required. Either the barrel or the cone can be increased in length to contain the vortex. However, cyclone barrels can be made too long. If the barrel is too long, the rotating spiral of solids along the wall can lose its momentum. When this happens, the solids along the wall can be reentrained into the rotating gas in the barrel, and cyclone efficiency will be reduced.

Hoffman et al. [*AIChE J.* 47(11): 2452–2460 (2001)] studied the effect of cyclone length on cyclone efficiency and showed that the efficiency of a cyclone increases with length. However, they also found that after a certain length, cyclone efficiency decreased. They reported that cyclone efficiency suddenly decreased after a certain cyclone length, which in their cyclone was at a length/diameter ratio of 5.65.

(Although many researchers employ this length/diameter ratio as a correlating parameter to make the length parameter dimensionless, it is likely that it is the actual length of the cyclone that is important.) Hoffman et al. stated that the probable reason for the sudden decrease in cyclone efficiency was the central vortex touching and turning on the cyclone cone. When this occurred, the efficiency collapsed, causing increased solids reentrainment.

Hoffman et al. also reported that cyclone pressure drop decreased with increasing cyclone length. This probably occurs for the same reason that cyclone pressure drop decreases with increasing cyclone loading. For long cyclones, the increased length of the cyclone wall results in a longer path for the gas to travel. This creates greater resistance to the flow of the gas in the cyclone (much as a longer pipe produces greater resistance to gas flow than a shorter pipe) that results in reducing the tangential velocity in the cyclone and, therefore, the cyclone pressure drop.

Commercial Equipment and Operations Simple cyclones are available in a wide variety of shapes ranging from long, slender units to short, large-diameter units. The body may be conical or cylindrical, and entrances may be involute or tangential and round or rectangular.

Some of the special types of commercial cyclones are shown in Fig. 17-43. In the Multiclone, a spiral motion is imparted to the gas by

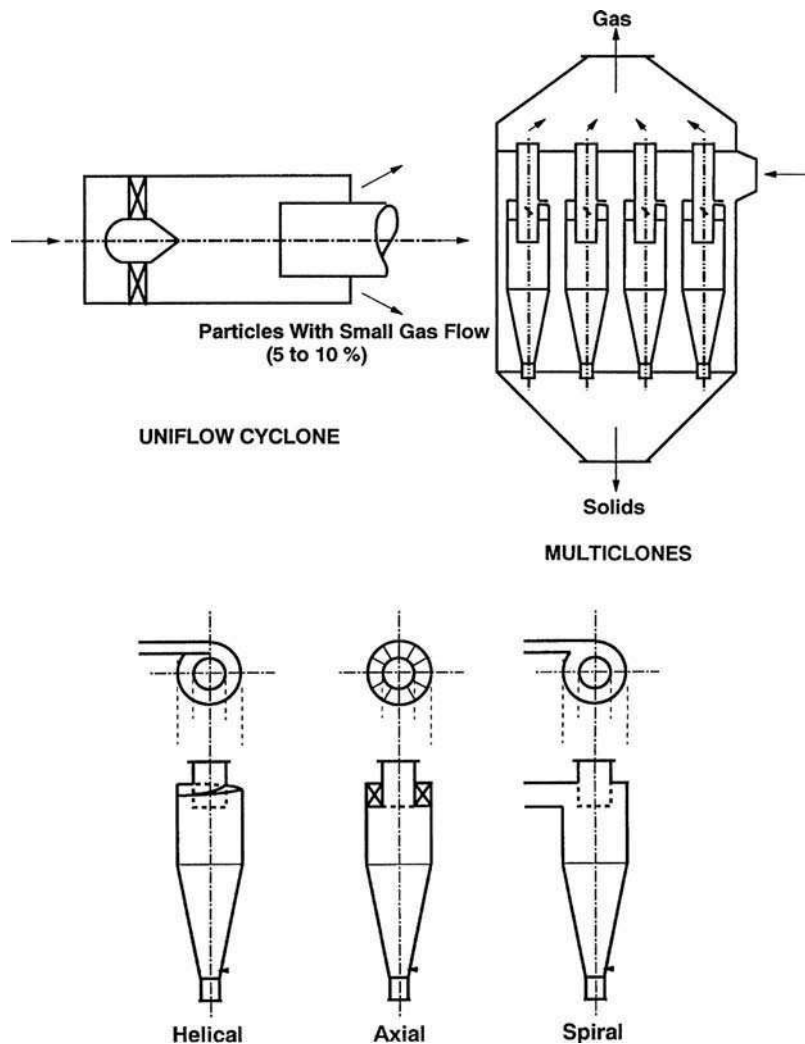


FIG. 17-43 Some commercial cyclone design alternatives. (Courtesy of PSRI, Chicago.)

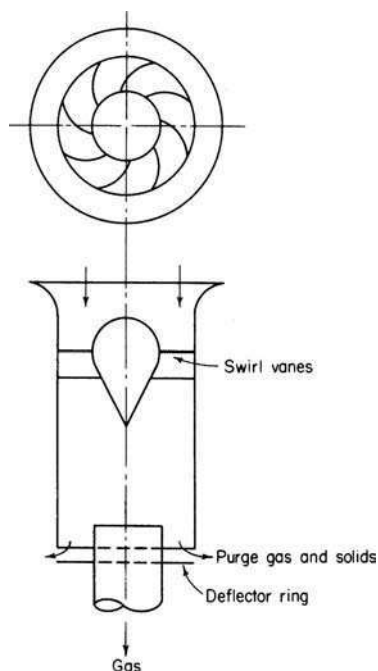


FIG. 17-44 Uniflow cyclone. [Ter Linden, *Inst. Mech. Eng. J.*, **160**, 233 (1949).]

annular vanes, and it is furnished in multiple units of 15.2- and 22.9-cm (6- and 9-in) diameter. Its largest field of application has been in the collection of fly ash from steam boilers. The tubes are commonly constructed of cast iron and other abrasion-resistant alloys. In addition to the conventional reverse-flow cyclones, some use is made of uniflow, or straight-through, cyclones, in which the gas and solids discharge at the same end (Fig. 17-44). These devices generally act as concentrators; the concentrated dust, together with the inlet gas, is discharged at the periphery, while the clean gas passes out through the center tube. The purge gas and concentrated dust enter a conventional cyclone for final separation. The straight-through cyclones are usually designed into multiple-tube units, or they can be used to try to minimize the residence time of the gas and solids to minimize unwanted reactions.

Most processes operate at high temperatures and/or high pressures. Therefore, it is important to know how cyclones operate at these conditions. Efficient cyclones are able to collect very small particles sizes. Therefore, cyclone efficiency is proportional to $1/D_{\text{pth}}$.

$$E_o \propto \frac{1}{D_{\text{pth}}} = \sqrt{\frac{\pi N_s U (\rho_p - \rho_g)}{9 \mu B_c}}$$

The effects of temperature and pressure manifest themselves in how they affect the gas density and gas viscosity. From the equation above, it can be seen that cyclone efficiency is theoretically related to gas density and gas viscosity as

$$E_o \propto \sqrt{\frac{\rho_p - \rho_g}{\mu}}$$

As pressure is increased, gas density will increase. However, the term $\rho_p - \rho_g$ does not change with increases in gas density because particle density is so much greater than the gas density (typically about 2000 kg/m³ versus approximately 20 kg/m³ at high pressure) that it dominates this term. Therefore, it is expected that gas density would have little or no effect on cyclone efficiency. Conversely, cyclone efficiency would be expected to decrease with system temperature because gas viscosity increases with increasing pressure.

Knowlton and Bachovchin [*Coal Processing Technol.* **4**:122–127 (1978)] studied the effect of pressure on cyclone performance and

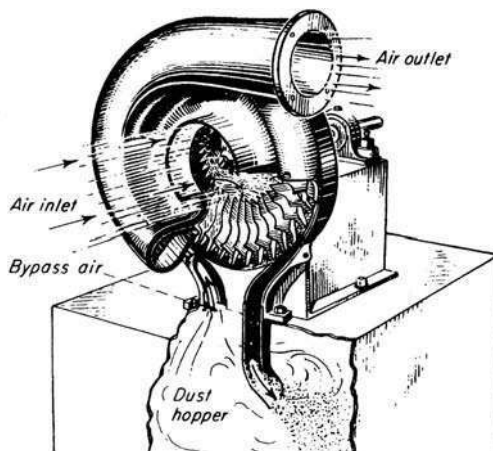


FIG. 17-45 Typical mechanical centrifugal separator. Type D Rotocyclone (cut-away view). (American Air Filter Co., Inc.)

found little change in overall cyclone efficiency with pressure over a pressure range from 0 to 55 barg. However, fractional efficiency curves from the same study showed that cyclone efficiency decreased with pressure for particle sizes less than about 20 to 25 μm . For particle sizes greater than about 25 μm there was no effect of pressure on cyclone efficiency.

The effect of temperature on cyclone efficiency was studied by both Parker et al. [*J. Environ. Sci. and Technol.* **15** (4):451 (1981)] and Patterson and Munz [*Canadian J. Chem. Eng.* **67**:321 (1989)]. Both studies showed that cyclone efficiency decreased with increasing gas viscosity. As with the studies at high pressure, Patterson and Munz (1989) reported that only the collection efficiency of particles less than about 10 μm was reduced because of operation at high temperature.

Mechanical Centrifugal Separators A number of collectors in which the centrifugal field is supplied by a rotating member are commercially available. In the typical unit shown in Fig. 17-45, the exhaustor or fan and dust collector are combined as a single unit. The blades are especially shaped to direct the separated dust into an annular slot leading to the collection hopper while the cleaned gas continues to the scroll.

Although no comparative data are available, the collection efficiency of units of this type is probably comparable with that of the single-unit high-pressure-drop-cyclone installation. The clearances are smaller and the centrifugal fields higher than in a cyclone, but these advantages are probably compensated for by the shorter gas path and the greater degree of turbulence with its inherent reentrainment tendency. The chief advantage of these units lies in their compactness, which may be a prime consideration for large installations or plants requiring a large number of individual collectors. Caution should be exercised when attempting to apply this type of unit to a dust that shows a marked tendency to build up on solid surfaces, because of the high maintenance costs that may be encountered from plugging and rotor unbalancing.

Particulate Scrubbers Wet collectors, or scrubbers, form a class of devices in which a liquid (usually water) is used to assist or accomplish the collection of dusts or mists. Such devices have been in use for well over 100 years, and innumerable designs have been or are offered commercially or constructed by users. Wet-film collectors logically form a separate subcategory of devices. They comprise inertial collectors in which a film of liquid flows over the interior surfaces, preventing reentrainment of dust particles and flushing away the deposited dust. Wetted-wall cyclones are an example [Stairmand, *Trans. Inst. Chem. Eng.* **29**, 356 (1951)]. Wet-film collectors have not been studied systematically but can probably be expected to perform much as do equivalent dry inertial collectors, except for the benefit of reduced reentrainment.

In particulate scrubbers, the liquid is dispersed into the gas as a spray, and the liquid droplets are the principal collectors for the dust particles. Depending on their design and operating conditions, particulate scrubbers can be adapted to collecting fine as well as coarse particles. Collection of particles by the drops follows the same principles illustrated in Fig. 17-35. Various investigations of the relative contributions of the various mechanisms have led to the conclusion that the predominant mechanism is inertial deposition. Flow-line interception is only a minor mechanism in the collection of the finer dust particles by liquid droplets of the sizes encountered in scrubbers. Diffusion is indicated to be a relatively minor mechanism for the particles larger than $0.1 \mu\text{m}$ that are of principal concern. Thermal deposition is negligible. Gravitational settling is ineffective because of the high gas velocities and short residence times used in scrubbers. Electrostatic deposition is unlikely to be important except in cases in which the dust particles or the water, or both, are being deliberately charged from an external power source to enhance collection. Deposition produced by Stefan flow can be significant when water vapor is condensing in a scrubber.

Despite numerous claims or speculations that wetting of dust particles by the scrubbing liquid plays a major role in the collection process, there is no unequivocal evidence that this is the case. The issue is whether wetting is an important factor in the adherence of a particle to a collecting droplet upon impact. From the body of general experience, it can be inferred that wettable particles probably are not collected much, if any, more readily than nonwettable particles of the same size. However, the available experimental techniques have not been adequate to permit any direct test to resolve the question. Changing from a wettable to a nonwettable test aerosol or from one scrubbing liquid to another is virtually certain to introduce other (and possibly unknown) factors into the scrubbing process. The most informative experimental studies appear to be some by Weber [Staub, English trans., **28**, 37 (November 1968); **29**, 12 (July 1969)], who bombarded single drops of various liquids with dust particles at different velocities and studied the behavior at impact by means of high-speed photography. Dust particles hitting the drops were invariably retained by the latter, regardless of their wettability by the liquid used.

The use of wetting agents in scrubbing water is equally controversial, and there has been no clear demonstration that it is beneficial.

A particulate scrubber may be considered as consisting of two parts: (1) a contactor stage, in which a spray is generated and the dust-laden gas stream is brought into contact with it; and (2) an entrainment separation stage, in which the spray and deposited dust particles are separated from the cleaned gas. These two stages may be separate or physically combined. The contactor stage may be of any form intended to bring about effective contacting of the gas and spray. The spray may be generated by the flow of the gas itself in contact with the liquid, by spray nozzles (pressure-atomizing or pneumatic-atomizing), by a motor-driven mechanical spray generator, or by a motor-driven rotor through which both gas and liquid pass.

Entrainment separation is accomplished with inertial separators, which are usually cyclones or impingement separators of various forms. If properly designed, these devices can remove virtually all droplets of the sizes produced in scrubbers. However, reentrainment of liquid can take place in poorly designed or overloaded separators.

Scrubber Types and Performance The diversity of particulate scrubber designs is so great as to defy any detailed and self-consistent system of classification based on configuration or principle of operation. However, it is convenient to characterize scrubbers loosely according to prominent constructional features, even though the modes of operation of different devices in a group may vary widely.

A relationship of power consumption to collection efficiency is characteristic of all particulate scrubbers. Attaining increased efficiency requires increased power consumption, and the power consumption required to attain a given efficiency increases as the particle size of the dust decreases. Experience generally indicates that the power consumption required to provide a specific efficiency on a given dust does not vary widely even with markedly different devices. The extent to which this generalization holds true has not been fully explored, but the known extent is sufficient to suggest that the underlying collection mechanism may be essentially the same in all types of particulate scrubbers.

Since some relationship of power consumption to performance appears to be a universal characteristic of particulate scrubbers, it is useful to characterize such devices broadly according to the source from which the energy is supplied to the gas-liquid contacting process. The energy may be drawn from (1) the gas stream itself, (2) the liquid stream, or (3) a motor driving a rotor. For convenience, devices in these classes may be termed, respectively, (1) gas-atomized spray scrubbers, (2) spray, or preformed-spray, scrubbers, and (3) mechanical scrubbers. In the spray scrubbers, all the energy may be supplied from the liquid, using a pressure nozzle, but some or all may be provided by compressed air or steam in a two-fluid nozzle or by a motor driving a spray generator.

Particulate scrubbers may also be classed broadly into low-energy and high-energy scrubbers. The distinction between the two classes is arbitrary, since the devices are not basically different and the same device may fall into either class depending on the amount of power it consumes. However, some differences in configuration are sometimes necessary to adapt a device for high-energy service. No specific level of power consumption is commonly agreed upon as the boundary between the two classes, but high-energy scrubbers may be regarded as those using sufficient power to give substantial efficiencies on sub-micrometer particles.

Scrubber Performance Models A number of investigators have made theoretical studies of the performance of venturi scrubbers and have sought to produce performance models, based on first principles, that can be used to design a unit for a given duty without recourse to experimental data other than the particle size and size distribution of the dust. Among these workers are Calvert [Am. Inst. Chem. Eng. J., **16**, 392 (1970); J. Air Pollut. Control Assoc., **24**, 929 (1974)], Boll [Ind. Eng. Chem. Fundam., **12**, 40 (1973)], Goel and Hollands [Atmos. Environ., **11**, 837 (1977)], and Yung et al. [Environ. Sci. Technol., **12**, 456 (1978)]. Comparatively few efforts have been made to model the performance of scrubbers of types other than the venturi, but a number of such models are summarized by Yung and Calvert (U.S. EPA-600/8-78-005b, 1978).

The various venturi-scrubber models embody a variety of assumptions and approximations. The solutions of the equations for particulate collection must in general be determined numerically, although Calvert et al. [J. Air Pollut. Control Assoc., **22**, 529 (1972)] obtained an explicit equation by making some simplifying assumptions and incorporating an empirical constant that must be evaluated experimentally; the constant may absorb some of the deficiencies in the model. Although other models avoid direct incorporation of empirical constants, use of empirical relationships is necessary to obtain specific estimates of scrubber collection efficiency. One of the areas of greatest uncertainty is the estimation of droplet size.

Most of the investigators have assumed the effective drop size of the spray to be the Sauter (surface-mean) diameter and have used the empirical equation of Nukiyama and Tanasawa [Trans. Soc. Mech. Eng., Japan, **5**, 63 (1939)] to estimate the Sauter diameter:

$$D_o = \frac{1920\sqrt{\sigma_L}}{V_o\sqrt{\rho_L/62.3}} + 75.4 \left(\frac{\mu_L}{\sqrt{\sigma_L\rho_L/62.3}} \right)^{0.45} \left(\frac{1000Q_L}{Q_C} \right)^{1.5} \quad (17-3)$$

where D_o = drop diameter, μm ; V_o = gas velocity, ft/s ; σ_L = liquid surface tension, dyn/cm ; ρ_L = liquid density, lb/ft^3 ; μ_L = liquid viscosity, cP ; Q_L = liquid flow rate, ft^3/s ; and Q_C = gas flow rate, ft^3/s .

The Nukiyama-Tanasawa equation, which is not dimensionally homogeneous, was derived from experiments with small, internal-mix pneumatic atomizing nozzles with concentric feed of air and liquid (Lapple et al., "Atomization: A Survey and Critique of the Literature," Stanford Res. Inst. Tech. Rep. No. 6, AD 821-314, 1967; Lapple, "Atomization," in McGraw-Hill Encyclopedia of Science and Technology, 5th ed., vol. 1, McGraw-Hill, New York, 1982, p. 558). The effect of nozzle size on the drop size is undefined. Even within the range of parameters for which the relationship was derived, the drop sizes reported by various investigators have varied by twofold to threefold from those predicted by the equation (Boll, op. cit.). The Nukiyama-Tanasawa equation has, nevertheless, been applied to large venturi and orifice scrubbers with configurations radically different from those of the atomizing nozzles for which the equation was originally developed.

Primarily because of the lack of adequate experimental techniques (particularly, the production of appropriate monodisperse aerosols), there has been no comprehensive experimental test of any of the venturi-scrubber models over wide ranges of design and operating variables. The models for other types of scrubbers appear to be essentially untested.

Contacting Power Correlation A scrubber design method that has achieved wide acceptance and use is based on correlation of the collection efficiency with the power dissipated in the gas-liquid contacting process, which is termed "contacting power." The method originated from an investigation by Lapple and Kamack [*Chem. Eng. Prog.*, **51**, 110 (1955)] and has been extended and refined in a series of papers by Semrau and coworkers [*Ind. Eng. Chem.*, **50**, 1615 (1958); *J. Air Pollut. Control Assoc.*, **10**, 200 (1960); **13**, 587 (1963); U.S. EPA-650/2-74-108, 1974; U.S. EPA-600/2-77-234, 1977; *Chem. Eng.*, **84**(20), 87 (1977); and "Performance of Particulate Scrubbers as Influenced by Gas-Liquid Contactor Design and by Dust Flocculation," EPA-600/9-82-005c, 1982, p. 43]. Other workers have made extensive independent studies of the correlation method [Walker and Hall, *J. Air Pollut. Control Assoc.*, **18**, 319 (1968)], and numerous studies of narrower scope have been made. The major conclusion from these studies is that the collection efficiency of a scrubber on a given dust is essentially dependent only on the contacting power and is affected to only a minor degree by the size or geometry of the scrubber or by the way in which the contacting power is applied. This contacting-power rule is strictly empirical, and the full extent of its validity has still not been explored. It has been best verified for the class of gas-atomized spray scrubbers, in which the contacting power is derived from the gas stream and takes the form of gas pressure drop. Tests of the equivalence of contacting power supplied from the liquid stream in pressure spray nozzles have been far less extensive and are strongly indicative but not yet conclusive. Evidence for the equivalence of contacting power from mechanically driven devices is also indicative but extremely limited in quantity.

Contacting power is defined as the power per unit of volumetric gas flow rate that is dissipated in gas-liquid contacting and is ultimately converted to heat. In the simplest case, in which all the energy is obtained from the gas stream in the form of pressure drop, the contacting power is equivalent to the friction loss across the wetted equipment, which is termed "effective friction loss," F_E . The pressure drop may reflect kinetic-energy changes rather than energy dissipation, and pressure drops that result solely from kinetic-energy changes in the gas stream do not correlate with performance. Likewise, any friction losses taking place across equipment that is operating dry do not contribute to gas-liquid contacting and do not correlate with performance. The gross power input to a scrubber includes losses in motors, drive shafts, fans, and pumps that obviously should be unrelated to scrubber performance.

The effective friction loss, or "gas-phase contacting power," is easily determined by direct measurements. However, the "liquid-phase contacting power," supplied from the stream of scrubbing liquid, and the "mechanical contacting power," supplied by a mechanically driven rotor, are not directly measurable; the theoretical power inputs can be estimated, but the portions of these quantities effectively converted to contacting power can only be inferred from comparison with gas-phase contacting power. Such data as are available indicate that the contributions of contacting power from different sources are directly additive in their relation to scrubber performance.

Contacting power is variously expressed in units of MJ/1000 m³ (SI), kWh/1000 m³ (meter-kilogram-second system), and hp/(1000 ft³/min) (U.S. customary). Relationships for conversion to SI units are

$$1.0 \text{ kWh}/1000 \text{ m}^3 = 3.60 \text{ MJ}/1000 \text{ m}^3$$

$$1.0 \text{ hp}/(1000 \text{ ft}^3/\text{min}) = 1.58 \text{ MJ}/1000 \text{ m}^3$$

The gas-phase contacting power P_G may be calculated from the effective friction loss by the following relationships:

SI units:

$$P_G = 1.0 F_E \quad (17-4)$$

where F_E = kPa.

U.S. customary units:

$$P_G = 0.1575 F_E \quad (17-5)$$

where F_E = in of water.

The power input from a liquid stream injected with a hydraulic spray nozzle may usually be taken as approximately equal to the product of the nozzle feed pressure p_F and the volumetric liquid rate. The liquid-phase contacting power P_L may then be calculated from the following formulas:

SI units:

$$P_L = 1.0 p_F (Q_L/Q_G) \quad (17-6)$$

where p_F = kPa gauge, and Q_L and Q_G = m³/s.

U.S. customary units:

$$P_L = 0.583 p_F (Q_L/Q_G) \quad (17-7)$$

where p_F = lbf/in² gauge, Q_L = gal/min, and Q_G = ft³/min.

The correlation of efficiency data is based on the total contacting power P_T , which is the sum of P_G , P_L , and any power P_M that may be supplied mechanically by a power-driven rotor.

In general, the liquid-to-gas ratio does not have an influence independent of contacting power on the collection efficiency of scrubbers of the venturi type. This is true at least of operation with liquid-to-gas ratios above some critical lower value. However, several investigations [Semrau and Lunn, "Performance of Particulate Scrubbers as Influenced by Gas-Liquid Contactor Design and by Dust Flocculation," EPA-600/9-82-005c, 1982, p. 43; and Muir et al., *Filtr. Sep.*, **15**, 332 (1978)] have shown that at low liquid-to-gas ratios relatively poor efficiencies may be obtained at a given contacting power. Such regions of operation are obviously to be avoided.

It has sometimes been asserted that multiple gas-liquid contactors in series will give higher efficiencies at a given contacting power than will a single contacting stage. However, there is little experimental evidence to support this contention. Lapple and Kamack (op. cit.) obtained slightly higher efficiencies with a venturi and an orifice in series than they did with a venturi alone. Muir and Mihisei [*Atmos. Environ.*, **13**, 1187 (1979)] obtained somewhat higher efficiencies on two redispersed dusts when using two venturis in series rather than one. The improvement obtained with two-stage scrubbing was greatest with the coarser of the two dusts and was relatively small with the finer dust. Flocculation or deflocculation of the dusts may have been responsible for some of the behavior encountered. Semrau et al. (EPA-600/2-77-234, 1977) compared the performance of a four-stage multiple-orifice contactor with that of a single-orifice contactor, using well-dispersed aerosols generated from ammonium fluorescein. The multiple-orifice contactor gave about the same efficiency as the single-orifice in the upper range of contacting power but lower efficiencies in the lower range. The deviations in performance in this case were probably characteristic of the particular multiple-orifice contactor rather than of multistage contacting as such.

Most scrubbers actually incorporate more than one stage of gas-liquid contacting even though these may not be identical (e.g., the contactor and the entrainment separator). The preponderance of evidence indicates that multiple-stage contacting is not inherently either more or less efficient than single-stage contacting. However, two-stage contacting may have practical benefits in dealing with abrasive or flocculated dusts.

Some investigators have proposed, mostly on the basis of mathematical modeling, to optimize the design of scrubbers to obtain a given efficiency with a minimum power consumption (e.g., Goel and Hollands, op. cit.). In fact, no optimum in performance appears to exist; so long as increased entrainment is prevented increased contacting power yields increased efficiency.

Scrubber Performance Curves The scrubber performance curve, which shows the relationship of scrubber efficiency to the contacting power, has been found to take the form

$$N_i = \alpha P_T^\gamma \quad (17-8)$$

where α and γ are empirical constants that depend primarily on the aerosol (dust or mist) collected. In a log-log plot of N_i versus P_T , γ is the slope of the performance curve and α is the intercept at $P_T = 1$.

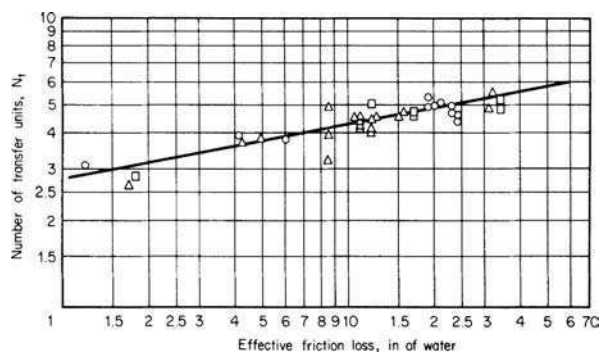


FIG. 17-46 Performance of pilot-plant venturi scrubber on fly ash. Liquid-to-gas ratio, gal/1000 ft³: ○, 10; △, 15; □, 20. (Raben, United States-U.S.S.R. Symposium on Fine-Particulate Emissions from Industrial Sources, San Francisco, 1974.)

Figure 17-46 shows such a performance curve for the collection of coal fly ash by a pilot-plant venturi scrubber (Raben "Use of Scrubbers for Control of Emissions from Power Boilers," United States-U.S.S.R. Symposium on Control of Fine-Particulate Emissions from Industrial Sources, San Francisco, 1974). The scatter in the data reflects not merely experimental errors but actual variations in the particle-size characteristics of the dust. Because the characteristics of an industrial dust vary with time, the scrubber performance curve necessarily must represent an average material, and the scatter in the data is frequently greater than is shown in Fig. 17-46. For best definition, the curve should cover as wide a range of contacting power as possible. Obtaining the data thus requires pilot-plant equipment with the flexibility to operate over a wide range of conditions. Because scrubber performance is not greatly affected by the size of the unit, it is feasible to conduct the tests with a unit handling no more than 170 m³/h (100 ft³/min) of gas.

A clear interpretation of γ , the slope of the curve, is still lacking. Presumably, it should be related to the particle-size distribution of the dust. Because scrubbing preferentially removes the coarser particles, the fraction of the dust removed (or the increment of N_t) per unit of contacting power should decrease as the contacting power and efficiency increase, so that the value of γ should be less than unity. In fact, the value of γ has been less than unity for most dusts. Nevertheless, some data in the literature have displayed values of γ greater than unity when plotted on the transfer-unit basis, indicating that the residual fraction of dust became more readily collectible as contacting power and efficiency increased. More recent studies by Semrau et al. (EPA-650/2-74-108 and EPA-600/2-77-237) have revealed performance curves having two branches (typified by Fig. 17-47), the lower having a slope greater than unity and the upper a slope less than unity. This suggests that had the earlier tests been extended into higher contacting-power ranges, performance curves with flatter slopes might have appeared in those ranges.

Among the aerosols that gave performance curves with $\gamma > 1$, the only obvious common characteristic was that a large fraction of each was composed of submicrometer particles.

Cut-Power Correlation Another design method, also based on scrubber power consumption, is the cut-power method of Calvert [*J. Air Pollut. Control Assoc.*, **24**, 929 (1974); *Chem. Eng.*, **84**(18), 54 (1977)]. In this approach, the cut diameter (the particle diameter for which the collection efficiency is 50 percent) is given as a function of the gas pressure drop or of the power input per unit of volumetric gas flow rate. The functional relationship is presented as a log-log plot of the cut diameter versus the pressure drop (or power input). In principle, the function could be constructed by experimentally determining scrubber performance curves for discrete particle sizes and then plotting the particle sizes against the corresponding pressure drops necessary to give efficiencies of 50 percent. In practice, Calvert and coworkers evidently have in most cases constructed the cut-power functions for various scrubbers by modeling (Yung and Calvert, U.S. EPA-600/8-78-005b, 1978). They show a variety of curves, whereas empirical studies have

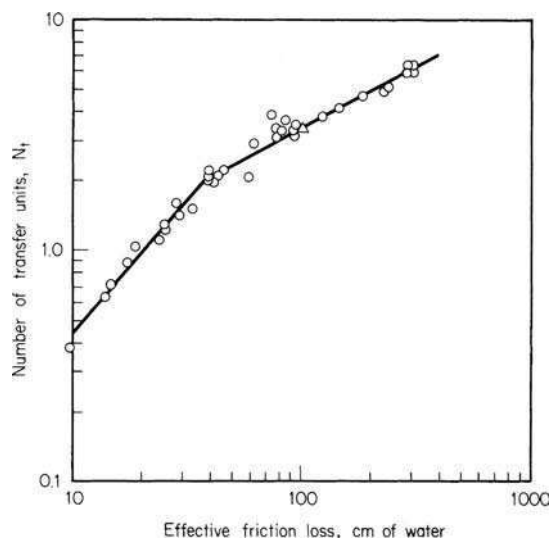


FIG. 17-47 Performance curve for orifice scrubber collecting ammonium fluorescein aerosol. (Semrau et al., EPA 600/2-77-237, 1977.)

indicated that different types of scrubbers generally have about the same performance at a given level of power consumption.

Condensation Scrubbing The collection efficiency of scrubbing can be increased by the simultaneous condensation of water vapor from the gas stream. Water-vapor condensation assists in particle removal by two entirely different mechanisms. One is the deposition of particles on cold-water droplets or other surfaces as the result of Stefan flow. The other is the condensation of water vapor on particles as nuclei, which enlarges the particles and makes them more readily collected by inertial deposition on droplets. Both mechanisms can operate simultaneously. However, for the buildup of particles by condensation to be effective, there must be adequate time for the particles to grow substantially before the principal gas-liquid-contacting operation takes place. Hence, if particle buildup is to be sought, the scrubber should be preceded by an appropriate gas-conditioning section. On the other hand, particle collection by Stefan flow can be induced simply by scrubbing the hot, humid gas with sufficient cold water to bring the gas below its initial dew point. Any practical method of inducing condensation on the dust particles will incidentally afford opportunities for the operation of the Stefan-flow mechanism. The hot gas stream must, of course, have a high initial moisture content, since the magnitude of the effects obtained is related to the quantity of water vapor condensed.

Although there is a considerable body of literature on particle collection by condensation mechanisms, most of it is either theoretical or, if experimental, treats basic phenomena in simplified cases. Few studies have been made to determine what performance may be expected from condensation scrubbing under practical conditions in industrial applications. In a series of studies, Calvert and coworkers investigated several types of equipment for condensation scrubbing, generally emphasizing the use of the condensation center effect to build up the particles for collection by inertial deposition (Calvert and Parker, EPA-600/8-78-005c, 1978). From early estimates, they predicted that a condensation scrubber would require only about one-third or less of the power required by a conventional high-energy scrubber. A subsequent demonstration-plant scrubber system consisted of a direct-contact condensing tower fed with cold water followed by a venturi scrubber fed with recirculated water (Chmielewski and Calvert, EPA-600/7-81-148, 1981). The condensation and particle buildup took place in the cooling tower. In operation on humidified iron-foundry-cupola gas, this system still required about 65 percent as much power as for conventional high-energy scrubbing.

Semrau and coworkers [*Ind. Eng. Chem.*, **50**, 1615 (1958); *J. Air Pollut. Control Assoc.*, **13**, 587 (1963); EPA-650/2-74-108, 1974] investigated

condensation scrubbing in pilot-plant studies in the field and, later, under laboratory conditions. Hot, humid gases were scrubbed directly with cold water under conditions that were favorable for the Stefan-flow mechanism but offered little or no opportunity for particle buildup. Some of the field studies indicated a contacting-power saving of as much as 50 percent for condensation scrubbing of Kraft-recovery-furnace fume. Laboratory tests on a predominantly submicrometer synthetic aerosol showed contacting-power savings of up to 40 percent with condensation scrubbing.

In the scrubbing of hot gases with high water content, condensation reduces contacting power and affords a direct power saving through the reduction of the gas volume by cooling and water-vapor condensation, but it incurs other costs for power and equipment for heat transfer and water cooling. However, condensation scrubbing may offer a net economic advantage if recovery of low-level heat is practical. It should also be advantageous when a hot gas must be not only cleaned but cooled and dehumidified as well; examples are the cleaning of blast-furnace gas for use as fuel and of SO_2 -bearing waste gases for feed to a sulfuric acid plant.

Entrainment Separation The entrainment separator is a critical element of a scrubber, since the collection efficiency of the scrubber depends on essentially total removal of the spray from the gas stream. The sprays generated in scrubbers are generally large enough in droplet size that they can be readily removed by properly designed inertial separators. Primary collection of the spray is seldom the critical limitation on separator performance, but reentrainment is a common problem. In dust scrubbers it is essential that the entrainment separator not be of a form readily subject to blockage by solids deposits and that it be readily cleared of deposits if they should occur. Cyclone separators are advantageous in this respect and are widely used with venturi contactors. However, they cannot readily be made integral with scrubbers of some other configurations, which can be more conveniently fitted with various forms of impingement separators. Although separator design can be important, the most common cause of reentrainment is simply the use of excessive gas velocities, and few data are available on the gas-handling

capacities of separators. In the absence of good data, there is a frequent tendency to underdesign separators in an effort to reduce costs.

Venturi Scrubbers The venturi scrubber is one of the most widely used types of particulate scrubbers. The designs have become generally standardized, and units are manufactured by a large number of companies. Venturi scrubbers may be used as either high- or low-energy devices but are most commonly employed as high-energy units. The units originally studied and used were designed to the proportions of the classical venturis used for metering, but since it was discovered that these proportions have no special merits, simpler and more practical designs have been adopted. Most "venturi" contactors in current use are in fact not venturis but variable orifices of one form or another. Any of a wide range of devices can be used, including a simple pipeline contactor. Although the venturi scrubber is not inherently more efficient at a given contacting power than other types of devices, its simplicity and flexibility favor its use. It is also useful as a gas absorber for relatively soluble gases, but because it is a cocurrent contactor, it is not well suited to absorption of gases having low solubilities.

Current designs for venturi scrubbers generally use the vertical downflow of gas through the venturi contactor and incorporate three features: (1) a "wet-approach" or "flooded-wall" entry section, to avoid dust buildup at a wet-dry junction; (2) an adjustable throat for the venturi (or orifice), to provide for adjustment of the pressure drop; and (3) a "flooded elbow" located below the venturi and ahead of the entrainment separator, to reduce wear by abrasive particles. The venturi throat is sometimes fitted with a refractory lining to resist abrasion by dust particles. The entrainment separator is commonly, but not invariably, of the cyclone type. An example of the "standard form" of venturi scrubber is shown in Fig. 17-48. The wet-approach

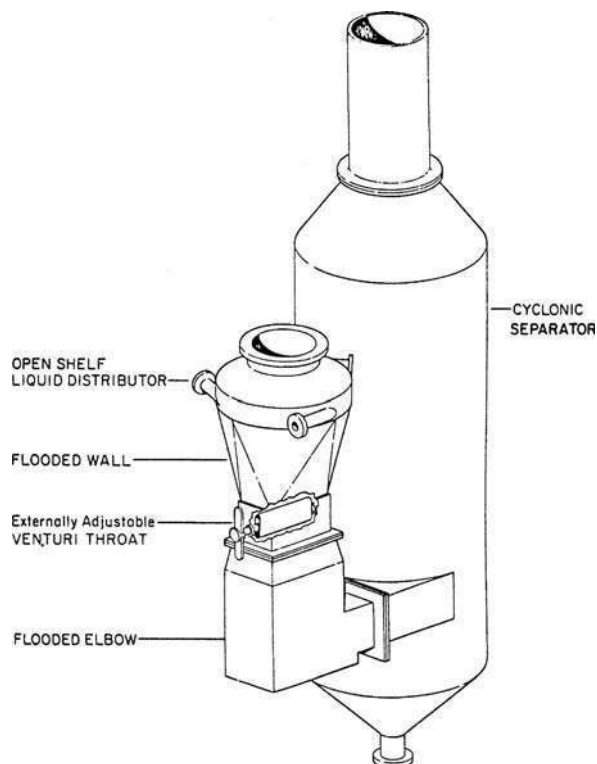


FIG. 17-48 Venturi scrubber. (Neptune AirPol.)

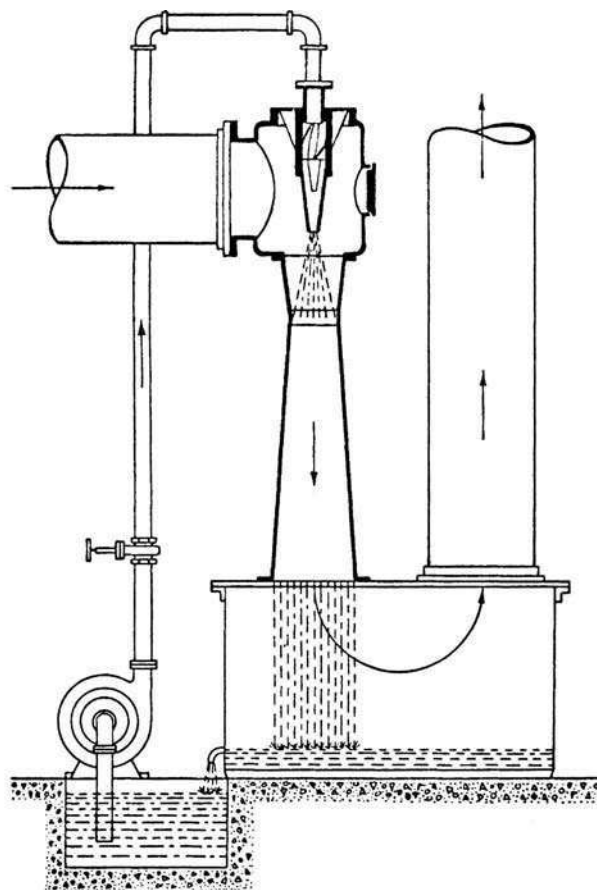


FIG. 17-49 Ejector-venturi scrubber. (Schutte & Koerting Division, Amtek, Inc.)

entry section has made practical the recirculation of slurries. Various forms of adjustable throats, which may be under manual or automatic control, permit maintaining a constant pressure drop and constant efficiency under conditions of varying gas flow.

Ejector-Venturi Scrubbers In the ejector-venturi scrubber (Fig. 17-49) the cocurrent water jet from a spray nozzle serves both to scrub the gas and to provide the draft for moving the gas. No fan is required, but the equivalent power must be supplied to the pump that delivers water to the ejector nozzle. The water must be supplied in sufficient volume and at high enough pressure to provide both adequate draft and enough contacting power for the required scrubbing operation. Considered as a gas pump, the ejector is not a very efficient device, but the dissipated energy that is not effective in pumping does serve in gas-liquid contacting. The energy equivalent to any gas pressure rise across the scrubber is not part of the contacting power (Semrau et al., EPA-600/2-77-234, 1974).

The ejector-venturi scrubber is widely used as a gas absorber, but the combinations of water pressure and flow rate that are sufficient to provide the required draft usually do not also yield enough contacting power to give high collection efficiency on submicrometer particles. Other types of ejectors have been employed to provide higher contacting-power levels. In one, superheated water is discharged through the nozzle, and part flashes to steam, increasing the mechanical energy available for scrubbing [Gardenier, *J. Air Pollut. Control Assoc.*, **24**, 954 (1974)]. Some units use two-fluid nozzles, with either compressed air or steam as the compressible fluid [Sparks, *J. Air Pollut. Control Assoc.*, **24**, 958 (1974)]. Most of the energy for gas movement and for atomizing the liquid and scrubbing the gas is derived from the compressed air or steam. In some ejector-venturi-scrubbers installations, part of the draft is supplied by a fan [Williams and Fuller, *TAPPI*, **60**(1), 108 (1977)].

Self-Induced Spray Scrubbers Self-induced spray scrubbers form a category of gas-atomized spray scrubbers in which a tube or a

duct of some other shape forms the gas-liquid-contacting zone. The gas stream flowing at high velocity through the contactor atomizes the liquid in essentially the same manner as in a venturi scrubber. However, the liquid is fed into the contactor and later recirculated from the entrainment separator section by gravity instead of being circulated by a pump as in venturi scrubbers. The scheme is well illustrated in Fig. 17-50a. A great many such devices using contactor ducts of various shapes, as in Fig. 17-50b, are offered commercially. Although self-induced spray scrubbers can be built as high-energy units and sometimes are, most such devices are designed for only low-energy service.

The principal advantage of self-induced spray scrubbers is the elimination of a pump for recirculation of the scrubbing liquid. However, the designs for high-energy service are somewhat more complex and less flexible than those for venturi scrubbers.

Spray Scrubbers Spray scrubbers consist of empty chambers of some simple form in which the gas stream is contacted with liquid droplets generated by spray nozzles. A common form is a spray tower, in which the gas flows upward through a bank or successive banks of spray nozzles. Similar arrangements are sometimes used in spray chambers with horizontal gas flow. Such devices have very low gas pressure drops, and all but a small part of the contacting power is derived from the liquid stream. The required contacting power is obtained from an appropriate combination of liquid pressure and flow rate. Most spray scrubbers are low-energy units. Collection of fine particles is possible but may require very high liquid-to-gas ratios, liquid feed pressures, or both. Plugging of small nozzles can be a persistent maintenance problem. Entrainment separators are necessary to prevent carryover of spray into the exit gas.

Nonatomizing reverse-jet froth scrubbers using large pluggage-resistant open nozzles allow effective use of higher contacting power to efficiently collect finer dust particles, as described in U.S. Patents 3,803,805, 4,056,371, and 4,057,404. Such scrubbers are provided by

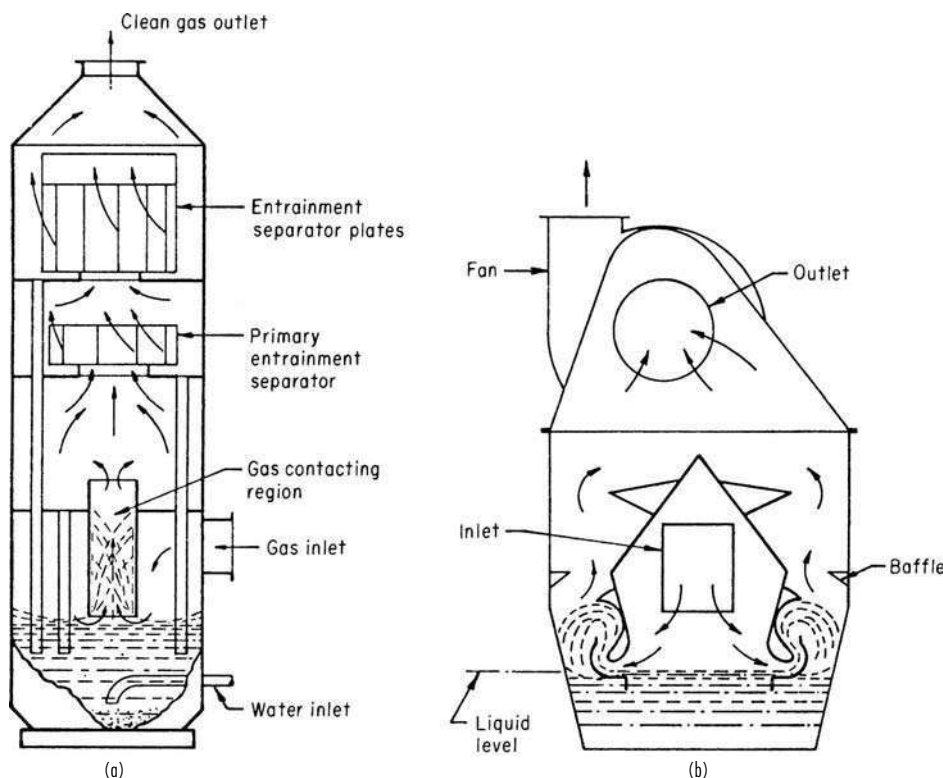


FIG. 17-50 Self-induced spray scrubbers. (a) Blaw-Knox Food & Chemical Equipment, Inc. (b) American Air Filter Co., Inc.

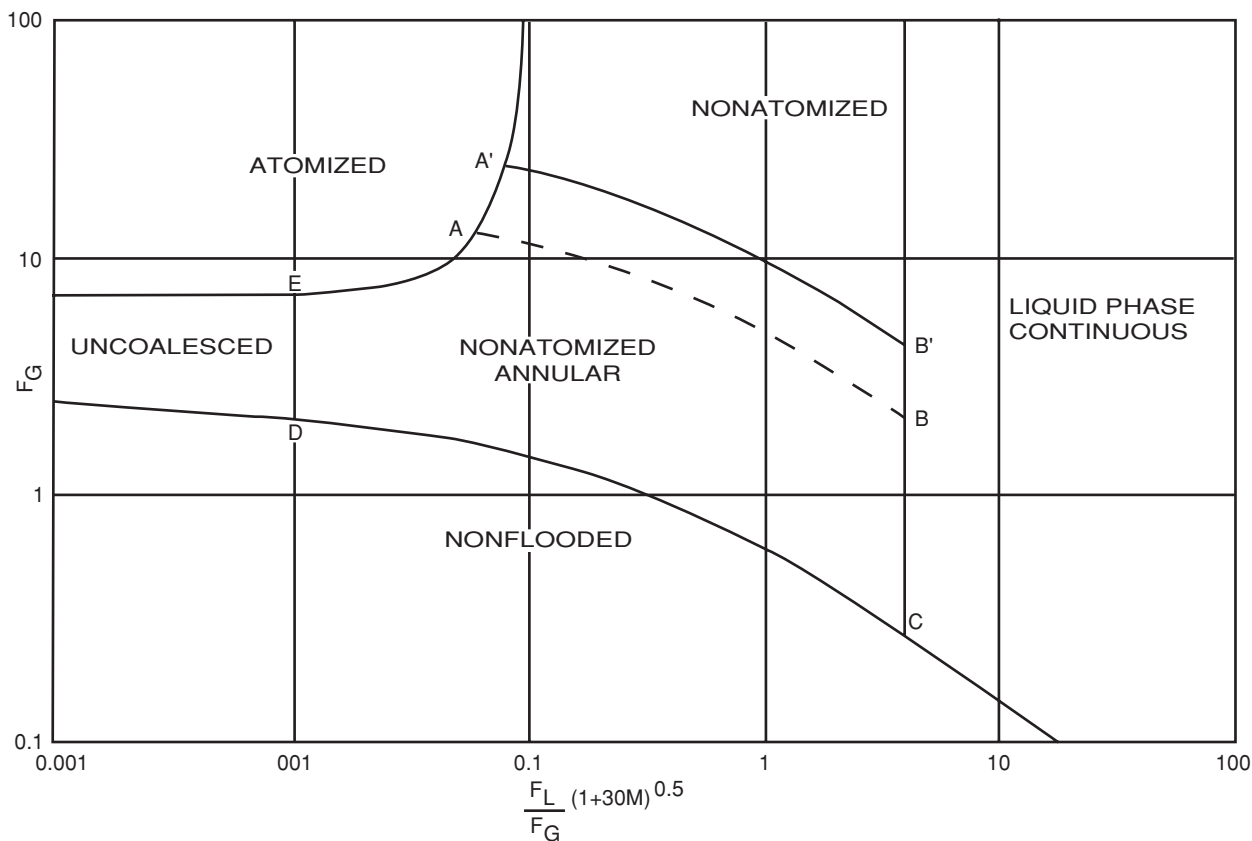


FIG. 17-51 Spray scrubbing flow regime map. (U.S. Patent 4,057,404.)

Monsanto Enviro-Chem under the Dyna-Wave® trademark; case histories can be downloaded from a web site (<http://www.enviro-chem.com/airpol/common/rjstop.html>). Nonatomizing froth scrubbing is described in those patents as occurring within defining boundaries on a new dimensionless velocity versus dimensionless liquid/gas ratio two-phase flow regime map, shown here as Fig. 17-51.

Cyclone Scrubbers The vessels of cyclone scrubbers are all in the form of cyclones, which provide for compact integral entrainment separation. The gas-liquid contacting section normally uses some sort of spray generator to disperse liquid throughout the gas. Performance is similar to other spray scrubbers.

Plate Towers Plate (tray) towers are countercurrent gas-atomized spray scrubbers using one or more plates for gas-liquid contacting. They are essentially the same as, if not identical to, the devices used for gas absorption and are frequently employed in applications in which gases are to be absorbed simultaneously with the removal of dust. Except possibly in cases in which condensation effects are involved, countercurrent operation is not significantly beneficial in dust collection.

The plates may be any of several types, including sieve, bubble-cap, and valve trays. Valve trays constitute multiple self-adjusting orifices that provide nearly constant gas pressure drop over considerable ranges of variation in gas flow. The gas pressure drop that can be taken across a single plate is necessarily limited, so that units designed for high contacting power must use multiple plates.

Plate towers are more subject to plugging and fouling than venturi-type scrubbers that have large passages for gas and liquid.

Packed-Bed Scrubbers Packed-bed scrubbers using gas-absorption-type tower packing may also be used for collecting soluble solids, but are not irrigated intensively enough to avoid plugging by deposits of

insoluble solids. Mobile-bed and fiber-bed scrubbers can be self-cleaning if they are adequately flushed.

Mobile-Bed Scrubbers Mobile-bed scrubbers (Fig. 17-52) are constructed with one or more beds of low-density spheres that are free to move between upper and lower retaining grids. The spheres are

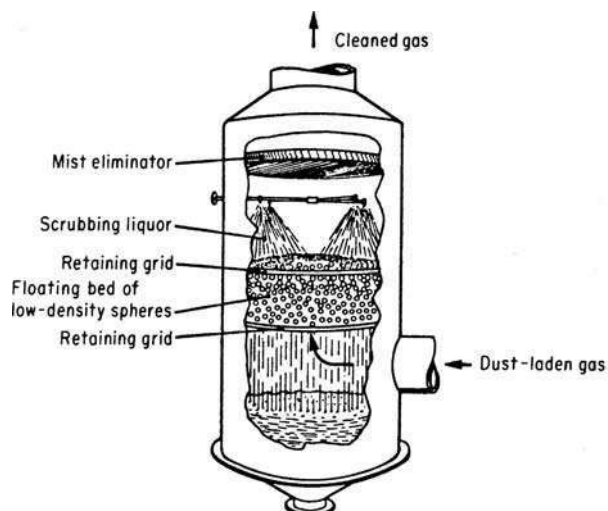


FIG. 17-52 Mobile-bed scrubber. (*Air Correction Division, UOP.*)

commonly 1.0 in (2.5 cm) or more in diameter and made from rubber or a plastic such as polypropylene. The plastic spheres may be solid or hollow. Gas and liquid flows are countercurrent, and the spherical packings are fluidized by the upward-flowing gas. The movement of the packings is intended to minimize fouling and plugging of the bed. Mobile-bed scrubbers were first developed for absorbing gases from gas streams that also carry solid or semisolid particles.

The spherical packings are too large to serve as effective targets for the deposition of fine dust particles. In dust-collection service, the packings actually serve as turbulence promoters, while the dust particles are collected primarily by the liquid droplets.

The gas pressure drop through the scrubber may be increased by increasing the gas velocity, the liquid-to-gas ratio, the depth of the bed, the density of the packings, and the number of beds in series. In an experimental study, Yung et al. (EPA-600/7-79-071, 1979) determined that the collection efficiency of a mobile-bed scrubber was dependent only on the gas pressure drop and was not influenced independently by the gas velocity, the liquid-to-gas ratio, or the number of beds except as these factors affected the pressure drop. Yung et al. also reported that the mobile-bed scrubber was less efficient at a given pressure drop than scrubbers of the venturi type, but without offering comparable experimental supporting evidence.

Fiber-Bed Scrubbers Fibrous-bed structures are sometimes used as gas-liquid contactors, with cocurrent flow of the gas and liquid streams. In such contactors, both scrubbing (particle deposition on droplets) and filtration (particle deposition on fibers) may take place. If only mists are to be collected, small fibers may be used, but if solid particles are present, the use of fiber beds is limited by the tendency of the beds to plug. For dust-collection service, the fiber bed must be composed of coarse fibers and have a high void fraction, so as to minimize the tendency to plug. The fiber bed may be made from metal or plastic fibers in the form of knitted structures, multiple layers of screens, or random-packed fibers. However, the bed must have sufficient dimensional stability so that it will not be compacted during operation.

Lucas and Porter (U.S. Patent 3,370,401, 1967) developed a fiber-bed scrubber in which the gas and scrubbing liquid flow vertically upward through a fiber bed (Fig. 17-53). The beds tested were composed of knitted structures made from fibers with diameters ranging from 89 to 406 μm . Lucas and Porter reported that the

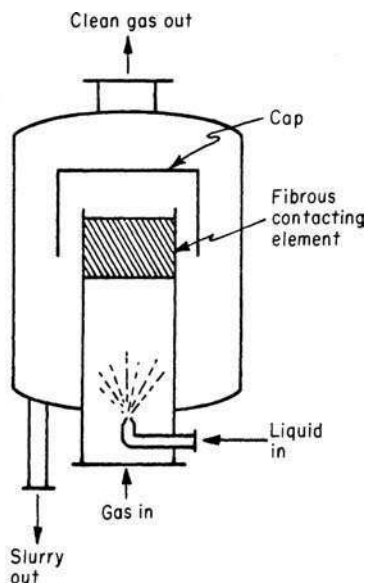


FIG. 17-53 Fibrous-bed scrubber. (Lucas and Porter, U.S. Patent 3,370,401, 1967.)

fiber-bed scrubber gave substantially higher efficiencies than did venturi-type scrubbers tested with the same dust at the same gas pressure drop. In similar experiments, Semrau (Semrau and Lunn, op. cit.) also found that a fiber-bed contactor made with random-packed steel-wool fibers gave higher efficiencies than an orifice contactor. However, there were indications that the fiber bed would have little advantage in the collection of submicrometer particles, presumably because of the large fiber size feasible for dust-collection service.

Alliger (U.S. Patents 3,659,402, 1972, and 3,905,788, 1975) describes fiber-bed structures which are not random, but are rather built up from flat mesh sheets offset angularly from one layer to the next and then compressed and bonded. Such bonded beds of relatively coarse hydrophobic fibers both are remarkably flushable, to prevent fouling by insoluble solids, and have surprisingly high collection efficiency per unit pressure drop for submicrometer particles, approaching that of irrigated fine hydrophobic fiber filters such as described by Fair (U.S. Patent 3,135,592, 1964) and Vosseller (U.S. Patent 3,250,059, 1966).

Electrically Augmented Scrubbers In some types of wet collectors, attempts are made to apply the electrostatic-deposition mechanism by charging the dust particles, the water droplets, or both. The objective is to combine in a scrubber high efficiency in collecting fine particles and the moderate power consumption characteristic of an electrical precipitator. Successful devices of this type have been essentially wet electrical precipitators and should properly be discussed in that category (see "Electrical Precipitators"). So far, there has been no clear demonstration of a device that combines the small size, compactness, and high efficiency of a high-energy scrubber with the relatively low power consumption of an electrical precipitator.

Mechanical Scrubbers Mechanical scrubbers comprise those devices in which a power-driven rotor produces the fine spray and the contacting of gas and liquid. As in other types of scrubbers, it is the droplets that are the principal collecting bodies for the dust particles. The rotor acts as a turbulence producer. An entrainment separator must be used to prevent carry-over of spray. Among potential maintenance problems are unbalancing of the rotor by buildup of dust deposits and abrasion by coarse particles.

The simplest commercial devices of this type are essentially fans upon which water is sprayed. The unit shown in Fig. 17-54 is adapted to light duty, and heavy dust loads are avoided to minimize buildup on the rotor.

Dry Scrubbing Dry scrubbing is an umbrella term used to associate several different unit operations and types of hardware that can

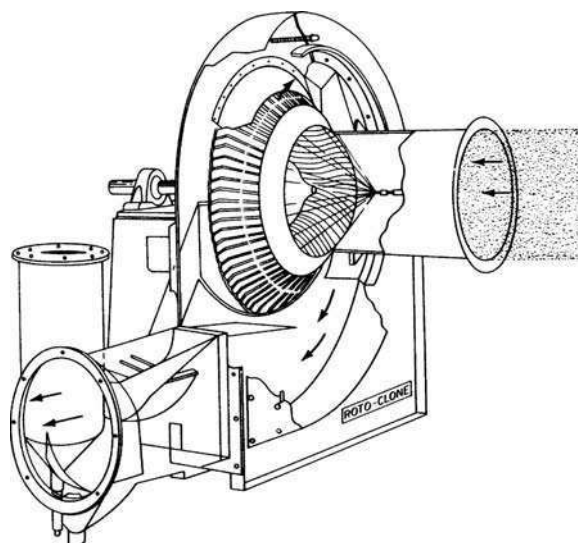


FIG. 17-54 Mechanical scrubber. (American Air Filter Co., Inc.)

be used in combinations to accomplish the unit process of dry scrubbing. They all utilize scrubbing, in which mass transfer takes place between the gas phase and an active liquidlike surface, and they all discharge the resulting products separately as a gas and a solidlike "dry" product for reuse or disposal.

It is helpful to recognize three phases in the development of systems using dry scrubbing. The first phase used mainly good contactors and relatively expensive sorbents, which were tailored to catch targeted compounds with high efficiency (>99 percent). The second phase used mainly minimal contactors and cheap sorbents in order to get enough sulfur dioxide capture (>60 percent) to justify using cheaper higher sulfur coal. The third phase is ongoing; it uses mainly better contactors, sorbents, and fabric filters in order to get not only high SO₂ capture (>90 percent), but also substantial capture of other regulated emissions such as metals and dioxins.

The principle of dry scrubbing was first recognized by Lamb and Wilson [*I&EC*, **11**(5), 420 (1919)] and Wilson [*I&EC*, **12**(10), 1000 (1920)] at MIT during their World War I program to develop better media for military gas masks. The preferred medium turned out to be a mixture of granular activated carbon with a new kind of porous granular hydrated lime, which they dubbed *activated soda-lime*. What activated it was a small but critical amount of sodium hydroxide, which was enough to maintain in use on French battlefields an equilibrium moisture film a few molecules thick on the surface of pores within lime particles. That film enhanced surface hydrolysis, diffusion, and neutralization of acid gases such as Cl₂ and SO₂, to the point that their removal became controlled by fast external gas-film diffusion, up to substantial utilization of the potential stoichiometric capacity. The better varieties of army activated soda-lime used in gas masks captured more than 99 percent of Cl₂ or SO₂ even after using up more than 10 percent of alkalinity. Gas-mask-canister sorbent beds typically used 8 × 14 mesh granules 10 cm deep, and tests were run with 1 percent acid gas at space velocities of 3000 to 7400 l/hr. Carbon dioxide was tested to higher penetrations; its capture was more than 50 percent even after using up more than 50 percent of available alkalinity.

With good dry scrubbing sorbents, the controlling resistance for gas cleaning is external turbulent diffusion, which also depends on energy dissipated by viscous and by inertial mechanisms. It turns out to be possible to correlate mass-transfer rate as a function of the friction factor [Ergun, *CEP* **48**(2):89 and **48**(5):227(1952)].

It turns out that packed beds much less than a hundred particles thick behave as if they were well-stirred due to the entrance effect. Although it may seem odd that a packed bed can behave as if well-stirred, it typically takes at least a 100-particle bed depth in order for a plug-flow concentration wave to develop.

Integration of the Ergun equation for well-stirred flow gives

$$\frac{C_2 - C_1}{C^\circ - C_2} = F_k \frac{\rho D_v (1 - \epsilon_v) L}{\mu \epsilon_s D_p} \quad (17-9)$$

Recognition of dry scrubbing as a mechanism useful in treating large gas flows first came in the 1960s as regulations were tightening on fly ash emissions from utility boilers. Dry scrubbing turns out to be the appropriate way to describe the interactions of sulfuric acid vapor with fly ash from pulverized coal combustion. On one hand, powdered sodium bicarbonate was added to some flue gases in order to protect fabric filter bags from attack by sulfuric acid films on ash from high sulfur coal combustion. On the other hand, sulfur trioxide was added to other flue gases in order to generate sulfuric acid films that would improve the performance of electrostatic precipitators collecting inherently high resistivity ash from low sulfur coal. From Eq. (17-9), it is readily apparent that the way to get more gas cleaning per unit pressure drop is to optimize the particle size and void fraction, and use relatively fewer particles operating at higher velocity. Such an arrangement runs out of stoichiometric capacity quickly, which leads naturally to transported bed contactors fed with fresh sorbent.

The actual name "dry scrubbing" was first publicized by Teller [U.S. Patent no. 3,721,066 (1973)]. He worked both with classical Army-type soda-lime and with his patented water-activated form of the alkaline feldspar nepheline syenite as a flow agent and feedstock sorbent for HF and SO₂ in hot, sticky fumes from glass melting furnaces. He

claimed capture of more than 99 percent of 180 ppm HF and SO₂ for more than 20 hours in a packed bed of 200 × 325 mesh hydrated nepheline syenite at 42,000/hr.

Teller [U.S. Patent no. 3,997,652 (1976)] also patented a number of "chromatographic sorbents" using semivolatile liquid coatings similar to mobile phases used in capillary columns for gas chromatographs. For simple acid gases, Teller generally used a flash-drying-type venturi injector and pneumatic transport system to disperse fresh sorbent into hot gas to be treated and to provide time for turbulent mixing. He then typically used a baghouse to collect the loaded sorbent for inclusion with other feeds into the glass melting system.

Aluminum producers were also early users of another type of dry scrubbing for the difficult cleaning problem of acid gases in hot, sticky fumes from aluminum reduction potlines. Bazhenov et al. [*Tsvetn. Met.—Russia*, (6) 44 (1975)] described the use of a fluidized bed of feedstock aluminum oxide to clean potline offgas in an arrangement where there was an integral freeboard baghouse. Loaded sorbent was purged from the bed for feed into the aluminum reduction system. In that anhydrous system there was no water film as such inside the pores, but there was high specific surface, having liquidlike mobile active sites with a strong affinity for binding fluoride. The potential for achieving significant gas cleaning at very high space velocities led to dry scrubbing being applied both for incremental desulfurization of flue gases from coal-fired utility boilers and for removal of trace acid gases from solid waste incineration. A number of different hardware arrangements were tested all the way to full scale in order to determine accurately the various effects that must be optimized in order to achieve minimum net cost to be carried by the rate-paying public.

It was apparent from the very earliest tests that control of thin moisture films on the surface of reactive particles was the key to success. The main three competing arrangements, as compared by Statnick et al. [4th Annual Pitt. Coal Conf. (1987)] involved slurry spray dryers, where lime and water were injected together, versus systems where the gas was humidified by water injection before or after injection of lime dry powder reagents. It turns out that there are tradeoffs among the costs of hardware, reagent, and water dispersion and reagent purchase and disposal. Systems where water evaporates in the presence of active particles are usually less expensive overall.

The evaporation lifetimes of pure water droplets are described by Marshall and Seltzer [*CEP*, **46**(10), 501 and **46**(11), 575 (1950)] and by Duffie and Marshall [*CEP*, **49**(8), 417 (1953) and **49**(9), 480 (1953)]. Figure 17-55 is easy to use to estimate the necessary spray dispersion. For typical humidifier conditions of 300°F gas inlet temperature and a 20°F approach to adiabatic saturation at the outlet, a 60-μm droplet will require about 0.9 s to evaporate, while a 110-μm droplet will require about 3 s. If the time available for drying is no more than 3 s, it therefore follows that the largest droplets in a humidifier spray should be no larger than 100 μm. If the material being dried contains solids, droplet top size will need to be smaller due to slower evaporation rates. Droplets this small not only require considerable expensive power to generate, but since they have inherent penetration distances of less than a meter, they require expensive dispersion arrangements to get good mixing into large gas flows without allowing damp, sticky particles to reach walls.

The economics seem to be better for systems where dry powdered fresh lime plus ground recycled lime is injected along with a relatively coarse spray which impinges on and dries out from the reagent, as described by Stouffer et al. [*I&EC Res.*, **28**(1):20 (1989)]. Withum et al. [9th Ann. Pitt. Coal Prep. Util. Env. Control Contractors Conf. (1993)] describes an advanced version of that system that has been further optimized to the point that it is competitive with wet limestone scrubbing for >90 percent flue gas desulfurization.

The most popular dry scrubbing systems for incinerators have involved the spray drying of lime slurries, followed by dry collection in electrostatic precipitators or fabric filters. Moller and Christiansen [Air Poll. Cont. Assoc. 84-9.5 (1984)] published data on early European technology. Moller et al. [U.S. Patent no. 4,889,698 (1989)] describe the newer extension of that technology to include both spray-dryer absorption and dry scrubbing with powdered, activated carbon injection. They claim greatly improved removal of mercury, dioxins, and NO_x.

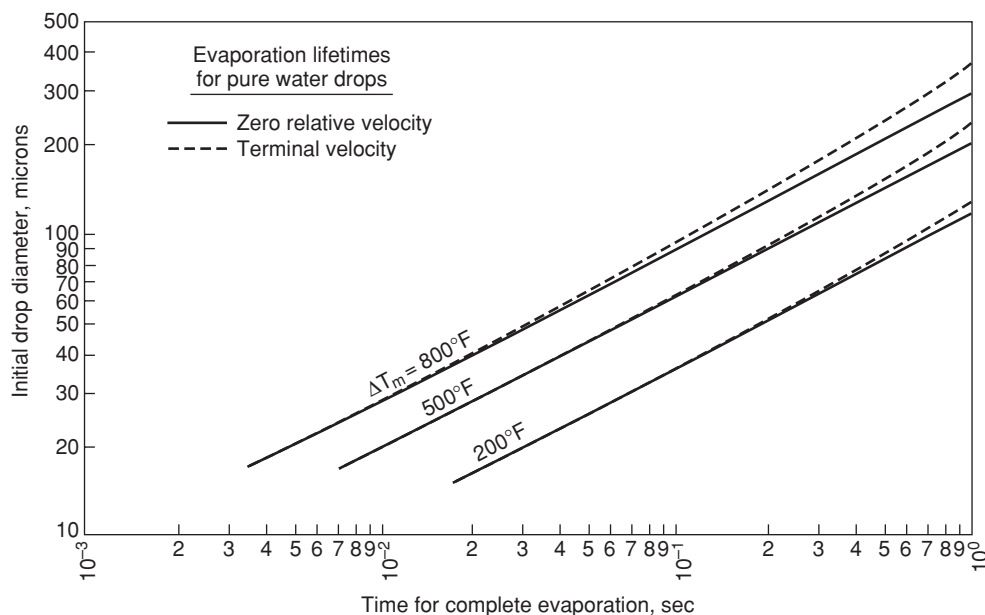


FIG. 17-55 Effect of drop diameter on time for complete evaporation of water drops.

Optimized modern dry scrubbing systems for incinerator gas cleaning are much more effective (and expensive) than their counterparts used so far for utility boiler flue gas cleaning. Brinckman and Maresca [ASME Med. Waste Symp. (1992)] describe the use of dry hydrated lime or sodium bicarbonate injection followed by membrane filtration as preferred treatment technology for control of acid gas and particulate matter emissions from modular medical waste incinerators, which have especially high dioxin emissions.

Kreindl and Brinckman [Air Waste Man. Assoc., paper 93-RP-154.04 (1993)] describe the three conventional flue-gas cleaning-process concepts (standard dry scrubbing, quasi-dry scrubbing, and wet scrubbing) as being inherently inadequate to meet the emerging European incinerator metal and organic emissions limits. Standard dry scrubbing can be upgraded by using powdered carbon along with lime, and switching to membrane filter media. Quasi-dry scrubbing can be upgraded by using powdered carbon along with lime spray drying [as also described by Moller (1989)], and switching to membrane filter media. Wet scrubbing can be upgraded by post-treatment following the wet scrubber with a circulating fluidized bed of granular activated carbon, again followed by a membrane filter.

Control of metal and organic emissions by dry scrubbing involves more than the simple acid gas neutralization discussed so far. Capture of metals other than mercury is mainly a matter of using a fabric filter having the capability of retaining unagglomerated submicron particles that are relatively enriched in toxic metals such as cadmium. Use of membrane-coated filter media is almost essential to do that well. Capture of mercury and dioxins is mainly achieved with activated carbon, which is more effective when used in combination with inhibitors for chlorine formation by the Deacon reaction. Karasek [U.S. Patent no. 4,793,270 (1988)] describes the inhibiting effects of sulfur compounds, which is why dioxins are rarely a problem in flue gases from coal-fired combustion. Naikwadi and Karasek [Chemosphere, 27(1-3), 335 (1993)] describe the potent inhibiting effect of amines. That is presumably why German experience is finding that dioxins are rarely a problem in flue gases that have undergone some form of ammonia deNO_x treatment, as described by Hahn [Chemosphere, 25(1-2), 57 (1992)]. Hahn, like Kreindl and Brinckman, also argues that classical wet scrubbers will need to be upgraded with modern dry scrubbers and fabric filters.

Maerker Sorbalit® (<http://www.maerker-umwelttechnik.de/englisch/Produkt.htm>) pioneered the use as a filter aid or precoat lime hydrated with activated carbon so as to adsorb heavy semivolatile polynuclear aromatic hydrocarbons under conditions which prevent their gradual chlorination to dioxins. The material is free-flowing and easy to disperse, and the little dust is nonflammable, unlike that of some plain activated carbons. Donau Carbon also produces an equivalent Desomix® material which is effective in entrained flow or precoat reactors (<http://www.dcffm.de/englisch/programm/flugstromverfahren.html>). Such filter precoat greatly decrease dioxin emissions and can capture mercury when impregnated with sulfur. A different approach also depending on sorption is to adsorb the heavy semivolatile polynuclear aromatic hydrocarbons onto an active oxidation catalyst where they can be destroyed in situ. Promoted catalysts similar to those used for ammonia deNO_x are generally effective. Such catalysts may be used either in traditional pellet beds following a baghouse or within a baghouse dispersed in a porous felted PTFE membrane filter fabric, e.g., W. L. Gore's Remedia® (U.S. Patent 5,620,669).

Emission control of heavy semivolatile polynuclear organic hydrocarbon soot from diesel engines is becoming required in many jurisdictions. That is generally accomplished by passing exhaust gases through a regenerable porous ceramic foam or fiber filter, so as to capture organics during cool smoky operation and then to burn them off during hotter clean operation. A typical porous wall tube array is shown in Fig. 17-56. Size distribution of soot in cool exhaust gas typically has a log-mean mobility diameter of 50 nm with a geometric standard deviation of 1.7. Given the small particle size and the short filter transit time, collection is clearly by diffusion, but it is not totally clear whether particles are yet fully agglomerated. There is very active ongoing research in this area, and the technology undoubtedly has not yet stabilized. Hardware which is already in use in Europe is summarized in the "VERT Filter List" of particle-trap systems which have been tested and approved for retrofitting diesel engines by the Swiss Agency for the Environment Forests and Landscape (SAEFL). The current list is available for download (http://www.umwelt-schweiz.ch/imperia/md/content/luft/fachgebiet/e/industrie/filterliste_01_03_04_e.pdf). The underlying technology is publicized by vendors and summarized by Mayer in many ongoing publications, such as his encyclopaedic article http://www.akpf.org/pub/2003_particle_traps.pdf.

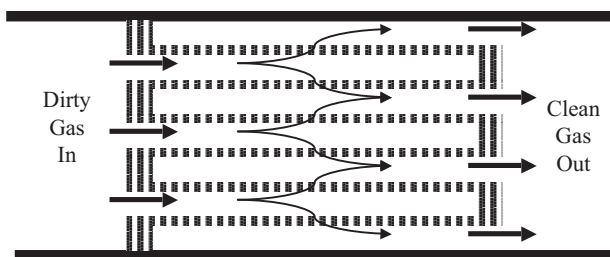


FIG. 17-56 Typical porous wall tube array diesel particulate filter.

Fabric Filters Fabric filters, commonly termed “bag filters” or “baghouses,” are collectors in which dust is removed from the gas stream by passing the dust-laden gas through a fabric of some type (e.g., woven cloth, felt, or porous membrane). These devices are “surface” filters in that dust collects in a layer on the surface of the filter medium, and the dust layer itself becomes the effective filter medium. The pores in the medium (particularly in woven cloth) are usually many times the size of the dust particles, so that collection efficiency is low until sufficient particles have been collected to build up a “pre-coat” in the fabric pores (Billings and Wilder, *Handbook of Fabric Filter Technology*, vol. I, EPA No. APTD-0690, NTIS No. PB-200648, 1979). During this initial period, particle deposition takes place mainly by inertial and flow-line interception, diffusion, and gravity. Once the dust layer has been fully established, sieving is probably the dominant deposition mechanism, penetration is usually extremely low except during the fabric-cleaning cycle, and only limited additional means remain for influencing collection efficiency by filter design. Filter design is related mainly to choices of gas filtration velocities and pressure drops and of fabric-cleaning cycles.

Because of their inherently high efficiency on dusts in all particle-size ranges, fabric filters have been used for collection of fine dusts and fumes for over 100 years. The greatest limitation on filter application has been imposed by the temperature limits of available fabric materials. The upper limit for natural fibers is about 90°C (200°F). The major new developments in filter technology that have been made since 1945 have followed the development of fabrics made from glass and synthetic fibers, which has extended the temperature limits to about 230 to 260°C (450 to 500°F). The capabilities of available fibers to resist high temperatures are still among the most severe limitations on the possible applications of fabric filters.

Gas Pressure Drops The filtration, or superficial face, velocities used in fabric filters are generally in the range of 0.3 to 3 m/min (1 to 10 ft/min), depending on the types of fabric, fabric supports, and cleaning methods used. In this range, gas pressure drops conform to Darcy’s law for streamline flow in porous media, in which the pressure drop is directly proportional to the flow rate. The pressure drop across the fabric and the collected dust layer may be expressed (Billings and Wilder, *op. cit.*) by

$$\Delta p = K_1 V_f + K_2 w V_f \quad (17-10)$$

where $\Delta p = \text{kPa}$, or in of water; $V_f = \text{superficial velocity through filter, m/min, or ft/min}$; $w = \text{dust loading on filter, g/m}^2$, or lbm/ft^2 ; and K_1 and K_2 are resistance coefficients for the “conditioned” fabric and the dust layer, respectively. The conditioned fabric is that fabric in which a relatively consistent dust load remains deposited in depth following cycles of filtration and cleaning. K_1 , expressed in units of $\text{kPa}/(\text{m/min})$ or in $\text{water}/(\text{ft/min})$, may be more than 10 times the value of the resistance coefficient for the original clean fabric. If the depth of the dust layer on the fabric is greater than about 0.2 cm ($1/16$ in), corresponding to a fabric dust loading on the order of 200 g/m^2 (0.04 lbm/ft^2), the pressure drop across the fabric (including the dust in the pores) is usually negligible relative to that across the dust layer.

The specific resistance coefficient for the dust layer K_2 was originally defined by Williams et al. [*Heat. Piping Air Cond.*, **12**, 259

TABLE 17-5 Specific Resistance Coefficients for Certain Dusts*

Dust	K_2 † for particle size less than						
	20 mesh	140 mesh	375 mesh	90 μm	45 μm	20 μm	2 μm
Granite	1.58	2.20				19.8	
Foundry	0.62	1.58	3.78			18.9	
Gypsum			6.30			27.3	
Feldspar			6.30				
Stone	0.96			6.30			
Lampblack							47.2
Zinc oxide							15.7†
Wood				6.30			
Resin (cold)		0.62				25.2	
Oats	1.58			9.60	11.0		
Corn	0.62		1.58	3.78	8.80		

*Data from Williams et al., *Heat. Piping Air Cond.*, **12**, 259–263 (1940).

$$\dagger K_2 = \frac{\Delta p_i}{V_f w} \cdot \frac{\text{in water}}{(\text{ft/min})(\text{lbm/ft}^2)}$$

NOTE: These data were obtained when filtering air at ambient conditions. For gases other than atmospheric air, the Δp_i values predicted from Table 17-5 should be multiplied by the actual gas viscosity divided by the viscosity of atmospheric air.

†Flocculated material not dispersed; size actually larger.

(1940)], who proposed estimating values of the coefficient by use of the Kozeny-Carman equation [Carman, *Trans. Inst. Chem. Eng. (London)*, **15**, 150 (1937)]. In practice, K_1 and K_2 are measured directly in filtration experiments. The K_1 and K_2 values can be corrected for temperature by multiplying by the ratio of the gas viscosity at the desired condition to the gas viscosity at the original experimental conditions. Values of K_2 determined for certain dusts by Williams et al. (*op. cit.*) are presented in Table 17-5.

Lapple (in Perry, *Chemical Engineers’ Handbook*, 3d ed., McGraw-Hill, New York, 1950) presents an alternative form of Eq. (17-10) in which the gas-viscosity term is explicit instead of being incorporated into the resistance coefficients:

$$\Delta p_i = K_c \mu V_f + K_d \mu w V_f \quad (17-11)$$

where $\Delta p_i = \text{in water}$, $\mu = \text{cP}$, $V_f = \text{ft/min}$, $w = \text{gr/ft}^2$, $K_c = \text{cloth resistance coefficient} = (\text{in water})/(\text{cP})(\text{ft/min})$, and $K_d = \text{dust-layer resistance coefficient} = (\text{in water})/(\text{cP})(\text{gr/ft}^2)(\text{ft/min})$. K_d may be expressed in the same units, using the Kozeny-Carman equation:

$$K_d = \frac{160.0(1 - \epsilon_v)}{\phi_s^2 D_p^2 \rho_s \epsilon_v^3} \quad (17-12)$$

where $D_p = \mu\text{m}$ and $\rho_s = \text{lbm/ft}^3$.

Data sufficient to permit reasonable predictions of K_d from Eq. (17-12) are seldom available. The range in the values of K_d that may be encountered in practice is illustrated in Fig. 17-57 in which available experimental determinations of K_d reported in the literature for a variety of dusts are plotted against particle size. In most cases no accurate particle-size data were reported, and the curves represent the estimated range of particle size involved. The data of Williams et al. (*op. cit.*) are related to a wide variety of dusts, and only the approximate limits enclosing these data are shown. Also included are curves predicted from Eq. (17-12) for specific values of ϕ_s , ρ_s , and ϵ_v . It is apparent from these curves that smaller particles tend toward higher values of ϵ_v , which is consistent with the observation that fine dusts (particularly those smaller than $10 \mu\text{m}$) have lower bulk densities than coarser fractions, apparently because of the effects of surface forces. For coarse dusts, K_d varies approximately inversely as the square of the particle diameter, which implies that the void fraction (or bulk density) does not change with particle size. However, for particles finer than $10 \mu\text{m}$, the value of K_d appears to become constant, increased voids compensating for the reduction in size. In addition to the increased voidages encountered with small particles, slip flow also contributes to the relative constancy of the experimental values of K_d for particle sizes under $5 \mu\text{m}$.

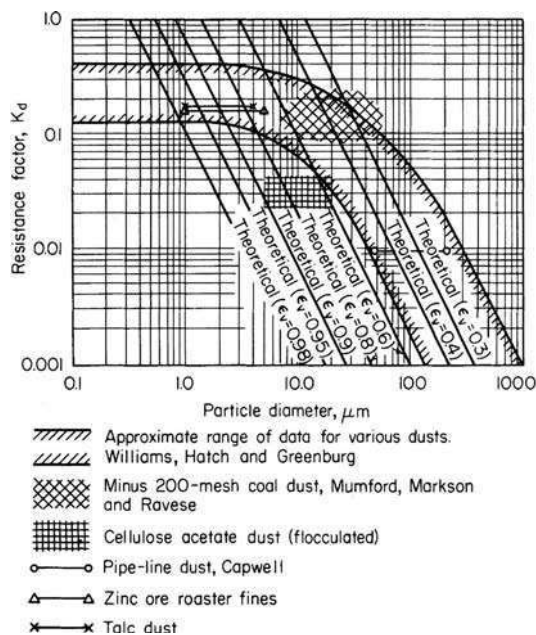


FIG. 17-57 Resistance factors for dust layers. Theoretical curves given are based on Eq. (20-78) for a shape factor of 0.5 and a true particle specific gravity of 2.0. [Williams, Hatch, and Greenburg, *Heat. Piping Air. Cond.*, **12**, 259 (1940); Mumford, Markson, and Ravese, *Trans. Am. Soc. Mech. Eng.*, **62**, 271 (1940); Capwell, *Gas*, **15**, 31 (August 1939)].

Because of the assumptions underlying its derivation, the Kozeny-Carman equation is not valid at void fractions greater than 0.7 to 0.8 (Billings and Wilder, op. cit.). In addition, in situ measurement of the void fraction of a dust layer on a filter fabric is extremely difficult and has seldom even been attempted. The structure of the layer is dependent on the character of the fabric surface as well as on the characteristics of the dust, whereas the application of Eq. (17-12) implicitly assumes that K_2 is dependent only on the properties of the dust. A smooth fabric surface permits the dust to become closely packed, leading to a relatively high value of K_2 . If the surface is napped or has numerous extended fibrils, the dust cake formed will be more porous and have a lower value of K_2 [Billings and Wilder, op. cit.; Snyder and Pring, *Ind. Eng. Chem.*, **47**, 960 (1955); and K. T. Semrau, unpublished data, SRI International, Menlo Park, Calif., 1952–1953].

Equation (17-10) indicates that for filtration at a given velocity the pressure drop is a linear function of the fabric dust loading w . In some cases, particularly with smooth-surfaced fabrics, this is at least approximately the case, but in other instances the function displays an upward curvature with increases in w , indicating compression of the dust layer, the fabric, or both, and a consequent increase in K_2 (Snyder and Pring, op. cit.; Semrau, op. cit.). Several investigations have shown K_2 to be increased by increases in the filtration velocity [Billings and Wilder, op. cit.; Spaite and Walsh, *Am. Ind. Hyg. Assoc. J.*, **24**, 357 (1968)]. However, the various investigators do not agree on the magnitude of the velocity effects. Billings and Wilder suggest assuming as an approximation that K_2 is directly proportional to the filtration velocity, but the actual relationship is probably dependent on the nature of the fabric and fabric surface, the characteristics of the dust, the dust loading on the fabric, and the pressure drop.

Clearly, the factors determining K_2 are far more complex than is indicated by a simple application of the Kozeny-Carman equation, and when possible, filter design should be based on experimental determinations made under conditions approximating those expected in the planned installation.

Types of Filters Current fabric-filter designs fall into three types, depending on the method of cleaning used: (1) shaker-cleaned,

(2) reverse-flow-cleaned, and (3) reverse-pulse-cleaned. The shaker-cleaned filter is the earliest form of bag filter (Fig. 17-58). The open lower ends of the bags are fastened over openings in the tube sheet that separates the lower dirty-gas inlet chamber from the upper clean-gas chamber. The bag supports from which the bags are suspended are connected to a shaking mechanism. The dirty gas flows upward into the filter bags, and the dust collects on the inside surfaces of the bags. When the gas pressure drop rises to a chosen upper limit as the result of dust accumulation, the gas flow is stopped and the shaker is operated, giving a whipping motion to the bags. The dislodged dust falls into the dust hopper located below the tube sheet. If the filter is to be operated continuously, it must be constructed with multiple compartments, so that the individual compartments can be sequentially taken off line for cleaning while the other compartments continue in operation (Fig. 17-59).

Shaker-cleaned filters are available as standard commercial units, although large baghouses for heavy-duty service are commonly custom-designed and -fabricated. The oval or round bags used in the standard units are usually 12 to 20 cm (5 to 8 in) in diameter and 2.5 to 5 m (8 to 17 ft) long. The large, heavy-duty baghouses may use bags up to 30 cm (12 in) in diameter and 9 m (30 ft) long. The bags must be made of woven fabrics to withstand the flexing and stretching involved in shaking. The fabrics may be made from natural fibers (cotton or wool) or synthetic fibers. Fabrics of glass or mineral fibers are generally too fragile to be cleaned by shaking and are usually used in reverse-flow-cleaned filters.

Large units (other than custom units) are usually built up of standardized rectangular sections in parallel. Each section contains on the order of 1000 to 2000 ft² of cloth, and the sections are assembled in the field to form a single filter housing. In this manner, the filter can be partitioned so that one or more sections at a time can be cut out of service for shaking or general maintenance.

Ordinary shaker-cleaned filters may be shaken every ¼ to 8 h, depending on the service. A manometer connected across the filter is useful in determining when the filter should be shaken. Fully automatic filters may be shaken as frequently as every 2 min, but bag maintenance will be greatly reduced if the time between shakings can be increased to 15 or 20 min without developing excessive pressure drop. Cleaning may be actuated automatically by a differential-pressure switch. It is essential that the gas flow through the filter be stopped when shaking in order to permit the dust to fall off. With very fine dust, it may even be necessary to equalize the pressure across the cloth [Mumford, Markson, and Ravese, *Trans. Am. Soc. Mech. Eng.*, **62**, 271 (1940)]. In practice this can be accomplished without interrupting the operation by cutting one section out of service at a time, as shown in Fig. 17-59. In automatic filters this operation involves closing the dampers, shaking the filter units either pneumatically or mechanically, sometimes with the addition of a reverse flow of cleaned gas through the filter, and lastly reopening the dampers. For compressed-air-operated automatic filters, this entire operation may take only 2 to 10 s. For ordinary mechanical filters equipped for automatic control, the operation may take as long as 3 min.

Equation (17-11) may be rewritten as

$$\Delta p_i = K_d \mu c_d V_f^2 t_m \quad (17-13)$$

where c_d = dust concentration in dirty gas, gr/ft³; and t_m = filtration time, min. This shows that the pressure drop due to dust accumulation varies as the square of the gas velocity through the filter. (The actual effect of velocity on pressure drop may be even greater in some instances.) Greater cloth area and reduced filtration velocity therefore afford substantial reductions in shaking frequency and in bag wear. Consequently, it is generally economical to be conservative in specifying cloth area. Shaker-cleaned filters are generally operated at filtration velocities of 0.3 to 2.5 m/min (1 to 8 ft/min) and at pressure drops of 0.5 to 1.5 kPa (2 to 6 in. water). For very fine dusts or high dust concentrations, filtration velocities should not exceed 1 m/min (3 ft/min). For fine fumes and dusts in heavy-duty installations, filtration velocities of 0.3 to 0.6 m/min (1 to 2 ft/min) have long been accepted on the basis of operating experience.

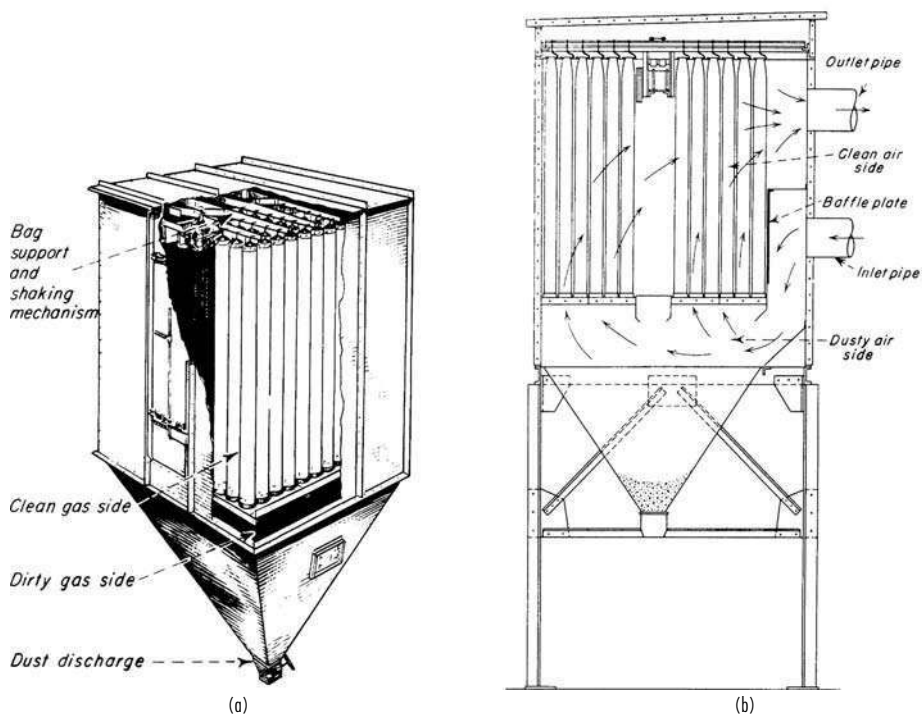


FIG. 17-58 Typical shaker-type fabric filters. (a) Buell Norblo (cutaway view). (b) Wheelabrator-Frye Inc. (sectional view).

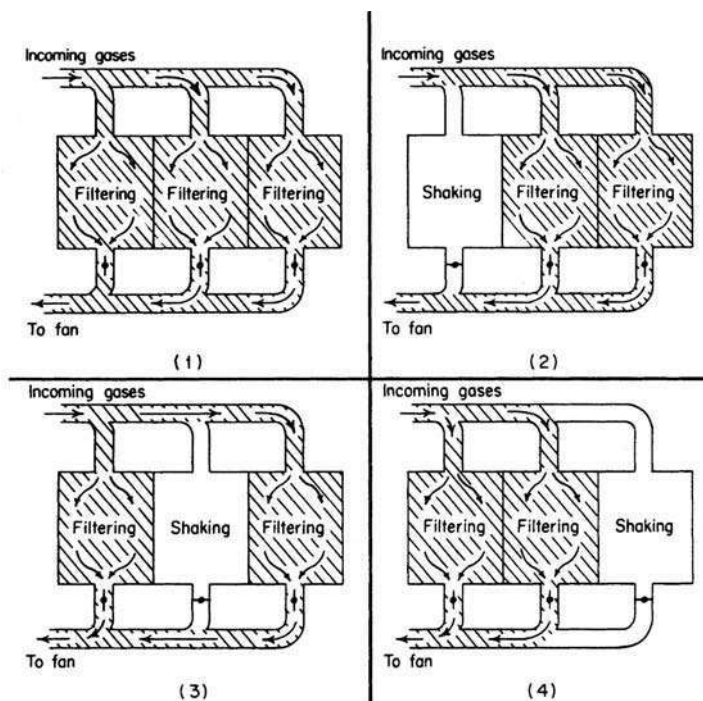


FIG. 17-59 Three-compartment bag filter at various stages in the cleaning cycle. (Wheelabrator-Frye Inc.)

Cyclone precleaners are sometimes used to reduce the dust load on the filter or to remove large hot cinders or other materials that might damage the bags. However, reducing the dust load on the filter by this means may not reduce the pressure drop, since the increase in K_2 produced by the reduction in average particle size may compensate for the decrease in the fabric dust loading.

In filter operation, it is essential that the gas be kept above its dew point to avoid water-vapor condensation on the bags and resulting plugging of the bag pores. However, fabric filters have been used successfully in steam atmospheres, such as those encountered in vacuum dryers. In such cases, the housing is generally steam-traced.

Reverse-flow-cleaned filters are generally similar to the shaker-cleaned filters except for the elimination of the shaker. After the flow of dirty gas has stopped, a fan is used to force clean gas through the bags from the clean-gas side. This flow of gas partly collapses the bags and dislodges the collected dust, which falls to the dust hopper. Rings are usually sewn into the bags at intervals along the length to prevent complete collapse, which would obstruct the fall of the dislodged dust. The principal applications of reverse-flow cleaning are in units using fiberglass fabric bags for dust collection at temperatures above 150°C (300°F). Collapsing and reinflation of the bags can be made sufficiently gentle to avoid putting excessive stresses on the fiberglass fabrics [Perkins and Imbalzano, "Factors Affecting the Bag Life Performance in Coal-Fired Boilers," 3d APCA Specialty Conference on the User and Fabric Filtration Equipment, Niagara Falls, N.Y., 1978; and Miller, *Power*, **125**(8), 78 (1981)]. As with shaker-cleaned filters, compartments of the baghouse are taken off line sequentially for bag cleaning. The gas for reverse-flow cleaning is commonly supplied in an amount necessary to give a superficial velocity through the bags of 0.5 to 0.6 m/min (1.5 to 2.0 ft/min), which is the same range as the filtration velocities frequently used.

In the reverse-pulse filter (frequently termed a reverse-jet filter), the filter bag forms a sleeve that is drawn over a wire cage, which is usually cylindrical (Fig. 17-60). The cage supports the fabric on the clean-gas side, and the dust is collected on the outside of the bag. A venturi nozzle is located in the clean-gas outlet from the bag. For cleaning, a jet of high-velocity air is directed through the venturi nozzle and into the bag, inducing a flow of cleaned gas to enter the bag and flow through the fabric to the dirty-gas side. The high-velocity jet is released in a sudden, short pulse (typical duration 100 ms or less) from a compressed-air line by a solenoid valve. The pulse of air and clean gas expands the bag and dislodges the collected dust. Rows of bags are cleaned in a timed sequence by programmed operation of the

solenoid valves. The pressure of the pulse is sufficient to dislodge the dust without cessation of the gas flow through the filter unit.

It has been a common practice to clean the bags on line (i.e., without stopping the flow of dirty gas into the filter), and reverse-pulse bag filters have been built without division into multiple compartments. However, investigations [Leith et al., *J. Air Pollut. Control Assoc.*, **27**, 636 (1977)] and experience have shown that, with online cleaning of reverse-pulse filters, a large fraction of the dust dislodged from the bag being cleaned may redeposit on neighboring bags rather than fall to the dust hopper. As a result, there is a growing trend to off-line cleaning of reverse-pulse filters. The baghouse is sectionalized so that the outlet-gas plenum serving the bags in a section can be closed off from the clean-gas exhaust, thereby stopping the flow of inlet gas through the bags. On the dirty-gas side of the tube sheet, the bags of the section are separated by partitions from the neighboring sections, where filtration is continuing. Sections of the filter are cleaned in rotation, as in shaker and reverse-flow filters.

Some manufacturers are using relatively low-pressure air (100 kPa, or 15 lbf/in², instead of 690 kPa, or 100 lbf/in²) and are eliminating the venturi tubes for clean-gas induction. Others have eliminated the separate jet nozzles located at the individual bags and use a single jet to inject a pulse into the outlet-gas plenum.

Reverse-pulse filters are typically operated at higher filtration velocities (air-to-cloth ratios) than shaker or reverse-flow filters designed for the same duty. Filtration velocities may range from 1 to 4.5 m/min (3 to 15 ft/min), depending on the dust being collected, but for most dusts the commonly used range is about 1.2 to 2.5 m/min (4 to 8 ft/min). The frequency of cleaning is also dependent on the nature and concentration of the dust, with the intervals between pulses varying from about 2 to 15 min.

The cleaning action of the pulse is so effective that the dust layer may be completely removed from the surface of the fabric. Consequently, the fabric itself must serve as the principal filter medium for at least a substantial part of the filtration cycle. Woven fabrics are unsuitable for such service, and felts of various types must be used. The bulk of the dust is still removed in a surface layer, but the felt ensures that an adequate collection efficiency is maintained until the dust layer has formed.

Filter Fabrics The cost of the filter bags represents a substantial part of the erected cost of a bag filter—typically 5 to 20 percent, depending on the bag material [Reigel and Bundy, *Power*, **121**(1), 68 (1977)]. The cost of bag repair and replacement is the largest component of the cost of bag-filter maintenance. Consequently, the proper choice of filter fabric is critical to both the technical performance and the economics of operating a filter. With the advent of synthetic fibers, it has become possible to produce fabrics having a wide range of properties. However, demonstrating the acceptability of a fabric still depends on experience with prolonged operation under the actual or simulated conditions of the proposed application. The choice of a fabric material for a given service is necessarily a compromise, since no single material possesses all the properties that may be desired. Following the choice of material, the type of fabric construction is critical.

Two principal types of fabric are adaptable to filter use: woven fabrics, which are used in shaker and reverse-flow filters; and felts, which are used in reverse-pulse filters. The felts made from synthetic fibers are needle felts (i.e., felted on a needle loom) and are normally reinforced with a woven insert. The physical properties and air permeabilities of some typical woven and felt filter fabrics are presented in Tables 17-6 and 17-7. The "air permeability" of a filter fabric is defined as the flow rate of air in cubic feet per minute (at 70°F, 1 atm) that will pass through 1 ft² of clean fabric under an applied differential pressure of ½ in. water. The resistance coefficient K_F of the clean fabric is defined by the equation in Table 17-6, which may be used to calculate the value of K_F from the air permeability. If Δp_i is taken as 0.5 in. water, μ as 0.0181 cP (the viscosity of air at 70°F and 1 atm), and V_f as the air permeability, then $K_F = 27.8/\text{air permeability}$.

Collection Efficiency The inherent collection efficiency of fabric filters is usually so high that, for practical purposes, the precise level has not commonly been the subject of much concern. Furthermore, for collection of a given dust, the efficiency is usually fixed by the choices of filter fabric, filtration velocity, method of cleaning, and

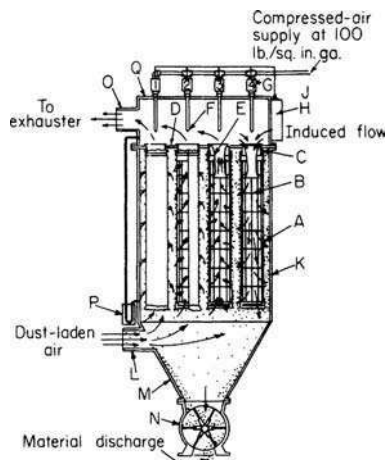


FIG. 17-60 Reverse-pulse fabric filter: (a) filter cylinders; (b) wire retainers; (c) collars; (d) tube sheet; (e) venturi nozzle; (f) nozzle or orifice; (g) solenoid valve; (h) timer; (j) air manifold; (k) collector housing; (l) inlet; (m) hopper; (n) air lock; (o) upper plenum. (Mikropul Division, U.S. Filter Corp.)

17-50 GAS-SOLID OPERATIONS AND EQUIPMENT

TABLE 17-6 Resistance Factors and Air Permeabilities for Typical Woven Fabrics

Cloth	Pore size, ^a in	Threads/in	Weight, oz/yd ²	Thread ^a diameter, in	K_F [†]	Air permeability, (ft ³ /min)/ft ² at $\Delta p_i = \frac{1}{2}$ in H ₂ O
Osnaburg cotton	0.01	32 × 28		0.02	0.51	55
Osnaburg cotton (soiled)‡		32 × 28			4.80	5.8
Drill cotton	0.01	65 × 40	5.28	0.01	0.093	300
Cotton§		46 × 56			1.39	20
Cotton§		104 × 68			1.54	18
Cotton sateen (unnaped)	0.007	96 × 56	6.88	0.009	0.27	103
Cotton sateen (unnaped)	0.005	96 × 64	8.23	0.01	0.88	32
Cotton sateen (unnaped)		96 × 60		0.012	1.63	17
Cotton sateen (unnaped)	0.004	96 × 56	10.2	0.011	1.12	25
Wool					0.25	111
Wool		40 × 50	11.5	0.014	0.33	84
Wool, white§		36 × 32			0.15	185
Wool, black§		28 × 30			0.25	110
Wool§		30 × 26			0.51	55
Vinyon§		37 × 37			0.12	23
Nylon tackle twill		72 × 196		0.010	0.66	42
Nylon sailcloth		130 × 130		0.007	1.66	17
Nylon§		37 × 37			1.74	16
Nylon§					3.71	7.5
Asbestos§					0.56	50
Orlon§		72 × 72			0.66	42
Orlon§		74 × 38			0.75	37
Orlon§					1.16	24
Orlon§					1.98	14
Smoothtex nickel screen		(300 mesh)			0.16	174
Glass		32 × 28		0.03	1.60	17
Dacron		60 × 40	5.8		0.84	33
Dacron		76 × 48	13.4		0.29	9.5
Teflon		76 × 70	8.7		1.39	20

^a Estimates based on microscopic examination.

[†] Measured with atmospheric air. This value will be constant only for streamline flow, which is the case for values of $\rho V_f/\mu$ of less than approximately 100.

$$K_F = \Delta p_i / \mu V_f$$

where Δp_i = pressure drop, in water; μ = gas viscosity, cP; V_f = superficial gas velocity through cloth, ft/min; and ρ = gas density, lb/ft³.

‡ Cloth, similar to previous one, that had been in service and contained dust in pores although free of surface accumulation.

§ Data from Pring, *Air Pollution*, McGraw-Hill, New York, 1952, p. 280.

cleaning cycle, leaving few if any controllable variables by which efficiency can be further influenced. Inefficiency usually results from bags that are poorly installed, torn, or stretched from excessive dust loading and pressure drop. Of course, certain types of fabrics may simply be unsuited for filtration of a particular dust, but usually this will soon become obvious.

Few basic studies of the efficiency of bag filters have been made. Increased dust penetration immediately following cleaning has been readily observed while the dust layer is being reestablished. However, field and laboratory studies have indicated that during the rest of the filtration cycle the effluent-dust concentration tends to remain

constant regardless of the inlet concentration [Dennis, *J. Air Pollut. Control Assoc.*, **24**, 1156 (1974)]. In addition, there has been little indication that the penetration is strongly related to dust-particle size, except possibly in the low-submicrometer range. These observations appear to be generally consistent with sieving being the principal collection mechanism.

Leith and First [*J. Air Pollut. Control Assoc.*, **27**, 534 (1977); **27**, 754 (1977)] studied the collection efficiency of reverse-pulse filters and concluded that once the dust cake has been established, "straight-through" penetration by dust particles that pass through the filter without being stopped is negligible by comparison with penetration by dust

TABLE 17-7 Physical Properties of Selected Felts for Reverse-Pulse Filters

Fiber	Weight, oz/yd ²	Thickness, in	Breaking strength, lbf/in width	Elongation, % to rupture	Air permeability, (ft ³ /min)/ft ² at $\Delta p_i = \frac{1}{2}$ in water	K_F
Wool	23.1	0.135			27.1	1.03
Wool	21.2	0.129			29.8	0.93
Orlon ^o	10.9	0.045	65	18	20–25	1.11–1.39
Orlon ^o	17.9	0.088	85	18	15–20	1.39–1.85
Orlon ^o	24	0.125	110	60	10–20	1.39–2.78
Acrilan ^o	17.9	0.075	100	22	15–20	1.39–1.85
Dynel ^o	24	0.125	60	80	30–40	0.70–0.93
Dacron ^o	17.9	0.080	125	22	15–20	1.39–1.85
Dacron ^o	9.9	0.250	20	150	200–225	0.11–0.14
Dacron ^o	24	0.125	175	80	20–30	0.93–1.39
Nylon ^o	24	0.125	100	100	30–40	0.70–0.93
Amel ^o	24	0.125	60	80	30–40	0.70–0.93
Teflon	15.6	0.053			82.5	0.34
Teflon	43.5	0.119			21.6	1.29

^o These data courtesy of American Felt Co.

that actually deposits initially and then "seeps" through the fabric to be reentrained into the exit air stream. They also noted that "pinholes" may form in the dust cake, particularly over pores between yarns in a woven fabric, and that particles may subsequently penetrate straight through at the pinholes. The formation of pinholes, or "cake puncture," had been observed earlier by Stephan et al. [*Am. Ind. Hyg. Assoc. J.*, **21**, 1 (1960)], but without measurement of the associated loss of collection efficiency. When a supported flat filter medium with extremely fine pores (e.g., glass-fiber paper, membrane filter) was used, no cake puncture took place even with very high pressure differentials across the cake. However, puncture did occur when a cotton-sateen filter fabric was used as the cake support. The formation of pinholes with certain combinations of dusts, fabrics, and filtration conditions was also observed by Kosciannowski et al. (EPA-600/7-78-056, 1978). Evidently puncture occurs when the local cake structure is not strong enough to maintain a bridge over the aperture represented by a large pore and the portion of the cake covering the pore is blown through the fabric. This suggests that formation of pinholes will be highly dependent on the strength of the surface forces between particles that produce flocculation of dusts. The seepage of a dust through a filter is probably also closely related to the strength of the surface forces.

Surface pores can be greatly reduced in size by coating what will become the dusty side of the filter fabric with a thin microporous membrane that is supported by the underlying fabric. That has the effect of decreasing the effective penetration, both by eliminating cake pinholes, and by preventing the seepage of dust that is dragged through the fabric by successive cleanings. A variety of different membrane-forming polymers can be used in compatible service. The most versatile and effective surface filtration membranes are microfibrillar Teflon as already described by Brinckman and Maresca [*ASME Med. Waste Symp.* (1992)] in the section on dry scrubbing.

Granular-Bed Filters Granular-bed filters may be classified as "depth" filters, since dust particles deposit in depth within the bed of granules. The granules themselves present targets for the deposition of particles by inertia, diffusion, flow-line interception, gravity, and electrostatic attraction, depending on the dust and filter characteristics and the operating conditions. Other deposition mechanisms are minor at most. Although it is physically possible under some circumstances for a dust layer to form on the inlet face of the filter, the practical limits of gas pressure drop will normally have been reached long before a surface dust layer can be established.

Granular-bed filters may be divided into three classes:

1. **Fixed-bed, or packed-bed, filters.** These units are not cleaned when they become plugged with deposited dust particles but are broken up for disposal or simply abandoned. If they are constructed from fine granules (e.g., sand particles), they may be designed to give high collection efficiencies on fine dust particles. However, if such a filter is to have a reasonable operating life, it can be used only on a gas containing a low concentration of dust particles.

2. **Cleanable granular-bed filters.** In these devices provisions are made to separate the collected dust from the granules either continuously or periodically, so that the units can operate continuously on gases containing moderate to high dust concentrations. The necessity for cleaning and recycling the granules generally restricts the practical lower granule size to about 3 to 10 mm. This in turn makes it difficult to attain high collection efficiencies on fine particles with granule beds of reasonable depth and gas pressure drop.

3. **Fluidized-bed filters.** Fluidized beds of granules have received considerable study on theoretical and experimental levels but have not been applied on a practical commercial scale.

Fixed Granular-Bed Filters Fixed-bed filters composed of granules have received considerable theoretical and experimental study [Thomas and Yoder, *AMA Arch. Ind. Health*, **13**, 545 (1956); **13**, 550 (1956); Knetting and Beeckmans, *Can. J. Chem. Eng.*, **52**, 703 (1974); Schmidt et al., *J. Air Pollut. Control Assoc.*, **28**, 143 (1978); Tardos et al., *J. Air Pollut. Control Assoc.*, **28**, 354 (1978); and Gutfinger and Tardos, *Atmos. Environ.*, **13**, 853 (1979)]. The theoretical approach is the same as that used in the treatment of deep-bed fibrous filters.

Fibers for filter applications can be produced with diameters smaller than it is practical to obtain with granules. Consequently, most concern with filtration of fine particles has been focused on fibrous-

bed rather than granular-bed filters. However, for certain specialized applications granular beds have shown some superior properties, such as greater dimensional stability. Granular-bed filters of special design (deep-bed sand filters) have been used since 1948 for removing radioactive particles from waste air and gas streams in atomic energy plants (Lapple, "Interim Report—200 Area Stack Contamination," U.S. AEC Rep. HDC-743, Oct. 11, 1948; Juvinall et al., "Sand-Bed Filtration of Aerosols: A Review of Published Information," U.S. AEC Rep. ANL-7683, 1970; and Burchsted et al., *Nuclear Air Cleaning Handbook*, U.S. ERDA 76-21, 1976). The filter characteristics needed included high collection efficiency on fine particles, large dust-holding capacity to give long operating life, and low maintenance requirements. The sand filters are as much as 2.7 m (9 ft) in depth and are constructed in graded layers with about a 2:1 variation in the granule size from one layer to the next. The air-flow direction is upward, and the granules decrease in size in the direction of the air flow. The bottom layer is composed of rocks about 5 to 7.5 cm (2 to 3 in) in diameter, and granule sizes in successive layers decrease to 0.3 to 0.6 mm (50 to 30 mesh) in the finest layer. With superficial face velocities of about 1.5 m/min (5 ft/min), gas pressure drops of clean filters have ranged from 1.7 to 2.8 kPa (7 to 11 in water). Collection efficiencies of up to 99.98 percent with a polydisperse dioctyl phthalate aerosol of 0.7- μ m mean diameter have been reported (Juvinall et al., op. cit.). Operating lives of 5 years or more have been attained.

Cleanable Granular-Bed Filters The principal objective in the development of cleanable granular-bed filters is to produce a device that can operate at temperatures above the range that can be tolerated with fabric filters. In some of the devices, the granules are circulated continuously through the unit, then are cleaned of the collected dust and returned to the filter bed. In others, the granular bed remains in place but is periodically taken out of service and cleaned by some means, such as backflushing with air.

A number of moving-bed granular filters have used cross-flow designs. One form of cross-flow moving-granular-bed filter, produced by the Combustion Power Company (Fig. 17-61), is currently in commercial use in some applications. The granular filter medium consists of 1/8- to 1/4-in (3- to 6-mm) pea gravel. Gas face velocities range from 30 to 46 m/min (100 to 150 ft/min), and reported gas pressure drops are in the range of 0.5 to 3 kPa (2 to 12 in water). The original form of the device [Reese, TAPPI, **60**(3), 109 (1977)] did not incorporate electrical augmentation. Collection efficiencies for submicrometer particles were low, and the electrical augmentation was added to correct the deficiency (Parquet, "The Electroscrubber Filter: Applications and Particulate Collection Performance," EPA-600/9-82-005c, 1982, p. 363). The electrostatic grid immersed in the bed of granules is charged to a potential of 20,000 to 30,000 V, producing an electric field between the grid and the inlet and outlet louvers that enclose the bed. No ionizing electrode is used to charge particles in the incoming gas; reliance is placed on the existence of natural charges on the dust particles. Individual dust particles commonly carry positive or negative charges even though the net charge on the dust as a whole is normally neutral. Depending on their charges, dust particles are attracted or repelled by the electrical field and are therefore caused to deposit on the rocks in the bed.

Self et al. ("Electrical Augmentation of Granular Bed Filters," EPA-600/9-80-039c, 1980, p. 309) demonstrated in theoretical studies and laboratory experiments that such an augmentation system should yield substantial increases in the collection efficiency for fine particles if the particles carry significant charges. Significant improvements in the performance of the Combustion Power units with electrical augmentation have been reported by the manufacturer (Parquet, op. cit.).

Another type of gravel-bed filter, developed by GFE in Germany, has had limited commercial application in the United States [Schueler, *Rock Prod.*, **76**(7), 66 (1973); **77**(11), 39 (1974)]. After precleaning in a cyclone, the gas flows downward through a stationary horizontal filter bed of gravel. When the bed becomes loaded with dust, the gas flow is cut off, and the bed is backflushed with air while being stirred with a double-armed rake that is rotated by a gear motor. The backflush air also flows backward through the cyclone, which then acts as a dropout chamber. Multiple filter units are constructed in parallel so that individual units can be taken off the line for cleaning. The dust dislodged

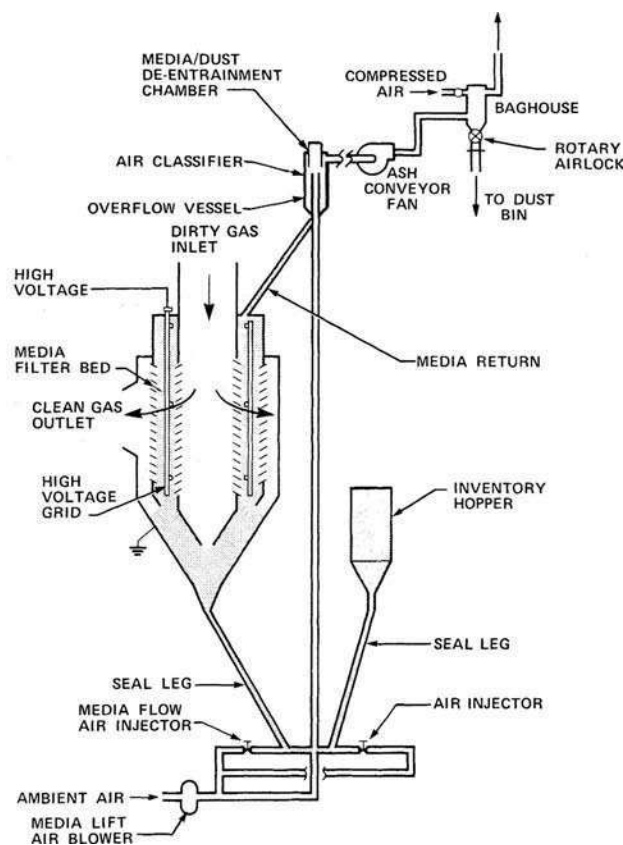


FIG. 17-61 Electrically augmented granular-bed filter. (Combustion Power Company.)

from the bed and carried by the backflush air is flocculated, and part is collected in the cyclone. The backflush air with the remaining suspended dust is cleaned in the other gravel-bed filter units that are operating on line. Performance tests made on one installation for the U.S. Environmental Protection Agency (EPA-600/7-78-093, 1978) did not give clear results but indicated that collection efficiencies were low on particles under $2\ \mu\text{m}$ and that some of the dust in the backflush air was redispersed sufficiently to penetrate the operating filter units.

Air Filters The types of equipment previously described are intended primarily for the collection of process dusts, whereas air filters comprise a variety of filtration devices designed for the collection of particulate matter at low concentrations, usually atmospheric dust. The difference in the two categories of equipment is not in the principles of operation but in the adaptations required to deal with the different quantities of dust. Process-dust concentrations may run as high as several hundred grams per cubic meter (or grains per cubic foot) but usually do not exceed $45\ \text{g/m}^3$ ($20\ \text{gr/ft}^3$). Atmospheric-dust concentrations that may be expected in various types of locations are shown in Table 17-8 and are generally below $12\ \text{mg/m}^3$ ($5\ \text{gr/1000 ft}^3$).

The most frequent application of air filters is in cleaning atmospheric air for building ventilation, which usually requires only moderately high collection-efficiency levels. However, a variety of industrial operations developed mostly since the 1940s require air of extreme cleanliness, sometimes for pressurizing enclosures such as clean rooms and sometimes for use in a process itself. Examples of applications include the manufacture of antibiotics and other pharmaceuticals, the production of photographic film, and the manufacture and assembly of semiconductors and other electronic devices. Air cleaning at the necessary efficiency levels is accomplished by the use of high-efficiency fibrous filters that have been developed since the 1940s.

TABLE 17-8 Average Atmospheric-Dust Concentrations*

$1\ \text{gr/1000 ft}^3 = 2.3\ \text{mg/m}^3$

Location	Dust concentration, gr/1000 ft ³
Rural and suburban districts	0.02–0.2
Metropolitan districts	0.04–0.4
Industrial districts	0.1–2.0
Ordinary factories or workrooms	0.2–4.0
Excessive dusty factories or mines	4.0–400

*Heating Ventilating Air Conditioning Guide, American Society of Heating, Refrigerating and Air-Conditioning Engineers, New York, 1960, p. 77.

Air filters are also used to protect internal-combustion engines and gas turbines by cleaning the intake air. In some locations and applications, the atmospheric-dust concentrations encountered are much higher than those normally encountered in air-conditioning service.

High-efficiency air filters are sometimes used for emission control when particulate contaminants are low in concentration but present special hazards; cleaning of ventilation air and other gas streams exhausted from nuclear plant operations is an example.

Air-Filtration Theory Current high-efficiency air- and gas-filtration methods and equipment have resulted largely from the development of filtration theory since about 1930 and particularly since the 1940s. Much of the theoretical advance was originally encouraged by the requirements of the military and atomic energy programs. The fibrous filter has served both as a practical device and as a model for theoretical and experimental investigation. Extensive reviews and new treatments of air-filtration theory and experience have been presented by Chen [*Chem. Rev.*, **55**, 595 (1955)], Dorman ("Filtration," in Davies, *Aerosol Science*, Academic, New York, 1966), Pich (*Theory of Aerosol Filtration by Fibrous and Membrane Filters*, in *ibid.*), Davies (*Air Filtration*, Academic, New York, 1973), and Kirsch and Stechkina ("The Theory of Aerosol Filtration with Fibrous Filters," in Shaw, *Fundamentals of Aerosol Science*, Wiley, New York, 1978). The theoretical treatment of filtration starts with the processes of dust-particle deposition on collecting bodies, as outlined in Fig. 17-35 and Table 17-2. All the mechanisms shown in Table 17-2 may come into play, but inertial deposition, flow-line interception, and diffusional deposition are usually dominant. Electrostatic precipitation may become a major mechanism if the collecting body, the dust particle, or both, are charged. Gravitational settling is a minor influence for particles in the size range of usual interest. Thermal precipitation is nil in the absence of significant temperature gradients. Sieving is a possible mechanism only when the pores in the filter medium are smaller than or approximately equal to the particle size and will not be encountered in fibrous filters unless they are loaded sufficiently for a surface dust layer to form.

The theoretical prediction of the efficiency of collection of dust particles by a fibrous filter consists of three steps (Chen, *op. cit.*):

1. Calculation of the target efficiency η_0 of an isolated fiber in an air stream having a superficial velocity the same as that in the filter
2. Determining the difference between the target efficiency of the isolated fiber and that of an individual fiber in the filter array η_i
3. Determining the collection efficiency of the filter η from the target efficiency of the individual fibers

The results of computations of η_0 for an isolated fiber are illustrated in Figs. 17-62 and 17-63. The target efficiency η_0 of an individual fiber in a filter differs from η_0 for two main reasons (Pich, *op. cit.*): (1) the average gas velocity is higher in the filter, and (2) the velocity field around the individual fibers is influenced by the proximity of neighboring fibers. The interference effect is difficult to determine on a purely theoretical basis and is usually evaluated experimentally. Chen (*op. cit.*) expressed the effect with an empirical equation:

$$\eta_i = \eta_0[1 + K_d(1 - \epsilon_v)] \quad (17-14)$$

This indicates that the target efficiency of the fiber is increased by the proximity of other fibers. The value of K_d averaged 4.5 for values of the void fraction ϵ_v , ranging from 0.90 to 0.99. Extending use of the equation to values of ϵ_v lower than 0.90 may result in large errors.

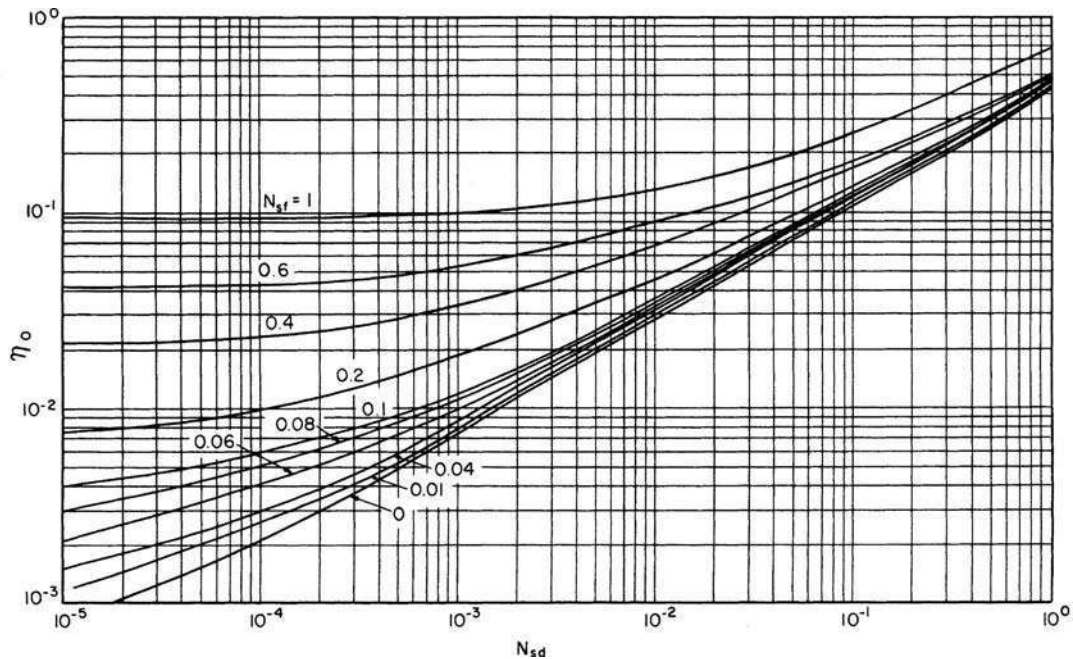


FIG. 17-62 Isolated fiber efficiency for combined diffusion and interception mechanism at $N_{Re} = 10^{-2}$. [Chen, Chem. Rev., **55**, 595 (1955).]

The collection efficiency of the filter may be calculated from the fiber target efficiency and other physical characteristics of the filter (Chen, op. cit.):

$$N_t = \frac{4\eta_t L(1 - \epsilon_v)}{\pi D_b \epsilon_v} \quad (17-15)$$

where D_b = fiber diameter and L = filter thickness. The derivation of Eq. (17-15) assumes that (1) η_t is the same throughout the filter, (2) all fibers are of the same diameter D_b , are cylindrical and are normal to the direction of the gas flow, (3) the fraction of the particles deposited in any one layer of fiber is small, and (4) the gas passing through the filter is essentially completely remixed after it leaves one layer of the filter and before it enters the next. The first assumption requires that Eq. (17-15) apply only for particles of a single size for which there are corresponding values of η_t , η , and N_t .

For filters of high porosity, ϵ_v approaches unity and Eq. (17-15) reduces to the expression used by Wong et al. [J. Appl. Phys., **27**, 161 (1956)] and Thomas and Lapple [Am. Inst. Chem. Eng. J., **7**, 203 (1961)]:

$$N_t = \frac{4\eta_t L(1 - \epsilon_v)}{\pi D_b} \quad (17-16)$$

The foregoing procedure is commonly employed in reverse to determine or confirm fiber target efficiencies from the experimentally determined efficiencies of fibrous filter pads.

Filtration theory assumes that a dust particle that touches a collector body adheres to it. This assumption appears to be valid in most cases, but evidence of nonadherence, or particle bouncing, has appeared in some instances. Wright et al. ("High Velocity Air Filters," WADC TR 55-457, ASTIA Doc. AD-142075, 1957) investigated the performance of fibrous filters at filtration velocities of 0.091 to 3.05 m/s (0.3 to 10 ft/s), using 0.3- μ m and 1.4- μ m supercooled liquid aerosols and a 1.2- μ m solid aerosol. The collection efficiencies agreed well with theoretical predictions for the liquid aerosols and apparently also for the solid aerosol at filtration velocities under 0.3 m/s (1 ft/s). But at filtration velocities above 0.3 m/s some of the solid particles failed to adhere. With a filter composed of 30- μ m glass fibers and a filtration velocity of 9.1 m/s (30 ft/s), there were indications that 90 percent of the solid aerosol particles striking a fiber bounced off.

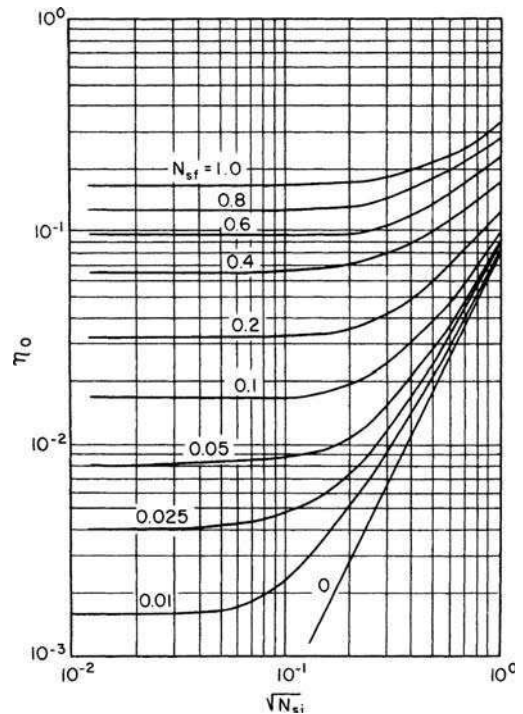


FIG. 17-63 Isolated fiber efficiency for combined inertia and interception mechanisms at $N_{Re} = 0.2$. [Chen, Chem. Rev., **55**, 595 (1955).]

Bouncing may be regarded as a defect in the particle-deposition process. However, particles that have been deposited in filters may subsequently be blown off and reentrained into the air stream (Corn, "Adhesion of Particles," in Davies, *Aerosol Science*, Academic, New York, 1966; and Davies, op. cit.).

The theories of filtration by a fibrous filter relate only to the initial efficiency of the clean filter in the "static" period of filtration before the deposition of any appreciable quantity of dust particles. The deposition of particles in a filter increases the number of targets available to intercept particles, so that collection efficiency increases as the filter loads. At the same time, the filter undergoes clogging and the pressure drop increases. No theory is available for dealing with the "dynamic" period of filtration in which collection efficiency and pressure drop vary with the loading of collected dust. The theoretical treatment of this filtration period is incomparably more complex than that for the "static" period. Investigators have noted that both the increase in collection efficiency and the increase in pressure drop are exponential functions of the loading of collected dust or are at least roughly so (Davies, op. cit.). Some empirical relationships have been derived for correlating data in particular instances.

The dust particles collected by a fibrous filter do not deposit in uniform layers on fibers but tend to deposit preferentially on previously deposited particles (Billings, "Effect of Particle Accumulation in Aerosol Filtration," Ph.D. dissertation, California Institute of Technology, Pasadena, 1966), forming chainlike agglomerates termed "dendrites." The growth of dendritic deposits on fibers has been studied experimentally [Billings, op. cit.; Bhutra and Payatakes, *J. Aerosol Sci.*, **10**, 445 (1979)], and Payatakes and coworkers [Payatakes and Tien, *J. Aerosol Sci.*, **7**, 85 (1976); Payatakes, *Am. Inst. Chem. Eng. J.*, **23**, 192 (1977); and Payatakes and Gradon, *Chem. Eng. Sci.*, **35**, 1083 (1980)] have attempted to model the growth of dendrites and its influence on filter efficiency and pressure drop.

Air-Filter Types Air filters may be broadly divided into two classes: (1) panel, or unit, filters; and (2) automatic, or continuous, filters. Panel filters are constructed in units of convenient size (commonly 20- by 20-in or 24- by 24-in face area) to facilitate installation, maintenance, and cleaning. Each unit consists of a cleanable or replaceable cell or filter pad in a substantial frame that may be bolted to the frames of similar units to form an airtight partition between the source of the dusty air and the destination of the cleaned air.

Panel filters may use either viscous or dry filter media. Viscous filters are so called because the filter medium is coated with a tacky liquid of high viscosity (e.g., mineral oil and adhesives) to retain the dust. The filter pad consists of an assembly of coarse fibers (now usually metal, glass, or plastic). Because the fibers are coarse and the media are highly porous, resistance to air flow is low and high filtration velocities can be used.

Dry filters are usually deeper than viscous filters. The dry filter media use finer fibers and have much smaller pores than the viscous media and need not rely on an oil coating to retain collected dust. Because of their greater resistance to air flow, dry filters must use lower filtration velocities to avoid excessive pressure drops. Hence, dry media must have larger surface areas and are usually pleated or arranged in the form of pockets (Fig. 17-64), generally sheets of cellulose pulp, cotton, felt, or spun glass.

Automatic filters are made with either viscous-coated or dry filter media. However, the cleaning or disposal of the loaded medium is essentially continuous and automatic. In most such devices the air passes horizontally through a movable filter curtain. As the filter loads with dust, the curtain is continuously or intermittently advanced to expose clean media to the air flow and to clean or dispose of the loaded medium. Movement of the curtain can be provided by a hand crank or a motor drive. Movement of a motor-driven curtain can be actuated automatically by a differential-pressure switch connected across the filter.

High-Efficiency Air Cleaning Air-filter systems for nuclear facilities and for other applications demanding extremely high standards of air purity require filtration efficiencies well beyond those attainable with the equipment described above. The *Nuclear Air Cleaning Handbook* (Burchsted et al., op. cit.) presents an extensive treatment of the requirements for and the design of such air-cleaning

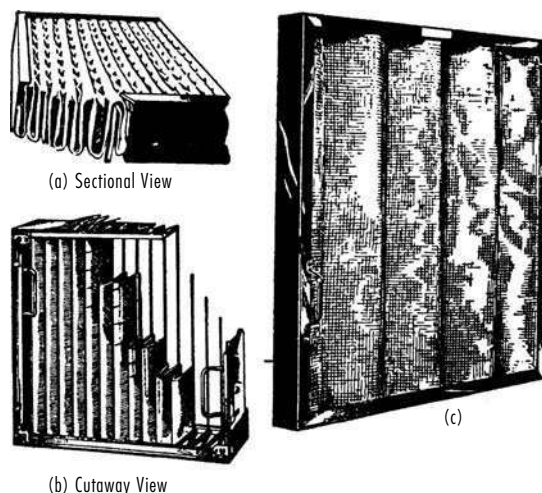


FIG. 17-64 Typical dry filters. (a) Throwaway type, Airplex (Davies Air Filter Corporation). (b) Replaceable medium type, Airmat PL-24, cutaway view (American Air Filter Co., Inc.). (c) Cleanable type, Amirglass sawtooth (Amirton Company).

facilities. Much of the material is pertinent to high-efficiency air-filter systems for applications to other than nuclear facilities.

HEPA (high-efficiency particulate air) filters were originally developed for nuclear and military applications but are now widely used and are manufactured by numerous companies. By definition, an HEPA filter is a "throwaway, extended-medium dry-type" filter having (1) a minimum particle-removal efficiency of not less than 99.97 percent for 0.3- μm particles, (2) a maximum resistance, when clean, of 1.0 in water when operated at rated air-flow capacity, and (3) a rigid casing extending the full depth of the medium (Burchsted et al., op. cit.). The filter medium is a paper made of submicrometer glass fibers in a matrix of larger-diameter (1- to 4- μm) glass fibers. An organic binder is added during the papermaking process to hold the fibers and give the paper added tensile strength. Filter units are made in several standard sizes (Table 17-9).

Because HEPA filters are designed primarily for high efficiency, their dust-loading capacities are limited, and it is common practice to use prefilters to extend their operating lives. In general, HEPA filters should be protected from (1) lint, (2) particles larger than 1 to 2 μm in diameter, and (3) dust concentrations greater than 23 mg/m^3 (10 $\text{gr}/1000 \text{ ft}^3$). Air filters used in nuclear facilities as prefilters and building-supply air filters are classified as shown in Table 17-10. The standard of the American Society of Heating, Refrigerating and Air-Conditioning Engineers (*Method of Testing Air Cleaning Devices Used in General Ventilation for Removing Particulate Matter*, ASHRAE 52-68, 1968) requires both a dust-spot (dust-stain) efficiency test made with atmospheric dust and a weight-arrestance test made with a synthetic test dust. A more precise comparison of the different groups of filters, based on removal efficiencies for particles of specific sizes, is presented in Table 17-11.

TABLE 17-9 Standard HEPA Filters*

Face dimensions, in	Depth, less gaskets, in	Design air-flow capacity at clean-filter resistance of 1.0 in water (standard ft^3/min)
24 × 24	11½	1000
24 × 24	5½	500
12 × 12	5½	125
8 × 8	5½	20
8 × 8	3¼	25

*Burchsted et al., *Nuclear Air Cleaning Handbook*, ERDA 76-21, Oak Ridge, Tenn., 1976.

TABLE 17-10 Classification of Common Air Filters*

Group	Efficiency	Filter type	Stain test efficiency, %	Arrestance, %
I	Low	Viscous impingement, panel type	<20†	40–80†
II	Moderate	Extended medium, dry type	20–60†	80–96†
III	High	Extended medium, dry type	60–98‡	96–99†
HEPA	Extreme	Extended medium, dry type	100§	100†

*Burchsted et al., *Nuclear Air Cleaning Handbook*, ERDA 76-21, Oak Ridge, Tenn., 1976.

†Test using synthetic dust.

‡Stain test using atmospheric dust.

§ASHRAE/52-68, American Society of Heating, Refrigerating and Air-Conditioning Engineers.

Table 17-12 presents the relative performance of Group I, II, and III filters with respect to air-flow capacity, resistance, and dust-holding capacity. The dust-holding capacities correspond to the manufacturers' recommended maximum allowable increases in air-flow resistance. The values for dust-holding capacity are based on tests with a synthetic dust and hence are relative. The actual dust-holding capacity in a specific application will depend on the characteristics of the dust encountered. In some instances it may be appropriate to use two or more stages of precleaning in air-filter systems to achieve a desired combination of operating life and efficiency. In very dusty locations, inertial devices such as multiple small cyclones may be used as first-stage separators.

Electrical Precipitators When particles suspended in a gas are exposed to gas ions in an electrostatic field, they will become charged and migrate under the action of the field. The functional mechanisms of electrical precipitation may be listed as follows:

1. Gas ionization
2. Particle collection
 - a. Production of electrostatic field to cause charging and migration of dust particles
 - b. Gas retention to permit particle migration to a collection surface
 - c. Prevention of reentrainment of collected particles
 - d. Removal of collected particles from the equipment

There are two general classes of electrical precipitators: (1) single-stage, in which ionization and collection are combined; (2) two-stage, in which ionization is achieved in one portion of the equipment, followed by collection in another. Various types in each class differ essentially in the details by which each function is accomplished.

The underlying theory presented in the following paragraphs assumes that the dust concentration is small, since only very incomplete evaluations for conditions of high dust concentration have been made.

Field Strength Whereas the applied potential or voltage is the quantity commonly known, it is the field strength that determines behavior in an electrostatic field. When the current flow is low (i.e., before the onset of spark or corona discharge), these are related by the following equations for two common forms of electrodes:

TABLE 17-11 Comparison of Air Filters by Percent Removal Efficiency for Various Particle Sizes*

Group	Efficiency	Removal efficiency, %, for particle size of			
		0.3 μm	1.0 μm	5.0 μm	10.0 μm
I	Low	0–2	10–30	40–70	90–98
II	Moderate	10–40	40–70	85–95	98–99
III	High	45–85	75–99	99–99.9	99.9
HEPA	Extreme	99.97 min	99.99	100	100

*Burchsted et al., *Nuclear Air Cleaning Handbook*, ERDA 76-21, Oak Ridge, Tenn., 1976.

TABLE 17-12 Air-Flow Capacity, Resistance, and Dust-Holding Capacity of Air Filters*

Group	Efficiency	Air-flow capacity, $\text{ft}^3/(\text{min} \cdot \text{ft}^2)$ of frontal area)	Resistance, in water		Dust-holding capacity, $\text{g}/(1000 \text{ ft}^3 \cdot \text{min})$ of air-flow capacity)
			Clean filter	Used filter	
I	Low	300–500	0.05–0.1	0.3–0.5	50–1000
II	Moderate	250–750	0.1–0.5	0.5–1.0	100–500
III	High	250–750	0.20–0.5	0.6–1.4	50–200

*Burchsted et al., *Nuclear Air Cleaning Handbook*, ERDA 76-21, Oak Ridge, Tenn., 1976.

Parallel plates:

$$\mathcal{E} = E/B_e \quad (17-17)$$

Concentric cylinders (wire-in-cylinder):

$$\mathcal{E} = \frac{E}{r \ln (D_t/D_d)} \quad (17-18)$$

The field strength is uniform between parallel plates, whereas it varies in the space between concentric cylinders, being highest at the surface of the central cylinder. After corona sets in, the current flow will become appreciable. The field strength near the center electrode will be less than given by Eq. (17-18) and that in the major portion of the clearance space will be greater and more uniform [see Eqs. (17-23) and (17-24)].

Potential and Ionization In order to obtain gas ionization it is necessary to exceed, at least locally, the electrical breakdown strength of the gas. Corona is the name applied to such a local discharge that fails to propagate itself. Sparking is essentially an advanced stage of corona in which complete breakdown of the gas occurs along a given path. Since corona represents a local breakdown, it can occur only in a nonuniform electrical field (Whitehead, *Dielectric Phenomena—Electrical Discharge in Gases*, Van Nostrand, Princeton, N.J., 1927, p. 40). Consequently, for parallel plates, only sparking occurs at a field strength or potential difference given by the empirical expressions

$$\mathcal{E}_s = \mathcal{E}_o k_p \left[1 + \left(\frac{K_o}{k_p B_e} \right) \right] \quad (17-19)$$

$$E_s = \mathcal{E}_o k_p B_e + K_o \mathcal{E}_o \quad (17-20)$$

For air in the range of $k_p B_e$ from 0.1 to 2, $\mathcal{E}_o = 111.2$ and $K_o = 0.048$. Thornton [*Phil. Mag.*, **28**(7), 666 (1939)] gives values for other gases. For concentric cylinders (Loeb, *Fundamental Processes of Electrical Discharge in Gases*, Wiley, New York, 1939; Peek, *Dielectric Phenomena in High-Voltage Engineering*, McGraw-Hill, New York, 1929; and Whitehead, op. cit.), corona sets in at the central wire when

$$\mathcal{E}_c = \mathcal{E}_o k_p \left(1 + \sqrt{\frac{K_o}{k_p D_d}} \right) \quad (17-21)$$

$$E_c = \left(\frac{\mathcal{E}_o k_p D_d}{2} \right) \left(1 + \sqrt{\frac{K_o}{k_p D_d}} \right) \ln \left(\frac{D_t}{D_d} \right) \quad (17-22)$$

For air approximate values are $\mathcal{E}_o = 110$, $K_o = 0.18$. Corona, however, will set in only if $(D_t/D_d) > 2.718$. If this ratio is less than 2.718, no corona occurs, and only sparking will result, following the laws given by Eqs. (17-21) and (17-22) (Peek, op. cit.).

In practice, precipitators are usually operated at the highest voltage practicable without sparking, since this increases both the particle charge and the electrical precipitating field. The sparking potential is generally higher with a negative charge on the discharge electrode and is less erratic in behavior than a positive corona discharge. It is the consensus, however, that ozone formation with a positive discharge is considerably less than with a negative discharge. For these reasons negative discharge is generally used in industrial precipitators, and a

TABLE 17-13 Sparking Potentials* (Small Wire Concentric in Pipe)

Pipe diameter, in	Sparking potential,† volts	
	Peak	Root mean square
4	59,000	45,000
6	76,000	58,000
9	90,000	69,000
12	100,000	77,000

*Data reported by Anderson in Perry, *Chemical Engineers' Handbook*, 2d ed., p. 1873, McGraw-Hill, New York, 1941.

†For gases at atmospheric pressure, 100°F, containing water vapor, air, CO₂, and mist, and negative-discharge-electrode polarity.

positive discharge is utilized in air-conditioning applications. In Table 17-13 are given some typical values for the sparking potential for the case of small wires in pipes of various sizes. The sparking potential varies approximately directly as the density of the gas but is very sensitive to the character of any material collected on the electrodes. Even small amounts of poorly conducting material on the electrodes may markedly lower the sparking voltage. For positive polarity of the discharge electrode, the sparking voltage will be very much lower. The sparking voltage is greatly affected by the temperature and humidity of the gas, as shown in Fig. 17-65.

Current Flow Corona discharge is accompanied by a relatively small flow of electric current, typically 0.1 to 0.5 mA/m² of collecting-electrode area (projected, rather than actual area). Sparking usually involves a considerably larger flow of current which cannot be tolerated except for occasional periods of a fraction of a second duration, and then only when suitable electrical controls are provided to limit the current. However, when suitable controls are provided, precipitators have been operated continuously with a small amount of sparking to ensure that the voltage is in the correct range to ensure corona. Besides disruptive effects on the electrical equipment and electrodes, sparking will result in low collection efficiency because of reduction in applied voltage, redispersion of collected dust, and current channeling. Although an exact calculation can be made for the current flow for a direct-current potential applied between concentric cylinders, the following simpler expression, based on the assumption of a constant space charge or ion density, gives a good approximation of corona current [Ladenburg, *Ann. Phys.*, **4**(5), 863 (1930)]:

$$I = \frac{8\lambda_i E(E - E_c)}{D_i^2 \ln(D_i/D_d)} \quad (17-23)$$

and the average space charge is given by (Whitehead, op. cit.)

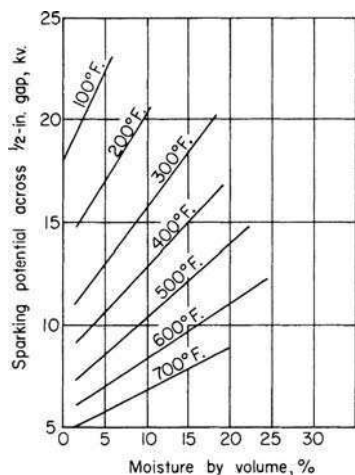


FIG. 17-65 Sparking potential for negative point-to-plane $\frac{1}{2}$ -in. (1.3-cm) gap as a function of moisture content and temperature of air at 1-atm (101.3-kPa) pressure. [Sproull and Nakada, *Ind. Eng. Chem.*, **43**, 1356 (1951).]

$$\sigma_{\text{avg}} = \frac{4(E - E_c)}{\pi D_i^2 \epsilon} \quad (17-24)$$

In the space outside the immediate vicinity of corona discharge, the field strength is sensibly constant, and an average value is given by

$$\mathcal{E} = \sqrt{2I/\lambda_i} \quad (17-25)$$

which applies if the potential difference is above the critical potential required for corona discharge so that an appreciable current flows.

Ionic mobilities are given by Loeb (*International Critical Tables*, vol. 6, McGraw-Hill, New York, 1929, p. 107). For air at 0°C, 760 mmHg, $\lambda_i = 624$ (cm/s)/(statV/cm) for negative ions. Positive ions usually have a slightly lower mobility. Loeb (*Basic Processes of Gaseous Electronics*, University of California Press, Berkeley and Los Angeles, 1955, p. 53) gives a theoretical expression for ionic mobility of gases which is probably good to within ± 50 percent:

$$\lambda_i = \frac{100.0}{k_p \sqrt{(\delta_g - 1)M}} \quad (17-26)$$

In general, ionic mobilities are inversely proportional to gas density. Ionic velocities in the usual electrostatic precipitator are on the order of 30.5 m/s (100 ft/s).

Electric Wind By virtue of the momentum transfer from gas ions moving in the electrical field to the surrounding gas molecules, a gas circulation, known as the "electric" or "ionic" wind, is set up between the electrodes. For conditions encountered in electrical precipitators, the velocity of this circulation is on the order of 0.6 m/s (2 ft/s). Also, as a result of this momentum transfer, the pressure at the collecting electrode is slightly higher than at the discharge electrode (Whitehead, op. cit., p. 167).

Charging of Particles [Deutsch, *Ann. Phys.*, **68**(4), 335 (1922); **9**(5), 249 (1931); **10**(5), 847 (1931); Ladenburg, op. cit.; and Mierdel, *Z. Tech. Phys.*, **13**, 564 (1932).] Three forces act on a gas ion in the vicinity of a particle: attractive forces due to the field strength and the ionic image; and repulsive forces due to the Coulomb effect. For spherical particles larger than 1- μ m diameter, the ionic image effect is negligible, and charging will continue until the other two forces balance according to the equation

$$N_o = \left(\frac{\zeta \mathcal{E} D_p^2}{4\epsilon} \right) \left(\frac{\pi \sigma \epsilon \lambda_i}{1 + \pi \sigma \epsilon \lambda_i t} \right) \quad (17-27)$$

The ultimate charge acquired by the particle is given by

$$N_o = \zeta \mathcal{E} D_p^2 / 4\epsilon \quad (17-28)$$

and is very nearly attained in a fraction of a second. For particles smaller than 1- μ m diameter, the initial charging will occur according to Eq. (17-27). However, owing to the ionic-image effect, the ultimate charge will be considerably greater because of penetration resulting from the kinetic energy of the gas ions. For charging times of the order encountered in electrical precipitation, the ultimate charge acquired by spherical particles smaller than about 1- μ m diameter may be approximated (± 30 percent) by the empirical expression

$$N_o = 3.4 \times 10^3 D_p T \quad (17-29)$$

Values of N_o for various sized particles are listed in Table 17-14 for 70°F, $\zeta = 2$, and $\mathcal{E} = 10$ statV/cm.

Particle Mobility By equating the electrical force acting on a particle to the resistance due to air friction, as expressed by Stokes' law, the particle velocity or mobility may be expressed by

1. For particles larger than 1- μ m diameter:

$$\lambda_p = \left(\frac{u_e}{\mathcal{E}_p} \right) = \frac{\zeta D_p \mathcal{E}_i K_m}{12\pi\mu} \quad (17-30)$$

2. For particles smaller than 1- μ m diameter:

$$\lambda_p = \left(\frac{u_e}{\mathcal{E}_p} \right) = \frac{360 K_m \epsilon T}{\mu} \quad (17-31)$$

For single-stage precipitators, $\mathcal{E}_i$ and $\mathcal{E}_p$ may be considered as essentially equal. It is apparent from Eq. (17-31) that the mobility in an electric field will be almost the same for all particles smaller than

TABLE 17-14 Charge and Motion of Spherical Particles in an Electric FieldFor $\zeta = 2$, and $\epsilon = \epsilon_i = \epsilon_p = 10$ statV/cm

Particle diam., μ	Number of elementary electrical charges, N_0	Particle migration velocity,* u_e , ft/sec
0.1	10	0.27
.25	25	.15
.5	50	.12
1.0	105	.11
2.5	655	.26
5.0	2,620	.50
10.0	10,470	.98
25.0	65,500	2.40

NOTE: To convert feet per second to meters per second, multiply by 0.3048.

about 1- μ m diameter, and hence, in the absence of reentrainment, collection efficiency should be almost independent of particle size in this range. Very small particles will actually have a greater mobility because of the Stokes-Cunningham correction factor. Values of u_e are listed in Table 17-14 for 70°F, $\zeta = 2$, and $\epsilon = \epsilon_i = \epsilon_p = 10$ statV/cm.

Collection Efficiency Although actual particle mobilities may be considerably greater than would be calculated on the basis given in the preceding paragraph because of the action of the electric wind in single-stage precipitators, the latter acts in a compensating fashion, and the overall effect of the electric wind is probably to provide an equalization of particle concentration between the electrodes similar to the action of normal turbulence (Mierdel, op. cit.). On this basis Deutsch (op. cit.) has derived the following equations for collection efficiency, the form of which had previously been suggested by Anderson on the basis of experimental data:

$$\eta = 1 - e^{-(u_e K_e / q)} = 1 - e^{-K_e} \quad (17-32)$$

For the concentric-cylinder (or wire-in-cylinder) type of precipitator, $K_e = 4L_e/D_i V_e$; for rod-curtain or wire-plate types, $K_e = L_e/B_e V_e$. Strictly speaking, Eq. (17-32) applies only for a given particle size, and the overall efficiency must be obtained by an integration process for a specific dust distribution, as described in the subsection "Cyclone Separators." However, over limited ranges of performance conditions, Eq. (17-32) has been found to give a good approximation of overall collection efficiency, with the term for particle migration velocity representing an empirical average value. Such values, calculated from overall collection-efficiency measurements, are given in Table 17-15 for specific installations.

For two-stage precipitators with close collecting-plate spacings (Fig. 17-76), the gas flow is substantially streamline, and no electric wind exists. Consequently, with reentrainment neglected, collection efficiency may be expressed as [Penny, *Electr. Eng.* **56**, 159 (1937)]

$$\eta = u_e L_e / V_e B_e \quad (17-33)$$

TABLE 17-15 Performance Data on Typical Single-Stage Electrical Precipitator Installations*

Type of precipitator	Type of dust	Gas volume, cu ft/min	Average gas velocity, ft/sec	Collecting electrode area, sq ft	Over-all collection efficiency, %	Average particle migration velocity, ft/sec
Rod curtain	Smelter fume	180,000	6	44,400	85	0.13
Tulip type	Gypsum from kiln	25,000	3.5	3,800	99.7	.64
Perforated plate	Fly ash	108,000	6	10,900	91	.40
Rod curtain	Cement	204,000	9.5	26,000	91	.31

*Research-Cottrell, Inc. To convert cubic feet per minute to cubic meters per second, multiply by 0.00047; to convert feet per second to meters per second, multiply by 0.3048; and to convert square feet to square meters, multiply by 0.0929.

which holds for values of $\eta \leq 1.0$. In practice, however, extraneous factors may cause the actual efficiency to approach a relationship of the type given by Eq. (17-32).

Application The theoretical considerations that have been expounded should be used only for order-of-magnitude estimates, since a number of extraneous factors may enter into actual performance. In actual installations rectified alternating current is employed. Hence the electric field is not fixed but varies continuously, depending on the waveform of the rectifier, although Schmidt and Anderson [*Electr. Eng.*, **57**, 332 (1938)] report that the waveform is not a critical factor. Allowances for high dust concentrations have not been fully studied, although Deutsch (op. cit.) has presented a theoretical approach. In addition, irregularities on the discharge electrode will result in local discharges. Such irregularities can readily result from dust incrustation on the discharge electrodes due to charging of particles with opposite polarity within the thin but appreciable flow or ionization layer surrounding this electrode. Very high dust loadings increase the potential difference required for corona and reduce the current due to the space charge of the particles. This tends to reduce the average particle charge and reduces collection efficiency. This can be compensated for by increasing the potential difference when high dust loadings are involved.

Several investigators have attempted to modify the basic Deutsch equation so that it would more nearly describe precipitator performance. Cooperman ("A New Theory of Precipitator Efficiency," Pap. 69-4, APCA meeting, New York, 1969) introduced correction factors for diffusional forces arising from variations in particle concentration along the precipitator length and also perpendicular to the collecting surface. Robinson [*Atmos. Environ.* **1**(3), 193 (1967)] derived an equation for collection efficiency in which two erosion or reentrainment terms are introduced.

An analysis of precipitator performance based on theoretical considerations was undertaken by the Southern Research Institute for the National Air Pollution Control Administration (Nichols and Oglesby, "Electrostatic Precipitator Systems Analysis," AIChE annual meeting, 1970). A mathematical model was developed for calculating the particle charge, electric field, and collection efficiency based on the Deutsch-Anderson equation. The system diagram is shown in Fig. 17-66. This system-analysis method, using high-speed computers, makes it possible to analyze what takes place in each increment of precipitator length. Collection efficiency versus particle size is computed for each 1 ft (0.3 m) of gas travel, and the inlet particle-size distribution is modified accordingly. Computed overall efficiencies compare well with measured values on three precipitators. The model assumes that field charging is the only charging mechanism. The authors considered the addition of several refinements to the program: the influence of diffusion charging; reentrainment effects due to rapping and erosion; and loss of efficiency due to maldistribution of gas, dust resistivity, and gas-property effects. The modeling technique appeared promising, but much more work was needed before it could be used for design. The same authors prepared a general treatise (Oglesby and Nichols, *A Manual of Electrostatic Precipitator Technology*, parts I and II, Southern Research Institute, Birmingham, Ala., U.S. Government Publications PB196360, 196381, 1970).

High-Pressure-High-Temperature Electrostatic Precipitation

In general, increased pressure increases precipitation efficiency, although a somewhat higher potential is required, because it reduces ion mobility and hence increases the potential required for corona and sparking. Increased temperature reduces collection efficiency because ion mobility is increased, lowering critical potentials, and because gas viscosity is increased, reducing migration velocities.

Precipitators have been operated at pressures up to 5.5 MPa (800 psig) and temperatures to 800°C.

The effect of increasing gas density on sparkover voltage has been investigated by Robinson [*J. Appl. Phys.*, **40**, 5107 (1969); *Air Pollution Control*, part 1, Wiley-Interscience, New York, 1971, chap. 5]. Figure 17-67 shows the effect of gas density on corona-starting and sparkover voltages for positive and negative corona in a pipe precipitator. The sparkover voltages are experimental and are given by the solid points. Experimental corona-starting voltages are given by the hollow points. The solid lines are corona-starting voltage curves

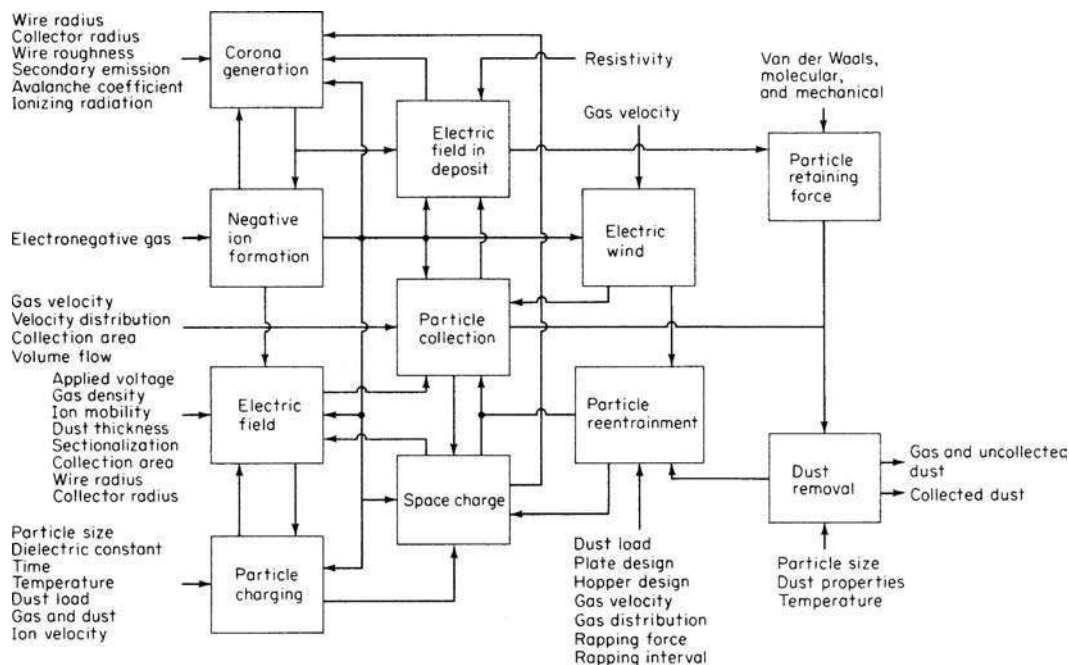


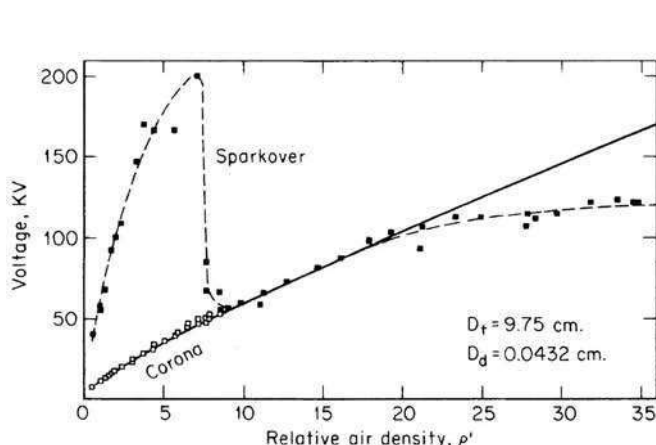
FIG. 17-66 Electrostatic-precipitator-system model. (Nichols and Oglesby, "Electrostatic Precipitator Systems Analysis," AIChE annual meeting, 1970.)

calculated from Eq. (17-33). This is an empirical relationship developed by Robinson.

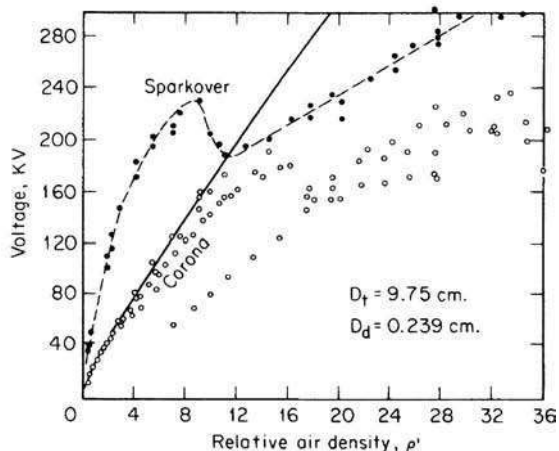
$$\frac{E_c}{\rho'} = A \frac{B}{\sqrt{D_d \rho'/2}} \quad (17-34)$$

E_c is the corona-starting field, kV/cm. ρ' is the relative gas density, equal to the actual gas density divided by the density of air at 25°C, 1 atm. D_d is the diameter of the ionizing wire, cm. A and B are constants which are characteristics of the gas. In dry air, $A = 32.2$ kV/cm and $B = 8.46$

kV/cm^{1/2}. Agreement between experimental and calculated starting voltages is good for the case of positive corona, but in the case of negative corona the calculated line serves as an upper limit for the data. This lower-than-expected starting-voltage characteristic of negative corona is confirmed by Hall et al. [Oil Gas J., 66, 109 (1968)] in a report of an electrostatic precipitator which removes lubricating-oil mist from natural gas at 5.5 MPa (800 psig) and 38°C (100°F). The use of electrostatic precipitators at elevated pressure is expected to increase, because the method requires very low pressure drop [approximately 69 Pa (0.1



(a) Positive Corona



(b) Negative Corona

FIG. 17-67 Corona-starting and sparkover voltages for coaxial wire-pipe electrodes in air (25°C). D_t and D_d are the respective pipe and wire diameters. The voltage is unvarying direct current. (Robinson, Air Pollution Control, part 1, Wiley-Interscience, New York, 1971, chap. 5.)

lb/in²). This results from the fact that the electric separation forces are applied directly to the particles themselves rather than to the entire mass of the gas, as in inertial separators. The use of electrostatic precipitators at temperatures up to 400°C is well developed for the power-house fly-ash application, but in the range of 600 to 800°C they are still in the experimental phase. The U.S. Bureau of Mines has tested a pilot-scale tubular precipitator for fly ash. See Shale [Air Pollut. Control Assoc. J., **17**, 159 (1967)] and Shale and Fasching (Operating Characteristics of a High-Temperature Electrostatic Precipitator, U.S. Bur. Mines Rep. 7276, 1969). It operated over a temperature range of 27 to 816°C (80 to 1500°F) and a pressure range of 552 kPa (35 to 80 psig). Initial collection efficiencies ranged from 90 to 98 percent at 793°C (1460°F), 552 kPa (80 psig), but continuous operation was not achieved because of excessive thermal expansion of internal parts.

Resistivity Problems Optimum performance of electrostatic precipitators is achieved when the electrical resistivity of the collected dust is sufficiently high to result in electrostatic pinning of the particles to the collecting surface, but not so high that dielectric breakdown of the dust layer occurs as the corona current passes through it. The optimum resistivity range is generally considered to be from 10^8 to 10^{10} Ω-cm, measured at operating conditions. As the dust builds up on the collecting electrode, it impedes the flow of current, so that a voltage drop is developed across the dust layer:

$$E_d = j\rho_d L_d \quad (17-35)$$

If E_d/L_d exceeds the dielectric strength of the dust layer, sparks occur in the deposit and form back-corona craters. Ions of both polarities are formed. Positive ions formed in the craters are attracted to the negatively charged particles in the gas stream, whose charge level is reduced so that collection efficiency decreases. Some of the positive ions neutralize part of the negative-space-charge cloud normally present near the wire, thereby increasing total current. Collection efficiency under these conditions will not correlate with total power input (Owens, E. I. du Pont de Nemours & Co. internal communication, 1971). Under normal conditions, collection efficiency is an exponential function of corona power (White, *Industrial Electrostatic Precipitation*, Addison-Wesley, Reading, Mass., 1963). With typical ion density in the range of $10^9/\text{cm}^3$, overall voltage gradient would be about 4000 V/cm, and current about 1 μA/cm². Dielectric breakdown of the dust layer (at about 10,000 V/cm) would therefore be expected for dusts with resistivities above 10^{10} Ω-cm.

Problems due to high resistivity are of great concern in fly-ash precipitation because air-pollution regulations require that coals have low (<1 percent) sulfur content. Figure 17-68 shows that the resistivity of low-sulfur coal ash exceeds the threshold of 10^{10} Ω-cm at common operating temperatures. This has resulted in the installation of a number of precipitators which have failed to meet guaranteed performance. This has occurred to an alarming extent in the United States but has also been encountered in Australia, where the sulfur content is typically 0.3 to 0.6 percent. Maartmann (Pap. EN-34F, 2d International Clean Air Congress, Washington, 1970) reports the installation of a number of precipitators which performed below guarantees, so that the Electricity Commission of New South Wales decided that each manufacturer wishing to bid on a new station must first make pilot tests to prove performance on the actual coal to be burned in that station. Problems of back corona and excessive sparking with low-sulfur coal usually require that the operating voltage be reduced. This reduces the migration velocity and leads to larger precipitators. Ramsdell (*Design Criteria for Precipitators for Modern Central Station Power Plants*, American Power Conference, Chicago, 1968) developed the curves in Fig. 17-69. They show the results of extensive field tests by the Consolidated Edison Co. In another paper ("Anti-pollution Program of Consolidated Edison Co. of New York," ASCE, May 13-17, 1968), Ramsdell traces the remarkable growth in the size of precipitators required for high efficiency on low-sulfur coals. The culmination of this work was the precipitator at boiler 30 at Ravenswood Station, New York. Resistivity problems were avoided by operating at high temperature [343°C (650°F)]. The mechanical (cyclone) collector was installed after the precipitator to clean up puffs due to rapping.

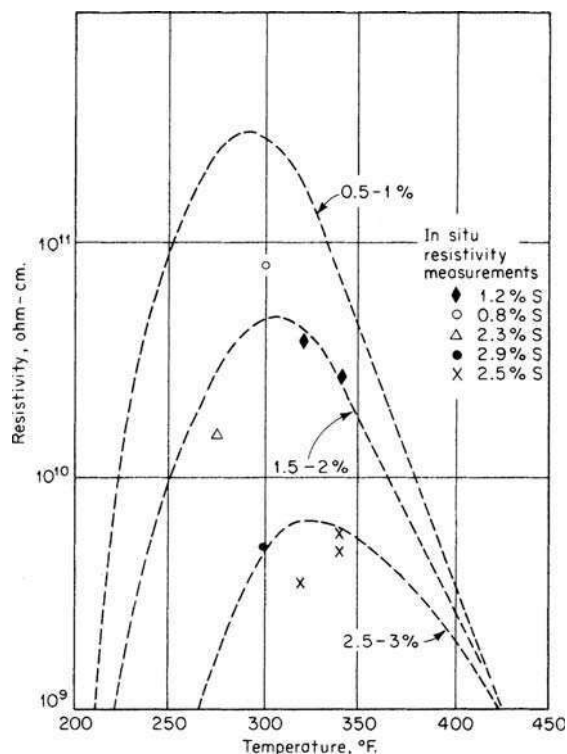


FIG. 17-68 Trends in resistivity of fly ash with variations in flue-gas temperature and coal sulfur content. °C = (°F - 32) × 5/9. (Oglesby and Nichols, A Manual of Electrostatic Precipitator Technology, part II, Southern Research Institute, Birmingham, Ala., 1970.)

Maartmann (op. cit.) agrees that sulfur content is important but feels that it should not be the sole criterion for the determination of collecting surface. He points to specific collecting-surface requirements as high as 500 ft²/(1000 ft³-min) for 95 percent collection efficiency with high-resistivity Australian ash.

Schmidt and Anderson (op. cit.) and Anderson [*Physics*, **3**, 23 (July 1932)] claim that resistivity of the collected dust may be a controlling factor which is very sensitive to moisture. They state that an increase in relative humidity of 5 percent may double the precipitation rate because of its effect on the conductivity of the collected dust layer.

Conditioning agents have been added to the flue gas to alter dust resistivity. Steam, sodium chloride, sulfur trioxide, and ammonia have all been successfully used. Research by Chittum and others [Schmidt, *Ind. Eng. Chem.*, **41**, 2428 (1949)] led to a theory of conditioning by alteration of the moisture-adsorption properties of dust surfaces. Chittum proposed that an intermediate chemical-adsorption film, which was strongly bound to the particle and which in turn strongly adsorbed water, would be an effective conditioner. This explains how acid conditioners, such as SO₃, help resistivity problems associated with basic dusts, such as many types of fly ash, whereas ammonia is a good additive for acidic dusts such as alumina. Moisture alone can be used as a conditioning agent. This is shown in Fig. 17-65. Moisture is beneficial in two ways: it reduces the electrical resistivity of most dusts (an exception is powdered sulfur, which apparently does not absorb water), and it increases the voltage which may safely be employed without sparking, as shown in Fig. 17-65.

Low resistivity can sometimes be a problem. If the resistivity is below 10^4 Ω-cm, the collected particles are so conductive that their charges leak to ground faster than they are replenished by the corona. The particles are no longer electrostatically pinned to the plate, and they may then be swept away and reentrained in the exit gas. The particles may even pick up positive charges from the collecting plate and

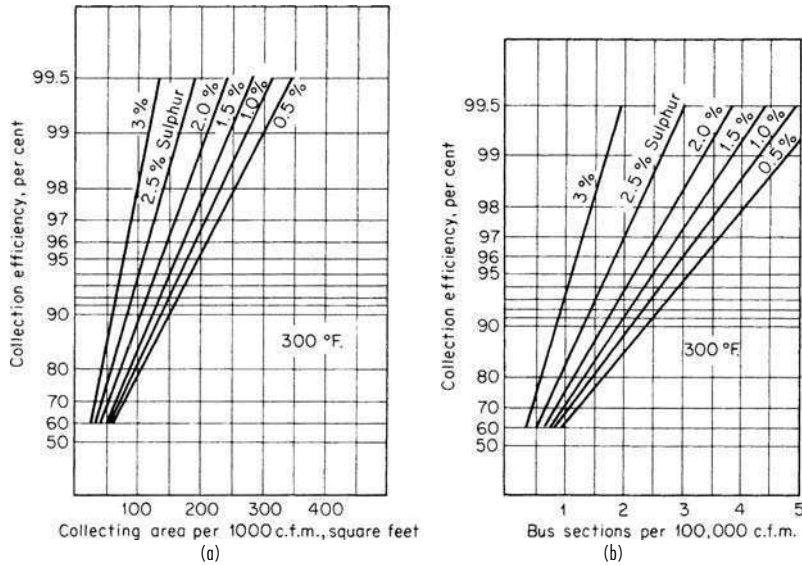


FIG. 17-69 Design curves for electrostatic precipitators for fly ash. Collection efficiency for various levels of percent sulphur in coal versus (a) specific collecting surface, and (b) bus sections per 100,000 ft³/min (4.7 m³/s). °C = (°F - 32) × 5/9. (Ramsdell, Design Criteria for Precipitators for Modern Central Station Power Plants, American Power Conference, Chicago, Ill., 1968.)

then be repelled. Low-resistivity problems are common with dusts of high carbon content and may also occur in fly-ash precipitators which handle the ash from high-sulfur coal and operate at low gas temperatures. Low resistivity in this case results from excessive condensation of electrically conductive sulfuric acid.

Single-Stage Precipitators The single-stage type of unit, commonly known as a Cottrell precipitator, is most generally used for dust or mist collection from industrial-process gases. The corona discharge is maintained throughout the precipitator and, besides providing initial ionization, serves to prevent redispersion of precipitated dust and recharges neutralized or discharged particle ions. Cottrell precipitators may be divided into two main classes, the so-called plate type (Fig. 17-70), in which the collecting electrodes consist of parallel plates, screens, or rows of rods, chains, or wires; and the pipe type (Fig. 17-71), in which the collecting electrodes consist of a nest of parallel pipes which may be square, round, or of any other shape. The discharge or precipitating electrodes in each case are wires or rods, either round or edged, which are placed midway between the collecting electrodes or in the center of the pipes and may be either parallel or perpendicular to the gas flow in the case of plate precipitators. When the collecting electrodes are screens or rows of rods or wires,

the gases are usually passed parallel to the plane of each but may also be passed through it. In pipe precipitators, the gas flow is generally vertical up through the pipe, although downflow is not unusual. The pipe-type precipitator is usually used for the removal of liquid particles and volatilized fumes [Beaver, op. cit.; and Cree, *Am. Gas J.*, **162**, 27 (March 1945)], and the plate type is used mainly on dusts. In the pipe type, the discharge electrodes are usually suspended from an insulated support and kept taut by a weight at the bottom. Cree (op. cit.) discusses the application of electrical precipitators to tar removal in the gas industry.

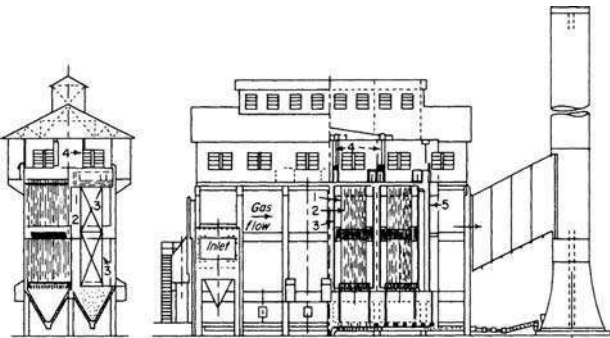


FIG. 17-70 Horizontal-flow plate precipitator used in a cement plant. (Western Precipitation Division, Joy Manufacturing Company.)

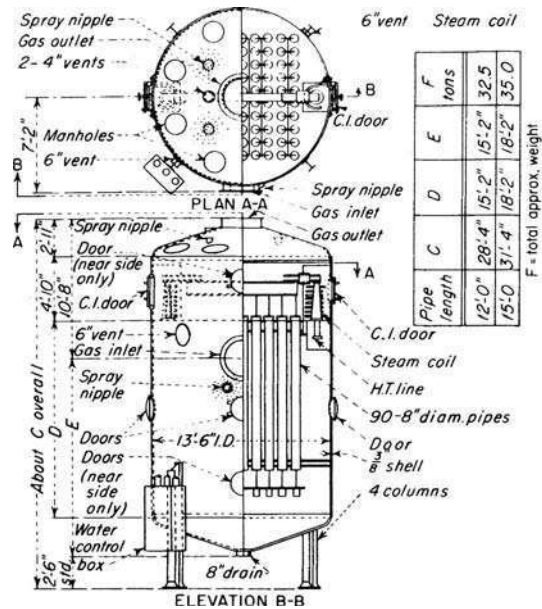


FIG. 17-71 Blast-furnace pipe precipitator. (Research-Cottrell, Inc.)

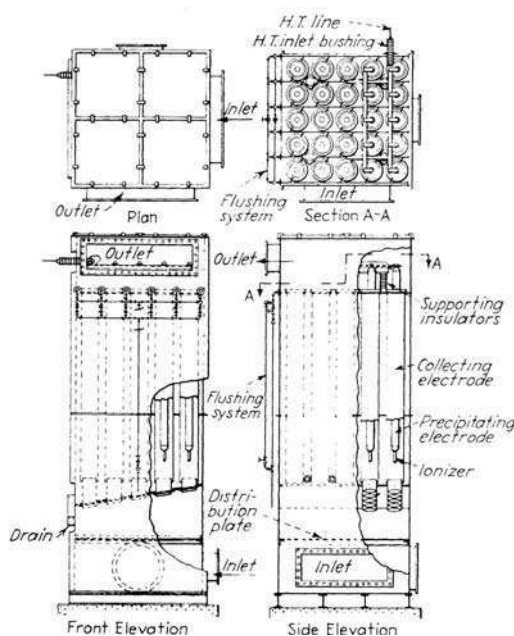


FIG. 17-72 Two-stage water-film pipe precipitator. (Western Precipitation Division, Joy Manufacturing Company.)

Rapping Except when liquid dispersoids are being collected or, in the case of film precipitators, when a liquid is circulated over the collecting-electrode surface (Fig. 17-72), thus continuously removing the precipitated material, the collected dust is dislodged from the

electrodes either periodically or continuously by mechanical rapping or scraping, which may be performed automatically or manually. Automatic rapping with either impact-type or vibrator-type rappers is common practice. White (op. cit.) recommends fairly continuous rapping with magnetic-impulse rappers. Rapping with excessive force leads to dust reentrainment and possible mechanical failure of the plates, while insufficient rapping leads to excessive dust buildup with poor electrical operation and reduced collection efficiency. Intermittent rapping at intervals of an hour or more causes heavy puffs of reentrained dust. Sproull [*Air Pollut. Control Assoc. J.*, **15**, 50 (1965)] reports the importance of electrode acceleration and shows that it varies with the type of dust, whether the electrode is rapped perpendicularly (normally) to the plate or parallel to it. Figure 17-73 shows the accelerations required for rapping normally to the plate. Difficult dusts may require as much as 100 G acceleration for 90 percent removal, and even higher accelerations are required when the vibrating force is applied in the plane of the plate.

Perforated-plate or rod-curtain precipitators are frequently rapped without shutting off the gas flow and with the electrodes energized. This procedure, however, results in a tendency for reentrainment of collected dust. Sectional or composite-plate collecting electrodes (sometimes known as hollow, pocket, or tulip electrodes) are used to minimize this tendency in the continuous removal of the precipitated material, provided that it is free-flowing. These are generally designed for vertical gas flow and comprise a collecting electrode containing a dead air space and provided with horizontal protruding slots that guide the dust into this space (see Fig. 17-74), although some types use horizontal flow.

The choice of size, shape, and type of electrode is based on economic considerations and is usually determined by the characteristics of the gas and suspended matter and by mechanical considerations such as flue arrangement, the available space, and previous experience with the electrodes on similar problems. The spacing between collecting electrodes in plate-type precipitators and the pipe diameter in pipe-type precipitators usually ranges from 15 to 38 cm (6 to 15 in.). The smaller the spacing, the lower the necessary voltage and overall

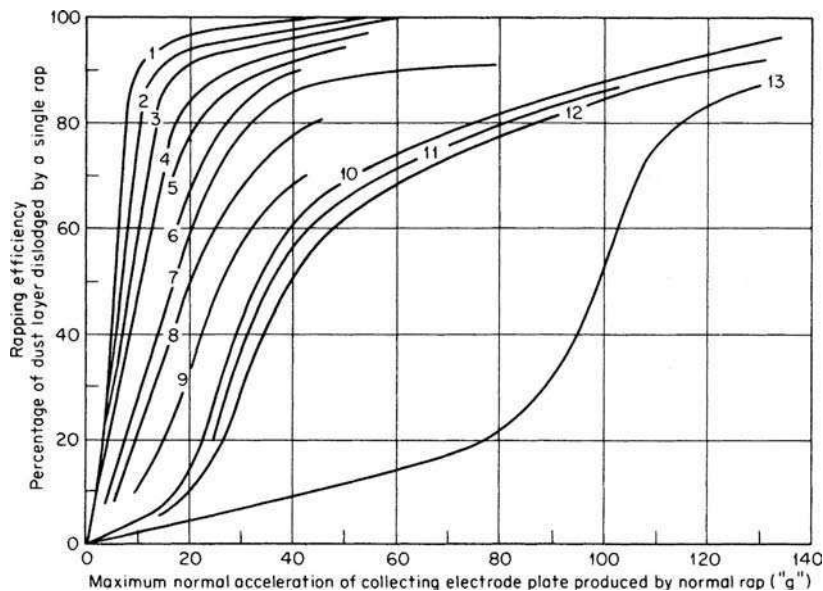


FIG. 17-73 Normal (perpendicular) rapping efficiency for various precipitated dust layers having about 0.03 g dust/cm² (0.2 g dust/in²) as a function of maximum acceleration in multiples of g. Curve 1, fly ash, 200 or 300°F, power off. Curve 2, fly ash, 70°F, power off; also 200 or 300°F, power on. Curve 3, fly ash, 70°F, power on. Curve 4, cement-kiln feed, 300°F, power off. Curve 5, cement dust, 300°F, power off. Curve 6, same as 5, except power on. Curve 7, cement-kiln feed, 300°F, power on. Curve 8, cement dust, 200°F, power off. Curve 9, same as 8, except power on. Curve 10, cement-kiln feed, 200°F, power off. Curve 11, same as 10, except at 70°F. Curve 12, cement-kiln feed, 200°F, power on. Curve 13, cement-kiln feed, 70°F, power on. °C = (°F - 32) × 5/9. [Sproull, *Air Pollut. Control Assoc. J.*, **15**, 50 (1965).]

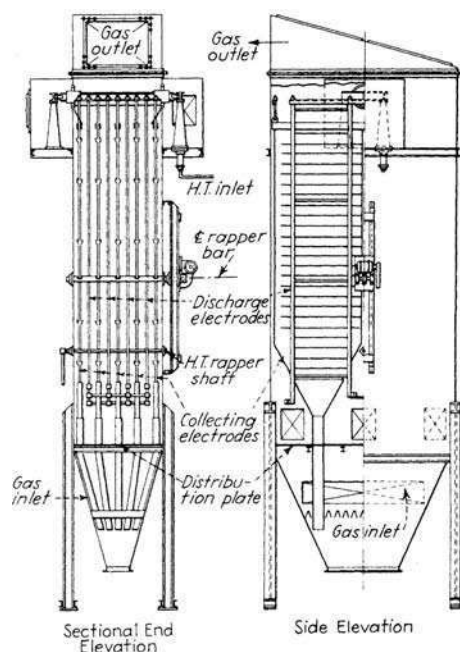


FIG. 17-74 Vertical-flow heavy-duty plate precipitator. (Western Precipitation Division, Joy Manufacturing Company.)

equipment size, but the greater the difficulties involved in maintaining proper alignment and resulting from disturbances due to collected material. Large spacings are usually associated with high dust concentration in order to minimize sparkover due to dust buildup. For very high dust concentrations, such as those encountered in fluid-catalyst plants, it is advantageous to use greater spacings in the first half of the precipitator than in the second half. Precipitators, especially of the plate type, are frequently built with groups of collecting electrodes in series in a common housing. Collecting electrodes are generally on the order of 0.9 to 1.8 m (3 to 6 ft) wide and 3 to 5.5 m (10 to 18 ft) high in plate-type precipitators and 1.8 to 4.6 m (6 to 15 ft) high in pipe types. It is essential for good collection efficiency that the gas be evenly distributed across the various electrode elements. Although this can be achieved by proper gas-inlet transitions and guide vanes, perforated plates or screens located on the upstream side of the electrodes are generally used for distribution. Perforated plates or screens located on the downstream side may be used in special cases.

Electrical precipitators are generally designed for collection efficiency in the range of 90 to 99.9 percent. It is essential, however, that the units be properly maintained in order to achieve the required collection efficiency. Electric power consumption is generally 0.2 to 0.6 kW/(1000 ft³·min) of gas handled, and the pressure drop across the precipitator unit is usually less than 124 Pa (0.5 in. water), ranging from 62 to 248 Pa (¼ to 1 in.) and representing primarily distributor and entrance-exit losses. Applied potentials range from 30,000 to 100,000 V. Gas velocities and retention times are generally in the range of 0.9 to 3 m/s (3 to 10 ft/s) and 1 to 15 s, respectively. Velocities are kept low in conventional precipitators to avoid reentrainment of dust. There are, however, precipitator installations on carbon black in which the precipitator acts to flocculate the dust so that it may be subsequently collected in multiple small-diameter cyclone collectors. By not attempting to collect the particles in the precipitator, higher velocities may be used with a correspondingly lower investment cost.

Power Supply Electrical precipitators are generally energized by rectified alternating current of commercial frequency. The voltage is stepped up to the required value by means of a transformer and then rectified. The rectifying equipment has undergone an evolution which began with the synchronous mechanical rectifier in 1904 and was

followed by mercury-vapor rectifiers in the 1920s; the first solid-state selenium rectifiers were introduced about 1939. Silicon rectifiers are the latest and most widely used type, since they provide high efficiency and reliability. Automatic controls commonly are tied to voltage, current, spark rate, or some combination of these parameters. Modern precipitators use control circuits similar to those on Fig. 17-75. A high-voltage silicon rectifier is used together with a saturable reactor and means for limiting current and controlling voltage and/or spark rate. One popular method adjusts the voltage to give a specified sparking frequency (typically 50 to 150 sparks per minute per bus section). Half-wave rectification is sometimes used because of its lower equipment requirements and power consumption. It also has the advantage of longer decay periods for sparks to extinguish between current pulses.

Electrode **insulators** must also be designed for a particular service. The properties of the dust or mist and gas determine their design as well as the physical details of the installation. Conducting mists require special allowances such as oil seals, energized shielding cups, or air bleeds. With saturated gas, steam coils are frequently used to prevent condensation on the electrodes.

Typical applications in the chemical field (Beaver, op. cit.) include detarring of manufactured gas, removal of acid mist and impurities in contact sulfuric acid plants, recovery of phosphoric acid mists, removal of dusts in gases from roasters, sintering machines, calciners, cement and lime kilns, blast furnaces, carbon-black furnaces, regenerators on fluid-catalyst units, chemical-recovery furnaces in soda and sulfate pulp mills, and gypsum kettles. Figure 17-74 shows a vertical-flow steel-plate-type precipitator similar to a type used for catalyst-dust collection in certain fluid-catalyst plants.

A development of interest to the chemical industry is the tubular precipitator of reinforced-plastic construction (Wanner, *Gas Cleaning Plant after TiO₂ Rotary Kilns*, technical bulletin, Lurgi Corp., Frankfurt, Germany, 1971). Tubes made of polyvinyl chloride plastic are reinforced on the outside with polyester-fiber glass. The use of modern economical materials of construction to replace high-maintenance materials such as lead has been long awaited for corrosive applications.

Electrical precipitators are probably the most versatile of all types of dust collectors. Very high collection efficiencies can be obtained regardless of the fineness of the dust, provided that the precipitators are given proper maintenance. The chief disadvantages are the high initial cost and, in some cases, high maintenance costs. Furthermore, caution must be exercised with dusts that are combustible in the carrier gas.

Two-Stage Precipitators In two-stage precipitators, corona discharge takes place in the first stage between two electrodes having a nonuniform field (see Fig. 17-76). This is generally obtained by a fine-wire discharge electrode and a large-diameter receiving electrode. In this stage the potential difference must be above that required for corona discharge. The second stage involves a relatively uniform electrostatic field in which charged particles are caused to migrate to a collecting surface. This stage usually consists of either alternately charged parallel plates or concentric cylinders with relatively close clearances compared with their diameters. The only voltage requirement in this stage is that no sparking occur, though higher voltages will result in increased collection efficiency. Since collection occurs in the absence of corona discharge, there is no way of recharging reentrained and discharged particles. Consequently, some means must be provided for avoiding reentrainment of particles from the collecting surface. It is also essential that there be sufficient time and mixing between the first and second stages to secure distribution of gas ions across the gas stream and proper charging of the dust particles.

A unit is available in which electrostatic precipitation is combined with a dry-air filter of the type shown in Fig. 17-64b. In another unit an electrostatic field is superimposed on an automatic filter. In this case the ionizer wires are located on the leading face of the unit, and the collecting electrodes consist of alternate stationary and rotating parallel plates. Cleaning in this case is automatic and continuous.

Although intended primarily for air-conditioning applications, these units have been successfully applied to the collection of relatively nonconducting mists such as oil. However, other process applications have been limited largely to experimental installations. The large cost advantage of these units over the Cottrell precipitator lies in the

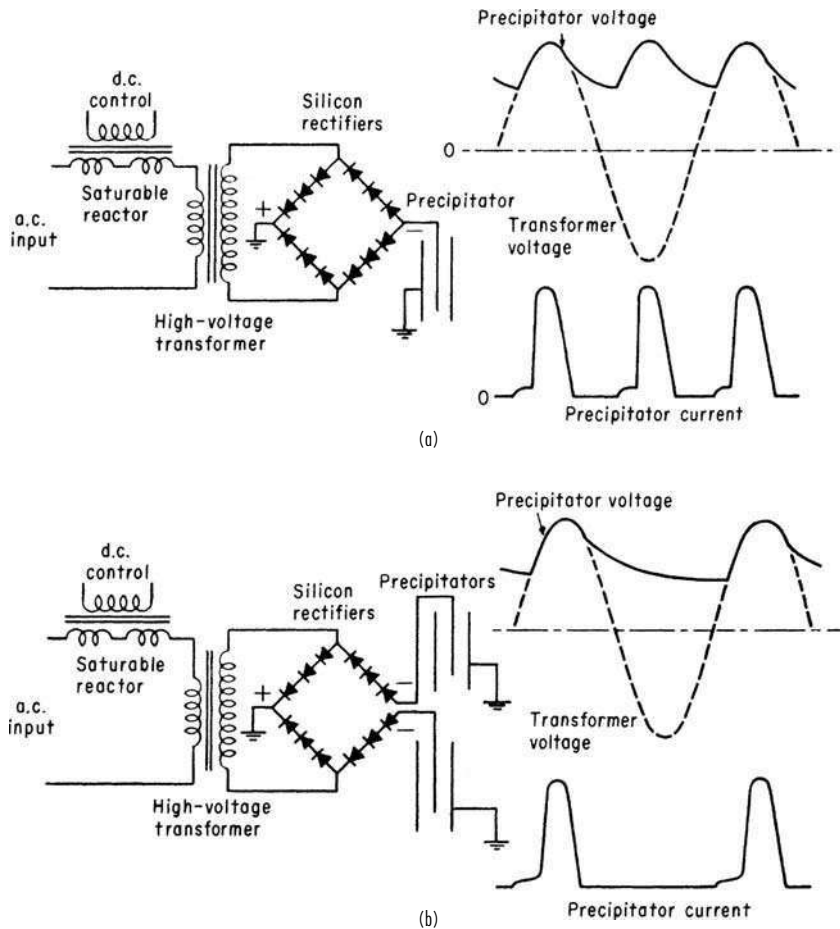


FIG. 17-75 Schematic circuits for silicon rectifier sets with saturable reactor control. (a) Full-wave silicon rectifier. (b) Half-wave silicon rectifier. (White, Industrial Electrostatic Precipitation, Addison-Wesley, Reading, Mass., 1963.)

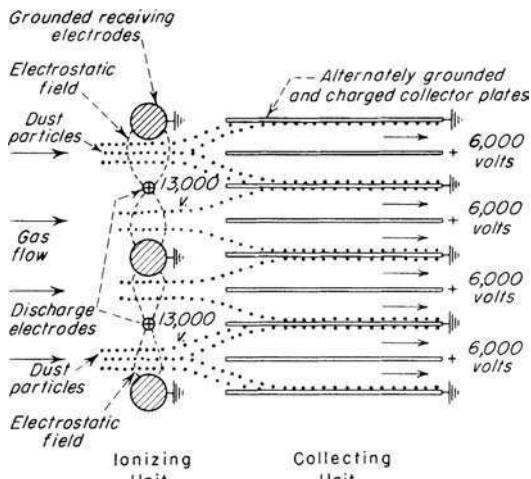


FIG. 17-76 Two-stage electrical-precipitation principle.

smaller equipment size made possible by the close plate spacing, in the lower power consumption due to the two-stage operation, and primarily in the mass production of standardized units. In process applications, the close plate spacing is objectionable because of the relatively high dust concentrations involved. Special material or weight requirements for the structural members may eliminate the mass-production advantage except for individual wide applications.

Alternating-Current Precipitators High-voltage alternating current may be employed for electrical precipitation. Corona discharge will result in a net rectification, provided that no spark gaps are used in series with the precipitator. However, the equipment capacity for a given efficiency is considerably lower than for direct current. In addition, difficulties due to induced high-frequency currents may be encountered. The simplicity of an ac system, on the other hand, has permitted very satisfactory adaptation for laboratory and sampling purposes [Drinker, Thomson, and Fitchet, *J. Ind. Hyg.*, **5**, 162 (September 1923)].

Some promising work with alternating current has been undertaken at the University of Karlsruhe. Lau [Staub, English ed., **29**, 10 (1969)] and coworkers found that ac precipitators operated at 50 Hz were more effective than dc precipitators for dusts with resistivities higher than $10^{11} \Omega\text{-cm}$. An insulating screen covering the collecting electrode permitted higher-voltage operation without sparkover.

SECTION 18

Liquid-Solid
Operations and
Equipment

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18-2 LIQUID-SOLID OPERATIONS AND EQUIPMENT

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Nomenclature

Symbol	Definition	SI units	U.S. customary units
c	Specific heat	J/(kg·K)	Btu/(lb·°F)
C	Constant		
C_o	Orifice coefficient	Dimensionless	Dimensionless
d_o	Orifice diameter	m	in
d_p	Drop diameter	m	ft
d_t	Pipe diameter	m	in
d_i	Tube diameter	m	ft
D	Impeller diameter	m	ft
D_a	Impeller diameter	m	ft
D_j	Diameter of jacketed vessel	m	ft
D_T	Tank diameter	m	ft
g	Acceleration	m/s ²	ft/s ²
g_c	Dimensional constant	$g_c = 1$ when using SI units	32.2 (ft·lb)/(lb·s ²)
h	Local individual coefficient of heat transfer, equals $dq/(dA)(\Delta T)$	J/(m ² ·s·K)	Btu/(h·ft ² ·°F)
H	Velocity head	m	ft
k	Thermal conductivity	J/(m·s·K)	(Btu·ft)/(h·ft ² ·°F)
L_p	Diameter of agitator blade	m	ft
N	Agitator rotational speed	s ⁻¹ , (r/s)	s ⁻¹ , (r/s)
N_{js}	Agitator speed for just suspension	s ⁻¹	s ⁻¹
N_{Re}	$D_a^2 N \rho / \mu$ impeller Reynolds number	Dimensionless	Dimensionless
N_p	Power number = $(g_p P) / \rho N^3 D_a^5$	Dimensionless	Dimensionless
N_Q	Impeller pumping coefficient = Q / ND_a^3	Dimensionless	Dimensionless
N_i	Impeller speed	s ⁻¹	s ⁻¹
N_t	Impeller speed	s ⁻¹	s ⁻¹
P	Power	(N·m/s)	ft·lb _f /s
Q	Impeller flow rate	m ³ /s	ft ³ /s
T	Tank diameter	m	ft
v	Average fluid velocity	m/s	ft/s
v'	Fluid velocity fluctuation	m/s	ft/s
V	Bulk average velocity	m/s	ft/s
Z	Liquid level in tank	m	ft
Greek Symbols			
γ	Rate of shear	s ⁻¹	s ⁻¹
Δp	Pressure drop across orifice		lb _f /ft ²
μ	Viscosity of liquid at tank temperature	Pa·s	lb/(ft·s)
μ	Stirred liquid viscosity	Pa·s	lb/(ft·s)
μ_b	Viscosity of fluid at bulk temperature	Pa·s	lb/(ft·s)
μ_c	Viscosity, continuous phase	Pa·s	lb/(ft·s)
μ_D	Viscosity of dispersed phase	Pa·s	lb/(ft·s)
μ_f	Viscosity of liquid at mean film temperature	Pa·s	lb/(ft·s)
μ_{wt}	Viscosity at wall temperature	Pa·s	lb/(ft·s)
ρ	Stirred liquid density	g/m ³	lb/ft ³
ρ	Density of fluid	kg/m ³	lb/ft ³
ρ_{av}	Density of dispersed phase	kg/m ³	lb/ft ³
ρ_c	Density	kg/m ³	lb/ft ³
σ	Interfacial tension	N/m	lb _f /ft
Φ_D	Average volume fraction of discontinuous phase	Dimensionless	Dimensionless

PHASE CONTACTING AND LIQUID-SOLID PROCESSING: AGITATION OF LOW-VISCOSITY PARTICLE SUSPENSIONS

GENERAL REFERENCES: Hamby, N., M. F. Edwards, and A. W. Neinow (eds.), *Mixing in the Process Industries*, Butterworth, Stoneham, Mass., 1986. Lo, T. C., M. H. I. Baird, and C. Hanson, *Handbook of Solvent Extraction*, Wiley, New York, 1983. Nagata, S., *Mixing: Principles and Applications*, Kodansha Ltd., Tokyo, Wiley, New York, 1975. Oldshue, J. Y., *Fluid Mixing Technology*, McGraw-Hill, New York, 1983. Tatterson, G. B., *Fluid Mixing and Gas Dispersion in Agitated Tanks*, McGraw-Hill, New York, 1991. Uhl, V. W., and J. B. Gray (eds.), *Mixing*, vols. I and II, Academic Press, New York, 1966; vol. III, Academic Press, Orlando, Fla., 1992. Ulbrecht, J. J., and G. K. Paterson (eds.), *Mixing of Liquids by Mechanical Agitation*, Gordon & Breach Science Publishers, New York, 1985.

PROCEEDINGS: *Fluid Mixing*, vol. I, Inst. Chem. Eng. Symp., Ser. No. 64 (Bradford, England), The Institute of Chemical Engineers, Rugby, England, 1984. *Mixing—Theory Related to Practice*, AIChE, Inst. Chem. Eng. Symp. Ser. No. 10 (London), AIChE and The Institute of Chemical Engineers, London, 1965. *Proc. First (1974), Second (1977), Third (1979), Fourth (1982), Fifth (1985), and Sixth (1988) European Conf. on Mixing*, N. G. Coles (ed.), (Cambridge, England) BHRA Fluid Eng., Cranfield, England. *Process Mixing, Chemical and Biochemical Applications*, G. B. Tatterson, and R. V. Calabrese (eds.), AIChE Symp. Ser. No. 286, 1992.

FLUID MIXING TECHNOLOGY

Fluid mixers cut across almost every processing industry including the chemical process industry; minerals, pulp, and paper; waste and water treating and almost every individual process sector. The engineer working with the application and design of mixers for a given process has three basic sources for information. One is published literature, consisting of several thousand published articles and several currently available books, and brochures from equipment vendors. In addition, there may be a variety of in-house experience which may or may not be cataloged, categorized, or usefully available for the process application at hand. Also, short courses are currently available in selected locations and with various course objectives, and a large body of experience and information lies in the hands of equipment vendors.

In the United States, it is customary to design and purchase a mixer from a mixing vendor and purchase the vessel from another supplier. In many other countries, it is more common to purchase the vessel and mixer as a package from one supplier.

In any event, the users of the mixer can issue a mechanical specification and determine the speed, diameter of an impeller, and power with in-house expertise. Or they may issue a process specification which describes the engineering purpose of the mixing operation and the vendor will supply a description of the mixer process performance as well as prepare a mechanical design.

This section describes fluid mixing technology and is referred to in other sections in this handbook which discuss the use of fluid mixing equipment in their various operating disciplines. This section does not describe paste and dough mixing, which may require planetary and extruder-type mixers, nor the area of dry solid-solid mixing.

It is convenient to divide mixing into five pairs (plus three triplets and one quadruplicate combination) of materials, as shown in Table 18-1. These five pairs are blending (miscible liquids), liquid-solid, liquid-gas, liquid-liquid (immiscible liquids), and fluid motion. There are also four other categories that occur, involving three or four phases. One concept that differentiates mixing requirements is the difference between physical criteria listed on the left side of Table 18-1, in which some degree of sampling can be used to determine the character of the mixture in various parts in the tank, and various definitions of mixing requirements can be based on these physical

descriptions. The other category on the right side of Table 18-1 involves chemical and mass-transfer criteria in which rates of mass transfer or chemical reaction are of interest and have many more complexities in expressing the mixing requirements.

The first five classes have their own mixing technologies. Each of these 10 areas has its own mixing technology. There are relationships for the optimum geometry of impeller types, D/T ratios, and tank geometry. They each often have general, overall mixing requirements and different scale-up relationships based on process definitions. In addition, there are many subclassifications, some of which are based on the viscosity of fluids. In the case of blending, we have blending in the viscous region, the transition region, and the turbulent region. Since any given mixer designed for a process may be required to do several different parts of these 10 categories, it must be a compromise of the geometry and other requirements for the total process result and may not optimize any one particular process component. If it turns out that one particular process requirement is so predominant that all the other requirements are satisfied as a consequence, then it is possible to optimize that particular process step. Often, the only process requirement is in one of these 10 areas, and the mixer can be designed and optimized for that one step only.

As an example of the complexity of fluid mixing, many batch processes involve adding many different materials and varying the liquid level over wide ranges in the tank, have different temperatures and shear rate requirements, and obviously need experience and expert attention to all of the requirements. Superimposing the requirements for sound mechanical design, including drives, fluid seals, and rotating shafts, means that the concepts presented here are merely a beginning to the overall, final design.

A few general principles are helpful at this point before proceeding to the examination of equipment and process details. For any given impeller geometry, speed, and diameter, the impeller draws a certain amount of power. This power is 100 percent converted to heat. In low-viscosity mixing (defined later), this power is used to generate a macro-scale regime in which one typically has the visual observation of flow pattern, swirls, and other surface phenomena. However, these flow patterns are primarily energy transfer agents that transfer the power down to the micro scale. The macro-scale regime involves the pumping capacity of the impeller as well as the total circulating capacity throughout the tank and it is an important part of the overall mixer design. The micro-scale area in which the power is dissipated does not care much which impeller is used to generate the energy dissipation. In contrast, in high-viscosity processes, there is a continual progress of energy dissipation from the macro scale down to the micro scale.

There is a wide variety of impellers using *fluidfoil principles*, which are used when flow from the impeller is predominant in the process requirement and macro- or micro-scale shear rates are a subordinate issue.

TABLE 18-1 Classification System for Mixing Processes

Physical	Components	Chemical, mass transfer
Blending	Blending	Chemical reactions
Suspension	Solid-liquid	Dissolving, precipitation
Dispersion	Gas-liquid	Gas absorption
	Solid-liquid-gas	
Emulsions	Liquid-liquid	Extraction
	Liquid-liquid-solid	
	Gas-liquid-liquid	
	Gas-liquid-liquid-solid	
Pumping	Fluid motion	Heat transfer

Scale-up involves selecting mixing variables to give the desired performance in both pilot and full scale. This is often difficult (sometimes impossible) using geometric similarity, so that the use of nongeometric impellers in the pilot plant compared to the impellers used in the plant often allows closer modeling of the mixing requirements to be achieved.

Computational fluid mixing allows the modeling of flow patterns in mixing vessels and some of the principles on which this is based in current techniques are included.

INTRODUCTORY FLUID MECHANICS

The fluid mixing process involves three different areas of viscosity which affect flow patterns and scale-up, and two different scales within the fluid itself: macro scale and micro scale. Design questions come up when looking at the design and performance of mixing processes in a given volume. Considerations must be given to proper impeller and tank geometry as well as the proper speed and power for the impeller. Similar considerations come up when it is desired to scale up or scale down, and this involves another set of mixing considerations.

If the fluid discharge from an impeller is measured with a device that has a high-frequency response, one can track the velocity of the fluid as a function of time. The velocity at a given point in time can then be expressed as an average velocity v plus fluctuating component v' . Average velocities can be integrated across the discharge of the impeller, and the pumping capacity normal to an arbitrary discharge plane can be calculated. This arbitrary discharge plane is often defined as the plane bounded by the boundaries of the impeller blade diameter and height. Because there is no casing, however, an additional 10 to 20 percent of flow typically can be considered as the primary flow from an impeller.

The velocity gradients between the average velocities operate only on larger particles. Typically, these larger-size particles are greater than 1000 μm . This is not a proven definition, but it does give a feel for the magnitudes involved. This defines macro-scale mixing. In the turbulent region, these macro-scale fluctuations can also arise from the finite number of impeller blades. These set up velocity fluctuations that can also operate on the macro scale.

Smaller particles see primarily only the fluctuating velocity component. When the particle size is much less than 100 μm , the turbulent properties of the fluid become important. This is the definition of the physical size for micro-scale mixing.

All of the power applied by a mixer to a fluid through the impeller appears as heat. The conversion of power to heat is through viscous shear and is approximately 2542 Btu/h/hp. Viscous shear is present in turbulent flow only at the micro-scale level. As a result, the power per unit volume is a major component of the phenomena of micro-scale mixing. At a 1- μm level, in fact, it doesn't matter what specific impeller design is used to supply the power.

Numerous experiments show that power per unit volume in the zone of the impeller (which is about 5 percent of the total tank volume) is about 100 times higher than the power per unit volume in the rest of the vessel. Making some reasonable assumptions about the fluid mechanics parameters, the root-mean-square (rms) velocity fluctuation in the zone of the impeller appears to be approximately 5 to 10 times higher than in the rest of the vessel. This conclusion has been verified by experimental measurements.

The ratio of the rms velocity fluctuation to the average velocity in the impeller zone is about 50 percent with many open impellers. If the rms velocity fluctuation is divided by the average velocity in the rest of the vessel, however, the ratio is on the order of 5 percent. This is also the level of rms velocity fluctuation to the mean velocity in pipeline flow. There are phenomena in micro-scale mixing that can occur in mixing tanks that do not occur in pipeline reactors. Whether this is good or bad depends upon the process requirements.

Figure 18-1 shows velocity versus time for three different impellers. The differences between the impellers are quite significant and can be important for mixing processes.

All three impellers are calculated for the same impeller flow Q and the same diameter. The A310 (Fig. 18-2) draws the least power and has the least velocity fluctuations. This gives the lowest micro-scale turbulence and shear rate. The A200 (Fig. 18-3) shows increased velocity

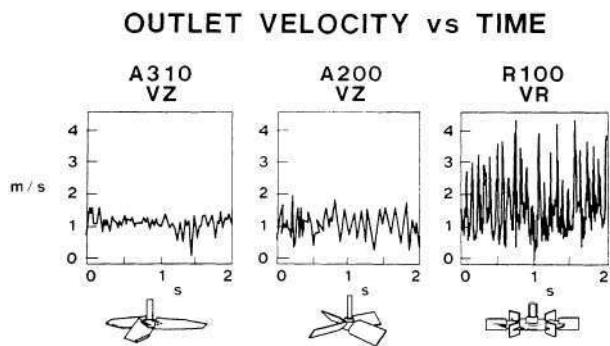


FIG. 18-1 Velocity fluctuations versus time for equal total pumping capacity from three different impellers.

fluctuations and draws more power. The R100 (Fig. 18-4) draws the most power and has the highest micro-scale shear rate. The proper impeller should be used for each individual process requirement.

Scale-up/Scale-down Two aspects of scale-up frequently arise. One is building a model based on pilot-plant studies that develop an understanding of the process variables for an existing full-scale mixing installation. The other is taking a new process and studying it in the pilot plant in such a way that pertinent scale-up variables are worked out for a new mixing installation.

There are a few principles of scale-up that can indicate which approach to take in either case. Using geometric similarity, the macro-scale variables can be summarized as follows:

- Blend and circulation times in the large tank will be much longer than in the small tank.



FIG. 18-2 An A310 impeller.

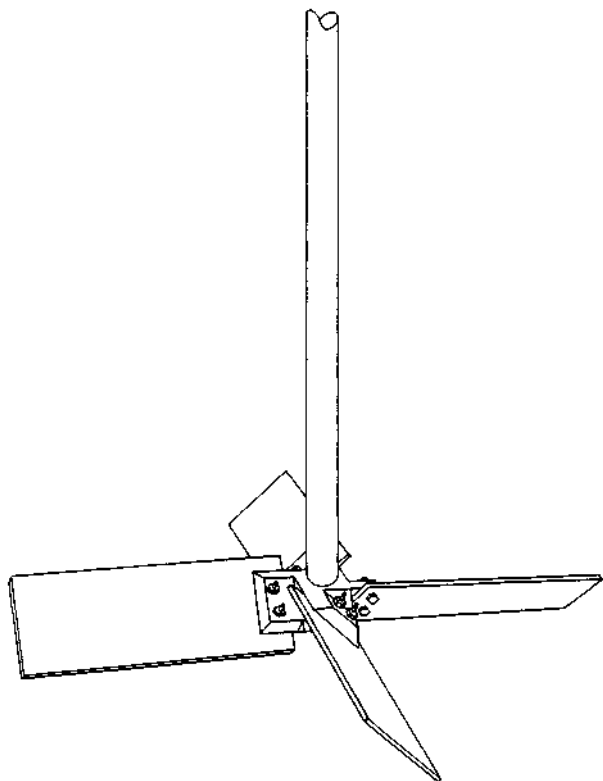


FIG. 18-3 Pitched-blade turbine.

- Maximum impeller zone shear rate will be higher in the larger tank, but the average impeller zone shear rate will be lower; therefore, there will be a much greater variation in shear rates in a full-scale tank than in a pilot unit.
- Reynolds numbers in the large tank will be higher, typically on the order of 5 to 25 times higher than those in a small tank.
- Large tanks tend to develop a recirculation pattern from the impeller through the tank back to the impeller. This results in a behavior similar to that for a number of tanks in a series. The net result is that the mean circulation time is increased over what would be predicted from the impeller pumping capacity. This also

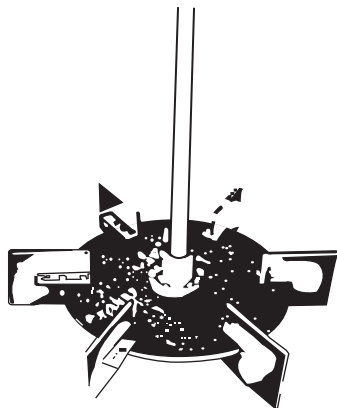


FIG. 18-4 Flat-blade turbine.

increases the standard deviation of the circulation times around the mean.

- Heat transfer is normally much more demanding on a large scale. The introduction of helical coils, vertical tubes, or other heat-transfer devices causes an increased tendency for areas of low recirculation to exist.
- In gas-liquid systems, the tendency for an increase in the gas superficial velocity upon scale-up can further increase the overall circulation time.

What about the micro-scale phenomena? These are dependent primarily on the energy dissipation per unit volume, although one must also be concerned about the energy spectra. In general, the energy dissipation per unit volume around the impeller is approximately 100 times higher than in the rest of the tank. This results in an rms velocity fluctuation ratio to the average velocity on the order of 10:1 between the impeller zone and the rest of the tank.

Because there are thousands of specific processes each year that involve mixing, there will be at least hundreds of different situations requiring a somewhat different pilot-plant approach. Unfortunately, no set of rules states how to carry out studies for any specific program, but here are a few guidelines that can help one carry out a pilot-plant program.

- For any given process, one takes a qualitative look at the possible role of fluid shear stresses. Then one tries to consider pathways related to fluid shear stress that may affect the process. If there are none, then this extremely complex phenomenon can be dismissed and the process design can be based on such things as uniformity, circulation time, blend time, or velocity specifications. This is often the case in the blending of miscible fluids and the suspension of solids.
- If fluid shear stresses are likely to be involved in obtaining a process result, then one must qualitatively look at the scale at which the shear stresses influence the result. If the particles, bubbles, droplets, or fluid clumps are on the order of 1000 μm or larger, the variables are macro scale and average velocities at a point are the predominant variable.

When macro-scale variables are involved, every geometric design variable can affect the role of shear stresses. They can include such items as power, impeller speed, impeller diameter, impeller blade shape, impeller blade width or height, thickness of the material used to make the impeller, number of blades, impeller location, baffle location, and number of impellers.

Micro-scale variables are involved when the particles, droplets, baffles, or fluid clumps are on the order of 100 μm or less. In this case, the critical parameters usually are power per unit volume, distribution of power per unit volume between the impeller and the rest of the tank, rms velocity fluctuation, energy spectra, dissipation length, the smallest micro-scale eddy size for the particular power level, and viscosity of the fluid.

- The overall circulating pattern, including the circulation time and the deviation of the circulation times, can never be neglected. No matter what else a mixer does, it must be able to circulate fluid throughout an entire vessel appropriately. If it cannot, then that mixer is not suited for the task being considered.

Qualitative and, hopefully, quantitative estimates of how the process result will be measured must be made in advance. The evaluations must allow one to establish the importance of the different steps in a process, such as gas-liquid mass transfer, chemical reaction rate, or heat transfer.

- It is seldom possible, either economically or timewise, to study every potential mixing variable or to compare the performance of many impeller types. In many cases, a process needs a specific fluid regime that is relatively independent of the impeller type used to generate it. Because different impellers may require different geometries to achieve an optimum process combination, a random choice of only one diameter of each of two or more impeller types may not tell what is appropriate for the fluid regime ultimately required.
- Often, a pilot plant will operate in the viscous region while the commercial unit will operate in the transition region, or alternatively, the pilot plant may be in the transition region and the commercial unit in the turbulent region. Some experience is required to estimate the difference in performance to be expected upon scale-up.
- In general, it is not necessary to model Z/T ratios between pilot and commercial units.

- In order to make the pilot unit more like a commercial unit in macro-scale characteristics, the pilot unit impeller must be designed to lengthen the blend time and to increase the maximum impeller zone shear rate. This will result in a greater range of shear rates than is normally found in a pilot unit.

MIXING EQUIPMENT

There are three types of mixing flow patterns that are markedly different. The so-called axial-flow turbines (Fig. 18-3) actually give a flow coming off the impeller of approximately 45° , and therefore have a recirculation pattern coming back into the impeller at the hub region of the blades. This flow pattern exists to an approximate Reynolds number of 200 to 600 and then becomes radial as the Reynolds number decreases. Both the R100 and A200 impellers normally require four baffles for an effective flow pattern. These baffles typically are $\frac{1}{2}$ of the tank diameter and width.

Radial-flow impellers include the flat-blade disc turbine, Fig. 18-4, which is labeled an R100. This generates a radial flow pattern at all Reynolds numbers. Figure 18-17 is the diagram of Reynolds number/power number curve, which allows one to calculate the power knowing the speed and diameter of the impeller. The impeller shown in Fig. 18-4 typically gives high shear rates and relatively low pumping capacity.

The current design of fluidfoil impellers includes the A310 (Fig. 18-2), as well as several other impellers of that type commonly referred to as *high-efficiency impellers*, *hydrofoil*, and other descriptive names to illustrate that they are designed to maximize flow and minimize shear rate. These impellers typically require two baffles, but are normally used with three, since three gives a more stable flow pattern. Since most industrial mixing processes involve pumping capacity and, to a lesser degree, fluid shear rate, the fluidfoil impellers are now used on the majority of the mixer installations. There is now an additional family of these fluidfoil impellers, which depend upon different solidity ratios to operate in various kinds of fluid mixing systems. Figure 18-5 illustrates four of these impellers. The solidity ratio is the ratio of total blade area to a circle circumscribing the impeller and, as viscosity increases, higher values of the solidity ratios are more effective in providing an axial flow pattern rather than a radial flow pattern. Also the A315-type provides an effective area of preventing gas bypassing through the hub of the impeller by having exceptionally wide blades. Another impeller of that type is the Prochem Maxflo T.

Small Tanks For tanks less than 1.8 m in diameter, the clamp or flanged mounted angular, off-center axial-flow impeller without baffles should be used for a wide range of process requirements (refer to Fig. 18-14). The impellers currently used are the fluidfoil type. Since small impellers typically operate at low Reynolds numbers, often in the transition region, the fluidfoil impeller should be designed to give good flow characteristics over a range of Reynolds numbers, probably on the order of 50 to 500. The Z/T ratio should be 0.75 to 1.5. The volume of liquid should not exceed 4 m^3 .

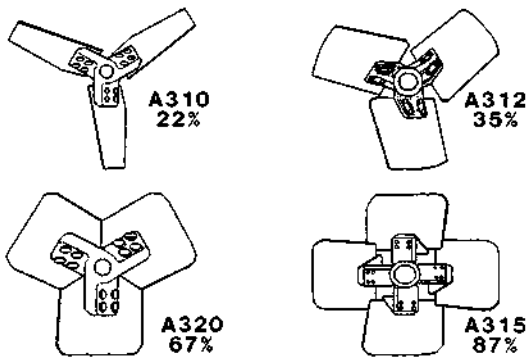


FIG. 18-5 The solidity ratio for four different impellers of the axial-flow fluidfoil type.

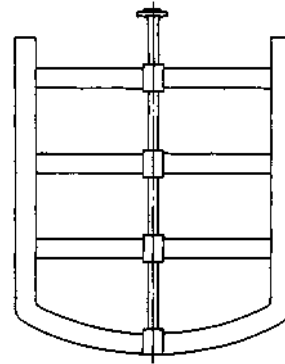


FIG. 18-6 Anchor impeller.

Close-Clearance Impellers There are two close-clearance impellers. They are the *anchor impeller* (Fig. 18-6) and the *helical impeller* (Fig. 18-7), which operate near the tank wall and are particularly effective in pseudoplastic fluids in which it is desirable to have the mixing energy concentrated out near the tank wall where the flow pattern is more effective than with the open impellers that were covered earlier.

Axial-Flow Impellers Axial-flow impellers include all impellers in which the blade makes an angle of less than 90° with the plane of

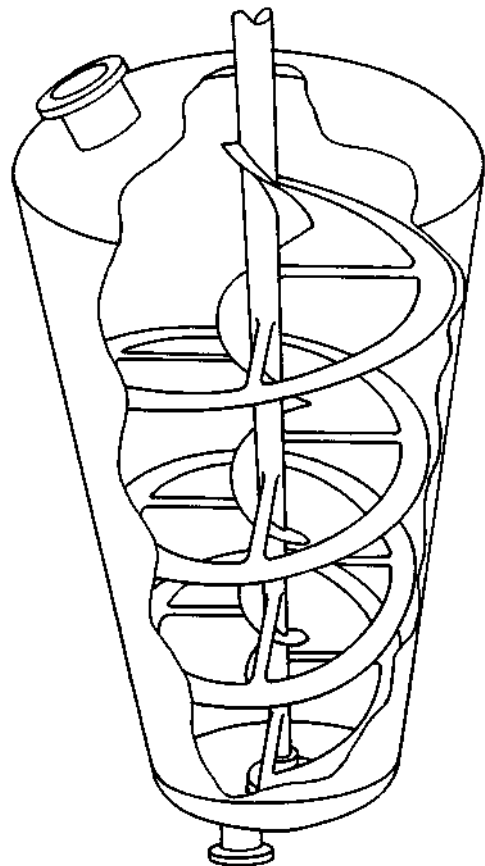


FIG. 18-7 Helical mixer for high-viscosity fluid.

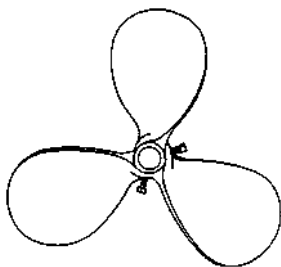


FIG. 18-8 Marine-type mixing impeller.

rotation. Propellers and pitched-blade turbines, as illustrated in Figs. 18-8 and 18-3, are representative axial-flow impellers.

Portable mixers may be clamped on the side of an open vessel in the angular, off-center position shown in Fig. 18-14 or bolted to a flange or plate on the top of a closed vessel with the shaft in the same angular, off-center position. This mounting results in a strong top-to-bottom circulation.

Two basic speed ranges are available: 1150 or 1750 r/min with direct drive and 350 or 420 r/min with a gear drive. The high-speed units produce higher velocities and shear rates (Fig. 18-9) in the impeller discharge stream and a lower circulation rate throughout the vessel than the low-speed units. For suspension of solids, it is common to use the gear-driven units, while for rapid dispersion or fast reactions the high-speed units are more appropriate.

Axial-flow impellers may also be mounted near the bottom of the cylindrical wall of a vessel as shown in Fig. 18-10. Such side-entering agitators are used to blend low-viscosity fluids [<0.1 Pa-s (100 cP)] or to keep slowly settling sediment suspended in tanks as large as some 4000 m^3 (10^6 gal). Mixing of paper pulp is often carried out by side-entering propellers.

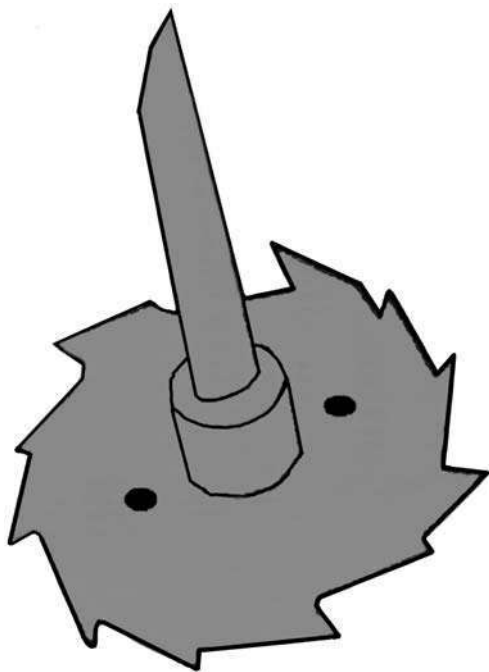


FIG. 18-9 High-shear-rate-impeller.

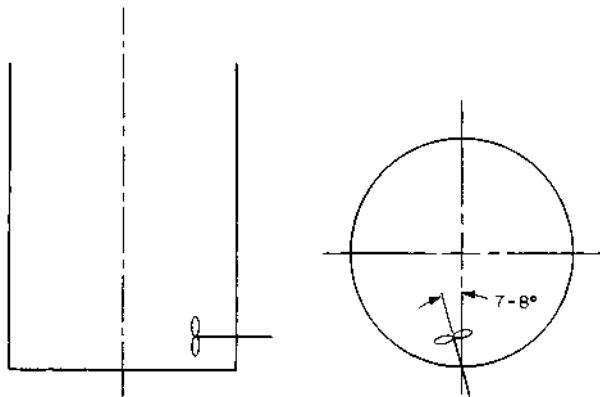


FIG. 18-10 Side-entering propeller mixer.

Pitched-blade turbines (Fig. 18-3) are used on top-entering agitator shafts instead of propellers when a high axial circulation rate is desired and the power consumption is more than 2.2 kW (3 hp). A pitched-blade turbine near the upper surface of liquid in a vessel is effective for rapid submergence of floating particulate solids.

Radial-Flow Impellers Radial-flow impellers have blades which are parallel to the axis of the drive shaft. The smaller multiblade ones are known as *turbines*; larger, slower-speed impellers, with two or four blades, are often called *paddles*. The diameter of a turbine is normally between 0.3 and 0.6 of the tank diameter. Turbine impellers come in a variety of types, such as curved-blade and flat-blade, as illustrated in Fig. 18-4. Curved blades aid in starting an impeller in settled solids.

For processes in which corrosion of commonly used metals is a problem, glass-coated impellers may be economical. A typical modified curved-blade turbine of this type is shown in Fig. 18-11.

Close-Clearance Stirrers For some pseudoplastic fluid systems stagnant fluid may be found next to the vessel walls in parts remote from propeller or turbine impellers. In such cases, an "anchor" impeller may be used (Fig. 18-6). The fluid flow is principally circular or helical (see Fig. 18-7) in the direction of rotation of the anchor. Whether substantial axial or radial fluid motion also occurs depends on the fluid viscosity and the design of the upper blade-supporting spokes. Anchor agitators are used particularly to obtain improved heat transfer in high-consistency fluids.

Unbaffled Tanks If a low-viscosity liquid is stirred in an unbaffled tank by an axially mounted agitator, there is a tendency for a swirling

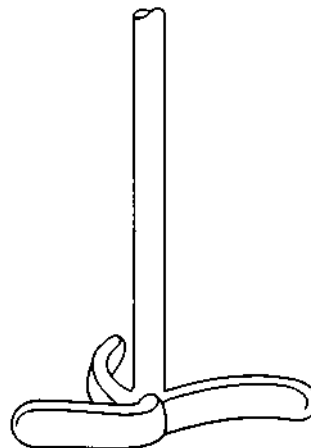


FIG. 18-11 Glass-steel impeller. (The Pfaunder Company.)

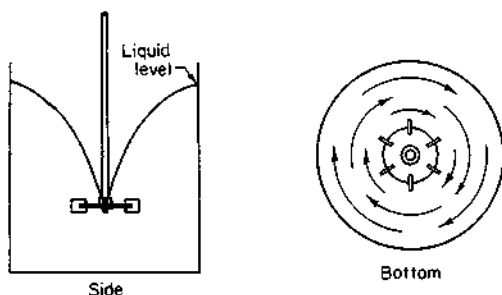


FIG. 18-12 Typical flow pattern for either axial- or radial-flow impellers in an unbaffled tank.

flow pattern to develop regardless of the type of impeller. Figure 18-12 shows a typical flow pattern. A vortex is produced owing to centrifugal force acting on the rotating liquid. In spite of the presence of a vortex, satisfactory process results often can be obtained in an unbaffled vessel. However, there is a limit to the rotational speed that may be used, since once the vortex reaches the impeller, severe air entrainment may occur. In addition, the swirling mass of liquid often generates an oscillating surge in the tank, which coupled with the deep vortex may create a large fluctuating force acting on the mixer shaft.

Vertical velocities in a vortexing low-viscosity liquid are low relative to circumferential velocities in the vessel. Increased vertical circulation rates may be obtained by mounting the impeller off center, as illustrated in Fig. 18-13. This position may be used with either turbines or propellers. The position is critical, since too far or too little off center in one direction or the other will cause greater swirling, erratic vortexing, and dangerously high shaft stresses. Changes in viscosity and tank size also affect the flow pattern in such vessels. Off-center mountings have been particularly effective in the suspension of paper pulp.

With axial-flow impellers, an angular off-center position may be used. The impeller is mounted approximately 15° from the vertical, as shown in Fig. 18-14.

The angular off-center position used with fluidfoil units is usually limited to impellers delivering 2.2 kW (3 hp) or less. The unbalanced fluid forces generated by this mounting can become severe with higher power.

Baffled Tanks For vigorous agitation of thin suspensions, the tank is provided with baffles which are flat vertical strips set radially along the tank wall, as illustrated in Figs. 18-15 and 18-16. Four baffles are almost always adequate. A common baffle width is one-tenth to one-twelfth of the tank diameter (radial dimension). For agitating

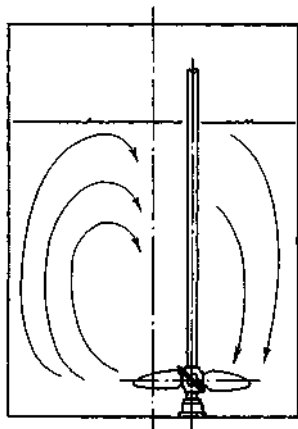


FIG. 18-13 Flow pattern with a paper-stock propeller, unbaffled; vertical off-center position.

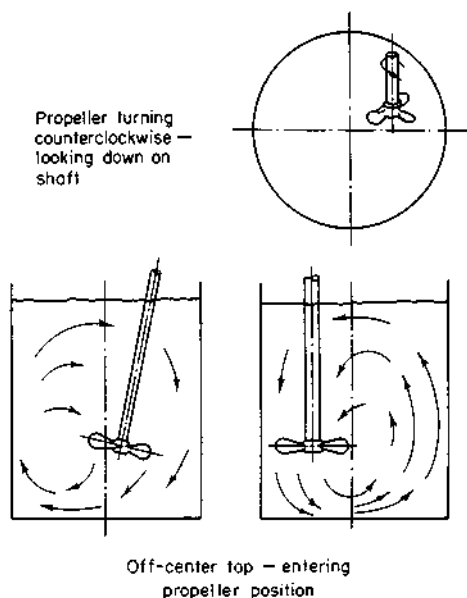


FIG. 18-14 Typical flow pattern with a propeller in angular off-center position without baffles.

slurries, the baffles often are located one-half of their width from the vessel wall to minimize accumulation of solids on or behind them.

For Reynolds numbers greater than 2000 baffles are commonly used with turbine impellers and with on-centerline axial-flow impellers. The flow patterns illustrated in Figs. 18-15 and 18-16 are quite different, but in both cases the use of baffles results in a large top-to-bottom circulation without vortexing or severely unbalanced fluid forces on the impeller shaft.

In the transition region [Reynolds numbers, Eq. (18-1), from 10 to 10,000], the width of the baffle may be reduced, often to one-half of standard width. If the circulation pattern is satisfactory when the tank is unbaffled but a vortex creates a problem, partial-length baffles may be used. These are standard-width and extend downward from the surface into about one-third of the liquid volume.

In the region of laminar flow ($N_{Re} < 10$), the same power is consumed by the impeller whether baffles are present or not, and they are seldom required. The flow pattern may be affected by the baffles, but not always advantageously. When they are needed, the baffles are usually placed one or two widths radially off the tank wall, to allow fluid to circulate behind them and at the same time produce some axial deflection of flow.

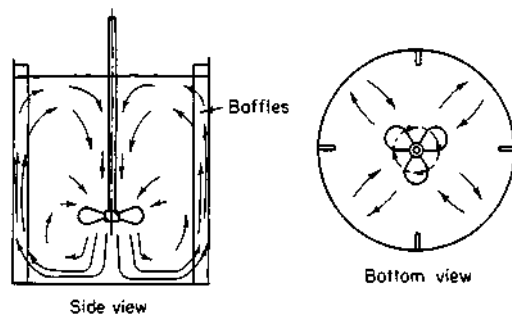


FIG. 18-15 Typical flow pattern in a baffled tank with a propeller or an axial-flow turbine positioned on center.

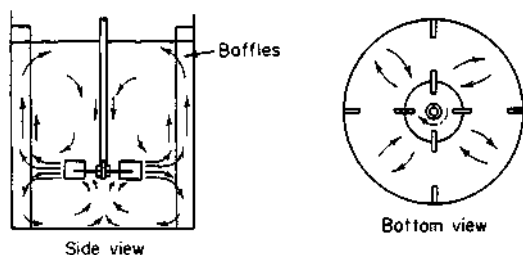


FIG. 18-16 Typical flow pattern in a baffled tank with a turbine positioned on center.

FLUID BEHAVIOR IN MIXING VESSELS

Impeller Reynolds Number The presence or absence of turbulence in an impeller-stirred vessel can be correlated with an impeller Reynolds number defined

$$N_{Re} = \frac{D_a^2 N \rho}{\mu} \quad (18-1)$$

where N = rotational speed, r/s ; D_a = impeller diameter, m (ft); ρ = fluid density, kg/m^3 (lb/ft^3); and μ = viscosity, $Pa \cdot s$ [$lb/(ft \cdot s)$]. Flow in the tank is turbulent when $N_{Re} > 10,000$. Thus viscosity alone is not a valid indication of the type of flow to be expected. Between Reynolds numbers of 10,000 and approximately 10 is a transition range in which flow is turbulent at the impeller and laminar in remote parts of the vessel; when $N_{Re} < 10$, flow is laminar only.

Not only is the type of flow related to the impeller Reynolds number, but also such process performance characteristics as mixing time, impeller pumping rate, impeller power consumption, and heat- and mass-transfer coefficients can be correlated with this dimensionless group.

Relationship between Fluid Motion and Process Performance Several phenomena which can be used to promote various processing objectives occur during fluid motion in a vessel.

1. Shear stresses are developed in a fluid when a layer of fluid moves faster or slower than a nearby layer of fluid or a solid surface. In laminar flow, the shear stress is equal to the product of fluid viscosity and velocity gradient or rate of shear. Under laminar-flow conditions, shear forces are larger than inertial forces in the fluid.

With turbulent flow, shear stress also results from the behavior of transient random eddies, including large-scale eddies which decay to small eddies or fluctuations. The scale of the large eddies depends on equipment size. On the other hand, the scale of small eddies, which dissipate energy primarily through viscous shear, is almost independent of agitator and tank size.

The shear stress in the fluid is much higher near the impeller than it is near the tank wall. The difference is greater in large tanks than in small ones.

2. Inertial forces are developed when the velocity of a fluid changes direction or magnitude. In turbulent flow, inertia forces are larger than viscous forces. Fluid in motion tends to continue in motion until it meets a solid surface or other fluid moving in a different direction. Forces are developed during the momentum transfer that takes place. The forces acting on the impeller blades fluctuate in a random manner related to the scale and intensity of turbulence at the impeller.

3. The interfacial area between gases and liquids, immiscible liquids, and solids and liquids may be enlarged or reduced by these viscous and inertia forces when interacting with interfacial forces such as surface tension.

4. Concentration and temperature differences are reduced by bulk flow or circulation in a vessel. Fluid regions of different composition or temperature are reduced in thickness by bulk motion in which velocity gradients exist. This process is called bulk diffusion or Taylor diffusion (Brodkey, in Uhl and Gray, op. cit., vol. 1, p. 48). The turbulent and molecular diffusion reduces the difference between these regions. In laminar flow, Taylor diffusion and molecular diffusion are the mechanisms of concentration- and temperature-difference reduction.

5. Equilibrium concentrations which tend to develop at solid-liquid, gas-liquid, or liquid-liquid interfaces are displaced or changed by molecular and turbulent diffusion between bulk fluid and fluid adjacent to the interface. Bulk motion (Taylor diffusion) aids in this mass-transfer mechanism also.

Turbulent Flow in Stirred Vessels Turbulence parameters such as intensity and scale of turbulence, correlation coefficients, and energy spectra have been measured in stirred vessels. However, these characteristics are not used directly in the design of stirred vessels.

Fluid Velocities in Mixing Equipment Fluid velocities have been measured for various turbines in baffled and unbaffled vessels. Typical data are summarized in Uhl and Gray, op. cit., vol. 1, chap. 4. Velocity data have been used for calculating impeller discharge and circulation rates but are not employed directly in the design of mixing equipment.

Impeller Discharge Rate and Fluid Head for Turbulent Flow When fluid viscosity is low and flow is turbulent, an impeller moves fluids by an increase in momentum from the blades which exert a force on the fluid. The blades of rotating propellers and turbines change the direction and increase the velocity of the fluids.

The pumping rate or discharge rate of an impeller is the flow rate perpendicular to the impeller discharge area. The fluid passing through this area has velocities proportional to the impeller peripheral velocity and velocity heads proportional to the square of these velocities at each point in the impeller discharge stream under turbulent-flow conditions. The following equations relate velocity head, pumping rate, and power for geometrically similar impellers under turbulent-flow conditions:

$$Q = N_Q N D_a^3 \quad (18-2)$$

$$H = \frac{N_p N^2 D_a^2}{N_Q g} \quad (18-3)$$

$$P = N_p \rho N^3 \left(\frac{D_a^5}{g_c} \right) \quad (18-4)$$

$$P = \frac{\rho H Q g}{g_c} \quad (18-5)$$

where Q = impeller discharge rate, m^3/s (ft^3/s); N_Q = discharge coefficient, dimensionless; H = velocity head, m (ft); N_p = power number, dimensionless; P = power, $(N \cdot m)/s$ [$(ft \cdot lbf)/s$]; g_c = dimensional constant, $32.2 (ft \cdot lb)/(lbf \cdot s^2)$ ($g_c = 1$ when using SI units); and g = gravitational acceleration, m/s^2 (ft/s^2).

The discharge rate Q has been measured for several types of impellers, and discharge coefficients have been calculated. The data of a number of investigators are reviewed by Uhl and Gray (op. cit., vol. 1, chap. 4). N_Q is 0.4 to 0.5 for a propeller with pitch equal to diameter at $N_{Re} = 10^5$. For turbines, N_Q ranges from 0.7 to 2.9, depending on the number of blades, blade-height-to-impeller-diameter ratio, and impeller-to-vessel-diameter ratio. The effects of these geometric variables are not well defined.

Power consumption has also been measured and correlated with impeller Reynolds numbers. The velocity head for a mixing impeller can be calculated, then, from flow and power data, by Eq. (18-3) or Eq. (18-5).

The velocity head of the impeller discharge stream is a measure of the maximum force that this fluid can exert when its velocity is changed. Such inertia forces are higher in streams with higher discharge velocities. Shear rates and shear stresses are also higher under these conditions in the smallest eddies. If a higher discharge velocity is desired at the same power consumption, a smaller-diameter impeller must be used at a higher rotational speed. According to Eq. (18-4), at a given power level $N \propto D_a^{-5/3}$ and $N D_a \propto D_a^{-2/3}$. Then, $H \propto D_a^{-4/3}$ and $Q \propto D_a^{4/3}$.

An impeller with a high fluid head is one with high peripheral velocity and discharge velocity. Such impellers are useful for (1) rapid reduction of concentration differences in the impeller discharge stream (rapid mixing), (2) production of large interfacial area and

small droplets in gas-liquid and immiscible-liquid systems, (3) solids deagglomeration, and (4) promotion of mass transfer between phases.

The impeller discharge rate can be increased at the same power consumption by increasing impeller diameter and decreasing rotational speed and peripheral velocity so that $N^3 D_a^5$ is a constant (Eq. 18-4). Flow goes up, velocity head and peripheral velocity go down, but impeller torque T_Q goes up. At the same torque, $N^2 D_a^5$ is constant, $P \propto D_a^{-5/2}$, and $Q \propto D_a^{1/2}$. Therefore, increasing impeller diameter at constant torque increases discharge rate at lower power consumption. At the same discharge rate, ND_a^3 is constant, $P \propto D_a^{-4}$, and $T_Q \propto D_a^{-1}$. Therefore, power and torque decrease as impeller diameter is increased at constant Q .

A large-diameter impeller with a high discharge rate is used for (1) short times to complete mixing of miscible liquid throughout a vessel, (2) promotion of heat transfer, (3) reduction of concentration and temperature differences in all parts of vessels used for constant-environment reactors and continuous averaging, and (4) suspension of particles of relatively low settling rate.

Laminar Fluid Motion in Vessels When the impeller Reynolds number is less than 10, the flow induced by the impeller is laminar. Under these conditions, the impeller drags fluid with it in a predominantly circular pattern. If the impeller blades curve back, there is a viscous drag flow toward the tips of these blades. Under moderate-viscosity conditions in laminar flow, centrifugal force acting on the fluid layer dragged in a circular path by the rotating impeller will move fluid in a radial direction. This centrifugal effect causes any gas accumulated behind a rotating blade to move to the axis of impeller rotation. Such radial-velocity components are small relative to tangential velocity.

For turbines at Reynolds numbers less than 100, toroidal stagnant zones exist above and below the turbine periphery. Interchange of liquid between these regions and the rest of the vessel is principally by molecular diffusion.

Suspensions of fine solids may have pseudoplastic or plastic-flow properties. When they are in laminar flow in a stirred vessel, motion in remote parts of the vessel where shear rates are low may become negligible or cease completely. To compensate for this behavior of slurries, large-diameter impellers or paddles are used, with $(D_a/D_T) > 0.6$, where D_T is the tank diameter. In some cases, for example, with some anchors, $D_a > 0.95 D_T$. Two or more paddles may be used in deep tanks to avoid stagnant regions in slurries.

In laminar flow ($N_{Re} < 10$), $N_p \propto 1/N_{Re}$ and $P \propto \mu N^2 D_a^3$. Since shear stress is proportional to rotational speed, shear stress can be increased at the same power consumption by increasing N proportionally to $D_a^{-3/2}$ as impeller diameter D_a is decreased.

Fluid circulation probably can be increased at the same power consumption and viscosity in laminar flow by increasing impeller diameter and decreasing rotational speed, but the relationship between Q , N , and D_a for laminar flow from turbines has not been determined.

As in the case of turbulent flow, then, small-diameter impellers ($D_a < D_T/3$) are useful for (1) rapid mixing of dry particles into liquids, (2) gas dispersion in slurries, (3) solid-particle deagglomeration, and (4) promoting mass transfer between solid and liquid phases. If stagnant regions are a problem, large impellers must be used and rotational speed and power increased to obtain the required results. Small continuous-processing equipment may be more economical than batch equipment in such cases.

Likewise, large-diameter impellers ($D_a > D_T/2$) are useful for (1) avoiding stagnant regions in slurries, (2) short mixing times to obtain uniformity throughout a vessel, (3) promotion of heat transfer, and (4) laminar continuous averaging of slurries.

Vortex Depth In an unbaffled vessel with an impeller rotating in the center, centrifugal force acting on the fluid raises the fluid level at the wall and lowers the level at the shaft. The depth and shape of such a vortex [Rieger, Dittl, and Novak, *Chem. Eng. Sci.*, **34**, 397 (1978)] depend on impeller and vessel dimensions as well as rotational speed.

Power Consumption of Impellers Power consumption is related to fluid density, fluid viscosity, rotational speed, and impeller diameter by plots of power number ($g_p P / \rho N^3 D_a^5$) versus Reynolds number ($D_a^2 N \rho / \mu$). Typical correlation lines for frequently used impellers operating in newtonian liquids contained in baffled cylindrical vessels are presented in Fig. 18-17. These curves may be used also

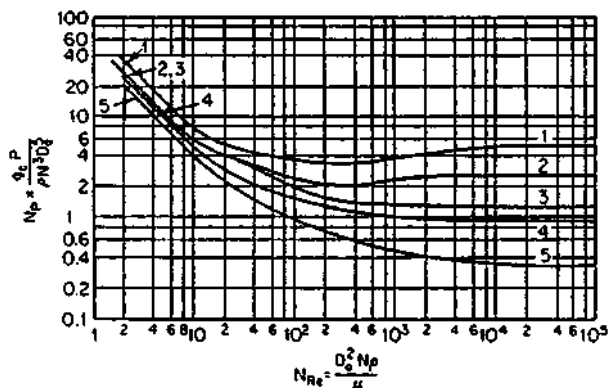


FIG. 18-17 Impeller power correlations: curve 1, six-blade turbine, $D_a/W_i = 5$, like Fig. 18-4 but with six blades, four baffles, each $D_T/12$; curve 2, vertical-blade, open turbine with six straight blades, $D_a/W_i = 8$, four baffles each $D_T/12$; curve 3, 45° pitched-blade turbine like Fig. 18-3 but with six blades, $D_a/W_i = 8$, four baffles, each $D_T/12$; curve 4, propeller, pitch equal to $2D_a$, four baffles, each $0.1D_T$, also same propeller in angular off-center position with no baffles; curve 5, propeller, pitch equal to D_a , four baffles each $0.1D_T$, also same propeller in angular off-center position as in Fig. 18-14 with no baffles. D_a = impeller diameter, D_T = tank diameter, g_p = gravitational conversion factor, N = impeller rotational speed, P = power transmitted by impeller shaft, W_i = impeller blade height, μ = viscosity of stirred liquid, and ρ = density of stirred mixture. Any set of consistent units may be used, but N must be rotations (rather than radians) per unit time. In the SI system, g_p is dimensionless and unity. [Curves 4 and 5 from Rushton, Costich, and Everett, *Chem. Eng. Prog.*, **46**, 395, 467 (1950), by permission; curves 2 and 3 from Bates, Fondy, and Corpstein, *Ind. Eng. Chem. Process Des. Dev.*, **2**, 310 (1963), by permission of the copyright owner, the American Chemical Society.]

for operation of the respective impellers in unbaffled tanks when the Reynolds number is 300 or less. When N_{Re} is greater than 300, however, the power consumption is lower in an unbaffled vessel than indicated in Fig. 18-17. For example, for a six-blade disc turbine with $D_T/D_a = 3$ and $D_a/W_i = 5$, $N_p = 1.2$ when $N_{Re} = 10^4$. This is only about one-fifth of the value of N_p when baffles are present.

Additional power data for other impeller types such as anchors, curved-blade turbines, and paddles in baffled and unbaffled vessels are available in the following references: Holland and Chapman, op. cit., chaps. 2, 4, Reinhold, New York, 1966; and Bates, Fondy, and Fenic, in Uhl and Gray, op. cit., vol. 1, chap. 3.

Power consumption for impellers in pseudoplastic, Bingham plastic, and dilatant non-newtonian fluids may be calculated by using the correlating lines of Fig. 18-17 if viscosity is obtained from viscosity-shear rate curves as described here. For a pseudoplastic fluid, viscosity decreases as shear rate increases. A Bingham plastic is similar to a pseudoplastic fluid but requires that a minimum shear stress be exceeded for any flow to occur. For a dilatant fluid, viscosity increases as shear rate increases.

The appropriate shear rate to use in calculating viscosity is given by one of the following equations when a propeller or a turbine is used (Bates et al., in Uhl and Gray, op. cit., vol. 1, p. 149):

For dilatant liquids,

$$\dot{\gamma} = 13N \left(\frac{D_a}{D_T} \right)^{0.5} \quad (18-6)$$

For pseudoplastic and Bingham plastic fluids,

$$\dot{\gamma} = 10N \quad (18-7)$$

where $\dot{\gamma}$ = average shear rate, s^{-1} .

The shear rate calculated from impeller rotational speed is used to identify a viscosity from a plot of viscosity versus shear rate determined with a capillary or rotational viscometer. Next N_{Re} is calculated, and N_p is read from a plot like Fig. 18-17.

DESIGN OF AGITATION EQUIPMENT

Selection of Equipment The principal factors which influence mixing-equipment choice are (1) the process requirements, (2) the flow properties of the process fluids, (3) equipment costs, and (4) construction materials required.

Ideally, the equipment chosen should be that of the lowest total cost which meets all process requirements. The total cost includes depreciation on investment, operating cost such as power, and maintenance costs. Rarely is any more than a superficial evaluation based on this principle justified, however, because the cost of such an evaluation often exceeds the potential savings that can be realized. Usually optimization is based on experience with similar mixing operations. Often the process requirements can be matched with those of a similar operation, but sometimes tests are necessary to identify a satisfactory design and to find the minimum rotational speed and power.

There are no satisfactory specific guides for selecting mixing equipment because the ranges of application of the various types of equipment overlap and the effects of flow properties on process performance have not been adequately defined. Nevertheless, what is frequently done in selecting equipment is described in the following paragraphs.

Top-Entering Impellers For vessels less than 1.8 m (6 ft) in diameter, a clamp- or flange-mounted, angular, off-center fluidfoil impeller with no baffles should be the initial choice for meeting a wide range of process requirements (Fig. 18-14). The vessel straight-side-height-to-diameter ratio should be 0.75 to 1.5, and the volume of stirred liquid should not exceed 4 m³ (about 1000 gal).

For suspension of free-settling particles, circulation of pseudoplastic slurries, and heat transfer or mixing of miscible liquids to obtain uniformity, a speed of 350 or 420 r/min should be stipulated. For dispersion of dry particles in liquids or for rapid initial mixing of liquid reactants in a vessel, an 1150- or 1750- r/min propeller should be used at a distance $D_T/4$ above the vessel bottom. A second propeller can be added to the shaft at a depth D_a below the liquid surface if the submergence of floating liquids or particulate solids is otherwise inadequate. Such propeller mixers are readily available up to 2.2 kW (3 hp) for off-center sloped-shaft mounting.

Propeller size, pitch, and rotational speed may be selected by model tests, by experience with similar operations, or, in a few cases, by published correlations of performance data such as mixing time or heat transfer. The propeller diameter and motor power should be the minimum that meets process requirements.

If agitation is required for a vessel less than 1.8 m (6 ft) in diameter and the same operations will be scaled up to a larger vessel ultimately, the equipment type should be the same as that expected in the larger vessel.

Axial-Flow Fluidfoil Impellers For vessel volumes of 4 to 200 m³ (1000 to 50,000 gal), a turbine mixer mounted coaxially within the vessel with four or more baffles should be the initial choice. Here also the vessel straight-side-height-to-diameter ratio should be 0.75 to 1.5. Four vertical baffles should be fastened perpendicularly to the vessel wall with a gap between baffle and wall equal to $D_T/24$ and a radial baffle width equal to $D_T/12$.

For suspension of rapidly settling particles, the impeller turbine diameter should be $D_T/3$ to $D_T/2$. A clearance of less than one-seventh of the fluid depth in the vessel should be used between the lower edge of the turbine blade tips and the vessel bottom. As the viscosity of a suspension increases, the impeller diameter should be increased. This diameter may be increased to 0.6 D_T and a second impeller added to avoid stagnant regions in pseudoplastic slurries. Moving the baffles halfway between the impeller periphery and the vessel wall will also help avoid stagnant fluid near the baffles.

As has been shown, power consumption is decreased and turbine discharge rate is increased as impeller diameter is increased at constant torque (in the completely turbulent regime). This means that for a stipulated discharge rate, more efficient operation is obtained (lower power and torque) with a relatively large impeller operating at a relatively low speed ($N \propto D_a^{-3}$). Conversely, if power is held constant, decreasing impeller diameter results in increasing peripheral velocity and decreasing torque. Thus at a stipulated power level the rapid, effi-

cient initial mixing of reactants identified with high peripheral velocity can be achieved by a relatively small impeller operating at a relatively high speed ($N \propto D_a^{-3/3}$).

For circulation and mixing to obtain uniformity, the impeller should be located at one-third of the liquid depth above the vessel bottom unless rapidly settling material or a need to stir a nearly empty vessel requires a lower impeller location.

Side-Entering Impellers For vessels greater than 4 m³ (1000 gal), a side-entering propeller agitator (Fig. 18-9) may be more economical than a top-mounted impeller on a centered vertical shaft. For vessels greater than 38 m³ (10,000 gal), the economic attractiveness of side-entering impellers increases. For vessels larger than 380 m³ (100,000 gal), units may be as large as 56 kW (75 hp), and two or even three may be installed in one tank. For the suspension of slow-settling particles or the maintenance of uniformity in a viscous slurry of small particles, the diameter and rotational speed of a side-entering agitator must be selected on the basis of model tests or experience with similar operations.

When abrasive solid particles must be suspended, maintenance costs for the submerged shaft seal of a side-entering propeller may become high enough to make this type of mixer an uneconomical choice.

Jet Mixers Continuous recycle of the contents of a tank through an external pump so arranged that the pump discharge stream appropriately reenters the vessel can result in a flow pattern in the tank which will produce a slow mixing action [Fossett, *Trans. Inst. Chem. Eng.*, **29**, 322 (1951)].

Large Tanks Most large vessels (over 4 m³) require a heavy-duty drive. About two-thirds of the mixing requirements industrially involve flow, circulation, and other types of pumping capacity requirements, including such applications as blending and solid suspension. There often is no requirement for any marked level of shear rate, so the use of the fluidfoil impellers is most common. If additional shear rate is required over what can be provided by the fluidfoil impeller, the axial-flow turbine (Fig. 18-3) is often used, and if extremely high shear rates are required, the flat-blade turbine (Rushton turbine) (Fig. 18-4) is required. For still higher shear rates, there is an entire variety of high-shear-rate impellers, typified by that shown in Fig. 18-10 that are used.

The fluidfoil impellers in large tanks require only two baffles, but three are usually used to provide better flow pattern asymmetry. These fluidfoil impellers provide a true axial flow pattern, almost as though there was a draft tube around the impeller. Two or three or more impellers are used if tanks with high D/T ratios are involved. The fluidfoil impellers do not vortex vigorously even at relatively low coverage so that if gases or solids are to be incorporated at the surface, the axial-flow turbine is often required and can be used in combination with the fluidfoil impellers also on the same shaft.

BLENDING

If the blending process is between two or more fluids with relatively low viscosity such that the blending is not affected by fluid shear rates, then the difference in blend time and circulation between small and large tanks is the only factor involved. However, if the blending involves wide disparities in the density of viscosity and surface tension between the various phases, then a certain level of shear rate may be required before blending can proceed to the required degree of uniformity.

The role of viscosity is a major factor in going from the turbulent regime, through the transition region, into the viscous regime and the change in the role of energy dissipation discussed previously. The role of non-newtonian viscosities comes into the picture very strongly since that tends to markedly change the type of influence of impellers and determines the appropriate geometry that is involved.

There is the possibility of misinterpretation of the difference between circulation time and blend time. Circulation time is primarily a function of the pumping capacity of the impeller. For axial-flow impellers, a convenient parameter, but not particularly physically accurate, is to divide the pumping capacity of the impeller by the cross-sectional area of the tank to give a superficial liquid velocity.

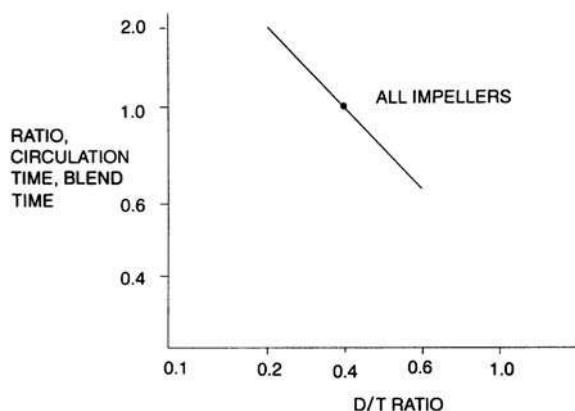


FIG. 18-18 Effect of D/T ratio on any impeller on the circulation time and the blend time.

This is sometimes used by using the total volume of flow from the impeller including entrainment of the tank to obtain a superficial liquid velocity.

As the flow from an impeller is increased from a given power level, there will be a higher fluid velocity and therefore a shorter circulation time. This holds true when dealing with any given impeller. This is shown in Fig. 18-18, which shows that circulation time versus D/T decreases. A major consideration is when increasing D/T becomes too large and actually causes the curve to reverse. This occurs somewhere around $0.45, \pm 0.05$, so that using impellers of D/T ratios of 0.6 to 0.8 is often counterproductive for circulation time. They may be useful for the blending or motion of pseudoplastic fluids.

When comparing different impeller types, an entirely different phenomenon is important. In terms of circulation time, the phenomena shown in Figs. 18-18 and 18-19 still apply with the different impellers shown in Fig. 18-5. When it comes to blending another factor enters the picture. When particles A and B meet each other as a result of shear rates, there has to be sufficient shear stress to cause A and B to blend, react, or otherwise participate in the process.

It turns out that in low-viscosity blending the actual result does depend upon the measuring technique used to measure blend time. Two common techniques, which do not exhaust the possibilities in reported studies, are to use an acid-base indicator and inject an acid or base into the system that will result in a color change. One can also put a dye into the tank and measure the time for color to arrive at uniformity. Another system is to put in a conductivity probe and inject a salt

or other electrolyte into the system. With any given impeller type at constant power, the circulation time will increase with the D/T ratio of the impeller. Figure 18-18 shows that both circulation time and blend time decrease as D/T increases. The same is true for impeller speed. As impeller speed is increased with any impeller, blend time and circulation time are decreased (Fig. 18-19).

However, when comparing different impeller types at the same power level, it turns out that impellers that have a higher pumping capacity will give decreased circulation time, but all the impellers, regardless of their pumping efficiency, give the same blend time at the same power level and same diameter. This means that circulation time must be combined with shear rate to carry out a blending experiment which involves chemical reactions or interparticle mixing (Fig. 18-20).

For other situations in low-viscosity blending, the fluid in tanks may become stratified. There are few studies on that situation, but Oldshue (op. cit.) indicates the relationship between some of the variables. The important difference is that blend time is inversely proportional to power, not impeller flow, so that the exponents are quite different for a stratified tank. This situation occurs more frequently in the petroleum industry, where large petroleum storage tanks become stratified either by filling techniques or by temperature fluctuations.

There is a lot of common usage of the terms *blend time*, *mixing time*, and *circulation time*. There are differences in concept and interpretation of these different "times." For any given experiment, one must pick a definition of *blend time* to be used. As an example, if one is measuring the fluctuation of concentration after an addition of material to the tank, then one can pick an arbitrary definition of blending such as reducing the fluctuations below a certain level. This often is chosen as a fluctuation equal to 5% of the original fluctuation when the feed material is added. This obviously is a function of the size of the probe used to measure these fluctuations, which often is on the order of 500 to 1000 μm .

At the micro-scale level, there really is no way to measure concentration fluctuations. Resort must be made to other qualitative interpretation of results for either a process or a chemical reaction study.

High-Viscosity Systems All axial-flow impellers become radial flow as Reynolds numbers approach the viscous region. Blending in the transition and low-viscosity system is largely a measure of fluid motion throughout the tank. For close-clearance impellers, the anchor and helical impellers provide blending by having an effective action at the tank wall, which is particularly suitable for pseudoplastic fluids.

Figure 18-21 gives some data on the circulation time of the helical impeller. It has been observed that it takes about three circulation times to get one blend time being the visual uniformity of a dye added to the material. This is a macro-scale blending definition.

Axial-flow turbines are often used in blending pseudoplastic materials, and they are often used at relatively large D/T ratios, from 0.5 to 0.7, to adequately provide shear rate in the majority of the batch particularly in pseudoplastic material. These impellers develop a flow

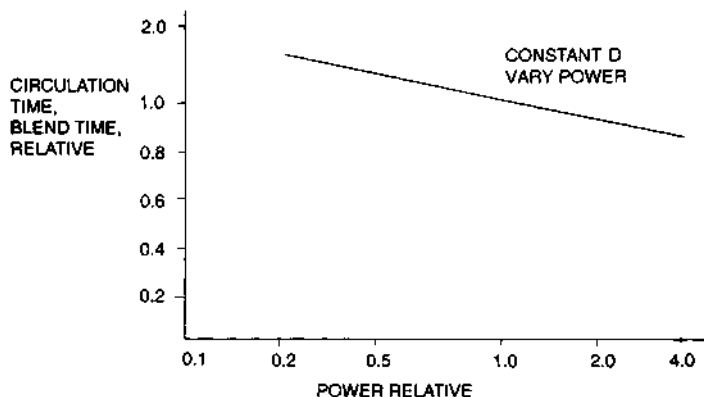


FIG. 18-19 Effect of impeller power for the same diameter on circulation time and blend time for a particular impeller.

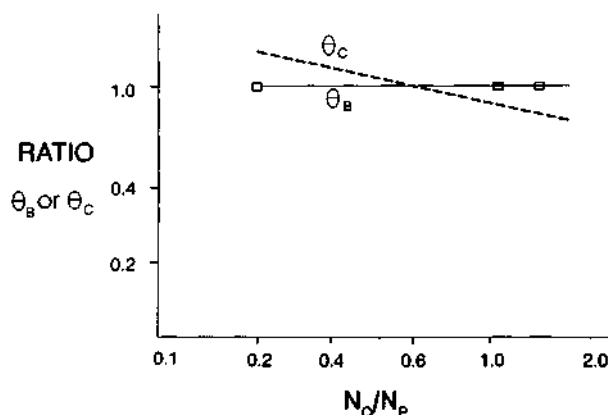


FIG. 18-20 At constant power and constant impeller diameter, three different impellers give the same blend time but different circulation times.

pattern which may or may not encompass an entire tank, and these areas of motion are sometimes referred to as *caverns*. Several papers describe the size of these caverns relative to various types of mixing phenomena. An effective procedure for the blending of pseudoplastic fluids is given in Oldshue (op. cit.).

Chemical Reactions Chemical reactions are influenced by the uniformity of concentration both at the feed point and in the rest of the tank and can be markedly affected by the change in overall blend time and circulation time as well as the micro-scale environment. It is possible to keep the ratio between the power per unit volume at the impeller and in the rest of the tank relatively similar on scale-up, but many details need to be considered when talking about the reaction conditions, particularly where they involve selectivity. This means that reactions can take different paths depending upon chemistry and fluid mechanics, which is a major consideration in what should be examined. The method of introducing the reagent stream can be projected in several different ways depending upon the geometry of the impeller and feed system.

Chemical reactions normally occur in the micro-scale range. In turbulent flow, almost all of the power dissipation occurs eventually in the micro-scale regime because that is the only place where the scale of the fluid fluctuations is small enough that viscous shear stress exists. At approximately 100 μm , the fluid does not know what type of impeller is used to generate the power; continuing down to 10 μm and, even further, to chemical reactions, the actual impeller type is not a major vari-

able as long as the proper macro-scale regime has been provided throughout the entire tank. The intensity of the mixing environment in the micro-scale regime can be related to a series of variables in an increasing order of complexity. Since all of the power is ultimately dissipated in the micro-scale regime, the power per unit volume throughout the tank is one measure of the overall measure of micro-scale mixing and the power dissipation at individual volumes in the tank is another way of expressing the influence. In general, the power per unit volume dissipated around an impeller zone can be 100 times higher than the power dissipated throughout the remainder of the tank.

The next level of complexity is to look at the rms velocity fluctuation, which is typically 50 percent of the mean velocity around the impeller zone and about 5 percent of the mean velocity in the rest of the vessel. This means that the feed introduction point for either a single reactant or several reactants can be of extreme importance. It seems that the selectivity of competing or consecutive chemical reactions can be a function of the rms velocity fluctuations in the feed point if the chemical reactants remain constant and involve an appropriate relationship to the time between the rms velocity fluctuations. There are three common ways of introducing reagents into a mixing vessel. One is to let them drip on the surface. The second is to use some type of introduction pipe to bring the material into various parts of the vessel. The third is to purposely bring them in and around the impeller zone. Generally, all three methods have to be tried before determining the effect of feed location.

Since chemical reactions are on a scale much below 1 μm , and it appears that the Komolgoroff scale of isotropic turbulence turns out to be somewhere between 10 and 30 μm , other mechanisms must play a role in getting materials in and out of reaction zones and reactants in and out of those zones. One cannot really assign a shear rate magnitude to the area around a micro-scale zone, and it is primarily an environment that particles and reactants witness in this area.

The next level of complexity looks at the kinetic energy of turbulence. There are several models that are used to study the fluid mechanics, such as the K ϵ model. One can also put the velocity measurements through a spectrum analyzer to look at the energy at various wave numbers.

In the viscous regime, chemical reactants become associated with each other through viscous shear stresses. These shear stresses exist at all scales (macro to micro) and until the power is dissipated continuously through the entire spectrum. This gives a different relationship for power dissipation than in the case of turbulent flow.

SOLID-LIQUID SYSTEMS

The most-used technique to study solid suspension, as documented in hundreds of papers in the literature, is called the *speed for just suspension*, N_{js} . The original work was done in 1958 by Zwietering and this is still the most extensive range of variables, although other investigators have added to it considerably.

This particular technique is suitable only for laboratory investigation using tanks that are transparent and well illuminated. It does not lend itself to evaluation of the opaque tanks, nor is it used in any study of large-scale tanks in the field. It is a very minimal requirement for uniformity, and definitions suggested earlier are recommended for use in industrial design.

Some Observations on the Use of N_{js} With D/T ratios of less than 0.4, uniformity throughout the rest of the tank is minimal. In D/T ratios greater than 0.4, the rest of the tank has a very vigorous fluid motion with a marked approach to complete uniformity before N_{js} is reached.

Much of the variation in N_{js} can be reduced by using P_{js} , which is the power in the just-suspended state. This also gives a better feel for the comparison of various impellers based on the energy requirement rather than speed, which has no economic relevance.

The overall superficial fluid velocity, mentioned earlier, should be proportional to the settling velocity of the solids if that were the main mechanism for solid suspension. If this were the case, the requirement for power if the settling velocity were doubled should be eight times. Experimentally, it is found that the increase in power is more nearly four times, so that some effect of the shear rate in macro-scale turbulence is effective in providing uplift and motion in the system.

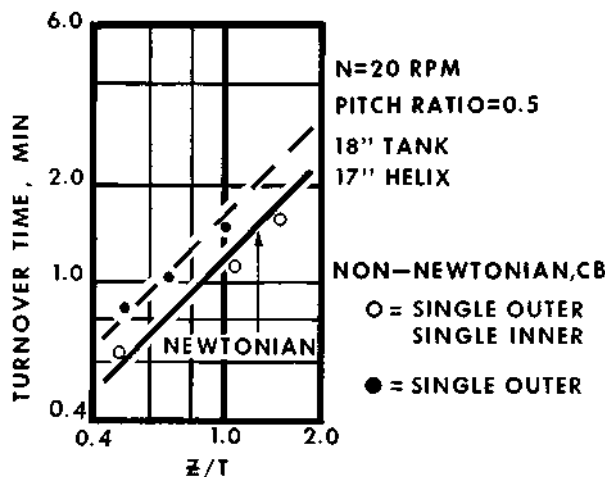


FIG. 18-21 Effect of impeller speed on circulation time for a helical impeller in the Reynolds number arranged less than 10.

Picking up the solids at the bottom of the tank depends upon the eddies and velocity fluctuations in the lower part of the tank and is a different criterion from the flow pattern required to keep particles suspended and moving in various velocity patterns throughout the remainder of the vessel. This leads to the variables in the design equation and a relationship that is quite different when these same variables are studied in relation to complete uniformity throughout the mixing vessel.

Another concern is the effect of multiple particle sizes. In general, the presence of fine particles will affect the requirements of suspension of larger particles. The fine particles act largely as a potential viscosity-increasing agent and give a similar result to what would happen if the viscosity of the continuous phase were increased.

Another phenomenon is the increase in power required with percent solids, which makes a dramatic change at approximately 40 percent by volume, and then dramatically changes again as we approach the ultimate weight percent of settled solids. This phenomenon is covered by Oldshue (op. cit.), who describes conditions required for mixing slurries in the 80 to 100 percent range of the ultimate weight percent of settled solids.

Solids suspension in general is not usually affected by blend time or shear-rate changes in the relatively low to medium solids concentration in the range from 0 to 40 percent by weight. However, as solids become more concentrated, the effect of solids concentration on power required gives a change in criterion from the settling velocity of the individual particles in the mixture to the apparent viscosity of the more concentrated slurry. This means that we enter into an area where the blending of non-newtonian fluid regions affects the shear rates and plays a marked role.

The suspension of a single solid particle should depend primarily on the upward velocity at a given point and also should be affected by the uniformity of this velocity profile across the entire tank cross section. There are upward velocities in the tank and there also must be corresponding downward velocities.

In addition to the effect of the upward velocity on a settling particle, there is also the random motion of the micro-scale environment, which does not affect large particles very much but is a major factor in the concentration and uniformity of particles in the transition and micro-scale size range.

Using a draft tube in the tank for solids suspension introduces another, different set of variables. There are other relationships that are very much affected by scale-up in this type of process, as shown in Fig. 18-22. Different scale-up problems exist whether the impeller is pumping up or down within the draft tube.

Solid Dispersion If the process involves the dispersion of solids in a liquid, then we may either be involved with breaking up agglomerates or possibly physically breaking or shattering particles that have a low cohe-

sive force between their components. Normally, we do not think of breaking up ionic bonds with the shear rates available in mixing machinery.

If we know the shear stress required to break up a particle, we can then determine the shear rate required from the machinery by various viscosities with the equation:

$$\text{Shear stress} = \text{viscosity} (\text{shear rate})$$

The shear rate available from various types of mixing and dispersion devices is known approximately and also the range of viscosities in which they can operate. This makes the selection of the mixing equipment subject to calculation of the shear stress required for the viscosity to be used.

In the equation referred to above, it is assumed that there is 100 percent transmission of the shear rate in the shear stress. However, with the slurry viscosity determined essentially by the properties of the slurry, at high concentrations of slurries there is a slippage factor. Internal motion of particles in the fluids over and around each other can reduce the effective transmission of viscosity efficiencies from 100 percent to as low as 30 percent.

Animal cells in biotechnology do not normally have tough skins like those of fungal cells and they are very sensitive to mixing effects. Many approaches have been and are being tried to minimize the effect of increased shear rates on scale-up. These include encapsulating the organism in or on microparticles and/or conditioning cells selectively to shear rates. In addition, traditional fermentation processes have maximum shear-rate requirements in which cells become progressively more and more damaged until they become motile.

Solid-Liquid Mass Transfer There is potentially a major effect of both shear rate and circulation time in these processes. The solids can either be fragile or rugged. We are looking at the slip velocity of the particle and also whether we can break up agglomerates of particles which may enhance the mass transfer. When the particles become small enough, they tend to follow the flow pattern, so the slip velocity necessary to affect the mass transfer becomes less and less available.

What this shows is that, from the definition of off-bottom motion to complete uniformity, the effect of mixer power is much less than from going to on-bottom motion to off-bottom suspension. The initial increase in power causes more and more solids to be in active communication with the liquid and has a much greater mass-transfer rate than that occurring above the power level for off-bottom suspension, in which slip velocity between the particles of fluid is the major contributor (Fig. 18-23).

Since there may well be chemical or biological reactions happening on or in the solid phase, depending upon the size of the process participants, macro- or micro-scale effects may or may not be appropriate to consider.

In the case of living organisms, their access to dissolved oxygen throughout the tank is of great concern. Large tanks in the fermentation industry often have a Z/T ratio of 2:1 to 4:1; thus, top-to-bottom blending can be a major factor. Some biological particles are facultative and can adapt and reestablish their metabolisms at different dissolved-oxygen levels. Other organisms are irreversibly destroyed by sufficient exposure to low dissolved-oxygen levels.

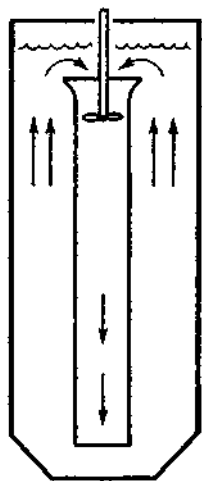


FIG. 18-22 Typical draft tube circulator, shown here for down-pumping mode for the impeller in the draft tube.

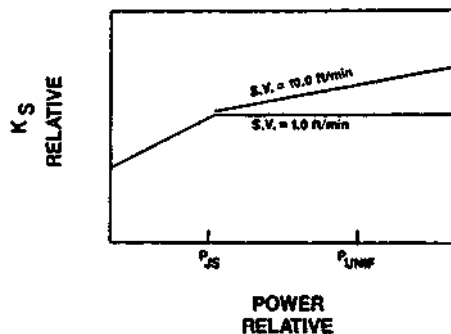


FIG. 18-23 Relative change in solid-liquid mass-transfer ratio with three different suspension levels, i.e., on-bottom motion, off-bottom motion, and complete uniformity.

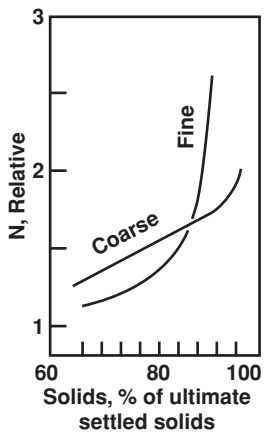


FIG. 18-24 Effect of percent solids on speed required for complete motion throughout the tank on two different grind sizes.

Leaching and Extraction of Mineral Values from High Concentration of Solids A uranium plant had 10 large slurry tanks for leaching and extraction (approximately 14 m in diameter and 14 m high). They had about 14,000-m³ capacity.

In a study designed to modify the leaching operation, it was desired to look at two different grind sizes of ore, one labeled fine grind and the other labeled coarse grind. Also, the effect of various mixer designs and power levels on the extraction efficiency to arrive at the overall economic optimum was examined. Figure 18-24 shows the results of a pilot study in which the impeller speed for a given impeller and tank geometry was measured for complete overall motion throughout the slurry for both the fine and coarse grinds at various weight percent solids. As can be seen in the figure, the fine material required lower horsepower at low weight percent solids while the coarse grind required less horsepower up near the ultimate settled solids weight percentage.

The interpretation is that at lower percent solids, the viscosity of the fine grind aided suspension whereas at higher percent solids, the higher viscosity of the fine material was detrimental to fluid mixing.

A mixing viscosimeter was used to measure the viscosity of the slurry. Figure 18-25 shows the viscosity of the fine and coarse slurries.

By combining the data from Figs. 18-24 and 18-25 into Fig. 18-26, it is seen that there is a correlation between the impeller speed required and the viscosity of the slurry regardless of whether the

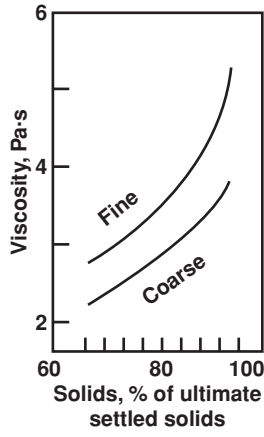


FIG. 18-25 Viscosity of fine grind and coarse grind at various weight percent solids.

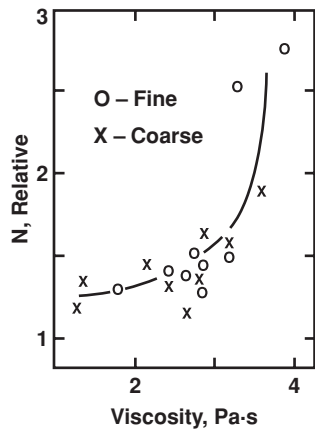


FIG. 18-26 Correlation of impeller speed vs. viscosity of the pulp including both fine and coarse grind experimental data points.

material was finely or coarsely ground. This illustrates that viscosity is a key parameter in the process design for solid-liquid slurries.

The overall process economics examined the extraction rate as a function of power, residence time, and grind size. The full-scale design possibilities were represented in the form of Table 18-2, which were accompanied by other charts that gave different heights of suspension in the tank for the three different particle size fractions: fine, medium, and coarse. These various combinations of power levels also gave various blending efficiencies and had different values of the effective residence time used in a system.

By calculating the residence times of the various solids in the tank and relating them to their corresponding extraction curves, the total uranium extraction for the entire train of mixers was estimated. The cost of the various mixer options, the production efficiency net result, and the cost of the installation and tank design could be combined to yield the economic optimum for the plant.

GAS-LIQUID SYSTEMS

Gas-Liquid Dispersion This involves physical dispersion of gas bubbles by the impeller, and the effect of gas flow on the impeller.

The observation of the physical appearance of a tank undergoing gas-liquid mass transfer can be helpful but is not a substitute for mass-transfer data on the actual process. The mixing vessel can have four regimes of visual comparisons between gas bubbles and flow patterns. A helpful parameter is the ratio between the power given up by the gas phase and the power introduced by the mixing impeller. In general, if the power in the gas stream (calculated as the expansion energy from the gas expanding from the sparging area to the top of the tank, shown in Fig. 18-27) is greater, there will be considerable blurring and entrainment of liquid drops by a very violent explosion of gas bubbles at the surface. If the power level is more than the expanding gas energy, then the surface action will normally be very coalescent and uniform by comparison, and the gas will be reasonably well distributed throughout the remainder of the tank. With power levels up to 10 to 100 times the

TABLE 18-2 Four Different Selections of Mixers with Different Mixing Characteristics on 14,000 ft³ of Leach Tanks

Proposal for Revised Installation			
Process factor	Relative torque	Motor kW	D/T (dual)
1.2	1.0	300	0.5
1.0	1.0	250	0.5
0.8	0.9	200	0.5
0.7	1.0	150	0.6

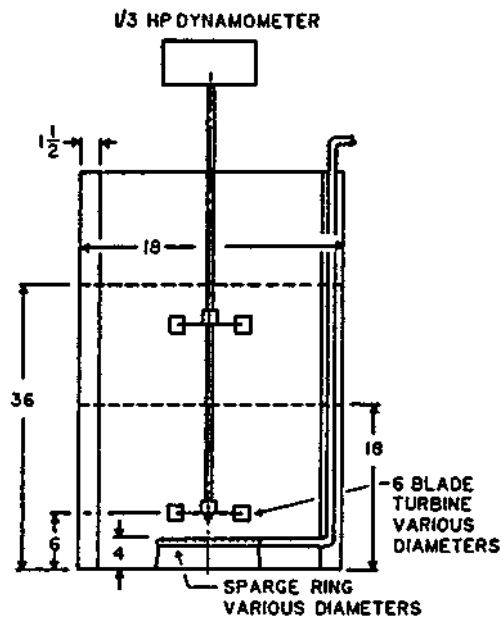


FIG. 18-27 Typical arrangement of Rushton radial-flow R100 flat-blade turbine with typical sparge ring for gas-liquid mass transfer.

gas energy, the impeller will cause a more uniform and vigorous dispersion of the gas bubbles and smaller gas bubbles in the vessel.

In the 1960s and before, most gas-liquid operations were conducted using flat-blade turbines as shown in Fig. 18-4. These impellers required input of approximately three times the energy in the gas stream before they completely control the flow pattern. This was usually the case, and the mass-transfer characteristics were comparable to what would be expected. One disadvantage of the radial-flow impeller is that it is a very poor blending device so blend time is very long compared to that in pilot-scale experiments and compared to the fluidfoil impeller types often used currently. Using curvature of the blades to modify the tendency of gas bubbles to streamline the back of the flat-blade turbine gives a different characteristic to the power drawn by the impeller at a given gas rate compared to no gas rate, but it seems to give quite similar mass transfer at power levels similar to those of the flat-blade design. In order to improve the blending and solid-suspension characteristics, fluidfoil impellers (typified by the A315, Fig. 18-28) have been introduced in recent years and they have many of the advantages and some of the disadvantages of the flat-blade turbine. These impellers typically have a very high solidity ratio, on the order of 0.85 or more, and produce a strong axial downflow at low gas rate. As the gas rate increases, the flow pattern becomes more radial due to the upflow of the gas counteracting the downward flow of the impeller.

Mass-transfer characteristics on large-scale equipment seem to be quite similar, but the fluidfoil impellers tend to release a larger-diameter bubble than is common with the radial-flow turbines. The blend time is one-half or one-third as long, and solid-suspension characteristics are better so that there have been notable improved process results with these impellers. This is particularly true if the process requires better blending and there is solid suspension. If this is not the case, the results from these impellers can be negative compared to radial-flow turbines.

It is very difficult to test these impellers on a small scale, since they provide better blending on a pilot scale where blending is already very effective compared to the large scale. Caution is recommended if it is desirable to study these impellers in pilot-scale equipment.

Gas-Liquid Mass Transfer Gas-liquid mass transfer normally is correlated by means of the mass-transfer coefficient $K_g a$ versus power level at various superficial gas velocities. The superficial gas velocity is



FIG. 18-28 An impeller designed for gas-liquid dispersion and mass transfer of the fluidfoil type, i.e., A315.

the volume of gas at the average temperature and pressure at the mid-point in the tank divided by the area of the vessel. In order to obtain the partial-pressure driving force, an assumption must be made of the partial pressure in equilibrium with the concentration of gas in the liquid. Many times this must be assumed, but if Fig. 18-29 is obtained in the pilot plant and the same assumption principle is used in evaluating the mixer in the full-scale tank, the error from the assumption is limited.

In the plant-size unit, Fig. 18-29 must be translated into a mass-transfer-rate curve for the particular tank volume and operating condition selected. Every time a new physical condition is selected, a different curve similar to that of Fig. 18-30 is obtained.

Typical exponents on the effect of power and gas rate on $K_g a$ tend to be around 0.5 for each variable, ± 0.1 .

Viscosity markedly changes the picture and, usually, increasing viscosity lowers the mass-transfer coefficient. For the common application of waste treating and for some of the published data on biological slurries, data for $k_L a$ (shown in Fig. 18-31) is obtained in the literature. For a completely new gas or liquid of a liquid slurry system, Fig. 18-29 must be obtained by an actual experiment.

Liquid-Gas-Solid Systems Many gas-liquid systems contain solids that may be the ultimate recipient of the liquid-gas-solid mass transfer entering into the process result. Examples are biological

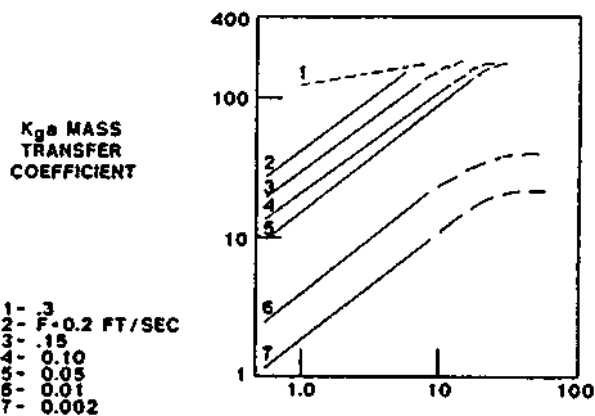


FIG. 18-29 Typical curve for mass transfer coefficient $K_g a$ as a function of mixer power and superficial gas velocity.

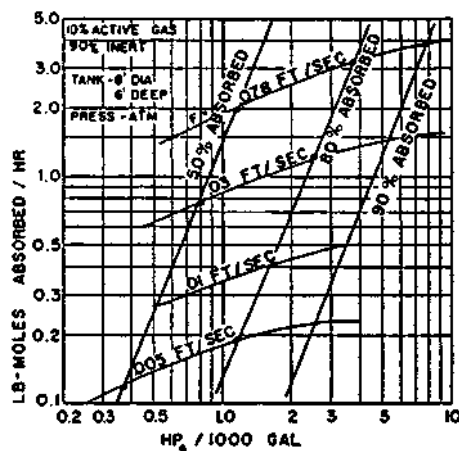


FIG. 18-30 Example of a specific chart to analyze the total mass-transfer rate in a particular tank under a process condition obtained from basic $K_L a$ data shown in Fig. 18-28.

processes in which the biological solids are the user of the mass transfer of the mixing-flow patterns, various types of slurries reactors in which the solids either are being reactive or there may be extraction or dissolving taking place, or there may be polymerization or precipitation of solids occurring.

Normally there must be a way of determining whether the mass-transfer rate with the solids is the key controlling parameter or the gas-liquid mass transfer rate.

In general, introduction of a gas stream to a fluid will increase the blend time because the gas-flow patterns are counterproductive to the typical mixer-flow patterns. In a similar vein, the introduction of a gas stream to a liquid-solid suspension will decrease the suspension uniformity because the gas-flow pattern is normally counterproductive to the mixer-flow pattern. Many times the power needed for the gas-liquid mass transfer is higher than the power needed for solid suspension, and the effect of the gas flow on the solid suspensions are of little concern. On the other hand, if power levels are relatively low and solid-suspension characteristics are critical—examples being the case of activated sludge reactors in the waste-treating field or biological solid reactors in the hydrometallurgical field—then the effect of the gas-flow pattern of the mixing system can be quite critical to the overall design.

Another common situation is batch hydrogenation, in which pure hydrogen is introduced to a relatively high pressure reactor and a decision must be made to recycle the unabsorbed gas stream from the top of the reactor or use a vortexing mode for an upper impeller to incorporate the gas from the surface.

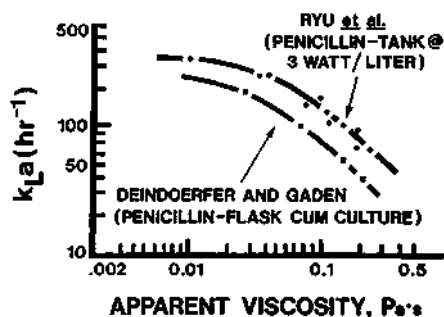


FIG. 18-31 Usually, the gas-liquid mass-transfer coefficient, $K_L a$, is reduced with increased viscosity. This shows the effect of increased concentration of microbial cells in a fermentation process.

Loop Reactors For some gas-liquid-solid processes, a recirculating loop can be an effective reactor. These involve a relatively high horsepower pumping system and various kinds of nozzles, baffles, and turbulence generators in the loop system. These have power levels anywhere from 1 to 10 times higher than the power level in a typical mixing reactor, and may allow the retention time to be less by a factor of 1 to 10.

LIQUID-LIQUID CONTACTING

Emulsions Almost every shear rate parameter affects liquid-liquid emulsion formation. Some of the effects are dependent upon whether the emulsion is both dispersing and coalescing in the tank, or whether there are sufficient stabilizers present to maintain the smallest droplet size produced for long periods of time. Blend time and the standard deviation of circulation times affect the length of time it takes for a particle to be exposed to the various levels of shear work and thus the time it takes to achieve the ultimate small particle size desired.

The prediction of drop sizes in liquid-liquid systems is difficult. Most of the studies have used very pure fluids as two of the immiscible liquids, and in industrial practice there almost always are other chemicals that are surface-active to some degree and make the prediction of absolute drop sizes very difficult. In addition, techniques to measure drop sizes in experimental studies have all types of experimental and interpretation variations and difficulties so that many of the equations and correlations in the literature give contradictory results under similar conditions. Experimental difficulties include dispersion and coalescence effects, difficulty of measuring actual drop size, the effect of visual or photographic studies on where in the tank you can make these observations, and the difficulty of using probes that measure bubble size or bubble area by light or other sample transmission techniques which are very sensitive to the concentration of the dispersed phase and often are used in very dilute solutions.

It is seldom possible to specify an initial mixer design requirement for an absolute bubble size prediction, particularly if coalescence and dispersion are involved. However, if data are available on the actual system, then many of these correlations could be used to predict relative changes in drop size conditions with changes in fluid properties or impeller variables.

STAGewise EQUIPMENT: MIXER-SETTLERS

Introduction Insoluble liquids may be brought into direct contact to cause transfer of dissolved substances, to allow transfer of heat, and to promote chemical reaction. This subsection concerns the design and selection of equipment used for conducting this type of liquid-liquid contact operation.

Objectives There are four principal purposes of operations involving the direct contact of immiscible liquids. The purpose of a particular contact operation may involve any one or any combination of the following objectives:

1. *Separation of components in solution.* This includes the ordinary objectives of liquid extraction, in which the constituents of a solution are separated by causing their unequal distribution between two insoluble liquids, the washing of a liquid with another to remove small amounts of a dissolved impurity, and the like. The theoretical principles governing the phase relationships, material balances, and number of ideal stages or transfer units required to bring about the desired changes are to be found in Sec. 15. Design of equipment is based on the quantities of liquids and the efficiency and operating characteristics of the type of equipment selected.

2. *Chemical reaction.* The reactants may be the liquids themselves, or they may be dissolved in the insoluble liquids. The kinetics of this type of reaction are treated in Sec. 4.

3. *Cooling or heating a liquid by direct contact with another.* Although liquid-liquid-contact operations have not been used widely for heat transfer alone, this technique is one of increasing interest. Applications also include cases in which chemical reaction or liquid extraction occurs simultaneously.

4. *Creating permanent emulsions.* The objective is to disperse one liquid within another in such finely divided form that separation

by settling either does not occur or occurs extremely slowly. The purpose is to prepare the emulsion. Neither extraction nor chemical reaction between the liquids is ordinarily sought.

Liquid-liquid contacting equipment may be generally classified into two categories: **stagewise** and **continuous (differential)** contact.

The function of a stage is to contact the liquids, allow equilibrium to be approached, and to make a mechanical separation of the liquids. The contacting and separating correspond to mixing the liquids, and settling the resulting dispersion; so these devices are usually called **mixer-settlers**. The operation may be carried out in batch fashion or with continuous flow. If batch, it is likely that the same vessel will serve for both mixing and settling, whereas if continuous, separate vessels are usually but not always used.

Mixer-Settler Equipment The equipment for extraction or chemical reaction may be classified as follows:

- I. Mixers
 - A. Flow or line mixers
 1. Mechanical agitation
 2. No mechanical agitation
 - B. Agitated vessels
 1. Mechanical agitation
 2. Gas agitation
- II. Settlers
 - A. Nonmechanical
 1. Gravity
 2. Centrifugal (cyclones)
 - B. Mechanical (centrifuges)
 - C. Settler auxiliaries
 1. Coalescers
 2. Separator membranes
 3. Electrostatic equipment

In principle, at least, any mixer may be coupled with any settler to provide the complete stage. There are several combinations which are especially popular. Continuously operated devices usually, but not always, place the mixing and settling functions in separate vessels. Batch-operated devices may use the same vessel alternately for the separate functions.

Flow or Line Mixers

Definition Flow or line mixers are devices through which the liquids to be contacted are passed, characterized principally by the very small time of contact for the liquids. They are used only for continuous operations or semibatch (in which one liquid flows continuously and the other is continuously recycled). If holding time is required for extraction or reaction, it must be provided by passing the mixed liquids through a vessel of the necessary volume. This may be a long pipe of large diameter, sometimes fitted with segmental baffles, but frequently the settler which follows the mixer serves. The energy for mixing and dispersing usually comes from pressure drop resulting from flow.

There are many types, and only the most important can be mentioned here. [See also Hunter, in Dunstan (ed.), *Science of Petroleum*, vol. 3, Oxford, New York, 1938, pp. 1779–1797.] They are used fairly extensively in treating petroleum distillates, in vegetable-oil refining, in extraction of phenol-bearing coke-oven liquors, in some metal extractions, and the like. Kalichevsky and Kobe (*Petroleum Refining with Chemicals*, Elsevier, New York, 1956) discuss detailed application in the refining of petroleum.

Jet Mixers These depend upon impingement of one liquid on the other to obtain a dispersion, and one of the liquids is pumped through a small nozzle or orifice into a flowing stream of the other. Both liquids are pumped. They can be used successfully only for liquids of low interfacial tension. See Fig. 18-32 and also Hunter and Nash [*Ind. Chem.*, **9**, 245, 263, 317 (1933)]. Treybal (*Liquid Extraction*, 2d ed., McGraw-Hill, New York, 1963) describes a more elaborate device. For a study of the extraction of antibiotics with jet mixers, see Anneskova and Boiko, *Med. Prom. SSSR*, **13**(5), 26 (1959). Insonation with ultrasound of a toluene-water mixture during methanol extraction with a simple jet mixer improves the rate of mass transfer, but the energy requirements for significant improvement are large [Woodle and Vilbrandt, *Am. Inst. Chem. Eng. J.*, **6**, 296 (1960)].

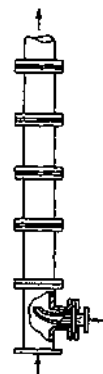


FIG. 18-32 Elbow jet mixer.

Injectors The flow of one liquid is induced by the flow of the other, with only the majority liquid being pumped at relatively high velocity. Figure 18-33 shows a typical device used in semibatch fashion for washing oil with a recirculated wash liquid. It is installed directly in the settling drum. See also Hampton (U.S. Patent 2,091,709, 1933), Sheldon (U.S. Patent 2,009,347, 1935), and Ng (U.S. Patent 2,665,975, 1954). Folsom [*Chem. Eng. Prog.*, **44**, 765 (1948)] gives a good review of basic principles. The most thorough study for extraction is provided by Kafarov and Zhukovskaya [*Zh. Prikl. Khim.*, **31**, 376 (1958)], who used very small injectors. With an injector measuring 73 mm from throat to exit, with 2.48-mm throat diameter, they extracted benzoic acid and acetic acid from water with carbon tetrachloride at the rate of 58 to 106 L/h, to obtain a stage efficiency $E = 0.8$ to 1.0. Data on flow characteristics are also given. Boyadzhiev and Elenkov [*Collect. Czech. Chem. Commun.*, **31**, 4072 (1966)] point out that the presence of surface-active agents exerts a profound influence on drop size in such devices.

Orifices and Mixing Nozzles Both liquids are pumped through constrictions in a pipe, the pressure drop of which is partly utilized to create the dispersion (see Fig. 18-34). Single nozzles or several in series may be used. For the orifice mixers, as many as 20 orifice plates

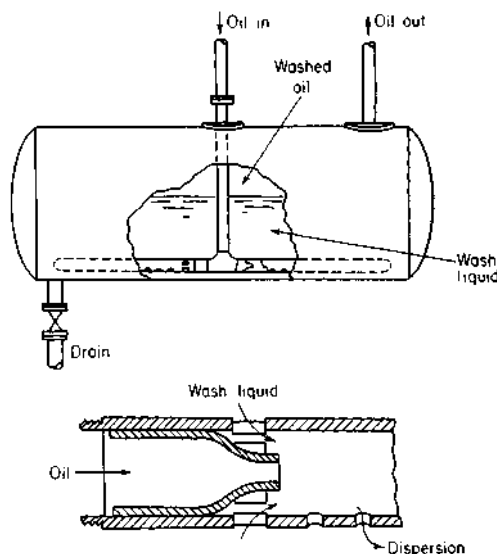


FIG. 18-33 Injector mixer. (Ayres, U.S. Patent 2,531,547, 1950.)

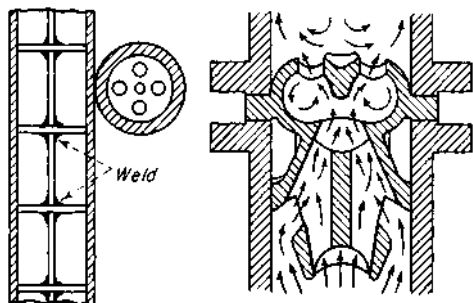


FIG. 18-34 Orifice mixer and nozzle mixer.

each with 13.8-kPa (2-lb/in²) pressure drop may be used in series [Morell and Bergman, *Chem. Metall. Eng.*, **35**, 211 (1928)]. In the Dualayer process for removal of mercaptans from gasoline, 258 m³/h (39,000 bbl/day) of oil and treating solution are contacted with 68.9-kPa (10-lb/in²) pressure drop per stage [Greek et al., *Ind. Eng. Chem.*, **49**, 1938 (1957)]. Holland et al. [*Am. Inst. Chem. Eng. J.*, **4**, 346 (1958); **6**, 615 (1960)] report on the interfacial area produced between two immiscible liquids entering a pipe (diameter 0.8 to 2.0 in) from an orifice, $\gamma_D = 0.02$ to 0.20, at flow rates of 0.23 to 4.1 m³/h (1 to 18 gal/min). At a distance 17.8 cm (7 in) downstream from the orifice,

$$a_{av} = \frac{0.179}{\sigma g_c} (C_O \Delta p)^{0.75} \left(\frac{\sigma \sqrt{g_c \rho_{av}}}{\mu_D} \right)^{0.158} \left[\left(\frac{d_i}{d_o} \right)^4 - 1 \right]^{0.117} \gamma_D^{0.878} \quad (18-8)$$

where a_{av} = interfacial surface, cm²/cm³; C_O = orifice coefficient, dimensionless; d_i = pipe diameter, in; d_o = orifice diameter, in; g_c = gravitational conversion factor, (32.2 lbm-ft)/(lbf-s²); Δp = pressure drop across orifice, lbf/ft²; μ_D = viscosity of dispersed phase, lbm/(ft-s); ρ_{av} = density of dispersed phase, lbm/ft³; and σ = interfacial tension, lbf/ft. See also Shiotsuka et al. [*Kagaku Kogaku*, **25**, 109 (1961)].

Valves Valves may be considered to be adjustable orifice mixers. In desalting crude petroleum by mixing with water, Hayes et al. [*Chem. Eng. Prog.*, **45**, 235 (1949)] used a globe-valve mixer operating at 110- to 221-kPa (16- to 32-lb/in²) pressure drop for mixing 66 m³/h (416 bbl/h) oil with 8 m³/h (50 bbl/h) water, with best results at the lowest value. Simkin and Olney [*Am. Inst. Chem. Eng. J.*, **2**, 545 (1956)] mixed kerosine and white oil with water, using 0.35- to 0.62-kPa (0.05- to 0.09-lb/in²) pressure drop across a 1-in gate valve, at 22-m³/h (10-gal/min) flow rate for optimum separating conditions in a cyclone, but higher pressure drops were required to give good extractor efficiencies.

Pumps Centrifugal pumps, in which the two liquids are fed to the suction side of the pump, have been used fairly extensively, and they offer the advantage of providing interstage pumping at the same time. They have been commonly used in the extraction of phenols from coke-oven liquors with light oil [Gollmar, *Ind. Eng. Chem.*, **39**, 596, 1947]; Carbone, *Sewage Ind. Wastes*, **22**, 200 (1950)], but the intense shearing action causes emulsions with this low-interfacial-tension system. Modern plants use other types of extractors. Pumps are useful in the extraction of slurries, as in the extraction of uranyl nitrate from acid-uranium-ore slurries [*Chem. Eng.*, **66**, 30 (Nov. 2, 1959)]. Shaw and Long [*Chem. Eng.*, **64**(11), 251 (1957)] obtain a stage efficiency of 100 percent ($E = 1.0$) in a uranium-ore-slurry extraction with an open impeller pump. In order to avoid emulsification difficulties in these extractions, it is necessary to maintain the organic phase continuous, if necessary by recycling a portion of the settled organic liquid to the mixer.

Agitated Line Mixer See Fig. 18-35. This device, which combines the features of orifice mixers and agitators, is used extensively in treating petroleum and vegetable oils. It is available in sizes to fit ½- to 10-in pipe. The device of Fig. 18-36, with two impellers in separate stages, is available in sizes to fit 4- to 20-in pipe.

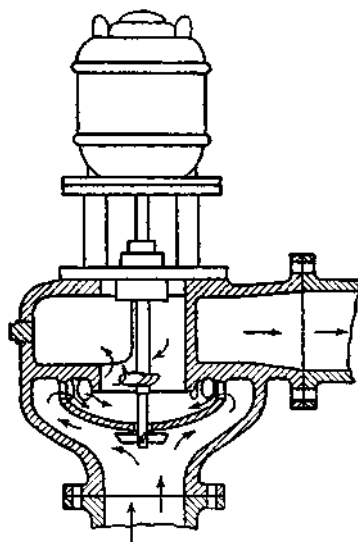


FIG. 18-35 Netco Corp. Flomix.

Packed Tubes Cocurrent flow of immiscible liquids through a packed tube produces a one-stage contact, characteristic of line mixers. For flow of isobutanol-water* through a 0.5-in diameter tube packed with 6 in of 3-mm glass beads, Leacock and Churchill [*Am. Inst. Chem. Eng. J.*, **7**, 196 (1961)] find

$$k_c a_{av} = c_1 L_c^{0.5} L_D \quad (18-9)$$

$$k_D a_{av} = c_2 L_c^{0.75} L_D^{0.75} \quad (18-10)$$

where $c_1 = 0.00178$ using SI units and 0.00032 using U.S. customary units; and $c_2 = 0.0037$ using SI units and 0.00057 using U.S. customary units. These indicate a stage efficiency approaching 100 percent. Organic-phase holdup and pressure drop for larger pipes similarly packed are also available [Rigg and Churchill, *ibid.*, **10**, 810 (1964)].

* Isobutanol dispersed: $L_D = 3500$ to 27,000; water continuous; $L_c = 6000$ to 32,000 in pounds-mass per hour-square foot (to convert to kilograms per second-square meter, multiply by 1.36×10^{-3}).

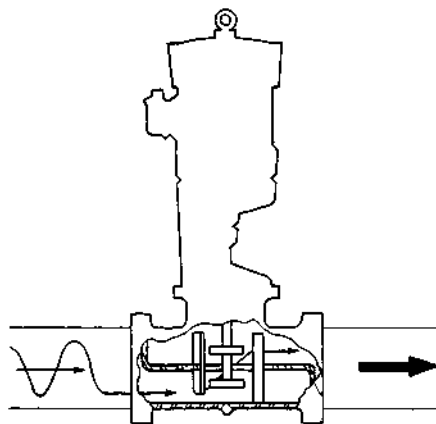


FIG. 18-36 Lightnin line blender. (Mixing Equipment Co., Inc., with permission.)

Pipe Lines The principal interest here will be for flow in which one liquid is dispersed in another as they flow cocurrently through a pipe (stratified flow produces too little interfacial area for use in liquid extraction or chemical reaction between liquids). Drop size of dispersed phase, if initially very fine at high concentrations, increases as the distance downstream increases, owing to coalescence [see Holland, loc. cit.; Ward and Knudsen, *Am. Inst. Chem. Eng. J.*, **13**, 356 (1967)]; or if initially large, decreases by breakup in regions of high shear [Sleicher, *ibid.*, **8**, 471 (1962); *Chem. Eng. Sci.*, **20**, 57 (1965)]. The maximum drop size is given by (Sleicher, loc. cit.)

$$\frac{d_{p,\max} \rho_c V^2}{\sigma_{gc}} \sqrt{\frac{\mu_c V}{\sigma_{gc}}} = C \left[1 + 0.7 \left(\frac{\mu_D V}{\sigma_{gc}} \right)^{0.7} \right] \quad (18-11)$$

where $C = 43$ ($d_t = 0.013$ m or 0.0417 ft) or 38 ($d_t = 0.038$ m or 0.125 ft), with $d_{p,av} = d_{p,\max}/4$ for high flow rates and $d_{p,\max}/13$ for low velocities.

Extensive measurements of the rate of mass transfer between *n*-butanol and water flowing in a 0.008-m (0.314-in) ID horizontal pipe are reported by Watkinson and Cavers [*Can. J. Chem. Eng.*, **45**, 258 (1967)] in a series of graphs not readily reproduced here. Length of a transfer unit for either phase is strongly dependent upon flow rate and passes through a pronounced maximum at an organic-water phase ratio of 0.5. In energy (pressure-drop) requirements and volume, the pipe line compared favorably with other types of extractors. Boyadzhiev and Elenkov [*Chem. Eng. Sci.*, **21**, 955 (1966)] concluded that, for the extraction of iodine between carbon tetrachloride and water in turbulent flow, drop coalescence and breakup did not influence the extraction rate. Yoshida et al. [*Coal Tar* (Japan), **8**, 107 (1956)] provide details of the treatment of crude benzene with sulfuric acid in a 1-in diameter pipe, $N_{Re} = 37,000$ to 50,000. Fernandes and Sharma [*Chem. Eng. Sci.*, **23**, 9 (1968)] used cocurrent flow downward of two liquids in a pipe, agitated with an upward current of air.

The pipe has also been used for the transfer of heat between two immiscible liquids in cocurrent flow. For hydrocarbon oil-water, the heat-transfer coefficient is given by

$$\frac{U a_{av} d_t^2}{v k_{to}} = \frac{\gamma_D N_{We,t}^{6/5}}{\frac{k_{to}}{0.415 k_{ic}} + \frac{k_{to}}{0.173 k_{id}}} \quad (18-12)$$

for $\gamma_D = 0$ to 0.2. Additional data for $\gamma_D = 0.4$ to 0.8 are also given. Data for stratified flow are given by Wilke et al. [*Chem. Eng. Prog.*, **59**, 69 (1963)] and Grover and Knudsen [*Chem. Eng. Prog.*, **51**, *Symp. Ser.* **17**, 71 (1955)].

Mixing in Agitated Vessels Agitated vessels may frequently be used for either batch or continuous service and for the latter may be sized to provide any holding time desired. They are useful for liquids of any viscosity up to 750 Pa·s (750,000 cP), although in contacting two liquids for reaction or extraction purposes viscosities in excess of 0.1 Pa·s (100 cP) are only rarely encountered.

Mechanical Agitation This type of agitation utilizes a rotating impeller immersed in the liquid to accomplish the mixing and dispersion. There are literally hundreds of devices using this principle, the major variations being found when chemical reactions are being carried out. The basic requirements regarding shape and arrangement of the vessel, type and arrangement of the impeller, and the like are essentially the same as those for dispersing finely divided solids in liquids, which are fully discussed in Sec. 18.

The following summary of operating characteristics of mechanically agitated vessels is confined to the data available on liquid-liquid contacting.

Phase Dispersed There is an ill-defined upper limit to the volume fraction of dispersed liquid which may be maintained in an agitated dispersion. For dispersions of organic liquids in water [Quinn and Sigloh, *Can. J. Chem. Eng.*, **41**, 15 (1963)],

$$\gamma_{D,\max} = \gamma' + \left(\frac{C}{N^3} \right) \quad (18-13)$$

where γ' is a constant, asymptotic value, and C is a constant, both depending in an unestablished manner upon the systems' physical properties and geometry. Thus, inversion of a dispersion may occur if the agitator speed is increased. With systems of low interfacial tension ($\sigma' = 2$ to 3 mN/m or 2 to 3 dyn/cm), γ_D as high as 0.8 can be maintained. Selker and Sleicher [*Can. J. Chem. Eng.*, **43**, 298 (1965)] and Yeh et al. [*Am. Inst. Chem. Eng. J.*, **10**, 260 (1964)] feel that the viscosity ratio of the liquids alone is important. Within the limits in which either phase can be dispersed, for *batch operation* of baffled vessels, that phase in which the impeller is immersed when at rest will normally be continuous [Rodger, Trice, and Rushton, *Chem. Eng. Prog.*, **52**, 515 (1956); Laity and Treybal, *Am. Inst. Chem. Eng. J.*, **3**, 176 (1957)]. With water dispersed, *dual emulsions* (continuous phase found as small droplets within larger drops of dispersed phase) are possible. In *continuous operation*, the vessel is first filled with the liquid to be continuous, and agitation is then begun, after which the liquid to be dispersed is introduced.

Uniformity of Mixing This refers to the gross uniformity throughout the vessel and not to the size of the droplets produced. For *unbaffled vessels, batch, with an air-liquid interface*, Miller and Mann [*Trans. Am. Inst. Chem. Eng.*, **40**, 709 (1944)] mixed water with several organic liquids, measuring uniformity of mixing by sampling the tank at various places, comparing the percentage of dispersed phase found with that in the tank as a whole. A power application of 200 to 400 W/m³ [(250 to 500 ft-lb)/(min-ft³)] gave maximum and nearly uniform performance for all. See also Nagata et al. [*Chem. Eng. (Japan)*, **15**, 59 (1951)].

For *baffled vessels operated continuously, no air-liquid interface*, flow upward, light liquid dispersed [Treybal, *Am. Inst. Chem. Eng. J.*, **4**, 202 (1958)], the average fraction of dispersed phase in the vessel $\gamma_{D,av}$ is less than the fraction of the dispersed liquid in the feed mixture, unless the impeller speed is above a certain critical value which depends upon vessel geometry and liquid properties. Thornton and Bouyatiotis [*Ind. Chem.*, **39**, 298 (1963); *Inst. Chem. Eng. Symp. Liquid Extraction*, Newcastle-upon-Tyne, April 1967] have presented correlations of data for a 17.8-cm (7-in) vessel, but these do not agree with observations on 15.2- and 30.5-cm (6- and 12-in) vessels in Treybal's laboratory. See also Kovalev and Kagan [*Zh. Prikl. Khim.*, **39**, 1513 (1966)] and Trambouze [*Chem. Eng. Sci.*, **14**, 161 (1961)]. Stemerding et al. [*Can. J. Chem. Eng.*, **43**, 153 (1965)] present data on a large mixing tank [15 m³ (530 ft³)] fitted with a marine-type propeller and a draft tube.

Drop Size and Interfacial Area The drops produced have a size range [Sullivan and Lindsey, *Ind. Eng. Chem. Fundam.*, **1**, 87 (1962); Sprow, *Chem. Eng. Sci.*, **22**, 435 (1967); and Chen and Middleman, *Am. Inst. Chem. Eng. J.*, **13**, 989 (1967)]. The average drop size may be expressed as

$$d_{p,av} = \frac{\sum n_i d_{pi}^3}{\sum n_i d_{pi}^2} \quad (18-14)$$

and if the drops are spherical,

$$a_{av} = \frac{6\gamma_{D,av}}{d_{p,av}} \quad (18-15)$$

The drop size varies locally with location in the vessel, being smallest at the impeller and largest in regions farthest removed from the impeller owing to coalescence in regions of relatively low turbulence intensity [Schindler and Treybal, *Am. Inst. Chem. Eng. J.*, **14**, 790 (1968); Vanderveen, U.S. AEC UCRL-8733, 1960]. Interfacial area and hence average drop size have been measured by light transmittance, light scattering, direct photography, and other means. Typical of the resulting correlations is that of Thornton and Bouyatiotis [*Inst. Chem. Eng. Symp. Liquid Extraction*, Newcastle-upon-Tyne, April 1967] for a 17.8-cm- (7-in-) diameter baffled vessel, six-bladed flat-blade turbine, $d_t = 6.85$ cm (0.225 ft), operated full, for organic liquids

($\sigma' = 8.5$ to 34, $\rho_D = 43.1$ to 56.4, $\mu_D = 1.18$ to 1.81) dispersed in water, in the absence of mass transfer, and under conditions giving nearly the vessel-average $d_{p,av}$:

$$\frac{d_{p,av}}{d_p^0} = 1 + 1.18\phi_D \left(\frac{\sigma_c^2 g_c^2}{d_p^0 \mu_c^2 g} \right) \left(\frac{\mu_c^4 g}{\Delta p \sigma_c^3 g_c^3} \right)^{0.62} \left(\frac{\Delta p}{\rho_c} \right)^{0.05} \quad (18-16)$$

where d_p^0 is given by

$$\frac{(d_p^0)^3 \rho_c^2 g}{\mu_c^2} = 29.0 \left(\frac{P^3 g_c^3}{v^3 \rho_c^3 \mu_c g^4} \right)^{-0.32} \left(\frac{\rho_c \sigma_c^3 g_c^3}{\mu_c^4 g} \right)^{0.14} \quad (18-17)$$

Caution is needed in using such correlations, since those available do not generally agree with each other. For example, Eq. (21-28) gives $d_{p,av} = 4.78(10^{-4})$ ft for a liquid pair of properties $a' = 30$, $\rho_c = 62.0$, $\rho_D = 52.0$, $\mu_c = 2.42$, $\mu_D = 1.94$, $\gamma_{D,av} = 0.20$ in a vessel $T = Z = 0.75$, a turbine impeller $d_i = 0.25$ turning at 400 r/min. Other correlations provide $3.28(10^{-4})$ [Thornton and Bouyatiotis, *Ind. Chem.*, **39**, 298 (1963)], $8.58(10^{-4})$ [Calderbank, *Trans. Inst. Chem. Eng. (London)*, **36**, 443 (1958)], $6.1(10^{-4})$ [Kafarov and Babinov, *Zh. Prikl. Khim.*, **32**, 789 (1959)], and $2.68(10^{-3})$ (Rushton and Love, paper at AIChE, Mexico City, September 1967). See also Vermeulen et al. [*Chem. Eng. Prog.*, **51**, 85F (1955)], Rodgers et al. [*ibid.*, **52**, 515 (1956)]; U.S. AEC ANL-5575 (1956)], Rodrigues et al. [*Am. Inst. Chem. Eng. J.*, **7**, 663 (1961)], Sharma et al. [*Chem. Eng. Sci.*, **21**, 707 (1966); **22**, 1267 (1967)], and Kagan and Kovalev [*Khim. Prom.*, **42**, 192 (1966)]. For the effect of absence of baffles, see Fick et al. (U.S. AEC UCRL-2545, 1954) and Schindler and Treybal [*Am. Inst. Chem. Eng. J.*, **14**, 790 (1968)]. The latter have observations during mass transfer.

Coalescence Rates The droplets coalesce and redisperse at rates that depend upon the vessel geometry, N , $\gamma_{D,av}$, and liquid properties. The few measurements available, made with a variety of techniques, do not as yet permit quantitative estimates of the coalescence frequency v . Madden and Damarell [*Am. Inst. Chem. Eng. J.*, **8**, 233 (1962)] found for baffled vessels that v varied as $N^{2.2} \gamma_{D,av}^{0.5}$, and this has generally been confirmed by Groothuis and Zuideweg [*Chem. Eng. Sci.*, **19**, 63 (1964)], Miller et al. [*Am. Inst. Chem. Eng. J.*, **9**, 196 (1963)], and Howarth [*ibid.*, **13**, 1007 (1967)], although absolute values of v in the various studies are not well related. Hillestad and Rushton (paper at AIChE, Columbus, Ohio, May 1966), on the other hand, find v to vary as $N^{0.73} \gamma_{D,av}$ for impeller Weber numbers $N_{We,i}$ below a certain critical value and as $N^{-3.5} \gamma_{D,av}^{1.58}$ for higher Weber numbers. The influence of liquid properties is strong. There is clear evidence [Groothuis and Zuideweg, loc. cit.; *Chem. Eng. Sci.*, **12**, 288 (1960)] that coalescence rates are enhanced by mass transfer from a drop to the surrounding continuum and retarded by transfer in the reverse direction. See also Howarth [*Chem. Eng. Sci.*, **19**, 33 (1964)]. For a theoretical treatment of drop breakage and coalescence and their effects, see Valentas and Amundsen [*Ind. Eng. Chem. Fundam.*, **5**, 271, 533 (1966); **7**, 66 (1968)], Gal-Or and Walatka [*Am. Inst. Chem. Eng. J.*, **13**, 650 (1967)], and Curl [*ibid.*, **9**, 175 (1963)].

In calculating the power required for mixers, a reasonable estimate of the average density and viscosity for a two-phase system is satisfactory.

Solids are often present in liquid streams either as a part of the processing system or as impurities that come along and have to be handled in the process. One advantage of mixers in differential contact equipment is the fact that they can handle slurries in one or both phases. In many industrial leaching systems, particularly in the minerals processing industry, coming out of the leach circuit is a slurry with a desired material involved in the liquid but a large amount of solids contained in the stream. Typically, the solids must be separated out by filtration or centrifugation, but there has always been a desire to try a direct liquid-liquid extraction with an immiscible liquid contact with this often highly concentrated slurry leach solution. The major problem with this approach is loss of organic material going out with the highly concentrated liquid slurry.

Data are not currently available on the dispersion with the newer fluidfoil impellers, but they are often used in industrial mixer-settler systems to maintain dispersion when additional residence time holdup is required, after an initial dispersion is made by a radial- or axial-flow turbine.

Recent data by Calabrese⁵ indicates that the sauter mean drop diameter can be correlated by equation and is useful to compare with other predictions indicated previously.

As an aside, when a large liquid droplet is broken up by shear stress, it tends initially to elongate into a dumbbell shape, which determines the particle size of the two large droplets formed. Then, the neck in the center between the ends of the dumbbell may explode or shatter. This would give a debris of particle sizes which can be quite different than the two major particles produced.

Liquid-Liquid Extraction The actual configuration of mixers in multistage mixer-settlers and/or multistage columns is summarized in Section 15. A general handbook on this subject is *Handbook of Solvent Extraction* by Lowe, Beard, and Hanson. This handbook gives a comprehensive review of this entire operation as well.

In the liquid-liquid extraction area, in the mining industry, coming out of the leach tanks is normally a slurry, in which the desired mineral is dissolved in the liquid phase. To save the expense of separation, usually by filtration or centrifugation, attempts have been made to use a resident pump extraction system in which the organic material is contacted directly with the slurry. The main economic disadvantage to this proposed system is the fact that considerable amounts of organic liquid are entrained in the aqueous slurry system, which, after the extraction is complete, are discarded. In many systems this has caused an economic loss of solvent into this waste stream.

LIQUID-LIQUID-SOLID SYSTEMS

Many times solids are present in one or more phases of a solid-liquid system. They add a certain level of complexity in the process, especially if they tend to be a part of both phases, as they normally will do. Approximate methods need to be worked out to estimate the density of the emulsion and determine the overall velocity of the flow pattern so that proper evaluation of the suspension requirements can be made. In general, the solids will behave as though they were a fluid of a particular average density and viscosity and won't care much that there is a two-phase dispersion going on in the system. However, if solids are being dissolved or precipitated by participating in one phase and not the other, then they will be affected by which phase is dispersed or continuous, and the process will behave somewhat differently than if the solids migrate independently between the two phases within the process.

FLUID MOTION

Pumping Some mixing applications can be specified by the pumping capacity desired from the impeller with a certain specified geometry in the vessel. As mentioned earlier, this sometimes is used to describe a blending requirement, but circulation and blending are two different things. The major area where this occurs is in draft tube circulators or pump-mix mixer settlers. In draft tube circulators (shown in Fig. 18-22), the circulation occurs through the draft tube and around the annulus and for a given geometry, the velocity head required can be calculated with reference to various formulas for geometric shapes. What is needed is a curve for head versus flow for the impeller, and then the system curve can be matched to the impeller curve. Adding to the complexity of this system is the fact that solids may settle out and change the character of the head curve so that the impeller can get involved in an unstable condition which has various degrees of erratic behavior depending upon the sophistication of the impeller and inlet and outlet vanes involved. These draft tube circulators often involve solids, and applications are often for precipitation or crystallization in these units. Draft tube circulators can either have the impeller pump up in the draft tube and flow down the annulus

or just the reverse. If the flow is down the annulus, then the flow has to make a 180° turn where it comes back at the bottom of the tank into the draft tube again. This is a very sensitive area, and special baffles must be used to carefully determine how the fluid will make this turn since many areas of constriction are involved in making this change in direction.

When pumping down the draft tube, flow normally makes a more troublefree velocity change to a flow going up the annulus. Since the area of the draft tube is markedly less than the area of the annulus, pumping up the draft tube requires less flow to suspend solids of a given settling velocity than does pumping down the draft tube.

Another example is to eliminate the interstage pump between mixing and settling stages in the countercurrent mixer-settler system. The radial-flow impeller typically used is placed very close to an orifice at the bottom of the mixing tank and can develop heads from 12 to 18 in. All the head-loss terms in the mixer and settler circuit have to be carefully calculated because they come very close to that 12- to 18-in range when the passages are very carefully designed and streamlined. If the mixing tank gets much above 10 ft in depth, then the heads have to be higher than the 12- to 18-in range and special designs have to be worked on which have the potential liability of increasing the shear rate acting on the dispersed phase to cause more entrainment and longer settling times. In these cases, it is sometimes desirable to put the mixer system outside the actual mixer tank and have it operate in a single phase or to use multiple impellers, each one of which can develop a portion of the total head required.

Heat Transfer In general, the fluid mechanics of the film on the mixer side of the heat transfer surface is a function of what happens at that surface rather than the fluid mechanics going on around the impeller zone. The impeller largely provides flow across and adjacent to the heat-transfer surface and that is the major consideration of the heat-transfer result obtained. Many of the correlations are in terms of traditional dimensionless groups in heat transfer, while the impeller performance is often expressed as the impeller Reynolds number.

The fluidfoil impellers (shown in Fig. 18-2) usually give more flow for a given power level than the traditional axial- or radial-flow turbines. This is also thought to be an advantage since the heat-transfer surface itself generates the turbulence to provide the film coefficient and more flow should be helpful. This is true to a limited degree in jacketed tanks (Fig. 18-37), but in helical coils (Fig. 18-38), the

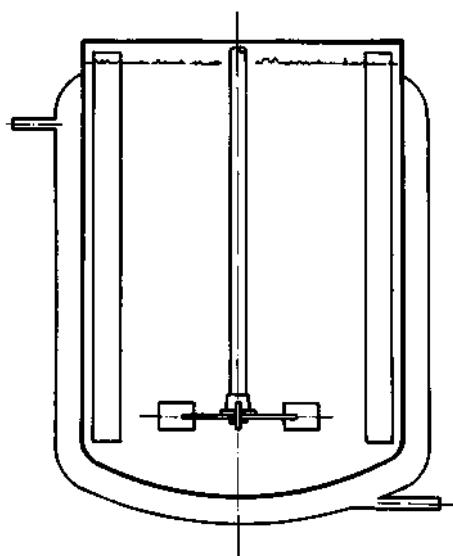


FIG. 18-37 Typical jacket arrangement for heat transfer.

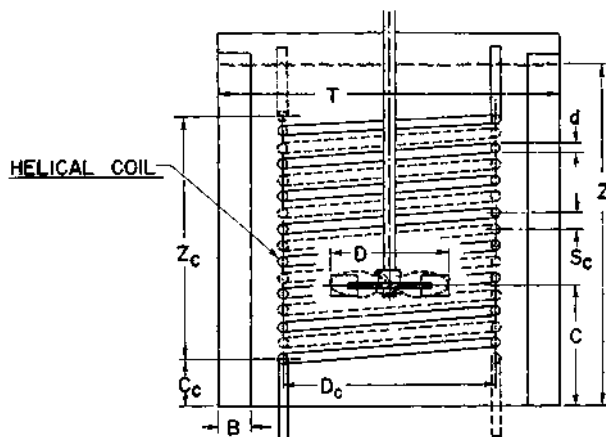


FIG. 18-38 Typical arrangement of helical coil at mixing vessel for heat transfer.

extreme axial flow of these impellers tends to have the first or second turn in the coil at the bottom of the tank blank off the flow from the turns above it in a way that (at the same power level) the increased flow from the fluidfoil impeller is not helpful. It best gives the same coefficient as with the other impellers and on occasion can cause a 5 to 10 percent reduction in the heat-transfer coefficient over the entire coil.

JACKETS AND COILS OF AGITATED VESSELS

Most of the correlations for heat transfer from the agitated liquid contents of vessels to jacketed walls have been of the form:

$$\frac{hD_j}{k} = a \left(\frac{L_p^2 N_r \rho}{\mu} \right)^b \left(\frac{c\mu}{k} \right)^{1/3} \left(\frac{\mu_b}{\mu_w} \right)^m \quad (18-18)$$

The film coefficient h is for the inner wall; D_j is the inside diameter of the mixing vessel. The term $L_p^2 N_r \rho / \mu$ is the Reynolds number for mixing in which L_p is the diameter and N_r the speed of the agitator. Recommended values of the constants a , b , and m are given in Table 18-3.

A wide variety of configurations exists for coils in agitated vessels. Correlations of data for heat transfer to helical coils have been of two forms, of which the following are representative:

$$\frac{hD_j}{k} = 0.87 \left(\frac{L_p^2 N_r \rho}{\mu} \right)^{0.62} \left(\frac{c\mu}{k} \right)^{1/3} \left(\frac{\mu_b}{\mu_w} \right)^{0.14} \quad (18-19)$$

TABLE 18-3 Values of Constants for Use in Eq. (18-18)

Agitator	a	b	m	Range of Reynolds number
Paddle ^a	0.36	$\frac{2}{3}$	0.21	300–3 × 10 ⁵
Pitched-blade turbine ^b	0.53	$\frac{2}{3}$	0.24	80–200
Disc, flat-blade turbine ^c	0.54	$\frac{2}{3}$	0.14	40–3 × 10 ⁵
Propeller ^d	0.54	$\frac{2}{3}$	0.14	2 × 10 ³ (one point)
Anchor ^b	1.0	$\frac{1}{2}$	0.18	10–300
Anchor ^b	0.36	$\frac{2}{3}$	0.18	300–40,000
Helical ribbon ^e	0.633	$\frac{1}{2}$	0.18	8–10 ⁵

^aChilton, Drew, and Jebens, *Ind. Eng. Chem.*, **36**, 510 (1944), with constant m modified by Uhl.

^bUhl, *Chem. Eng. Progr., Symp. Ser.* **17**, **51**, 93 (1955).

^cBrooks and Su, *Chem. Eng. Progr.*, **55**(10), 54 (1959).

^dBrown et al., *Trans. Inst. Chem. Engrs. (London)*, **25**, 181 (1947).

^eGluz and Pavlushenko, *J. Appl. Chem. U.S.S.R.*, **39**, 2323 (1966).

where the agitator is a paddle, the Reynolds number range is 300 to 4×10^5 [Chilton, Drew, and Jebens, *Ind. Eng. Chem.*, **36**, 510 (1944)], and

$$\frac{hD_o}{k} = 0.17 \left(\frac{L_p^2 N_p \rho}{\mu} \right)^{0.67} \left(\frac{c\mu}{k} \right)^{0.37} \left(\frac{L_p}{D_j} \right)^{0.1} \left(\frac{D_o}{D_j} \right)^{0.5} \quad (18-20)$$

where the agitator is a disc flat-blade turbine, and the Reynolds number range is 400 to $(2)(10^5)$ [Oldshue and Gretton, *Chem. Eng. Prog.*, **50**, 615 (1954)]. The term D_o is the outside diameter of the coil tube.

The most comprehensive correlation for heat transfer to vertical baffle-type coils is for a disc flat-blade turbine over the Reynolds number range 10^3 to $(2)(10^6)$:

$$\frac{hD_o}{k} = 0.09 \left(\frac{L_p^2 N_p \rho}{\mu} \right)^{0.65} \left(\frac{c\mu}{k} \right)^{1/3} \left(\frac{L_p}{D_j} \right)^{1/3} \left(\frac{2}{n_b} \right)^{0.2} \left(\frac{\mu}{\mu_f} \right)^{0.4} \quad (18-21)$$

where n_b is the number of baffle-type coils and μ_f is the fluid viscosity at the mean film temperature [Dunlop and Rushton, *Chem. Eng. Prog. Symp. Ser. 5*, **49**, 137 (1953)].

Chapman and Holland (*Liquid Mixing and Processing in Stirred Tanks*, Reinhold, New York, 1966) review heat transfer to low-viscosity fluids in agitated vessels. Uhl ["Mechanically Aided Heat Transfer," in Uhl and Gray (eds.), *Mixing: Theory and Practice*, vol. I, Academic, New York, 1966, chap. V] surveys heat transfer to low- and high-viscosity agitated fluid systems. This review includes scraped-wall units and heat transfer on the jacket and coil side for agitated vessels.

LIQUID-LIQUID-GAS-SOLID SYSTEMS

This is a relatively unusual combination, and one of the more common times it exists is in the fermentation of hydrocarbons with aerobic microorganisms in an aqueous phase. The solid phase is a microorganism which is normally in the aqueous phase and is using the organic phase for food. Gas is supplied to the system to make the fermentation aerobic. Usually the viscosities are quite low, percent solids is also modest, and there are no special design conditions required when this particular gas-liquid-liquid-solid combination occurs. Normally, average properties for the density or viscosity of the liquid phase are used. In considering that the role the solids play in the system is adequate, there are cases of other processes which consist of four phases, each of which involves looking at the particular properties of the phases to see whether there are any problems of dispersion, suspension, or emulsification.

COMPUTATIONAL FLUID DYNAMICS

There are several software programs that are available to model flow patterns of mixing tanks. They allow the prediction of flow patterns based on certain boundary conditions. The most reliable models use accurate fluid mechanics data generated for the impellers in question and a reasonable number of modeling cells to give the overall tank flow pattern. These flow patterns can give velocities, streamlines, and localized kinetic energy values for the systems. Their main use at the present time is to look at the effect of making changes in mixing variables based on doing certain things to the mixing process. These programs can model velocity, shear rates, and kinetic energy, but probably cannot adapt to the actual chemistry of diffusion or mass-transfer kinetics of actual industrial process at the present time.

Relatively uncomplicated transparent tank studies with tracer fluids or particles can give a similar feel for the overall flow pattern. It is important that a careful balance be made between the time and expense of calculating these flow patterns with computational fluid dynamics compared to their applicability to an actual industrial process. The future of computational fluid dynamics appears very encouraging and a

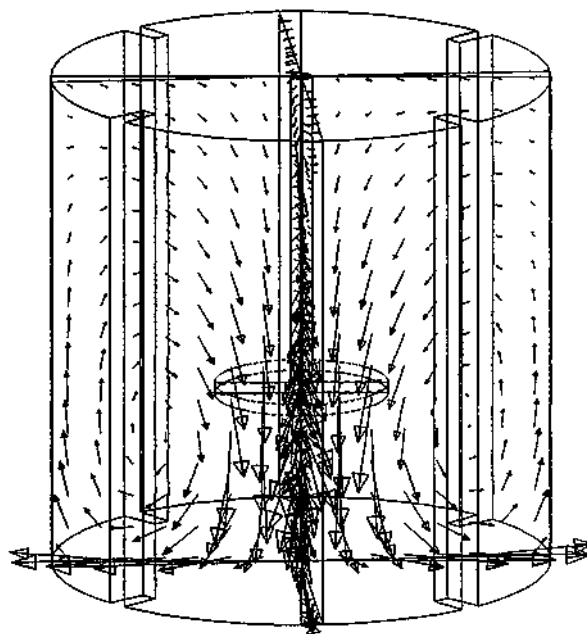


FIG. 18-39 Laser scan.

reasonable amount of time and effort put forth in this regard can yield immediate results as well as potential for future process evaluation.

Figures 18-39, 18-40, and 18-41 show some approaches. Figure 18-39 shows velocity vectors for an A310 impeller. Figure 18-40 shows contours of kinetic energy of turbulence. Figure 18-41 uses a particle trajectory approach with neutral buoyancy particles.

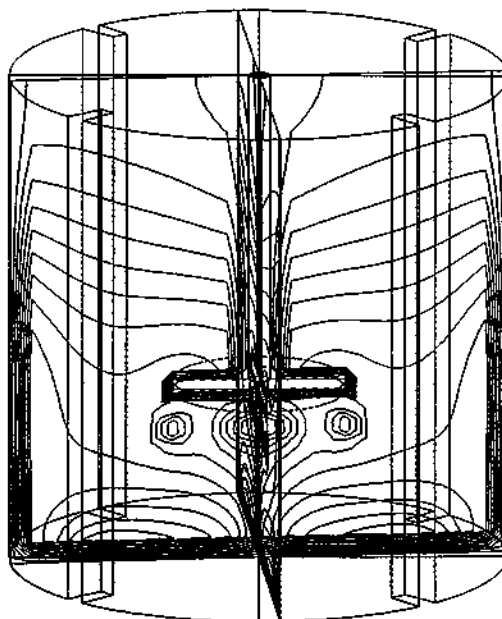


FIG. 18-40 Laser scan.

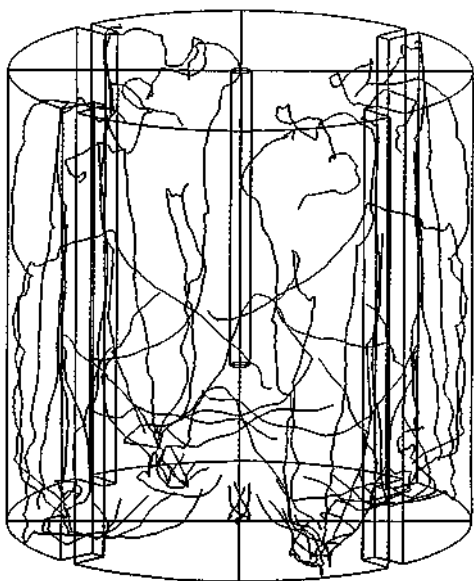


FIG. 18-41 Laser scan.

Numerical fluid mechanics can define many of the fluid mechanics parameters for an overall reactor system. Many of the models break up the mixing tank into small microcells. Suitable material and mass-transfer balances between these cells throughout the reactor are then made. This can involve long and massive computational requirements. Programs are available that can give reasonably acceptable models of experimental data taken in mixing vessels. Modeling the three-dimensional aspect of a flow pattern in a mixing tank can require a large amount of computing power.

Most modeling codes are a time-averaging technique. Depending upon the process, a time-dependent technique may be more suitable. Time-dependent modeling requires much more computing power than does time averaging.

GENERAL REFERENCES

1. J. Y. Oldshue, "Mixing '89," *Chemical Engineering Progress*, **85**(5): 33-42 (1989).
2. J. C. Middleton, *Proc. 3d European Conf. on Mixing*, 4/89, BHRA, pp. 15-36.
3. J. Y. Oldshue, T. A. Post, R. J. Weetman, "Comparison of Mass Transfer Characteristics of Radial and Axial Flow Impellers," *BHRA Proc. 6th European Conf. on Mixing*, 5/88.
4. A. W. Neimow, B. Buckland, R. J. Weetman, *Mixing XII Research Conference*, Potosi, Mo., 8/89.
5. R. Calabrese et al., *AIChE J.* **32**: 657, 677 (1986).
6. T. N. Zwietering, *Chemical Engineering Science*, **8**(3): 244-253 (1958).
7. J. Y. Oldshue, *Chemical Engineering Progress*, "Mixing of Slurries Near the Ultimate Settled Solids Concentration," **77**(5): 95-98 (1981).

MIXING OF VISCOUS FLUIDS, PASTES, AND DOUGHS

GENERAL REFERENCES: Paul, E. L., V. A. Atiemo-Obeng, and S. M. Kresta (eds.), *Handbook of Industrial Mixing, Science and Practice*, Wiley, Hoboken, N.J., 2004. Harnby, N., M. F. Edwards, and A. W. Nienow (eds.), *Mixing in the Process Industries*, 2d ed., Butterworth-Heinemann, Boston, 1992. Oldshue, J. Y., *Fluid Mixing Technology*, McGraw-Hill, New York, 1983. Ottino, J. M., *The Kinematics of Mixing: Stretching, Chaos, and Transport*, Cambridge University Press, New York, 1999. Tatterson, G. B., *Fluid Mixing and Gas Dispersion in Agitated Tanks*, McGraw-Hill, New York, 1991. Zlokarnik, M., *Stirring, Theory and Practice*, Wiley-VCH, New York, 2001.

INTRODUCTION

Even the definition of mixing for viscous fluids, pastes, and doughs is complicated. While mixing can be defined simply as increasing or maintaining uniformity, the devices that cause mixing to take place may also accomplish deagglomeration, dispersion, extrusion, heat transfer, or other process objectives. Fluids with viscosities greater than 10 Pa·s (10,000 cP) can be considered *viscous*. However, non-newtonian fluid properties are often as important in establishing mixing requirements. Viscous fluids can be polymer melts, polymer solutions, and a variety of other high-molecular-weight or low-temperature materials. Many polymeric fluids are shear thinning. Pastes are typically formed when particulate materials are wetted by a fluid to the extent that particle-particle interactions create flow characteristics similar to those of viscous fluids. The particle-particle interactions may cause shear thickening effects. Doughs have the added characteristic of elasticity. Viscous materials often exhibit a combination of other non-newtonian characteristics, such as a yield stress.

One common connection between viscous fluids, pastes, and doughs is the types of equipment used to mix or process them. While often designed for a specific process objective or a certain fluid characteristic, most types of viscous mixing equipment have some common characteristics. The nature of all viscous materials is their resistance to flow. This resistance is usually overcome by a mixer that will eventually contact or directly influence all the material in a container, particularly material near the walls or in corners. Small clearances between rotating and stationary parts of a mixer create regions

of high local shear. Intermeshing blades or stators prevent material from rotating as a solid mass. Such equipment provides greater control of fluid motion than equipment used for low-viscosity fluids, but typically at greater cost and complexity.

The one failure common to all mixing equipment is any region of stagnant material. With a shear thinning material, the relative motion between a rotating mixer blade and adjacent fluid will reduce the local viscosity. However, away from the mixer blade, shear will decrease and the viscosity will increase, leading to the possibility of stagnation. With a shear thickening material, high shear near a mixer blade will result in high viscosity, which may reduce either local relative motion or the surrounding bulk motion. Yield stress requires some minimum shear stress to accomplish any motion at all. Viscoelastic characteristics cause motion normal to the applied stresses. Thus all major non-newtonian characteristics reduce effective mixing and increase the possibility of local stagnation.

Blade shape and mixing action can have significant impacts on the mixing process. A scraping action is often necessary to promote heat transfer or prevent adhesion to equipment surfaces. A smearing action can improve dispersion. A combination of actions is necessary to accomplish the random or complicated pattern necessary for complete mixing. No one mixing effect or equipment design is ideal for all applications.

Because of high viscosity, the mixing Reynolds number ($N_{Re} = D^2 N \rho / \mu$, where D is impeller diameter, N is rotational speed, ρ is density, and μ is viscosity) may be less than 100. At such viscous conditions, mixing occurs because of laminar shearing and stretching. Turbulence is not a factor, and complicated motion is a direct result of the mixer action. The relative motion between moving parts of the mixer and the walls of the container or other mixer parts creates both shear and bulk motion. The shear effectively creates thinner layers of nonuniform material, which diminishes striations or breaks agglomerates to increase homogeneity. Bulk motion redistributes the effects of the stretching processes throughout the container.

Often as important as or more important than the primary viscosity is the relative viscosity of fluids being mixed. When a high-viscosity material is added to a low-viscosity material, the shear created by the

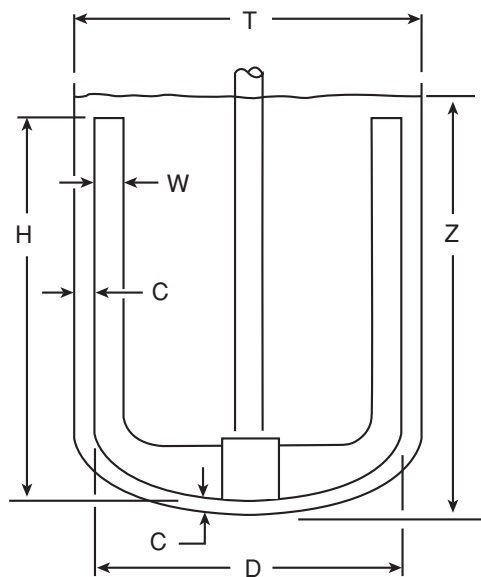


FIG. 18-42 Anchor impeller with nomenclature.

low-viscosity material may not be sufficient to stretch and interact with the high-viscosity material. When a low-viscosity material is added to a high-viscosity material, the low-viscosity material may act as a lubricant, thus allowing slippage between the high-viscosity material and the mixer surfaces. Viscosity differences can be orders of magnitude different. Density differences are smaller and typically less of a problem in viscous mixing.

Besides mixing fluids, pastes, and doughs, the same equipment may be used to create those materials. Viscous fluids such as polymers can be created by reaction from low-viscosity monomers in the same equipment described for viscous mixing. Pastes may be created by either the addition of powders to liquids or the removal of liquids from slurries, again using the same type of equipment as for bulk mixing. Doughs are usually created by the addition of a powder to liquid and the subsequent hydration of the powder. The addition process itself becomes a mixer application, which may fall somewhere between low-viscosity and high-viscosity mixing, but often including both types of mixing.

BATCH MIXERS

Anchor Mixers Anchor mixers are the simplest and one of the more common types of high-viscosity mixers (Fig. 18-42). The diameter of the anchor D is typically 90 to 95 percent of the tank diameter T . The result is a small clearance C between the rotating impeller and the tank wall. Within this gap the fluid is sheared by the relative motion between the rotating blade and the stationary tank wall. The shear near the wall typically reduces the buildup of stagnant material and promotes heat transfer. To reduce buildups further, flexible or spring-loaded scrapers, typically made of polymeric material, can be mounted on the rotating blades to move material physically away from the wall.

The benefits of an anchor mixer are limited by the fact that the vertical blades provide very little fluid motion between the top and bottom of the tank. Ingredient additions at the surface of the fluid may make many rotations before gradually being spread and circulated to the bottom of the tank. To promote top-to-bottom fluid motion, angled blades on the anchor or helical ribbon blades, described in the next subsection, make better mixers for uniform blending. Significant viscosity differences between fluids may extend mixing times to unacceptable limits with the basic anchor.

Anchor mixers may be used in combination with other types of mixers, such as turbine mixers, high-shear mixers, or rotor-stator mixers, which were described in the previous subsection. Such mixers can be placed on a vertical shaft midway between the anchor shaft and blade. A secondary mixer can promote top-to-bottom motion and also limit bulk rotation of the fluid. A stationary baffle is sometimes placed between the anchor shaft and rotating blade to limit fluid rotation and enhance shear.

A dimensionless group called the power number is commonly used to predict the power required to rotate a mixing impeller. The *power number* is defined as $P/(\rho N^3 D^5)$, where P is power, ρ is fluid density, N is rotational speed, and D is impeller diameter. To be dimensionless, the units of the variables must be coherent, such as SI metric; otherwise appropriate conversion factors must be used. The conversion factor for common engineering units gives the following expression for power number:

$$N_p = \frac{1.524 \times 10^{13} P}{\text{sp gr } N^3 D^5} \quad (18-22)$$

where P is power in horsepower, sp gr is fluid specific gravity based on water, N is rotational speed in rpm, and D is impeller diameter in inches. The power number is an empirically measured value that describes geometrically similar impellers. Power number is a function of Reynolds number, which accounts for the effects of fluid properties. Impeller Reynolds number, as defined earlier, is another dimensionless group. A conversion factor is needed for common engineering units:

$$N_{\text{Re}} = \frac{10.4 D^2 N \text{ sp gr}}{\mu} \quad (18-23)$$

where D is the impeller diameter in inches, N is rotational speed in rpm, sp gr is specific gravity based on water, and μ is viscosity in centipoise.

Power can be calculated by rearranging the definition of power number; see the following example. A value for the appropriate power number must be obtained from empirically derived data for geometrically similar impellers. Power number correlations for anchor impellers are shown in Fig. 18-43. The typical anchor impellers have two vertical arms with a blade width W equal to one-tenth of the impeller diameter D , and the arm height H equal to the impeller diameter D . Correlations are shown for typical impellers 95 and 90 percent of the tank diameter. The clearance C is one-half of the difference between the impeller diameter and the tank diameter, or 2.5 and 5.0 percent of the tank diameter for the respective correlations. An additional correlation is shown for an anchor with three vertical arms and a diameter equal to 95 percent of the tank diameter. The correlation for a three-arm impeller which anchors 90 percent of the tank diameter is the same as that for the typical anchor 95 percent of the tank diameter.

The power number and corresponding power of an anchor impeller are proportional to the height of the vertical arm. Thus, an anchor with a height H equal to 75 percent of the impeller diameter would have a power number equal to 75 percent of the typical values shown in Fig. 18-43. Similarly, a partially filled tank with a liquid level Z that covers only 75 percent of the vertical arm will also have a power number that is 75 percent of the typical correlation value. The addition of scrapers will increase the power requirement for an anchor impeller, but the effect depends on the clearance at the wall, design of the scrapers, processed material, and many other factors. Correlations are not practical or available.

Unfortunately, the power number only provides a relationship between impeller size, rotational speed, and fluid properties. The power number does not tell whether a mixer will work for an application. Successful operating characteristics for an anchor mixer usually depend on experience with a similar process or experimentation in a pilot plant. Scale-up of pilot-plant experience is most often done for a geometrically similar impeller and equal tip (peripheral) speed.

Helical Ribbon Mixers Helical ribbon mixers (Fig. 18-44), or simply helix mixers, have major advantages over the anchor mixer, because they force strong top-to-bottom motion even with viscous

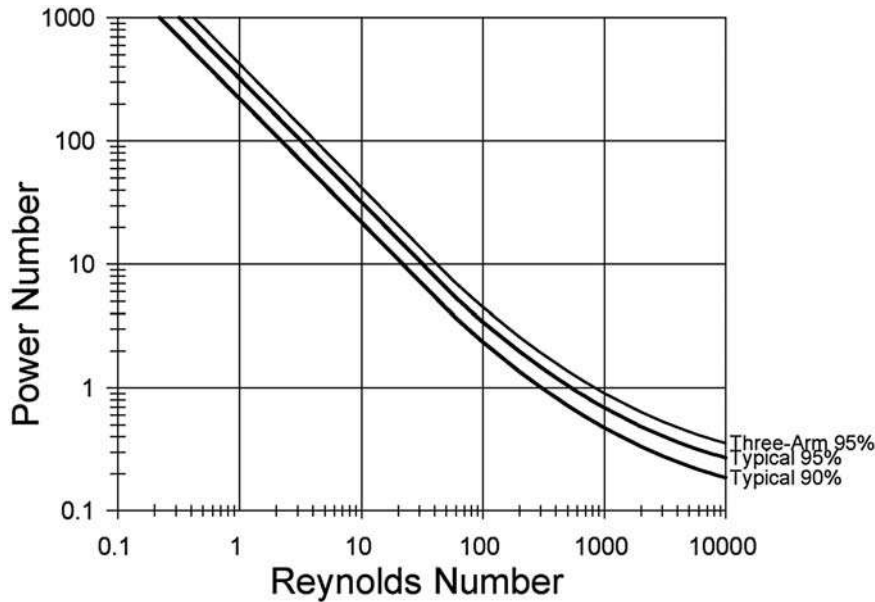


FIG. 18-43 Power numbers for anchor impellers: typical two-arm impeller anchors 95 percent of tank diameter T and 90 percent of T ; three-arm impeller anchors 95 percent of T ; and three-arm impeller anchors 90 percent of T , similar to two-arm impeller that anchors 95 percent of T .

materials. These impellers are some of the most versatile mixing impellers, but also some of the most expensive. Besides having a formed helical shape, the blades must be rolled the hard way with the thick dimension normal to the direction of the circular rolled shape. Helical ribbon mixers will work with most viscous fluids up to the lim-

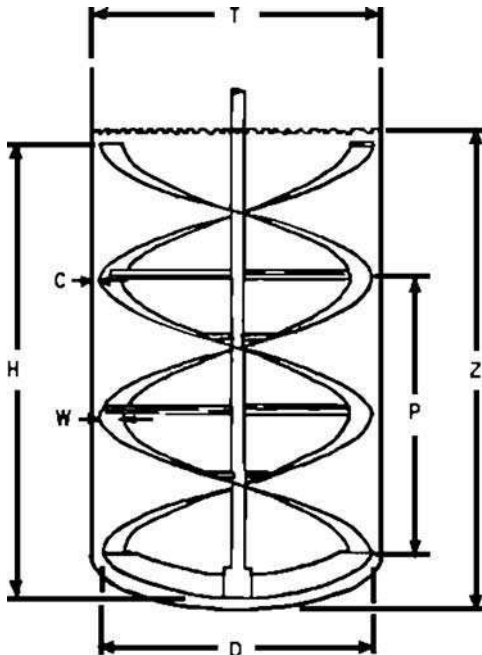


FIG. 18-44 Helical ribbon impeller with nomenclature.

its of a flowable material, as high as 4,000,000 cP or more depending on non-newtonian characteristics. While not cost-effective for low-viscosity materials, they will adequately mix, and even suspend solids, in low-viscosity liquids. These characteristics make helical ribbon mixers effective for batch processes, such as polymerization or other processes beginning with low-viscosity materials and changing to high-viscosity products. Helical ribbon mixers will even work with heavy pastes and flowable powders. Usually the helix pumps down at the tank wall with fluids and up at the wall with pastes or powders.

The helical ribbon power numbers are a function of Reynolds number similar to the correlations for anchor impellers. Figure 18-45 shows correlations for some typical helical ribbon power numbers. The upper curve is for a double-flight helix with the blade width W equal to one-tenth the impeller diameter D , the pitch P equal to the impeller diameter, and the impeller diameter at 95 percent of the tank diameter T . The height H for this typical helix is equal to the impeller diameter and pitch, not 15 times the pitch, as shown in Fig. 18-45. A second curve shows the power number correlation for a helical ribbon impeller that is 90 percent of the tank diameter. The curve marked "Single 90%" is for a single flight helix, 90 percent of the tank diameter. Each ribbon beginning at the bottom of the impeller and spiraling around the axis of the impeller is called a flight. Single-flight helices are theoretically more efficient, but a partially filled tank can cause imbalanced forces on the impeller. The correlation for a 95 percent diameter single-flight helix is the same as the correlation for the double-flight 90 percent diameter helix.

Example 1: Calculate the Power for a Helix Impeller Calculate the power required to rotate a double-flight helix impeller that is 57 in in diameter, 57 in high, with a 57-in pitch operating at 30 rpm in a 60-in-diameter tank. The tank is filled 85 percent full with a 100,000-cP fluid, having a 1.05 specific gravity.

$$N_{Re} = \frac{10.4 D^2 N \text{ sp gr}}{\mu} = \frac{10.4(57)^2(30)(1.05)}{100,000} = 10.6$$

Referring to Fig. 18-45, the power number N_p for the full-height helix impeller is 27.5 at $N_{Re} = 10.6$. At 85 percent full, the power number is $0.85 \times 27.5 = 23.4$. Power can be calculated by rearranging Eq. (18-22).

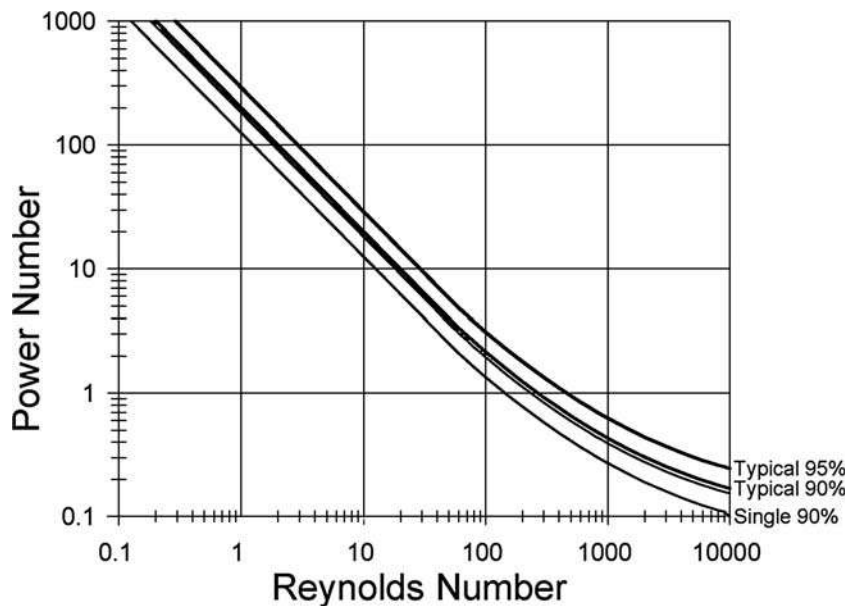


FIG. 18-45 Power numbers for helical-ribbon impeller: typical double-flight helixes 95 percent of tank diameter T and 90 percent of T ; single-flight helix 90 percent of T ; single-flight 95 percent of T similar to double-flight 90 percent of T .

$$P = \frac{N_P \text{ sp gr } N^3 D^5}{1.524 \times 10^{13}} = \frac{23.4(1.05)(30)^3(57)^5}{1.524 \times 10^{13}} = 26.2 \text{ hp}$$

Helical ribbon mixers can also be formed to fit in conical bottom tanks. While not as effective at mixing as in a cylindrical tank, the conical bottom mixer can force material to the bottom discharge. By more effectively discharging, a higher yield of the product can be obtained.

Planetary Mixers A variation on the single anchor mixer is essentially a double anchor mixer with the impellers moving in a planetary pattern. Each anchor impeller rotates on its own axis, while the pair of intermeshing anchors also rotates on the central axis of the tank. The intermeshing pattern of the two impellers gives a kneading action with blades alternately wiping each other. The rotation around the central axis also creates a scraping action at the tank wall and across the bottom. With successive rotations of the impellers, all the tank contents can be contacted directly. A typical planetary mixer is shown in Fig. 18-46.

The intimate mixing provided by the planetary motion means that the materials need not actively flow from one location in the tank to another. The rotating blades cut through the material, creating local shear and stretching. Even thick pastes and viscoelastic and high-viscosity fluids can be mixed with planetary mixers. The disadvantage of poor top-to-bottom motion still exists with conventional planetary mixers. However, some new designs offer blades with a twisted shape to increase vertical motion.

To provide added flexibility and reduce batch-to-batch turnaround or cross-contamination, a change-can feature is often available with planetary and other multishaft mixers. The container (can) in a change-can mixer is a separate part that can be rapidly exchanged between batches. Batch ingredients can even be put in the can before it is placed under the mixing head. Once the mixing or processing is accomplished, the container can be removed from the mixer and taken to another location for packaging and cleaning. After one container is removed from the mixer and the blades of the impeller are cleaned, another batch can begin processing. Because the cans are relatively inexpensive compared with the cost of the mixer head, a change-can mixer can be better utilized and processing costs can be reduced.



FIG. 18-46 Planetary mixer. (Charles Ross & Son Company.)

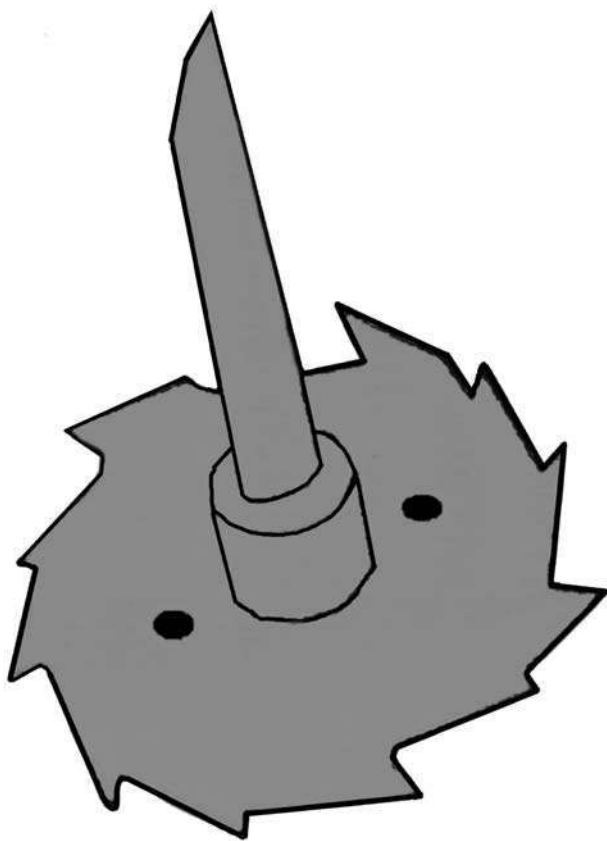


FIG. 18-47 High-shear impeller.

Double- and Triple-Shaft Mixers The planetary mixer is an example of a double shaft mixer. However, many different combinations of mixing actions can be achieved with multi-shaft mixers. One variation on planetary motion involves replacing one anchor-style impeller with a high-shear impeller similar to the one shown in Fig. 18-47. The high-shear mixer can be used to incorporate powdered material effectively or create a stable emulsion leading to a final batch of viscous paste or fluid.

Many types of multishaft mixers do not require planetary motion. Instead the mixers rely on an anchor-style impeller to move and shear material near the tank wall, while another mixer provides a different type of mixing. The second or third mixer shafts may have a pitched-blade turbine, hydrofoil impeller, high-shear blade, rotor-stator mixer, or other type of mixer. The combination of multiple impeller types adds to the flexibility of the total mixer. Many batch processes involve different types of mixing over a range of viscosities. Some mixer types provide the top-to-bottom motion that is missing from the anchor impeller alone.

Double-Arm Kneading Mixers A double-arm kneader consists of two counter-rotating blades in a rectangular trough with the bottom formed like two overlapping or adjacent half-cylinders (Fig. 18-48). The blades are driven by gearing at one or both ends. The older-style kneaders emptied through a door or valve at the bottom. Those mixers are still used where complete discharge or thorough cleaning between batches is not essential. More commonly, double-arm kneaders are tilted for discharge. The tilting mechanism may be manual, mechanical, or hydraulic, depending on the size of the mixer and weight of the material.

A variety of blade shapes have evolved for different applications. The mixing action is a combination of bulk movement, shearing,

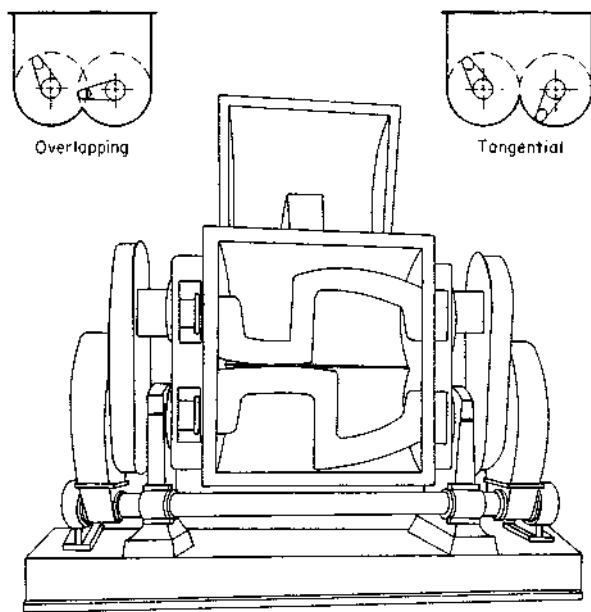


FIG. 18-48 Double-arm kneader. (APV Baker Invenstys.)

stretching, folding, dividing, and recombining. The material being mixed is also squeezed and stretched against the blades, bottom, and sidewalls of the mixer. Clearances may be as close as 1 mm (0.04 in). Rotation is usually such that the material is drawn down in the center between the blades and up at the sidewalls of the trough. Most of the blades are pitched to cause end-to-end motion.

The blades can be tangential or overlapping. Tangential blades can run at different speeds with the advantages of faster mixing caused by changes in the relative position of the blades, greater heat-transfer surface area per unit volume, and less tendency for the material to ride above the blades. Overlapping blades can reduce the buildup of material sticking to the blades.

Because the materials most commonly mixed in kneaders are very viscous, often elastic or rubbery materials, a large amount of energy must be applied to the mixer blades. All that energy is converted to heat within the material. Often the material begins as a semisolid mass, with liquid or powder additives, and the blending process both combines the materials and heats them to create uniform bulk properties.

The blade design most commonly used is the sigma blade (Fig. 18-49a). The sigma-blade mixer can start and operate with either liquids or solids, or a combination of both. Modifications to the blade faces have been introduced to increase particular effects, such as shredding or wiping. The sigma blades can handle elastic materials and readily discharge materials that do not stick to the blades. The sigma blades are easy to clean, even with sticky materials.

The dispersion blade in Fig. 18-49b was developed to provide higher compressive shear than the standard sigma blade. The blade shape forces material against the trough surface. The compressive action is especially good for dispersing fine particles in a viscous material. Rubbery materials have a tendency to ride the blades, and a dispersion blade is frequently used to keep the material in the mixing zone.

Multiwiping overlapping (MWOL) blades (Fig. 18-49c), are commonly used for mixtures that start tough and rubberlike. The blade shape initially cuts the material into small pieces before plasticating it.

The single-curve blade (Fig. 18-49d), was developed for incorporating fiber reinforcement into plastics. In this application the individual fibers, e.g., glass, must be wetted with the polymer without undue fiber breakage.

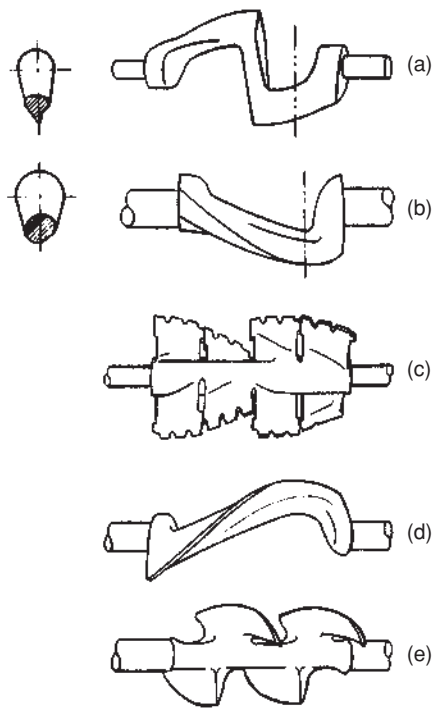


FIG. 18-49 Agitator blades for double-arm kneader: (a) Sigma; (b) dispersion; (c) multiwiping; (d) single-curve; (e) double-naben. (APV Baker Incensys.)

Many other designs have been developed for specific applications. The double-naben blade (Fig. 18-49e), is good for mixtures which “ride,” meaning they form a lump that bridges across the sigma blade. Figure 18-50 provides a guide for some typical applications of double-arm mixers. Individual formulations may require more power.

Screw-Discharge Batch Mixers A variant of the sigma-blade mixer has an extrusion-discharge screw located at the center of the

Character of Material	Material and Blade Type	Net horsepower per gal.	Blade interaction	
			Tangential	Overlapping
Semi-solid	Enamel (dispersion)	1.0	x	x
	Floor tile (sigma)	0.6	x x x x	
	Flush colors (sigma)	0.4	x x x x	
	Brake lining (sigma)	0.2	x	x
	Adhesive (sigma)	0.10	x	x
Plastic mass	Toothpaste (dispersion)	0.06	x	x
	Molding (mod. sigma)	0.04	x	x
	Cookie dough (sigma)	0.02	x	x
Paste	Epoxy resin (sigma)	0.01	x	x
	Dog food (sigma)	0.006	x	x
	Gravy base (sigma)		x	x

FIG. 18-50 Typical application and power for double-arm kneaders. To convert horsepower per gallon to kilowatts per cubic meter, multiply by 197.3. [Parker, Chem. Eng. 72(18): 125 (1965); excerpted by special permission of the copyright owner, McGraw-Hill, Inc.]

trough, just below the rotating blades. During the mixing cycle the screw moves the material within the reach of the mixing blades, thus accelerating the mixing process. At discharge time, the screw extrudes the finished material through a die opening in the end of the machine. The discharge screw is driven independently of the mixer blades.

INTENSIVE MIXERS

Banbury Mixers The dominant high-intensity mixer, with power input up to 6000 kW/m³ (30 hp/gal), is the Banbury mixer made by Farrel Co. (Fig. 18-51). It is used primarily in the plastics and rubber industries. The batch charge of material is forced into the mixing chamber by an air-operated ram at the top of the mixer. The clearance between the rotors and the walls is extremely small. The mixing action takes place in that small gap. The rotors of the Banbury mixer operate at different speeds, so one rotor can drag material against the rear of the other and thus clean ingredients from behind and between the rotors.

The extremely high power consumption of these machines, which operate at speeds of 40 rpm or less, requires large-diameter shafts. The combination of heavy shafts, stubby blades, close clearances, and

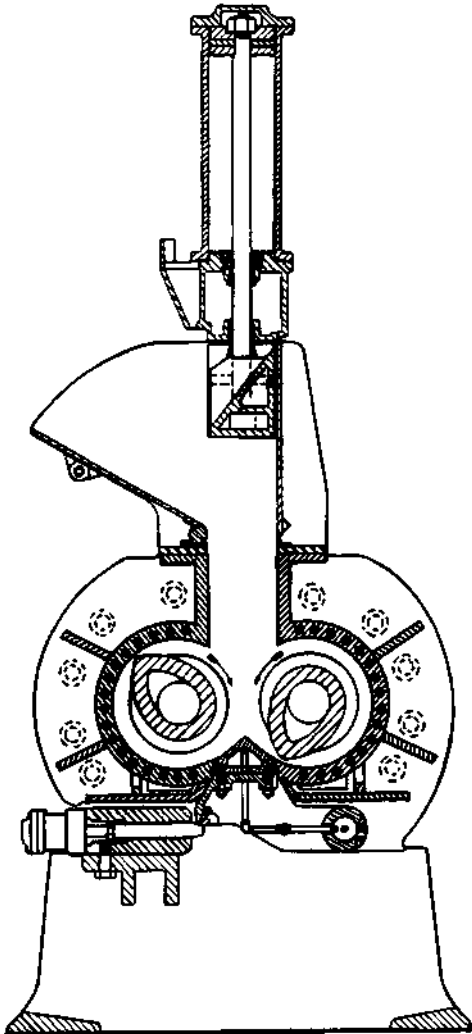


FIG. 18-51 Banbury mixer. (Farrel Co.)

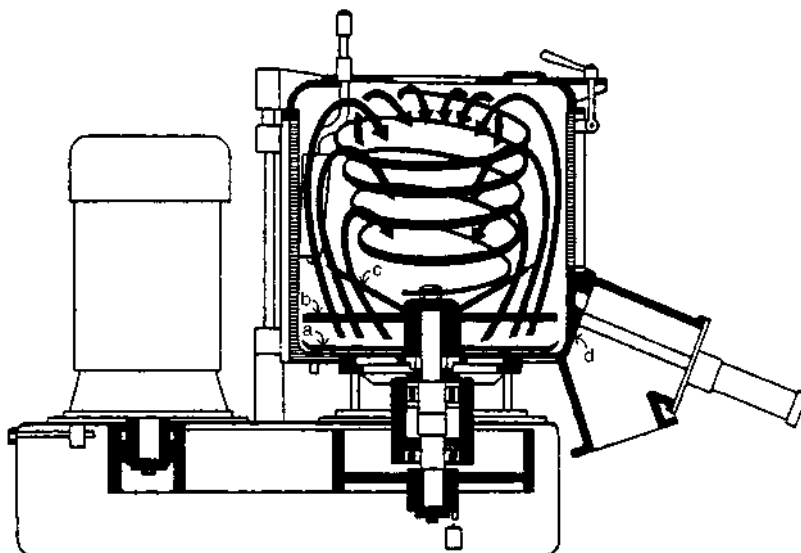


FIG. 18-52 High-intensity mixer: (a) bottom scraper; (b) fluidizing tool; (c) horn tool; (d) flush-mounted discharge valve. (*Henschel Mixers America, Inc.*)

a confined charge limits the Banbury mixer to small batches. The production rate is increased as much as possible by using powerful drives and rotating the blades at the highest speed that the material can tolerate without degradation. The heat added by the high-power input often limits operating conditions because of temperature limits on the material being mixed. Equipment is available from laboratory size to a mixer that can handle a 450-kg (1000-lb) charge and applying 2240 kW (3000 hp).

High-Intensity Mixers Mixers, such as the one shown in Fig. 18-52, combine a high-shear zone with a fluidized vortex for mixing of pastes and powders. Blades at the bottom of the vessel scoop the material upward with peripheral speeds of about 40 m/s (130 ft/s). The high-shear stresses between the blade and the bowl, along with blade impact, easily reduce agglomerates and create an intimate dispersion of powders and liquids. Because the energy input is high, 200 kW/m³ (8 hp/ft³), even powdery material can heat rapidly.

These mixers are particularly suited for the rapid mixing of powders and granules with liquids, for dissolving resins or solids in liquids, or for removal of volatiles from pastes under a vacuum. Scale-up is usually based on constant peripheral speed of the impeller.

Roll Mills Roll mills can provide extremely high localized shear while retaining extended surface area for temperature control. A typical roll mill has two parallel rolls mounted in a heavy frame with provisions for accurately regulating the pressure and distance between the rolls. Since one pass between the rolls does only a little blending, the mills are usually used as a series of mixers. Only a small amount of material is in the high-shear zone at a time, thus allowing time and exposure for cooling.

To increase the shearing action, the rolls are usually operated at different speeds. The material passing between the rolls can be returned to the feed by the rotation of the rolls. If the rolls are at different temperatures, the material will usually stick to the hotter roll and return to the feed point as a thick layer.

At the end of a period of batch mixing, heavy materials may be discharged by simply dropping from between the rolls. Thin, lighter mixes may be removed by a scraper bar pressing against the descending surface of one of the rolls. Roll mixers are used primarily for preparing color pastes for inks, paints, and coatings. A few applications in heavy-duty blending of rubber stocks use corrugated rolls for masticating the material.

Miscellaneous Batch Mixers Many mixers used for solids blending (Sec. 19 of seventh edition) are suitable for liquid-solids blending.

Some solids processing applications involve the addition of liquids, and the same blenders may transition from dry powders to cohesive pastes.

Ribbon blenders typically have multiple helical ribbons with opposing pitches operating in a horizontal trough with a half-cylinder bottom. These mixers can be used for wetting or coating a powder. The final product may have a paste consistency, but must remain at least partially flowable for removal from the blender.

Plowshare mixers have plow-shaped blades mounted at the ends of arms on a horizontal rotating shaft in a cylindrical chamber. The shaft rotates at a sufficient speed to toss the material into the free space in the vessel. The angled surfaces of the plow-shaped blades provide additional intermixing and blending in the bed of solids. High-speed (3600-rpm) chopper blades mounted in the lower side of the mixing chamber can disperse fine particles or break agglomerates. Mixers are available in sizes from 0.03- to 30-m³ (1.0- to 1000-ft³) working capacity. Plowshare mixers can be used for either batch or continuous processing.

Conical mixers are also known as Nauta mixers (Fig. 18-53). Material placed in the conical bin is lifted by the rotation of the helical screw, which in turn is rotated around the wall of the cone. The lifting actions of the screw combined with motion around the cone provide bulk mixing for flowable dry powders, paste materials, and even viscous fluids. The specific energy input is relatively small, and the large volume of the mixers can even provide storage capacity. The mixers may have multiple screws, tapered screws, and high-speed dispersers for different applications. At constant speed, both the mixing time and power scale up with the square root of volume. Sizes from 0.1 to 20 m³ (3.3 to 700 ft³) are available.

Pan mullers are the modern industrial equivalent of the traditional mortar and pestle. Typical mullers have two broad wheels (M1 and M2) on an axle (Fig. 18-54). The mixer rotates about the approximate midpoint of the axle, so that the wheels both rotate and skid over the bottom of the mixing chamber (A). Plow blades (P1 and P2), which rotate with the mixer, push material from the center (T) and walls (C) of the mixing chamber into the path of the rollers. The mixing action combines both crushing and shearing to break lumps or agglomerates and evenly distribute moisture.

Mullers can be used if the paste is not too fluid or sticky. The main application of muller mixers is now in the foundry industry to mix small amounts of moisture and binder with sand for both core and molding sand. Muller mixers also handle such diverse materials as

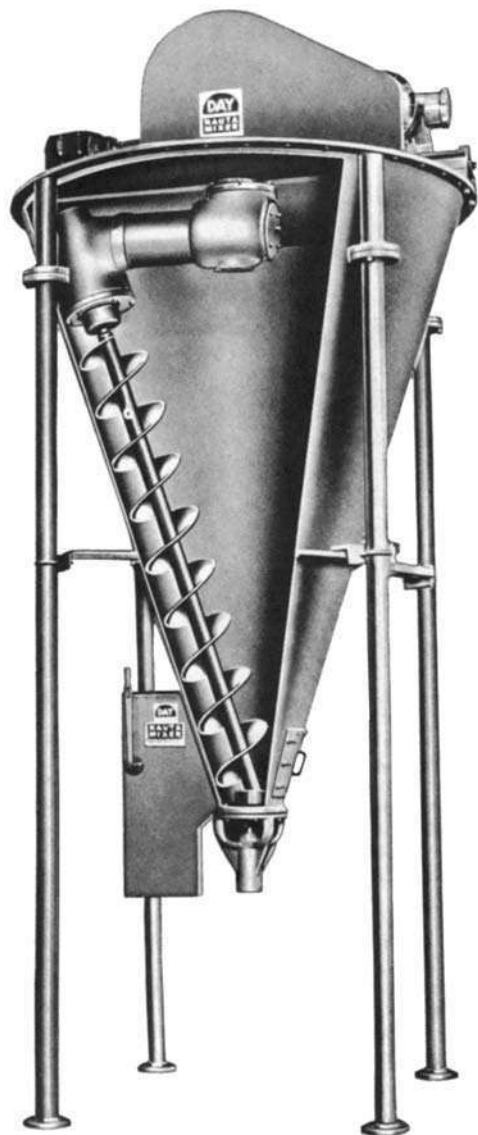


FIG. 18-53 Day Nauta conical mixer. (Littleford Day, Inc.)

clay, storage-battery paste, welding-rod coatings, and chocolate coatings. Standard muller mixers range in capacity from 0.01 to 1.7 m³ (0.4 to 60 ft³), with power requirements from 0.2 to 56 kW (1/3 to 75 hp).

A continuous muller design employs two intersecting and communicating chambers, each with its own mullers and plows. At the point of intersection of the two chambers, the outside plows give an approximately equal exchange of material from one chamber to the other. Material builds in the first chamber until the feed rate and the discharge rate of the material are equal. The quantity of material in the muller is regulated by adjusting the outlet gate.

CONTINUOUS MIXERS

Some batch mixers previously described can be modified for continuous processing. Product uniformity may be limited because of broad residence time distributions. If ingredients can be accurately metered,

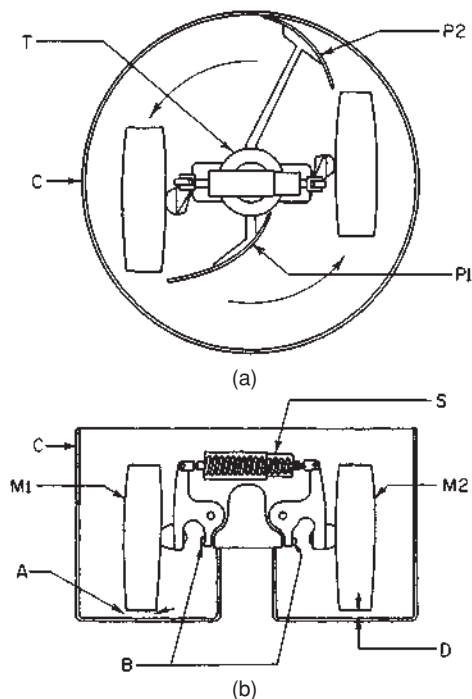


FIG. 18-54 Pan muller: (a) plan view; (b) sectional elevation. [Bullock, Chem. Eng. Prog. 51: 243 (1955); by permission.]

which can be a problem with powdered or viscous materials, several continuous mixers are available. Continuous mixers often consist of a closely fitting agitator element rotating within a stationary housing.

Single-Screw Extruders The use of extruders, like the one shown in Fig. 18-55, is widespread in the plastic industries. The quality and utility of the product often depend on the uniformity of additives, stabilizers, fillers, etc. A typical extruder combines the process functions of melting the base resin, mixing in additives, and developing the pressure required for shaping the product into pellets, sheet, or profiles. Dry ingredients, sometimes premixed in a batch blender, are fed into the feed throat where the channel depth is deepest. As the root diameter of the screw is increased, the plastic is melted by a combination of friction and heat transfer from the barrel. Shear forces can be very high, especially in the melting zone. The mixing is primarily a laminar shear action.

Single-screw extruders can be built with a long length-to-diameter ratio to permit sufficient space and residence time for a sequence of process operations. Capacity is determined by diameter, length, and power. Most extruders are in the 25- to 200-mm-diameter range. Larger units have been made for specific applications, such as polyethylene homogenization. Mixing enhancers (Fig. 18-56) are used to provide both elongation and shearing action to enhance dispersive (axial) and distributive (radial) mixing.

The maximum power (P in kilowatts) supplied for single-screw extruders varies with the screw diameter (D in millimeters) approximately as

$$P = 5.3 \times 10^{-3} D^{2.25} \quad (18-24)$$

The energy required for most polymer mixing applications is from 0.15 to 0.30 kWh/kg (230 to 460 Btu/lb).

Twin-Screw Extruders Two screws in a figure-eight barrel have the advantage of interaction between the screws plus action between the screws and the barrel. Twin-screw extruders are used to melt continuously, mix, and homogenize different polymers and additives. Twin-screw extruders can also be used to provide the intimate

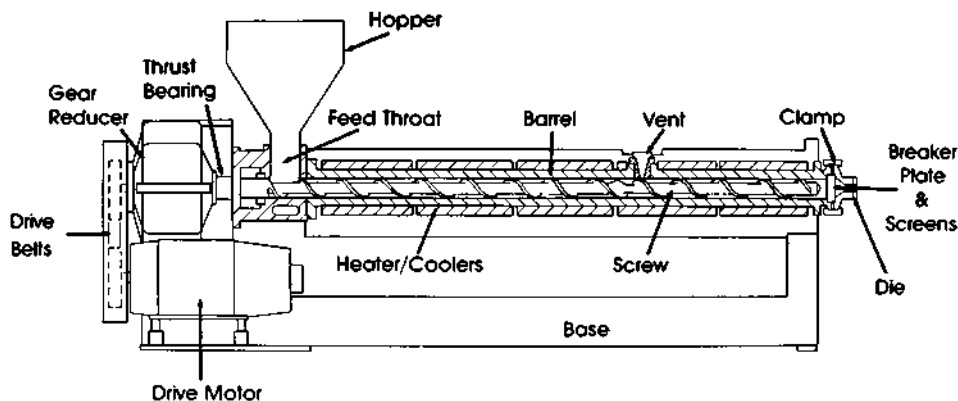


FIG. 18-55 Single-screw extruder. (Davis Standard.)

mixing needed to carry out chemical reactions in high-viscosity materials. The screws can be either tangential or intermeshing, with the latter either corotating or counterrotating. Tangential designs allow variability in the channel depth and permit longer lengths.

The most common twin-screw extruder is the counterrotating intermeshing type. The counterrotating intermeshing screws provide a dispersive milling action between the screws and the ability to generate pressure efficiently. The two keyed or splined shafts are fitted with pairs of slip-on kneading or conveying elements, as shown in Fig. 18-57. Each pair of kneading paddles causes an alternating compression and expansion effect that massages the contents and provides a combination of shearing and elongational mixing actions. The arrays of elements can

be varied to provide a wide range of mixing effects. The barrel sections are also segmented to allow for optimum positioning of feed ports, vents, barrel valves, etc. The barrels may be heated electrically or with oil or steam and cooled with air or water.

Counterrotating twin-screw extruders are available in diameters ranging from 15 to 300 mm (0.5 to 12 in), with length-to-diameter ratios up to 50 and throughput capacities to 7 kg/s (55,000 lb/h). Screw speeds can be as high as 8 r/s (500 rpm) in small production extruders. Residence times for melting are usually less than 120 s (2 min).

Farrel Continuous Mixer The Farrel mixer (Fig. 18-58) consists of rotors similar in cross section to the Banbury batch mixer. The first section of the rotor acts as a screw conveyor, moving the feed ingredients into the mixing section. The mixing action is a combination of intensive shear between the rotor and chamber wall, kneading between the rotors, and a rolling action within the material itself. The amount and quality of mixing are controlled by adjustment of speed, feed rate, and discharge orifice opening. Mixers are available with chamber volumes up to 0.12 m^3 (4.2 ft³). With speeds to 3.3 r/s (200 rpm), the power range is from 5 to 2200 kW (7.5 to 3000 hp).

Miscellaneous Continuous Mixers Because of the diversity of material properties and process applications involving viscous fluids, pastes, and doughs, the types of mixers are almost as diverse.

Trough-and-screw mixers usually consist of a single rotor or twin rotors that continually turn the feed material over as it progresses toward the discharge end of the mixer. Some mixers have been designed with extensive heat-transfer surface area. The continuous-screw, **Holo-Flite processor** (Fig. 18-59) is used primarily for heat transfer, since the hollow screws provide extended surfaces without creating much shear. Two or four screws may be used.

Another type of trough-and-screw mixer is the **AP Conti paste mixer**, shown in Fig. 18-60. These self-cleaning mixers are particularly appropriate when the product being handled goes through a sticky stage, which could plug the mixer or foul the heat-transfer surfaces.

Pug mills have one or two shafts fitted with short heavy paddles, mounted in a cylinder or trough holding the material to be processed. In the two-shaft mills the shafts are parallel and may be either horizontal or vertical. The paddles may or may not intermesh. Clearances are wide, so considerable mass mixing takes place. Unmixed or partially mixed ingredients are fed at one end of the machine, which is usually totally enclosed. Liquid may be added to the material entering the mixer. The paddles push the material forward as they cut through it. The action of the paddles carries the material toward the discharge end of the mixer. The product may discharge through one or two open ports or through extrusion nozzles. The nozzles create roughly shaped continuous strips of material. Automatic cutters may be used to make blocks or pellets from the strips. Pug mills are most often used for mixing mineral or clay products.

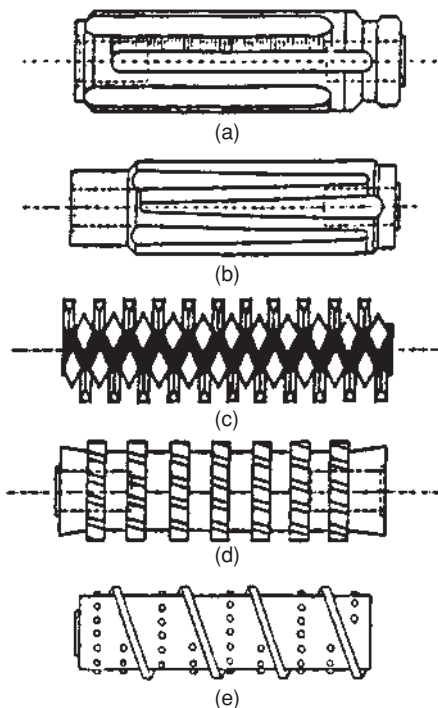


FIG. 18-56 Mixing enhancers for single-screw extruders: (a) Maddock, straight; (b) Maddock, tapered; (c) pineapple; (d) gear; (e) pin.

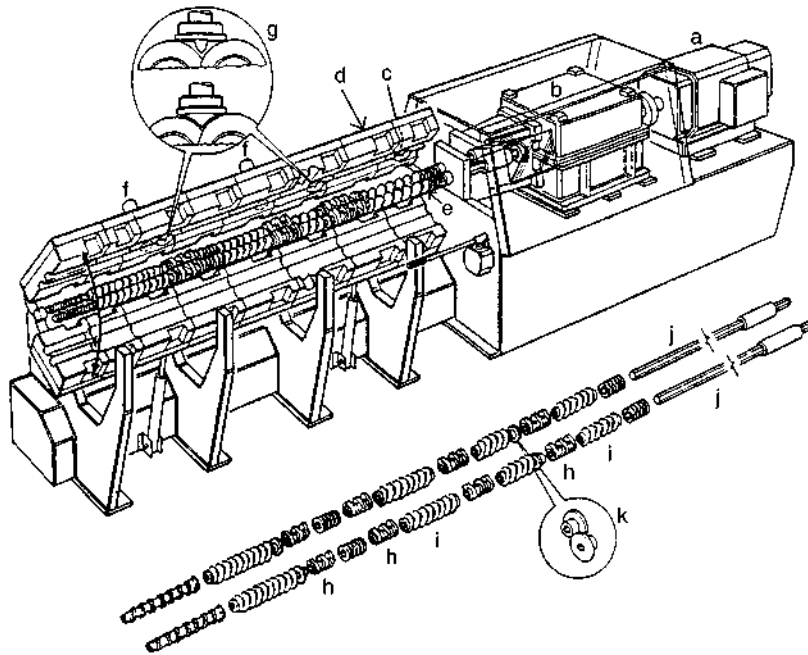


FIG. 18-57 Intermeshing corotating twin-screw extruder: (a) drive motor; (b) gearbox; (c) feed port; (d) barrel; (e) assembled rotors; (f) vent; (g) barrel valve; (h) kneading paddles; (i) conveying screws; (j) splined shafts; (k) blister rings. (APV Chemical Machinery, Inc.)

Motionless mixers are an alternative to rotating impeller mixers. Motionless or static mixers use stationary shaped elements inside pipes or conduits to divide, divert, twist, and recombine flowing material. The dividing, stretching, and recombining processes lead to thinner and thinner striations in viscous materials to achieve uniformity.

The twisted-element mixers, such as the **Kenics static mixer** (Fig. 18-61), create 2^n layers in n divisions. Each element twists the flow, moving material from the center to the wall and from the wall to the center. The twisting also stretches striations having different properties and reorients the material before the next division. The following element

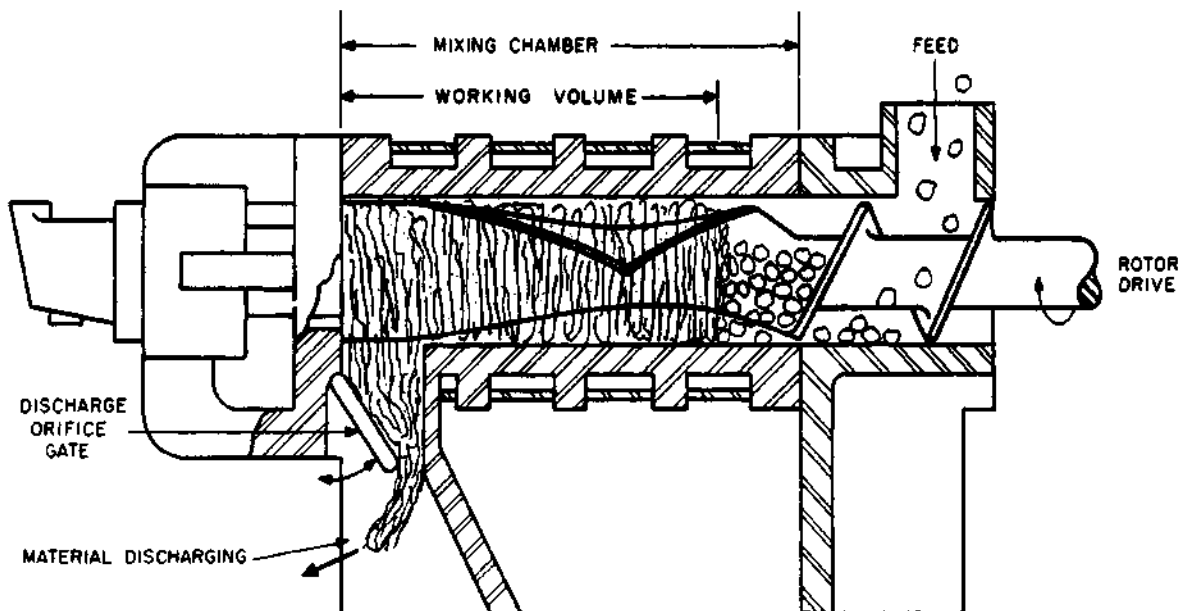


FIG. 18-58 Farrel continuous mixer. (Farrel Co.)

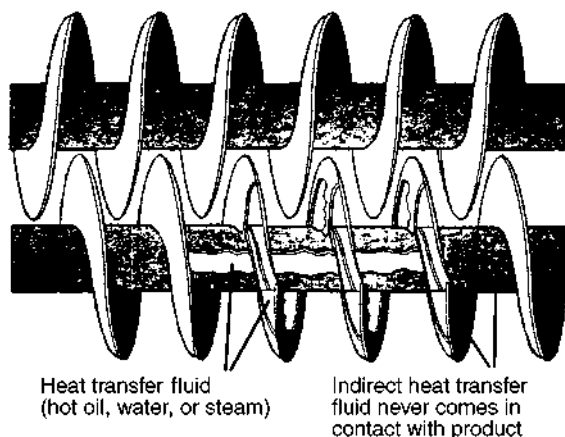


FIG. 18-59 Holo-Flite Processor. (Metso Minerals.)

twists the divided material in the opposite direction. The more viscous the material, the more mixing elements are required for uniformity.

Other motionless designs, such as the **Sulzer static mixer** (Fig. 18-62), accomplish mixing by making multiple divisions at each element transition. The flowing material follows a wavy path to stretch and distort the striations. The number of divisions and distorted paths causes more rapid mixing, but at the expense of a greater pressure drop per unit length of the mixer.

The power required to accomplish mixing in a motionless mixer is provided by the pump used to force the fluid through the mixer. The pressure drop through a motionless mixer is usually expressed as a multiplier K of the open pipe loss or as a valve coefficient C_v . The value of the multiplier is strongly dependent on the detail geometry of the mixer, but is usually available through information from the supplier. Fluid properties are taken into account by the value of the Reynolds

number for the open pipe. Motionless mixers are usually sized to match the diameter of the connecting pipe. Pumping adjustments are made when necessary to handle the increased pressure drop.

Because motionless mixers continuously interchange fluid between the walls and the center of the conduit, they also provide good heat transfer, especially with the twisted-element style of mixers. Sometimes, high-viscosity heat exchange is best accomplished with a static mixer.

Distributive (radial) mixing is usually excellent; dispersive (axial) mixing is often poor. The result can be a good plug-flow mixer or reactor, with corresponding benefits and limitations.

PROCESS DESIGN CONSIDERATIONS

Scale-up of Batch Mixers While a desirable objective of scale-up might be equal blending uniformity in equal time, practicality dictates that times for blending are longer with larger batches. Scale-up of many processes and applications can be successfully done by holding constant the peripheral speed of the rotating element in the mixer. Equal peripheral speed, often called *equal tip speed*, essentially means that the maximum velocity in the mixer remains constant.

Perhaps one of the most difficult concepts to grasp about viscous mixing is that, unlike in turbulent mixing, greater mixer speed does not always translate to better mixing results. If a rotating mixer blade cuts through a viscous fluid or heavy paste too quickly, the stretching process that reduces striation thickness does not take place throughout the material. At high rotational speeds, rapid shearing between a blade tip and the wall or housing may take place, but flow to create bulk motion may not have time to occur. Thus, slower speeds may actually give better mixing results.

With geometric similarity, equal tip speed means that velocity gradients are reduced and blend times become longer. However, power per volume is also reduced, and viscous heating problems are likely to be more controllable. With any geometric scale-up, the surface-to-volume ratio is reduced, which means that any internal heating, whether by viscous dissipation or chemical reaction, becomes more difficult to remove through the surface of the vessel.

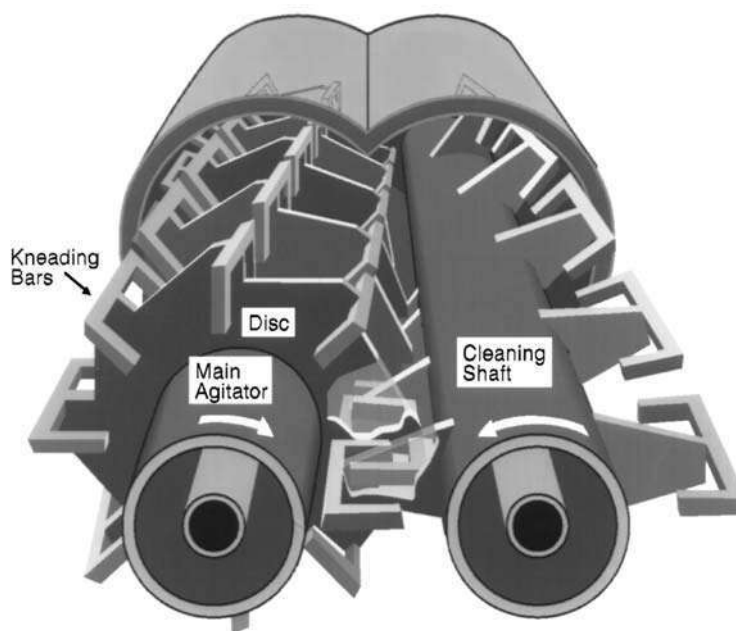


FIG. 18-60 AP Conti paste mixer. (LIST, Inc.)



FIG. 18-61 Kenics static mixer. (Chemineer, Inc.)

In many applications the blend time is closely related to the actual number of revolutions made by the mixing device. Thus if mixing were successfully accomplished in 5 min at 60 rpm in a small mixer, the same uniformity could be achieved in 10 min at 30 rpm in a larger mixer. Other factors, such as the rate of heating, could limit scale-up and mixing times.

The physical properties of a paste are difficult to define because a combination of yield stress, shear dependence, time dependence, and even elasticity may be present. Further, many process applications involve the formation or modification of the physical properties. To relate accurately specific material properties to mixing characteristics or power requirements can be extremely difficult. Actual observation and measurement in small-scale equipment or comparison with similar existing processes may be the only practical way of predicting successful operating conditions. Power measurements in small-scale equipment are often essential to predict large-scale conditions and may form the basis for operating production equipment.

Scale-up of Continuous Mixers While geometric similarity may be practical for most batch mixers, changes in the length-to-diameter ratio or other geometry may be necessary with continuous mixers. The most common problem is heat generation by friction and heat removal by surface transfer.

In single-screw extruders, e.g., Fig. 18-55, channel depth in the flights cannot be increased in proportion to the screw diameter because the distribution of heat generated by friction at the barrel wall requires more time as the channel depth becomes greater. With constant retention time, therefore, a nonhomogeneous product would be discharged from a geometrically similar large-scale extruder.

As the result of the departure from geometric similarity, the throughput rate of single-screw extruders scales up with the diameter to 2.0 to 2.5 power, instead of the diameter cubed, at constant length-to-diameter and screw speed. The throughput rates of twin-screw extruders (Fig. 18-57) and the Farrel continuous mixer (Fig. 18-58) are scaled up with the diameter to about the 2.6 power.

The extent of axial dispersion through a continuous mixer can be characterized either by an axial diffusion coefficient or by analogy to a number of well-mixed stages in series. Retention time can control the performance of a mixing system. As the number of apparent stages increases, the greater is the assurance that all the material will have the required residence time. Under conditions requiring uniform retention time, the feed streams must enter at the correct ratio on a time scale much shorter than the average residence time of the mixer. Otherwise, variations in the feed will appear as changes in the product. Different types of continuous mixers have different degrees of axial dispersion. Thus, appropriate feed conditions must be considered. Single-screw extruders have an equivalent number of stages equal to approximately one-half the length-to-diameter ratio.

HEATING AND COOLING MIXERS

Heat Transfer Pastes and viscous fluids are often heated or cooled by heat transfer through the walls of the mixing container or hollow mixing arms. A uniform temperature throughout the bulk material is almost as important for good heat transfer as a large heat-transfer surface to mixer volume ratio. Bulk temperature uniformity

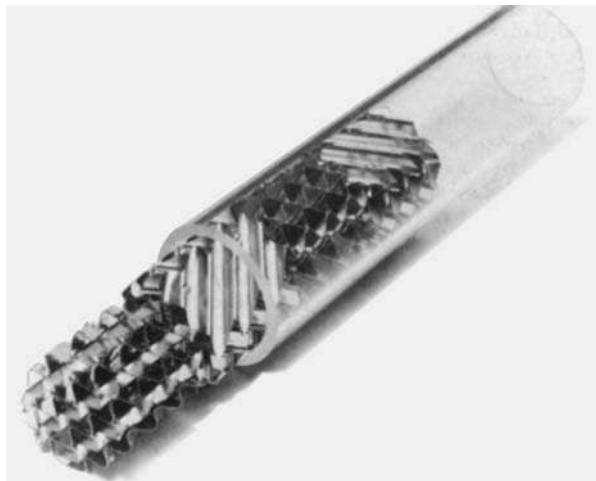


FIG. 18-62 Sulzer static mixer. (Sulzer Chemtech.)

will maximize the temperature-difference driving force for heat transfer. Surface area is a direct factor in overall heat transfer. Effective motion near the surface promotes convection over conduction for better heat transfer. Most mixers for pastes or viscous fluids have some sort of scraper or close-clearance device to move stagnant material away from heat-transfer surfaces.

Typical overall heat-transfer coefficients are between 20 and 200 $\text{J}/(\text{m}^2 \cdot \text{s} \cdot \text{K})$ [4 to 35 $\text{Btu}/(\text{h} \cdot \text{ft}^2 \cdot ^\circ\text{F})$].

Heating Methods Steam heating is widely used because it is economical, safe, and easily controlled. The mixer shell must be designed to withstand both the positive pressure of steam and a vacuum caused when the steam condenses. Transfer liquid heating, using water, oil, special organic liquids, or molten salts, permits good temperature control and provides insurance against overheating the process material. Jackets for transfer liquids are usually baffled to provide good circulation. Higher temperatures can be achieved without the heavy vessel construction required by steam pressures.

Electric heating requires that the elements be electrically insulated from the vessel, while still providing good thermal contact. The heaters must be designed for uniform heating to avoid creating hot spots. Temperature control can be precise, maintenance costs low, but utility costs can be very high for large mixers. Electrical heating may be excluded when flammable vapors or dusts are present.

Friction or viscous heating develops rapidly in some mixers, such as a Banbury mixer. The first temperature rise may be beneficial in softening the materials and accelerating chemical reactions. Because energy inputs can be high, higher temperatures detrimental to the products may develop rapidly. So cooling may be required during other portions of a process.

Cooling Methods Air cooling with air blown over external surfaces or external fins may be sufficient for some mixers. Evaporation of excess water or solvent under a vacuum or ambient pressure provides good cooling. A small amount of evaporation produces a large amount of cooling. However, removing too much solvent may damage the product. Some mixers are cooled by circulation of water or refrigerants through jackets or hollow agitators. With viscous fluids, lower temperatures near the cooled surfaces increase viscosity and make heat transfer more difficult.

EQUIPMENT SELECTION

The most common and sometimes the only available approach is by analogy. Many companies manufacture similar products, either of their own or those of competitors. With similar products, both good and bad features of existing or typical mixing equipment need to be considered carefully. Some types of mixing equipment are commonly

used throughout certain industries. Sometimes existing equipment can be adapted to a new process. Otherwise, new equipment will be needed.

If new equipment is needed, laboratory or pilot-plant studies are recommended. Often unique product features involve unusual or special fluid properties, which makes prediction of mixer performance almost impossible. The objective is to find potentially suitable equipment and test available mixers. Most equipment vendors have equipment to rent or a demonstration laboratory to test their mixers.

The following list provides some characteristics of a new process that must be considered:

1. List all materials in the process and describe their characteristics.
 - a. Method of delivery to the mixer: bags, drums, tote sacks, bulk, pipeline, etc.
 - b. Storage and/or weighing requirements at the mixer
 - c. Physical form of the material
 - d. Specific gravity and bulk characteristics
 - e. Particle size or size range
 - f. Viscosity
 - g. Melting, boiling, or degradation point
 - h. Corrosive properties
 - i. Abrasive characteristics
 - j. Toxicity
 - k. Fire or explosion hazards
 - l. Irritant characteristics, to skin, eyes, or lungs
 - m. Sensitivity of materials when exposed to air, moisture, or heat
2. List pertinent information related to production.
 - a. Quantity to be produced per batch
 - b. Formulation and order of addition
 - c. Analysis required
 - d. Cleaning requirements between batches or products
 - e. Preceding and/or following process steps
 - f. Any changes in physical state during process

- g. Any chemical reactions—exothermic or endothermic
- h. Temperature requirements
- i. Physical form of final product
- j. Removal of product from mixer—pumping or gravity flow through piping, chute, or dumping
3. Describe the controlling features of the finished product.
 - a. Degree of uniformity: solution, aggregates, particle size, etc.
 - b. Stability of emulsion or dispersion
 - c. Ultimate color requirements
 - d. Uniformity of active ingredients, as in a pharmaceutical product
 - e. Degree of moisture content control

Preparation and Addition of Materials To ensure product quality and productivity, ingredient preparation is important. Order of addition, method and rate of addition, and even preprocessing must be considered.

Some finely powdered materials, such as carbon black or silica, contain a lot of air. If possible, such materials should be compacted, wetted, or agglomerated before addition to the mixture. Air bubbles can be extremely difficult to remove from viscous materials. Holding the product under a vacuum may help release some air or trapped gases. The presence of air in the product may make packaging difficult and may even cause eventual degradation of the product.

Critical ingredients, such as vulcanizers, antioxidants, surfactants, and active agents, are often present in small proportions. If these materials form lumps or aggregates, milling or screening of the materials may be necessary to ensure a uniform product. If small ingredients are soluble in liquid ingredients, adding them as a solution may improve blending. Master batching small quantities of an ingredient into part of a major ingredient often simplifies mixing and makes a more uniform product.

Additional considerations, such as automatic weighing, feed control, liquid metering, and automatic control, may be essential for continuous processes.

CRYSTALLIZATION FROM SOLUTION

GENERAL REFERENCES: *AIChE Testing Procedures: Crystallizers*, American Institute of Chemical Engineers, New York, 1970; *Eaporators*, 1961. Bennett, *Chem. Eng. Prog.*, **58**(9), 76 (1962). Buckley, *Crystal Growth*, Wiley, New York, 1951. Campbell and Smith, *Phase Rule*, Dover, New York, 1951. De Jong and Jancic (eds.), *Industrial Crystallization*, North-Holland Publishing Company, Amsterdam, 1979. "Crystallization from Solution: Factors Influencing Size Distribution," *Chem. Eng. Prog. Symp. Ser.*, **67**(110), (1971). Mullin (ed.), *Industrial Crystallization*, 4th ed., Butterworth-Heinemann, Boston, 2001. Mersmann (ed.), *Crystallization Technology Handbook*, Marcel Dekker, New York, 1995. Jancic and Grootscholten, *Industrial Crystallization*, D. Reidel Publishing, Boston, 1984. Jones, *Crystallization Process Systems*, Butterworth-Heinemann, Boston, 2002. Genck, *Chem. Eng. Prog.* **100** (10), 26 (2004). Newman and Bennett, *Chem. Eng. Prog.*, **55**(3), 65 (1959). Palermo and Larson (eds.), "Crystallization from Solutions and Melts," *Chem. Eng. Prog. Symp. Ser.*, **65**(95), (1969). Randolph (ed.), "Design, Control and Analysis of Crystallization Processes," *Am. Inst. Chem. Eng. Symp. Ser.*, **76**(193), (1980). Randolph and Larson, *Theory of Particulate Processes*, Academic, New York, 2d ed., 1988. Seidell, *Solubilities of Inorganic and Metal Organic Compounds*, American Chemical Society, Washington, 1965.

Crystallization is important as an industrial process because of the number of materials that are and can be marketed in the form of crystals. Its wide use is due to the highly purified and favorable form of a chemical solid which can be obtained from relatively impure solutions in a single processing step. In terms of energy requirements, crystallization requires much less energy for separation than do distillation and other commonly used methods of purification. In addition, it can be performed at relatively low temperatures and on a scale which varies from a few grams up to thousands of tons per day.

Crystallization may be carried out from a vapor, from a melt, or from a solution. Most of the industrial applications of the operation involve crystallization from solutions. Nevertheless, crystal solidification of metals is basically a crystallization process, and much theory has been developed in relation to metal crystallization. This topic is highly specialized, and is outside the scope of this subsection, which is limited to crystallization from solution.

PRINCIPLES OF CRYSTALLIZATION

Crystals A crystal may be defined as a solid composed of atoms or molecules arranged in an orderly, repetitive array. The interatomic distances in a crystal of any definite material are constant and are characteristic of that material. Because the pattern or arrangement of the atoms or molecules is repeated in all directions, there are definite restrictions on the kinds of symmetry that crystals can possess.

There are five main types of crystals, and these types have been arranged into seven crystallographic systems based on the crystal interfacial angles and the relative length of its axes. The treatment of the description and arrangement of the atomic structure of crystals is the science of **crystallography**. The material in this discussion will be limited to a treatment of the growth and production of crystals as a unit operation.

Solubility and Phase Diagrams Equilibrium relations for crystallization systems are expressed in the form of solubility data which are plotted as phase diagrams or solubility curves. Solubility data are ordinarily given as parts by weight of anhydrous material per 100 parts by weight of total solvent. In some cases these data are reported as parts by weight of anhydrous material per 100 parts of solution. If water of crystallization is present in the crystals, this is indicated as a separate phase. The concentration is normally plotted as a function of temperature and has no general shape or slope. It can also be reported as a function of pressure, but for most materials the change in solubility with change in pressure is very small. If there are two components in solution, it is common to plot the concentration of these two components on the X and Y axes and represent the solubility by isotherms. When three or more components are present, there are various techniques for depicting the solubility and phase relations in both three-dimension and two-dimension models. For a description of these techniques, refer to Campbell and Smith (loc. cit.). Shown in Fig. 18-63 is a phase diagram for magnesium sulfate in water. The line *p-a*

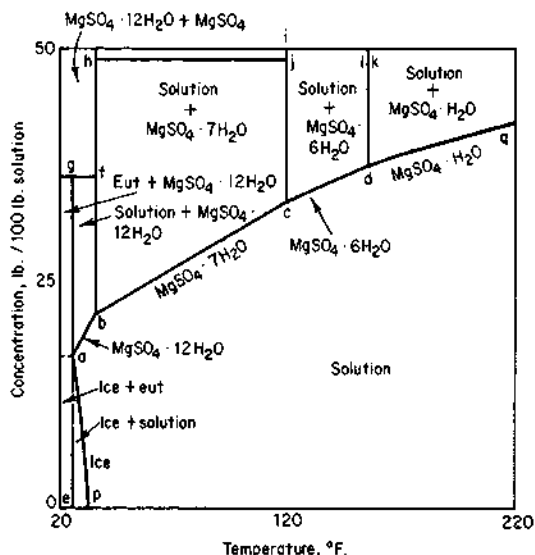


FIG. 18-63 Phase diagram, $\text{MgSO}_4\text{-H}_2\text{O}$. To convert pounds to kilograms, divide by 2.2; $K = (^\circ\text{F} + 459.7)/1.8$.

represents the freezing points of ice (water) from solutions of magnesium sulfate. Point *a* is the eutectic, and the line *a-b-c-d-q* is the solubility curve of the various hydrates. Line *a-b* is the solubility curve for $\text{MgSO}_4\cdot 12\text{H}_2\text{O}$, *b-c* is the solubility curve for $\text{MgSO}_4\cdot 7\text{H}_2\text{O}$, *c-d* is the solubility curve for $\text{MgSO}_4\cdot 6\text{H}_2\text{O}$, and *d-q* is the portion of the solubility curve for $\text{MgSO}_4\cdot \text{H}_2\text{O}$.

As shown in Fig. 18-64, the mutual solubility of two salts can be plotted on the *X* and *Y* axes with temperatures as isotherm lines. In the example shown, all the solution compositions corresponding to 100°C with solid-phase sodium chloride present are shown on the line *DE*. All the solution compositions at equilibrium with solid-phase KCl at 100°C are shown by the line *EF*. If both solid-phase KCl and NaCl are present, the solution composition at equilibrium can only be represented by point *E*, which is the invariant point (at constant pressure). Connecting all the invariant points results in the mixed-salt line. The locus of this line is an important consideration in making phase separations.

There are numerous solubility data in the literature; the standard reference is by Seidell (loc. cit.). Valuable as they are, they nevertheless must be used with caution because the solubility of compounds

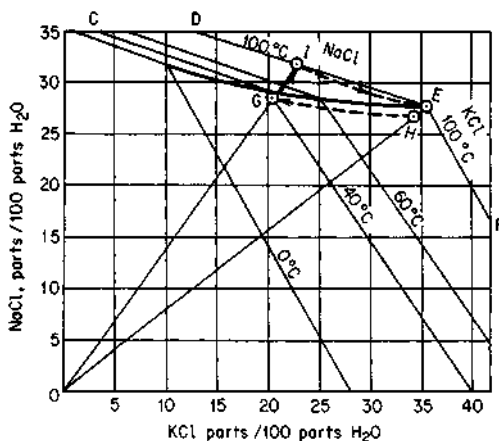


FIG. 18-64 Phase diagram, $\text{KCl-NaCl-H}_2\text{O}$. $K = ^\circ\text{C} + 273.2$.

is often influenced by pH and/or the presence of other soluble impurities which usually tend to depress the solubility of the major constituents. While exact values for any system are frequently best determined by actual composition measurements, the difficulty of reproducing these solubility diagrams should not be underestimated. To obtain data which are readily reproducible, elaborate pains must be taken to be sure the system sampled is at equilibrium, and often this means holding a sample at constant temperature for a period of from 1 to 100 h. While the published curves may not be exact for actual solutions of interest, they generally will be indicative of the shape of the solubility curve and will show the presence of hydrates or double salts.

Heat Effects in a Crystallization Process The heat effects in a crystallization process can be computed by two methods: (1) a heat balance can be made in which individual heat effects such as sensible heats, latent heats, and the heat of crystallization can be combined into an equation for total heat effects; or (2) an enthalpy balance can be made in which the total enthalpy of all leaving streams minus the total enthalpy of all entering streams is equal to the heat absorbed from external sources by the process. In using the heat-balance method, it is necessary to make a corresponding mass balance, since the heat effects are related to the quantities of solids produced through the heat of crystallization. The advantage of the enthalpy-concentration-diagram method is that both heat and mass effects are taken into account simultaneously. This method has limited use because of the difficulty in obtaining enthalpy-concentration data. This information has been published for only a few systems.

With compounds whose solubility increases with increasing temperature there is an absorption of heat when the compound dissolves. In compounds with decreasing solubility as the temperature increases, there is an evolution of heat when solution occurs. When there is no change in solubility with temperature, there is no heat effect. The solubility curve will be continuous as long as the solid substance of a given phase is in contact with the solution, and any sudden change in the slope of the curve will be accompanied by a change in the heat of solution and a change in the solid phase. Heats of solution are generally reported as the change in enthalpy associated with the dissolution of a large quantity of solute in an excess of pure solvent. Tables showing the heats of solution for various compounds are given in Sec. 2.

At equilibrium the heat of crystallization is equal and opposite in sign to the heat of solution. Using the heat of solution at infinite dilution as equal but opposite in sign to the heat of crystallization is equivalent to neglecting the heat of dilution. With many materials the heat of dilution is small in comparison with the heat of solution and the approximation is justified; however, there are exceptions. Relatively large heat effects are usually found in the crystallization of hydrated salts. In such cases the total heat released by this effect may be a substantial portion of the total heat effects in a cooling-type crystallizer. In evaporative-type crystallizers the heat of crystallization is usually negligible when compared with the heat of vaporizing the solvent.

Yield of a Crystallization Process In most cases the process of crystallization is slow, and the final mother liquor is in contact with a sufficiently large crystal surface so that the concentration of the mother liquor is substantially that of a saturated solution at the final temperature in the process. In such cases it is normal to calculate the yield from the initial solution composition and the solubility of the material at the final temperature. If evaporative crystallization is involved, the solvent removed must be taken into account in determining the final yield. If the crystals removed from solution are hydrated, account must be taken of the water of crystallization in the crystals, since this water is not available for retaining the solute in solution. The yield is also influenced in most plants by the removal of some mother liquor with the crystals being separated from the process. Typically, with a product separated on a centrifuge or filter, the adhering mother liquor would be in the range of 2 to 10 percent of the weight of the crystals.

The actual yield may be obtained from algebraic calculations or trial-and-error calculations when the heat effects in the process and any resultant evaporation are used to correct the initial assumptions on calculated yield. When calculations are made by hand, it is generally

preferable to use the trial-and-error system, since it permits easy adjustments for relatively small deviations found in practice, such as the addition of wash water, or instrument and purge water additions. The following calculations are typical of an evaporative crystallizer precipitating a hydrated salt. If SI units are desired, kilograms = pounds $\times 0.454$; $K = (^{\circ}\text{F} + 459.7)/1.8$.

Example 2: Yield from a Crystallization Process A 10,000-lb batch of a 32.5 percent MgSO_4 solution at 120°F is cooled without appreciable evaporation to 70°F . What weight of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ crystals will be formed (if it is assumed that the mother liquor leaving is saturated)?

From the solubility diagram in Fig. 18-56 at 70°F the concentration of solids is 26.3 lb MgSO_4 per 100-lb solution.

The mole weight of MgSO_4 is 120.38.

The mole weight of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ is 246.49.

For calculations involving hydrated salts, it is convenient to make the calculations based on the hydrated solute and the "free water."

$$0.325 \text{ weight fraction} \times \frac{246.94}{120.38} = 0.6655 \text{ MgSO}_4 \cdot 7\text{H}_2\text{O in the feed solution}$$

$$0.263 \times \frac{246.94}{120.38} = 0.5385 \text{ MgSO}_4 \cdot 7\text{H}_2\text{O in the mother liquor}$$

Since the free water remains constant (except when there is evaporation), the final amount of soluble $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ is calculated by the ratio of $\frac{0.538 \text{ lb MgSO}_4 \cdot 7\text{H}_2\text{O}}{(1 - 0.538) \text{ lb free water}}$

	Total	$\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$	Free water	$\frac{\text{MgSO}_4 \cdot 7\text{H}_2\text{O}}{\text{Free water}}$
Feed	10,000	6655	3345	1.989
Mother liquor	7249	3904*	3345	1.167
Yield	2751	2751		

$$*3345 \times (0.538/0.462) = 3904$$

A formula method for calculation is sometimes used where

$$P = R \frac{100W_0 - S(H_0 - E)}{100 - S(R - 1)}$$

where P = weight of crystals in final magma, lb
 R = mole weight of hydrate/mole weight of anhydrous = 2.04759
 S = solubility at mother-liquor temperature (anhydrous basis) in lb per 100 lb solvent. $[0.263/(1 - 0.263)] \times 100 = 35.68521$
 W_0 = weight of anhydrous solute in the original batch. $10,000(0.325) = 3250 \text{ lb}$
 H_0 = total weight of solvent at the beginning of the batch. $10,000 - 3250 = 6750 \text{ lb}$
 E = evaporation = 0
 $P = 2.04 \frac{(100)(3250) - 35.7(6750)}{100 - 35.7(2.04 - 1)} = 2751 \text{ lb}$

Note that taking the difference between large numbers in this method can increase the chance for error.

Fractional Crystallization When two or more solutes are dissolved in a solvent, it is often possible to (1) separate these into the pure components or (2) separate one and leave the other in the solution. Whether or not this can be done depends on the solubility and phase relations of the system under consideration. Normally alternative 2 is successful only when one of the components has a much more rapid change in solubility with temperature than does the other. A typical example which is practiced on a large scale is the separation of KCl and NaCl from water solution. A phase diagram for this system is shown in Fig. 18-64. In this case the solubility of NaCl is plotted on the Y axis in parts per 100 parts of water, and the solubility of KCl is plotted on the X axis. The isotherms show a marked decrease in solubility for each component as the amount of the other is increased. This is typical for most inorganic salts. As explained earlier, the mixed-salt line is CE , and to make a separation of the solutes into the pure components it is necessary to be on one side of this line or the other. Normally a 95 to 98 percent approach to this line is possible. When evaporation occurs during a cooling or concentration process, this can be represented by movement away from the origin on a straight line through the origin. Dilution by water is represented by movement in the opposite direction.

A typical separation might be represented as follows: Starting at E with a saturated brine at 100°C a small amount of water is added to dissolve any traces of solid phase present and to make sure the solids precipitated initially are KCl. Evaporative cooling along line HG results in the precipitation of KCl. During this evaporative cooling, part of the water evaporated must be added back to the solution to prevent the coprecipitation of NaCl. The final composition at G can be calculated by the $\text{NaCl/KCl/H}_2\text{O}$ ratios and the known amount of NaCl in the incoming solution at E . The solution at point G may be concentrated by evaporation at 100°C . During this process the solution will increase in concentration with respect to both components until point I is reached. Then NaCl will precipitate, and the solution will become more concentrated in KCl, as indicated by the line IE , until the original point E is reached. If concentration is carried beyond point E , a mixture of KCl and NaCl will precipitate.

Example 3: Yield from Evaporative Cooling Starting with 1000 lb of water in a solution at H on the solubility diagram in Fig. 18-64, calculate the yield on evaporative cooling and concentrate the solution back to point H so the cycle can be repeated, indicating the amount of NaCl precipitated and the evaporation and dilution required at the different steps in the process.

In solving problems of this type, it is convenient to list the material balance and the solubility ratios. The various points on the material balance are calculated by multiplying the quantity of the component which does not precipitate from solution during the transition from one point to another (normally the NaCl in cooling or the KCl in the evaporative step) by the solubility ratio at the next step, illustrated as follows:

Basis. 1000 lb of water at the initial conditions.

Solution component	KCl	NaCl	Water	Solubility ratios		
				KCl	NaCl	Water
H	343	270	1000	34.3	27.0	100
$G(a)$	194	270	950	20.4	28.4	100
KCl yield	149					
Net evaporation			50			
$I(b)$	194	270	860	22.6	31.4	100
$E(c)$	194	153	554	35.0	27.5	100
NaCl yield		117				
Evaporation			306			
Dilution			11			
H'	194	153	565	34.3	27.0	-100

The calculations for these steps are:

- 270 lb NaCl (100 lb water/28.4 lb NaCl) = 950 lb water
950 lb water (20.4 lb KCl/100 lb water) = 194 lb KCl
- 270 lb NaCl (100 lb water/31.4 lb NaCl) = 860 lb water
860 lb water (22.6 lb KCl/100 lb water) = 194 lb KCl
- 194 lb KCl (100 lb water/35.0 lb KCl) = 554 lb water
554 lb water (27.5 lb NaCl/100 lb water) = 153 lb NaCl

Note that during the cooling step the maximum amount of evaporation which is permitted by the material balance is 50 lb for the step shown. In an evaporative-cooling step, however, the actual evaporation which results from adiabatic cooling is more than this. Therefore, water must be added back to prevent the NaCl concentration from rising too high; otherwise, coprecipitation of NaCl will occur.

Inasmuch as only mass ratios are involved in these calculations, kilograms or any other unit of mass may be substituted for pounds without affecting the validity of the example.

Although the figures given are for a step-by-step process, it is obvious that the same techniques will apply to a continuous system if the fresh feed containing KCl and NaCl is added at an appropriate part of the cycle, such as between steps G and I for the case of dilute feed solutions.

Another method of fractional crystallization, in which advantage is taken of different crystallization rates, is sometimes used. Thus, a solution saturated with borax and potassium chloride will, in the absence of borax seed crystals, precipitate only potassium chloride on rapid cooling. The borax remains behind as a supersaturated solution, and the potassium chloride crystals can be removed before the slower borax crystallization starts.

Crystal Formation There are two steps involved in the preparation of crystal matter from a solution. The crystals must first form and

then grow. The formation of a new solid phase either on an inert particle in the solution or in the solution itself is called **nucleation**. The increase in size of this nucleus with a layer-by-layer addition of solute is called **growth**. The growth process involves two steps, diffusion of the solute to the crystal interface followed by incorporation of the same into the lattice. One of these will control depending on factors such as the degree of agitation and temperature. Nucleation can be classified as primary or secondary. The former usually occurs at high supersaturation and does not involve product crystals. Secondary nucleation involves nuclei generation from product crystals by contact with the agitator, with the crystallizer internals and with one another. Each system has a metastable zone where growth is encouraged in the presence of supersaturation. Secondary nucleation can occur within the zone. Both nucleation and crystal growth have supersaturation as a common driving force. Unless a solution is supersaturated, crystals can neither form nor grow. Supersaturation refers to the quantity of solute present in solution compared with the quantity which would be present if the solution were kept for a very long period of time with solid phase in contact with the solution. The latter value is the equilibrium solubility at the temperature and pressure under consideration. The supersaturation coefficient can be expressed

$$S = \frac{\text{parts solute}/100 \text{ parts solvent}}{\text{parts solute at equilibrium}/100 \text{ parts solvent}} \geq 1.0 \quad (18-25)$$

Solutions vary greatly in their ability to sustain measurable amounts of supersaturation. With some materials, such as sucrose, it is possible to develop a supersaturation coefficient of 1.4 to 2.0 with little danger of nucleation. With some common inorganic solutions such as sodium chloride in water, the amount of supersaturation which can be generated stably is so small that it is difficult or impossible to measure.

Certain qualitative facts in connection with supersaturation, growth, and the yield in a crystallization process are readily apparent. If the concentration of the initial solution and the final mother liquor are fixed, the total weight of the crystalline crop is also fixed if equilibrium is obtained. The particle-size distribution of this weight, however, will depend on the relationship between the two processes of nucleation and growth. Considering a given quantity of solution cooled through a fixed range, if there is considerable nucleation initially during the cooling process, the yield will consist of many small crystals. If only a few nuclei form at the start of the crystallization (or seeds are added) and the resulting yield occurs uniformly on these nuclei or seeds without significant secondary nucleation, a crop of large uniform crystals will result. Obviously, many intermediate cases of varying nucleation rates and growth rates can also occur, depending on the nature of the materials being handled, the rate of cooling, agitation, and other factors.

When a process is continuous, nucleation frequently occurs in the presence of a seeded solution by the combined effects of mechanical stimulus and nucleation caused by supersaturation (heterogeneous nucleation). If such a system is completely and uniformly mixed (i.e., the product stream represents the typical magma circulated within the system) and if the system is operating at steady state, the particle-size distribution has definite limits which can be predicted mathematically with a high degree of accuracy, as will be shown later in this section.

Geometry of Crystal Growth Geometrically a crystal is a solid bounded by planes. The shape and size of such a solid are functions of the interfacial angles and of the linear dimension of the faces. As the result of the constancy of its interfacial angles, each face of a growing or dissolving crystal, as it moves away from or toward the center of the crystal, is always parallel to its original position. This concept is known as the "principle of the parallel displacement of faces." The rate at which a face moves in a direction perpendicular to its original position is called the translation velocity of that face or the rate of growth of that face.

From the industrial point of view, the term **crystal habit** or **crystal morphology** refers to the relative sizes of the faces of a crystal. The crystal habit is determined by the internal structure and external influences on the crystal such as the growth rate, solvent used, and impurities present during the crystallization growth period. The crystal habit of commercial products is of very great importance. Long,

needlelike crystals tend to be easily broken during centrifugation and drying. Flat, platelike crystals are very difficult to wash during filtration or centrifugation and result in relatively low filtration rates. Complex or twinned crystals tend to be more easily broken in transport than chunky, compact crystal habits. Rounded or spherical crystals (caused generally by attrition during growth and handling) tend to give considerably less difficulty with caking than do cubical or other compact shapes.

Internal structure (unit cell) can be different in crystals that are chemically identical. This is called **polymorphism**. Polymorphs can vary substantially in physical and chemical properties such as bioavailability and solubility. They can be identified by analytical techniques such as X-ray diffraction, infrared, Raman spectro, and microscopic techniques. For the same internal structure, very small amounts of foreign substances will often completely change the crystal habit. The selective adsorption of dyes by different faces of a crystal or the change from an alkaline to an acidic environment will often produce pronounced changes in the crystal habit. The presence of other soluble anions and cations often has a similar influence. In the crystallization of ammonium sulfate, the reduction in soluble iron to below 50 ppm of ferric ion is sufficient to cause significant change in the habit of an ammonium sulfate crystal from a long, narrow form to a relatively chunky and compact form. Additional information is available in the patent literature and Table 18-4 lists some of the better-known additives and their influences.

Since the relative sizes of the individual faces of a crystal vary between wide limits, it follows that different faces must have different translational velocities. A geometric law of crystal growth known as the **overlapping principle** is based on those velocity differences: in growing a crystal, only those faces having the lowest translational velocities survive; and in dissolving a crystal, only those faces having the highest translational velocities survive.

For example, consider the cross sections of a growing crystal as in Fig. 18-65. The polygons shown in the figure represent varying stages in the growth of the crystal. The faces marked *A* are slow-growing faces (low translational velocities), and the faces marked *B* are fast-growing (high translational velocities). It is apparent from Fig. 18-65 that the faster *B* faces tend to disappear as they are overlapped by the slower *A* faces.

It has been predicted that crystal habit or crystal morphology was related to the internal structure based on energy considerations and speculated that it should be possible to predict the growth shape of crystals from the slice energy of different flat faces. One can predict the calculated attachment energy for various crystal species. Recently computer programs have been developed that predict crystal morphology from attachment energies. These techniques are particularly useful in dealing with organic or molecular crystals and rapid progress in this area is being made by companies such as Molecular Simulations of Cambridge, England.

Purity of the Product If a crystal is produced in a region of the phase diagram where a single-crystal composition precipitates, the crystal itself will normally be pure provided that it is grown at relatively low rates and constant conditions. With many products these purities approach a value of about 99.5 to 99.8 percent. The difference between this and a purity of 100 percent is generally the result of small pockets of mother liquor called inclusions trapped within the crystal. Although frequently large enough to be seen with an ordinary microscope, these inclusions can be submicroscopic and represent dislocations within the structure of the crystal. They can be caused by either attrition or breakage during the growth process or by slip planes within the crystal structure caused by interference between screw-type dislocations and the remainder of the crystal faces. To increase the purity of the crystal beyond the point where such inclusions are normally expected (about 0.1 to 0.5 percent by volume), it is generally necessary to reduce the impurities in the mother liquor itself to an acceptably low level so that the mother liquor contained within these pockets will not contain sufficient impurities to cause an impure product to be formed. It is normally necessary to recrystallize material from a solution which is relatively pure to surmount this type of purity problem.

In addition to the impurities within the crystal structure itself, there is normally an adhering mother-liquid film left on the surface of the

TABLE 18-4 Some Impurities Known to Be Habit Modifiers

Material crystallized	Additive(s)	Effect	Concentration	References
Ba(NO ₃) ₂	Mg, Te ⁺⁴	Helps growth	—	1
CaSO ₄ ·2H ₂ O	Citric, succinic, tartaric acids	Helps growth	Low	
	Sodium citrate	Forms prisms	—	5
CuSO ₄ ·5H ₂ O	H ₂ SO ₄	Chunky crystals	0.3%	5
KCl	K ₄ Fe(CN) ₆	Inhibits growth, dendrites	1000 ppm	4
	Pb, Bi, Sn ⁺² , Ti, Zr, Th, Cd, Fe, Hg, Mg	Helps growth	Low	1
KClO ₄	Congo red (dye)	Modifies the 102 face	50 ppm	6
K ₂ CrO ₄	Acid magenta (dye)	Modifies the 010 face	50 ppm	6
KH ₂ PO ₄	Na ₂ B ₄ O ₇	Aids growth	—	1
KNO ₂	Fe	Helps growth	Low	1
KNO ₃	Acid magenta (dye)	Tabular crystals	—	7
	Pb, Th, Bi	Helps growth	Low	1
K ₂ SO ₄	Acid magenta (dye)	Forms plates	2000 ppm	6
	Cl, Mn, Mg, Bi, Cu, Al, Fe	Helps growth	Low	1
	Cl ₃	Reduces growth rate	1000 ppm	4
	(NH ₄) ₃ Ce(NO ₃) ₆	Reduces growth rate	1000 ppm	4
LiCl·H ₂ O	Cr-Mn ⁺² , Sn ⁺² , Co, Ni, Fe ⁺³	Helps growth	Low	1
MgSO ₄ ·7H ₂ O	Borax	Aids growth	5%	1
Na ₂ B ₄ O ₇ ·10H ₂ O	Sodium oleate	Reduces growth & nuc.	5 ppm	
	Casein, gelatin	Promotes flat crystals	—	2, 5
	NaOH, Na ₂ CO ₃	Promotes chunky crystals	—	
Na ₂ CO ₃ ·H ₂ O	SO ₄ ⁼	Reduces L/D ratio	0.1–1.0%	Canadian Patent 812,685
	Ca ⁺² and Mg ⁺²	Increase bulk density	400 ppm	U.S. Patent 3,459,497
NaCO ₃ ·NaHCO ₃ ·2H ₂ O	D-40 detergent	Aids growth	20 ppm	U.S. Patent 3,233,983
NaCl	Na ₄ Fe(CN) ₆ , CdBr	Forms dendrites	100 ppm	4
	Pb, Mn ⁺² , Bi, Sn ⁺² , Ti, Fe, Hg	Helps growth	Low	1
	Urea, formamide	Forms octahedra	Low	2
	Tetraalkyl ammon. salts	Helps growth & hardness	1–100 ppm	U.S. Patent 3,095,281
	Polyethylene-oxy compounds	Helps growth & hardness	—	U.S. Patent 3,000,708
NaClO ₃	Na ₂ SO ₄ , NaClO ₄	Tetrahedrons	—	3
NaNO ₃	Acid green (dye)	Flattened rhombahedra	—	7
Na ₂ SO ₄	NH ₄ SO ₄ @ pH 6.5	Large single crystals	Low	
	CdCl ₂	Inhibits growth	1000 ppm	4
	Alkyl aryl sulfonates	Aids growth	—	2
	Calgon	Aids growth	100 ppm	
NH ₄ Cl	Mn, Fe, Cu, Co, Ni, Cr	Aid growth	Low	1
	Urea	Forms octahedra	—	5
NH ₄ ClO ₄	Azurine (dye)	Modifies the 102 face	22 ppm	6
NH ₄ F	Ca	Helps growth	Low	1
(NH ₄)NO ₃	Acid magenta (dye)	Forms 010 face plates	1%	6
(NH ₄) ₂ HPO ₄	H ₂ SO ₄	Reduces L/D ratio	7%	
NH ₄ H ₂ PO ₄	Fe ⁺³ , Cr, Al, Sn	Helps growth	Traces	1
(NH ₄) ₂ SO ₄	Cr ⁺³ , Fe ⁺³ , Al ⁺³	Promotes needles	50 ppm	
	H ₂ SO ₄	Promotes needles	2–6%	U.S. Patent 2,092,073
	Oxalic acid, citric acid	Promotes chunky crystals	1000 ppm	U.S. Patent 2,228,742
	H ₃ PO ₄ , SO ₂	Promotes chunky crystals	1000 ppm	
ZnSO ₄ ·7H ₂ O	Borax	Aids growth	—	1
Adipic acid	Surfactant-SDBS	Aids growth	50–100 ppm	2
Fructose	Glucose, difructose	Affects growth	—	8
L-asparagine	L-glutamic acid	Affects growth	—	8
Naphthalene	Cyclohexane (solvent)	Forms needles	—	2
	Methanol (solvent)	Forms plates	—	
Pentaerythritol	Sucrose	Aids growth	—	1
	Acetone (solvent)	Forms plates	—	2
Sodium glutamate	Lysine, CaO	Affects growth	—	8
Sucrose	Raffinose, KCl, NaBr	Modify growth rate	—	
Urea	Biuret	Reduces L/D & aids growth	2–7%	
	NH ₄ Cl	Reduces L/D & aids growth	5–10%	

1. Gillman, *The Art and Science of Growing Crystals*, Wiley, New York, 1963.
2. Mullin, *Crystallization*, Butterworth, London, 1961.
3. Buckley, *Crystal Growth*, Wiley, New York, 1961.
4. Phoenix, L., *British Chemical Engineering*, vol. II, no. 1 (Jan. 1966), pp. 34–38.
5. Garrett, D. E., *British Chemical Engineering*, vol. I, no. 12 (Dec. 1959), pp. 673–677.
6. Buckley, *Crystal Growth*, (Faraday Soc.) Butterworths, 1949, p. 249.
7. Butchart and Whetstone, *Crystal Growth*, (Faraday Soc.) Butterworths, 1949, p. 259.
8. Nyvlt, J., *Industrial Crystallization*, Verlag Chemie Publishers, New York, 1978, pp. 26–31.

crystal after separation in a centrifuge or on a filter. Typically a centrifuge may leave about 2 to 10 percent of the weight of the crystals as adhering mother liquor on the surface. This varies greatly with the size and shape or habit of the crystals. Large, uniform crystals from low-viscosity mother liquors will retain small quantities of mother liquor, while nonuniform or small crystals crystallized from viscous solutions will retain a considerably larger proportion. Comparable

statements apply to the filtration of crystals, although normally the amounts of mother liquor adhering to the crystals are considerably larger. It is common practice when crystallizing materials from solutions which contain appreciable quantities of impurities to wash the crystals on the centrifuge or filter with either fresh solvent or feed solution. In principle, such washing can reduce the impurities quite substantially. It is also possible in many cases to reslurry the crystals in

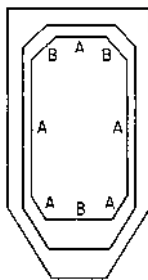


FIG. 18-65 Overlapping principle.

fresh solvent and recentrifuge the product in an effort to obtain a longer residence time during the washing operation and better mixing of the wash liquors with the crystals. Mother liquor inclusions and residual moisture after drying can present caking problems.

Coefficient of Variation One of the problems confronting any user or designer of crystallization equipment is the expected particle-size distribution of the solids leaving the system and how this distribution may be adequately described. Most crystalline-product distributions plotted on arithmetic-probability paper will exhibit a straight line for a considerable portion of the plotted distribution. In this type of plot the particle diameter should be plotted as the ordinate and the cumulative percent on the log-probability scale as the abscissa.

It is common practice to use a parameter characterizing crystal-size distribution called the coefficient of variation. This is defined as follows:

$$CV = 100 \frac{PD_{16\%} - PD_{84\%}}{2PD_{50\%}} \quad (18-26)$$

where CV = coefficient of variation, as a percentage

PD = particle diameter from intercept on ordinate axis at percent indicated

In order to be consistent with normal usage, the particle-size distribution when this parameter is used should be a straight line between approximately 10 percent cumulative weight and 90 percent cumulative weight. By giving the coefficient of variation and the mean particle diameter, a description of the particle-size distribution is obtained which is normally satisfactory for most industrial purposes. If the product is removed from a mixed-suspension crystallizer, this coefficient of variation should have a value of approximately 50 percent (Randolph and Larson, op. cit., chap. 2).

Crystal Nucleation and Growth

Rate of Growth Crystal growth is a layer-by-layer process, and since growth can occur only at the face of the crystal, material must be transported to that face from the bulk of the solution. Diffusional resistance to the movement of molecules (or ions) to the growing crystal face, as well as the resistance to integration of those molecules into the face, must be considered. As discussed earlier, different faces can have different rates of growth, and these can be selectively altered by the addition or elimination of impurities.

If L is a characteristic dimension of a crystal of selected material and shape, the rate of growth of a crystal face that is perpendicular to L is, by definition,

$$G \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta L}{\Delta t} = \frac{dL}{dt} \quad (18-27)$$

where G is the growth rate over time interval t . It is customary to measure G in the practical units of millimeters per hour. It should be noted that growth rates so measured are actually twice the facial growth rate.

The delta L law. It has been shown by McCabe [*Ind. Eng. Chem.*, **21**, 30, 112 (1929)] that all geometrically similar crystals of the same

material suspended in the same solution grow at the same rate if growth rate is defined as in Eq. (18-27). The rate is independent of crystal size, provided that all crystals in the suspension are treated alike. This generalization is known as the delta L law. Although there are some well-known exceptions, they usually occur when the crystals are very large or when movement of the crystals in the solution is so rapid that substantial changes occur in diffusion-limited growth of the faces.

It is emphasized that the delta L law does not apply when similar crystals are given preferential treatment based on size. It fails also when surface defects or dislocations significantly alter the growth rate of a crystal face. Nevertheless, it is a reasonably accurate generalization for a surprising number of industrial cases. When it is, it is important because it simplifies the mathematical treatment in modeling real crystallizers and is useful in predicting crystal-size distribution in many types of industrial crystallization equipment.

Important exceptions to McCabe's growth-rate model have been noted by Bramson, by Randolph, and by Abegg. These are discussed by Canning and Randolph, *Am. Inst. Chem. Eng. J.*, **13**, 5 (1967).

Nucleation The mechanism of crystal nucleation from solution has been studied by many scientists, and their work suggests that—in commercial crystallization equipment, at least—the nucleation rate is the sum of contributions by (1) primary nucleation and (2) nucleation due to contact between crystals and (a) other crystals, (b) the walls of the container, and (c) the impeller. If B^0 is the net number of new crystals formed in a unit volume of solution per unit of time,

$$B^0 = B_{ss} + B_{cc} + B_{ci} \quad (18-28)$$

where B_{ci} is the rate of nucleation due to crystal-impeller contacts, B_{cc} is that due to crystal-crystal contacts, and B_{ss} is the primary nucleation rate due to the supersaturation driving force. The mechanism of the last-named is not precisely known, although it is obvious that molecules forming a nucleus not only have to coagulate, resisting the tendency to redissolve, but also must become oriented into a fixed lattice. The number of atoms or molecules required to form a stable crystal nucleus has been variously estimated at from 80 to 100 (with ice), and the probability that a stable nucleus will result depends on many factors such as activation energies and supersaturation. In commercial crystallization equipment, in which supersaturation is low and agitation is employed to keep the growing crystals suspended, the predominant mechanism is contact nucleation or, in extreme cases, attrition.

In order to treat crystallization systems both dynamically and continuously, a mathematical model has been developed which can correlate the nucleation rate to the level of supersaturation and/or the growth rate. Because the growth rate is more easily determined and because nucleation is sharply nonlinear in the regions normally encountered in industrial crystallization, it has been common to assume

$$B^0 = k_s s^b \quad (18-29)$$

where s , the supersaturation, is defined as $(C - C_s)$, C being the concentration of the solute and C_s its saturation concentration; and the exponent b and dimensional coefficient k are values characteristic of the material.

While Eq. (18-29) has been popular among those attempting correlations between nucleation rate and supersaturation, it has become common to use a derived relationship between nucleation rate and growth rate by assuming that

$$G = k' s^g \quad (18-30)$$

whence, in consideration of Eq. (18-29),

$$B^0 = k'' G^i \quad (18-31)$$

where the dimensional coefficient k' and exponent g are characteristic of the material and the conditions of crystallization and $k'' = k/(k')^i$ with $i = b/g$, a measure of the relative dependence of B^0 and G on supersaturation. Feeling that a model in which nucleation depends only on supersaturation or growth rate is simplistically deficient, some have proposed that contact nucleation rate is also a power function of slurry density and that

$$B^0 = k_n G^i M_T^j \quad (18-32)$$

where M_T is the density of the crystal slurry, g/L.

Although Eqs. (18-31) and (18-32) have been adopted by many as a matter of convenience, they are oversimplifications of the very complex relationship that is suggested by Eq. (18-28); Eq. (18-32) implicitly and quite arbitrarily combines the effects of homogeneous nucleation and those due to contact nucleation. They should be used only with caution.

In work pioneered by Clontz and McCabe [*Chem. Eng. Prog. Symp. Ser.*, **67**(110), 6 (1971)] and subsequently extended by others, contact nucleation rate was found to be proportional to the input of energy of contact, frequency of contact, as well as being a function of contact area and supersaturation. This observation is important to the scaling up of crystallizers. At the laboratory or bench scale, particle contact frequency with the agitator is high while in commercial equipment the contact energy input is higher at the impeller but the contact frequency is less. Scale-up modeling of a crystallizer, therefore, must include its mechanical characteristics as well as the physiochemical driving force.

Nucleation and Growth From the preceding, it is clear that no analysis of a crystallizing system can be truly meaningful unless the simultaneous effects of nucleation rate, growth rate, heat balance, and material balance are considered. The most comprehensive treatment of this subject is by Randolph and Larson (op. cit.), who developed a mathematical model for continuous crystallizers of the mixed-suspension or circulating-magma type [*Am. Inst. Chem. Eng. J.*, **8**, 639 (1962)] and subsequently examined variations of this model that include most of the aberrations found in commercial equipment. Randolph and Larson showed that when the total number of crystals in a given volume of suspension from a crystallizer is plotted as a function of the characteristic length as in Fig. 18-66, the slope of the line is usefully identified as the crystal population density, n :

$$n = \lim_{\Delta L \rightarrow 0} \frac{\Delta N}{\Delta L} = \frac{dN}{dL} \quad (18-33)$$

where N = total number of crystals up to size L per unit volume of magma. The population density thus defined is useful because it characterizes the nucleation-growth performance of a particular crystallization process or crystallizer.

The data for a plot like Fig. 18-67 are easily obtained from a screen analysis of the total crystal content of a known volume (e.g., a liter) of magma. The analysis is made with a closely spaced set of testing sieves (or intervals for a particle counter), the cumulative number of particles smaller than each sieve in the nest being plotted against the aperture dimension of that sieve. The fraction retained on each sieve is weighed, and the mass is converted to the equivalent number of particles by dividing by the calculated mass of a particle whose dimension is the arithmetic mean of the mesh sizes of the sieve on which it is retained and the sieve immediately above it.

In industrial practice, the size-distribution curve usually is not actually constructed. Instead, a mean value of the population density for any sieve fraction of interest (in essence, the population density of the

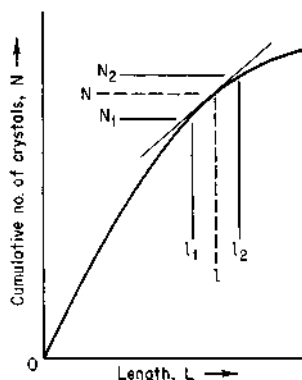


FIG. 18-66 Determination of the population density of crystals.

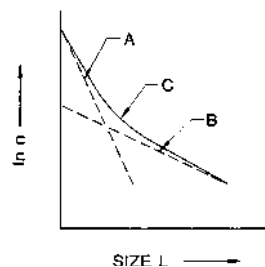


FIG. 18-67 Population density of crystals resulting from Bujacian behavior.

particle of average dimension in that fraction) is determined directly as $\Delta N/\Delta L$, ΔN being the number of particles retained on the sieve and ΔL being the difference between the mesh sizes of the retaining sieve and its immediate predecessor. It is common to employ the units of $(\text{mm} \cdot \text{L})^{-1}$ for n .

For a steady-state crystallizer receiving solids-free feed and containing a well-mixed suspension of crystals experiencing negligible breakage, a material-balance statement yields negligible agglomeration and breakage to a particle balance (the Randolph-Larson general-population balance); in turn, it simplifies to

$$\frac{dn}{dL} + \frac{n}{Gt} = 0 \quad (18-34)$$

if the delta L law applies (i.e., G is independent of L) and the draw-down (or retention) time is assumed to be invariant and calculated as $t = V/Q$. Integrated between the limits n^0 , the population density of nuclei (for which L is assumed to be zero), and n , that of any chosen crystal size L , Eq. (18-34) becomes

$$\int_{n^0}^n \frac{dn}{n} = - \int_0^L \frac{dL}{Gt} \quad (18-35)$$

$$\ln n = \frac{-L}{Gt} + \ln n^0 \quad (18-36a)$$

or

$$n = n^0 e^{-L/Gt} \quad (18-36b)$$

It can be shown that

$$B^0 = n^0 G \quad (18-36c)$$

A plot of $\ln n$ versus L is a straight line whose intercept is $\ln n^0$ and whose slope is $-1/Gt$. (For plots on base-10 log paper, the appropriate slope correction must be made.) Thus, from a given product sample of known slurry density and retention time it is possible to obtain the nucleation rate and growth rate for the conditions tested if the sample satisfies the assumptions of the derivation and yields a straight line. A number of derived relations which describe the nucleation rate, size distribution, and average properties are summarized in Table 18-5.

If a straight line does not result (Fig. 18-67), at least part of the explanation may be violation of the delta L law (Canning and Randolph, loc. cit.). The best current theory about what causes size-dependent growth suggests what has been called growth dispersion or "Bujacian behavior" [Mullen (ed.), op. cit., p. 254]. In the same environment different crystals of the same size can grow at different rates owing to differences in dislocations or other surface effects. The graphs of "slow" growers (Fig. 18-67, curve A) and "fast" growers (curve B) sum to a resultant line (curve C), concave upward, that is described by Eq. (18-37) (Randolph, in deJong and Jancic, op. cit., p. 254):

$$n = \sum \frac{B_i^0}{G_i} e^{(-L/G_i t)} \quad (18-37)$$

Equation (18-34) contains no information about the crystallizer's influence on the nucleation rate. If the crystallizer is of a mixed-suspension, mixed-product-removal (MSMPR) type, satisfying the criteria for Eq. (18-34), and if the model of Clontz and McCabe is valid, the contribution to the nucleation rate by the circulating pump

TABLE 18-5 Common Equations for Population-Balance Calculations

Name	Symbol	Units	Systems without fines removal	Systems with fines removal		References
				Fines stream	Product stream	
Drawdown time (retention time)	t	h	$t = V/Q$	$t_F = V_{\text{liquid}}/Q_F$	$t = V/Q$	
Growth rate	G	mm/h	$G = dL/dt$	$G = dL/dt$	$G = dL/dt$	
Volume coefficient	K_v	1/no. (crystals)	$K_v = \frac{\text{volume of one crystal}}{L^3}$	$K_v = \frac{\text{volume of one crystal}}{L^3}$	$K_v = \frac{\text{volume of one crystal}}{L^3}$	
Population density	n	No. (crystals)/mm	$n = dN/dL$	$n = dN/dL$	$n = dN/dL$	1
Nuclei population density	n^o	No. (crystals)/mm	$n^o = K_M M^i G^{i-1}$			2
Population density	n	No. (crystals)/mm	$n = n^o e^{-L/Gt}$	$n_F = n^o e^{-L/Gt_F}$	$n = n^o e^{-L/Gt_F} e^{-L/Gt}$	1, 3
Nucleation rate	B_0	No. (crystals)/h	$B_0 = G n^o = K_M M^i G^i$	$B_0 = G_n^o$		4
Dimensionless length	x	None	$x = \frac{L}{Gt}$	$x_F = \frac{L}{Gt_F}, L_0 \rightarrow L_f$	$x = \frac{L}{Gt}, L_f \rightarrow L$	1
Mass/unit volume (slurry density)	M_T	g/L	$M_t = K_v \rho \int_0^\infty n L^3 dL$ $M_t = K_v \rho 6 n^o (Gt)^4$	$M_{t_F} = K_v \rho \int_0^{L_f} n^o e^{-L/Gt_F} L^3 dL$	$M_T = K_v \rho \int_{L_f}^\infty n^o e^{-L/Gt_F} e^{-L/Gt} L^3 dL$	1
Cumulative mass to x Total mass	W_x	None	$W_x = 1 - e^{-x} \left(\frac{x^3}{6} + \frac{x^2}{2} + x + 1 \right)$	$W_F = \frac{e^{-x} (x^3 + 3x^2 + 6x + 6) - 6}{e^{-x} (x_c^3 + 3x_c^2 + 6x_c + 6) - 6}$	$W = \frac{6K_v \rho n^o e^{-L_c/Gt_F} Gt^4 \left[1 - e^{-x} \left(\frac{x^3}{6} + \frac{x^2}{2} + x + 1 \right) \right]}{\text{Slurry density } M, \text{ g/L}}$ when $L_c \approx 0$, compared with L_a	5
Dominant particle	L_d	mm	$L_d = 3Gt$			
Average particle, weight	L_a	mm	$L_a = 3.67Gt$			6
Total number of crystals	N_T	No./L	$N_T = \int_0^\infty n dL$	$N_F = \int_0^{L_f} n_F dL$	$N_T = \int_{L_f}^\infty n dL$	1, 3

1. Randolph and Larson, *Am. Inst. Chem. Eng. J.*, **8**, 639 (1962).
2. Timm and Larson, *Am. Inst. Chem. Eng. J.*, **14**, 452 (1968).
3. Larson, private communication.
4. Larson, Timm, and Wolff, *Am. Inst. Chem. Eng. J.*, **14**, 448 (1968).
5. Larson and Randolph, *Chem. Eng. Prog. Symp. Ser.*, **65**(95), 1 (1969).
6. Schoen, *Ind. Eng. Chem.*, **53**, 607 (1961).

can be calculated [Bennett, Fiedelman, and Randolph, *Chem. Eng. Prog.*, **69**(7), 86 (1973)]:

$$B_e = K_e \left(\frac{I^2}{P} \right) \rho G \int_0^\infty n L^4 dL \quad (18-38)$$

where I = tip speed of the propeller or impeller, m/s
 ρ = crystal density, g/cm³
 P = volume of crystallizer/circulation rate (turnover), m³/(m³/s) = s

Since the integral term is the fourth moment of the distribution (m_4), Eq. (18-38) becomes

$$B_e = K_e \rho G \left(\frac{I^2}{P} \right) m_4 \quad (18-39)$$

Equation (18-39) is the general expression for impeller-induced nucleation. In a fixed-geometry system in which only the speed of the circulating pump is changed and in which the flow is roughly proportional to the pump speed, Eq. (18-39) may be satisfactorily replaced with

$$B_e = K_e'' \rho G (S_R)^3 m_4 \quad (18-40)$$

where S_R = rotation rate of impeller, r/min. If the maximum crystal-impeller impact stress is a nonlinear function of the kinetic energy, shown to be the case in at least some systems, Eq. (18-40) no longer applies.

In the specific case of an MSMPR exponential distribution, the fourth moment of the distribution may be calculated as

$$m_4 = 4! n^0 (Gt)^5 \quad (18-41)$$

Substitution of this expression into Eq. (18-39) gives

$$B_e = k n^0 G (S_R)^3 L_D^5 \quad (18-42)$$

where $L_D = 3Gt$, the dominant crystal (mode) size.

Equation (18-42) displays the competing factors that stabilize secondary nucleation in an operating crystallizer when nucleation is due mostly to impeller-crystal contact. Any increase in particle size produces a fifth-power increase in nucleation rate, tending to counteract the direction of the change and thereby stabilizing the crystal-size distribution. From dimensional argument alone the size produced in a mixed crystallizer for a (fixed) nucleation rate varies as $(B^0)^{1/3}$. Thus, this fifth-order response of contact nucleation does not wildly upset the crystal size distribution but instead acts as a stabilizing feedback effect.

Nucleation due to crystal-to-crystal contact is greater for equal striking energies than crystal-to-metal contact. However, the viscous drag of the liquid on particle sizes normally encountered limits the velocity of impact to extremely low values. The assumption that only the largest crystal sizes contribute significantly to the nucleation rate by crystal-to-crystal contact permits a simple computation of the rate:

$$B_e = K_e \rho G m_i^2 \quad (18-43)$$

where m_i = the fourth, fifth, sixth, or higher moments of the distribution.

A number of different crystallizing systems have been investigated by using the Randolph-Larson technique, and some of the published growth rates and nucleation rates are included in Table 18-6. Although the usefulness of these data is limited to the conditions tested, the table gives a range of values which may be expected, and it permits resolution of the information gained from a simple screen analysis into the fundamental factors of growth rate and nucleation rate. Experiments may then be conducted to determine the independent effects of operation and equipment design on these parameters.

Although this procedure requires laborious calculations because of the number of samples normally needed, these computations and the determination of the best straight-line fit to the data are readily programmed for digital computers.

Example 4: Population Density, Growth, and Nucleation Rate

Calculate the population density, growth, and nucleation rates for a crystal sample of urea for which there is the following information. These data are from Bennett and Van Buren [*Chem. Eng. Prog. Symp. Ser.*, **65**(95), 44 (1969)].

Slurry density = 450 g/L
 Crystal density = 1.335 g/cm³
 Drawdown time $t = 3.38$ h

Shape factor $k_v = 1.00$

Product size:

-14 mesh, +20 mesh	4.4 percent
-20 mesh, +28 mesh	14.4 percent
-28 mesh, +35 mesh	24.2 percent
-35 mesh, +48 mesh	31.6 percent
-48 mesh, +65 mesh	15.5 percent
-65 mesh, +100 mesh	7.4 percent
-100 mesh	2.5 percent

n = number of particles per liter of volume

14 mesh = 1.168 mm, 20 mesh = 0.833 mm, average opening 1.00 mm

Size span = 0.335 mm = ΔL

$$n_{20} = \frac{(450 \text{ g/L})(0.044)}{(1.335/1000) \text{ g/mm}^3 (1.00^3 \text{ mm}^3/\text{particle})(0.335 \text{ mm})(1.0)}$$

$$n_{20} = 44,270$$

$$\ln n_{20} = 10,698$$

Repeating for each screen increment:

Screen size	Weight, %	k_v	$\ln n$	L , average diameter, mm
100	7.4	1.0	18.099	0.178
65	15.5	1.0	17.452	0.251
48	31.6	1.0	16.778	0.356
35	24.2	1.0	15.131	0.503
28	14.4	1.0	13.224	0.711
20	4.4	1.0	10.698	1.000

Plotting $\ln n$ versus L as shown in Fig. 18-68, a straight line having an intercept at zero length of 19.781 and a slope of -9.127 results. As mentioned in discussing Eq. (18-27), the growth rate can then be found.

$$\text{Slope} = -1/Gt \text{ or } -9.127 = -1/[G(3.38)]$$

or $G = 0.0324$ mm/h

and $B_0 = G n^0 = (0.0324)(e^{19.781}) = 12.65 \times 10^6 \frac{n^0}{L \cdot h}$

and $L_n = 3.67(0.0324)(3.38) = 0.40$ mm

An additional check can be made of the accuracy of the data by the relation

$$M_T = 6k_i \rho n^0 (Gt)^4 = 450 \text{ g/L}$$

$$M_T = (6)(1.0) \frac{1.335 \text{ g/cm}^3}{1000 \text{ mm}^3/\text{cm}^3} e^{19.781} [(0.0324)(3.38)]^4$$

$$M_T = 455 \text{ g/L} \approx 450 \text{ g/L}$$

Had only the growth rate been known, the size distribution of the solids could have been calculated from the equation

$$W_f = 1 - e^{-x} \left(\frac{x^3}{6} + \frac{x^2}{2} + x + 1 \right)$$

where W_f is the weight fraction up to size L and $x = L/Gt$.

$$x = \frac{L}{(0.0324)(3.38)} = \frac{L}{0.1095}$$

Screen size	L , mm	x	W_f^*	Cumulative % retained 100 (1 - W_f)	Measured cumulative % retained
20	0.833	7.70	0.944	5.6	4.4
28	0.589	5.38	0.784	21.6	18.8
35	0.417	3.80	0.526	47.4	43.0
48	0.295	2.70	0.286	71.4	74.6
65	0.208	1.90	0.125	87.5	90.1
100	0.147	1.34	0.048	95.2	97.5

*Values of W_f as a function of x may be obtained from a table of Wick's functions.

Note that the calculated distribution shows some deviation from the measured values because of the small departure of the actual sample from the theoretical coefficient of variation (i.e., 47.5 versus 50 percent).

The critical value of i , which is defined in Eq. (18-31) as the ratio of b/g or the relative dependence of nucleation and growth on supersaturation, can be determined by a few extra experiments. This is done by varying the residence time of the crystals (changing feed rate) while maintaining everything else constant. The

TABLE 18-6 Growth Rates and Kinetic Equations for Some Industrial Crystallized Products

Material crystallized	G , m/s $\times 10^8$	Range t , h	Range M_T , g/L	Temp., °C	Scale °	Kinetic equation for B_0 no./L-s)	References†
(NH ₄) ₂ SO ₄	1.67	3.83	150	70	P	$B_0 = 6.62 \times 10^{-25} G^{0.82} \rho^{-0.92} m_2^{3.05}$	Bennett and Wolf, <i>AIChE</i> , SFC, 1979.
(NH ₄) ₂ SO ₄	0.20	0.25	38	18	B	$B_0 = 2.94(10^{10})G^{1.03}$	Larsen and Mullen, <i>J. Crystal Growth</i> 20 : 183 (1973).
(NH ₄) ₂ SO ₄	—	0.20	—	34	B	$B_0 = 6.14(10^{-11})S_R^{7.84}M_T^{0.98}G^{1.22}$	Youngquist and Randolph, <i>AIChE J.</i> 18 : 421 (1972).
MgSO ₄ ·7H ₂ O	3.0–7.0	—	—	25	B	$B_0 = 9.65(10^{12})M_T^{0.67}G^{1.24}$	Sikdar and Randolph, <i>AIChE J.</i> 22 : 110 (1976).
MgSO ₄ ·7H ₂ O	—	—	Low	29	B	$B_0 = f(N, L^4, N^{4.2}, S^{2.5})$	Ness and White, <i>AIChE Symposium Series</i> 153, vol. 72, p. 64.
KCl	2–12	—	200	32	P	$B_0 = 7.12(10^{39})M_T^{0.14}G^{4.99}$	Randolph et al., <i>AIChE J.</i> 23 : 500 (1977).
KCl	3.3	1–2	100	37	B	$B_0 = 5.16(10^{23})M_T^{0.91}G^{2.77}$	Randolph et al., <i>Ind. Eng. Chem. Proc. Design Dev.</i> 20 : 496 (1981).
KCl	0.3–0.45	—	50–147	25–68	B	$B_0 = 5 \times 10^{-3} G^{2.78} (M_T TIP^2)^{1.2}$	Qian et al., <i>AIChE J.</i> 33 (10): 1690 (1987).
KCr ₂ O ₇	1.2–9.1	0.25–1	14–42	—	B	$B_0 = 7.33(10^4)M_T^{0.6}G^{0.5}$	Desari et al., <i>AIChE J.</i> 20 : 43 (1974).
KCr ₂ O ₇	2.6–10	0.15–0.5	20–100	26–40	B	$B_0 = 1.59(10^{-3})S_R^3 M_T G^{0.48}$	Janse, Ph.D. thesis, Delft Technical University, 1977.
KNO ₃	8.13	0.25–0.050	10–40	20	B	$B_0 = 3.85(10^{16})M_T^{0.5}G^{2.06}$	Juraszek and Larson, <i>AIChE J.</i> 23 : 460 (1977).
K ₂ SO ₄	—	0.03–0.17	1–7	30	B	$B_0 = 2.62(10^3)S_R^{2.5}M_T^{0.5}G^{0.54}$	Randolph and Sikdar, <i>Ind. Eng. Chem. Fund.</i> 15 : 64 (1976).
K ₂ SO ₄	2–6	0.25–1	2–20	10–50	B	$B_0 = 4.09(10^6) \exp\left(\frac{10900}{RT}\right) M_T G^{0.5}$	Jones, Budz, and Mullin, <i>AIChE J.</i> 33 : 12 (1986).
K ₂ SO ₄	0.8–1.6	—	—	—	B	$\frac{G}{G_0} = 1 + 2L^{2/3} (L \text{ in } \mu\text{m})$	White, Bendig, and Larson, <i>AIChE Mtg.</i> , Washington, D.C., Dec. 1974.
NaCl	4–13	0.2–1	25–200	50	B	$B_0 = 1.92(10^{10})S_R^2 M_T G^2$	Asselbergs, Ph.D. thesis, Delft Technical University, 1978.
NaCl	—	0.6	35–70	55	P	$B_0 = 8 \times 10^{10} N^2 G^2 M_T$	Grootscholten et al., <i>Chem. Eng. Design</i> 62 : 179 (1984).
NaCl	0.5	1–2.5	70–190	72	P	$B_0 = 1.47(10^{23})\left(\frac{I^2}{P}\right)m_4^{0.84}G^{0.98}$	Bennett et al., <i>Chem. Eng. Prog.</i> 69 (7): 86 (1973).
Citric acid	1.1–3.7	—	—	16–24	B	$B_0 = 1.09(10^{10})m_4^{0.084}G^{0.84}$	Sikdar and Randolph, <i>AIChE J.</i> 22 : 110 (1976).
Fructose	0.1–0.25	—	—	50	B	—	Shiau and Berglund, <i>AIChE J.</i> 33 : 6 (1987).
Sucrose	—	—	—	80	B	$B_0 = 5 \times 10^6 N^{0.7} M_T^{0.3} G^{0.4}$	Berglund and deJong, <i>Separations Technology</i> 1 : 38 (1990).
Sugar	2.5–5	0.375	50	45	B	$B_0 = 4.38(10^6)M_T^{1.01}(\Delta C - 0.5)^{1.42}$	Hart et al., <i>AIChE Symposium Series</i> 193, vol. 76, 1980.
Urea	0.4–4.2	2.5–6.8	350–510	55	P	$B_0 = 5.48(10^{-1})M_T^{3.87}G^{1.66}$	Bennett and Van Buren, <i>Chem. Eng. Prog. Symposium Series</i> 95 (7): 65 (1973).
Urea	—	—	—	3–16	B	$B_0 = 1.49(10^{-31})S_R^{2.3}M_T^{1.07}G^{-3.54}$	Lodaya et al., <i>Ind. Eng. Chem. Proc. Design Dev.</i> 16 : 294 (1977).

°B = bench scale; P = pilot plant.

†Additional data on many components are in Garside and Shah, *Ind. Eng. Chem. Proc. Design Dev.*, **19**, 509 (1980).

B^0 and G values are determined at each residence time, and a plot of $\ln B^0$ versus $\ln G$ should yield a straight line of slope i . High values of i indicate a propensity to nucleate versus grow and dictate the need to ensure low values of supersaturation.

Had sufficient data indicating a change in n^0 for various values of M_T at constant G been available, a plot of $\ln n^0$ versus $\ln M_T$ at corresponding G 's would permit determination of the power j .

Crystallizers with Fines Removal In Example 4, the product was from a forced-circulation crystallizer of the MSMR type. In many cases, the product produced by such machines is too small for commercial use; therefore, a separation baffle is added within the crystallizer to permit the removal of unwanted fine crystalline material from the magma, thereby controlling the population density in the machine so as to produce a coarser crystal product. When this is done, the product sample plots on a graph of $\ln n$ versus L as shown in line P , Fig. 18-69. The line of steepest slope, line F , represents the particle-size distribution of the fine material, and samples which show this distribution can be taken from the liquid leaving the fines-separation baffle. The prod-

uct crystals have a slope of lower value, and typically there should be little material present smaller than L_f , the size which the baffle is designed to separate. However, this is not to imply that there are no fines in the product stream. The effective nucleation rate for the product material is the intersection of the extension of line P to zero size.

As long as the largest particle separated by the fines-destruction baffle is small compared with the mean particle size of the product, the seed for the product may be thought of as the particle-size distribution corresponding to the fine material which ranges in length from zero to L_f , the largest size separated by the baffle.

The product discharged from the crystallizer is characterized by the integral of the distribution from size L_f to infinity:

$$M_T = k_v \rho \int_{L_f}^{\infty} n^0 \exp(-L_f/Gt_f) \exp(L/GT) L^3 dL \quad (18-44)$$

The integrated form of this equation is shown in Table 18-5.

For a given set of assumptions it is possible to calculate the characteristic curves for the product from the crystallizer when it is operated

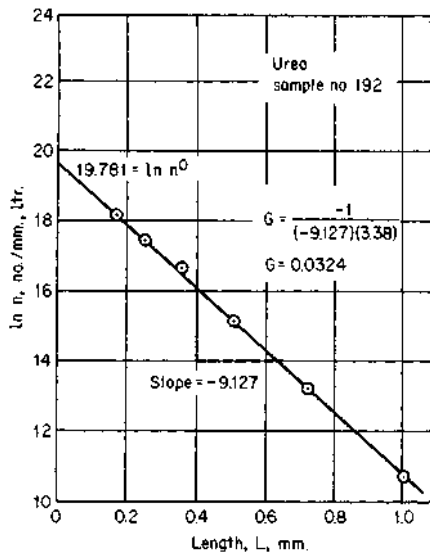
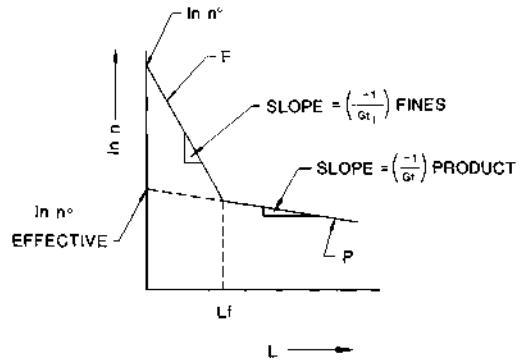


FIG. 18-68 Population density plot for Example 4.

at various levels of fines removal as characterized by L_f . This has been done for an ammonium sulfate crystallizer in Fig. 18-70. Also shown in that figure is the actual size distribution obtained. In calculating theoretical size distributions in accordance with the Eq. (18-44), it is assumed that the growth rate is a constant, whereas in fact larger values of L_f will interact with the system driving force to raise the growth rate and the nucleation rate. Nevertheless, Fig. 18-70 illus-

FIG. 18-69 Plot of Log N against L for a crystallizer with fines removal.

trates clearly the empirical result of the operation of such equipment, demonstrating that the most significant variable in changing the particle-size distribution of the product is the size removed by the baffle. Conversely, changes in retention time for a given particle-removal size L_f make a relatively small change in the product-size distribution. Jancic and Grootscholten (op. cit., p. 318) have found that the size enlargement is dependent on the fines size, the relative kinetic order i , and the rate of flow to the fines circuit versus product flow.

It is implicit that increasing the value of L_f will raise the supersaturation and growth rate to levels at which mass nucleation can occur, thereby leading to periodic upsets of the system or cycling [Randolph, Beer, and Keener, *Am. Inst. Chem. Eng. J.*, **19**, 1140 (1973)]. That this could actually happen was demonstrated experimentally by Randolph, Beckman, and Kraljevich [*Am. Inst. Chem. Eng. J.*, **23**, 500 (1977)], and that it could be controlled dynamically by regulating the

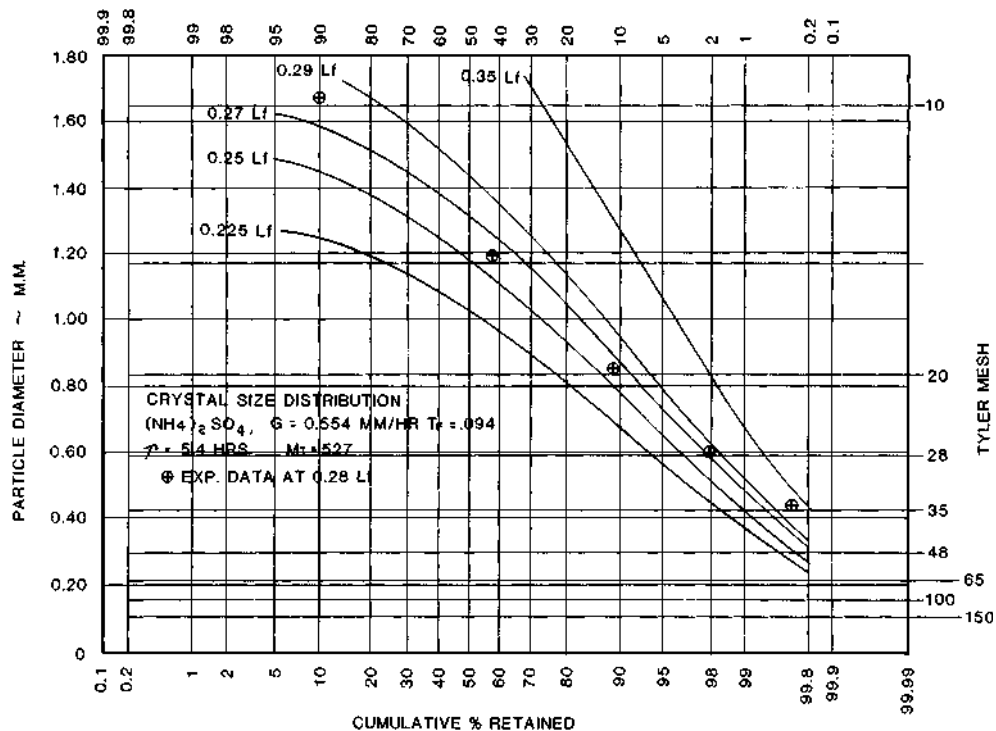


FIG. 18-70 Calculated product-size distribution for a crystallizer operation at different fine-crystal-separation sizes.

fines-destruction system was shown by Beckman and Randolph [ibid., (1977)]. Dynamic control of a crystallizer with a fines-destruction baffle and fine-particle-detection equipment employing a light-scattering (laser) particle-size-measurement instrument is described in U.S. Patent, 4,263,010 and 5,124,265.

CRYSTALLIZATION EQUIPMENT

Whether a vessel is called an evaporator or a crystallizer depends primarily on the criteria used in arriving at its sizing. In an evaporator of the salting-out type, sizing is done on the basis of vapor release. In a crystallizer, sizing is normally done on the basis of the volume required for crystallization or for special features required to obtain the proper product size. In external appearance, the vessels could be identical. Evaporators are discussed in Sec. 11. Genck (loc. cit., 2004) provides a detailed discussion of guidelines for crystallizer selection and operation.

In the discussion which follows, crystallization equipment has been classified according to the means of suspending the growing product. This technique reduces the number of major classifications and segregates those to which Eq. (18-34) applies.

Mixed-Suspension, Mixed-Product-Removal Crystallizers
This type of equipment, sometimes called the circulating-magma crystallizer, is by far the most important in use today. In most commercial equipment of this type, the uniformity of suspension of product solids within the crystallizer body is sufficient for the theory [Eqs. (18-34) to (18-36c)] to apply. Although a number of different varieties and features are included within this classification, the equipment operating with the highest capacity is the kind in which the vaporization of a solvent, usually water, occurs.

Although surface-cooled types of MSMR crystallizers are available, most users prefer crystallizers employing vaporization of solvents or of refrigerants. The primary reason for this preference is that heat transferred through the critical supersaturating step is through a boiling-liquid-gas surface, avoiding the troublesome solid deposits that can form on a metal heat-transfer surface. In this case very low LMTDs are required to stay within the metastable zone to promote growth and reduce scaling. The result is multipass, large-surface-area heat exchangers.

Forced-Circulation Evaporator-Crystallizer This crystallizer is shown in Fig. 18-71. Slurry leaving the body is pumped through a circulating pipe and through a tube-and-shell heat exchanger, where its temperature increases by about 2 to 6°C (3 to 10°F). Since this heating is done without vaporization, materials of normal solubility should produce no deposition on the tubes. The heated slurry, returned to the body by a recirculation line, mixes with the body slurry and raises its temperature locally near the point of entry, which causes boiling at the liquid surface. During the consequent cooling and vaporization to achieve equilibrium between liquid and vapor, the supersaturation which is created causes growth on the swirling body of suspended crystals until they again leave via the circulating pipe. Severe vortexing must be eliminated to ensure that the supersaturation is relieved. The quantity and the velocity of the recirculation, the size of the body, and the type and speed of the circulating pump are critical design items if predictable results are to be achieved. A further discussion of the parameters affecting this type of equipment is given by Bennett, Newman, and Van Buren [*Chem. Eng. Prog.*, **55**(3), 65 (1959); *Chem. Eng. Prog. Symp. Ser.*, **65**(95), 34, 44 (1969)].

If the crystallizer is not of the evaporative type but relies only on *adiabatic evaporative cooling* to achieve the yield, the heating element is omitted. The feed is admitted into the circulating line after withdrawal of the slurry, at a point sufficiently below the free-liquid surface to prevent flashing during the mixing process.

FC units typically range from 2 to 20 ft in diameter. They are especially useful for high evaporation loads. For example, a unit used to evaporate water at 380 mmHg can typically be designed to handle 250 to 300 lb/(h·ft²). Other than allowing one to adjust the residence time or slurry density, the FC affords little opportunity to change the size distribution.

Draft-Tube-Baffle (DTB) Evaporator-Crystallizer Because mechanical circulation greatly influences the level of nucleation within the crystallizer, a number of designs have been developed that

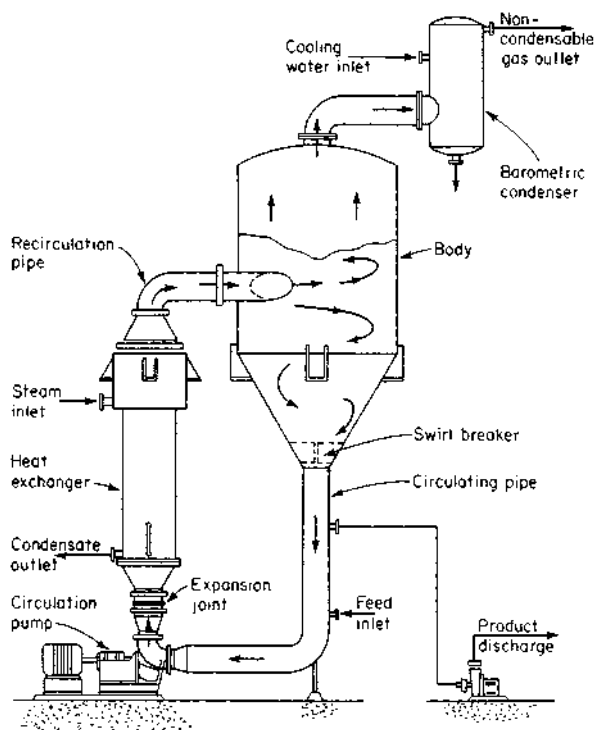


FIG. 18-71 Forced-circulation (evaporative) crystallizer. (Swenson Process Equipment, Inc.)

use circulators located within the body of the crystallizer, thereby reducing the head against which the circulator must pump. This technique reduces the power input and circulator tip speed and therefore the rate of nucleation. A typical example is the draft-tube-baffle (DTB) evaporator-crystallizer (Swenson Process Equipment, Inc.) shown in Fig. 18-72. The suspension of product crystals is maintained by a large, slow-moving propeller surrounded by a draft tube within the body. The propeller directs the slurry to the liquid surface so as to prevent solids from short-circuiting the zone of the most intense supersaturation. Slurry which has been cooled is returned to the bottom of the vessel and recirculated through the propeller. At the propeller, heated solution is mixed with the recirculating slurry.

The design of Fig. 18-72 contains a fines-destruction feature comprising the settling zone surrounding the crystallizer body, the circulating pump, and the heating element. The heating element supplies sufficient heat to meet the evaporation requirements and to raise the temperature of the solution removed from the settler so as to destroy any small crystalline particles withdrawn. Coarse crystals are separated from the fines in the settling zone by gravitational sedimentation, and therefore this fines-destruction feature is applicable only to systems in which there is a substantial density difference between crystals and mother liquor.

This type of equipment can also be used for applications in which the only heat removed is that required for adiabatic cooling of the incoming feed solution. When this is done and the fines-destruction feature is to be employed, a stream of liquid must be withdrawn from the settling zone of the crystallizer and the fine crystals must be separated or destroyed by some means other than heat addition—for example, either dilution or thickening and physical separation.

In some crystallization applications it is desirable to increase the solids content of the slurry within the body above the natural make, which is that developed by equilibrium cooling of the incoming feed solution to the final temperature. This can be done by withdrawing a stream of mother liquor from the baffle zone, thereby thickening the

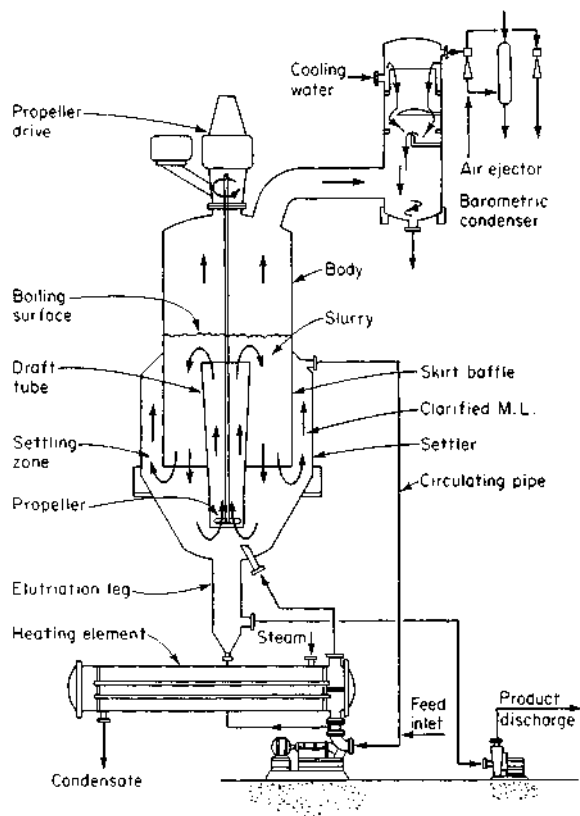


FIG. 18-72 Draft-tube-baffle (DTB) crystallizer. (Swenson Process Equipment, Inc.)

slurry within the growing zone of the crystallizer. This mother liquor is also available for removal of fine crystals for size control of the product.

Draft-Tube (DT) Crystallizer This crystallizer may be employed in systems in which fines destruction is not needed or wanted. In such cases the baffle is omitted, and the internal circulator is sized to have the minimum nucleating influence on the suspension.

In DTB and DT crystallizers the circulation rate achieved is generally much greater than that available in a similar forced-circulation crystallizer. The equipment therefore finds application when it is necessary to circulate large quantities of slurry to minimize supersaturation levels within the equipment. In general, this approach is required to obtain long operating cycles with material capable of growing on the walls of the crystallizer. The draft-tube and draft-tube-baffle designs are commonly used in the production of granular materials such as ammonium sulfate, potassium chloride, photographic hypo, and other inorganic and organic crystals for which product in the range 8 to 30 mesh is required.

Surface-Cooled Crystallizer For some materials, such as sodium chlorate, it is possible to use a forced-circulation tube-and-shell exchanger in direct combination with a draft-tube-crystallizer body, as shown in Fig. 18-73. Careful attention must be paid to the temperature difference between the cooling medium and the slurry circulated through the exchanger tubes. In addition, the path and rate of slurry flow within the crystallizer body must be such that the volume contained in the body is "active." That is to say, crystals must be so suspended within the body by the turbulence that they are effective in relieving supersaturation created by the reduction in temperature of the slurry as it passes through the exchanger. Obviously, the circulating pump is part of the crystallizing system, and careful attention must be paid to its type and its operating parameters to avoid undue nucleating influences.

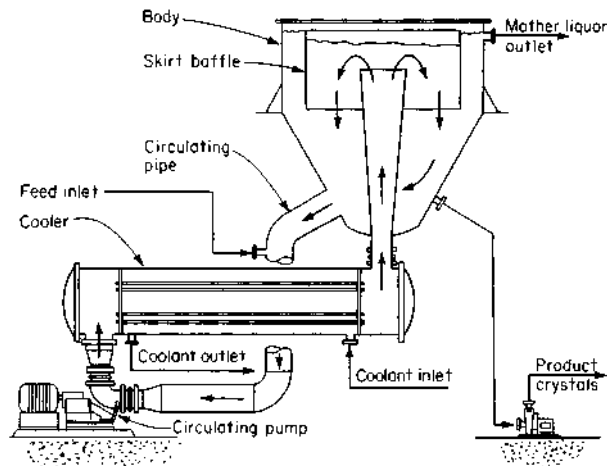


FIG. 18-73 Forced-circulation baffle surface-cooled crystallizer. (Swenson Process Equipment, Inc.)

The use of the internal baffle permits operation of the crystallizer at a slurry consistency other than that naturally obtained by the cooling of the feed from the initial temperature to the final mother-liquor temperature. The baffle also permits fines removal and destruction.

With most inorganic materials this type of equipment produces crystals in the range 30 to 100 mesh. The design is based on the allowable rates of heat exchange and the retention required to grow the product crystals.

Direct-Contact-Refrigeration Crystallizer For some applications, such as the freezing of ice from seawater, it is necessary to go to such low temperatures that cooling by the use of refrigerants is the only economical solution. In such systems it is sometimes impractical to employ surface-cooled equipment because the allowable temperature difference is so small (under 3°C) that the heat-exchanger surface becomes excessive or because the viscosity is so high that the mechanical energy put in by the circulation system requires a heat-removal rate greater than can be obtained at reasonable temperature differences. In such systems, it is convenient to admix the refrigerant with the slurry being cooled in the crystallizer, as shown in Fig. 18-74, so that the heat of vaporization of the refrigerant cools the slurry by direct contact. The successful application of such systems requires that the refrigerant be relatively immiscible with the mother liquor and be capable of separation, compression, condensation, and subsequent recycle into the crystallizing system. The operating pressures and temperatures chosen have a large bearing on power consumption.

This technique has been very successful in reducing the problems associated with buildup of solids on a cooling surface. The use of direct-contact refrigeration also reduces overall process-energy requirements, since in a refrigeration process involving two fluids a greater temperature difference is required on an overall basis when the refrigerant must first cool some intermediate solution, such as calcium chloride brine, and that solution in turn cools the mother liquor in the crystallizer.

Equipment of this type has been successfully operated at temperatures as low as -59°C (-75°F).

Reaction-Type Crystallizers In chemical reactions in which the end product is a solid-phase material such as a crystal or an amorphous solid the type of equipment described in the preceding subsections or shown in Fig. 18-75 may be used. By mixing the reactants in a large circulated stream of mother liquor containing suspended solids of the equilibrium phase, it is possible to minimize the driving force created during their reaction and remove the heat of reaction through the vaporization of a solvent, normally water. Depending on the final particle size required, it is possible to incorporate a fines-destruction baffle as shown in Fig. 18-75 and take advantage of the control over particle size afforded by this technique. In the case of ammonium sulfate crystallization from ammonia gas and concentrated sulfuric acid, it is necessary to vaporize water to remove the heat of reaction, and this water so

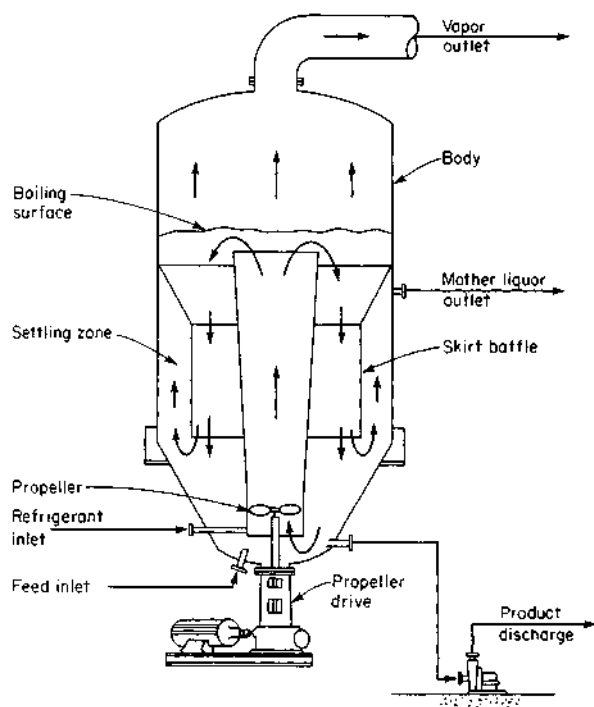


FIG. 18-74 Direct-contact-refrigeration crystallizer (DTB type). (Svenson Process Equipment, Inc.)

removed can be reinjected after condensation into the fines-destruction stream to afford a very large amount of dissolving capability.

Other examples of this technique are where a solid material is to be decomposed by mixing it with a mother liquor of a different composition, as shown in Fig. 18-76. Carnallite ore ($\text{KCl} \cdot \text{MgCl}_2 \cdot 4\text{H}_2\text{O}$) can be added to a mother liquor into which water is also added so that decomposition of the ore into potassium chloride (KCl) crystals and magnesium chloride-rich mother liquor takes place. Circulated slurry in the draft tube suspends the product crystals as well as the incoming ore particles until the ore can decompose into potassium chloride crystals and mother liquor. By taking advantage of the fact that water must be added to the process, the fines-bearing mother liquor can be removed behind the baffle and then water added so that the finest particles are dissolved before being returned to the crystallizer body.

Other examples of this technique involve neutralization reactions such as the neutralization of sulfuric acid with calcium chloride to result in the precipitation of gypsum.

Mixed-Suspension, Classified-Product-Removal Crystallizers Many of the crystallizers just described can be designed for classified-product discharge. Classification of the product is normally done by means of an elutriation leg suspended beneath the crystallizing body as shown in Fig. 18-72. Introduction of clarified mother liquor to the lower portion of the leg fluidizes the particles prior to discharge and selectively returns the finest crystals to the body for further growth. A relatively wide distribution of material is usually produced unless the elutriation leg is extremely long. Inlet conditions at the leg are critical if good classifying action or washing action is to be achieved.

If an elutriation leg or other product-classifying device is added to a crystallizer of the MSMPR type, the plot of the population density versus L is changed in the region of largest sizes. Also the incorporation of an elutriation leg destabilizes the crystal-size distribution and under some conditions can lead to cycling. To reduce cycling, fines destruction is usually coupled with classified product removal. The theoretical treatment of both the crystallizer model and the cycling relations is discussed by Randolph, Beer, and Keener (loc. cit.).

Although such a feature can be included on many types of classified-suspension or mixed-suspension crystallizers, it is most common to use this feature with the forced-circulation evaporative-crystallizer and the DTB crystallizer.

Classified-Suspension Crystallizer This equipment is also known as the **growth** or **Oslo crystallizer** and is characterized by the production of supersaturation in a circulating stream of liquor. Supersaturation is developed in one part of the system by evaporative cooling or by cooling in a heat exchanger, and it is relieved by passing the liquor through a fluidized bed of crystals. The fluidized bed may be contained in a simple tank or in a more sophisticated vessel arranged for a pronounced classification of the crystal sizes. Ideally this equipment operates within the metastable supersaturation field described by Miers and Isaac, *J. Chem. Soc.*, 1906, 413.

In the **evaporative crystallizer** of Fig. 18-77, solution leaving the vaporization chamber at *B* is supersaturated slightly within the metastable zone so that new nuclei will not form. The liquor contacting the bed at *E* relieves its supersaturation on the growing crystals and leaves through the circulating pipe *F*. In a cooling-type crystallization hot feed is introduced at *G*, and the mixed liquor flashes when it reaches the vaporization chamber at *A*. If further evaporation is required to produce the driving force, a heat exchanger is installed between the circulating pump and the vaporization changer to supply the heat for the required rate of vaporization.

The transfer of supersaturated liquor from the vaporizer (point *B*, Fig. 18-77) can cause salt buildup in the piping and reduction of the operating cycle in equipment of this type. The rate of buildup can be reduced by circulating a thin suspension of solids through the vaporizing chamber; however, the presence of such small seed crystals tends to rob the supersaturation developed in the vaporizer, thereby lowering the efficiency of the recirculation system.

The decrease in temperature due to flashing is typically less than 4 to 6°F, and the increase in solute concentration in the circulating liquor is often around 1 to 3 g/L solvent. Care must be taken to ensure that the liquid velocities in the tapered cross-section of the lower body allow classification of the solids. One must know the settling rates and morphologies of the crystals for proper design and operation. An unclassified operation will perform as an FC unit.

The Oslo crystallizer is best suited for use with compounds with high settling velocities such as greater than 20 to 40 mm/s. If the crystals have high settling rates, the larger particles will settle out quickly. Crystals with low settling velocities require large cross-sectional areas which implies large crystallizers and low crystal production rates.

The suggested productivities for concentration driving forces depend on the settling velocities of the crystals. For a given ΔC , the higher the settling velocity, the higher the allowable crystallizer productivity. For example, for a change in concentration of 2 g/L, the recommended productivity increases from 125 to 250 kg/(h·m³) as the settling rate increases from 20 to 30 mm/s.

An **Oslo surface-cooled crystallizer** is illustrated in Fig. 18-78. Supersaturation is developed in the circulated liquor by chilling in the cooler *H*. This supersaturated liquor is contacted with the suspension of crystals in the suspension chamber at *E*. At the top of the suspension chamber a stream of mother liquor *D* can be removed to be used for fines removal and destruction. This feature can be added on either type of equipment. Fine crystals withdrawn from the top of the suspension are destroyed, thereby reducing the overall number of crystals in the system and increasing the particle size of the remaining product crystals.

Scraped-Surface Crystallizer A number of crystallizer designs employing direct heat exchange between the slurry and a jacket or double wall containing a cooling medium have been developed. The heat-transfer surface is scraped or agitated in such a way that the deposits cannot build up. The scraped-surface crystallizer provides an effective and inexpensive method of producing slurry in equipment which does not require expensive installation or supporting structures. At times these units are employed to provide auxiliary cooling capacity for existing units.

Double-Pipe Scraped-Surface Crystallizer This type of equipment consists of a double-pipe heat exchanger with an internal agitator fitted with spring-loaded scrapers that wipe the wall of the inner pipe.

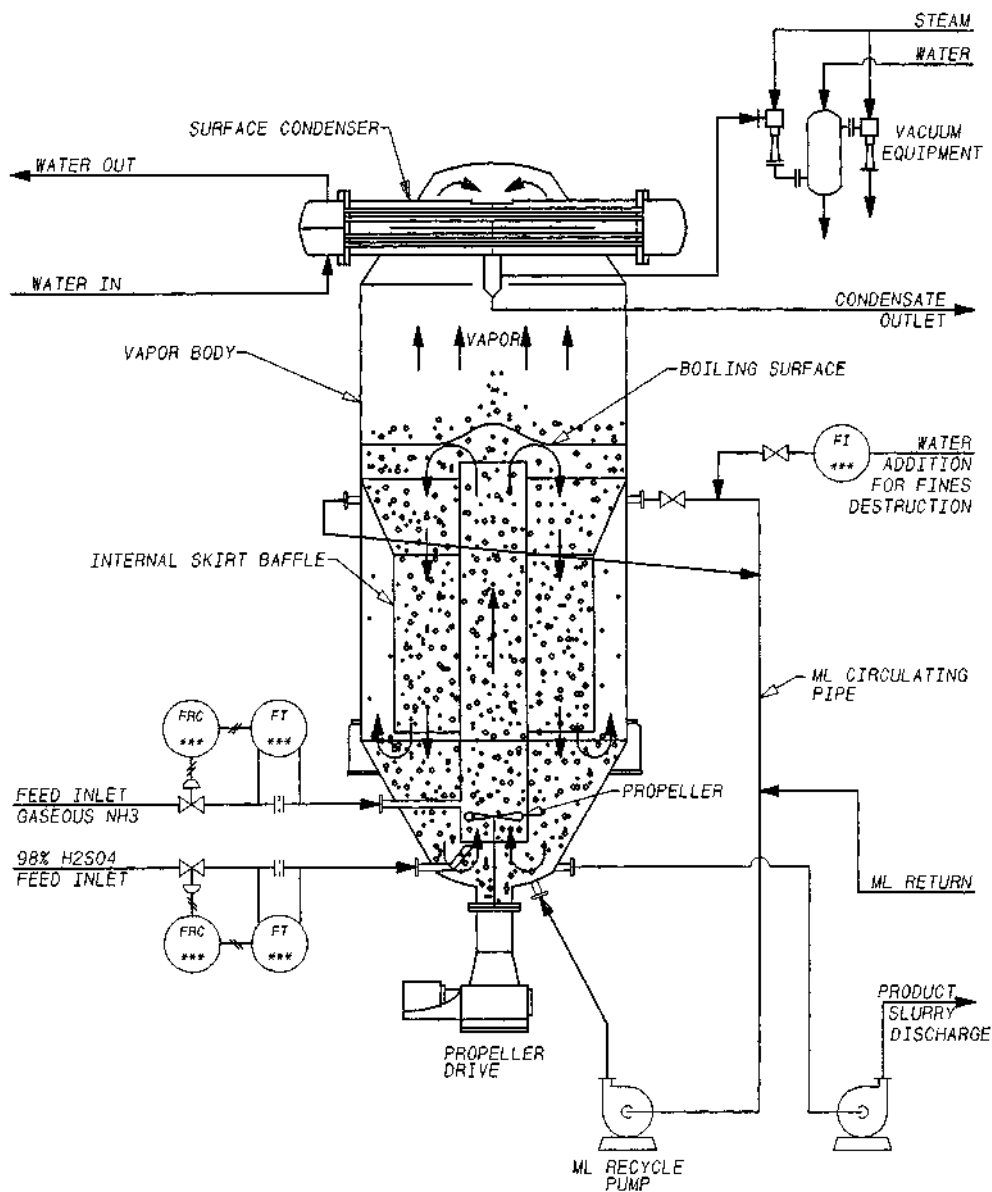


FIG. 18-75 Swenson reaction type DTB crystallizer. (Swenson Process Equipment, Inc.)

The cooling liquid passes between the pipes, this annulus being dimensioned to permit reasonable shell-side velocities. The scrapers prevent the buildup of solids and maintain a good film coefficient of heat transfer. Crystal growth is in the bulk of the liquid. The equipment can be operated in a continuous or in a recirculating batch manner.

Such units are generally built in lengths to above 12 m (40 ft). They can be arranged in parallel or in series to give the necessary liquid velocities for various capacities. Heat-transfer coefficients have been reported in the range of 170 to 850 W/(m²·K) [30 to 150 Btu/(h·ft²·°F)] at temperature differentials of 17°C (30°F) and higher [Garrett and Rosenbaum, *Chem. Eng.*, **65**(16), 127 (1958)]. Equipment of this type is marketed as the **Votator** and the **Armstrong crystallizer**.

Batch Crystallization Batch crystallization has been practiced longer than any other form of crystallization in both atmospheric

tanks, which are either static or agitated, as well as in vacuum or pressure vessels. It is widely practiced in the pharmaceutical and fine chemical industry or in those applications where the capacity is very small. This supersaturation can be generated by a number of modes including antisolvent addition, cooling, evaporation, pH adjustment, and chemical reaction.

A typical batch process involves charging the crystallizer with concentrated or near-saturated solution, producing supersaturation by means of a cooling temperature profile or evaporation profile and seeding the batch in the metastable zone or by allowing spontaneous nucleation to occur. The final mother liquor temperature and concentration is achieved by a time-dependent profile and the batch is then held for ripening followed by transferring the same to downstream processing such as centrifuging, filtration, and drying.

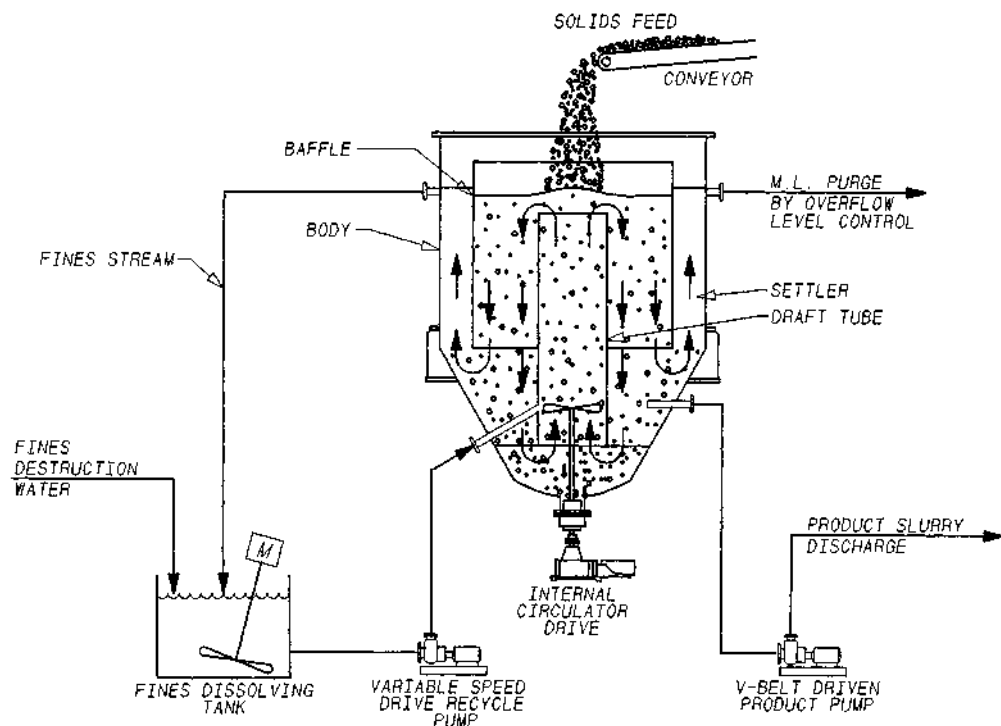


FIG. 18-76 Swenson atmospheric reaction-type DTB crystallizer. (Swenson Process Equipment, Inc.)

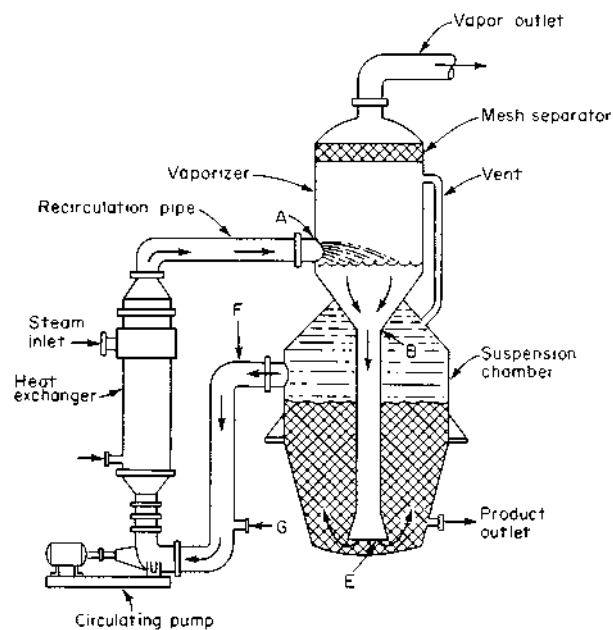


FIG. 18-77 OSLO evaporative crystallizer.

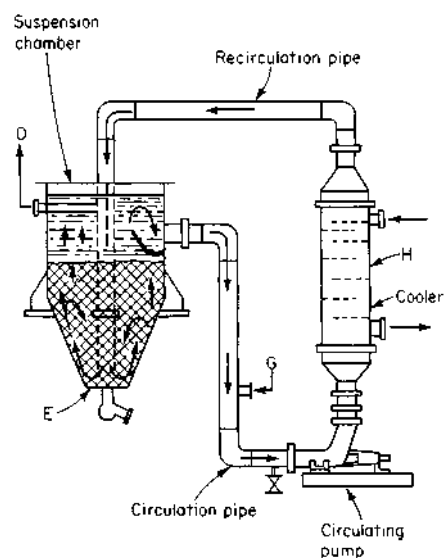


FIG. 18-78 OSLO surface-cooled crystallizer.

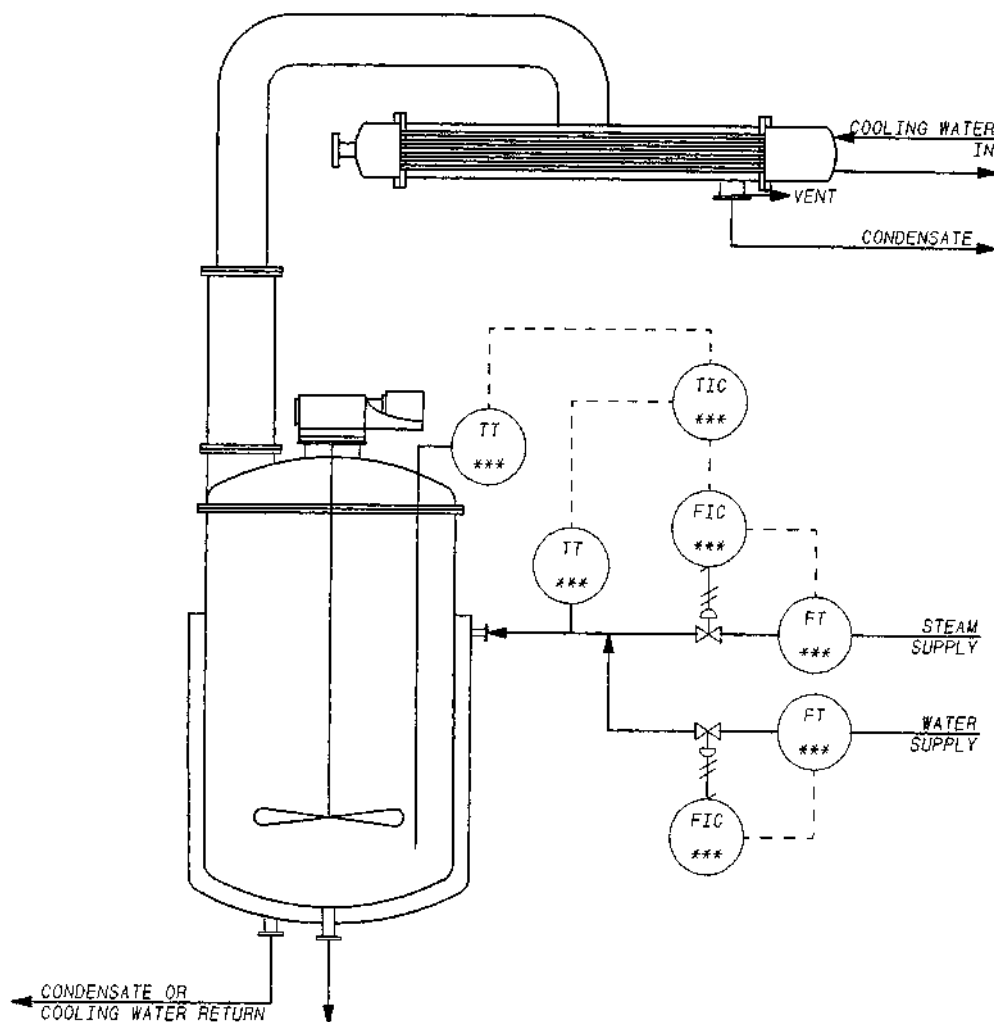


FIG. 18-79 Typical agitated batch crystallizer. (Swenson Process Equipment, Inc.)

Control of a batch crystallizer is critical to achieve the desired size distribution. It is necessary to have some means for determining when the initial solution is supersaturated so that seed of the appropriate size, quantity, and habit may be introduced into the batch. Following seeding, it is necessary to limit the cooling or evaporation in the batch to that which permits the generated supersaturation to be relieved on the seed crystals. This means that the first cooling or evaporation following seeding must be at a very slow rate, which is increased nonlinearly in order to achieve the optimum batch cycle and product properties. Frequently, such controls are operated by cycle timers or computers so as to achieve the required conditions. Shown in Fig. 18-79 is a typical batch crystallizer comprising a jacketed closed tank with top-mounted agitator and feed connections. The tank is equipped with a short distillation column and surface condenser so that volatile materials may be retained in the tank and solvent recycled to maintain the batch integrity. Provisions are included so that the vessel may be heated with steam addition to the shell or cooling solution circulated through the jacket so as to control the temperature. Tanks of this type are intended to be operated with

a wide variety of chemicals under both cooling and solvent evaporation conditions.

A detailed discussion of crystallization practice is provided by Genck in the following articles: Genck, *Chem. Eng.*, **104**(11), 94 (1997); Genck, *Chem. Eng.*, **107**(8), 90 (2000), and Genck, *Chem. Eng. Progress*, **99**(6), 36 (2003).

Recompression Evaporation-Crystallization In all types of crystallization equipment wherein water or some other solvent is vaporized to produce supersaturation and/or cooling, attention should be given to the use of mechanical vapor recompression, which by its nature permits substitution of electrical energy for evaporation and solvent removal rather than requiring the direct utilization of heat energy in the form of steam or electricity. A typical recompression crystallizer flowsheet is shown in Fig. 18-80, which shows a single-stage evaporative crystallizer operating at approximately atmospheric pressure. The amount of heat energy necessary to remove 1 kg of water to produce the equivalent in crystal product is approximately 550 kilocalories. If the water evaporated is compressed by a mechanical compressor of high efficiency

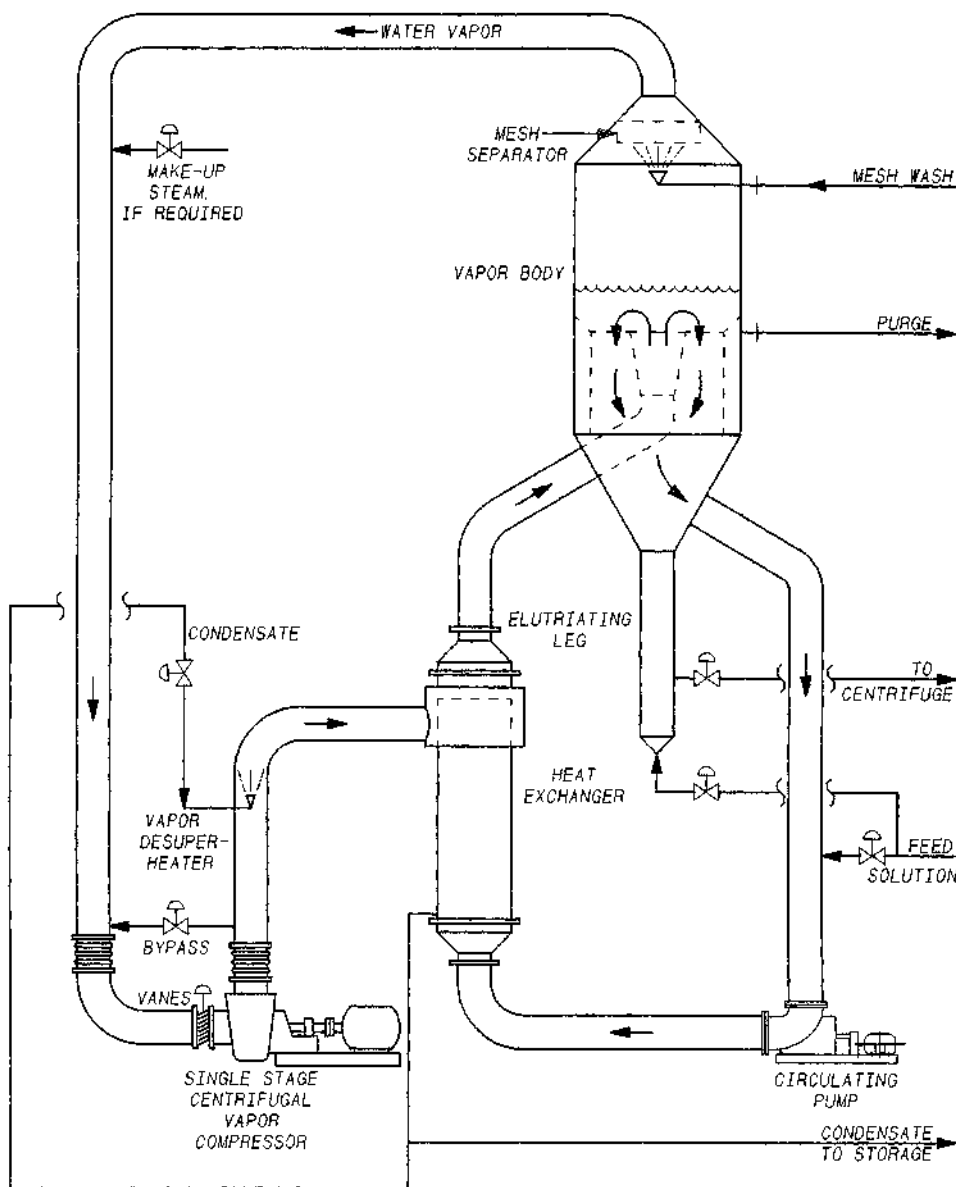


FIG. 18-80 Swenson single-stage recompression evaporator. (Swenson Process Equipment, Inc.)

to a pressure where it can be condensed in the heat exchanger of the crystallizer, it can thereby supply the energy needed to sustain the process. Then the equivalent power for this compression is about 44 kilocalories (Bennett, *Chem. Eng. Progress*, 1978, pp. 67-70).

Although this technique is limited economically to those large-scale cases where the materials handled have a relatively low boiling point elevation and in those cases where a significant amount of heat is required to produce the evaporation for the crystallization step, it nevertheless offers an attractive technique for reducing the use of heat energy and substituting mechanical energy or electrical energy in those cases where there is a cost advantage for doing so. This technique finds many applications in the crystallization of sodium sulfate,

sodium carbonate monohydrate, and sodium chloride. Shown in Fig. 18-81 is the amount of vapor compressed per kilowatt-hour for water vapor at 100°C and various ΔT s. The amount of water vapor compressed per horsepower decreases rapidly with increasing ΔT and, therefore, normal design considerations dictate that the recompression evaporators have a relatively large amount of heat-transfer surface so as to minimize the power cost. Often this technique is utilized only with the initial stages of evaporation where concentration of the solids is relatively low and, therefore, the boiling-point elevation is negligible. In order to maintain adequate tube velocity for heat transfer and suspension of crystals, the increased surface requires a large internal recirculation within the crystallizer body, which consequently lowers the supersaturation in the fluid pumped through the tubes.

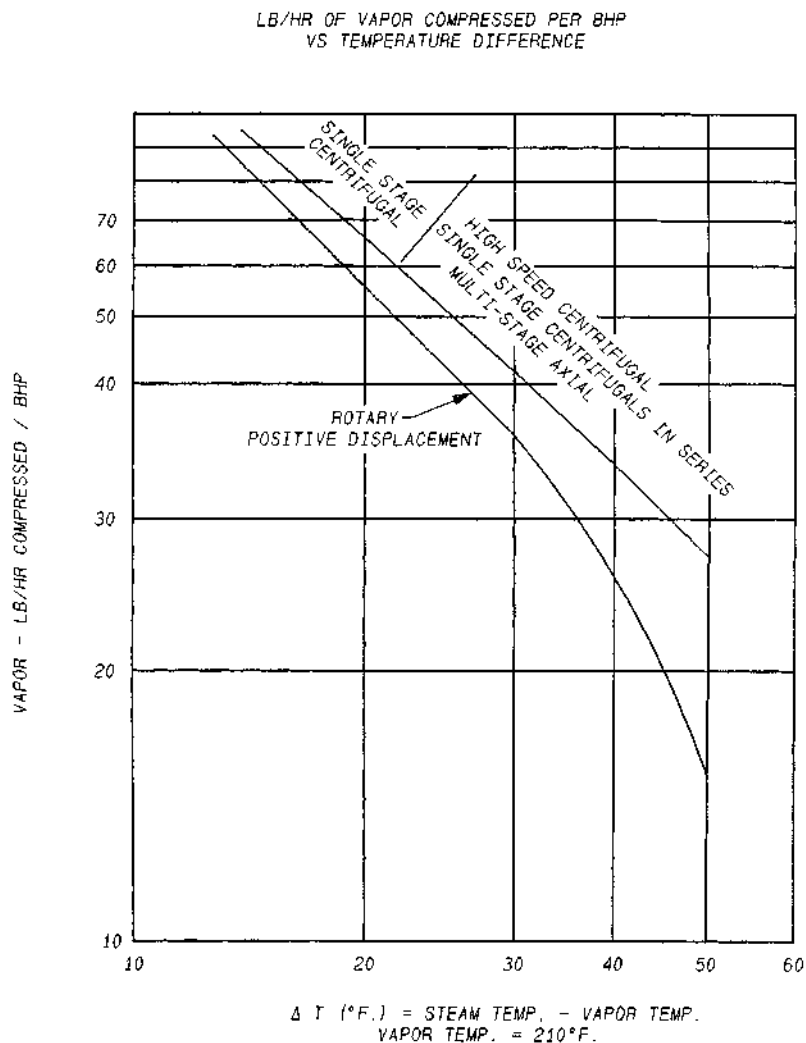


FIG. 18-81 Recompression evaporator horsepower as a function of overall ΔT .

One benefit of this design is that with materials of flat or inverted solubility, the use of recompression complements the need to maintain low ΔT 's to prevent fouling of the heat-transfer surface.

INFORMATION REQUIRED TO SPECIFY A CRYSTALLIZER

The following information regarding the product, properties of the feed solution, and required materials of construction must be available before a crystallizer application can be properly evaluated and the appropriate equipment options identified. Is the crystalline material being produced a hydrated or an anhydrous material? What is the solubility of the compound in water or in other solvents under consideration, and how does this change with temperature? Are other compounds in solution which coprecipitate with the product being crystallized, or do these remain in solution, increasing in concentration until some change in product phase occurs? What will be the influence of impurities in the solution on the crystal habit, growth, and nucleation rates? What are the physical prop-

erties of the solution and its tendency to foam? What is the heat of crystallization of the product crystal? What is the production rate, and what is the basis on which this production rate is computed? What is the tendency of the material to grow on the walls of the crystallizer? What materials of construction can be used in contact with the solution at various temperatures? What utilities will be available at the crystallizer location, and what are the costs associated with the use of these utilities? Is the final product to be blended or mixed with other crystalline materials or solids? What size of product and what shape of product are required to meet these requirements? How can the crystalline material be separated from the mother liquor and dried? Are there temperature requirements or wash requirements which must be met? How can these solids or mixtures of solids be handled and stored without undue breakage and caking? Is polymorphism an issue?

Another basic consideration is whether crystallization is best carried out on a batch basis or on a continuous basis. The present tendency in most processing plants is to use continuous equipment

whenever possible. Continuous equipment permits adjusting of the operating variables to a relatively fine degree in order to achieve the best results in terms of energy usage and product characteristics. It allows the use of a smaller labor force and results in a continuous utility demand, which minimizes the size of boilers, cooling towers, and power-generation facilities. It also minimizes the capital investment required in the crystallizer and in the feed-storage and product-liquor-storage facilities.

Materials that have a tendency to grow readily on the walls of the crystallizer require periodic washout, and therefore an otherwise continuous operation would be interrupted once or even twice a week for the removal of these deposits. The impact that this contingency may have on the processing-equipment train ahead of the crystallizer must be considered.

A batch operation usually has economic application only on small scale, or when multiple products are produced in common facilities.

CRYSTALLIZER OPERATION

Crystal growth is a layer-by-layer process, and the retention time required in most commercial equipment to produce crystals of the size normally desired is often on the order of 2 to 6 h. Growth rates are usually limited to less than 1 to 2 $\mu\text{m}/\text{min}$. On the other hand, nucleation in a supersaturated solution can be generated in a fraction of a second. The influence of any upsets in operating conditions, in terms of the excess nuclei produced, is very short-term in comparison with the total growth period of the product removed from the crystallizer. A worst-case scenario for batch or continuous operation occurs when the explosion of nuclei is so severe that it is impossible to grow an acceptable crystal size distribution, requiring redissolution or washout of the system. In a practical sense, this means that steadiness of operation is much more important in crystallization equipment than it is in many other types of process equipment.

It is to be expected that six to nine retention periods will pass before the effects of an upset will be damped out. Thus, the recovery period may last from 12 to 54 h.

The **rate of nuclei formation** required to sustain a given product size decreases exponentially with increasing size of the product. Although when crystals in the range of 100 to 50 mesh are produced, the system may react quickly, the system response when generating large crystals in the 14-mesh size range is quite slow. This is because a single pound of 150-mesh seed crystals is sufficient to provide the total number of particles in a ton of 14-mesh product crystals. In any system producing relatively large crystals, nucleation must be carefully controlled with respect to all internal and external sources. Particular attention must be paid to preventing seed crystals from entering with the incoming feed stream or being returned to the crystallizer with recycle streams of mother liquor coming back from the filter or centrifuge.

Experience has shown that in any given body operating at a given production rate, control of the magma (slurry) density is important to the control of crystal size. Although in some systems a change in slurry density does not result in a change in the rate nucleation, the more general case is that an increase in the magma density increases the product size through reduction in nucleation and increased retention time of the crystals in the growing bed. The reduction in supersaturation at longer retention times together with the increased surface area at higher percent solids appears to be responsible for the larger product.

A reduction in the magma density will generally increase nucleation and decrease the particle size. This technique has the disadvantage that crystal formation on the equipment surfaces increases because lower slurry densities create higher levels of supersaturation within the equipment, particularly at the critical boiling surface in a vaporization-type crystallizer.

High levels of supersaturation at the liquid surface or at the tube walls in a surface-cooled crystallizer are the dominant cause of wall salting. Although some types of crystallizers can operate for several months continuously when crystallizing KCl or $(\text{NH}_4)_2\text{SO}_4$, most

machines have much shorter operating cycles. Second only to control of particle size, the extension of operating cycles is the most difficult operating problem to be solved in most installations.

In the forced-circulation-type crystallizer (Fig. 19-71) primary control over particle size is exercised by the designer in selecting the circulating system and volume of the body. From the operating standpoint there is little that can be done to an existing unit other than supply external seed, classify the discharge crystals, or control the slurry density. Nevertheless, machines of this type are frequently carefully controlled by these techniques and produce a predictable and desirable product-size distribution.

When crystals cannot be grown sufficiently large in forced-circulation equipment to meet product-size requirements, it is common to employ one of the designs that allow some influence to be exercised over the population density of the finer crystals. In the DTB design (Fig. 18-76) this is done by regulating the flow in the circulating pipe so as to withdraw a portion of the fines in the body in the amount of about 0.05 to 0.5 percent by settled volume. The exact quantity of solids depends on the size of the product crystals and on the capacity of the fines-dissolving system. If the machine is not operating stably, this quantity of solids will appear and then disappear, indicating changes in the nucleation rate within the circuit. At steady-state operation, the quantity of solids overflowing will remain relatively constant, with some solids appearing at all times. Should the slurry density of product crystals circulated within the machine rise to a value higher than about 50 percent settled volume, large quantities of product crystals will appear in the overflow system, disabling the fines-destruction equipment. Too high a circulating rate through the fines trap will produce this same result. Too low a flow through the fines circuit will remove insufficient particles and result in a smaller product-size crystal. To operate effectively, a crystallizer of the type employing fines-destruction techniques requires more sophisticated control than does operation of the simpler forced-circulation equipment.

The classifying crystallizer (Fig. 18-77) requires approximately the same control of the fines-removal stream and, in addition, requires control of the fluidizing flow circulated by the main pump. This flow must be adjusted to achieve the proper degree of fluidization in the suspension chamber, and this quantity of flow varies as the crystal size varies between start-up operation and normal operation. As with the draft-tube-baffle machine, a considerably higher degree of skill is required for operation of this equipment than of the forced-circulation type.

While most of the industrial designs in use today are built to reduce the problems due to excess nucleation, it is true that in some crystallizing systems a deficiency of seed crystals is produced and the product crystals are larger than are wanted or required. In such systems nucleation can be increased by increasing the mechanical stimulus created by the circulating devices or by seeding through the addition of fine crystals from some external source.

CRYSTALLIZER COSTS

Because crystallizers can come with such a wide variety of attachments, capacities, materials of construction, and designs, it is very difficult to present an accurate picture of the costs for any except certain specific types of equipment, crystallizing specific compounds. This is illustrated in Fig. 18-82, which shows the prices of equipment for crystallizing two different compounds at various production rates, one of the compounds being produced in two alternative crystallizer modes. Installed cost (including cost of equipment and accessories, foundations and supporting steel, utility piping, process piping and pumps, electrical switchgear, instrumentation, and labor, but excluding cost of a building) will be approximately twice these price figures.

Most crystallization equipment is custom-designed, and costs for a particular application may vary greatly from those illustrated in Fig. 18-82. Realistic estimation of installation costs also requires reference to local labor rates, site-specific factors, and other case specifics.

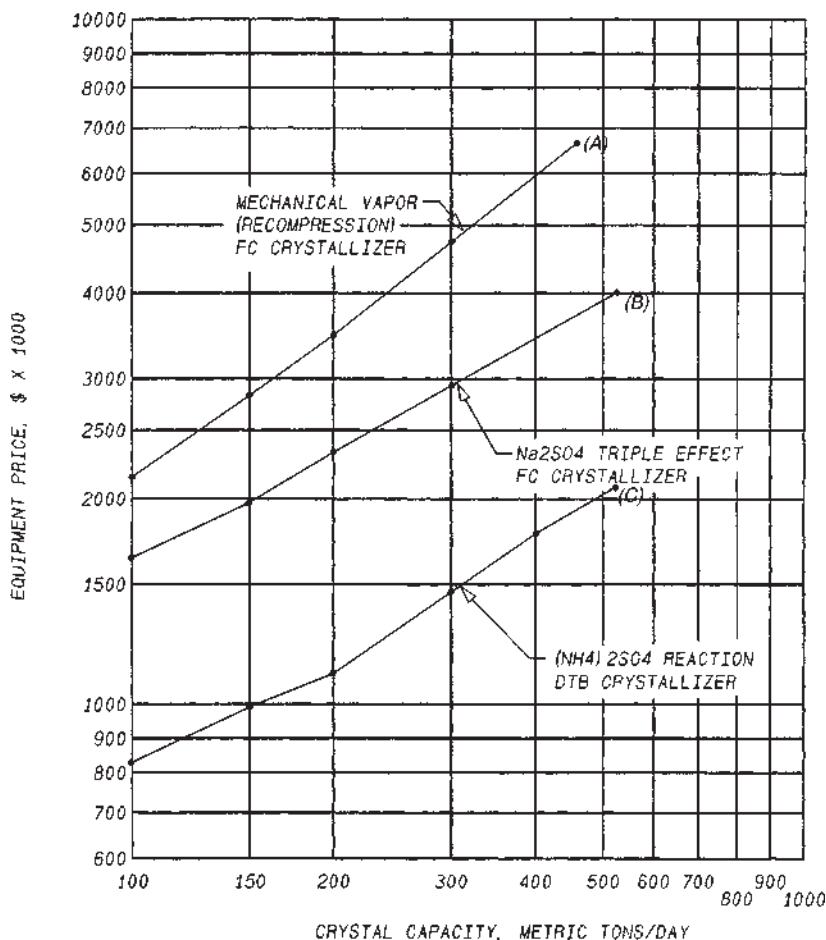


FIG. 18-82 Equipment prices, FOB point of fabrication, for typical crystallizer systems. Prices are for crystallizer plus accessories including vacuum equipment (2005). (A), (B) Na₂SO₄ production from Glauber's salt. Melting tank included. (C) Reaction of NH₃ + H₂SO₄ to make (NH₄)₂SO₄.

LEACHING

GENERAL REFERENCES: Coulson and Richardson, *Chemical Engineering*, 5th ed., vol. 2, Butterworth-Heinemann Publisher, 2002, chap. 10, "Leaching," pp. 502–541. Prabhudesai in Schweitzer, *Handbook of Separation Techniques for Chemical Engineers*, 3d ed., McGraw-Hill, New York, 1996, sec. 5.1. Wilkes, *Fluid Mechanics for Chemical Engineers*, Prentice-Hall, 1999. Wakeman, "Extraction (Liquid-Solid)" in *Kirk-Othmer Encyclopedia of Chemical Technology*, 4th ed., vol. 10, Wiley, New York, 1993, p. 186. McCabe, Smith, and Harriott, *Unit Operations of Chemical Engineering*, 7th ed., McGraw-Hill, New York, 2005. Harriott, *Chemical Reactor Design*, Marcel Dekker, 2003, pp. 89–99. Mular, Halbe, and Barratt, *Mineral Processing Plant Design, Practice, and Control*, vols. 1 and 2, Society for Mining, Metallurgy, and Exploration, Inc., 2002. Section on reactors in annual issues of *Chemical Engineering Buyers' Guide*.

DEFINITION

Leaching is the removal of a soluble fraction, in the form of a solution, from an insoluble, usually permeable, solid phase with which it

is associated. Leaching generally involves selective dissolution with or without diffusion; in the extreme case of simple washing, it requires only displacement (with some mixing) of one interstitial liquid by another with which it is miscible. The soluble constituent may be solid or liquid, and it may be incorporated within, chemically combined with, adsorbed upon, or bound mechanically in the pore structure of the insoluble material. Sometimes, the insoluble phase may be massive and porous, but usually it is particulate; the particles may be openly porous, cellular with selectively permeable cell walls, or surface-activated.

By convention, elution of a surface-adsorbed solute is treated as a special case of adsorption, rather than leaching. The washing of filter cakes is also excluded.

Due to its great breadth of application and its importance to some ancient processes, leaching is known by many names including *extraction*, *solid-liquid extraction*, *lixivation*, *percolation*, *infusion*, *washing*,

and *decantation-settling*. If the stream of solids being leached is densified by settling, it is often called *underflow* and hydrometallurgists may refer to it as *pulp*. Oil seed processors may refer to the solids as *marc*. The liquid stream containing the leached solute is called *overflow*, *extract*, *solution*, *lixivate*, *leachate*, or *miscella*.

Mechanism Leaching may simply result from the solubility of a substance in a liquid, or it may be enabled by a chemical reaction. The rate of transport of solvent into the mass to be leached, or of the soluble fraction into the solvent, or of extract solution out of the insoluble material, or of some combination of these rates may influence overall leaching kinetics, as may an interfacial resistance or a chemical reaction rate.

Inasmuch as the overflow and underflow streams are not immiscible phases but streams based on the same solvent, the concept of equilibrium for leaching is not the one applied in other mass-transfer separations. If the solute is not adsorbed on the inert solid, true equilibrium is reached only when all the solute is dissolved and distributed uniformly throughout the solvent in both underflow and overflow (or when the solvent is uniformly saturated with the solute, a condition never encountered in a properly designed extractor). The practical interpretation of leaching equilibrium is the state in which the overflow and underflow liquids are of the same composition; on a y - x diagram, the equilibrium line will be a straight line through the origin with a slope of unity. It is customary to calculate the number of ideal (equilibrium) stages required for a given leaching task and to adjust the number by applying a stage efficiency factor, although local efficiencies, if known, can be applied stage by stage.

Usually, however, it is not feasible to establish a stage or overall efficiency or a leaching rate index (e.g., overall coefficient) without testing small-scale models of likely apparatus. In fact, the results of such tests may have to be scaled up empirically, without explicit evaluation of rate or quasi-equilibrium indices.

Methods of Operation Leaching systems are distinguished by operating cycle (batch, continuous, or multibatch intermittent); by direction of streams (cocurrent, countercurrent, or hybrid flow); by staging (single-stage, multistage, or differential-stage); and by method of contacting (sprayed percolation, immersed percolation, or solids dispersion). In general, descriptors from all four categories must be assigned to stipulate a leaching system completely (e.g., the Bollman-type extractor is a continuous hybrid-flow multistage sprayed percolator).

Whatever the mechanism and the method of operation, it is clear that the leaching process will be favored by increased surface per unit volume of solids to be leached and by decreased radial distances that must be traversed within the solids, both of which are favored by decreased particle size. Fine solids, on the other hand, cause slow percolation rate, difficult solids separation, and possible poor quality of solid product. The basis for an optimum particle size is established by these characteristics.

LEACHING EQUIPMENT

There are two primary categories of contacting method according to which leaching equipment is classified: (1) leaching may be accomplished by percolation and (2) the particulate solids may be dispersed into a liquid phase and then separated from it. Each may be operated in a batch or continuous manner. Materials that disintegrate during leaching are treated in the second class of equipment. An important exception to this classification is in-situ leaching, as discussed below.

Percolation *Heap leaching*, as shown in Fig. 18-83 (see Mular et al. pp. 1571–1630, loc. cit.) is very widely applied to the ores of copper and precious metals, but percolation is also conducted on a smaller scale in batch tanks or vats and in continuous or dump extractors. In the heap leaching of low-grade oxidized gold ores, for instance, a dilute alkaline solution of sodium cyanide is distributed over a heap of ore that typically has been crushed finer than 1 in and the fines agglomerated with the addition of Portland cement at conveyor transfer points. Heap leaching of very low-grade gold ores and many oxide copper ores is conducted on run-of-mine material. Heap leaching is the least expensive form of leaching. In virtually all cases, an impervious polymeric membrane is installed before the heap is constructed.

In situ leaching, depicted in Fig. 18-84, depends on the existing permeability of a subsurface deposit containing minerals or compounds that are to be dissolved and extracted. Holes (“wells”) are drilled into the rock or soil surrounding the deposit and are lined with tubing that is perforated at appropriate depth intervals. The leaching solution is pumped down the injection wells and flows through the deposit or “formation,” and the “pregnant” solution is extracted from production wells, treated for solute recovery, reconstituted, and reinjected. In situ leaching is used for extraction of halite (NaCl) and uranium, as well as for the removal of toxic or hazardous constituents from contaminated soil or groundwater.

Batch Percolators The *batch tank* is not unlike a big nutsche filter; it is a large circular or rectangular tank with a false bottom. The solids to be leached are dumped into the tank to a uniform depth. They are sprayed with solvent until their solute content is reduced to an economic minimum and are then excavated. Countercurrent flow of the solvent through a series of tanks is common, with fresh solvent entering the tank containing most nearly exhausted material. So-called *vat leaching* was practiced in oxide copper ore processing prior to 1980, and the vats were typically 53 by 20 by 5.5 m (175 by 67 by 18 ft) and extracted about 8200 Mg (9000 U.S. tons) of ore on a 13-day cycle. Some tanks operate under pressure, to contain volatile solvents or increase the percolation rate. A series of pressure tanks operating with countercurrent solvent flow is called a **diffusion battery**.

Continuous Percolators Coarse solids are also leached by percolation in moving-bed equipment, including single-deck and multideck

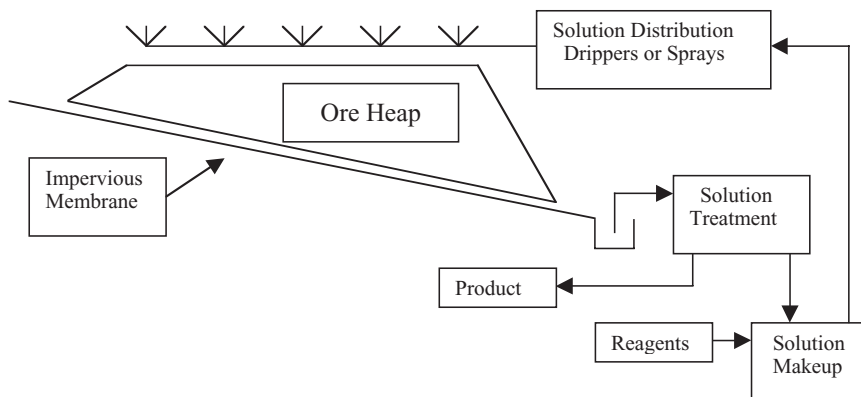


FIG. 18-83 Heap leaching for copper or precious metals.

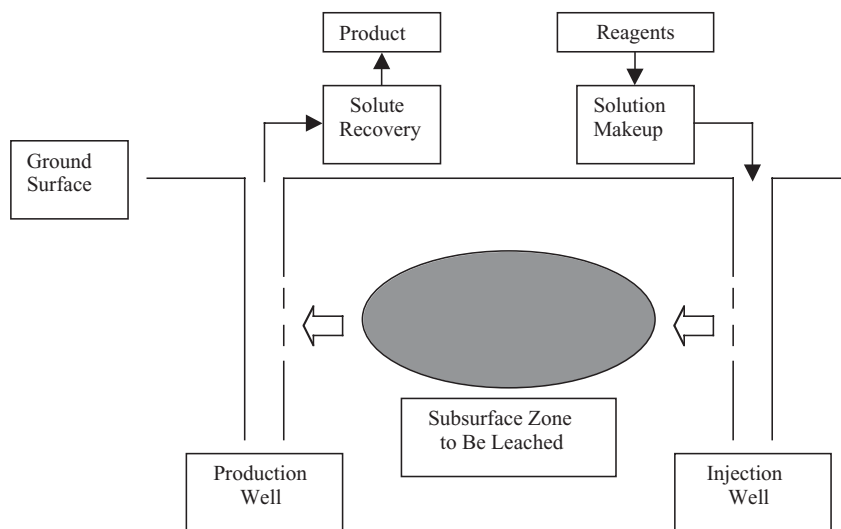


FIG. 18-84 In situ leaching.

rake classifiers, bucket-elevator contactors, and horizontal-belt conveyors.

The **Bollman-type extractor** shown in Fig. 18-85 is a bucket-elevator unit designed to handle about 2000 to 20,000 kg/h (50 to 500 U.S. tons/day) of flaky solids (e.g., soybeans). Buckets with perforated bottoms are held on an endless moving belt. Dry flakes, fed into the descending buckets, are sprayed with partially enriched solvent ("half miscella") pumped from the bottom of the column of ascending buckets. As the buckets rise on the other side of the unit, the solids are sprayed with a countercurrent stream of pure solvent. Exhausted flakes are dumped from the buckets at the top of the unit into a pad-

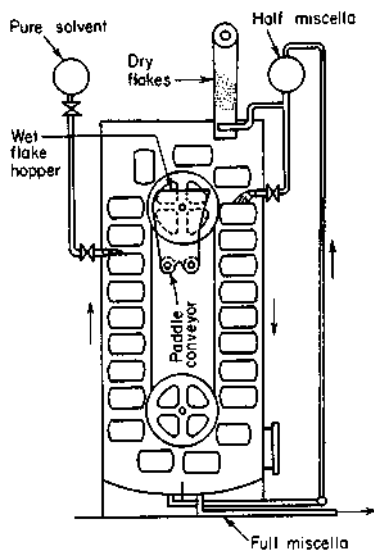


FIG. 18-85 Bollman-type extractor. (McCabe, Smith, and Harriott, *Unit Operations of Chemical Engineering*, 5th ed., p. 616. Copyright 1993 by McGraw-Hill, Inc. and used with permission.)

dle conveyor; enriched solvent, the "full miscella," is pumped from the bottom of the casing. Because the solids are unagitated and because the final miscella moves cocurrently, the Bollman extractor permits the use of thin flakes while producing extract of good clarity. It is only partially a countercurrent device, however, and it sometimes permits channeling and consequent low stage efficiency. Perhaps for this reason, it is being displaced in the oil extraction industry by horizontal basket, pan, or belt percolators (Schwartzberg, loc. cit.).

In the **horizontal-basket design**, illustrated by the **Rotocel extractor** (Fig. 18-86), walled compartments in the form of annular sectors with liquid-permeable floors revolve about a central axis. The compartments successively pass a feed point, a number of solvent sprays, a drainage section, and a discharge station (where the floor opens to discharge the extracted solids). The discharge station is circumferentially contiguous to the feed point. Countercurrent extraction is achieved by feeding fresh solvent only to the last compartment before dumping occurs and by washing the solids in each preceding compartment with the effluent from the succeeding one. The Rotocel is simple and inexpensive, and it requires little headroom. This type of equipment is made by a number of manufacturers. Horizontal table and tilting-pan vacuum filters, of which it is the gravity counterpart, are used as extractors for leaching processes involving difficult solution-residue separation. Detailed descriptions of the Bollman-type and Rotocel extractors are presented on pp. 765 and 766 of McCabe, et al., 7th ed., loc. cit.

The **endless-belt percolator** (Wakeman, loc. cit.) is similar in principle, but the successive feed, solvent spray, drainage, and dumping stations are linearly rather than circularly disposed. Examples are the **de Smet belt extractor** (uncompartmented) and the **Lurgi frame belt** (compartmented), the latter being a kind of linear equivalent of the Rotocel. Horizontal-belt vacuum filters, which resemble endless-belt extractors, are sometimes used for leaching.

The **Kennedy extractor** (Fig. 18-87), also requiring little headroom, operates substantially as a percolator that moves the bed of solids through the solvent rather than the conventional opposite. It comprises a nearly horizontal line of chambers through each of which in succession the solids being leached are moved by a slow impeller enclosed in that section. There is an opportunity for drainage between stages when the impeller lifts solids above the liquid level before dumping them into the next chamber. Solvent flows countercurrently from chamber to chamber. Because the solids are subjected to mechanical action somewhat more intense than in other types of

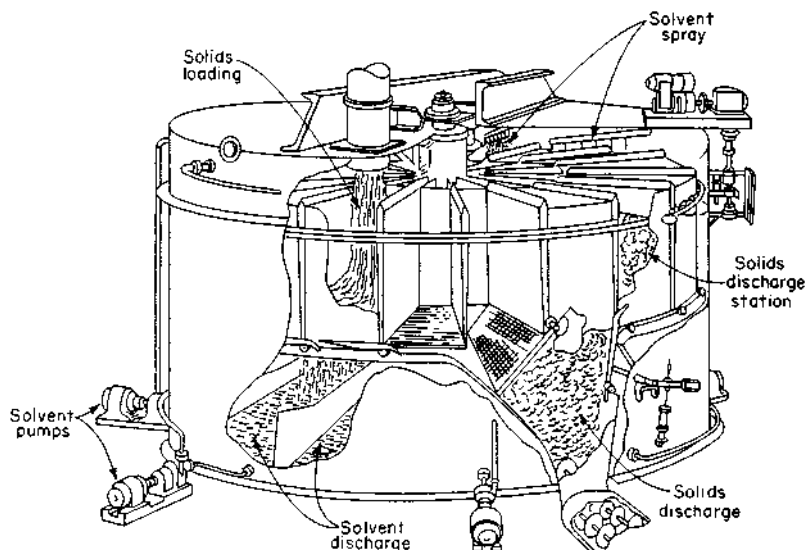


FIG. 18-86 Rotocel extractor. [Rickles, Chem. Eng. 72(6): 164 (1965). Used with permission of McGraw-Hill, Inc.]

continuous percolator, the Kennedy extractor is now little used for fragile materials such as flaked oil seeds.

Dispersed-Solids Leaching Equipment for batch leaching of fine solids in a liquid suspension is now confined mainly to batch tanks with rotating impellers. For a detailed discussion of all aspects of the suspension of solid particles in fluids, refer to agitation of particle suspensions at the beginning of this subsection.

Batch Stirred Tanks Tanks agitated by coaxial impellers (turbines, paddles, or propellers) are commonly used for batch dissolution of solids in liquids and may be used for leaching fine solids. Insofar as the controlling rate in the mass transfer is the rate of transfer of material into or from the interior of the solid particles rather than the rate of transfer to or from the surface of particles, the main function of the agitator is to supply unexhausted solvent to the particles while they reside in the tank long enough for the diffusive process to be completed. The agitator does this most efficiently if it just gently circulates the solids across the tank bottom or barely suspends them above the bottom. However, if the slurry contains particles having significantly different settling velocities, it is usually necessary to introduce sufficient mixing power to ensure full suspension of all particles. Failure to do so will result in an accumulation of the larger or denser particles unless provision is made to drain the settled material continuously.

The leached solids must be separated from the extract by settling and decantation or by external filters, centrifuges, or thickeners, all of which are treated elsewhere in Sec. 18. The difficulty of solids-extract

separation and the fact that a batch stirred tank provides only a single equilibrium stage are its major disadvantages.

Pachuca Tanks Air-agitated Pachuca tanks were widely used in mineral processing until the 1960s when the industry concluded that mechanical agitation was more economical and more effective for solids suspension. A description of Pachuca tanks can be found in previous editions of *Perry's Handbook*.

Impeller-agitated ("Stirred") tanks Often called continuous stirred-tank reactors (CSTRs), they can be operated singly or in series. Figure 18-89 illustrates three tanks in series, each with a mechanical agitator. Nearly all stirred tanks are equipped with vertical baffles to prevent swirling and ineffective energy utilization by the agitator. Advancing of slurry from one stage to the next may be by overflow if successive stages are lower, or interstage pumps may be used.

Autoclaves Autoclaves, as shown in Fig. 18-90, are closed, usually multi-compartmented, vessels often designed for operation at pressures in excess of 600 psig (40 bar) and temperatures of 600°F or higher. The purpose of some autoclaves is simply to effect aqueous oxidation, e.g., of organic wastes or sulfide minerals. In the latter case, an example is oxidation of pyrite, followed by cyanide leaching of precious metals under ambient conditions. Other autoclaves are designed to effect leaching, as in the case of sulfuric acid leaching of nickel and cobalt from lateritic ores. The feed stream is preheated by steam from the flash cooling tower(s) and delivered to the autoclave by one or

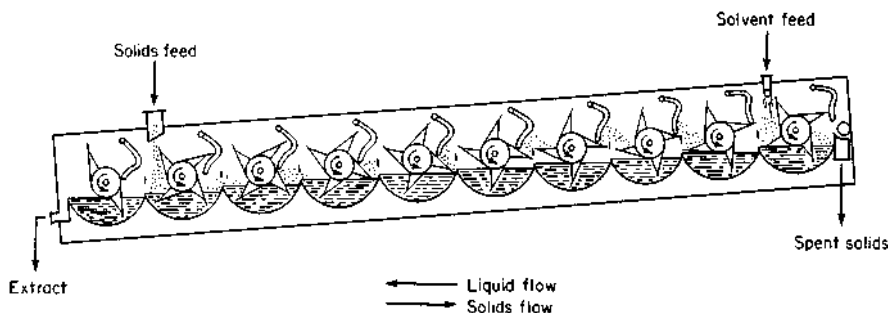


FIG. 18-87 Kennedy extractor. (Vulcan Cincinnati, Inc.)

more positive displacement pumps, usually of the piston diaphragm type. If oxidation is required, oxygen may be used instead of air to reduce operating pressure or to improve kinetics. Flash cooling of the autoclave product is usually accomplished in one or more pressure reduction vessels with abrasion-resistant nozzles and targets.

Continuous Dispersed-Solids Leaching

Vertical-plate extractor. Exemplified by the **Bonotto extractor** (Fig. 18-88), this consists of a column divided into cylindrical compartments by equispaced horizontal plates. Each plate has a radial opening staggered 180° from the openings of the plates immediately above and below it, and each is wiped by a rotating radial blade. Alternatively, the plates may be mounted on a coaxial shaft and rotated past stationary blades. The solids, fed to the top plate, thus are caused to fall to each lower plate in succession. The solids fall as a curtain into solvent which flows upward through the tower. They are discharged by a screw conveyor and compactor. Like the Bollman extractor, the Bonotto has been virtually displaced by horizontal belt or tray percolators for the extraction of oil seeds.

Gravity sedimentation tanks. Operated as thickeners, these tanks can serve as continuous contacting and separating devices in which fine solids may be leached continuously. A series of such units properly connected permit true continuous countercurrent washing of fine solids. If appropriate, a mixing tank may be associated with each thickener to improve the contact between the solids and liquid being fed to that stage. Gravity sedimentation thickeners are described under "Gravity Sedimentation Operations." Of all continuous leaching equipment, gravity thickeners require the most area, and they are limited to relatively fine solids.

Impeller-agitated ("stirred") tanks. Often called continuous stirred-tank reactors (CSTR), they can be operated singly or in series. Figure 18-89 illustrates three tanks in series, each with a mechanical agitator. Nearly all stirred tanks are equipped with vertical baffles to prevent swirling and ineffective energy utilization by the agitator. Advancing of slurry from one stage to the next may be by overflow if successive stages are lower, or interstage pumps may be used.

Autoclaves. Autoclaves, as shown in Fig. 18-90, are closed, usually multicompartimented, vessels often designed for operation at pressures in excess of 600 psig (40 bars) and temperatures of 600°F or higher. The purpose of some autoclaves is simply to effect aqueous oxidation, e.g., of organic wastes or sulfide minerals. In the latter case, an example is oxidation of pyrite, followed by cyanide leaching of precious metals under ambient conditions. Other autoclaves are designed to effect leaching, as in the case of sulfuric acid leaching of nickel and cobalt from lateritic ores. The feed stream is preheated by steam from the flash cooling tower(s) and delivered to the autoclave by one or more

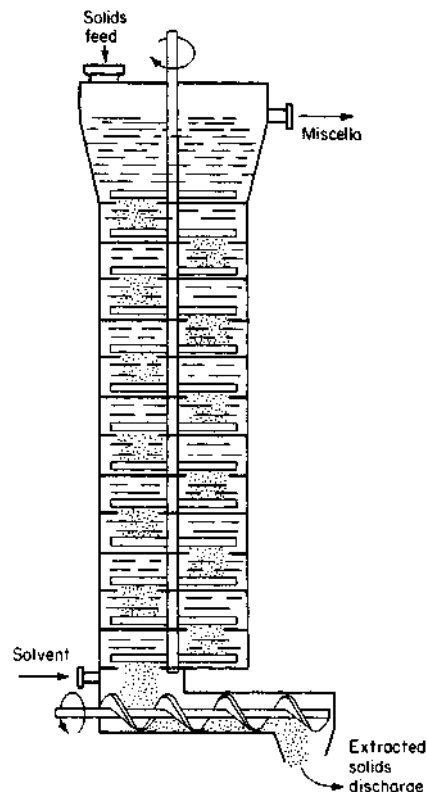


FIG. 18-88 Bonotto extractor. [Rickles, Chem. Eng. 72(6): 163 (1965); copyright 1965 by McGraw-Hill, Inc., New York. Excerpted with special permission of McGraw-Hill.]

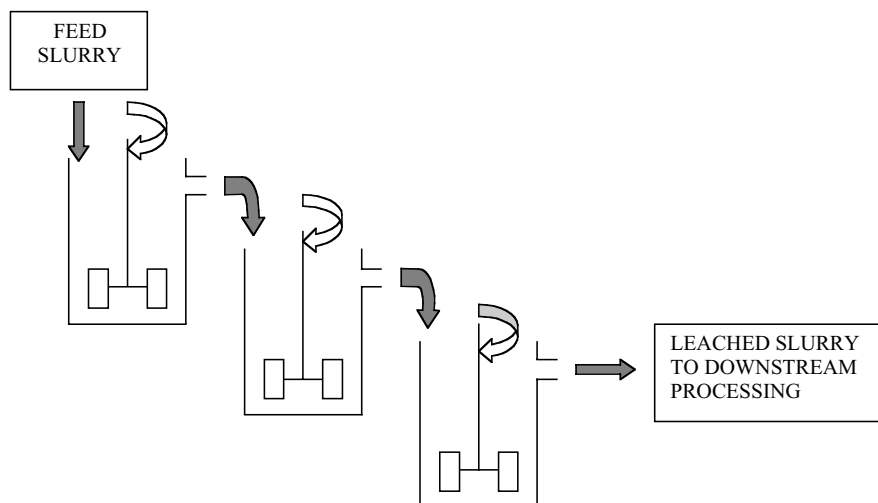


FIG. 18-89 Stirred tanks, three in series, with gravity overflow.

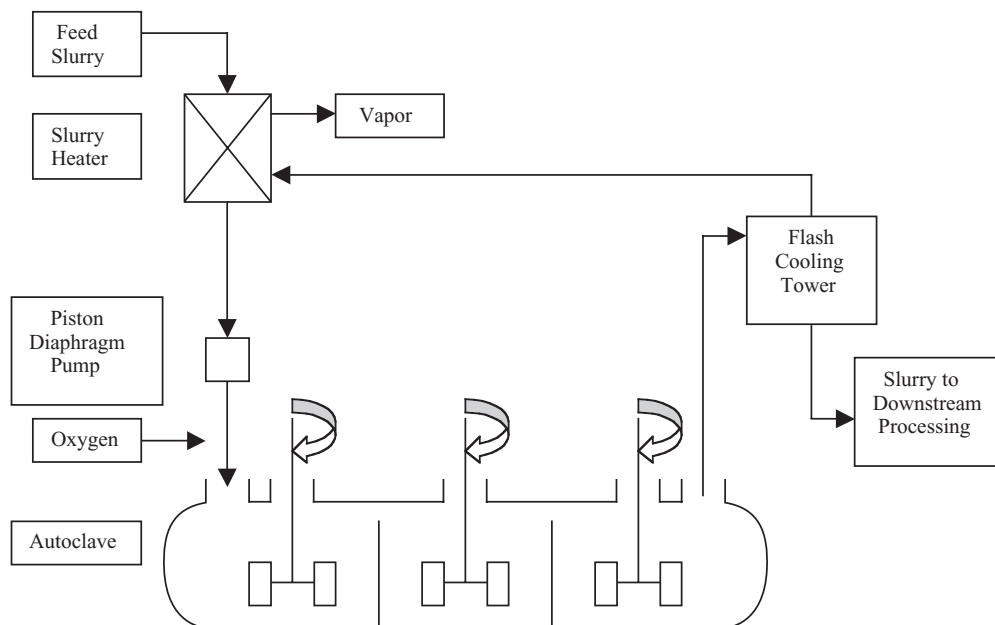


FIG. 18-90 Three-compartment autoclave.

positive displacement pumps, usually of the piston diaphragm type. If oxidation is required, oxygen may be used instead of air to reduce operating pressure or to improve kinetics. Flash cooling of the autoclave product is usually accomplished in one or more pressure-reduction vessels with abrasion-resistant nozzles and targets.

Screw-Conveyor Extractors One type of continuous leaching equipment, employing the screw-conveyor principle, is strictly speaking neither a percolator nor a dispersed-solids extractor. Although it is often classed with percolators, there can be sufficient agitation of the

solids during their conveyance by the screw that the action differs from an orthodox percolation.

The **Hildebrandt total-immersion extractor** is shown schematically in Fig. 18-91. The helix surface is perforated so that solvent can pass through countercurrently. The screws are so designed to compact the solids during their passage through the unit. The design offers the obvious advantages of countercurrent action and continuous solids compaction, but there are possibilities of some solvent loss and feed overflow, and successful operation is limited to light, permeable solids.

A somewhat similar but simpler design uses a horizontal screw section for leaching and a second screw in an inclined section for washing, draining, and discharging the extracted solids.

In the **De Danske Sukkerfabriker**, the axis of the extractor is tilted to about 10° from the horizontal, eliminating the necessity of two screws at different angles of inclination.

Sugar-beet cossettes are successfully extracted while being transported upward in a vertical tower by an arrangement of inclined plates or wings attached to an axial shaft. The action is assisted by staggered guide plates on the tower wall. The shell is filled with water that passes downward as the beets travel upward. This configuration is employed in the **BMA diffusion tower** (Wakeman, loc. cit.).

Schwartzberg (loc. cit.) reports that screw-conveyor extractors, once widely employed to extract flaked oil seeds, have fallen into disuse for this application because of their destructive action on the fragile seed flakes.

Tray Classifier A hybrid like the screw-conveyor classifier, the tray classifier rakes pulp up the sloping bottom of a tank while solvent flows in the opposite direction. The solvent is forced by a baffle to the bottom of the tank at the lower end before it overflows. The solids must be rugged enough to stand the stress of raking.

SELECTION OR DESIGN OF A LEACHING PROCESS*

At the heart of a leaching plant design at any level—conceptual, preliminary, firm engineering, or whatever—is unit-operations and process design of the extraction unit or line. The major aspects that are particular for the leaching operation are the selection of process

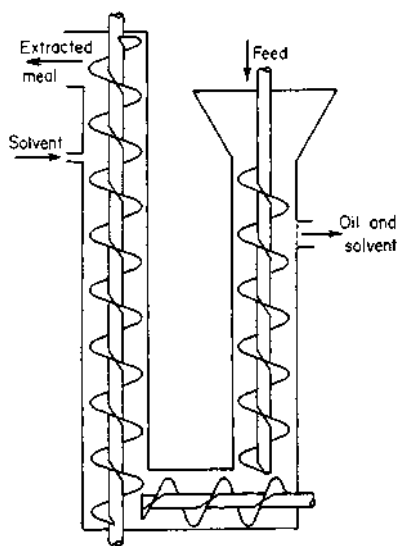


FIG. 18-91 Hildebrandt extractor. (McCabe, Smith, and Harriott, Unit Operations of Chemical Engineering, 5th ed., p. 616. Copyright 1993 by McGraw-Hill, Inc. and used with permission.)

*Portions of this subsection are adaptations from the still-pertinent article by Rickles (loc. cit.).

and operating conditions and the sizing of the extraction equipment.

Process and Operating Conditions The major parameters that must be fixed or identified are the solvent to be used, the temperature, the terminal stream compositions and quantities, leaching cycle (batch or continuous), contact method, and specific extractor choice.

Choice of Solvent The solvent selected will offer the best balance of a number of desirable characteristics: high saturation limit and selectivity for the solute to be extracted, capability to produce extracted material of quality unimpaired by the solvent, chemical stability under process conditions, low viscosity, low vapor pressure, low toxicity and flammability, low density, low surface tension, ease and economy of recovery from the extract stream, and price. These factors are listed in an approximate order of decreasing importance, but the specifics of each application determine their interaction and relative significance, and any one can control the decision under the right combination of process conditions.

Temperature The temperature of the extraction should be chosen for the best balance of solubility, solvent-vapor pressure, solute diffusivity, solvent selectivity, and sensitivity of product. In some cases, temperature sensitivity of materials of construction to corrosion or erosion attack may be significant.

Terminal Stream Compositions and Quantities These are basically linked to an arbitrary given: the production capacity of the leaching plant (rate of extract production or rate of raw-material purification by extraction). When options are permitted, the degree of solute removal and the concentration of the extract stream chosen are those that maximize process economy while sustaining conformance to regulatory standards.

Leaching Cycle and Contact Method As is true generally, the choice between continuous and intermittent operation is largely a matter of the size and nature of the process of which the extraction is a part. The choice of a percolation or solids-dispersion technique depends principally on the amenability of the extraction to effective, sufficiently rapid percolation.

Type of Reactor The specific type of reactor that is most compatible (or least incompatible) with the chosen combination of the preceding parameters seldom is clearly and unequivocally perceived without difficulty, if at all. In the end, however, that remains the objective. As is always true, the ultimate criteria are reliability and profitability.

Extractor-Sizing Calculations For any given throughput rate (which fixes the cross-sectional area and/or the number of extractors), the size of the units boils down to the number of stages required, actual or equivalent. In calculation, this resolves into determination of the number of ideal stages required and application of appropriate stage efficiencies. The methods of calculation resemble those for other mass-transfer operations (see Secs. 13, 14, and 15), involving equilibrium data and contact conditions, and based on material balances. They are discussed briefly here with reference to countercurrent contacting.

Software Packages Since the late 1990s, increasing use has been made of software developed for modeling and simulation of all types of unit operations, including leaching. Packages currently available for mineral processing applications can be found, for instance, in Mular et al., loc. cit., pp. 479 and 495. Monthly issues of *Chemical Engineering Progress (CEP)* usually contain a page entitled *Software* that announces new packages for various applications, and the same publication usually contains a summary each year of all packages of use to chemical engineers and mineral processors.

Composition Diagrams In its elemental form, a leaching system consists of three components: inert, insoluble solids; a single non-adsorbed solute, which may be liquid or solid; and a single solvent.^{*} Thus, it is a ternary system, albeit an unusual one, as already mentioned, by virtue of the total mutual "insolubility" of two of the phases and the simple nature of equilibrium.

The composition of a typical system is satisfactorily presented in the form of a diagram. Those diagrams most frequently employed are a right-triangular plot of mass fraction of solvent against mass fraction of solute (Fig. 18-92a) and a plot suggestive of a Ponchon-Savarit dia-

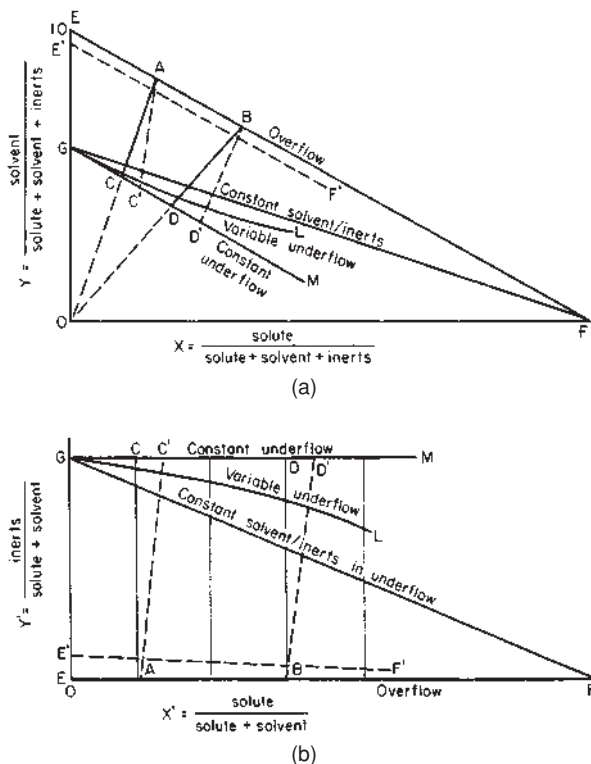


FIG. 18-92 Composition diagrams for leaching calculations: (a) right-triangular diagram; (b) modified Ponchon-Savarit diagram.

gram, with inerts taking the place of enthalpy (Fig. 18-92b). A third diagram, less frequently used, is a modified McCabe-Thiele plot in which the overflow solution (inerts-free) and the underflow solution (traveling out of a stage with the inerts) are treated as pseudo phases, the mass fraction of solute in overflow, y , being plotted against the mass fraction of solute in underflow, x . (An additional representation, the equilateral-triangular diagram frequently employed for liquid-liquid ternary systems, is seldom used because the field of leaching data is confined to a small portion of the triangle.)

With reference to Fig. 18-92 (both graphs), EF represents the locus of overflow compositions for the case in which the overflow stream contains no inert solids. $E'F'$ represents the overflow streams containing some inert solids, either by entrainment or by partial solubility in the overflow solution. Lines GF , GL , and GM represent the loci of underflow compositions for the three different conditions indicated on the diagram. In Fig. 18-92a, the constant underflow line GM is parallel to EF , the hypotenuse of the triangle, whereas GF passes through the right-hand vertex representing 100 percent solute. In Fig. 18-92b, underflow line GM is parallel to the abscissa, and GF passes through the point on the abscissa representing the composition of the clear solution adhering to the inert solids.

Compositions of overflow and underflow streams leaving the same stage are represented by the intersection of the composition lines for those streams with a tie line (AC , AC' , BD , BD'). Equilibrium tie lines (AC , BD) pass through the origin (representing 100 percent inerts) in Fig. 18-92a, and are vertical (representing the same inert-free solution composition in both streams) in Fig. 18-92b. For nonequilibrium conditions with or without adsorption or for equilibrium conditions with selective adsorption, the tie lines are displaced, such as AC' and BD' . Point C' is to the right of C if the solute concentration in the overflow solution is less than that in the underflow solution adhering to the solids. Unequal concentrations in the two solutions indicate insufficient contact time and/or preferential adsorption of one of the components on the inert solids. Tie lines such as AC' may be considered as

^{*}The solubility of the inert, adsorption of solute on the inert, and complexity of solvent and extracted material can be taken into account if necessary. Their consideration is beyond the scope of this treatment.

"practical tie lines" (i.e., they represent actual rather than ideal stages) if data on underflow and overflow composition have been obtained experimentally under conditions simulating actual operation, particularly with respect to contact time, agitation, and particle size of solids.

The illustrative construction lines of Fig. 18-92 have been made with the assumption of constant underflow. In the more realistic case of variable underflow, the points C, C', D, D' would lie along line GL . Like the practical tie lines, GL is a representation of experimental data.

Algebraic Computation This method starts with calculation of the quantities and compositions of all the terminal streams, using a convenient quantity of one of the streams as the basis of calculation. Material balance and stream compositions are then computed for a terminal ideal stage at either end of an extraction battery (i.e., at point A or point B in Fig. 18-92), using equilibrium and solution-retention data. Calculations are repeated for each successive ideal stage from one end of the system to the other until an ideal stage which corresponds to the desired conditions is obtained. Any solid-liquid extraction problem can be solved by this method.

For certain simplified cases it is possible to calculate directly the number of stages required to attain a desired product composition for a given set of feed conditions. For example, if equilibrium is attained in all stages and if the underflow mass rate is constant, both the equilibrium and operating lines on a modified McCabe-Thiele diagram are straight, and it is possible to calculate directly the number of ideal stages required to accommodate any rational set of terminal flows and compositions (McCabe, Smith, and Harriott, op. cit.):

$$N = \frac{\log [(y_b - x_b)/(y_a - x_a)]}{\log [(y_b - y_a)/(x_b - x_a)]} \quad (18-45)$$

Even when the conditions of equilibrium in each stage and constant underflow obtain, Eq. (18-45) normally is not valid for the first stage because the unextracted solids entering that stage usually are not premixed with solution to produce the underflow mass that will leave. This is easily rectified by calculating the exit streams for the first stage and using those values in Eq. (18-45) to calculate the number of stages required after stage 1.

Graphical Method This method of calculation is simply a diagrammatic representation of all the possible compositions in a leaching system, including equilibrium values, on which material balances across ideal (or, in some cases, nonideal) stages can be evaluated in the graphical equivalent of the stage-by-stage algebraic computation. It normally is simpler than the hand calculation of the algebraic solution, and it is viewed by many as helpful because it permits visualization of the process variables and their effect on the operation. Any of the four types of composition diagrams described above can be used, but modified Ponchon-Savarit or right-triangular plots (Fig. 18-92) are most convenient for leaching calculations.

The techniques of graphical solution, in fact, are not unlike those for distillation and absorption (binary) problems using McCabe-Thiele, Ponchon-Savarit, and right-triangular diagrams and are similar to those described in Sec. 15 for solvent-extraction (ternary) systems. More detailed explanations of the application of the several graphical conventions to leaching are presented by: Coulson and Richardson, right triangle; Rickles, modified Ponchon-Savarit; McCabe, Smith, and Harriott, modified McCabe-Thiele; and Schwartzberg, equilateral ternary diagram; all in the publications cited as general references. (See also Treybal, *Mass Transfer Operations*, 3d ed., McGraw-Hill, New York, 1980.)

GRAVITY SEDIMENTATION OPERATIONS

GENERAL REFERENCES: Albertson, *Fluid/Particle Sep. J.*, **7**, 1S (1994). Jewell, Fourie, and Lord, *Paste and Thickened Tailings—A Guide*, pp. 49–79, Australian Centre for Geomechanics, 2002. Mular, Halbe, and Barratt, *Mineral Processing Plant Design, Practice, and Control*, vol. 2, pp. 1295–1312 and 2164–2173, SME, 2002. Sankey and Payne, *Chemical Reagents in the Mineral Processing Industry*, p. 245, SME, 1985. Schweitzer, *Handbook of Separation Techniques for Chemical Engineers*, 2d ed., pp. 4-121 to 4-147, McGraw-Hill, 1988. Wilhelm and Naide, *Min. Eng.* (Littleton, Colo.), 1710 (1981).

Sedimentation is the partial separation or concentration of suspended solid particles from a liquid by gravity settling. This field may be divided into the functional operations of thickening and clarification. The primary purpose of thickening is to increase the concentration of suspended solids in a feed stream, while that of clarification is to remove a relatively small quantity of suspended particles and produce a clear effluent. These two functions are similar and occur simultaneously, and the terminology merely makes a distinction between the primary process results desired. Generally, thickener mechanisms are designed for the heavier-duty requirements imposed by a large quantity of relatively concentrated pulp, while clarifiers usually will include features that ensure essentially complete suspended-solids removal, such as greater depth, special provision for coagulation or flocculation of the feed suspension, and greater overflow-weir length.

CLASSIFICATION OF SETTLEABLE SOLIDS AND THE NATURE OF SEDIMENTATION

The types of sedimentation encountered in process technology will be greatly affected not only by the obvious factors—particle size, liquid viscosity, solid and solution densities—but also by the characteristics of the particles within the slurry. These properties, as well as the process requirements, will help determine both the type of equipment which will achieve the desired ends most effectively and the testing methods to be used to select the equipment.

Figure 18-93 illustrates the relationship between solids concentration, interparticle cohesiveness, and the type of sedimentation that may exist. "Totally discrete" particles include many mineral particles (usually greater in diameter than 20 μm), salt crystals, and similar substances that have little tendency to cohere. "Flocculent" particles generally will include those smaller than 20 μm (unless present in a dispersed state owing to surface charges), metal hydroxides, many chemical precipitates, and most organic substances other than true colloids.

At low concentrations, the type of sedimentation encountered is called

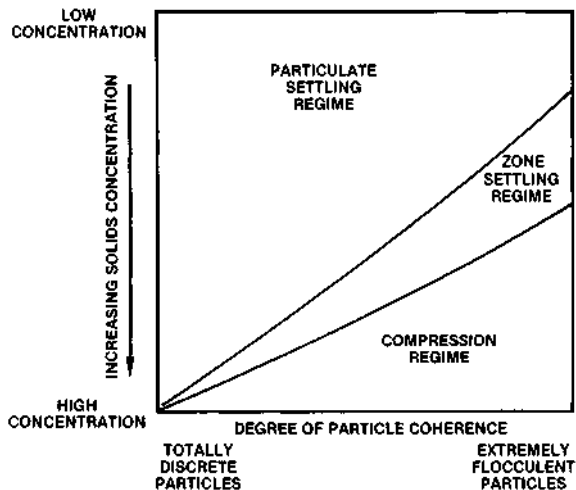


FIG. 18-93 Combined effect of particle coherence and solids concentration on the settling characteristics of a suspension.

particulate settling. Regardless of their nature, particles are sufficiently far apart to settle freely. Faster-settling particles may collide with slower-settling ones and, if they do not cohere, continue downward at their own specific rate. Those that do cohere will form flocules of a larger diameter that will settle at a rate greater than that of the individual particles.

There is a gradual transition from particulate settling into the zone-settling regime, where the particles are constrained to settle as a mass. The principal characteristic of this zone is that the settling rate of the mass, as observed in batch tests, will be a function of its solids concentration (for any particular condition of flocculation, particle density, etc.).

The solids concentration ultimately will reach a level at which particle descent is restrained not only by hydrodynamic forces but also partially by mechanical support from the particles below; therefore, the weight of particles in mutual contact can influence the rate of sedimentation of those at lower levels. This compression, as it is termed, will result in further solids concentration because of compaction of the individual flocules and partial filling of the interfloc voids by the deformed flocules. Accordingly, the rate of sedimentation in the compression regime is a function of both the solids concentration and the depth of pulp in this particular zone. As indicated in Fig. 18-93, granular, nonfloculent particles may reach their ultimate solids concentration without passing through this regime.

As an illustration, coarse-size (45 μm) the aluminum oxide trihydrate particles produced in the Bayer process would be located near the extreme left of Fig. 18-93. These solids settle in a particulate manner, passing through a zone-settling regime only briefly, and reach a terminal density or ultimate solids concentration without any significant compressive effects. At this point, the solids concentration may be as much as 80 percent by weight. The same compound, but of the gelatinous nature it has when precipitated in water treatment as aluminum hydroxide, would be on the extreme right-hand side of the figure. This flocculent material enters into a zone-settling regime at a low concentration (relative to the ultimate concentration it can reach) and gradually thickens. With sufficient pulp depth present, preferably aided by gentle stirring or vibration, the compression-zone effect will occur; this is essential for the sludge to attain its maximum solids concentration, around 10 percent. Certain fine-size (1- to 2- μm) precipitates of this compound will possess characteristics intermediate between the two extremes.

A feed stream to be clarified or thickened can exist at any state represented within this diagram. As it becomes concentrated owing to sedimentation, it may pass through all the regimes, and the settling rate in any one may be the size-determining factor for the required equipment.

Sedimentation Testing To design and size sedimentation equipment, reference information from similar applications is preferred. Data from full-scale sedimentation equipment, operating in the application under consideration, are always a first choice for sizing new equipment. However, quite often the application under question deviates sufficiently from reference installations. The characteristics of the feed stream for the new application (i.e., solids characteristics, particle size, viscosities, pH, use of flocculants, etc.) must be identical to the existing application. It is also necessary to know how close to "capacity" the existing equipment is operating. If the feed characteristics and operating conditions are different for the application under question, bench- or pilot-scale testing is recommended to size and design a new sedimentation unit.

To properly design and size sedimentation equipment, several pieces of information are required. Some information is unique to the job site (application, feed rate, etc.), while other data are supplied from similar references or from test work. Site-specific information from the plant site includes

- Application – objectives (underflow, TSS, hardness, etc.)
- Feed rate—design and maximum
- Feed characteristics—solids concentration, chemistry
- Site-specific requirements: seismic zone, weather-related specifications, local mechanical design codes, and the user's preferred design specifications
- Local operating practices

In the event testing is required to design either a thickener or a clarifier, the testing must be structured to produce all or some of the following information:

- Feed stream characteristics
- Chemical treatment (type, solution concentration, dose, etc.): coagulants and flocculants (organic or inorganic); acid/base for treatment and pH correction
- Coagulation and flocculation (mixing time, energy requirements, solids concentration)
- Expected sedimentation objectives: underflow slurry density or concentration; overflow solids concentration (suspended solids and/or turbidity); chemical treatment for soluble components (i.e., hardness, metals, anions, pH, etc.)
- Vessel area and depth
- Settled solids rheology (for raking mechanism design and drive torque specification)

There are three basic approaches to testing for sedimentation equipment:

- **Batch bench-scale settling tests** The most common procedure requires a relatively small amount of sample tested in a controlled environment using laboratory equipment under static conditions.
- **Semicontinuous bench-scale tests** Laboratory pumps are used which pump feed slurry and chemicals into settling cylinders from which overflow liquor and underflow slurry are continuously collected.
- **Continuous piloting** A small-diameter thickener or clarifier of the same design as the full-scale equipment being considered is used.

TESTING COMMON TO CLARIFIERS AND THICKENERS

Feed Characterization Sample characterization is necessary for both thickening and clarification testing. Without these data included in the basis of design, the sizing and predicted performance cannot be validated for the specified feed stream. Characterization requires the following measurements as a minimum:

- General chemical makeup of the solids and liquor phases
- Feed solids concentration
- Particle size distribution—include coarse (+100 μm) and fine (–20 μm) particle diameters
- Particle specific gravity
- Liquid specific gravity
- Liquid-phase dissolved materials concentration
- Temperature
- pH

Coagulant and/or Flocculant Selection Coagulants and flocculants are widely used to enhance the settling rate which reduces thickener and clarifier size and improves overflow clarity and/or underflow slurry density. The terms *coagulation* and *flocculation* are sometimes used interchangeably; however, each term describes separate functions in the particle agglomeration process.

Coagulation is a preconditioning step that may be required to destabilize the solids suspension to allow complete flocculation to occur in clarification applications. Flocculation is the bridging and binding of destabilized solids into larger particles. As particle size increases, settling rate generally increases. The science of flocculation is not discussed here but can be found in numerous texts and literature which are readily available from flocculant vendors.

Both coagulation and flocculation are typically considered in designing clarifiers, whereas flocculation is normally the only step in designing thickeners.

Coagulants may be either organic such as polyelectrolytes or inorganic such as alum. Coagulants can be used alone or in conjunction with flocculants to improve the performance of the flocculant or reduce the quantity of the flocculant required. In some systems, where a flocculant has been used in an upstream process, a coagulant may be needed to allow additional flocculant to be effective.

There are two primary types of flocculants:

- **Natural flocculants** Starch, guar, and other natural materials have historically been used for sedimentation flocculation, but have been replaced by more effective synthetic polymers.
- **Synthetic polymeric flocculants** There are hundreds of synthetic polymers available developed for specific applications.

Because of the many available flocculants, a screening program is necessary to choose an effective flocculant. The choice of flocculant can be narrowed by considering the following:

- Prior experience with flocculants on the feed stream under evaluation is always a good source of data.
- Test one each of the major types of flocculant charge: anionic, non-ionic, and cationic.
- Test one each of the synthetic polymer length: long chain, short chain.

The purpose of the screening test is to select a coagulant or flocculant whose generic type will most likely be effective in plant operation, and therefore, suitable for clarifier or thickener testing. Although a thickener or clarifier may be started up on the flocculant selected in the testing, it is very common to conduct further tests on the full-scale machine to further optimize dosage or flocculant type. The flocculant manufacturer can be a source of great assistance in both the testing and the full-scale optimization of flocculant use.

Coagulant or flocculant solutions should be made up according to the manufacturer's instructions and used within the shelf life recommended. The solution concentration recommended for testing is typically more dilute than the "neat" concentration so that the viscosity is lower to make dispersion more rapid during testing.

In the screen tests, each coagulant or flocculant is added to the beaker samples of representative slurry or liquor in a dropwise fashion, while the sample is mixed with a spatula, stirrer, or 3-6 jar stirrer mechanism. The amount of coagulant or flocculant required to initiate floc particle formation is noted along with relevant notes as to the size of the floc, capture of fines, resultant liquor clarity, and stability of the floc structure. The dosage is typically noted in g/t solids if the sample is primarily solids (thickener design), or in mg/L liquor if the sample is primarily for clarification and the solids concentration is low.

TESTING SPECIFIC TO CLARIFICATION

Detention Test This test utilizes a 1- to 4-L beaker or similar vessel. The sample is placed in the container, flocculated by suitable means if required, and allowed to settle. Small samples for suspended-solids analysis are withdrawn from a point approximately midway between liquid surface and settled solids interface, taken with sufficient care that settled solids are not resuspended. Sampling times may be at consecutively longer intervals, such as 5, 10, 20, 40, and 80 min.

The suspended-solids concentration can be plotted on log-log paper as a function of the sampling (detention) time. A straight line usually will result, and the required static detention time t to achieve a certain suspended-solids concentration C in the overflow of an ideal basin can be taken directly from the graph. If the plot is a straight line, the data are described by the equation

$$C = Kt^m \quad (18-46)$$

where the coefficient K and exponent m are characteristic of the particular suspension.

Should the suspension contain a fraction of solids which can be considered "unsettleable," the data are more easily represented by using the so-called second-order procedure. This depends on the data being reasonably represented by the equation

$$Kt = \frac{1}{C - C_\infty} - \frac{1}{C_0 - C_\infty} \quad (18-47)$$

where C_∞ is the unsettleable-solids concentration and C_0 is the concentration of suspended solids in the unsettled (feed) sample. The residual-solids concentration remaining in suspension after a sufficiently long detention time (C_∞) must be determined first, and the data then plotted on linear paper as the reciprocal concentration function $1/(C - C_\infty)$ versus time.

Bulk Settling Test After the detention test is completed, a bulk settling test is done to determine the maximum overflow rate. This is done by carrying out a settling test in which the solids are first concentrated to a level at which zone settling just begins. This is usually marked by a very diffuse interface during initial settling. Its rate of descent is measured with a graduated cylinder of suitable size, preferably at least 1 L, and the initial straight-line portion of the settling curve is used for specifying a bulk-settling rate. The design overflow rate generally should not exceed half of the bulk settling rate. From the two clarifier tests, detention time and bulk settling rate, the more

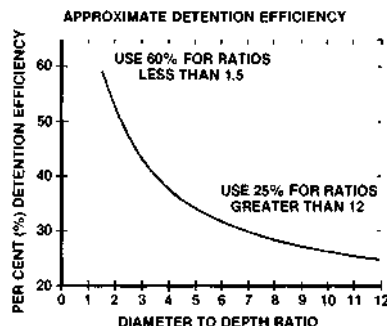


FIG. 18-94 Efficiency curve for scale-up of batch clarification data to determine nominal detention time in a continuous clarifier.

conservative results will govern the size of the clarifier.

Clarification with Solids Recycle In many instances, the rate of clarification is enhanced by increasing the solids concentration in the flocculation zone of the clarifier. This is done in a full-scale operation by internally or externally recycling previously settled solids into the flocculation zone where they are mixed with fresh, coagulated feed. The higher population of solids improves the flocculation efficiency and clarification rate.

To conduct these tests, a sample of feed is first treated at the chemical dosages and mixing intensity determined in the screening tests and flocculated according to the screening test. The solids are allowed to settle, and the supernatant is carefully decanted. The settled solids are then transferred to a new fresh sample, and tests are conducted again, using the same chemical dosages and mixing intensity. Recycle can continue with subsequent tests until the suspended solids in the sample can have concentrations of 1, 2, 3, and 5 g/L. Bulk settling rate, suspended solids, and other effluent parameters are measured with each test until an optimal treatment scenario is found.

In some suspensions, very fine colloidal solids are present and are very difficult to coagulate. In these cases, it is typically necessary to adjust for coagulation mixing intensity and time to obtain coagulated solids that are more amenable to flocculation.

Detention Efficiency Conversion from the ideal basin sized by detention-time procedures to an actual clarifier requires the inclusion of an efficiency factor to account for the effects of turbulence and nonuniform flow. Efficiencies vary greatly, being dependent not only on the relative dimensions of the clarifier and the means of feeding but also on the characteristics of the particles. The curve shown in Fig. 18-94 can be used to scale up laboratory data in sizing circular clarifiers. The static detention time determined from a test to produce a specific effluent solids concentration is divided by the efficiency (expressed as a fraction) to determine the nominal detention time, which represents the volume of the clarifier above the settled pulp interface divided by the overflow rate. Different diameter-depth combinations are considered by using the corresponding efficiency factor. In most cases, area may be determined by factors other than the bulk-settling rate, such as practical tank-depth limitations.

TESTING SPECIFIC TO THICKENING

Optimization of Flocculation Conditions After a flocculant type is selected, the next step is to conduct a range of tests using the selected flocculant, to gather data on the effects of feed slurry solids concentrations on flocculant dosage and settling rate. There are a range of solids concentrations for which flocculation effectiveness is maximized, resulting in improved settling characteristics. Operating within this feed solids range results in smaller equipment sizes, higher underflow slurry densities, better overflow liquor clarity, and lower flocculant dosages.

The tests are conducted using a series of samples prepared at solids concentrations decreasing incrementally in concentration from the expected thickener feed concentration. Typically, the samples are prepared in 250- to 500-mL graduated cylinders which give some distance

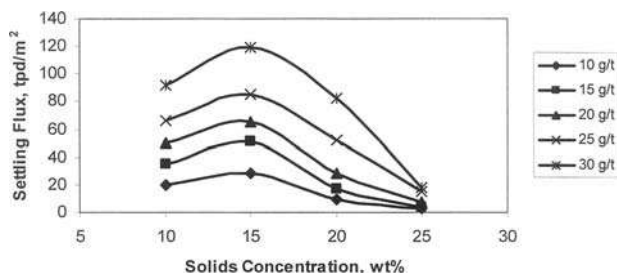


FIG. 18-95 Data showing that slurry solids concentration affects flocculation efficiency, thus improving solids settling flux.

to measure the settling rate more accurately. For some very fine solids samples (e.g., alumina red mud, clays, leached nickel laterites, etc.), it is recommended to also check a sample diluted to 2 to 3 wt % solids. Begin adding the flocculant solution dropwise; make notes on the dosage at which flocculation begins and the settling velocity. Continue adding flocculant incrementally and noting the floc structure, fines capture, liquor clarity, and settling velocity. Once the settling velocity remains constant for a few tests, sample testing can be stopped. From the tests, the plot shown in Fig. 18-95 can be drawn and the results used to set conditions for the larger and final tests for sizing the thickening equipment. The test procedure for the design tests should be structured to span the optimum solids concentration and two points slightly higher and lower. The flocculant dosage should be checked at the optimum and at dosages slightly higher and slightly lower than that determined in the above tests.

Determination of Thickener Basin Area The area requirements for thickeners frequently are based on the solids flux rates measured in the zone-settling regime. Theory holds that, for any specific sedimentation condition, a critical concentration will exist in the thickener which will limit the solids throughput rate. As the concentration in this critical zone represents a steady-state condition, its depth in the settling bed of solids may vary, responding to changes in the feed rate, underflow withdrawal rate, or flocculant dosage. In thickeners operating at relatively high solids retention times and/or low throughput rates, this zone generally does not exist.

Many batch-test methods which are based on determining the solids flux rate at this critical concentration have been developed. Most methods recognize that as the solids enter compression, thickening behavior is no longer a function only of solids concentration. Hence, these methods attempt to utilize the "critical" point dividing these two zones and size the area on the basis of the settling rate of a layer of pulp at this concentration. The difficulty lies in discerning where this point is located on the settling curve.

Many procedures have been developed, but two have been more widely used: the Coe and Clevenger approach and the Kynch method as defined by Talmage and Fitch (op. cit.).

The former requires measurement of the initial settling rate of a pulp at different solids concentrations varying from feed to final underflow value. The area requirement for each solids concentration tested is calculated by equating the net overflow rate to the corresponding interfacial settling rate, as represented by the following equation for the unit area:

$$\text{Unit area} = \frac{1/C_i - 1/C_u}{v_i} \quad (18-48)$$

where C_i is the solids concentration at the interfacial settling velocity v_i and C_u is the underflow concentration, both concentrations being expressed in terms of mass of solids per unit volume of slurry. Using kg/L for the concentrations and m/day for the settling velocity yields a unit area value in $\text{m}^2/(\text{ton}/\text{day})$.

These unit area values, plotted as a function of the feed concentration, will describe a maximum value that can be used to specify the thickener design unit area for the particular underflow concentration C_u employed in Eq. (18-48).

The method is applicable for *unfloculated* pulps or those in which the ionic characteristics of the solution produce a flocculent structure. If polymeric flocculants are used, an approach based on the Kynch theory is preferred. In this method, the test is carried out at the optimum feed solids and flocculant dose (as determined in tests described earlier) and continued until underflow concentration is achieved in the cylinder. The flocculant solution should be added to the slurry under conditions which promote rapid dispersion and uniform, complete mixing with a minimum of shear. In cylinder tests, this is accomplished by simultaneously injecting and mixing flocculant with the slurry, using an apparatus consisting of a syringe, a tube, and an inverted rubber stopper. The rubber stopper, having a diameter approximately 75 percent that of the cylinder diameter, provides sufficient turbulence as it is moved gently up and down through the slurry to cause good blending of the flocculant and pulp. To determine the unit area, Talmage and Fitch (op. cit.) proposed an equation derived from a relationship equivalent to that shown in Eq. (18-49):

$$\text{Unit area} = \frac{t_u}{C_0 H_0} \quad (18-49)$$

where t_u is the time, days; C_0 is the initial solids concentration in the feed, t/m^3 ; and H_0 is the initial height of the slurry in the test cylinder, m. The term t_u is taken from the intersection of a tangent to the curve at the critical point and a horizontal line representing the depth of pulp at underflow concentration. There are various means for selecting this critical point, all of them empirical, and the unit area value determined cannot be considered precise. The review by Pearse (op. cit.) presents many of the different procedures used in applying this approach to laboratory settling test data.

Two other approaches avoid using the critical point by computing the area requirements from the settling conditions existing at the underflow concentration. The Wilhelm and Naide procedure (op. cit.) applies zone-settling theory (Kynch) to the entire thickening regime. Tangents drawn to the settling curve are used to calculate the settling velocity at all concentrations obtained in the test. This permits construction of a plot (Fig. 18-96) showing unit area as a function of underflow concentration.

A second, "direct" approach which yields a similar result, since it also takes compression into account, utilizes the value of settling time t_s taken from the settling curve at a particular underflow concentration. This value is used to solve the Talmage and Fitch equation (18-49) for unit area.

Compression bed depth will have a significant effect on the overall settling rate (increasing compression zone depth reduces unit area). Therefore, in applying either of these two procedures it is necessary to run the test in a vessel having an average bed depth close to that expected in a full-scale thickener. This requires a very large sample, and it is more convenient to carry out the test in a cylinder having a volume of 1 to 4 liters. The calculated unit area value from this test can be extrapolated to full-scale depth by carrying out similar tests at

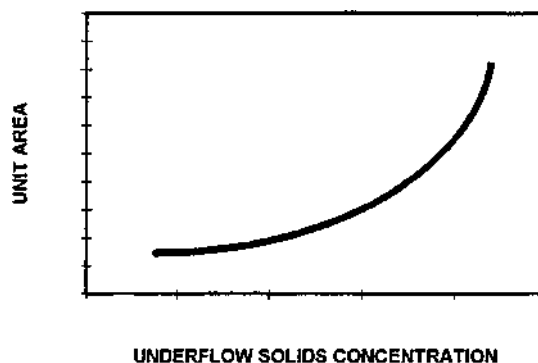


FIG. 18-96 Characteristic relationship between thickener unit area and underflow solids concentration (fixed flocculant dosage and pulp depth).

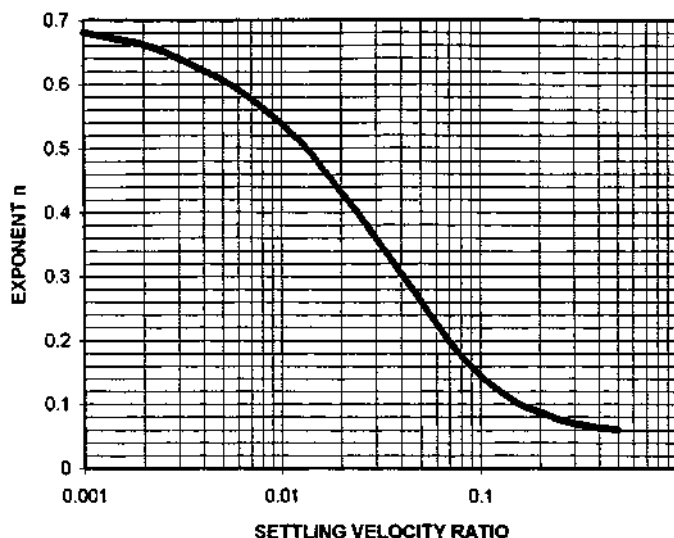


FIG. 18-97 Depth correction factor to be applied to unit areas determined by Wilhelm-Naide and "direct" methods. Velocity ratio calculated using tangents to settling curve at a particular settled solids concentration and at start of test.

different depths to determine the effect on unit area. Alternatively, an empirical relationship can be used which is effective in applying a depth correction to laboratory cylinder data over normal operating ranges. The unit area calculated by either the Wilhelm and Naide approach or the direct method is multiplied by a factor equal to $(h/H)^n$, where h is the average depth of the pulp in the cylinder, H is the expected full-scale compression zone depth, usually taken as 1 m, and n is the exponent calculated from Fig. 18-97. For conservative design purposes, the minimum value of this factor that should be used is 0.25.

It is essential to use a slow-speed (approximately 0.1 r/min) picket rake in all cylinder tests to prevent particle bridging and allow the sample to attain the underflow density which is obtainable in a full-scale thickener.

Continuously operated, small-scale or pilot-plant thickeners, ranging from 75 mm diameter by 400 mm depth to several meters in diameter, are also effectively used for sizing full-scale equipment. This approach requires a significantly greater volume of sample, such as would be available in an operating installation or a pilot plant. Continuous units and batch cylinders will produce equivalent results if proper procedures are followed with either system.

Thickener-Basin Depth The pulp depth required in the thickener will be greatly affected by the role that compression plays in determining the rate of sedimentation. If the zone-settling conditions define the area needed, then depth of pulp will be unimportant and can be largely ignored, as the "normal" depth found in the thickener will be sufficient. On the other hand, with the compression zone controlling, depth of pulp will be significant, and it is essential to measure the sedimentation rate under these conditions. This is true for the new deep-bed, high-density thickeners.

To determine the compression-zone requirement in a thickener, a test should be run in a deep cylinder in which the average settling pulp depth approximates the depth anticipated in the full-scale basin. The average density of the pulp in compression is calculated and used in Eq. (18-50) to determine the required compression-zone volume:

$$V = \frac{\theta_c(\rho_s - \rho_l)}{\rho_s(\rho_s - \rho_l)} \quad (18-50)$$

where V is the volume, m^3 , required per ton of solids per day; θ_c is the compression time, days, required in the test to reach underflow concentration; and ρ_s , ρ_l , ρ_{sl} , are the densities of the solids, liquid, and

slurry (average), respectively, ton/m^3 . This value divided by the average depth of the pulp during the period represents the unit area defined by compression requirements. If it exceeds the value determined from the zone-settling tests, it is the quantity to be used.

The side depth of the thickener is determined as the sum of the depths needed for the compression zone and for the clear zone. Normally, 1.5 to 2 m of clear liquid depth above the expected pulp level in a thickener will be sufficient for stable, effective operation. When the location of the pulp level cannot be predicted in advance or it is expected to be relatively low, a thickener sidewall depth of 2 to 3 m is usually safe. Greater depth may be used in order to provide better clarity, although in most thickener applications the improvement obtained by this means will be marginal.

Scale-up Factors Factors used in thickening will vary, but, typically, a 1.2 to 1.3 multiplier applied to the unit area calculated from laboratory data is sufficient if proper testing procedures have been followed and the samples are representative.

Torque Requirements Sufficient torque must be available in the raking mechanism of a full-scale thickener to allow it to move through the slurry and assist solids movement to the underflow outlet. Granular, particulate solids that settle rapidly and reach a terminal solids concentration without going through any apparent compression or zone-settling region require a maximum raking capability, as they must be moved to the outlet solely by the mechanism. At the other end of the spectrum, extremely fine materials, such as clays and precipitates, require a minimum of raking, for most of the solids may reach the underflow outlet hydrodynamically. The rakes prevent a gradual buildup of some solids on the bottom, however, and the gentle stirring action from the rake arm often aids the thickening process. As the underflow concentration approaches its ultimate limit, the consistency will increase greatly, resulting in a higher raking requirement and an increase in torque.

For most materials, the particle size lies somewhere between these two extremes, and the torques required in two properly designed thickeners of the same size but in distinct applications can differ greatly. Unfortunately, test methods to specify torque from small-scale tests are of questionable value, since it is difficult to duplicate actual conditions. Manufacturers of sedimentation equipment select torque ratings from experience with similar substances and will recommend a torque capability on this basis. Definitions of *operating torque* vary with the manufacturer, and the user should ask the supplier to specify the *B-10 life* for bearings and to reference appropriate mechanical

standards for continuous operation of the selected gear set at specific torque levels. This will provide guidelines for plant operators and help avoid premature failure of the mechanism. Abnormal conditions above the normal operating torque are inevitable, and a thickener should be provided with sufficient torque capability for short-term operation at higher levels in order to ensure continuous performance.

Underflow Pump Requirements Many suspensions will thicken to a concentration higher than that which can be handled by conventional slurry pumps. Thickening tests should be performed with this in mind, for, in general, the unit area to produce the maximum concentration that can be pumped is the usual design basis. Determination of this ultimate pumpable concentration is largely a judgmental decision requiring some experience with slurry pumping; however, the behavior of the thickened suspension can be used as an approximate guide to pumpability. The supernatant should be decanted following a

test and the settled solids repulped in the cylinder to a uniform consistency. Repulping is done easily with a rubber stopper fastened to the end of a rigid rod. If the bulk of the repulped slurry can be poured from the cylinder when it is tilted 10 to 30° above the horizontal, the corresponding thickener underflow can be handled by most types of slurry pumps. But if the slurry requires cylinder shaking or other mechanical means for its removal, it should be diluted to a more fluid condition, if conventional pump systems are to be employed. The best suggestion is to consult with pump experts and suppliers who have amassed large databases for a wide range of materials.

THICKENERS

The primary function of a continuous thickener is to concentrate suspended solids by gravity settling so that a steady-state material balance

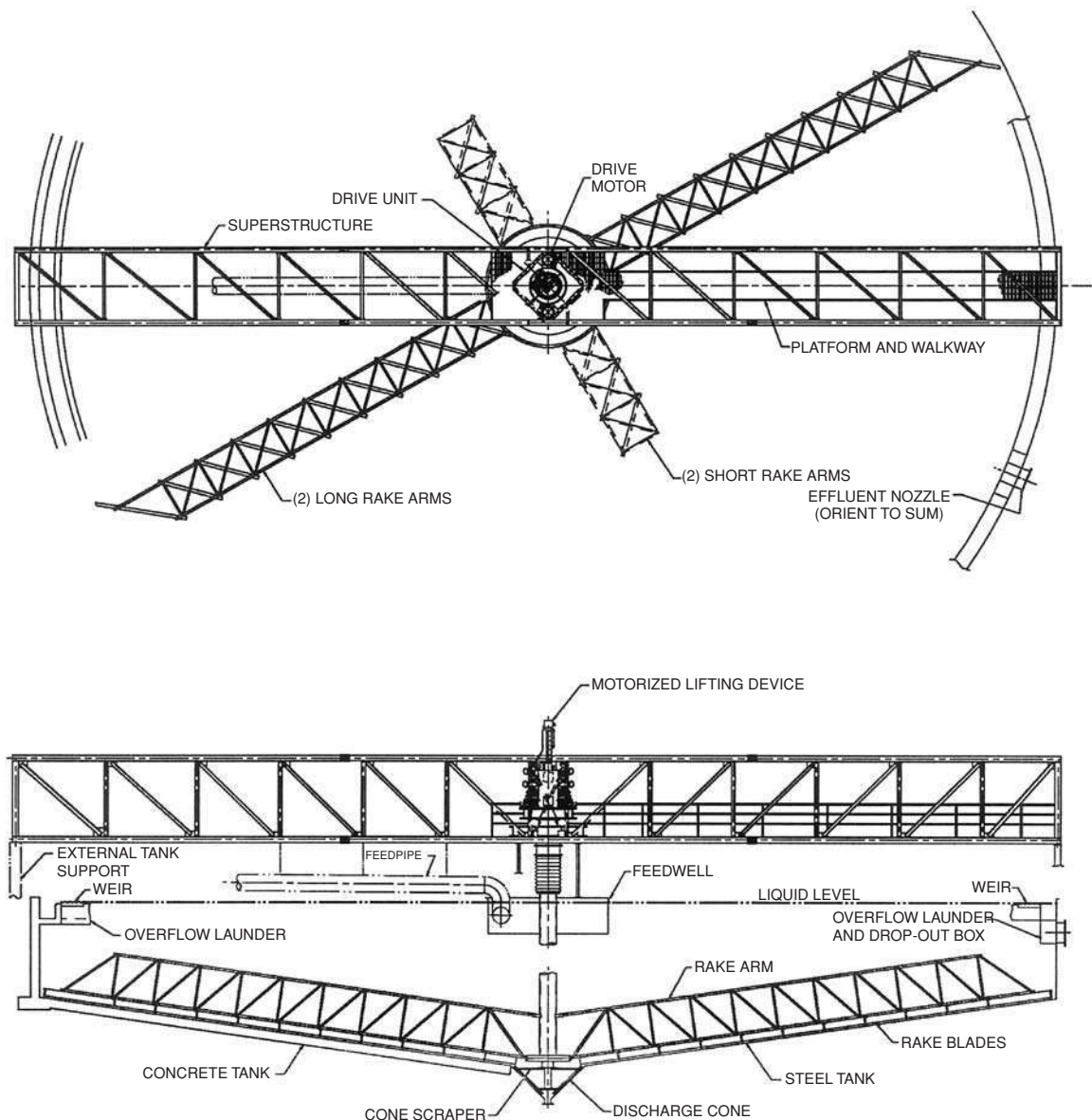


FIG. 18-98 Unit thickener bridge-mounted mechanism. (Dorr-Oliver EIMCO.)

is achieved, solids being withdrawn continuously in the underflow at the rate they are supplied in the feed. Normally, an inventory of pulp is maintained in order to achieve the desired concentration. This volume will vary somewhat as operating conditions change; on occasion, this inventory can be used for storage of solids when feed and underflow rates are reduced or temporarily suspended.

A thickener has several basic components: a tank to contain the slurry, feed piping and a feedwell to allow the feed stream to enter the tank, a rake mechanism to assist in moving the concentrated solids to the withdrawal points, an underflow solids-withdrawal system, and an overflow launder. The basic design of a bridge-supported thickener mechanism is illustrated in Fig. 18-98.

Thickener Types Flocculants are commonly used in thickeners, and this practice has resulted in thickener classifications as *conventional*, *high-rate*, *ultrahigh-rate*, or *high-density*. These designations can be confusing in that they imply sharp distinction between each type, which is *not* the case.

High-Rate Thickeners The greater capacity expected from a high-rate thickener is due solely to the effective use of flocculant to maximize throughput. In most applications there is a threshold dosage and feed solids concentration at which a noticeable increase in capacity begins to occur, as shown in Fig. 18-99. This effect will continue up to a limit, at which point the capacity will be a maximum unless a lower underflow solids concentration is accepted, as illustrated in Fig. 18-95. Since flocculant is usually added to a thickener in either the feed line or the feedwell, there are a number of proprietary feedwell designs which are used in high-rate thickeners to help optimize flocculation. Deaeration systems may be included in some cases to avoid air entrainment in the flocculated slurry. The other components of these units are not materially different from those of a conventional thickener.

Ultrahigh-Rate Thickeners This type of thickener uses a tall, deep tank with a steep bottom cone and may be used with or without a raking mechanism. This combines the functions of a thickener (to

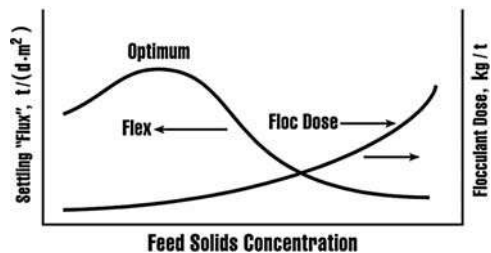


FIG. 18-99 Settling flux curve.

provide a dense underflow) and a clarifier (to provide a clear overflow or supernatant) but is considerably taller. It is generally one-half to one-third the diameter of a conventional or high-rate thickener. Figure 18-100 illustrates the internals of these units, showing the use of dewatering cones, whose function is similar to that of the lamella inclined plates of the tilted-plate thickeners.

High-Density Thickeners Thickeners can be designed to produce underflows having very high apparent viscosity, permitting disposal of waste slurries at a concentration that avoids segregation of fines and coarse particles or formation of a free-liquid pond on the surface of the deposit. This practice is applied in *dry-stacking* systems and underground *paste-fill* operations for disposal of mine tailings and similar materials. The thickener mechanism generally will require a special rake design and provide a torque capability much higher than normal for that particular diameter thickener (Fig. 18-101). Underflow slurries will be at a higher concentration than for conventional or high-rate thickeners, being 5 to 10 percent lower than vacuum filter cake from the same material. Special pumping requirements are necessary if the slurry is to be

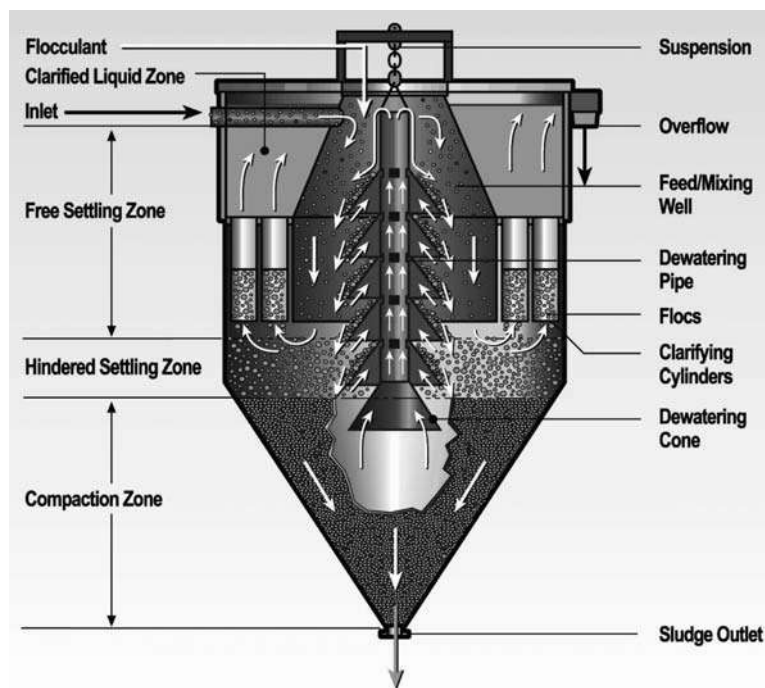


FIG. 18-100 Ultrahigh-rate thickener. (Dorr-Oliver EIMCO.)

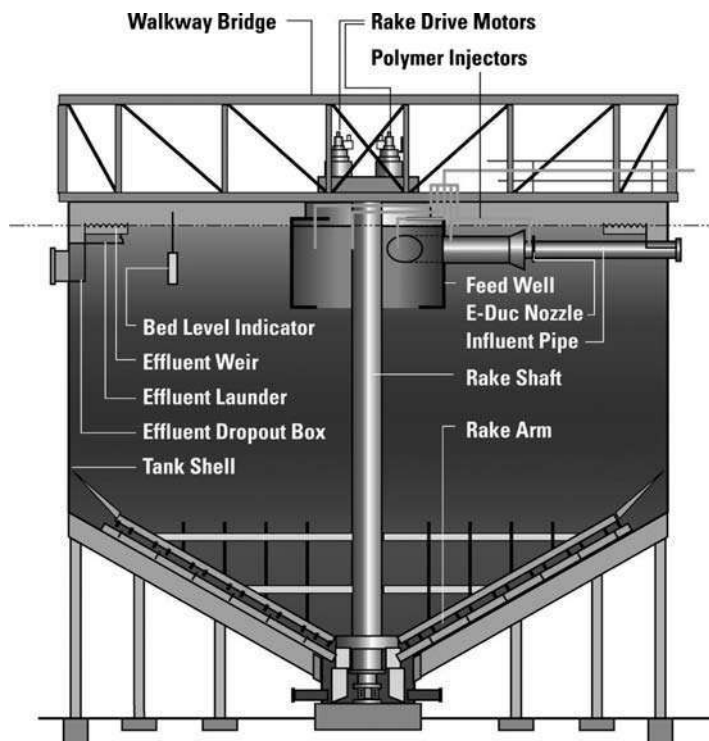


FIG. 18-101 Deep Cone™ paste thickener. (Courtesy of Dorr-Oliver EIMCO.)

transported a significant distance, with line pressure drop typically in the range of 3 to 4 kPa/m of pipeline.

Design Features There are four classes of thickeners, each differentiated by its drive mechanism: (1) bridge-supported, (2) center-column supported, (3) traction drives, and (4) without drives. The diameter of the tank will range from 2 to 150 m (6.5 to 492 ft), and the support structure often is related to the size required. These classes are described in detail in the subsection “Components and Accessories for Sedimentation Units.”

Operation When operated correctly, thickeners require a minimum of attention and, if the feed characteristics do not change radically, can be expected to maintain design performance consistently. In this regard, it is usually desirable to monitor feed and underflow rates and solids concentrations, flocculant dosage rate, and pulp interface level, preferably with dependable instrumentation systems. Process variations are then easily handled by changing the principal operating controls—underflow rate and flocculant dose—to maintain stability.

Starting up a thickener is usually the most difficult part of the operation, and there is more potential for mechanical damage to the mechanism at this stage than at any other time. In general, two conditions require special attention at this point: underflow pumping and mechanism torque. If possible, the underflow pump should be in operation as soon as feed enters the system, recirculating underflow slurry at a reduced rate if the material is relatively fine or advancing it to the next process step (or disposal) if the feed contains a considerable quantity of coarse solids, e.g., more than 20 percent + 75 μm particles. At this stage of the operation, coarse solids separate from the pulp and produce a difficult raking and pumping situation. Torque can rise rapidly if this material accumulates faster than it is removed. If the torque reaches a point where the automatic control system raises the rakes, it is usually preferable to reduce or cut off the feed completely until the torque drops and the rakes are returned to the lowest position. As the fine fraction of the feed slurry begins to thicken and accumulate in the basin, providing both buoyancy and fluidity, torque will drop and

normal feeding can be continued. This applies whether the thickener tank is empty at start-up or filled with liquid. The latter approach contributes to coarse-solids raking problems but at the same time provides conditions more suited to good flocculation, with the result that the thickener will reach stable operation much sooner.

As the solids inventory in the thickener reaches a normal level—usually about 0.5 to 1.0 m below the feedwell outlet—with underflow slurry at the desired concentration, the torque will reach a normal operating range. Special note should be made of the torque reading at this time. Subsequent higher torque levels while operating conditions remain unchanged can almost always be attributed to island formation, and corrective action can be taken early, before serious problems develop. *Island* is the name given to a mass of semisolidified solids that have accumulated on or in front of the rakes, often as a result of excessive flocculant use. This mass usually will continue to grow in size, eventually producing a torque spike that can shut down the thickener and often resulting in lower underflow densities than would otherwise be achievable.

An island is easily detected, usually by the higher-than-normal, gradually increasing torque reading. Probing the rake arms near the thickener center with a rigid rod will confirm this condition—the mass is easily distinguished by its cohesive, claylike consistency. At an early stage, the island is readily removed by raising the rakes until the torque drops to a minimum value. The rakes are then lowered gradually, a few centimeters at a time, so as to shave off the mass of solids and discharge this gelled material through the underflow. This operation can take several hours, and if island formation is a frequent occurrence, the procedure should be carried out on a regular basis, typically once a day, preferably with an automatic system to control the entire operation.

Stable thickener performance can be maintained by carefully monitoring operating conditions, particularly the pulp interface level and the underflow rate and concentration. As process changes occur, the pulp level can vary; regulation of the underflow pumping rate will

keep the level within the desired range. If the underflow varies in concentration, this can be corrected by adjusting the flocculant dosage. Response will not be immediate, of course, and care should be taken to make only small step changes at any one time. Procedures for use of automatic control are described in the section on instrumentation.

CLARIFIERS

Continuous clarifiers generally are employed with dilute suspensions, principally industrial process streams and domestic municipal wastes, and their primary purpose is to produce a relatively clear overflow. They are basically identical to thickeners in design and layout except that they employ a mechanism of lighter construction and a drive head with a lower torque capability. These differences are permitted in clarification applications because the thickened pulp produced is smaller in volume and appreciably lower in suspended solids concentration, owing in part to the large percentage of relatively fine (smaller than $10\ \mu\text{m}$) solids. The installed cost of a clarifier, therefore, is approximately 5 to 10 percent less than that of a thickener of equal tank size, as given in Fig. 18-106.

Rectangular Clarifiers Rectangular clarifiers are employed primarily in municipal water and waste treatment plants, as well as in certain industrial plants, also for waste streams. The raking mechanism employed in many designs consists of a chain-type drag, although suction systems are used for light-duty applications. The drag moves the deposited pulp to a sludge hopper located on one end by means of scrapers fixed to endless chains. During their return to the sludge raking position, the flights may travel near the water level and thus act as skimming devices for removal of surface scum. Rectangular clarifiers are available in widths of 2 to 10 m (6 to 33 ft). The length is generally 3 to 5 times the width. The larger widths have multiple raking mechanisms, each with a separate drive.

This type of clarifier is used in applications such as preliminary oil-water separations in refineries and clarification of waste streams in steel mills. When multiple units are employed, common walls are possible, reducing construction costs and saving on floor space. Overflow clarities, however, generally are not as good as with circular clarifiers, due primarily to reduced overflow weir length for equivalent areas.

Circular Clarifiers Circular units are available in the same three basic types as single-compartment thickeners: bridge, center-column, and peripheral-traction. Because of economic considerations, the bridge-supported type is limited generally to tanks less than 40 m in diameter.

A circular clarifier often is equipped with a surface-skimming device, which includes a rotating skimmer, scum baffle, and scum-box assembly. In sewage and organic-waste applications, squeegees normally are provided for the rake-arm blades, as it is desirable that the bottom be scraped clean to preclude accumulation of organic solids, with resultant septicity and flotation of decomposing material.

Center-drive mechanisms are also installed in square tanks. This mechanism differs from the standard circular mechanism in that a hinged corner blade is provided to sweep the corners which lie outside the path of the main mechanism.

Clarifier-Thickener Clarifiers can serve as thickeners, achieving additional densification in a deep sludge sump adjacent to the center that extends a short distance radially and provides adequate retention time and pulp depth to compact the solids to a high density. Drive mechanisms on this type of clarifier usually must have higher torque capability than would be supplied on a standard clarifier.

Industrial Waste Secondary Clarifiers Many plants which formerly discharged organic wastes to the sewer have turned to using their own treatment facilities in order to reduce municipal treatment plant charges. For organic wastes, the waste-activated sludge process is a preferred approach, using an aeration basin for the bio-oxidation step and a secondary clarifier to produce a clear effluent and to concentrate the biomass for recycling to the basin. To produce an acceptable effluent and achieve sufficient concentration of the low-density solids that make up the biomass, certain design criteria must be followed. Typical design parameters include the following:

Feed pipe velocity: $\leq 1.2\ \text{m/s}$.

Energy-dissipating feed entry velocity (tangential): $\leq 0.5\ \text{m/s}$.

Downward velocity from feedwell: $\leq 0.5\text{--}0.75$ (peak) m/min.

Feedwell depth: Entry port depth $+1\text{--}3\ \text{m}$.

Tank depth: typically 3–5 m.

Radial velocity below feedwell: $\leq 90\%$ of downward velocity.

Overflow rate can range between 0.68 and 2.0 m/h depending on the application. Consult an equipment supplier and manual of practices for recommended overflow rates for specific applications.

Inclined-Plate Clarifiers Lamella or inclined-plate separators have achieved increased use for clarification. They contain a multiplicity of plates inclined at 45 to 60° from the horizontal. Various feed methods are employed so that the influent passes into each inclined channel. The geometry of the plates results in the solids having to settle only a short distance in each channel before sliding down the base to the collection zone beneath the plates. The clarified liquid passes in the opposite direction beneath the ceiling of each channel to the overflow connection.

The area that is theoretically available for separation is equal to the sum of the projected areas of all channels on the horizontal plane. Figure 18-102 shows the horizontally projected area A_s of a single channel in a clarifier of unit width. For a settling length L and width W , inclined at angle α , the horizontally projected area A_s can be calculated as

$$A_s = LW \cos \alpha \quad (18-51)$$

Multiply this area by the total number of plates in the clarifier to calculate the total clarification area available. However, α must be larger than the angle of repose of the sludge so that it will slide down the plate, and the most common range is 55 to 60° . Plate spacing must be large enough to accommodate the opposite flows of liquid and sludge while reducing interference and preventing plugging and to provide enough residence time for the solids to settle to the bottom plate. Usual X values are 50 to $75\ \text{mm}$ (2 to $3\ \text{in}$).

Many different designs are available, the major difference among them being in feed-distribution methods and plate configurations. Operating capacities range from 0.2 to $1.2\ \text{gpm/ft}^2$ projected horizontal area.

The principal advantage of the inclined-plate clarifier is the increased solids recirculation capacity per unit of plane area. Major disadvantages are an underflow solids concentration that generally is lower than in other gravity clarifiers and difficulty of cleaning when scaling or deposition occurs. The lower underflow composition is due primarily to the reduced compression-zone volume relative to the large settling area. When flocculants are employed, flocculating equipment and tankage preceding the separator are required, as the design does not permit internal flocculation.

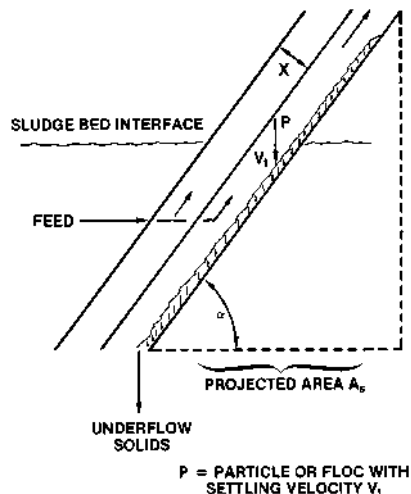


FIG. 18-102 Basic concept of the inclined-plate type of clarifier.

The **ultrahigh-rate** (rakeless) thickener uses internal cones to achieve the tilted-plate effect. The design allows internal flocculation. The tank is tall, with a 60° bottom cone, providing sludge compression height and volume, resulting in a high-density underflow.

Solids-Contact Clarifiers When desirable, mixing, flocculation, and sedimentation all may be accomplished in a single tank. Of the various designs available, those employing mechanically assisted mixing in the reaction zone are the most efficient. They generally permit the highest overflow rate at a minimum chemical dosage while producing the best effluent quality. The unit illustrated in Fig. 18-103a consists of a combination dual drive which has a low-speed rake mechanism and a high-rate low-shear turbine located in the top portion of the center well for internal solids recirculation. The influent, dosed with chemicals, is contacted with previously settled solids in a recirculation draft tube within the reaction well by means of the pumping action of the turbine. Owing to the higher concentration of solids being recirculated, chemical reactions are more rapid, and flocculation is improved. Outside of the feed well the flocculated particles settle to the bottom and are raked to the center to be used again in the recirculation process. When particles are too heavy to be circulated up through the draft tube (as in the case of metallurgical pulps), a modified design (see Fig. 18-103) using external recirculation of a portion of the thickened underflow is chosen. These units employ a special mixing impeller in a feed well with a controlled outlet.

Solids-contact clarifiers are advantageous for clarifying turbid waters or slurries that require coagulation and flocculation for the removal of bacteria, suspended solids, or color. Applications include softening water by lime addition; clarifying industrial-process streams, sewage, and industrial wastewaters; tertiary treatment for removal of phosphates, BOD₅, and turbidity; and silica removal from produce water, cooling tower makeup, and geothermal brines.

COMPONENTS AND ACCESSORIES FOR SEDIMENTATION UNITS

Sedimentation systems consist of a collection of components, each of which can be supplied in a number of variations. The basic components are the same, whether the system is for thickening or clarifying: tank, drive-support structure, drive unit and lifting device, rake structure, feedwell, overflow arrangement, underflow arrangement, instrumentation, and flocculation facilities.

Tanks Tanks or basins are constructed of such materials as steel, concrete, wood, compacted earth, plastic sheeting, and soil cement. The selection of the materials of construction is based on environmental cost, availability, topography, water table, ground conditions, climate, operating temperature, and chemical-corrosion resistance. Typically, industrial tanks up to 30 m (100 ft) in diameter are made of steel. Concrete generally is used in municipal applications and in larger industrial applications. Extremely large units employing earthen basins with impermeable liners have proved to be economical.

Drive-Support Structures There are three basic drive mechanisms. These are (1) the bridge-supported mechanism, (2) the center-column-supported mechanism, and (3) the traction-drive thickener containing a center-column-supported mechanism with the driving arm attached to a motorized carriage at the tank periphery.

Bridge-Supported Thickeners These thickeners (Fig. 18-98) are common in diameters up to 30 m, the maximum being about 45 m (150 ft). They offer the following advantages over a center-column-supported design: (1) ability to transfer loads to the tank periphery; (2) ability to give a denser and more consistent underflow concentration with the single draw-off point; (3) a less complicated lifting device; (4) fewer structural members subject to mud accumulation; (5) access to the drive from both ends of the bridge; and (6) lower cost for units smaller than 30 m in diameter.

Center-Column-Supported Thickeners These thickeners are usually 20 m (65 ft) or more in diameter. The mechanism is supported by a stationary steel or concrete center column, and the raking arms are attached to a driving cage which rotates around the center column.

Traction Thickeners These thickeners are most adaptable to tanks larger than 60 m (200 ft) in diameter. Maintenance generally is

less difficult than with other types of thickeners, which is an advantage in remote locations. The drive may be supported on the concrete wall (the wall would be a structural member) or supported outside the wall on the ground (a standard tank wall could be used). Disadvantages of the traction thickener are that (1) no practical lifting device can be used, (2) operation may be difficult in climates where snow and ice are common, and (3) the driving-torque effort must be transmitted from the tank periphery to the center, where the heaviest raking conditions occur.

The rakeless ultrahigh-rate thickeners use elevated tanks up to 20-m diameter. Advantages are no drive, high throughput rate, and the small footprint. Disadvantages are the height of the elevated tank.

Drive Assemblies The drive assembly is the key component of a sedimentation unit. The drive assembly provides (1) the force to move the rakes through the thickened pulp and to move settled solids to the point of discharge, (2) the support for the mechanism which permits it to rotate, (3) adequate reserve capacity to withstand upsets and temporary overloads, and (4) a reliable control which protects the mechanism from damage when a major overload occurs.

Drives usually have steel or iron main spur gears mounted on bearings, alloy-steel pinions, or a planetary gear. Direct-drive hydraulic systems are also employed. The gearing components preferably are enclosed for maximum service life. The drive typically includes a torque-measuring system with torque indicated on the mechanism and often transmitted to a remote indicator. If the torque becomes excessive, it can automatically activate such safeguards against structural damage as sounding an alarm, raising the rakes, and stopping the drive.

Rake-Lifting Mechanisms These should be provided when abnormal thickener operation is probable. Abnormal thickener operation or excessive torque may result from insufficient underflow pumping, surges in the solids feed rate, excessive amounts of large particles, sloughing of solids accumulated between the rakes and the bottom of the tank or on structural members of the rake mechanism, or miscellaneous obstructions falling into the thickener. The lifting mechanism may be set to raise the rakes automatically when a specific torque level (e.g., 40 percent of design) is encountered, continuing to lift until the torque returns to normal or until the maximum lift height is reached. Generally, corrective action must be taken to eliminate the cause of the upset. Once the torque returns to normal, the rake mechanism is lowered slowly to "plow" gradually through the excess accumulated solids until these are removed from the tank.

Motorized rake-lifting devices typically are designed to allow for a vertical lift of the rake mechanism of up to 90 cm (3 ft).

The cable arm design uses cables attached to a truss above or near the liquid surface to move the rake arms, which are hinged to the drive structure, allowing the rakes to raise when excessive torque is encountered. A major advantage of this design is the relatively small surface area of the raking mechanism, which reduces the solids accumulation and downtime in applications in which scaling or island formation can occur.

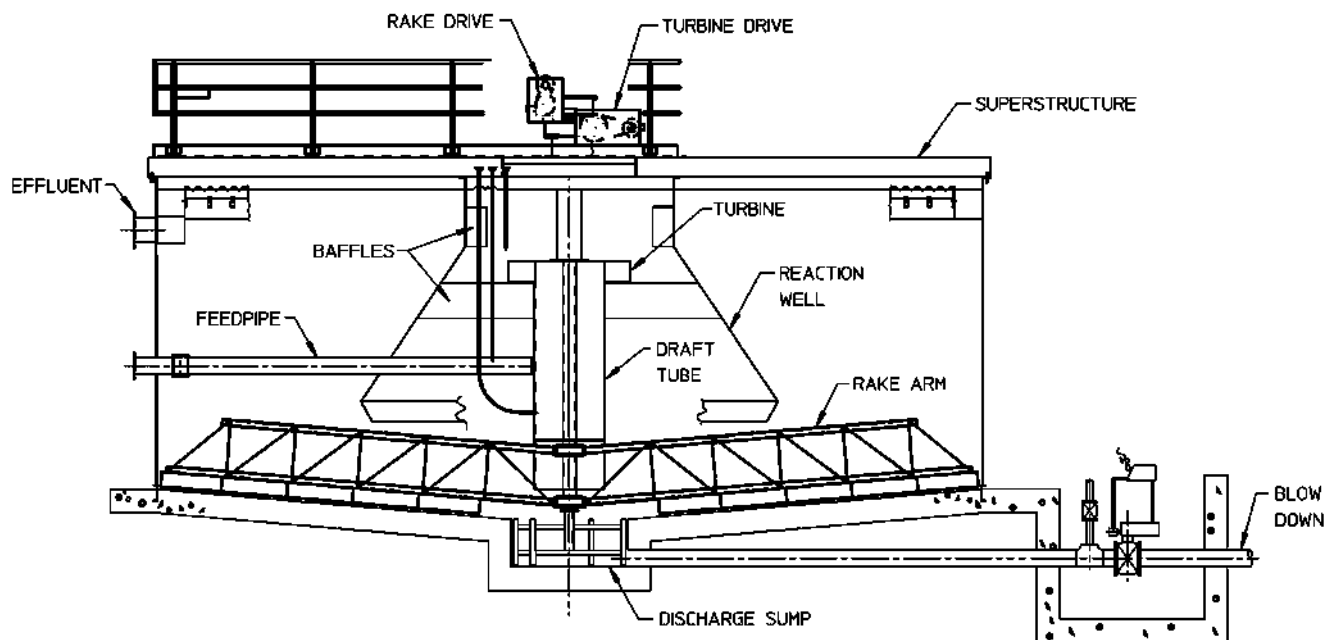
One disadvantage of this or any hinged-arm or other self-lifting design is that there is very little lift at the center, where the overload usually occurs. A further disadvantage is the difficulty of returning the rakes to the lowered position in settlers containing solids that compact firmly.

Rake Mechanism The rake mechanism assists in moving the settled solids to the point of discharge. It also aids in thickening the pulp by disrupting bridged floccules, permitting trapped fluid to escape and allowing the floccules to become more consolidated. Rake mechanisms are designed for specific applications, usually having two long rake arms with an option for two short rake arms for bridge-supported and center-column-supported units. Traction units usually have one long arm, two short arms, and one intermediate arm.

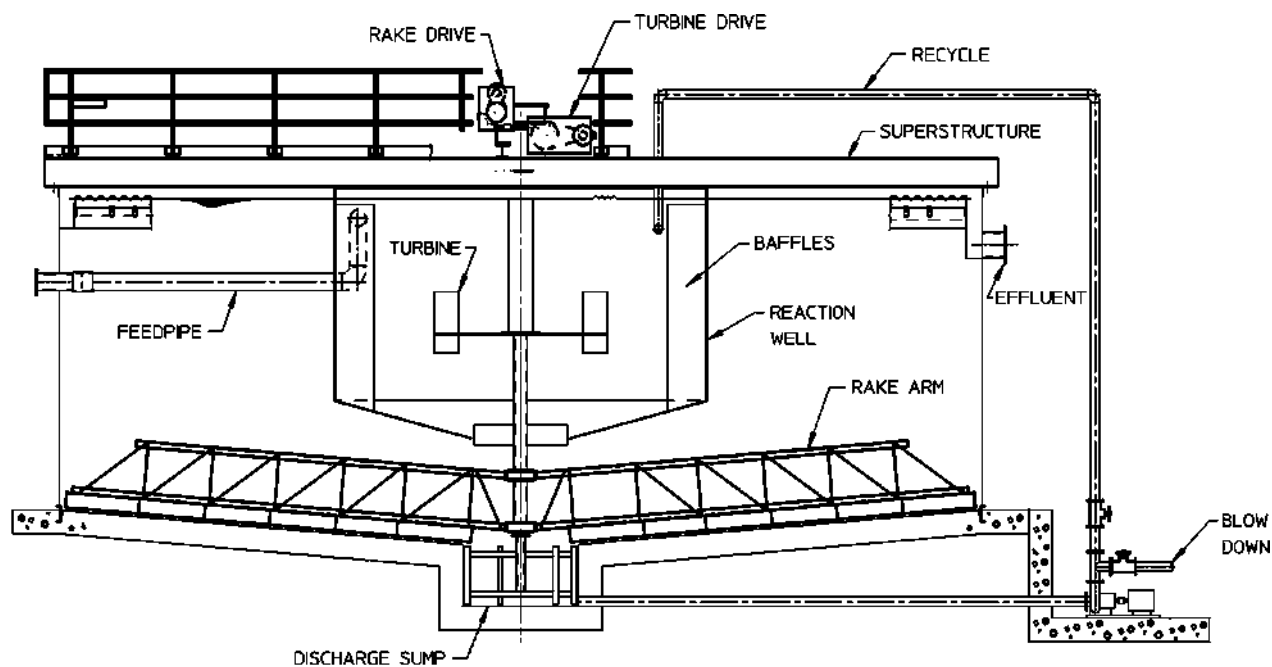
Figure 18-104 illustrates types of rake-arm designs. The conventional design typically is used in bridge-supported units, while the dual-slope design is used for units of larger diameter.

Rake blades can have attached spikes or serrated bottoms to cut into solids that have a tendency to compact. Lifting devices typically are used with these applications.

Rake-speed requirements depend on the type of solids entering the thickener. Peripheral speed ranges used are, for slow-settling solids, 3 to 8 m/min (10 to 25 ft/min); for fast-settling solids, 8 to 12 m/min



(a)



(b)

FIG. 18-103 Solids-contact reactor clarifiers. (Dorr-Oliver EIMCO.)